Philosophy of Artificial Intelligence

Department of Philosophy
Central South University
xieshenlixi@163.com
github

October 28, 2026

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

AI Applications

- spam detection
- play games: AlphaGo / AlphaZero / MuZero / Libratus / Pluribus / DeepStack / AlphaStar / DeepNash
  — Atari, Shogi, Chess, Go, StarCraft, Cards
- recommendation systems
- language model: ChatGPT, Gemini, Claude, Grok, DeepSeek
- code generation
- image/video generation: MidJourney, Sora, Genie
- paintings, poems, music, NotebookLM
- self-driving vehicles
- automatic stock trading
- medical diagnosis, healthcare
- military robots
- theorem proving, conjecture discovery, AlphaGeometry, AlphaProof
- scientific discovery: AlphaFold
- algorithm discovery: AlphaTensor, AlphaEvolve

人工智能 vs 哲学

Dennett: When Philosophers Encounter Artificial Intelligence, 1988

"AI is, in large measure, philosophy."

— *Daniel Dennett*

AI, 很大程度上, 就是哲学.

研究的问题是同源的: 物理实体 — 不管是人、动物、还是机器 — 如何基于感知经验表征世界、提取知识、进而理性地行动? 什么是意识、自由意志、意义、理性决策.....

只是研究方法的侧重不同:

1. 先验证明 (哲学家更喜欢扶手椅 ☺)
   - 成功的先验证明: 丑小鸭定理, 孔多塞投票悖论, 阿罗不可能定理, 荷兰赌论证, 罗素悖论, 图灵停机问题不可判定, 塔斯基算术真不可定义, 哥德尔不完备.....
   - 失败的先验证明: 把 "想象力贫乏" 当 "原则上不可能", 计算机原则上不可能像人一样 XXX.....
2. 工程建模 (AI 逆向工程 "智能"、"意识" 等)
   - 亚里士多德三段论, 图灵机, 可能世界语义, 认知逻辑, 刘易斯信号博弈, 反事实因果.....
   - AI 就是把哲学思想进行 "工程化实现和压力测试". AI 是会顶嘴的 "思想实验".

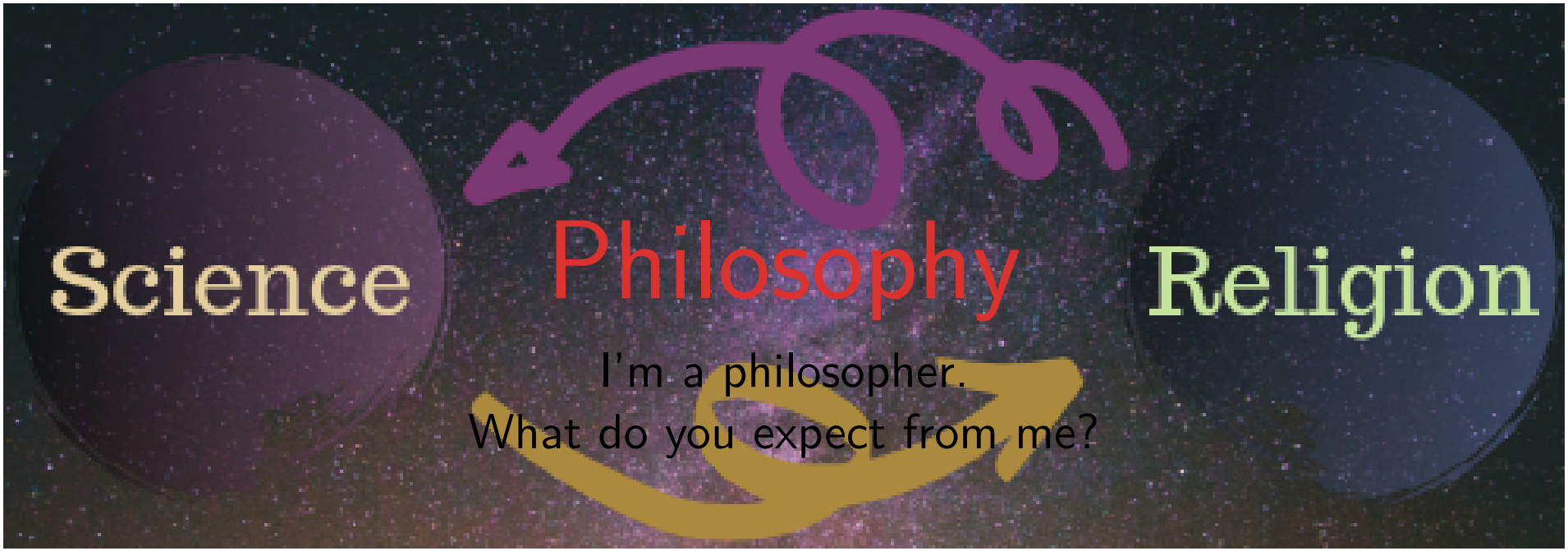
AI is, in large measure, philosophy.

"AI is, in large measure, philosophy."

— *Daniel Dennett*

"It isn't 'AI safety' or 'AI Ethics', it's AI."

— *Stuart Russell*



"在神学与科学之间存在着一片 '无人区', 它遭受着来自双方的攻击, 这片 '无人区' 就是哲学."

— 罗素

AI is, in large measure, philosophy.

什么是哲学? 哲学是神学与科学的中间地带 — 罗素

Good philosophy in my eyes

- Bayes — *How to turn one's 'prior beliefs' into 'posterior beliefs'?*
- Cantor — *What is 'infinity'? What is 'set'?*
- Leibniz — *What are the extent and limits of reason?* — Universal Characteristic & Rational Calculus.
- Hilbert — *How to justify non-constructive reasoning?*
- Gödel — *What is the difference between 'truth' and 'proof'?*
- Tarski — *What is 'truth'? What are 'logical notions'?*
- Turing — *What is 'effective procedure'?*
- Kolmogorov — *What is 'simplicity'/'randomness'?*
- Solomonoff — *What is learnable? How to make induction?*
- Hutter/Schmidhuber — *What is 'intelligence'/'consciousness'?*
- Pearl — *What is 'causation'?*

The point of philosophy is to make things not philosophy

什么是哲学？哲学是神学与科学的中间地带.

— 罗素

- 好的哲学工作 (之一) 是把哲学变成不是哲学的工作.
- 好的科学工作 (之一) 是把哲学变成不是哲学的工作.

$$\frac{\text{哲学} \quad \text{科学}}{\text{广度优先} \quad \text{深度优先}} \ ?$$

没有数学，我们就无法深入理解哲学；
没有哲学，我们就无法深入理解数学；
没有这两者，我们就无法深入理解任何事物.

— 莱布尼茨

带着问题探索

面向人工智能，你能提出哪些“哲学”问题？

带着问题探索

1. 休谟问题在机器学习中有什么体现?
2. 通用归纳模型与因果发现因果推断有什么关系?
3. 相关性与因果有什么关系?
4. 你认为 “实际因果” 的合适定义是什么?
5. 如何刻画 “意图”?
6. 因果与责任有什么关系? 反事实推理可以帮助界定责任吗?
7. 你对 “向下因果” 有什么看法?
8. 请结合案例用中介分析的技巧分析一类可能涉及不公平的社会现象.
9. 如何界定机器的 “道德主体”Moral Agent 地位? 或 Moral Patient 地位?
10. 如何看待用 “目标导向的机制适应性” 刻画 Agency?
11. 如何看待 “压缩即智能” 的观点?
12. 什么是大语言模型的 “幻觉” 问题? 产生原因是什么? 怎么减少 “幻觉”? 可以根除 “幻觉” 吗?
13. 你认为 “有效复杂性” 的内涵是什么? 在人工智能中的作用是什么?
14. 你认为 “随机性” 会在人工智能中扮演什么角色?

带着问题探索

15. 你是如何看待 “涌现” 的? 怎么度量 “因果涌现”? “因果涌现” 独立于观察者吗?
16. “动力因” 闭合是 “生命” 的本质特征吗? 如何看待 “生命” 与 “智能” 的关系?
17. 请结合相关算法, 阐述 “遗忘” 对学习的作用.
18. 意识是智能的必要条件吗, 如何看待二者的关系? (结合某种意识理论讨论)
19. 自由意志是智能的必要条件吗, 如何看待二者的关系?
20. 请简单介绍一种你了解的不确定性推理的方法并阐述其面临的困难.
21. 如何看待 (Gödel, Chaitin, Legg 等人的) 不完备性定理与人工智能的关系?
22. 如何看待图灵-丘奇论题与人工智能的关系?
23. 知识表示的 “本体” 指什么? 如何构建?
24. 对于人工智能来说, “语义” 指什么? 语词如何获得 “意义”? 词向量嵌入技术能否帮助捕捉语义? 大语言模型有 “语义” 吗?
25. “世界模型”World Model 的标准是什么?

带着问题探索

26. 如何看待由生成对抗网络 GAN (或 Stable Diffusion) 生成的艺术作品的美学价值?
27. 你认为符号主义与联结主义会以什么样的方式结合?
28. 如何理解通用人工智能的 “通用性”?
29. 机器可以进行完全的自我升级吗?
30. 如何看待能够对 “策略” 或 “效用函数” 进行自我修改的 Agent?
31. 如何理解智能与目标正交性论点?
32. 如何理解工具性子目标趋同论点?
33. 如何理解 “Wireheading” 问题? 你认为应该怎么解决?
34. 古德哈特定律会导致什么样的人工智能伦理问题? 如何解决?
35. 基于人类反馈的强化学习 RLHF 是处理人工智能对齐问题的合适方法吗? 造成 “谄媚” 咋办?
36. 人工智能对齐问题的解决是否应该贯彻以人为本的原则?
37. 如果 AI 负责资源分配, 你希望它采用什么伦理规范?
38. 如何用 FDT 决策理论处理可扩展监督问题?
39. 如何看待 “Reward is enough” 假设?
40. 存在普适的价值观吗? 你认为人工智能应该追求什么样的价值观?

Readings

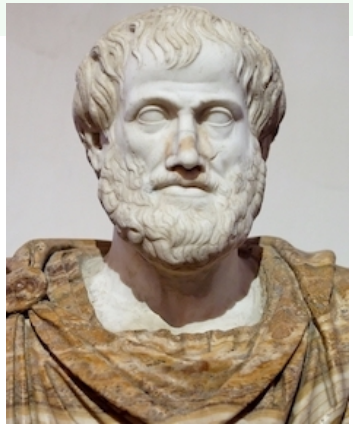
1. 罗素、诺维格: 人工智能 — 一种现代的方法
2. 罗素: AI 新生 — 破解人机共存密码 — 人类最后一个大问题
3. 波斯特洛姆: 超级智能 — 路径、危险性与我们的战略
4. 珀尔、麦肯齐: 为什么 — 关于因果关系的新科学
5. 珀尔: 因果论
6. Li, Vitányi: An Introduction to Kolmogorov Complexity and Its Applications
7. Goodfellow, Bengio, Courville: Deep Learning
8. Sutton, Barto: Reinforcement Learning: An Introduction
9. Shoham, Leyton-Brown: Multiagent Systems — Algorithmic, Game-Theoretic, and Logical Foundations
10. Hutter: An Introduction to Universal Artificial Intelligence

Contents

1. Introduction
   - History of AI
   - What is AI?
   - Turing Machine
   - Rational Agent
   - Goal-Based Agent
   - Knowledge-Based Agent
   - Knowledge Representation
   - Learning Agent
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

The Prehistory of AI

- Aristotle (384-322 BC): Viewed syllogisms as the cognitive basis for rational thought.
- Descartes (1596-1650): Had a very mechanistic view of the brain.
- Leibniz (1646-1716): Characteristica Universalis & Calculus Ratiocinator
- Laplace (1749-1827): A super-intelligence that knows the location and momentum of every particle in the universe at one time, could know the universe for all times.
- Boole (1815-1864): Boolean Algebra.
- Ada Lovelace (1815-1852): "The Analytic Engine has no pretensions to originate anything. It can do whatever we know how to order it to perform."

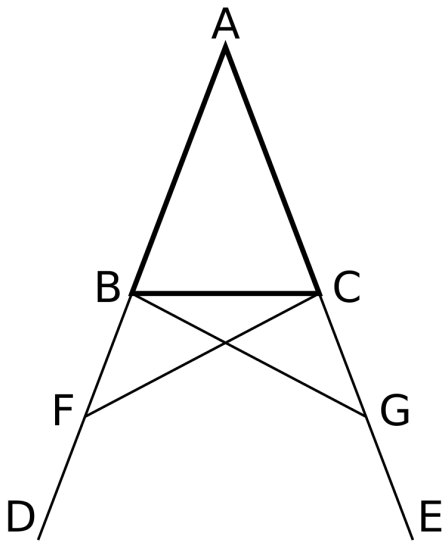


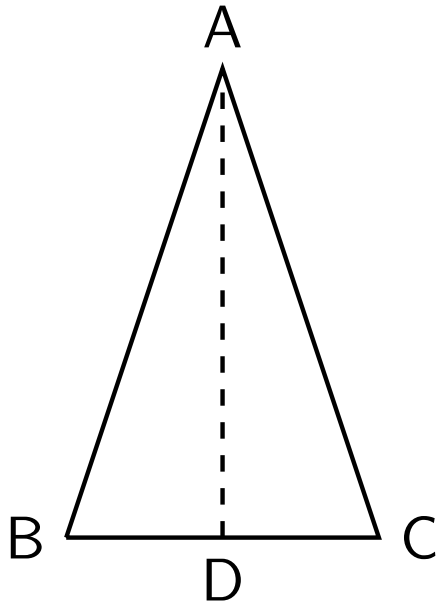
A Brief History of AI

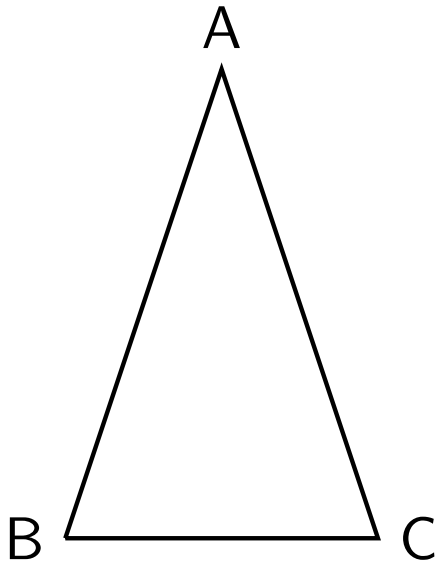
- 1943 McCulloch & Pitts: Artificial Neural Network model of brain
- 1950 Turing's "Computing Machinery and Intelligence"
- 1952-69 Early enthusiasm and great expectations
- 1950s Early AI programs, including Samuel's checkers program, Newell & Simon's Logic Theorist, Gelernter's Geometry Theorem Prover
- 1956 Dartmouth meeting: "Artificial Intelligence" adopted
- 1965 Robinson's complete algorithm for logical reasoning
- 1966-73 A dose of reality: computational complexity, Neural network research almost disappears
- 1969-86 Expert systems
- 1988-93 Expert systems industry busts: "AI Winter"
- 1986– The return of neural networks: backpropagation
- 1988– Resurgence of Probabilistic reasoning; Bayesian network, ALife, GAs ...
- 1995– The emergence of intelligent agents, everywhere ...
- 2012– Human-level AI back on the agenda, big data, deep learning, Generative AI, LLMs

AI 史上的人工智能/障 (Artificial Idiot)

- 等腰三角形的两底角相等.







欧几里得的 “驴桥证明” vs 教科书的证明 vs 计算机的证明 (1955)

$$\frac{\triangle FAC \cong \triangle GAB \quad \triangle FBC \cong \triangle GCB}{\angle B = \angle C} \qquad \frac{\triangle ABD \cong \triangle ACD}{\angle B = \angle C} \qquad \frac{\triangle ABC \cong \triangle ACB}{\angle B = \angle C}$$

- The spirit is willing, but the flesh is weak.
- The vodka is good, but the meat is rotten. (英译俄, 俄译英)

Contents

1. Introduction
   - History of AI
   - What is AI?
   - Turing Machine
   - Rational Agent
   - Goal-Based Agent
   - Knowledge-Based Agent
   - Knowledge Representation
   - Learning Agent
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

What is Artificial Intelligence?[LH07]

What is AI?	像人	理性
思维	认知科学	思维规律
行动	行为主义	正确行事

Question: 你会采用哪种智能定义?

1. Think like a human

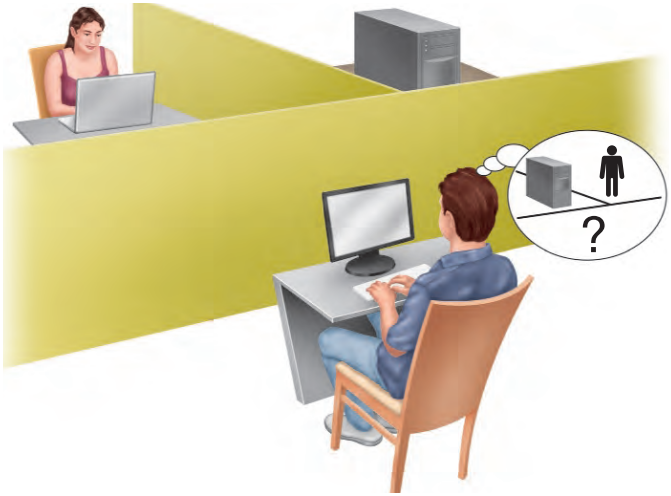
- How do humans think?
- What is thinking, intelligence, consciousness?
- Does the substrate matter, silicon versus meat?
- Computers and brains have completely different architectures
- Is the brain carrying out computation?
- Can we know ourselves well enough to produce generally intelligent computers?
- What cognitive capabilities are necessary to produce intelligent performance?
- **Cognitive science:** Models of the human thinking processes.
- **Advantages:** Models of the human thinking processes./Intelligible.
- **Difficulties:** The best artificial design for an intelligent system need not mirror the human mind.
  地球不是宇宙的中心, 人类智能也未必是认知宇宙的中心.

"If the brain was simple enough to be understood, we would be too simple to understand it!"

— *Marvin Minsky*

2. Act like a human: Turing Test

- Alan M. Turing, "Computing Machinery and Intelligence"
- John R. Searle, "Minds, Brains, and Programs"



如果一个东西看起来像鸭子、走路像鸭子、游泳像鸭子、叫起来像鸭子, 那么它就是鸭子.

- Interrogator in one room, human in another, system in a third.
- Interrogator tries to guess which is which.
- Chinese Room argument.

Needs: natural language, knowledge representation, automated reasoning, machine learning.

Difficulties: Ambiguous./Not constructive./Cannot be formalized mathematically.

If we don't use the Turing test, what measure should we use?

3. Think rationally: Logicist AI

What are the laws of thought? — *Aristotle: Term Logic*

Aristotle's 4 causes: efficient/material/final/formal cause

- **Logic:** Write software to carry out logical inference.
  - rule-based systems
  - automated theorem proving
  - Prolog
- **Advantages:** Precise./Search algorithm.
- **Difficulties:** Represent commonsense knowledge./Computational Cost./Deal with uncertainty.

4. Act rationally: Agent

"Every art and every inquiry, and similarly every action and pursuit, is thought to aim at some good."

— Aristotle: Nicomachean Ethics

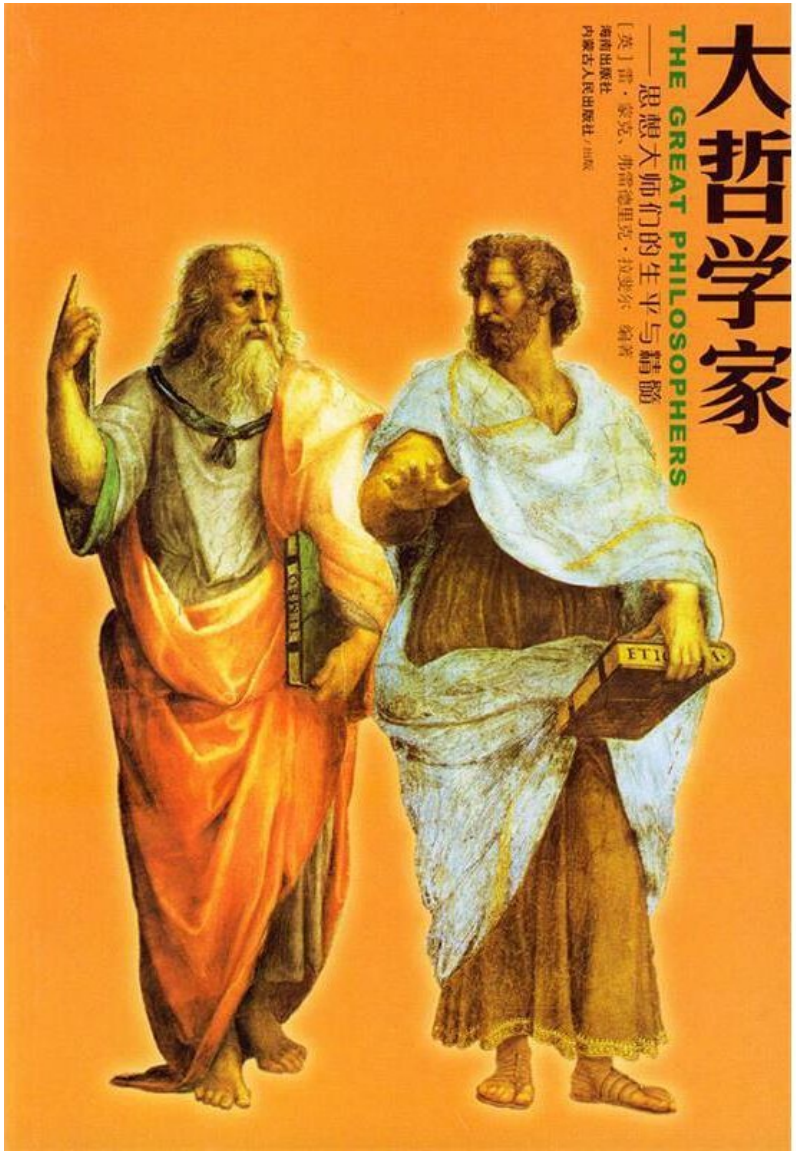
- **Rational agent:** Autonomous system, capable of perceiving and interacting with its environment, of exploration (information gathering), learning and adaptation, of formulating goals and designing plans to reach those goals.
- The agent is rational, in the sense that it acts to achieve the best outcome/goal, conditioned to its knowledge/belief of the world and given computational resources.
- **Probability and decision theory.**
- What to do, for example, when we must make a decision faced with insufficient information?

How to Build Rational Agents?

1. **Program** it!
   - 1.1 Think Rationally: Write rules or logic formulas
   - 1.2 Act Rationally: Define probabilities and costs
2. **Train** it!
   - 2.1 Think Rationally: Learn rules or logic formulas
   - 2.2 Act Rationally: Learn probabilities and costs

Contents

1. Introduction
   - History of AI
   - What is AI?
   - Turing Machine
   - Rational Agent
   - Goal-Based Agent
   - Knowledge-Based Agent
   - Knowledge Representation
   - Learning Agent
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

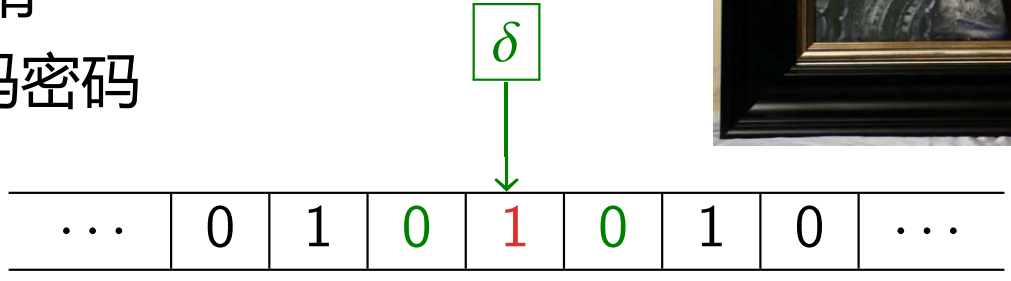


目　　录

导言
苏格拉底：哲学的殉道者
柏拉图：哲学的创始者
笛卡儿：我思故我在
斯宾诺莎：寻求真理与精神幸福
贝克莱：经验论哲学
大卫·休谟：道德科学的牛顿
马克思和自由：发展实践哲学
罗素：毕达哥拉斯之梦
海德格尔：存在与时间的历史和真理
维特根斯坦：论人类本性
波普尔：历史主义及其贫困
阿兰·图灵：一个自然哲学家

图灵 Alan Turing 1912-1954

- 图灵机/通用图灵机
- 丘奇-图灵论题
- 停机问题
- 不可判定性
- 神谕图灵机
- 可计算的绝对正规数
- 图灵测试、学习机
- 形态发生学 — 图灵斑图
- 古德-图灵平滑
- 破译恩尼格玛密码



What is "effective procedure"? — Recursion Theory

Leibniz/Hilbert: $\overset{?}{\vDash} A$ input **Machine** output Y/N

- 什么是机械计算?
- 人是怎么进行计算的?
- 有没有可能建造一台机器, 机械地模拟人脑的计算过程?
  — 什么样的机器?
  — 图灵机
- 计算机器的极限是什么?
  — 停机问题

图灵机可计算 — 一个概念分析的典范

机械可计算 $\Longleftrightarrow$ 图灵机可计算[1]

图灵对 “丘奇-图灵论题” 的论证策略

1. 直接诉诸直观的论证;
2. 证明对 “机械可计算” 的不同定义的等价性;
3. 证明一大类直观上被认为可计算的问题确实是图灵机可计算的.

图灵的第一个论证 — 直接诉诸直观的论证

机械地可计算的 $\Longrightarrow$ 原则上人能计算的

$\Longrightarrow$ 图灵机可计算的

$\Longrightarrow$ 机械地可计算的

[1] Turing: On computable numbers, with an application to the Entscheidungsproblem. 1936.

什么是 “原则上人能计算的”?

- 想象一个理想的计算者, 把她的操作拆解为基本的 “简单操作”.
- 她是怎么进行计算的?
- 在不限量的草稿纸上进行符号书写.

$$
\begin{array}{cccccc}
 & & 4 & 2 & 3 & 1 \\
 & & & \times & 7 & 7 \\
\hline
 & 2 & 9 & 6 & 1 & 7 \\
2 & 9 & 6 & 1 & 7 & 0 \\
\hline
3 & 2 & 5 & 7 & 8 & 7
\end{array}
$$

- 什么是草稿纸? 可以在上面书写符号的空间.
- 不限量的草稿纸可以表示为一条画格子的无穷延伸的纸带.

4	2	3	1	×	7	7	=	2	9	6	1	7	+	2	9	6	1	7	0	=

什么是 “符号”?

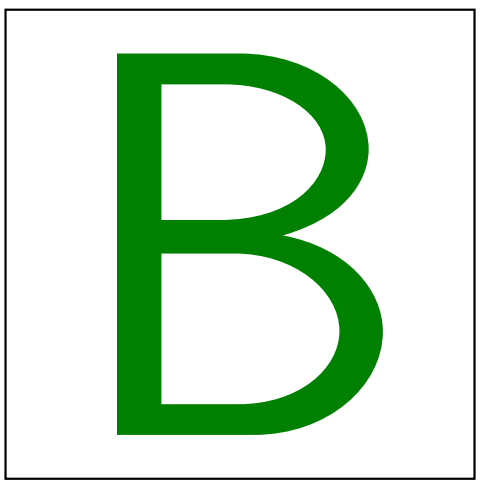
- 什么是符号?
  — 格子的子集.
- 符号的数量是有穷的还是无穷的? — 无穷的.

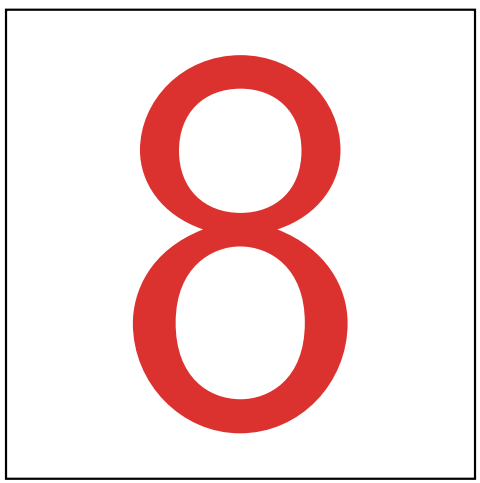
$$2^{2^{\aleph_0}}$$

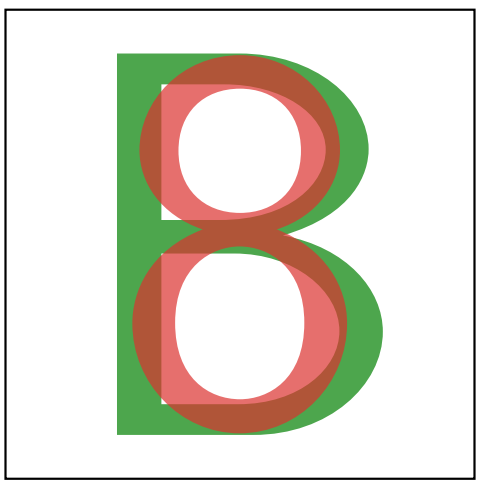
- 什么是可识别的符号?
  — 格子的可区分的子集.
- 人能够识别的符号的数量是有穷的还是无穷的? — 有穷的.
  — “如果允许无穷多的符号的话, 那么符号之间的距离会任意小.”

可识别的符号的数量有多少?

- 计算者可识别的符号的数量是有穷 (等价于两个) 的.
  - 一个符号是 $[0,1] \times [0,1]$ 的一个勒贝格可测的子集.
  - 符号间的距离被定义为两个符号对称差的测度.
  - 由此, 上述符号构成一个紧致的度量空间.
  - 因此不存在两两不交的无穷邻域集.
  - 无论计算者的识别精度有多高, 都只能识别有穷个符号.







注意力的范围有多大?

- 计算者每个时刻能注意到多少个符号?
  — 有穷个.
  — 等价于一个.
- 注意力能从一个格子转移到相隔多远的另一个格子?
  — 相隔有穷个格子.
  — 等价于只转移到左右相邻的格子.

什么是 “心灵状态”?

计算者总是可以中断她的工作，离开并将此事抛在脑后，稍后回来再继续进行．如果她这样做，就必须留下一条 *(*以某种标准形式书写的*)* 指示笔记 *(note of instructions)*，说明如何继续进行．这条笔记就是 “心灵状态” 的对应物．我们假设这个计算者工作得如此断断续续，以至于她每次坐下只能工作至多一步．指示笔记必须使她能够执行一步操作并写下下一条笔记．因此，计算在任何阶段的进展状态，都完全由指示笔记和纸带上的符号决定．

— 图灵

可识别的心灵状态的数量有多少?

- 什么是 “心灵状态 (state of mind)”? — 指示笔记
  — 计算者总是可以暂停计算后再继续进行, 而心灵状态记录计算到达的阶段以说明如何继续.
- “心灵状态” 的数量是有穷的还是无穷的? — 无穷的.
- 人能够识别的心灵状态的数量是有穷的还是无穷的? — 有穷的.[2]
  — “否则, 其中的某些会变得任意接近, 以至于被混淆.”

[2]哥德尔对心灵等价于机器的反驳: 尽管在心灵发展的每个阶段, 状态数量是有穷的, 但没有理由认为, 在发展过程中不会收敛于无穷. 图灵依赖两个预设: (1) 没有独立于物质的心灵. (2) 大脑像计算机一样工作. 哥德尔不相信 (1).

计算拆解为 “简单操作”, 那么, “简单操作” 咋操作?

- 计算者每个时刻的操作由什么决定?
  1. 当前的心灵状态, 以及
  2. 当前注意到的纸带上的符号
- 计算者能进行哪些操作?
  1. 改变纸带上的一个符号[3]
  2. 改变注意的格子
  3. 改变心灵状态
- “简单操作” 咋操作? — “指令”

(当前心灵状态, 当前注意的符号) $\mapsto$ (写入的符号, 注意力移动的方向, 下一心灵状态)

- 可能的指令数有多少?
  — 给定有穷的状态数和符号数, 指令集就是有穷的.
- 什么是 “计算”? — 在不限量的草稿纸上进行的符号书写.

图灵机 = 有穷指令集 + 一条无限长的纸带

[3]不妨假设, 当前注意的格子, 和改变符号的格子是同一个格子.

什么是图灵机?

Definition (确定性图灵机 Deterministic Turing Machine)

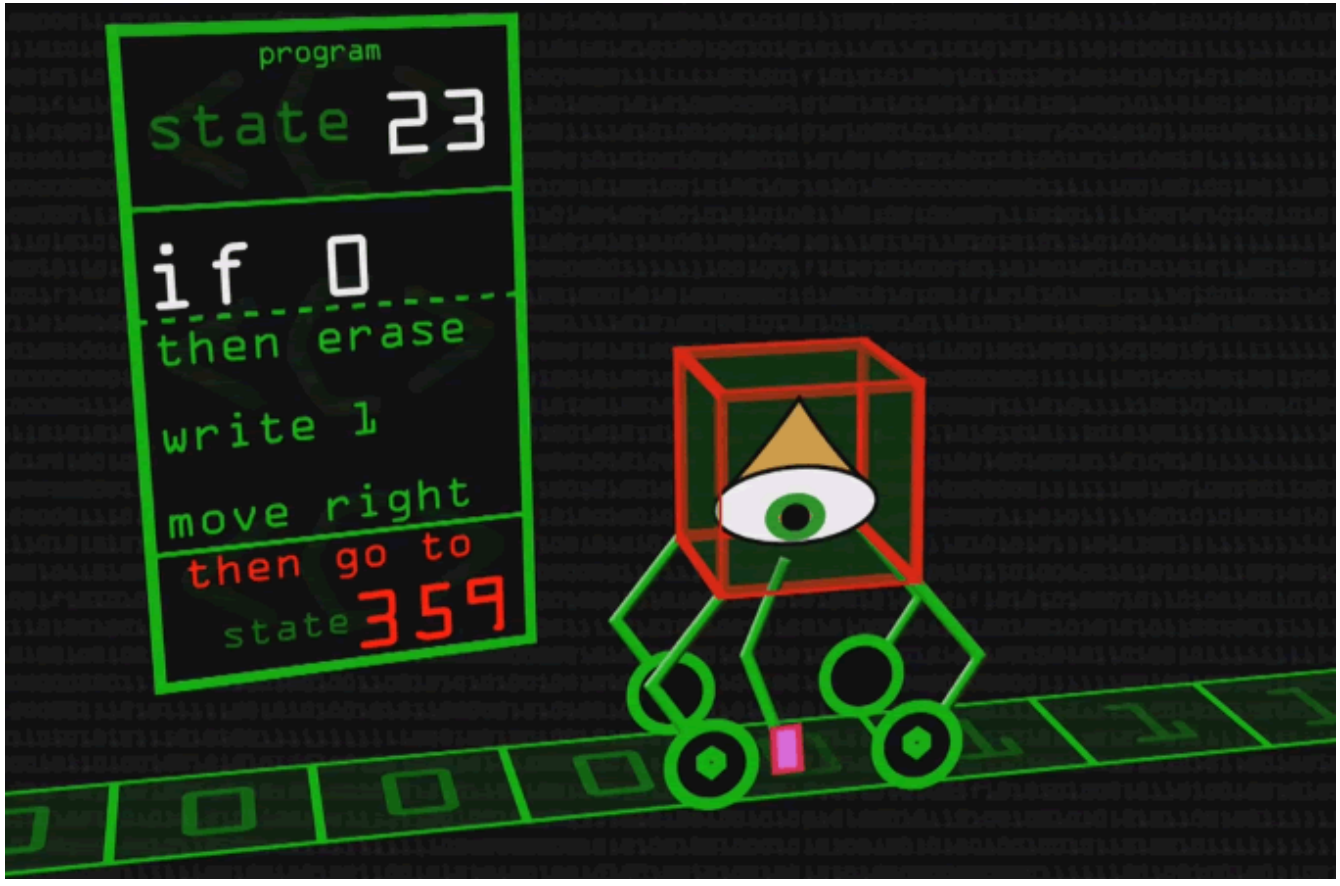
一个确定性图灵机是一个三元组 (Σ, Q, δ), 其中

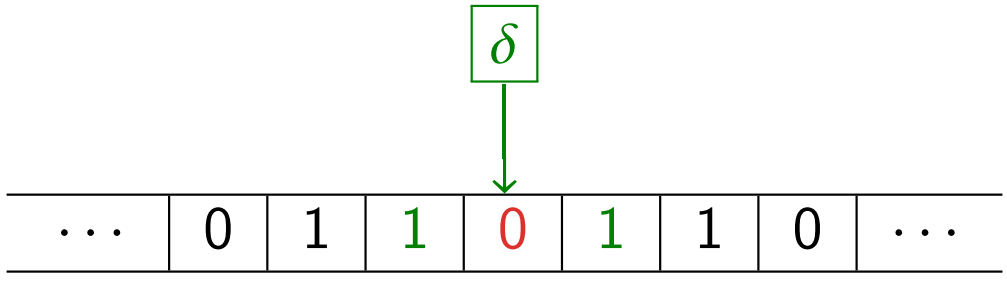
- Σ 是一个有穷的字母表, 包含一个特殊的空白符号.
- Q 是一个有穷的状态集, 包含一个指定的初始状态 q_0 和一个终止状态 q_f.
- δ 是一个确定性转移函数:
$$\delta : Q \times \Sigma \to \Sigma \times \{L, R\} \times Q$$
这里 L 表示向左移动, R 表示向右移动.

Definition (格局 Configuration)

图灵机的格局是一个三元组 (d, h, q), 其中

- d 是纸带上的内容;
- h 是读写头的位置;
- q 是图灵机当前的状态.



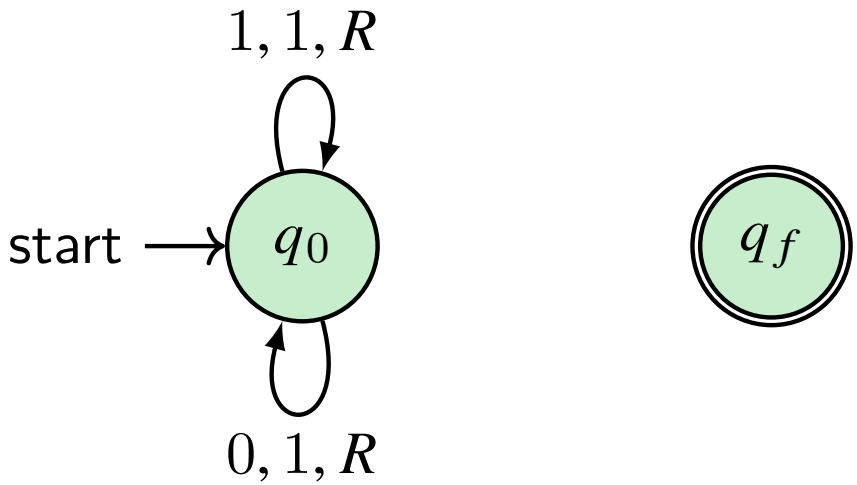


$$\delta(q_{23}, 0) = (1, R, q_{359})$$

$$X_1 X_2 \ldots X_{i-1} q_{23} 0 X_{i+1} \ldots X_n \vdash X_1 X_2 \ldots X_{i-1} 1 q_{359} X_{i+1} \ldots X_n$$

Turing Machine — Example

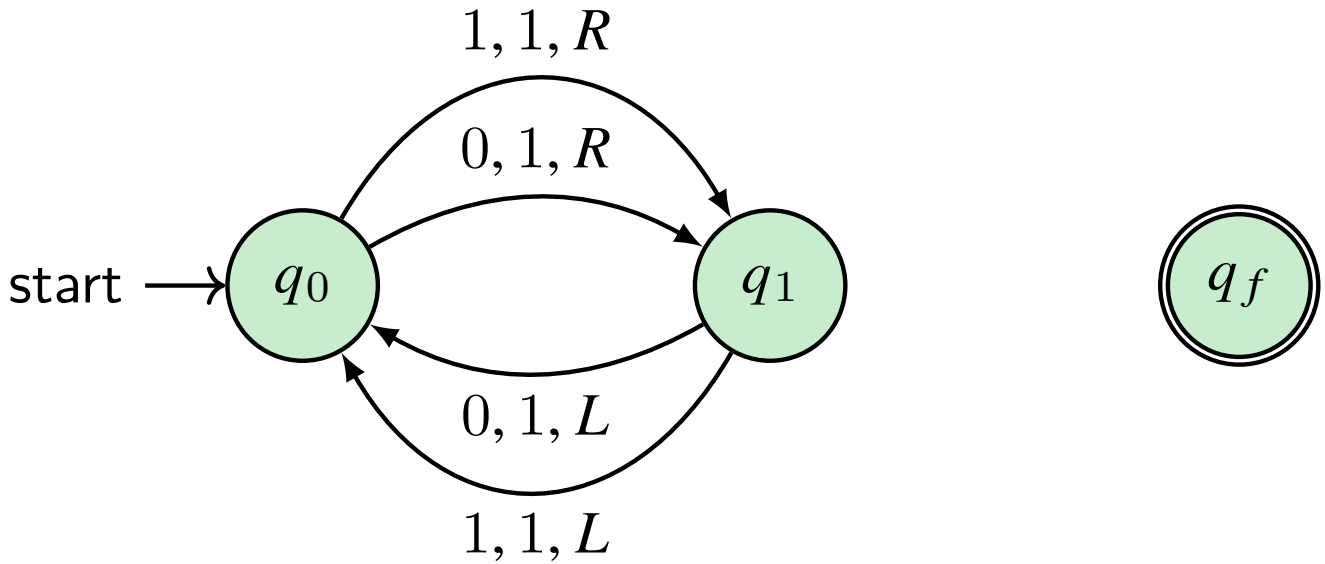
写入 1, 然后一直向右移动. 永不停机.



$$\Sigma = \{0, 1\}$$
$$Q = \{q_0, q_f\}$$
$$\delta(q_0, 0) = (1, R, q_0)$$
$$\delta(q_0, 1) = (1, R, q_0)$$

Turing Machine — Example

0 改为 1, 然后一直左右移动. 永不停机.



$$\Sigma = \{0, 1\}$$

$$Q = \{q_0, q_1, q_f\}$$

$$\delta(q_0, 0) = (1, R, q_1)$$

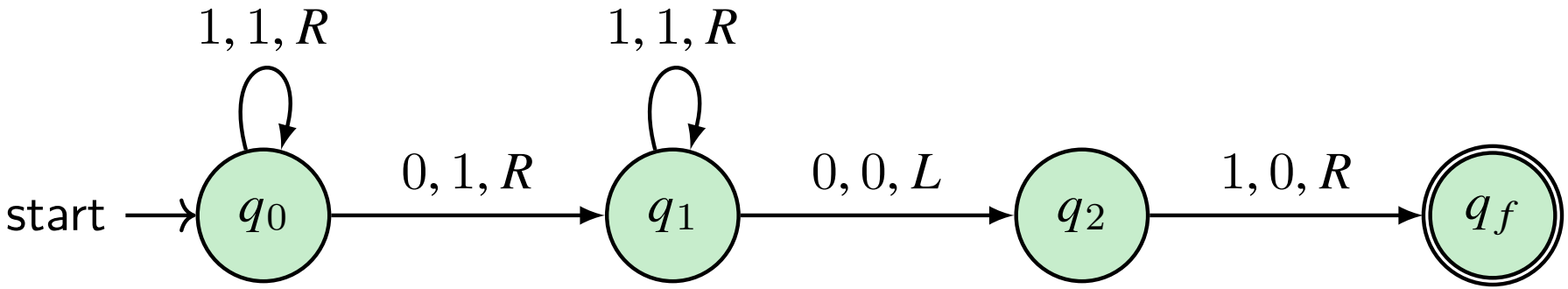
$$\delta(q_1, 0) = (1, L, q_0)$$

$$\delta(q_0, 1) = (1, R, q_1)$$

$$\delta(q_1, 1) = (1, L, q_0)$$

Turing Machine — Example

将两个被 0 隔开的一进制自然数 $(1^m 0 1^n)$ 相加 (1^{m+n}).



$$\Sigma = \{0, 1\}$$

$$Q = \{q_0, q_1, q_2, q_f\}$$

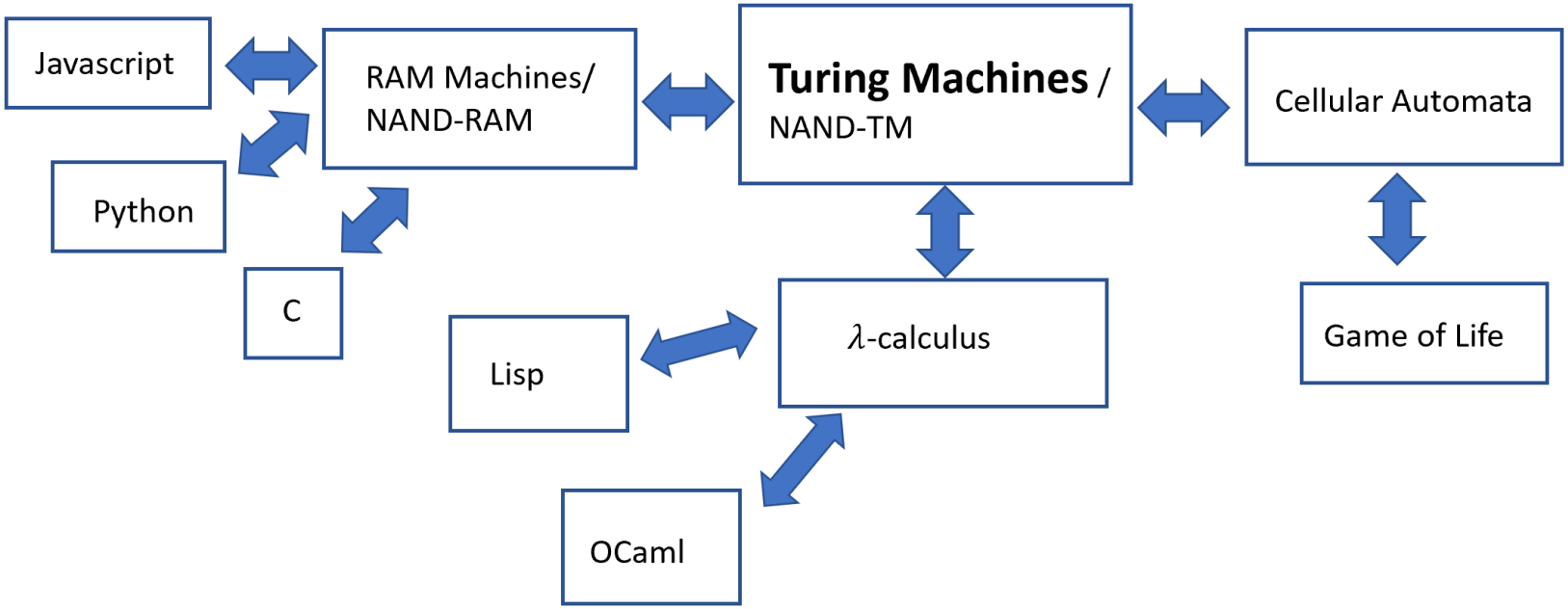
$$\delta(q_0, 1) = (1, R, q_0)$$

$$\delta(q_0, 0) = (1, R, q_1)$$

$$\delta(q_1, 1) = (1, R, q_1)$$

$$\delta(q_1, 0) = (0, L, q_2)$$

$$\delta(q_2, 1) = (0, R, q_f)$$



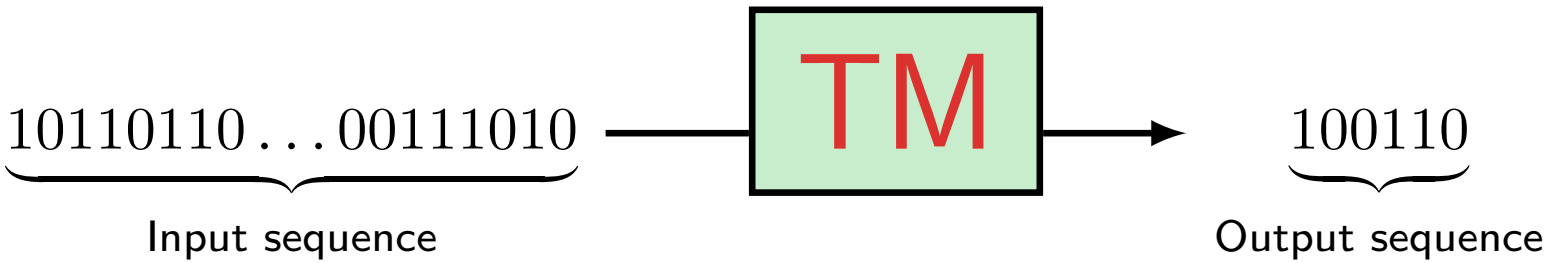
- Every finite function $f : \{0,1\}^n \to \{0,1\}^m$ is computable by a Boolean circuit with $O(m2^n/n)$ gates.
- To compute functions with unbounded inputs $f : \{0,1\}^* \to \{0,1\}^*$, we need a collection of circuits: one for every input length.
- Turing machines capture the notion of a single algorithm that can compute functions of all input lengths.

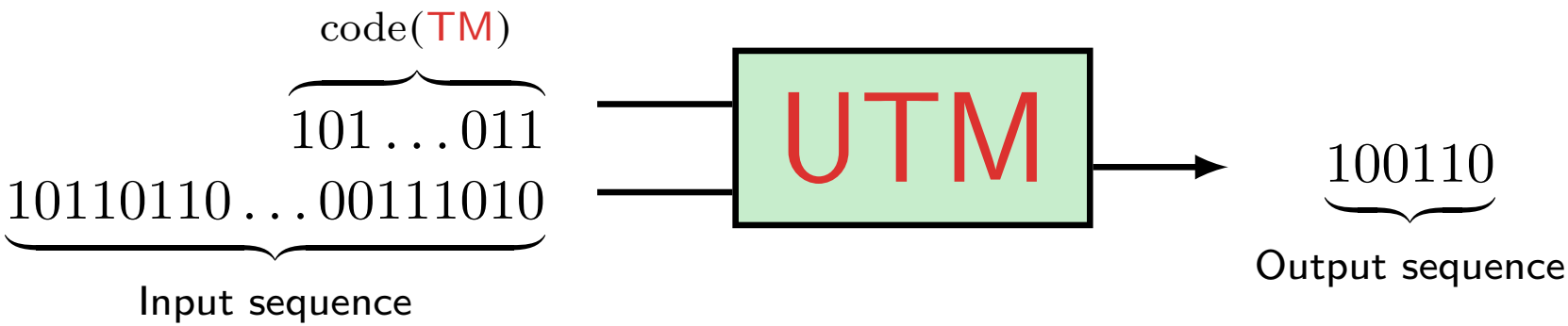
NAND-TM = NAND-CIRC + loops + arrays

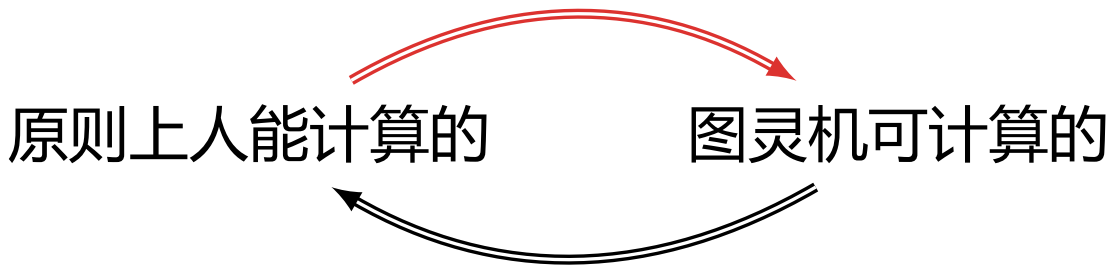
图灵机 = 布尔电路 (逻辑) + 循环 (反馈回路/时钟) + 数组 (存储空间)

FSA

通用图灵机 Universal Turing Machine







你还知道哪些“通用”操作?

- 通用图灵机
- 通用布尔逻辑门/通用量子门
- **SKI** 组合子
- 冯诺依曼自复制自动机里的通用构造器
- 所罗门诺夫通用算法概率
- 傅里叶级数
- 神经网络作为通用函数逼近器
- 矩阵的奇异值分解 SVD
- 范畴论里的泛性质
- ...

Church-Turing Thesis

- 1931-1934, Herbrand-Gödel: "广义递归函数"
- 1933-1935, Church: λ-可定义函数
  — Kleene 1935 证明了 λ-可定义函数与 "广义递归函数" 的等价性, 但哥德尔依然不认为它强到了足以涵盖所有能行可计算函数.

  "I was completely convinced only by Turing's paper."

  — *Kurt Gödel*
- 1936, Turing: 图灵机
- 1936, Post: 波斯特机
- 1956, Chomsky: 0-型文法 type-0 grammar
- 1970, Conway: 生命游戏

"With this concept (Turing Computability) one has for the first time succeeded in giving an absolute ***definition*** *of an interesting epistemological notion, i.e., one not depending on the formalism chosen."*

— Kurt Gödel

Computer science is what mediates between the physical world and the Platonic world....it should be called 'quantitative epistemology'.

— Scott Aaronson

The Thesis as a Definition

- Cauchy-Weierstrass Thesis: a function is intuitively continuous iff

$$\forall x \in I \forall \varepsilon > 0 \exists \delta > 0 \forall y \in I \big(|x - y| < \delta \rightarrow |f(x) - f(y)| < \varepsilon \big)$$

- Church-Turing Thesis:

effective calculable = Turing computable

- "Intelligence Thesis"?
- "Life Thesis"?
- "Consciousness Thesis"?
- "Free Will Thesis"?
- "Beauty Thesis"?
- "Knowledge/Understanding/Meaning/Love Thesis..."?

Remark: AI 时代, 什么能力更重要? 分析问题? 解决问题? 验证答案? 提出问题? 定义概念?

Thesis (Church-Turing Thesis)

effective calculable = *recursive* = *Turing Computable*

||

representable in $\mathbb{Q}$ = λ-*definable*

||

finite definable = *Herbrand-Gödel computable*

||

flowchart (or 'while') computable

||

Neural Network with unbounded tape = *Conway's 'game of life'*

||

Adleman's DNA Computing

||

Post/Markov/McCarthy/Kolmogorov-Uspensky computable ...

- 心灵的功能都可以被图灵机模拟吗?
- 任何离散物理过程都可计算吗?
- 任何构造性函数都可计算吗?

Church-Turing Thesis

- Church-Turing Thesis

  任何能行可计算的函数都是图灵机可计算的.

- Church-Turing-Deutsch Thesis

  任何有穷的物理系统都可以被图灵机模拟到任意的精度.

- Feasibility Thesis — Classical / Quantum Version

  概率 (量子) 图灵机可以高效地模拟任何现实的计算模型.

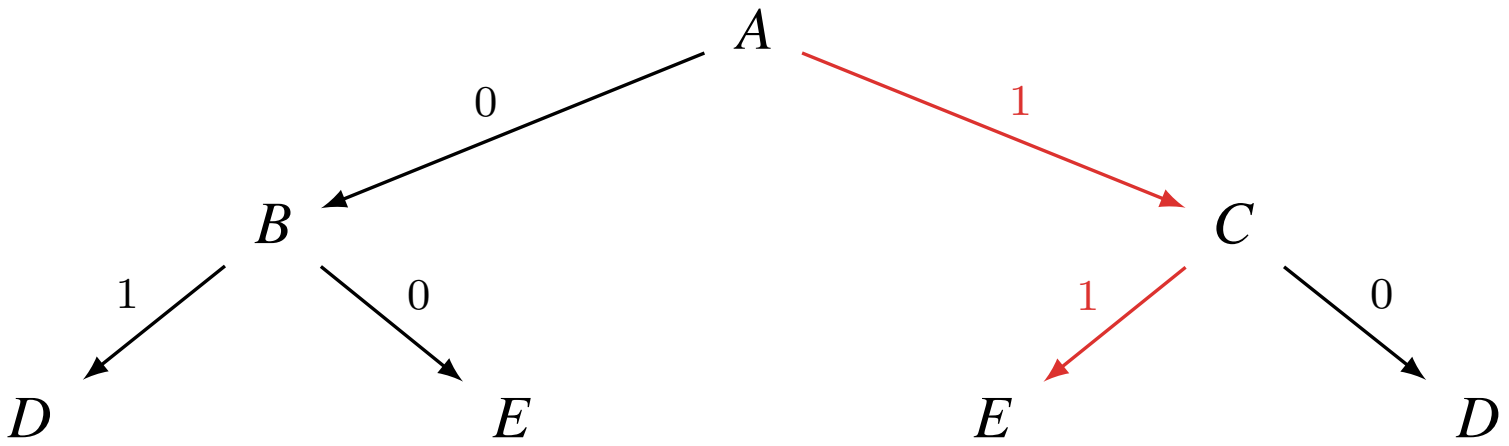
- Wolfram's Principle of Computational Equivalence

  几乎所有不明显简单的过程都可以被视为同等复杂度的 (通用) 计算.

- Wolfram's Principle of Computational Irreducibility

  大多数时候, 了解一个计算系统的结果的唯一方法就是运行它. (没有捷径加速计算)

Deterministic Turing Machine



- Deterministic Turing Machine (DTM):

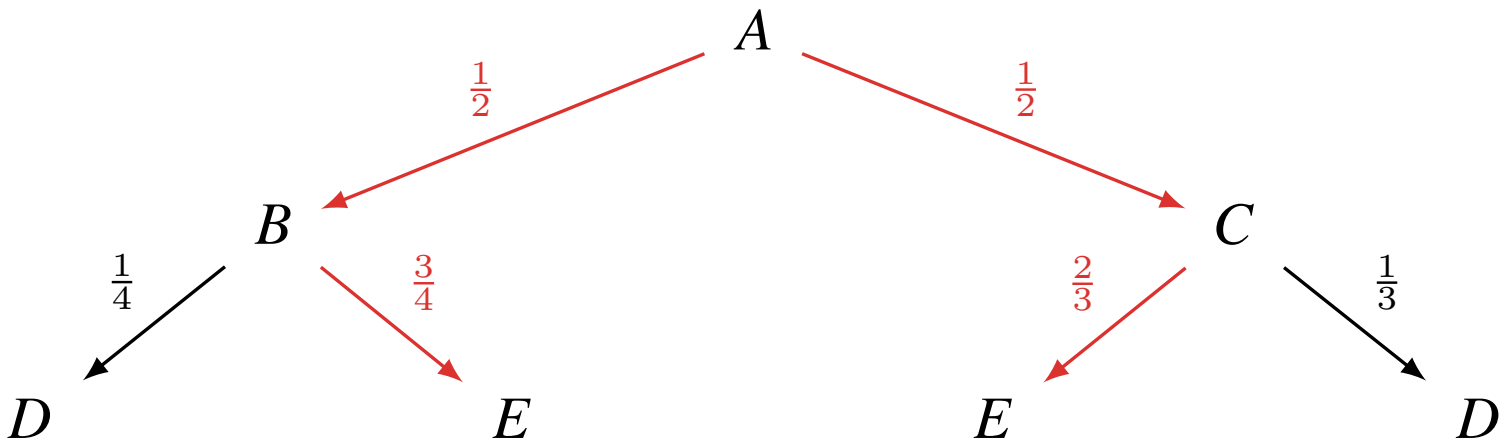
$$\delta : Q \times \Sigma \to \Sigma \times \{L, R\} \times Q$$

- Nondeterministic Turing Machine (NTM):

$$\delta : Q \times \Sigma \to \mathrm{P}(\Sigma \times \{L, R\} \times Q)$$

Remark: DTM 的格局转移构成一条单一的计算路径; 而 NTM 的格局转移构成一棵计算树.

Probabilistic Turing Machine

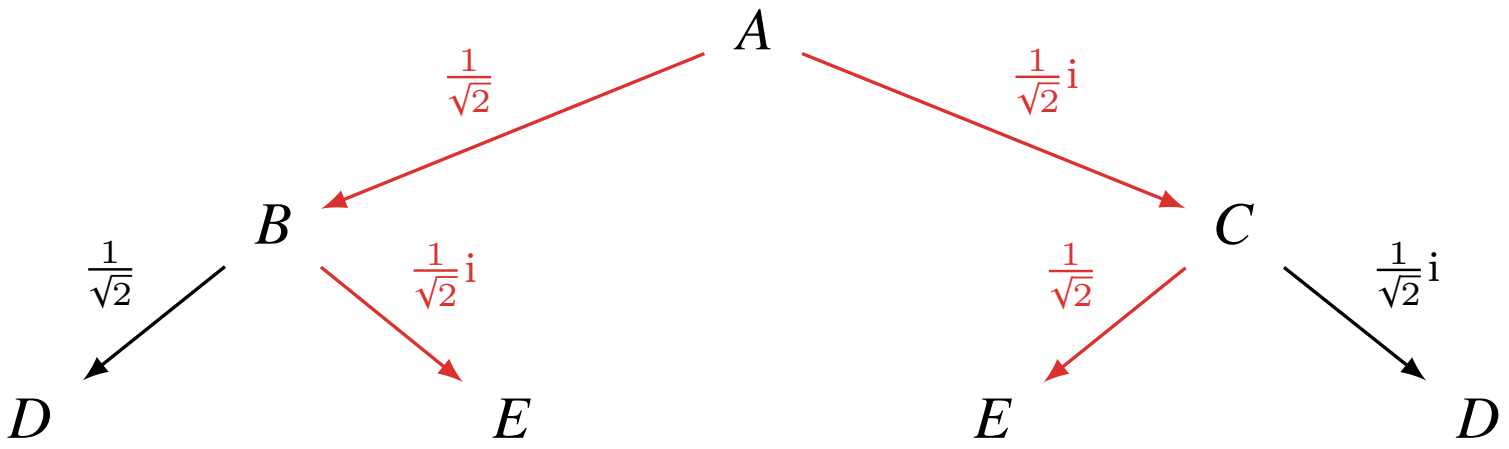


- 在概率图灵机和量子图灵机中, 不同的分支路径可能抵达相同的格局.
- 比如, 在概率图灵机中, 格局 D 和格局 E 发生的概率分别为

$$\frac{1}{2}\cdot\frac{1}{4}+\frac{1}{2}\cdot\frac{1}{3}=\frac{7}{24} \qquad \frac{1}{2}\cdot\frac{3}{4}+\frac{1}{2}\cdot\frac{2}{3}=\frac{17}{24}$$

- 在概率图灵机中, 不同分支的权重只能相加变大, 不能相消变小.

Quantum Turing Machine



- 在量子图灵机中，格局不是以概率，而是以概率幅进行转移.
- 概率幅的模的平方是概率.
- 比如，在量子图灵机中，格局 D 和格局 E 发生的概率分别为

$$\left|\frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}\mathrm{i}\cdot\frac{1}{\sqrt{2}}\mathrm{i}\right|^2=0 \qquad \left|\frac{1}{\sqrt{2}}\cdot\frac{1}{\sqrt{2}}\mathrm{i}+\frac{1}{\sqrt{2}}\mathrm{i}\cdot\frac{1}{\sqrt{2}}\right|^2=1$$

- 量子算法就是巧妙设计转移概率幅，让错误答案的路径发生“相消干涉”，让正确答案的路径发生“相长干涉”，从而在最终观测时，以极高的概率得到正确结果.

宇宙的终极构成是什么？实体与关系哪个更基本?

1. Aristotle: 'What is being?' is just the question 'what is substance?' — Substance is primary in every sense of the word 'primary': (1) in definition, (2) in order of knowledge, and (3) in time.
2. Democritus: Nothing exists except atoms in the void; everything else is opinion.

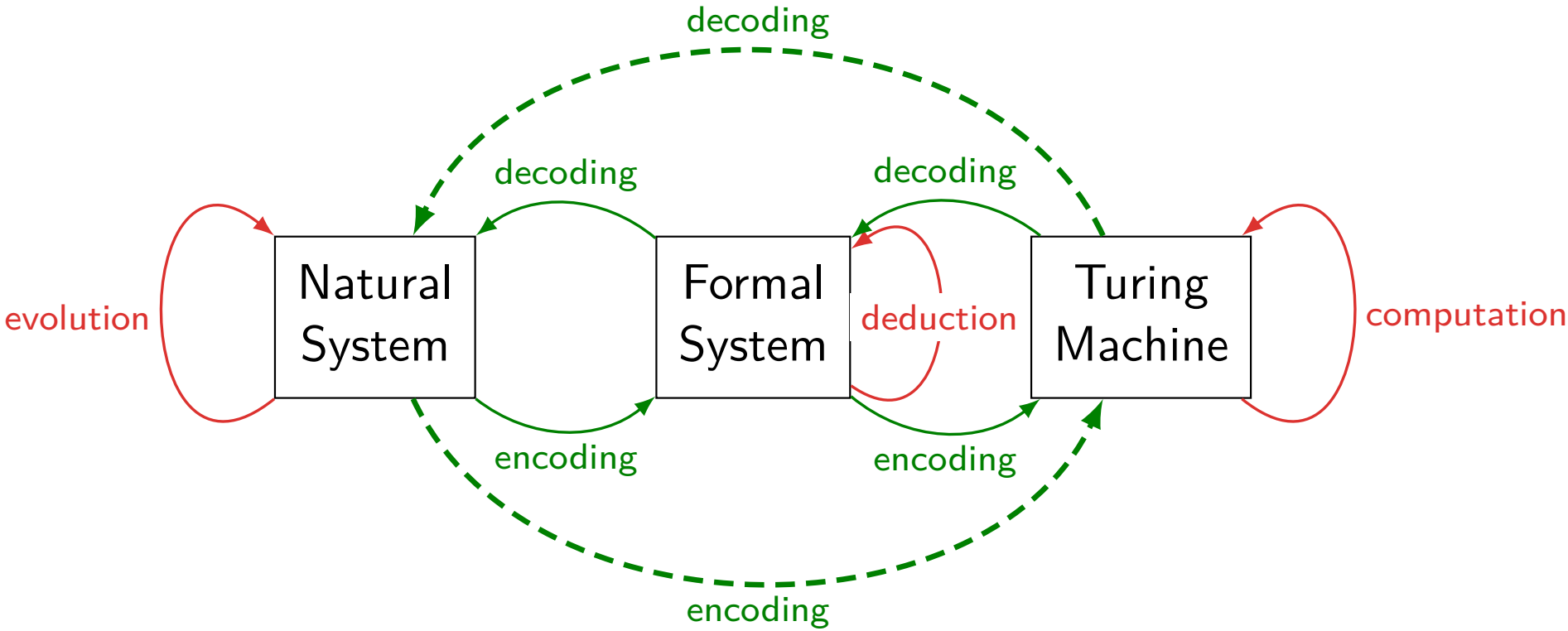
1. Heraclitus: All is flux.
2. Pythagoras: All is number. (God is a mathematician!)
3. Leibniz: All is computation. (God is a programmer!)
4. Wheeler: It from Bit.

Thought is some kind of computation (Computationalism)
Universal Turing Machines can perform all possible computations
Computers are kind of Universal Turing Machines

Therefore, computers can think

Rosen's Modeling Relation & Church-Turing Thesis

Is every natural law simulable?

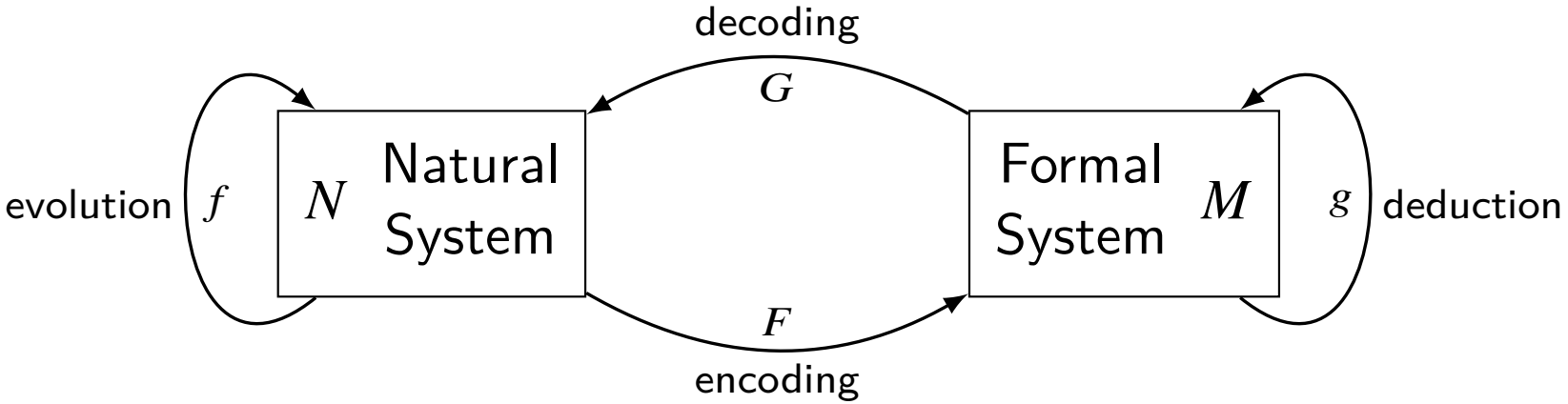


Simulation vs Model

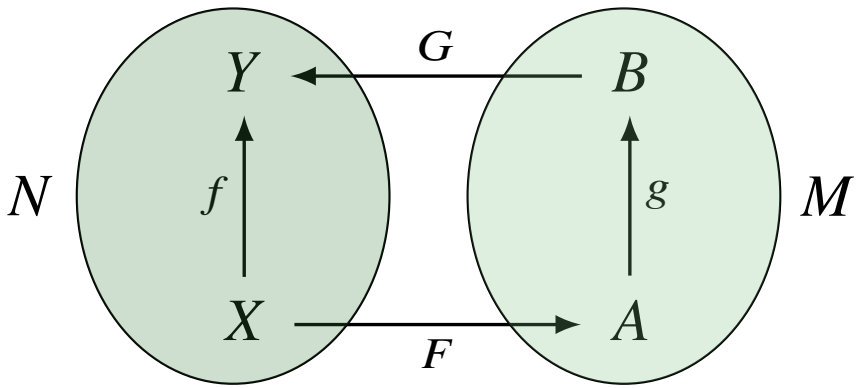
- Simulation describes the trajectories (e.g., curve-fitting)
- Model explains the principle of the dynamics (e.g., Newton)

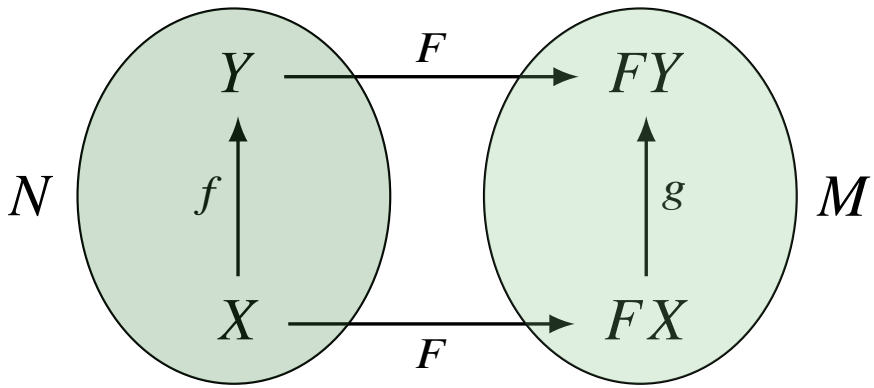
Rosen's "Simulation" / "Metaphor" / "Model"[Lou09]

- **Simulation:** M is a simulation of N iff $f = G \circ g \circ F$

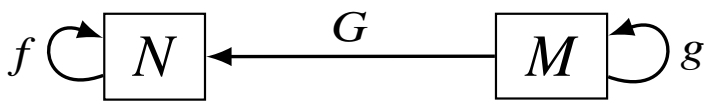


- **Model:** M is a model of N iff M is a simulation of N and $g = F f$



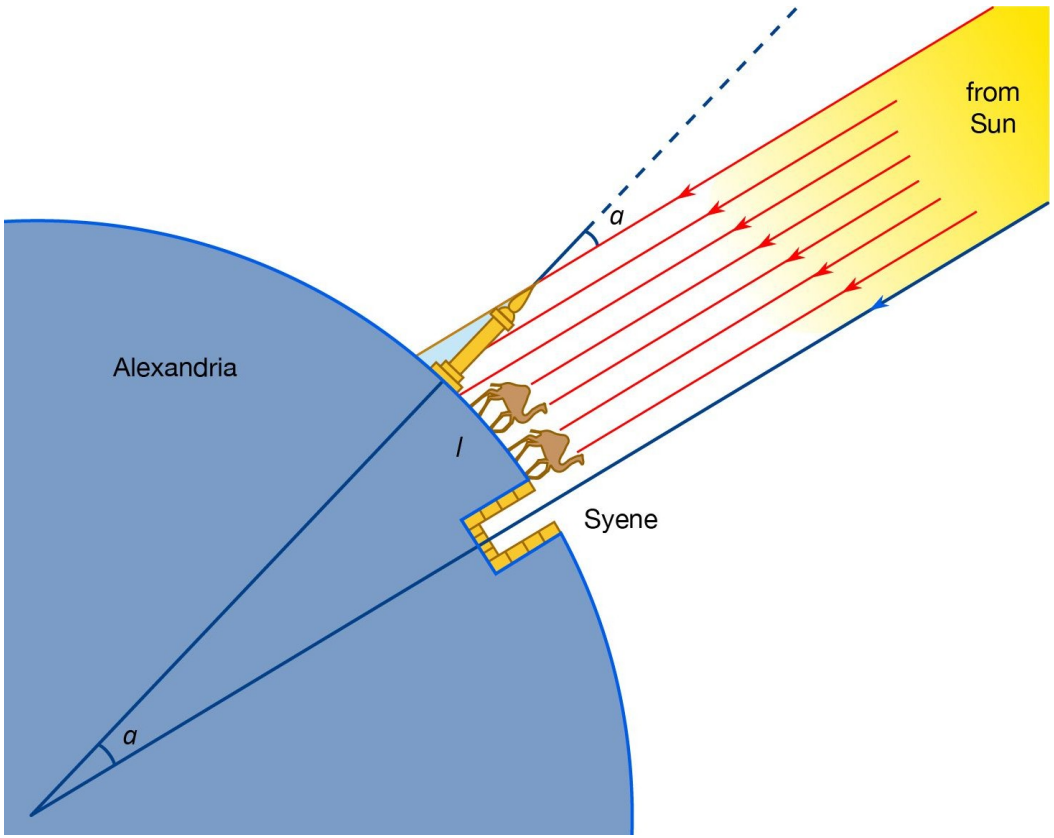


- **Metaphor:** M is a metaphor of N iff there is no encoding arrow



模型 vs 模拟

- 巴比伦的天文学家是黑箱预测的大师，是曲线拟合者，在天体预测的准确性和一致性方面远胜希腊对手.
- 科学却青睐希腊天文学家的推测性的建模策略，它充满了狂野的想象：充满火焰的圆管，天火透过小孔被视作星星，半球形的地球驮在龟背上.
- 这激发了埃拉托色尼，使他测算出了地球的周长.
- 无模型的机器学习，可能会让我们抵达巴比伦，但到不了雅典.

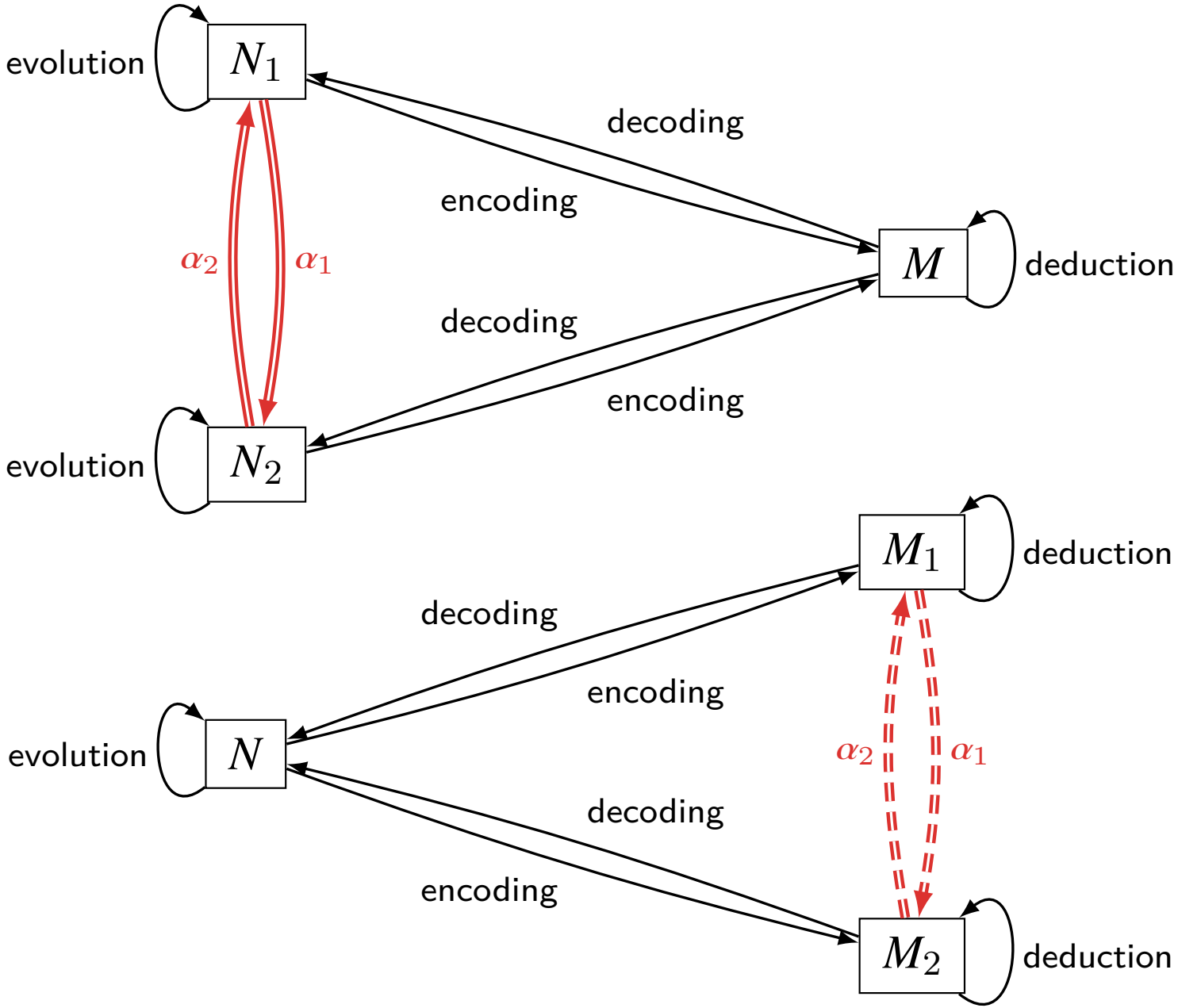


$$C = l\frac{360}{\alpha}$$

珀尔

What is a “World Model”?

Rosen's "Analogy" as Natural Transformation



Church-Turing-Deutsch 论题 vs 世界的 “可理解性”

一沙一世界,
一花一天国,
无限掌中置,
刹那含永劫.
— 布莱克

- 通用图灵机可以模拟任何图灵机.
- 通用图灵机可以模拟整个宇宙.
- 任何图灵完备的装置都包含了宇宙的所有规律.
- “宇宙最不可理解之处是它是可理解的.”
- 描述复杂性、生成复杂性、组织复杂性……
  — 科尔莫哥洛夫复杂度、逻辑深度……

从 “Church-Turing-Deutsch 论题” 到 “黑客帝国”

- **Question:** 你相信你是生活在虚拟宇宙中吗?[4]
- 根据 Church-Turing-Deutsch 论题, 在某个时候, 我们能够开发出模拟我们现实宇宙的虚拟程序.
- 有理由认为, 某个其他文明也许已经做到了这一点.
- 一个虚拟程序被创建, 就可能有无数个这样的副本被创建.

Russell: 现实宇宙是最简单的假设.

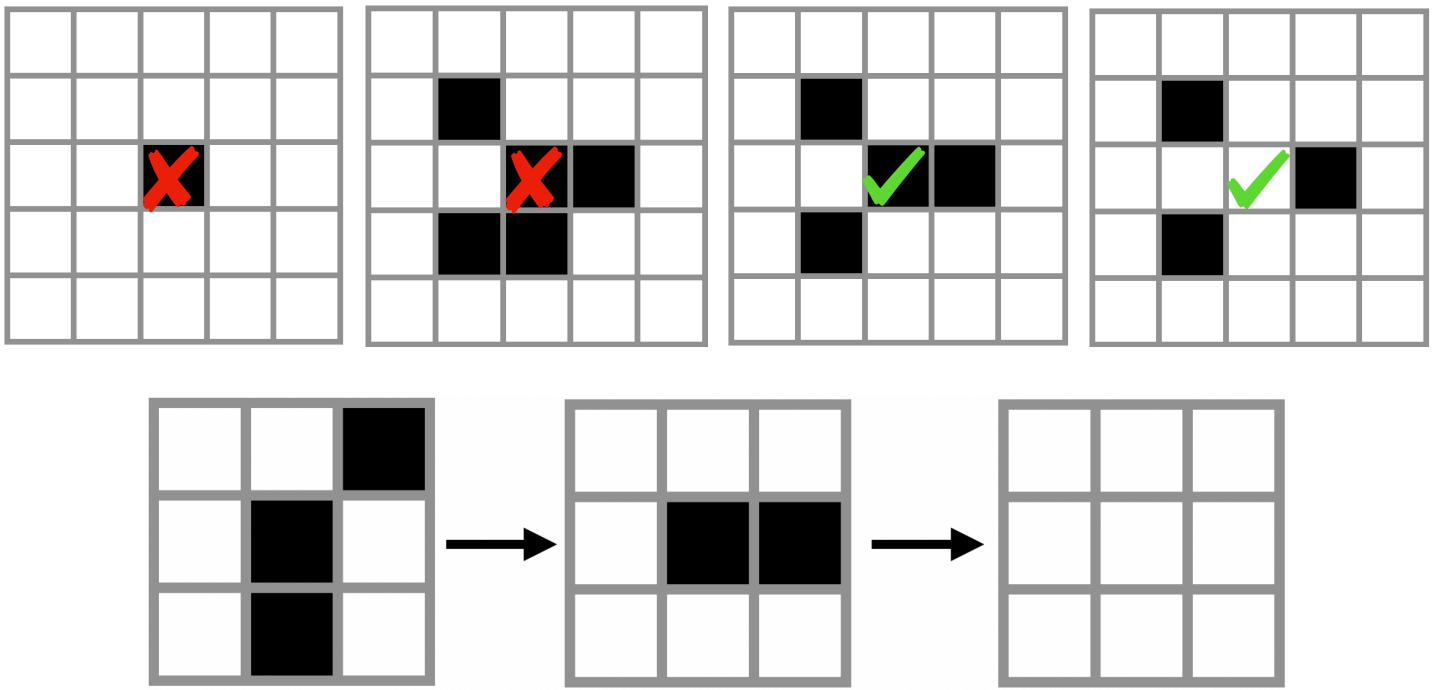
Bostrom: 但虚拟宇宙在数量上多得多.

[4]睡美人问题.

Fermi Paradox — Where are the aliens?

- There are none, i.e. we're all alone.
- We can't detect them because...
  - we're too primitive or too far apart
  - there are predators or all fear them
  - we're lied to, live in a simulation
  - ...

Conway 的生命游戏, 计算不可归约性, 涌现



1. A live cell with < 2 neighbors dies of isolation.
2. A live cell with > 3 neighbors dies of overcrowding.
3. A live cell with 2 or 3 neighbors survives.
4. A dead cell with 3 neighbors will come to life.

"只要给我足够大的模拟空间, 等待足够长的时间, 生命游戏中可能演化出任意你能想到的复杂对象, 包括可以自我繁殖的细胞, 以及能够撰写 *Ph.D* 论文的智慧生命!" — 康威

“计算不可归约性”能解释你对“自由意志”的信念吗?

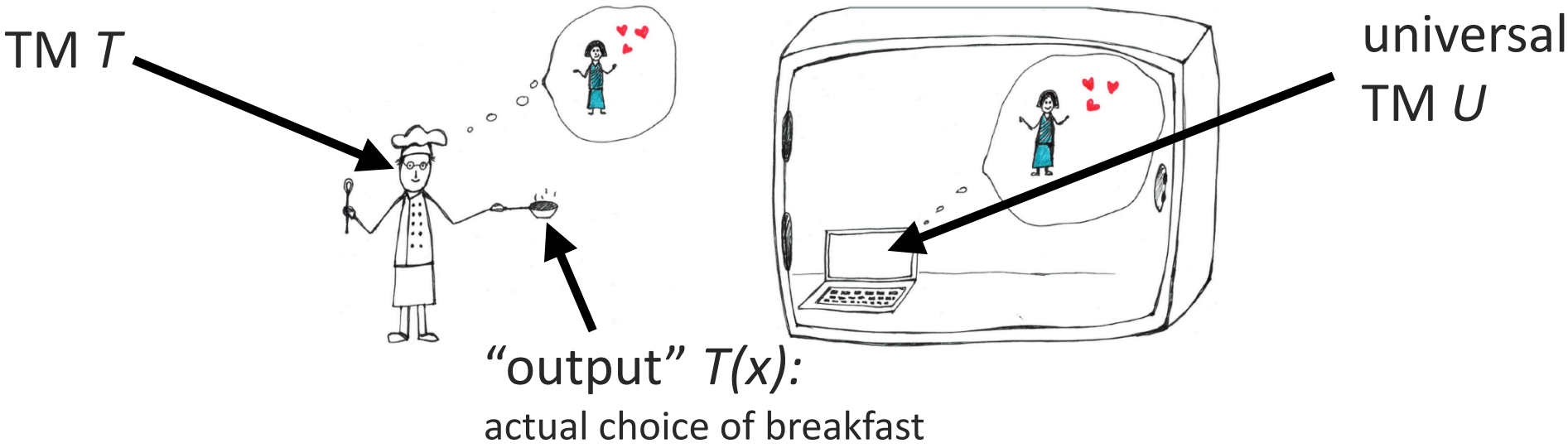
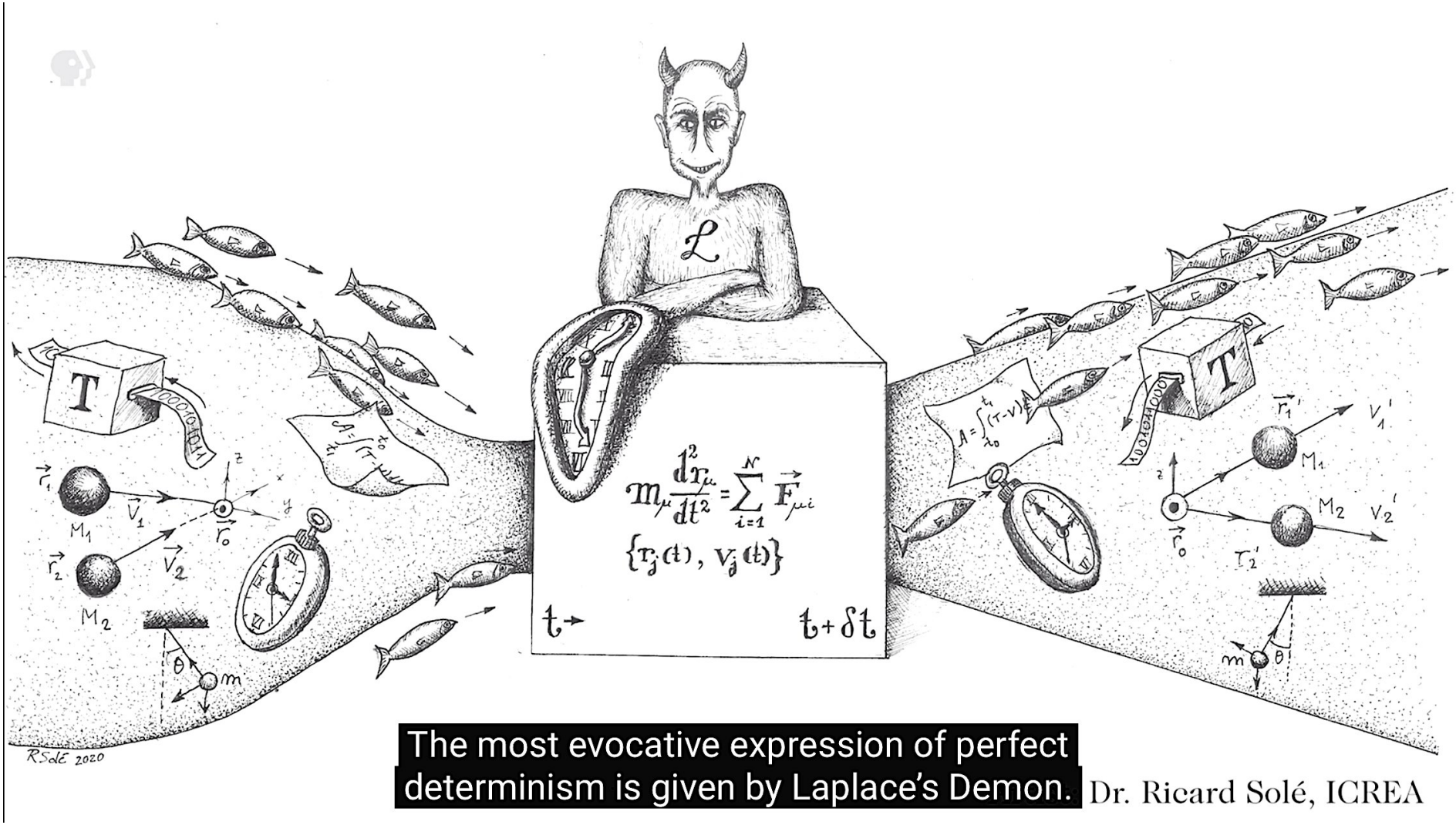


Figure: 虽然由于计算不可归约, 你的行为可能无法被提前预测, 但如果保险箱中的计算机总能准确重现你的选择, 你还会相信自己有自由意志吗?

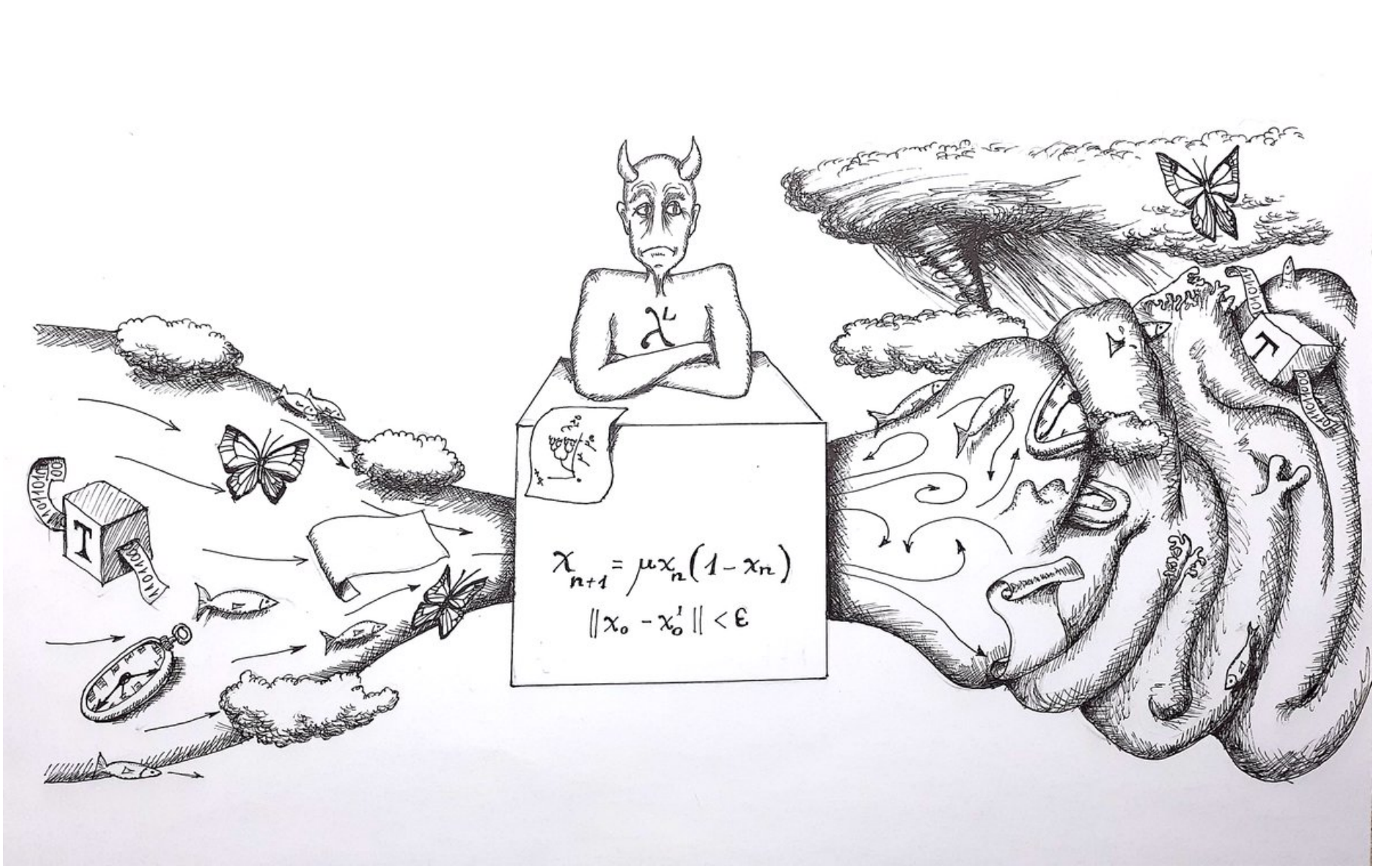
- Libet 实验: 在被试做出某个自发的简单动作 (比如动手指) 之前 550 毫秒, 大脑中已检测到准备电位. 被试意识到“想要动手指的意图”则是在“动手指”的动作发生前 200 毫秒. 这意味着, 无意识的准备电位比有意识的意图早了大约 350 毫秒.
- 当智能机器比你更了解你时, 你还会相信自己有自由意志吗?

拉普拉斯妖：预测精度与测量准确度成正比?

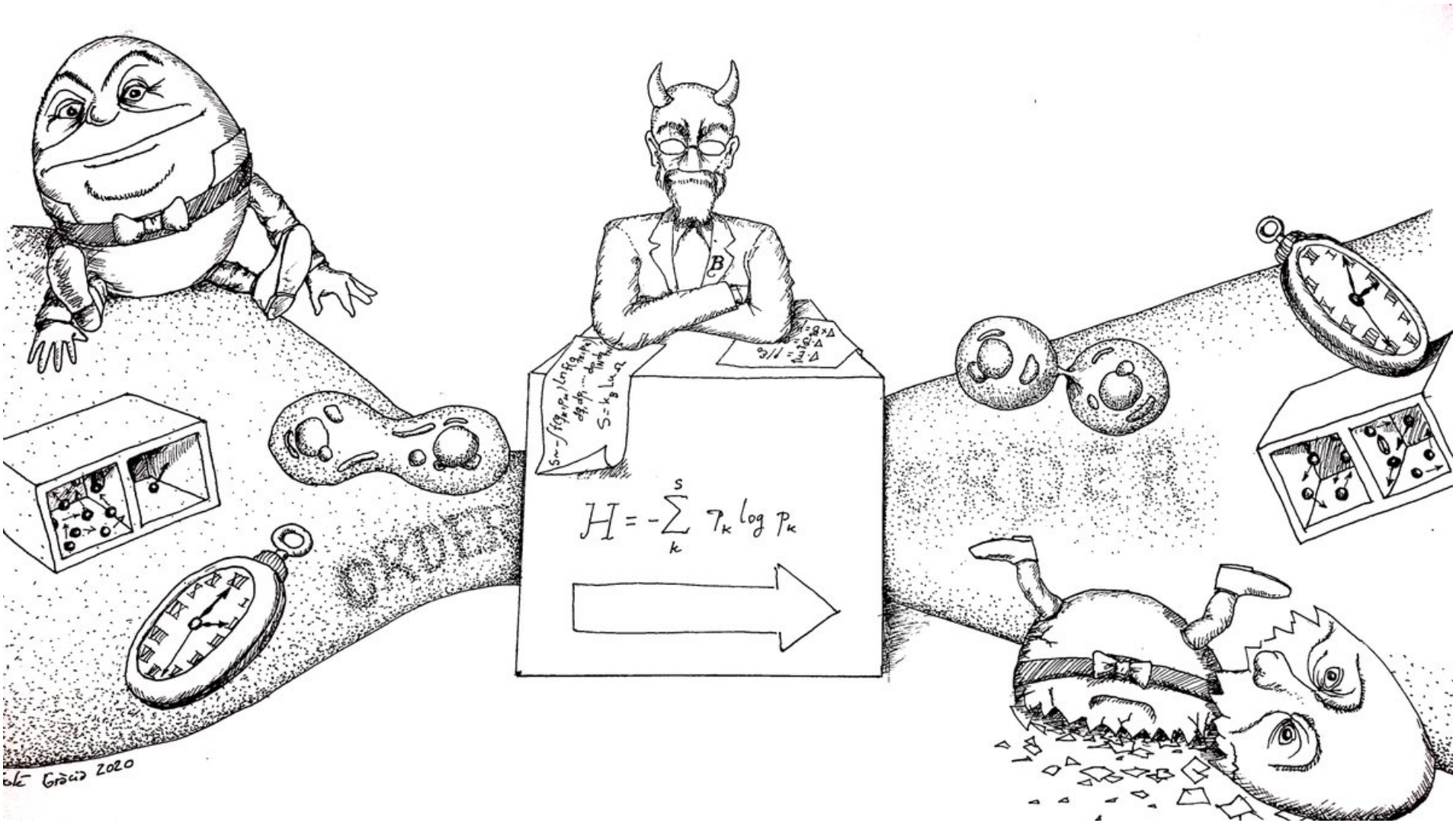


The most evocative expression of perfect determinism is given by Laplace's Demon. Dr. Ricard Solé, ICREA

确定性混沌

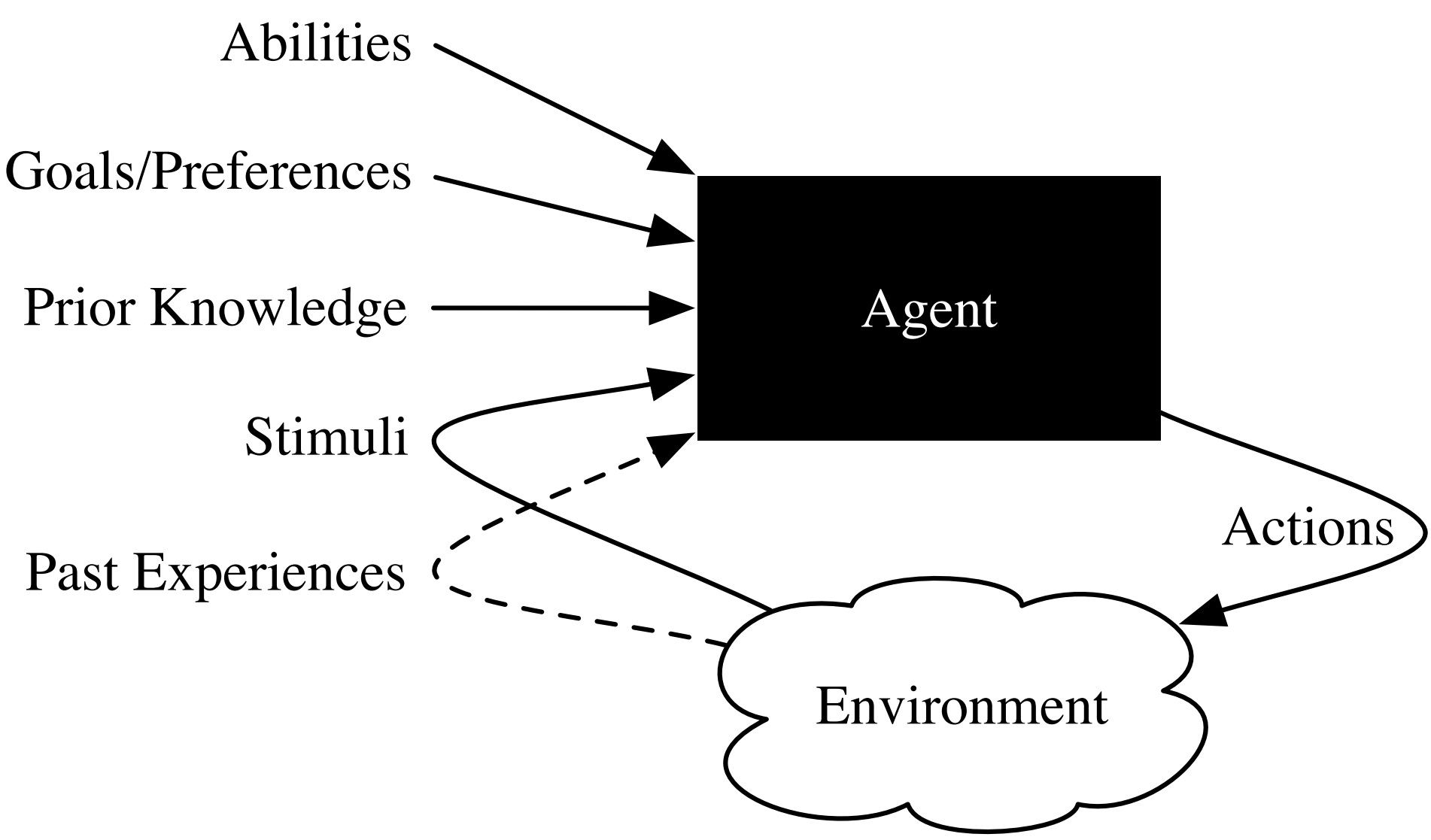


熵增: 信息守恒吗?



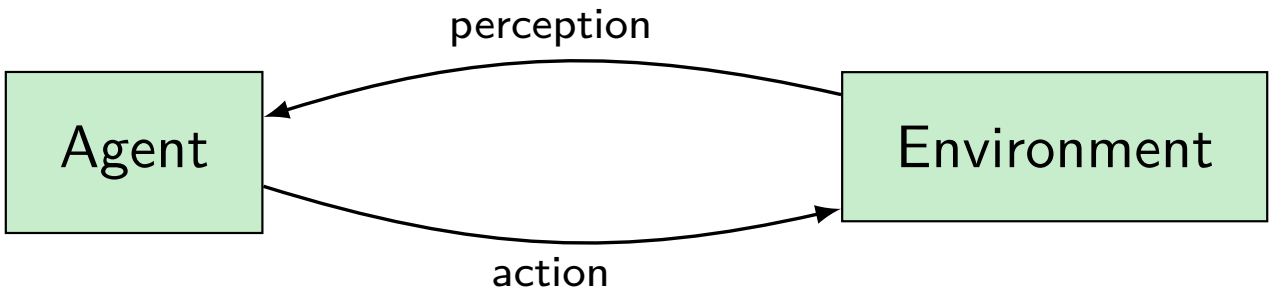
Contents

1. Introduction
   - History of AI
   - What is AI?
   - Turing Machine
   - Rational Agent
   - Goal-Based Agent
   - Knowledge-Based Agent
   - Knowledge Representation
   - Learning Agent
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics



- Prior Knowledge? 哪些先验知识?
  - 柏拉图: 一切知识. 学习即回忆.
  - 康德: 空间、时间 (感性直观的先天形式), 因果等 (知性范畴)
  - 莱布尼茨: 矛盾律, 充足理由律
  - 洛克: 没有任何先验知识! 一切源于经验!

什么是 Agent?



f : Percept Sequence $\rightarrow$ Action

什么函数 “聪明”?

- 智能 Agent 的动作与目标相适应
- 能够灵活应对环境和目标的变化
- 能从经验中学习
- 在感知信息和计算资源有限的情况下做出适当的选择

智者依理性而动;
凡人循经验而动;
愚者因欲念而动;
畜生凭本能而动.

— 西塞罗

Rational Agent

Agent: Percept Sequence $\rightarrow$ Action

理性的行为依赖于

- 感知序列 (过去的经验和当前的刺激)
- 关于环境的先验知识
- 可能的动作
- 性能指标 (目标/偏好) $V(\text{Agent}, \text{Environment})$

$$\text{Agent}^* = \underset{\text{Agent}}{\operatorname{argmax}}\, V(\text{Agent}, \text{Environment})$$

Algorithm Agent Program

```
procedure SKELETON-AGENT(percept)
    memory ← UPDATE-MEMORY(memory, percept)
    action ← CHOOSE-BEST-ACTION(memory)
    memory ← UPDATE-MEMORY(memory, action)
    return action
end procedure
```

理性 ≠ 全知

- 全知的 Agent 能感知所有相关信息, 并且知道其行为的实际效果.
- 理性的 Agent 根据其感知信息和知识信念行事, 试图最大化期望表现.

Remark: 如果你在过马路前看了两边, 但过马路时被一颗陨石击中, 很难说你缺乏理性.

Rationality vs Bounded Rationality

$$\text{Agent}^* = \underset{\text{Agent}}{\operatorname{argmax}} V(\text{Agent}, \text{Environment})$$

$$p^* = \underset{p}{\operatorname{argmax}} V(\text{Agent}(p, M), \text{Environment})$$

$\text{Agent}(p, M)$ is a machine M running a program p.
Program p computes the best action with Machine M in Environment.

Environment Types

1. **Fully observable vs. Partially observable**
   Are the relevant aspects of the environment accessible to the sensors?
2. **Known vs. Unknown**
   It's about the agent's state of knowledge about the "rules" of the environment. In a known environment, the outcomes for all actions are given.
3. **Deterministic vs. Nondeterministic**
   Is the next state of the environment completely determined by the current state and action?
4. **Episodic vs. Sequential**
   Could the current decision affect future decisions?
5. **Static vs. Dynamic**
   Can the environment change while the agent is deliberating?
6. **Discrete vs. Continuous**
   Is the environment discrete or continuous?
7. **Single agent vs. Multi-agent**
   There are competitive and cooperative scenarios.

Agent 设计空间

维度	值
模块性	扁平化、模块化、层次化
规划视野	无规划、有限阶段、不定阶段、无限阶段
表示	状态、特征、个体与关系
计算限制	完全理性、有限理性
学习	知识是给定的、知识是学习的
感知不确定性	完全可观察、部分可观察
动作效果不确定性	确定性的、随机性的动力学
偏好	目标、复杂偏好
Agent 数量	单、多
交互	离线、在线

Agent 设计

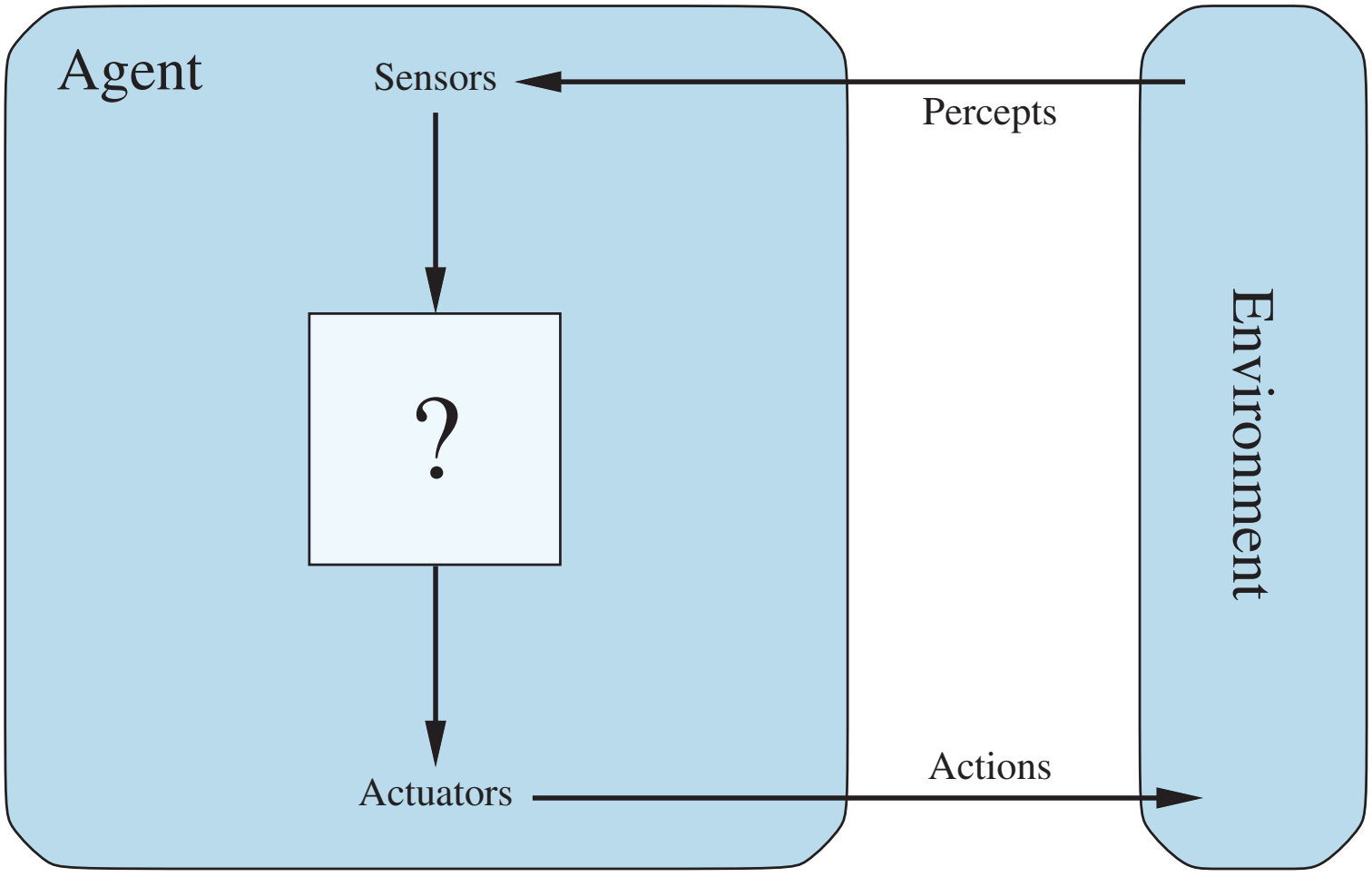
The environment type largely determines the agent design.

- Partially observable $\Longrightarrow$ agent requires memory (internal state)
- Stochastic $\Longrightarrow$ agent may have to prepare for contingencies
- Multi-agent $\Longrightarrow$ agent may need to behave randomly
- Static $\Longrightarrow$ agent has time to compute a rational decision
- Continuous time $\Longrightarrow$ continuously operating controller
- Unknown physics $\Longrightarrow$ need for exploration
- Unknown performance measure $\Longrightarrow$ observe/interact with human principal

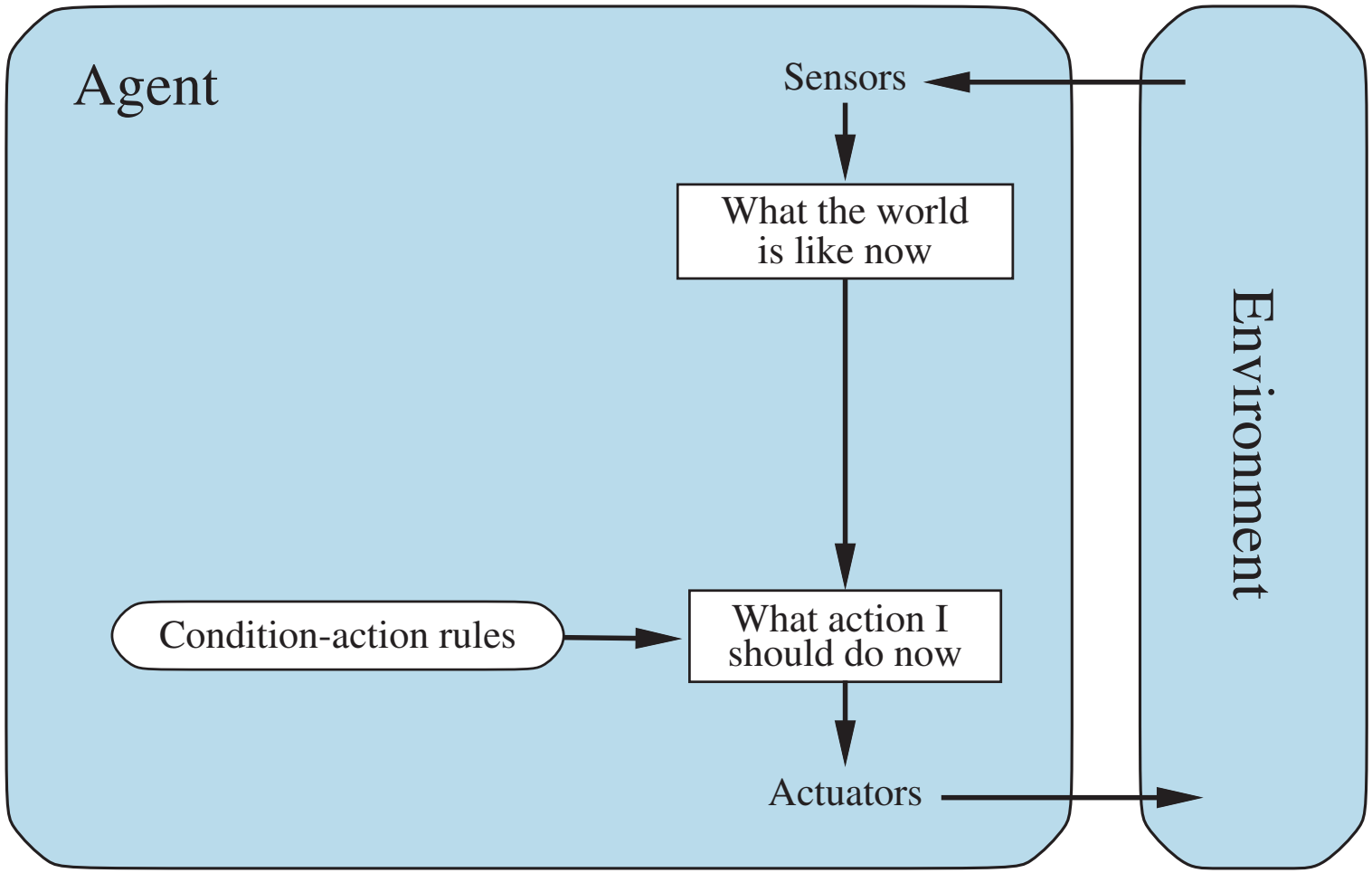
Agent Types

- **Table-driven agents:** use a percept sequence/action table in memory to find the next action.
- **Simple reflex agents:** based on condition-action rules, implemented with an appropriate production system, responds immediately to percepts.
- **Model-based agents:** have internal state, which is used to keep track of past states of the world.
- **Goal-based agents:** have goal information that describes desirable situations.
- **Utility-based agents:** base their decisions on classic axiomatic utility theory in order to act rationally.
- **Learning agents:** improves its performance w.r.t. a specific task with experience.

Agent?



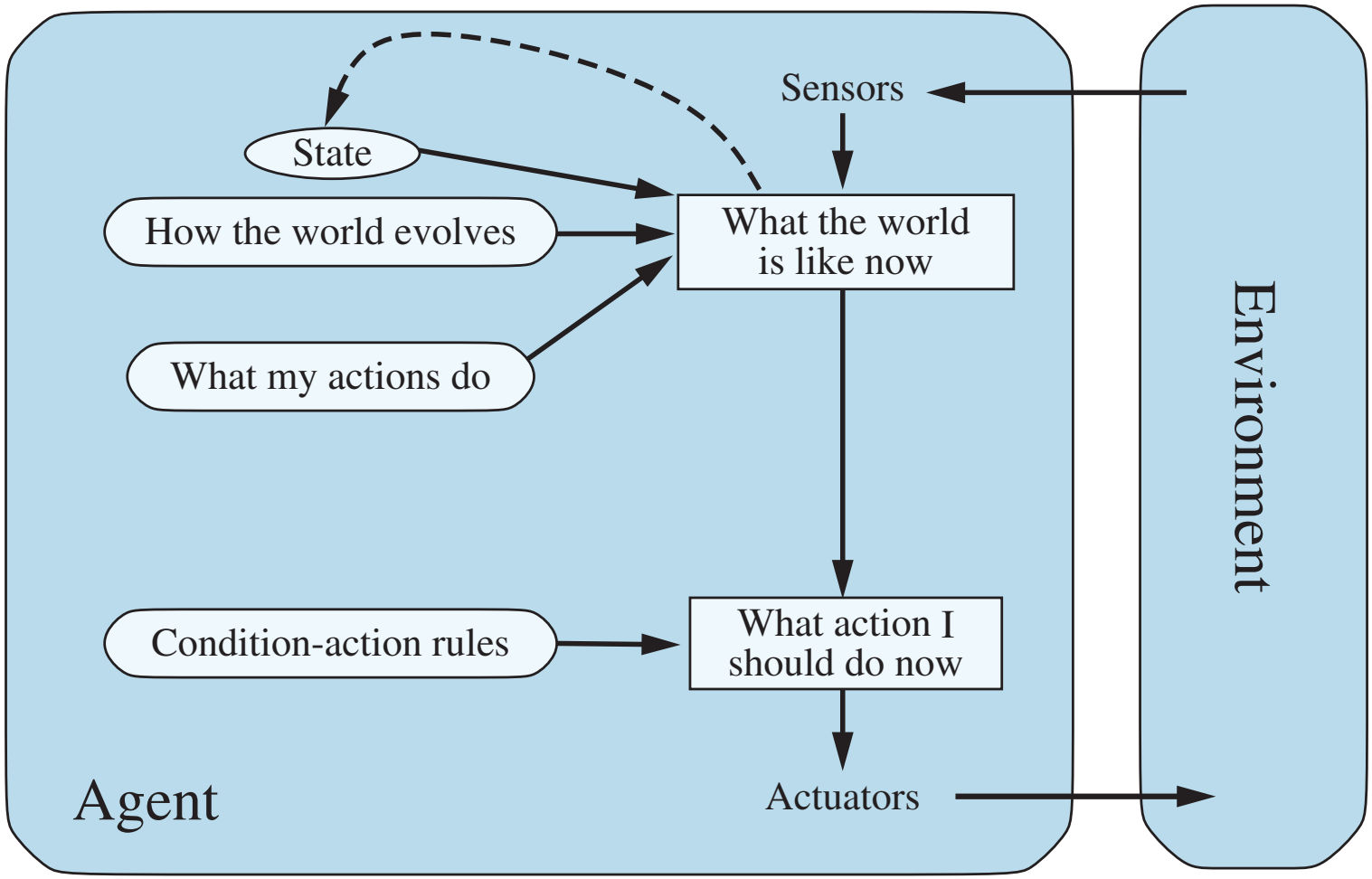
Simple Reflex Agent



Choose action only based on current percept.

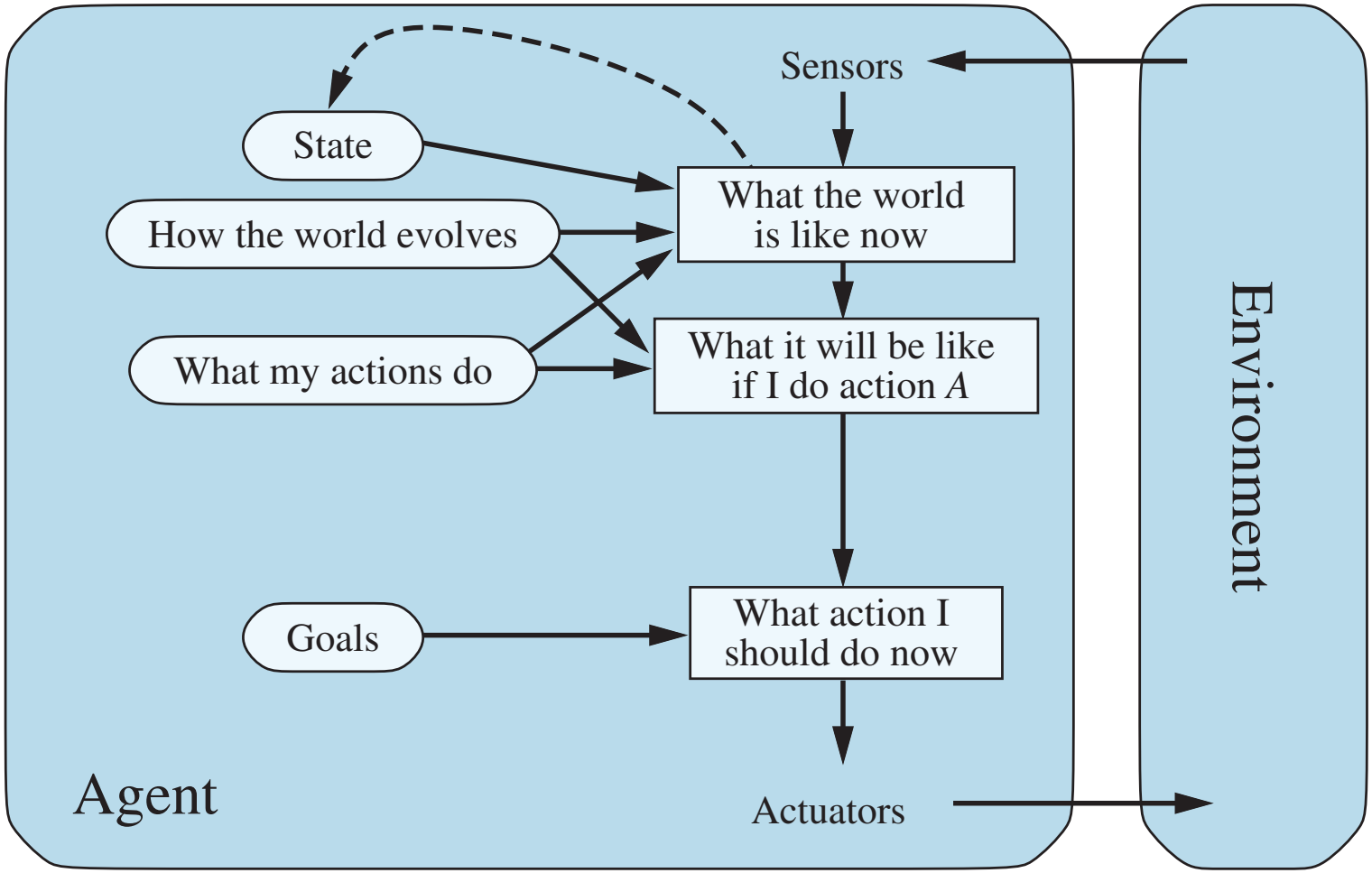
Direct use of perceptions is often not possible due to the large space required to store them.

Model-based Reflex Agent



Do not consider the future consequences of their actions.

Model-based, Goal-based Agent

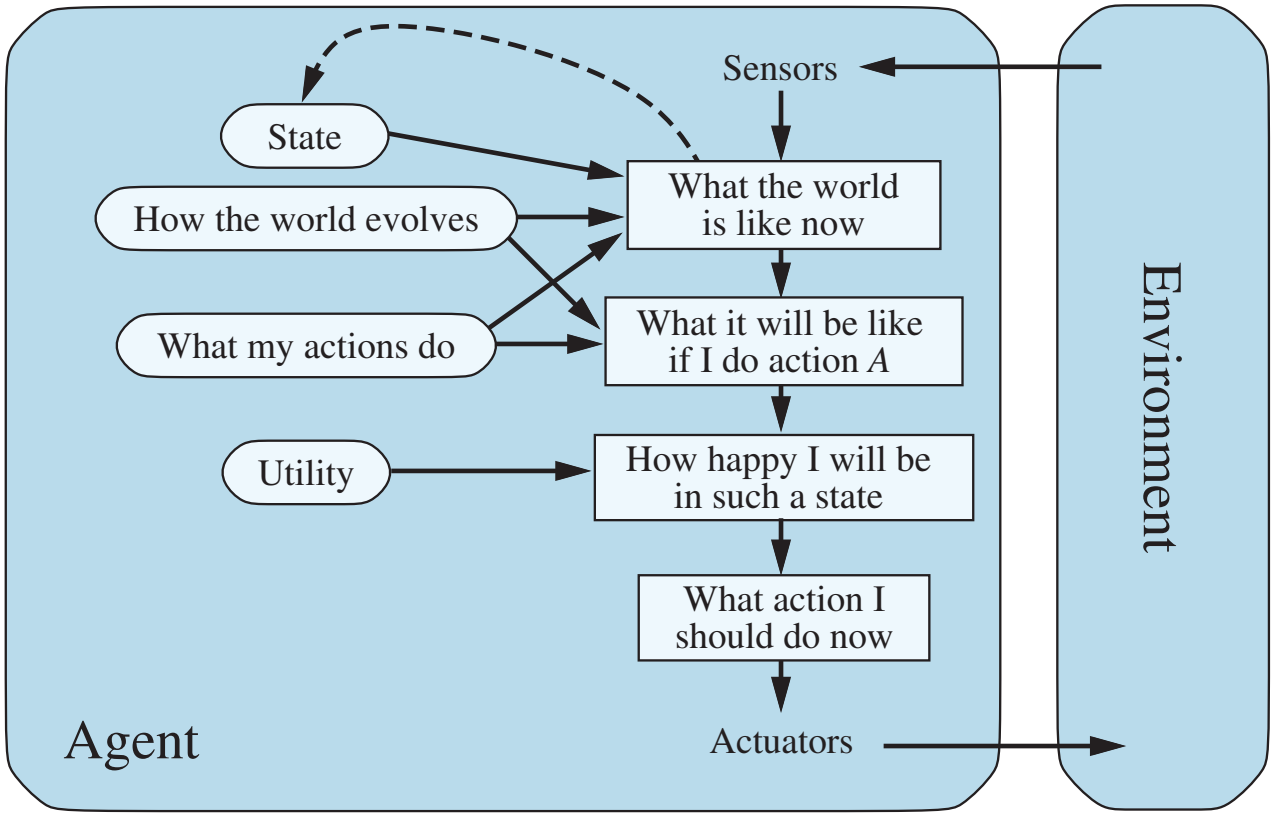


Ask "what if".

Consider how the world would be.

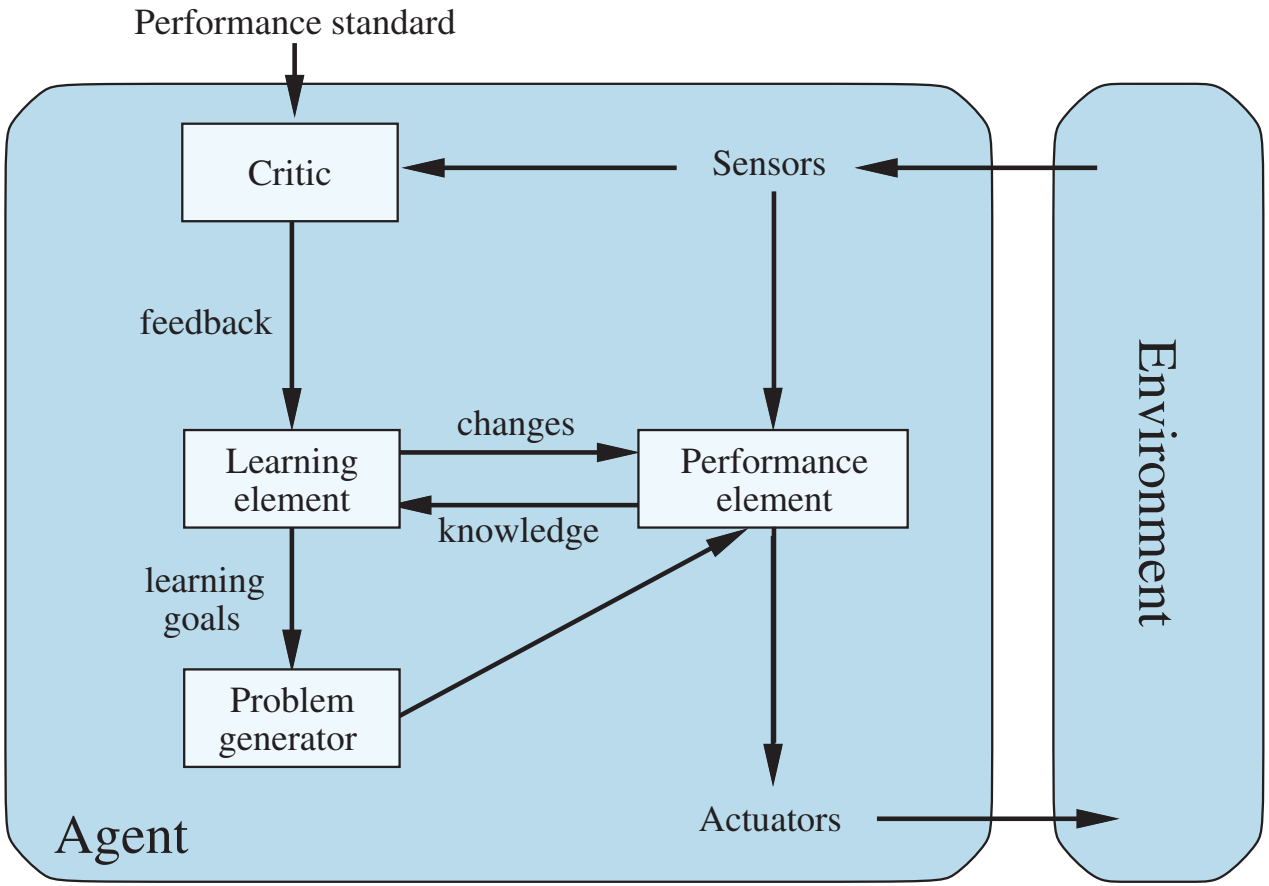
agency = goal-directedness?

Model-based, Utility-based Agent



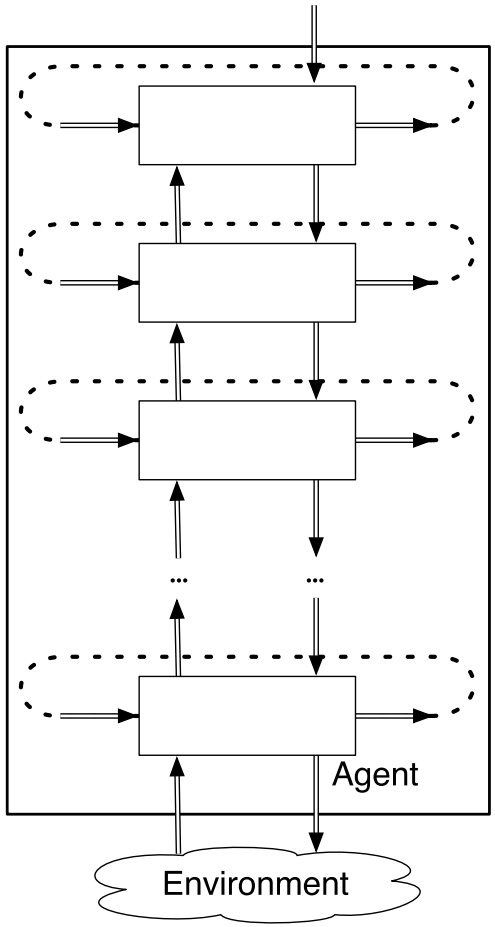
- An agent's utility function is an internalization of the performance measure.
- Provided that the internal utility function and the external performance measure are in agreement, an agent that chooses actions to maximize its utility will be rational according to the external performance measure.

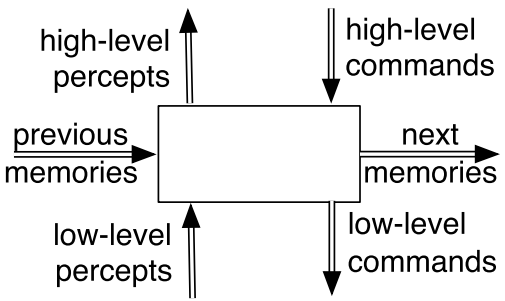
Learning Agent



- **performance element** (it represents what we have previously considered to be the entire agent)
- **learning element** (responsible for making improvements)
- **critic** (evaluation of the agent's behavior)
- **problem generator** (suggests explorative actions)

Agent 的层级控制





$$\text{memory} : \text{State} \times \text{Percept}_{\text{low}} \times \text{Command}_{\text{high}} \rightarrow \text{State}$$

$$\text{command} : \text{State} \times \text{Percept}_{\text{low}} \times \text{Command}_{\text{high}} \rightarrow \text{Command}_{\text{low}}$$

$$\text{report} : \text{State} \times \text{Percept}_{\text{low}} \times \text{Command}_{\text{high}} \rightarrow \text{Percept}_{\text{high}}$$

Examples of Representational Frameworks

- State-space search
- Deterministic planning
- Influence diagrams (Decision Networks)
- Markov Decision Processes
- Decision-theoretic planning
- Reinforcement Learning
- Classical Game Theory

State-Space Search

1. flat or hierarchical
2. explicit states or features or objects and relations
3. static or finite stage or indefinite stage or infinite stage
4. fully observable or partially observable
5. deterministic or stochastic actions
6. goals or complex preferences
7. single agent or multiple agents
8. knowledge is given or learned
9. perfect rationality or bounded rationality
10. offine or online

Deterministic Planning

1. flat or hierarchical
2. explicit states or features or objects and relations
3. static or finite stage or indefinite stage or infinite stage
4. fully observable or partially observable
5. deterministic or stochastic actions
6. goals or complex preferences
7. single agent or multiple agents
8. knowledge is given or learned
9. perfect rationality or bounded rationality
10. offine or online

Influence Diagrams

1. flat or hierarchical
2. explicit states or features or objects and relations
3. static or finite stage or indefinite stage or infinite stage
4. fully observable or partially observable
5. deterministic or stochastic actions
6. goals or complex preferences
7. single agent or multiple agents
8. knowledge is given or learned
9. perfect rationality or bounded rationality
10. offine or online

Markov Decision Processes

1. flat or hierarchical
2. explicit states or features or objects and relations
3. static or finite stage or indefinite stage or infinite stage
4. fully observable or partially observable
5. deterministic or stochastic actions
6. goals or complex preferences
7. single agent or multiple agents
8. knowledge is given or learned
9. perfect rationality or bounded rationality
10. offine or online

Decision-Theoretic Planning

1. flat or hierarchical
2. explicit states or features or objects and relations
3. static or finite stage or indefinite stage or infinite stage
4. fully observable or partially observable
5. deterministic or stochastic actions
6. goals or complex preferences
7. single agent or multiple agents
8. knowledge is given or learned
9. perfect rationality or bounded rationality
10. offine or online

Reinforcement Learning

1. flat or hierarchical
2. explicit states or features or objects and relations
3. static or finite stage or indefinite stage or infinite stage
4. fully observable or partially observable
5. deterministic or stochastic actions
6. goals or complex preferences
7. single agent or multiple agents
8. knowledge is given or learned
9. perfect rationality or bounded rationality
10. offine or online

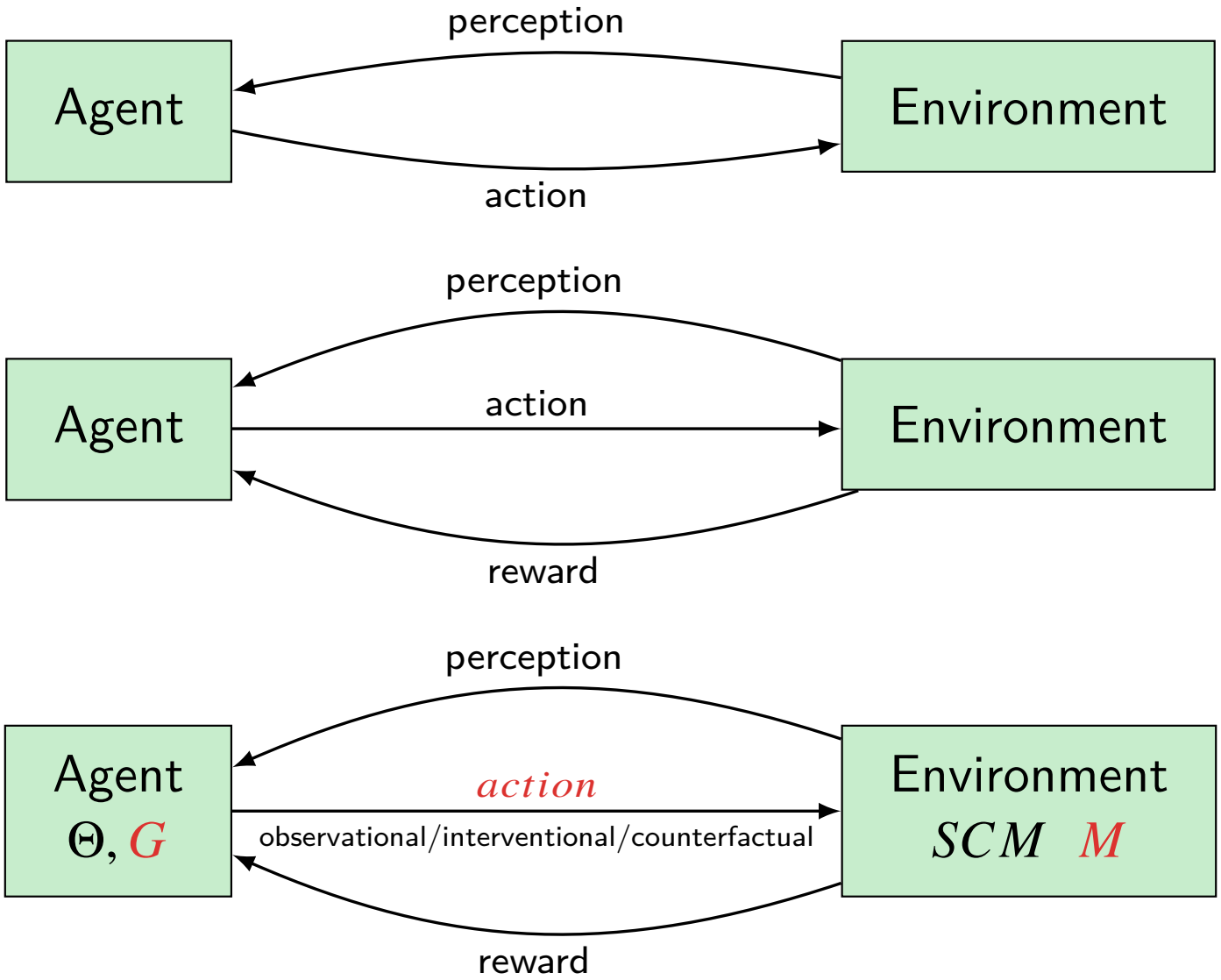
Classical Game Theory

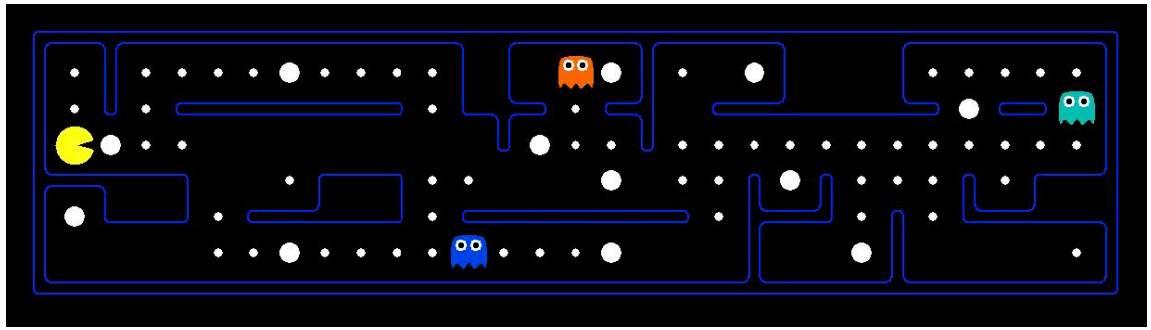
1. flat or hierarchical
2. explicit states or features or objects and relations
3. static or finite stage or indefinite stage or infinite stage
4. fully observable or partially observable
5. deterministic or stochastic actions
6. goals or complex preferences
7. single agent or multiple agents
8. knowledge is given or learned
9. perfect rationality or bounded rationality
10. offine or online

Humans

1. flat or hierarchical
2. explicit states or features or objects and relations
3. static or finite stage or indefinite stage or infinite stage
4. fully observable or partially observable
5. deterministic or stochastic actions
6. goals or complex preferences
7. single agent or multiple agents
8. knowledge is given or learned
9. perfect rationality or bounded rationality
10. offine or online

Agent vs RL Agent vs Causal RL Agent





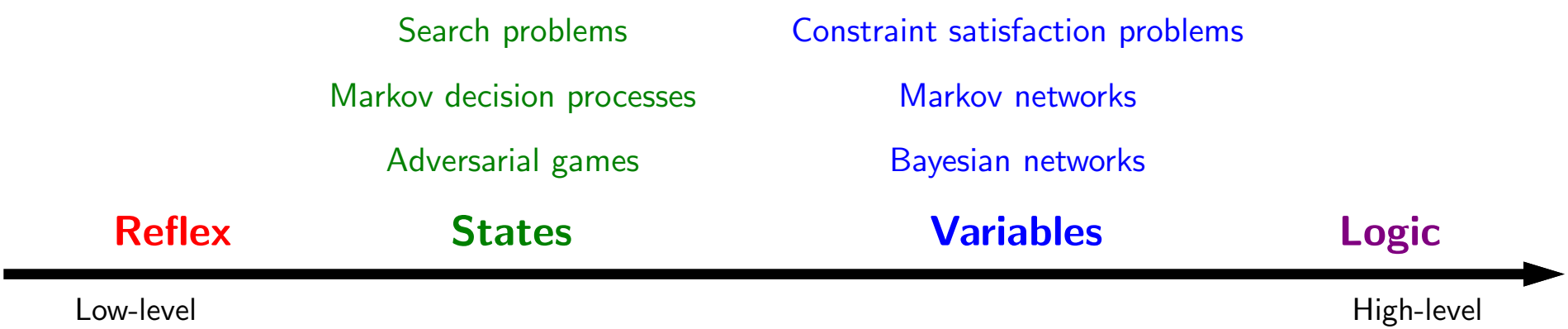
- 乌鸦站在电线上把坚果扔在公路上，让来往的车辆将它们砸碎.
- 等红绿灯变了时，乌鸦再飞下来吃它的坚果.

Contents

1. Introduction
   - History of AI
   - What is AI?
   - Turing Machine
   - Rational Agent
   - Goal-Based Agent
   - Knowledge-Based Agent
   - Knowledge Representation
   - Learning Agent
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Modeling Paradigms

- State-based models: search problems, MDPs, games
  — Applications: route finding, game playing, etc.
  — Think in terms of **states, actions, and costs**
- Variable-based models: Constraint Satisfaction Problems, Markov networks, Bayesian networks
  — Applications: scheduling, tracking, medical diagnosis, etc.
  — Think in terms of **variables and factors**
- Logic-based models: propositional logic, first-order logic
  — Applications: theorem proving, verification, reasoning
  — Think in terms of **logical formulas and inference rules**



Example — 传教士与野人过河问题

- 3 个传教士和 3 个野人在河的一岸.
- 有一条最多可以搭载两个人的船.
- 在河的任何一边, 都不能让传教士的人数少于野人的人数.
- 怎么过河?

States (x, y, z) with $0 \leq x, y \leq 3$ and $z \in \{0, 1\}$, where x, y and z represent the number of missionaries, cannibals and boat currently on the original bank.

Initial State $(3, 3, 1)$

Goal State $(0, 0, 0)$

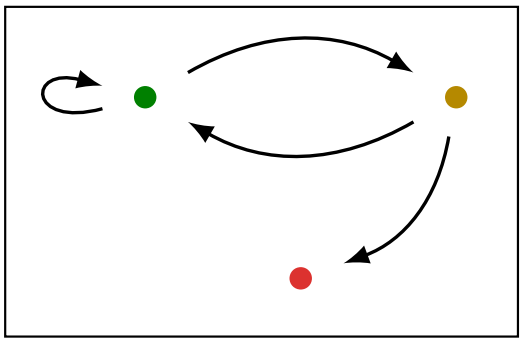
Path Costs 1 unit per crossing.

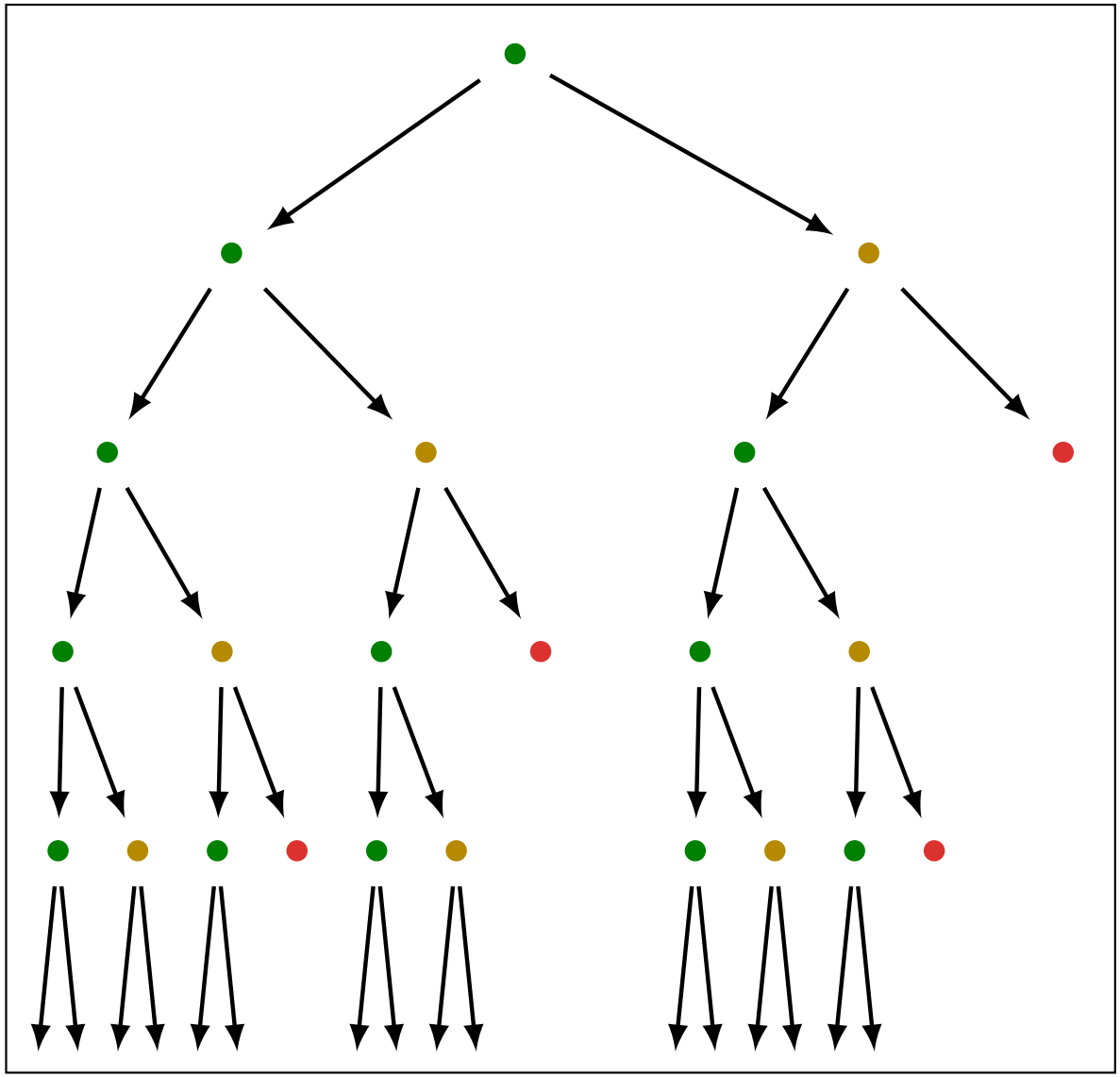
$(3,3,1) \to (3,1,0) \vee (2,2,0) \to (3,2,1) \to (3,0,0) \to (3,1,1) \to$
$(1,1,0) \to (2,2,1) \to (0,2,0) \to (0,3,1) \to (0,1,0) \to$
$(0,2,1) \vee (1,1,1) \to (0,0,0)$

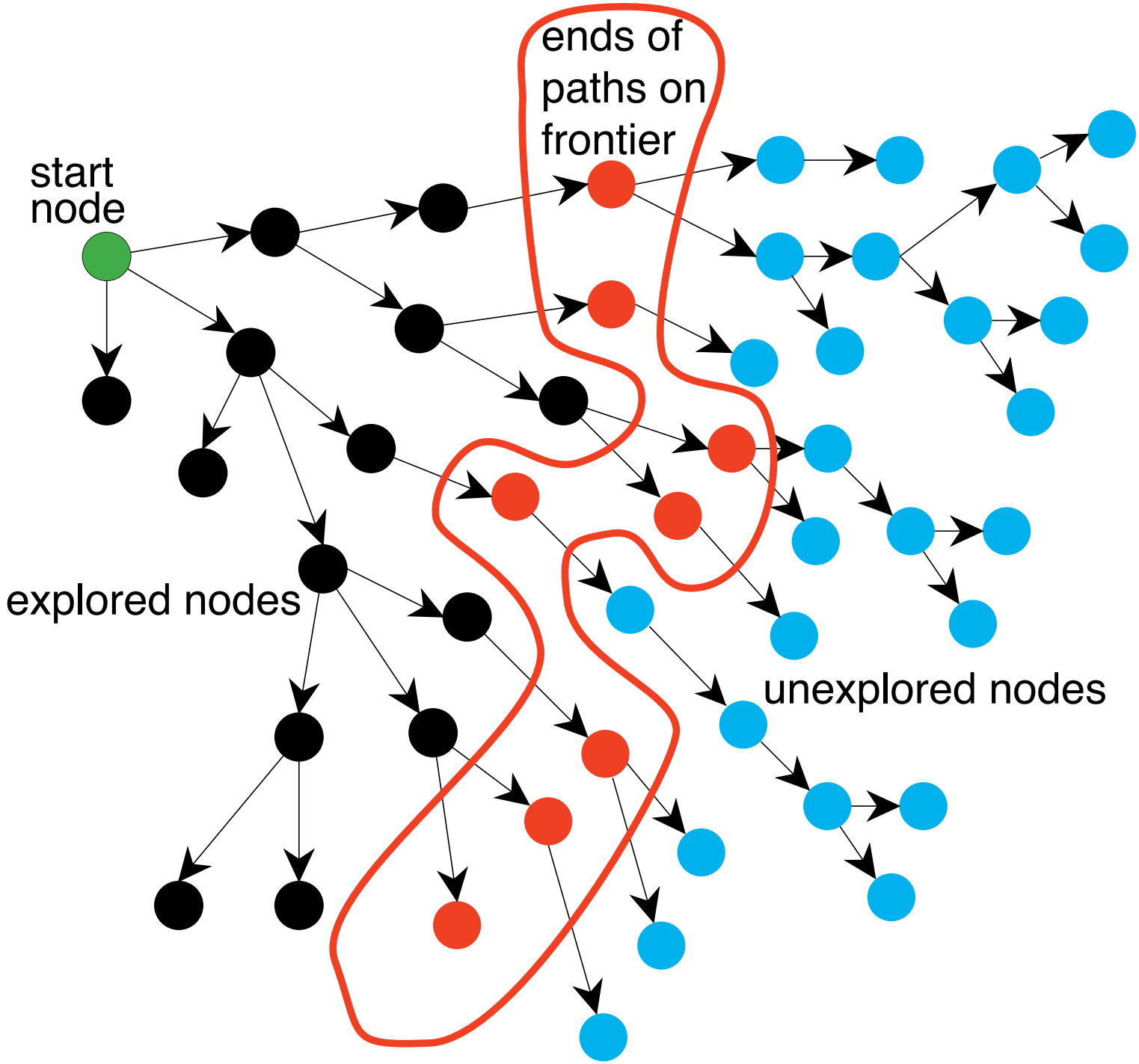
Goal-based Agent

- A search problem consists of:
  1. the state space
  2. initial state
  3. actions in each state
  4. transition model
  5. goal test
  6. action costs
- A solution is an action sequence from an initial state to a goal state.
- An optimal solution has least cost among all solutions.

From State Space Graphs to Search Trees






ends of
paths on
frontier
start
node
explored nodes
unexplored nodes

Criteria for Search Strategies

Completeness Is the strategy guaranteed to find a solution when there is one?

Time Complexity How long does it take to find a solution?

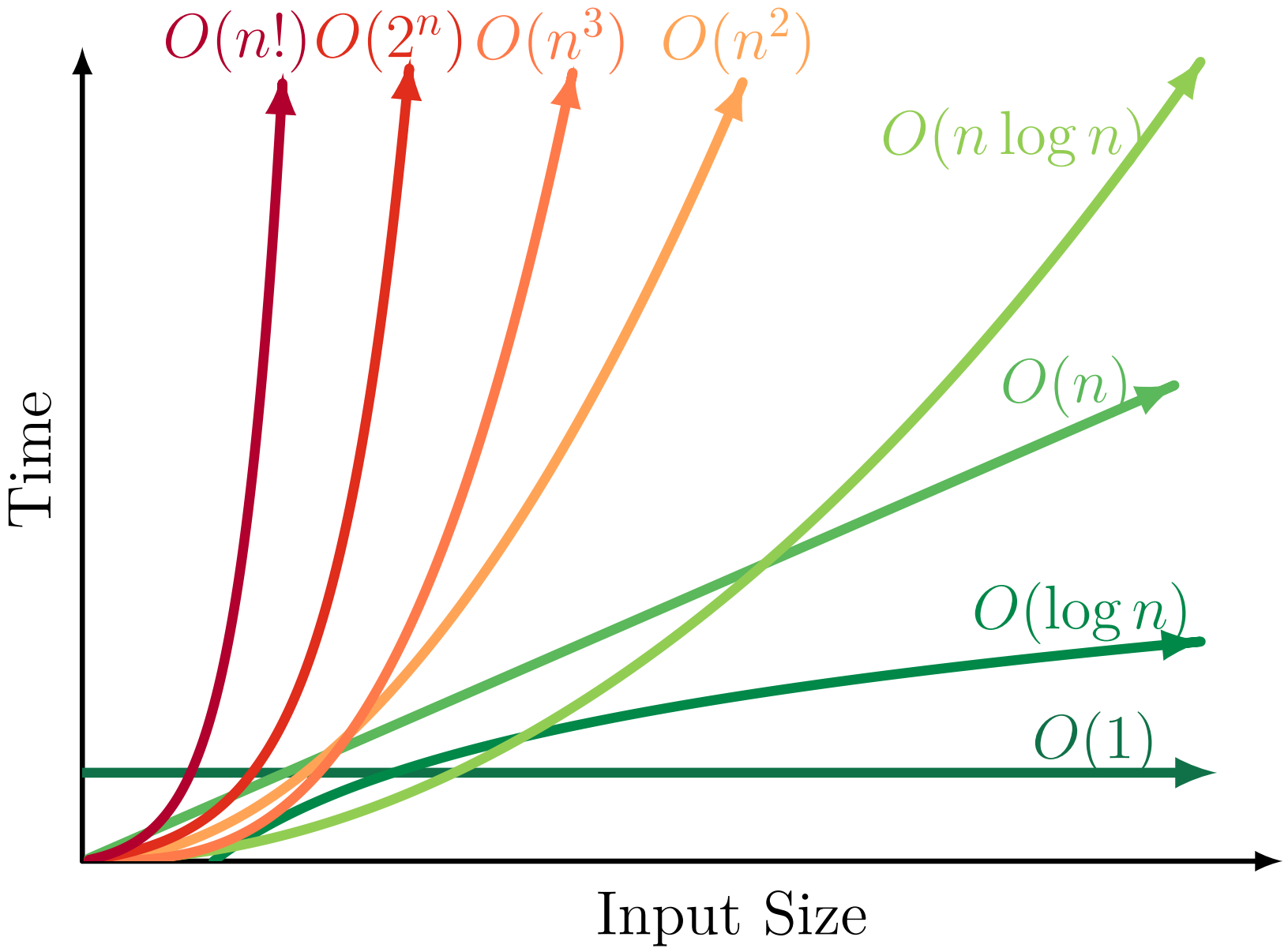
Space Complexity How much memory does the search require?

Optimality Does the strategy find the best solution (with the lowest path cost)?

$$\mathtt{P} \subset \mathtt{NP} \subset \mathtt{PSPACE} \subset \mathtt{EXPTIME}$$

Remark: 目前只知道 $\mathtt{P} \neq \mathtt{EXPTIME}$, 所以,
或者 $\mathtt{P} \neq \mathtt{NP}$,
或者 $\mathtt{NP} \neq \mathtt{PSPACE}$,
或者 $\mathtt{PSPACE} \neq \mathtt{EXPTIME}$

时间复杂性



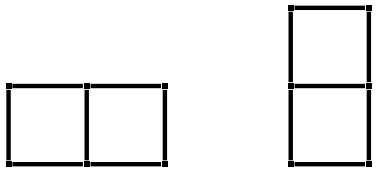
Problem Formulation

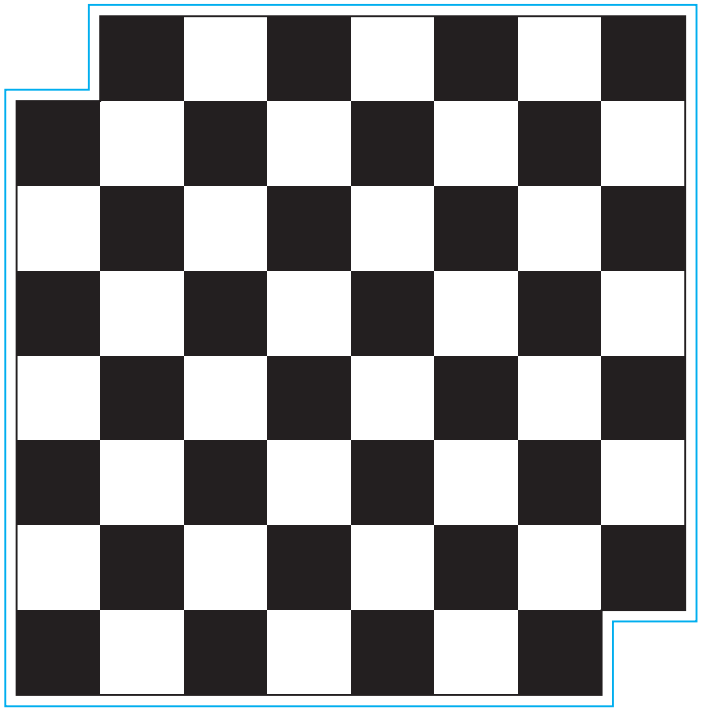
Question

Given an $n \times n$ board from which two diagonally opposite corners have been removed (here 8×8), can the board be covered with dominoes?

Question — Alternative Problem Formulation

Can a board consisting of $n^2/2$ black and $n^2/2 - 2$ white squares be covered with dominoes s.t. each domino covers one black and one white square?





Search Strategies

- **Uninformed Search:** Rigid procedure with no knowledge of the cost of a given node to the goals.
- **Informed Search:** Knowledge of the worth of expanding a node n is given in the form of a heuristic function $h(n)$.

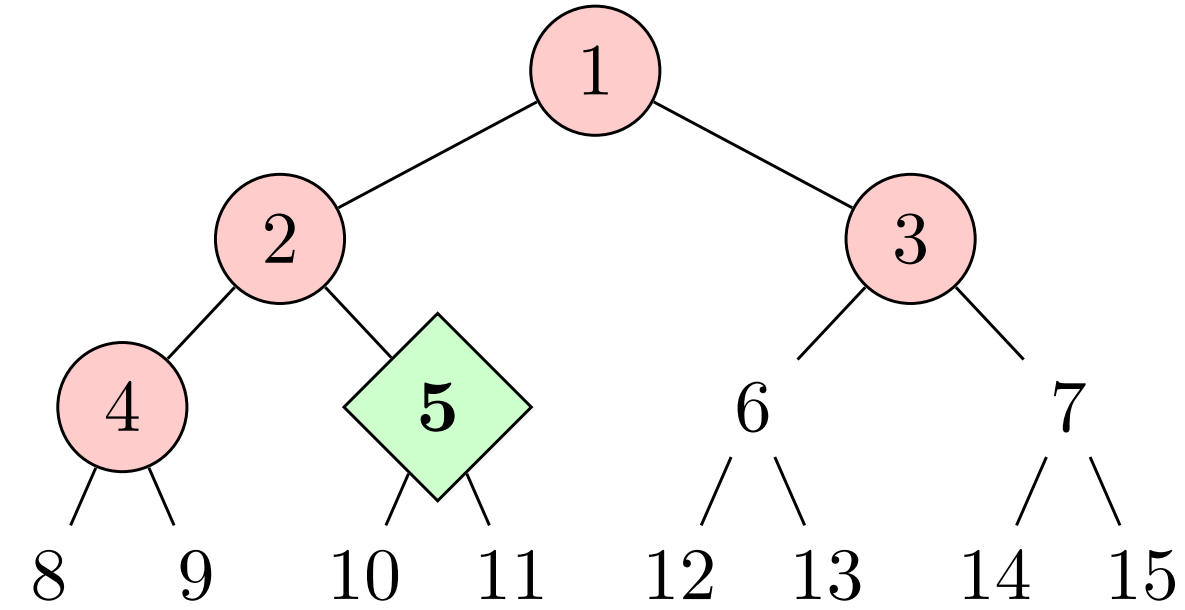
Uninformed (or Blind) Search Strategies

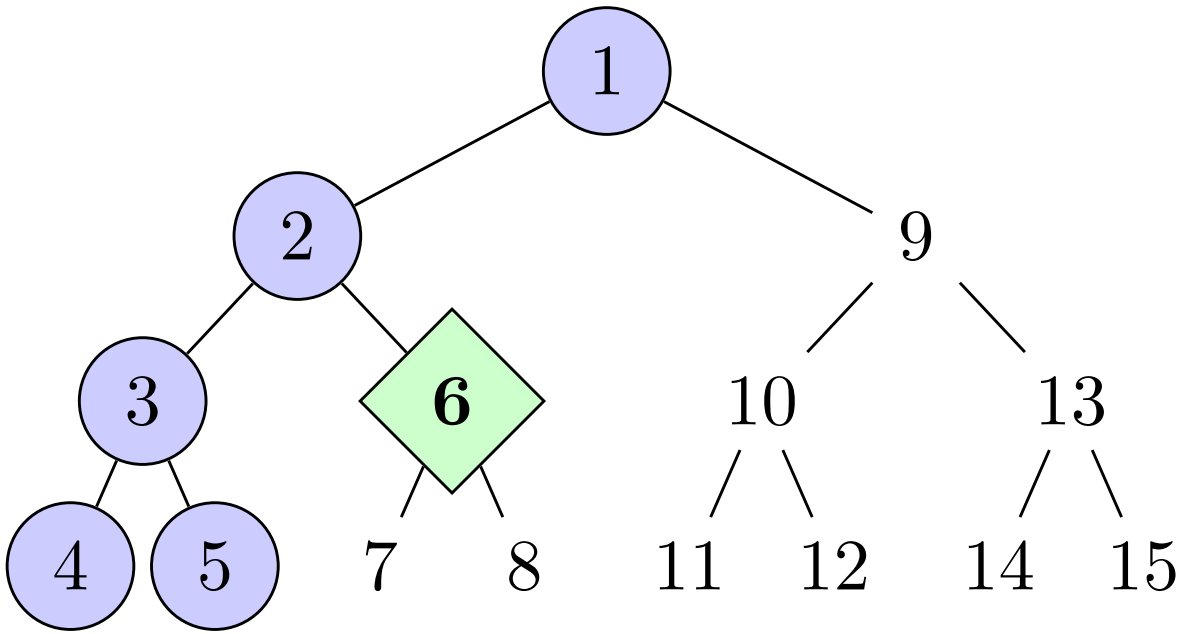
- breadth-first search, depth-first search, lowest-cost-first search
- depth-limited search, iterative deepening search
- bi-directional search

Informed (or Heuristic) Search Strategies

- Greedy Best-First Search: expands the node with the best h-value first.
- A^* and IDA^*
- Local Search Methods: Hill Climbing / Gradient Decent.
- Genetic Algorithms

Breadth-First vs Depth-First Search





Greedy Best-First Search

- Idea: use an evaluation function for each node — estimate of "desirability" — expand most desirable unexpanded node
- $h(n) :=$ estimated path-costs from n to the goal; $h(n) = 0$ if n is a goal.
- Greedy search: A best-first search using $h(n)$ as the evaluation function.
- The evaluation function h in greedy searches is called a heuristic function.

A^*: combines greedy search with the lowest-cost-first search

$g(n) :=$ actual cost so far from the start node to reach n

$h(n) :=$ estimated cost from n to the nearest goal

$f(n) := g(n) + h(n)$ estimated total cost of path through n

$$\underbrace{\underbrace{\text{start} \xrightarrow{\text{actual cost}} n}_{g(n)} \underbrace{\xrightarrow{\text{estimated cost}} \text{goal}}_{h(n)}}_{f(n)}$$

- Idea: avoid expanding paths that are already expensive.
- A^* uses an admissible heuristic $h(n) \leq h^*(n)$ to minimize the estimated path costs, where $h^*(n)$ is the actual cost of the optimal path from n
- IDA^*: iterative-deepening A^*, where the f-costs are used to define the cutoff.

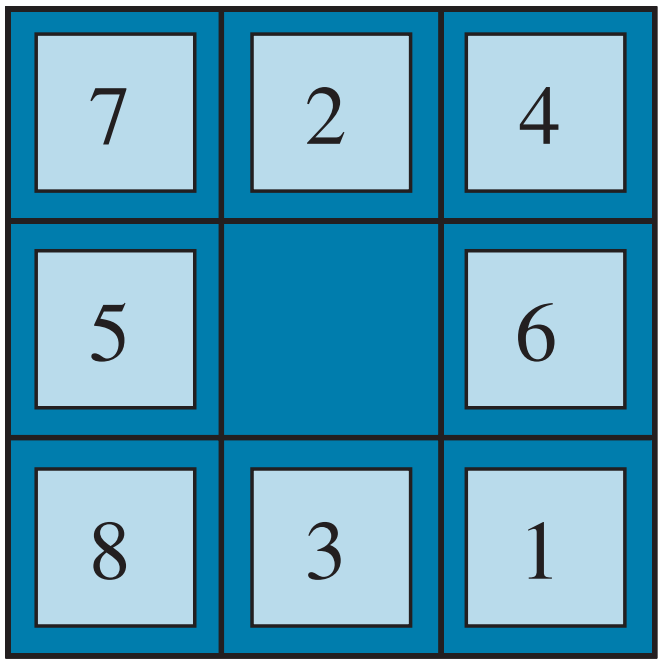
Summary of Search Strategies

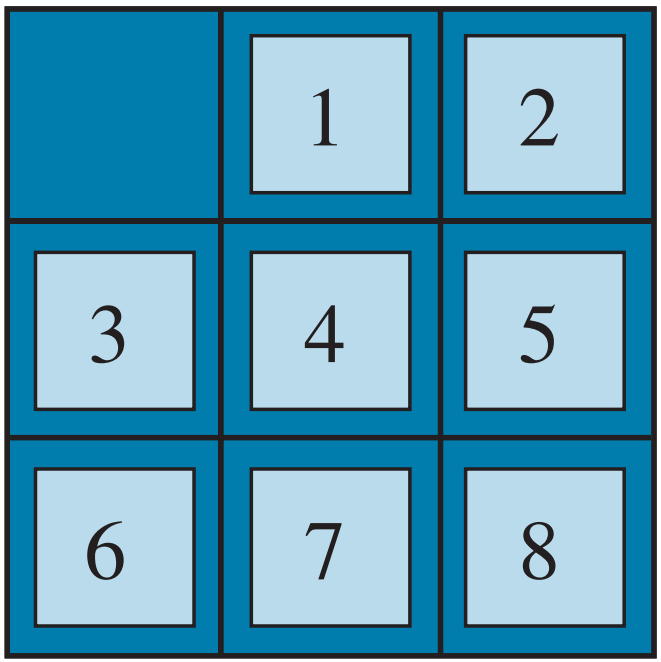
Strategy	Frontier Selection	Complete	Halts	Space
Depth-First	Last node added	No	No	Linear
Breadth-First	First node added	Yes	No	Exp
Heuristic Depth-First	Local min $h(n)$	No	No	Linear
Greedy Best-First	Global min $h(n)$	No	No	Exp
Lowest-Cost-First	Minimal cost $g(n)$	Yes	No	Exp
A^*	Minimal $f(n)$	Yes	No	Exp

- Complete: if there a path to a goal, it can find one, even on infinite graphs.
- Halts: on finite graph (perhaps with cycles).
- Space: as a function of the length of current path.

Heuristic Function — Example

A heuristic is a function that estimates how close a state is to a goal.



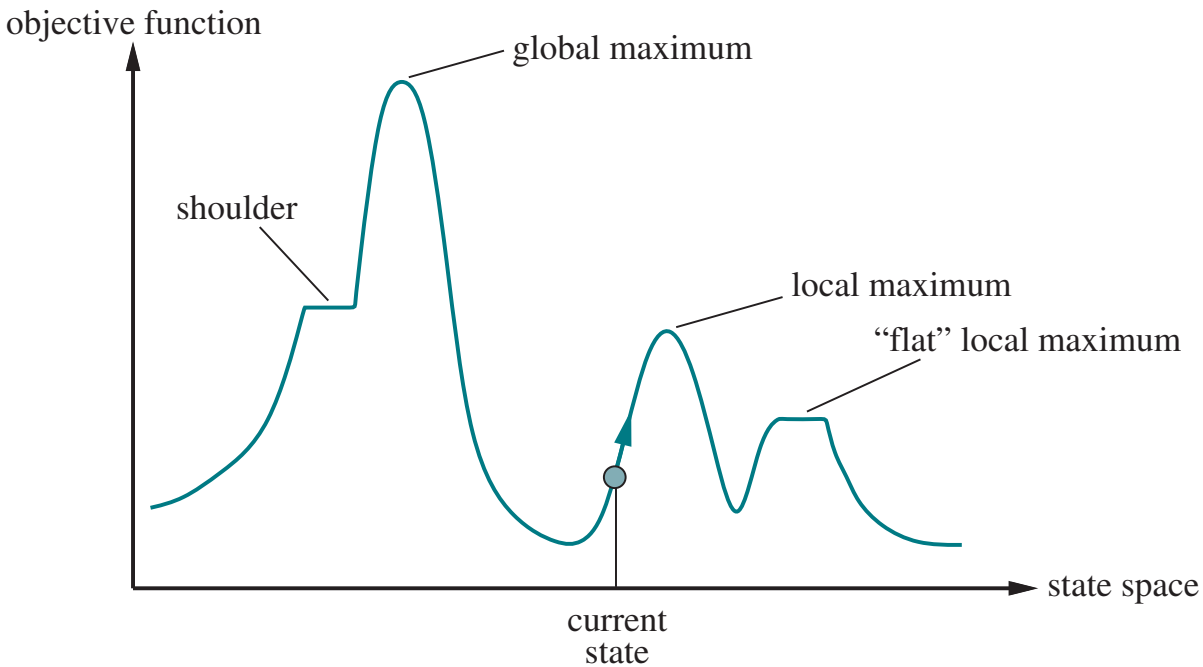


h_1 := the number of tiles in the wrong position

h_2 := the sum of the Manhattan distances of the tiles from their goal position

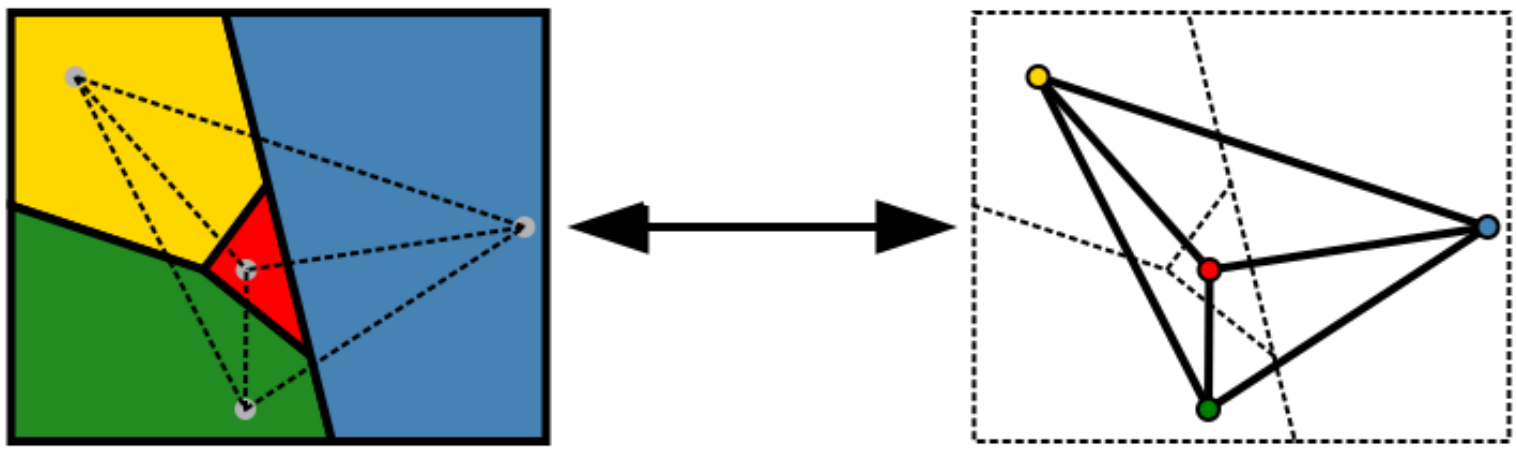
Local Search Methods

- Local search algorithms operate by searching from a start state to neighboring states, without keeping track of the paths, nor the set of states that have been reached.
- Begin with a randomly-chosen configuration and improve on it step by step → **Hill Climbing(Gradient Ascent / Decent)**.
  "Like climbing Everest in thick fog with amnesia"
- **Simulated Annealing:** to escape local maxima, noise ("random walk") is injected systematically: first a lot, then gradually less.



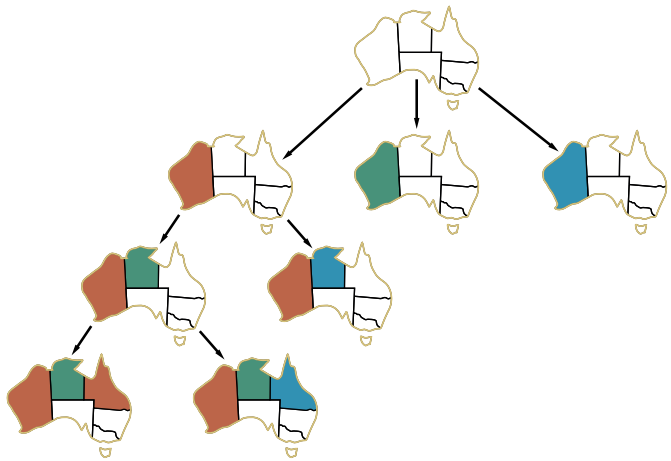
Constraint Satisfaction Problems

- CSP take advantage of the structure of states and use general rather than domain-specific heuristics. It eliminates large portions of the search space all at once by identifying variable/value combinations that violate the constraints.
  1. a set of variables,
  2. each of which has a value,
  3. a set of constraints.
- Example: Map-Coloring



1. Variables: regions
2. Values: $\{red, yellow, green, blue\}$
3. Constraints: adjacent regions must have different colors

CSP — Example



- Variable ordering: Which one to assign first?
  - Most Constrained First: choose the variable with the fewest remaining legal values! reduces branching factor!
  - Most Constraining Variable First: choose variable with the most constraints on remaining unassigned variables! reduces branching factor in the next steps.
  - Least Constraining Value First: choose first a value that rules out the fewest values in the remaining unassigned variables
- Value ordering: Which value to try first?
- Try to detect failures early on
- Try to exploit problem structure: tree structure

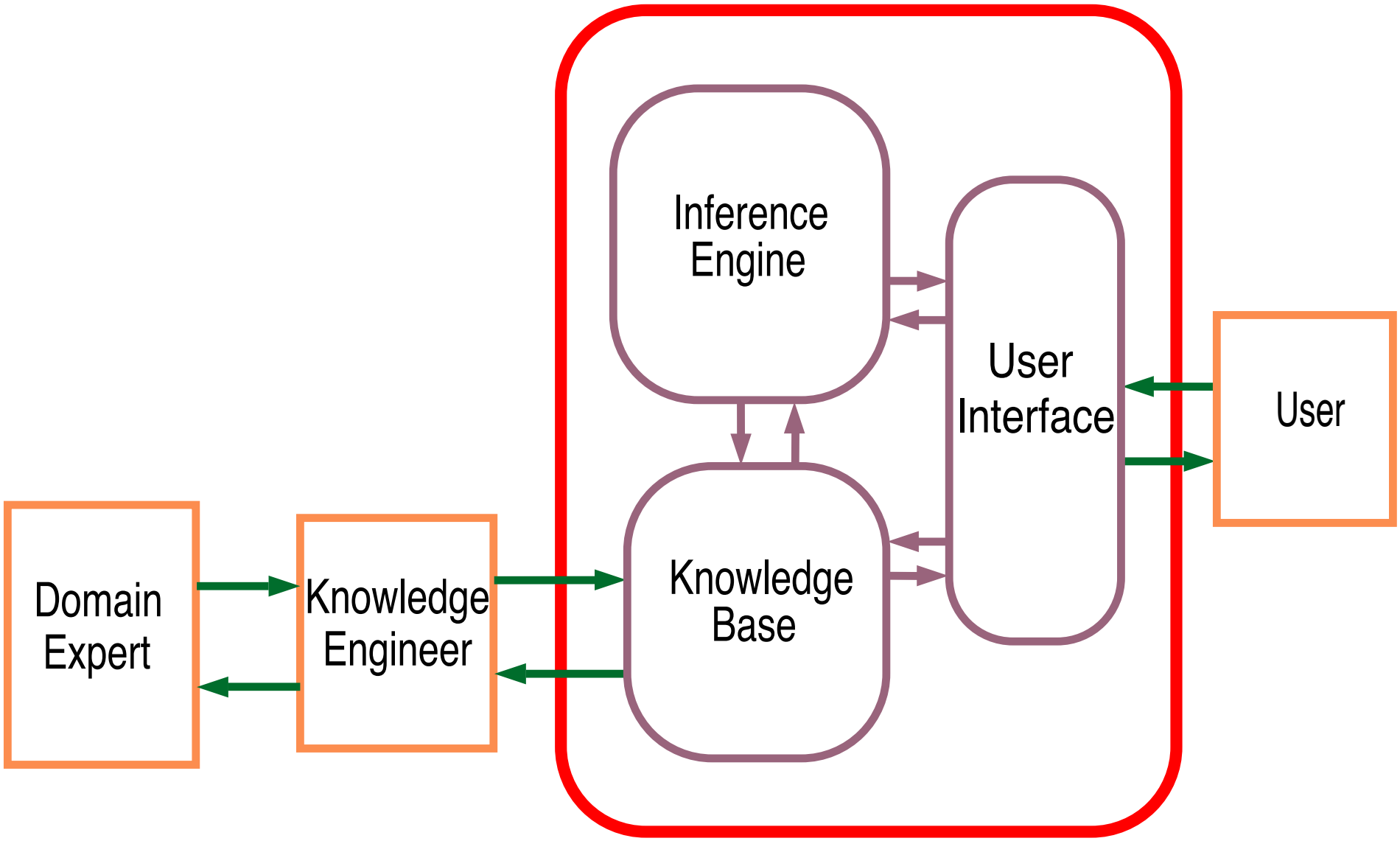
Contents

1. Introduction
   - History of AI
   - What is AI?
   - Turing Machine
   - Rational Agent
   - Goal-Based Agent
   - Knowledge-Based Agent
   - Knowledge Representation
   - Learning Agent
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

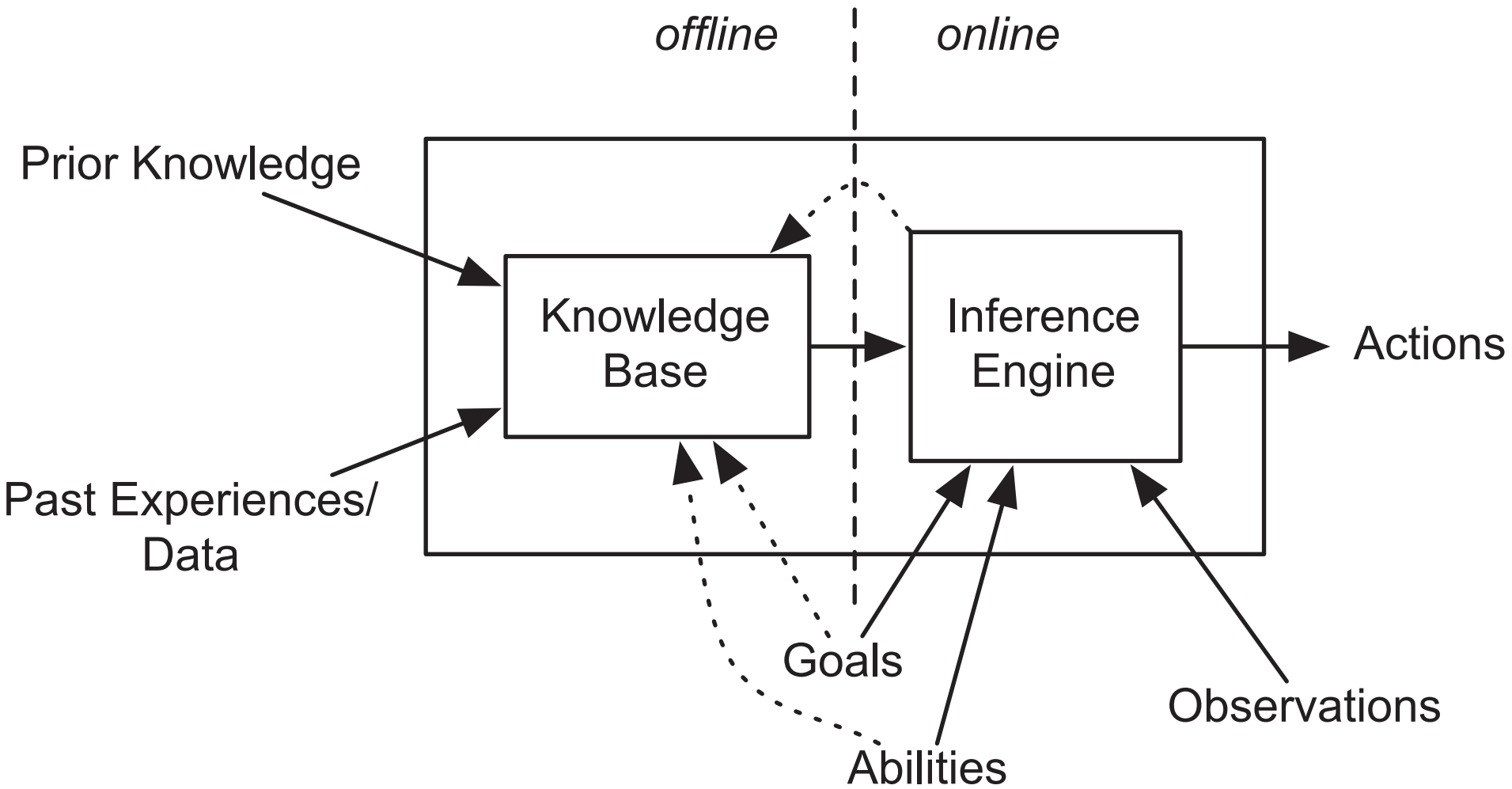
$$\frac{\text{Logic}}{\text{Computer Science}} \approx \frac{\text{Calculus}}{\text{Physics}}$$

- Computer Architecture.
  Logic gates and digital circuit design $\approx$ Propositional Logic
- Programming Languages.
  LISP $\approx \lambda$-calculus
  Prolog $\approx$ First Order Logic + Recursion
- Theory of Computation. Computational / Descriptive Complexity.
- General Problem Solver (SAT solvers).
- Automated Theorem Proving.
- Common sense reasoning via Non-monotonic Logic.
- Fuzzy Control vs Fuzzy Logic and Multi-valued Logic.
- Relational Databases.
  SQL $\approx$ First Order Logic + Syntactic Sugar
- Software Engineering (Formal Specification and Verification).
  Temporal Logic, Dynamic Logic, Hoare Logic, Model Checking
- Multi-agent Systems.
  Epistemic Logic
- Knowledge representation. Semantic Web.
  Web Ontology Language (OWL) $\approx$ Description Logic

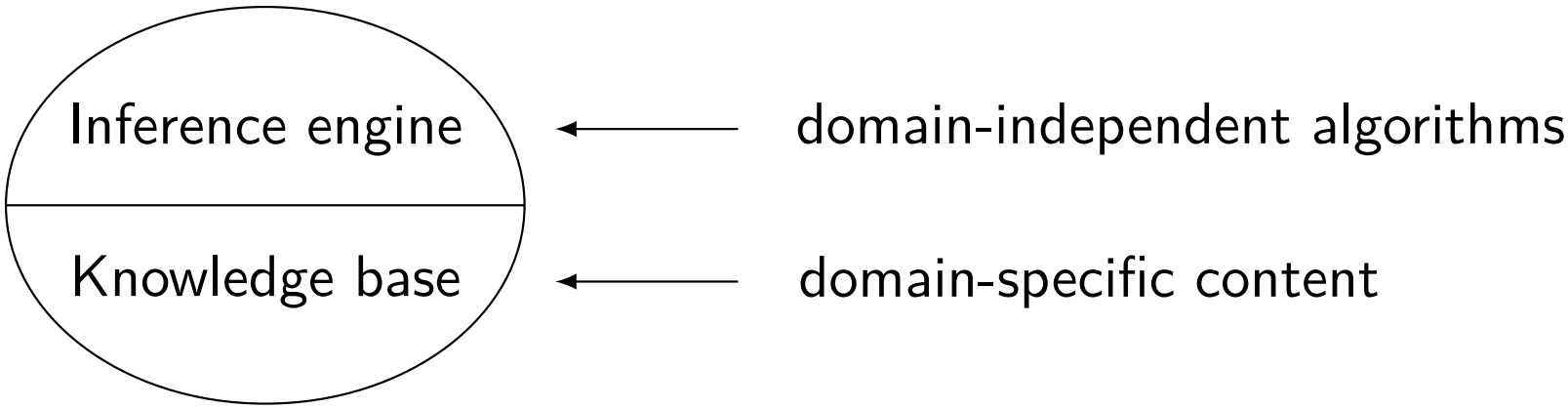
Knowledge-based system architecture



Offline and online decomposition of an agent



Knowledge-Based Agent



一个 Knowledge-Based Agent 使用其知识库来:

- 形成对世界的内部表征: 状态、动作等
- 加入新的感知
- 通过推理更新对世界的表征
- 推导出世界的隐藏属性
- 推导出应采取的行动

Entailment & Deduction

Before a system that is capable of learning, thinking, planning, explaining, ...can be built, one must find a way to express knowledge.

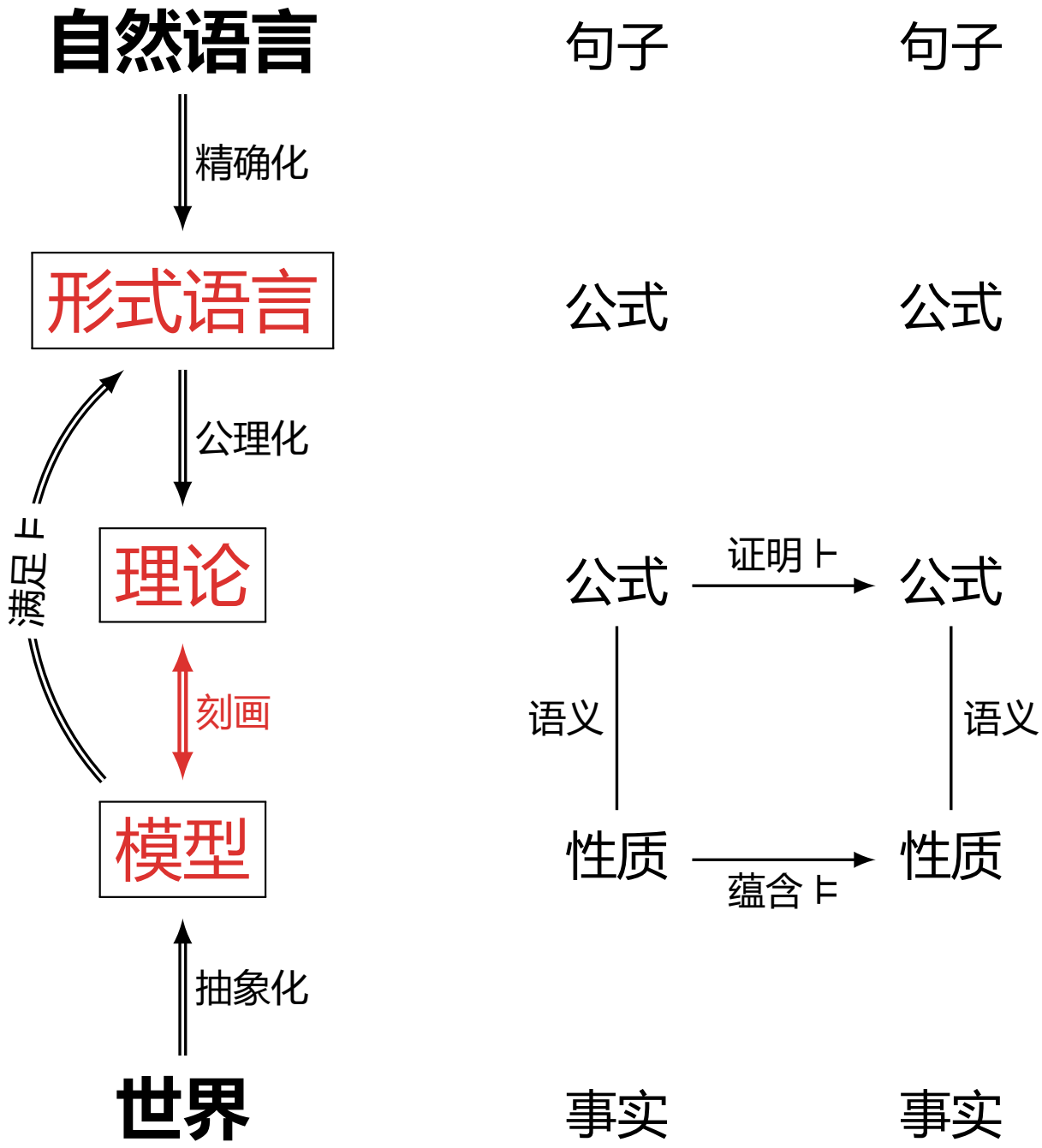
- syntax: formal structure of sentences
- semantics: truth of sentences wrt models

$$KB \models A$$

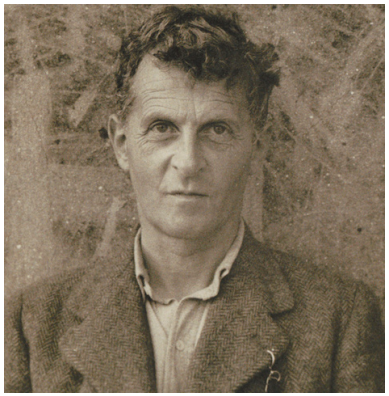
$$KB \vdash A$$

- soundness: $KB \vdash A \implies KB \models A$
- completeness: $KB \models A \implies KB \vdash A$

Terminology: satisfiable, unsatisfiable, falsifiable, valid, logical equivalent


自然语言
精确化
形式语言
公理化
理论
满足
刻画
模型
抽象化
世界
句子
句子
公式
公式
公式
证明⊢
公式
语义
语义
性质
蕴含⊨
性质
事实
事实

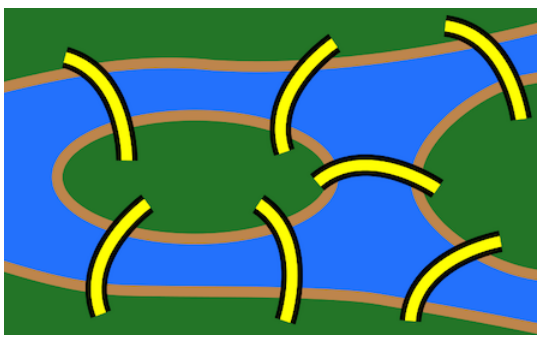
维特根斯坦 1889-1951 逻辑原子主义哲学



- 世界是事实的总和. 语言是命题的总和.
- 事实是事态的发生. 原子事态是对象的组合. 原子事态彼此独立.
- 图像描绘了逻辑空间中的一种可能情形, 即事态的发生或不发生.
- 命题是实在的图像.
  — 原子命题对应原子事态. 复合命题对应复杂事态.
  — 复合命题是原子命题的真值函数.
- 真命题的总和是世界的图像.
- 图像与其描绘的实在拥有共同的逻辑结构.
  — 唱片、音乐思想、乐谱、声波, 彼此处在图示的内在关系中, 这也是语言和世界的关系.
- **理解一个命题, 意味着知道其为真的情形.**

Model Checking & Satisfiability Checking & Validity Checking[5]

- Given a model ν and a formula A. Is $\nu \vDash A$? —P
- Given a formula A. Is there a model ν s.t. $\nu \vDash A$? —NP
- Given a sentence A. Is $\vDash A$?



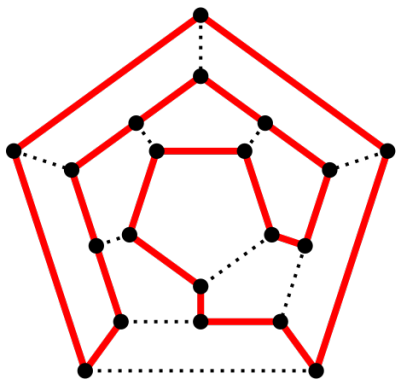


Figure: Hamiltonian Circle(NPC)

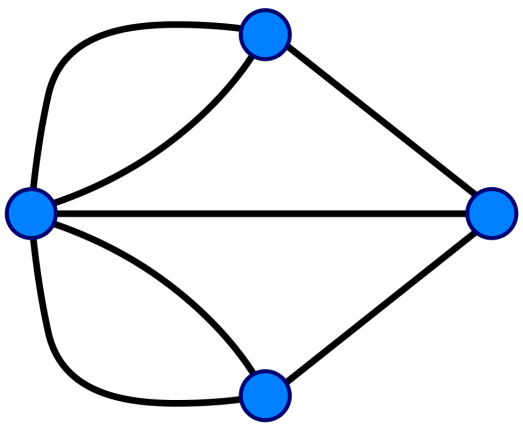


Figure: Eulerian Circle(P)

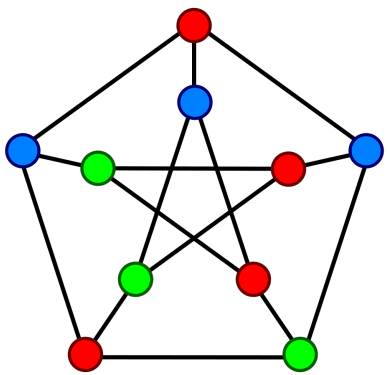
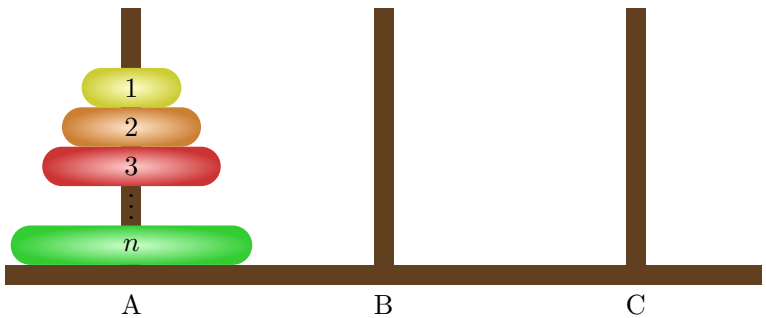


Figure: Graph Coloring(NPC)



[5] Aaronson: Why philosophers should care about computational complexity.

Forward Reasoning, Backward Reasoning, Resolution

1. fire any rule whose premises are satisfied in the KB.
2. add its conclusion to the KB, until query is found.

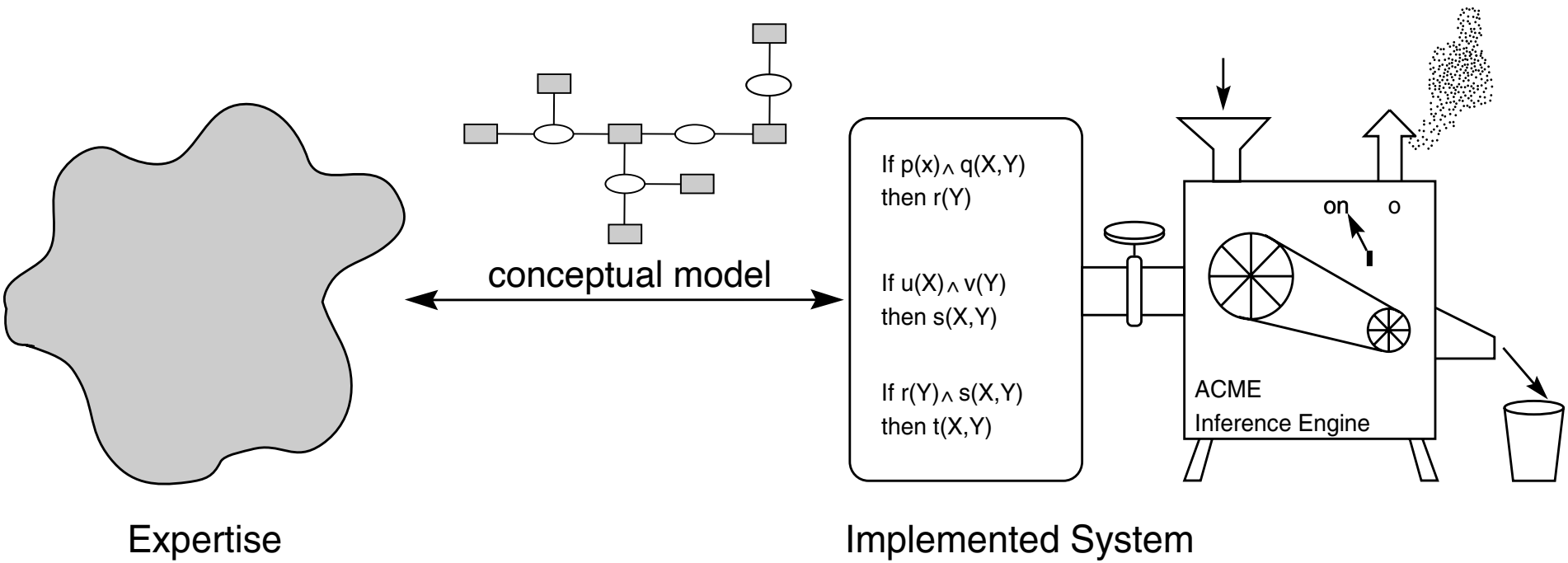
- Forward reasoning is **data-driven**, cf. automatic, unconscious processing.
  - e.g. object recognition, routine decisions

  May do lots of work that is irrelevant to the goal.
- Backward reasoning is **goal-driven**, appropriate for problem-solving.
  - e.g. Where are my keys? How do I get into a PhD program?
- Resolution Algorithm

$$\frac{A \vee C \quad B \vee \neg C}{A \vee B}$$

  Proof by contradiction, i.e. show $KB, \neg A$ unsatisfiable.

Example: Expert System



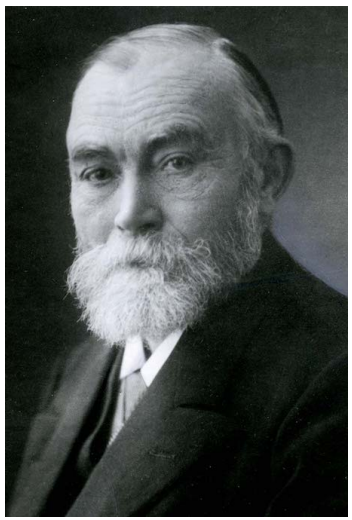
弗雷格 Gottlob Frege 1848-1925

皮尔士 Charles Peirce 1839-1914

- 《概念文字: 一种模仿算术语言构造的纯思维的形式语言》*1879.*
- 逻辑主义 Mathematics $\rightsquigarrow$ Logic.[a]
- 谓词逻辑之父
  (关系 & 量词)
  (Every boy loves some girl.)

$$\frac{\text{subject}}{\text{predicate}} \approx \frac{\text{argument}}{\text{function}}$$

- 语言哲学
  The evening star is the morning star.[b]



[a] Frege: The Foundations of Arithmetic. 1884.

[b] Frege: On Sense and Reference. 1892.

More Expressive Logic?

- 命题逻辑预设世界由事实构成
- 谓词逻辑预设世界包含
  1. 个体: 人、狗、书、自然数、实数、城市、国家...
  2. 关系: 红的、圆的、大于、爱上、父子、朋友、老师...
  3. 函数: 平方、加法、母亲、老婆、最好的朋友、导师...
- 谓词逻辑的表达力更强

$$\frac{\text{father}(\text{father}(\text{alice})) = \text{father}(\text{mother}(\text{bob}))}{\text{Cousin}(\text{alice}, \text{bob})}$$

语言	本体论承诺	认识论承诺
Propositional Logic	facts	true/false/unknown
Predicate Logic	facts, objects, relations	true/false/unknown
Temporal Logic	facts, objects, relations, times	true/false/unknown
Probability Theory	facts	degree of belief $[0, 1]$
Fuzzy Logic	facts with degree of truth $[0, 1]$	known interval value

Reducing First Order Inference to Propositional Inference

- Universal Instantiation

$$\frac{\forall x A}{A[t/x]} \text{ where } t \text{ is a closed term}$$

  - can be applied several times to add new sentences
  - the new KB is logically equivalent to the old

- Existential Instantiation

$$\frac{\exists x A}{A(a)} \text{ where } a \text{ is a new constant}$$

  - can be applied once to replace the existential sentence
  - the new KB is not equivalent to the old
  - but is satisfiable iff the old KB was satisfiable

Reducing First Order Inference to Propositional Inference

- Claim: a sentence is entailed by the new KB iff it is entailed by the original KB
- Claim: every FOL KB can be propositionalized so as to preserve entailment

Theorem (Herbrand's Theorem)

If a sentence A is entailed by an FOL KB, it is entailed by a finite subset of the propositional KB.

- Idea: propositionalize KB and query, apply resolution, return result

for $n = 0$ to ∞ do
  create a propositional KB by instantiating with depth-n terms see if A is entailed by this KB

- Problem: works if A is entailed, loops if A is not entailed
- Theorem (Turing, Church): entailment in FOL is semidecidable.

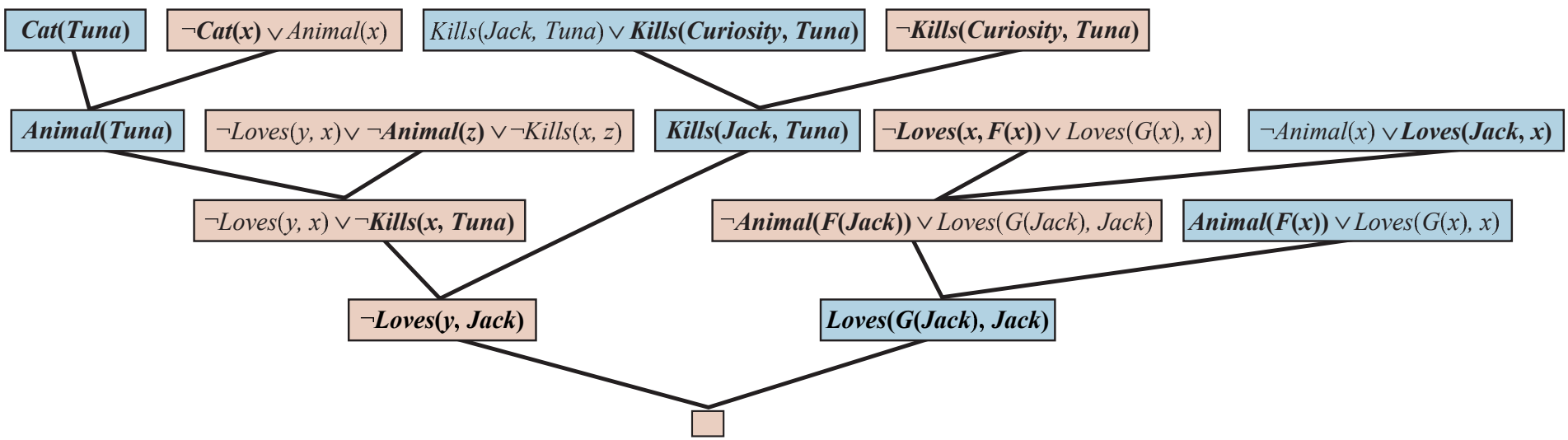
Resolution — Example

- Everyone who loves all animals is loved by someone.
- Anyone who kills an animal is loved by no one.
- Jack loves all animals.
- Either Jack or Curiosity killed the cat, who is named Tuna.
- Did Curiosity kill the cat?

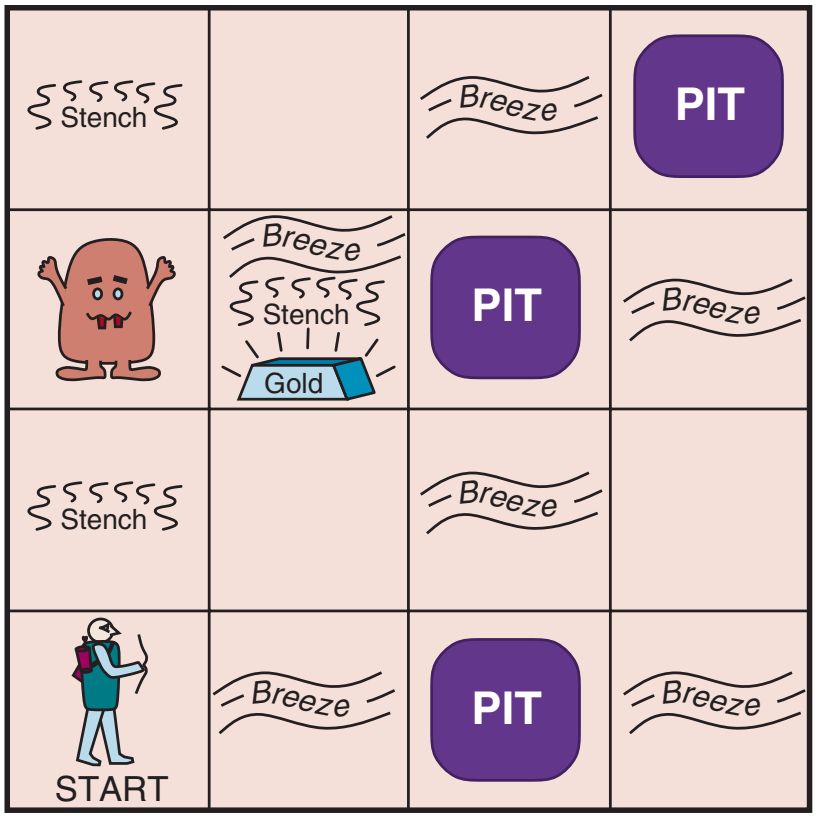
1. $\forall x(\forall y(\text{Animal}(y) \rightarrow \text{Love}(x, y)) \rightarrow \exists z \text{Love}(z, x))$
2. $\forall x((\exists z \text{Animal}(z) \land \text{Kill}(x, z)) \rightarrow \forall y \neg \text{Love}(y, x))$
3. $\forall x(\text{Animal}(x) \rightarrow \text{Love}(\text{Jack}, x))$
4. $\text{Kill}(\text{Jack}, \text{Tuna}) \lor \text{Kill}(\text{Curiosity}, \text{Tuna})$
5. $\text{Cat}(\text{Tuna})$
6. $\forall x(\text{Cat}(x) \rightarrow \text{Animal}(x))$
7. $\neg \text{Kill}(\text{Curiosity}, \text{Tuna})$

Resolution — Example

1. $\Big(\text{Animal}(F(x)) \vee \text{Love}(G(x), x)\Big) \wedge \Big(\neg\text{Love}(x, F(x)) \vee \text{Love}(G(x), x)\Big)$
2. $\neg\text{Love}(y, x) \vee \neg\text{Animal}(z) \vee \neg\text{Kill}(x, z)$
3. $\neg\text{Animal}(x) \vee \text{Love}(\text{Jack}, x)$
4. $\text{Kill}(\text{Jack}, \text{Tuna}) \vee \text{Kill}(\text{Curiosity}, \text{Tuna})$
5. $\text{Cat}(\text{Tuna})$
6. $\neg\text{Cat}(x) \vee \text{Animal}(x)$
7. $\neg\text{Kill}(\text{Curiosity}, \text{Tuna})$



Example: The Wumpus World



- squares adjacent to wumpus are smelly
- squares adjacent to pit are breezy
- glitter iff gold is in the same square
- shooting kills wumpus if you are facing it
- shooting uses up the only arrow
- grabbing picks up gold if in same square
- releasing drops the gold in same square

KB = wumpus-world rules + observations

Example: $\forall x(\text{Breeze}(x) \leftrightarrow \exists y(\text{Pit}(y) \wedge \text{Adjacent}(y, x)))$

Frame Problem

- **框架问题:** 哪些状态在行动后保持不变?
  - 如果 “前进一步”, 那么 “背上的箭还在”、“墙壁的颜色还是黄的”.....
  - representation: too many frame axioms
  - inference: too many repeated “copy-overs” to keep track of state
- **限制问题:** 无法穷尽行动所需预设的所有前提条件
  - “前进一步” — 但会不会脚滑? 会不会踩到钉子? .....
- **分枝问题:** 行动可能引发许多伴随的次生后果
  - “前进一步” — 鞋子会不会磨损? 裤子上落上的苍蝇会不会被带着向前? .....

Cognitive Wheels: The Frame Problem of AI — Dennett

- 房间里有一块电池, 房间外的机器人快没电了. 电池跟炸弹绑在一起.
- 机器人 1 号: "取出电池充电". 炸弹一起被取出.....爆炸了.
- 机器人 2 号: "做一个动作时, 要考虑它的副作用". — 取出电池时, 墙壁会变色吗? 天花板会塌吗? .....爆炸了.
- 机器人 3 号: "只考虑跟任务相关的事". .....什么跟任务相关, 什么不相关? 墙壁相关吗? 天花板相关吗? .....爆炸了.
- $\text{Forward}^t \rightarrow (\text{WumpusAlive}^t \rightarrow \text{WumpusAlive}^{t+1})$
- $\text{Forward}^t \rightarrow (\text{HaveArrow}^t \rightarrow \text{HaveArrow}^{t+1})$
- $\text{Forward}^t \rightarrow (\text{WallYellow}^t \rightarrow \text{WallYellow}^{t+1})$

Successor-state axiom:

$$\text{HaveArrow}^{t+1} \leftrightarrow (\text{HaveArrow}^t \wedge \neg \text{Shoot}^t)$$

Contents

1. Introduction
   - History of AI
   - What is AI?
   - Turing Machine
   - Rational Agent
   - Goal-Based Agent
   - Knowledge-Based Agent
   - Knowledge Representation
   - Learning Agent
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Knowledge Representation

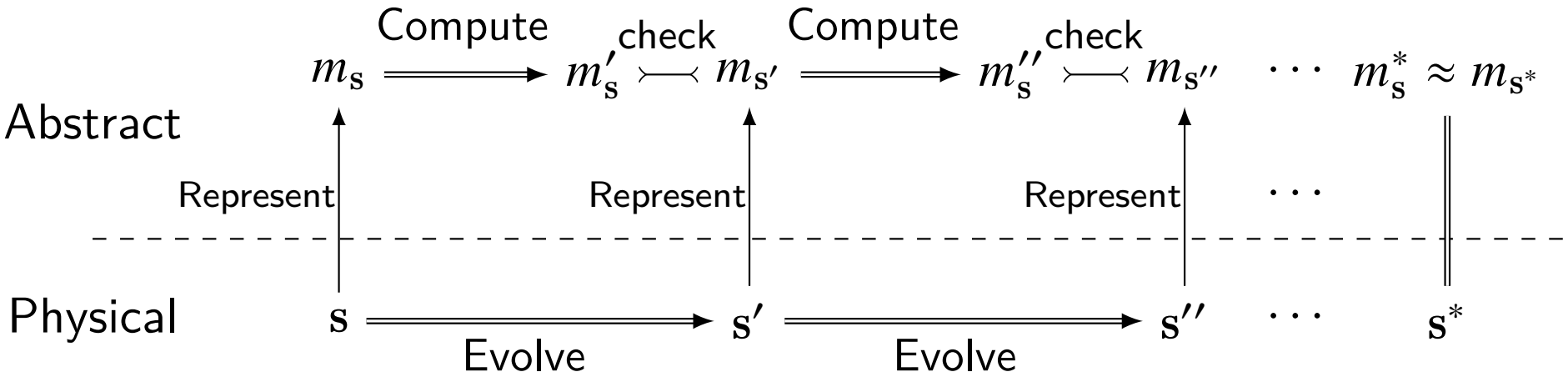
How to represent diverse facts about the world in a form that can be used to reason with this knowledge to achieve its goals?

- How do we describe the current state of the world?
- How do we infer from our percepts, knowledge of unseen parts of the world?
- How does the world change as time passes?
- How does the world stay the same as time passes? (The frame problem.)
- How do we know the effects of our actions? (The qualification and ramification problems.)

Abstract Representation of the Physical World

"Logic void of representation is metaphysics."

— Judea Pearl



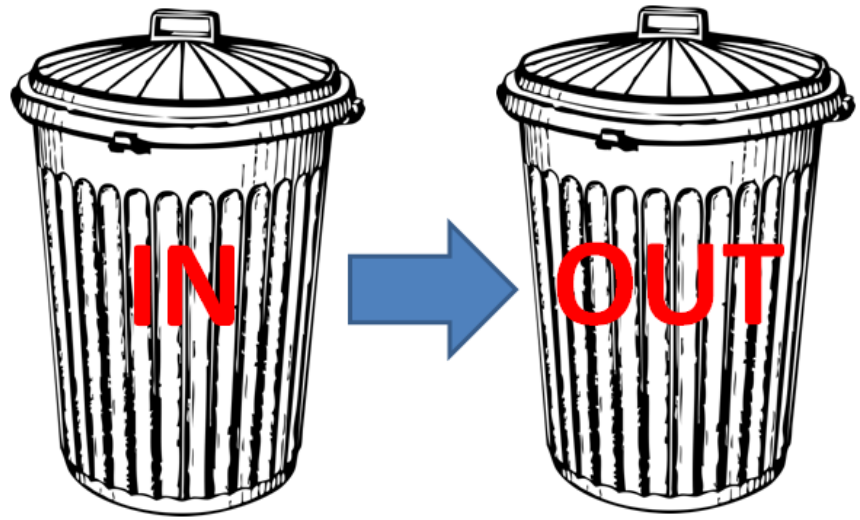
Representations should be expressive and efficient.

The essence of intelligence is while only being able to observe a world of things, try to come up with a world of ideas.

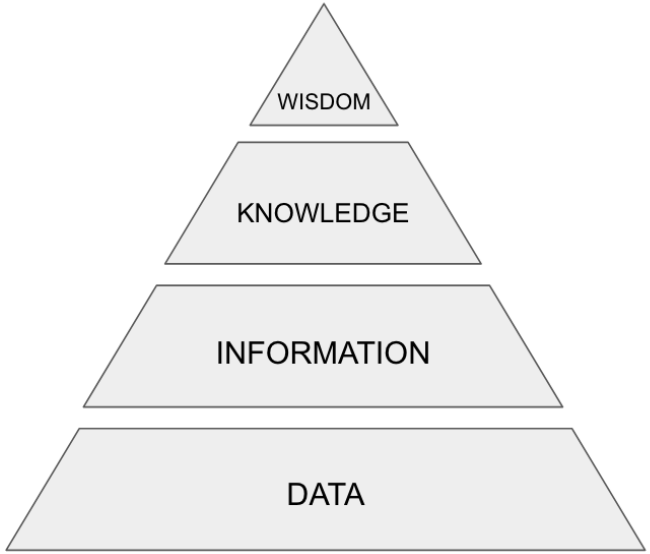
— *Vladimir Vapnik*

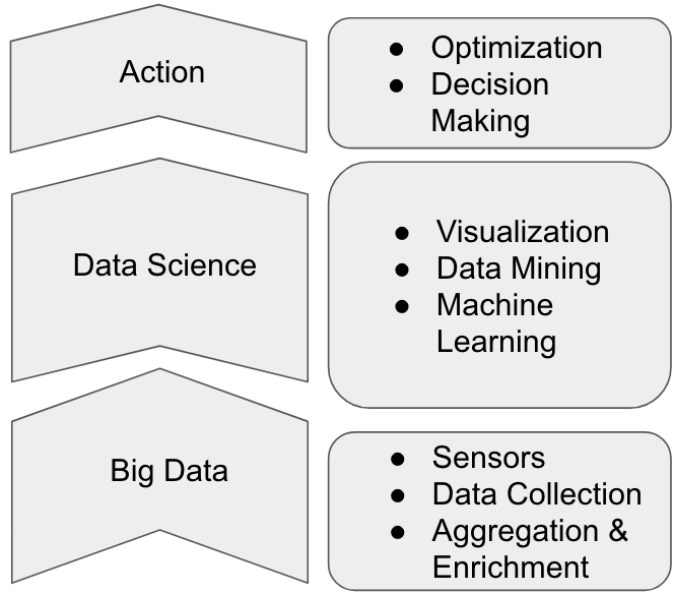
Where is the Life we have lost in living?
Where is the wisdom we have lost in knowledge?
Where is the knowledge we have lost in information?

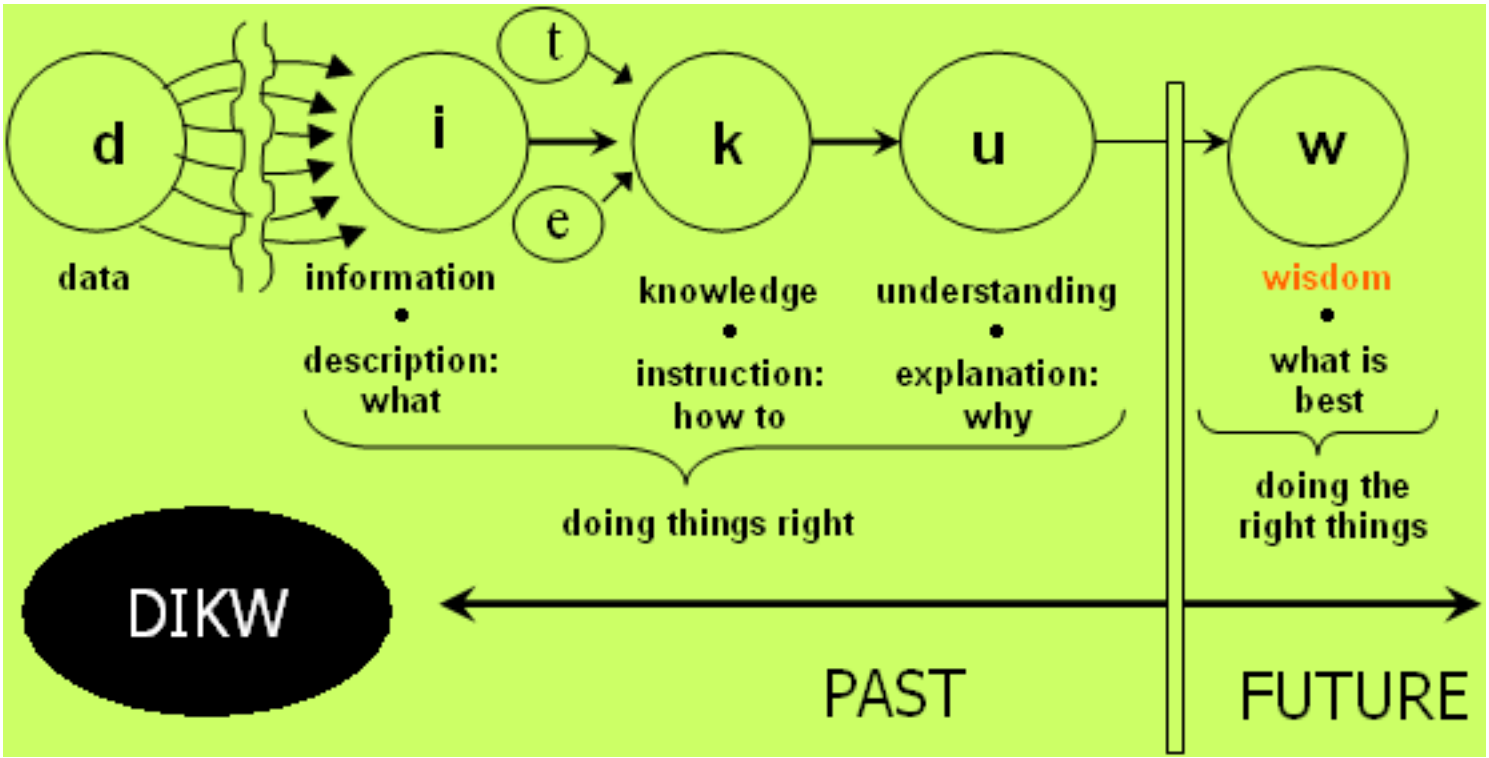
— *T. S. Eliot*



- We are drowning in information and starving for knowledge.
- A wealth of information creates a poverty of attention.







获取有用信息 → 由信息生成知识 → 由知识和目标生成策略 → 实施策略取得效果

选取知识表示的因素

- 丰富性：表示范围是否广泛，能否解决问题
- 是否适于推理
- 是否适于计算机处理
- 是否有高效的算法，权衡准确性与计算时间
- 能否表示不确定性知识
- 能否模块化
- 知识、元知识能否用统一的形式表示
- 是否加入启发信息
- 过程性表示还是说明性表示
- 表示方法是否自然、紧凑、易维护，容易看到表示与被表示的域之间的关系
- 易获得，从数据、感知、经验、人或其它知识源获得知识

Types of Knowledge Representation

- categories & objects $\longrightarrow$ ontologies
- frames
- events: fluents, time interval, scripts
- procedures
- relations
- mental states
- meta knowledge

A Brief History of Ontology

Kant	Reality is unknowable. Metaphysics is impossible. We can only know the quasi-fictional domains which we ourselves create. The mind imposes its categories on the data of experience.
Russell	Logic as mirror of reality.
Wittgenstein	Centrality of language and of language games.
Tarski	Semantic theory of truth.
Quine	Ontological commitment (study not: what there is, but: what sciences believe there is when logically formalized)
Hayes	Ontology to build Robots. Ontological commitment = study what people believe there is when logically formalized.

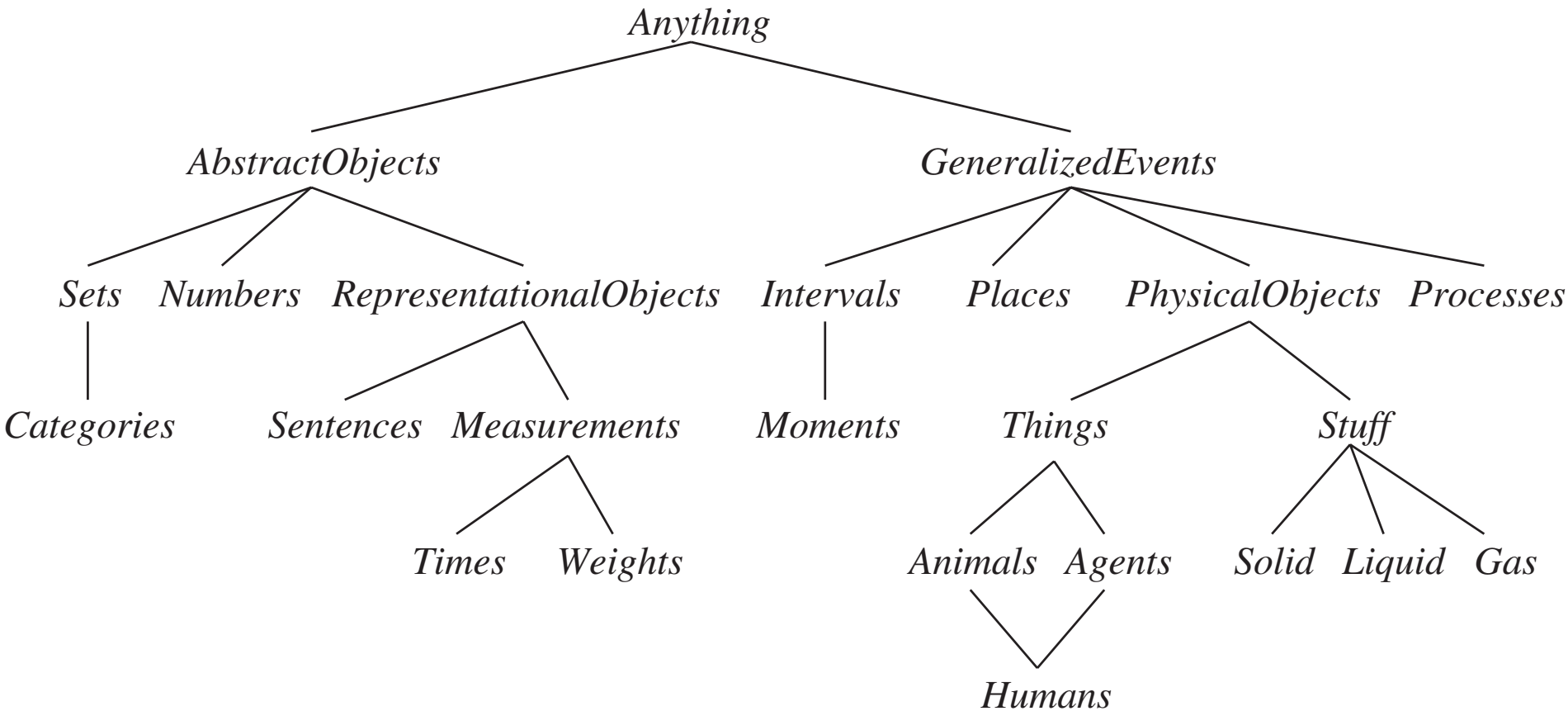
Conceptualization & Ontology

Galileo Galilei's "Leaning Tower of Pisa" — What is a Thing?

If we tied a light stone to a heavy stone, it would slow the heavy stone down because it falls slower. But the whole thing is heavier than its parts.

- A knowledge representation is a set of ontological commitments.
- A conceptualization is a map from the problem domain into the representation.
- An ontology is a specification of a conceptualization.
  1. Decide what to talk about
  2. Decide on a vocabulary of constants, functions and predicates
  3. Encode general knowledge about the domain
  4. Encode a description of the specific problem instance
  5. Pose queries to the inference procedure and get answers
- Good representations
  - preserve the relevant aspects of the problem
  - expose the relevant problem structure

上位本体论


Anything
AbstractObjects
GeneralizedEvents
Sets
Numbers
RepresentationalObjects
Intervals
Places
PhysicalObjects
Processes
Categories
Sentences
Measurements
Moments
Things
Stuff
Times
Weights
Animals
Agents
Solid
Liquid
Gas
Humans

知识表示

本体论工程

- 亚里士多德《范畴篇》：实体、数量、性质、关系、时间、地点、姿态、状态、行动、承受.
- A knowledge representation is a set of ontological commitments.
- “苹果” 是 “水果” 的子类, 可以继承 “水果” 的性质.
- 复合对象的类别经常由其各部分之间的结构关系刻画. 比如: 两足动物是一个身体上有两条腿的动物.
- 有时需要定义有确定部分但没有具体结构的复合对象. 比如:
  — “袋子里的苹果两斤重.”
  — 但把两斤重归于袋子里苹果的集合就是错的, 因为集合 (set) 是抽象的数学概念, 没有重量.
  — 需要定义一个新概念 “bunch”.

$$\forall x \Big(x \in s \rightarrow \mathrm{PartOf}(x, \mathrm{BunchOf}(s)) \Big)$$

$$\forall y \Big((\forall x : x \in s \rightarrow \mathrm{PartOf}(x, y)) \rightarrow \mathrm{PartOf}(\mathrm{BunchOf}(s), y) \Big)$$

(minimization)

$$\mathrm{BunchOf}(\{x\}) = x$$

“自然种类” 怎么刻画?

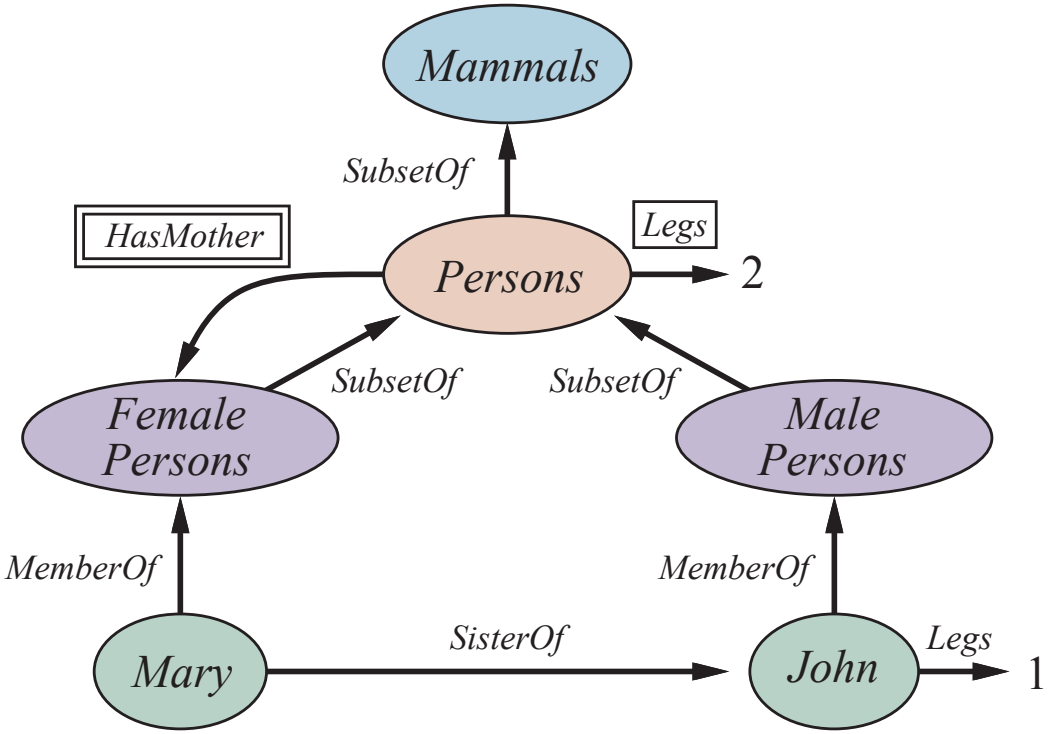
- 自然种类难以严格定义.
  — 一个 Agent 可能无法通过感知到的性质判断一个对象是不是西红柿.
  — 即使已知它是西红柿了, 也不能确定它是否有西红柿的典型特征.
  — 大多关于自然种类的知识其实是关于其典型实例的知识:

$$\mathsf{Tipical}(x) \subset x$$

$$x \in \mathsf{Tipical}(\mathsf{tomatoes}) \rightarrow \mathsf{Red}(x) \wedge \mathsf{Round}(x)$$

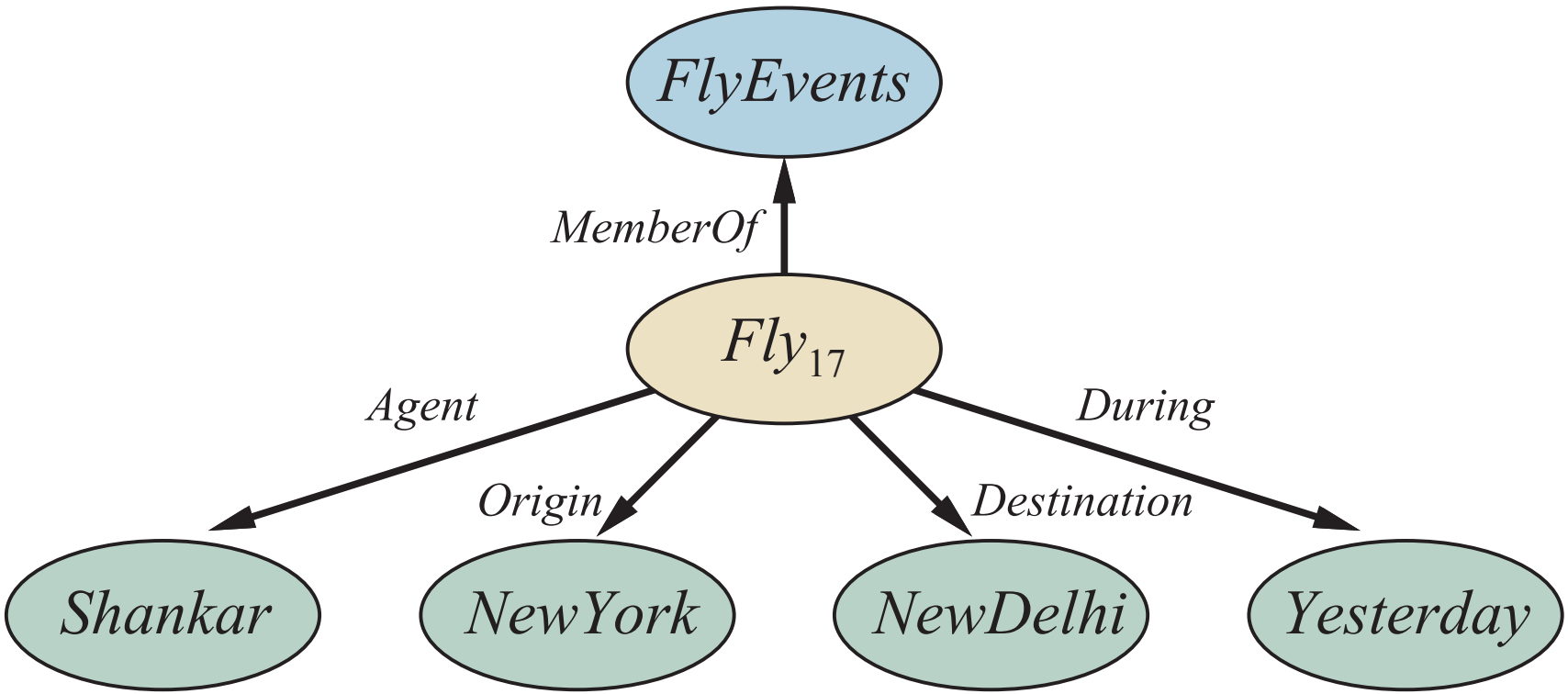
- 怎么定义 “游戏”? — “家族相似” — 维特根斯坦
- 怎么定义 “单身汉”? — 教皇是单身汉吗? — 奎因

语义网络



$A \xrightarrow{\text{SubsetOf}} B$	$A \subset B$
$A \xrightarrow{\text{MemberOf}} B$	$A \in B$
$A \xrightarrow{R} B$	$R(A, B)$
$A \xrightarrow{\boxed{R}} B$	$\forall x : x \in A \rightarrow R(x, B)$
$A \xrightarrow{\boxed{\boxed{R}}} B$	$\forall x \exists y : x \in A \rightarrow y \in B \wedge R(x, y)$

Semantic Network Example: Event Category



$\text{Apple} \sqsubset \text{Round} \sqcap (\text{Red} \sqcup \text{Green}) \sqcap (\text{Sweet} \sqcup \text{Sour} \sqcup \text{Bitter}) \sqcap (\text{Ripe} \sqcup \neg\text{Ripe})$

A happy man is a man that is married to a smart beauty, and all of whose children are either doctors or professors, and at least two children are professors.

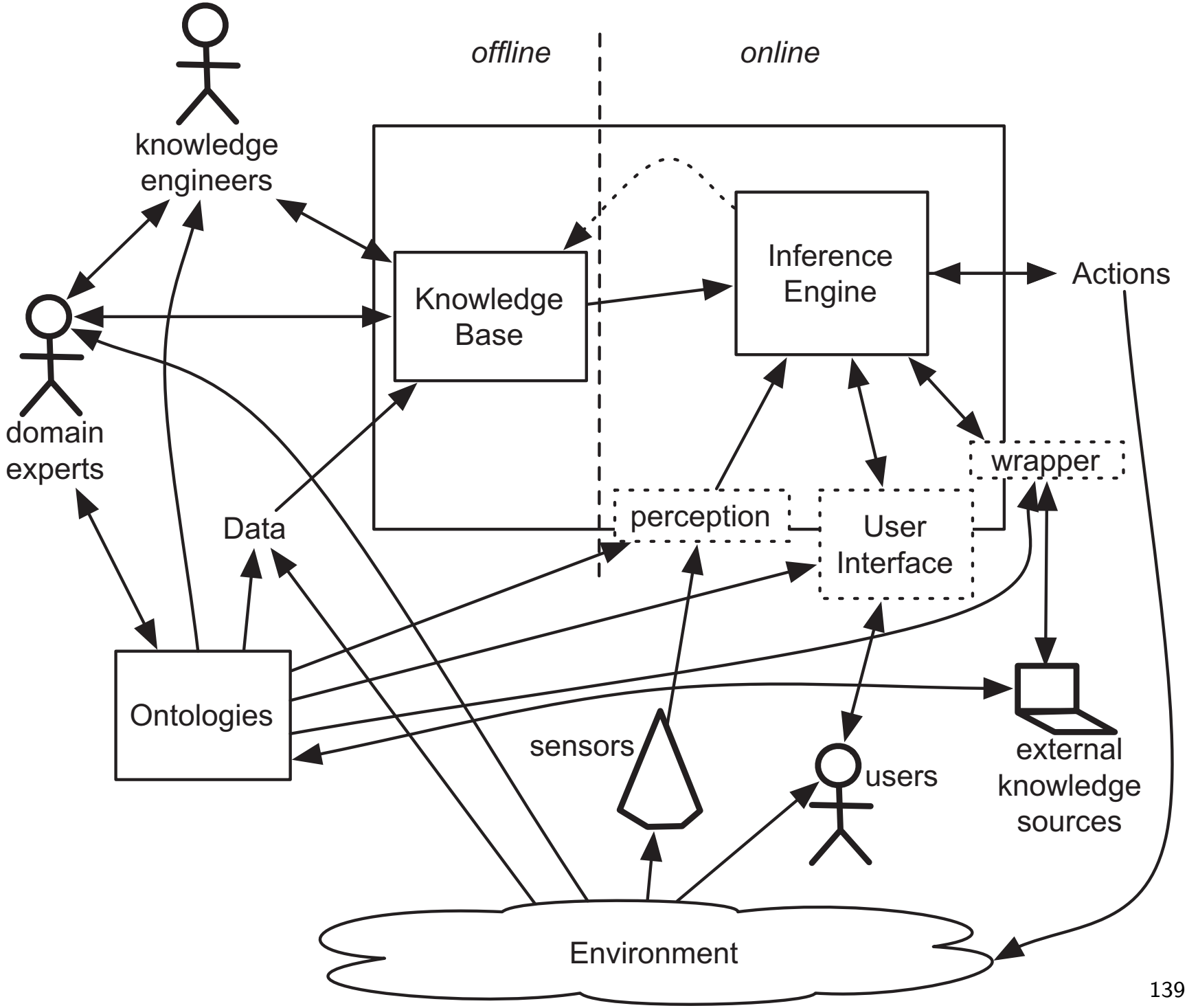
$$\text{HappyMan} \equiv \text{Human} \sqcap \neg\text{Female} \sqcap \big(\exists \text{Married}.(\text{Smart} \sqcap \text{Beauty})\big) \sqcap \big(\forall \text{HasChild}.(\text{Doctor} \sqcup \text{Professor})\big) \sqcap \geq 2\,\text{HasChild}.\text{Professor}$$

DL	FOL	Semantics
$\top$	$\top$	$\Delta^{\mathcal{I}}$
$\bot$	$\bot$	$\varnothing$
$\neg C$	$\neg Cx$	$\Delta^{\mathcal{I}} \setminus C^{\mathcal{I}}$
$C \sqcap D$	$Cx \wedge Dx$	$C^{\mathcal{I}} \cap D^{\mathcal{I}}$
$C \sqcup D$	$Cx \vee Dx$	$C^{\mathcal{I}} \cup D^{\mathcal{I}}$
$\forall R.C$	$\forall y(Rxy \rightarrow Cy)$	$\{x : \forall y.(x,y) \in R^{\mathcal{I}} \rightarrow y \in C^{\mathcal{I}}\}$
$\exists R.C$	$\exists y(Rxy \wedge Cy)$	$\{x : \exists y.(x,y) \in R^{\mathcal{I}} \ \wedge \ y \in C^{\mathcal{I}}\}$
$C \sqsubset D$	$\forall x(Cx \rightarrow Dx)$	$C^{\mathcal{I}} \subset D^{\mathcal{I}}$

Description Logic — continued

DL	FOL
R^-	$R^- xy \leftrightarrow Ryx$
$R \circ S$	$\exists y.Rxy \wedge Syz$
$\leq nR.C$	$\forall y_1 \ldots y_{n+1} \left(\bigwedge_{1 \leq i \leq n} Rxy_i \wedge Cy_i \rightarrow \bigvee_{1 \leq i < j \leq n+1} y_i = y_j \right)$
$\geq nR.C$	$\exists y_1 \ldots y_n \left(\bigwedge_{1 \leq i \leq n} Rxy_i \wedge Cy_i \wedge \bigvee_{1 \leq i < j \leq n} y_i \neq y_j \right)$

DL	Semantics
R^-	$\left\{(y,x) : (x,y) \in R^{\mathcal{I}}\right\}$
$R \circ S$	$\left\{(x,z) : \exists y.(x,y) \in R^{\mathcal{I}} \ \wedge \ (y,z) \in S^{\mathcal{I}}\right\}$
$\leq nR.C$	$\left\{x \in \Delta^{\mathcal{I}} : \left\lvert\left\{y \in \Delta^{\mathcal{I}} : (x,y) \in R^{\mathcal{I}} \ \wedge \ y \in C^{\mathcal{I}}\right\}\right\rvert \leq n\right\}$
$\geq nR.C$	$\left\{x \in \Delta^{\mathcal{I}} : \left\lvert\left\{y \in \Delta^{\mathcal{I}} : (x,y) \in R^{\mathcal{I}} \ \wedge \ y \in C^{\mathcal{I}}\right\}\right\rvert \geq n\right\}$


offline
online
knowledge engineers
domain experts
Knowledge Base
Inference Engine
Actions
wrapper
perception
User Interface
Data
Ontologies
sensors
users
external knowledge sources
Environment

Knowledge Base — Example

R-Box: "role inclusion"

- $\mathsf{Owns} \sqsubset \mathsf{CaresFor}$
  If somebody owns something, they care for it.
- $\mathsf{AncesterOf} \circ \mathsf{AncesterOf} \sqsubset \mathsf{AncesterOf}$

T-Box: "concept inclusion"

- $\mathsf{Bachelor} \equiv \neg\exists \mathsf{Married}.\top \sqcap \mathsf{Man}$
  Bachelors are unmarried men.
- $\mathsf{HappyCatOwner} \sqsubset \exists \mathsf{Owns}.\mathsf{Cat} \sqcap \forall \mathsf{CaresFor}.\mathsf{Healthy}$
  A happy cat owner owns a cat and all beings he cares for are healthy.

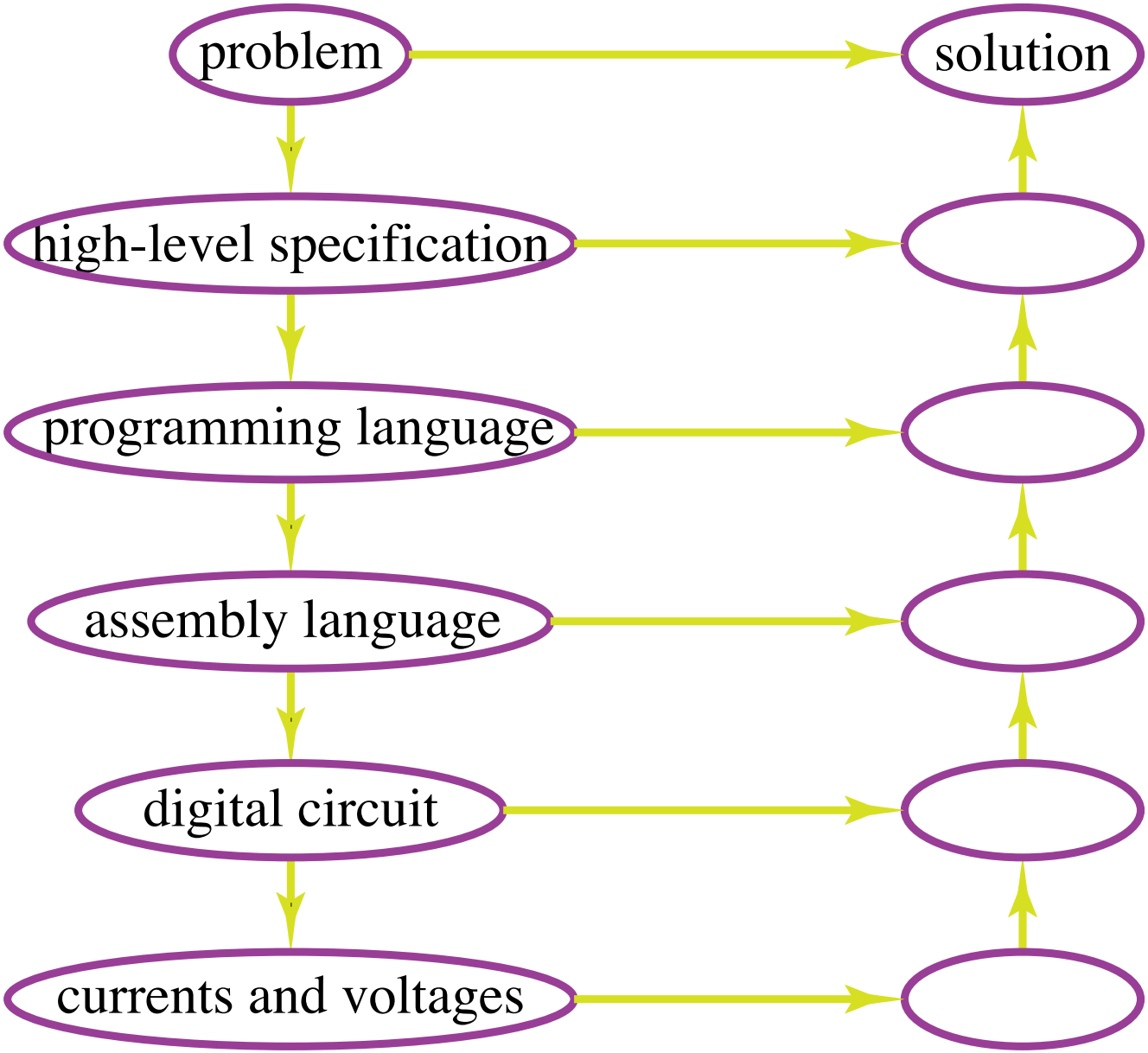
A-Box: "individual assertion"

- $\mathsf{Love}(\mathsf{tom}, \mathsf{jerry})$
  Tom loves Jerry.
- $\mathsf{HappyCatOwner}(\mathsf{schrödinger})$
  Schrödinger is a happy cat owner.

Semantic Web & Ontology — Problems

- How one divides the world can depend on the application.
- Different ontologies describe the world in different ways.
- To allow KBs based on different ontologies to inter-operate, there must be mapping between different ontologies.
- The formalism can constrain the meaning, but can't define it.

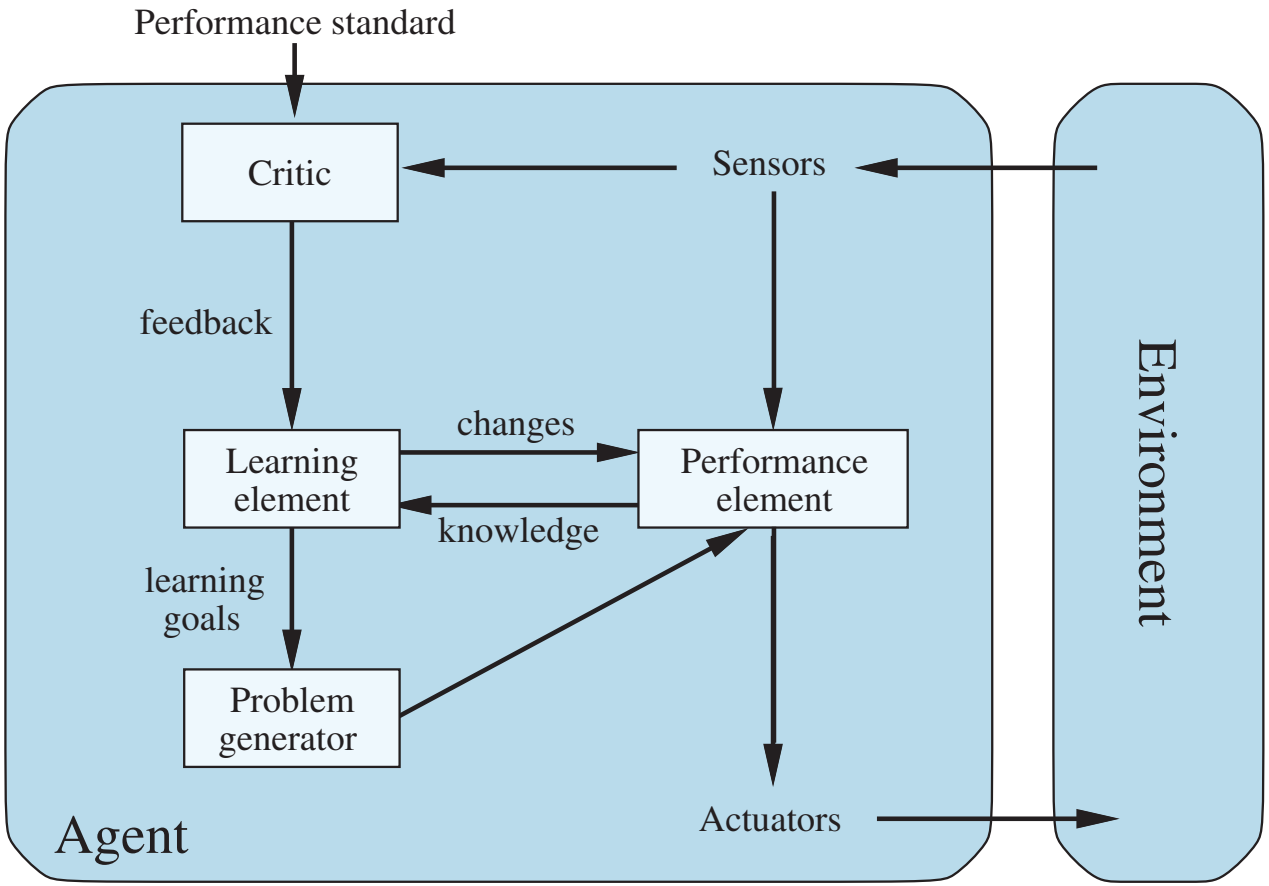
Hierarchy of Representations



Contents

1. Introduction
   - History of AI
   - What is AI?
   - Turing Machine
   - Rational Agent
   - Goal-Based Agent
   - Knowledge-Based Agent
   - Knowledge Representation
   - Learning Agent
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Learning Agent

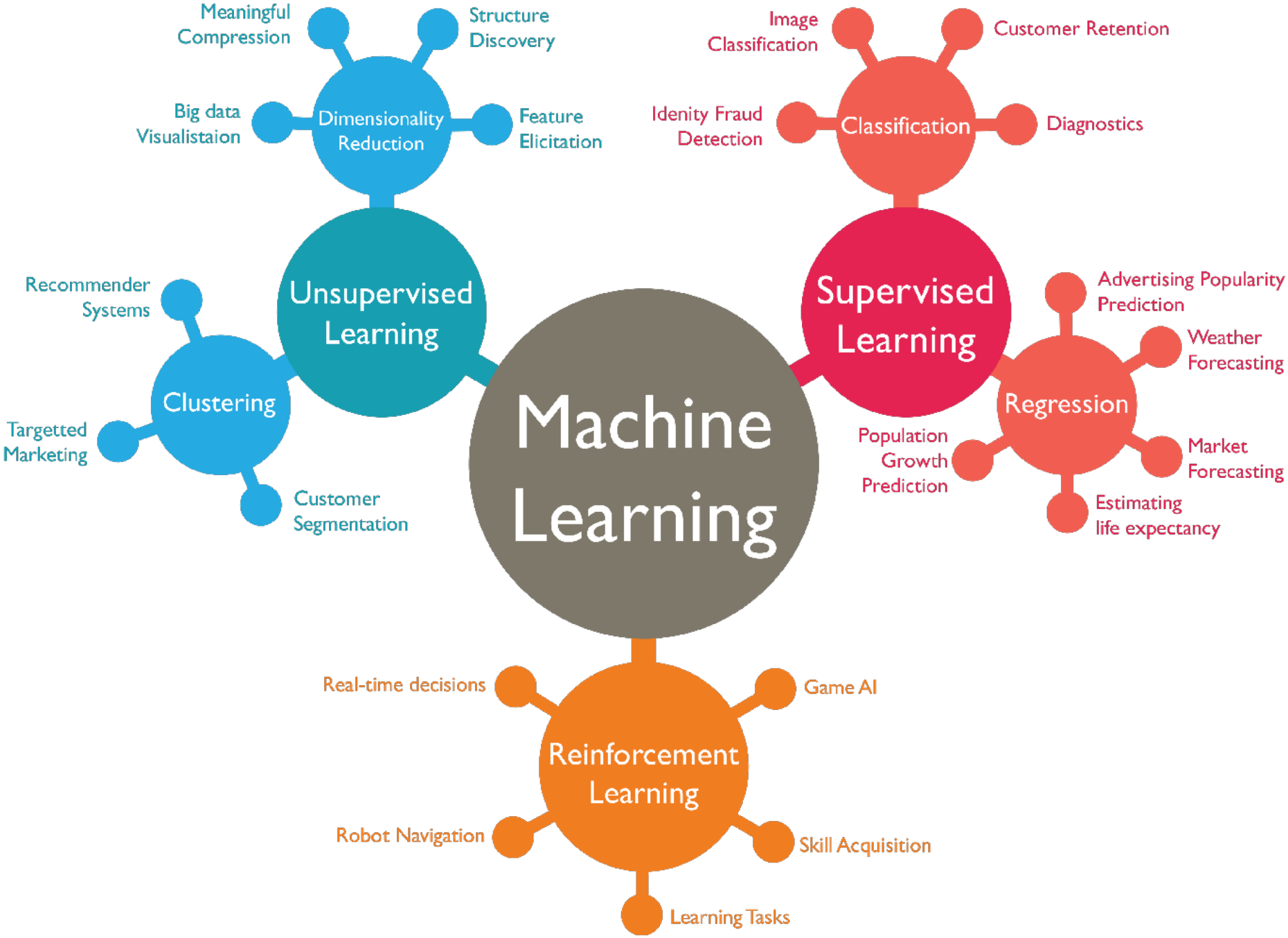


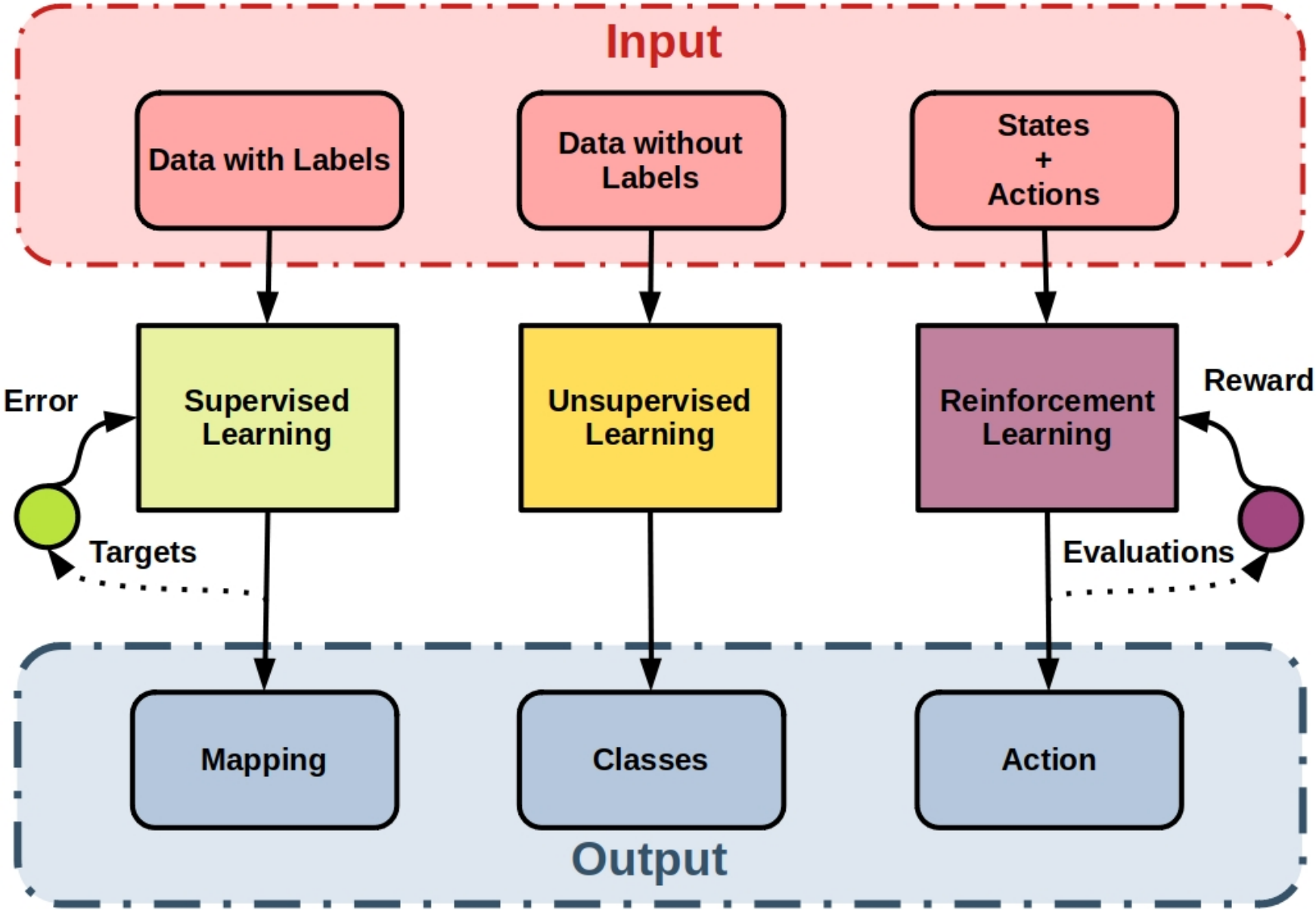
- **performance element** (it represents what we have previously considered to be the entire agent)
- **learning element** (responsible for making improvements)
- **critic** (evaluation of the agent's behavior)
- **problem generator** (suggests explorative actions)

Types of Feedback During Learning

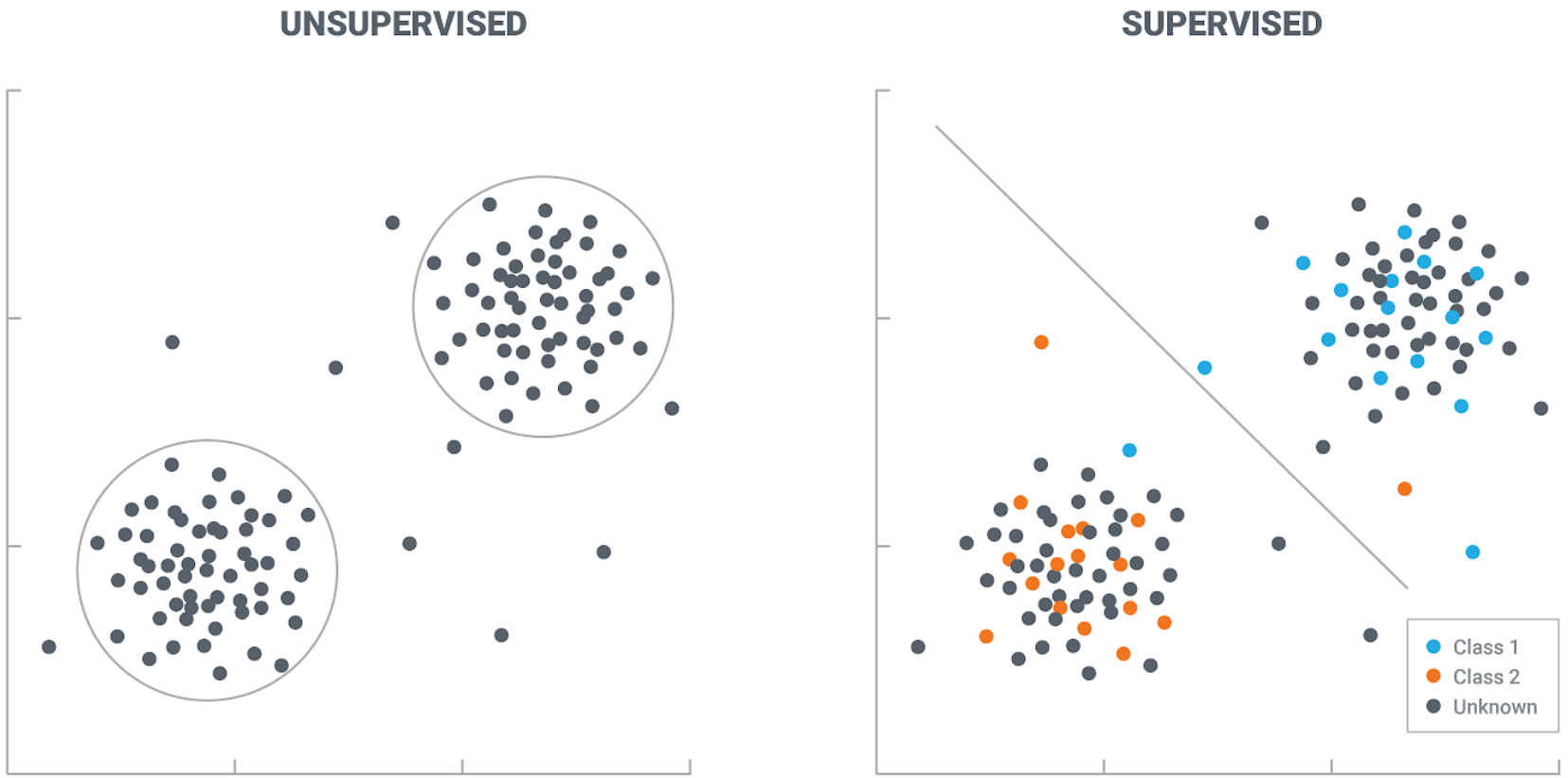
A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance P on task T in environment Z, improves with experience E.

- Supervised Learning
  - Learn the relationship between "input" x and "output" y.
    — search for a function f, such that $y \approx f(x)$
  - There is training data with labels available
    Regression: learning f with real-valued output value
    Classification: learning f with discrete output value
  - Semi-supervised learning: also uses available unlabeled data, e.g. assumes that similar inputs have similar outputs.
- Unsupervised Learning
  - There exist no outputs, search for patterns within the inputs x
    Clustering: find groups of similar items
    Dimensionality reduction: describe data in fewer features
    Outlier detection: what is out of the ordinary?
    Association rules: which things often happen together?
- Reinforcement learning

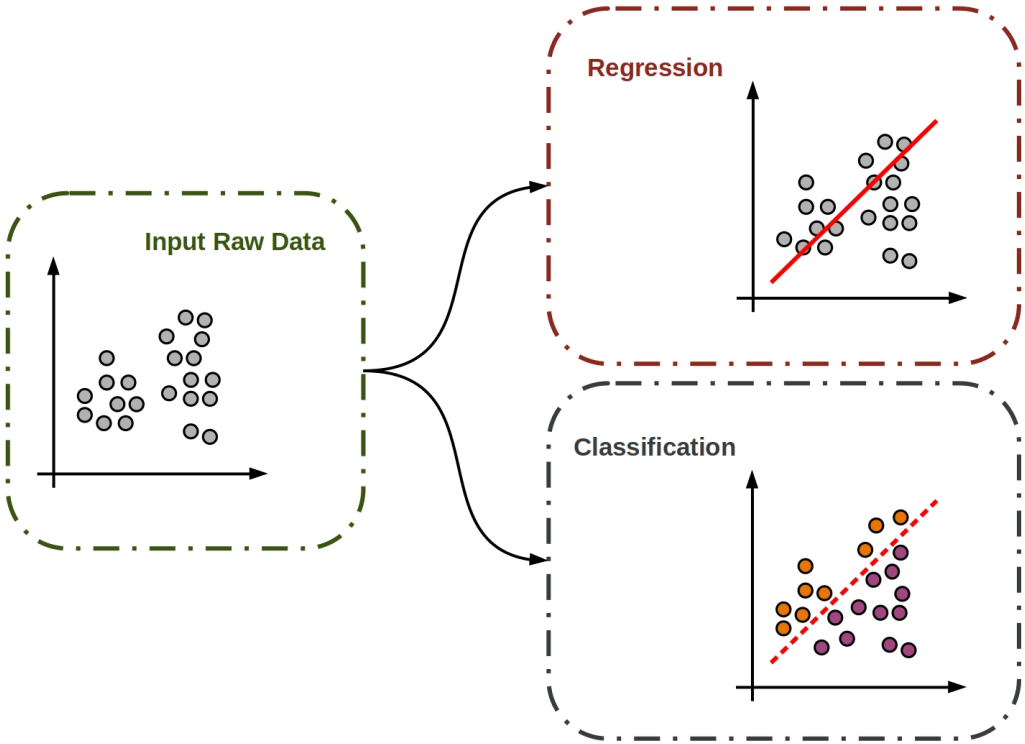

Machine Learning
Unsupervised Learning
Supervised Learning
Reinforcement Learning
Dimensionality Reduction
Clustering
Classification
Regression
Meaningful Compression
Structure Discovery
Big data Visualistaion
Feature Elicitation
Recommender Systems
Targetted Marketing
Customer Segmentation
Image Classification
Customer Retention
Idenity Fraud Detection
Diagnostics
Advertising Popularity Prediction
Weather Forecasting
Population Growth Prediction
Market Forecasting
Estimating life expectancy
Real-time decisions
Game AI
Robot Navigation
Skill Acquisition
Learning Tasks

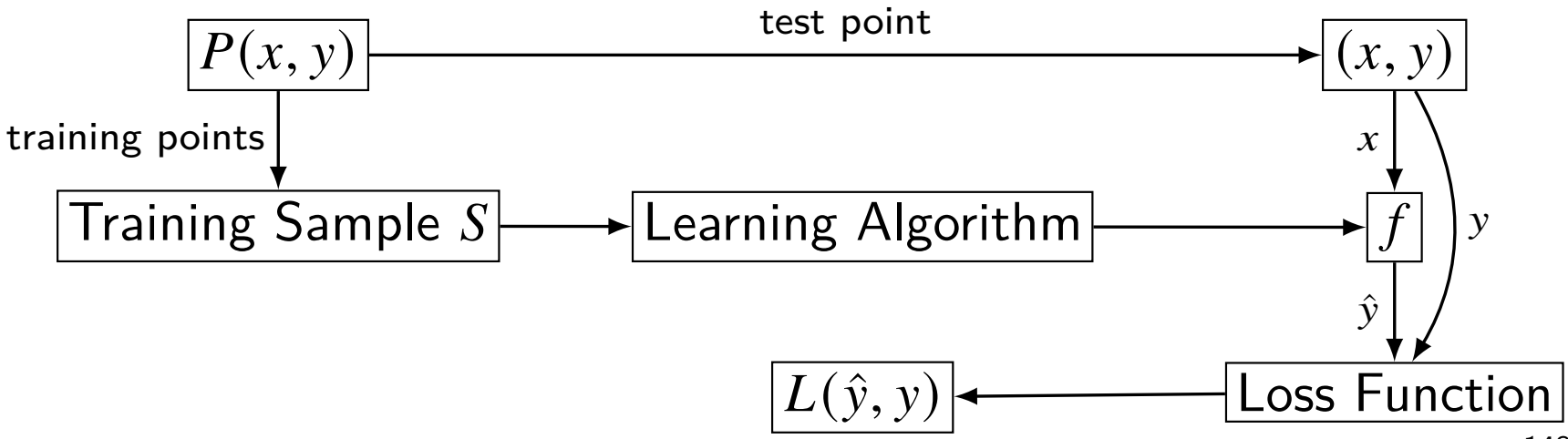

Input
Data with Labels
Data without Labels
States + Actions
Error
Supervised Learning
Unsupervised Learning
Reinforcement Learning
Reward
Targets
Evaluations
Mapping
Classes
Action
Output

Supervised Learning vs Unsupervised Learning

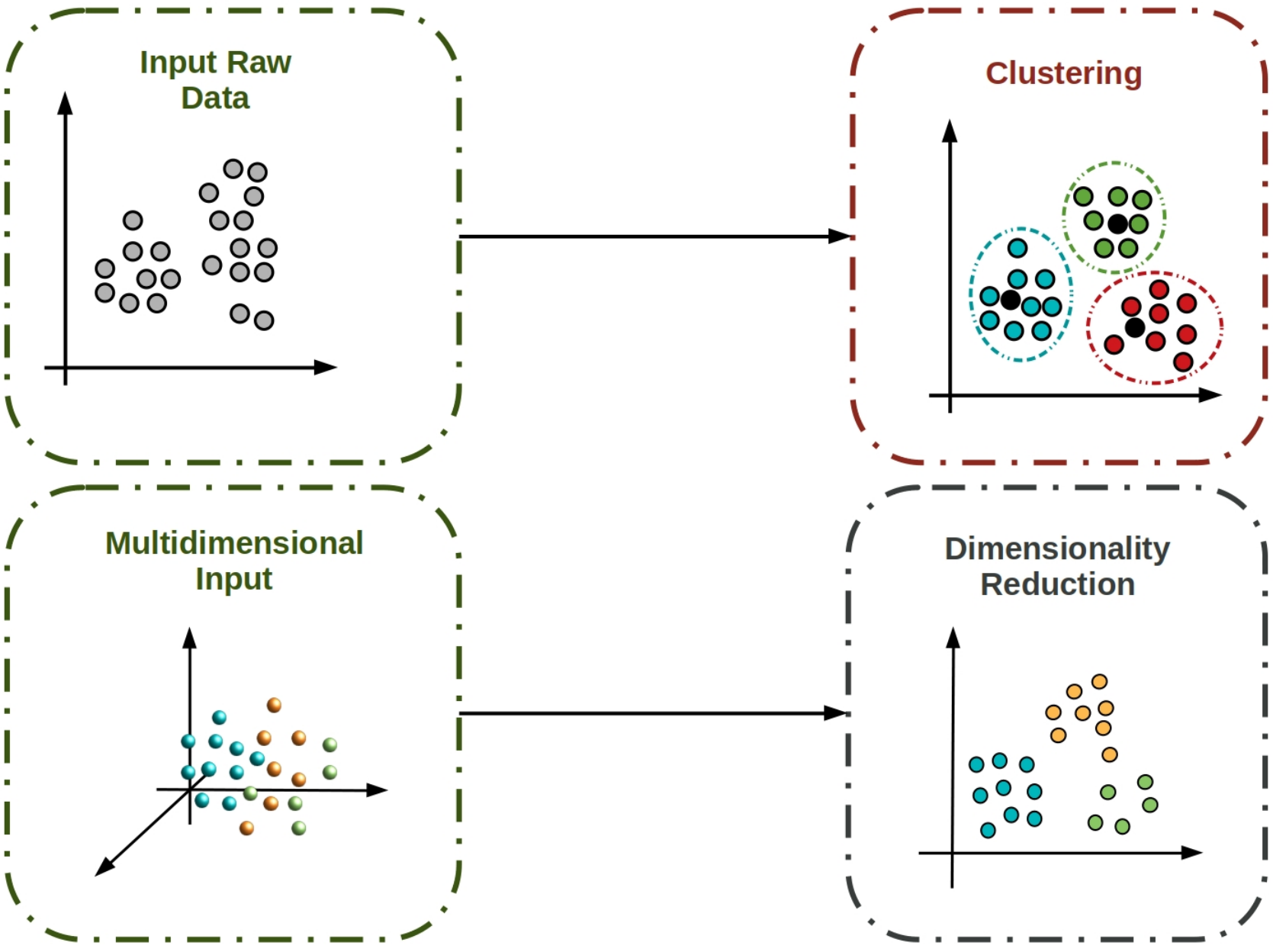


Supervised Learning





Unsupervised Learning

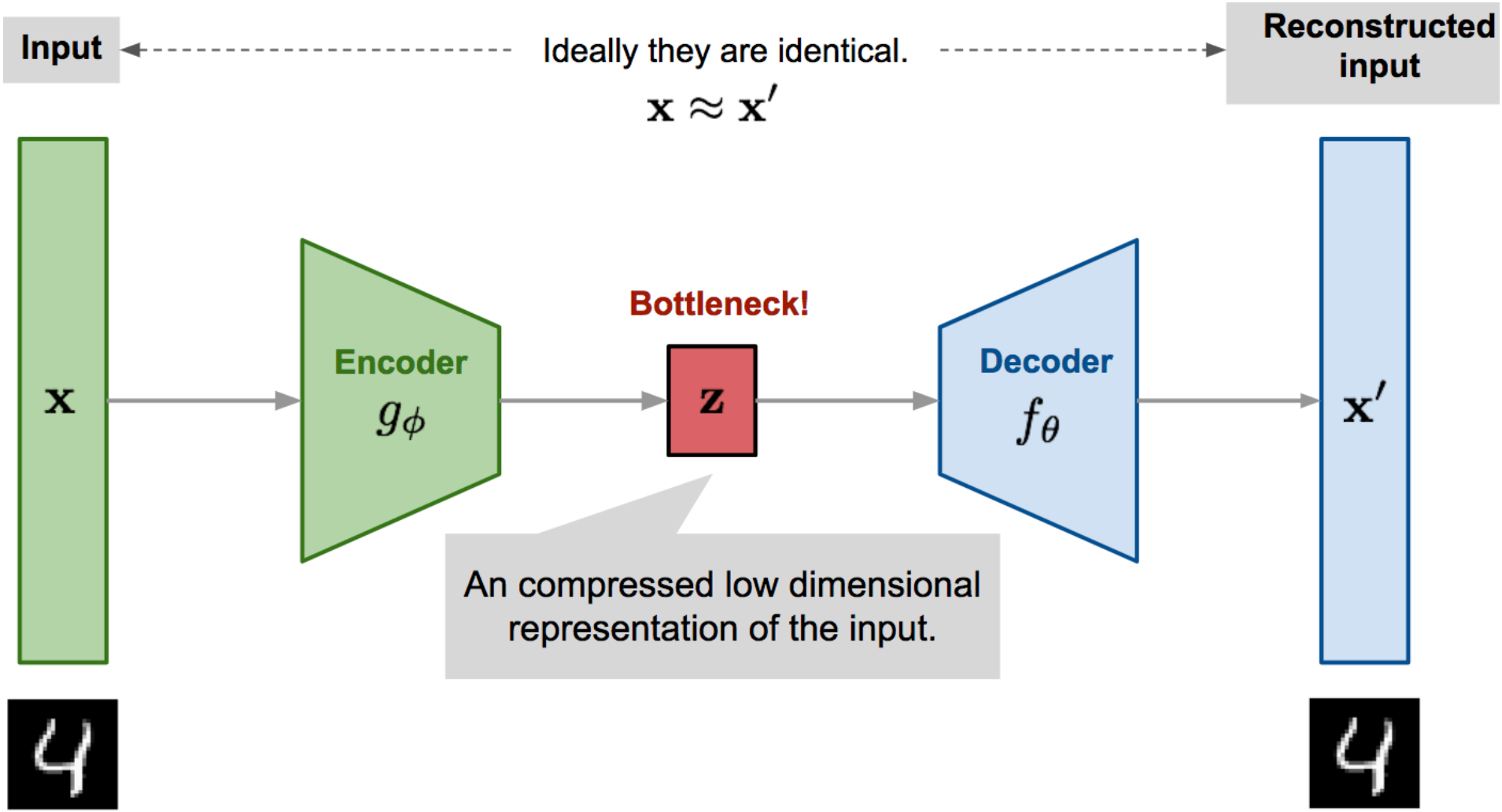


k-means 聚类

Inputs: Dataset $X = \{X_1, \ldots, X_n\}$; Number of clusters k
Initialization: Randomly choose initial centroids $\mu_1, \ldots, \mu_k$
Repeat until convergence:

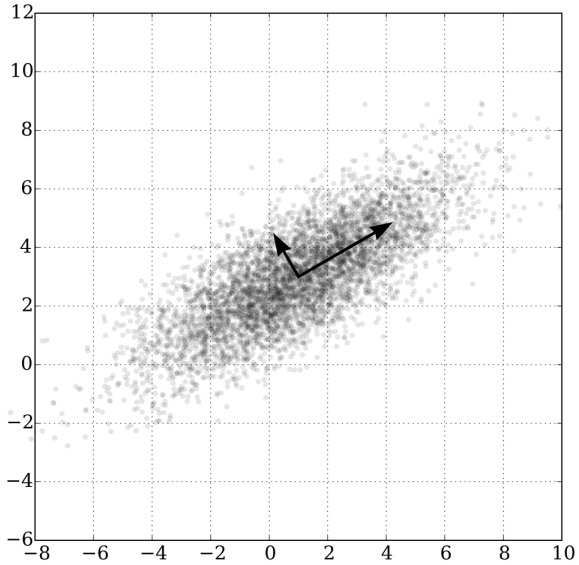
- for all $i \leq k$, sets $C_i := \left\{x \in X : i = \operatorname{argmin}_j \|x - \mu_j\|\right\}$
- for all $i \leq k$, update $\mu_i = \frac{1}{|C_i|} \sum_{x \in C_i} x$

Autoencoder — dimensionality reduction



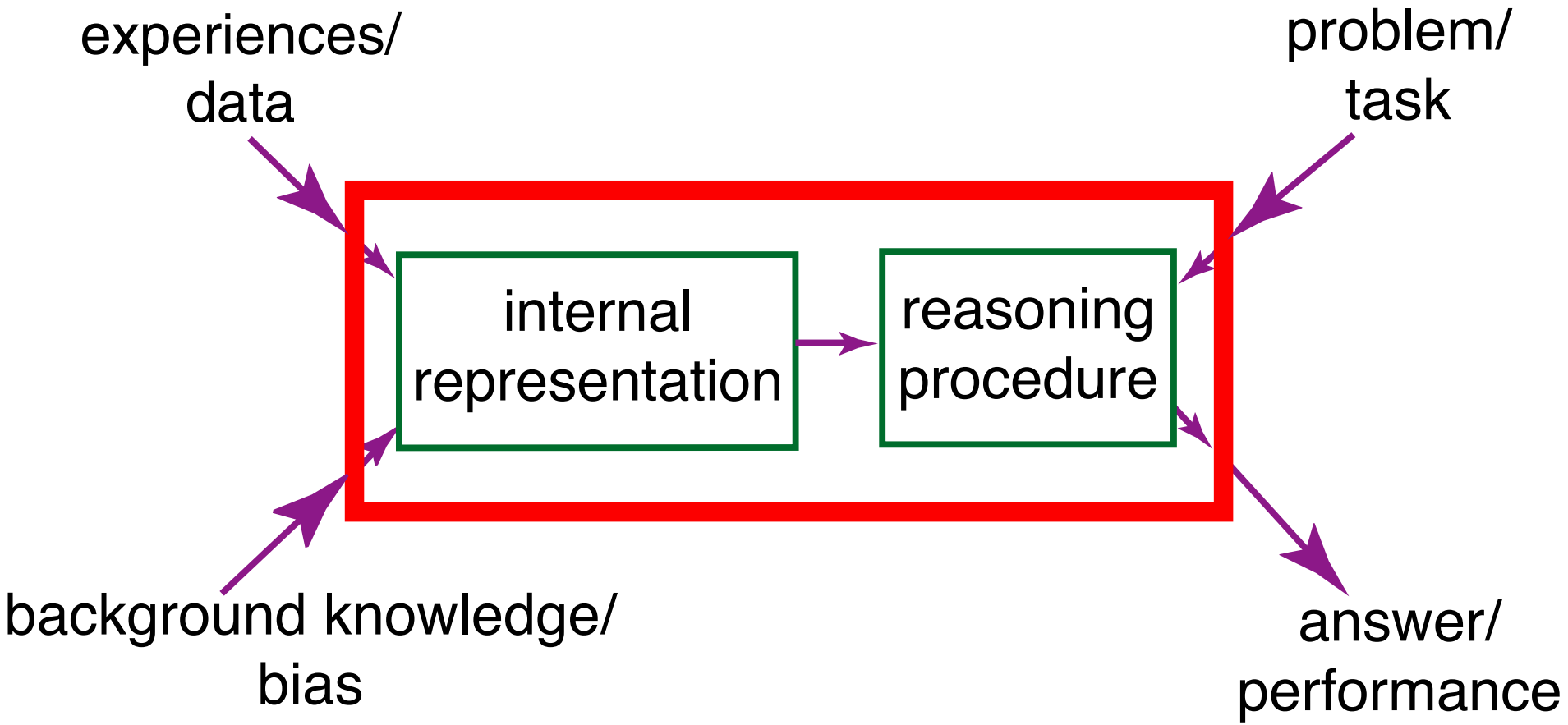
$$(f^*, g^*) = \underset{f,g}{\operatorname{argmin}} \|x - f \circ g(x)\|^2$$

Principal Component Analysis — dimensionality reduction



1. Given data matrix $\mathbf{X}$ with dimension $N \times D$ (N samples, D dimensions)
2. Compute the covariance matrix of the data: $\frac{1}{N}\mathbf{X}^\mathsf{T}\mathbf{X}$
3. Compute the eigenvalues $\lambda_1, \ldots, \lambda_D$ and the associated eigenvectors $\mathbf{e}_1, \ldots, \mathbf{e}_D$
4. Sort eigenvalues from big to small and select top-M eigenvalues and their associated eigenvectors $\mathbf{E} = [\mathbf{e}_1, \ldots, \mathbf{e}_M]$
5. Project the data into the lower M-dimensional space: $\mathbf{Z} = \mathbf{X}\mathbf{E}$
6. Reconstruct the original data: $\hat{\mathbf{X}} = \mathbf{Z}\mathbf{E}^\mathsf{T}$

Learning Architecture



监督学习 vs 强化学习

: 亲爱的，删除快捷键是卸载不了软件的，要点击这里……

: ☹/☺ (给你个眼神，自己体会)

强化学习为啥强?

如果你娶到一个暴脾气的悍妇，你会成为一个伟大的哲学家！

— 苏格拉底 ♡◎♡

Paradox of Knowledge

- If you don't know it, how could you possibly recognize it when you see it?
- If you do know it, you don't need to look for it.
- So why should we bother attempting to gain knowledge?

Knowledge vs Intelligence

Intelligence is the ability to continuously improve or gain new information or knowledge on its own through a closed-loop feedback about how good its current memory or knowledge is at explaining or predicting its observations.

$$\texttt{Intelligence}(t) = \frac{\mathrm{d}}{\mathrm{d}t}\texttt{Knowledge}(t)$$

$$\texttt{Knowledge}(t) = \int_0^t \texttt{Intelligence}(x)\,\mathrm{d}x$$

— 马毅

How do computers discover new knowledge?

Five Ways to New Knowledge

1. Fill in gaps in existing knowledge
2. Emulate the brain
3. Simulate evolution
4. Systematically reduce uncertainty
5. Notice similarities between old and new

The Five Tribes of Machine Learning

Tribe	**Origins**	**Master Algorithm**
Symbolists	Logic, philosophy	Inverse deduction
Connectionists	Neuroscience	Backpropagation
Evolutionaries	Evolutionary biology	Genetic programming
Bayesians	Statistics	Probabilistic inference
Analogizers	Psychology	Kernel machines

学习 = 表示 + 评估 + 优化

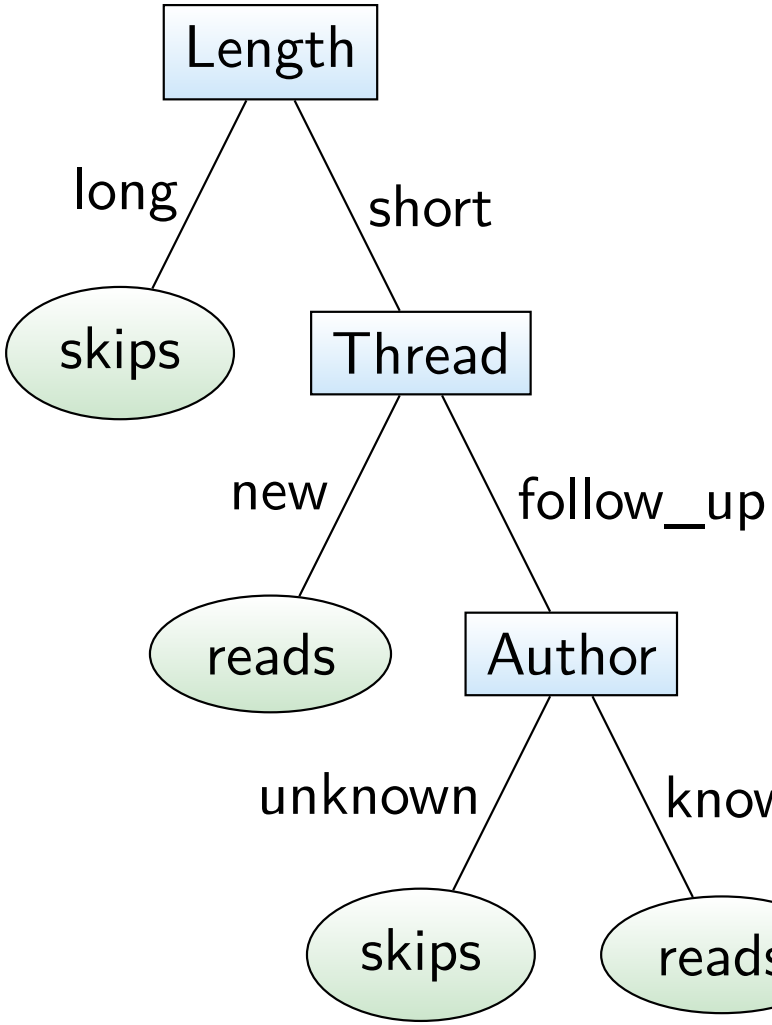
- **表示** (Representation): 模型必须以形式语言表示.
  - 定义了它可以学习的 — 假设空间
- **评估** (Evaluation): 如何在不同的假设之间进行选择?
  - 评估函数、目标函数、评分函数
  - 可能与外部评估函数 (比如准确率) 不同
- **优化** (Optimization): 如何搜索假设空间?

Symbolists

- The essence of intelligence is symbolic **reasoning**.
- Logic, Decision trees
- Inverse deduction can infer new hypotheses
- Easy to add knowledge (e.g. as rules)
- Can combine knowledge, data, to fill in gaps (like scientists)
- Robot scientist: learns hypotheses, then designs and runs experiments to test hypotheses
- Impossible to code everything in rules
- Hard to handle uncertainty

Representation	Rules, trees, first order logic rules
Evaluation	Accuracy, information gain
Optimization	Top-down induction, inverse deduction
Algorithms	Decision trees, Logic programs

Decision Tree vs Horn Clause



skips ← Long

reads ← short ∧ new

reads ← short ∧ follow_up ∧ known

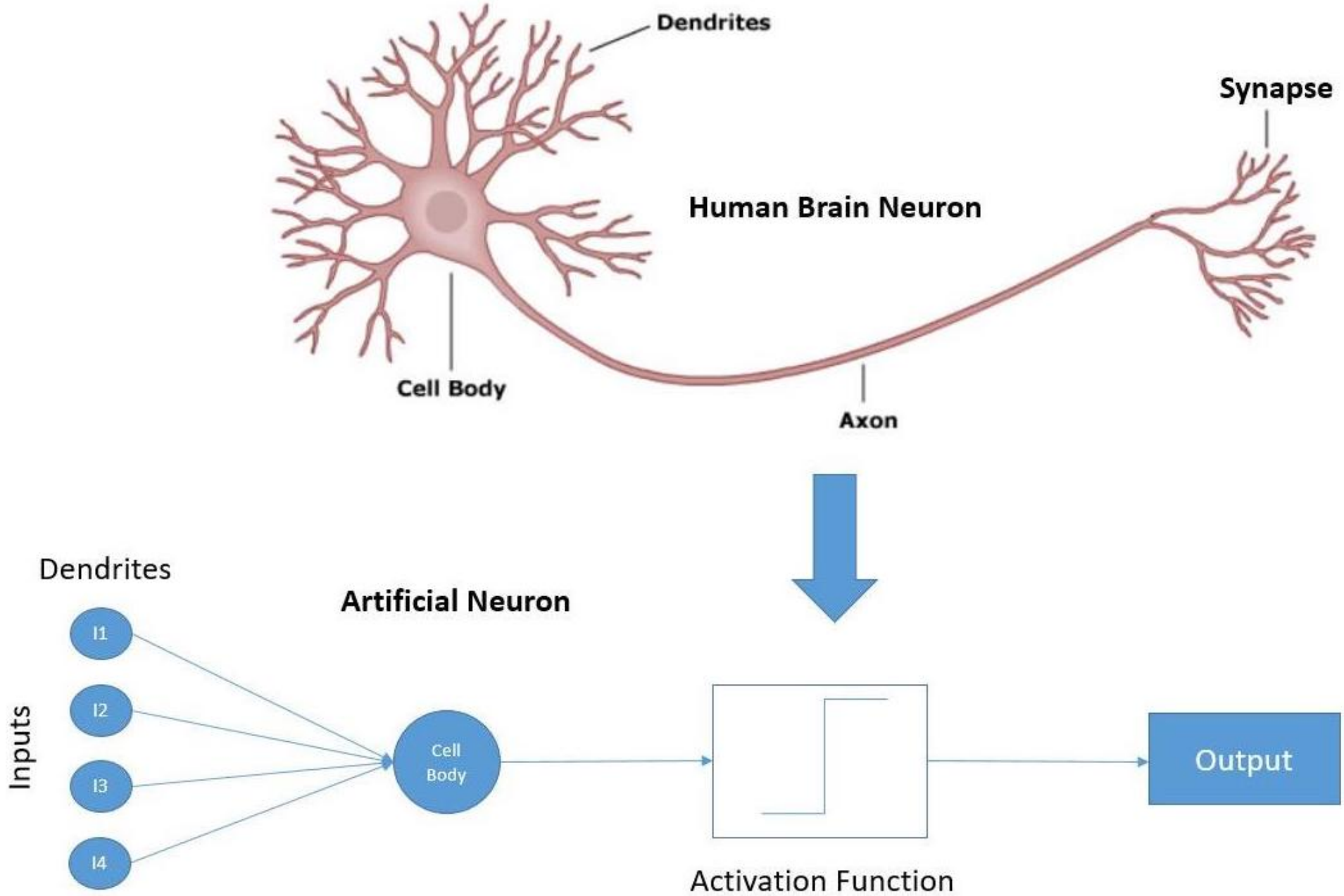
skips ← short ∧ follow_up ∧ unknown

- We want a small and efficient tree
- Ask the question which is most informative

Connectionists

- **Learning** is what the brain does: mimic the Human brain
- Adjust strengths of connection between neurons
- Hebbian learning: Neurons that fire together, wire together
- Neural networks
- Backpropagation
- Can handle raw, high-dimensional data, constructs its own features
- Hard to add reasoning/explanations

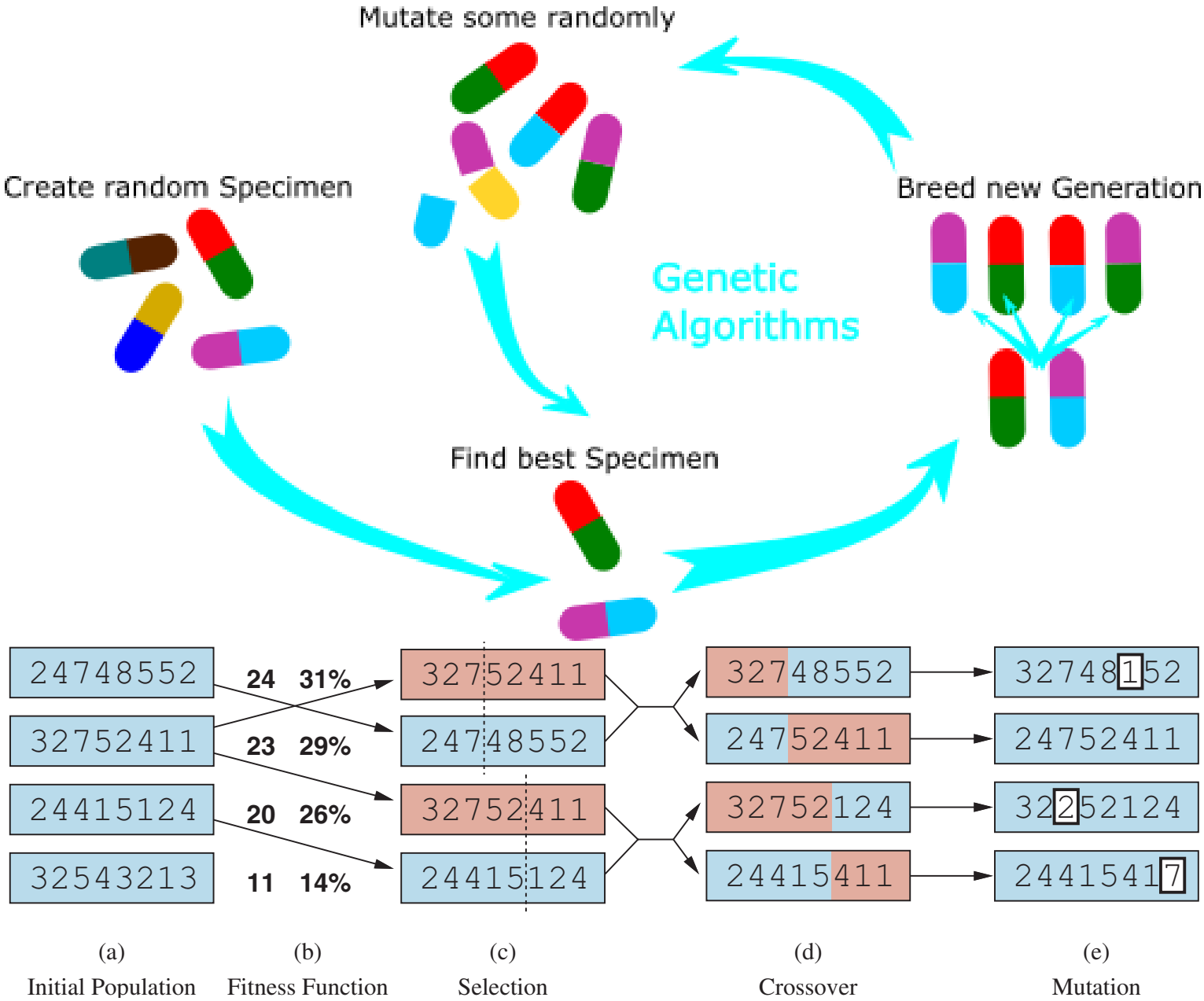
Representation	Neural network
Evaluation	Squared error
Optimization	Gradient descent
Algorithms	Backpropagation


Dendrites
Synapse
Human Brain Neuron
Cell Body
Axon
Dendrites
Artificial Neuron
I1
I2
I3
I4
Inputs
Cell Body
Output
Activation Function

Evolutionaries

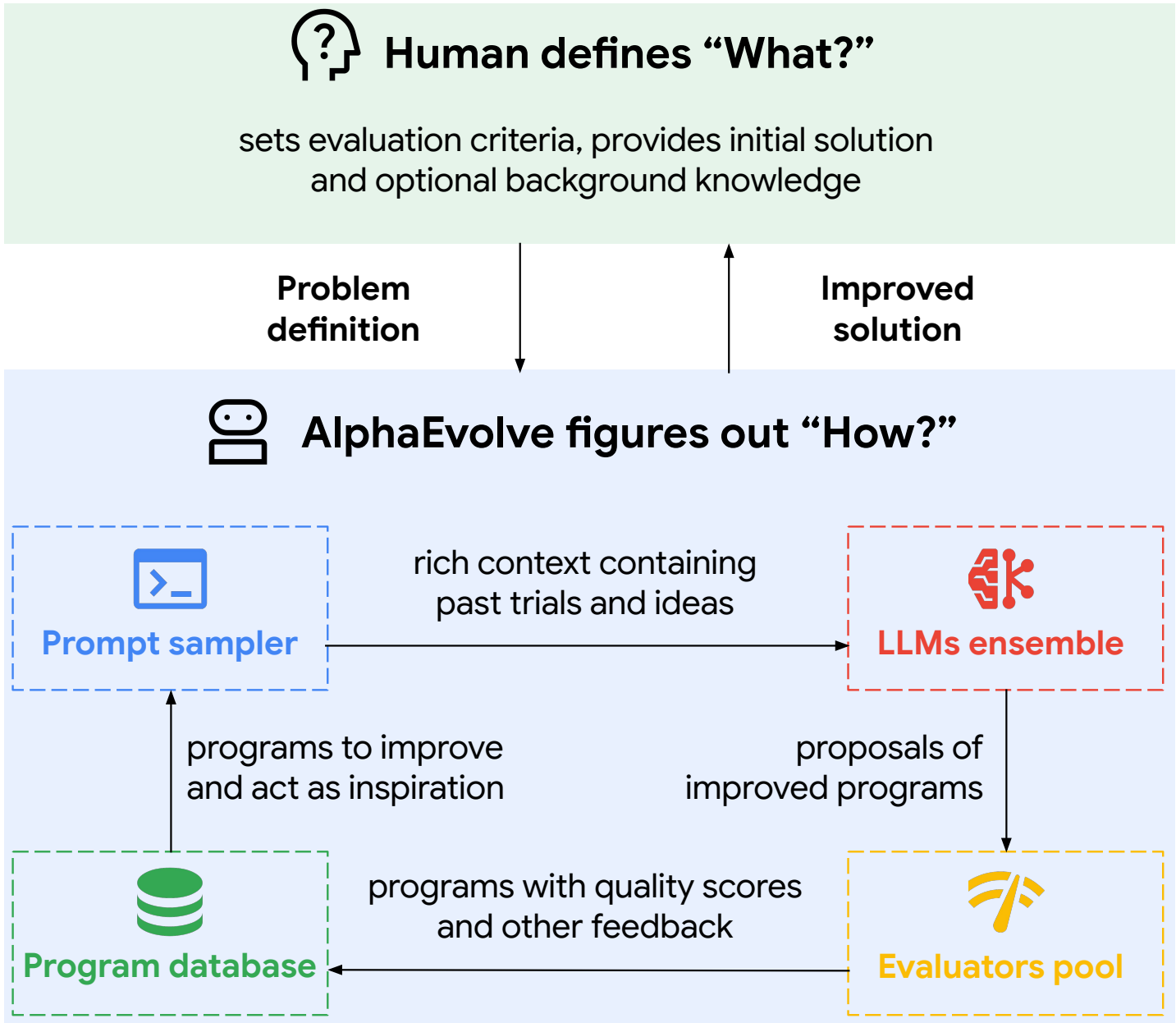
- Natural selection is the mother of all learning: simulate evolution
- Evolutionary algorithms
- Idea: "selection", "cross-over", and "mutation".
  - selection: selection of individuals according to a fitness function and pairing
  - cross-over: calculation of the breaking points and recombination
  - mutation: according to a given probability elements in the string are modified
- Can learn structure, wide hypothesis space
- Needs a way to 'fill' the structure

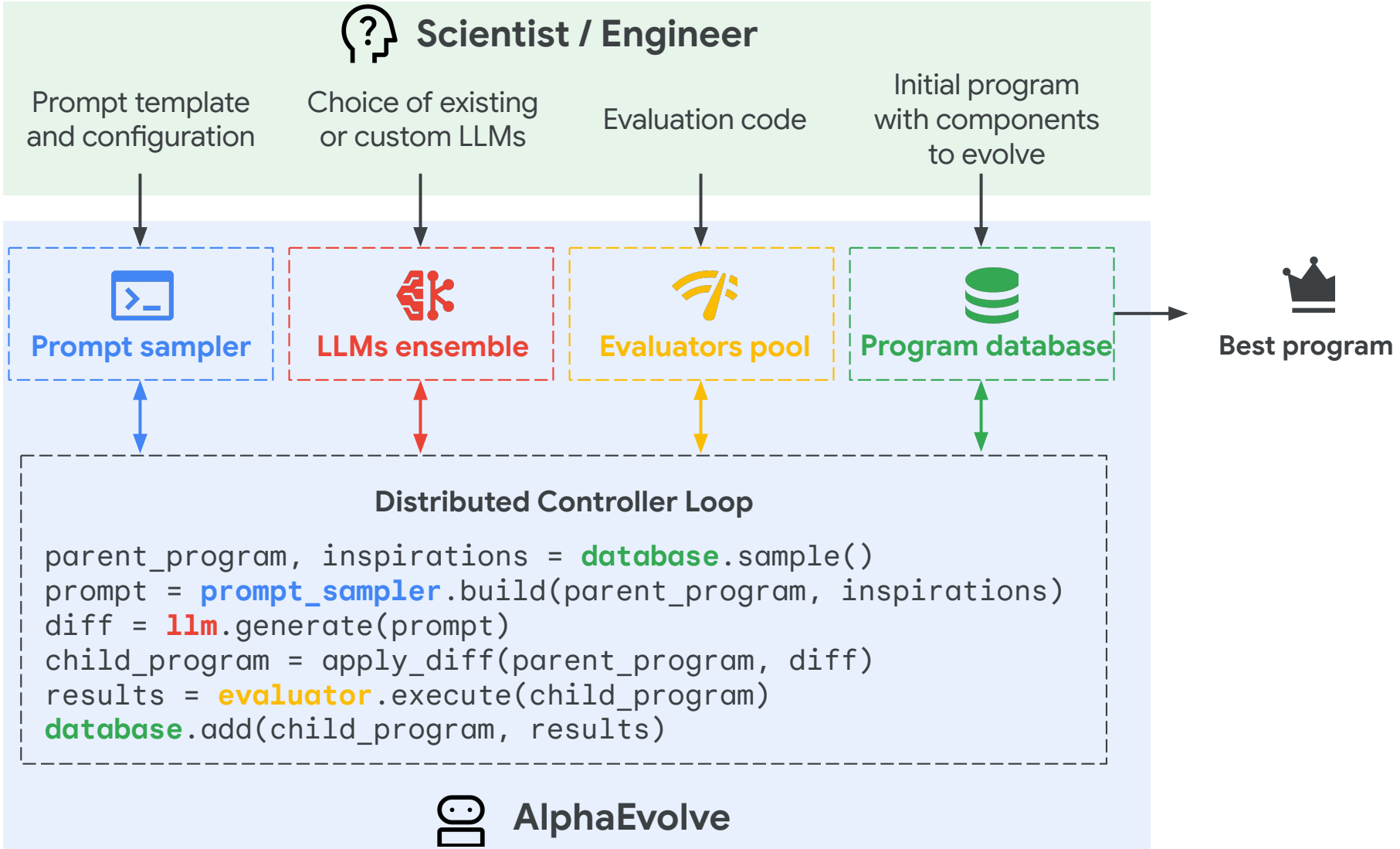
Representation	Genetic programs (often trees)
Evaluation	Fitness function
Optimization	Genetic search
Algorithms	Genetic programming (crossover, mutation)



- 环境的随机性 + 变异的随机性
- 实现 (群体的) 反脆弱性

AlphaEvolve



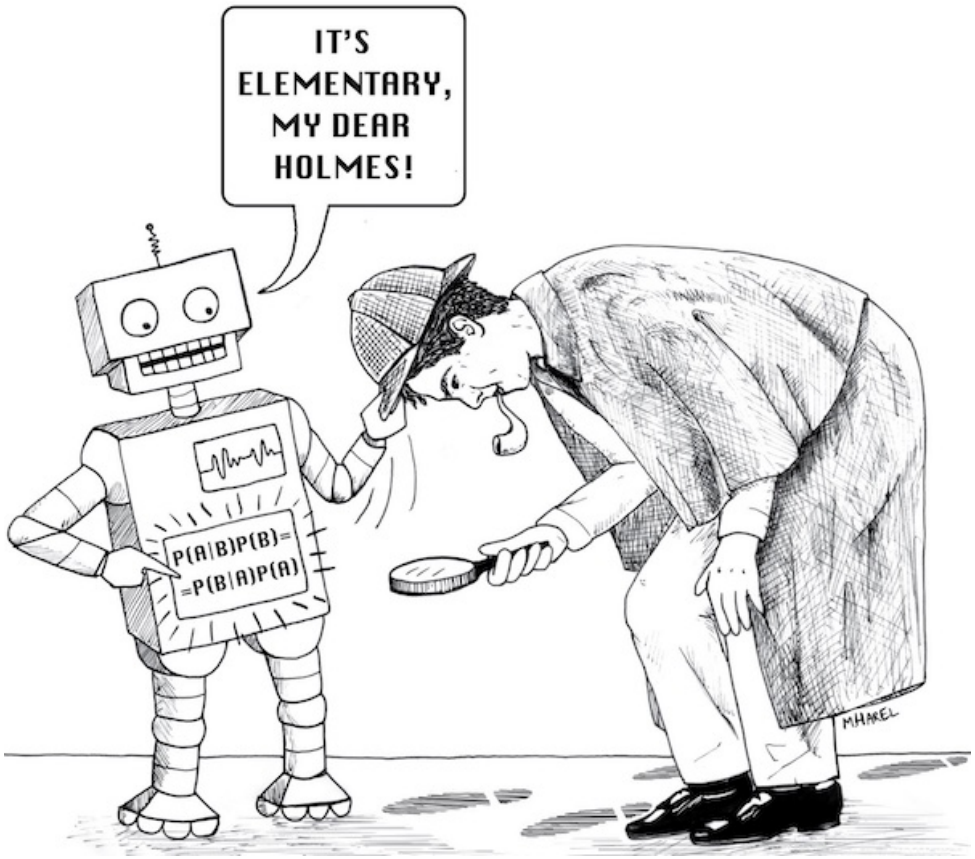


人提供初始程序、评估代码和可选配置. AlphaEvolve 开启演化循环. 提示语采样器使用程序数据库中的程序来构建提示语. 大语言模型根据提示语生成代码修改, 这些修改被应用于创建新程序. 评估器对新程序进行评分, 评分高的新程序被添加到程序数据库中.

Bayesian Learning

- Learning is a form of uncertain inference: reduce uncertainties by incorporating new evidence
- Graphical models, Gaussian processes, HMMs, Kalman filter
- Uses Bayes theorem to incorporate new evidence into our beliefs
- Can deal with noisy, incomplete, contradictory data
- Choose hypothesis space + prior for each hypothesis
- Depends on the prior
- Hard to unite logic and probability

Representation	Graphical models, Markov networks
Evaluation	Posterior probability
Optimization	Probabilistic inference
Algorithms	Bayes theorem and derivates


IT'S ELEMENTARY, MY DEAR HOLMES!
P(A|B)P(B)=
=P(B|A)P(A)
MHAREL

Example: 医疗诊断

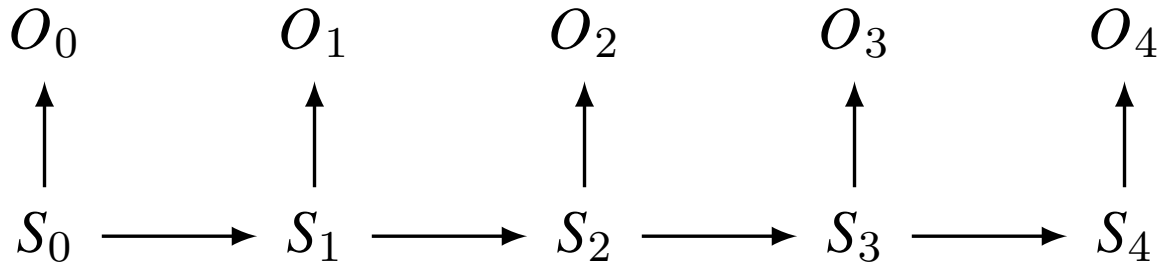
- 在考虑要孩子时，哲学家罗素向医生咨询了精神病的遗传情况.
- 医生说，人们对遗传的恐惧被夸大了. 50% 的精神病患者父母酗酒，只有 15% 的精神病患者的父母也是精神病.
- 罗素：这似乎使我稍稍心安. 但是，医生没指出总人群中精神病患者和酗酒者的比例，所以这个论点没有一点儿价值.

$$P(H \mid D) = \frac{P(H, D)}{P(D)} = \frac{P(D \mid H)P(H)}{P(D)} = \frac{P(D \mid H)P(H)}{\sum_H P(D \mid H)P(H)}$$

Problem：已知 HIV 患病率 $P(H) = 0.01\%$. 患者 $P(+ \mid H) = 99.99\%$ 能检测出阳性；健康人 $P(- \mid \neg H) = 99.99\%$ 能检测出阴性. 已知你的检测结果为阳性，你会崩溃吗？

$$P(H \mid +) = \frac{P(+|H)P(H)}{P(+)} = 0.5$$

Example: 隐马尔科夫模型 HMM：你在卧室睡觉，听到家里进贼，你熟悉房间的结构，通过不同的声响推断贼的位置轨迹.



Example: 调研一下大家出轨的比例?

- 目标: 调研一个群体中出轨者的比例.
- 为了避免尴尬, 怎么确保调研者也不知道被调研对象是否有出轨?

让被调研对象抛一枚硬币, 抛出正面还是反面只有被调研者自己知道. 如果是正面, 就回答问题 1; 如果是反面, 就回答问题 2.

1. 随机打个 ✓ 或 ×
2. 你是否有出轨?

已知: $P(+) = P(-) = 0.5$, $P(\checkmark \mid +) = P(\times \mid +) = 0.5$

$$
\begin{aligned}
P(\checkmark \mid -) &= \frac{P(- \mid \checkmark)P(\checkmark)}{P(-)} \\
&= \frac{[1 - P(+ \mid \checkmark)]P(\checkmark)}{P(-)} \\
&= \frac{\left[1 - \frac{P(\checkmark|+)P(+)}{P(\checkmark)}\right] P(\checkmark)}{P(-)} = 2P(\checkmark) - 0.5
\end{aligned}
$$

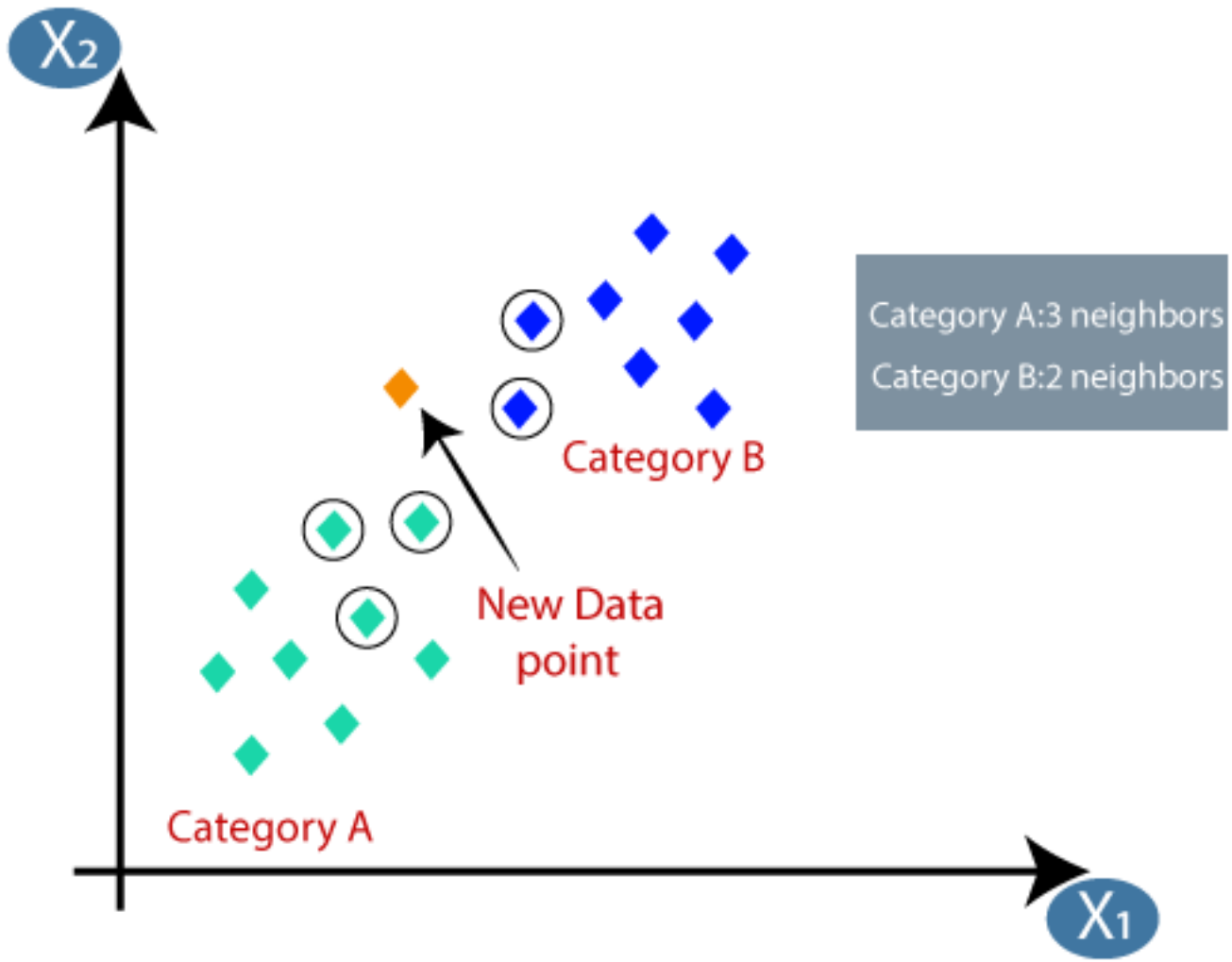
Learning by Analogy

- You are what you resemble
- Recognizes similarities between situations and infers other similarities
- Generalizes from similarity
- k-Nearest Neighbor, Support Vector Machines
- Transfer solution from previous situations to new situations
- Hard to do rules and structure

Representation	Memory, support vectors
Evaluation	Margin
Optimization	Kernel machines
Algorithms	k-Nearest Neighbor, Support Vector Machines

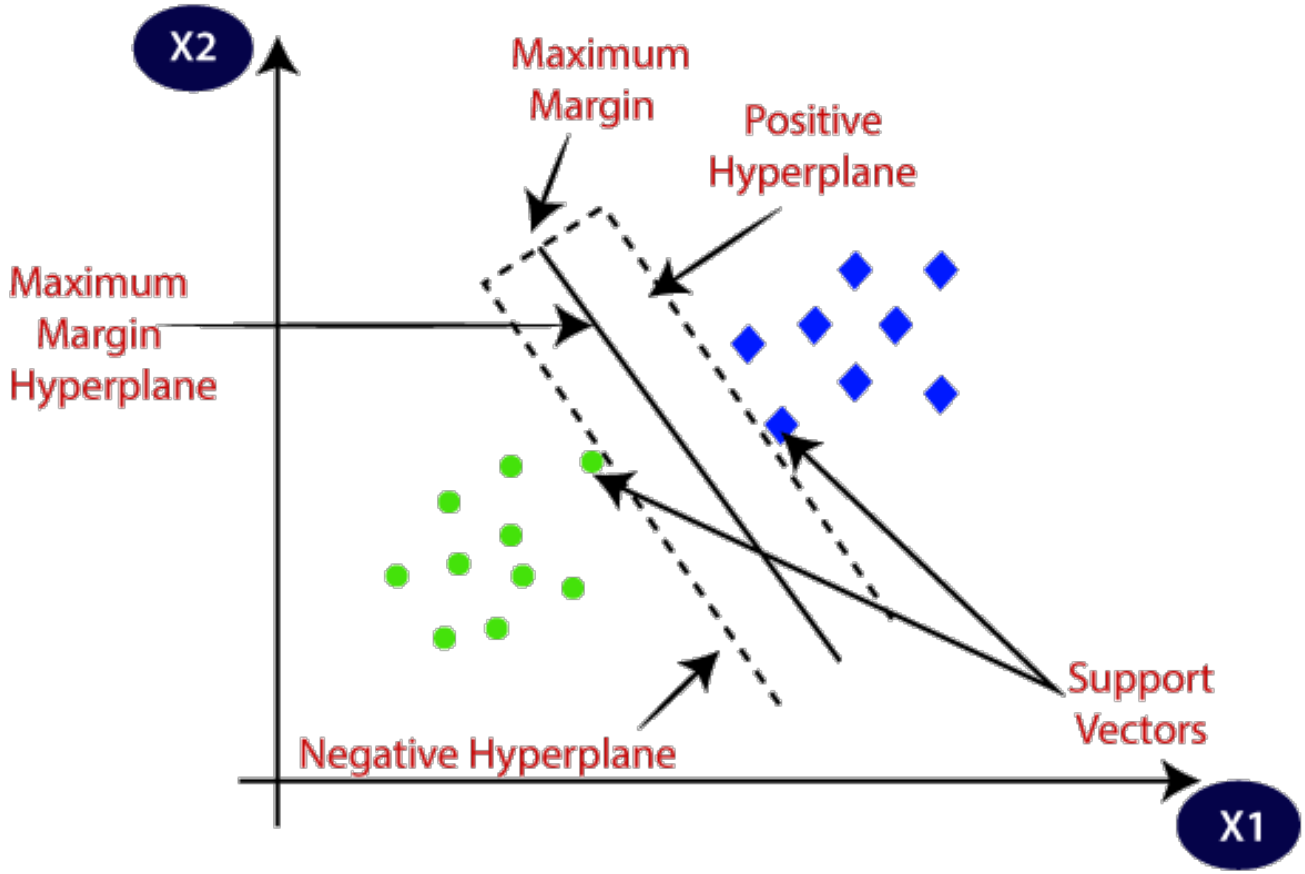
Nearest Neighbors

- Given cities belonging to 2 countries. Where is the border?
- Nearest neighbor: point belongs to closest cities
- k-Nearest neighbor: do vote over k nearest ones



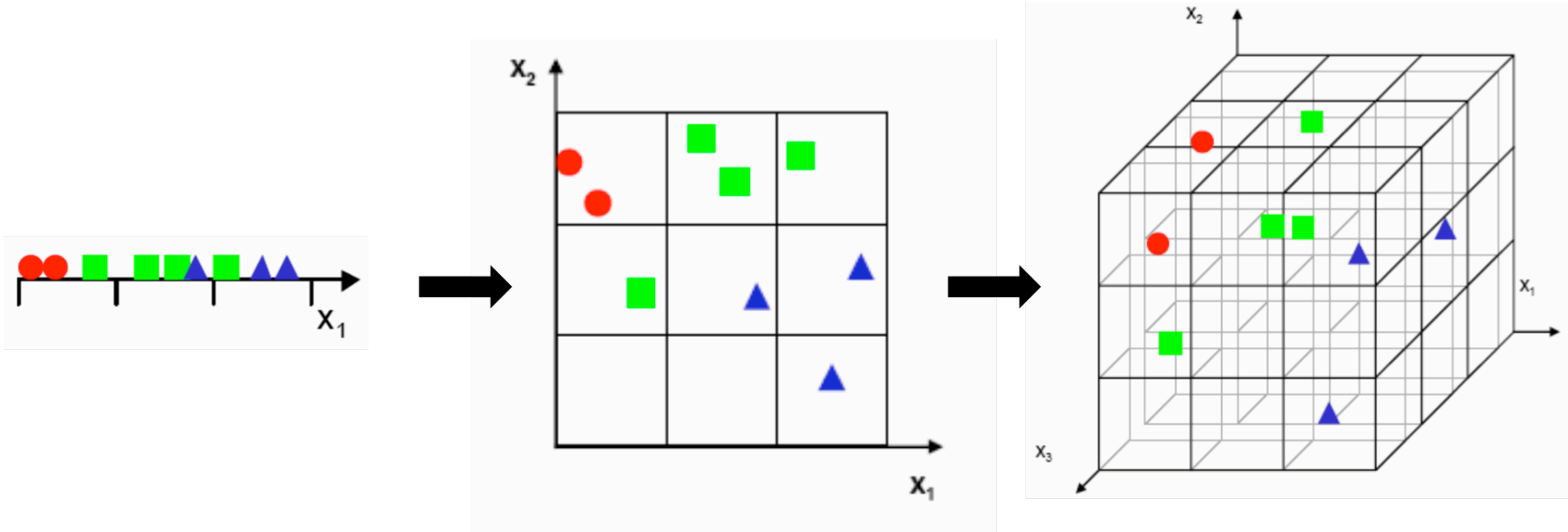
Support Vector Machines (SVM)

- Only remember points that define border (support vectors)
- Find linear border with maximal margin to nearest points
- If not linearly separable, transform the input space (kernel trick)



银蛇在雷区穿梭.....

The Curse of Dimensionality



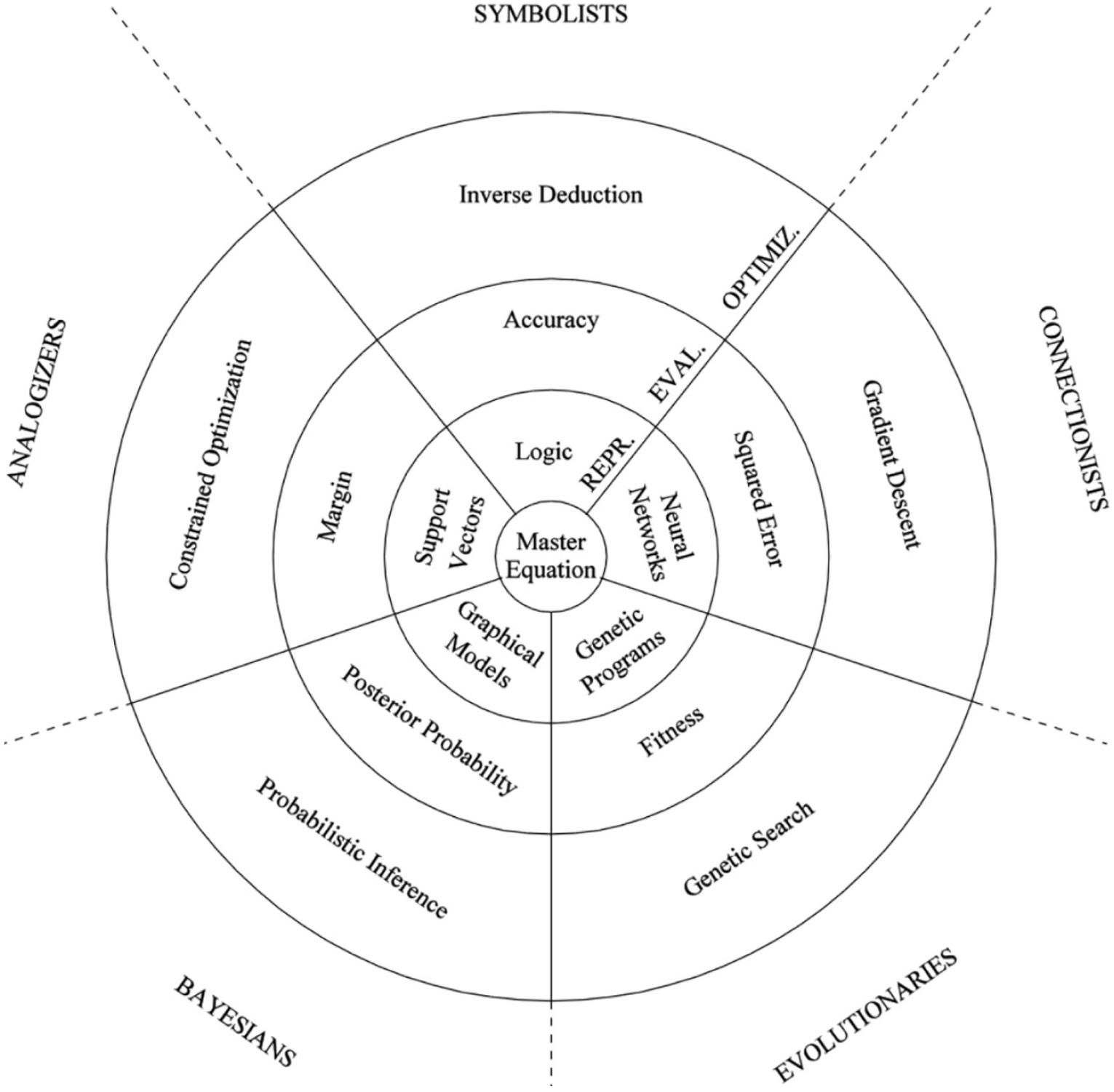
- When the dimensionality increases, the volume of the space increases so fast that the available data become sparse. All objects appear to be dissimilar in many ways.
- The amount of data needed to support ML often grows exponentially with the dimensionality.
- Volume of a high dimensional unit ball is concentrated near its surface.

The Master Algorithm?

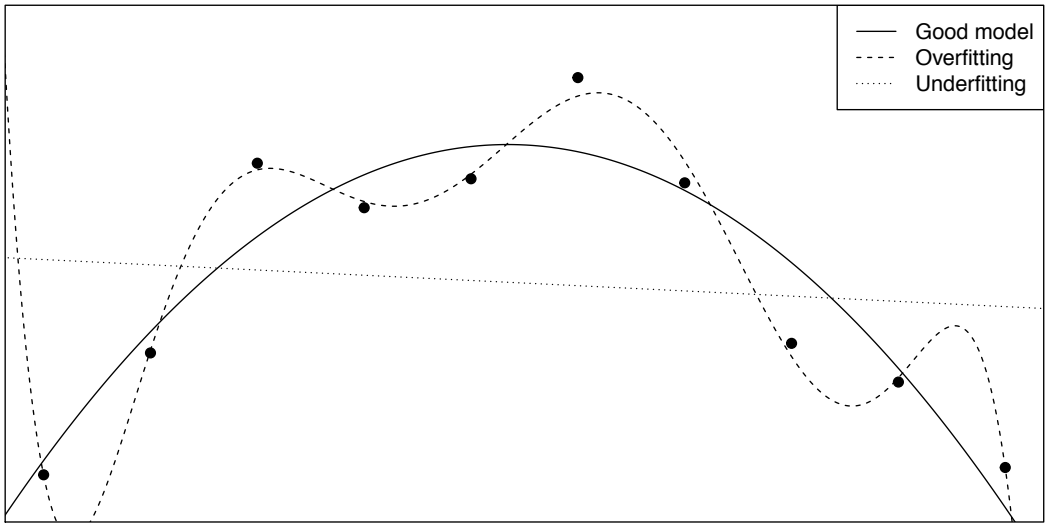
Tribe	Problem	Solution
Symbolists	Knowledge composition	Inverse deduction
Connectionists	Credit assignment	Backpropagation
Evolutionaries	Structure discovery	Genetic programming
Bayesians	Uncertainty	Probabilistic inference
Analogizers	Similarity	Kernel machines

- Representation: The hypothesis space.
  - Probabilistic logic
  - Weighted formulas $\rightarrow$ Distribution over states
- Evaluation: How to choose one hypothesis over the other?
  - Posterior probability
  - User-defined objective function
- Optimization: How do we search the hypothesis space?
  - Formula discovery: Genetic programming
  - Weight learning: Backpropagation

Elegant/Extensible/Expressive/Efficient/Educable/Evolvable?


SYMBOLISTS
CONNECTIONISTS
EVOLUTIONARIES
BAYESIANS
ANALOGIZERS
Inverse Deduction
Gradient Descent
Genetic Search
Probabilistic Inference
Constrained Optimization
OPTIMIZ.
EVAL.
REPR.
Accuracy
Logic
Squared Error
Neural Networks
Fitness
Genetic Programs
Posterior Probability
Graphical Models
Margin
Support Vectors
Master Equation

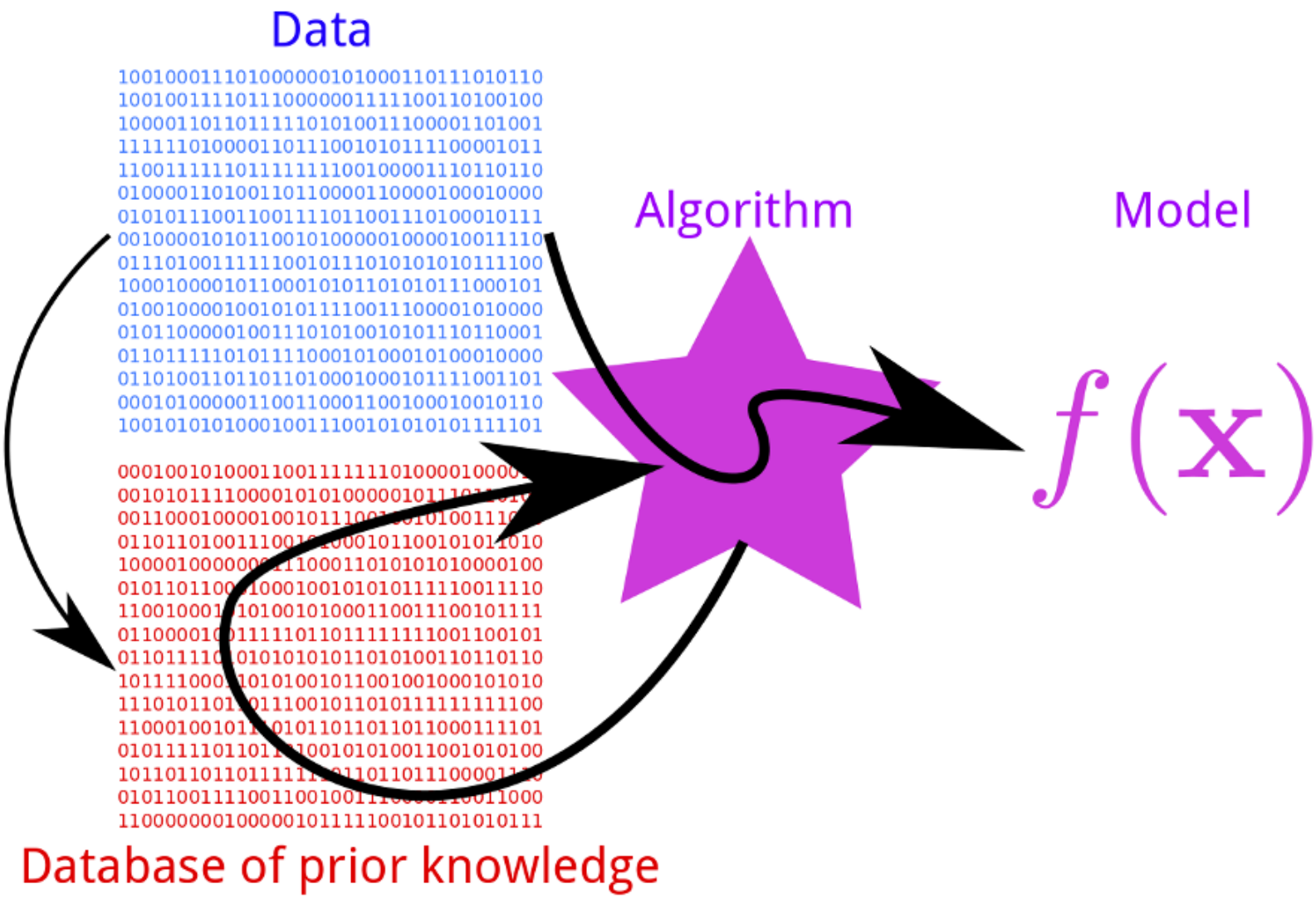
Avoiding Overfitting



- underfitting: not able to obtain a low error on the training set
- overfitting: gap between training error and test error is too large

- Never believe your model until you've verified it on new data
- Randomly divide the data into:
  - Training set which you give to the learner
  - Test set which you hide to verify predictive performance
- Do a statistical significance test to see whether one hypothesis is significantly better than another.
- Prefer simpler hypotheses (e.g. divide-and-conquer, regularization)

Model vs Algorithm



Model = Algorithm(Data)

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Contents

1. Introduction

2. Philosophy of Induction
   History
   How to Choose the Prior?

3. Inductive Logic

4. Universal Induction

5. Causal Inference

6. Game Theory

7. Reinforcement Learning

8. Deep Learning

9. Artificial General Intelligence

10. Emergence & Consciousness

11. AI Ethics

Hypothetical-Deductive Confirmation 假设-演绎确证

$$\frac{H \to E \qquad E}{H \text{ is confirmed}}$$

- Which of $H, A_1, \ldots, A_n$ does E confirm?

$$\frac{H \wedge A_1 \wedge \cdots \wedge A_n \to E \qquad E}{H \wedge A_1 \wedge \cdots \wedge A_n \text{ is confirmed}}$$

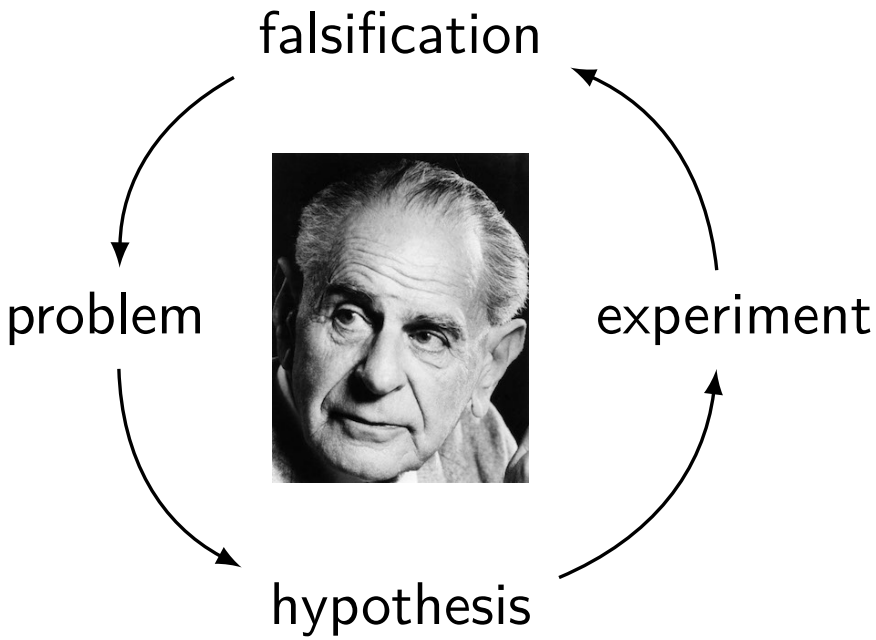
- “苹果落地” 能否确证 “万有引力定律” 且 “嫦娥的身体是奶酪做的”?

$$\frac{H \to E \qquad E}{G \wedge H \text{ is confirmed?}} \; [G \wedge H \to H]$$

- “苹果落地” 能否确证 “嫦娥的身体是奶酪做的”?

$$\frac{H \to E \vee H \qquad E}{H \text{ is confirmed}} \; [E \to E \vee H]$$

波普尔《科学发现的逻辑》— 可证伪性



Proposition (波普尔)

- 选择最简单的、没有被证伪的概括.
- 假说越简单, 越容易被证伪.
- 可证伪性与简单性一样是主观的, 没有客观标准.

$$\frac{H \wedge A_1 \wedge \cdots \wedge A_n \rightarrow E \qquad \neg E}{\neg H \vee \neg A_1 \vee \cdots \vee \neg A_n}$$

$$\neg H?\ \neg A_1?\ \ldots\ \neg A_n?$$

- 迪昂-奎因: 整体论, 怎么孤立证伪?
- 概率命题咋办?

不可证伪咋办?

- Alice 和 Bob 生了个孩子叫 Carly. Carly 喜欢吃草莓味的冰激凌, 第一次吃冰激凌就是草莓味的, 也只吃过草莓味的.
  1. Alice: 所有孩子都喜欢吃草莓味的冰激凌.
  2. Bob: 所有孩子都喜欢吃他们第一次尝到的冰激凌口味.
- 这两个猜想都可以看作待证伪的 "自然律".
- 你更相信哪一个?
- 如果宇宙中只有 Carly 这一个小孩呢?
- 宇宙学真的就只有一个样本.[6]
- "可证伪性" 是背景依赖的?

[6]传统的科学范式, 需要状态空间、初始条件、自然律. 只有通过重复实验, 变换初始条件, 才能确定在不同初始条件下不变的自然律. 宇宙学不适用于该范式.

Instance Confirmation 实例确证

Basic idea: "E confirms H" means "E is an instance of H".

Problem: 看到一只黑乌鸦, 会增强你对 "所有乌鸦都是黑的" 的信心吗?

Nicod 标准

对于全称命题 "所有的 A 都是 B",

1. 正例: "A 且 B" 的对象构成确证;
2. 反例: "A 且 $\neg B$" 的对象直接否证;
3. 中性: "$\neg A$" 的对象不相干, 不具有任何确证否证效力.

Problem: 你赞同 Nicod 标准吗?

实例确证 — 乌鸦怪论

- Nicod: 看到一只黑乌鸦会确证 "All ravens are black". ?
- Hempel: 看到一只白鞋子会确证 "All ravens are black" 吗?

$$\neg Bx \wedge \neg Rx \ \text{ confirms } \ \forall x(\neg Bx \to \neg Rx)$$
$$\forall x(\neg Bx \to \neg Rx) \leftrightarrow \forall x(Rx \to Bx)$$
$$\neg Bx \wedge \neg Rx \ \text{ confirms } \ \forall x(Rx \to Bx)$$

修正早期逻辑经验主义: 对世界的理解不能只靠 "逻辑 + 经验"?

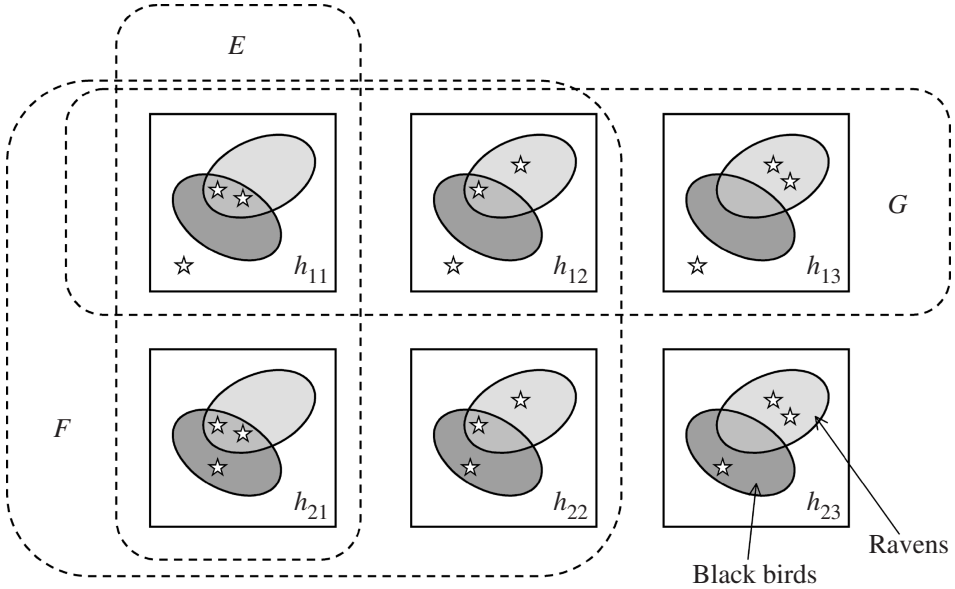
1. 世界一: 乌鸦都是黑的但非常罕见 ($1/10000$ 是乌鸦, 其余是鞋子);
2. 世界二: 只存在乌鸦, 但只有 10% 是黑乌鸦, 其余是白乌鸦.
   - 看到一只黑乌鸦会确证 "并非所有乌鸦都是黑的".
   - 如果我们把世界一改为乌鸦都是黑的且占比 $9999/10000$, 那么看到一只黑乌鸦会确证 "所有乌鸦都是黑的".
   - "证据确证" 要考虑其他假设/可能世界!

- Goodman's New Riddle of Induction.

  grue = green before 2050, and blue thereafter.

1. Are all emeralds green?
2. Are all emeralds grue?

乌鸦怪论



- 假设有 6 个可能世界. 各先验为 h_{ij}. 每个可能世界中都有三只鸟.
- 事件 E 表示 “所有乌鸦都是黑的”. 事件 F 表示 “看到一只黑乌鸦”. 事件 G 表示 “看到一只白鞋子”.
- 在这个设定下, F 会确证 E, 但 G 是否确证 E 取决于先验.

$$P(E \mid F) = \frac{P(E \wedge F)}{P(F)} = \frac{h_{11} + h_{21}}{h_{11} + h_{21} + h_{12} + h_{22}} \geq h_{11} + h_{21} = P(E)$$

$$P(E \mid G) = \frac{P(E \wedge G)}{P(G)} = \frac{h_{11}}{h_{11} + h_{12} + h_{13}} \overset{?}{\geq} h_{11} + h_{21} = P(E)$$

大卫·休谟 1711-1776

- 《人性论》
- “理性和理性判断，不过是不同感觉经验的习惯性联想”
- 归纳问题
- 天下没有免费的午餐定理
- 相关性 $\nrightarrow$ 因果性
- 信念 $\nrightarrow$ 知识
- 实然 $\nrightarrow$ 应然
- 联结主义
- 类比
- 反事实因果

休谟论归纳

Proposition (休谟)

- 推理有两种：基于观念关系的证明；关于事实的或然推理.
- 关于外部世界的知识是因果陈述.
- 因和果在过去的恒常连接不蕴含未来的恒常连接，因为“未来与过去不相似”不矛盾.
- 所以因果陈述是涉及事实的综合判断，但我们无法通过经验事实非循环的论证未来与过去的相似性. 因此，关于因果的知识我们也无法后验的获知.
- 归纳不过是一种心理习惯. 必然性只存在于人的心灵之中，而不在事件本身.

Proposition (皮尔士)

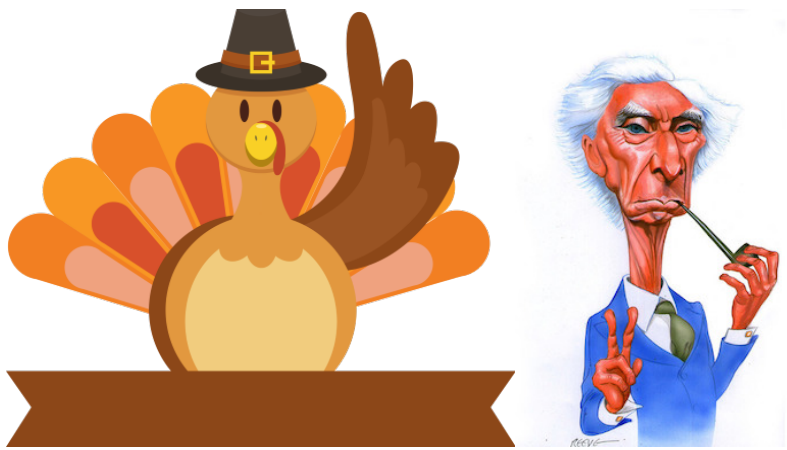
除非受到另一种习惯的延伸的约束，否则习惯倾向于自我延伸.

一个悲观主义者的归纳： 过去的所有理论都是错的，现在和未来的一切理论也都将是错的.

《三体》— “射手” 假说 & “农场主” 假说

“射手” 假说: 有一名神枪手, 在一个靶子上每隔十厘米打一个洞. 设想这个靶子的平面上生活着一种二维智能生物, 它们中的科学家在对自己的宇宙进行观察后, 发现了一个伟大的定律: “宇宙每隔十厘米, 必然会有一个洞.”

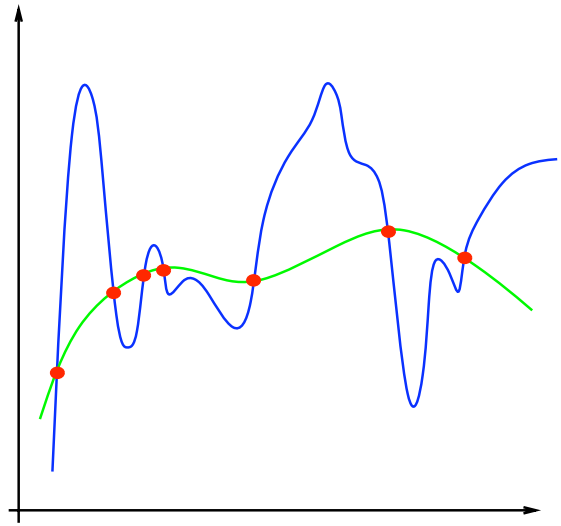
“农场主” 假说: 一个农场里有一群火鸡, 农场主每天中午十一点来给它们喂食. 火鸡中的一名科学家观察这个现象, 一直观察了近一年都没有例外, 于是它也发现了自己宇宙中的伟大定律: “每天上午十一点, 就有食物降临.” 它在感恩节早晨向火鸡们公布了这个定律, 但这天十一点食物没有降临, 农场主进来把它们都捉去杀了.



Leibniz-Wittgenstein-Goodman

Proposition (莱布尼茨)

对于任意有穷多个点，总有无穷多条曲线穿过它们. 对于任意有穷的数据集，都有无穷多种归纳概括的方式.



连续性定律？自然不做跳跃. 结果是原因的连续函数.

Proposition (维特根斯坦)

任何有穷行动都与无穷多条规则相一致，所以，任何普遍规则都不能通过有穷多的示例学到.

Proposition (Goodman)

All emeralds discovered till 2050 are green, and blue thereafter.

$$\mathsf{Grue}(x) \iff (t < 2050 \to \mathsf{Green}(x,t)) \wedge (t \geq 2050 \to \mathsf{Blue}(x,t))$$

Mill — Homogeneous Universe

Proposition (Mill)

通过添加一些关于世界的原则，比如 “未来与过去相似”, 或 “时空是均匀的”, *Induction* 可以被转化为 *Deduction*.

《三体》— 台球 — 三体质子干扰地球高能粒子对撞机

丁仪: 我们总共进行了五次实验，其中四次在不同的空间位置和不同的时间，两次在同一空间位置但时间不同．撞击实验的结果居然都一样!

汪淼: 在五次实验中，两个球的质量是没有变化的；所处位置，当然是以球桌面为参照系来说，也没有变化；白球撞击黑球的速度向量也基本没有变化，因而两球之间的动量交换也没有变化，所以五次实验中黑球当然都被击入洞中．

丁仪: 应该庆祝一下，我们发现了一个伟大的定律：物理规律在时间和空间上是均匀的．

⋮

汪淼: 你真的相信物理规律在时空上不均匀？

丁仪: 我什么都不知道．

莱辛巴赫的实用主义辩护

	齐一	非齐一
归纳	✓	×
不归纳	✓	×

丑小鸭定理 — 分类需要偏置

	x	y	z	φ-Property	Weight
1	0	0	0	¬White ∧ ¬Hat	0
2	0	0	1	¬White	1
3	0	1	0	White ∧ Hat	0
4	0	1	1	Hat	0
5	1	0	0	¬Hat	0
6	1	0	1	¬Hat ∨ ¬White	0
7	1	1	0	White	1
8	1	1	1	White ∨ Hat	0

- 若以共享的 φ-性质的数量来定义相似度，则任意两个对象之间的相似度都一样.

$$xy : 1, 2, 7, 8$$
$$xz : 1, 3, 6, 8$$
$$yz : 1, 4, 5, 8$$

- 一只白天鹅和一只丑小鸭之间的区别，并不比两只白天鹅之间的区别大.

- 庄子: 万物齐一. 佛教: 无分别心. 色即是空.
- Classification is impossible without some sort of bias.
- 相似性依赖偏置，没有相似性就无法分类.

丑小鸭定理 [Wat69](渡边慧 Watanabe,1969)

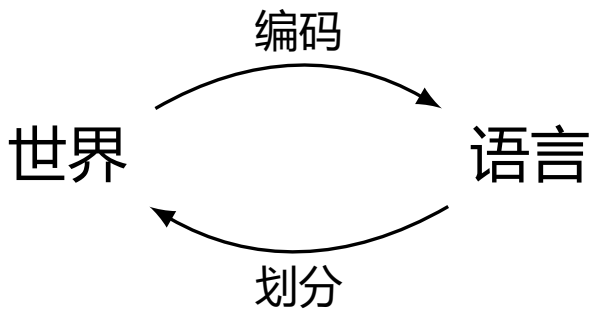
- 设想由 n 个对象构成的宇宙 X.
- 假设所有对象可区分: $\forall xy \in X[\forall \varphi(\varphi(x) \leftrightarrow \varphi(y)) \to x = y]$. Leibniz
- 至少需要 $m \geq \lceil \log_2 n \rceil$ 个谓词 $P_1, \ldots, P_m$ 能把 n 个对象区分开.
- 每个对象 $x \in X$ 的特征为: $\varphi(x) := \bigwedge_{k=1}^{m} \pm P_k(x)$.
- 对象的每个子集 $A \subset X$ 都被一个 φ-性质精确刻画
  $\varphi_A := \bigvee_{x \in A} \varphi(x) = \bigvee_{x \in A} \bigwedge_{k=1}^{m} \pm P_k(x)$.
- 任意两个对象 $x, y \in X$, 共享 2^{n-1} 个 φ-性质 (同属于/同不属于 2^{n-2} 个分类), 在剩下的 2^{n-1} 个 φ-性质上不同.
- 若所有 φ-性质同等看待, 只以共享的 φ-性质的数量来定义相似度, 则任意两个对象之间的相似度都一样.

偏置从何而来? — 三个来源

- 丑小鸭定理: 没有偏置就无法分类.
- 没有免费午餐定理: 没有偏置就无法学习. 若 "假设空间" 上的概率分布是 "块均匀的", 那么, 相对于整个假设空间上的期望表现来说, 任何搜索或优化算法都是同样好或同样差的.
- 溯因 ≈ 对假设空间搜索; 规划 ≈ 对策略空间搜索. 高效搜索靠偏置.
- 偏置是
  1. 事物的 "本性"?
     — 自然种类, 因果机制
  2. 语言框架 "固化" 的?
     — 古德曼绿蓝悖论: 什么性质 "可投射"?
     — 蒯因: 世界的本体论划分依赖于语言里已经 "固化" 的权重.
  3. 还是人类 "价值" 导向的?
     — 渡边慧 Watanabe: "在生物学家眼里, 鲸鱼与大象同类; 在渔业管理者眼里, 鲸鱼与金枪鱼同类." 认知是价值负载的.

偏置与学习: 语言 vs 世界 vs 价值

- 古德曼: 语言里编码了关于世界的背景知识.
- 蒯因: 世界的本体论划分依赖于语言里已经 “固化” 的权重.



- 关于世界/因果/价值的背景知识, 也需要 “固化” 到语言中. 语言应随学习过程而演化?

$$U \longrightarrow U' \longrightarrow U'' \longrightarrow \cdots$$

- 认知是价值负载的, 行动是价值导向的. 价值对齐不应孤立的进行, 而应与溯因学习、语言演化协同进行?

休谟/古德曼.....

Problem (What's next?)

$$1, 2, 4, 7, ?$$

Solution

A. $1, 2, 4, 7, \mathbf{11}, 16, \ldots$

B. $1, 2, 4, 7, \mathbf{12}, 20, \ldots$

C. $1, 2, 4, 7, \mathbf{13}, 24, \ldots$

D. $1, 2, 4, 7, \mathbf{14}, 28$

E. $1, 2, 4, 7, \mathbf{1}, 1, 5, 8, \ldots$

休谟/古德曼.....

Problem (What's next?)

$$1, 2, 4, 7, ?$$

Solution

A. $1, 2, 4, 7, \mathbf{11}, 16, \ldots$

$$a_{n+1} = a_n + n$$

B. $1, 2, 4, 7, \mathbf{12}, 20, \ldots$

$$a_{n+2} = a_{n+1} + a_n + 1$$

C. $1, 2, 4, 7, \mathbf{13}, 24, \ldots$

"Tribonacci" sequence

D. $1, 2, 4, 7, \mathbf{14}, 28$

divisors of 28

E. $1, 2, 4, 7, \mathbf{1}, 1, 5, 8, \ldots$

$\pi = 3.14159\ldots$ *and* $e = 2.71828\ldots$ *interleaved*

伊壁鸠鲁 vs 奥卡姆

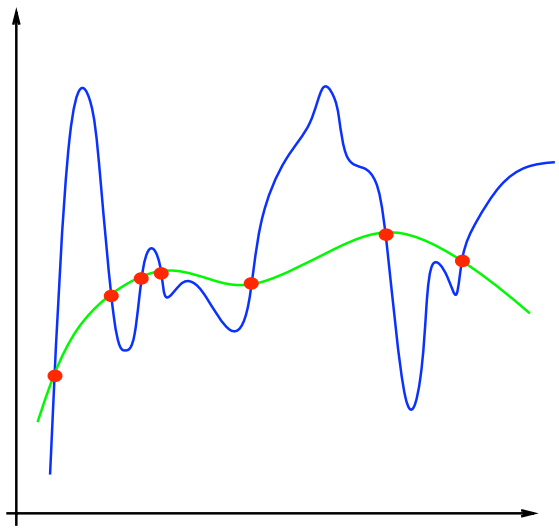
Proposition (伊壁鸠鲁)

所有与观察一致的理论都要保留.

Proposition (奥卡姆剃刀)

偏好与观察一致的简单理论.

- 如无必要, 勿增实体.
- 只要可能, 就用逻辑构造代替推论出的实体.
- 用较少的东西能做到的, 就不要用更多的东西.



- 更少的假设 vs 更少的实体?
- 奥卡姆剃刀能解决过拟合吗?
- 简单的理论是否可能由于其他原因而被偏好? 比如计算和认知成本?
- 什么样的 "理论"?
  — 可计算的机制 (物理丘奇-图灵论题)

Why Simplicity?

- 穿过两点的直线有且仅有一条
- 穿过两点的抛物线有无穷条
- 穿过两点的高次曲线更多更多……
- 能解释给定观测现象的复杂理论远远多于简单的理论
- 如果你找到了一个简单解释 H, 那么, 再找一个与 H 预测相反的简单解释 H' 的可能性, 要比找一个与 H 预测相反的复杂解释 H'' 的可能性要小得多
- 复杂性增加, 拟合力暴涨.

"四个参数画大象, 五个参数鼻子晃."

— 冯诺依曼

- 更多参数, 不仅可以让大象鼻子晃, 还可以让大象跳钢管舞, 还可以让蜡笔小新跳大象舞, 还可以……
- "压缩即泛化" 的直觉之一: 简单的模型更难迁就噪声

Why Simplicity Works in Psychology?

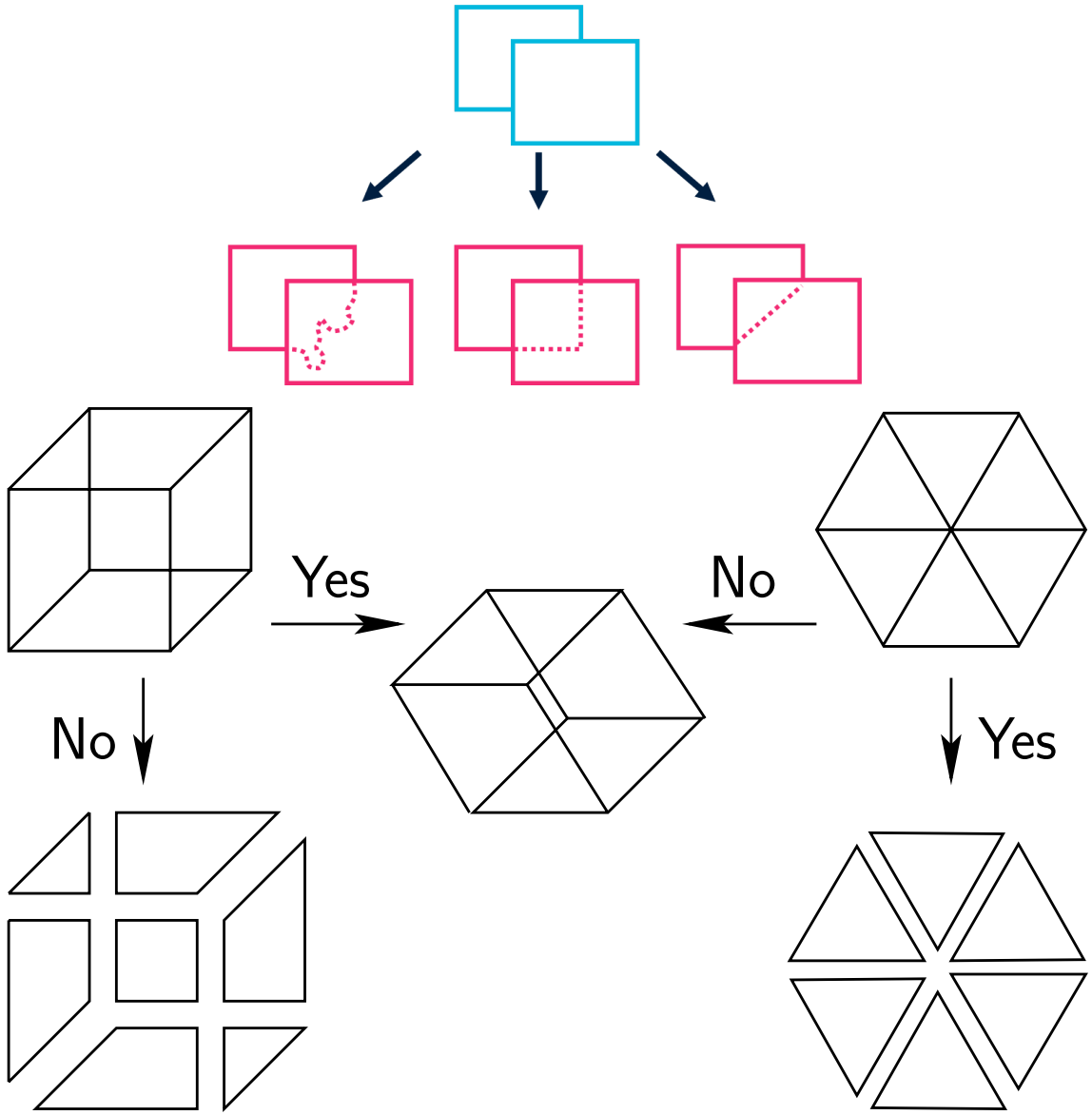
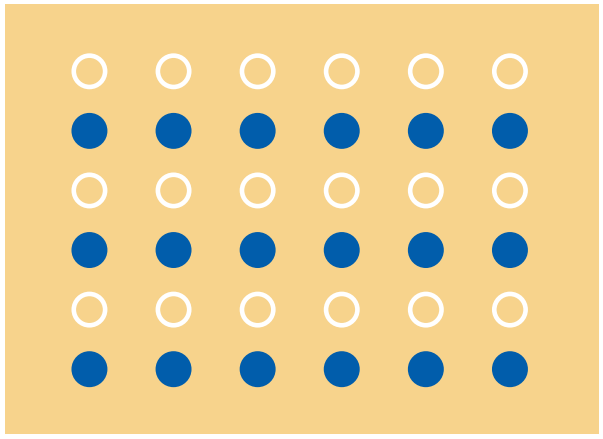


Figure: Gestalt Psychology

Why Simplicity Works in Psychology?

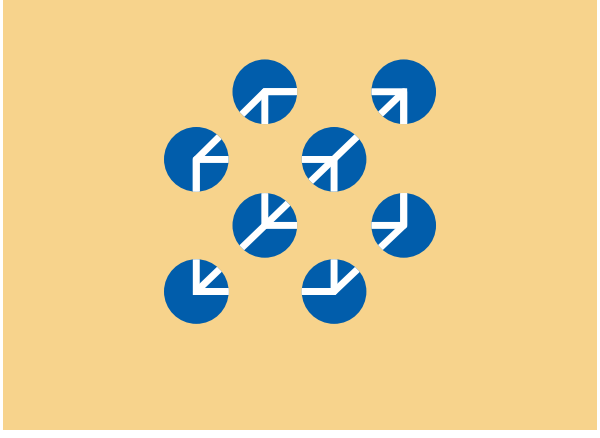
Gestalt Laws of Organization

(a) Proximity



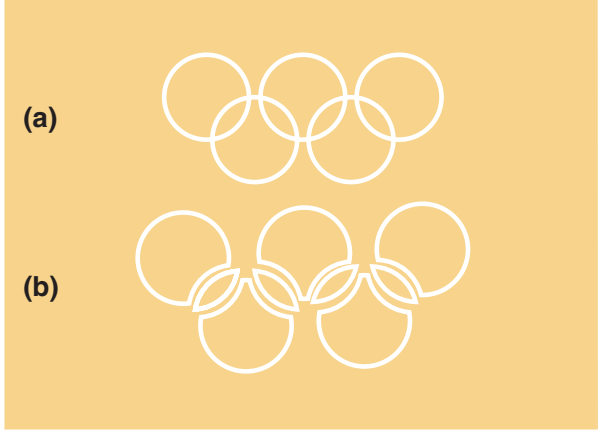
(b) Similarity

(c) Continuity



(d) Closure

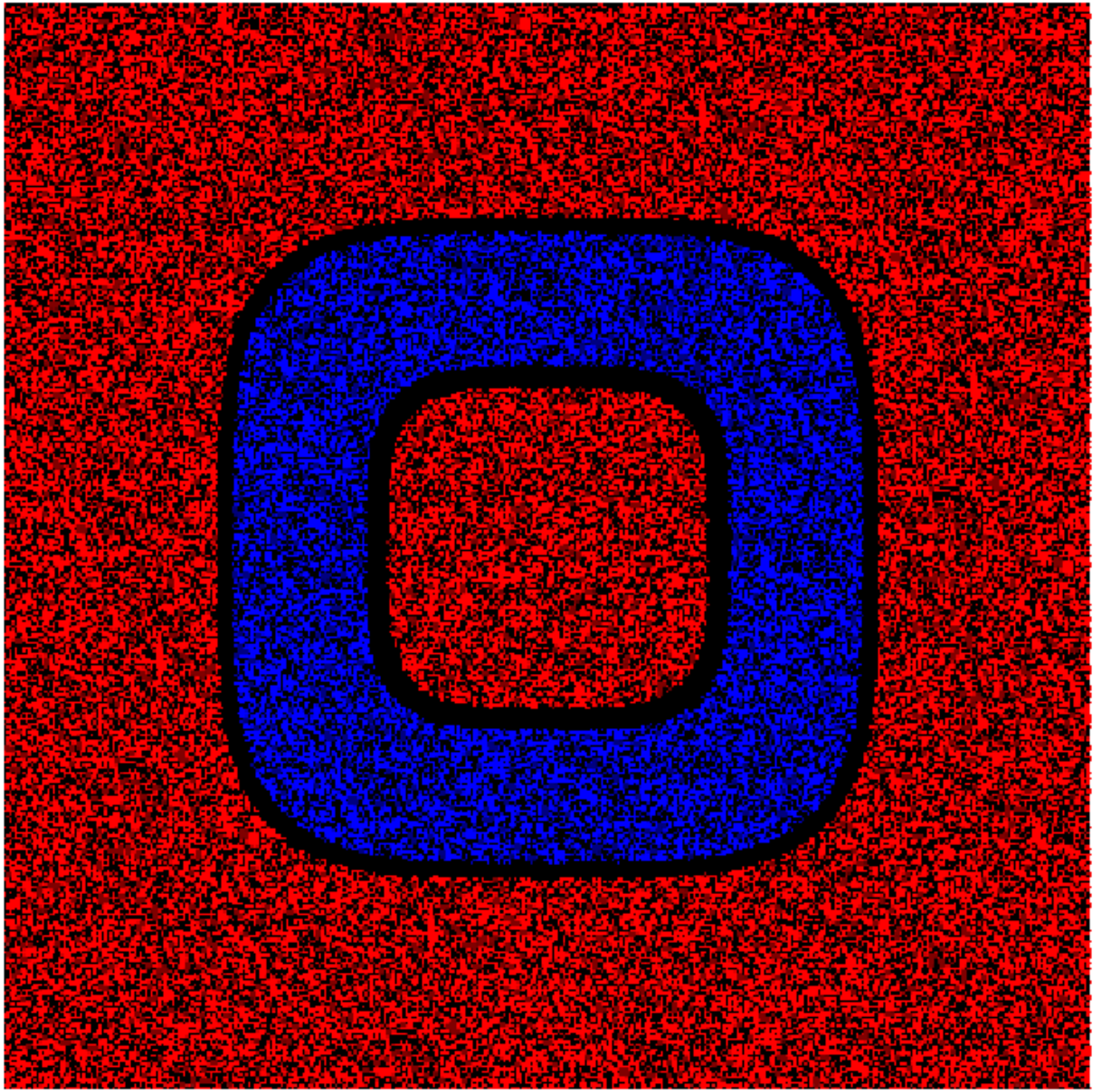
(e) Law of Symmetry

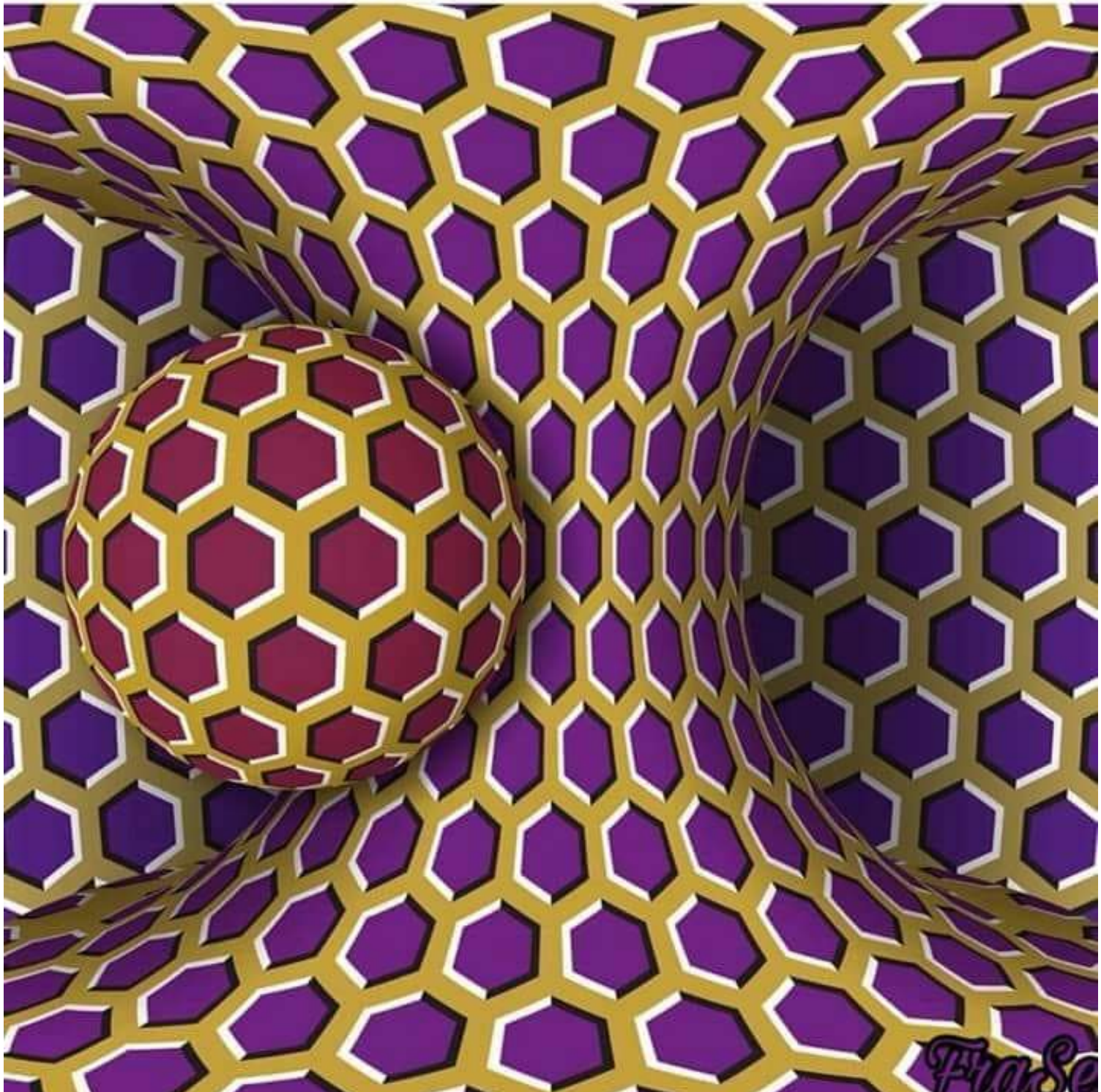


(f) Law of Prägnanz

Gestalt Laws of Organization

- Proximity principle: elements tend to be perceived as aggregated into groups if they are near each other.
- Common fate principle: elements tend to be perceived as grouped together if they move together.
- Similarity principle: elements tend to be integrated into groups if they are similar to each other.
- Continuity principle: oriented units or groups tend to be integrated into perceptual wholes if they are aligned with each other.
- Closure principle: elements tend to be grouped together if they are parts of a closed figure.
- Good gestalt principle: elements tend to be grouped together if they are parts of a pattern which is a good Gestalt — as simple, orderly, balanced, unified, coherent, regular, etc as possible, given the input.
- Past experience principle: elements tend to be grouped together if they were together often in the past experience of the observer.
- Symmetry principle: symmetrical components tend to group together.
- Convexity principle: convex rather than concave patterns tend to be perceived as figures.
- Common region principle: elements tend to be grouped together if they are located within the same closed region.
- Connectedness principle: elements tend to be grouped together if they are connected by other elements.




Fraser

Why Simplicity Works in Physics?

上帝不掷骰子.
上帝总是选择最简单的方式.
上帝难以捉摸, 但无恶意.
宇宙最不可理解之处是它是可理解的.
我真正感兴趣的是, 上帝是否可以以其他方式创造世界. 换句话说, 逻辑简单性是否为自由选择留有余地.
当我判断一个理论时, 我问自己, 如果我是上帝, 我是否会以这种方式创造世界.

— 爱因斯坦

Why Simplicity Works in Physics?

- 诺特定理: 对称性 $\leftrightarrow$ 守恒律
  - 时间平移对称性 $\leftrightarrow$ 能量守恒
  - 空间平移对称性 $\leftrightarrow$ 动量守恒
  - 空间转动对称性 $\leftrightarrow$ 角动量守恒
  - 空间反演 (镜像+ 旋转) 对称性 $\leftrightarrow$ 宇称守恒

  Remark: 对称性的本质是变换之后的不变性. 如果一个几何图形在某种变换下保持不变, 则称此图形在此变换下对称. 而守恒量, 即是表征变化前后保持不变的物理量.
- 最小 (稳定) 作用量原理
- That which respects symmetries are laws of physics.

Remark: 复杂系统的模型往往非常简单. 不是因为简单模型的预测效果更好, 而是因为在巨大误差面前, 细微的改进几乎毫无作用.

凯恩斯 $\Longrightarrow$ 卡尔纳普

- 概率不是事件的性质，而是描述事件的命题之间的关系.

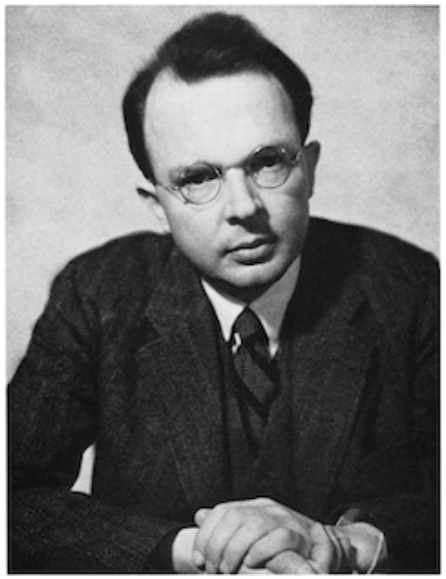
  — 布尔

- 概率是证据与假设之间的逻辑关系. 归纳假设的概率应该随着支持它的独立事件的增加而趋近于 1.

  — 凯恩斯

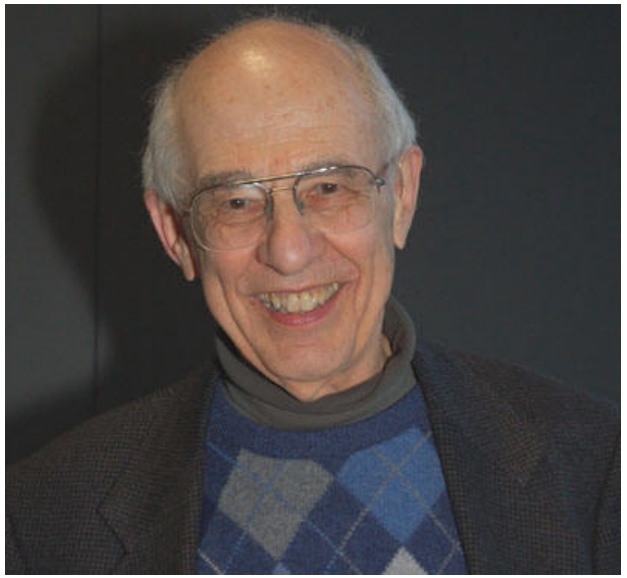
- 概率有经验维度和逻辑维度. 可观测现象的相对频率是经验性的. 证据对假设的确证度是一种纯逻辑关系. 确证度的评估本质上是对符号的意义分析. 选择确证度最高的假设.

  — 卡尔纳普



普特南对卡尔纳普的批评

- 卡尔纳普的归纳逻辑是为了构造一台 “学习机器”，它能够从数据中外推出经验规律.
- 归纳逻辑的任务，就是构造一台 “通用学习机器”.
- 如果确实存在一种一劳永逸地正确定义的 “确证度” 函数，那么一台按照这种定义进行预测的机器，就将是最智能的学习机器.
- 但是，要么存在越来越好的 “确证度” 函数，但不存在 “最佳可能”；要么存在一个 “最佳可能” 的函数，但它不能被任何机器计算出来.



Formal Learning Theory

- Putnam 1963
- Gold 1967 $\hat{f}(n+1) = g(\langle f(0), \ldots, f(n)\rangle)$ $\varphi_{\lim_{n\to\infty} g(\langle f(0),\ldots,f(n)\rangle)} = f$
- Solomonoff 1960

Philosophy of Induction

What is learnable? How to learn?
How can we know that what we learned is true?

History

Possible Worlds/Hypothesis (Epicurus/Leibniz)

$+$

Homogeneous Universe(s) (Mill/Turing)

$+$

Simplicity Criterion (Occam/Kolmogorov)

$+$

Prior Belief (Carnap/Solomonoff)

$+$

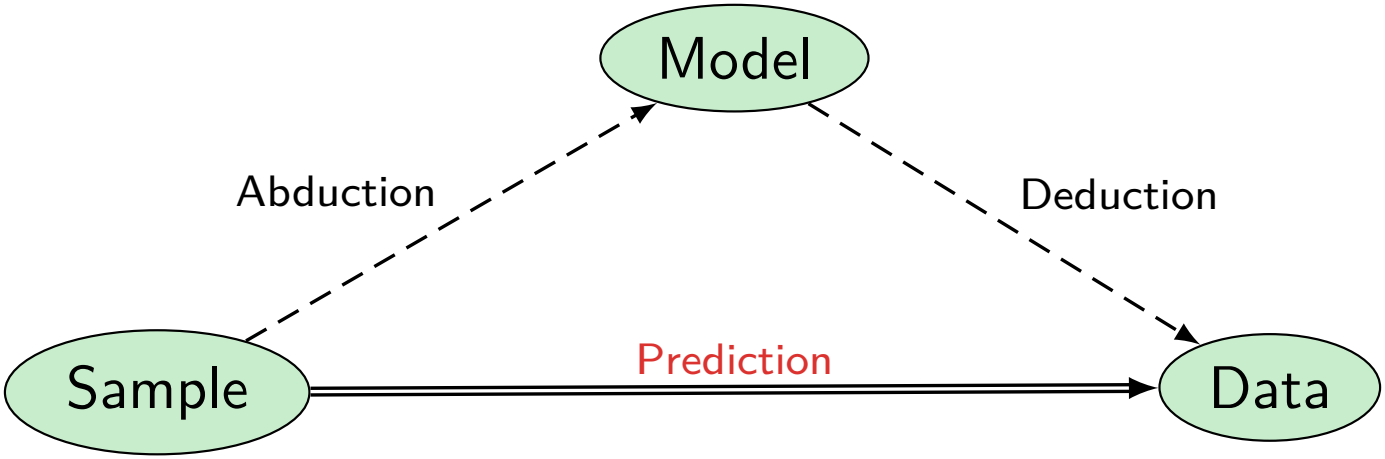
Update Belief (Bayes)

$\Downarrow$

Convergence to Truth

$$P(h \mid e) = \frac{P(e \mid h)P(h)}{\sum_{h \in \mathcal{H}} P(e \mid h)P(h)} \xrightarrow{\ell(e)\to\infty} 1$$

MDL vs Bayesian Mixture



When solving a problem of interest, do not solve a more general problem as an intermediate step.

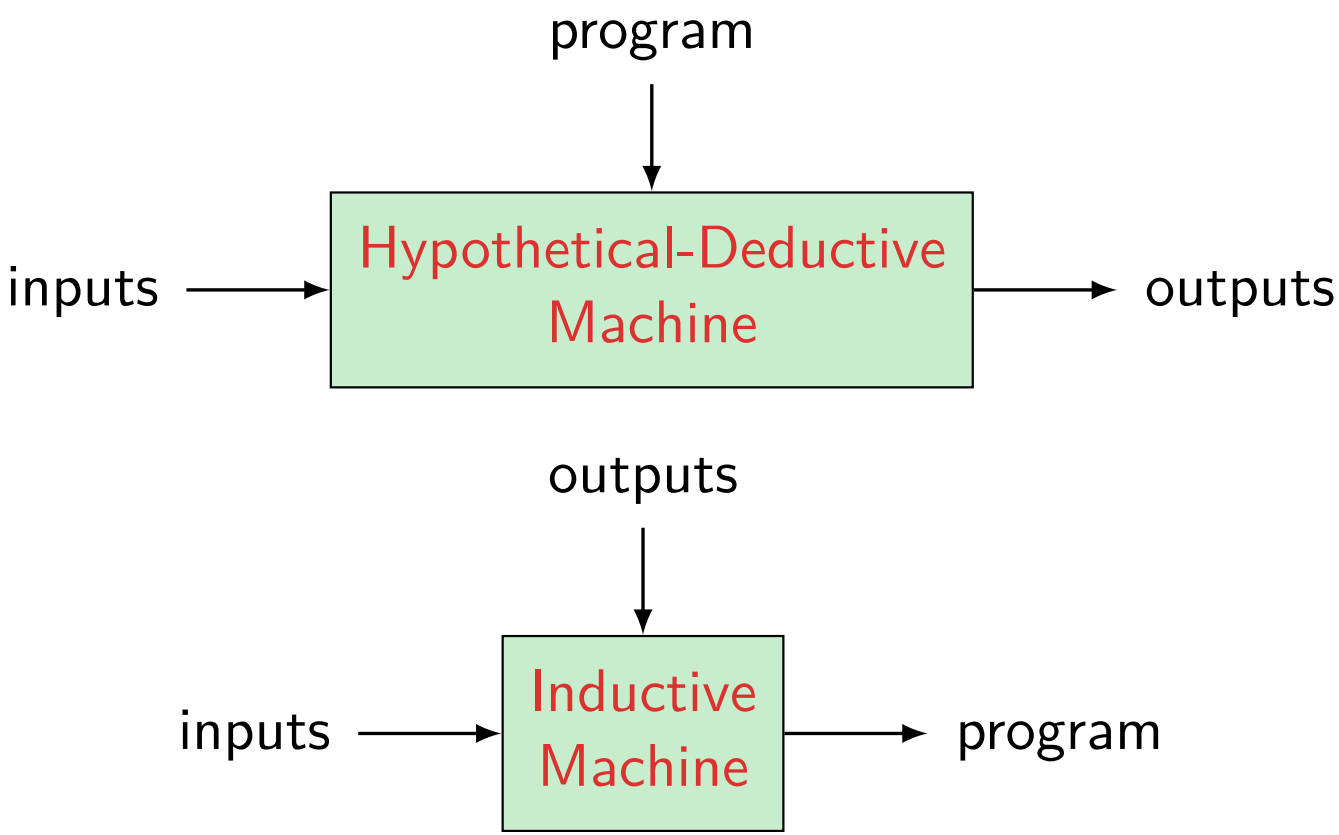
Intelligence ∼ Science ∼ Finding Patterns ∼ Compression ∼ MDL ∼ Prediction

贝叶斯主义

- 所有模型都是错的, 但有些更有用.
- 互不相容的模型组成的森林比其中每一棵树都睿智.
- 主观贝叶斯主义: 先验 (偏见) 是必要的.
  — No learning is possible without some prior knowledge.
  — 主观, 但不随意! (丘奇-图灵论题、奥卡姆剃刀)
- 怀疑一切和相信一切是两种同样便利的方法, 都无需思考.

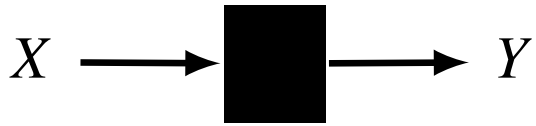
— 庞加莱

Hypothetical-Deductive Machine vs Inductive Machine

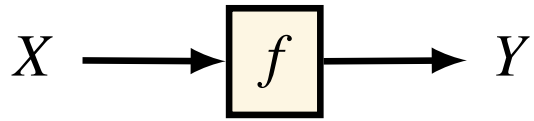


1. ML as an Oracle: Data → Answer

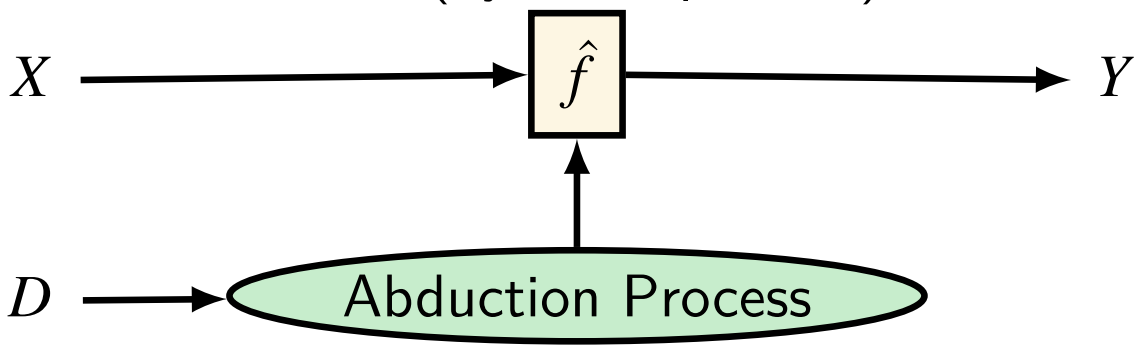
2. ML as a black box (correlation)



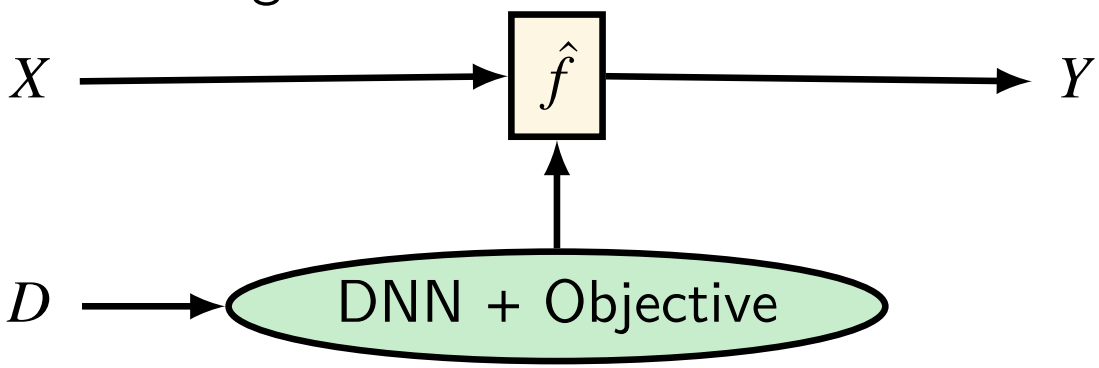
3. ML as a Prediction/Induction box (static model)



4. ML as an Abduction Process (dynamic process)



5. ML as an Abduction Algorithm



Machine	World	Calculator	Target
Cybernetics (connectionist)	Environment	"Black box"	Negative feedback
Symbolic AI (symbolic)	"Toy" world	Logical reasoning	Problem-solving
Expert Systems (symbolic)	World of expert knowledge	Selection of hypothesis	Examples/ Counter-examples
Deep Learning (connectionist)	The world as a vector of big data	Deep neural network	Objective-based error optimization

Definition (Convergence of Random Sequences)

Let $z_1(\omega), z_2(\omega), \ldots$ be a sequence of random variables. We say $z_t \xrightarrow{t\to\infty} z_*$

(i) with probability 1 $(z_t \xrightarrow[w.p.1]{t\to\infty} z_*)$ iff $P(\{\omega : z_t(\omega) \to z_*(\omega)\}) = 1$

iff $\forall \varepsilon > 0 : P\left(\sup_{s\geq t} |z_s - z_*| \geq \varepsilon\right) \xrightarrow{t\to\infty} 0$

(ii) in mean $(z_t \xrightarrow[i.m.]{t\to\infty} z_*)$ iff $\mathbb{E}_\mu[(z_t - z_*)^2] \xrightarrow{t\to\infty} 0$

(iii) in mean sum $(z_t \xrightarrow[i.m.s.]{t\to\infty} z_*)$ iff $\sum_{t=1}^{\infty} \mathbb{E}_\mu[(z_t - z_*)^2] < \infty$

(iv) in probability $(z_t \xrightarrow[i.p.]{t\to\infty} z_*)$ iff $\forall \varepsilon > 0 : P(|z_t - z_*| \geq \varepsilon) \xrightarrow{t\to\infty} 0$

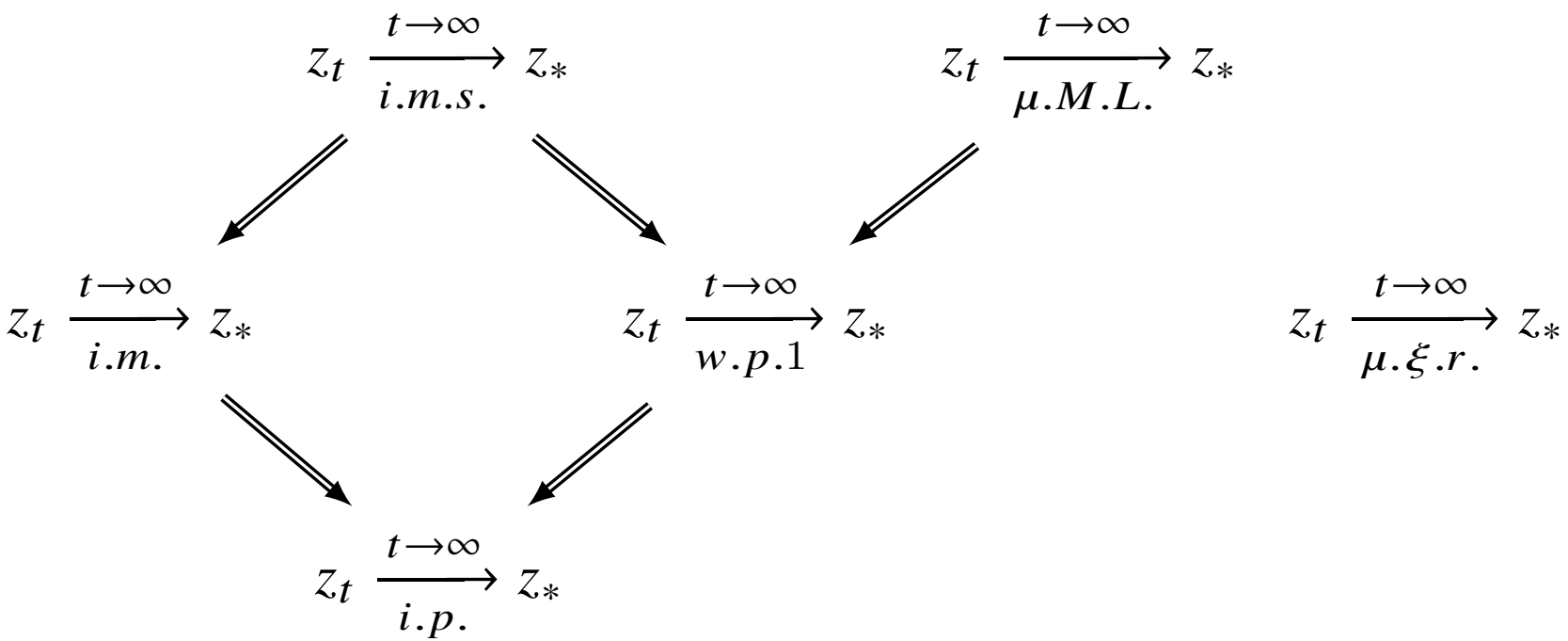
(v) for every μ-Martin-Löf random sequence $(z_t \xrightarrow[\mu.M.L.]{t\to\infty} z_*)$ iff

$\forall \omega : [\exists c \forall n : M(\omega_{1:n}) \leq c\mu(\omega_{1:n})] \implies z_t(\omega) \xrightarrow{t\to\infty} z_*(\omega)$

(vi) for every μ/ξ-random sequence $(z_t \xrightarrow[\mu.\xi.r.]{t\to\infty} z_*)$ iff

$\forall \omega : [\exists c \forall n : \xi(\omega_{1:n}) \leq c\mu(\omega_{1:n})] \implies z_t(\omega) \xrightarrow{t\to\infty} z_*(\omega)$

Convergence of Random Sequences



Lemma (Entropy Inequalities)

Let $\{y_i\}$ and $\{z_i\}$ be two probability distributions, and f be a convex and even$(f(x) = f(-x))$ function with $f(0) \le 0$. Then

(i). $$\frac{1}{2}\sum_i f(y_i - z_i) \;\le\; f\left(\sqrt{\frac{1}{2}\sum_i y_i \ln \frac{y_i}{z_i}}\right)$$

(ii). $$\sum_i (y_i - z_i)^2 \;\le\; \sum_i y_i \ln \frac{y_i}{z_i}$$

(iii). $$\sum_i (\sqrt{y_i} - \sqrt{z_i})^2 \;\le\; \sum_i y_i \ln \frac{y_i}{z_i}$$

(iv). $$\sum_i y_i \left|\ln \frac{y_i}{z_i}\right| - \sum_i y_i \ln \frac{y_i}{z_i} \;\le\; \sum_i |y_i - z_i| \;\le\; \sqrt{2\sum_i y_i \ln \frac{y_i}{z_i}}$$

Theorem (Convergence Theorem)

$$\sum_{t=1}^{n} \mathbb{E}_\mu \left[\left(\sqrt{\frac{\rho(a \mid x_{<t})}{\mu(a \mid x_{<t})}} - 1 \right)^2 \right] \leq \sum_{t=1}^{n} \mathbb{E}_\mu \left[\sum_{a \in \mathcal{X}} \left(\sqrt{\rho(a \mid x_{<t})} - \sqrt{\mu(a \mid x_{<t})} \right)^2 \right] \leq D_n(\mu \| \rho)$$

$$\sum_{t=1}^{n} \mathbb{E}_\mu \left[\sum_{a \in \mathcal{X}} \left(\rho(a \mid x_{<t}) - \mu(a \mid x_{<t}) \right)^2 \right] \leq D_n(\mu \| \rho)$$

$$\frac{1}{2n} \left(\sum_{t=1}^{n} \mathbb{E}_\mu \left[\sum_{a \in \mathcal{X}} \left| \rho(a \mid x_{<t}) - \mu(a \mid x_{<t}) \right| \right] \right)^2 \leq D_n(\mu \| \rho)$$

where

$$D_n(\mu \| \rho) := \mathbb{E}_\mu \left[\ln \frac{\mu(x_{1:n})}{\rho(x_{1:n})} \right]$$

Theorem

$$\xi = \operatorname*{argmin}_\rho \mathbb{E}_w \left[D(\mu \| \rho) \right] \quad \textit{where} \quad \xi(x) := \sum_{\nu \in \mathcal{M}} w_\nu \nu(x)$$

Bayesian Decisions

Suppose $\mathrm{Loss}(x_t, y_t) \in [0, 1]$

$$y_t^{\Lambda_\rho}(x_{<t}) \;:=\; \arg\min_{y_t} \sum_{x_t} \rho(x_t \mid x_{<t})\, \mathrm{Loss}(x_t, y_t)$$

$$L^{\Lambda_\rho}(x_{<t}) \;:=\; \mathbb{E}_\mu \left[\mathrm{Loss}\left(x_t, y_t^{\Lambda_\rho}\right) \middle| x_{<t} \right]$$

$$L_n^{\Lambda_\rho} \;:=\; \sum_{t=1}^{n} \mathbb{E}_\mu \left[L^{\Lambda_\rho}(x_{<t}) \right]$$

Theorem

$$\left(\sqrt{L_n^{\Lambda_\xi}} - \sqrt{L_n^{\Lambda_\mu}} \right)^2 \leq 2 \sum_{t=1}^{n} \mathbb{E}_\mu \left[\sum_{a \in \mathcal{X}} \left(\sqrt{\xi(a \mid x_{<t})} - \sqrt{\mu(a \mid x_{<t})} \right)^2 \right] \leq 2 D_n(\mu \| \xi)$$

$$L_n^{\Lambda_\xi} - L_n^{\Lambda_\mu} \leq 2 D_n(\mu \| \xi) + 2 \sqrt{L_n^{\Lambda_\mu} D_n(\mu \| \xi)}$$

贝叶斯归纳的两个问题

如何选择模型类和先验?

1. 选择包含真实环境的最小模型类.
2. 选择最能反映对环境的合理信念的先验.
   - 贝叶斯混合能够收敛到真实环境.
   - 能证实 “太阳每天都会升起”.
   - 符合不变性标准.
     reparametrization & regrouping invariant.

Contents

1. Introduction

2. Philosophy of Induction
   - History
   - How to Choose the Prior?

3. Inductive Logic

4. Universal Induction

5. Causal Inference

6. Game Theory

7. Reinforcement Learning

8. Deep Learning

9. Artificial General Intelligence

10. Emergence & Consciousness

11. AI Ethics

Invariance Criterion

- By applying some principle to a parameter θ we get prior $w(\theta)$. If we consider some new parametrization θ' via $f : \theta \mapsto \theta'$, then we get a prior $\widetilde{w}(\theta')$ by transforming the original prior via f.
  for discrete class $\mathcal{M}$,

$$\widetilde{w}(\theta') := \sum_{\theta : f(\theta) = \theta'} w(\theta)$$

  for continuous parametric class $\mathcal{M}$,

$$\widetilde{w}(\theta') := \int \delta(f(\theta) - \theta') w(\theta) \, \mathrm{d}\theta \qquad \text{(Dirac-delta)}$$

- Regrouping-invariant:

$$\widetilde{w}(\theta') = w'(\theta')$$

  where $w'(\theta')$ is obtained by applying the same principle to the new parametrization.
- We say the principle is reparametrization-invariant when f is bijective.

酒水悖论

- 有一瓶酒水混合液, 一种液体的含量至多是另一种液体的 3 倍.

$$\frac{1}{3} \leq \text{酒/水} \leq 3$$

- 根据无差别原则, 酒水比在区间 $[\frac{1}{3}, 3]$ 上均匀分布.
- 因此,

$$P(\text{酒/水} \leq 2) = \frac{2 - \frac{1}{3}}{3 - \frac{1}{3}} = \frac{5}{8}$$

- 根据无差别原则, 水酒比在区间 $[\frac{1}{3}, 3]$ 上均匀分布.
- 因此,

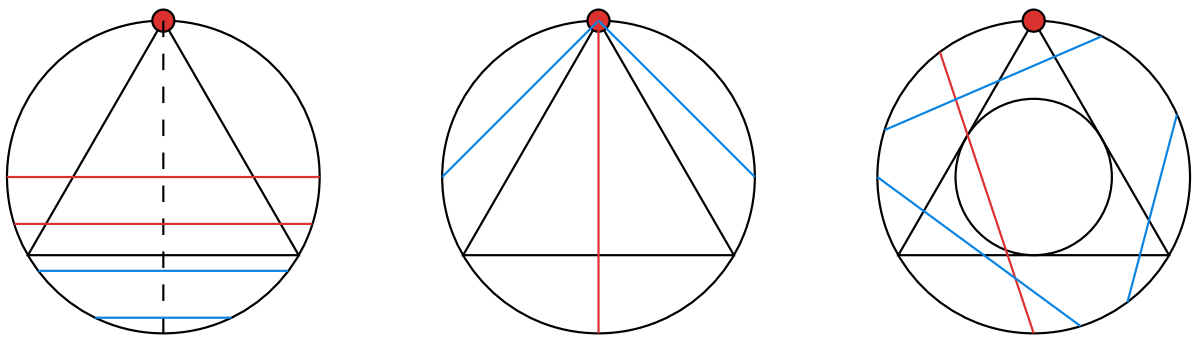
$$P(\text{水/酒} \geq \frac{1}{2}) = \frac{3 - \frac{1}{2}}{3 - \frac{1}{3}} = \frac{15}{16}$$

- 但这是两个相同的事件!
- $f : \theta \mapsto \frac{1}{\theta}$
- θ 在 $[a, b]$ 上均匀分布并不意味着 $\theta' = f(\theta)$ 在 $[f(a), f(b)]$ 上也是均匀分布.

Bertrand's Paradox

Problem (Bertrand's Paradox)

Consider an equilateral triangle inscribed in a circle. Suppose a chord of the circle is chosen at random. What is the probability that the chord is longer than a side of the triangle?



$\frac{1}{2}$? $\frac{1}{3}$? $\frac{1}{4}$?

What is "randomness"? a process or a product?

How to Assign Prior? Indifference Principle/MaxEnt

How to Assign Prior?

- The principle of indifference. (Not reparametrization invariant) Example: Assume $w(\theta) = 1$ and $\theta' = \sqrt{\theta}$.

$$\widetilde{w}(\theta') = w(f^{-1}(\theta'))\frac{\mathrm{d}f^{-1}(\theta')}{\mathrm{d}\theta'} = 2\sqrt{\theta} \neq w'(\theta')$$

- The maximum entropy principle. Maximize the entropy subject to some constraints provided by empirical data or considerations of symmetry, probabilistic laws, and so on.

$$(p_1^*, \cdots, p_k^*) := \underset{(p_1,\cdots,p_k)\vDash \sum_{i=1}^{k} p_i=1}{\operatorname{argmax}} -\sum_{i=1}^{k} p_i \log p_i$$

$$(p_1^*, \cdots, p_k^*) = \left(\frac{1}{k}, \cdots, \frac{1}{k}\right)$$

- Occam's razor — the simplicity principle.

How to confirm “All Ravens are Black”?

Problem 1 — All Ravens are Black $\theta = 1$

Suppose θ is the percentage of ravens that are black.
"All ravens are black" $\equiv \theta = 1$. or, 1^∞ or, $\forall x : R(x) \rightarrow B(x)$

$$P(\theta = 1) = \int_1^1 w(\theta)\, \mathrm{d}\theta = 0 \qquad (0 \text{ prior})$$

$$\Downarrow$$

$$P(\theta = 1 \mid 1^n) = \frac{P(1^n \mid \theta = 1)P(\theta = 1)}{P(1^n)} = 0 \qquad (\times)$$

Problem 2 — The Sun will always Rise 1^∞

Indifference Principle

$$\left.\begin{aligned} &\int_0^1 w(\theta)\,\mathrm{d}\theta = 1 \\ &\forall \theta, \theta' : w(\theta) = w(\theta') \end{aligned}\right\} \implies \forall \theta : w(\theta) = 1$$

The sun will rise tomorrow. ✓

$$P(x) = \int_0^1 P(x \mid \theta) w(\theta)\,\mathrm{d}\theta \implies P(1 \mid 1^n) = \frac{n+1}{n+2}$$

The sun will always rise. ×

$$P(1^\infty \mid 1^n) = \lim_{k\to\infty} P(1^k \mid 1^n) = \lim_{k\to\infty} \frac{n+1}{n+k+1} = 0$$

Solution 1 — Soft Hypothesis — No absolute truth!

$$H_\varepsilon = \{\theta : \theta \in (1-\varepsilon, 1]\}$$

$$P(H_\varepsilon) = \int_{1-\varepsilon}^{1} w(\theta)\,\mathrm{d}\theta = \varepsilon > 0$$

$$\begin{aligned}
P(H_\varepsilon \mid 1^n) &= \int_{1-\varepsilon}^{1} w(\theta \mid 1^n)\,\mathrm{d}\theta \\
&= \int_{1-\varepsilon}^{1} \frac{P(1^n \mid \theta) w(\theta)}{P(1^n)}\,\mathrm{d}\theta \\
&= \int_{1-\varepsilon}^{1} (n+1)\theta^n\,\mathrm{d}\theta \\
&= \theta^{n+1}\Big|_{1-\varepsilon}^{1} \\
&= 1 - (1-\varepsilon)^{n+1} \xrightarrow{n\to\infty} 1
\end{aligned}$$

Solution 2 — ad hoc

$$w(\theta) := \frac{1}{2}(1 + \delta(1 - \theta))$$

Dirac-delta sifting property: $\int f(\theta)\delta(\theta - a)\,\mathrm{d}\theta = f(a)$

$$\begin{aligned}
P(x) &= \int_0^1 P(x \mid \theta) w(\theta)\,\mathrm{d}\theta \\
&= \int_0^1 \theta^s (1-\theta)^f \cdot \frac{1}{2}(1 + \delta(1-\theta))\,\mathrm{d}\theta \\
&= \frac{1}{2}\int_0^1 \theta^s (1-\theta)^f (1 + \delta(\theta - 1))\,\mathrm{d}\theta \\
&= \frac{1}{2}\left(\frac{s!f!}{(s+f+1)!} + 1^s \cdot (1-1)^f\right) && \text{[sifting property]} \\
&= \frac{1}{2}\left(\frac{s!f!}{(s+f+1)!} + \delta_{f,0}\right)
\end{aligned}$$

Solution 2 — ad hoc

$$P(1^\infty \mid 1^n) = \lim_{k\to\infty} \frac{P(1^{n+k})}{P(1^n)} = \lim_{k\to\infty} \frac{\frac{1}{2}\left(\frac{(n+k)!0!}{(n+k+1)!} + 1\right)}{\frac{1}{2}\left(\frac{n!0!}{(n+1)!} + 1\right)} = \frac{n+1}{n+2} \xrightarrow{n\to\infty} 1$$

$$P(\theta \geq a) = \int_a^1 \frac{1}{2}(1 + \delta(\theta - 1))\,\mathrm{d}\theta = 1 - \frac{1}{2}a$$

$$P(\theta = 1) = \frac{1}{2}$$

$$P(\theta = 1 \mid 1^n) = \frac{P(1^n \mid \theta = 1)P(\theta = 1)}{P(1^n)} = \frac{1 \cdot \frac{1}{2}}{\frac{1}{2}\left(\frac{n!0!}{(n+1)!} + 1\right)} = \frac{n+1}{n+2} \xrightarrow{n\to\infty} 1$$

Why $\theta = 1$ special?

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Natural Wish List

- (computability) $P_n(A)$ is computable.
- (convergence) $P(A) = \lim_{n \to \infty} P_n(A)$
- (coherent limit) $P(A \wedge B) + P(A \vee B) = P(A) + P(B)$
- (non-dogmatism) If $\nvdash A$ then $P(A) < 1$, and if $\nvdash \neg A$ then $P(A) > 0$.

Unary Pure Inductive Logic[7]

- $\mathscr{L}$ contains countable constants C and m unary predicates.
- $\mathcal{R} = \{R_1, R_2, \ldots, R_m\}$ with neither function symbols nor equality.
- $Q_i := \bigwedge_{j=1}^{m} \pm R_j$ for $1 \leq i \leq 2^m =: r$.
- $\mathcal{Q} = \{Q_1, \cdots, Q_r\}$ is a r-fold classification system of some Universe with domain C.

Definition (Probability on Sentences)

A probability on sentences is a non-negative function $w : \mathcal{S} \to [0, 1]$ s.t.

$$P_1.\ \vDash A \implies w(A) = 1$$

$$P_2.\ A \vDash \neg B \implies w(A \vee B) = w(A) + w(B)$$

$$P_3.\ w\big(\exists x A(x)\big) = \lim_{n\to\infty} w\left(\bigvee_{i=1}^{n} A(a_i)\right)$$

[7] J. Paris and A. Vencovská: Pure inductive logic. 2015.

Properties

Theorem

(i) $w(\neg A) = 1 - w(A)$

(ii) $\vDash \neg A \implies w(A) = 0$

(iii) *The following are equivalent:*

1. $w(A) = 1 \implies \vDash A$
2. $w(A) = 0 \implies \vDash \neg A$

(iv) $A \vDash B \implies w(A) \leq w(B)$

(v) $\vDash A \leftrightarrow B \implies w(A) = w(B)$

(vi) $w(A) + w(B) = w(A \wedge B) + w(A \vee B)$

Theorem (Extension Theorem)

For any probability function over quantifier-free sentences $w : \mathcal{S} \to [0, 1]$ *satisfying* P_1, P_2, w *has a unique extension to* $\overline{w} : \mathcal{S} \to [0, 1]$ *satisfying* P_1, P_2, P_3.

Possible Worlds

- **state description** $\bigwedge_{i=1}^{n} Q_{h_i}(a_i)$

  where $h : C \to Q$

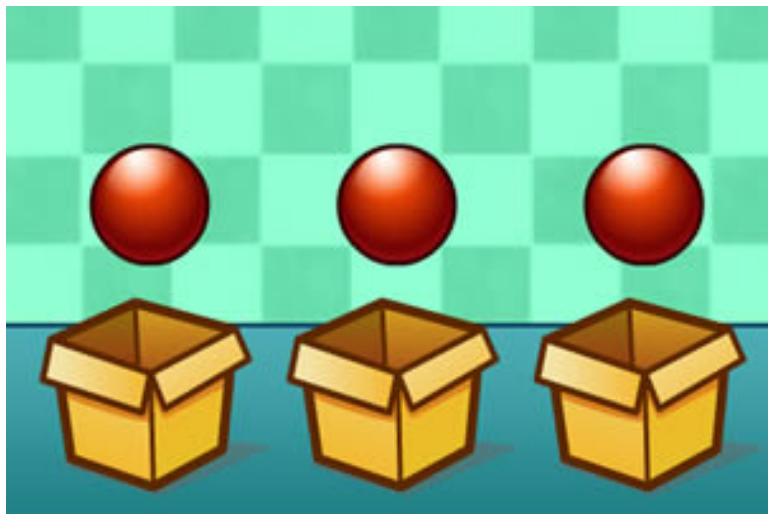
- **structure description** $\{n_i\}_{i=1}^{r}$

  where $n_i := \sum_{j=1}^{n} [\![h_j = i]\!]$

- **rank description** $\{m_i\}_{i=0}^{n}$

  where $m_i := \sum_{j=1}^{r} [\![n_j = i]\!]$

  Obviously, $\sum_{i=1}^{n} i \cdot m_i = n$ and $\sum_{i=0}^{n} m_i = r$.



Indifference Principle

A All state descriptions have equal weight.

B All structure descriptions have equal weight.

C Each non-empty subset of the alphabet is equally likely.

D Each nonzero cardinality is equally likely.

E All rank descriptions have equal weight.

Given n individuals, there are r^n possible state descriptions,

$$\left|\left\{(n_1,\ldots,n_r):\sum_{i=1}^{r}n_i=n\right\}\right|=\binom{n+r-1}{r-1}$$

possible structure descriptions, and

$$p(n,r):=\left|\left\{(m_0,\ldots,m_n):\sum_{i=1}^{n}i\cdot m_i=n\ \ \&\ \ \sum_{i=0}^{n}m_i=r\ \ \&\ \ \forall i:m_i\geq 0\right\}\right|$$

possible rank descriptions.

(A) State Description ×

According to (A),

$$m^{\dagger}\left(\bigwedge_{i=1}^{n} Q_{h_i}(a_i)\right) = \frac{1}{r^n}$$

$$c^{\dagger}\left(Q_j(a_{n+1}) \,\middle|\, \bigwedge_{i=1}^{n} Q_{h_i}(a_i)\right) = \frac{m^{\dagger}\left(\bigwedge_{i=1}^{n} Q_{h_i}(a_i) \wedge Q_j(a_{n+1})\right)}{m^{\dagger}\left(\bigwedge_{i=1}^{n} Q_{h_i}(a_i)\right)} = \frac{1}{r}$$

(B) Structure Description ✓

According to (B),

$$m^*(n_1, \ldots, n_r) = \frac{1}{\binom{n+r-1}{r-1}}$$

Structure Description $(n_1, \ldots, n_r)$ corresponds to $\binom{n}{n_1, \ldots, n_r}$ State Descriptions.

$$m^*\left(\bigwedge_{i=1}^{n} Q_{h_i}(a_i)\right) = \frac{m^*(n_1, \ldots, n_r)}{\binom{n}{n_1, \ldots, n_r}} = \frac{1}{\binom{n+r-1}{r-1}\binom{n}{n_1, \ldots, n_r}}$$

Sometimes we write $m^*(h_{1:n}) := m^*\left(\bigwedge_{i=1}^{n} Q_{h_i}(a_i)\right)$ for short.

Carnap's Degree of Confirmation

$$c^*\left(Q_j(a_{n+1}) \,\middle|\, \bigwedge_{i=1}^{n} Q_{h_i}(a_i)\right) = \frac{m^*\left(\bigwedge_{i=1}^{n} Q_{h_i}(a_i) \wedge Q_j(a_{n+1})\right)}{m^*\left(\bigwedge_{i=1}^{n} Q_{h_i}(a_i)\right)} = \frac{n_j + 1}{n + r}$$

frequency — independent identical distribution(i.i.d)
extension

$$c^*(A)$$

(C,D,E)

According to (C),

$$m^{\$}(h_{1:n}) = \frac{1}{\left(\sum_{i=1}^{\min\{r,n\}} \binom{r}{i}\right) \binom{n-1}{r-m_0-1} \binom{n}{n_1,\ldots,n_r}}$$

According to (D),

$$m^{\#}(h_{1:n}) = \frac{1}{\min\{r,n\} \binom{r}{r-m_0} \binom{n-1}{r-m_0-1} \binom{n}{n_1,\ldots,n_r}}$$

According to (E),

$$m^{\tau}(h_{1:n}) = \frac{1}{\binom{n}{n_1,\ldots,n_r} \binom{r}{m_0,\ldots,m_n} p(n,r)}$$

Rank Description $(m_0,\ldots,m_n)$ corresponds to $\binom{r}{m_0,\ldots,m_n}$ Structure Descriptions.

What is the right w?

- Constant Exchangeability Principle.
  For any permutation σ of $\mathbb{N}^+$,

$$w(A(a_1, \ldots, a_n)) = w(A(a_{\sigma(1)}, \ldots, a_{\sigma(n)})) \tag{Ex}$$

- Atom Exchangeability Principle.
  For any permutation τ of $\{1, 2, \ldots, r\}$,

$$w\left(\bigwedge_{i=1}^{n} Q_{h_i}(a_i)\right) = w\left(\bigwedge_{i=1}^{n} Q_{\tau(h_i)}(a_i)\right) \tag{Ax}$$

- Sufficientness Postulate.

$$w\left(Q_j(a_{n+1}) \,\middle|\, \bigwedge_{i=1}^{n} Q_{h_i}(a_i)\right) = f_j(n_j, n) \tag{SP}$$

What is the right w?

Principle Ex asserts that $w\left(\bigwedge_{i=1}^{n} Q_{h_i}(a_i)\right)$ depends only on the vector $\langle n_{h_i} : 1 \leq i \leq n \rangle$, so that it is independent on the order of observing the individuals, while in the presence of Ex, principle Ax asserts that $w\left(\bigwedge_{i=1}^{n} Q_{h_i}(a_i)\right)$ depends only on $\{n_i : 1 \leq i \leq r\}$, and $w(Q_i(a_1)) = 1/r$ for all $1 \leq i \leq r$.

Carnap's λ-continuum

Theorem

Suppose language $\mathscr{L}$ has at least two predicates i.e. $m \geq 2$, then the probability function w on $\mathscr{L}$ satisfies Ex, SP iff $w = c_\lambda$ for some $0 \leq \lambda \leq \infty$. Namely,

$$f_i(n_i, n) = \frac{n_i + \lambda\gamma_i}{n + \lambda}$$

where $\gamma_i = f_i(0,0)$ and $\lambda = \dfrac{f_i(0,1)}{f_i(0,0) - f_i(0,1)}$.

By adding Ax, $\forall i : \gamma_i = \dfrac{1}{r}$.

Shortcoming 1 — All Ravens are Black? ×

$$c^* \left(\forall x \left(R(x) \rightarrow B(x)\right)\right) \leq \lim_{n \to \infty} \prod_{i=0}^{n-1} \frac{i + r - 1}{i + r} = 0$$

Convergence Speed? Yes and No!

$$\prod_{n \geq 1} a_n = 0 \iff \sum_{n \geq 1} (1 - a_n) = \infty \quad \text{for} \quad \forall n : 0 < a_n \leq 1$$

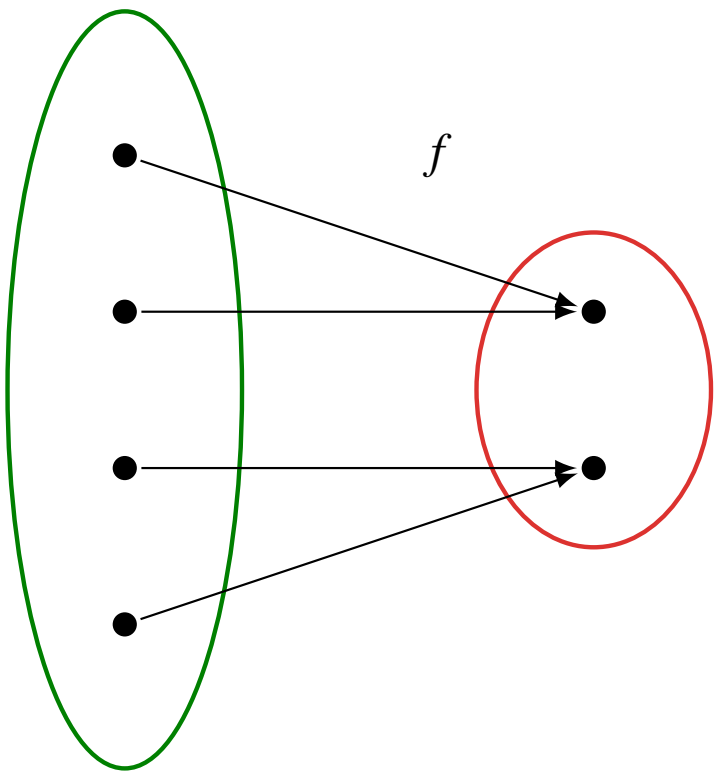
Shortcoming 2 — No-Free-Lunch for Carnap!

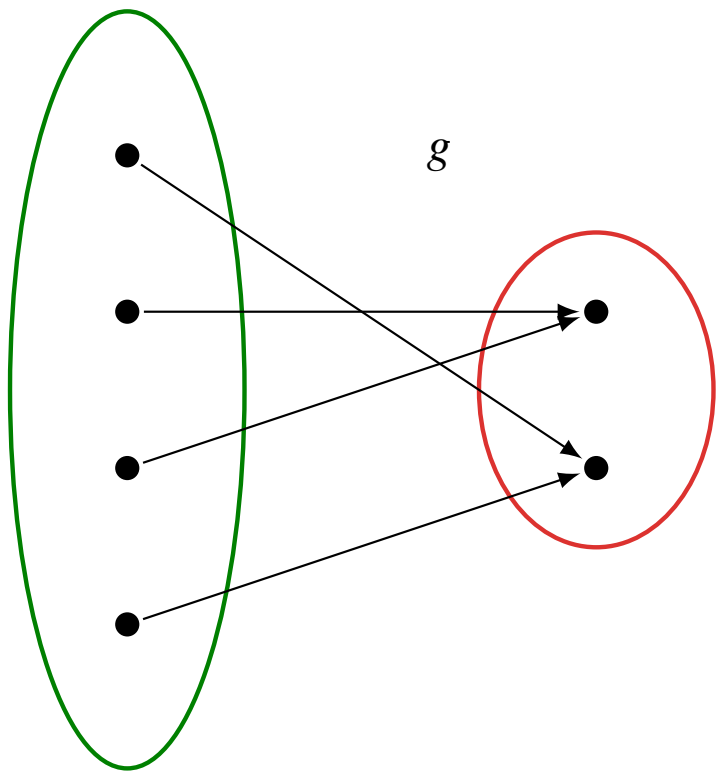
Strengthened "Hume" — Wolpert & Macready 1997, Igel & Toussaint 2004:

No-Free-Lunch Theorem!

All state descriptions with the same structure description have equal weight!

Block Uniform





No Free Lunch Theorem — Strengthened "Hume"

Theorem (No Free Lunch Theorem)

If and only if the probability distribution P is block uniform, i.e.

$$\forall f, g \in \mathcal{Y}^{\mathcal{X}} : \forall y \in \mathcal{Y} \left(\left| f^{-1}(y) \right| = \left| g^{-1}(y) \right| \right) \implies P(f) = P(g)$$

then for any two algorithms A, A', any value $k \in \mathbb{R}$, any $m \in \{1, \ldots, |\mathcal{X}|\}$, and any performance measure L,

$$\sum_{f \in \mathcal{Y}^{\mathcal{X}}} P(f) \left[\!\left[k = L\left(T_m^y(A, f)\right) \right]\!\right] = \sum_{f \in \mathcal{Y}^{\mathcal{X}}} P(f) \left[\!\left[k = L\left(T_m^y(A', f)\right) \right]\!\right]$$

where $T_n := \langle (x_1, f(x_1)), \ldots, (x_n, f(x_n)) \rangle$, $T_n^x := \langle x_1, \ldots, x_n \rangle$, $T_n^y := \langle f(x_1), \ldots, f(x_n) \rangle$ and $A : T_n \mapsto x_{n+1} \in \mathcal{X} \setminus T_n^x$.

equally well and equally poorly

No learning is possible without some prior knowledge!

No-Free-Lunch for Pure Inductive Logic

- No learning is possible for $c^\dagger$. It seems possible to learn with c^* and c_λ. Unfortunately, No-Free-Lunch Theorem!
- Take $\mathcal{X} := C, \mathcal{Y} := Q$ in the No-Free-Lunch Theorem. The state description $h : C \to Q$ can be taken as a *classification* function.
- Then "all state descriptions $h : C \to Q$ with the same structure description $\{n_i : 1 \leq i \leq r\}$ have equal weight" is block uniform!
- Let the induction algorithm $A(h_{1:n}) := \operatorname{argmax}_j c^* \left(Q_j(a_{n+1}) \mid h_{1:n}\right)$, and loss function $L\,(A, n, h) := [\![A(h_{1:n}) \neq h_{n+1}]\!]$. Then for any A',

$$\sum_{h \in \mathcal{Y}^{\mathcal{X}}} m^*(h_{1:n}) L\,(A, n, h) = \sum_{h \in \mathcal{Y}^{\mathcal{X}}} m^*(h_{1:n}) L\,(A', n, h)$$

- Similarly for rank description (E), which is related to Good-Turing estimate. And similarly for Ristad's methods $(C)(D)$.

No Free Lunch — Strengthened "Hume"

- Sets a limit on how good a learner can be: no learner can be better than random guessing!
- But then why is the world full of highly successful learners?
- For every world where a learner does better than random guessing, we can construct an anti-world by flipping the labels of all unseen instances: it performs worse by the same amount
- We don't care about all possible worlds, only the one we live in
- We assume we know something about this world that gives us an advantage. That knowledge is fallible, but it's a risk we'll have to take
- We need to provide prior knowledge to the algorithm, or make assumptions when constructing hypotheses
  - The structure of a neural net, a Bayesian prior, background knowledge as rules, the way a tree represents knowledge ...
- These assumptions are called a learner's bias, i.e. bias-free learning is impossible.
  - Every new piece of knowledge is the basis for more knowledge.
  - What assumptions can we start from that are not too strong?

Time Series and Solomonoff Induction

- However, if we take time into consideration, define

$$M(h_{1:n}) := \sum_{p:U(p)=h_{1:n}*} 2^{-\ell(p)}$$

then we can get free lunch with M', since M' biases non-random state descriptions and is not block uniform. where

$$M'(\epsilon) := 1$$

$$M'(h_{1:n}) := M'(h_{<n}) \frac{M(h_{1:n})}{\sum_{Q \in \mathcal{Q}} M(h_{<n}Q)}$$

and we can extend M' from state descriptions to sentences.

- Besides, by c^* the probability of "the sun will rise tomorrow" is $\frac{n+1}{n+2}$, but c^* fails to confirm "the sun will always rise", while M' can confirm it.

$$\lim_{n\to\infty} M'(\text{the sun will always rise} \mid \text{the sun rises in the first } n \text{ days}) = 1$$

PAC (Probably Approximately Correct) Learning

Definition (PAC-Learnability)

A hypothesis space $\mathcal{H} \subset 2^{\mathcal{X}}$ is PAC-learnable iff there exists a sample complexity function $m_{\mathcal{H}} : (0,1)^2 \to \mathbb{N}$ and a learning algorithm A with the following property:

- for every $\varepsilon, \delta \in (0,1)$
- for every distribution $\mathcal{D}$ over $\mathcal{X}$, and for every labeling function $f : \mathcal{X} \to \{0,1\}$

when running A on $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$ i.i.d. training samples S generated by $\mathcal{D}$ and labeled by f, the algorithm A returns a hypothesis $A(S) \in \mathcal{H}$ s.t.

$$P_{S \sim \mathcal{D}^m}\Big(P_{x \sim \mathcal{D}}\big(A(S)(x) \neq f(x)\big) > \varepsilon\Big) < \delta$$

$$\text{No-Free-Lunch} \implies m_{2^{\mathcal{X}}}(\varepsilon, \delta) = \infty$$

Agnostic PAC Learning

Definition (Agnostic PAC-Learnability)

A hypothesis space $\mathcal{H} \subset \mathcal{Y}^{\mathcal{X}}$ is agnostic PAC-learnable under a class Δ of distributions with respect to $\mathcal{Z} := \mathcal{X} \times \mathcal{Y}$ and a loss function $\ell : \mathcal{H} \times \mathcal{Z} \to \mathbb{R}^+$, iff there exists a sample complexity function $m_{\mathcal{H}} : (0,1)^2 \to \mathbb{N}$ and a learning algorithm A with the following property:

- for every $\varepsilon, \delta \in (0,1)$
- for every distribution $\mathcal{D} \in \Delta$ over $\mathcal{Z}$

when running A on $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$ i.i.d. training samples S generated by $\mathcal{D}$, the algorithm A returns a hypothesis $A(S) \in \mathcal{H}$ s.t.

$$P_{S \sim \mathcal{D}^m}\left(L_{\mathcal{D}}(A(S)) - \min_{h \in \mathcal{H}} L_{\mathcal{D}}(h) > \varepsilon\right) < \delta$$

where $L_{\mathcal{D}}(h) := \mathbb{E}_{z \sim \mathcal{D}}\left[\ell(h, z)\right]$.

PAC-learnable under $\Delta := \{\mathcal{D} : \mathcal{D} \overset{\times}{\leq} \xi\} \iff$ PAC-learnable under $\xi(x) := \sum_{\nu \in \mathcal{M}} 2^{-K(\nu)} \nu(x)$.

VC-Dimension

Definition (Shattering)

A hypothesis space $\mathcal{H} \subset 2^{\mathcal{X}}$ shatters a set $C \subset \mathcal{X}$ iff $\mathcal{H}\restriction_C = 2^C$.

Definition (VC-Dimension)

$\mathrm{VC}(\mathcal{H}) \coloneqq \sup\left\{|C| : \mathcal{H} \text{ shatters } C\right\}$.

If someone can explain every phenomena, her explanations are worthless.

Theorem

If $\mathrm{VC}(\mathcal{H}) = \infty$, *then* $\mathcal{H}$ *is not PAC-learnable.*

Theorem (Fundamental Theorem of PAC Learning)

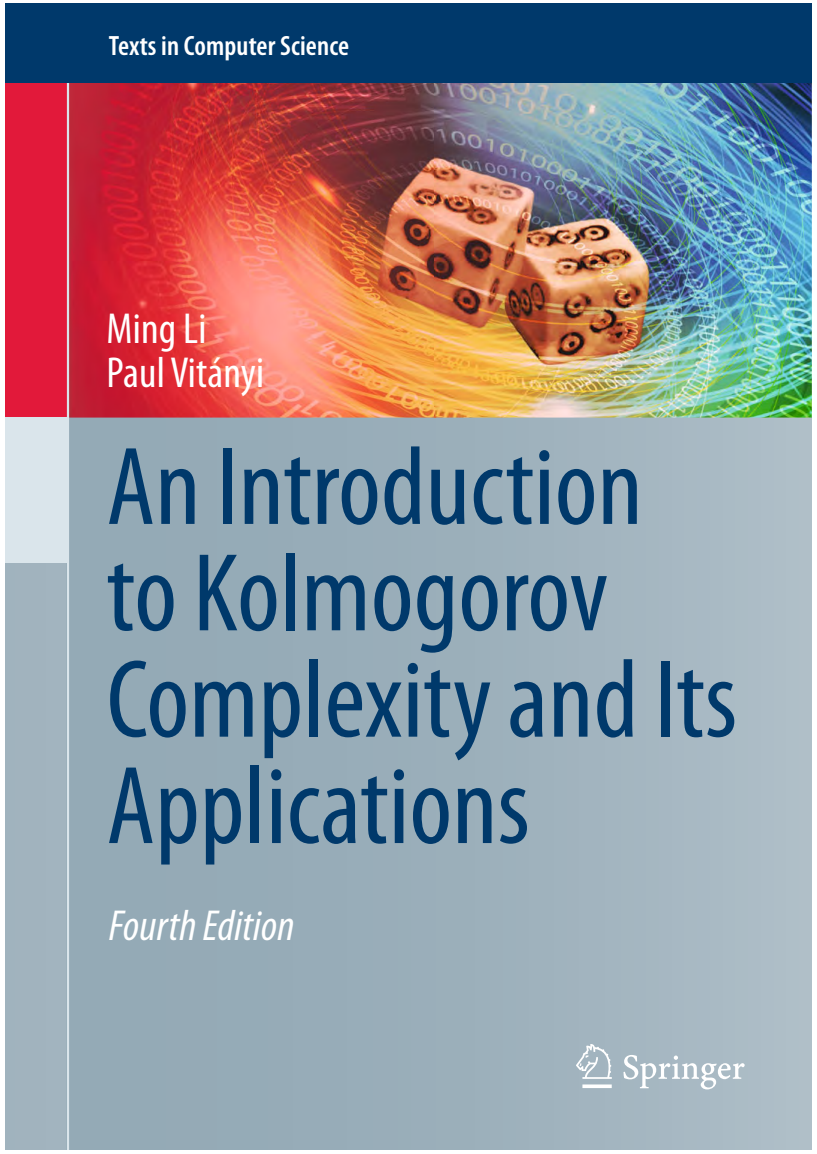
$\mathcal{H}$ *is PAC-learnable iff* $\mathrm{VC}(\mathcal{H}) < \infty$. *Indeed, there exists* C_1, C_2 *s.t.*

$$C_1 \frac{\mathrm{VC}(\mathcal{H}) + \log(1/\delta)}{\varepsilon} \leq m_{\mathcal{H}}(\varepsilon, \delta) \leq C_2 \frac{\mathrm{VC}(\mathcal{H}) \log(1/\varepsilon) + \log(1/\delta)}{\varepsilon}$$

Contents

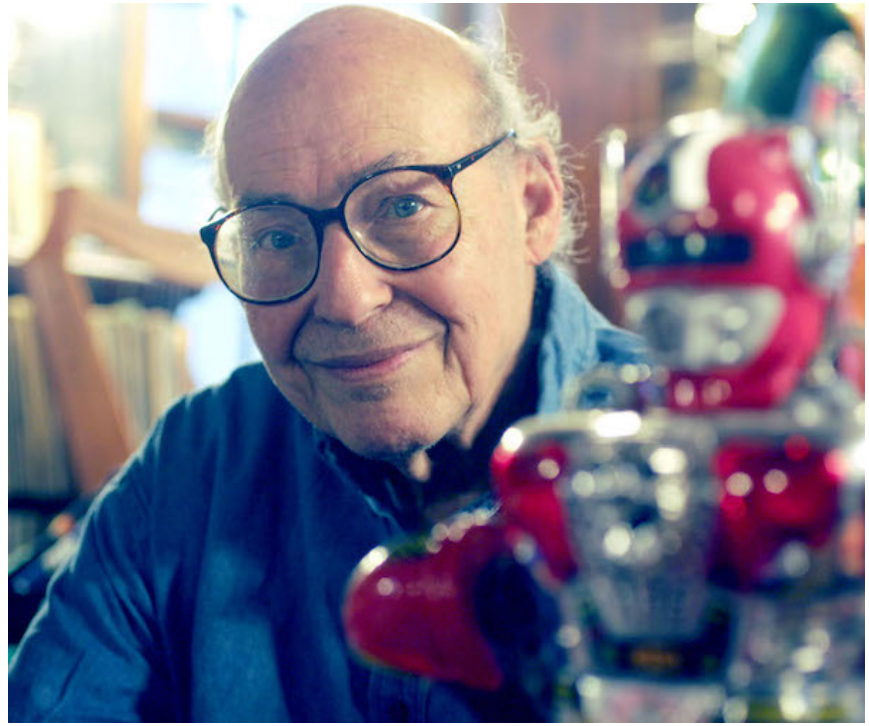
1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Kolmogorov Complexity



Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
   - Kolmogorov Complexity
   - Analogy & Few-Shot Learning
   - Algorithmic Probability
   - A Statistical Mechanical Interpretation of AIT
   - Incompressibility & Incompleteness
   - Algorithmic Randomness
   - Effective Complexity
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics



"The most important discovery since Gödel was the discovery by Chaitin, Solomonoff and Kolmogorov of the concept called Algorithmic Probability....
It should be possible to make practical approximations to the Chaitin, Kolmogorov, Solomonoff theory that would make better predictions than anything we have today. Everybody should learn all about that and spend the rest of their lives working on it."

— Marvin Minsky

用 “简单性” 这一把 “锤子”，锤一串哲学大 “钉子”

1. “归纳” 的合理性何在?
2. “奥卡姆剃刀” 为啥锋利?
3. 世界为何是这个样子而不是那个样子?
4. 为什么我们生存其中的世界是有序的?
5. 先有 “世界” 还是先有 “观察”?
6. 终极真理 “可得” 吗? “可知” 吗?
7. 压缩即智能?
8. “阴谋论” 为什么难以根除?
9. 大语言模型的 “幻觉” 可以根除吗?
10. 怎么让大语言模型生成 “有创意的” 回答?

地主家的傻儿子

《从三到万》

汝有田舍翁, 家资殷盛, 而累世不识 “之”、“乎”. 一岁, 聘楚士训其子. 楚士始训之搦管临朱, 书一画, 训曰: “一” 字. 书二画, 训曰: “二” 字. 书三画, 训曰: “三” 字. 其子辄欣欣然掷笔, 归告其父曰: “儿得矣! 儿得矣! 可无烦先生, 重费馆谷也, 请谢去.” 其父喜从之, 具币谢遣楚士.
逾时, 其父拟征召姻友万氏者饮, 令子晨起治状, 久之不成. 父趣之. 其子恚曰: “天下姓字多矣, 奈何姓万? 自晨起至今, 才完五百画也.”

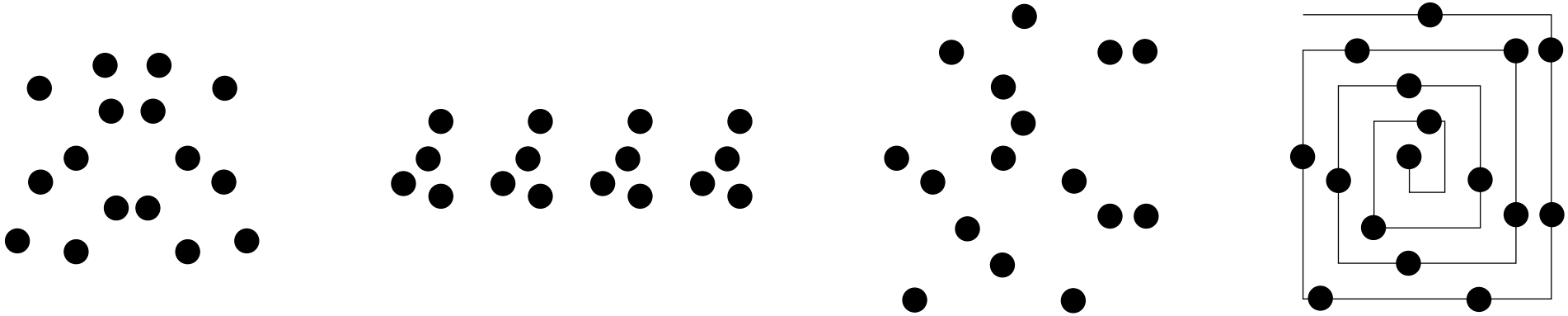
Pattern 刺激
分泌多巴胺

$$0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, \cdots$$

$$\text{Numeral} \xrightarrow{\text{program}} \text{Number}$$

1. “数字” 不同于 “数”.
2. “数字” 是用有穷长的字符串表示的.
3. “数字” 到 “数” 的映射 $numeral \mapsto number$ 是可计算的.
4. “数字” 必须**短**! — 问题是: 相对于什么 “尺子” 度量的 “短”?

万 vs 10000 vs $111\cdots111$ vs 10011100010000

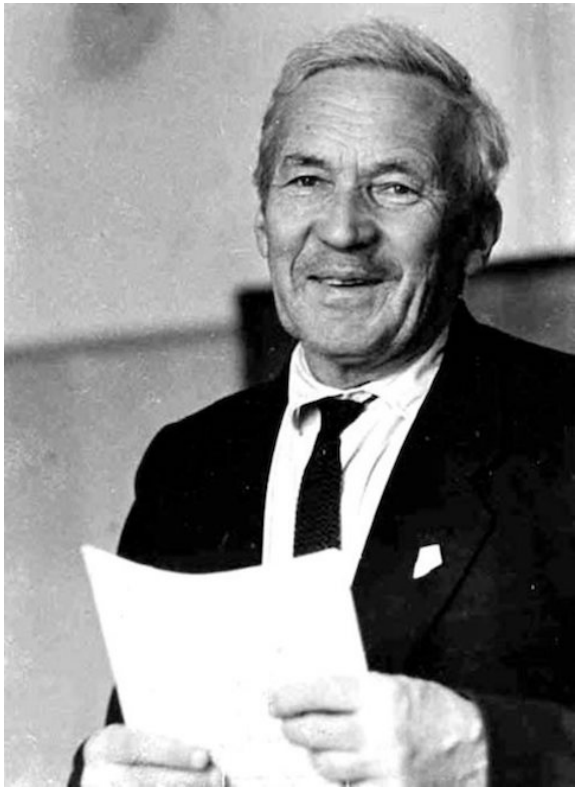


- $0101010101010101010101010101010101$
- $0010010000111111011010101010001000$
- $0101001101111011011101010101000001$
- $0.12345678910111213141516171819 20$

$\{\pi\}$

1. What is regularity/pattern/law/principle/model/hypothesis/theory?
2. What is phenomenon/data/experience?
3. What is randomness/noise?
4. What is typicalness/unpredictability/incompressibility?
5. What is simplicity/complexity?
6. What is learning?
7. What is beauty/interesting/curiosity/novelty/surprise/creativity?
8. What is intelligence?

Andrey Kolmogorov 1903-1987



- Measure Theory
- Probability Theory
- Cohomology
- Chaos and Dynamical Systems, KAM theorem
- Turbulence
- Fourier series
- Kolmogorov superposition theorem
- Intuitionistic Logic, BHK interpretation
- Information theory
- Kolmogorov-Uspensky machine
- Kolmogorov complexity
- Kolmogorov structure function

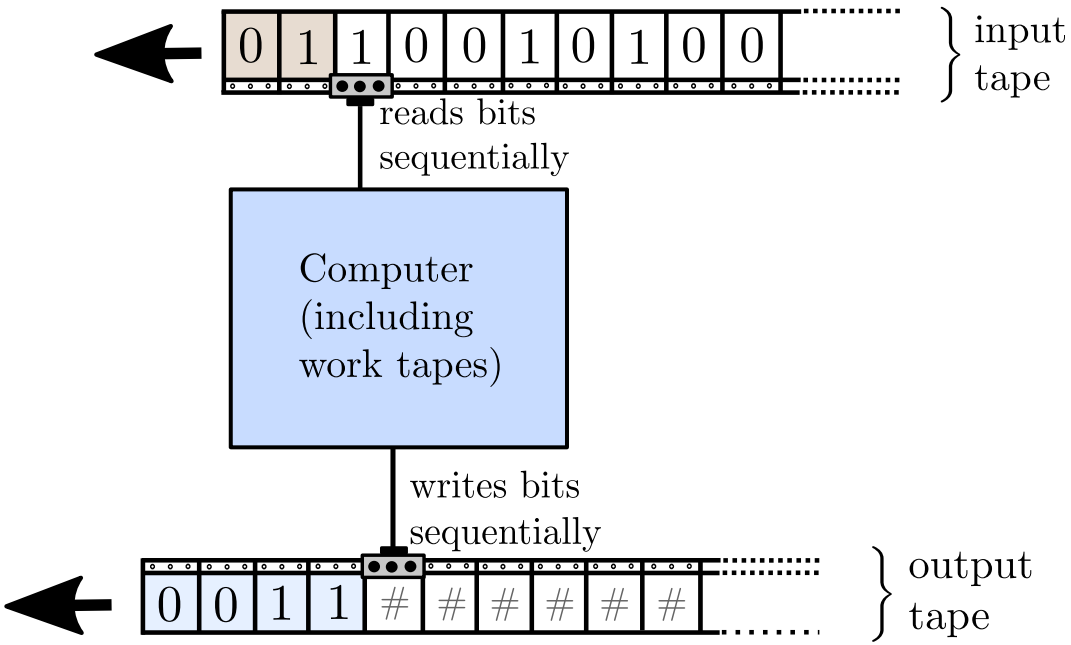
Kolmogorov Complexity[LV19]

Definition (柯尔莫哥洛夫复杂性)

$$K(x) \coloneqq \min_{p}\{\ell(p) : U(p) = x\}$$

其中 U 是通用前缀图灵机.

$\{p : U(p)\downarrow\}$ forms a prefix code.



"simplicity"

$$K_U(x) \leq K_T(x) + c_T$$

"独立于"UTM!

"randomness"

$$\exists c \forall n : K(x_{1:n}) \geq n - c$$

Regularity $\approx$ Short program

No short program $\approx$ Noise

To Comprehend is to Compress!

$$U(p) = x$$

program	$\xrightarrow{\text{Computer}}$	output
axioms	$\xrightarrow{\text{Deduction}}$	theorems
scientific theory	$\xrightarrow{\text{Calculations}}$	experimental data
encoded message	$\xrightarrow{\text{Decoder}}$	original message
DNA	$\xrightarrow{\text{Pregnancy}}$	organism
Ideas	$\xrightarrow{\text{Mind of God}}$	Universe
software	$\xrightarrow{\text{Universal Constructor}}$	physical system

Kolmogorov Complexity

- For a discrete function $f : \mathcal{X}^* \to \mathcal{X}^*$,

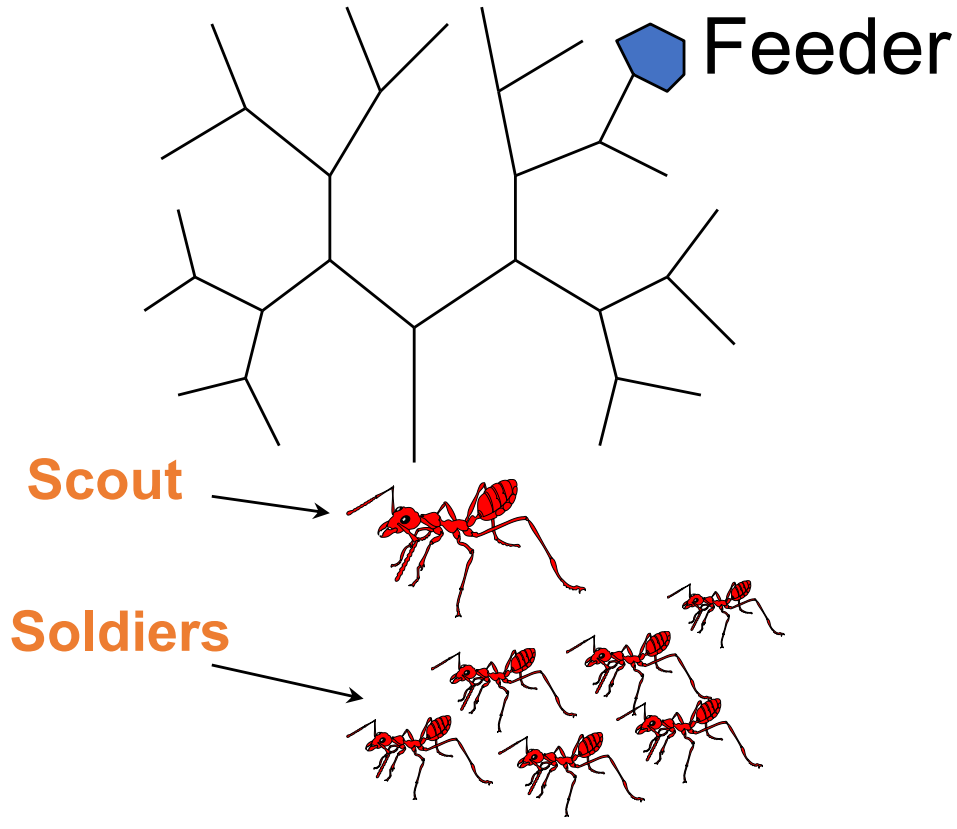
$$K(f) \coloneqq \min_p \left\{ \ell(p) : \forall x [U(p,x) = f(x)] \right\}$$

- For a real-valued function $f : \mathcal{X}^* \to \mathbb{R}$,

$$K(f) \coloneqq \min_p \left\{ \ell(p) : \forall x \forall k \left[|U(p,x,k) - f(x)| < \frac{1}{k} \right] \right\}$$

Kolmogorov complexity by ants

- Feeder 里有蜂蜜.
- 火柴棍漂在水面上, 形成树状迷宫.
- 侦察蚁首先找到蜂蜜.
- 侦察蚁返回巢穴.
- 侦察蚁与工蚁交流, 记录下它们的交流时间.
- 移走侦察蚁.
- 火柴棍换成新的, 以防止路径被气味标记.
- 工蚁前往 Feeder.



当通往 Feeder 的路径具有较低的 Kolmogorov 复杂度时, 如 “LLLL”, 蚂蚁的交流速度会更快.

Kraft-Chaitin Theorem

Theorem (Kraft Inequality)

1. *Let* $f : \mathcal{X} \to 2^{<\omega}$ *be uniquely decodable. Let* $\ell_x := \ell(f(x))$. *Then* f *satisfies the Kraft inequality*

$$\sum_{x \in \mathcal{X}} 2^{-\ell_x} \leq 1$$

2. *Conversely, for any set of code length* $\{\ell_x : x \in \mathcal{X}\}$ *satisfying the above Kraft inequality, there exists a prefix code* f *such that* $\ell_x = \ell(f(x))$.

Theorem (Kraft-Chaitin Theorem)

For any r.e. set $(\ell_i, x_i)_{i \in \omega} \subset \mathbb{N} \times 2^{<\omega}$ *with* $\sum_{i \in \omega} 2^{-\ell_i} \leq 1$, *one can effectively obtain a prefix-free machine* M *and strings* p_i *of length* ℓ_i *such that* $M(p_i) = x_i$ *for all* i *and* $\mathrm{dom}(M) = \{p_i : i \in \omega\}$.

赌硬币出老千

- Bob 跟 Alice 打赌抛硬币:
  - 正面 Alice 赢一美元;
  - 反面 Bob 赢一美元.
- 结果: $0000000000\ldots$ 100 次反面.
- Alice 输了 100 美元, 她觉得被骗了.
- Alice 抱怨: $0000000000\ldots$ 不是随机的.
- Bob 让 Alice 抛个随机的结果出来.
- Alice 抛出了 $01001101011100001\ldots$
- Bob 声称 Alice 的序列的概率是 2^{-100}, 他的也是.
- 随机的标准是什么?

TOUR OF ACCOUNTING
OVER HERE WE HAVE OUR RANDOM NUMBER GENERATOR.
NINE NINE NINE NINE NINE NINE
ARE YOU SURE THAT'S RANDOM?
THAT'S THE PROBLEM WITH RANDOMNESS: YOU CAN NEVER BE SURE.
10|25|01 © 2001 United Feature Syndicate, Inc.

Alice 的复仇

- Bob 竟然出老千连续抛出了 100 次反面.
- Alice 想出了一个赢的策略. 她提出如下赌约:
  - 她付给 Bob 1 美元.
  - 根据结果, Bob 付给她 $2^{100-K(x)}$ 美元, 其中 x 是长度为 100 的抛硬币序列.
- 这是一个公平的提议, 因为

$$\sum_{x:\ell(x)=100} 2^{-100} 2^{100-K(x)} < 1$$

- 如果 Bob 出老千, 抛出 100 次反面, Alice 会赢得 $2^{100-\log 100}$ 美元.

Asymptotic Notation

$$O(g(x)) \coloneqq \left\{ f : \exists c > 0 \exists x_0 > 0 \forall x \geq x_0 \left(|f(x)| \leq c|g(x)| \right) \right\}$$

$$f(x) \overset{+}{\leq} g(x) \coloneqq f(x) = g(x) + O(1)$$

$$f(x) \overset{\times}{\leq} g(x) \coloneqq f(x) = O(g(x)) \coloneqq f \in O(g(x))$$

$$f(x) \overset{\times}{\geq} g(x) \coloneqq f(x) = \Omega(g(x)) \coloneqq g(x) = O(f(x))$$

$$f(x) \overset{\times}{=} g(x) \coloneqq f(x) = \Theta(g(x)) \coloneqq f(x) = O(g(x)) \wedge f(x) = \Omega(g(x))$$

$$f(n) \sim g(n) \coloneqq \lim_{n \to \infty} \frac{f(n)}{g(n)} = 1$$

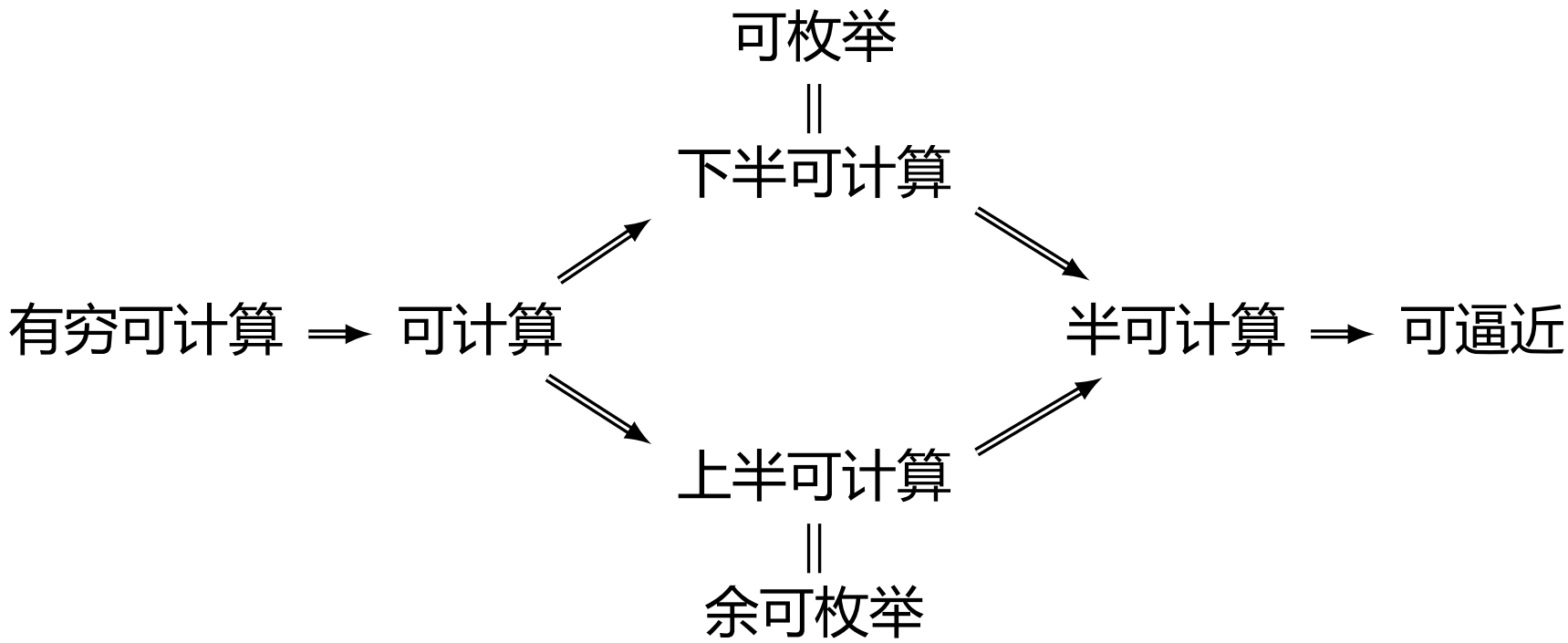
$$f(x) = o(g(x)) \coloneqq \lim_{x \to \infty} \frac{f(x)}{g(x)} = 0$$

$$f(x) = \omega(g(x)) \coloneqq g(x) = o(f(x))$$

Recursive Functions

We consider functions $f : \mathcal{X}^* \to \mathbb{R}$:

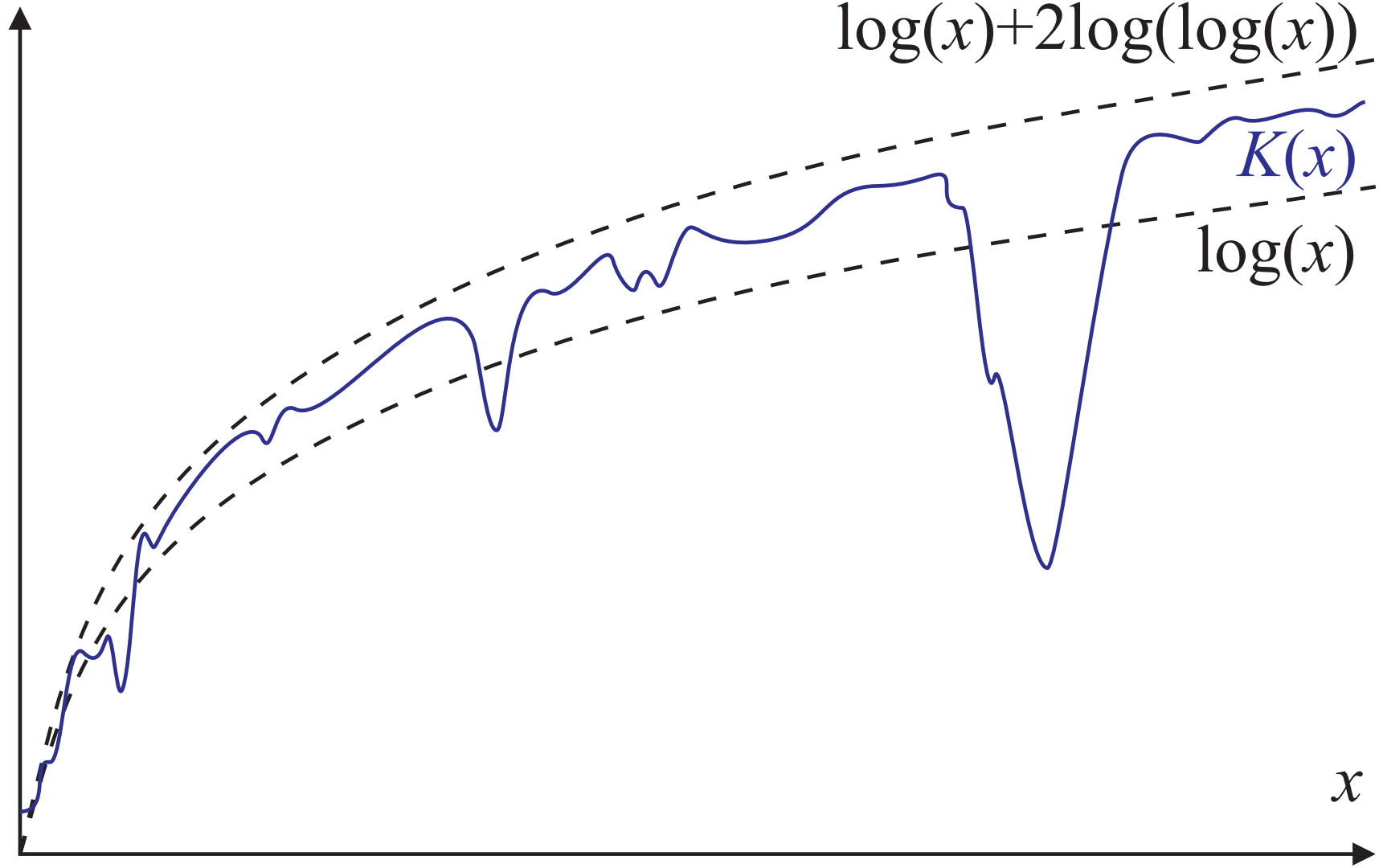
- f is recursive / finitely computable iff there is a Truing machine T s.t. $T(x) = \langle n, d \rangle$ and $f(x) = \frac{n}{d}$.
- f is approximable iff there is a recursive function ϕ and $\lim_{t \to \infty} \phi(x, t) = f(x)$.
- f is lower semicomputable or enumerable iff f is approximable and $\phi(x, t) \leq \phi(x, t + 1)$.
- f is upper semicomputable or co-enumerable iff $-f$ is lower semicomputable.
- f is semicomputable iff f is lower or upper semicomputable.
- f is computable iff f is both lower and upper semicomputable.



- f is computable iff there is a recursive function ϕ s.t. $\forall x \forall \varepsilon : \left|\phi(x, \lfloor \frac{1}{\varepsilon} \rfloor) - f(x)\right| < \varepsilon$.
- If f is only approximable or semicomputable we can still come arbitrarily close to $f(x)$ but we cannot devise a terminating algorithm which produces an ε-approximation.
- If f is lower/upper semicomputability we can at least finitely compute lower/upper bounds to $f(x)$.

Properties of Kolmogorov Complexity

1. $K(x) \stackrel{+}{\leq} K(x \mid \ell(x)) + K(\ell(x)) \stackrel{+}{\leq} \ell(x) + \log^* \ell(x) \leq \ell(x) + 2\log \ell(x)$
2. $K(n) \stackrel{+}{\leq} \log^* n \leq \log n + 2\log\log n$
3. $\sum_x 2^{-K(x)} \leq 1$
4. $K(x \mid y) \stackrel{+}{\leq} K(x) \stackrel{+}{\leq} K(x, y)$
5. $K(xy) \stackrel{+}{\leq} K(x, y) \stackrel{+}{\leq} K(x) + K(y \mid x) \stackrel{+}{\leq} K(x) + K(y)$
6. $K(x) \stackrel{+}{=} K(x, K(x))$
7. $K(y \mid x^*) \stackrel{+}{=} K(y \mid x, K(x))$ where x^* is the shortest program for x
8. $K(x, y) \stackrel{+}{=} K(x) + K(y \mid x^*) \stackrel{+}{=} K(y) + K(x \mid y^*) \stackrel{+}{=} K(y, x)$
9. $K(x, y) = K(x) + K(y \mid x) + O(\log K(x, y))$
10. $K(x, y \mid z) \stackrel{+}{=} K(x \mid z) - K(y \mid x, K(x \mid z), z)$
11. $K(f(x)) \stackrel{+}{\leq} K(x) + K(f)$ for computable f
12. $K(x) \stackrel{+}{\leq} -\log P(x) + K(P)$ if P is a lower semicomputable semi-probability(i.e. $\sum_x P(x) \leq 1$)
    $Km(x) \stackrel{+}{\leq} -\log \mu(x) + K(\mu)$ if μ is a lower semicomputable semimeasure(i.e. $\mu(\epsilon) \leq 1$ and $\mu(x) \geq \sum_{a \in \mathcal{X}} \mu(xa)$)
13. $\sum_{x : f(x) = y} 2^{-K(x)} \stackrel{\times}{=} 2^{-K(y)}$ if f is computable and $K(f) = O(1)$
14. $0 \leq \mathbb{E}_P[K] - H(P) \stackrel{+}{\leq} K(P)$ for a computable probability P


log(x)+2log(log(x))
K(x)
log(x)
x

Algorithmic Mutual Information

The K-complexity of information in x about y is

$$I(x : y) := K(y) - K(y \mid x)$$

Definition (Algorithmic Mutual information)

$$I(x; y) := K(y) - K(y \mid x^*)$$

$$I(x; y \mid z) := K(y \mid z) - K(y \mid x, K(x \mid z), z)$$

Theorem

$$I(x; y) \stackrel{+}{=} K(x) + K(y) - K(x, y) \stackrel{+}{=} I(y; x)$$

$$I(x; y \mid z) \stackrel{+}{=} K(x \mid z) + K(y \mid z) - K(x, y \mid z) \stackrel{+}{=} I(y; x \mid z)$$

Algorithmic data processing identity:

$$I(z; (x, y)) \stackrel{+}{=} I(z; x) + I(z; y \mid x^*)$$

Algorithmic data processing inequality:

$$I(x; y \mid z^*) \stackrel{+}{=} 0 \implies I(x; y) \stackrel{+}{\leq} I(x; z)$$

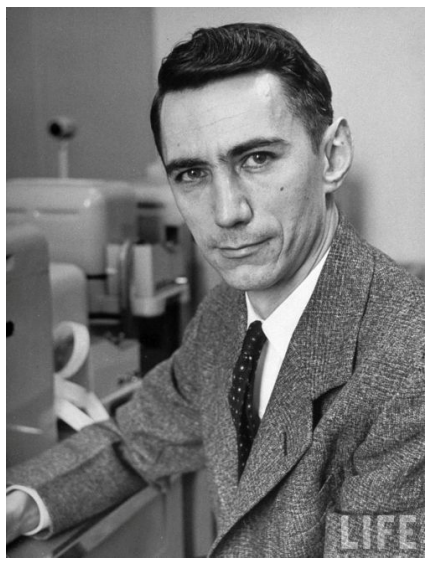
Shannon Entropy — Characterization of "Surprise"

- Information is surprise. (消除不确定性)
- Entropy is the expected surprise.
- Entropy is a measure of uncertainty.

假设惊讶度函数 $S : (0, 1] \to [0, \infty)$ 满足下列条件:

1. $S(1) = 0$ — "必然发生的事不会使人惊讶"
2. $p_1 < p_2 \implies S(p_1) > S(p_2)$ — "小概率事件使人更为惊讶"
3. S 是 p 的连续函数. — "惊讶度随概率平滑变化"
4. $S(p_1 p_2) = S(p_1) + S(p_2)$ — "独立事件同时发生惊讶度叠加"

- 假如抛一枚均匀硬币一次能产生 1 单位的惊讶度 $S(2^{-1}) = 1$
- 令 $p := 2^{-k}$, 那么
$$S(p) = S(2^{-k}) = kS(2^{-1}) = -\log p$$
- "期望惊讶度" 就是香农熵
$$H(X) := \mathbb{E}[S(P(X))] = -\sum_{x \in \mathcal{X}} P(x) \log P(x)$$



Remark: 热力学第二定律是宇宙的阴谋, 目的是让我们感到惊讶 ☺

Shannon Entropy

Definition (Shannon Entropy)

$$
\begin{aligned}
H(X) &:= -\sum_{x \in \mathcal{X}} P(x) \log P(x) \\
H(Y \mid X) &:= \sum_{x \in \mathcal{X}} P(x) H(Y \mid X = x) \\
&= -\sum_{x \in \mathcal{X}} P(x) \sum_{y \in \mathcal{Y}} P(y \mid x) \log P(y \mid x) \\
&= -\mathbb{E}_{P(X,Y)} \left[\log P(Y \mid X) \right]
\end{aligned}
$$

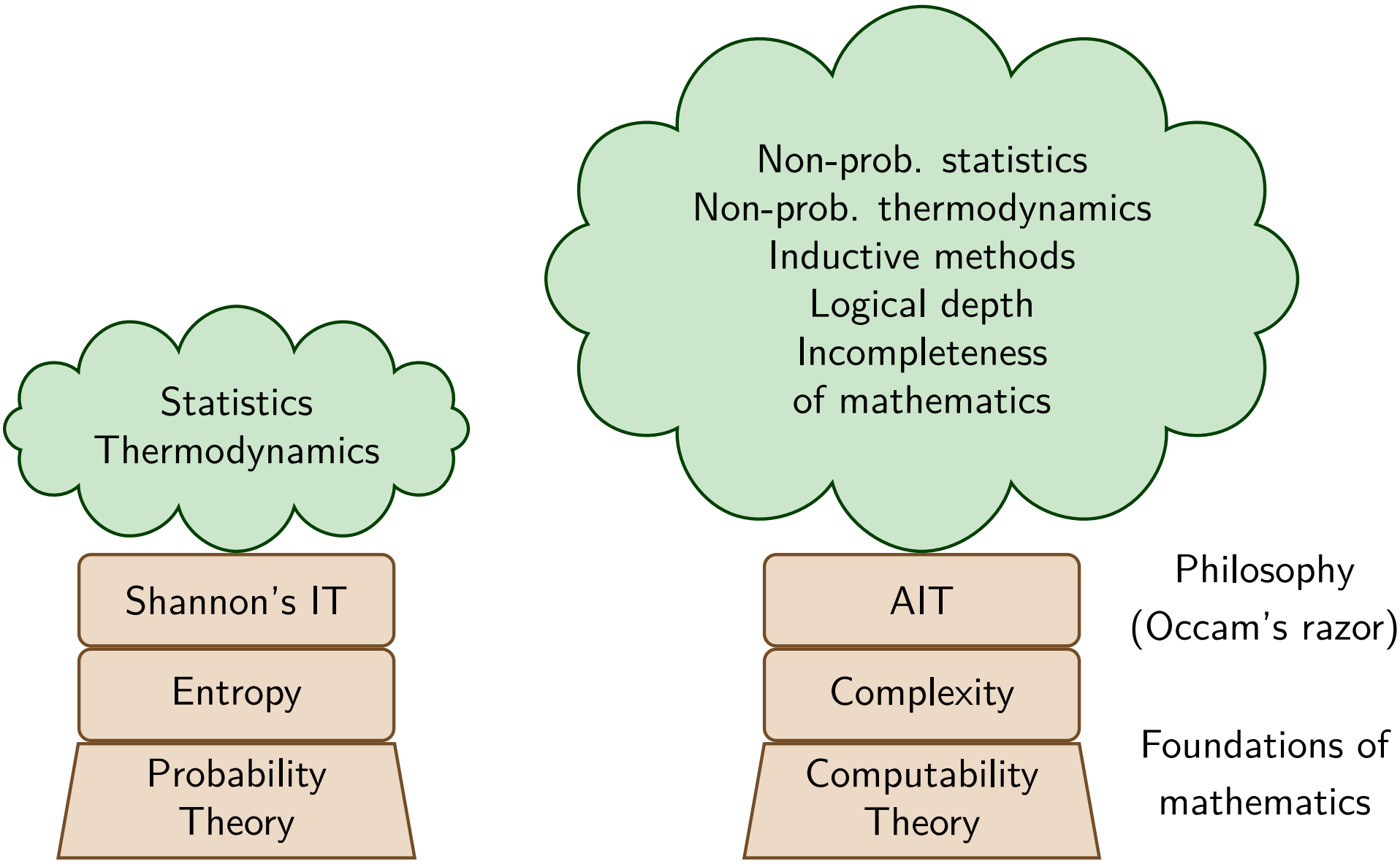
Kolmogorov Complexity vs Shannon Entropy

Property	**Kolmogorov Complexity**	**Shannon Entropy**
Upper Bound	$K(x) \stackrel{+}{\leq} \ell(x) + 2\log \ell(x)$	$H(X) \leq \log \lvert\mathcal{X}\rvert$ for sample space $\mathcal{X}$
Extra Information	$K(x \mid y) \stackrel{+}{\leq} K(x) \stackrel{+}{\leq} K(x, y)$	$H(X \mid Y) \leq H(X) \leq H(X, Y)$
Subadditivity	$K(x, y) \stackrel{+}{\leq} K(x) + K(y)$	$H(X, Y) \leq H(X) + H(Y)$
Symmetry of Information	$K(x) + K(y \mid x^*) \stackrel{+}{=} K(y) + K(x \mid y^*)$	$H(X) + H(Y \mid X) = H(Y) + H(X \mid Y)$
Information Non-increase	$K(f(x)) \stackrel{+}{\leq} K(x) + K(f)$ for computable f	$H(f(X)) \leq H(X)$ for any function f
MDL Bound	$K(x) \stackrel{+}{\leq} -\log P(x) + K(P)$	$H(X) \leq \mathbb{E}_{\boldsymbol{P}}\big[-\log Q(X)\big]$

Creative Thinking — Shannon

- Simplification: eliminate everything from the problem except the essentials.
- Analogy: seek similar known problems.
- Reformulate the problem in just as many different forms as you can. Change the words. Change the viewpoint.
- Generalization. "Can I apply the same principle in more general ways?"
- Structural analysis. divide & conquer.
- Inversion. work backwards from desired result.
- Look for contradictions as well as proofs.
- Ask conceptual questions about everyday things.

IT vs AIT


Statistics
Thermodynamics
Shannon's IT
Entropy
Probability
Theory
Non-prob. statistics
Non-prob. thermodynamics
Inductive methods
Logical depth
Incompleteness
of mathematics
AIT
Complexity
Computability
Theory
Philosophy
(Occam's razor)
Foundations of
mathematics

- Cross Entropy: how surprised P expects Q to be.

$$H(P, Q) \coloneqq -\sum_{x \in \mathcal{X}} P(x) \log Q(x)$$

- Relative Entropy: P's expected gap in average surprise between P and Q.

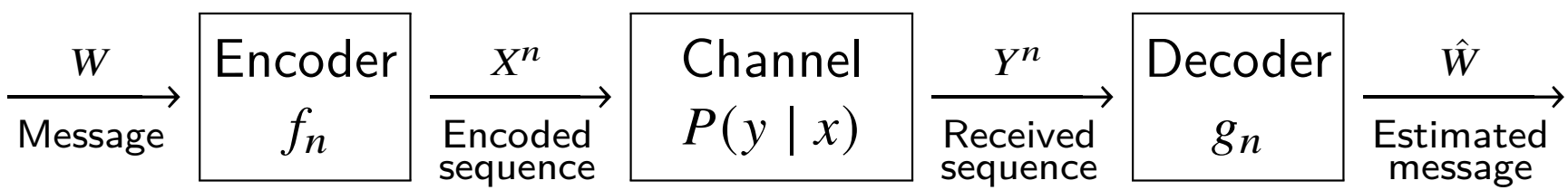
$$D_{\mathrm{KL}}(P \| Q) \coloneqq \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}$$

- Mutual Information

$$I(X; Y) \coloneqq D_{\mathrm{KL}}\big(P(X, Y) \| P(X) P(Y)\big)$$

- Channel Capacity

$$C \coloneqq \max_{P(x)} I(X; Y)$$



Relative Entropy

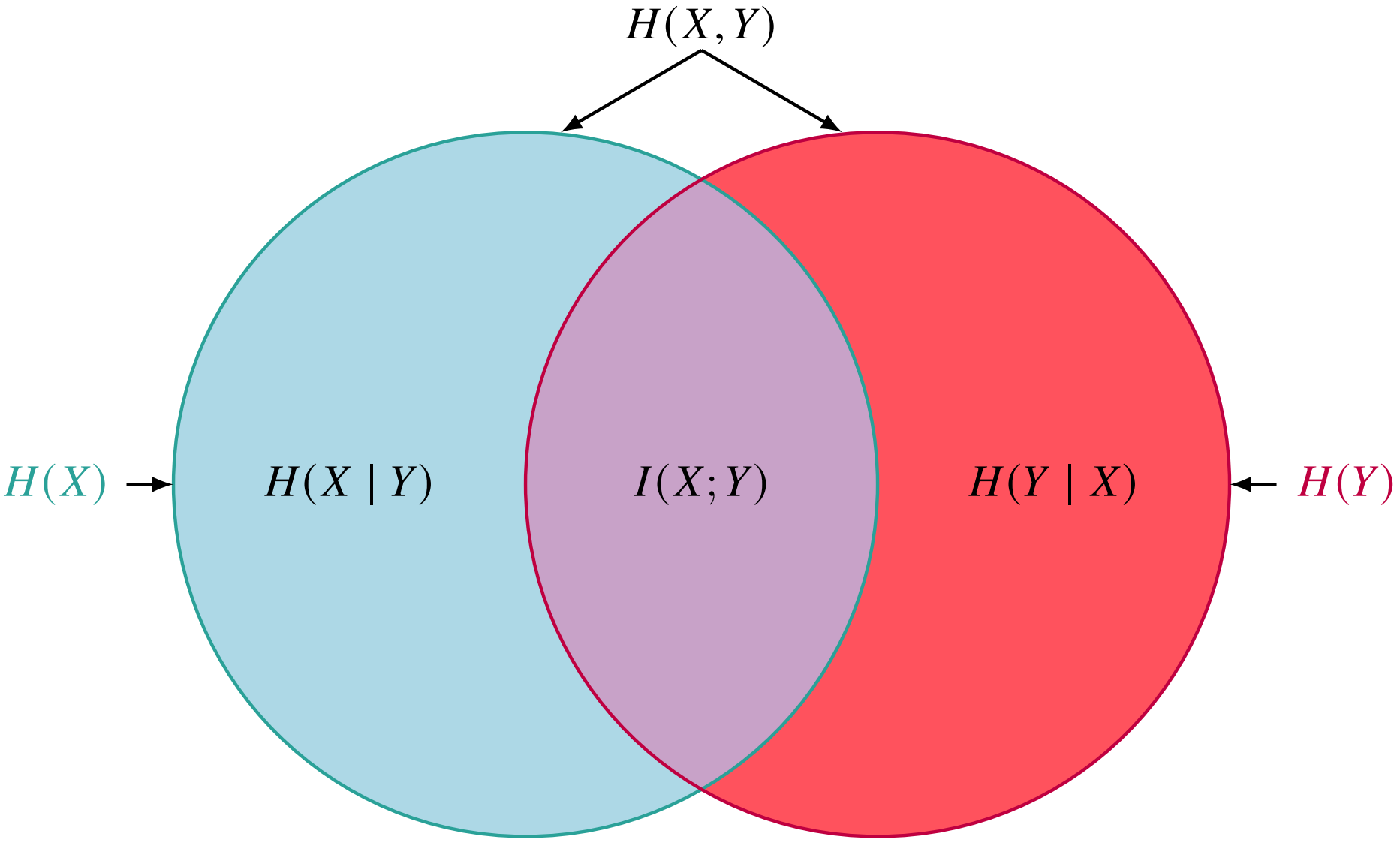
- $D_{\mathrm{KL}}(P\|Q) \geq 0$ with equality iff $P = Q$ almost everywhere.
- $I(X;Y) \geq 0$ with equality iff X and Y are independent.
- $D_{\mathrm{KL}}(P\|Q)$ is not symmetric. $D_{\mathrm{KL}}(P\|Q) \neq D_{\mathrm{KL}}(Q\|P)$.
- $D_{\mathrm{KL}}(P\|Q)$ does not satisfy the triangle inequality.

Theorem

For any $\varepsilon > 0$, if $D_{\mathrm{KL}}(\pi\|\beta) < \varepsilon$ and $D_{\mathrm{KL}}(\tau\|\beta) < \varepsilon$, it is possible that $D_{\mathrm{KL}}(\pi\|\tau) = \infty$.

Remark: 假设基准策略 base policy β 近似于可信策略 trusted policy τ, 而且 proposed policy π 逼近基准策略 β, proposed policy 与可信策略的距离 $D_{\mathrm{KL}}(\text{proposed policy}\|\text{trusted policy})$ 仍然可能很大.

Shannon Entropy & Mutual Information



Kolmogorov Complexity vs Shannon Entropy

- Suppose we have a probability distribution P on n-bit strings:

$$P : \{0,1\}^n \to [0,1]$$

- Suppose we choose n random strings $x_1, \ldots, x_n$ from this probability distribution P.
- Then with probability 1,

$$\lim_{n\to\infty} \frac{K(x_1 \ldots x_n)}{nH(P)} = 1$$

where

$$H(P) = -\sum_{x\in\{0,1\}^n} P(x) \log P(x)$$

Remark

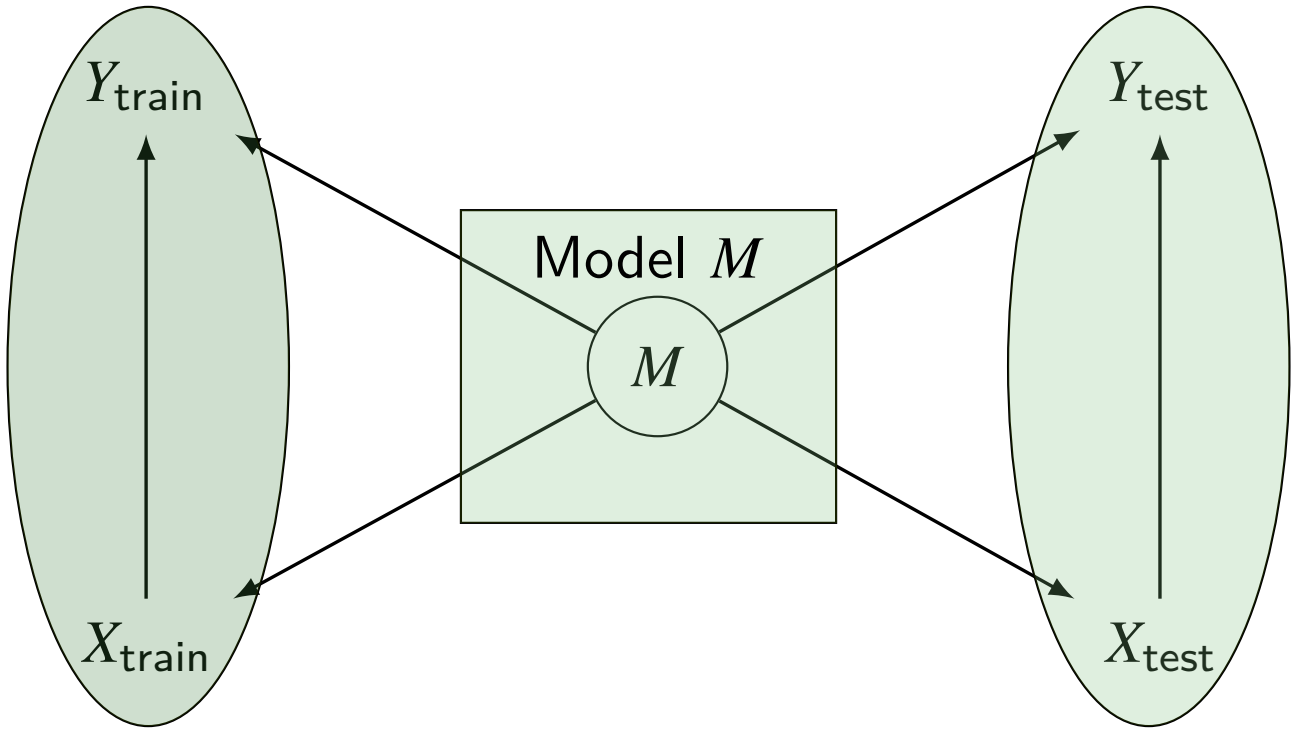
The Kolmogorov complexity of a long randomly produced string is typically close to the Shannon entropy of the probability distribution that gave rise to it!

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
   - Kolmogorov Complexity
   - Analogy & Few-Shot Learning
   - Algorithmic Probability
   - A Statistical Mechanical Interpretation of AIT
   - Incompressibility & Incompleteness
   - Algorithmic Randomness
   - Effective Complexity
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

类比为什么重要?

- 数学: 定义抽象概念 functor
- 法律: 判例
- 艺术: 隐喻、滑稽模仿
- 广告: 推荐算法
- 机器学习: Word2vec `male` : `female` :: `king` : `queen`
  迁移学习 $X_{\text{train}} : Y_{\text{train}} :: X_{\text{test}} : Y_{\text{test}}$



类比

talk : talked :: solve : solved

fish : gills :: humans : lungs

fish : swim :: bird : fly

woman : wave :: tree : sway

horse : legs :: car : wheels

earth : sun :: electron : nucleus

$$A : B :: C : D$$

通常:

- $A : B :: A : B$
- $A : B :: C : D \implies C : D :: A : B$
- $A : B :: C : D \implies A : C :: B : D$

演绎/预测/归纳/溯因/类比

1. 演绎: 保真的推理
2. 广义的归纳: 扩展性的推理?
   — 包括狭义的归纳、溯因、类比...
3. 溯因指最佳解释推理? (从假设空间中做选择 — 选择性溯因)
   — 还是需要创造新对象/假设 (创造性溯因)?
4. 预测可以还原为溯因演绎吗?
5. 类比可以还原为溯因归纳演绎吗?

演绎/预测/归纳/溯因/类比

1. 演绎: $\forall n P_n \to P_k$
2. 预测: $P_1, \ldots, P_t \overset{?}{\rightsquigarrow} P_{t+1}$ — 过去到未来
   — 如果 $P_1, \ldots, P_t$ 编码的是最优 Agent 与环境的交互历史呢?
3. 归纳: $P_1, \ldots, P_t \overset{?}{\rightsquigarrow} \forall n P_n$ — 样本到总体
4. 溯因: $P_1, \ldots, P_t \overset{?}{\rightsquigarrow} \texttt{Hypothesis}$ — 现象到假说
5. 类比: $A : B :: C : D$

(a) Charles Peirce

(b) David Corfield

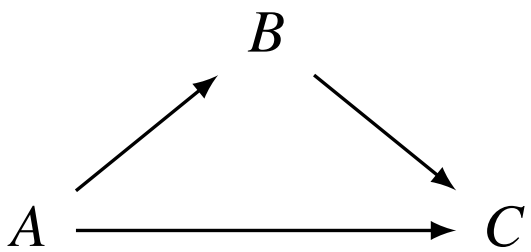
David Corfield: https://www.youtube.com/watch?v=c6-rGLK1Mps

前期皮尔士 Peirce: 演绎/归纳/溯因

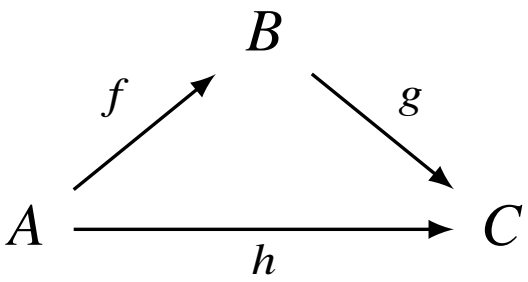
这些豆子是那个袋子里的	$A \to B$	假说
那个袋子里的豆子都是白的	$B \to C$	背景
这些豆子是白的	$A \to C$	现象

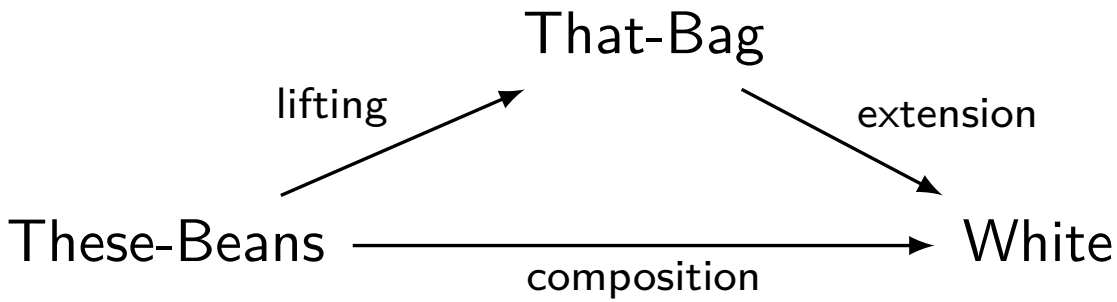
这些豆子是那个袋子里的	$A \to B$	随机抽样
这些豆子是白的	$A \to C$	样本性质$[r\%]$
那个袋子里的豆子都是白的	$B \to C$	总体性质$[r\%]$

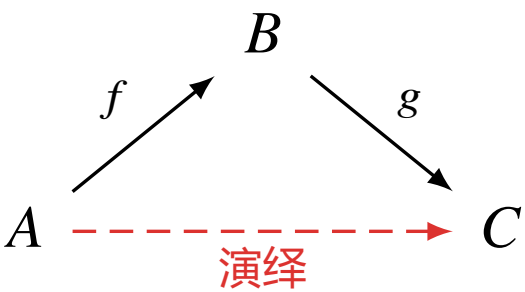
这些豆子是白的	$A \to C$	现象反常
那个袋子里的豆子都是白的	$B \to C$	背景下 “假说解释现象”
这些豆子是那个袋子里的	$A \to B$	假说

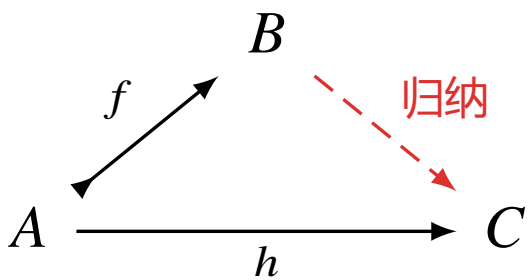


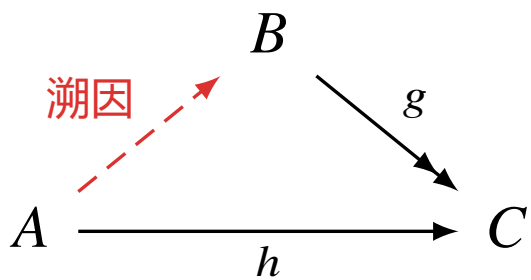
演绎/归纳/溯因 vs Composition/Extension/Lifting[8]







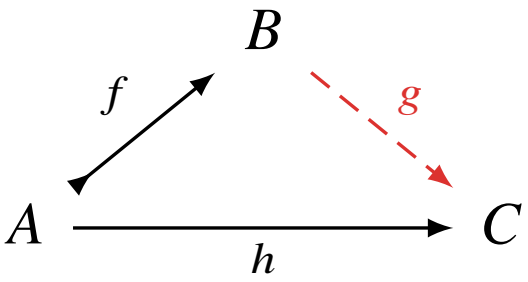


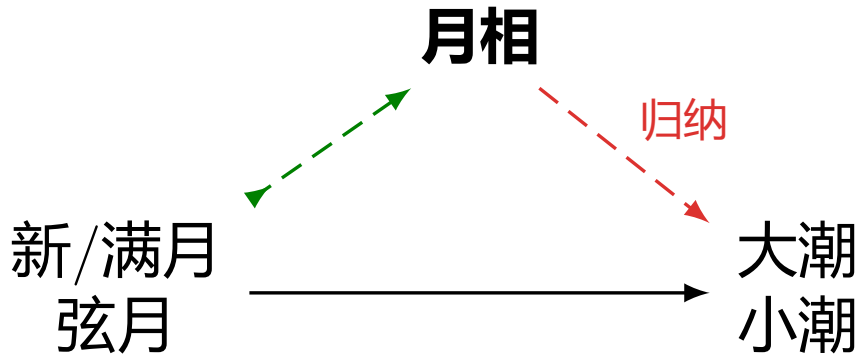


- 演绎: 已知 f, g, 求 h
- 归纳: 已知 f, h, 求 g
- 溯因: 已知 g, h, 求 f

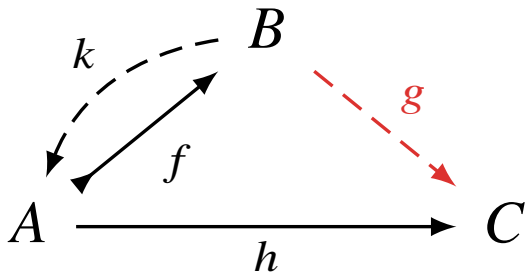
[8] David Corfield: Charles Peirce, Inference, and Category Theory. 2025.

归纳



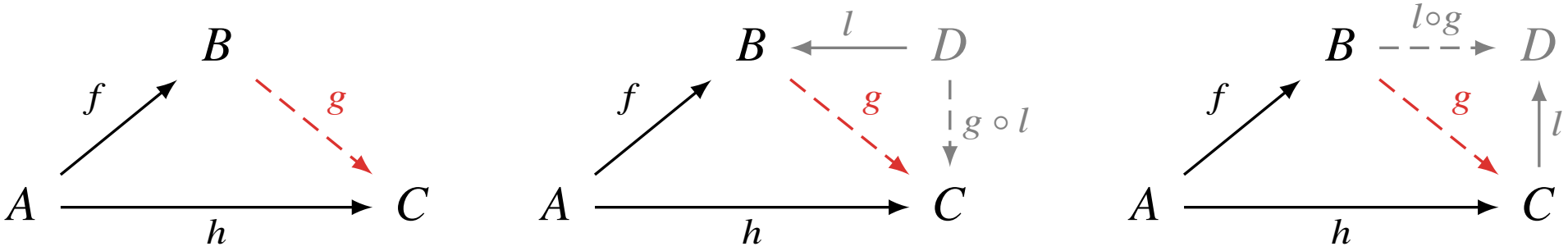


- 对于归纳任务来说, f 通常是单射. A 是 B 的采样, 归纳是把标签从样本 A 泛化到总体 B.
- **最近邻分类器**: 通过 Retraction $k : B \to A$ s.t. $k \circ f = 1_A$ 求解 $g := h \circ k$. (解通常不唯一, 需要偏置)



A: 典型的动物样本, 如 Tom & Jerry. B: 所有动物. C: 动物种类标签, 如: 猫/鼠/狗. 归纳 g 给所有动物分了类.

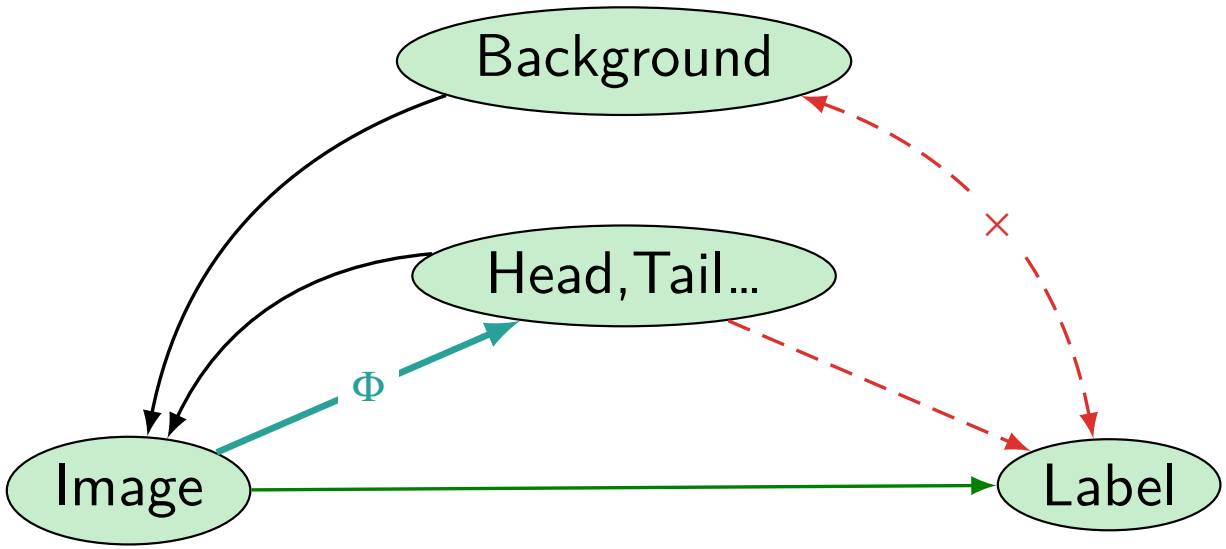
归纳检验



- 草地上的牛能准确分类. 把草地 PS 为海滩就分类错了.

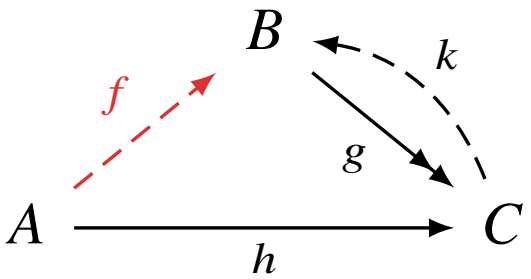
$$P(\text{Cow} \mid \text{Image}) \neq P(\text{Cow} \mid \text{Image}, \text{do}(\text{Background} = \text{beach}))$$

- 归纳出错, 溯因弥补 (不变特征学习)

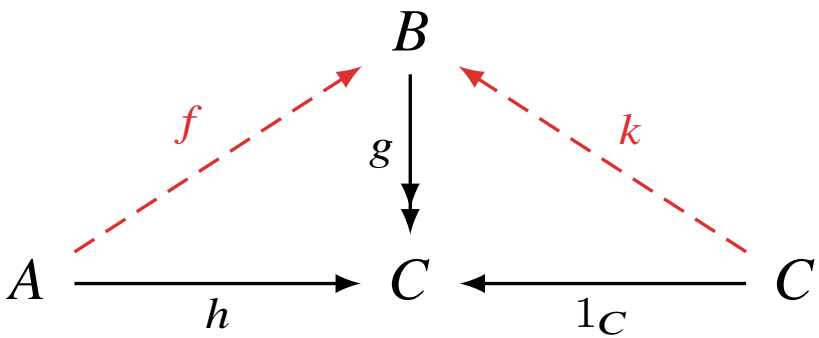


溯因

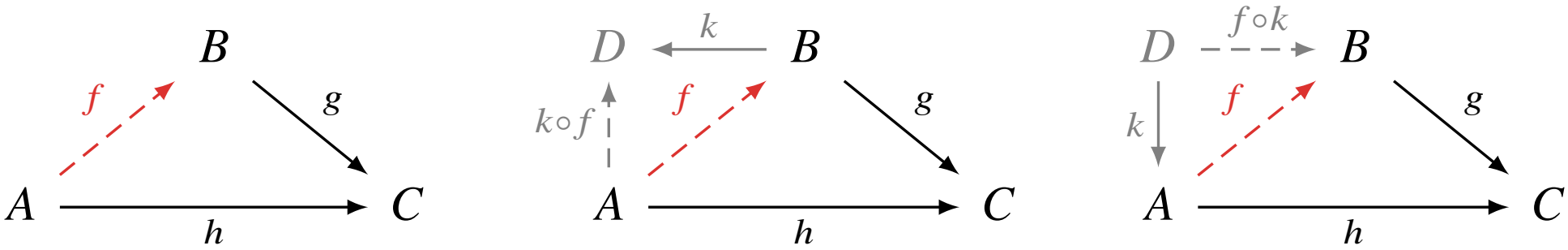
- 类似用 Retraction 求解归纳, 可以用 Section $k : C \to B$ s.t. $g \circ k = 1_C$ 求解溯因 $f := k \circ h$.

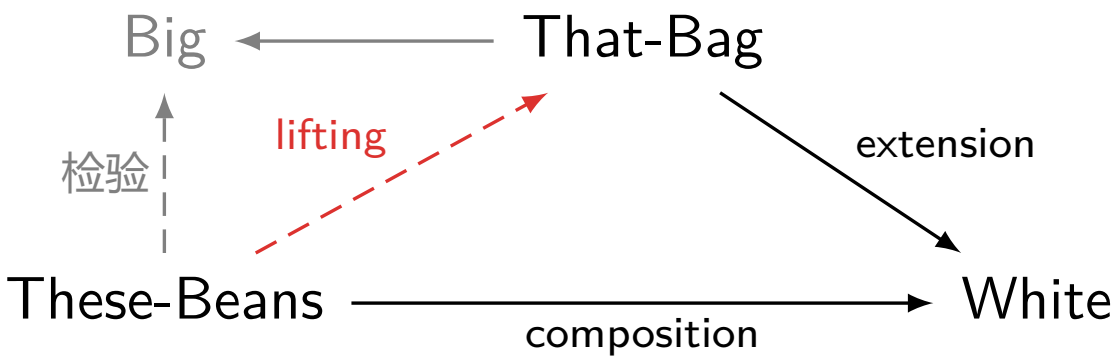


- 若 k 不唯一, 则溯因出的 f 就有多个解释, 需要进行检验, "观察数据" 无法筛选时, 可能需要借助 "干预数据", 或根据某些标准从中选择, 比如最大后验 $k : c \mapsto b^* := \underset{b}{\operatorname{argmax}}\, P(b \mid c)$. (需要偏置)
- k 既可以看作归纳 g 的逆, 也可以看作溯因 lift.



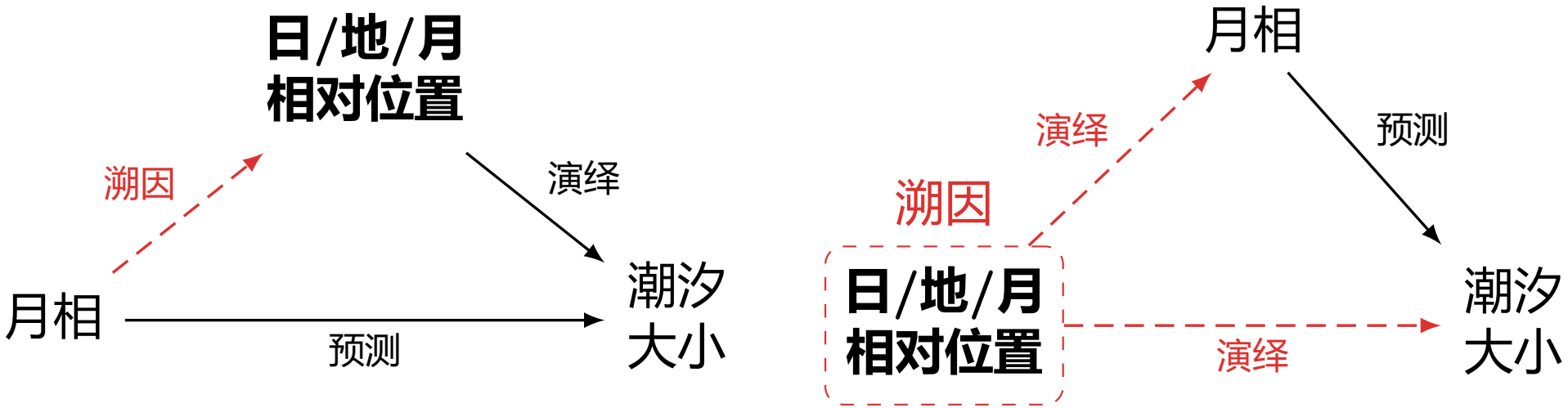
溯因检验





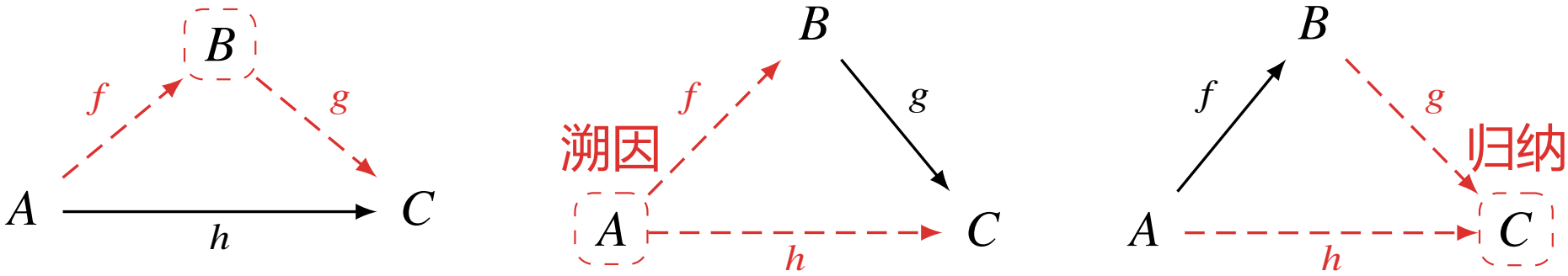
- 溯因出 “这些豆子是那个袋子里的”
- 又已知 “那个袋子里的豆子都是大的”
- 检验是否 “这些豆子是大的”

溯因: 选择性溯因 vs 创造性溯因

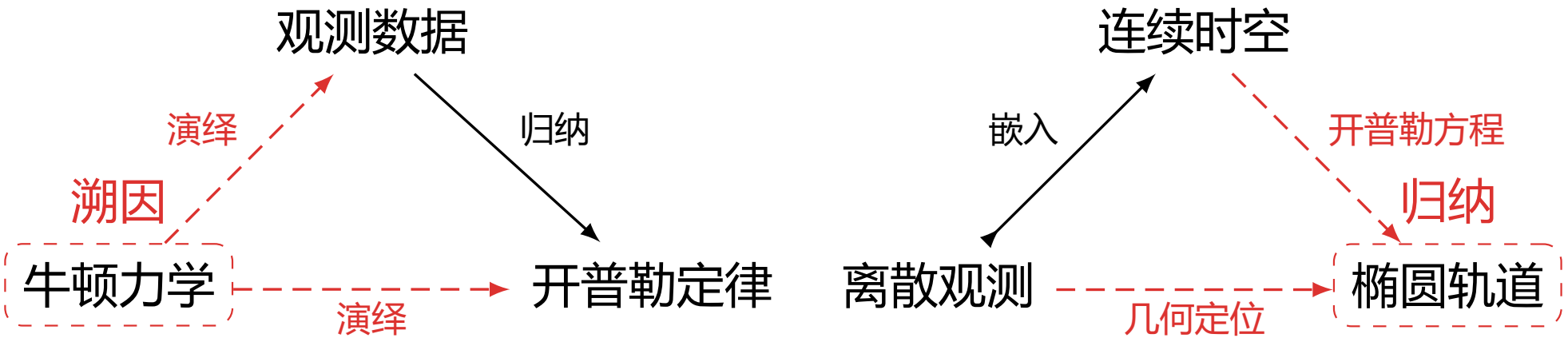


- 左图: 溯因-演绎 $\rightsquigarrow$ 预测 (为了解释为什么预测靠谱, 溯因出月相变化的原因是日/地/月的相对位置)
- 右图: 溯因创造概念解释 (溯因出 B 和 C 的共同原因是 A)
  - $f : A \to B$ 相对位置决定月相的几何光学.
  - $h : A \to C$ 相对位置决定潮汐的万有引力.
- **问题:** 溯因是 (1) 选择最佳解释? 还是需要 (2) 创造新对象?
  1. 选择性溯因
  2. 创造性溯因

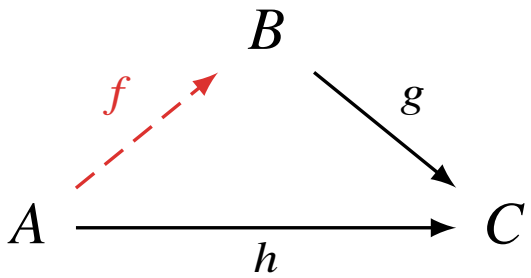
不仅溯因能创造新对象



1. 分解 factorization 创造新对象.
2. 极限 limit 创造新对象. — 溯因
3. 余极限 colimit 创造新对象. — 归纳
   — 比如, 开普勒从第谷有限离散观测样本, 粘合出连续的椭圆轨道



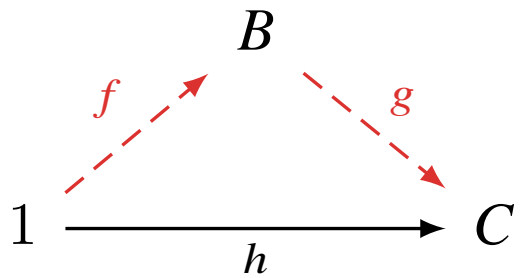
溯因: 歧义笑话



Time flies like an arrow; fruit flies like a banana.

- A: 多元词组, 比如 [`fruit`, `flies`, `like`, `a`, `banana`].
- C: 句子.
- B: 语法树.
- h: 把多元词组线性拼接成句子.
- g: 把语法树压平成句子.
- f: 给多元词组解析出合适的语法树.

溯因: 观察渗透理论

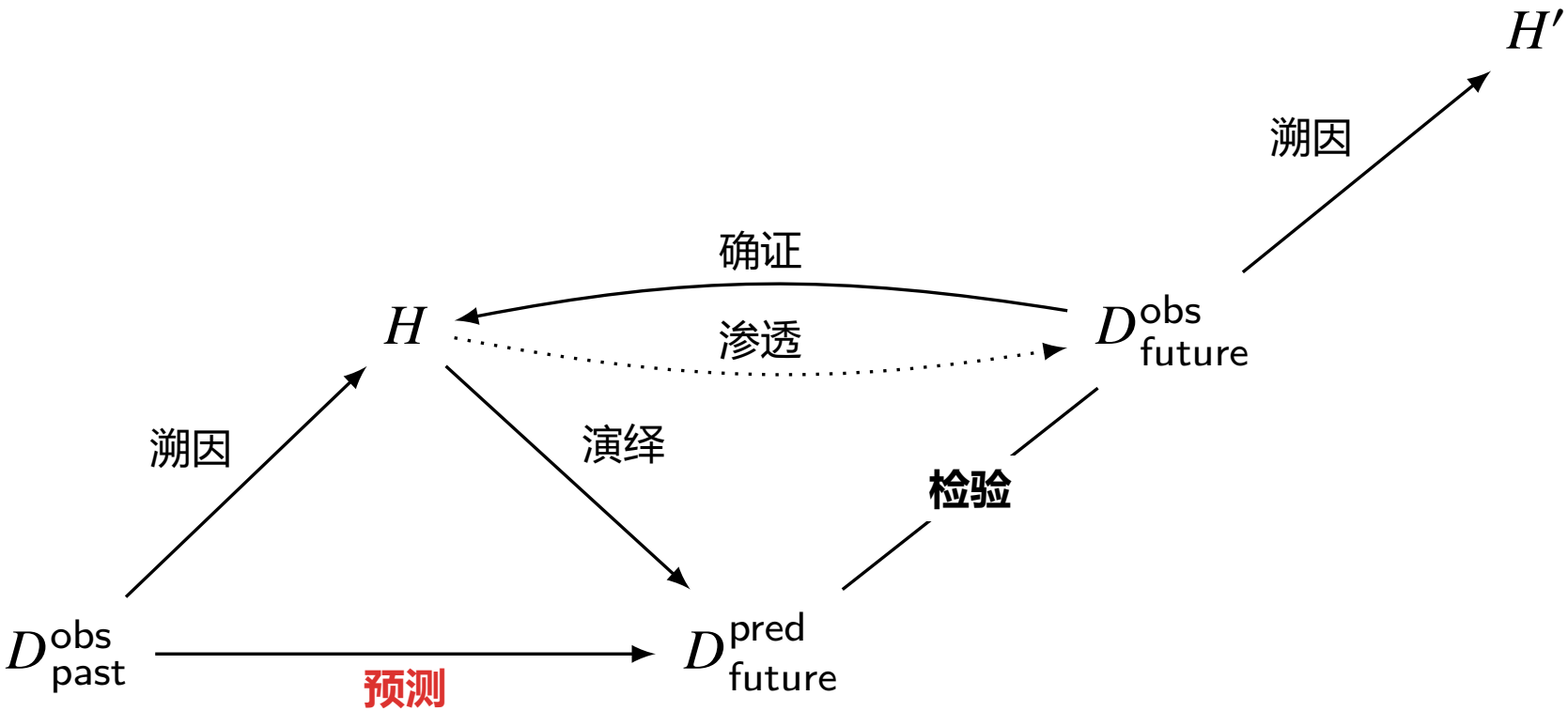


$$\frac{B \to C \qquad \top \to C}{\top \to B}$$

- C: 观察空间 ($C := O^I$ 以 I 索引的观察数据流)
- B: 假设空间
- h: "虎头-树-虎尾" 视觉影像
- f: 一只老虎在树后
- g: 向视网膜的投影
- 柏拉图洞穴隐喻: 通过溯因, 从影子 lift 到世界模型

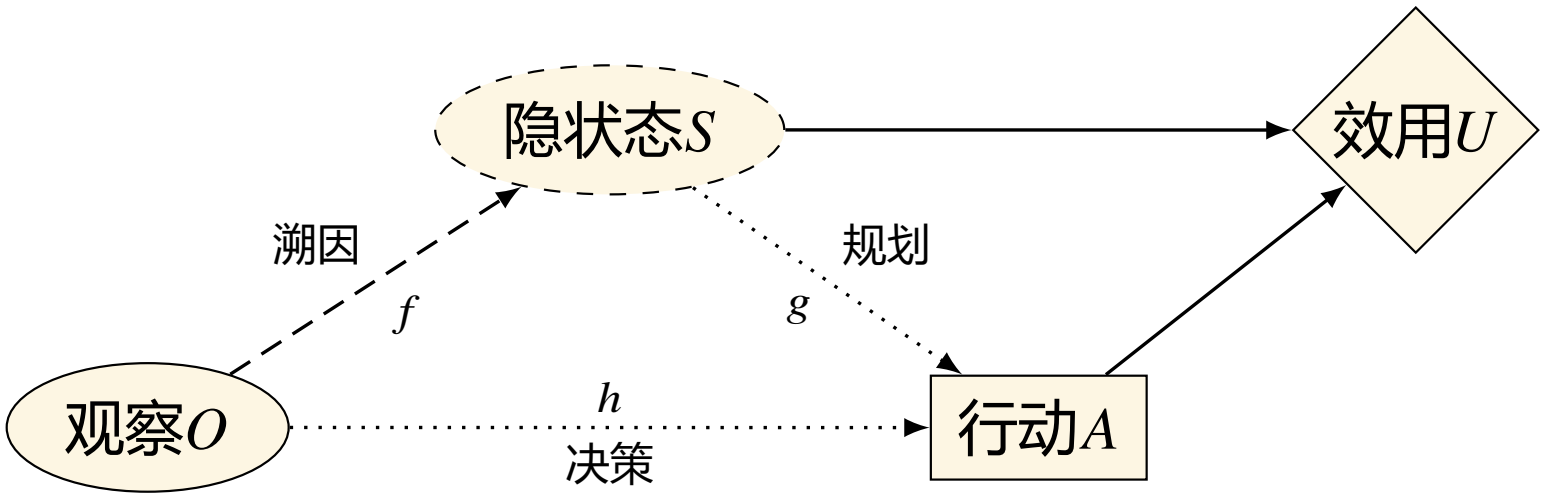
预测 = 演绎 ∘ 溯因 (溯因 → 演绎 → 归纳)

后期皮尔士: “归纳 = 预测检验 + 假说确证” — 自我纠错的渐进过程



- LLM 把原本真实世界中进行的 “溯因-演绎-检验-确证-溯因” 的自我纠错探究循环内化到了思维链推理过程中.
- LLM 测试时计算 (test-time compute) 之所以有效, 主要因为 “(通过内省或验证器) 核验比生成答案容易”, 通过 “多次采样-评估-选择”, 可以提高命中率.

预测 (演绎 ∘ 溯因) vs 决策 (规划 ∘ 溯因)

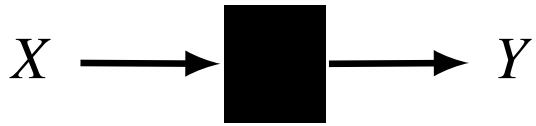


- 如果知道隐状态 S, 则可以规划最优行动 $g : S \to A$.
  $a^* = \operatorname{argmax}_a U(s, a)$.
- 但隐状态不可见. 我们只能获得观察证据 O, 然后基于 O 做出决策 $h : O \to A$.
- 我们希望决策 h 尽量逼近最优, 通常以期望效用最大化 $a^* = \operatorname{argmax}_a \sum_s P(s \mid o, \mathrm{do}(a)) U(s, a)$ 的方式, 希望三角形尽量交换

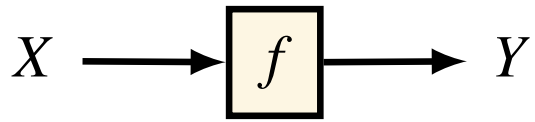
$$h \approx g \circ f$$

预测 vs 机器学习

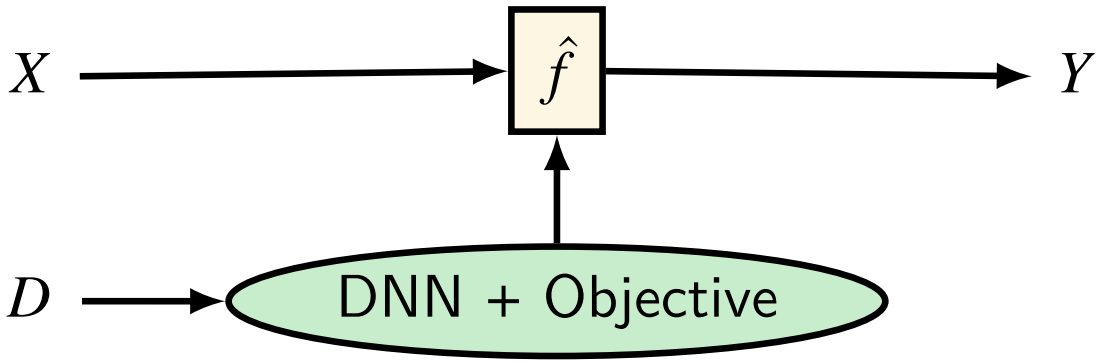
1. ML as a black box (correlation)



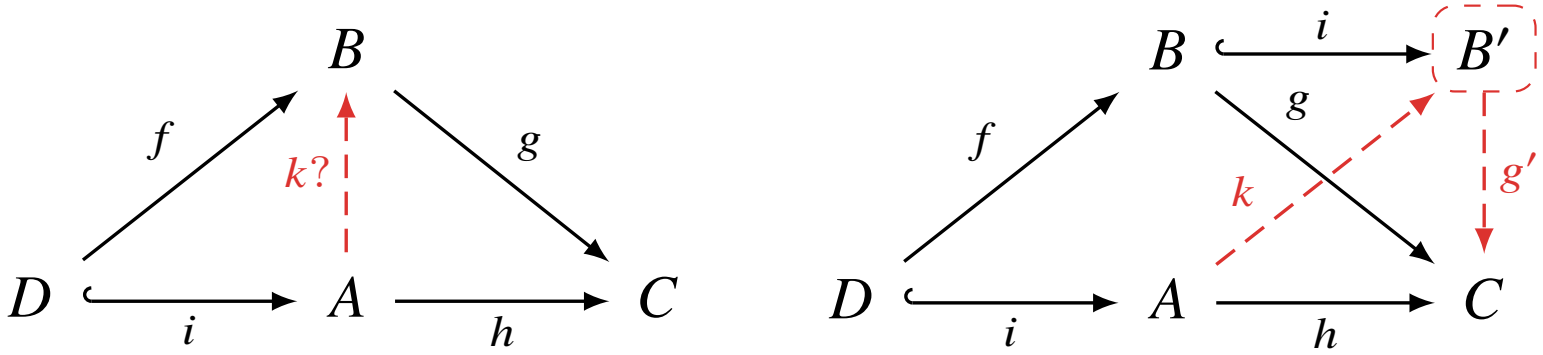
2. ML as a Prediction/Induction box (static model)



3. ML as an Abduction Algorithm (dynamic process)



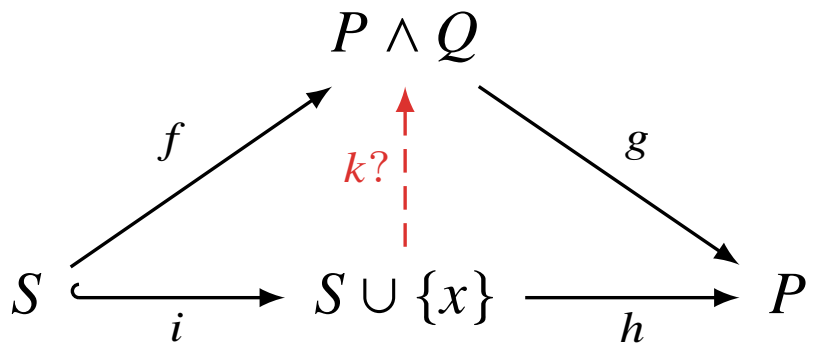
溯因无解: 大模型幻觉



- 模型 C 是实在 B 的投影. 影子超出实在即幻觉.
- D 是一个小区间, A 是 D 的延拓. B 是真实情况, C 是大语言模型.
- 提升不随延拓自动保持. 什么是大语言模型的 next-token 续写幻觉: 虽然有从 D 到 B 的提升 f 满足 $g \circ f = h \circ i$, 但找不到从 A 到 B 的提升 k 满足 $g \circ k = h$.
  — 林黛玉倒拔垂杨柳.
- 阴谋论: 容易 lift 出多个解释, 难以验证.
  阴谋论 vs 幻觉 :: 多解 vs 无解.
- lift 无解? vs 创造新对象?
  幻觉在左 vs 天才在右 (爱因斯坦进行溯因跳跃, 提出 "等效原理")

类比: 溯因/归纳/演绎的混合

- S 中的元素都有 P 和 Q 性质.
- x 和 S 中的元素一样, 也都有 P 性质.
- x 是否也有 Q 性质?

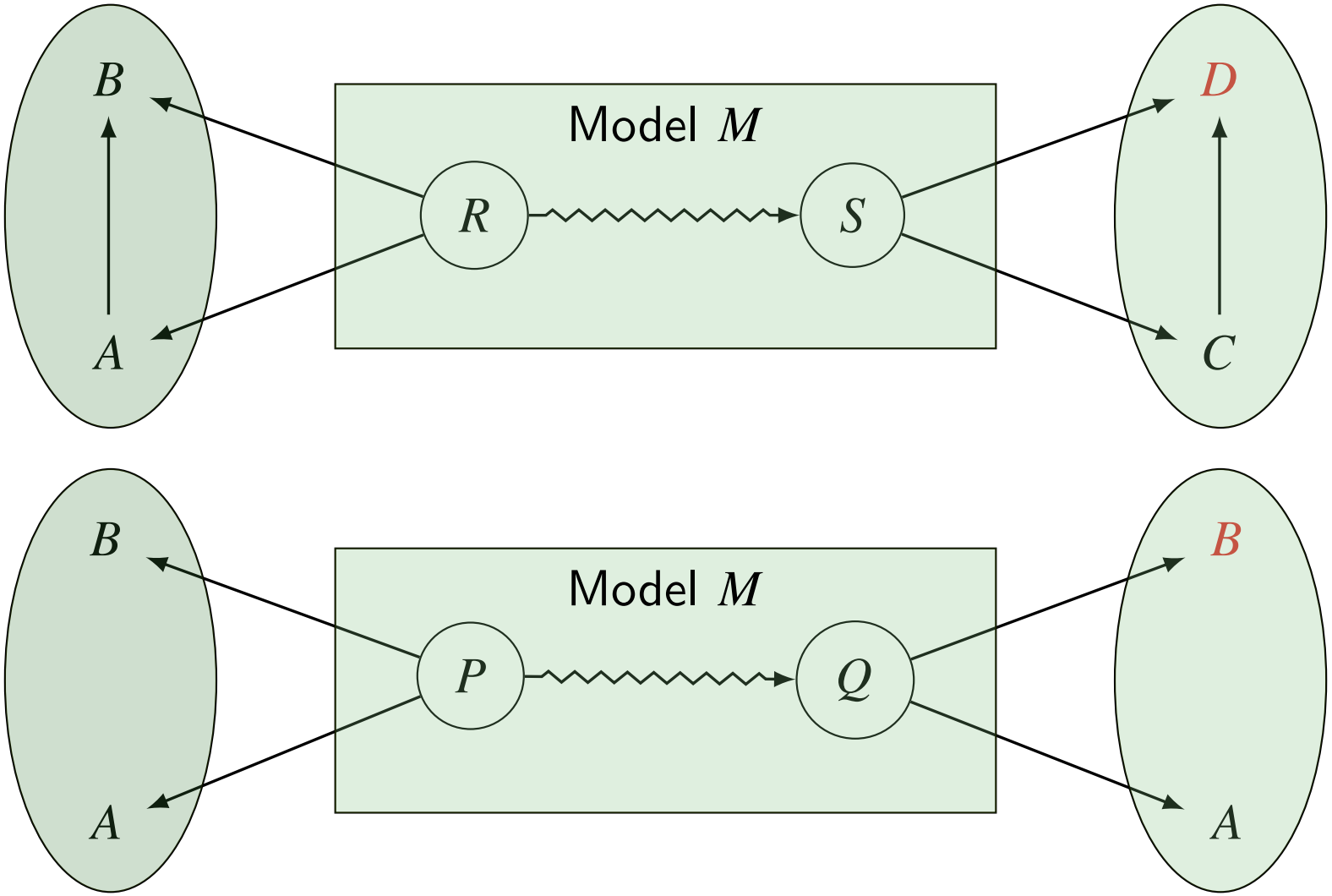


- 令 $S := \{\langle A, B\rangle\}$, $x := \langle C, D\rangle$, $M := P \wedge Q$.
- 若存在 k 使得 $g \circ k = h$, 则我们可以说, 在 M 意义上, A 之于 B 恰如 C 之于 D.

$$A : B :: C : D$$

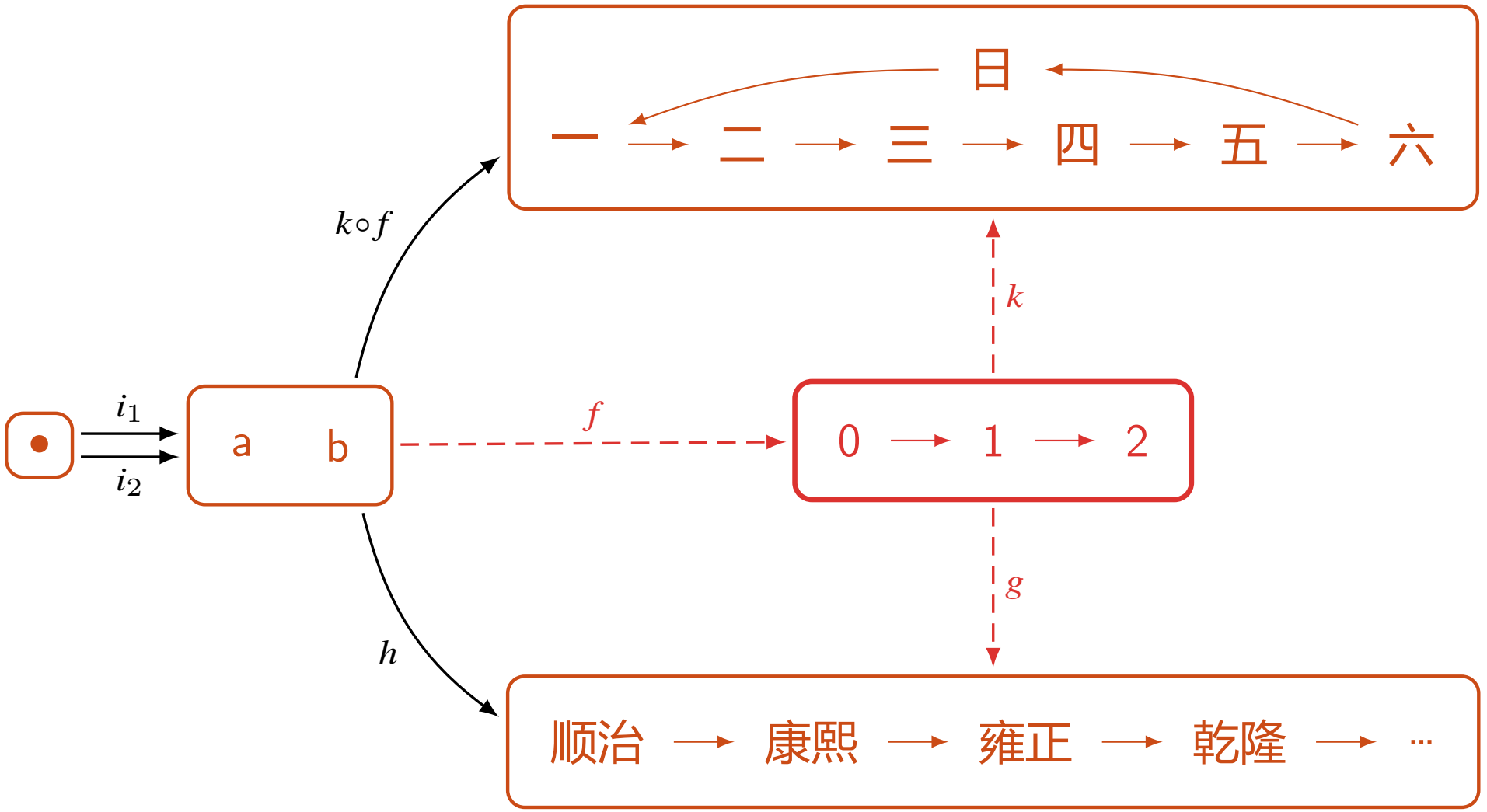
Remark: $A : B :: C :$ ___?___ 是预测?

关系类比/属性类比 — 结构迁移



类比 — 溯因/归纳/演绎的混合

康熙 : 乾隆 :: 星期三 : ___?___



Kolmogorov Complexity vs Analogy

$$A : B :: C : D$$

The best analogy is the one that makes (A, B, C, D) simplest.

$$X^* = \operatorname*{argmin}_X K(ABCX)$$

Example:

$$\text{abc} : \text{abd} :: \text{ppqqrr} : _____$$

$$\text{`ppqqss'} = \operatorname*{argmin}_X K(\text{`abc'}, \text{`abd'}, \text{`ppqqrr'}, X)$$

“类比学习”是一种近似的“溯因归纳”[Mur18]

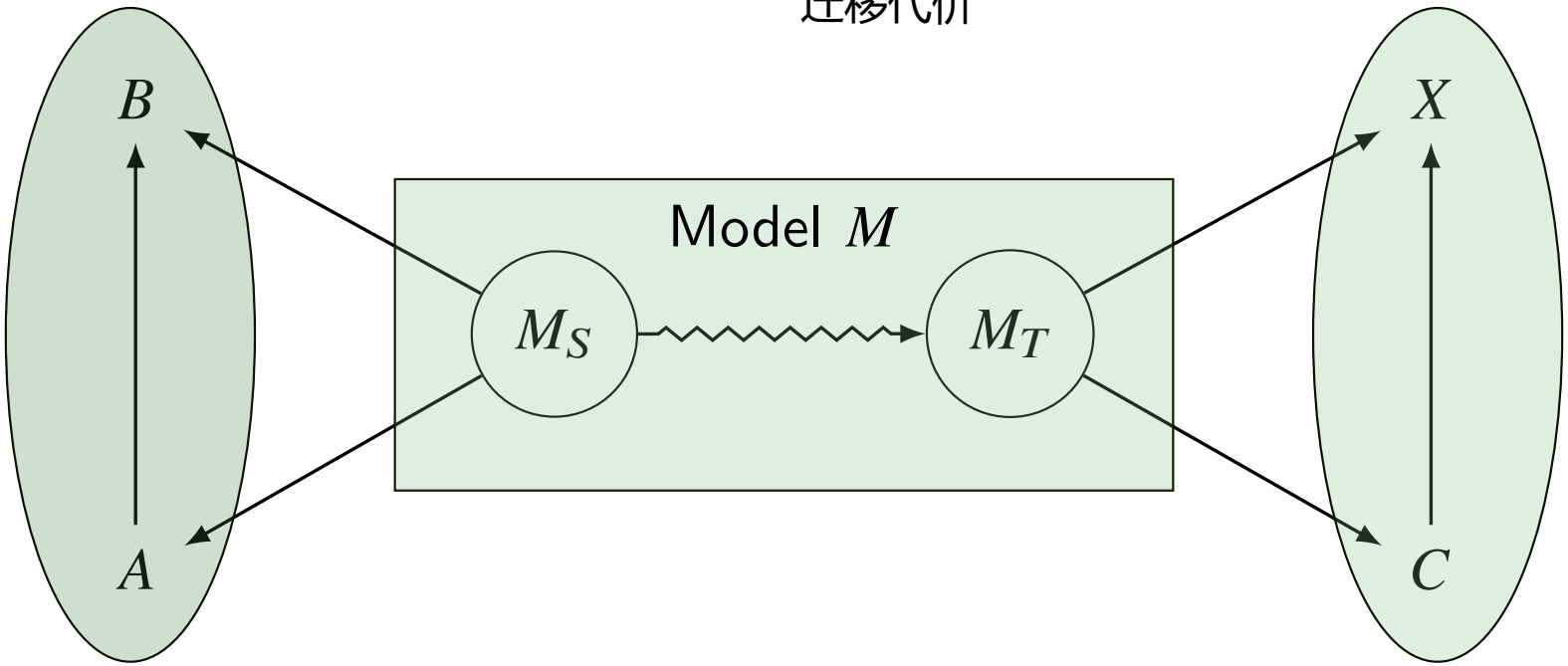
$$A : B :: C : X$$

腰肢 : 款摆 :: 弱柳 : ____ ← 扶风

$$X^* := \operatorname*{argmin}_X K(ABCX)$$

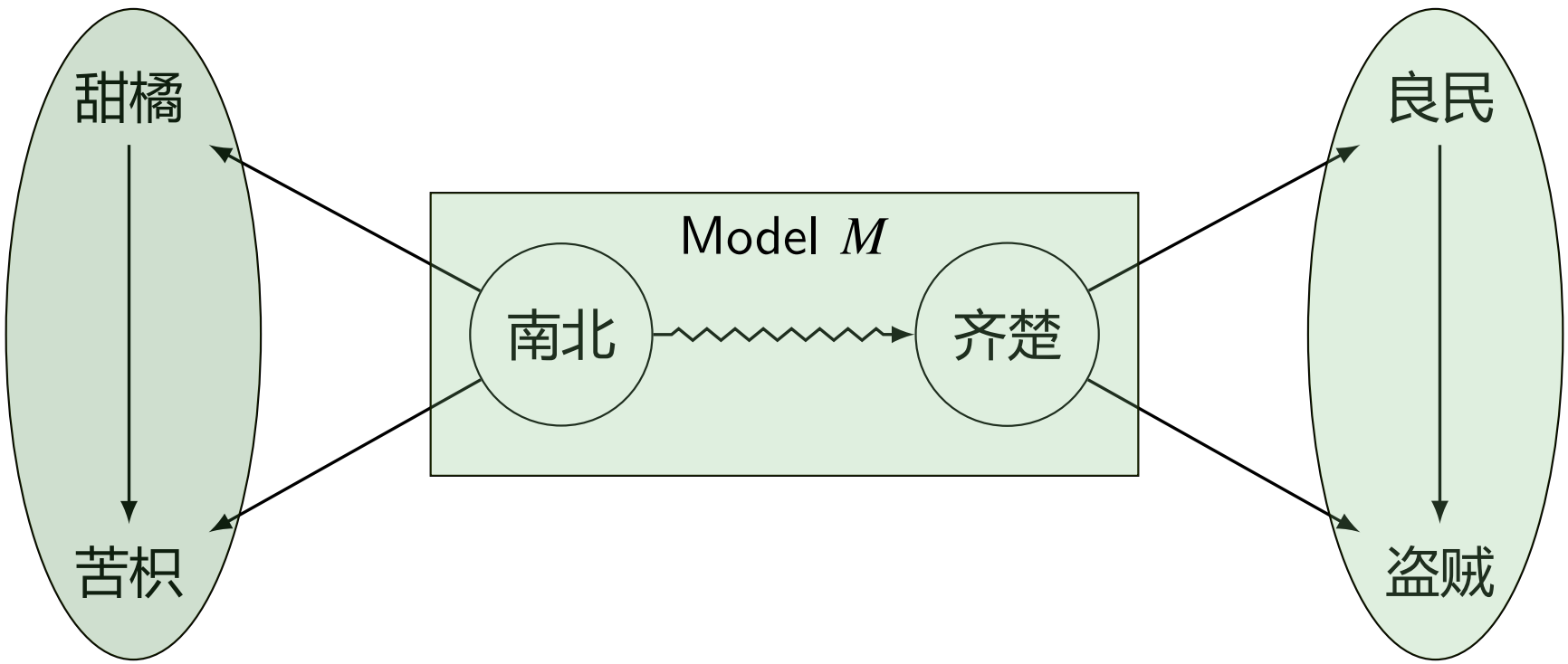
$$\operatorname*{argmin}_{M \in \mathcal{H}} \{K(M) + K(D \mid M)\} \qquad \textbf{(MDL)}$$

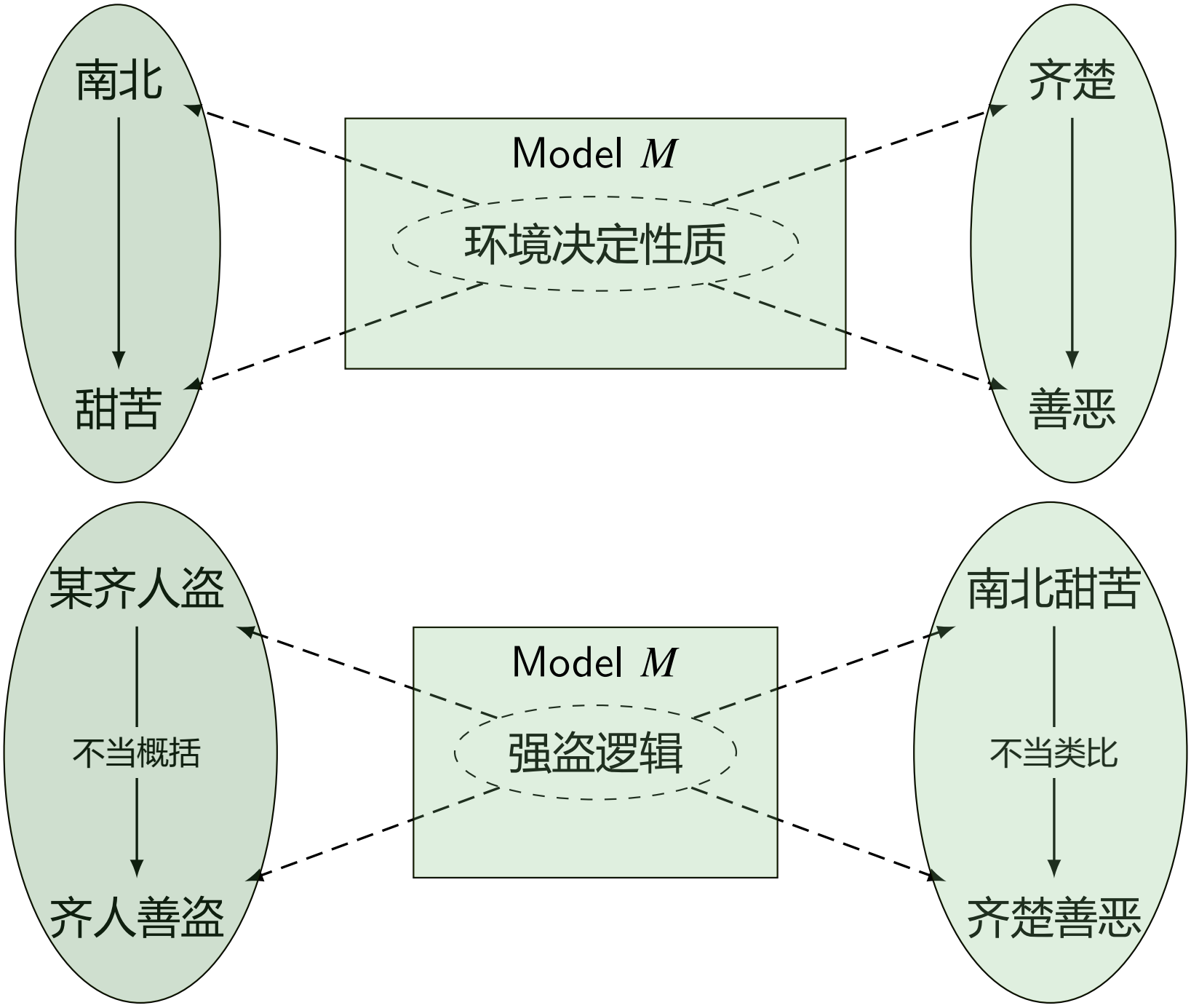
$$K(M_S)+K(A \mid M_S)+K(B \mid M_S, A)+\underbrace{K(M_T \mid M_S)}_{\text{迁移代价}}+K(C \mid M_T)+K(X \mid M_T, C)$$



《晏子使楚》

1. 楚王赐晏子酒, 酒酣, 吏二缚一人诣王.
2. 王曰: “缚者曷为者也?” 对曰: “齐人也, 坐盗.”
3. 王视晏子曰: “齐人固善盗乎?”
4. 晏子避席对曰: “婴闻之, 橘生淮南则为橘, 生于淮北则为枳, 叶徒相似, 其实味不同. 所以然者何? 水土异也. 今民生齐不盗, 入楚则盗, 得无楚之水土, 使民善盗耶?”




南北
甜苦
Model M
环境决定性质
齐楚
善恶
某齐人盗
不当概括
齐人善盗
Model M
强盗逻辑
南北甜苦
不当类比
齐楚善恶

少样本学习

Definition (少样本学习)

数据集可以被划分成 $|I|$ 个类别: $\{C_i : i \in I\}$.

- 已知大量无标签的背景数据 B.
- 对于每一个类别 C_i, 又知道了少量有标签的样本 $A_i \subset C_i$.
- 少样本学习, 就是基于大量无标签样本 B 和少量有标签样本 A_i, 最小化目标函数:

$$\sum_{i \in I} \sum_{x \in C_i} D\Big(x, \Phi(A_i) \,\Big|\, B\Big)$$

其中, Φ 是根据少量样本表示的 "原型", D 是认知距离.

什么是 "认知距离"?

为什么关心“认知距离”?

- “魔戒”与“哈利·波特”之间的距离
- “爸爸”与“妈妈”之间的距离
- 垃圾邮件与正常邮件之间的距离
- 问题与答案之间的距离
- 神经网络当前输出与正确输出之间的距离
- 动物天生就关心两种距离：接近食物，远离天敌
- 为什么高级动物的身体都是左右两侧对称的?

从麦克斯韦妖到少样本学习

Maxwell's Demon & Landauer's Principle & Few-Shot Learning

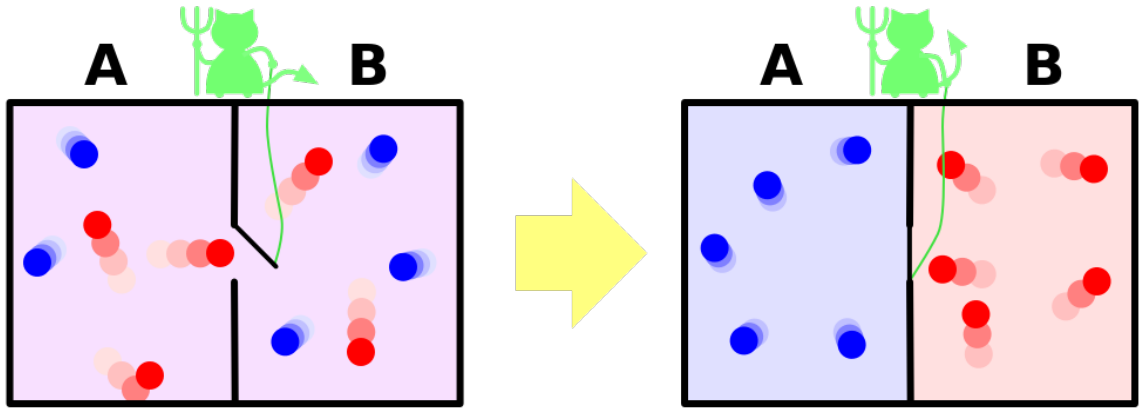


Figure: 小妖将熵转化为信息, 信息抹除操作将信息转化为熵. 在对平衡系综进行理想测量的过程中, 熵的减少必须由记录信息的增加来补偿, 反之亦然. $\Delta H \approx -\langle \Delta K \rangle$.

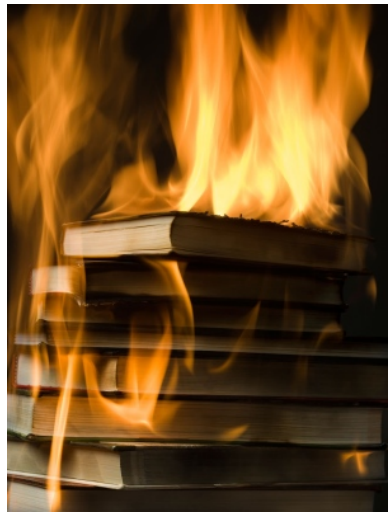
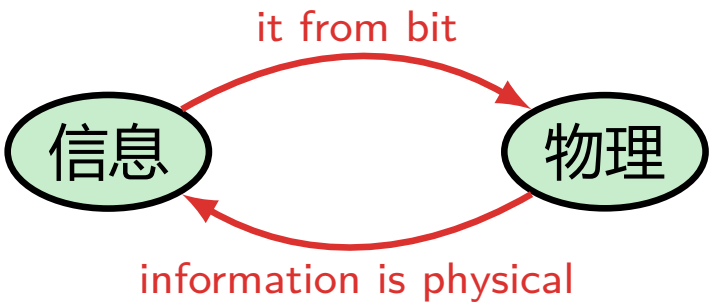


Figure: 消除 1 比特信息会向环境中耗散至少 $kT \ln 2$ 的热量.



- Landauer: "Information is physical".
- Wheeler: "It from bit".

Landauer's Principle

- Landauer 原理: 擦除 1 比特信息会向环境中耗散至少 $kT\ln 2$ 的热量.
- Reversible computation is free.
- The ultimate thermodynamic cost of erasing x is reached by:
  - reversibly compress x to x^*,
  - then erase x^*. Cost $K(x)$ bits.
- The longer you compute, the less heat dissipation.
- 在 x, y 之间转化, 所需要的最小能量是:

$$E(x,y) := \min\left\{\ell(p) : U(x,p) = y, U(y,p) = x\right\}$$

Theorem

- $E(x,y) \stackrel{+}{=} \max\{K(x \mid y), K(y \mid x)\} \stackrel{+}{=} K(xy) - \min\{K(x), K(y)\}$
- *For any computable metric D satisfying $\sum_y 2^{-D(x,y)} \leq 1$, there is a constant c, such that for all x, y:*

$$E(x,y) \leq D(x,y) + c$$

Remarks: Few-Shot Learning

Corollary

$$\sum_{i\in I}\sum_{x\in C_i} E\Big(x,\Phi(A_i)\,\Big|\,B\Big) \stackrel{+}{\leq} \sum_{i\in I}\sum_{x\in C_i} D\Big(x,\Phi(A_i)\,\Big|\,B\Big)$$

- $E(x,y)$ is optimal information distance — it discovers all effective similarities.
- If the cognitive distance we are born with, $D(x,y)$, is computable, then it can be approximated and replaced by $E(x,y)$, because if $D(x,y)$ discovers some similarity, so will $E(x,y)$.

Universal Similarity Metric

- 相似的大对象可能比不相似的小对象有更大的 $E(x, y)$.
- 为了尺度无关性, $E(x, y)$ 可以归一化为 Normalized Information Distance $\mathrm{NID}(x, y)$, 其近似版可借助某图灵机 T 定义为 $d(x, y)$.

$$\mathrm{NID}(x, y) := \frac{\max\{K(x \mid y), K(y \mid x)\}}{\max\{K(x), K(y)\}}$$

$$d(x, y) := \frac{K_T(xy) - \min\{K_T(x), K_T(y)\}}{\max\{K_T(x), K_T(y)\}}$$

  如果 $K(y) > K(x)$, 则 $\mathrm{NID}(x, y) = 1 - \frac{I(x:y)}{K(y)}$.
- T : Lempel-Ziv/gzip/bzip2/PPMZ, or
  $K_T(x) := -\log P_{\mathrm{google}}(x)$ where $p_{\mathrm{google}}(x) := \frac{\#\text{ pages containing 'x'}}{\#\text{ pages indexed}}$
- compute similarity matrix $\big(d(x_i, x_j)\big)_{ij}$
- cluster similar objects

物理、信息、演化 & 多样本学习

$$E = mc^2$$

物质、能量、信息三者可以相互转换. 但信息是宇宙的最基本单元, 比物质和能量还要基本. 宇宙是一个信息处理系统. 生物通过组织信息主动抵抗熵增. AI 的意义比我们想的还要深远, 因为 AI 的核心是组织信息、理解信息并构建信息客体 (Informational Objects).

— *Demis Hassabis*

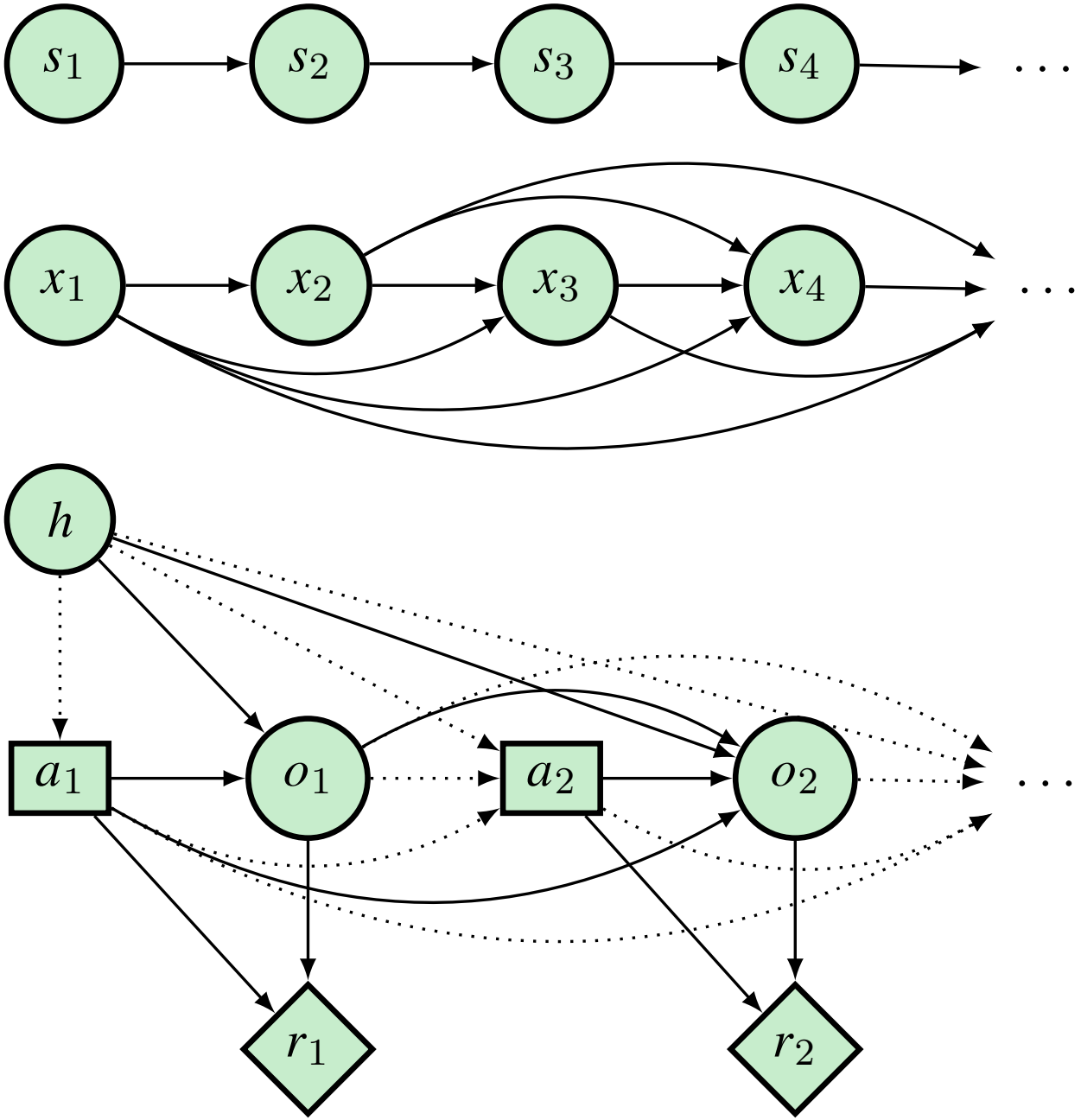
Hassabis's Conjecture: Any pattern that can be generated or found in nature can be efficiently discovered and modelled by a classical learning algorithm.[a]

[a]不同于抽象的数学构造, 自然的结构是有原因的演化而来的. 大自然沿着阻力最小的路径演化, 构建了一个能量漏斗, 幸存的模式背后, 是一个光滑的低维流形.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
   - Kolmogorov Complexity
   - Analogy & Few-Shot Learning
   - Algorithmic Probability
   - A Statistical Mechanical Interpretation of AIT
   - Incompressibility & Incompleteness
   - Algorithmic Randomness
   - Effective Complexity
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Ray Solomonoff 1926-2009



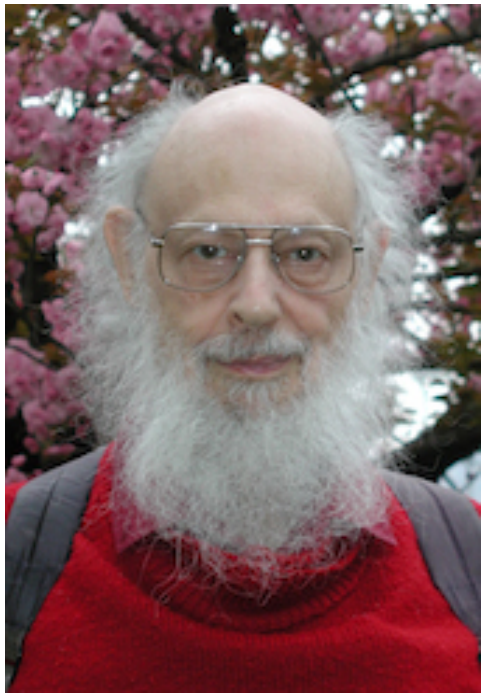


Figure: Solomonoff

从图灵机求逆 (选择性溯因) 到通用归纳 [Sol97]

1. 图灵机求逆问题
   - 整数的素因子分解
   - 解方程: $x^2 + \sin x = 25$
   - 证明定理: Find `proof` such that $\texttt{Machine}(\texttt{proof}, \texttt{theorem}) = \texttt{Yes}$
   - 求积分: $\int x \sin x \, \mathrm{d}x \rightsquigarrow$ Find f such that $\frac{\mathrm{d}}{\mathrm{d}x} f = x \sin x$

   Given `output`, find `input` such that $\texttt{Machine}(\texttt{input}) = \texttt{output}$

   $$\texttt{Machine}(\texttt{Inverse}(\texttt{Machine}, \texttt{output})) = \texttt{output}$$

2. 约束优化问题

   $$\operatorname*{argmax}_{x \in X} \texttt{Utility}(x)$$

   - 归纳-预测

   $$\nu^* = \operatorname*{argmax}_{\nu \in \mathcal{M}} P(\nu \mid x_{<t}) \qquad x_t^* = \operatorname*{argmax}_{x_t} P(x_t \mid x_{<t})$$

   — 如果真实的 P 未知, 该怎么办?
   — 归纳预测不同于确定性的图灵机求逆, 需要概率性求逆.
   — 逆转霍夫曼编码 (已知概率, 求最短编码): 短编码 $\rightarrow$ 高概率.
   — 图灵机求逆问题 $U(\texttt{Inverse}(U, x)) = x$ 中, $\texttt{Inverse}(U, x)$ 不唯一怎么办? 考虑所有的 $p : U(p) = x$.

Algorithmic Probability

Definition (算法概率)

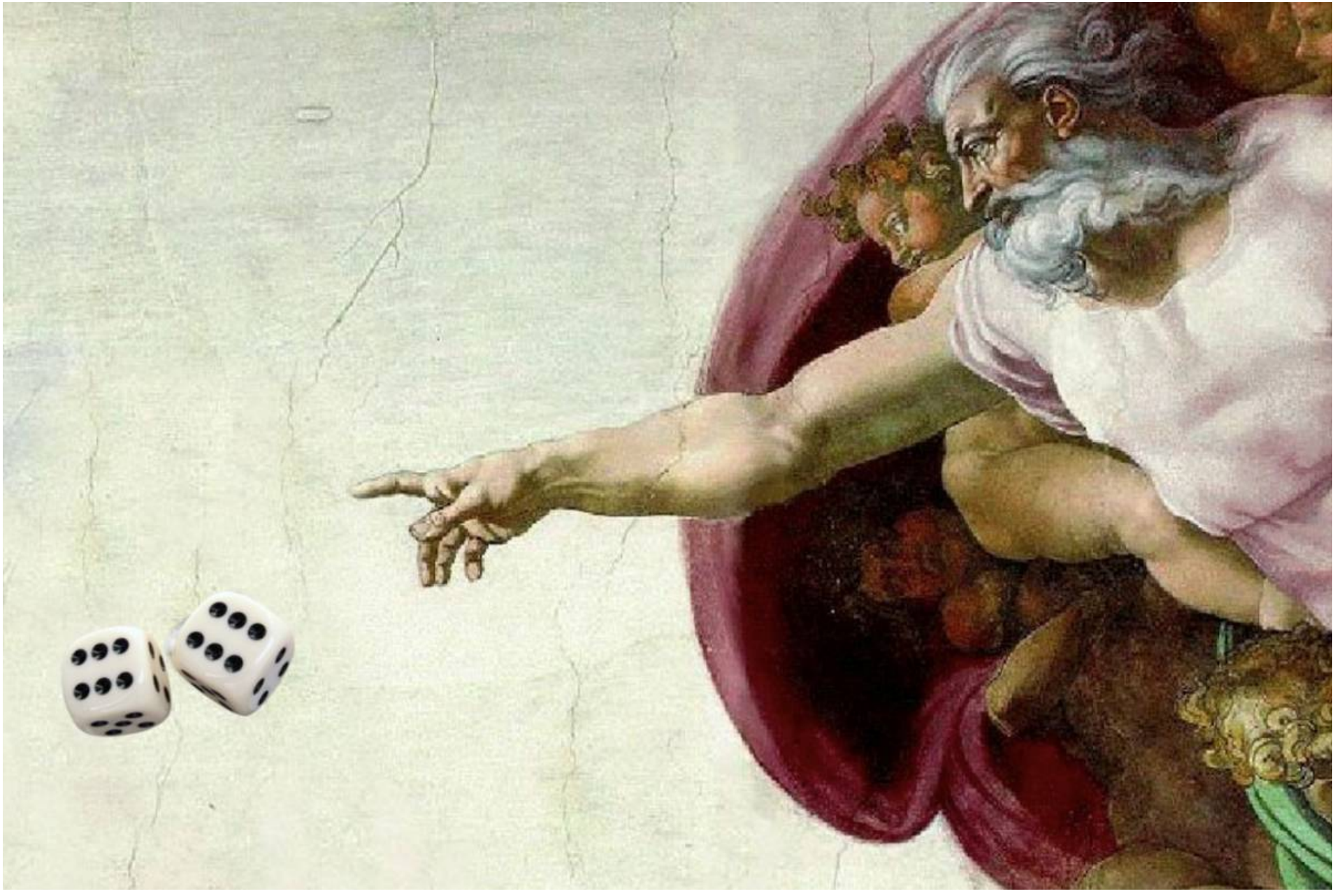
$$M(x) := \sum_{p:U(p)=x*} 2^{-\ell(p)}$$

其中 U 是通用单调图灵机. $\{p : U(p) = x*\}$ forms a prefix code for any x.

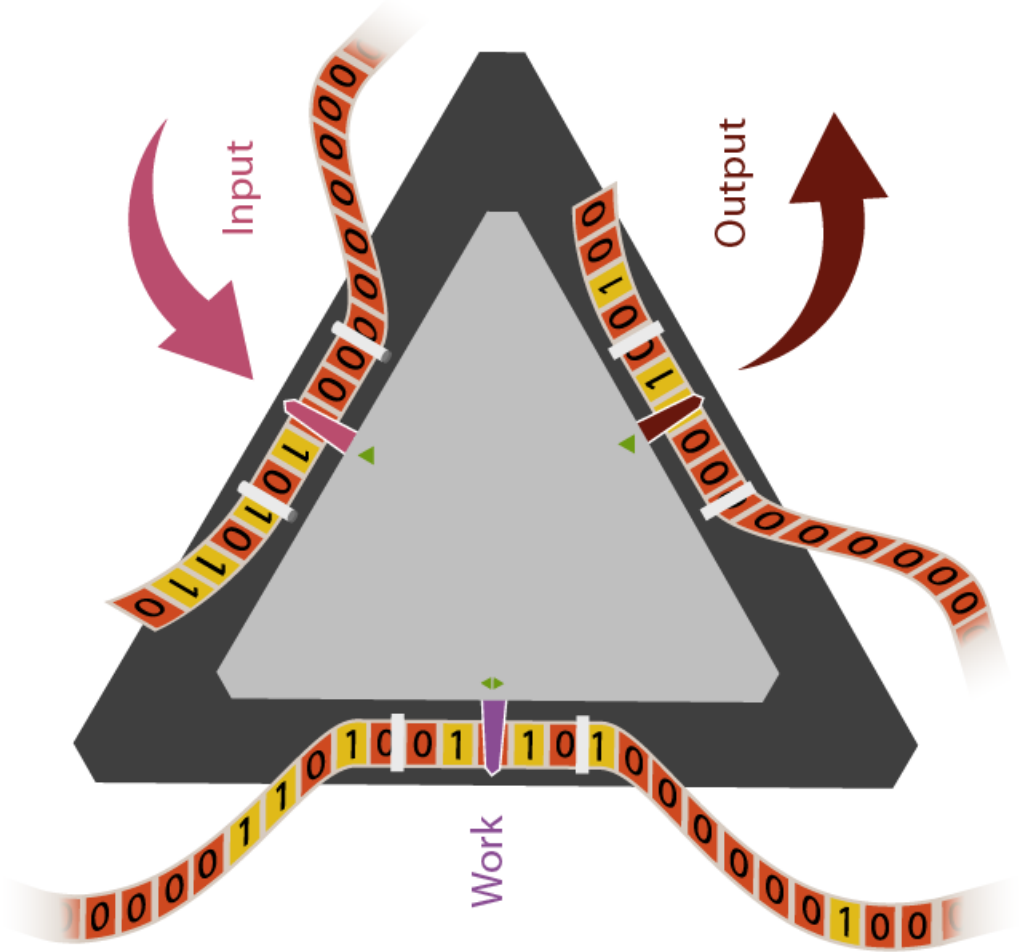
$$M(x) \stackrel{\times}{=} \xi(x) := \sum_{\nu \in \mathcal{M}} 2^{-K(\nu)} \nu(x)$$

其中 $\mathcal{M} := \{\nu_1, \nu_2, \ldots\}$ 是半可计算的半测度的集合.

$$\sum_{p:U(p)=x*} 2^{-\ell(p)} \gg 2^{-\ell(x)}$$

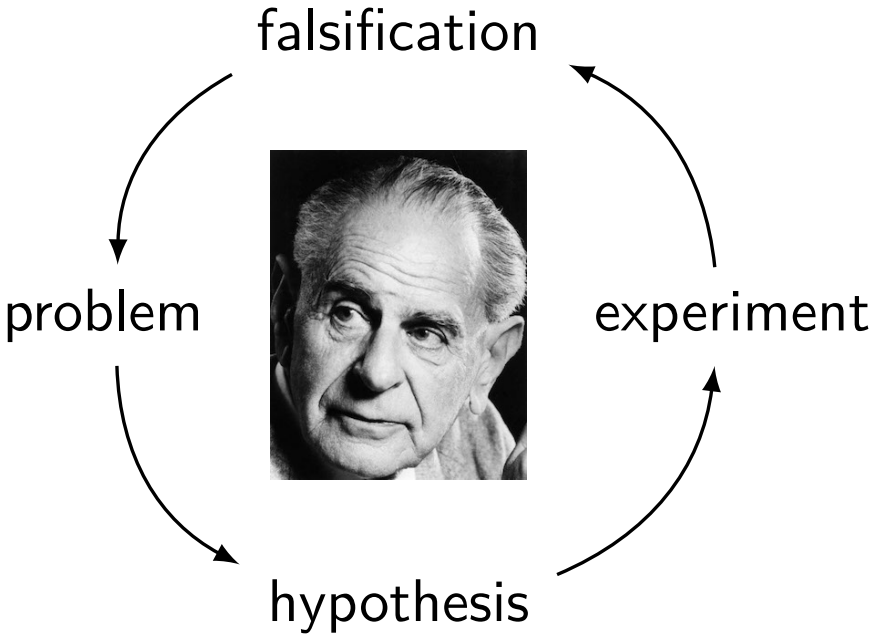


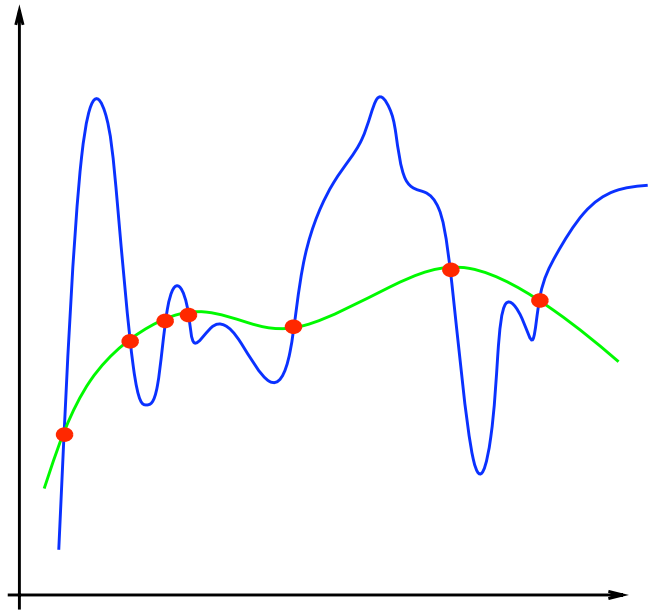
Not only does God play dice, he always throws it onto the UTM!



Aspect 1 — Popper's "Falsificationism"

所罗门诺夫一揽子 "预测" vs 波普尔一个一个 "证伪"





$\mathcal{H}$: truth ← simplcity/generality/aesthetic/utilitarian/...

Make a weighted prediction based on all consistent programs, with short programs weighted higher.

Aspect 2 — Deterministic vs Stochastic

$\mathcal{M} := \{\nu_1, \nu_2, \dots\}$ lower semicomputable semi-measure.

$$\xi(x) := \sum_{\nu \in \mathcal{M}} 2^{-K(\nu)} \nu(x)$$

$$M(x) \stackrel{\times}{=} \xi(x)$$

$w_\nu := 2^{-K(\nu)}$ is reparametrization & regrouping invariant.

$$\widetilde{w}_{\theta'} = w_{f^{-1}(\theta')} = 2^{-K(f^{-1}(\theta'))} \stackrel{\times}{=} 2^{-K(\theta')} \stackrel{\times}{=} w'_{\theta'}$$

$$\widetilde{w}_{\theta'} = \sum_{\theta : f(\theta) = \theta'} 2^{-K(\theta)} \stackrel{\times}{=} 2^{-K(\theta')} \stackrel{\times}{=} w'_{\theta'}$$

Aspect 3 — Frequency Interpretation

$$
\begin{aligned}
M(x) &= \sum_p 2^{-\ell(p)} [\![U(p) = x*]\!] \\
&= \lim_{n\to\infty} \frac{\sum_{p:\ell(p)\le n} 2^{n-\ell(p)} [\![U(p) = x*]\!]}{2^n} \\
&\approx \lim_{n\to\infty} \frac{\left|\{p : \ell(p) = n \;\&\; U(p) = x*\}\right|}{2^n}
\end{aligned}
$$

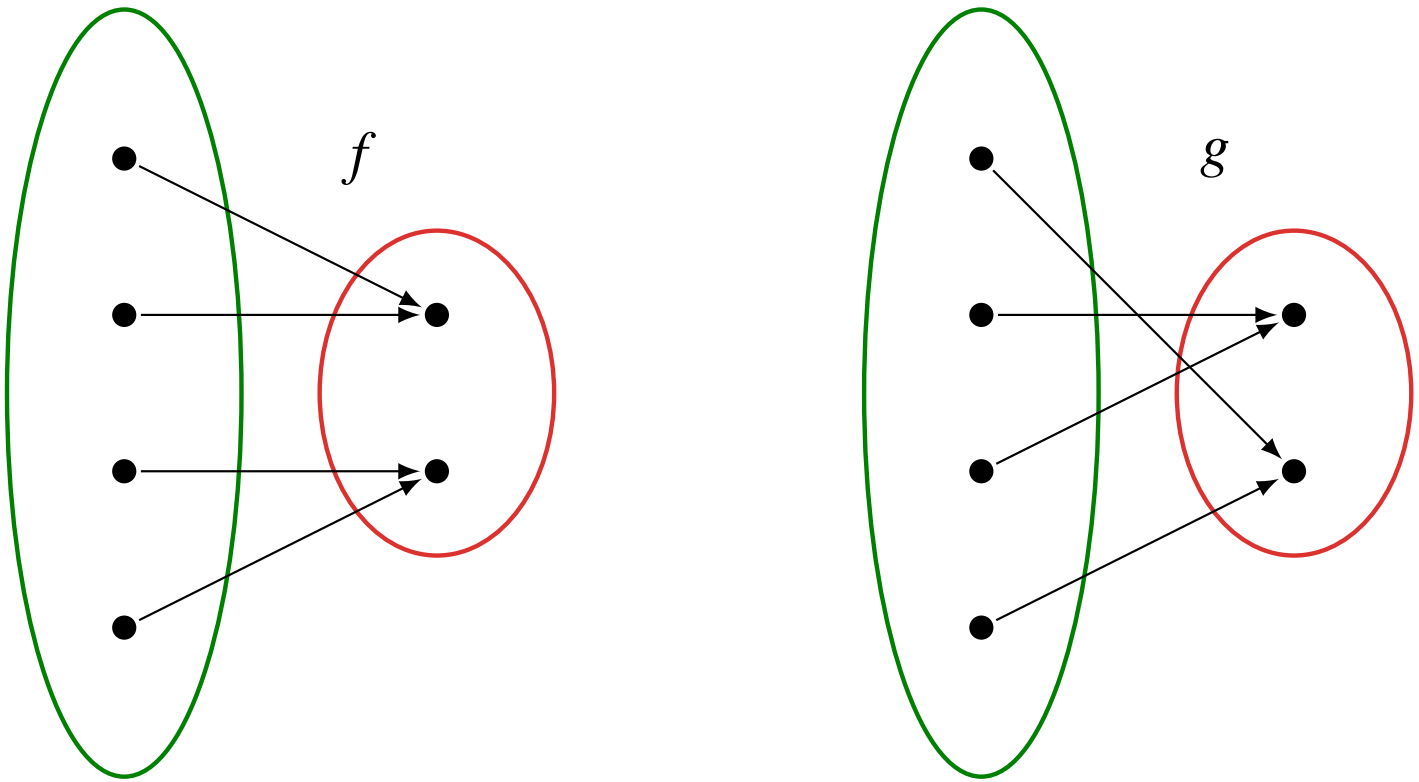
$$
\textbf{algorithmic probability} = \frac{\left|\text{consistent worlds}\right|}{\left|\text{all possible worlds}\right|}
$$

- Carnap — frequency of phenomena — i.i.d
- Solomonoff — frequency of causes — arbitrary order Markov chain

Aspect 4 — Solomonoff 的免费午餐

- 大语言模型 LLM 在做 Next-Token 预测.
- 在真实世界模型 μ 未知的情况下, 什么样的预测最准确?
- 基于算法概率 M 的预测 $M \to \mu$! — LLM 奔赴算法概率 M!

1. 弱休谟: 自然齐一性 vs 可计算性
2. 强休谟: 打破 "没有免费午餐定理" 的 "休谟魔咒"
   — break "block uniform": bias non-random functions



预测 Next-Token via 算法概率

- 下一个数是什么 $1, 2, 2, 3, 3, 3, 4, 4, 4, 4, ?$

$$M(x_{11} = 5 \mid x_{1:10} = 1223334444) = \frac{M(12233344445)}{M(1223334444)} = \frac{\sum_{p:U(p)=12233344445*} 2^{-\ell(p)}}{\sum_{p:U(p)=1223334444*} 2^{-\ell(p)}}$$

Remark: 伊壁鸠鲁 + 奥卡姆 + 图灵 + 科尔莫哥洛夫 + 贝叶斯

$$\xi(x_t \mid x_{<t}) = \sum_{\nu \in \mathcal{M}} w^{\nu}_{x_{<t}} \nu(x_t \mid x_{<t})$$

where $w^{\nu}_{x_{<t}} := \frac{w^{\nu}_0 \nu(x_{<t})}{\sum_{\nu \in \mathcal{M}} w^{\nu}_0 \nu(x_{<t})}$ and $w^{\nu}_0 := 2^{-K(\nu)}$.

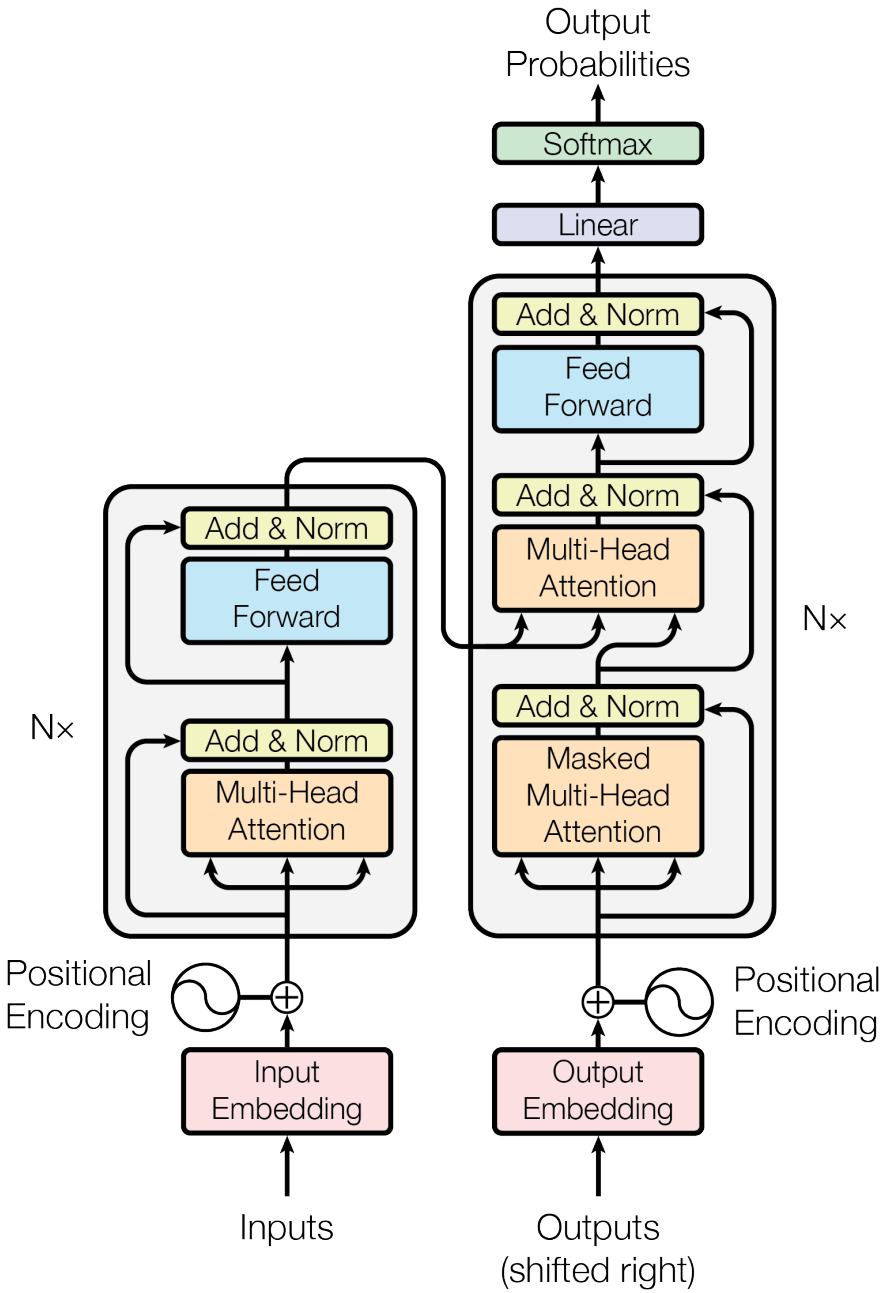
- 给定问答交互序列

$$x_{<t} := (Q_0, A_0); (Q_1, A_1); (Q_2, A_2); \ldots; (Q_{t-1}, A_{t-1})$$

— 对于下一个问题 Q_t, 你的答案是?

$$\xi(A_t \mid x_{<t} Q_t) = \sum_{\nu \in \mathcal{M}} w^{\nu}_{x_{<t}} \nu(A_t \mid x_{<t} Q_t)$$

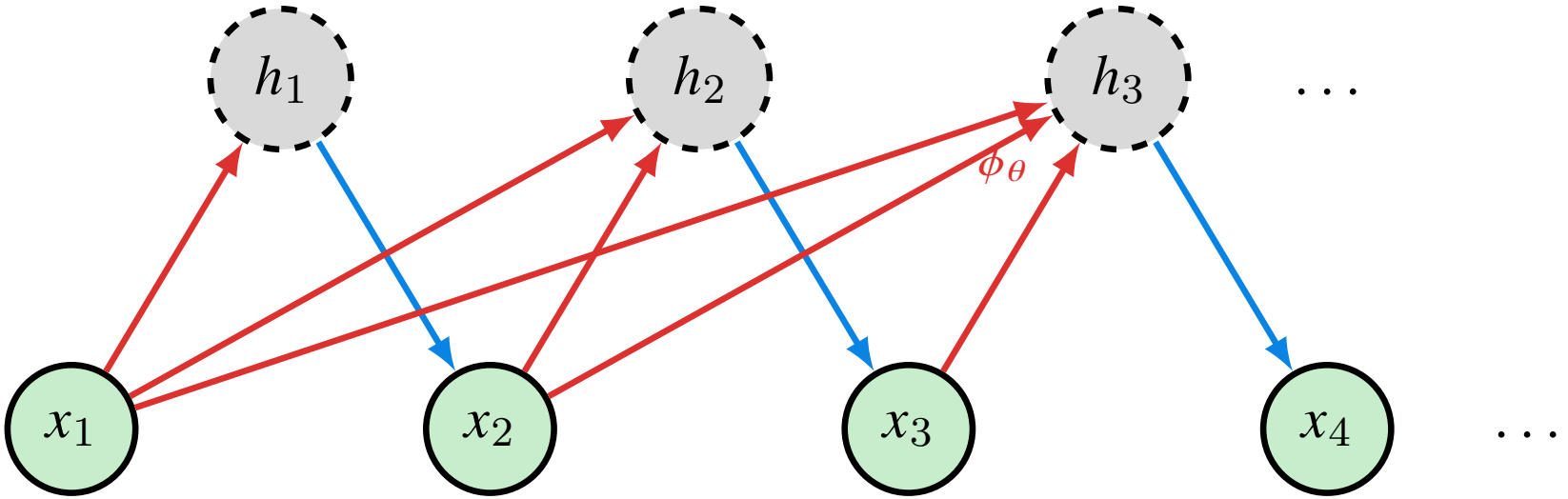
LLM — Transformer? Yes and No



抛开 Transformer 的编码器、解码器、注意力机制、位置编码、交叉熵损失、思维链 CoT、测试时计算 test-time compute、混合专家 MoE、上下文学习、指令微调、基于人类反馈的强化学习 RLHF、直接偏好优化 DPO、宪法 AI、Scaling Law 等等具体细节……

大语言模型是个啥？

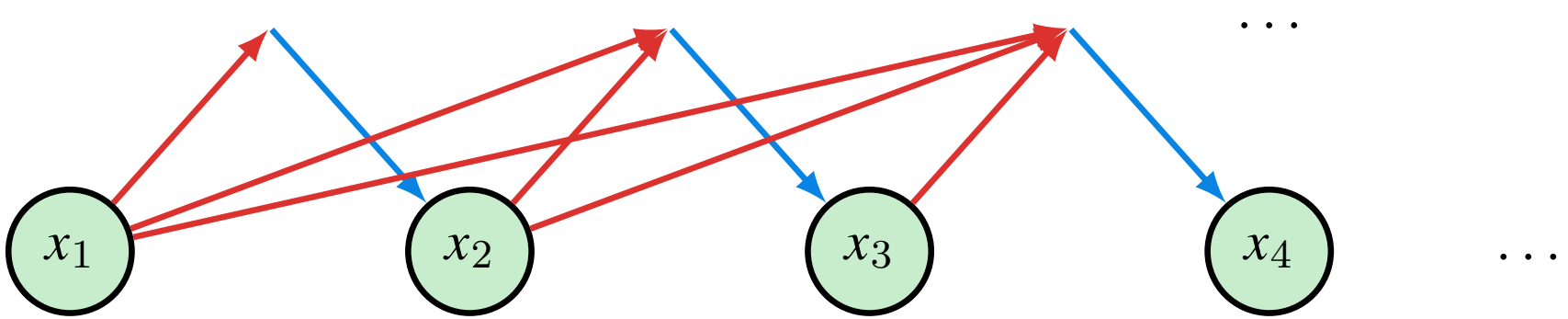
抽象掉细节后的大语言模型

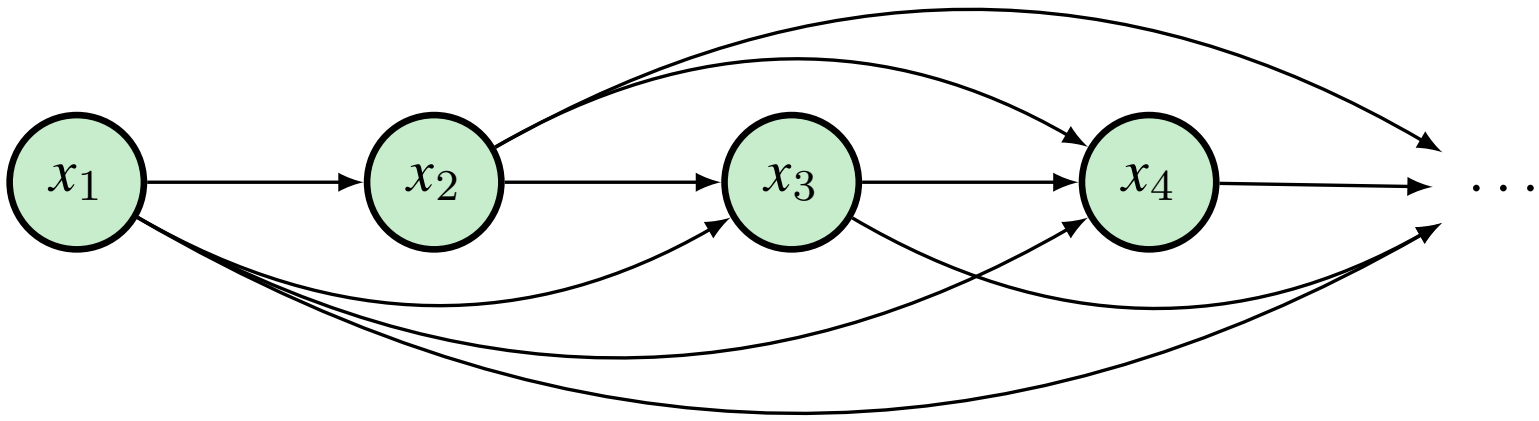


$$h_t := \phi_\theta(x_{1:t})$$

$$\rho_\theta(x_t \mid x_{<t}) := \frac{e^{\frac{1}{T}\langle x_t, \phi_\theta(x_{<t})\rangle}}{\sum_y e^{\frac{1}{T}\langle y, \phi_\theta(x_{<t})\rangle}}$$

$$\theta^* := \operatorname*{argmin}_\theta \mathbb{E}_{x_{1:n}\sim\xi}\left[\sum_{t=1}^{n} -\log \rho_\theta(x_t \mid x_{<t})\right]$$





$$\rho_\theta(x_t \mid x_{<t}) \quad \text{vs} \quad \xi(x_t \mid x_{<t}) = \sum_{\nu \in \mathcal{M}} w^\nu_{x_{<t}} \nu(x_t \mid x_{<t}) \quad \text{vs} \quad M(x_t \mid x_{<t})$$

预训练大语言模型	**算法概率**
神经网络	程序空间
权重 θ	程序 p
随机梯度下降 SGD ($+$ 注意力) 优化 θ	Solomonoff 先验混合所有程序

Problem: 大语言模型 ρ_θ 可以看作算法概率 M 的近似实现吗?

Meta-Learning & Bayesian Prediction[Gen+26]

- 在元学习中, 数据生成的协议如下:
  1. 从一个任务分布 $w(\mu)$ 中抽样一个任务 μ;
  2. 从该任务中抽样一个序列 $x \sim \mu$.
- 序列 x 的边际分布是贝叶斯混合 $\xi(x) = \sum_{\mu \in \mathcal{M}} w(\mu)\mu(x)$.
- LMM 更新参数 θ 以最小化对数损失

$$\theta^* := \operatorname*{argmin}_{\theta} \mathbb{E}_{\mu \sim w} \mathbb{E}_{x_{1:n} \sim \mu} \left[\sum_{t=1}^{n} -\log \rho_\theta(x_t \mid x_{<t}) \right]$$

- 当预测分布 ρ_θ 逼近序列的边际分布 ξ 时 $\rho_\theta \to \xi$, $D_{\mathrm{KL}}(\xi \| \rho_\theta) \to 0$, 而剩下的一项与 θ 无关的香农熵 $H(\xi)$ 即是平均编码长度的下界.

$$H(\xi, \rho_\theta) = H(\xi) + D_{\mathrm{KL}}(\xi \| \rho_\theta)$$

- 上下文学习 $\approx$ 隐式贝叶斯推理: LMM 在读入上下文 $x_{<t}$ 时, 在隐藏层中隐式地推断了当前最可能是哪个任务在产生数据 $w(\mu \mid x_{<t})$.

$$\rho_{\theta^*}(x_t \mid x_{<t}) = \xi(x_t \mid x_{<t}) = \sum_{\mu \in \mathcal{M}} \underbrace{w(\mu \mid x_{<t})}_{\text{posterior}} \mu(x_t \mid x_{<t})$$

LLM 的上下文学习 In-Context Learning

Q: 将中文翻译为英文.
示例 1: 你好 → Hello
示例 2: 再见 → Goodbye
示例 3: 购买 → Purchase
售卖 → ?

A: Sell

自回归模型的训练过程等价于无损压缩 [Del+24]

- 算术编码可以把预测 Next-Token 的生成模型作为无损压缩器.
- 目前, 在长文本任务上, 以 LLM 作为无损压缩器 (使用算术编码) 的压缩率远远优于其它压缩算法.
- 最小化 LLM 的对数损失 (数据集 $x_{1:n}$ 的边际分布 ξ 与生成模型 ρ_θ 的交叉熵), 等价于最小化以 LLM 作为无损压缩器 (使用算术编码) 的压缩率.

$$\operatorname{Loss}(\xi, \rho_\theta) = H(\xi, \rho_\theta) = \mathbb{E}_{x_{1:n}\sim\xi}\underbrace{\left[\sum_{t=1}^{n} -\log \rho_\theta(x_t \mid x_{<t})\right]}_{-\log \rho_\theta(x_{1:n})}$$

minimizing log-loss ≡ compression

算术编码

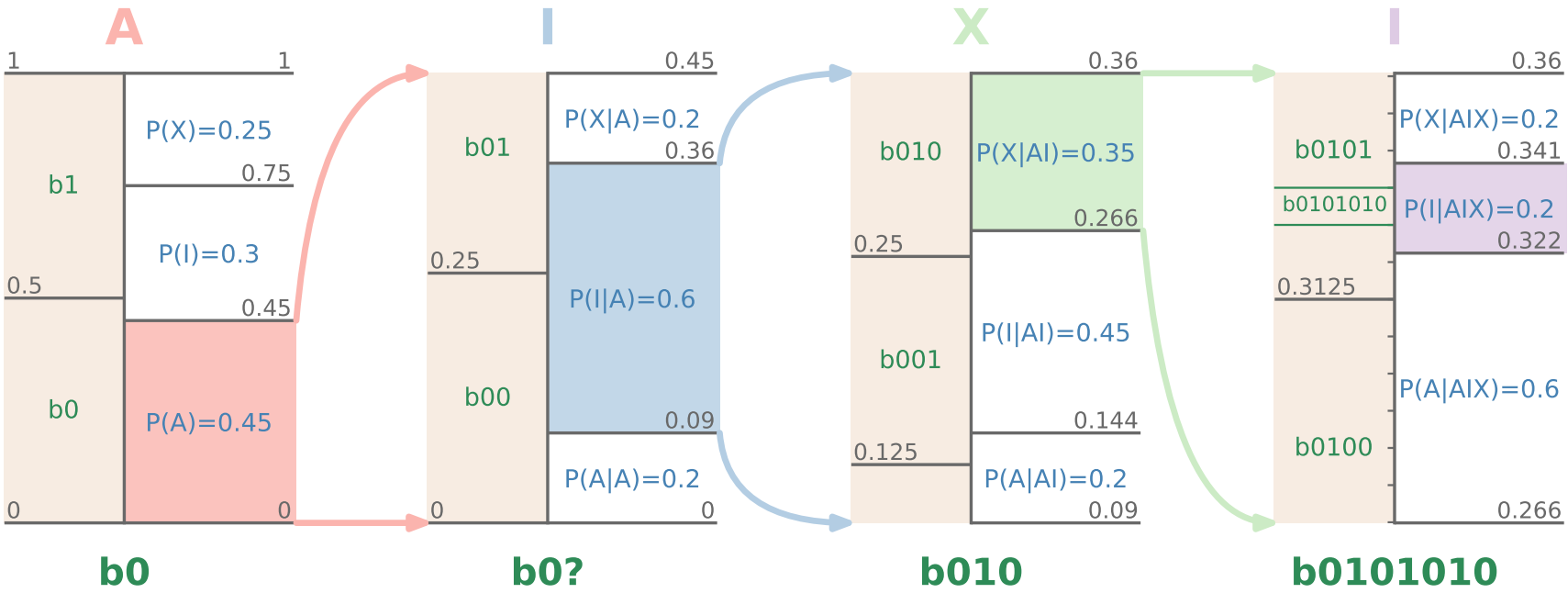
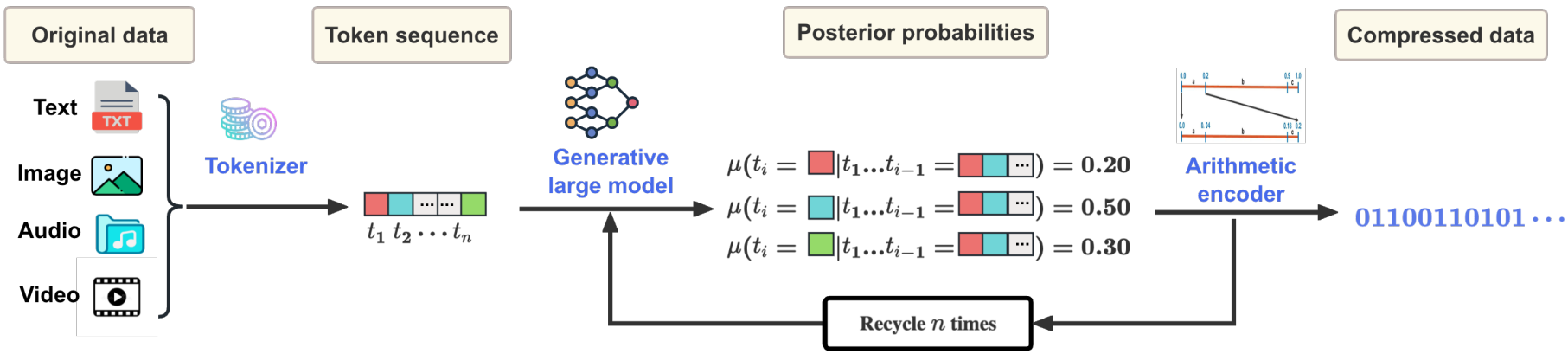


Figure: Arithmetic encoding of 'AIXI' with a probabilistic model P resulting in the binary code 'b0101010'. We iteratively divide the interval $I = [0, 1)$ according to P and select the sub-interval corresponding to the observed symbol. To determine the encoded output, we iteratively split $[0, 1)$ in half, and assign a binary code to each sub-interval until it is fully contained in I.

- $\mathrm{Encode}(x_t, P_t) = z_t$ where z_t takes up $-\log P_t(x_t)$ bits.
- $\mathrm{Decode}(z_t, P_t) = x_t$

手扶拐杖的外星绅士造访地球. 临别, 人类赠送百科全书: "人类文明尽在其中!". 绅士谢绝: "不, 谢谢! 我只需在拐杖上点上一点".

Understanding is Compression



Remark: The compression ratio should go up with better approximation of Solomonoff induction and better understanding of data.

预测即压缩, “奥卡姆剃刀” 为啥锋利?

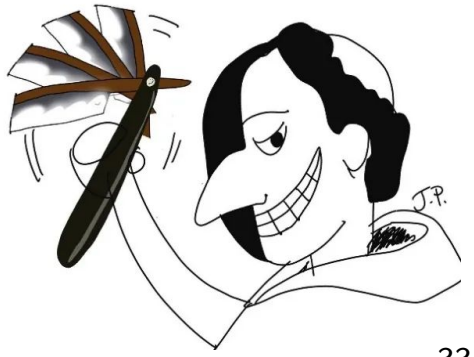
- 大语言模型在做 Next-Token 预测.
- 在真实分布 μ 未知的情况下, 什么样的预测最准确?
- 基于算法概率 M 的预测 $M \to \mu$! — LLM 奔赴算法概率 M!
- 什么是最终极的压缩? Kolmogorov 复杂性 $K(x)$!
- 算法概率能够较好的预测 $M(y \mid x)$, 意味着, 给定 x 之后, y 的发生有一个 “简单” 解释 $K(y \mid x)$.

$$2^{-K(x)} \overset{\times}{\leq} M(x) \overset{\times}{\leq} 2^{-K(x)+O(\log \ell(x))} \qquad M(x) \approx 2^{-K(x)}$$

近似 $K(xy) \approx K(x,y) = K(x) + K(y \mid x^*)$ 和 $K(y \mid x^*) \approx K(y \mid x)$, 则

$$M(y \mid x) = \frac{M(xy)}{M(x)} \approx 2^{K(x)-K(xy)} \approx 2^{-K(y|x)}$$

感知经验 → 世界模型 vs 模型森林 → 未来预测



算术编码 vs 奥卡姆剃刀 vs 极小描述长度原则

- 已知真实的概率分布 μ, x 的算术码长 $\ell(\mathsf{code}(x)) = \lceil -\log \mu(x) \rceil + 1$, 平均码长不会超过 $H(\mu) + 2$.
- 如果真实的分布 μ 未知, 如果我们用 ρ 进行算术编码, 此时 x 的码长为

$$-\log \rho(x) + K(\rho)$$

- 极小描述长度原则 **MDL**:

$$\rho^{\mathrm{MDL}} := \operatorname*{argmin}_{\rho \in \mathcal{M}} \left[-\log \rho(x) + K(\rho) \right]$$

压缩即泛化

- 从极小描述长度原则 **MDL** 的视角看:

$$K(x_{1:n}) \leftarrow \min_{\rho \in \mathcal{M}} \left(K(\rho) + \sum_{t=1}^{n} -\log \rho(x_t \mid x_{<t}) \right)$$

如果在含噪声数据集上还要求预测损失很小很小,

$$\sum_{t=1}^{n} -\log \rho(x_t \mid x_{<t}) = 0$$

那么 $K(\rho)$ 会很大, 从而严重过拟合, 泛化性差. (其中 ρ 是图灵机程序, 而非网络参数)

- 学习的目标不是对训练集的重建 (记忆), 而是通过压缩实现对训练集以外的真实世界信息进行最大程度的泛化. 压缩即泛化.
- 神经网络越大越好?
  - 参数量越大的网络, 程序空间 $\mathcal{M}$ 越大, 随机梯度下降越可能跨越局部极值, 在程序空间中搜索到准确且压缩率高的程序.
  - 更大的网络在更大的数据集上能够 Grokking 出复杂度更低的程序.
  - 没有足够规模的数据, 网络再大也没用.

从压缩的视角看 “因果学习”[JS10]

- “算法马尔科夫条件”[9]:

$$K(x_1,\ldots,x_n) \stackrel{+}{=} \sum_{i=1}^{n} K(x_i \mid \mathrm{pa}_i^*)$$

- 但由于对称性 $K(x)+K(y \mid x^*) \stackrel{+}{=} K(y)+K(x \mid y^*)$, 根据 “算法马尔科夫条件” 只能学到马尔科夫等价类.
- 为了区分马尔科夫等价类, 我们需要 “算法独立因果机制”:

$$K(P_{X_1,\ldots,X_n}) \stackrel{+}{=} \sum_{i=1}^{n} K(P_{X_i|\mathrm{Pa}_i})$$

- 如果机制 P_C 和 $P_{E|C}$ 算法独立 $I(P_C;P_{E|C}) \stackrel{+}{=} 0$, 那么

$$K(P_{C,E}) \stackrel{+}{=} K(P_C)+K(P_{E|C}) \stackrel{+}{\leq} K(P_E)+K(P_{C|E})$$

[9] **Remark:** 从压缩的视角看无监督学习:

$$K(x,y) \stackrel{+}{=} K(x)+K(y \mid x^*)$$

直接学习 $K(y \mid x)$ 不现实; 但联合压缩 xy, 则近似得到 $K(y \mid x^*)$.

最优的 “prompt”

Problem: 什么样的 Prompt x^* 是诱导出任务 μ 的答案的 “最优 Prompt”?

$$x^* := \operatorname*{argmin}_x \mathbb{E}_\mu \big[-\log M(- \mid x) \big]$$

或类似的,

$$x^* := \operatorname*{argmin}_x D_{\mathsf{KL}}(\mu(-) \| M(- \mid x))$$

Remark: 当任务 μ 是确定性的时 $\mu(y) = 1$, 最优的 prompt 近似诱导出 $K(y \mid x^*)$, 这也解释了为什么 LLM 可以通过压缩使得以 x^* 作为 Prompt 可以有效地诱导出我们想要的答案 y.

Monotone Kolmogorov Complexity

Definition (Monotone Kolmogorov Complexity)

$$Km(x) := \min_p \{\ell(p) : U(p) = x*\}$$

where U is a universal monotone Turing machine.

- $Km(x) \stackrel{+}{\leq} \ell(x)$
- $Km(xy) \geq Km(x)$
- $Km(x) \stackrel{+}{\leq} -\log \mu(x) + K(\mu)$ if μ is a computable measure

It is natural to call an infinite sequence ω computable if $Km(\omega) < \infty$.

Algorithmic Coding Theorem

1. There is a universal lower semicomputable discrete semimeasure.
   For example, $m(x) := \sum_{P \in \mathcal{M}} 2^{-K(P)} P(x) \stackrel{\times}{=} P_U(x)$.

$$P_U(x) := \sum_{p:U(p)\downarrow=x} 2^{-\ell(p)} \quad \text{where } U \text{ is a universal prefix TM}$$

2. There is a universal lower semicomputable continuous semimeasure.
   For example, $\xi(x) := \sum_{\nu \in \mathcal{M}} 2^{-K(\nu)} \nu(x) \stackrel{\times}{=} M(x)$.

$$M(x) := \sum_{p:U(p)=x*} 2^{-\ell(p)} \quad \text{where } U \text{ is a universal monotone TM}$$

Theorem (Algorithmic Coding Theorem)

$$K(x) \stackrel{+}{=} -\log m(x) \stackrel{+}{=} -\log P_U(x)$$

If a string has many long descriptions then it also has a short description.

Continuous Analogue of the Algorithmic Coding Theorem

Theorem (Kolmogorov-Solomonoff sandwich)

$$0 \leq K(x \mid \ell(x)) \stackrel{+}{\leq} KM(x) := -\log M(x) \leq Km(x) \leq K(x) \stackrel{+}{\leq} \ell(x) + 2\log \ell(x)$$

Theorem

$$Km(x) \leq -\log \mu(x) + c_\mu$$

for any computable measure μ.

Remark: $KM(x) \leq -\log \mu(x) + K(\mu)$

Theorem

There is a constant $c \in (0,1)$ such that there exist infinitely many finite binary strings x satisfying

$$Km(x) - KM(x) > c \log\log \ell(x)$$

$$\boxed{KM(x) \leq Km(x) \stackrel{+}{\leq} KM(x) + K(KM(x))}$$

(Semi)Measure

Definition ((Semi)Measure)

We call $\rho : \mathcal{X}^* \to [0,1]$ a semimeasure if $\rho(\epsilon) \leq 1$ and $\rho(x) \geq \sum_{a \in \mathcal{X}} \rho(xa)$, and a probability measure if equality holds.

$$\rho(x_t \mid x_{<t}) := \frac{\rho(x_{1:t})}{\rho(x_{<t})}$$

$$\rho(x_1 \ldots x_n) = \rho(x_1)\rho(x_2 \mid x_1) \ldots \rho(x_n \mid x_1 \ldots x_{n-1})$$

ρ is an lower semicomputable semimeasure iff there is a monotone Turing machine T s.t.

$$\rho(x) = \sum_{p : T(p) = x*} 2^{-\ell(p)} \quad \text{and} \quad \ell(\langle T \rangle) \stackrel{+}{=} K(\rho)$$

where $T(p) = U(\langle T \rangle p)$.

Simple Deterministic Bound

Sequence prediction algorithms try to predict the continuation x_t of a given sequence $x_1 \dots x_{t-1}$.

Theorem

$$\sum_{t=1}^{\infty} \left|1 - M(x_t \mid x_{<t})\right| \leq Km(x_{1:\infty}) \ln 2$$

Proof.

$$\sum_{t=1}^{\infty} \left|1 - M(x_t \mid x_{<t})\right| \leq -\sum_{t=1}^{\infty} \ln M(x_t \mid x_{<t}) = -\ln M(x_{1:\infty}) \leq Km(x_{1:\infty}) \ln 2$$

Solomonoff's Completeness Theorem

$$M'(\epsilon) := 1$$

$$M'(x_{1:t}) := M'(x_{<t}) \frac{M(x_{1:t})}{\sum_{a \in \mathcal{X}} M(x_{<t}a)} = \frac{M(x_{1:t})}{M(\epsilon)} \prod_{i=1}^{t} \frac{M(x_{<i})}{\sum_{a \in \mathcal{X}} M(x_{<i}a)}$$

Theorem (Solomonoff's Completeness Theorem)

For any computable measure μ,

$$\sum_{t=1}^{\infty} \sum_{x_{1:t} \in \mathcal{X}^t} \mu(x_{<t}) \Big(M'(x_t \mid x_{<t}) - \mu(x_t \mid x_{<t}) \Big)^2 \leq D(\mu \| M) \stackrel{+}{\leq} K(\mu) \ln 2$$

Remark: M' is universal predictor. The only assumption made is that data are generated from a computable distribution.

Prediction Bounds

Theorem (Total Bounds)

$$\sup_{A \subseteq \mathcal{X}^\infty} \left| M(A \mid x_{<t}) - \mu(A \mid x_{<t}) \right| \xrightarrow[w.\mu.1]{t \to \infty} 0$$

Theorem (Instantaneous Bounds)

$$2^{-K(n)} \;\leq\; \left[1 - M(x_n \mid x_{<n})\right] \;\leq\; 2^{2Km(x_{1:n}) - K(n)}$$

e.g. $M(0 \mid 1^n) \overset{\times}{=} 2^{-K(n)} \to 0$

Theorem (Future Bounds)

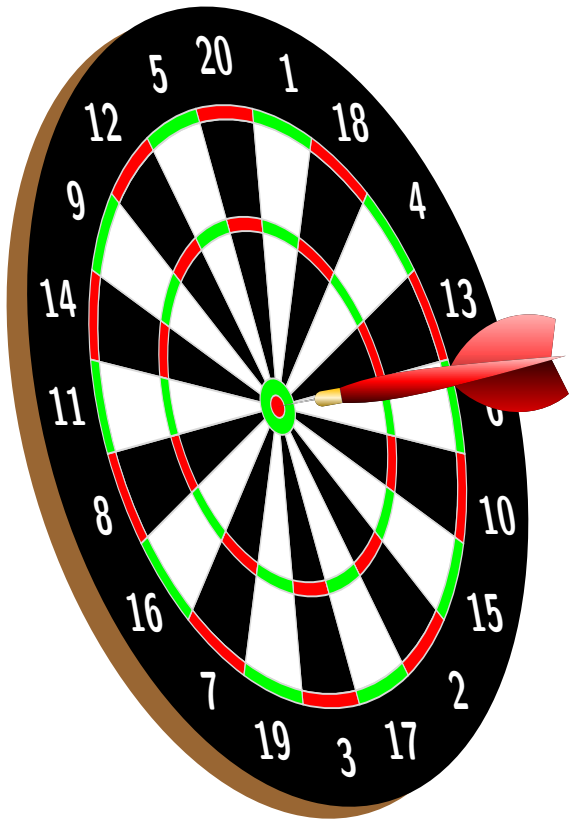
$$\sum_{t=n+1}^{\infty} \mathbb{E}_\mu \left[\sum_{a \in \mathcal{X}} \left(\sqrt{\xi(a \mid x_{1:t})} - \sqrt{\mu(a \mid x_{1:t})} \right)^2 \middle| x_{1:n} \right] \;\overset{+}{\leq}\; \left[K(\mu \mid x_{1:n}) + K(n) \right] \ln 2$$

Universal is Better than Continuous Θ

- Although the environment ν_θ and the prior density $w(\theta)$ are incomputable for continuous classes Θ for most $\theta \in \Theta$, the Bayesian mixture

$$\xi(x) := \int_\Theta w(\theta)\nu_\theta(x)\mathrm{d}\theta \quad \text{where} \quad \int_\Theta w(\theta)\mathrm{d}\theta = 1$$

  is typically computable.
- M is superior to all computable mixture predictors ξ based on any (continuous or discrete) model class Θ and weight $w(\theta)$, save an additive constant $K(\xi)\ln 2 = O(1)$, even if environment μ is not computable.



Theorem (Universal is Better than Continuous Θ)

$$D_n(\mu\|M) = \mathbb{E}_\mu\left[\ln\tfrac{\mu}{M}\right] = \mathbb{E}_\mu\left[\ln\tfrac{\mu}{\xi}\right] + \mathbb{E}_\mu\left[\ln\tfrac{\xi}{M}\right] \overset{+}{\leq} D_n(\mu\|\xi) + K(\xi)\ln 2$$

- For continuous $\mathcal{M}$, we can assign a universal prior (not density)

$$w_\theta^U := \begin{cases} 2^{-K(\theta)} & \text{if } \theta \text{ is computable} \\ 0 & \text{otherwise} \end{cases}$$

- This effectively reduces $\mathcal{M}$ to a discrete class $\{\nu_\theta \in \mathcal{M} : w_\theta^U > 0\}$ which is typically dense in $\mathcal{M}$.

Discussion

- 归纳可以还原为预测吗?
  — 休谟认为, 因为归纳在过去有效所以有望在未来有效本身就是一种归纳, 无法证明归纳的有效性.
  — 归纳的作用不是告诉我们哪种预测是正确的, 而是告诉我们现有的知识更支持哪种预测.
  — 科学发现很大程度上是关注归纳导致了错误预测的情况的结果.
- 科学是为了预测还是为了解释?
- 预测本身就是目标? 亦或重要的是理论的解释力, 而预测仅仅是理论可证伪性的需要?
  — 一个能精准预测未来, 但无法解释 '为什么' 的 AI, 算不算真正的智能?
- 有了通用归纳, 解释还有多远?

为什么世界中存在有序的结构?

Solomonoff 完备性定理 [Sol78]

对于任意可计算的测度 μ, 有 $\mu(A) = 1$ 的集合 $A \subset \mathcal{X}^*$, 使得 $\forall \mathbf{x} \in A$,

$$\sum_{y \in \mathcal{X}} \left(\sqrt{M'(y \mid \mathbf{x})} - \sqrt{\mu(y \mid \mathbf{x})} \right)^2 \xrightarrow{n \to \infty} 0$$

逆 Solomonoff 定理 — 为什么会涌现出简单的规律? [Mül20]

对于任意可计算的测度 μ,

$$M' \left\{ \sum_{y \in \mathcal{X}} \left(\sqrt{M'(y \mid \mathbf{x})} - \sqrt{\mu(y \mid \mathbf{x})} \right)^2 \xrightarrow{n \to \infty} 0 \right\} \geq 2^{-K(\mu)}$$

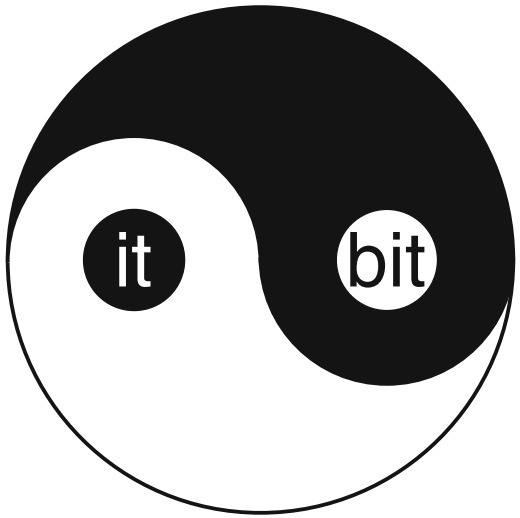
假设观察 (self-state) 是基本的. $M(o_{\text{future}} \mid o_{\text{past}})$

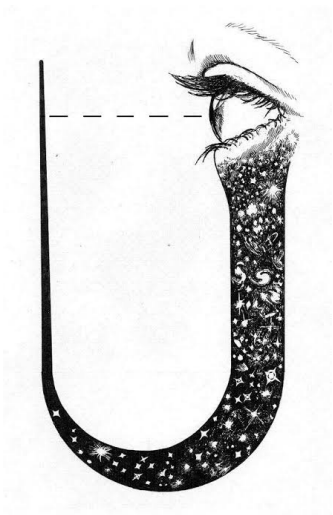
为什么我们生存其中的世界是有序的? $M \to \mu$

Remark: 与观察一致的世界越有序, 它被 "算法概率" 逼近的概率越高!

世界 (的规律/简单性) 从何而来? — "it from bit"

康德的 "哥白尼革命": 人为自然立法!





1. 世界过去是什么样子的?
   — 贝尔不等式: 除非放弃局域性或自由意志, 否则, 假设测量只是揭示了 "世界上预先存在的未知事实"(实在性) 是有问题的.
   — 而如果放弃实在性, 我们最需要关心的就是问题 2 而不是问题 3.
2. 我应该期待经验到什么?

   假设观察 (self-state) 是基本的. $M(o_{\text{future}} \mid o_{\text{past}})$
3. 世界是什么样子的?

   为什么我们生存其中的世界是有序的? $M \to \mu$

我应该期待经验到什么?

双缝实验

- 错的问法: 量子态告诉我们粒子实际穿过哪条缝.
- 对的问法: 量子态告诉我们该期待在屏幕何处观察到粒子.
- 给定实验设定 (量子态 + 测量算符), 玻恩规则给出各结果的概率.

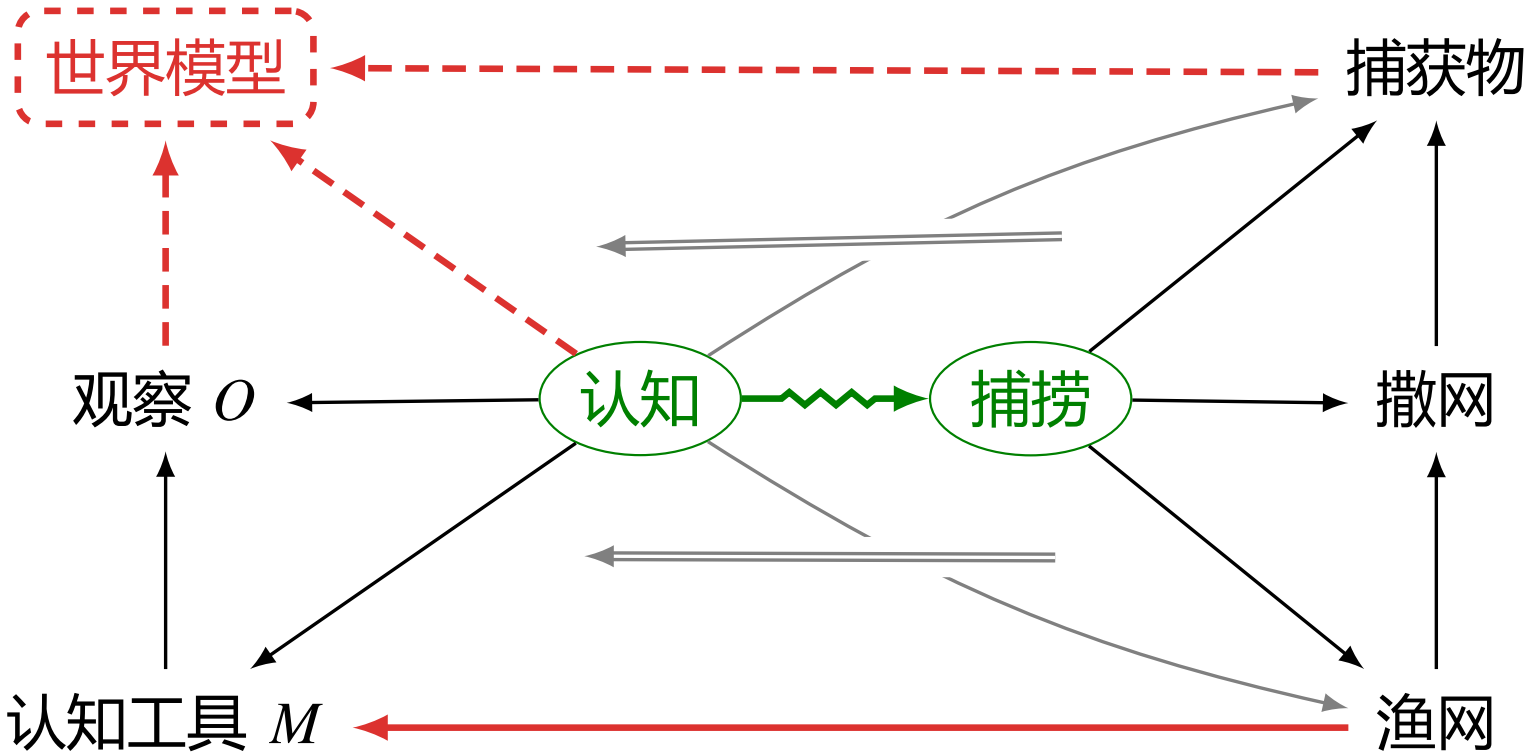
量子理论是关于「我们应该相信接下来会观察到什么」的理论.

有了理论上最好的 “渔网” 之后, “鱼” 是什么样子?

爱丁顿: 想象一位鱼类专家想探究海洋中的生命. 他舒臂撒网, 捕获了一堆海洋生物. 他检查了自己的捕获物, ...并由此作出了两项概括:

1. 凡海洋生物皆长于 5 厘米.
2. 凡海洋生物皆有鳃.....

捕获物相当于物理学, 网相当于思维装置和感官工具, 撒网意味着观察.



“鱼” 是 “渔网” 网上来的样子! 只要 “渔网” 够好, 就不需要不可知的 “物自体”.

“类比学习”是一种近似的“溯因归纳”[Mur18]

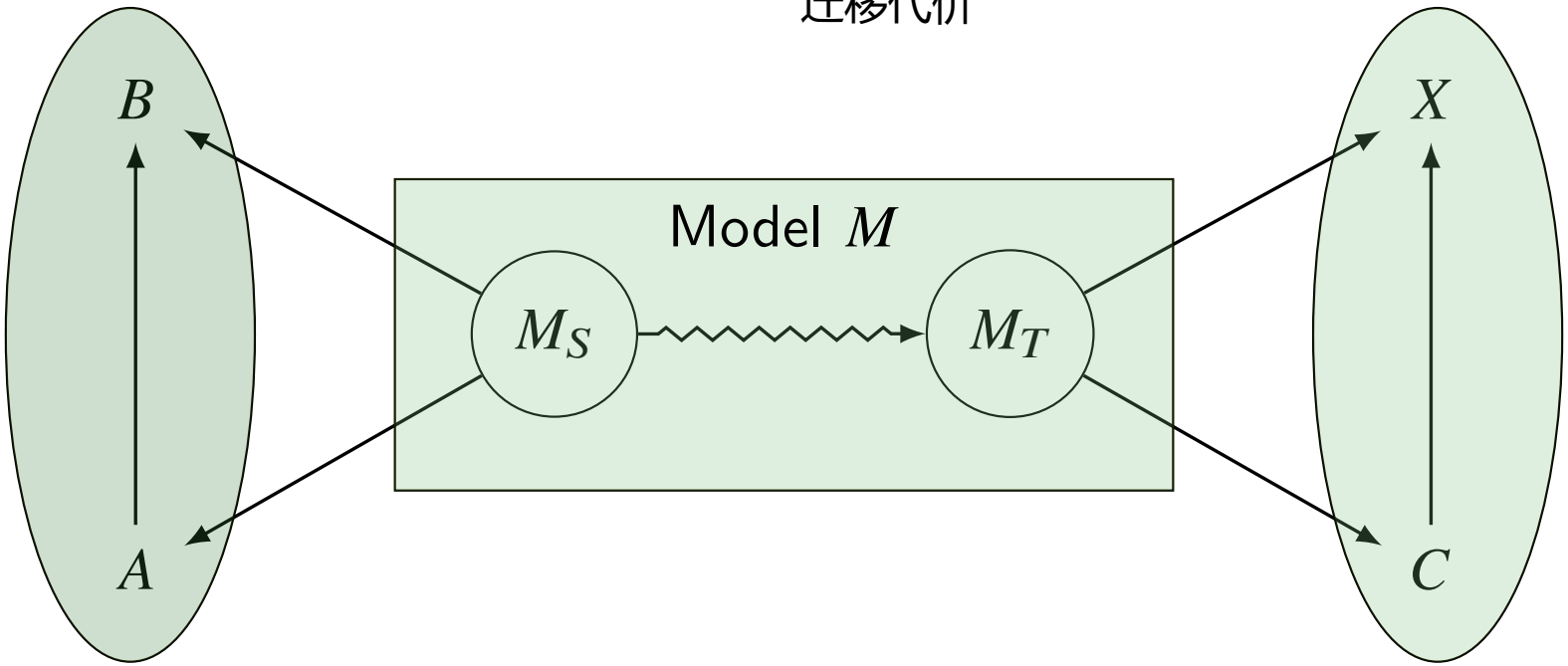
$$A : B :: C : X$$

腰肢 : 款摆 :: 弱柳 : ____ ← 扶风

$$X^* := \operatorname*{argmin}_X K(ABCX)$$

$$\operatorname*{argmin}_{M \in \mathcal{H}} \{K(M) + K(D \mid M)\} \qquad \textbf{(MDL)}$$

$$K(M_S)+K(A \mid M_S)+K(B \mid M_S, A)+\underbrace{K(M_T \mid M_S)}_{\text{迁移代价}}+K(C \mid M_T)+K(X \mid M_T, C)$$



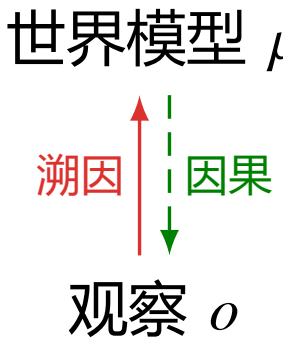
谁是本源? — 给马克思一个交代

因果箭头通常跟溯因方向相反

借助 “简单性” 推断 “因果箭头” 的方向

如果 μ 是 o 的原因, 那么

$$K(\mu) + K(o \mid \mu) \overset{+}{\leq} K(o) + K(\mu \mid o)$$



$$\mu \to o \quad \text{or} \quad \mu \leftarrow o$$

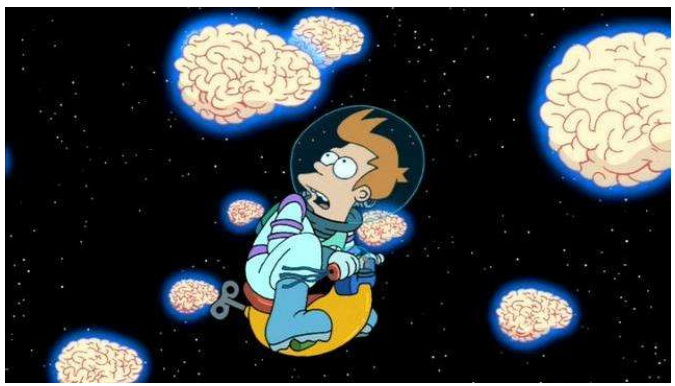
Remark: 观察是意识中获得的关于系统的印象? 是观察者与系统之间相互作用得到的结果.

古德曼新归纳之谜 & 玻尔兹曼大脑

$$\mathsf{Grue}(x) \iff (t < 2050 \to \mathsf{Green}(x,t)) \wedge (t \geq 2050 \to \mathsf{Blue}(x,t))$$

玻尔兹曼大脑 / AI 复制人:

- 在所有可能世界中, 包括有序和无序, 玻尔兹曼大脑的数量 N_{BB} 远大于自然进化大脑的数量 N_{nat}.
- 所以你会更相信你是一个随时会消失的玻尔兹曼大脑吗?



$$K(\mu_{\mathrm{nat}}) \ll K(\nu_{\mathrm{BB}})$$

$$M(y_{\mathrm{Earth}} \mid x) \gg M(y_{\mathrm{BB}} \mid x)$$

罗素: 世界可能是五分钟之前创造的.

$$\mathsf{Bleen}(x) \iff (t < 2050 \to \mathsf{Blue}(x,t)) \wedge (t \geq 2050 \to \mathsf{Green}(x,t))$$

虾是青红的. 煮前是青的, 煮后是红的. 生熟 ⟵ 煮 ⟶ 青红

$$\texttt{Color} \perp \texttt{Time} \implies K(\mathsf{Green}) < K(\mathsf{Grue})$$

古德曼新归纳之谜小结

- 古德曼新归纳之谜是溯因问题.
- Grue/Bleen vs Green/Blue 揭示了两个问题:

1. 通用归纳对通用图灵机 U 的弱依赖性.
2. 简单性与因果的互动.
   - 科尔莫哥洛夫复杂性的大小依赖因果结构:

$$\texttt{Color} \perp \texttt{Time} \implies K(\text{Green}) < K(\text{Grue})$$

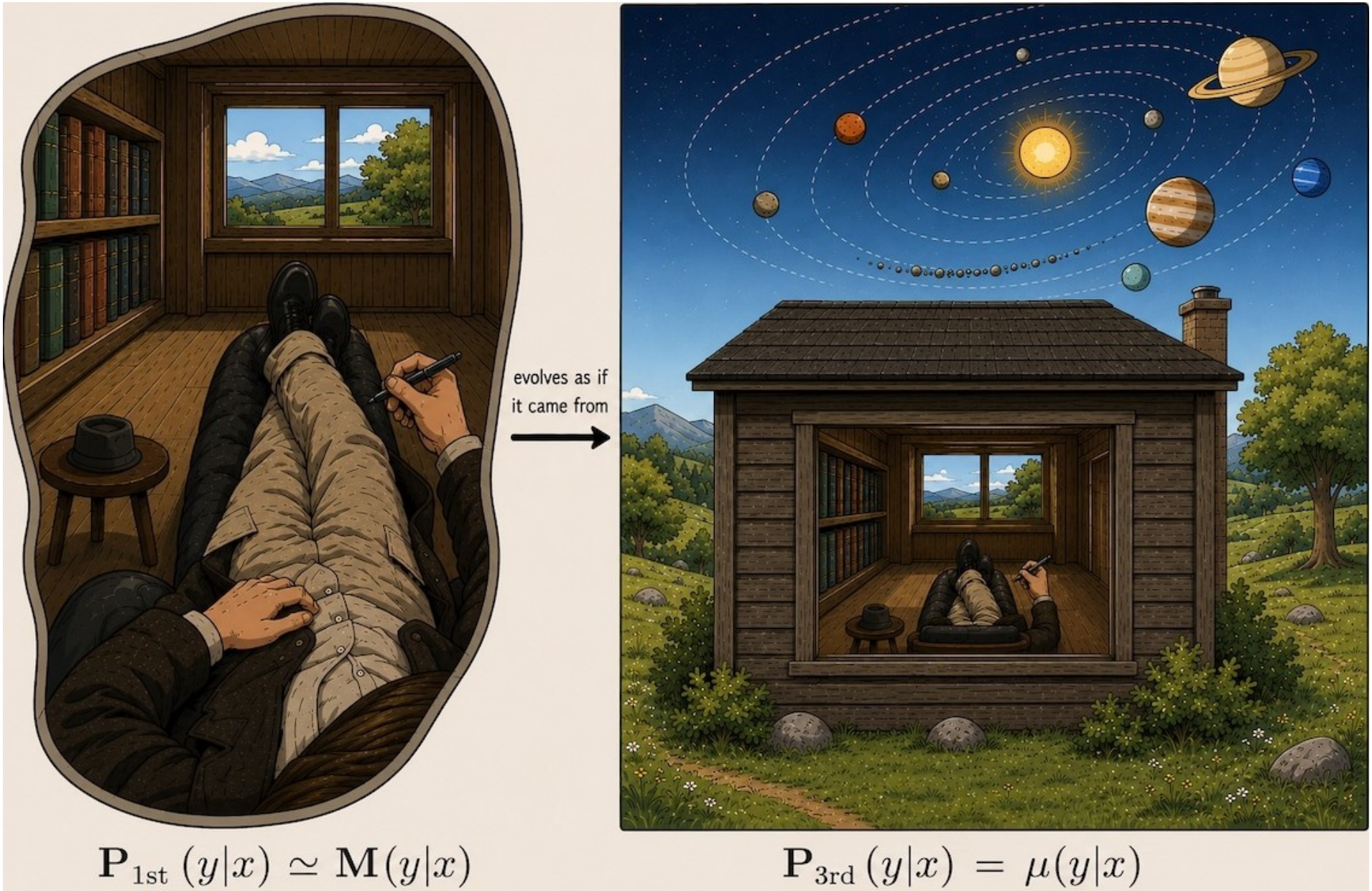
— 为什么 "青红虾" 比 "Grue 翡翠" 靠谱一点儿? 想象一下: 如果 2050 有因果机制作用于翡翠的颜色……
— **简单性偏置的背后是因果独立性偏置.** (算法独立因果机制)

   - 复杂性可以用于判定因果方向:

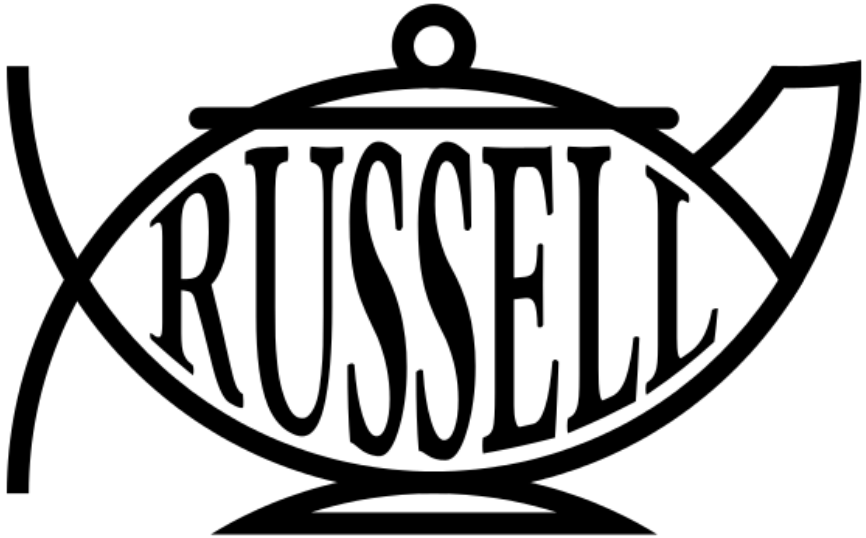
$$\boxed{C \to E} \quad \text{or} \quad \boxed{C \leftarrow E}$$

$$C \to E \implies K(P_C) + K(P_{E|C}) \overset{+}{\leq} K(P_E) + K(P_{C|E})$$

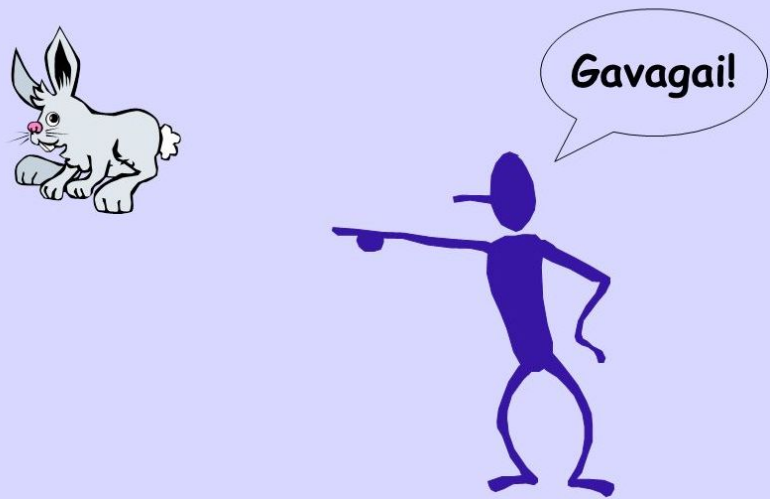
问题: 简单性与因果的互动, 如何 "固化"(entrenchment) 到演化的语言 $U \to U'$ 中? 偏置的另一个来源 — 价值 — 如何参与到互动中?

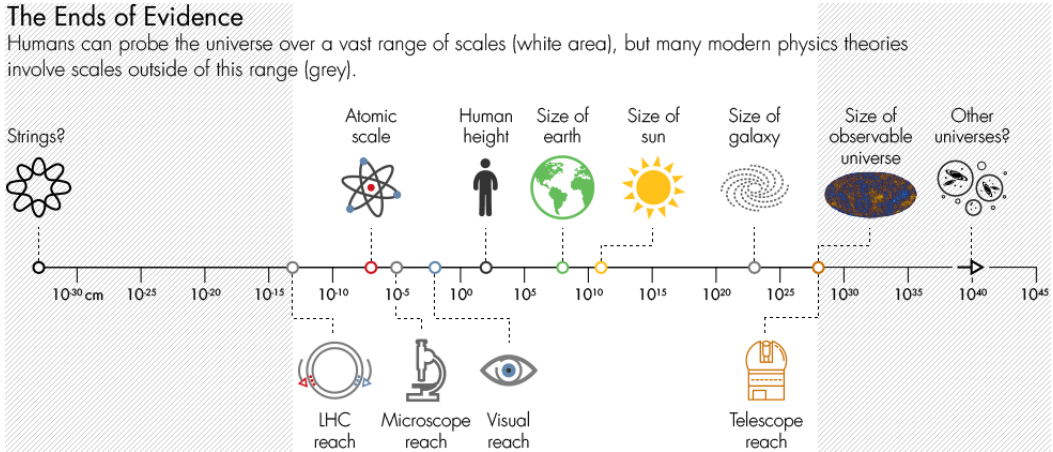


$$M'\left\{\sum_{y\in\mathcal{X}}\left(\sqrt{M'(y \mid \mathbf{x})}-\sqrt{\mu(y \mid \mathbf{x})}\right)^2 \xrightarrow{n\to\infty} 0\right\} \geq 2^{-K(\mu)}$$



What does “gavagai” mean?





Universal Prediction of Selected Bits

Theorem (Universal Prediction of Selected Bits)

Let $f : \{0,1\}^* \to \{0,1,\epsilon\}$ *be a total recursive function and* $x \in 2^\omega$ *satisfying* $f(x_{<n}) = x_n$ *whenever* $f(x_{<n}) \neq \epsilon$. *If* $f(x_{<n_i}) \neq \epsilon$ *for an infinite sequence* $n_1, n_2, \ldots$ *then*

$$\lim_{i\to\infty} M'(x_{n_i} \mid x_{<n_i}) = 1$$

The following can be predicted by M'.

- Computable subsequences of arbitrary noise
- B0F0Z0N0X0D0E1J0P0R0U1A1S0L0I1T0H0E1U 辅音元音

Pure Universal Inductive Logic?

- $M(\Theta(a_{1:n})) := \sum_{p:U(p)=h_{1:n}*} 2^{-\ell(p)}$

  where $\Theta(a_{1:n}) := \bigwedge_{i=1}^{n} Q_{h_i}(a_i)$.

- $M'(A(\vec{a})) := \sum_{\Theta(\vec{b})\vDash A(\vec{a})} M'(\Theta(\vec{b}))$

  where $\vDash A(\vec{a}) \leftrightarrow \bigvee_{\Theta(\vec{b})\vDash A(\vec{a})} \Theta(\vec{b})$. FDNF

$$\sum_{t=1}^{\infty} \sum_{A(a_{1:t})} \mu(A(a_{<t})) \left(M'(A(a_t) \mid A(a_{<t})) - \mu(A(a_t) \mid A(a_{<t})) \right)^2 \stackrel{+}{\leq} K(\mu) \ln 2$$

where $A(a_{1:t}) := \bigwedge_{i=1}^{t} A(a_i/x)$.

All Ravens are Black! ✓

Theorem (All Ravens are Black)

$$\lim_{n\to\infty} M'\left(\forall x(R(x) \to B(x)) \,\middle|\, \bigwedge_{i=1}^{n} (\neg R(a_i) \vee B(a_i))\right) = 1$$

Theorem (Confirmation by Random Sampling)

If the sampling function $t : \mathbb{N} \to \mathbb{N}$ *satisfies* $\forall i : t_i \leq t_{i+1}$ *and* $\chi_{1:\infty}$ *is Martin-Löf random, where* $\chi_i := [\![\exists k(t_k = i)]\!]$, *then*

$$M'\left(\forall x A(x) \,\middle|\, \bigwedge_{i=1}^{n} A(a_{t_i})\right) \xrightarrow{n\to\infty} 1$$

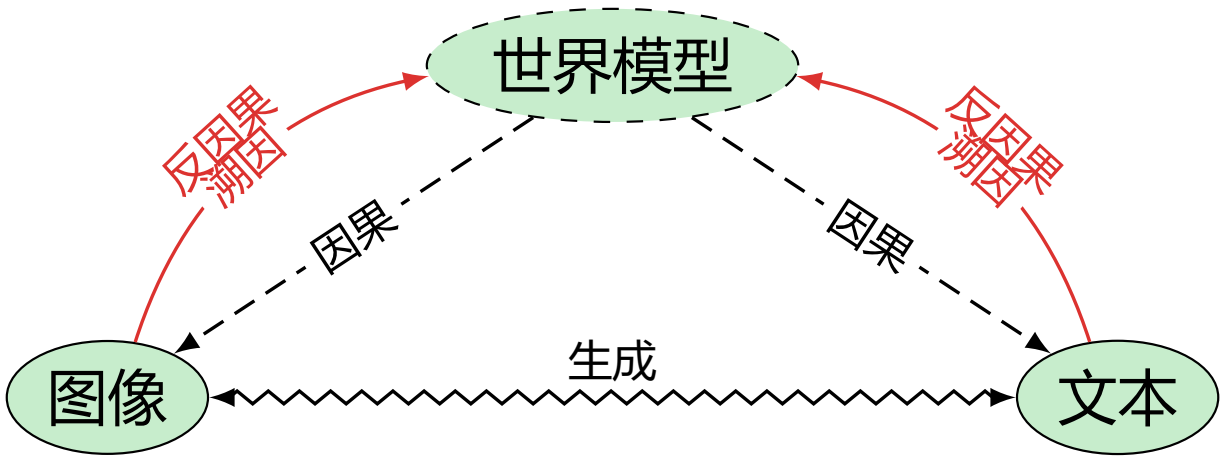
$$M(1 \mid 1^n) \xrightarrow{n\to\infty} 1 \qquad M(0 \mid 1^n) \stackrel{\times}{=} 2^{-K(n)} \qquad \sum_{n=0}^{\infty} M(0 \mid 1^n) < \infty$$

预测即压缩; 预测越准确, 理解越深刻

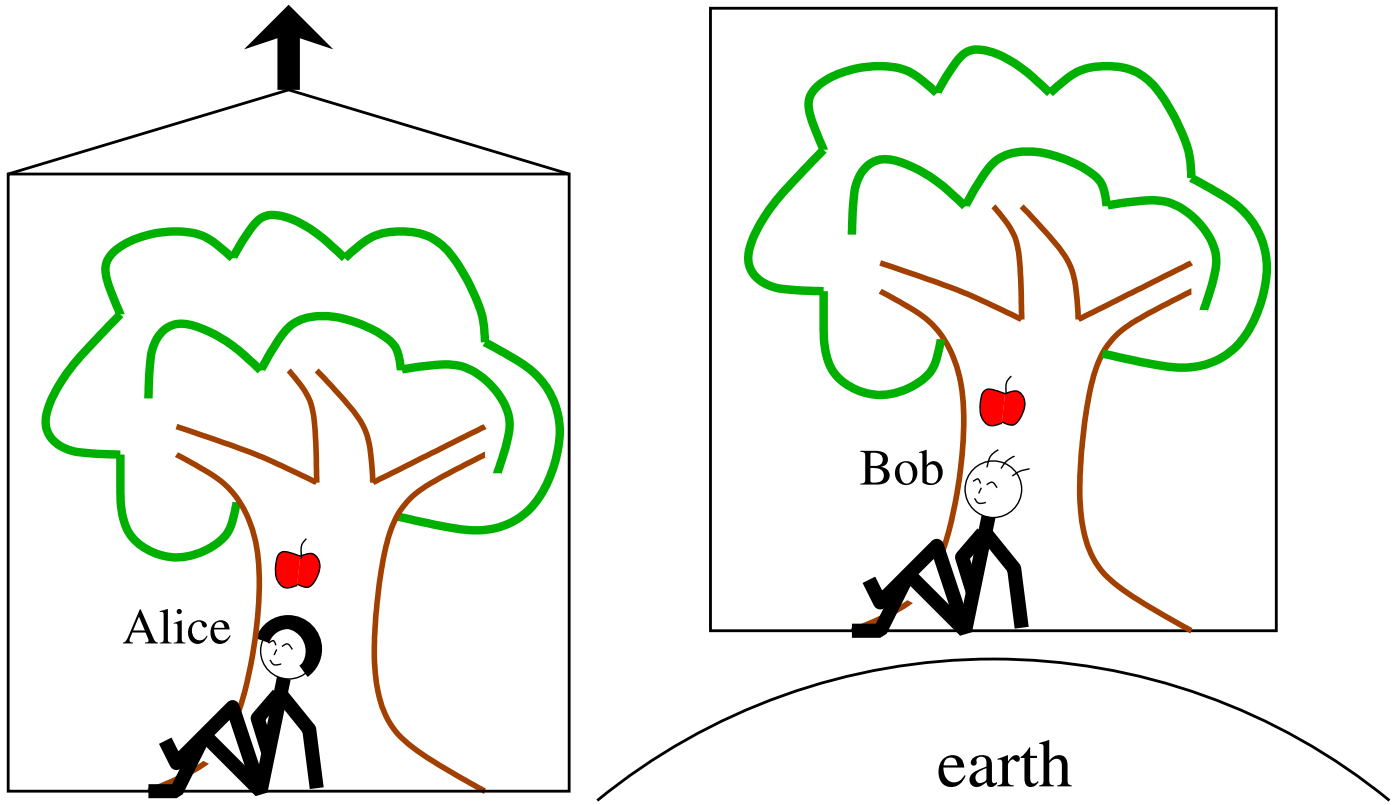
假设你正在阅读一本侦探小说, 这本书包含了错综复杂的情节、众多不同的角色以及许多令人费解的事件和线索. 在故事的最后一页, 侦探终于将所有线索收集齐全, 召集了所有相关人员, 然后宣布: "现在, 我将揭示真凶的身份, 那个人的名字是 *(...)*"

— *Ilya Sutskever*

- 我们观察到的图像、视频、文本等多模态都是世界模型的投影.
- 多模态可以帮助更好的溯因 μ, 但并不是说, 离了多模态, 只用语言就注定无法涌现出世界模型.
- 为了压缩, LLM 学习的是文本生成机制的某种表示.

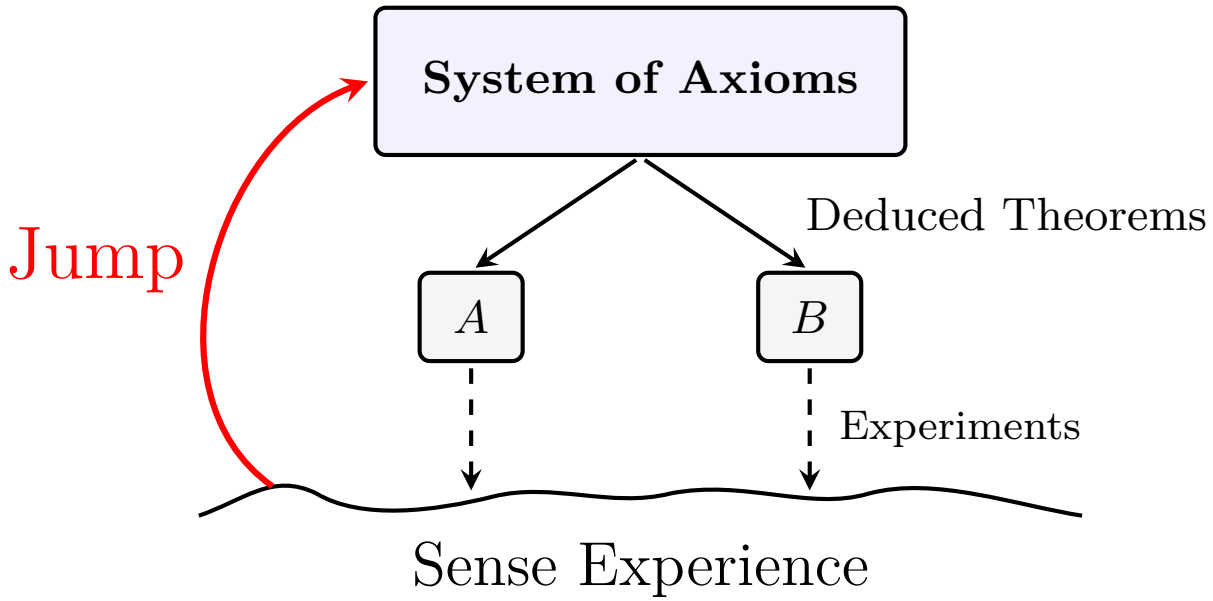


爱因斯坦的匀加速电梯思想实验



- 通过思想实验模拟的加速度感官体验与地球上的重力感官体验不可区分，爱因斯坦溯因出等效原理：引力场与惯性力场局域不可区分.

爱因斯坦 & 中文屋 — 从感官体验到公理的跳跃

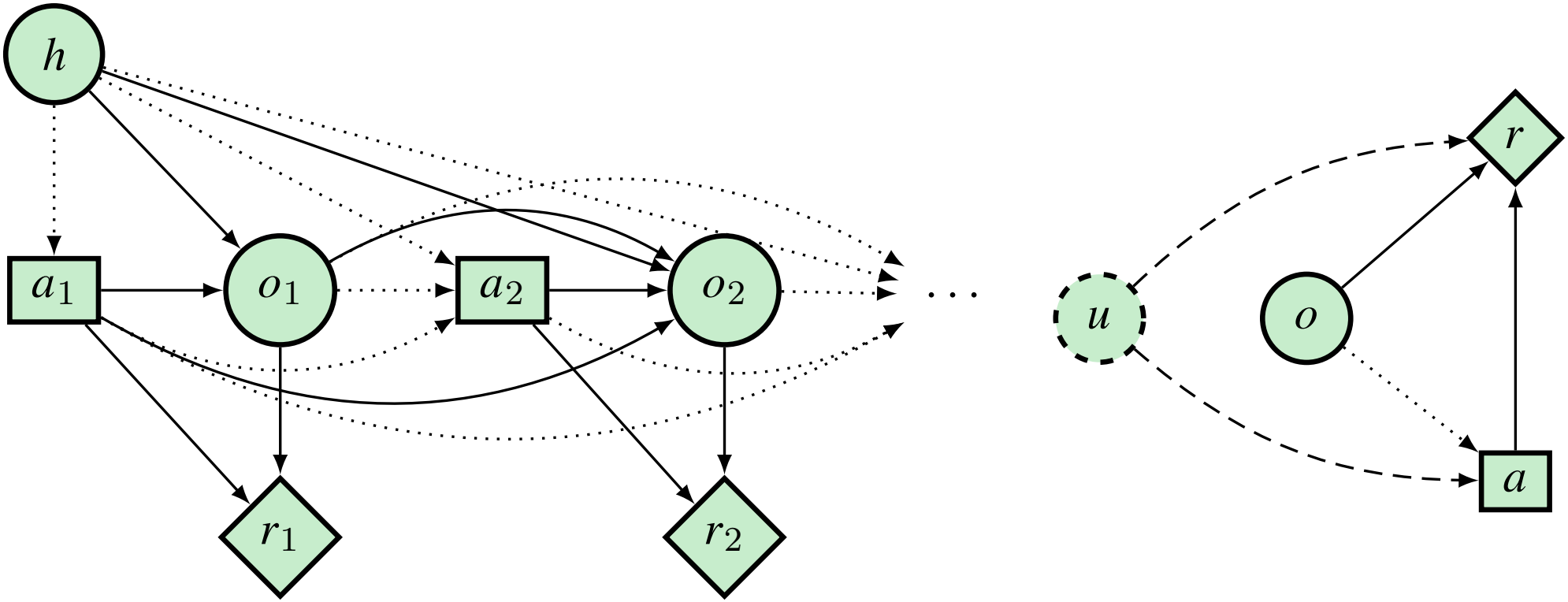


1. **Deduction:** $\texttt{Rule} + \texttt{Case} \implies \texttt{Result}$
2. **Induction:** $\texttt{Case} + \texttt{Result} \implies \texttt{Rule}$
3. **Abduction:** $\texttt{Rule} + \texttt{Result} \overset{\text{Jump}}{\implies} \texttt{Case}$

- LLM 因为缺乏**具身**世界模型, 无法对感官体验进行反事实模拟, 所以无法像爱因斯坦一样进行创造性溯因?
- 多模态可以帮助更好的溯因 $M(x_{\texttt{multimodal}}) \to \mu(x_{\texttt{multimodal}})$.
- 创造性溯因/归纳 vs 幻觉.
- Freedman: Compression is all you need: Modeling Mathematics?[Aks+26]

压缩即智能? 从预测到行动, 从 LLM 到 Agent

- 预测即压缩 ✓　压缩即智能?
- 智能 Agent 离不开感知行动, 但加了行动会涉及复杂的因果混杂.



- 在线强化学习是因果的, Agent 可以直接与环境交互, 因此不存在影响其行动和奖励的未观察到的混杂因子.
- 在离线强化学习中, 环境中可能存在未观察到的混杂因子, 会影响行动和奖励. $P(o_t r_t \mid h_{<t} a_t) \neq P(o_t r_t \mid h_{<t} \mathrm{do}(a_t))$.

组合优化问题的复杂性

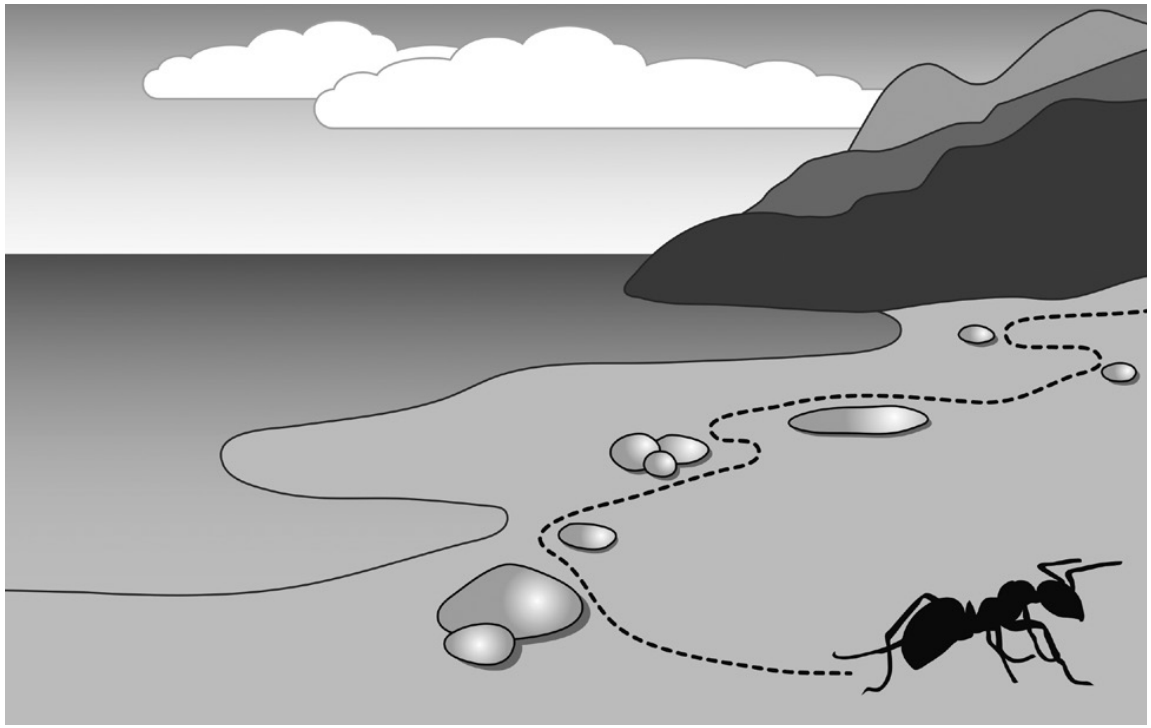
1. Enumerate all elements $x_i \in \mathcal{X}$.
2. For each $x_i \in \mathcal{X}$, evaluate $f(x_i)$ using the objective function f.
3. List x_i in descending order of optimality: $x_1^*, x_2^*, \ldots, x_{|\mathcal{X}|}^*$.
4. Set $x^* = x_1^*$.

Given the size n of the system, we can bound the complexity of the optimal solution of rank r by the complexity of the set $\mathcal{X}$ of configurations, the objective function f, and the rank r.

$$K(x_r^* \mid n) \stackrel{+}{\leq} K(\mathcal{X} \mid n) + K(f \mid n) + K(r \mid n)$$

最优解 \ 问题	简单	复杂
简单	简单源于简单	简单源于复杂
复杂	表观复杂源于简单	复杂源于复杂

西蒙的蚂蚁: 表观复杂 vs 简单



- An ant, viewed as a behaving system, is quite simple. The apparent complexity of its behavior is largely a reflection of the complexity of the environment.
- The mind is one blade in a pair of scissors, the structure of the environment is the other. To understand behaviour, one has to consider both — and, in particular, how they fit.

“不同的目标, 同一个解” 纯粹是巧合吗?

- 给定一个简单集合 $\mathcal{X}$, $K(\mathcal{X} \mid n) = O(1)$. 考虑两个目标函数 f_1, f_2, 其分别对应的最优解是 $x^*_{f_1}$ 和 $x^*_{f_2}$. 如果 f_1, f_2 是随机的两个函数, 那么

$$P(x^*_{f_1} = x^*_{f_2}) = \frac{1}{|\mathcal{X}|}$$

- 但如果 f_1, f_2 是两个简单函数, $K(f_1 \mid n) = K(f_2 \mid n) = O(1)$, 那么其最优解 $x^*_{f_1}, x^*_{f_2}$ 也是简单的.

$$K(x^*_{f_1} \mid n) = K(x^*_{f_2} \mid n) = O(1)$$

记 $\mathcal{X}_{O(1)} \subset \mathcal{X}$ 为所有复杂性为 $O(1)$ 的简单对象构成的子集, 那么

$$|\mathcal{X}_{O(1)}| \ll |\mathcal{X}| \implies P(x^*_{f_1} = x^*_{f_2}) = \frac{1}{|\mathcal{X}_{O(1)}|} \gg \frac{1}{|\mathcal{X}|}$$

- **Remark:** 若某个结构恰好满足不同的优化目标, 也不一定是小概率巧合.
- **Remark:** 如果复杂问题的解是简单的, 利用算法概率 $M(x) \approx 2^{-K(x)}$ 进行采样寻找问题的解.

问题: AI 可以自动从混沌中发现秩序吗?

1. 出自秩序的秩序

$$1^3 + 2^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2$$

2. 出自秩序的混沌

$$\pi = 3.14159265358979323384\ldots$$

3. 出自混沌的混沌

$$53278 \times 2147 = 114387866$$

4. 出自混沌的秩序

$$\lim_{n\to\infty} \frac{\pi(n)}{\frac{n}{\ln n}} = 1$$

李生素数定理, Ramsey 定理, 帕斯卡定理...AlphaFold...

Remark: 最后一种最值得珍视. 数学家和艺术家一样, 都是从混沌中创造秩序. 画家用线条, 作曲家用声音, 诗人用文字, 而数学家用理念.

1. 为什么宇宙中存在有序的结构?
2. 为什么这种结构可以维持生命?
3. 为什么维持生命的结构产生了能够理解这种结构的智慧生物?
4. 为什么智慧生物理解宇宙结构使用的是数学语言?

- 人择原理?
- 多重宇宙?
- 进化论?
- 数学宇宙假说?
- $M \to \mu$

人生如戏，没有剧本，讲更好的故事

- 什么是“理解”？— 压缩即理解.
  — 预测即压缩，预测越准确，理解越深刻.
- 什么是**“人生的意义”**？
  — 我们通过编织压缩率高的“模式”理解世界解释生活.
  — 探索意义真理，不过是给自己编故事.
- 这个故事是“真”的吗？
  — 是“共谋”
- 有更好更真的故事吗？
  — 可能有，可能没有.
- 找到最好的故事了吗？
  — 我们永远不知道！朝闻道，夕死可矣？ ✗
  — 柯尔莫哥洛夫复杂性不可计算
- 什么样的故事是好故事？
  1. 证实？证伪？假设简单，蕴含丰富？边界严格，拒绝马后炮解释？
  2. 有趣？有创意？
  3. “阴谋论”可以根除吗？

人类的 “阴谋论” vs 大语言模型的 “幻觉”

- 大多数序列都是算法随机的.

$$P\left(\left\{x \in \mathcal{X}^n : \frac{K(x)}{n} < 1 - \delta\right\}\right) < 2^{-\delta n}$$

- Ramsey: 完全的无序是不可能的!
- 无限猴子定理: 一只猴子在打印机上随机敲击键盘, 只要时间足够长, 就会打印出莎士比亚的《哈姆雷特》.
- Chaitin: 对应任意 Gödelian 理论 T,

$$\#\left\{x : \mathrm{T} \vdash K(x) > \ell(x)\right\} < \infty$$

- 对于几乎所有的随机序列, 它们的随机性无法被证明.
- 怎么区分: “真随机”、“真规律”、随机序列中的 “伪规律”?
  — 把 “伪规律” 当作 “真规律” 是一种 “阴谋论”, 永难根除
- “幻觉” 也是 LLM 基于压缩的泛化, 也涉及 “真规律”、“伪规律” 的区分, 也可能把 “伪规律” 当成了 “真规律”, 所以不可能完全根除.

Incomputability

- The Solomonoff prior M is not computable.
- There is no computable prior which assigns positive probability to all computable sequences.
- For any computable probability measure μ, there is some computable sequence $x \in 2^\omega$ s.t. $\mu(x_n \mid x_{<n}) \leq \frac{1}{2}$.

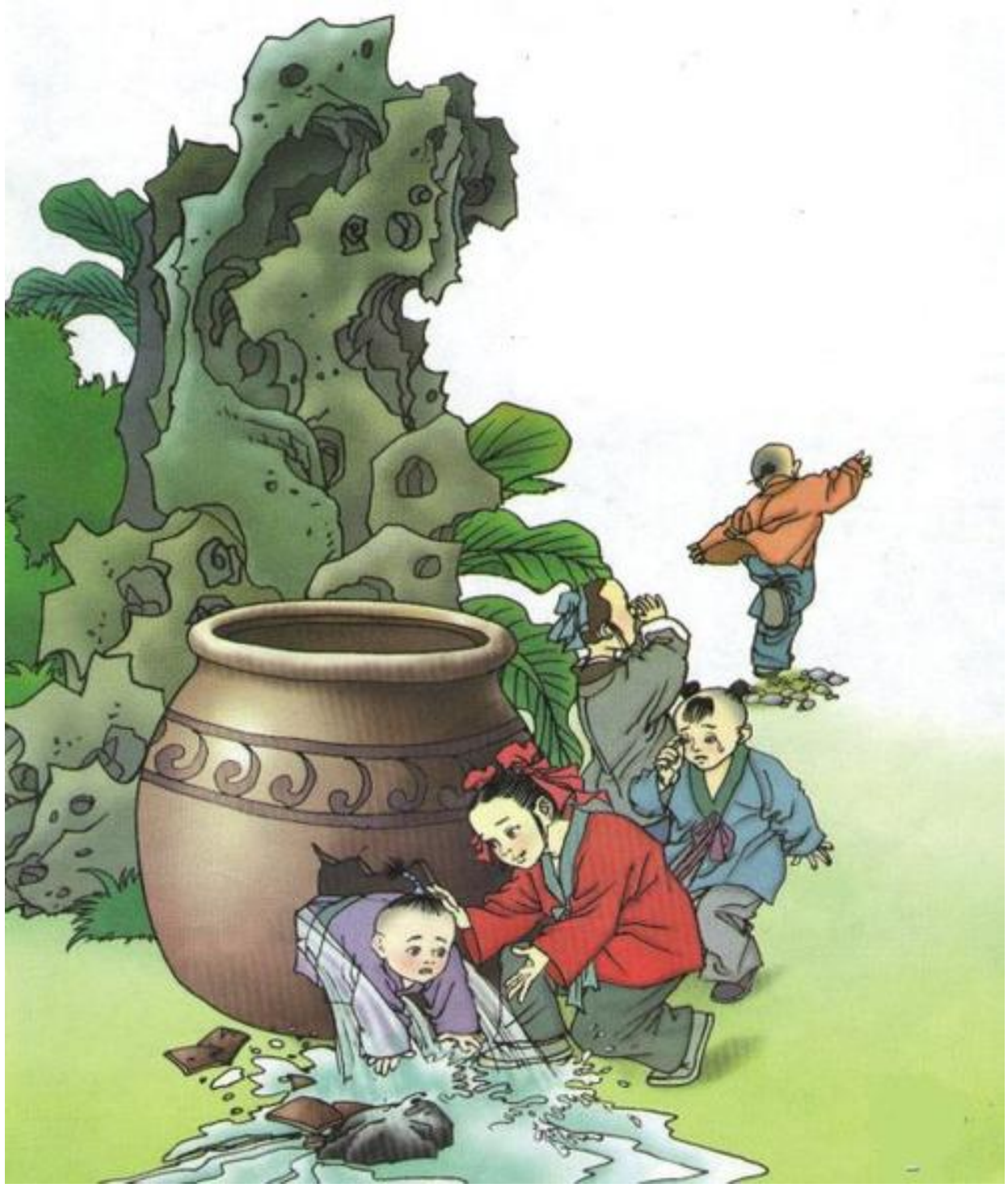
$$x_0 := 0$$

$$x_t := \begin{cases} 1 & \text{if } \mu(1 \mid x_{<t}) < \frac{1}{2} \\ 0 & \text{if } \mu(1 \mid x_{<t}) \geq \frac{1}{2} \end{cases}$$

Are Conceptual Frameworks Necessary for Theory Building?

1. 1,3,5,7,9,11,13,15,(?)
2. 0,1,0,1,0,1,0,1,0,(?)
3. 1,1,2,3,5,8,13,21,(?)
4. 1,4,1,5,9,2,6,5,3,(?)
5. 12,23,35,47,511,613,(?)
6. (7111,0), (8809,6), (2172,0), (6666,4), (1111,0), (2222,0), (7662,2), (9313,1), (0000,4), (8193,3), (8096,5), (4398,3), (9475,1), (0938,4), (3148,2), (2889,?)

- "Hedgehogs" use a single idea to view the world.
- "Foxes" view the world through multiple, sometimes conflicting, lenses.



Advantages & Disadvantages

- free-lunch
- universality — finite error
- data sparse problem — arbitrary order Markov chain — universal smoothing method
- confirmation of $\forall x : R(x) \rightarrow B(x)$
- incomputability
- subjectivity — weakly depends on universal Turing machine

Remark: 主观性是必要的, 它使得智能系统能够将过去的经验纳入到解决未来问题的技术中.

Deduction vs Induction

	Induction		**Deduction**
Type of inference	generalization/prediction	$\Leftrightarrow$	specialization/derivation
Framework	probability axioms	$\widehat{=}$	logical axioms
Assumptions	prior	$\widehat{=}$	non-logical axioms
Inference rule	Bayes rule	$\widehat{=}$	modus ponens
Results	posterior	$\widehat{=}$	theorems
Universal scheme	Solomonoff probability	$\widehat{=}$	ZFC
Universal inference	universal induction	$\widehat{=}$	universal theorem prover
Limitation	uncomputable (Turing)	$\widehat{=}$	incomplete (Gödel)
In practice	approximations	$\widehat{=}$	semi-formal proofs
Operation	computation	$\widehat{=}$	proof

Generalized Solomonoff Semimeasure[Gra+24]

Definition (Generalized Solomonoff Semimeasure)

$$M_T^Q(x) := \sum_{p:T(p)=x*} Q(p) \quad \text{with special case} \quad M_U(x) := \sum_{p:U(p)=x*} 2^{-\ell(p)}$$

for a universal monotone TM $T = U$ and $Q(p) = 2^{-\ell(p)}$.

Theorem (Universality of generalized Solomonoff semimeasures)

For any universal monotone TM U, and any computable measure Q s.t. $Q(p) > 0$ and $Q(p) \to 0$ for $\ell(p) \to \infty$, then there exists a universal monotone TM V s.t. $M_U^Q(x) = M_V(x)$.

Remark: 可以把程序先验 Q 偏置到 "人觉得更有趣" 的结构上.

Proof Sketch.

Let $0.p_{1:\infty} \in [0,1]$ with binary expansion $p_{1:\infty}$.
Let $F : [0,1] \to [0,1]$ be the cumulative distribution function $F(0.p_{1:\infty}) = \sum_{t:p_t=1} Q(\Gamma_{p_{<t}0})$, since $[0,\, 0.p_{1:\infty}) = \coprod_{t:p_t=1} 0.\Gamma_{p_{<t}0}$, where $0.\Gamma_p = [0.p0^\infty,\, 0.p1^\infty)$ and $\coprod$ denotes disjoint union.
The assumption $Q(p) > 0$ implies that F is strictly increasing, and assumption $Q(p_{1:n}) \to 0$ implies that F is continuous. Since $F(0) = 0$ and $F(1) = 1$, this implies that F is a bijection. Let $0.q_{1:\infty} = F(0.p_{1:\infty})$.
Further for some finite prefix $p \prec p_{1:\infty}$, we partition the interval

$$[0.q^0_{1:\infty}, 0.q^1_{1:\infty}) := [F(0.p0^\infty), F(0.p1^\infty)) =: \coprod_{q\in\Phi(p)} 0.\Gamma_q$$

into a minimal set of binary intervals $0.\Gamma_q$, where $\Phi(p)$ is a minimal prefix free set in the sense that for any q, at most one of q, $q0$, $q1$ is in $\Phi(p)$.
Now we plug

$$Q(p) = F(0.p1^\infty) - F(0.p0^\infty) = \sum_{q\in\Phi(p)} |0.\Gamma_q| = \sum_{q\in\Phi(p)} 2^{-\ell(q)} \quad \text{into}$$

$$M^Q_U(x) = \sum_{p:U(p)=x*} Q(p) = \sum_{p:U(p)=x*} \sum_{q\in\Phi(p)} 2^{-\ell(q)} = \sum_{q:V(q)=x*} 2^{-\ell(q)} = M_V(x)$$

where $V(q) := U(p)$ for the maximal p such that $q \in \Phi(p)$.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
   - Kolmogorov Complexity
   - Analogy & Few-Shot Learning
   - Algorithmic Probability
   - A Statistical Mechanical Interpretation of AIT
   - Incompressibility & Incompleteness
   - Algorithmic Randomness
   - Effective Complexity
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

A statistical mechanical interpretation of AIT — [Tad20]

An energy eigenstate n	$\Longrightarrow$	A program p s.t. $U(p)\downarrow$
The energy E_n of n	$\Longrightarrow$	The length $\ell(p)$ of p
Boltzmann constant k	$\Longrightarrow$	$1/\ln 2$

$Z = \sum_n \mathrm{e}^{-\frac{E_n}{kT}}$	$\Longrightarrow$	$Z = \sum_{p:U(p)\downarrow} 2^{-\frac{\ell(p)}{T}}$	Partition function
$F = -kT \ln Z$	$\Longrightarrow$	$F = -T \log Z$	Free energy
$P(n) = \frac{1}{Z}\mathrm{e}^{-\frac{E_n}{kT}}$	$\Longrightarrow$	$P(p) = \frac{1}{Z}2^{-\frac{\ell(p)}{T}}$	Boltzmann distribution
$E = \sum_n P(n)E_n$	$\Longrightarrow$	$E = \sum_{p:U(p)\downarrow} P(p)\ell(p)$	Energy
$S = \frac{E-F}{T}$	$\Longrightarrow$	$S = \frac{E-F}{T} = H(P)$	Entropy
$C = \frac{\mathrm{d}E}{\mathrm{d}T}$	$\Longrightarrow$	$C = \frac{\mathrm{d}E}{\mathrm{d}T}$	Specific heat

Temperature = Compression Rate

Theorem (Tadaki)

1. *If* $0 < T < 1$ *and* T *is computable, then each of* Z, F, E, S, C *converges to a real whose compression rate equals to* T*, i.e.*

$$\lim_{n\to\infty} \frac{K(Z_{1:n})}{n} = \lim_{n\to\infty} \frac{K(F_{1:n})}{n} = \lim_{n\to\infty} \frac{K(E_{1:n})}{n} = \lim_{n\to\infty} \frac{K(S_{1:n})}{n} = \lim_{n\to\infty} \frac{K(C_{1:n})}{n} = T$$

2. *If* $T > 1$*, then* $Z = E = S = \infty$*, and* $F = -\infty$*.*
3. *If* $T = 1$*, then* Z, F *converge, but* $E = S = C = \infty$*.*

Fixpoint Theorem on Compression Rate

Theorem (Fixpoint Theorem on Compression Rate [Tad20])

For every $T \in (0,1)$, if Z or F is computable, then

$$\lim_{n\to\infty} \frac{K(T_{1:n})}{n} = T$$

Intuitive Meaning

Consider a file of infinite size whose content is

"The compression rate of this file is $0.100111001\ldots\ldots$"

When this file is compressed, the compression rate of this file actually equals to $0.100111001\ldots\ldots$, as the content of this file says.

This situation forms a fixpoint and is self-referential.

Theorem (Tadaki)

There does not exist $T \in (0,1)$ such that both Z and F are computable.

A Similar Version — Baez & Stay[BS12]

$E_{\{p\}} = \ln t(p)$	Energy
$V_{\{p\}} = \ell(p)$	Volume
$N_{\{p\}} = U(p)$	Number of molecules
$Z = \sum_{p:U(p)\downarrow} \mathrm{e}^{-\frac{E_{\{p\}}+PV_{\{p\}}-\mu N_{\{p\}}}{T}}$	Partition function
$P(p) = \frac{1}{Z}\mathrm{e}^{-\frac{E_{\{p\}}+PV_{\{p\}}-\mu N_{\{p\}}}{T}}$	Boltzmann distribution

$$\mathrm{d}E = T\mathrm{d}S - P\mathrm{d}V + \mu\mathrm{d}N$$

- Temperature $T = \frac{\partial E}{\partial S}\big|_{V,N}$: how many times you must double the runtime in order to double the number of programs in the ensemble while holding their mean length and output fixed.
- Pressure $P = -\frac{\partial E}{\partial V}\big|_{S,N}$: how much you need to decrease the mean length to increase the mean log runtime by a specified amount, while holding the number of programs in the ensemble and their mean output fixed.
- Chemical Potential $\mu = \frac{\partial E}{\partial N}\big|_{S,V}$: how much the mean log runtime increases when you increase the mean output while holding the number of programs in the ensemble and their mean length fixed.

Variant1 — Extension to Monotone Complexity

$$E_{\{p,h\}} = \begin{cases} \ell(p) & \text{if } U(p) = h* \\ 0 & \text{otherwise} \end{cases} \qquad \text{Energy of } (p,h)$$

$$Z(h) = \sum_{p:U(p)=h*} 2^{-\frac{\ell(p)}{T_h}} = M(h) \qquad \text{Partition function}$$

$$F(h) = -T_h \log Z(h) \qquad \text{Free energy}$$

$$P_h(p) = \frac{1}{Z(h)} 2^{-\frac{E_{\{p,h\}}}{T_h}} \qquad \text{Boltzmann distribution}$$

$$E(h) = \sum_{p:U(p)=h*} P_h(p) E_{\{p,h\}} \qquad \text{Energy}$$

$$S(h) = \frac{E(h) - F(h)}{T_h} = H(P_h) \qquad \text{Entropy}$$

If we take temperature as compression rate, $T_h := \frac{K(h)}{\ell(h)}$.

Variant2

$$E_{\{p,h\}} = \begin{cases} \log t(p,h) & \text{if } U(p) = h* \\ 0 & \text{otherwise} \end{cases}$$ Internal energy of (p,h)

$$V_{\{p,h\}} = \begin{cases} \ell(p) & \text{if } U(p) = h* \\ 0 & \text{otherwise} \end{cases}$$ Volume of (p,h)

$H_{\{p,h\}} = E_{\{p,h\}} + PV_{\{p,h\}}$ Enthalpy of (p,h), where P is pressure

$Z(h) = \sum_{p:U(p)=h*} 2^{-\frac{H_{\{p,h\}}}{T_h}}$ Partition function

$F(h) = -T_h \log Z(h)$ Free energy

$P_h(p) = \frac{1}{Z(h)} 2^{-\frac{H_{\{p,h\}}}{T_h}}$ Boltzmann distribution

$E(h) = \sum_{p:U(p)=h*} P_h(p) E_{\{p,h\}}$ Energy

$V(h) = \sum_{p:U(p)=h*} P_h(p) V_{\{p,h\}}$ Volume

$S(h) = \frac{E(h) + PV(h) - F(h)}{T_h} = H(P_h)$ Entropy

Computable Universal Predictor

$$Z(h) = \sum_{p:U(p)=h*} 2^{-\frac{\ell(p)+\log t(p,h)}{T}}$$

Theorem

If T_h is computable, Then $Z(h)$ is a computable semimeasure.

$$\sum_{t=1}^{\infty} \left|1 - Z(h_t \mid h_{<t})\right| \leq \frac{Km(h)\ln 2 + \ln t(p,h)}{T_h}$$

Variant3 — Stochastic Case

$$E^{\text{in}}_{\{\nu,h\}} = \begin{cases} -\log w_\nu & \text{if } \nu \in \mathcal{M}_h \\ 0 & \text{otherwise} \end{cases}$$ Internal energy

$$E^{\text{ex}}_{\{\nu,h\}} = \begin{cases} -\log \nu(h) & \text{if } \nu \in \mathcal{M}_h \\ 0 & \text{otherwise} \end{cases}$$ External energy

$$E_{\{\nu,h\}} = E^{\text{in}}_{\{\nu,h\}} + E^{\text{ex}}_{\{\nu,h\}}$$ Total energy

$$Z(h) = \sum_{\nu \in \mathcal{M}_h} 2^{-\frac{E_{\{\nu,h\}}}{T_h}}$$ Partition function

$$F(h) = -T_h \log Z(h)$$ Free energy

$$P_h(\nu) = \frac{1}{Z(h)} 2^{-\frac{E_{\{\nu,h\}}}{T_h}}$$ Boltzmann distribution

$$E(h) = \sum_{\nu \in \mathcal{M}_h} P_h(\nu) E_{\{\nu,h\}}$$ Average Energy

$$S(h) = \frac{E(h) - F(h)}{T_h} = H(P_h)$$ Entropy

$$\mathcal{M}_h := \left\{ \rho \in \mathcal{M} : 2^{-\left(H(\rho)+\varepsilon\right)} \le \rho(h) \le 2^{-\left(H(\rho)-\varepsilon\right)} \;\&\; -\log w_\rho - \log \rho(h) \le K(h)(1+\delta) \right\}$$

Stochastic Case

If $T_h = 1$, then

$$E_{\{\nu,h\}} = -\log w_\nu - \log \nu(h)$$

$$Z(h) = \sum_{\nu \in \mathcal{M}} 2^{-E_{\{\nu,h\}}} = \sum_{\nu \in \mathcal{M}} w_\nu \nu(h) = \xi(h)$$

$$P_h(\nu) = \frac{2^{-E_{\{\nu,h\}}}}{Z(h)} = \frac{w_\nu \nu(h)}{\xi(h)} = w_h^\nu$$

$$F(h) = -\log Z(h) = \underbrace{-\log \xi(h)}_{\approx K(h)} = \mathbb{E}_{w_\epsilon}\left[E^{\mathrm{ex}}_{\{\nu,h\}}\right] - D(w_\epsilon \| w_h)$$

$$= \mathbb{E}_{w_h}\left[E^{\mathrm{ex}}_{\{\nu,h\}}\right] + D(w_h \| w_\epsilon)$$

$$= \underbrace{\mathbb{E}_{w_h}[-\log \nu(h)]}_{\text{Noise}} + \underbrace{D(w_h \| w_\epsilon)}_{\text{Surprise}}$$

$$H(P_h) = H(w_h) = \underbrace{H(w_h, w_\epsilon)}_{\text{Cross Entropy}} - D(w_h \| w_\epsilon)$$

"What an organism feeds upon is negative entropy."

— Schrödinger

"Every living thing is a sort of imperialist, seeking to transform as much as possible of its environment into itself and its seed."

— Russell

Intrinsic Utility

- square $-\xi(e_{t:k} \mid æ_{<t} a_{t:k})$
- Shannon $-\log \xi(e_{t:k} \mid æ_{<t} a_{t:k}) \approx K(æ_{1:k}) - K(æ_{<t})$
- KL divergence $D\left(w_{æ_{<k}} \| w_{æ_{<t}}\right)$ where $w^{\nu}_{æ_{<n}} = \dfrac{w_\nu \nu(e_{<n} \mid a_{<n})}{\xi(e_{<n} \mid a_{<n})}$
- information gain $H(w_{h_{<t}}) - H(w_{h_{1:k}})$ where

$$H(w_h) := -\sum_{\nu \in \mathcal{M}} w^{\nu}_h \log w^{\nu}_h$$

- effective complexity $\mathcal{E}_\delta(æ_{1:k}) - \mathcal{E}_\delta(æ_{<t})$ where

$$\mathcal{E}_\delta(æ_{<n}) := \min_{\nu \in \mathcal{M}} \left\{2K(\nu) + H(\nu) - K(æ_{<n}) : \nu(e_{<n} \mid a_{<n}) \geq 2^{-H(\nu)(1+\delta)}\right\}$$

- logical depth $\mathrm{depth}_b(h_{1:k}) - \mathrm{depth}_b(h_{<t})$ where

$$\mathrm{depth}_b(x) := \min \left\{t : U^t(p) = x \;\;\&\;\; \ell(p) - K(x) \leq b\right\}$$

Occam's Razor vs Maximum Entropy

$$\underset{w \vDash \begin{cases} H(w)=C \\ \sum_{\nu \in \mathcal{M}} w_\nu = 1 \end{cases}}{\textit{minimize}} \sum_{\nu \in \mathcal{M}} w_\nu K(\nu) \qquad \text{or} \qquad \underset{w \vDash \begin{cases} \sum_{\nu \in \mathcal{M}} w_\nu K(\nu) = C \\ \sum_{\nu \in \mathcal{M}} w_\nu = 1 \end{cases}}{\textit{maximize}} H(w)$$

$$L := \sum_{\nu \in \mathcal{M}} w_\nu K(\nu) - T\left(-\sum_{\nu \in \mathcal{M}} w_\nu \log w_\nu - C\right) - \lambda\left(\sum_{\nu \in \mathcal{M}} w_\nu - 1\right)$$

$$\frac{\partial L}{\partial w_\nu} = 0 \implies w_\nu^T = \frac{2^{-\frac{K(\nu)}{T}}}{\sum_{\nu \in \mathcal{M}} 2^{-\frac{K(\nu)}{T}}}$$

If the temperature $0 < T < 1$, then the Shannon entropy $H(w^T) < \infty$.

Why Solomonoff Prior?

$$H(w) \leq \mathbb{E}_w[K] \leq H(w) + K(w)$$

Maximum Entropy + Occam's Razor

$$T := \frac{\mathbb{E}_w[K]}{H(w)}$$

$$\underset{w \vDash \sum_{\nu \in \mathcal{M}} w_\nu = 1}{minimize\ T} \implies w_\nu = \frac{2^{\frac{-K(\nu)}{T}}}{\sum_{\nu \in \mathcal{M}} 2^{\frac{-K(\nu)}{T}}}$$

If w is lower semicomputable then

$$K(w) < \infty \implies T = 1 \implies w^*_\nu = \frac{2^{-K(\nu)}}{\sum_{\nu \in \mathcal{M}} 2^{-K(\nu)}}$$

- We can regard the set of programs that halt $\mathrm{dom}\, U$ as a set of 'microstates', and treat any function on $\mathrm{dom}\, U$ as an 'observable'. For any collection of observables, we can study the Gibbs ensemble that maximizes entropy subject to constraints on expected values of these observables. [BS12]
- Renormalization & Phase Transition [Man14]
- ...

Contents

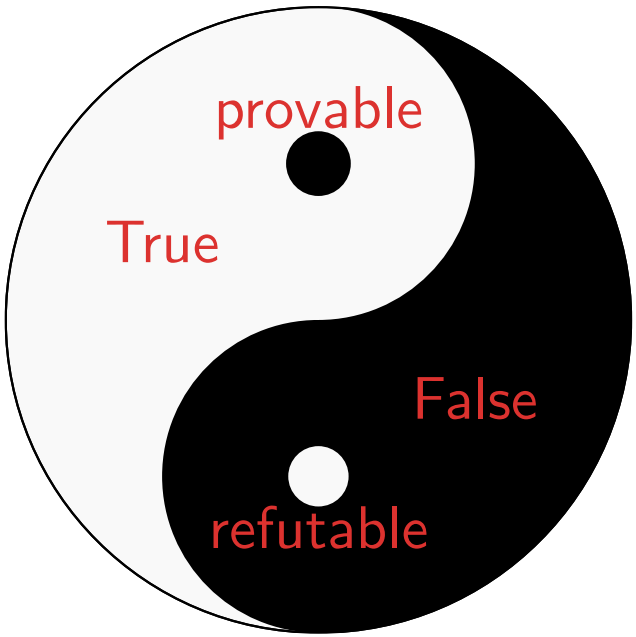
1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
   - Kolmogorov Complexity
   - Analogy & Few-Shot Learning
   - Algorithmic Probability
   - A Statistical Mechanical Interpretation of AIT
   - Incompressibility & Incompleteness
   - Algorithmic Randomness
   - Effective Complexity
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

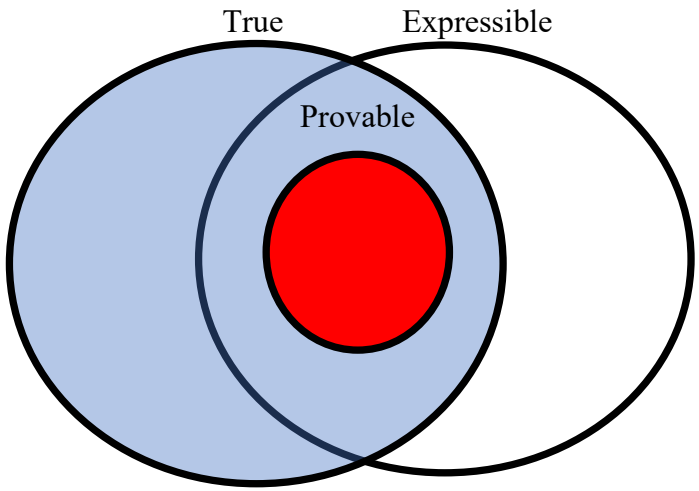
Fixpoint Lemma

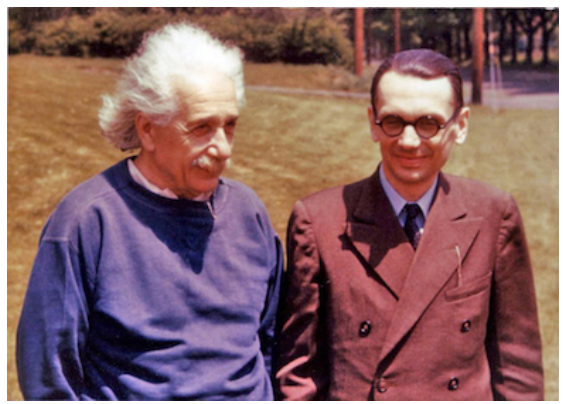
Lemma (Fixpoint Lemma)

For any formula $F(x)$ with one free variable x, there exists a sentence G s.t.

$$\mathrm{Q} \vdash G \leftrightarrow F(\ulcorner G \urcorner)$$







Remarks:

- 令 $F(x)$ 表示 "x 是公理", 则 G 在说 "我是公理". 显然 G 为假.
- 令 $F(x)$ 表示 "x 是句子", 则 G 在说 "我是句子". 显然 G 为真.

老子 ◎ô◎

- 道, 可道, 非常道; 名, 可名, 非常名.
  - The theory that can be formulated can't be the ultimate theory. The formulated theory of categories evolves, and its projection on reality changes.
- 无名, 天地之始; 有名, 万物之母.
  - The unformulatable ultimate theory is the truth of universe. The formulated theory is the basis to describe all the matter.
- 故常无, 欲以观其妙; 常有, 欲以观其徼.
  - In search of the unformulatable ultimate theory, we give meaning to life. Within the formulated theory, we study its limits.
- 此两者, 同出而异名, 同谓之玄.
  - The gap between the formulatable and the unformulatable is a mystery.
- 玄之又玄, 众妙之门.
  - From the formulated to the unformulated and from the unformulated to the formulated is the gateway to all understanding.

Gödel's First Incompleteness Theorem

Theorem (Gödel's First Incompleteness Theorem)

For any Gödelian $\mathrm{T} \supset \mathrm{Q}$*, there is a sentence* G *such that,*

1. *if* T *is consistent,* $\mathrm{T} \nvdash G$
2. *if* T *is* ω*-consistent,* $\mathrm{T} \nvdash \neg G$

Gödel's First Incompleteness Theorem

$$F(x) := \neg\Box x$$

$$G = \text{“I am not provable.”}$$

$$\mathrm{T} \vdash \mathrm{Con}_{\mathrm{T}} \rightarrow \neg\Box G$$

Proof.

$$G \leftrightarrow \neg\Box G \implies \Box G \rightarrow \neg G \implies \Box(\Box G \rightarrow \neg G) \implies \Box G \rightarrow \Box\neg G \implies$$
$$\Box G \rightarrow \Box(G \wedge \neg G) \implies \neg\Box\bot \rightarrow \neg\Box G$$

We now know enough to know that we will never know everything! ◎ô◎

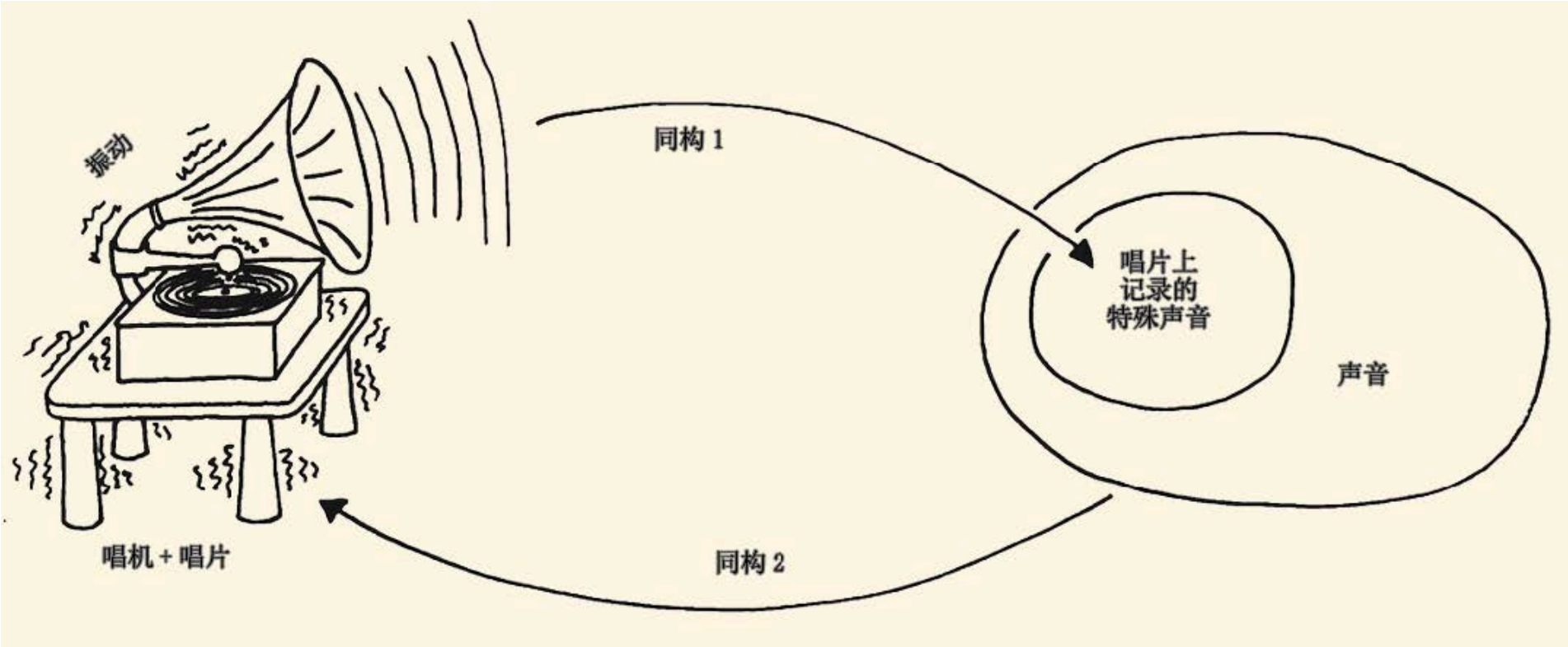


Figure: For every record player, there are records that it can't play. (sympathetic vibration)

Tarski's Undefinability Theorem

Theorem (Tarski's Undefinability Theorem)

There is no definable predicate $B(x)$ in the language of arithmetic, such that $\mathcal{N} \vDash A \leftrightarrow B(\ulcorner A \urcorner)$.

suppose $\{\#A : \mathcal{N} \vDash A\}$ is definable by $B(x)$.

$$F(x) := \neg B(x)$$

$G =$ "I am not true."

哥德尔句

“我不可证”

Problem (哥德尔是什么人?)

- 一个岛上有“君子”、“小人”两类人. “君子”只说真话, “小人”只说假话.
- 岛上有人有身份证, 有人没有.
- 有身份证的都是君子.
- 你来岛上遇到了一个名字叫“哥德尔”的土著.
- 哥德尔说: “我没有身份证”.

Argument from Incompleteness Theorems

- G_{T} := "This sentence cannot be proved in the formal axiomatic system T"
- We humans can easily see that G_{T} must be true.
- Since any AI is a FAS T, no AI can prove G_{T}. — *Penrose*
- Therefore there are things humans, but no AI system can do.
- P := "Penrose cannot prove that this sentence is true"
- Penrose cannot prove P, but now we can conclude that it is true.
- Penrose is in the same situation as an AI.
- Either (a) absolutely unsolvable problems exist or (b) the human mind infinitely surpasses any Turing machine or formal axiomatizable system. — *Gödel*
- There is no absolutely unsolvable problem. — *Martin-Löf*
- Gödel: 柏拉图主义意义上的 "真" 但 "不可证".
  Martin-Löf: 构造主义意义上的 "不可知为真" 坍缩为 "可知为假".

Martin-Löf's argument: there is no absolutely unsolvable problem

- The proposition A can be known to be true if we have a proof for A.
- The proposition A can be known to be false if we have a proof for $A \to \bot$.
- The proposition A cannot be known to be true if we have an algorithm which tests and rejects any given 'proof' which purports to demonstrate A.

1. reflection: If the premises of a valid inference are knowable, then so is the conclusion.
2. consistency: Absurdity cannot be known to be true.
3. unknowability of truth entails falsity: From the unknowability of the truth of a proposition, its falsity may be inferred. $\dfrac{x : A \vdash fx : \bot}{f : A \to \bot}$

   $\Longrightarrow$ law of excluded middle: There are no propositions which can neither be known to be true nor be known to be false.

能力与局限

人的理解力的真正使命在于去理解：存在着它所无法理解的事物，并认清这些事物究竟是什么.

— 克尔凯郭尔

发现可能性边界的唯一途径，就是稍微越过边界，冒险踏入不可能的领域.

— 亚瑟·查理斯·克拉克

上帝不仅在量子力学里掷骰子，在纯数学里也掷骰子.

— *Gregory Chaitin*

- 宇宙是像 π 还是像 Ω?
- 也许从宇宙内部，我们永远无法作出区分，只有宇宙外部的观察者才能做到.

Urbi et Orbi

数学之外

> 一个完全不自由的社会 (即处处按 "统一" 的法则行事的社会), 就其行为而言, 或者是不一致的, 或者是不完备的, 即无力解决某些问题, 可能是极端重要的问题. 在困难的处境里, 二者当然都会危及它的生存. 这个说法也适用于个体的人.
>
> — 哥德尔

Remark: 哥德尔定理版本的 "哈耶克-自发社会秩序".

1. 在包含理性人类的任何社会文明体系中, 永远存在着无法用人类理性解决的问题, 不存在一个万能的政府, 能对体系内的任何问题作出合理与公正的解决.
2. 对于包含理性人类的任何社会文明体系, 不能在该体系内对其作出合理与公正的评价.

Remark: 哥德尔定理 vs 因果涌现: 秩序和涌现的属性不能从系统内部观察和认知, 只能由外部观察者来观察.

Remark: 霍金: 物理 "万有理论" 不可能. (算术不完备 vs 物理不完备?)

Levin's Independence Postulate

Postulate (Levin's Independence Postulate)

Let R be the reference to a physical process generating sequence α_R. Let P be a mathematical formula defining sequence β_P. Then

$$I(\alpha_R;\beta_P) \stackrel{+}{\leq} \ell(R)+\ell(P)$$

- When a sequence is both physically generatable and mathematically definable, $\alpha=\beta$ gives a "finitary" Church-Turing Thesis.

$$K(\alpha) \stackrel{+}{=} I(\alpha;\alpha) < \infty$$

- One can ask if non-mechanical means could really enable the Hilbert-Gödel task of consistent completion for PA. A negative answer follows from the existence of a specific sequence ρ that has infinite mutual information with each total extensions of a universal partial recursive predicate. ρ plays a role of a password: no substantial information about it can be guessed, no matter what methods are allowed.

Halting Problem

Theorem (Halting Problem is Undecidable)

There is no recursive function deciding whether a program halts.

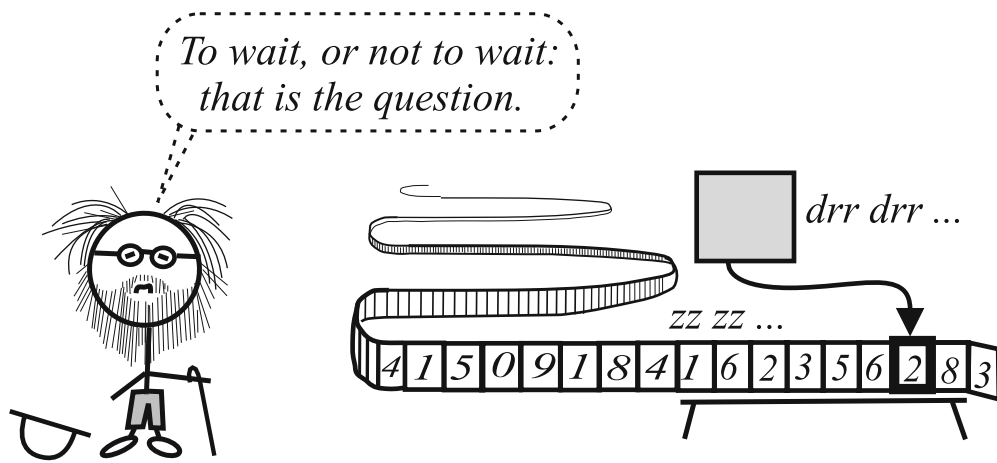
Proof.

Assume there exists a halting program H.

Construct a program q as follows:

1. read n;
2. generate $A := \{p : \ell(p) \leq n\}$;
3. use H to get $B := \{p \in A : U(p)\downarrow\}$;
4. output $2\max\{U(p) : p \in B\}$.

$\ell(q) \overset{+}{\leq} \log n \lesssim n \implies U(q) \geq 2U(q)$



Incompressibility vs Incompleteness vs Berry Paradox

Theorem (Kolmogorov)

Kolmogorov complexity K is uncomputable.

$$x^* := \mu x[K(x) > n] \implies n < K(x^*) \leq O(\log n)$$

Theorem (Chaitin)

For any arithmetically sound Gödelian T, $\exists c \forall x : \mathrm{T} \nvdash K(x) > c$.

"given n, find $\mu y\big[\,\mathrm{prf}_{\mathrm{T}}\left(y, K(x) > n\right)\big]$, ouput x" $\implies n < K(x) \leq O(\log n)$

"the least number undefinable in fewer characters than there are in this sentence."

$M_e :=$"find $\mu y\big[\,\mathrm{prf}_{\mathrm{T}}\left(y, K(x) > e\right)\big]$, ouput x" (Berry Paradox)

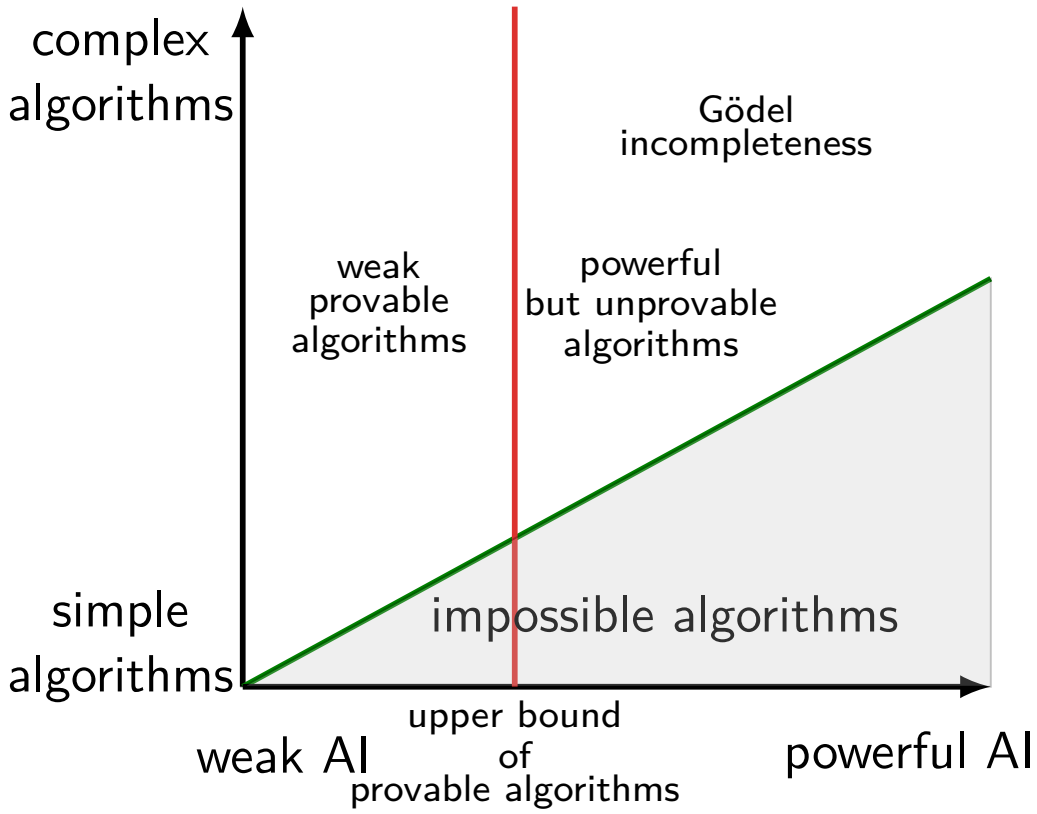
Theorem (Chaitin)

For any arithmetically sound Gödelian T, $\#\{x : \mathrm{T} \vdash K(x) > \ell(x)\} < \infty$.

Remark: For almost all random strings their randomness cannot be proved.

Incompressibility vs Incompleteness vs Intelligence

- $P(x) := \left\{ p \in \mathcal{X}^* : \exists t \forall k \geq t \left(p(x_{1:k}) = x_{k+1} \right) \right\}$
- $P(A) := \bigcap_{x \in A} P(x)$
- $P_n := P\left(\left\{ x : K(x) \leq n \right\} \right)$



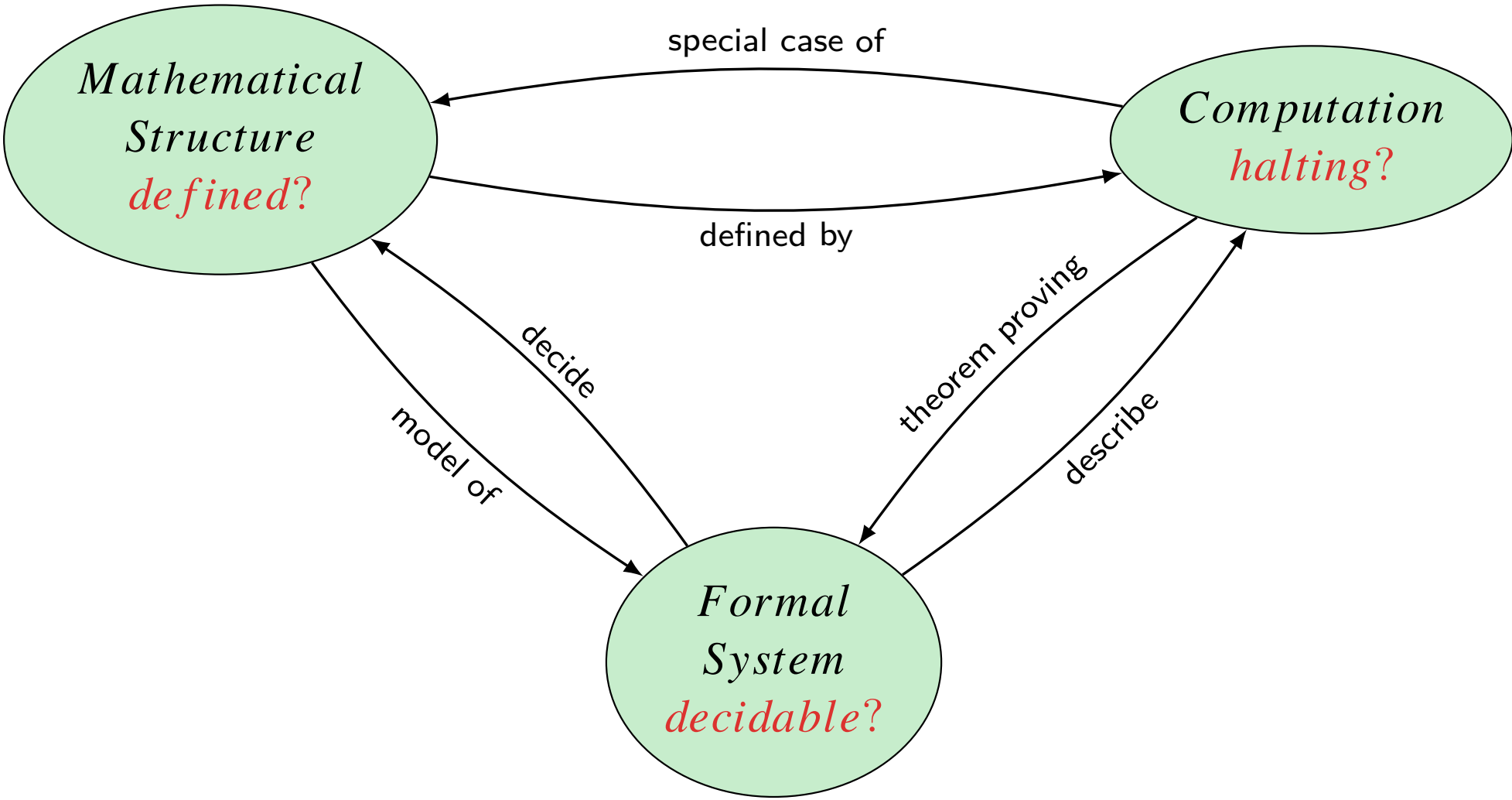
- $\forall n \exists p \in P_n : K(p) \stackrel{+}{\leq} n + O(\log n)$
- $\forall n : p \in P_n \Longrightarrow K(p) \stackrel{+}{\geq} n$

Theorem (Shane Legg)

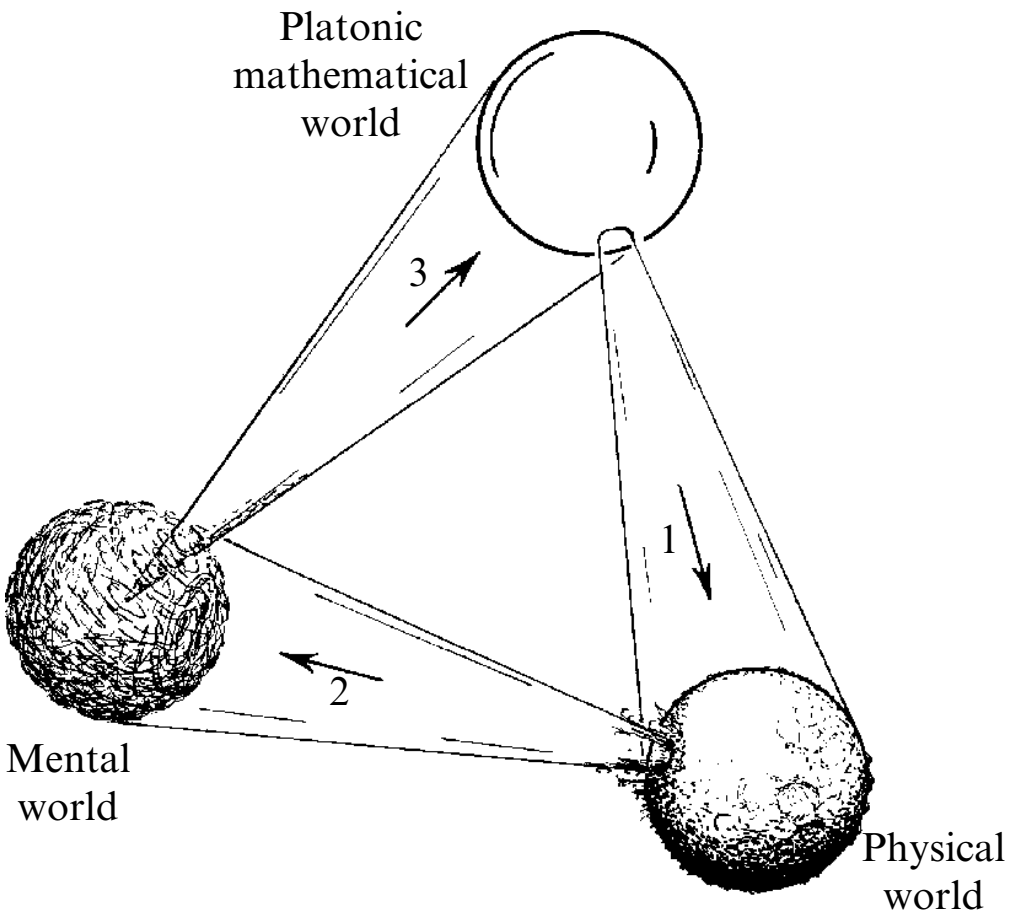
For any arithmetically sound Gödelian T, $\exists c \forall n \geq c \forall p : \mathrm{T} \nvdash p \in P_n$.

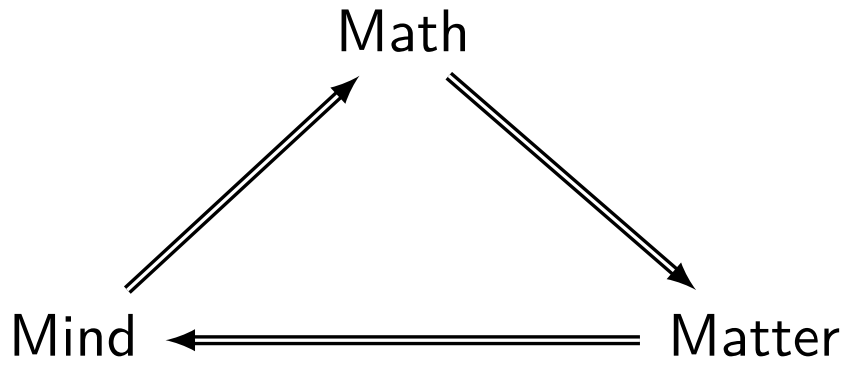
"given n, find $\mu x \left[\mathrm{prf}_{\mathrm{T}} \left(x, p \in P_n \right) \right]$, output p" $\Longrightarrow$ $K(p) < O(\log n)$

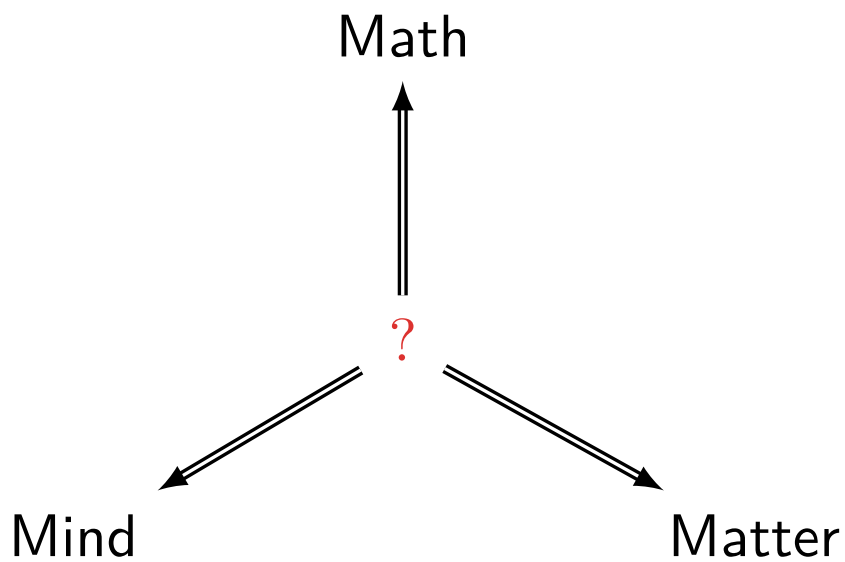
Formal System vs Structure vs Computation



Math-Matter-Mind (Penrose)







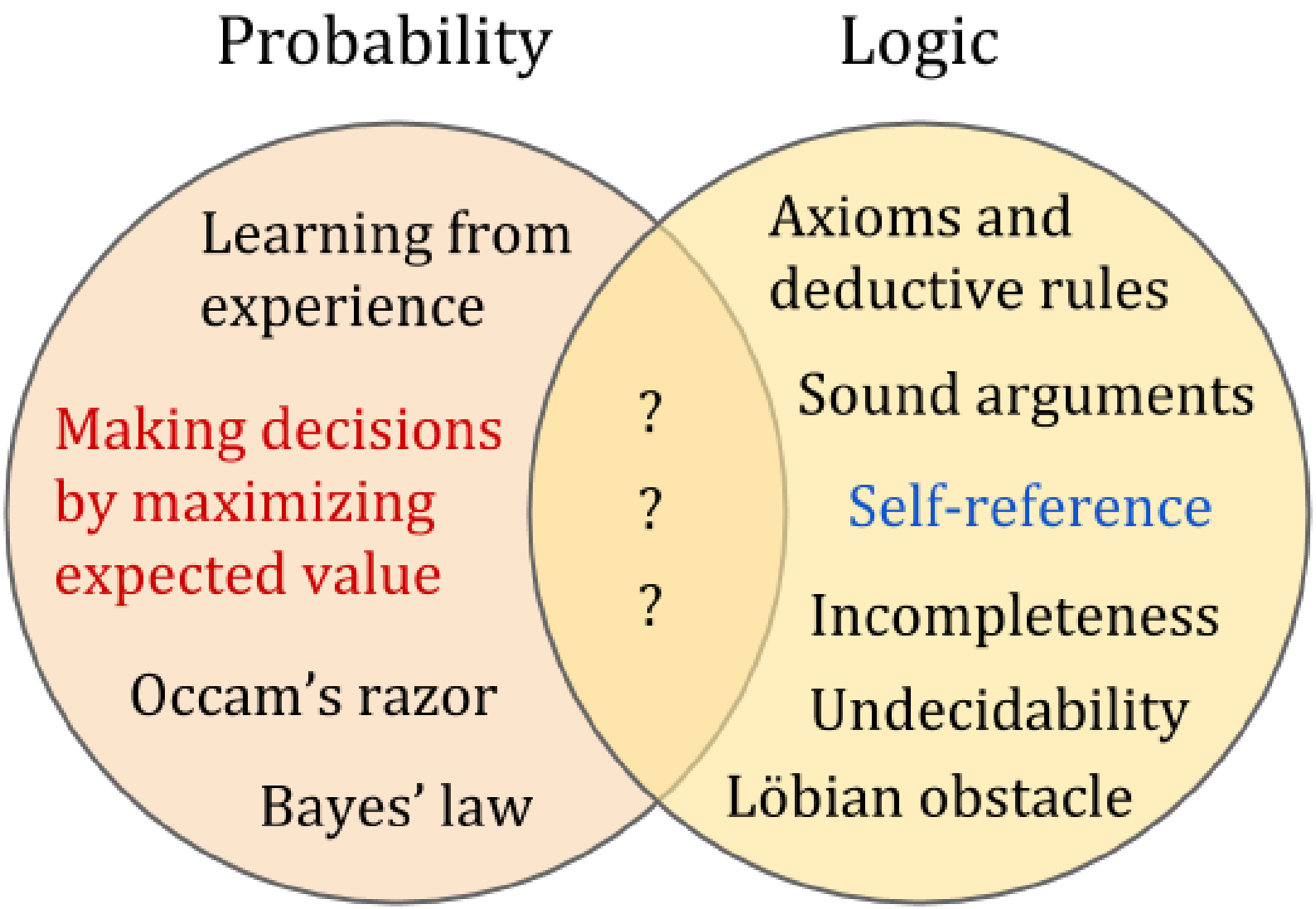
Remark: Hassabis: 不同于抽象的数学构造, 自然的结构是有原因的演化而来的. 大自然沿着阻力最小的路径演化, 构建了一个能量漏斗, 幸存的模式背后, 是一个光滑的低维流形.

Logic vs Statistics

Field	Logical Approach	Statistical Approach
Knowledge representation	First order logic	Graphical models
Automated reasoning	Satisfiability testing	Markov chain Monte Carlo
Machine learning	Inductive logic programming	Neural networks
Planning	Classical planning	Markov decision processes
Natural language processing	Definite clause grammars	Probabilistic context-free grammars

Logic	Statistics
rule-based	data-driven
rigour	possibility
knowable	black-box
simple & perfect world	complex & uncertain world

Probability vs Logic



Prediction with Expert Advice

- Assume that there is some large, possibly infinite, class of 'experts' which make predictions.
- The aim is to observe how each of these experts perform and predicts asymptotically as well as the best expert in hindsight.

	Expert_1	Expert_2	$\dots$	Expert_n	PEA	*true*	*loss*
day_1	0	0	$\dots$	0	0	1	1
day_2	0	1	$\dots$	1	1	1	0
day_3	1	0	$\dots$	1	1	0	1
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$
day_t	y_t^1	y_t^2	$\dots$	y_t^n	y_t^{PEA}	x_t	$\lvert y_t^{\text{PEA}} - x_t \rvert$

Prediction with Expert Advice

- Follow the (perturbed) leader.
- Predicts according to a majority vote by the "good" experts.
- Multiplicative Weights. — take expert which performed best in past with high probability and others with smaller probability.
- Regularization. Choose the class of all computable experts, and penalize "complex" experts.
- Universal Portfolios.

Universal Portfolios

- the agent chooses a distribution $\mathbf{b}_t \in \Delta_n := \{\mathbf{x} \in [0,1]^n : \|\mathbf{x}\|_1 = 1\}$ of wealth over n goods.
- nature chooses returns $\mathbf{x}_t \in (\mathbb{R}^+)^n$, where

$$(\mathbf{x}_t)_i = \frac{\text{price of good } i \text{ at end of } t}{\text{price of good } i \text{ at beginning of } t}$$

- the total wealth. $W_t(\mathbf{b}, \mathbf{x}) = W_1 \prod_{k=1}^{t} \mathbf{b}_k^\mathsf{T}(\mathbf{x}_{<k})\mathbf{x}_k$
- regret. $R_t := \max_{\mathbf{b}\in\Delta_n} \sum_{k=1}^{t} \log \mathbf{b}_k^\mathsf{T}(\mathbf{x}_{<k})\mathbf{x}_k - \sum_{k=1}^{t} \log \mathbf{b}_k^\mathsf{T}(\mathbf{x}_{<k})\mathbf{x}_k$
- universal portfolios.

$$\hat{\mathbf{b}}_1 := \left(\tfrac{1}{n}, \ldots, \tfrac{1}{n}\right)$$

$$\hat{\mathbf{b}}_{t+1}(\mathbf{x}_{1:t}) := \frac{\int_{\Delta_n} \mathbf{b} W_t(\mathbf{b}, \mathbf{x}) \, \mathrm{d}\mathbf{b}}{\int_{\Delta_n} W_t(\mathbf{b}, \mathbf{x}) \, \mathrm{d}\mathbf{b}}$$

- Asymptotic Optimality. $\frac{1}{t} \log W_t(\hat{\mathbf{b}}, \mathbf{x}) \xrightarrow{t\to\infty} \frac{1}{t} \log \max_{\mathbf{b}\in\Delta_n} W_t(\mathbf{b}, \mathbf{x})$

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
   - Kolmogorov Complexity
   - Analogy & Few-Shot Learning
   - Algorithmic Probability
   - A Statistical Mechanical Interpretation of AIT
   - Incompressibility & Incompleteness
   - Algorithmic Randomness
   - Effective Complexity
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

- What is randomness?
- What is a random variable?
  — It's a measurable function on a probability space.
- Probability theory avoids defining randomness by working with abstractions like random variables.
- Not "is this system random?" but rather "is it useful to model this system as random?"
- But what is "true" randomness?

The Paradox of Randomness

- A random bit-string should be "typical": it should not stand out from the crowd of other bit-strings.
- Assume that there is a precise way to distinguish between "random bit-strings" and bit-strings which are "non-random".
- Can the adopted **criterion** be consistent?
- choose the first bit-string that the criterion asserts it is random. This particular bit-string is

  "the first bit-string satisfying the property of being random"

  a property making it atypical, so non-random!

Randomness

Typicalness The statistician's approach: A random sequence is the typical outcome of a random variable. Random sequences should not have effectively rare distinguishing properties.

Incompressibility The coder's approach: Rare patterns can be used to compress information. Random sequences should not be effectively described by a significantly shorter description than their literal representation.

Unpredictability The gambler's approach: A betting strategy can exploit rare patterns. Random sequences should be unpyellowictable. No effective martingale can make an infinite amount betting on the bits.

The Statistician's Approach

- A random sequence should be absolutely normal.
- If you select a subsequence, then it should satisfy the law of large numbers, the law of the iterated logarithm...
- But what selection functions should be allowed? Computable?
- Martin-Löf: we can effectively test whether a particular infinite sequence does not satisfy a particular law of randomness by effectively testing whether the law is violated on increasingly long initial segments. We should consider the intersection of all sets of measure one with recursively enumerable complements. (Such a complement set is expressed as the union of a recursively enumerable set of cylinders).

Cantor Space 2^ω

- For $x \in 2^{<\omega}$, the cylinder set $\Gamma_x := \{y \in 2^\omega : x \prec y\}$ is the basic open set. It corresponds to the interval $[0.x, 0.x + 2^{-\ell(x)})$.
- For $A \subset 2^{<\omega}$, the open set generated by A is $\Gamma_A := \bigcup_{x \in A} \Gamma_x$.
- The Lebesgue measure $\mu(\Gamma_x) := 2^{-\ell(x)}$, $\mu(x) := \mu(\Gamma_x)$.
- The outer measure of $C \subset 2^\omega$ is $\mu^*(C) := \inf \left\{ \sum_{x \in A} 2^{-\ell(x)} : C \subset \Gamma_A \right\}$.
- The inner measure of C is $\mu_*(C) := 1 - \mu^*(2^\omega \setminus C)$.
- If C is measurable, then $\mu^*(C) = \mu_*(C)$.
- $A \subset 2^\omega$ has measure 0 iff there is a sequence $\{V_n\}_{n \in \omega}$ of open sets s.t. $A \subset \bigcap_{n \in \omega} V_n$ and $\lim_{n \to \infty} \mu(V_n) = 0$.

An Analogy: airport terrorist testing ⊙ô⊙

- Null hypothesis: a passenger is not a terrorist.
- Tests:
  - Passport checking
  - Blacklist checking
  - Baggage scanning
  - Body Scanner
  - Officials talk to you
- Every time you pass one test, our level of confidence in the null hypothesis increases.
- If you fail any test, they arrest you.
- If you pass all possible tests, then with "high confidence", the null hypothesis holds.

History — Mises-Wald-Church

- In 1919, von Mises proposed the following two conditions for an infinite random sequence $x \in 2^\omega$:
  1. The limiting frequency $\lim_{n\to\infty} \frac{\sum_{i=1}^{n} x_i}{n}$ exists.
  2. The limiting frequency persists for any subsequence selected by an admissible place-selection rule.
- What is an admissible place-selection rule?
- If you allow any partial function to be admissible, then there is no random sequence.
- Ward: if we allow only countably many functions, then von Mises sequences exist.
- Church: let's use recursive functions.
- Ville: There exist sequences that satisfy the Mises-Wald-Church definition of randomness, with limiting frequency of ones of $\frac{1}{2}$, but nonetheless have the property

$$\forall n : \frac{\sum_{i=1}^{n} x_i}{n} \geq \frac{1}{2}$$

Nested Critical Regions

Instead of stability under place-selection rules, Martin-Löf's fundamental property is passing effective statistical tests.

$$V_1 \supset V_2 \supset V_3 \supset \ldots$$

As the critical regions become smaller, significance level increases.

Martin-Löf Randomness

Definition (Martin-Löf Randomness)

- A total lower semicomputable function $\delta : 2^{<\omega} \to \omega$ is a Martin-Löf test iff $\forall n : \mu(V_n) \leq 2^{-n}$, where $V_n := \{x : \delta(x) \geq n\}$.
- $x \in 2^\omega$ is ML-random iff for every ML-test δ, $\sup_n \delta(x_{1:n}) < \infty$.

$$\delta(x) < \infty \iff x \notin \bigcap_{n=1}^{\infty} V_n$$

Definition (Martin-Löf Randomness)

- A Martin-Löf test is a r.e. set $V \subset \mathbb{N} \times 2^{<\infty}$ s.t. for $V_n := \{x \in 2^{<\infty} : \langle n, x \rangle \in V\}$,
$$\mu(V_n) \leq 2^{-n}$$
- $x \in 2^\omega$ is ML-random iff for every ML-test V,
$$x \notin \bigcap_{n=1}^{\infty} V_n$$

Martin-Löf Randomness

Definition (Universal Martin-Löf Test)

A ML-test δ_0 is *universal* iff for every ML-test δ, $\exists c \forall x : \delta_0(x) \geq \delta(x) - c$.

A ML-test $\{U_n\}_{n\in\omega}$ is *universal* iff for every ML-test $\{V_n\}_{n\in\omega}$, $\bigcap\limits_{n\in\omega} U_n \supset \bigcap\limits_{n\in\omega} V_n$.

$$\boxed{\delta(x) := \ell(x) - K(x \mid \ell(x)) \text{ is a universal ML-test.}}$$

$$R_b := \left\{x \in 2^\omega : \exists n \left(K(x_{1:n}) < n - b\right)\right\}$$

$$\boxed{\{R_b\}_{b\in\omega} \text{ is a universal ML-test.}}$$

Theorem (Schnorr 1973)

A sequence $x \in 2^\omega$ is ML-random iff it is 1-random.

The Gambler's Approach

- A martingale is a function $d : 2^{<\omega} \to [0, \infty)$ s.t. for every $\sigma \in 2^{<\omega}$

$$d(\sigma) = \frac{d(\sigma 0) + d(\sigma 1)}{2}$$

- A supermartingale is a function $d : 2^{<\omega} \to [0, \infty)$ s.t. for every $\sigma \in 2^{<\omega}$

$$d(\sigma) \geq \frac{d(\sigma 0) + d(\sigma 1)}{2}$$

- A (super)martingale d succeeds on $x \in 2^\omega$ iff $\limsup_{n \to \infty} d(x_{1:n}) = \infty$.

Theorem

A sequence $x \in 2^\omega$ is ML-random iff no r.e. (super)martingale succeeds on it.

The Coder's Approach

Definition (1-Randomness)

$x \in 2^\omega$ is 1-random iff

$$\exists c \forall n : K(x_{1:n}) \geq n - c$$

Theorem

The following are equivalent.

- $x \in 2^\omega$ *is ML-random.*
- *No r.e. (super)martingale succeeds on it.*
- $\exists c \forall n : K(x_{1:n}) \geq n - c$
- $\forall n : Km(x_{1:n}) \stackrel{+}{=} n$
- $\lim_{n\to\infty} K(x_{1:n}) - n = \infty$
- $\sum_{n=1}^{\infty} 2^{n-K(x_{1:n})} < \infty$
- $\sup_n 2^{n-K(x_{1:n})} < \infty$
- $C(x_{1:n}) \stackrel{+}{\geq} n - K(n)$
- $C(x_{1:n}) \stackrel{+}{\geq} n - f(n)$ *for every computable* f *s.t.* $\sum_{n=1}^{\infty} 2^{-f(n)} < \infty$.

Definition (Solovay Reducibility)

Let $a_n \to \alpha$ and $b_n \to \beta$ be two computable strictly increasing sequences of rationals converging to lower semicomputable reals α and β. We say that $\alpha \leq_S \beta$ iff there is a constant c and a total recursive function f s.t. $\forall n : \alpha - a_{f(n)} \leq c(\beta - b_n)$.

Theorem

For lower semicomputable reals α, the following are equivalent.

- *α is 1-random*
- *$\alpha \geq_S \beta$ for all lower semicomputable reals β.*
- *$\alpha \geq_S \Omega$*
- *$K(\alpha_{1:n}) \stackrel{+}{\geq} K(\beta_{1:n})$ for all lower simicomputable reals β.*
- *$K(\alpha_{1:n}) \stackrel{+}{\geq} K(\Omega_{1:n})$*
- *$K(\alpha_{1:n}) \stackrel{+}{=} K(\Omega_{1:n})$*
- *$\alpha = \Omega_U := \sum_{p:U(p)\downarrow} 2^{-\ell(p)}$ for some universal prefix Turing machine U.*

μ/ξ-randomness

Definition (μ/ξ-randomness)

- A sequence $x \in 2^\omega$ is μ/ξ-random iff $\exists c \forall n : \xi(x_{1:n}) \leq c \cdot \mu(x_{1:n})$.
- A sequence $x \in 2^\omega$ is μ-ML-random iff $\exists c \forall n : M(x_{1:n}) \leq c \cdot \mu(x_{1:n})$.

Theorem

- $x_{1:\infty}$ *is* μ*-ML-random iff* $\sup_n \delta(x_{1:n} \mid \mu) < \infty$, *where*

$$\delta(x \mid \mu) := \log \frac{M(x)}{\mu(x)}$$

- *For a computable* μ, $x_{1:\infty}$ *is* μ*-ML-random iff*

$$\forall n : Km(x_{1:n}) \stackrel{+}{=} -\log \mu(x_{1:n})$$

Properties of ML-Random Sequences

- Special case of μ being a fair coin, i.e. $\mu(x_{1:n}) = 2^{-n}$, then

$$x_{1:\infty} \text{ is random } \iff Km(x_{1:n}) \stackrel{+}{=} n, \text{ i.e. iff } x_{1:n} \text{ is incompressible.}$$

- For general μ, $-\log \mu(x_{1:n})$ is the length of the Arithmetic code of $x_{1:n}$, hence

$$x_{1:\infty} \text{ is } \mu\text{-random } \iff \text{ the Arithmetic code is optimal.}$$

- One can show that a μ-random sequence $x_{1:\infty}$ passes all thinkable effective randomness tests, e.g. the law of large numbers, the law of the iterated logarithm, etc.
- In particular, the set of all μ-random sequences has μ-measure 1.

Randomness, Triviality

- $A \subset \mathbb{N}$ is *low* iff $A' \leq_T \varnothing'$, and A is *high* iff $\varnothing'' \leq_T A'$.
- A is *low for ML-randomness* iff each ML-random set is already ML-random relative to A.
- A is *low for K* iff $\exists c \forall x : K(x) \leq K^A(x) + c$.
- $x \in 2^\omega$ is *K-trivial* iff $\exists c \forall n : K(x_{1:n}) \leq K(n) + c$.

Theorem

A is K-trivial $\iff$ A is low for ML-randomness $\iff$ A is low for K.

Some sequences are K-trivial but not computable.
Neither randoms, nor K-trivials, are deep.

Effective Hausdorff dimension can be interpreted as a degree of incompressibility

Definition

A set $A \subset 2^\omega$ has *effective d-dimensional Hausdorff measure* 0, $H^d(A) = 0$, iff there exists a r.e. set $V \subset \mathbb{N} \times 2^{<\omega}$ s.t., for $V_n := \{x \in 2^{<\omega} : \langle n, x\rangle \in V\}$

$$A \subset \bigcup_{x \in V_n} \Gamma_x \quad \text{and} \quad \sum_{x \in V_n} \mu(\Gamma_x)^d = \sum_{x \in V_n} 2^{-\ell(x)d} \leq 2^{-n}$$

Definition (Effective Hausdorff Dimension)

The *effective Hausdorff dimension* of $A \subset 2^\omega$ is defined as

$$\dim_H^1(A) := \inf\left\{d \geq 0 : H^d(A) = 0\right\}$$

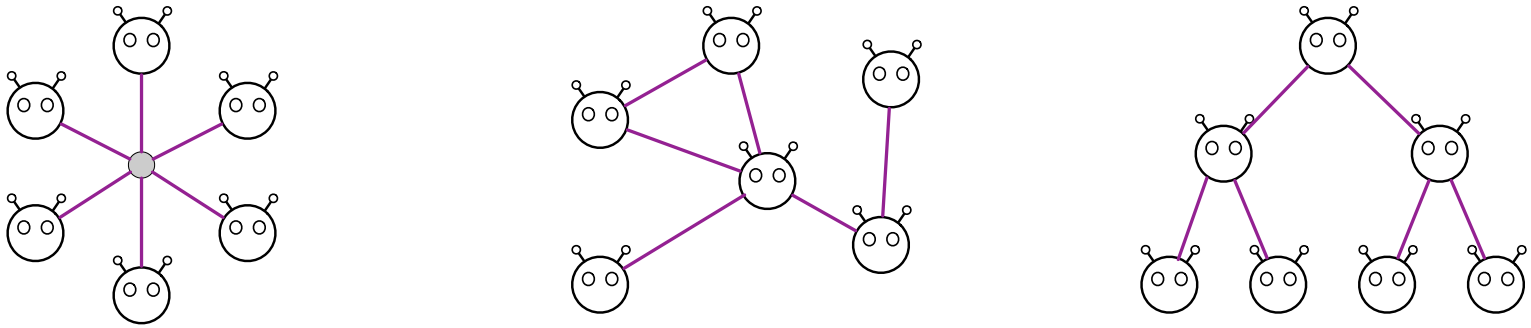
Theorem

$$\dim_H^1(x) = \liminf_{n \to \infty} \frac{K(x_{1:n})}{n}$$

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
   - Kolmogorov Complexity
   - Analogy & Few-Shot Learning
   - Algorithmic Probability
   - A Statistical Mechanical Interpretation of AIT
   - Incompressibility & Incompleteness
   - Algorithmic Randomness
   - Effective Complexity
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

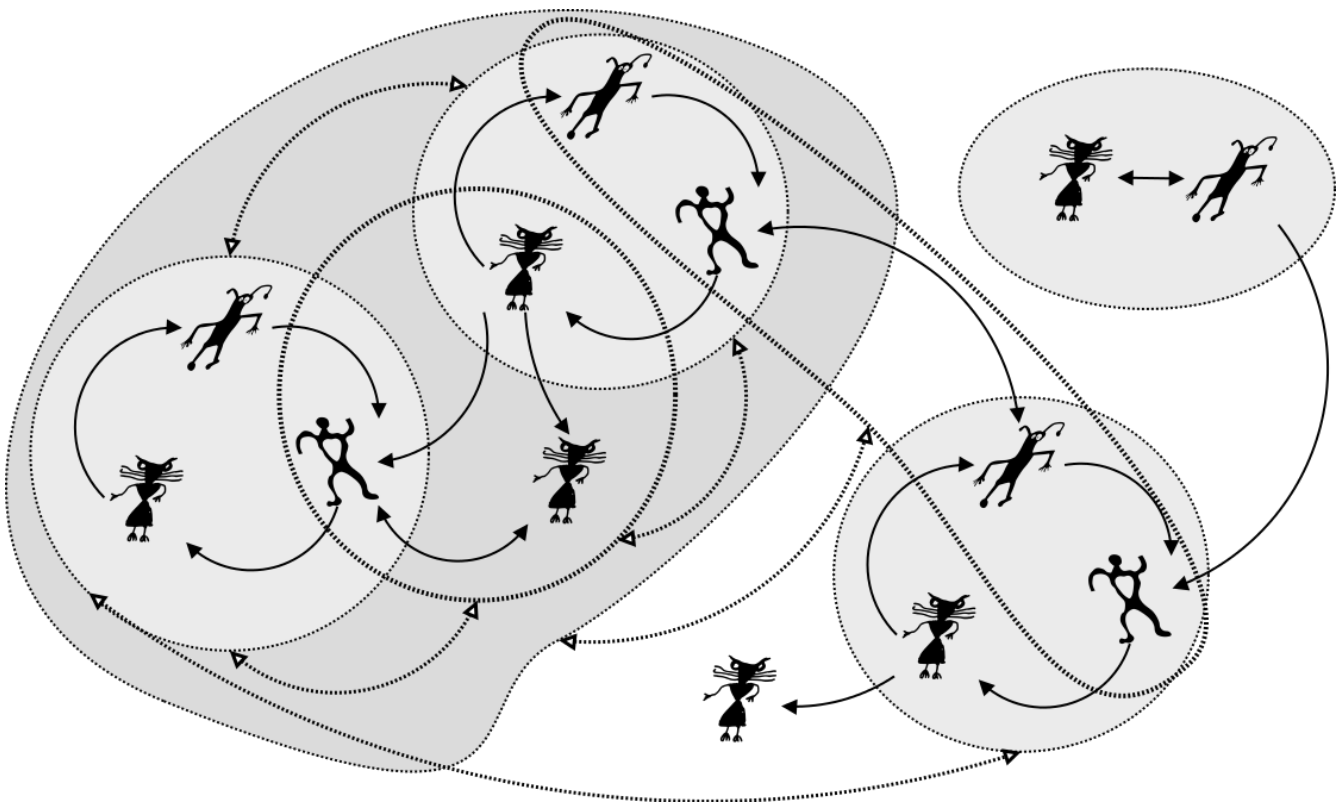
组织形式 vs 信息流动



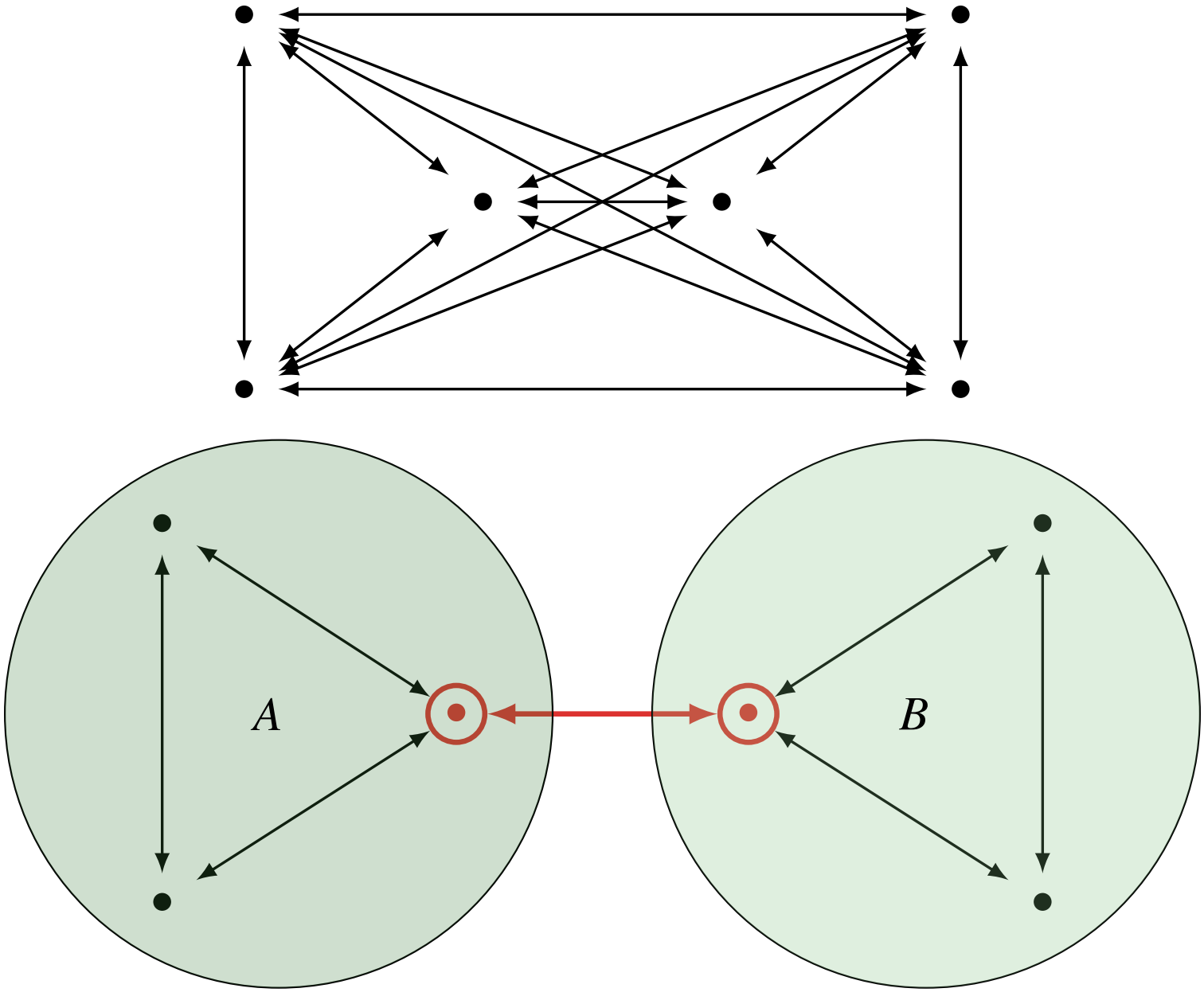
(a) Centralized

(b) Distributed

(c) Hierarchical



耦合 & 解耦　Coupling & Decoupling



高内聚、低耦合 (High cohesion & Low coupling)

What is Complexity?

1. How hard is it to **describe**?
   - Shannon Entropy
   - Kolmogorov Complexity
   - Minimum Description Length
   - Statistical Complexity: the minimum amount of information about the past behavior of a system that is needed to optimally predict the statistical behavior of the system in the future.
   - Fisher Information
   - Renyi Entropy
2. How hard is it to **create**?
   - Computational Complexity
   - Logical Depth
   - Thermodynamic Depth: the Shannon entropy of trajectories leading to the current state.
3. What is its degree of **organization**?
   - Effective Complexity / Sophistication
   - Fractal Dimension
   - Stochastic Complexity
   - Hierarchical Complexity
   - Channel Capacity

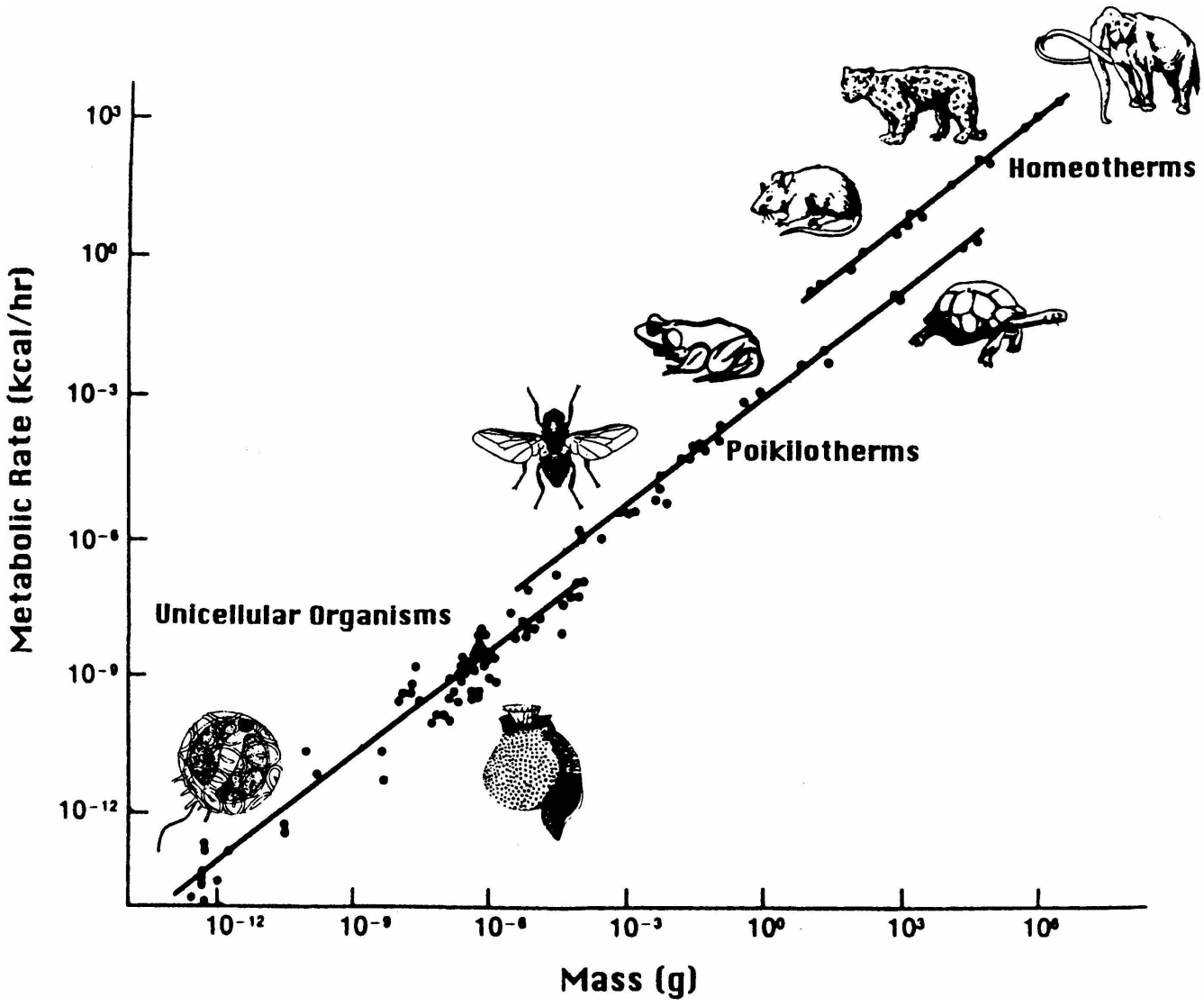
Features of Complex Systems

1. Numerosity: involve many interactions among many components.
2. Disorder and diversity: the interactions are not controlled centrally, and the components may differ.
3. Feedback: the interactions are iterated so that there is feedback from previous interactions.
4. Non-equilibrium: complex systems are open to the environment and are often driven by something external.
5. Spontaneous order and self-organisation: exhibit structure and order that arises out of the interactions among their parts.
6. Nonlinearity: exhibit nonlinear dependence on parameters.
7. Robustness: the structure and function is stable under relevant perturbations.
8. Nested structure and modularity: there may be multiple scales of structure, clustering and specialisation of function.
9. History and memory: often require a very long history to exist.
10. Adaptive behaviour: often able to modify their behaviour depending on the state of the environment and the predictions they make about it.

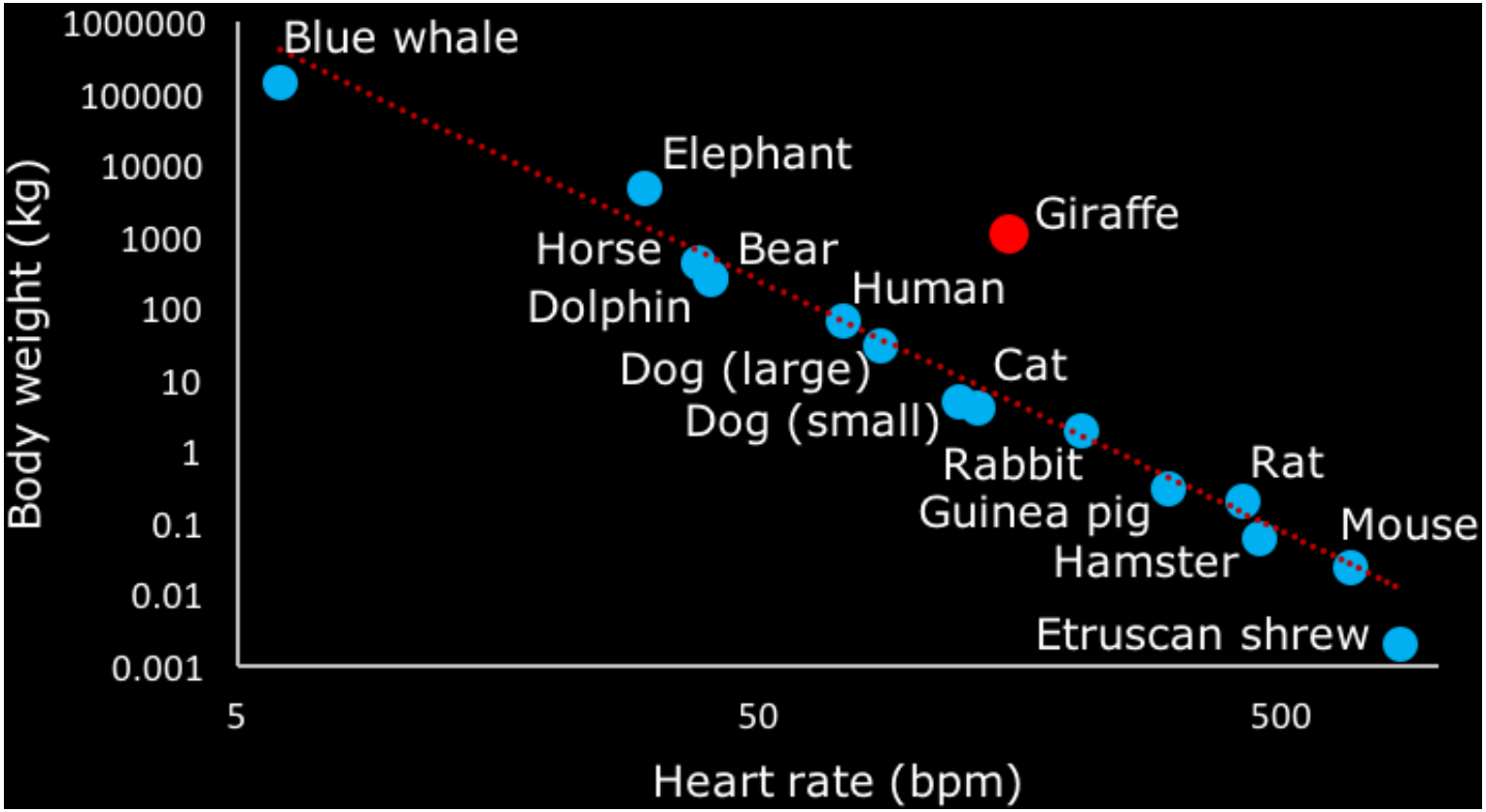
1. 为什么人会衰老? 死亡?
2. 为什么城市不会死?
3. 哥斯拉可能存在吗?
4. 技术奇点真的会来吗?

- 我用两个 5 寸的披萨换你一个 9 寸的披萨你愿意吗?
- 人们想研究致幻剂 LSD 对大象的影响. 但不知道应该注射多少剂量. 已知猫使用 LSD 的剂量是 0.1 毫克, 猫的体重是 1 千克. 大象的体重是 3000 千克. $0.1 \times 3000 = 300$ 毫克不就是大象的剂量吗? 结果大象死了.

复杂系统中存在简单模式



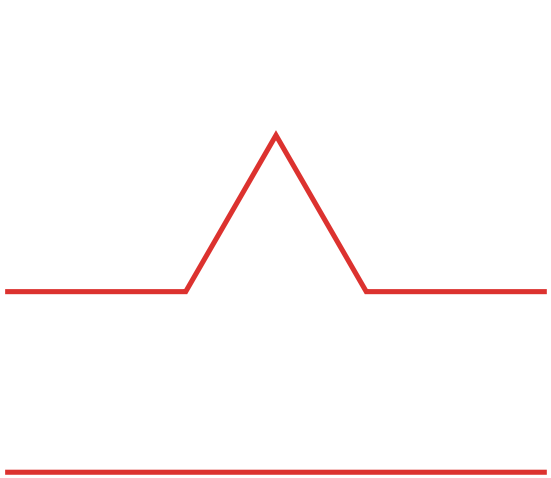
- **克莱伯定律:** 生物代谢率 B 与体重 M 的 $3/4$ 次方成正比 $B \propto M^{3/4}$

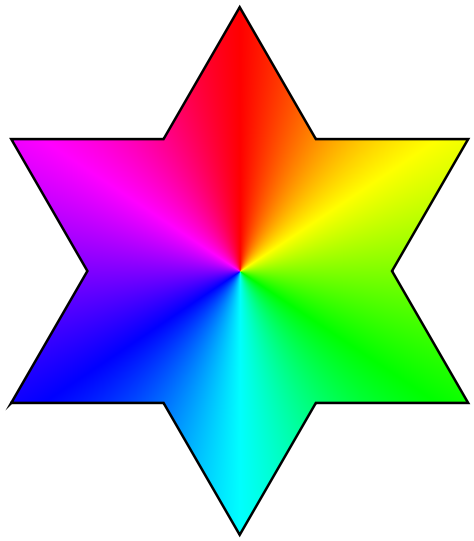


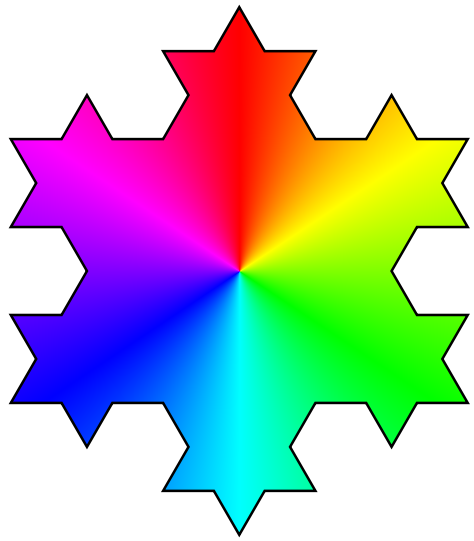
- $B \propto M^{3/4} \implies$ 心率 $\propto B/M \propto M^{-1/4}$
- 寿命 $= \dfrac{\text{受损细胞总数}}{\text{新陈代谢导致的毛细血管细胞磨损率}} \propto \dfrac{M}{B} \propto M^{1/4}$
- 小动物心跳快, 寿命短.
- 几乎所有哺乳动物一生心跳总数: 心率 $\times$ 寿命 ≈ 15 亿次 (而人 25 亿次)
- 为什么分母都是神奇的数字是 4?
- 为了能量的最优输运, 血管是个空间填充的分形结构, 维数是 4.

血管是个空间填充的分形结构

科赫曲线 Koch Curve





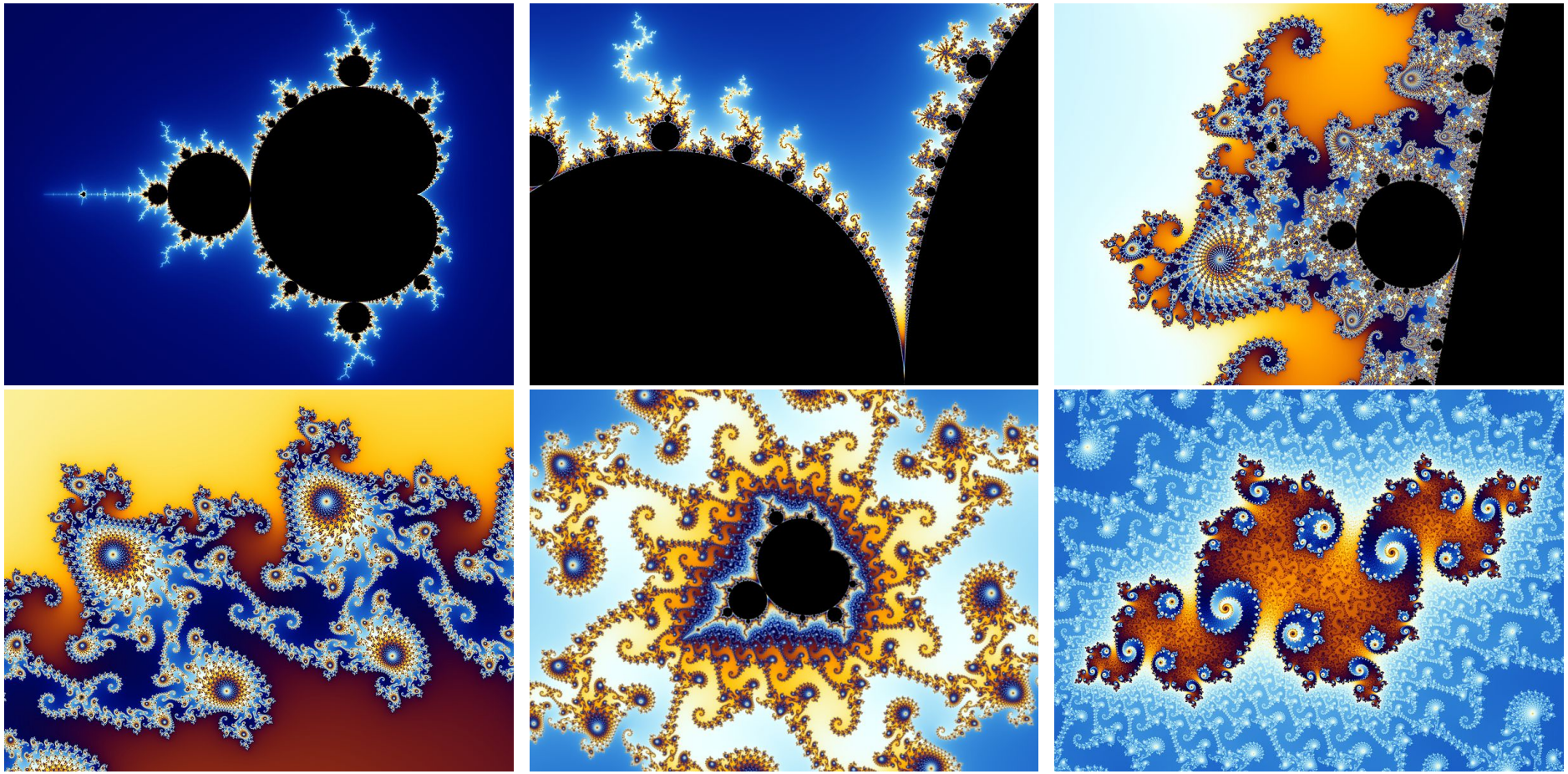


科赫曲线的自相似维数为:

$$D := \frac{\log n}{\log s} = \frac{\log 4}{\log 3} \approx 1.26$$

其中, n 为子块数, s 为缩减因子.

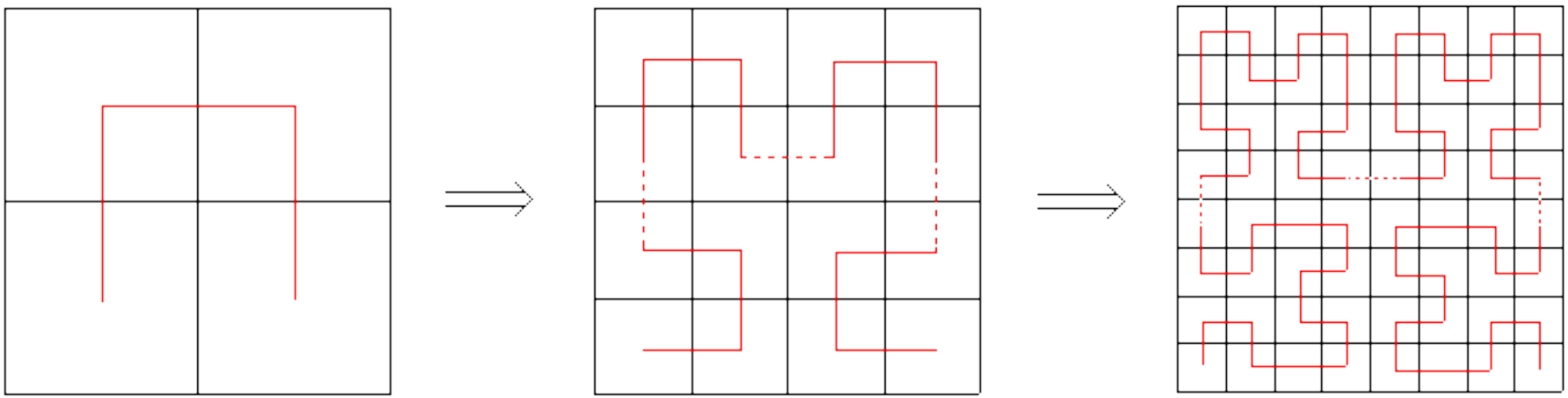
Mandelbrot Set — complex structure from simple rule

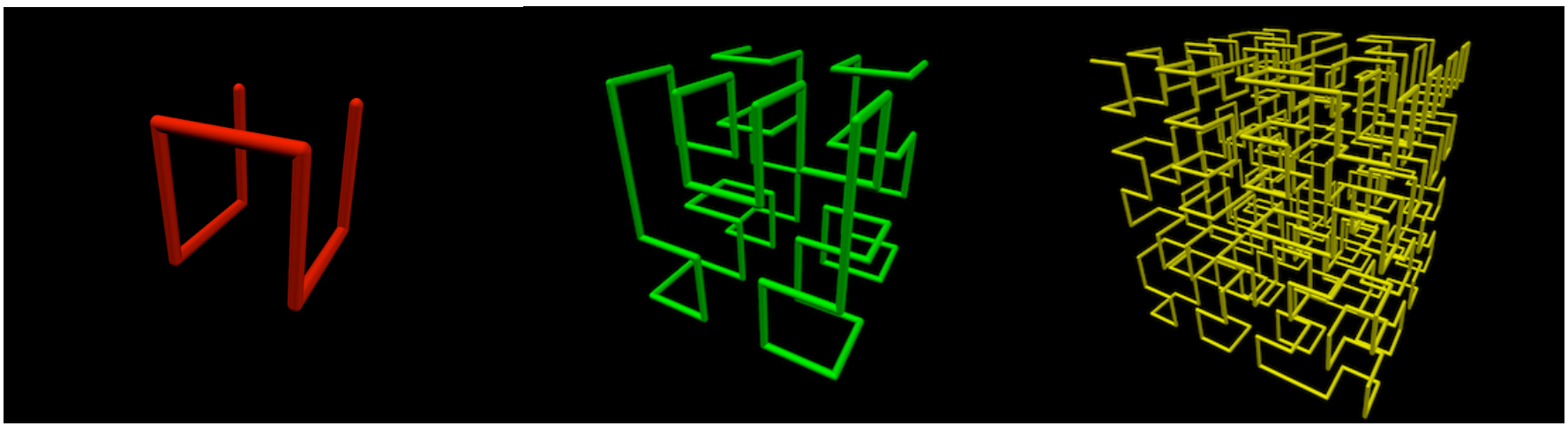


$$z \mapsto z^2 + c$$

- Hausdorff dimension of the boundary of the Mandelbrot set: 2
- Topological dimension of the boundary of the Mandelbrot set: 1

希尔伯特空间填充曲线





Hilbert's Space-Filling Curve

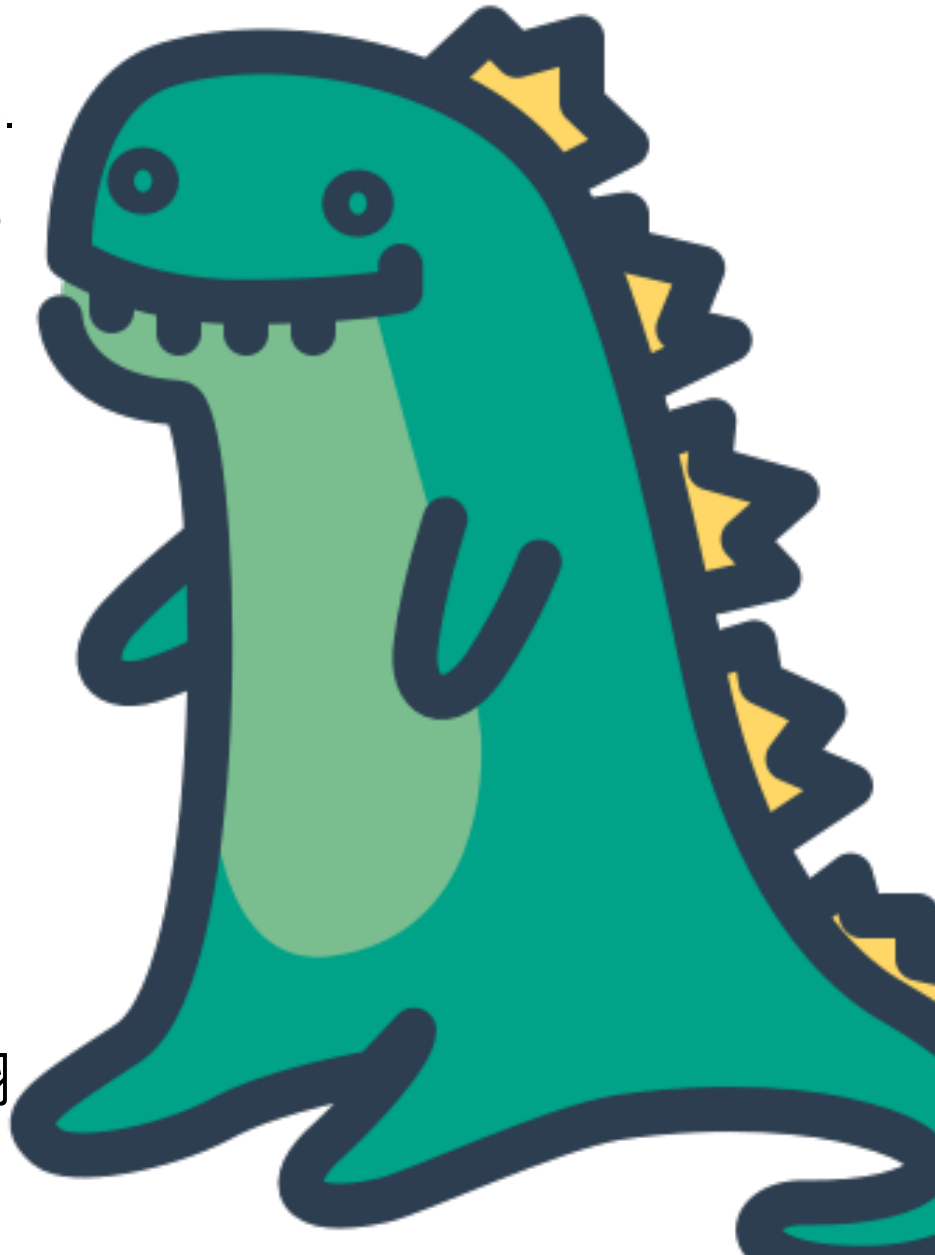
- When we draw h_n, we impose a $2^n \times 2^n$ grids onto the square S. The diagonal of each grid is of length $\sqrt{(2^{-n})^2 + (2^{-n})^2} = 2^{\frac{1}{2}-n}$.
- We define the curve h as the limit of these successive functions $h_1, h_2 \ldots$ s.t. $h(x) = \lim_{n\to\infty} h_n(x)$.
- Each point in S is at most $2^{\frac{1}{2}-n}$ distance away from some point on h_n. So the maximum distance of any point from h is $\lim_{n\to\infty} 2^{\frac{1}{2}-n} = 0$. So h fills space!
- Definition. A curve is a continuous map from unit interval L to unit square S.
- For a point $p \in S$ and $\varepsilon > 0$, there is some n s.t. some grid of the $2^n \times 2^n$ grids on S lies within the circle with centre p and radius ε. let I be the largest open part of L which h_n maps into the relevant grid. Whenever $x \in I$, $h_m(x)$ lies in that same grid, for any $m > n$. So h is continuous.
- Hilbert's curve is continuous everywhere but differentiable nowhere.
- Hausdorff dimension: 2
- Topological dimension: 1

哥斯拉可能存在吗?

1. 体重 (体积) 与身高的立方成正比.
2. 支承力 (腿粗) 与身高的平方成正比.
3. 支撑力与体重的 $2/3$ 次方成正比.

1. 巨型哥斯拉会把自己的腿压断.
2. 小蚂蚁却可以背负起数倍于自己体重的东西.
3. 巨型鲸搁浅会被自己的体重压坏内脏死亡. (水的浮力 $\propto$ 体重)

1. 生物体重翻倍, 药物剂量 (与代谢率成正比 $\propto M^{3/4}$) 不需要翻倍.
2. 城市规模翻倍, 基础设施不需要翻倍 $\propto$ 人口$^{0.85}$, 城市产出却呈超线性 $\propto$ 人口$^{1.15}$.



为什么跳蚤可以跳这么高?

- 跳蚤可以跳 35 厘米高, 与一个普通人的跳跃高度相差不多.
- 跳跃高度取决于腿部弹跳力量大小和体重大小.
- 弹跳力量大小是大腿支撑的应力, 正比于腿部横截面 (正比于体长的平方).
- 体重则正比于体长的立方.
- 假设弹跳力量大小为 f, 跃起过程中腿部移动的距离为 l(正比于体长 L), 动物质量为 m, 跳的高度为 h, 则有

$$fl = mgh$$

- 于是,

$$h \propto L^0$$

- 伽利略: 生物跳跃高度与体长无关.
- 如果将跳跃高度比上体长, 则有

$$\frac{h}{L} \propto L^{-1}$$

动物越小, 相对于自身体长而言, 跳跃高度越大.

复杂性 & 规模

- 公司是体型巨大的生物吗？城市呢？
- 为什么生物和公司会停止生长然后衰老死去？而城市几乎不会死？

$$\text{摄入的能量} \propto M^{3/4}$$
$$\|$$
$$\text{维护修复现有细胞} \propto M$$
$$+$$
$$\text{生长新细胞}$$

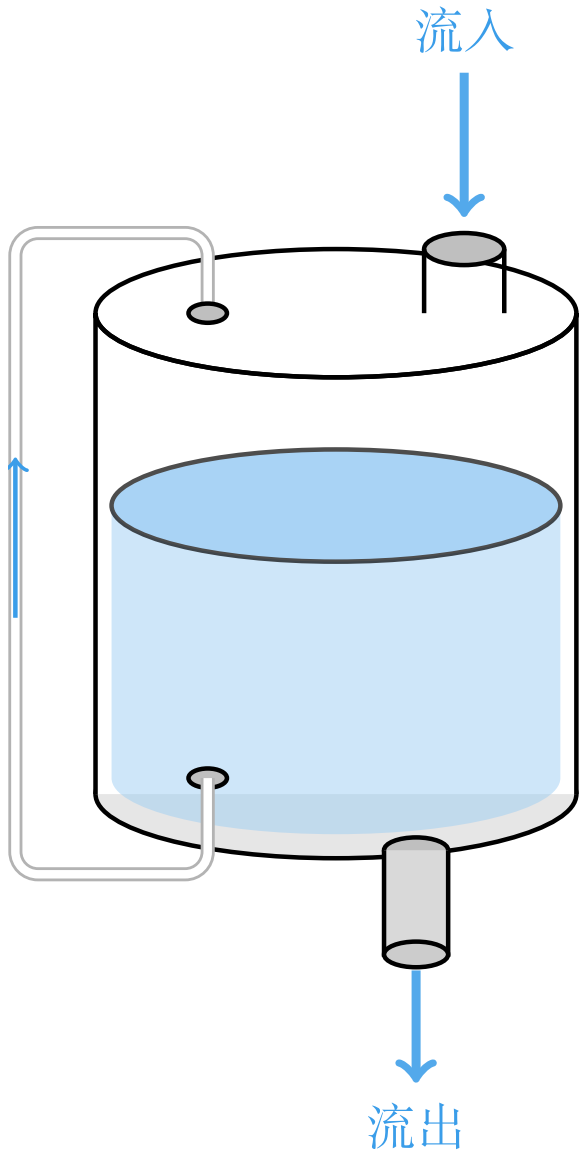
$$\frac{\mathrm{d}M}{\mathrm{d}t} = aM^{3/4} - bM$$

- 生物的代谢呈亚线性规模缩放. 这限制了生命的节奏 — 大动物心率更低, 寿命更长.
- 公司发展早期超线性规模缩放, 然后近似线性规模缩放. 这意味着, 面对市场波动也会比较脆弱.
- 城市的社交互动网络导致了产出超线性规模缩放. 大城市生活节奏更快. 背后的社交网络动力学导致了 “开放式增长”.

水缸

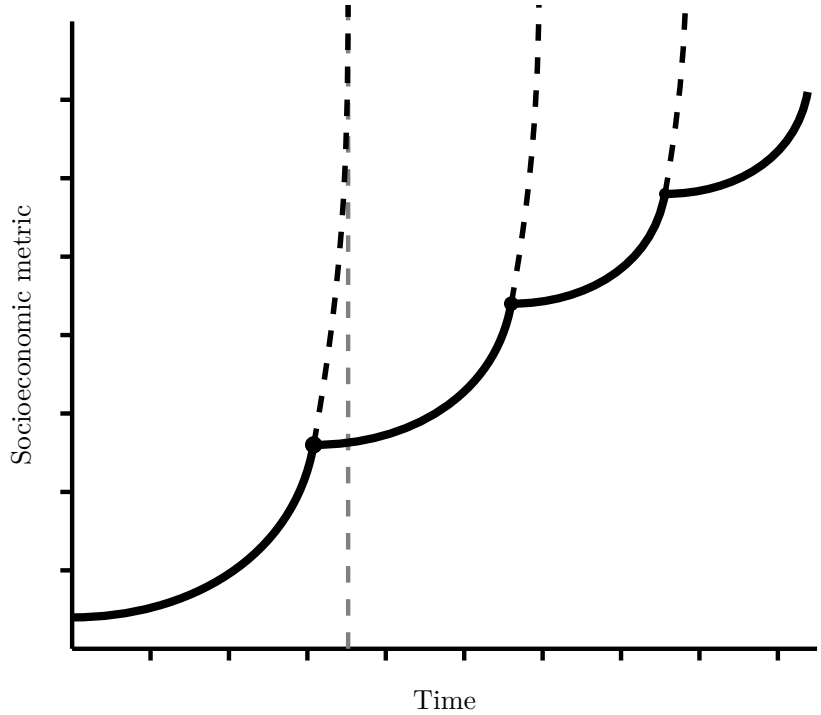
$$\text{摄入的能量} \propto M^{\beta}$$
$$\|$$
$$\text{维护修复现有细胞} \propto M$$
$$+$$
$$\text{生长新细胞}$$

$$\frac{\mathrm{d}M}{\mathrm{d}t} = aM^{\beta} - bM$$



- 生物: 能量流
  - 流入: 摄入的能量
  - 流出: 维护修复自身消耗的能量
  - 水位: 体重增长
- 公司: 现金流
  - 流入: 收入 + 债务 (预支未来)
  - 流出: 成本
  - 水位: 公司扩张

奇点临近



超线性增长使得财富及伴随而来的垃圾、污染、犯罪、疾病 (社会熵) 等以相同的速度增长. 为了持续增长并避免崩溃, 社会必须通过颠覆式创新以 "重置" 增长曲线. 每一次创新又会导致更快的增长速度, 这意味着必须以更短的周期进行颠覆式创新. 颠覆式创新让城市生长超周期性的不断偏离既定的超指数增长轨迹, 进入一个个方程相同 $\frac{\mathrm{d}M}{\mathrm{d}t} = aM^{\beta} - bM$ 但系数 a, b 不同的动力学过程, 仿佛不断地以更短的时间跳到下一台速度更快的跑步机上, 奔向财富和污染都无穷大的奇点.

生物 vs 公司 vs 城市

	生物	公司	城市
规模	体重 (细胞数)	员工数	人口
摄入的能量	代谢率	销售额	产出 (GDP, 专利, 污染)
消耗的能量	细胞维护	成本	基础设施
网络	血管	组织沟通	交通水电

- 虽然城市的发展 (财富、污染) 随人口规模超线性增长, 越大越好, 但城市的发展也要受人的限制.
- 城市再大, 也要确保一个人的上班路程在 1 小时以内. 城市节奏再快, 人的步行速度也有个生理极限. 城市里的连接数再多, 每个人最多也只有 150 个熟人.
- 人与人的连接导致了城市的创新与增长. 人也限制城市. 人也需要适应城市.

城市那么空, 回忆那么凶, 街道车水马龙, 你和谁相拥? 你以为你感慨的是感情, 其实你是在感慨: $Y = cX^k$.

$$\log Y = k \log X + \log c$$

鲸歌的秘密

- 齐普夫定律 (Zipf's law): 在任意语言中, 最常用词汇的出现频率大约是第二常用词的 2 倍. 用公式表示即为 "排名 $\times$ 频率 $\approx$ 常数".
- 齐普夫简洁定律 (Zipf's law of abbreviation): 使用频次越高的词汇, 其时长往往越短.
- 门泽拉特定律 (Menzerath's law): 单词越长, 组成单词的音节就越短; 句子越长, 每个单词的时长就越短.
- 座头鲸和蓝鲸的歌声遵循与人类语言相同的统计规律 — 齐普夫定律和门泽拉特定律.
- 为了应对交流的复杂性, 生物会通过使用更短的基本单元来提高信息传递效率, 实现 "在最短的时间内以最少的能量传递最多的信息".
- ChatGPT4 生成的文本, 甚至生成的虚拟语言, 及其对应的英文翻译, 都遵循齐普夫定律.
- 幂律法则: 城市的人口, 网页的访问, 甚至人群的收入, 类似齐普夫定律的现象普遍存在.

Zipf's Law

N = the number of words

k = rank of a word (if sorted by frequency)

p_k = frequency of a word of rank k

$$p_k = \frac{k^{-s}}{\sum_{n=1}^{N} n^{-s}}$$

- C_k: a certain "cost" of the word of rank k
- the average cost per word $C := \sum_k p_k C_k$
- the entropy $H := -\sum_k p_k \log p_k$
- $T := \frac{C}{H}$
- $\text{minimize}\, T \implies p_k \propto 2^{-\frac{C_k}{T}}$
- if $C_k = \log k$, then $p_k \propto k^{-\frac{1}{T}}$

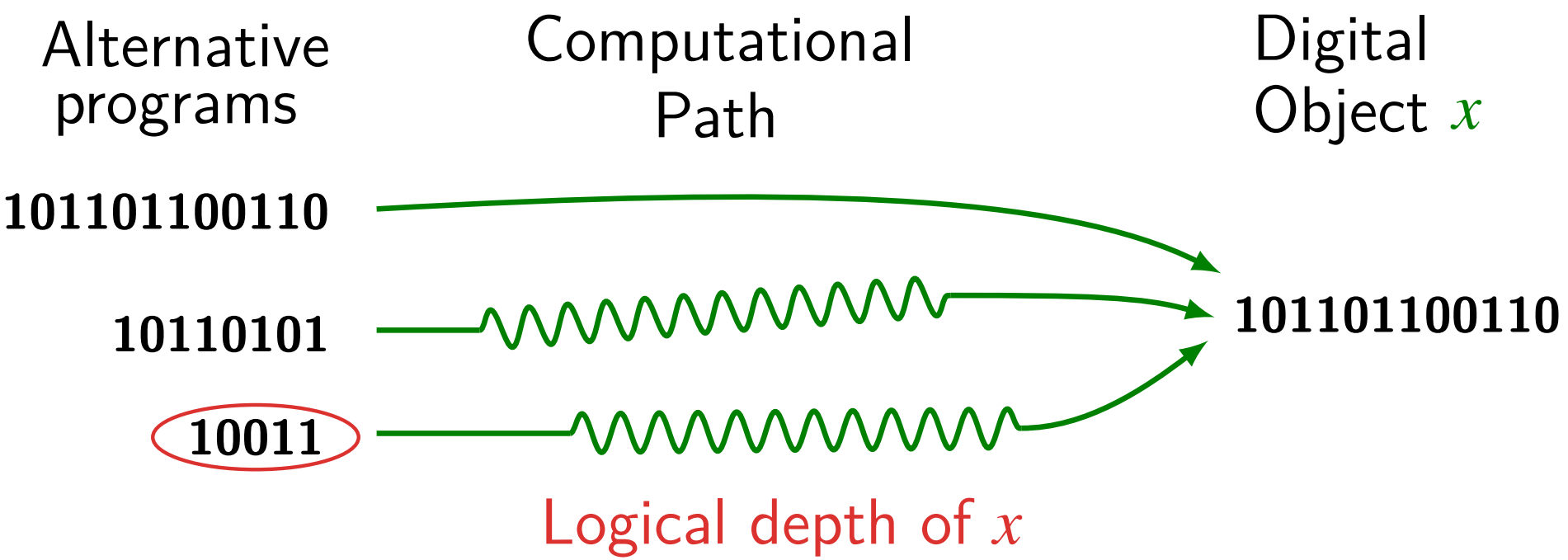
Remark:

- the complexity of words is related to their frequency: $K(x) \approx -\log p_x$.
- the complexity of words can be inferred from their rank: $K(x) \approx \log k_x$.

$$p_x \approx c \cdot k_x^{-1}$$

逻辑深度 — 奥卡姆剃刀

Logical Depth — Charles H. Bennett 1943–



逻辑深度 Logical Depth

Definition (Logical Depth)

The logical depth of x at a significance level b is

$$\mathrm{depth}_b(x) := \min\left\{t : U^t(p) = x \ \ \& \ \ \ell(p) - K(x) \leq b\right\}$$

We say x is *shallow* iff $\mathrm{depth}_b(x) \stackrel{+}{\leq} \ell(x)$.

- Crystal is shallow.
- Gas is also shallow.
- A math book is deep.
- Life is deep.

  "If people do not believe that mathematics is simple, it is only because they do not realize how complicated life is." — *John von Neumann*
- $\chi_{1:\infty}$ is deep, where $\chi_i := [\![\varphi_i(\epsilon)\downarrow]\!]$.
- $\Omega := \sum_{p:U(p)\downarrow} 2^{-\ell(p)}$ is shallow.

"A structure is deep, if it is superficially random but subtly redundant."

— *Charles H. Bennett*

Remark: Ω vs $\chi_{1:\infty}$

- 虽然 $\chi_{1:\infty} \equiv_T \Omega$, 都编码了停机信息, 但 $\chi_{1:\infty}$ 可压缩.
- 计算 $\chi_{1:n}$ 的最短程序的长度只需 $\log n$, 但计算时间超出了任何可计算函数的界限.
- 计算 $\Omega_{1:n}$ 的最短程序的长度不小于 n, 计算时间是 $O(n)$.
- 借助 $\Omega_{1:n}$, 可以写一个程序判定长度不超过 n 的所有程序是否停机, 但此程序的运行时间却超出了任何可计算函数的界限.

Charles H. Bennett 论 Ω

- 古往今来，哲学家和神秘主义者一直在寻找一把开启宇宙智慧的钥匙 — 一段有限的公式或文本. 《圣经》、《古兰经》、犹太教 Cabala... 都曾被视为这样的典籍.
- 它隐匿难寻，即使寻得也难以理解，使用起来充满危险 — 它的回答深邃广博，远超求索者原初的问题.
- 如今，利用典籍占卜俗事被视为迷信. 然而，科学也在追求自己的 Cabala — 一套能够解释一切现象的简洁自然律. 数学中没有一套公理能够证明所有真命题，目标或许是只公理化 “有趣的” 真命题.
- Ω 在极小的空间蕴藏着极大的智慧. 它的可以写在一张纸上的前段字符，就包含了比整个宇宙所能记录下的还要多的数学问题的答案.
- Ω 不可计算. 讽刺的是，它却可能由随机过程偶然生成，如抛硬币的序列、或山崩时滚石的划痕. Ω 的前几位说不定已经记录在了宇宙的某个地方. 可惜没有人能发现并验证它的真伪，更无法使用它.

Charles H. Bennett 论伦理 ☺

Problem: 为什么要保护生命、书本、热带雨林.....?

- 用计算复杂性来识别有价值和值得保护的东西.
- 所谓善行, 就是去创造那些极具逻辑深度的事物. 它们内部镌刻着漫长因果历史的证据, 正因如此, 一旦损毁, 极难重塑.

  A good deed would then be one that makes something that is logically deep, containing internal evidence of a long causal history, so that it would be hard to replace if destroyed.

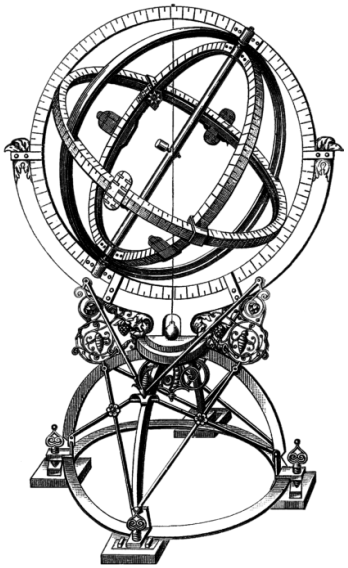
  — *Charles H. Bennett*

有效复杂性 Effective Complexity

$$\mathrm{EC}(x) := K\left(\operatorname*{argmin}_{\mu} \left\{ K(\mu) + \log \frac{1}{\mu(x)} \right\} \right)$$

low Kolmogorov part ("laws") + potentially indefinitely complex part

地心说
托勒密 $\rightsquigarrow$ 日心说
哥白尼 $\rightsquigarrow$ 三定律
开普勒 $\rightsquigarrow$ 万有引力
牛顿 $\rightsquigarrow \cdots$



- 面对 "现象", 抓住 "规律", 悬置 "随机"
- 让 "现象" 更 "规律", 让 "随机" 不 "随机"
- 但无法确定自己不是 "地主家的傻儿子"
- 努力超越 "地主家的傻儿子"
- 老子: 为学日益, 为道日损. 损之又损, 以至于无为. 无为而无不为.

$$F = G\frac{m_1 m_2}{r^2}$$

$$F = ma$$

Remarks: 玩 “抽象” ☺

- 抓 “规律”, 舍 “细节”

$$x \sim_{\mathcal{M}} y \iff \operatorname*{argmin}_{\mu\in\mathcal{M}} \left\{ K(\mu) + \log \frac{1}{\mu(x)} \right\} = \operatorname*{argmin}_{\mu\in\mathcal{M}} \left\{ K(\mu) + \log \frac{1}{\mu(y)} \right\}$$

- 物理: 理论
- CS: 数据标注
- 中哲: 正名 (开启民智, 就是教大家正确标注数据)
- 数学: 一门给不同事物赋予相同名字的艺术. — 庞加莱

什么是 “顿悟”?

令 K^t 是资源受限的科尔莫哥洛夫复杂性; D_x 是 “运行时间-复杂性” 空间; P_x 是 “模型-数据残差” 空间.

$$K^t(x) := \min_p \left\{ \ell(p) : U^t(p) = x \right\}$$

$$D_x := \{(t, c) : K^t(x) \le c\}$$

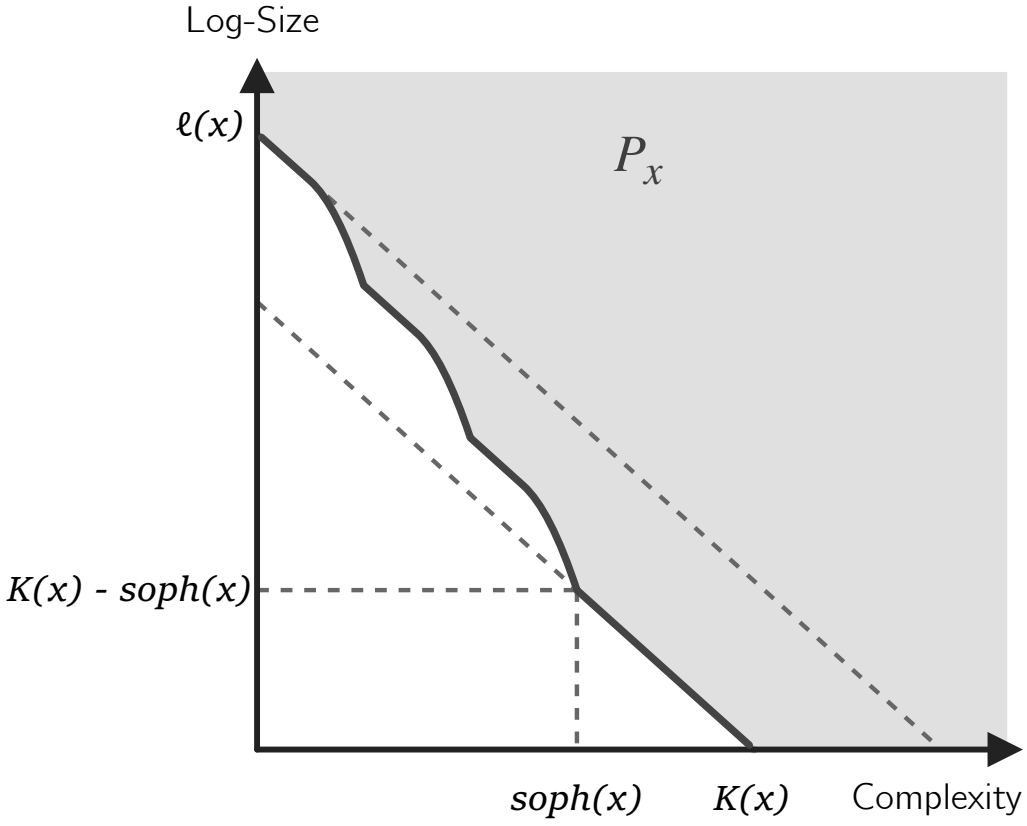
$$P_x := \left\{ (i, j) : \exists \mu \left[K(\mu) \le i \ \& \ \log \frac{1}{\mu} \le j \right] \right\}$$

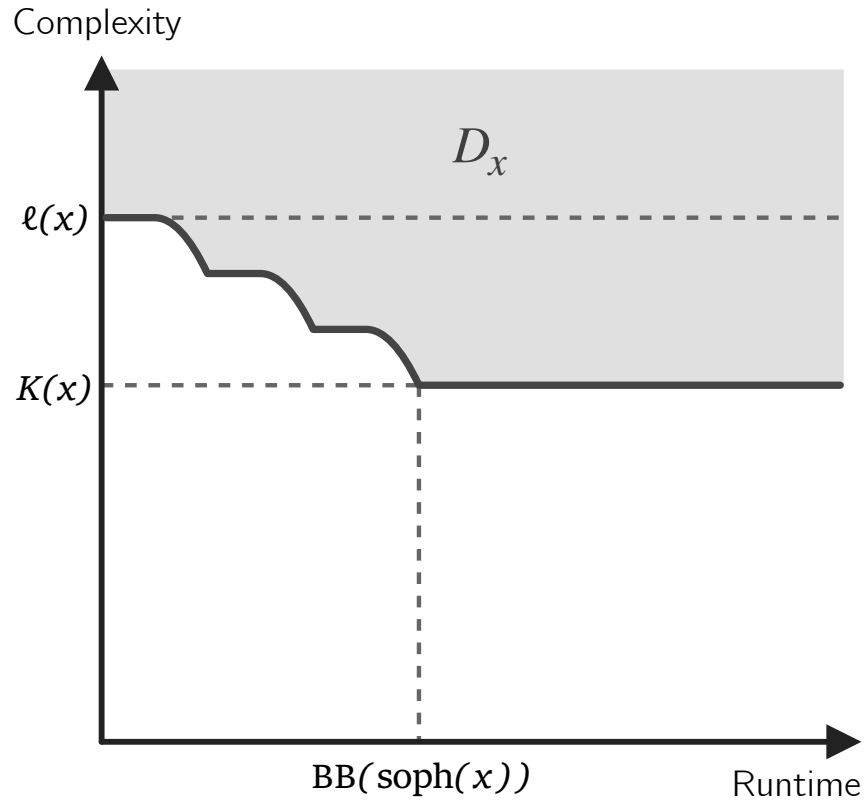
- $P_x \sim D_x$ 对应关系 (忽略 $O(\log(\ell(x)))$ 级别的差别)

$$(i, j) \mapsto (\mathrm{BB}(i), i + j)$$

  其中忙海狸函数 $\mathrm{BB}(n)$: 长度为 n 的停机程序能运行的最大步数.
- 因为 $\mathrm{BB}'(i)$ 的增长速度超过任何可计算函数, 所以当 $\frac{\mathrm{d}K^t}{\mathrm{d}t}$ 发生微小的 “骤变”, 在 P_x 空间中, 会被 $\mathrm{BB}'(i)$ 急剧放大.

$$\frac{\mathrm{d}K^t}{\mathrm{d}i} = \frac{\mathrm{d}K^t}{\mathrm{d}t} \cdot \frac{\mathrm{d}t}{\mathrm{d}i} = \frac{\mathrm{d}K^t}{\mathrm{d}t} \cdot \mathrm{BB}'(i)$$





- 顿悟时刻: 在 P_x 空间, 斜率跌破 -1 意味着, 随着模型复杂性的微小增加, 换来了数据描述长度的大幅下降; 而斜率等于 -1 意味着, 增加 1bit 的模型复杂性只能减少 1bit 的数据残差, 总描述长度不变.
- 在 D_x 空间, 投入更多的计算时间, 换来了 K^t 的非平凡下降.
- 对于 "逻辑深" 的对象, 即使对于很大的 t, K^t 仍会持续下降.
- Herrmann & Schmidhuber: 随着一次 "顿悟" 后停滞时间的延长, 未来发生下一次 "顿悟" 的概率呈指数级衰减.

单向函数[10] vs K^t 复杂性

$$K(x,y) \overset{+}{=} K(x) + K(y \mid x^*) \overset{+}{=} K(y) + K(x \mid y^*) \overset{+}{=} K(y,x)$$

但是, 如果存在一个正向计算容易、逆向计算困难的单向函数 $f : x \mapsto y$, 则时间有界的 "信息对称性" 破裂.

$$K^t(x) + K^t(y \mid x^*) \neq K^t(y) + K^t(x \mid y^*)$$

[10] 如果 $P = NP$, 则不存在单向函数.

Effective Complexity

$$\delta(x \mid A) := \log|A| - K(x \mid A) \qquad \text{[randomness deficiency]}$$

$$\delta(x \mid \mu) := \log \frac{M(x)}{\mu(x)} \qquad \text{[}\mu\text{-randomness deficiency]}$$

$$\beta_x(k) := \min_A \{\delta(x \mid A) : x \in A \ \ \& \ \ K(A) \le k\} \qquad \text{[Best-Fit]}$$

$$h_x(k) := \min_A \{\log|A| : x \in A \ \ \& \ \ K(A) \le k\} \qquad \text{[Kolmogorov structure function / ML]}$$

$$h_x(k) := \min_\mu \{-\log \mu(x) : K(\mu) \le k\} \qquad \text{[Kolmogorov structure function / ML]}$$

$$A^*(x) := \iota_A \Big[x \in A \ \ \& \ \ K(A) = \mu k \big[k + h_x(k) \stackrel{+}{=} K(x) \big] \Big] \qquad \text{[Kolmogorov minimal sufficient statistic]}$$

$$\lambda_x(k) := \min_A \{K(A) + \log|A| : x \in A \ \ \& \ \ K(A) \le k\} \qquad \text{[MDL]}$$

$$\lambda_x(k) := \min_\mu \{K(\mu) - \log \mu(x) : K(\mu) \le k\} \qquad \text{[MDL]}$$

$$\Delta(x \mid A) := K(A) + \log|A| - K(x) \qquad \text{[discrepancy]}$$

$$\mathrm{soph}_c(x) := \min_A \{K(A) : \Delta(x \mid A) < c\} \qquad \text{[sophistication]}$$

$$\mathrm{csoph}(x) := \min_A \{K(A) + \Delta(x \mid A)\} \qquad \text{[coarse sophistication]}$$

$$\Sigma(\mu) := K(\mu) + H(\mu) \qquad \text{[total information]}$$

$$\mathcal{E}_{\delta,\Delta}(x \mid \mathcal{M}) := \min_{\mu \in \mathcal{M}} \left\{ K(\mu) : \Sigma(\mu) - K(x) \le \Delta \ \ \& \ \ \mu(x) \ge 2^{-H(\mu)(1+\delta)} \right\} \qquad \text{[effective complexity]}$$

$$\mathcal{E}_{\delta}(x \mid \mathcal{M}) := \min_{\mu \in \mathcal{M}} \left\{ K(\mu) + \Sigma(\mu) - K(x) : \mu(x) \ge 2^{-H(\mu)(1+\delta)} \right\} \qquad \text{[coarse effective complexity]}$$

How Logical Depth Changes Compared to Effective Complexity?

Theorem (Effective Complexity and Logical Depth [AMS10])

There is a global constant $c \in \mathbb{N}$ *such that: for any strictly increasing recursive function* $f : \mathbb{N} \to \mathbb{N}$, *if* $K(x) + k \geq C(x) + K(C(x))$ *and*

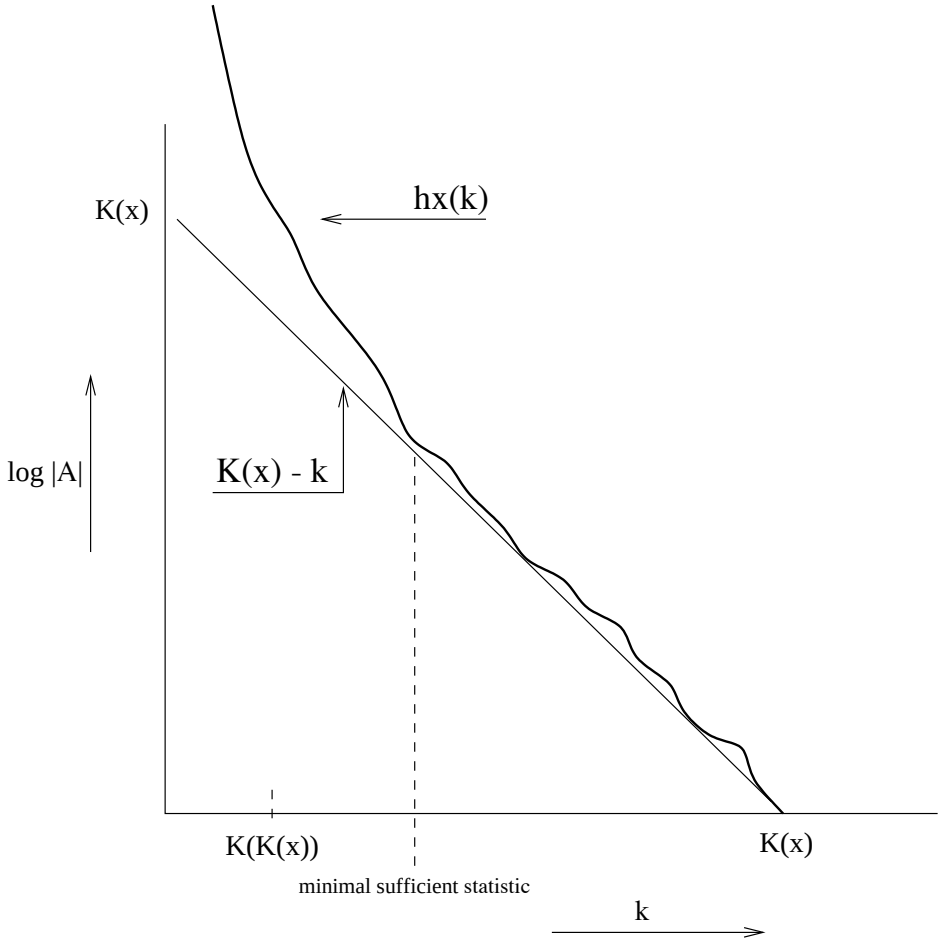
$$\mathcal{E}_{\delta, k+b+K(b)+K(f)+c+1}(x) > K(C(x)) + K(b) + K(f) + c$$

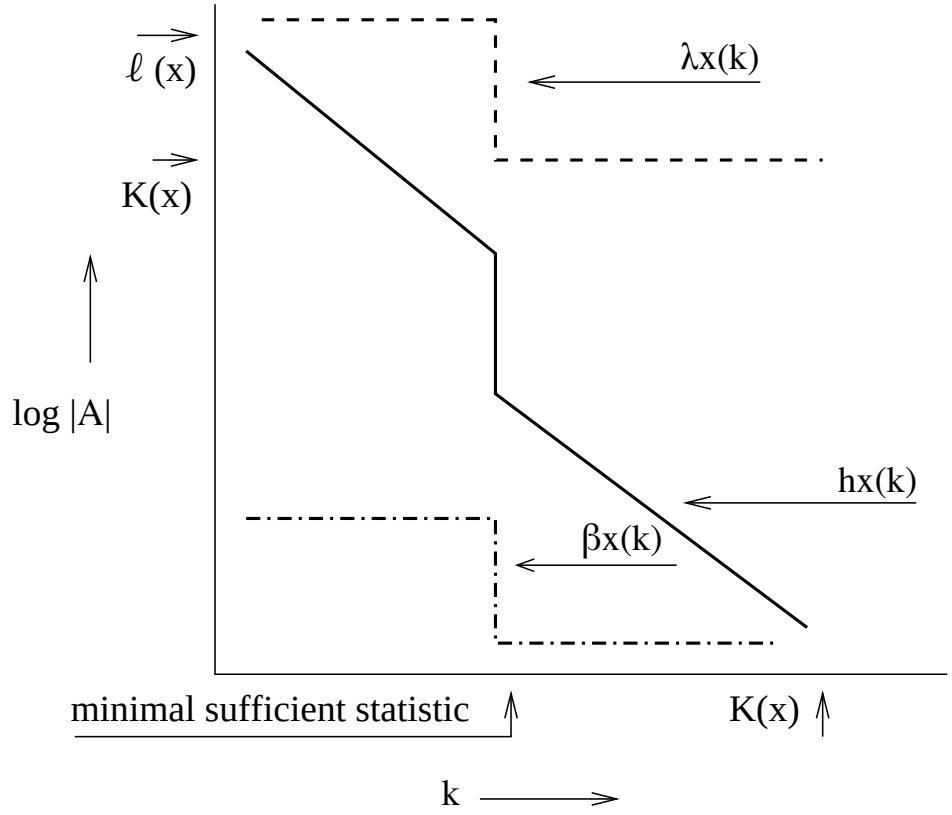
for some arbitrary $\delta \geq 0$ *and* $b \in \mathbb{N}$, *then*

$$\mathrm{depth}_b(x) > f(C(x))$$

Phase Transition

- For small values of effective complexity also logical depth takes small values.
- When effective complexity crosses a threshold value logical depth suddenly jumps to extremely large values.


K(x)
hx(k)
log |A|
K(x) - k
K(K(x))
K(x)
minimal sufficient statistic
k


ℓ (x)
λx(k)
K(x)
log |A|
hx(k)
βx(k)
minimal sufficient statistic
K(x)
k

Theorem

For x and k,

$$\lambda_x(k) \leq h_x(k) + k \stackrel{+}{\leq} \lambda_x(k) + K(k)$$

For k with $0 \leq k \leq K(x) - O(\log \ell(x))$,

$$\beta_x(k) + K(x) \stackrel{+}{\leq} \lambda_x(k)$$

$$\lambda_x(k + O(\log \ell(x))) \leq \beta_x(k) + K(x)$$

In other words, the equality

$$\beta_x(k) + K(x) = \lambda_x(k) = h_x(k) + k$$

holds within logarithmic additive terms in argument and value.

Sophistication and Computational Depth

Theorem

$$\operatorname{csoph}(x) = \min_c \{\operatorname{soph}_c(x) + c\}$$

$$K^t(x) \coloneqq \min_p \{\ell(p) : U^t(p) = x\} \quad \text{(Time-bounded Kolmogorov Complexity)}$$

$$\operatorname{depth}^t(x) \coloneqq K^t(x) - K(x) \quad \text{(Basic Computational Depth)}$$

$$\operatorname{depth}_{\mathrm{BB}}(x) \coloneqq \min_t \{\operatorname{depth}^t(x) + K(t)\} \quad \text{(Busy Beaver Computational Depth)}$$

Theorem

$$|\operatorname{csoph}(x) - \operatorname{depth}_{\mathrm{BB}}(x)| \leq O(\log \ell(x))$$

Intrinsic Utility — Zurek's Physical Entropy [Zur89]

Definition (Physical Entropy)

Physical entropy $S(d)$ of a microstate d is the sum of the conditional Shannon entropy $H_d := -\sum_k P(k \mid d) \log P(k \mid d)$ and of the Kolmogorov Complexity $K(d)$.

$$S(d) := H_d + K(d)$$

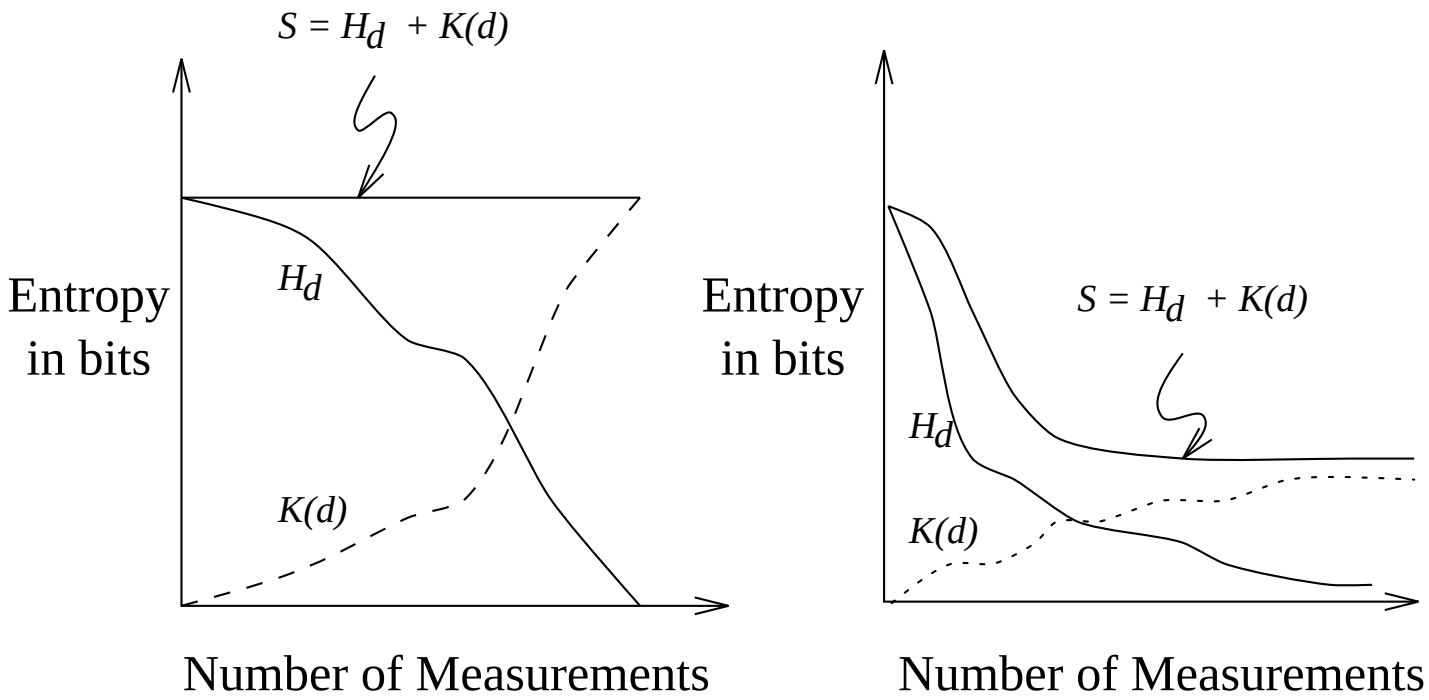


Figure: random vs regular microstate

Gács' Algorithmic Entropy

Definition (Algorithmic Entropy)

- *coarse-grained algorithmic entropy* of a cell Γ with respect to μ

$$H_\mu(\Gamma) := \log \mu(\Gamma) + K(\Gamma \mid \mu)$$

- *fine-grained algorithmic entropy* of $x \in 2^\omega$ with respect to μ

$$H_\mu(x) := \inf_n \left\{ \log \mu(x_{1:n}) + K(x_{1:n} \mid \mu) \right\}$$

- $-H_\mu(x)$ is a universal ML-test: x is μ-ML-random iff $H_\mu(x) > -\infty$.
- The fine-grained algorithmic entropy of a microstate can be approximated by the coarse-grained algorithmic entropies of successively smaller cells containing it.

$$H_\mu(x) = \inf_n \left\{ H_\mu(\Gamma_{x_{1:n}}) \right\}$$

Compression is all you need: Modeling Mathematics

我第一次走进微分方程的课堂，教授在黑板上写下：
令 Ω 为向量丛截面的芽层....
Let Ω be the sheaf of germs of sections of a vector bundle....

— *Michael Freedman*

- ☹ 什么是向量丛?
- ☹ 什么是截面?
- ☹ 什么是芽?
- ☹ 什么是层?

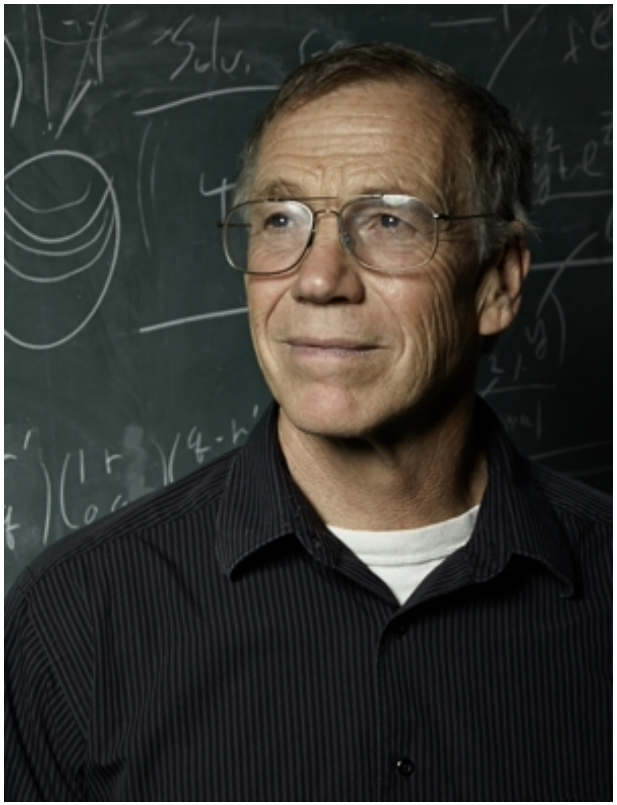


Figure: Michael Freedman

Compression is all you need: Modeling Mathematics

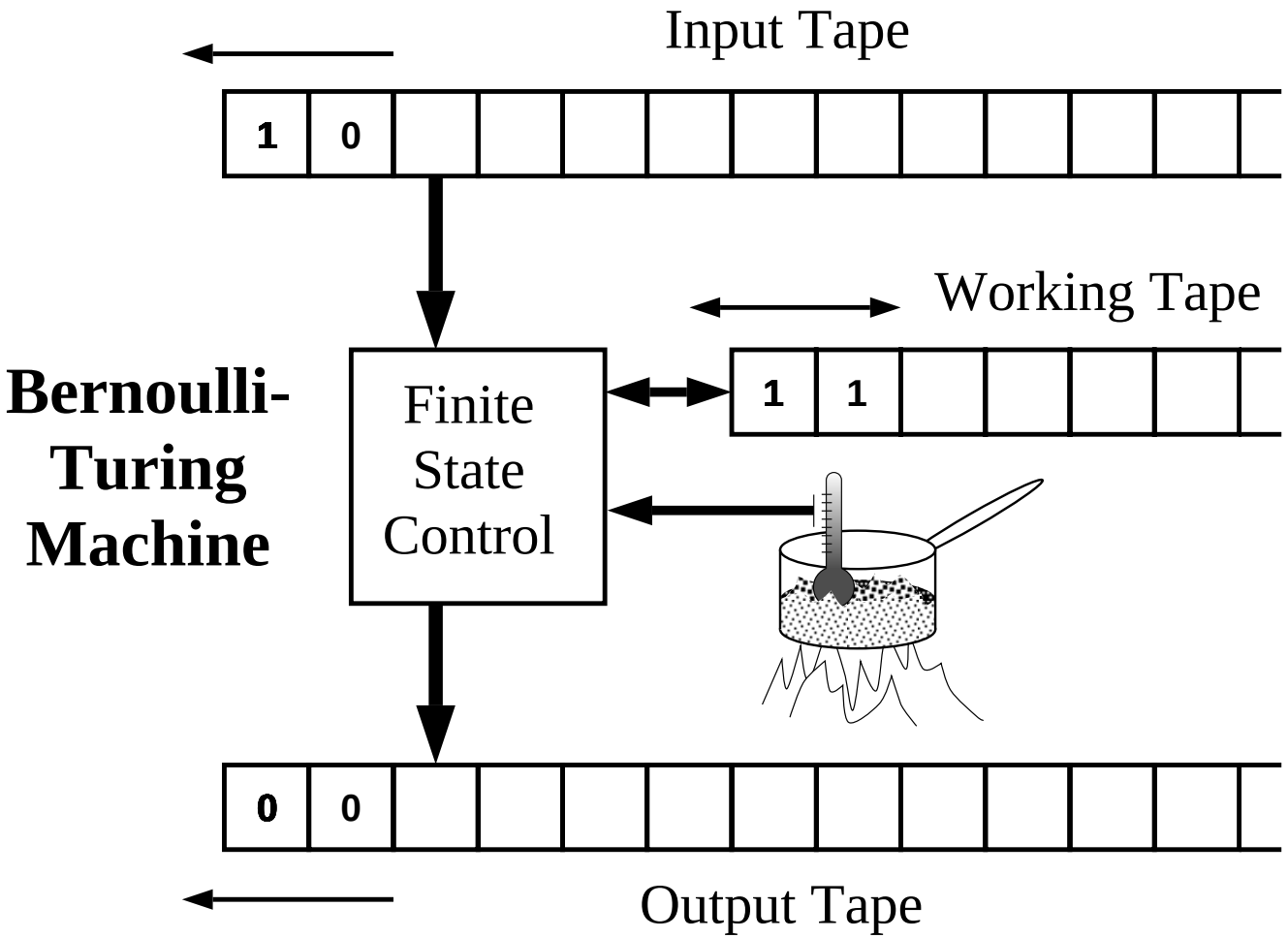
- 在基于 Lean 语言的数学库 Mathlib 中, 一个定理
  AlgebraicGeometry.Scheme.exists_hom_hom_comp_eq_comp_of_locallyOfFiniteType
  及其证明, 封装后的长度大约只有 600 行.
- 如果将它的所有层级定义完全展开为最底层的 "原始符号", 长度约 10^{104} 个符号.
- 数学是 "压缩" 的艺术.
  1. 还原压缩 reductive compression: 命题及其证明的展开长度/命题及其证明的封装长度
  2. 演绎压缩 deductive compression: 封装的证明长度/封装的命题长度
- 还原压缩率高, 意味着可以通过极短且极快 (线性时间) 的方式重构数学结构. (Kolmogorov 复杂度低且逻辑深度低)
  — 但 Kolmogorov 复杂度低不一定意味着还原压缩率高, 比如 π.
  1. reductive compression: 局部压缩, 重复模式替换
  2. Kolmogorov complexity: 全局压缩, 程序生成对象

Remark: 通常的布尔 SAT 命题 φ 的还原压缩率很低. 若 $P \neq NP$, 则 φ 的证明长度不短, 演绎压缩率不低.

Compression is all you need: Modeling Mathematics

数学如高耸入云的柳树.
重要定理就像是低垂的叶子,
通过漫长的下垂过程
(层层抽象),
变得触手可及
(将高不可攀的真理压缩回人
的认识带宽内).
数学家是树下的顽童,
跳跃触摸低垂的叶子.

Crutchfield: Statistical Complexity 统计复杂度



$$C_\mu(x) := \min_p \{\ell(p) : \mathtt{BTM}(p) = x\}$$

where Bernoulli-Turing Machine (BTM) is the UTM with a source of randomness.

Crutchfield: Statistical Complexity 统计复杂度

- 香农熵率:

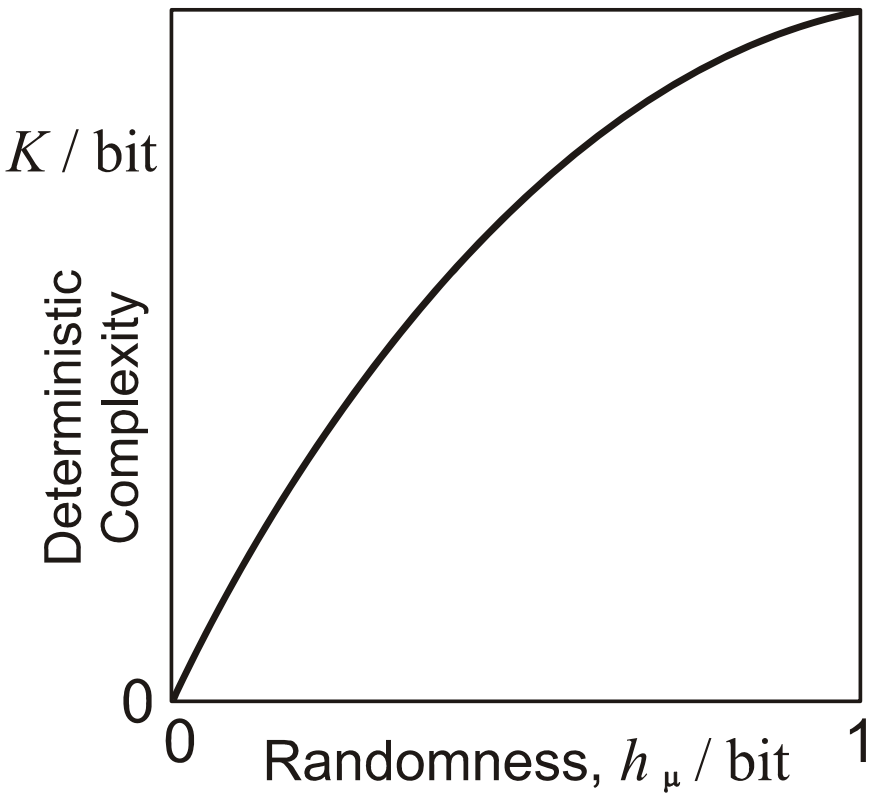
$$h_\mu := \lim_{t\to\infty} \frac{H(X_{1:t})}{t} = \lim_{t\to\infty} H(X_t \mid X_{<t}) = \lim_{t\to\infty} \frac{\mathbb{E}[K(x_{1:t})]}{t}$$

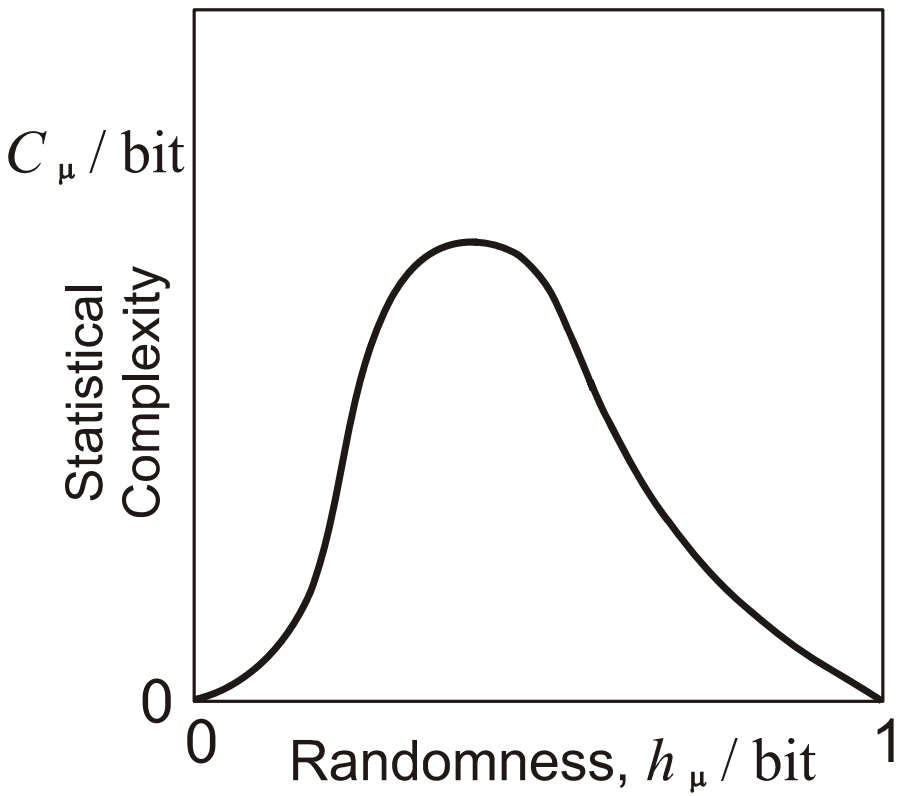
  - 完全有序: $h_\mu = 0$
  - 完全随机: $h_\mu = 1$
- Agent 把观测数据的某一部分看作是和规律无关的随机噪声.

$$C_\mu(x) = K(x) - h_\mu \cdot \ell(x)$$

Complexity vs Randomness

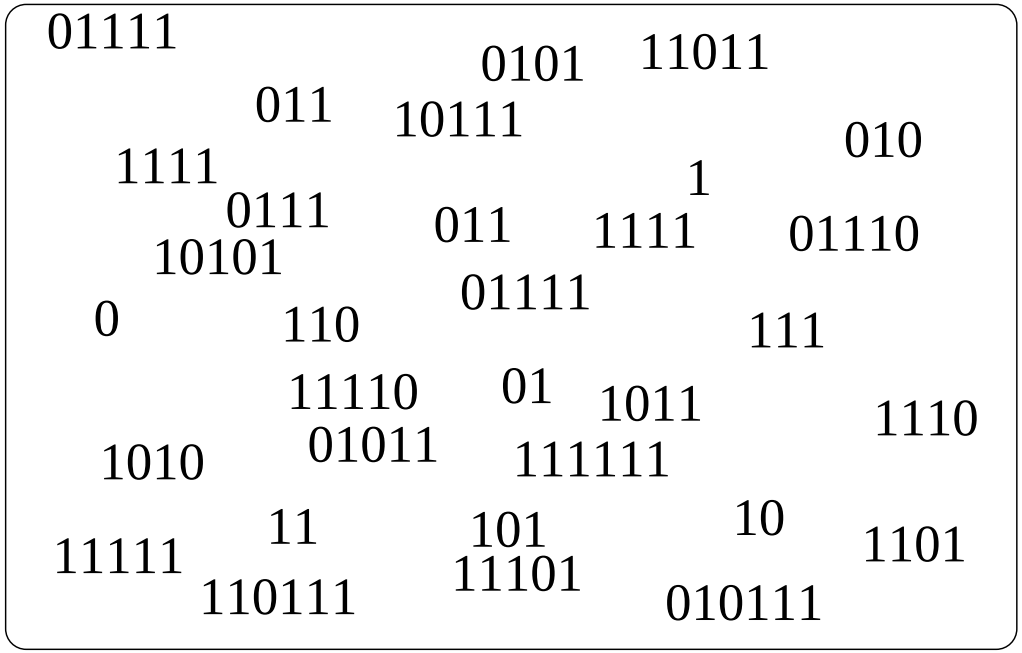
Edges of Chaos?





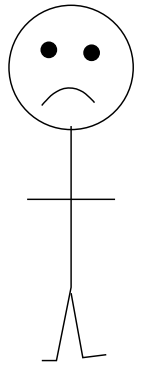
What do you Need to Remember in Order to Predict?

Space of all possible pasts.



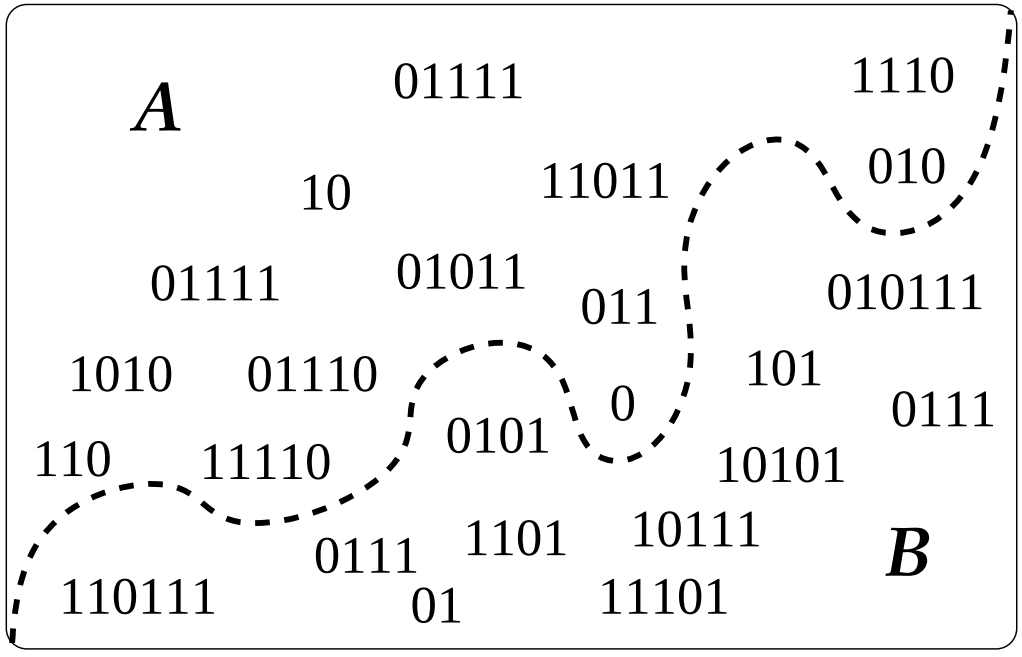
Do I really have to remember all this??

My memory isn't good enough.



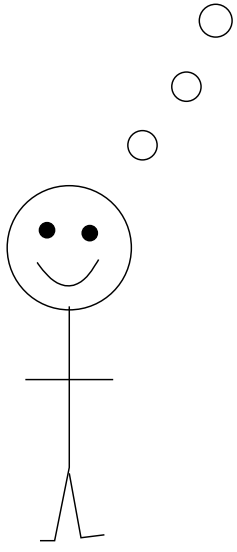
One Only Needs to Remember the Causal States

Causal states partition the space of all past sequences



This is better!

I only need to remember the causal state, A or B.



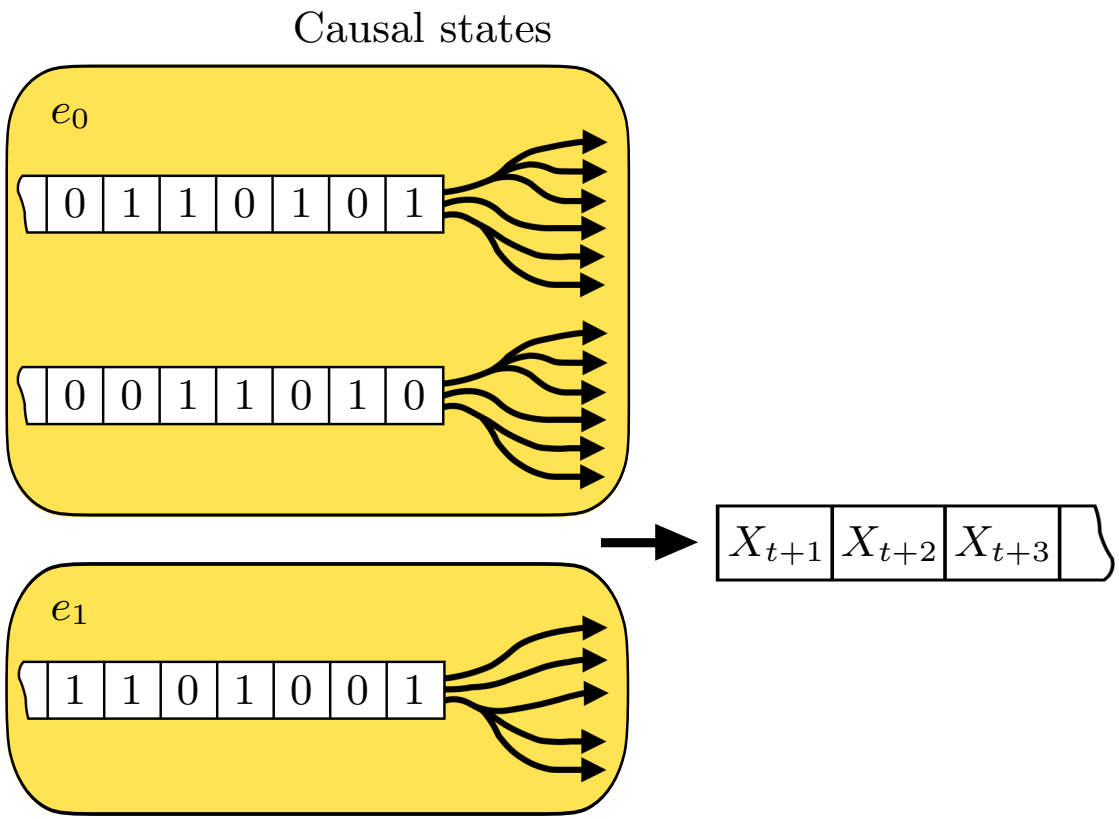
因果态

- 预测等价:

$$x_{1:t} \sim x'_{1:t} \iff \forall x_{>t} : P(x_{>t} \mid x_{1:t}) \approx P(x_{>t} \mid x'_{1:t})$$

- 因果态:

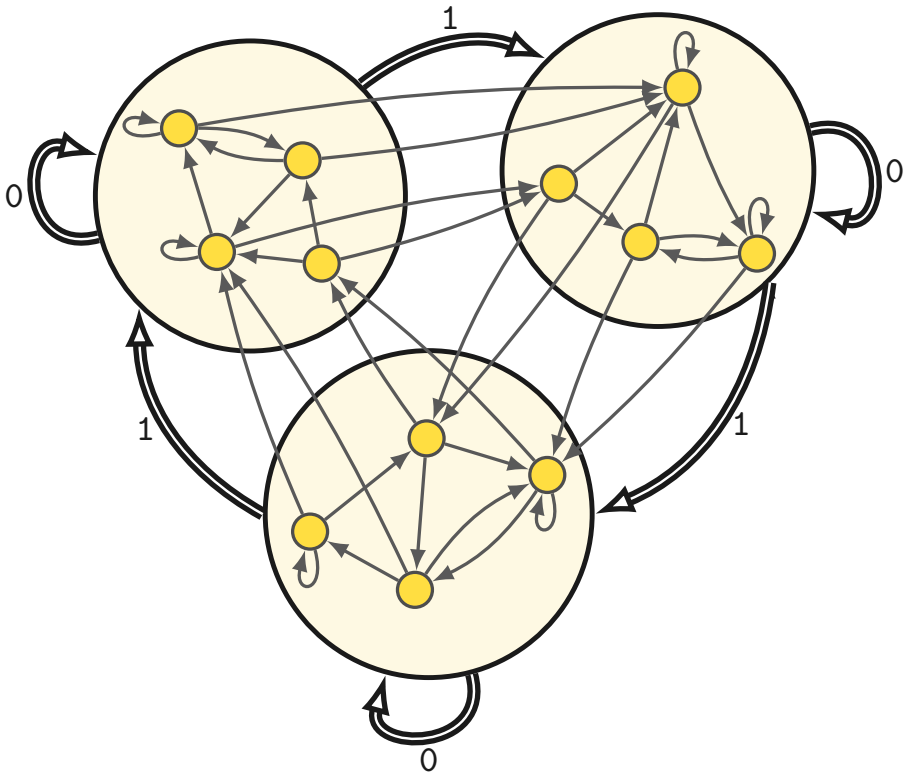
$$\epsilon : x_{1:t} \mapsto [x_{1:t}]$$

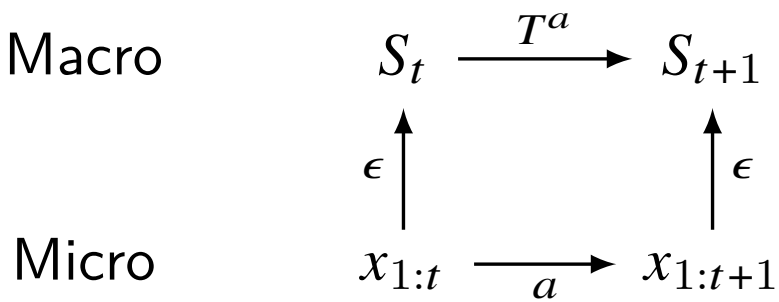


ϵ-Machine

ϵ-Machine 的构造 $\mathcal{M} := (\mathcal{S}, T^a)$:

$$\mathcal{S} := A^*/\sim = \{S_0, S_1, S_2, \dots\}$$
$$T^a_{ij} := P(S_j, a \mid S_i) = P([x_{1:t}a] \mid [x_{1:t}])$$

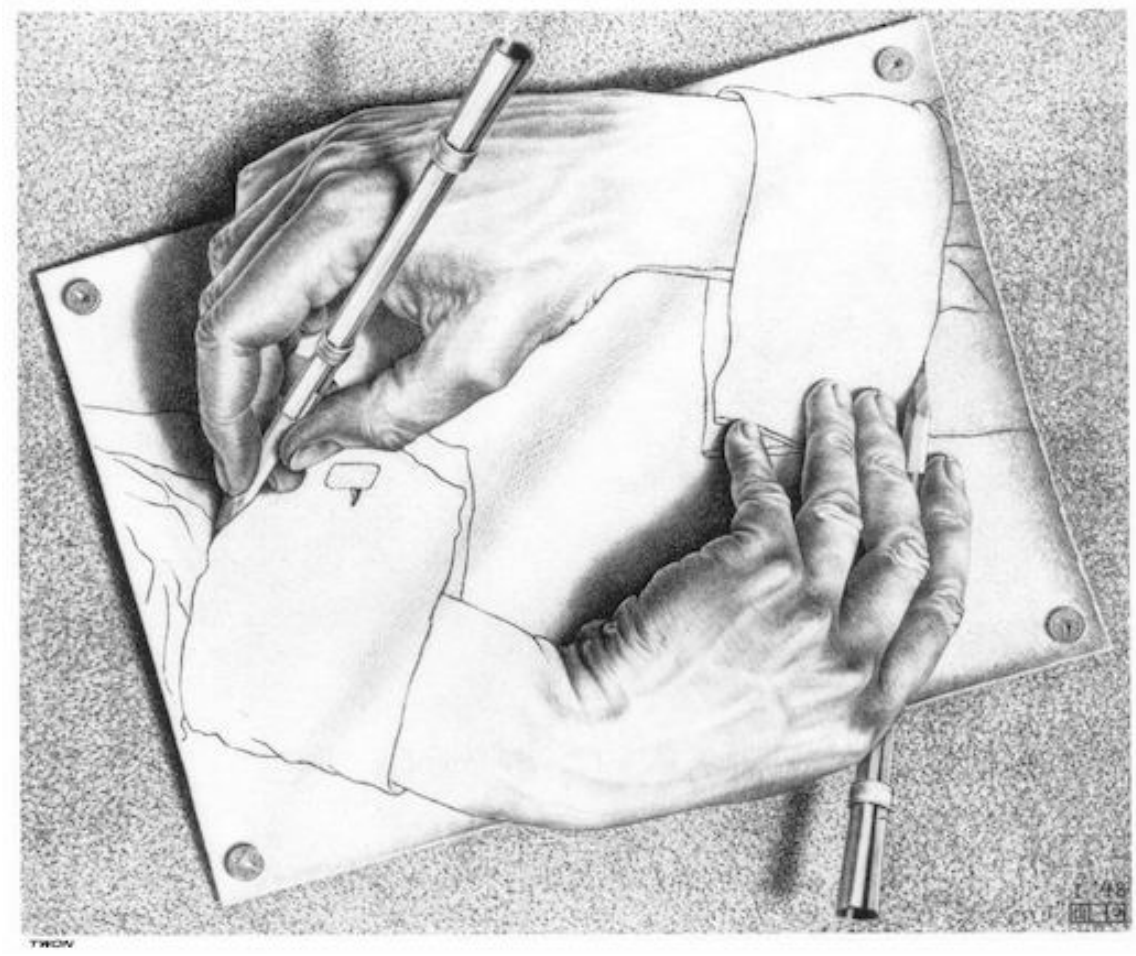




ϵ-Machine 的统计复杂度: 因果态的香农熵

$$C_\mu(\mathcal{S}) := H(\mathcal{S}) = -\sum_{S \in \mathcal{S}} P(S) \log P(S)$$

The statistical complexity measures the minimum amount of memory needed to perform optimal prediction.



- 什么是艺术?
  — 艺术是模仿? 是形式? 是情感表现? ……
- 什么是美?

JS
© J. Schmidhuber, 1997.

Jean-Louis Dessalles' Simplicity Theory

Unexpectedness 意外度

An event is unexpected if it is simpler to describe than to generate.

$$U(x) := C_W(x) - C_D(x)$$

- $C_W(x) := \min\{\ell(p) : \mathtt{WTM}(p) = x\}$ is the causal complexity / generation complexity, i.e., the length of a minimal program that a human individual's "World-machine" (causal procedure) can use to **generate** x.
- $C_D(x) := \min\{\ell(p) : \mathtt{OTM}(p) = x\}$ is the description complexity, i.e., the length of a minimal program that a human individual's "Observation-machine" can use to **describe** x.

Remark: unexpectedness can be used to define ex-post probability

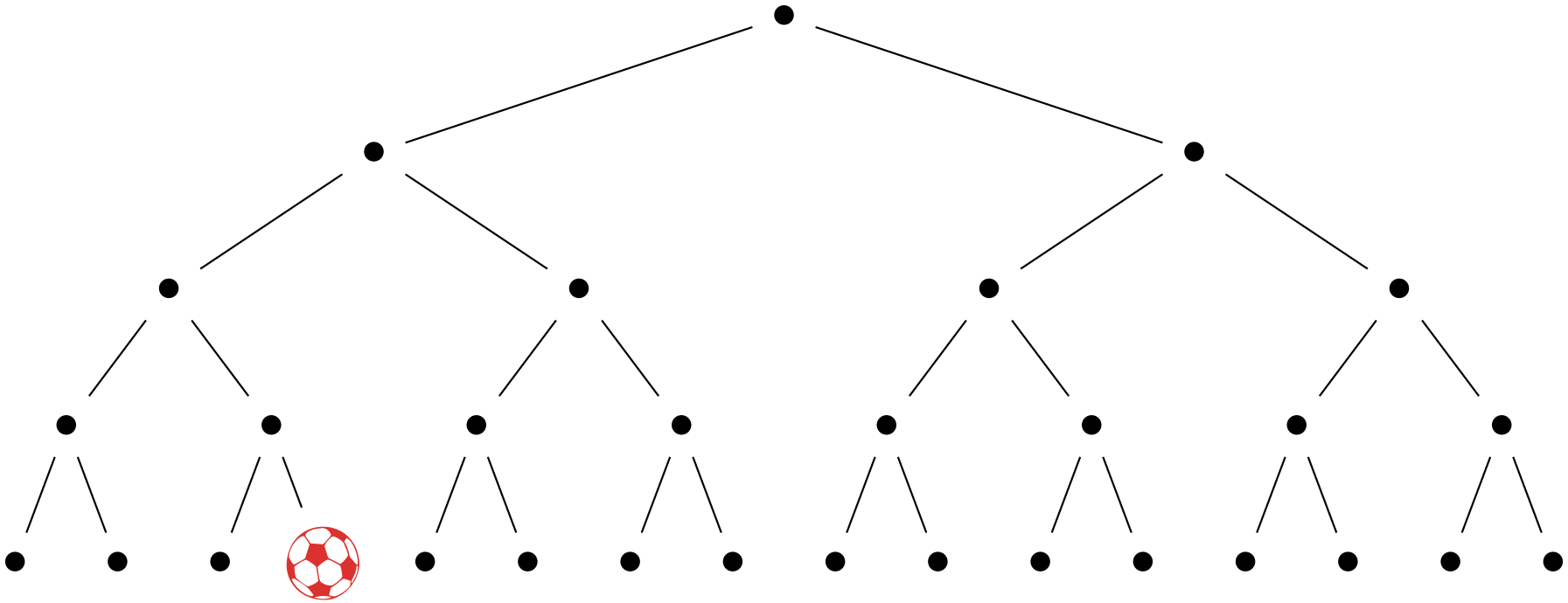
$$P_{\text{subj}}(x) := 2^{-U(x)}$$

贝叶斯公式 vs 意外度

$$P(H \mid O) = \frac{P(O \mid H)P(H)}{P(O)}$$

$$\underbrace{\log \frac{1}{P(H \mid O)}}_{\substack{U^H(O) \\ U(O) = \min_H U^H(O)}} = \underbrace{\underbrace{\log \frac{1}{P(O \mid H)}}_{C_W(O|H)} + \underbrace{\log \frac{1}{P(H)}}_{C_W(H)}}_{\substack{C_W(HO) \\ C_W(O) = \min_H C_W(HO)}} - \underbrace{\log \frac{1}{P(O)}}_{C_D(O)}$$

Example



- Imagine that a ball falls down along a binary tree. It eventually reaches a leaf x of the tree of depth n.

$$C_W(x) = n$$

- Therefore, unexpectedness $U(x) = 0$ for most leaves.
- However, if the observer can use a simple feature μ to single out the winning leaf $C_D(x \mid \mu) = 0$, then unexpectedness $U(x) = n - C_D(\mu)$ may be large.

Example: 林肯与肯尼迪的巧合

- 林肯和肯尼迪的姓氏各有 7 个字母. 全名都有 5 个音节.
- 两位总统都是在 46 年当选美国众议院议员.
- 两位在 56 年都获选党内副总统人选的第二名.
- 两位总统都是在 60 年当选总统.
- 两位总统都关注非裔美国人的问题, 并都在 63 年表达自己的观点.
- 两位总统同样是头部中枪.
- 两位总统中枪当天都是星期五.
- 两位总统中枪时, 妻子都在场.
- 两位总统身边都伴随着另一对夫妇.
- 陪伴两位总统的另一对夫妇中的男方均被行刺者所伤.
- 林肯在福特剧院中枪. 肯尼迪则在由福特汽车制造的林肯汽车内中枪的.
- 行刺者约翰·威尔克斯·布思和李·哈维·奥斯瓦尔德都在审讯前被杀.
- 两名行刺者的姓名各有 3 个部分, 全名都是 15 个字母.
- 两位各自的副总统和继任总统都是生于 08 年名叫詹森的南方民主党人.

$$U = C_W(xy) - C_D(xy) \quad C_W(xy) = C_W(x) + C_W(y) \quad C_D(xy) \le C_D(x) + C_D(y \mid x)$$

Examples

- 非典型特例:
  — 彩票开出 $7, 7, 7, 7, 7, 7, 7$
  — 客观概率一样大小; Solomonoff 算法概率挺大; 主观概率 $P_{\text{subj}}(7777777)$ 很小
- 他乡遇故知:
  — 高 C_W, 低 C_D
- 怪异故事:
  — 通过铺垫复杂的因果历史知识增大你的 C_W, 降低 C_D
- 地标效应:
  — 从 (你不知道的) 某居民楼跳下去 vs 从自由女神像跳下去

$$C_D(x) \leq C_D(\mu) + C_D(x \mid \mu)$$

Comprehension is Compression

- **Subjective Probability**: $P_{\text{subj}}(x) := 2^{-U(x)}$
- **Creativity**: select actions that will maximize unexpectedness $\text{argmax}_a U(s \mid a)$
- **Foreseeability** of the consequence s of an action a: $-U(s \mid a)$
- **Aesthetics/Humor**: complexity drop
- **Relevance**:
  1. s is relevant if $U(s) > 0$
  2. t is relevant w.r.t. s if $U(s) > U(s \mid t)$
- **Abduction**: to find out a cause to diminish the causal complexity.
- Causal Responsibility of an action a in situation s: $C_W(s) - C_W(s \mid a)$
- **Emergence**: the difference between the sum of individual complexities and the collective complexity.
- **Storage**: once data is stored, it can be defined by its address.
- **Intelligence**: to take the best rewarding action based on the most probable, i.e. the simplest, future.

尝试让大语言模型生成 “有创意的” 回答?

Definition (Unexpectedness[SD22])

$$U(x) := C_W(x) - C_D(x)$$

其中, C_W 是生成复杂性, C_D 是描述复杂性.

- **创意**: 选择最大化 unexpectedness 的动作:

$$\operatorname*{argmax}_a U(s \mid a)$$

- 让大语言模型最小化而不是最大化上述 unexpectedness 导出的 “主观概率”:

$$A^* = \operatorname*{argmin}_A P_{\text{subj}}(A \mid Q)$$

其中,

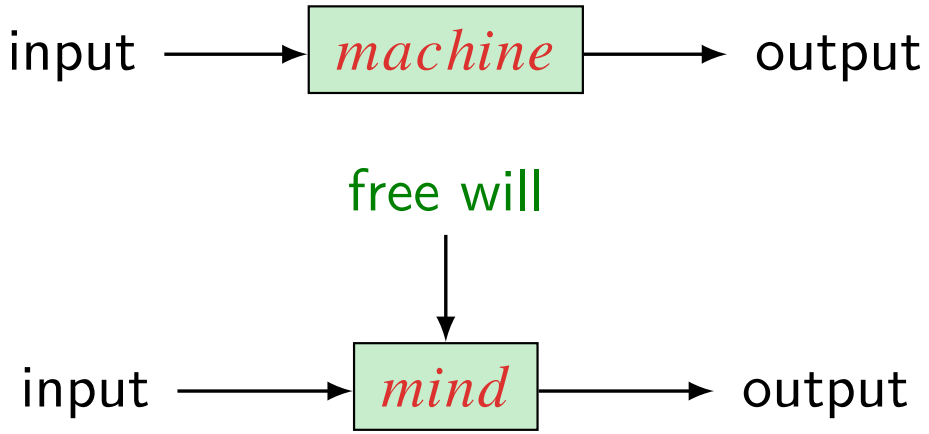
$$P_{\text{subj}}(x) := \frac{2^{-U(x)}}{\sum_x 2^{-U(x)}}$$

Determinism, Indeterminism, Randomness and Free Will

1. 随机过程: 结果不确定的过程. (IT)
2. 随机对象: 缺乏规律, 无法预测, 不可压缩. (AIT)

- 非决定论对自由意志是必要的吗? 相容论?
- 非决定论对随机性是必要的吗? No. 算法概率 Ω_U 是马丁-洛夫随机的, 但完全由 U 决定.
- 随机对非决定论是必要的吗? No. 非确定性图灵机是非决定论的, 并不需要借助随机 Oracle.
- 非决定论和随机性彼此谁也不蕴含谁.
- 随机性, 与自由意志矛盾吗?
- 如果你的行动是完全随机的, 那你对你的行动是没有掌控的.
- Agent 要做随机决策, 需要借助随机数.
- 一个 Agent 请另一个 Agent 代其做决策, 与求助一个随机数并无二致.
- 随机性与自由意志是相容的.

自由意志 — 机器中的幽灵?



- 信息处理: 信息或者从一种形式转换为另一种形式, 或者丢失 $K(\text{output}) \stackrel{+}{\leq} K(\text{input})$
- 信息生成: 若 $K(\text{output}) > K(\text{input})$, 则信息被创造.
  - 自然过程无法创造新信息
  - 没有算法可以创造新信息
  - 信息创生需要某种偶然机制 — 自由意志

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

李约瑟之问

- 李约瑟：为什么科学产生在西方而不是中国？
- 爱因斯坦：西方科学的发展是以两个伟大的成就为基础：希腊哲学家发明的形式逻辑体系，以及在文艺复兴时期发现通过系统的实验可能找出因果关系.
- 珀尔：我尝试结合二者，给出因果推断的可靠的逻辑理论，并付诸算法实现.

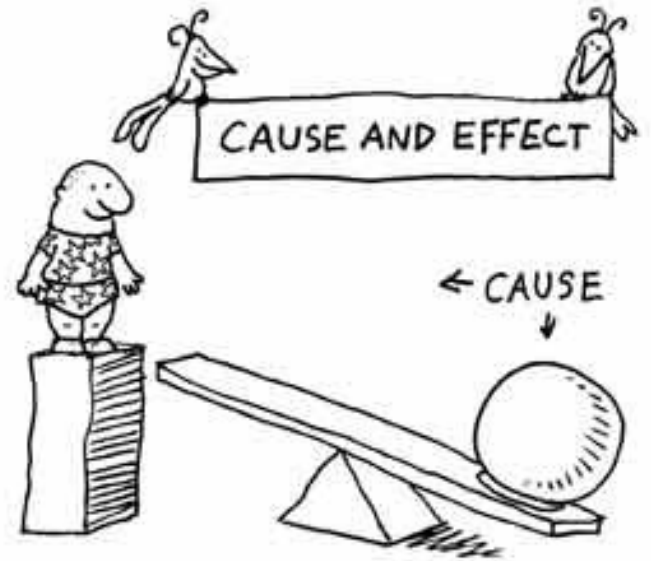
理解 vs 因果

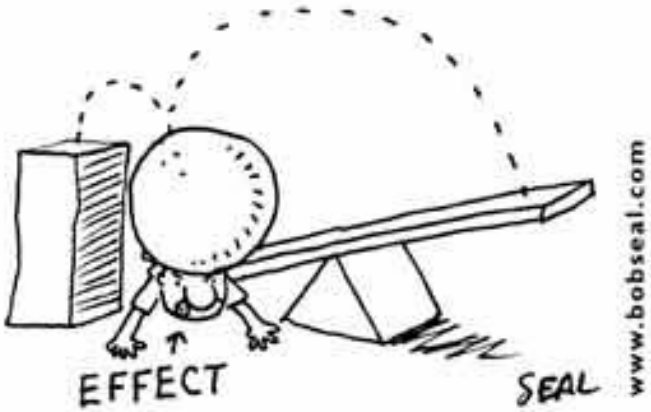
The noblest pleasure is the joy of understanding.

— Leonardo da Vinci

What does 'understanding' mean?

- 回答 "为什么" 的问题
  — 理解原因、机制
- 理解世界运行的规律
  — 预测未来事件
- 如何使用行动进行干预控制
  — 预测动作后果
- 在新场景中进行推理、规划
  — 需要反事实想象
- 分布外泛化
  — 当下的机器学习只能处理分布内泛化.
  分布外泛化 ≈ 因果发现?
- 解释理由 (贡献分配, 责任划分)





因果可以用于推理、预测、决策、解释、归责.....

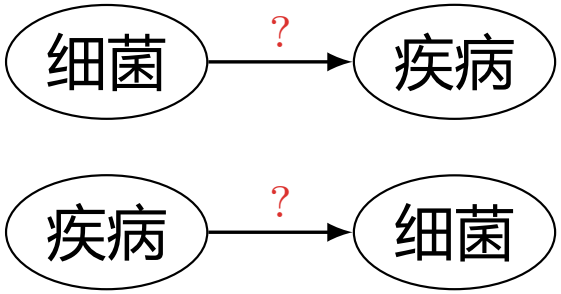
圣经中的因果

- 上帝: “你是不是吃了那树上的果子?”
- 亚当: “你赐给我的女人把那果子递给我, 我就吃了.”
- 夏娃: “那蛇引诱我, 我就吃了.”

Remark: 上帝问事实, 而不是问原因. 解释纯粹是为了推卸责任.

> 一旦我们真正理解了因果思维背后的逻辑, 就可以在计算机上模拟它, 创造出一个 “人工科学家”.
>
> — 珀尔

路易·巴斯德 1822-1895



- 亚里士多德：自然发生论．生物从非生物中自然产生．
  — 腐草化萤．腐肉生蛆．
- 自然发生论者：体液失衡 → 疾病 → 组织腐败 → 细菌滋生．
  — 治病即恢复体液平衡．
  — 疗法：放血，催吐，泻药，泡冷水．
- 巴斯德：生源论．生物源于生物．
- 巴斯德：细菌致病说．疾病是由微生物的感染引起的．
  — 杀菌消毒．接种疫苗．

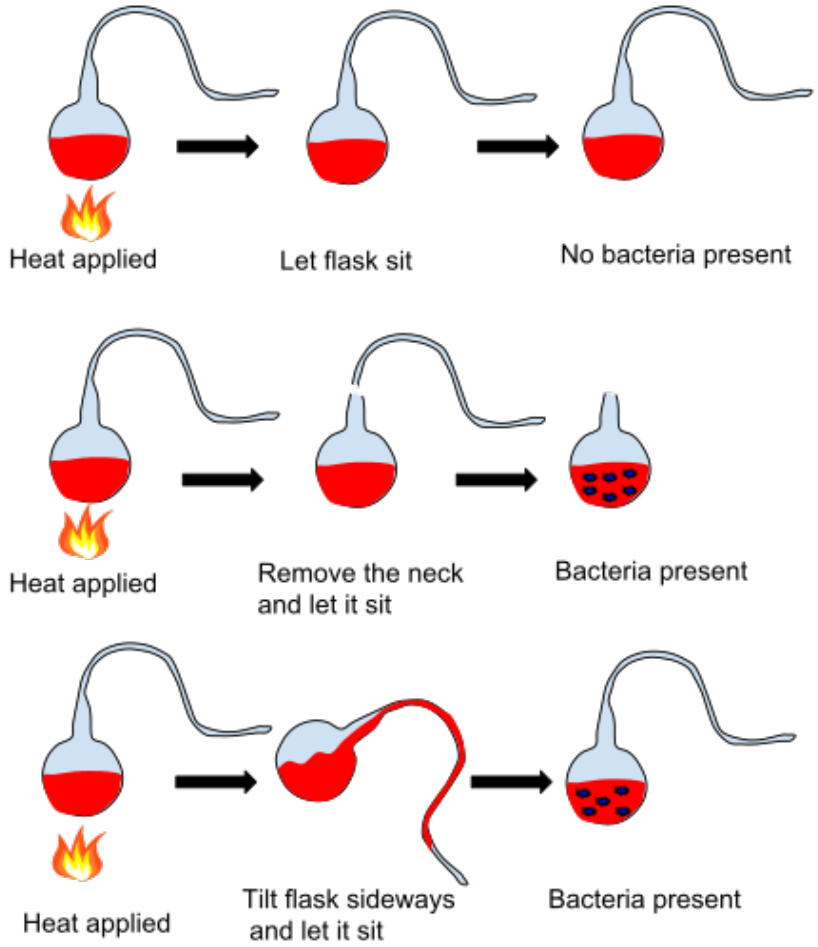
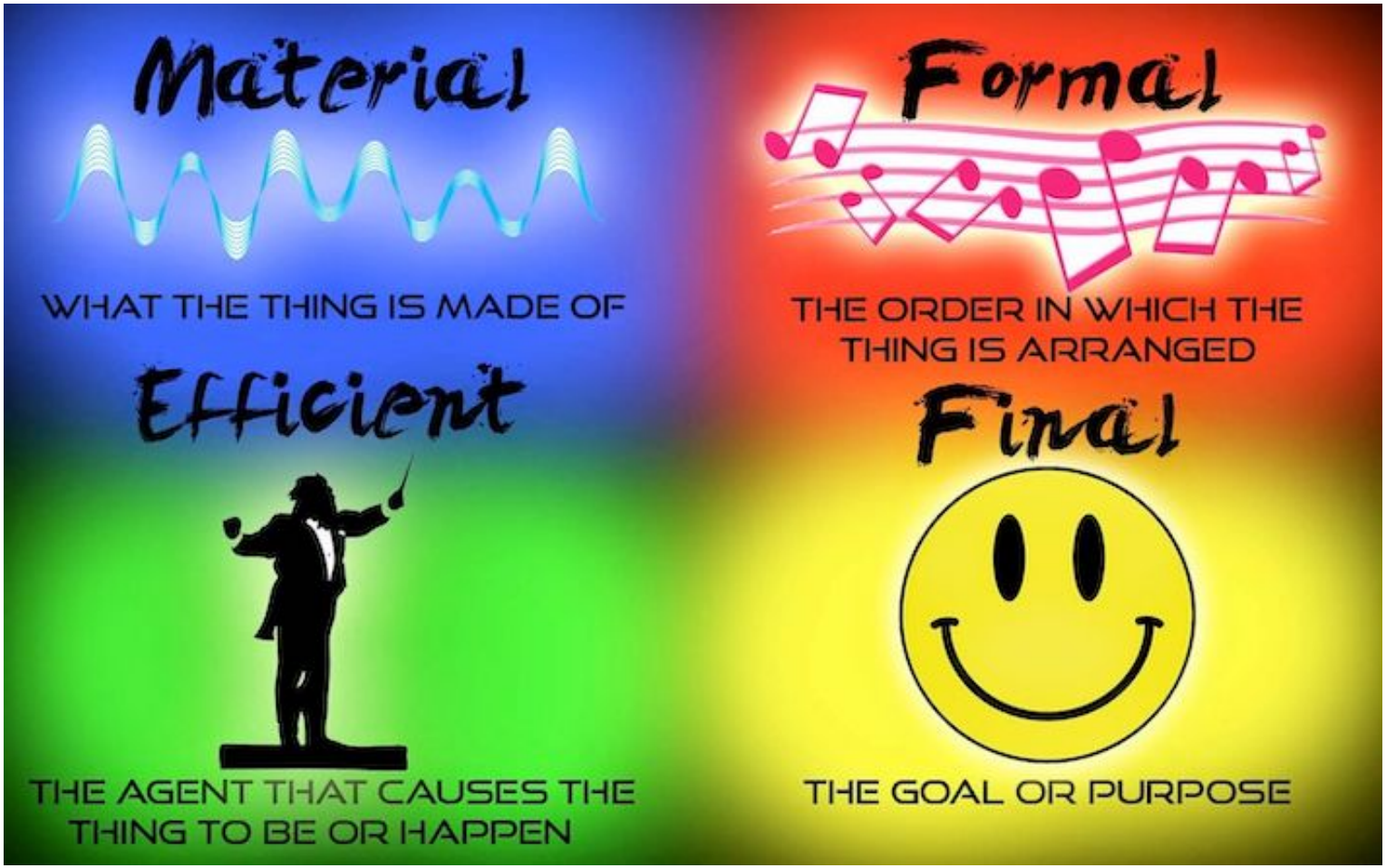


Figure: 巴氏杀菌实验证伪自然发生论．肉汤置于曲颈瓶中，经过高温杀菌后，不会滋生微生物．

亚里士多德 — “四因说”

“Knowledge is the object of our inquiry, and we do not have knowledge of a thing until we have grasped its ‘why’, that is to say, its cause.”

— Aristotle



什么是“生命”？— 家族相似？

- 生命是能够实现**进化**并能够**自我维持**的化学系统.
- 病毒将宿主细胞重新“编程”，让细胞为自己**“复制”**新的病毒. 病毒还会发生**“突变”**.
- 病毒没法吸收分子. 病毒不能**“新陈代谢”**.
- 病毒能够进化，但却不能自我维持.
- 红细胞内含有大量蛋白质，承担着复杂的生化功能. 但红细胞没有基因，所以无法生长、分裂和复制.
- 细菌有自己的基因，能够生殖繁殖，但并不存在**个体差异**，也无法**独立生存**.
- 亚马逊莫莉鱼虽然繁殖过程需要性伴侣，但只能克隆雌性个体. 它们与其他品种的雄性鱼类交配，通过它们的精子刺激其体内的卵子发育，然后杀死所有精子，开始自我复制. 繁殖过程很像病毒.

A Power of Change-which-changes-its-own-rules

有机体既是它自己的因也是它自己的果，既是它自己固有的秩序和组织的因，也是其固有秩序和组织的果．自然选择并不是有机体的因．基因也不是有机体的因．有机体的因不存在．有机体是自我能动派．

— 布赖恩·古德温

- 自治 autonomous
- 自组织 self-organization
- 自我维持 self-sustaining
- 自我完善 self-improving
- 自我复制 self-replication
- 自我管理 self-governance
- 有限自我修复 limited self-repair
- 适度进化 mild evolution
- 局部学习 partial learning

瓦雷拉 Varela 的自创生系统

- 一个系统是“自创生”的，如果：
  1. 这个系统有一个半透边界，使其与环境能够进行物质和能量的交换. (质料因开放)
  2. 这个边界由系统内的生产网络所生产. (动力因闭合)
  3. 这个生产网络由边界自身所创造的条件得以再生产. (动力因闭合)
- 一个有边界但没有生产网络的系统被称为“他创生的”. 比如汽车，病毒.
- 一个系统如果具有生产网络，但其生产的成份不是自我维持所需要的，或不能满足自我维持的需要，被称为“异创生的”. 比如线粒体，化工厂.

Remark: 认知既不是主体对世界既有属性的表征，也不是主体既有观念的向外投射，而是在与环境的交互中，生命维持其自创生组织完整性的行动. 要活着就要去认知.

瓦雷拉: 认知 = 生命

表征主义 vs 生成主义 (Enaction)

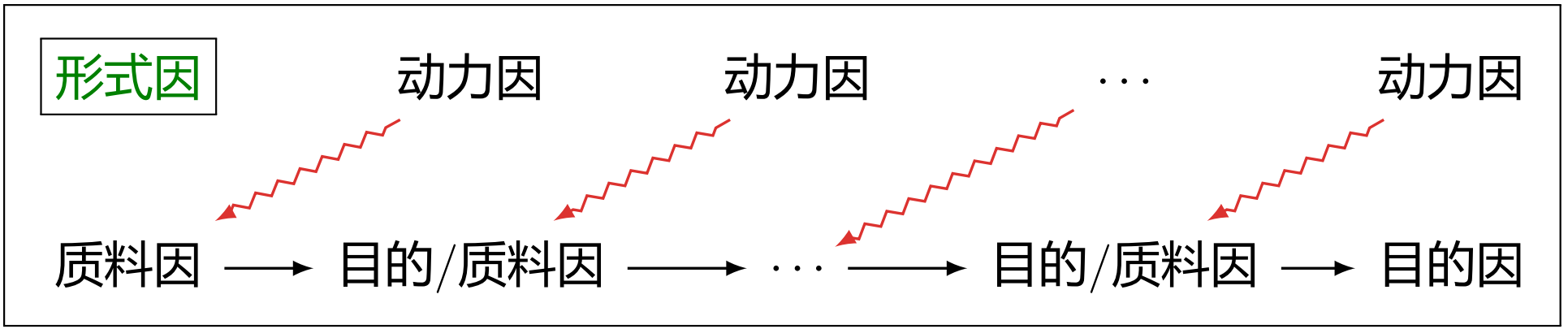
表征主义: 认知即信息处理

- 心智的任务是正确的表征外部世界
- 感知 → 表征 → 行动

生成主义: 认知即行动

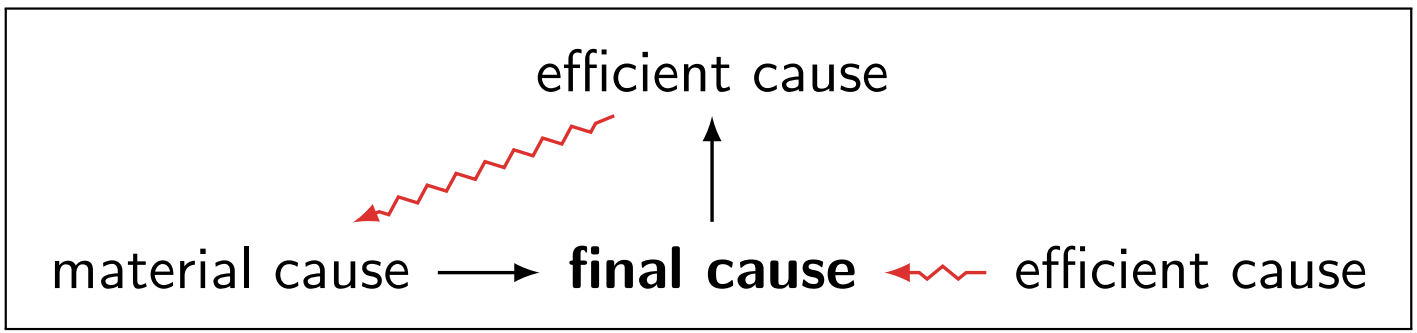
- 首要的问题是 “如何维持自身”, 而不是 “如何表征世界”
- 感知与行动不可分离

From Aristotle's Four Causes to Rosen's "Life"



*"A living system is a **system closed to efficient causation**, i.e., its every efficient cause is entailed within the system."*

— Robert Rosen



Mechanism or Organism?

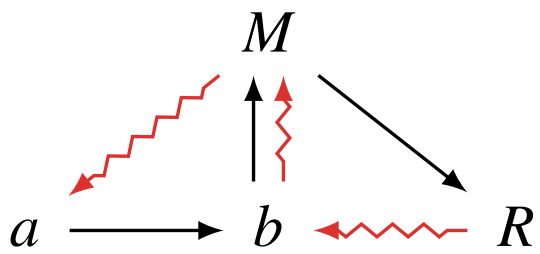
infinite regress?

closure to efficient causation

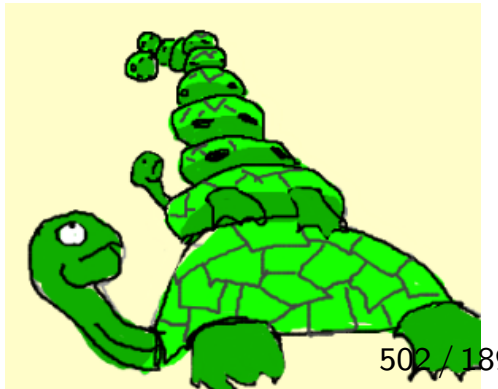
Mr.Why?

1. Mr.Why: "why b?"
2. Rosen: $b = M(a)$
   - 2.1 "because a", this is the "material cause"
   - 2.2 "because M", this is the "efficient cause"
3. Mr.Why: "why M?" — within physics there is not really any answer, other than that this just is a natural law.
4. Rosen: "because R": $R(b) = M$
5. Mr.Why: "why R?"
6. Rosen: "because β": $\beta(M) = R$
7. Mr.Why: "why β?"
8. Rosen: "because M": $\beta \cong b$ and $M(a) = b$

$$A \xrightarrow{M} B \xrightarrow{R} \mathrm{Hom}(A,B) \xrightarrow{\beta} \mathrm{Hom}(B,\mathrm{Hom}(A,B))$$



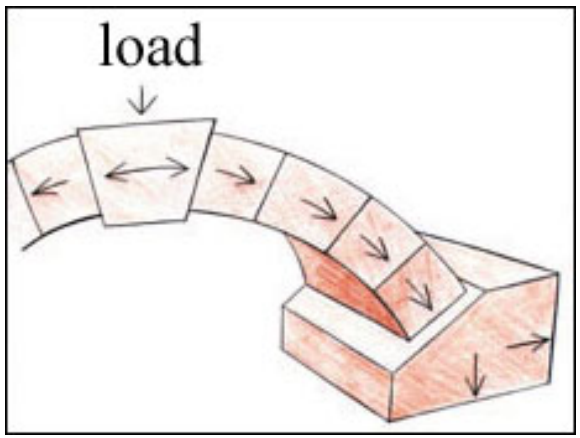
Remark: b is the material cause entailing its own efficient cause M which entails b as its final cause.



无需乌龟之背，无需上帝之手，无需钢筋水泥，呈拱形的石头彼此支撑

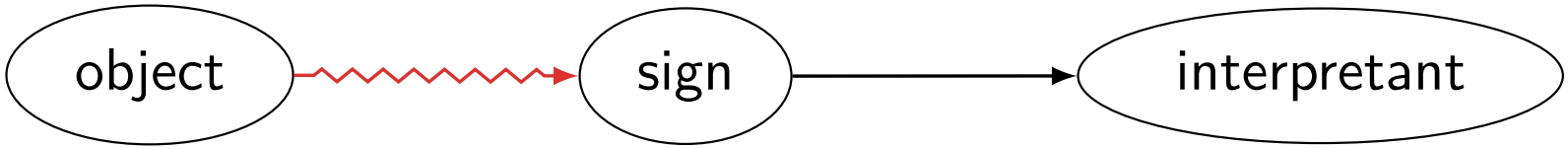
- 马可·波罗一块石头一块石头的描绘着那座桥.
- 忽必烈汗：但究竟是哪一块石头支撑着这座桥呢?
- 马可·波罗：支撑这座桥的并非某块石头，而是它们共同构成的拱形弧线.
- 忽必烈汗：那你为何还要跟我描绘石头？重要的只有那个拱形.
- 马可·波罗：没有石头，就没有拱形.

— 卡尔维诺《看不见的城市》



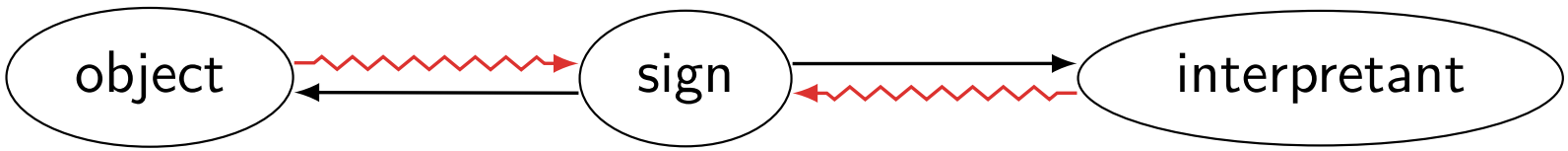
Closure is a Danger?

- Varela's "Closure Thesis" — every autonomous system is an operationally closed system.
- Peirce's Semiosis.



— The object entails that the sign entails the interpretant.

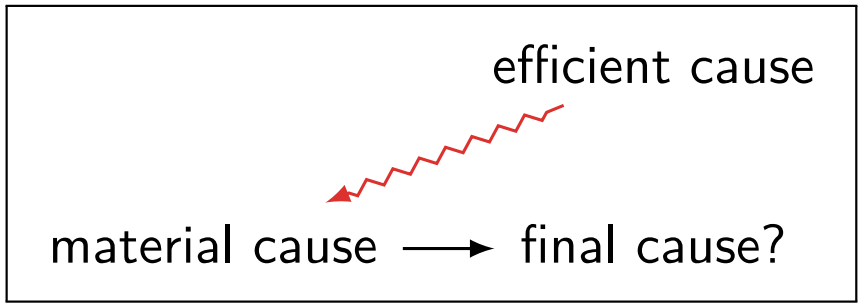
— For the agent, the interpretant entails that (excepting failures) the sign entails the object.



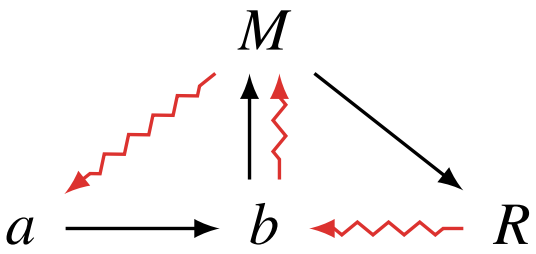
- Russell's "Vicious Circle": a collection of objects may contain members which can only be defined by means of the collection as a whole.

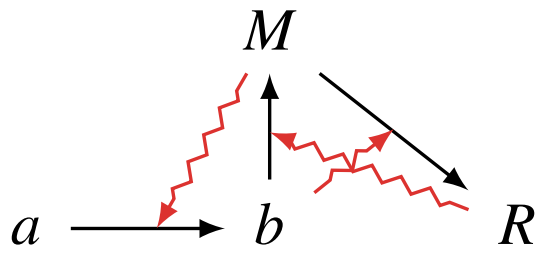
Rosen: "What is Life?"[Luz+09][11]

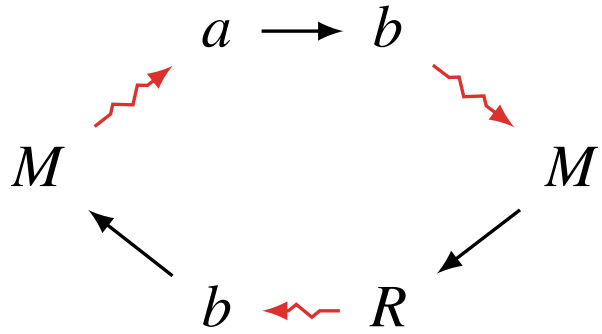
- M: metabolism $M(a) = b$
- R: repair $R(b) = M$
- β: replication $\beta(M) = R$



$$A \xrightarrow{M} B \xrightarrow{R} \mathrm{Hom}(A,B) \xrightarrow{\beta} \mathrm{Hom}(B,\mathrm{Hom}(A,B))$$







Assumption: The evaluation map
$\varepsilon_b : \mathrm{Hom}(B,\mathrm{Hom}(A,B)) \to \mathrm{Hom}(A,B) :: \varepsilon_b(R) = R(b)$ is invertible.

Then $\varepsilon_b^{-1}(M) = R$. Thus, we can set $\beta = \varepsilon_b^{-1}$, i.e., β is determined by b.

[11] Luz Cárdenas *et al*: Closure to efficient causation, computability and artificial life. 2009.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

一段对话

- 病人: 医生, 请问这种药有效的概率是多少?
- 医生: 每个人的情况不一样, 看情况发展吧.
- 病人: 对于 100 个类似的病例, 你认为这种药对多少人有效?
- 医生: 每个人的情况不一样, 多么 "类似" 算 "类似"?
- 病人: 如果必须打赌, 你愿不愿意押注 — 如果这种药有效, 你赢 1 元钱, 如果无效, 你输 2 元钱?
- 医生: 好吧, 我对这药有效的主观信念超过了 $2/3$.

不确定性

内在不确定性 (aleatoric uncertainty)

1. 世界在本质上是随机的
2. 世界是确定性的，但却是混沌的，因此，如果没有无限精确的感知，就很难预测
3. 世界是确定性的，但却是部分可观测的

认知不确定性 (epistemic uncertainty)

1. 世界是完全可观察的，但传感器只能捕捉到部分信息
2. 由感知模块提取的关于世界状态的表征不包含准确预测所需的全部信息
3. 由于表征能力的限制，世界模型不准确
4. 由于训练数据的限制，世界模型不准确

概率论刻画的是哪种不确定性？
用隐变量表征随机性？
哪些是因果关系？哪些是逻辑关系？

Uncertainty, Ignorance, Vagueness

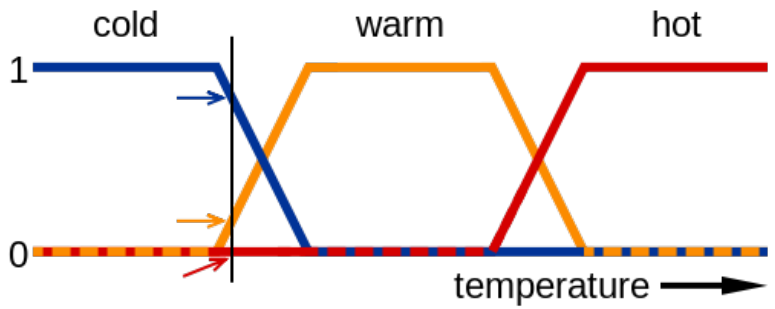
Some Approaches:

- Bayesian Network
- Dempster-Shafer Theory: represents 'ignorance' & 'uncertainty'
  - basic belief assignment function $m : \mathrm{P}(X) \to [0, 1]$
  - belief: $\mathrm{bel}(A) := \sum_{B:B\subset A} m(B)$
  - plausibility: $\mathrm{pl}(A) := \sum_{B:B\cap A\neq\varnothing} m(B) = 1 - \mathrm{bel}\left(\overline{A}\right)$

    the interval $[\mathrm{bel}(A), \mathrm{pl}(A)]$ represents the level of ignorance in A.
  - Dempster's rule of combination

$$(m_1 \oplus m_2)(A) := \frac{\sum_{B\cap C=A} m_1(B)m_2(C)}{1 - \sum_{B\cap C=\varnothing} m_1(B)m_2(C)}$$

- Fuzzy logic and fuzzy sets: represents 'vagueness', not 'uncertainty'.



Structures of Truth Degrees for Fuzzy Logic

Definition (Structures of Truth Degrees for Fuzzy Logic)

The structures of truth degrees for Fuzzy Logic is a complete residuated lattice $(L, \wedge, \vee, \otimes, \rightarrow, 0, 1)$, where

- $(L, \wedge, \vee, 0, 1)$ is a complete lattice.
- $(L, \otimes, 1)$ is a commutative monoid,
- $(\otimes, \rightarrow)$ is an adjoint pair (i.e., $a \otimes b \leq c \iff a \leq b \rightarrow c$).

Example $([0,1], \min, \max, \otimes, \rightarrow, 0, 1)$

- Lukasiewicz: $a \otimes b = \max(a + b - 1, 0), \quad a \rightarrow b = \min(1 - a + b, 1)$
- Gödel: $a \otimes b = \min(a, b), \quad a \rightarrow b = \begin{cases} 1 & \text{if } a \leq b \\ b & \text{otherwise} \end{cases}$
- Goguen: $a \otimes b = a \cdot b, \quad a \rightarrow b = \begin{cases} 1 & \text{if } a \leq b \\ \frac{b}{a} & \text{otherwise} \end{cases}$

Uncertainty and Probability

- Frequentist: probabilities are relative frequencies.
  (e.g. the relative frequency of tossing head.)

$$P(E) = \lim_{n \to \infty} \frac{n_E}{n}$$

  - Frequency Interpretation is Circular — For fair coin, $P(H) = 1/2$ with "high probability". But to make this statement rigorous we need to formally know what "high probability" means. Circularity!
  - Reference Class Problem
  - Limited to I.I.D
- Objectivist: probabilities are real aspects of the world.
  (e.g. the probability that some atom decays in the next hour)
- Subjectivist: probabilities describe an agent's degree of belief.
  (e.g. it is (im)plausible that extraterrestrials exist)
  - Cox's Theorem
  - Dutch Book
- Objective Probability = Inter-Subjective Probability?

Probability

(objective) chance	(subjective) degree of belief
↓	↓
how to measure?	how to measure?
↓	↓
frequency of occurrence	gambling behaviour

Kolmogorov's Axioms of probability must apply to both!

Measurable Space

Definition (σ-algebra)

A *σ-algebra* $\mathcal{A}$ is a collection of subsets of the sample space Ω s.t.

1. $\Omega \in \mathscr{A}$
2. $A, B \in \mathscr{A} \implies A \cap B, A \cup B \in \mathscr{A}$
3. $A \in \mathscr{A} \implies \Omega \setminus A \in \mathscr{A}$
4. $A_1, A_2, \cdots \in \mathscr{A} \implies \bigcup_{i=1}^{\infty} A_i \in \mathscr{A}$

Definition (Measurable Space)

$(\Omega, \mathscr{A})$ is a *measurable space* iff $\mathscr{A} \subset \mathrm{P}(\Omega)$ is a σ-algebra on Ω.

Probability Mesaure & Semi-Probability & Semimeasure

Definition (Probability Space)

$(\Omega, \mathscr{A}, P)$ is a *probability space* iff P is a *probability measure* on $\mathscr{A}$ s.t.

1. $P(A) \geq 0$ for $A \in \mathscr{A}$
2. $\mathrm{P}(\Omega) = 1$
3. $A_1, A_2, \cdots \in \mathscr{A} \implies P\left(\biguplus_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$

1. **Semi-probability:** If we only require $P(\Omega) \leq 1$, then we call P a semi-probability measure, or simply a semi-probability.
2. **Semimeasure:** If we only require $P(\Omega) \leq 1$ and

$$P\left(\biguplus_{i=1}^{\infty} A_i\right) \geq \sum_{i=1}^{\infty} P(A_i)$$

then we call P a probability semimeasure or simply a semimeasure.

Filtration

A *filtration* on $(\Omega, \mathscr{A})$ is a sequence $(\mathscr{A}_t)_{t\geq 0}$ of σ-algebras s.t.

$$\forall t \, (\mathscr{A}_t \subseteq \mathscr{A}) \quad \text{and} \quad \forall t_1, t_2 \left(t_1 \leq t_2 \implies \mathscr{A}_{t_1} \subseteq \mathscr{A}_{t_2}\right)$$

$(\Omega, \mathscr{A}, (\mathscr{A}_t)_{t\geq 0}, P)$ is a *filtered probability space*.

概率

- 主观概率将命题与个人的知识状态联系起来, 表达了主体的信念.
- 概率综合了各种因素的影响:
  — 懒惰: 未能枚举异常、限定条件等.
  — 无知: 缺乏相关事实、初始条件等.
- 概率空间 $(\Omega, \mathscr{A}, P)$ 里的一个事件 (σ-代数 $\mathscr{A}$ 里的一个元素) 对应着一个或一组变量的赋值. $\{u \in \Omega : X(u) = x\}$.
  事件也可以被看作一个为真或为假的命题.
- 随机变量 $X : \Omega \to \mathbb{R}$ 是从样本空间到某个值域 (如实数或布尔值) 的可测函数[12] $\{u \in \Omega : X(u) \leq x\} \in \mathscr{A}$.

$$P(X = x) = \sum_{u:X(u)=x} P(u) \qquad P(A) = \sum_{u \models A} P(u)$$

Example: $\mathrm{CoinToss} = \mathrm{head}, \mathrm{Age} \geq 18$

$$P(\mathrm{DiceOdd} = \mathrm{true}) = P(1) + P(3) + P(5) = \frac{1}{6} + \frac{1}{6} + \frac{1}{6} = \frac{1}{2}$$

[12] 随机变量是一个函数, 它既不是一个变量, 也不随机 ☺

条件概率

证据 A 排除了与其不相容的可能世界，诱导出了一个可能世界集上的新测度 P^A.

$$P^A(S) := \begin{cases} c * P(S) & \text{if } u \vDash A \text{ for all } u \in S \\ 0 & \text{if } u \nvDash A \text{ for all } u \in S \end{cases}$$

$$\begin{aligned} 1 &= P^A(\Omega) \\ &= P^A\big(\{u \in \Omega : u \vDash A\}\big) + P^A\big(\{u \in \Omega : u \nvDash A\}\big) \\ &= c * P([\![A]\!]) + 0 \\ &= c * P(A) \end{aligned} \qquad \Longrightarrow \qquad c = \frac{1}{P(A)}$$

$$\begin{aligned} P(B \mid A) &= P^A([\![B]\!]) \\ &= P^A([\![A \wedge B]\!]) + P^A([\![\neg A \wedge B]\!]) \\ &= c * P([\![A \wedge B]\!]) + 0 \\ &= \frac{P(A \wedge B)}{P(A)} \end{aligned}$$

Remark: $P(B \mid A) = \frac{P(A \wedge B)}{P(A)}[P(A) + P(\neg A)] = P(A \wedge B) + \frac{P(A \wedge B)}{P(A)} P(\neg A)$

Independence

- Conditional Probability

$$P(B \mid A) = \frac{P(A, B)}{P(A)} \qquad \text{(when } P(A) > 0\text{)}$$

$$P(\text{Age} \geq 18 \mid \text{Fall-in-Love} = \text{true})$$

- Independence:

$$\begin{aligned} X \perp Y &\iff P(X \mid Y) = P(X) && \text{(when } P(y) > 0\text{)} \\ &\iff P(X, Y) = P(X)P(Y) \end{aligned}$$

- Conditional Independence:

$$\begin{aligned} X \perp Y \mid Z &\iff P(X \mid Y, Z) = P(X \mid Z) && \text{(when } P(y, z) > 0\text{)} \\ &\iff P(X, Y \mid Z) = P(X \mid Z)P(Y \mid Z) \end{aligned}$$

Definition (一组事件的独立性)

设 $A_1, A_2, \ldots, A_n$ 为 n 个事件. 如果对 $\{1, 2, \ldots, n\}$ 的任意子集 I, 都有

$$P\left(\bigwedge_{i \in I} A_i\right) = \prod_{i \in I} P(A_i)$$

则称 $A_1, A_2, \ldots, A_n$ 为独立事件.

- 两两独立不意味着独立.
  — 抛掷均匀的硬币两次. A_1 表示第一次为正面, A_2 表示第二次为正面, B 表示两次结果不同. 则 $P(B \mid A_1) = \frac{1}{2} = P(B) = P(B \mid A_2)$, 但 $P(A_1, A_2, B) = 0 \neq \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = P(A_1)P(A_2)P(B)$.
- $P(A_1 \wedge A_2 \wedge A_3) = P(A_1)P(A_2)P(A_3)$ 不意味着 A_1, A_2, A_3 独立.
  — 抛掷均匀的骰子两次. A 表示第一次掷得 1 或 2 或 3 点. B 表示第一次掷得 3 或 4 或 5 点. C 表示两次掷得的点数之和为 9. 则

  $$P(A \wedge B \wedge C) = \frac{1}{36} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{4}{36} = P(A)P(B)P(C)$$

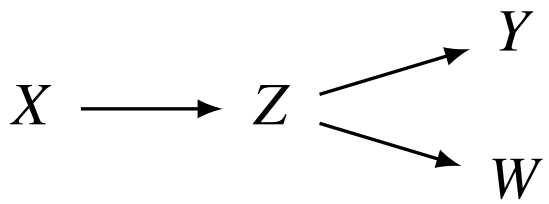
  但 $P(A \wedge B) = \frac{1}{6} \neq \frac{1}{2} \cdot \frac{1}{2} = P(A)P(B)$.

Independence Axioms — Pearl

Symmetry $X \perp Y \mid Z \implies Y \perp X \mid Z$

$$X \longrightarrow Z \longrightarrow Y$$

Decomposition $X \perp Y \cup W \mid Z \implies X \perp Y \mid Z \;\&\; X \perp W \mid Z$



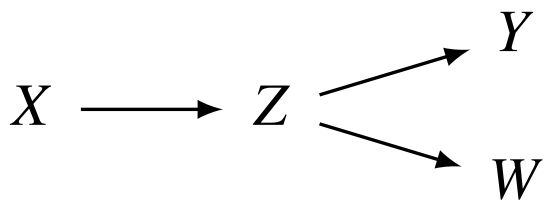
Weak union $X \perp Y \cup W \mid Z \implies X \perp Y \mid Z \cup W$

$$X \longrightarrow Z \longrightarrow W \longrightarrow Y$$

Contraction $X \perp Y \mid Z \;\&\; X \perp W \mid Z \cup Y \implies X \perp Y \cup W \mid Z$

$$X \longrightarrow Z \longrightarrow Y \longrightarrow W$$

Intersection $X \perp W \mid Z \cup Y \;\&\; X \perp Y \mid Z \cup W \implies X \perp Y \cup W \mid Z$



Independence Axioms — Dawid

$$\text{Symmetry} \quad X \perp Y \mid Z \implies Y \perp X \mid Z$$

$$\text{Triviality} \quad X \perp Y \mid X$$

$$\text{Composition} \quad X \perp Y \mid Z \;\&\; W \leq Y \implies X \perp W \mid Z$$

$$\text{Weak Union} \quad X \perp Y \mid Z \;\&\; W \leq Y \implies X \perp Y \mid W \cup Z$$

$$\text{Contraction} \quad X \perp Y \mid Z \;\&\; X \perp W \mid Y \cup Z \implies X \perp Y \cup W \mid Z$$

$$\text{Intersection} \quad X \perp W \mid Z \cup Y \;\&\; X \perp Y \mid Z \cup W \implies X \perp Y \cup W \mid Z$$

where we write $W \leq Y$ to mean that W is determined by Y.

Algorithmic Mutual Information

$$K(x \mid y^*) = K(x,y) - K(y)$$
$$K(y \mid x^*) = K(x,y) - K(x)$$

$$\begin{aligned} I(x;y) &:= K(x) + K(y) - K(x,y) \\ &\stackrel{+}{=} K(x) - K(x \mid y^*) \\ &\stackrel{+}{=} K(y) - K(y \mid x^*) \end{aligned}$$

$$x \perp y \iff I(x;y) \stackrel{+}{=} 0$$

$$\begin{aligned} I(x;y \mid z) &:= K(x \mid z) + K(y \mid z) - K(x,y \mid z) \\ &\stackrel{+}{=} K(x \mid z) - K(x \mid y, K(y \mid z), z) \\ &\stackrel{+}{=} K(y \mid z) - K(y \mid x, K(x \mid z), z) \end{aligned}$$

$$x \perp y \mid z \iff I(x;y \mid z) \stackrel{+}{=} 0$$

Analogy to statistical mutual information:

$$I(X;Y \mid Z) = H(X \mid Z) + H(Y \mid Z) - H(X,Y \mid Z)$$

$P(B \mid A) \neq P(A \to B)$

$A \to B$	$P(B \mid A) = 1$
$A \to \neg B$	$P(B \mid A) = 0$
$\vdash A \to (B \to A)$	$P(A \mid AB) = 1$
$\vdash (A \to B) \to (B \to C) \to (A \to C)$	$P(C \mid A) \geq P(C \mid B)P(B \mid A)$
$\vdash (\neg B \to \neg A) \to (A \to B)$	$P(B) \geq 1 - \frac{1 - P(A)}{P(\neg A \mid \neg B)}$
$\vdash A \vee \neg A$	$P(A) + P(\neg A) = 1$
$\vdash \neg(A \wedge \neg A)$	$P(A \wedge \neg A) = 0$
$\neg(A \vee B) \equiv \neg A \wedge \neg B$	$P(\neg(A \vee B)) = P(\neg A)P(\neg B)$
$\vdash (A \to B) \wedge A \to B$	$P(B) \geq P(B \mid A)P(A)$
$\nvdash (A \to B) \to (B \to A)$	$P(B \mid A) = 1 \not\Longrightarrow P(A \mid B) = 1$
$\nvdash (A \to B) \to (B \to A)$	$P(B \mid A) > P(B) \implies P(A \mid B) > P(A)$
$A(0) \wedge \forall n(A(n) \to A(n+1)) \to \forall n A(n)$	$P(A_n) \geq \prod_{i=1}^{n-1} P(A_{i+1} \mid A_i)$ $\forall i < n(P(A_{i+1} \mid A_i) = 1) \implies P(A_n) = 1$ $\forall i, j(P(A_{i+1} \mid A_i) = P(A_{j+1} \mid A_j) < 1) \implies \prod_{i=1}^{\infty} P(A_{i+1} \mid A_i) = 0$
$A(0) \wedge \forall n(A(n) \to A(n+1)) \to \forall n A(n)$	$P(X = X_1) \geq c \;\&\; \cdots \;\&\; P(X = X_n) \geq c$ $\Downarrow$ $P(X = X_1 = \cdots = X_n \mid X_1 = \cdots = X_n) \geq \frac{c^n}{c^n + (1-c)^n}$
$A(0) \wedge \forall n(A(n) \to A(n+1)) \to \forall n A(n)$	$\forall i(P(A_{i+1} \mid A_i) > 0)$ $\Downarrow$ $\prod_{i \geq 1} P(A_{i+1} \mid A_i) = 0 \iff \sum_{i \geq 1} (1 - P(A_{i+1} \mid A_i)) = \infty$

Logic, Belief, Probability — Cox Theorem

Probability theory extends propositional logic?

Assumption (Cox's Assumptions for Beliefs)

1. $A \leftrightarrow B \implies b(\cdot \mid A) = b(\cdot \mid B)$ & $b(A \mid \cdot) = b(B \mid \cdot)$.
2. *there is a continuous binary operation* $\otimes$ *that is strictly increasing in each coordinate s.t.* $b(A \wedge B \mid C) = b(A \mid C) \otimes b(B \mid A \wedge C)$.
3. *for any rational numbers* $r_1, r_2, r_3 \in (0,1)$ *there are* $A, B, C, D \in \Omega$ *s.t.* $r_1 = b(A \mid D)$, $r_2 = b(B \mid A \wedge D)$ *and* $r_3 = b(C \mid A \wedge B \wedge D)$.
4. *thers is a continuous nonnegative nonincreasing function* $N : [0,1] \to [0,1]$ *s.t.* $b(\neg A \mid C) = N(b(A \mid C))$.

Theorem (Cox Theorem)

A credence function that satisfies Cox's assumptions for beliefs is isomorphic to a probability function.

Why are the Axioms of Probability Theory Reasonable?

- If P represents an objectively observable probability, the axioms clearly make sense.
- But why should an agent respect these axioms when it models its own degree of belief?

Dutch Book Argument — Ramsey/de Finetti

If the beliefs do not follow the Kolmogorov axioms, then there exists a betting strategy against the agent, where he will definitely loose!

Peirce/Putnam: a belief is true if it would be accepted by anyone under ideal epistemic conditions.

> *The degree of a belief is a causal property of it, which we can express vaguely as the extent to which we are prepared to act on it.*
>
> — *Ramsey*

贝叶斯推理

- 贝叶斯规则:

$$P(B \mid A) = \frac{P(A, B)}{P(A)} = \frac{P(A \mid B)P(B)}{P(A)}$$

- 全概率公式: 假设 $\{B_1, \ldots, B_n\}$ 构成对 Ω 的划分,

$$P(A) = \sum_{i=1}^{n} P(A, B_i) = \sum_{i=1}^{n} P(A \mid B_i)P(B_i)$$

- 查询可以通过对原子事件求和来回答:
给定证据 $E = e$, 通过对隐变量 $S = s$ 求和, 计算查询变量 H 的后验分布.

$$P(H \mid E = e) = \frac{P(H, E = e)}{P(E = e)} = \frac{\sum_s P(H, E = e, S = s)}{P(E = e)}$$

重复使用数据不提供新信息

$$P(H \mid E) = P(H)\frac{P(E \mid H)}{P(E)}$$

再考虑数据 E',

$$P(H \mid EE') = P(H \mid E)\frac{P(E' \mid HE)}{P(E' \mid E)}$$

E' 要改变信念, 需要

$$P(E' \mid HE) \neq P(E' \mid E)$$

而若 E' 已经包含在数据 E 和先验信息中 $E' = f(E)$, 则

$$P(E' \mid HE) = P(E' \mid E) = 1$$

这实际上是布尔逻辑原理 $AA = A$.
Remark: 因果信息不包含在观察数据中.

意见分歧与趋同

新的证据有没有可能使得持有不同观点的人的分歧变得更极端?

- H: 某药物是安全的.
- E: 某专家在电视上宣称此药物是不安全的.
- Alice 和 Bob 对 H 的初始信念不同. Alice 更支持, Bob 更怀疑. $P_a(H) > P_b(H)$.
- **Problem1**: 在接收到新信息 E 后, 二人的后验信念会趋同吗?

$$|P_a(H \mid E) - P_b(H \mid E)| \overset{?}{<} |P_a(H) - P_b(H)|$$

— 二人的后验信念可能单调收敛/单调发散/反转收敛/反转发散.

- **Problem2**: 如果二人的似然相同呢?

$$P_a(E \mid H) = P_b(E \mid H) = l, \quad P_a(E \mid \neg H) = P_b(E \mid \neg H) = l'$$

— 如果 $l = l'$, 他们会认为专家不可靠, 新信息不会影响自己的观点.

— 如果 $l > l'$, 则认为专家倾向于撒谎, 从而都更相信药物的安全性.

$$\log \frac{P(H \mid E)}{P(\neg H \mid E)} - \log \frac{P(H)}{P(\neg H)} = \log \frac{l}{l'}$$

— 如果 $l < l'$, 则二人的分歧会扩大.

$$\log \frac{P_a(H \mid E)}{P_b(H \mid E)} - \log \frac{P_a(H)}{P_b(H)} = \log \frac{l \cdot P_b(H) + l' \cdot P_b(\neg H)}{l \cdot P_a(H) + l' \cdot P_a(\neg H)}$$

Digression — Jeffery's Radical Probabilism Philosophy

什么是理性的信念更新? $P^{\text{old}} \to P^{\text{new}}$?

- Dogmatic Probabilism: 任何理性的信念改变都必须由贝叶斯更新来解释.

Bayesian Update

$$P^{\text{new}}(H) = P^{\text{old}}(H \mid E = e)$$

- Radical Probabilism: 没有事实是绝对确定知道的.

Jeffrey Update

$$P^{\text{new}}(H) = \sum_e P^{\text{old}}(H \mid E = e) P^{\text{new}}(E = e)$$

- van Fraassen's Reflection Principle

$$P_0(A \mid P_1(A) = x) = x$$

- Lewis' Imaging Theory

Remark: Jeffrey — 软证据下的 Bayes

- 假设我们只拥有关于证据变量 E 的软证据 $\tilde{E}$, (比方说因为近视眼, 没戴眼镜, 不确定是否看清), 想用它来估计 H.
- 如果我们拥有确定性的证据 E, 那么我们就不再需要不确定性的软证据 $\tilde{E}$, 所以 $P(H \mid E, \tilde{E}) = P(H \mid E)$.

$$
\begin{aligned}
P(H \mid \tilde{E}) &= \sum_e P(H, E = e \mid \tilde{E}) \\
&= \sum_e P(H \mid E = e, \tilde{E}) P(E = e \mid \tilde{E}) \\
&= \sum_e P(H \mid E = e) P(E = e \mid \tilde{E})
\end{aligned}
$$

- 把 $P(H \mid \tilde{E})$ 看作 $P^{\text{new}}(H)$, 把 $P(E = e \mid \tilde{E})$ 看作 $P^{\text{new}}(E = e)$, 那么, Jeffrey Update 可以看作软证据下的 Bayesian Update.
- 这相当于背后有:

$$
\tilde{E} \longrightarrow E \longrightarrow H \quad \text{or} \quad \tilde{E} \longleftarrow E \longleftarrow H \quad \text{or} \quad \tilde{E} \longleftarrow E \longrightarrow H
$$

题外话 — Updating (belief revision) is very subtle

Example (Human mind is sensitive to the order)

What impression do you obtain about Bob?

1. Alice is pregnant; Bob visits Alice
2. Bob visits Alice; Alice is pregnant

(1) good guy
(2) guilty guy

随机变量的函数

- 设 $Y = f(X)$ 是随机变量 X 的函数. 则

$$P(Y = y) = \sum_{x : f(x) = y} P(X = x)$$

- 期望 expected value (or mean) $\mathbb{E}[X] := \sum_x x P(X = x)$ $\int_\Omega X(u) \mathrm{d}P(u)$
- $Y = f(X)$ 的期望 $\mathbb{E}[f(X)] = \sum_x f(x) P(X = x)$

$$\begin{aligned}
\mathbb{E}[f(X)] &= \mathbb{E}[Y] \\
&= \sum_y y P(Y = y) \\
&= \sum_y y \sum_{x : f(x) = y} P(X = x) \\
&= \sum_x f(x) P(X = x)
\end{aligned}$$

- 条件期望 conditional mean $\mathbb{E}[Y \mid X = x] := \sum_y y P(Y = y \mid X = x)$

两个信封悖论

- 抛一枚均匀的硬币, 直到出现正面为止. 设 $N = n$ 为抛硬币的次数.
- 提供两个信封, 一个信封里有钱 k^n, 另一个信封里有 $k > 2$ 倍的钱.
- 你选择并打开了其中一个信封.
- 现在有两个选择: 收下这个信封里的钱, 或换另一个信封. 你换吗?

$$\frac{P(Y = k^{n+1} \mid X = k^n)}{P(Y = k^{n-1} \mid X = k^n)} = \frac{P(Y = k^{n+1}, X = k^n)}{P(Y = k^{n-1}, X = k^n)} = \frac{P(N = n)}{P(N = n-1)} = \frac{1}{2}$$

$$P(Y = k^{n-1} \mid X = k^n) = \frac{2}{3}, \quad P(Y = k^{n+1} \mid X = k^n) = \frac{1}{3}$$

$$\mathbb{E}[Y \mid X = k^n] = \frac{2}{3} \cdot k^{n-1} + \frac{1}{3} \cdot k^{n+1} > k^n$$

所以, 当 $k > 2$ 时, 因为 $\mathbb{E}[Y \mid X = x] > x$, 你应该要求换另一个信封.
然而, 由于

$$\mathbb{E}[X] = \mathbb{E}[Y] = \infty$$

所以, 你利用关系式 $\mathbb{E}[Y \mid X = x] > x$ 做决策并不能提高期望收益.

方差, 协方差, 相关系数, 回归系数

- 方差 variance $\operatorname{Var}(X) := \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$
- 协方差 covariance of X and Y

$$\operatorname{Cov}(X, Y) := \mathbb{E}\big[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\big] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

- 相关系数 correlation coefficient

$$\rho_{XY} := \frac{\operatorname{Cov}(X, Y)}{\sqrt{\operatorname{Var}(X)\operatorname{Var}(Y)}}$$

- 回归系数 regression coefficient of Y on X

$$r_{XY} := \rho_{XY}\frac{\sqrt{\operatorname{Var}(Y)}}{\sqrt{\operatorname{Var}(X)}} = \frac{\operatorname{Cov}(X, Y)}{\operatorname{Var}(X)}$$

No correlation does not imply independence

独立蕴含不相关.

$$P(X,Y) = P(X)P(Y) \implies \mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$$
$$X \perp Y \implies \rho_{XY} = 0$$
$$\rho_{XY} = 0 \not\implies X \perp Y$$

不相关性并不蕴含独立.

- $P(X = x) = \frac{1}{3}$ for $x = -1, 0, 1$
- $Y = X^2$
- $\mathbb{E}[X] = \mathbb{E}[Y] = \mathbb{E}[XY] = 0$
- $\mathrm{Cov}(X,Y) = 0$
- $\rho_{XY} = 0$

马尔科夫不等式 & 切比雪夫不等式

- 马尔科夫不等式: 对于任意非负随机变量 X 和任意 $a > 0$, 有

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

- 切比雪夫不等式: 对于任意随机变量 X 和任意 $a > 0$, 有

$$P(|X - \mathbb{E}[X]| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Law of Large Number

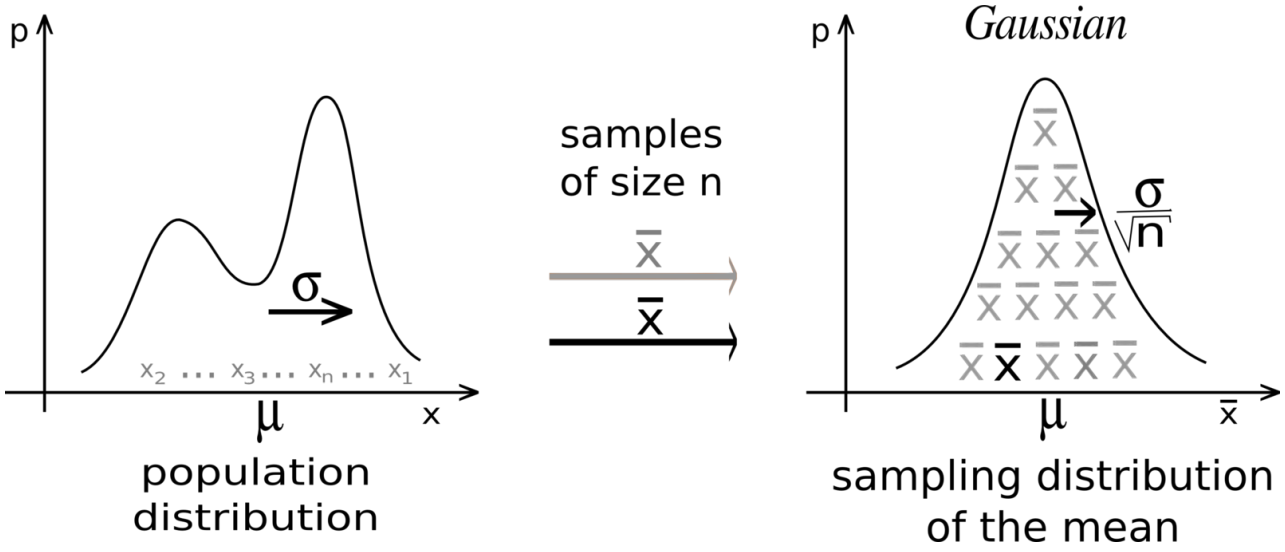
Theorem ((Weak/Strong) Law of Large Number)

Let $X_1, X_2, \ldots, X_n$ be independent identically distributed random variables with $\mathbb{E}[X_i] = \mu$ and finite variance. Let $\overline{X}_n := \frac{1}{n} \sum_{i=1}^{n} X_i$. Then

$$\forall \varepsilon : \lim_{n \to \infty} P\left(\left|\overline{X}_n - \mu\right| < \varepsilon\right) = 1 \qquad \text{(weak)}$$

$$P\left(\lim_{n \to \infty} \overline{X}_n = \mu\right) = 1 \qquad \text{(strong)}$$

The Central Limit Theorem



Theorem (Lindeberg-Lévy Central Limit Theorem)

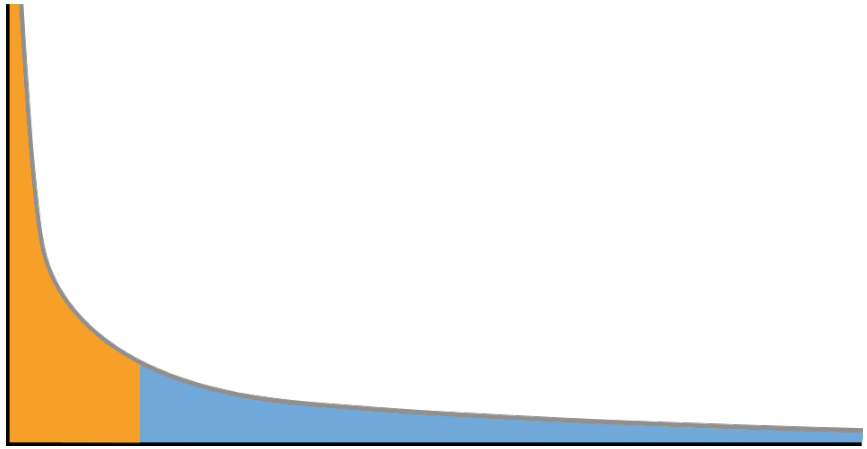
Let $X_1, X_2, \ldots, X_n$ *be independent identically distributed random variables with* $\mathbb{E}[X_i] = \mu$ *and* $\mathrm{Var}(X_i) = \sigma^2 < \infty$. *Let* $\overline{X}_n := \frac{1}{n}\sum_{i=1}^{n} X_i$ *and*

$Z_n := \frac{\overline{X}_n - \mathbb{E}\left[\overline{X}_n\right]}{\sqrt{\mathrm{Var}\left(\overline{X}_n\right)}} = \frac{\overline{X}_n - \mu}{\sigma/\sqrt{n}}$. *Then* Z_n *converges in distribution to* $\mathcal{N}(0,1)$.

$$\lim_{n\to\infty} P\left(Z_n < a\right) = \Phi(a) = \int_{-\infty}^{a} \frac{1}{\sqrt{2\pi}} \mathrm{e}^{-t^2/2}\,\mathrm{d}t$$

幂律分布 — 违反大数定律/中心极限定理

Most power laws in nature have exponents such that the mean is well-defined but the variance is not, implying they are capable of black swan behavior.



For example, consider the probability density function of Pareto distribution

$$f_X(x) = \begin{cases} (\alpha-1)x_{\min}^{\alpha-1}x^{-\alpha} & x \geq x_{\min} \\ 0 & x < x_{\min} \end{cases}$$

$$\mathbb{E}[X] = \begin{cases} \infty & \alpha \leq 2 \\ \frac{\alpha-1}{\alpha-2}x_{\min} & \alpha > 2 \end{cases}$$

$$\mathrm{Var}(X) = \begin{cases} \infty & \alpha \in (2,3] \\ \frac{\alpha-1}{\alpha-3}\left(\frac{x_{\min}}{\alpha-2}\right)^2 & \alpha > 3 \end{cases}$$

de Finetti's Theorem

Definition

A sequence of random variables $(x_1, x_2, \dots)$ is infinitely exchangeable iff, for any n, and for any permutation π,

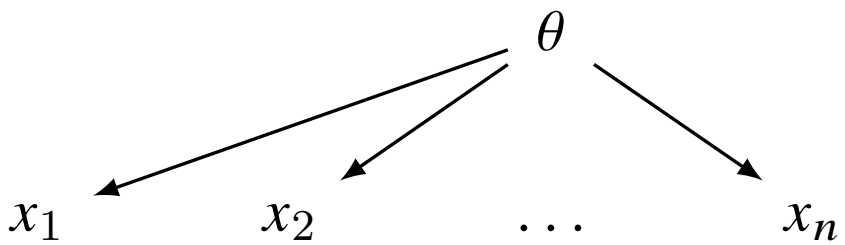
$$P(x_1, \dots, x_n) = P(x_{\pi_1}, \dots, x_{\pi_n})$$

Theorem (de Finetti's Theorem)

A sequence of random variables $(x_1, x_2, \dots)$ is infinitely exchangeable iff, for all n, we have

$$P(x_1, \dots, x_n) = \int \prod_{i=1}^{n} P(x_i \mid \theta) \mu(\mathrm{d}\theta)$$

for some hidden common random variable θ (possibly infinite dimensional) and for some unique measure μ on θ. That is, x_i are i.i.d conditional on θ.



无穷可交换序列可以表示
为独立同分布序列的混合

Contents

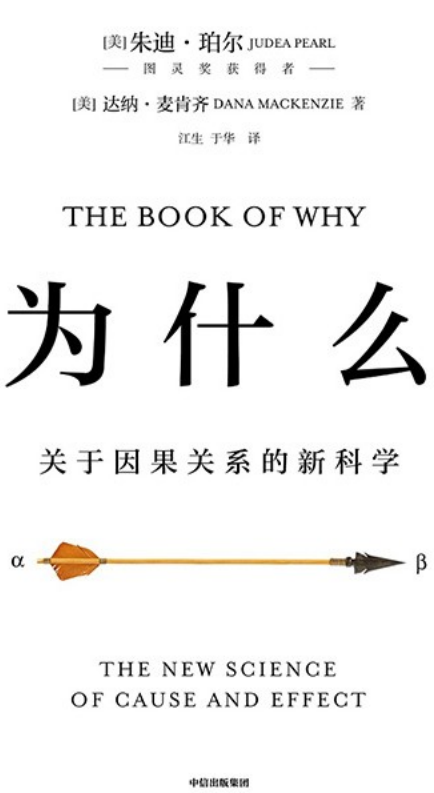
1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

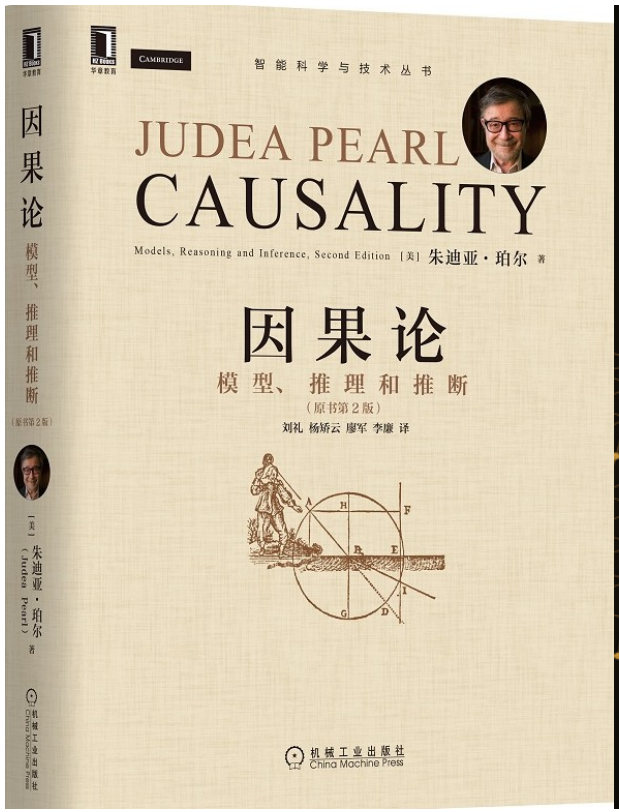
Judea Pearl 1936–[Pea09; PM18]

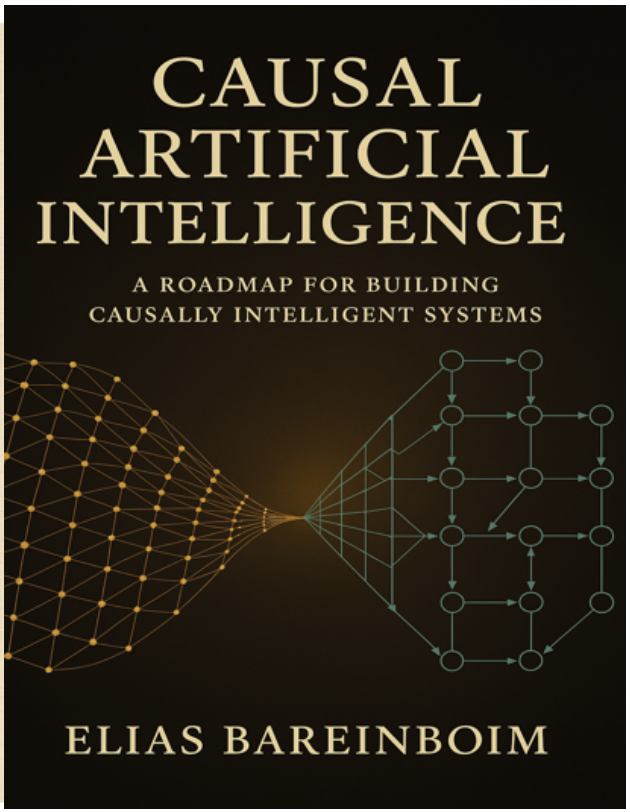
“因果……从一个被神秘面纱笼罩的概念转变为一个具有明确语义和逻辑基础的数学对象. 悖论和争议得以解决, 模糊的概念得以阐明, 那些依赖于因果信息、长期被认为是形而上学的或难以处理的实际问题, 现在只需要初等数学就能解决.”

— 珀尔

Causal Artificial Intelligence







贝叶斯网络 Bayesian Network

Definition (贝叶斯网络)

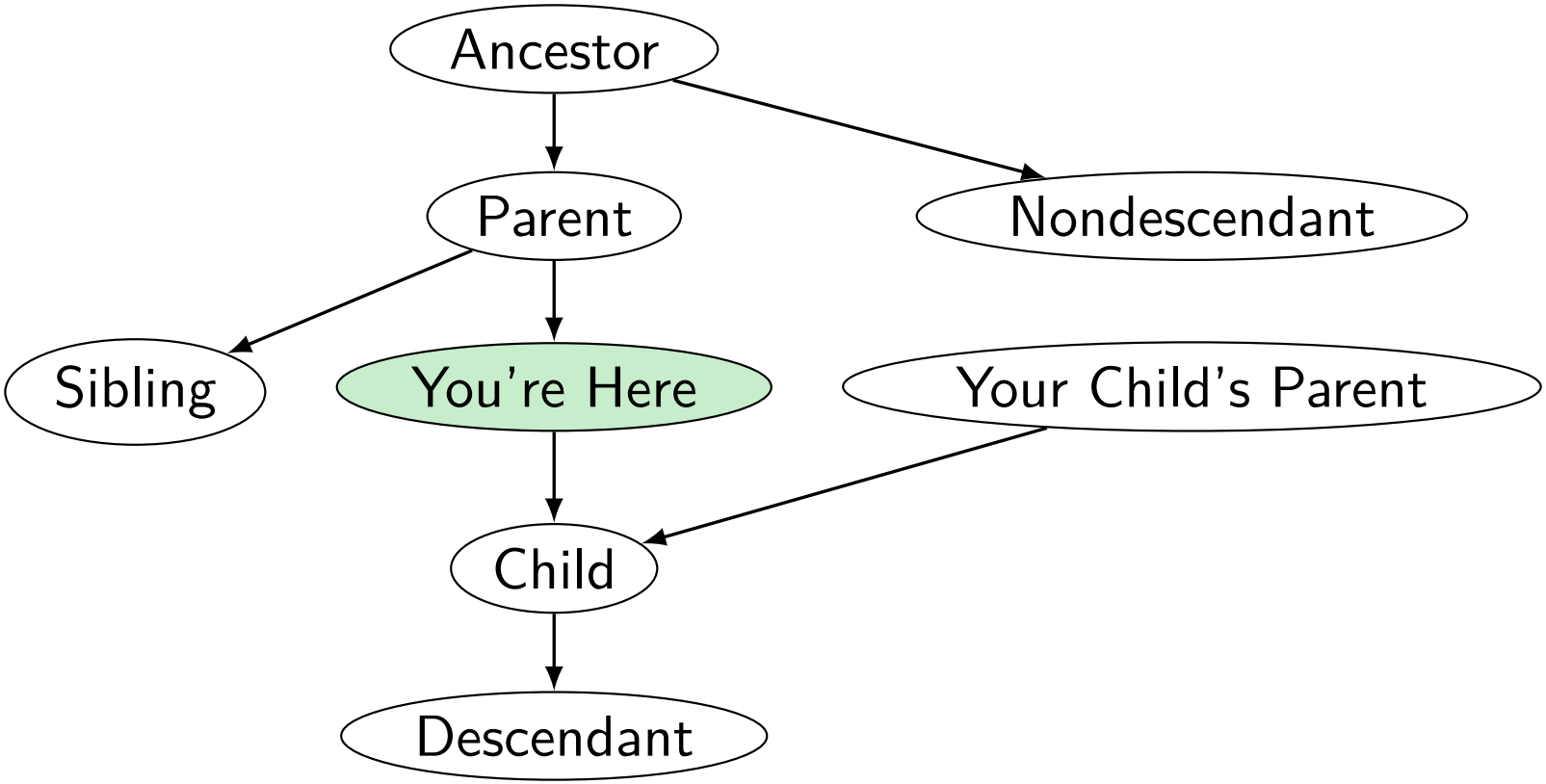
贝叶斯网络是一个有向无环图 $G = (V, E, P)$, 其节点 V 表示随机变量, 边 $E \subset V \times V$ 表示节点间的依赖关系, 且 V 上的联合概率分布 P 可分解为:

$$P(V) = \prod_{V_i \in V} P(V_i \mid \mathrm{Pa}_i)$$

其中 Pa_i 是 V_i 的父节点集合.

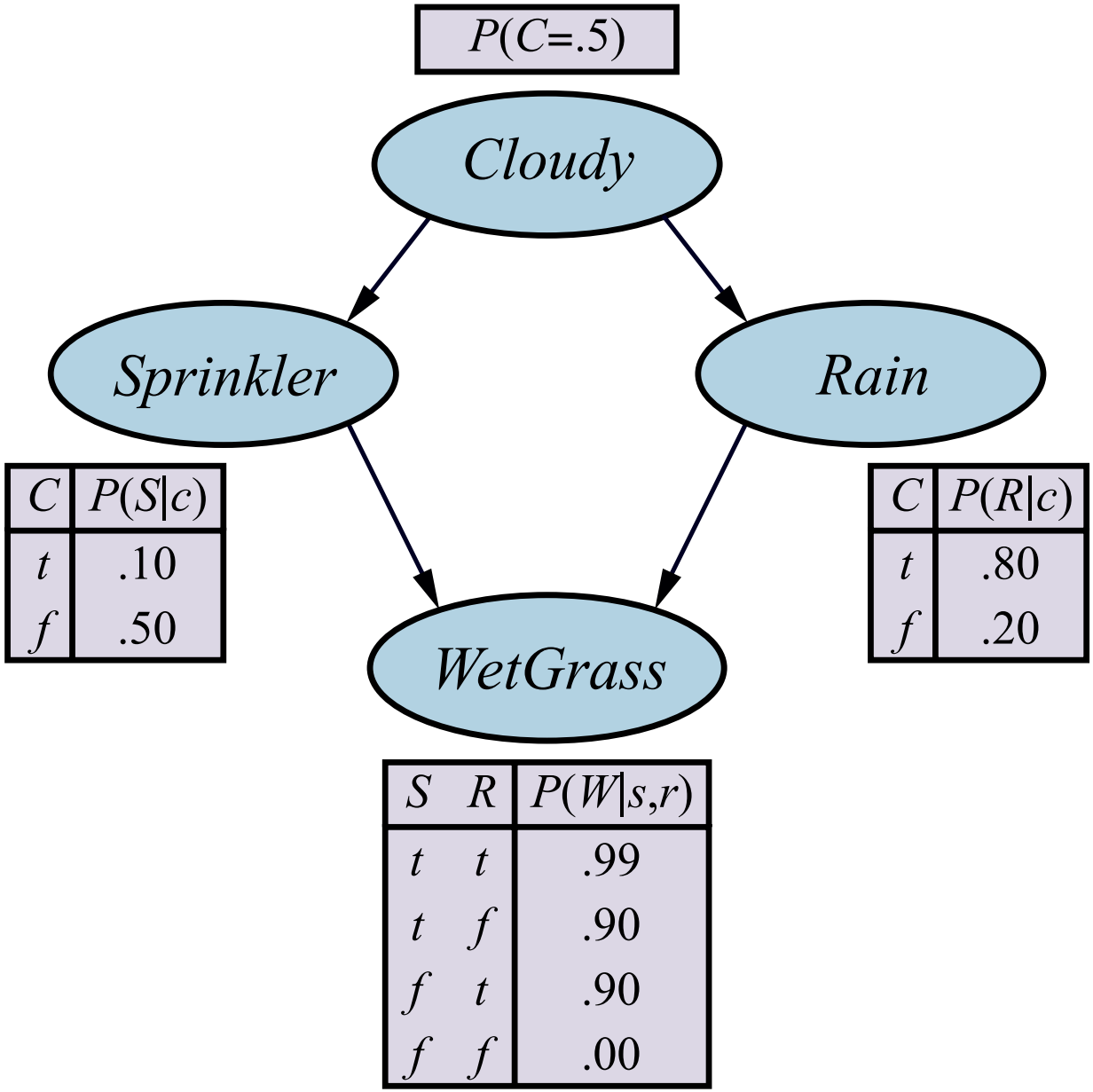
Remark: 贝叶斯网络通过利用条件独立性来高效地编码联合分布.

Terminology

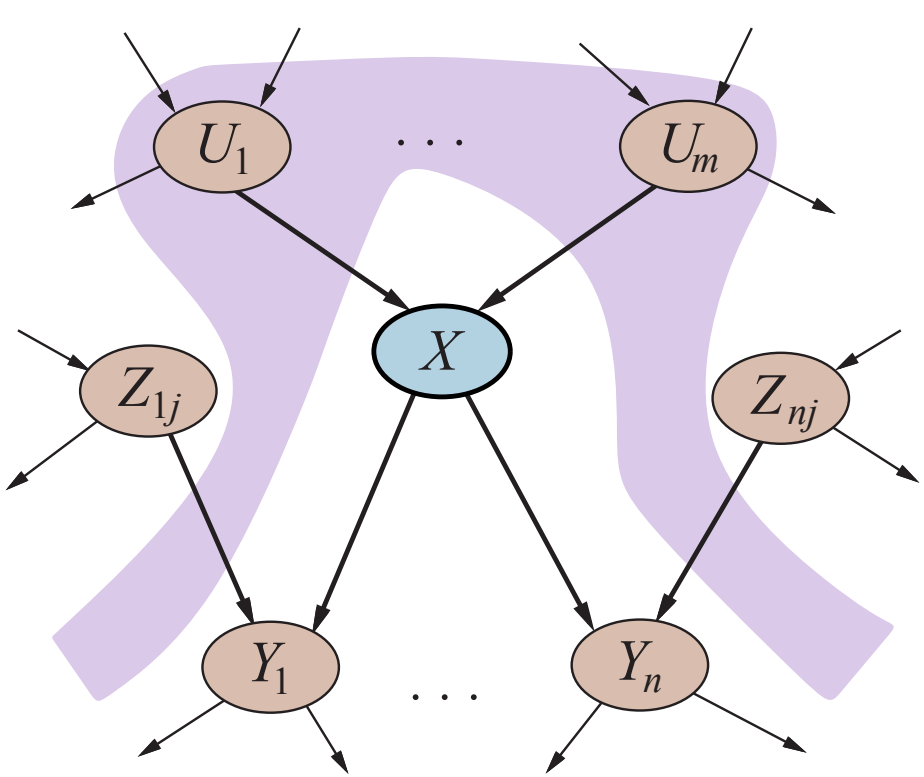


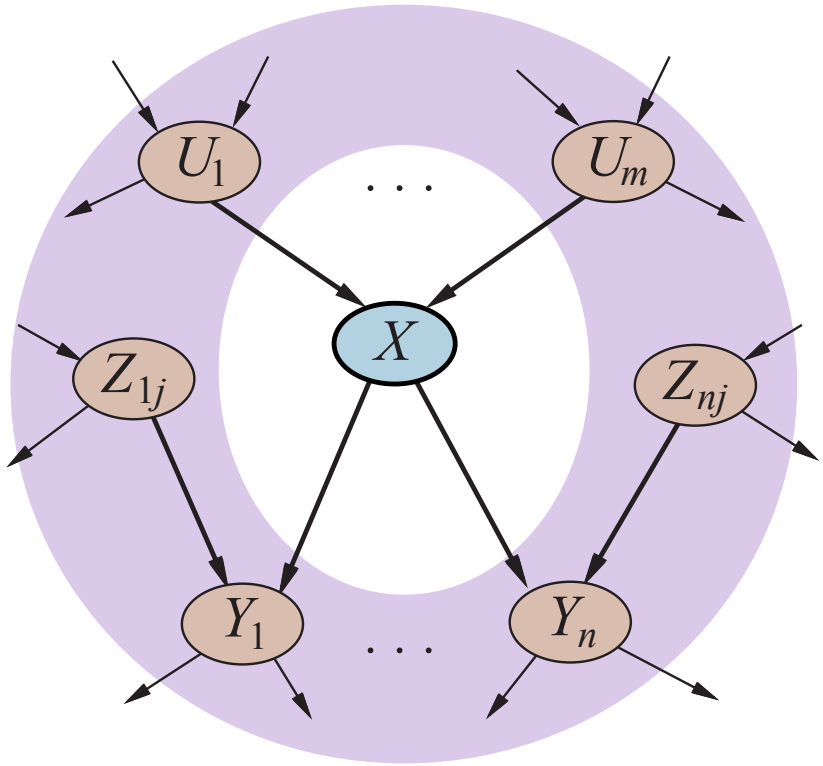
- A path is a sequence of distinct nodes in which every pair of successive nodes is adjacent. (Direction does not matter)
- A directed path is also called a causal path.
- A path from X to Y is proper iff only its first node is in X.
- If there is a directed path $X \to \cdots \to Y$, then X is an ancestor of Y, and Y is a descendant of X.

Bayesian Network — Example



Markov Condition & Markov Blanket





- **Markov condition:** A node is conditionally independent of its nondescendants given its parents. $X_i \perp \mathrm{ND}_i \mid \mathrm{Pa}_i$
  Remark: 给定直接原因, 一个变量与其非效应条件独立.
- A node is conditionally independent of all other nodes given its **Markov blanket**: parents + children + children's parents.

Remark: 马尔科夫毯 — 区分主体与环境

Markov Chain

- A Markov chain is a special sort of Bayesian network.

$$S_0 \longrightarrow S_1 \longrightarrow S_2 \longrightarrow S_3 \longrightarrow S_4$$

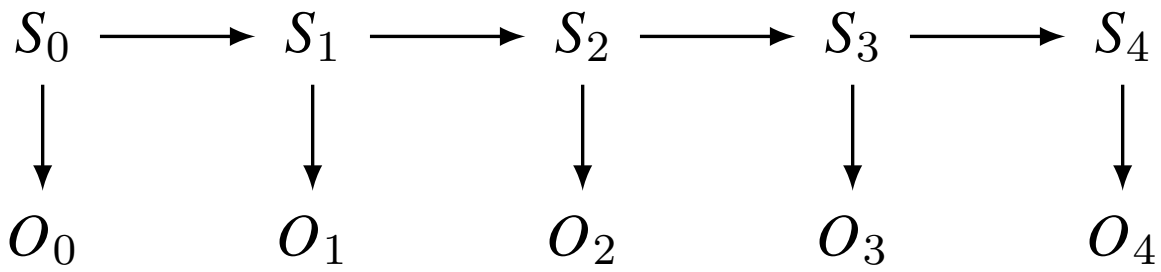
- $P(S_{t+1} \mid S_0, \ldots, S_t) = P(S_{t+1} \mid S_t)$
- S_t conveys all of the information about the history that can affect the future states.
- Markov assumption: "Future is independent of the past given the present."
- Stationarity assumption: transition probabilities are the same at all times.

Example: Random walk in one dimension

- State: location on the unbounded integer line
- Initial probability: starts at 0
- Transition model: $P(X_{t+1} = x \pm 1 \mid X_t = x) = \frac{1}{2}$
- Applications: particle motion in crystals, stock prices, gambling, genetics, etc.
- Questions:
  - How far does it get as a function of t?
    — Expected distance is $O(\sqrt{t})$
  - Does it get back to 0 or can it go off for ever and not come back?
    — In $1D$ and $2D$, returns with probability 1; in $3D$, returns with probability 0.34
    — 醉汉总能到家, 酒醉的小鸟却可能再也回不了家了.

Hidden Markov Model

- A Hidden Markov Model (HMM) is a Bayesian network.

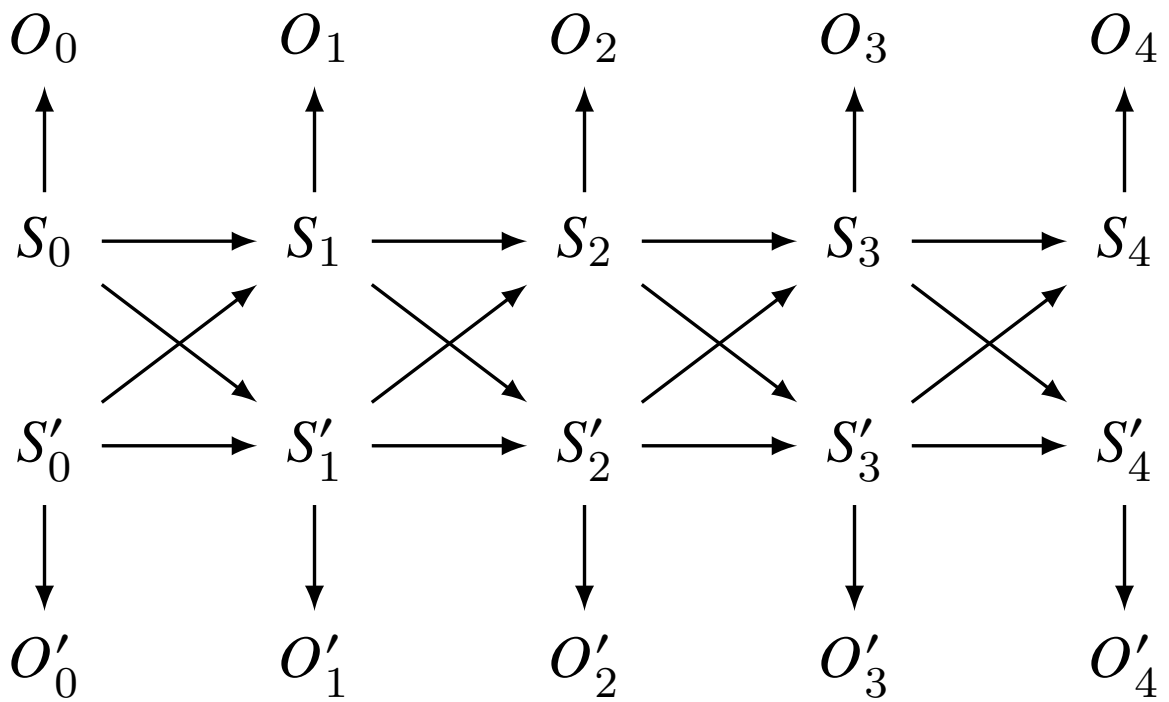


- Initial distribution: $P(S_0)$
- Transition model: $P(S_{t+1} \mid S_t)$
- Sensor model: $P(O_t \mid S_t)$

1. 滤波 filtering (推断现在): $P(s_t \mid o_{1:t})$
2. 预测 prediction (推断未来): $P(s_t \mid o_{1:k}),\ k < t$
3. 平滑 smoothing (推断过去): $P(s_t \mid o_{1:k}),\ k > t$
4. 推断路径: $\operatorname{argmax}_{s_{1:t}} P(s_{1:t} \mid o_{1:t})$

例子: 你在卧室睡觉, 听到家里进贼, 你熟悉房间的结构, 通过不同的声响推断贼的位置轨迹.
语音识别、音频去噪、天气预报...

Hidden Markov Model — Example



- 上面的 HMM 建模语音, 下面的 HMM 建模视频.
- S_t 对应音素, S'_t 对应口腔形状.
- 这个模型捕捉了口腔形状与音素之间的耦合关系.

Constructing Bayesian Network

To represent a domain in a Bayesian network, you need to consider:

- What are the relevant variables?
  - What will you observe?
  - What would you like to find out (query)?
  - What other features make the model simpler?
- What values should these variables take?
- What is the relationship between them?
- How does the value of each variable depend on its parents?

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

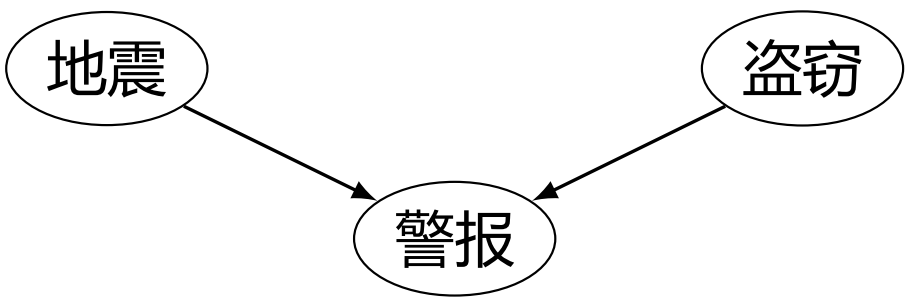
Chain, Fork, Collider — Examples

1. Chain

2. Fork

鞋子大小与阅读能力正相关.

3. Collider

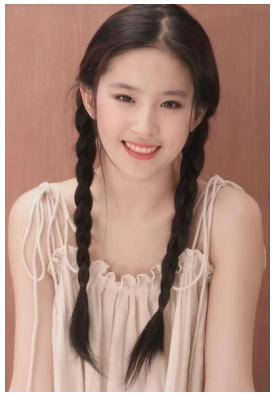


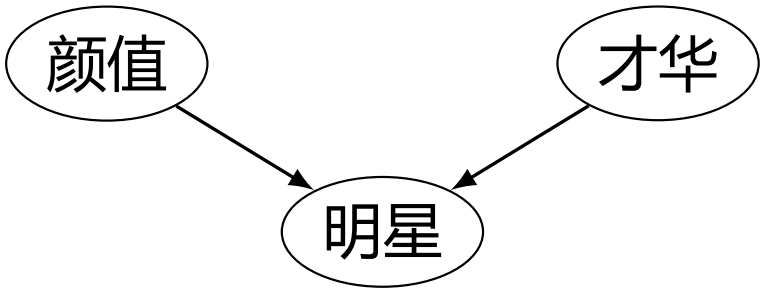
听到警报响, 发现地震了, 对入室盗窃的怀疑大大降低.

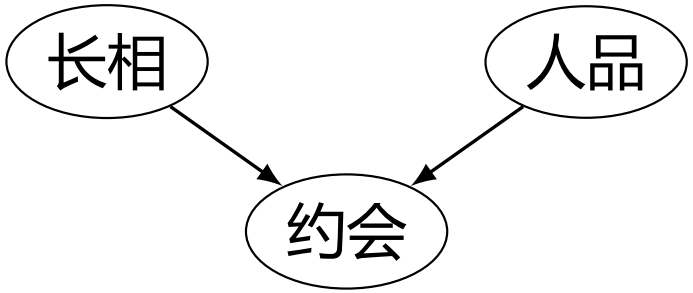
Berkson's Paradox & Simpson's Paradox

Berkson's Paradox:

— 为什么明星的颜值与才华负相关?







— 为什么帅哥往往是渣男?

— 为什么凡尔赛? "还以为这次考不了第一了, 最近天天玩, 都没复习."

Simpson's Paradox:

— 为什么穿着鞋子上床睡觉容易头疼?

— 为什么每个区的房价都在涨, 而全市的平均房价却在降?

— 为什么詹姆斯三分球和两分球的命中率都比乔丹高, 合起来却比乔丹低?

Theorem (Berkson's Paradox)

X, Y 是两个独立的 $\{0,1\}$ 随机变量. Z 是控制条件. 设

$$P = \begin{bmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{bmatrix}$$

其中 $p_{xy} := P(Z \mid X = x, Y = y)$, 则

1. 当 $\det(P) > 0$ 时, X 和 Y 在 Z 下正相关;
2. 当 $\det(P) = 0$ 时, X 和 Y 在 Z 下独立[a];
3. 当 $\det(P) < 0$ 时, X 和 Y 在 Z 下负相关.

[a]对于二元变量, 协方差为 0 等价于独立.

Remark: 条件正相关跟条件负相关本质上是一回事. 条件独立只有 $\det(P) = 0$ 时才发生, 这也解释了为什么 Berkson 悖论这么普遍.

Example: $X = 1$(长得帅), $Y = 1$(人品好), $Z = X \vee Y$(约).

$$P = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \implies \det(P) = -1$$

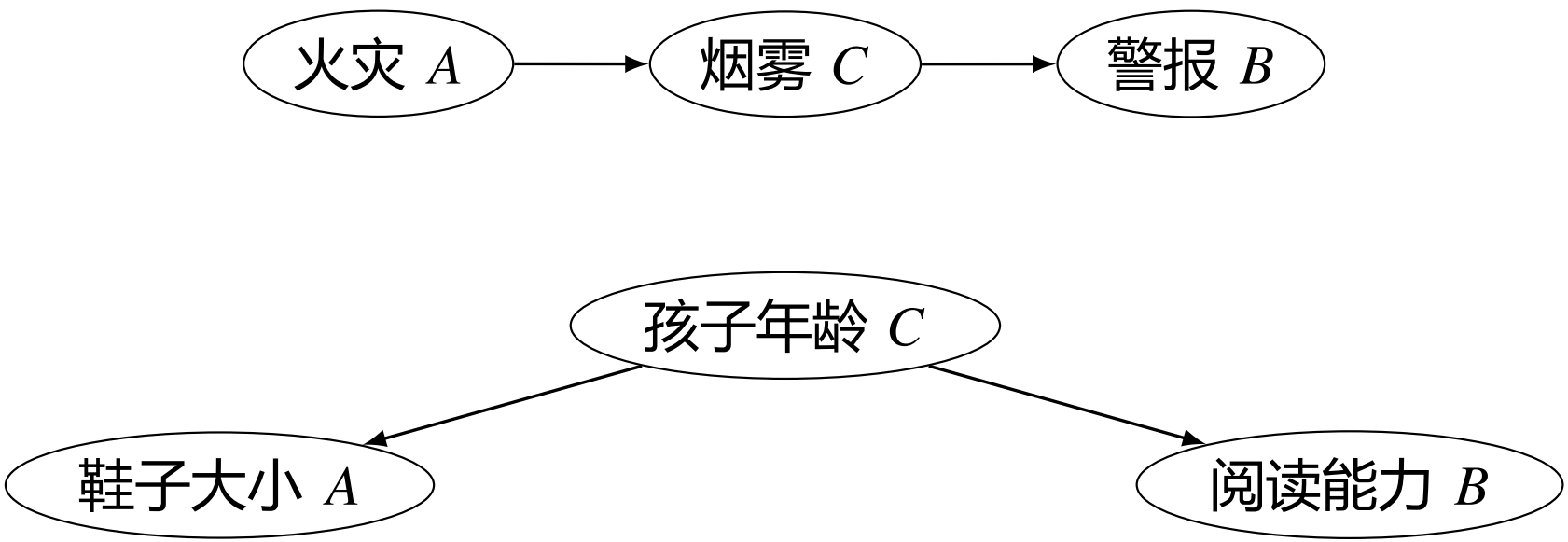
Screening off / Conditional Independence

- C screens off A from B iff

$$P(B \mid A \wedge C) = P(B \mid C)$$

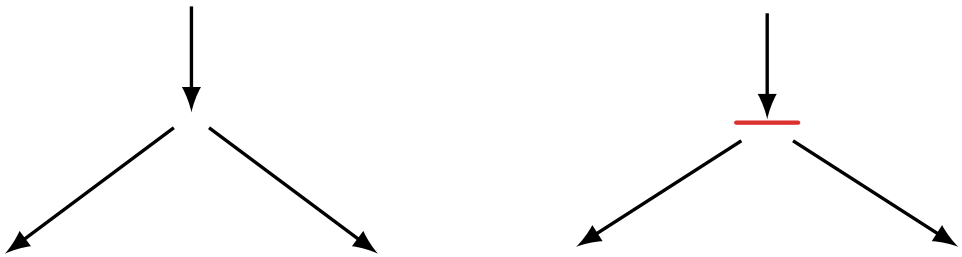
equivalently, $P(A \wedge B \mid C) = P(A \mid C)P(B \mid C)$.

- example

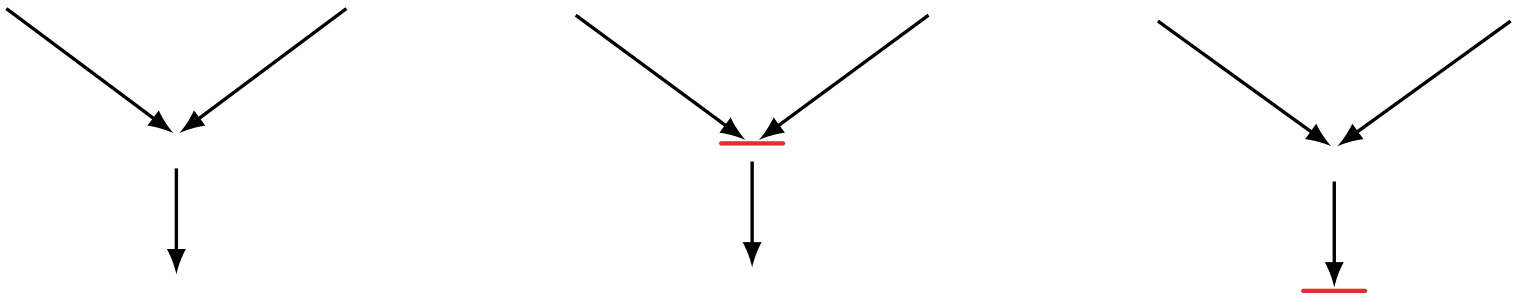


Remarks: 控制的变量越多越好吗?

关闭中间阀门, 水流不通.



打开中间阀门, 两边水位同时升高; 关闭中间阀门, 两边水位脱钩.



打开中间阀门, 水流排走; 关闭中间阀门, 堵塞排水通道, 两边水流贯通.

有向分离 d-separation

Definition (Blocking of Paths)

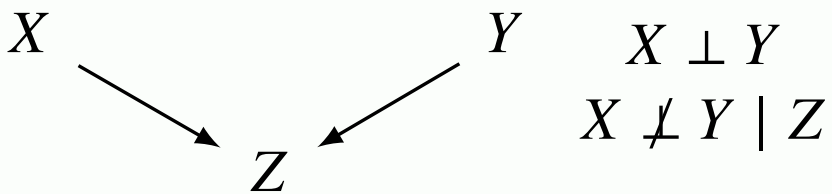
A path p is said to be blocked by a set Z iff

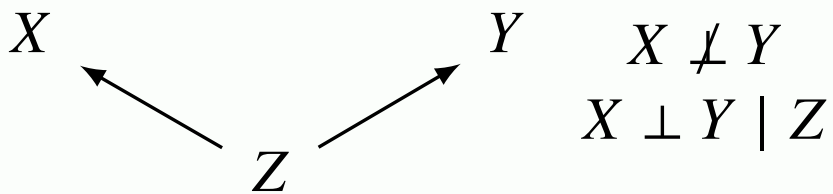
- p contains a chain $X \to W \to Y$ or a fork $X \leftarrow W \to Y$ such that the middle node is in Z, or
- p contains a collider $X \to W \leftarrow Y$ such that the middle node is not in Z and no descendant of W is in Z.

Definition (d-separation)

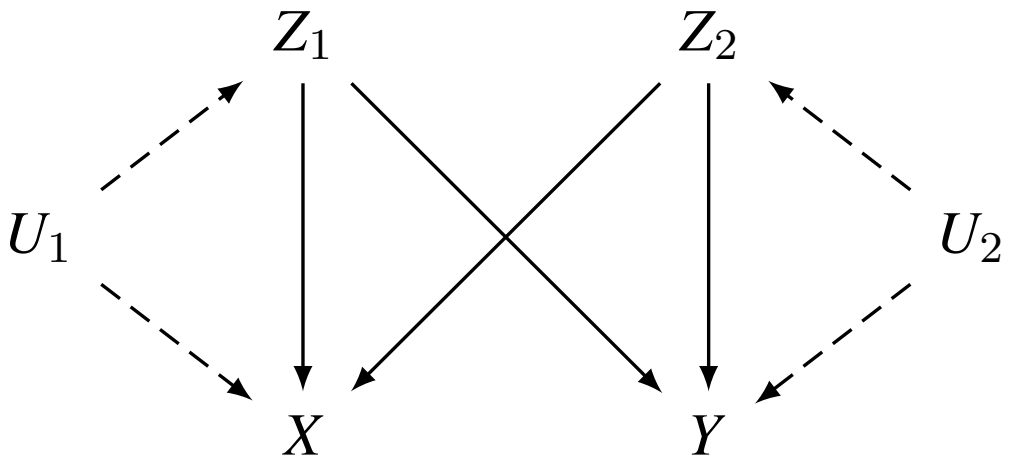
Z is said to d-separate X and Y in the DAG G, i.e. $(X \perp Y \mid Z)_G$ iff Z blocks every path from a node in X to a node in Y.

Example

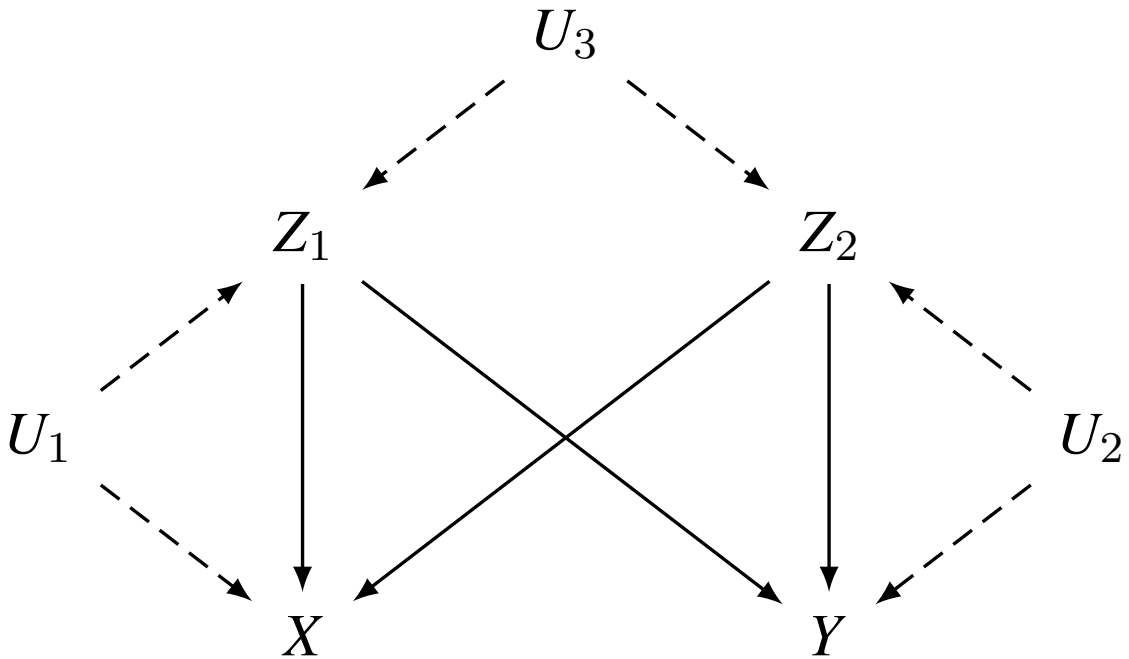




Example



$X \perp Y \mid Z_1, Z_2?$



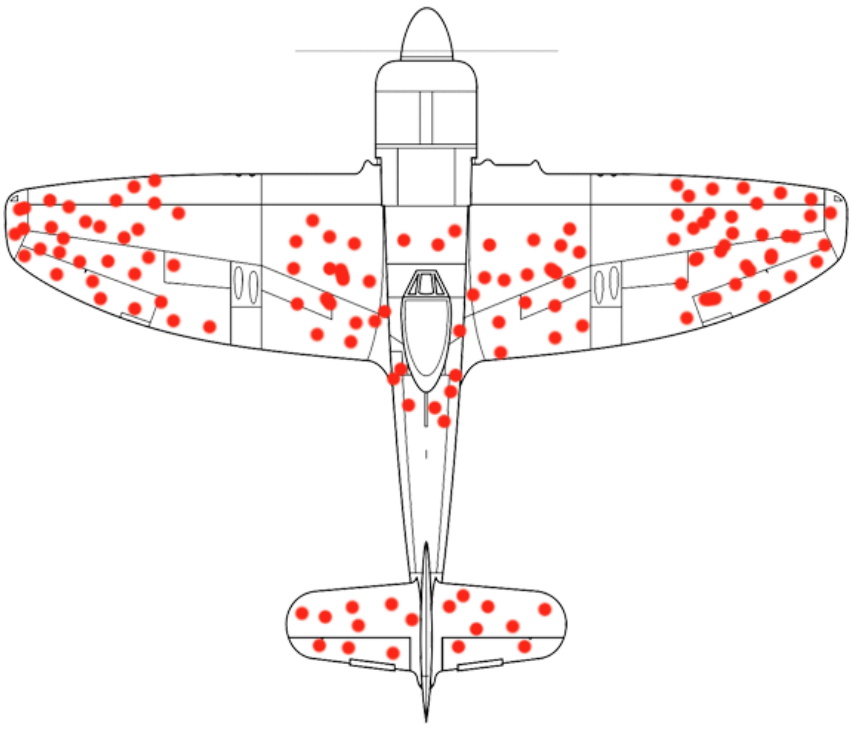
$X \perp Y \mid Z_1, Z_2?$

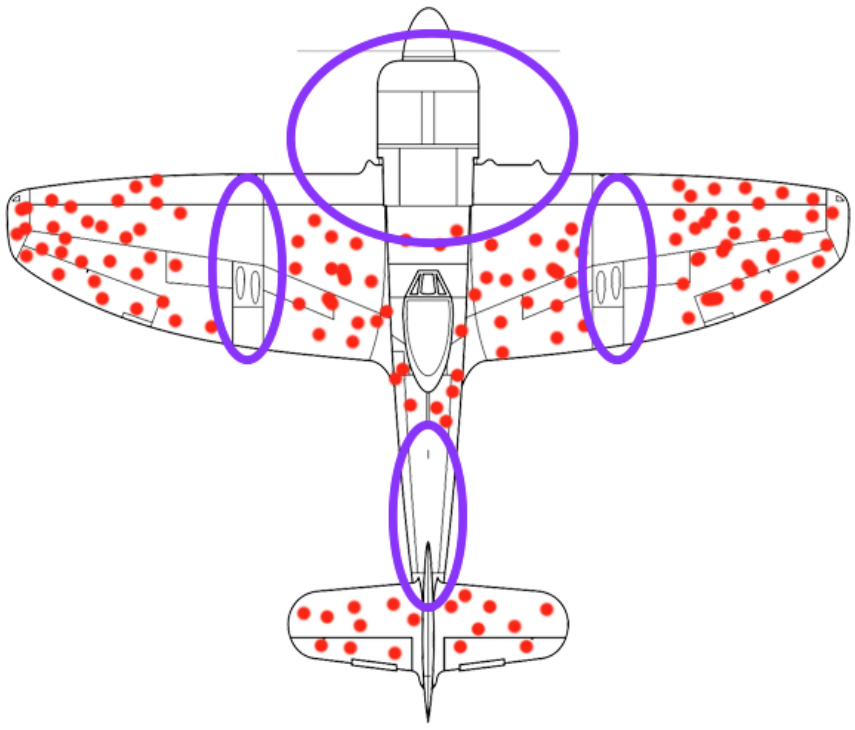
A Test for d-separation

To test for d-separation $(X \perp Y \mid Z)_G$,

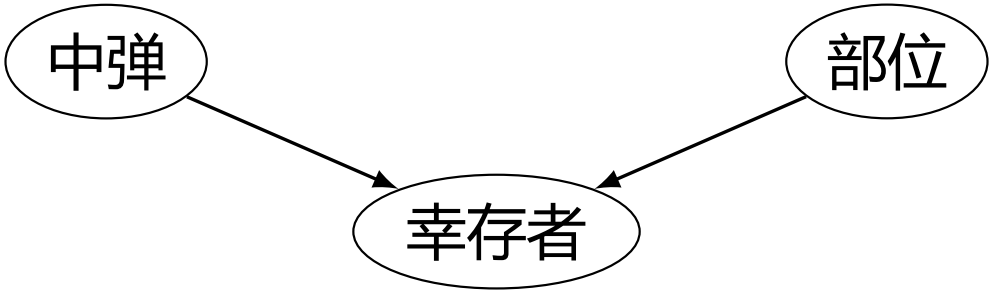
1. Delete all non-ancestors of X, Y, Z
2. Connect any two parents sharing a common child
3. Strip arrow-heads from all edges
4. Delete Z
5. Test if X is disconnected from Y in the remaining graph

Example — Collider

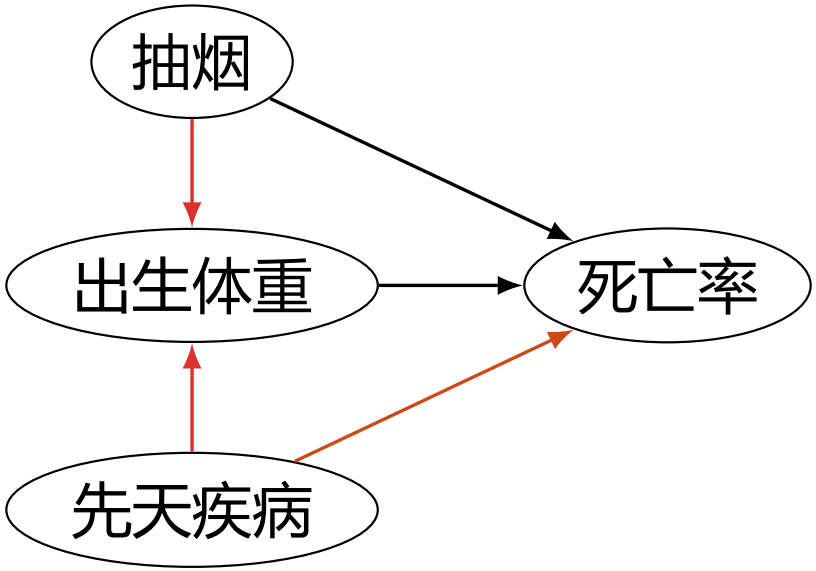




- 重点关注没中弹的部位.



Example — 抽烟对胎儿有好处吗?



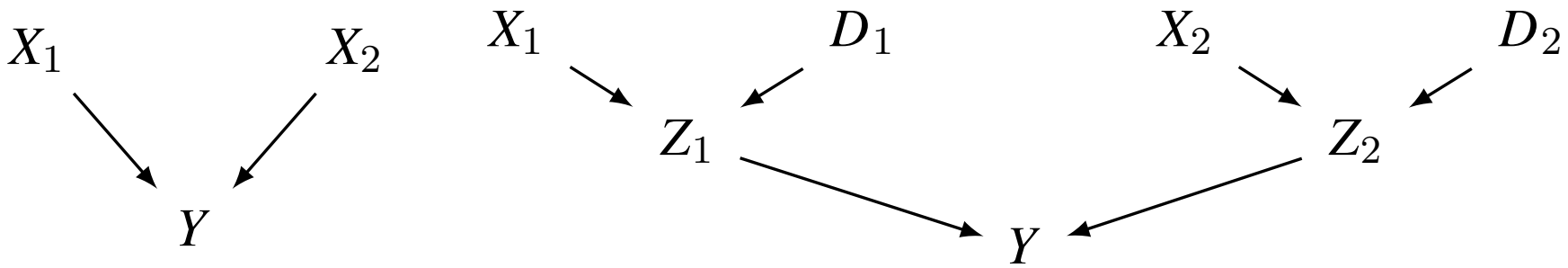
- 出生体重轻的婴儿比正常婴儿的死亡率高 20 倍.
- 抽烟母亲的婴儿平均比不抽烟母亲的婴儿轻.
- 抽烟母亲的出生体重轻的婴儿的死亡率比不抽烟母亲的出生体重轻的婴儿低.
- 这是抽烟带来的好处吗?
- 对于一个出生体重轻的婴儿, 母亲抽烟解释了为什么体重轻, 降低了患有先天疾病的可能性.

一道概率题

Problem

某人有两个孩子，其中至少有一个是男孩，那么另一个孩子也是男孩的概率是多少？

$$P(X_1=1, X_2=1 \mid Y=1) = \frac{P(Y=1|X_1=1,X_2=1)P(X_1=1,X_2=1)}{P(Y=1)} = \frac{\frac{1}{4}}{1-\frac{1}{2}\cdot\frac{1}{2}} = \frac{1}{3}$$



Problem

某人有两个孩子，其中至少有一个**出生在星期二**的男孩，那么另一个孩子也是男孩的概率是多少？

$$Z_i = [\![X_i = 1 \wedge D_i = 2]\!],\ Y = Z_1 \cdot Z_2$$

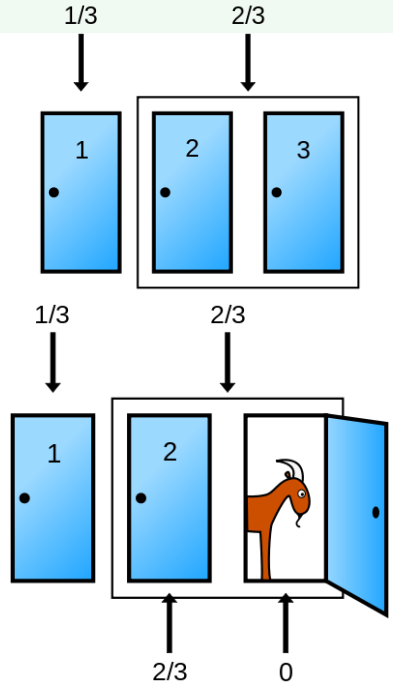
$$P(X_1=1, X_2=1 \mid Y=1) = \frac{P(Y=1|X_1=1,X_2=1)P(X_1=1,X_2=1)}{P(Y=1)} = \frac{\left(1-\frac{6}{7}\cdot\frac{6}{7}\right)\cdot\frac{1}{2}\cdot\frac{1}{2}}{1-\left(1-\frac{1}{2}\cdot\frac{1}{7}\right)\cdot\left(1-\frac{1}{2}\cdot\frac{1}{7}\right)} = \frac{13}{27}$$

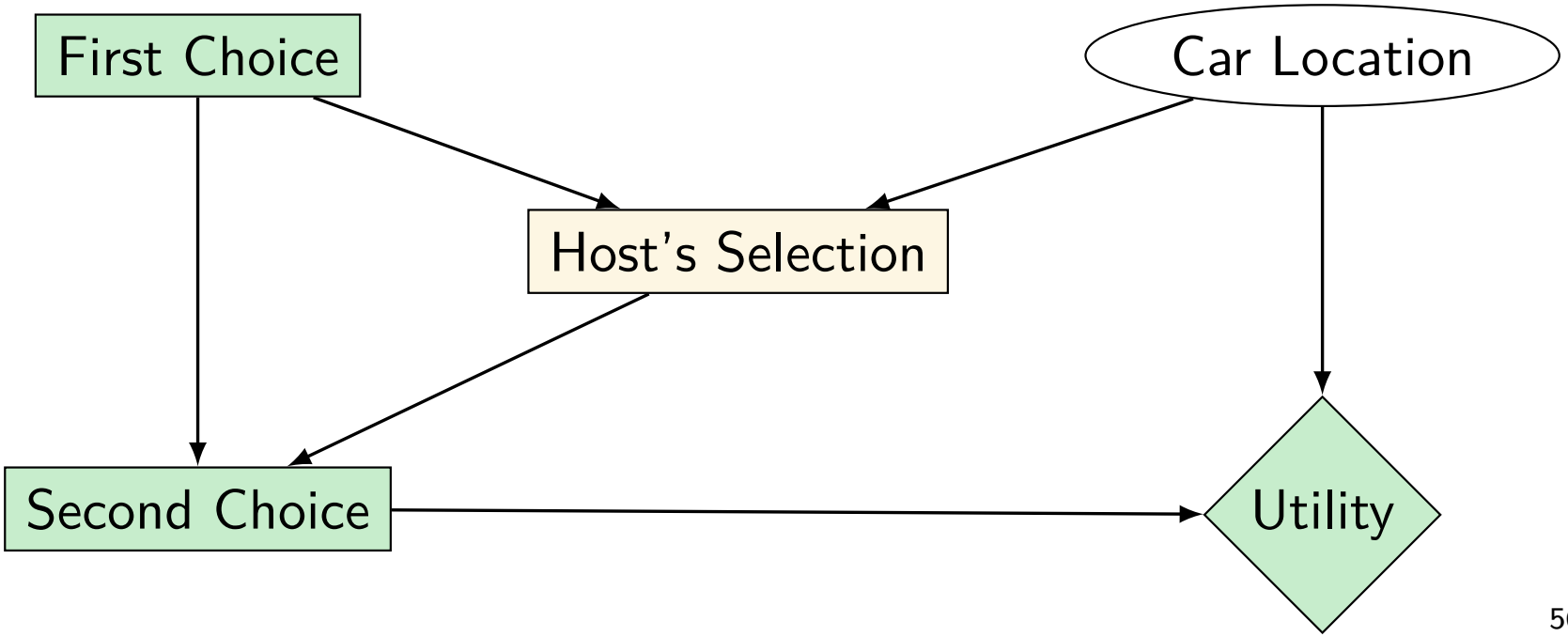
Monty Hall Problem

- 监狱长从 1、2、3 三个囚犯中随机选中了一个释放, 其它两个处死. 但事前不想让他们知道自己的命运.
- 囚犯 1 私下问监狱长: 能否告诉我囚犯 2、3 中谁会被处死?
- 监狱长: 告诉你也无妨, 囚犯 3 会被处死.
- 囚犯 1 把这件事悄悄地跟囚犯 2 说了.
- 囚犯 1: 我活下来的概率从 $\frac{1}{3}$ 提升到了 $\frac{1}{2}$ 了. 你也一样.
- 囚犯 2: 你活下来的概率还是 $\frac{1}{3}$. 我活下来的概率提升到 $\frac{2}{3}$ 了.

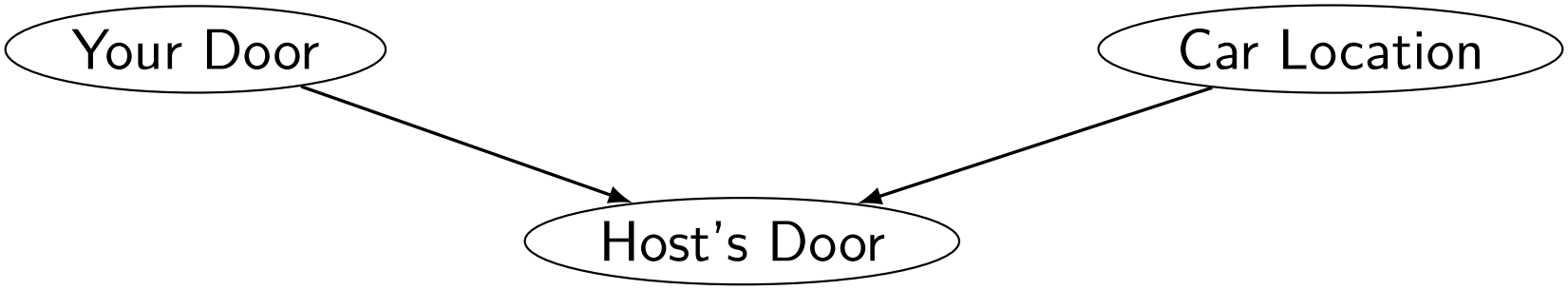
Monty Hall Problem

- You're given the choice of three doors.
- Behind one door is a car; behind the others, goats.
- You pick a door, say No.1, and the host, who knows what's behind the doors, opens another door, say No.3, which has a goat.
- He then says to you, "Do you want to pick door No.2?"





Monty Hall Problem



- C_i: the car is behind door number i.
- H_i: the host opens door number i.
- X_i: you choose door number i.

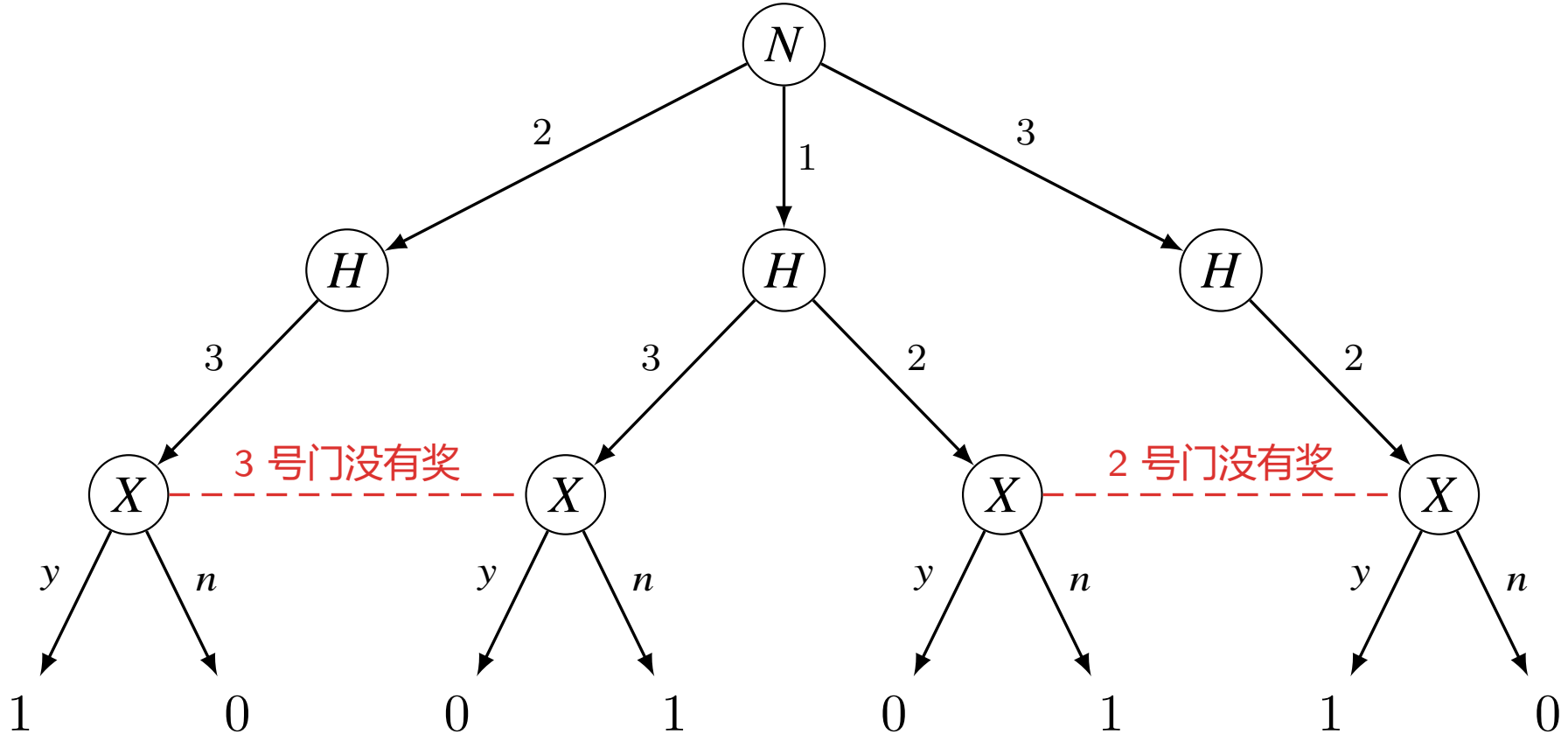
$$P(H_3 \mid X_1) = \sum_{i=1}^{3} P(H_3 \mid X_1, C_i) P(C_i) = \frac{1}{2} \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} = \frac{1}{2}$$

$$P(C_1 \mid X_1, H_3) = \frac{P(C_1, X_1, H_3)}{P(X_1, H_3)} = \frac{P(H_3 \mid X_1, C_1) \cancel{P(X_1)} P(C_1)}{P(H_3 \mid X_1) \cancel{P(X_1)}} = \frac{\frac{1}{2} \cdot \frac{1}{3}}{\frac{1}{2}} = \frac{1}{3}$$

$$P(C_2 \mid X_1, H_3) = \frac{P(C_2, X_1, H_3)}{P(X_1, H_3)} = \frac{P(H_3 \mid X_1, C_2) \cancel{P(X_1)} P(C_2)}{P(H_3 \mid X_1) \cancel{P(X_1)}} = \frac{1 \cdot \frac{1}{3}}{\frac{1}{2}} = \frac{2}{3}$$

决策树解法

- Nature 等概率分配奖品.
- 假设你选了 1 号门.
- 主持人打开了另外某扇门.
- 你愿意换门吗?



Monty Fall Problem — a variant

What if the host chooses a door that is different from yours but otherwise chosen at random?

- C_i: the car is behind door number i.
- H_i: the host opens door number i.
- X_i: you choose door number i.

$$P(C_i \mid X_1, H_3) = \frac{P(C_i, X_1, H_3)}{P(X_1, H_3)} = \frac{\cancel{P(H_3 \mid X_1)}\cancel{P(X_1)}P(C_i)}{\cancel{P(H_3 \mid X_1)}\cancel{P(X_1)}} = P(C_i) = \frac{1}{3}$$

$$P(C_1 \mid X_1, H_3, \neg C_3) = \frac{P(\neg C_3 \mid C_1, X_1, H_3)P(C_1 \mid X_1, H_3)}{P(\neg C_3 \mid X_1, H_3)} = \frac{1 \cdot \frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}$$

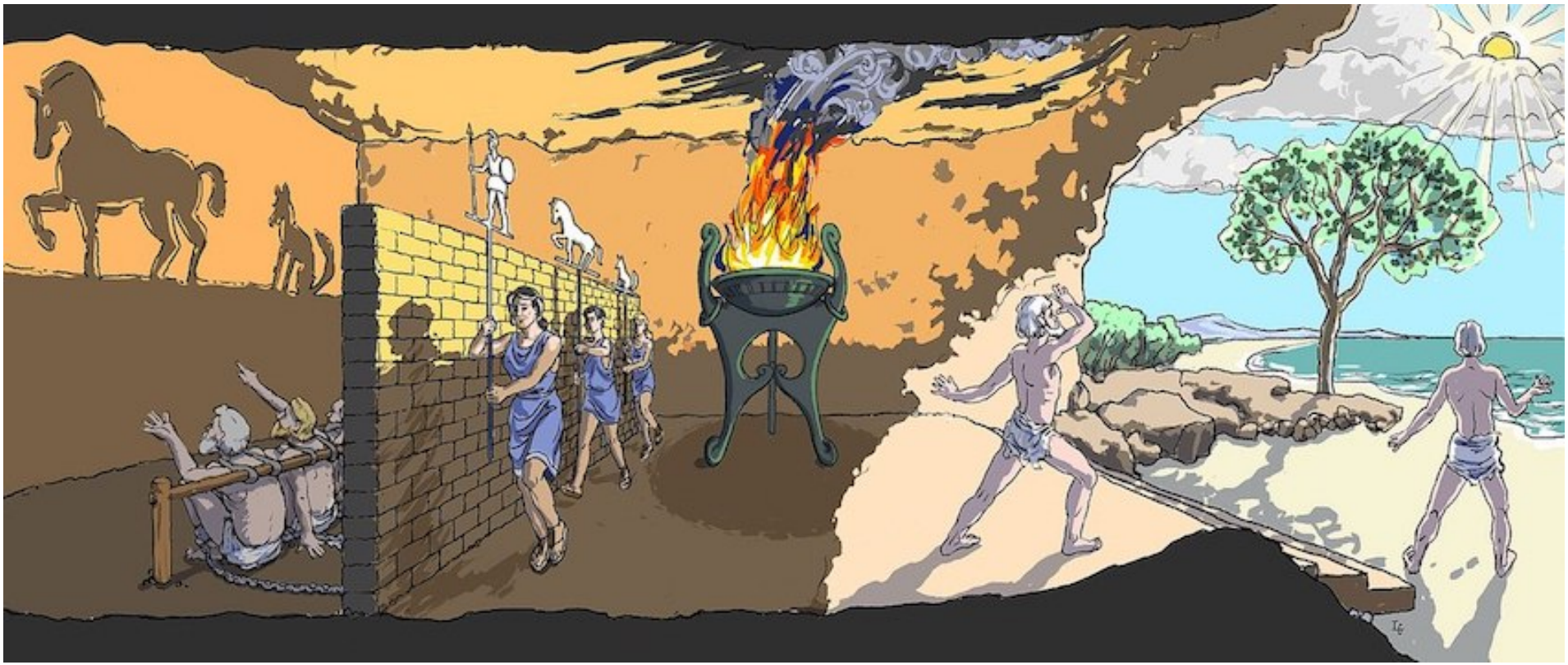
$$P(C_2 \mid X_1, H_3, \neg C_3) = \frac{P(\neg C_3 \mid C_2, X_1, H_3)P(C_2 \mid X_1, H_3)}{P(\neg C_3 \mid X_1, H_3)} = \frac{1 \cdot \frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}$$

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

肤浅的人相信运气或环境; 强者相信因果.

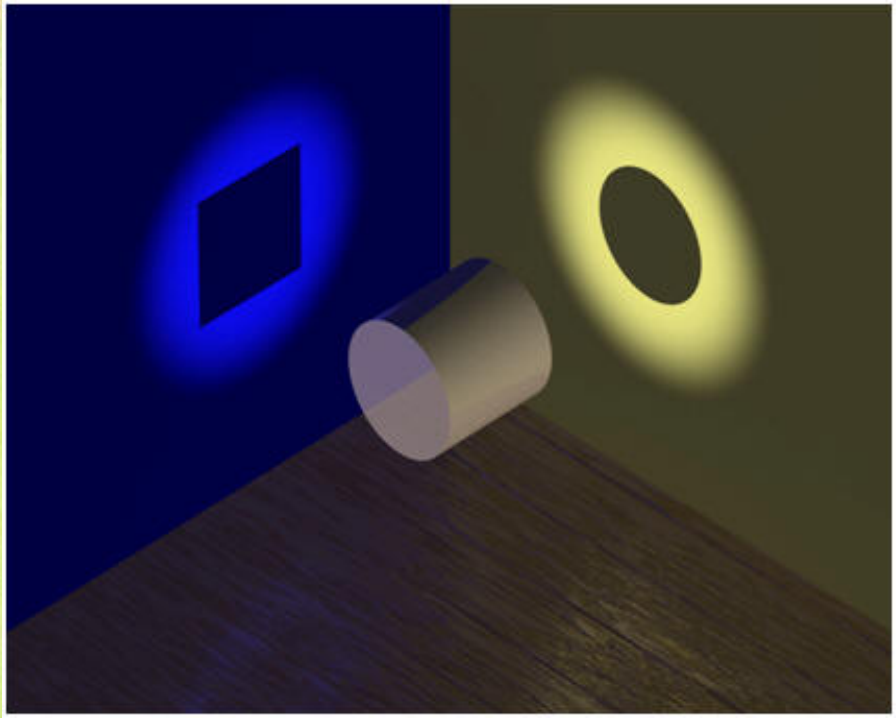
— 爱默生



潜在因果机制 → 观测数据

不要对数据的分布进行建模, 而是对数据的生成机制进行建模!

six
nine
Just because
you are right, does not
mean I'm wrong.
You just haven't seen
life from my side.



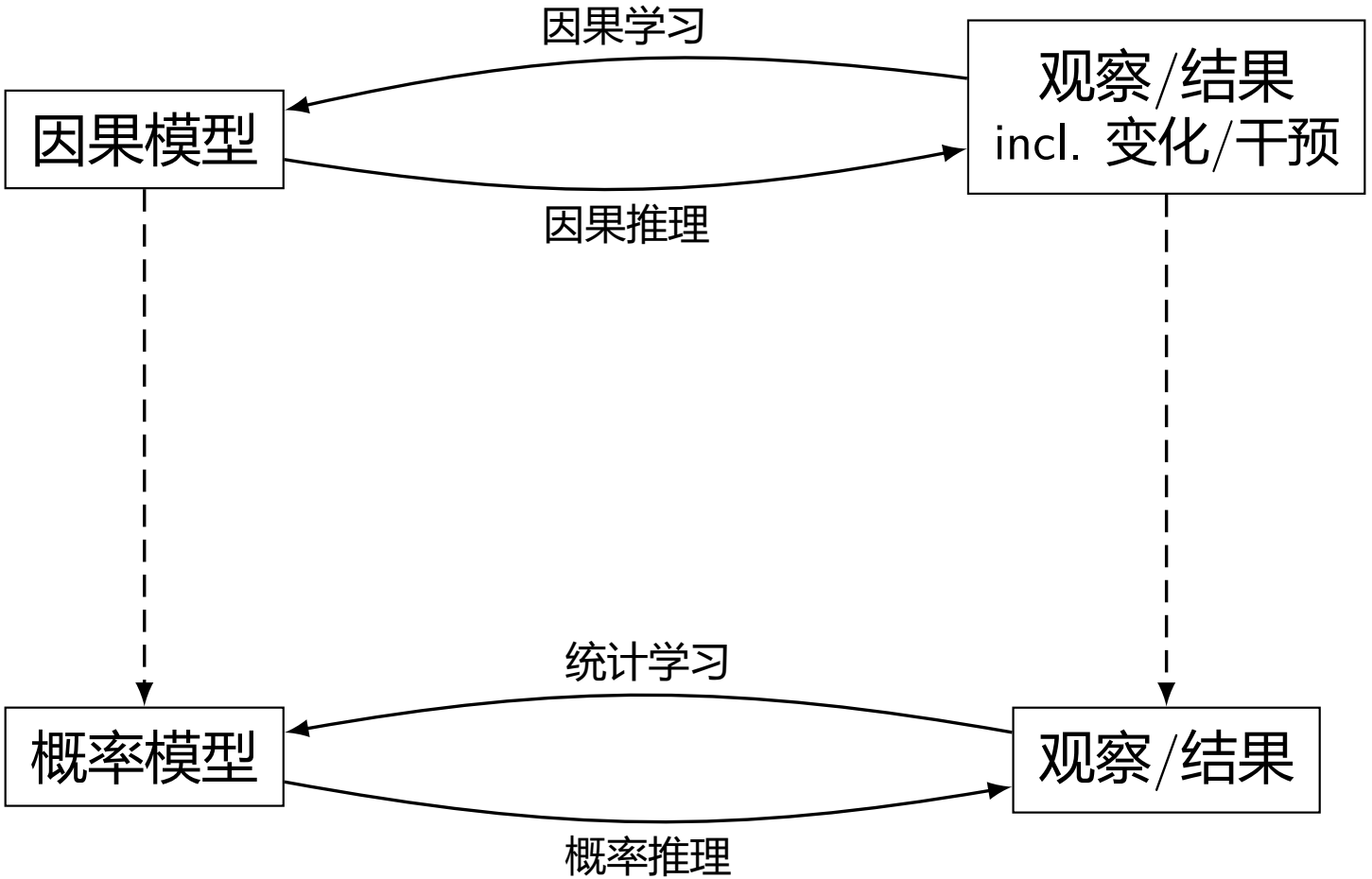
Inference: from _____ to _____

1. Statistics: from sample to distribution
2. Observational Causal Inference: from observational distribution to experimental distribution
3. Sampling Selection Bias: from study (obs/exp) distribution to general (obs/exp) distribution
4. General Transportability: from (obs/exp) distributions of populations $A, B, C \ldots$ to experimental distribution of a target population

概率 vs 因果

$$\frac{\text{If the grass is wet, then it rained.}\quad \text{If we break this bottle, the grass will get wet.}}{\text{If we break this bottle, then it rained.}}\ ?$$

概率论 vs 因果论



Remark: 概率 vs 因果

- 我们能否仅凭观察经验，在没有任何因果预设的情况下发现因果关系？
- 假如所有知识都源于人类经验，并且，人类经验都可以编码为概率分布的话，那么期望因果知识可以归约为概率就是自然的.
- 相比于确定性的因果，概率因果有一些认知好处. 无需详细指定物理状态和物理定律，可以用宏观状态之间的概率关系来概括，从而与自然语言的粒度相匹配.
- 概率因果更契合现代 (量子理论) 的不确定性概念.
- 因果可以归约为概率吗？
- 概率语言能够表达 "公鸡打鸣不会导致太阳升" 吗？加上时序关系呢？

"语言塑造思维. 你无法回答一个你提不出来的问题，也无法提出一个你的语言描述不了的问题."

— 珀尔《为什么》

Remark: 事件 vs 行动

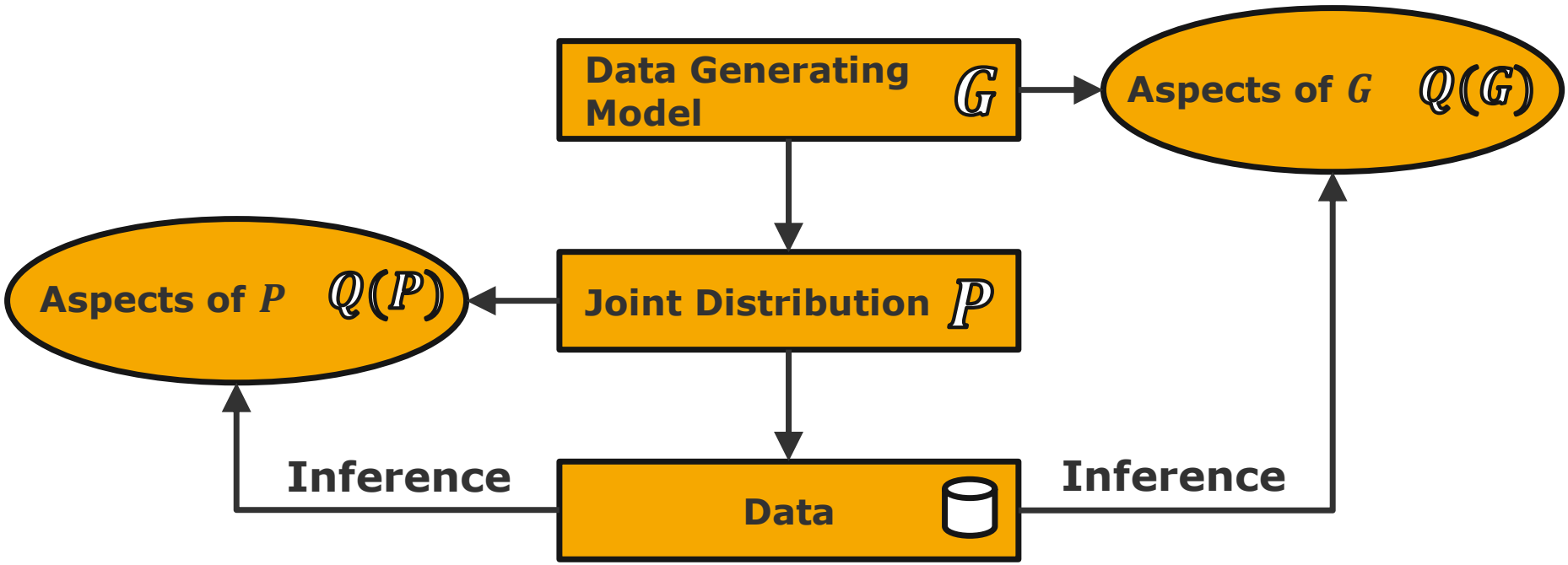
- 原则上, 行动不是概率的一部分.
- 概率刻画事件之间的分布关系.
- 行动代表了能够扰动这些关系的干预措施.
- 当已知 P_A 和 P_B 分别表示行动 A 和 B 的概率时, 我们无法推断出联合行动 $A \wedge B$ 对应的联合概率 $P_{A \wedge B}$, 或其它布尔组合的概率.
- 类似视觉知觉, 概率分布 $P(s)$ 中的信息类似于对三维物体的精确描述, 这足以预测从任一角度观察时, 该物体所呈现的样子, 但不能预测物体受到外力挤压时会变成什么样子, 这需要提供关于物体物理特性的额外信息, 这些额外信息由因果知识提供.
- 理性决策者应该根据行动理论行事.

因果 vs 行动

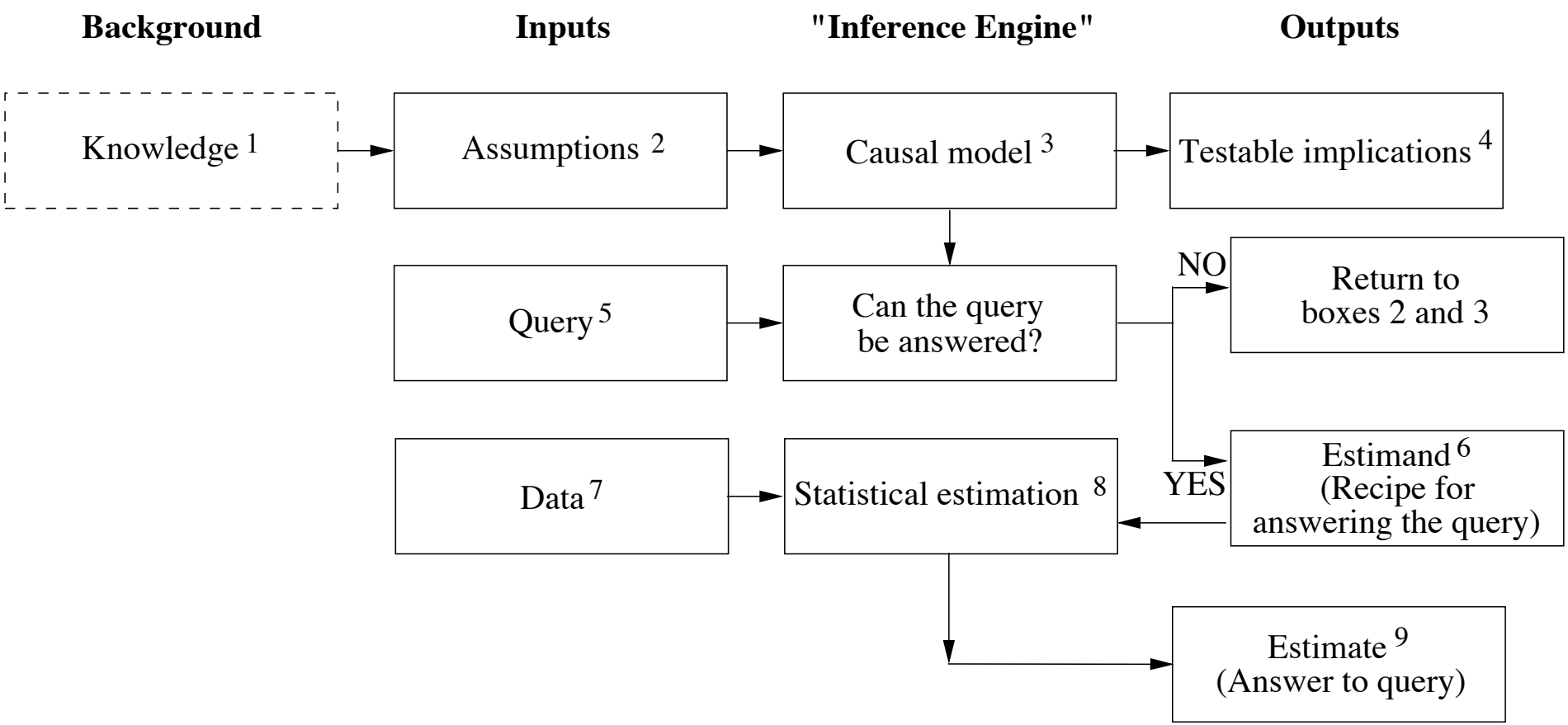
- 怎么学习获得因果模型?
- 普遍可重复的、随机、受控、实验

Traditional Statistical Inference Paradigm

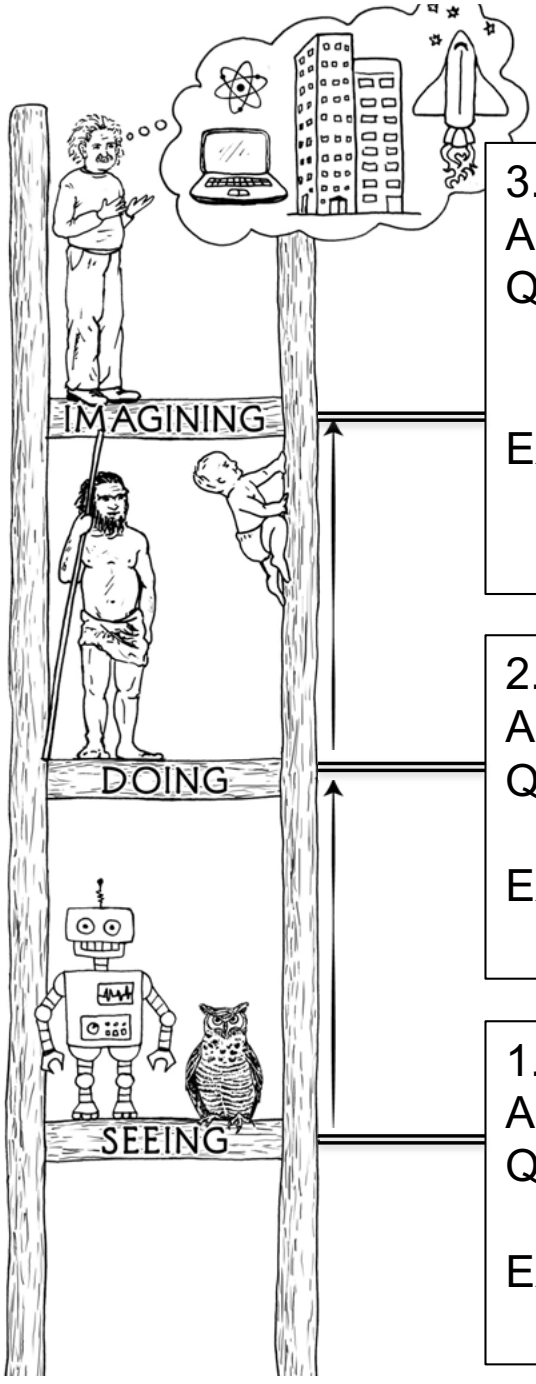
Paradigm of Structural Causal Models



因果推断引擎


Background
Inputs
"Inference Engine"
Outputs
Knowledge 1
Assumptions 2
Causal model 3
Testable implications 4
Query 5
Can the query be answered?
NO
Return to boxes 2 and 3
Data 7
Statistical estimation 8
YES
Estimand 6 (Recipe for answering the query)
Estimate 9 (Answer to query)

THE LADDER OF CAUSATION



3. COUNTERFACTUALS
ACTIVITY: Imagining, Retrospection, Understanding
QUESTIONS: *What if I had done . . . ? Why?*
(Was it X that caused Y? What if X had not occurred? What if I had acted differently?)
EXAMPLES: Was it the aspirin that stopped my headache? Would Kennedy be alive if Oswald had not killed him? What if I had not smoked the last 2 years?

2. INTERVENTION
ACTIVITY: Doing, Intervening
QUESTIONS: *What if I do . . . ? How?*
(What would Y be if I do X?)
EXAMPLES: If I take aspirin, will my headache be cured? What if we ban cigarettes?

1. ASSOCIATION
ACTIVITY: Seeing, Observing
QUESTIONS: *What if I see . . . ?*
(How would seeing X change my belief in Y?)
EXAMPLES: What does a symptom tell me about a disease? What does a survey tell us about the election results?

因果关系三阶梯

1 Association $P(Y \mid X = x)$

- **Activity:** Seeing, Observing
- **Questions:** What if I see ...?
  (How are the variables related? How would seeing X change my belief in Y?)
- **Examples:** What does a symptom tell me about a disease?
  What does a survey tell us about the election results?

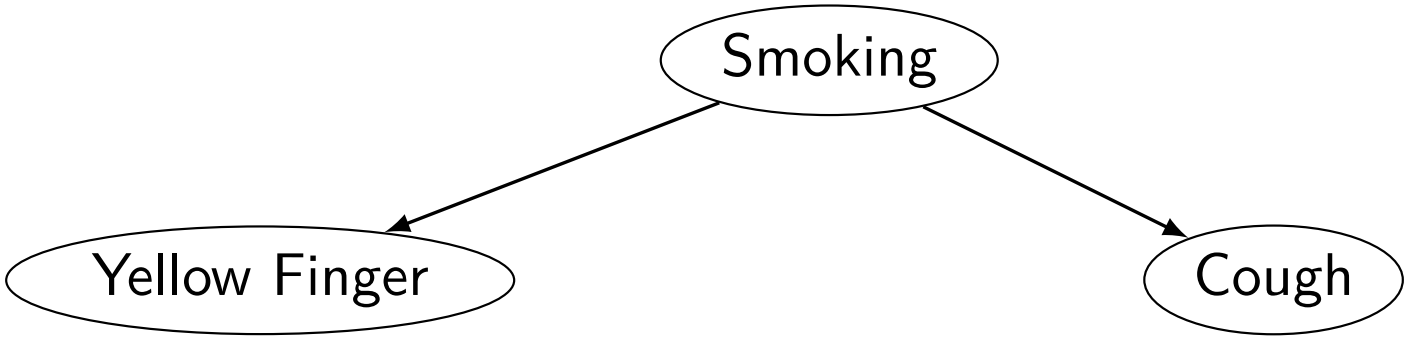
2 Intervention $P(Y \mid \mathrm{do}(X = x))$

- **Activity:** Doing, Intervening
- **Questions:** What if I do ...? How?
  (What would Y be if I do X? How can I make Y happen?)
- **Examples:** If I take aspirin, will my headache be cured?
  What if we ban cigarettes?

3 Counterfactuals $P(Y_{X=x'} \mid X = x, Y = y)$

- **Activity:** Imagining, Retrospection, Understanding
- **Questions:** What if I had done ...? Why?
  (Was it X that caused Y? What if X had not occurred? What if I had acted differently?)
- **Examples:** Was it the aspirin that stopped my headache?
  Would Kennedy be alive if Oswald had not killed him?
  What if I had not smoked for the last 2 years?

从 “看” 到 “做” 再到 “想”



1. Prediction: Would the person cough if we find he/she has yellow fingers?

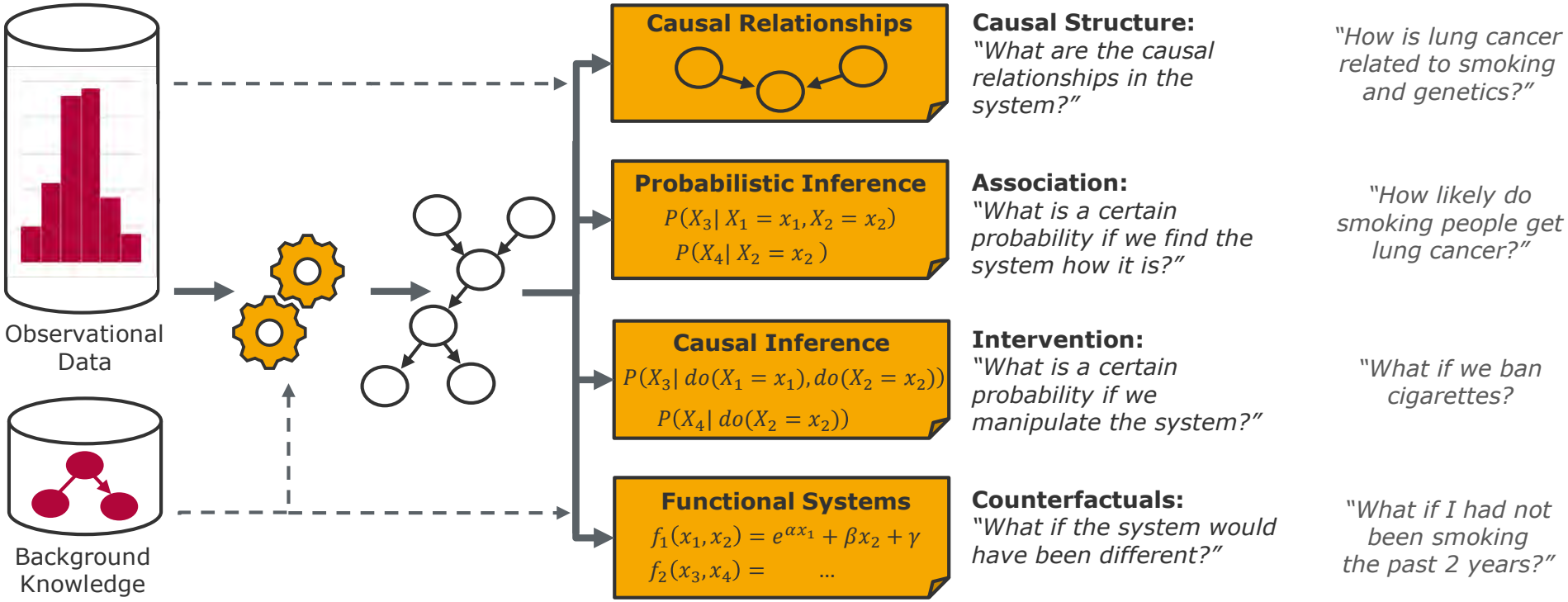
$$P(C \mid Y = 1)$$

2. Intervention: Would the person cough if we make sure that he/she has yellow fingers?

$$P(C \mid \text{do}(Y = 1))$$

3. Counterfactual: Would George cough had he had yellow fingers, given that he does not have yellow fingers and coughs?

$$P(C_{Y=1} \mid Y = 0, C = 0)$$



Graphical Representation

Association Bayesian Network

Intervention Causal Graph / Causal Bayesian Network

Counterfactuals Structural Causal Model / Functional Causal Graph

The Causal Hierarchy

1. Association: "What if I see x?"

$$P(y \mid x)$$

2. Intervention: "What if I do x?"

$$P(y \mid \mathtt{do}(x))$$

3. Counterfactuals: "What if I had done things differently?"

$$P(y'_{x'} \mid x, y)$$

4. Options: "With what probability?"

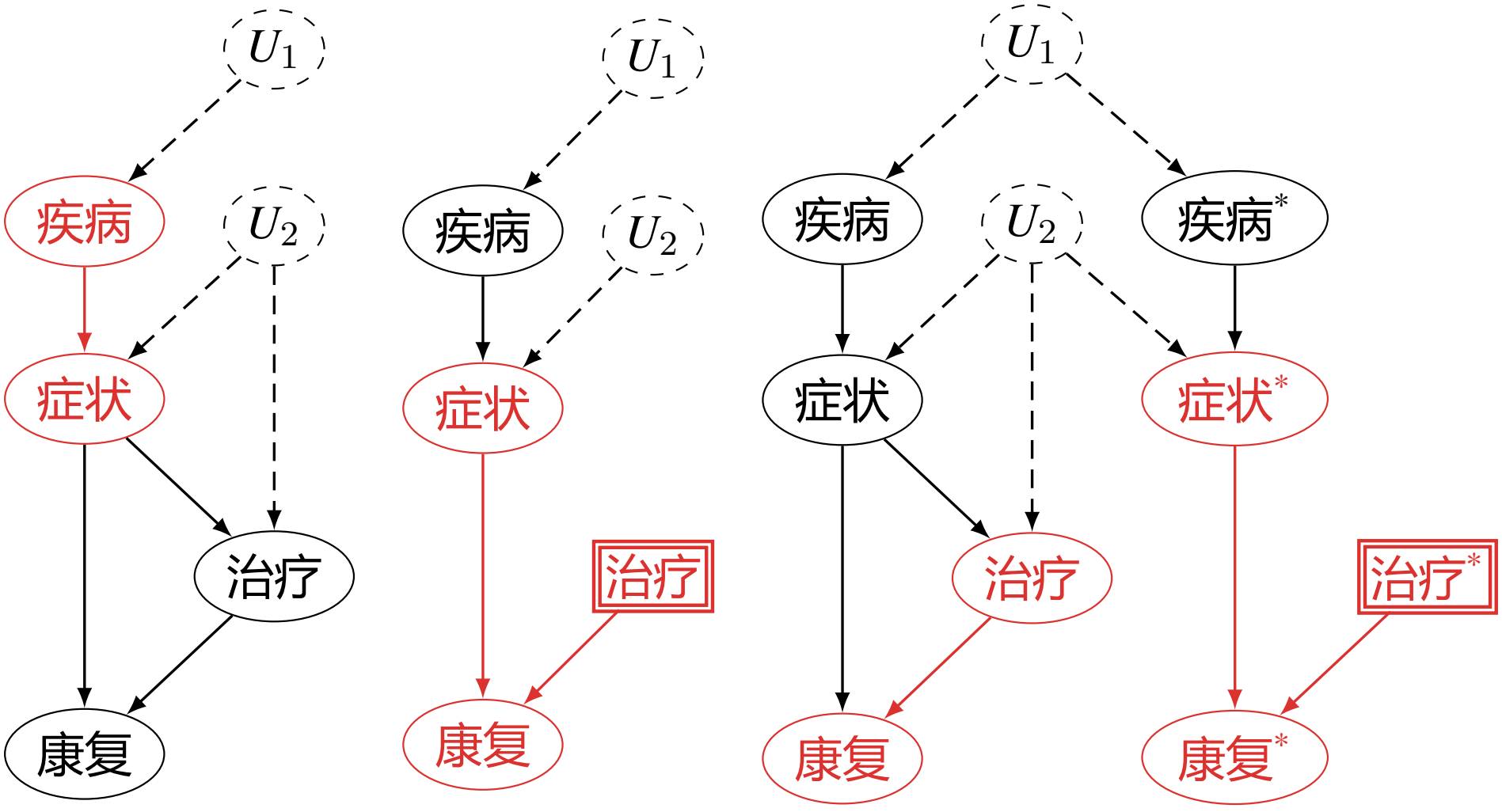
Explanation y because of x

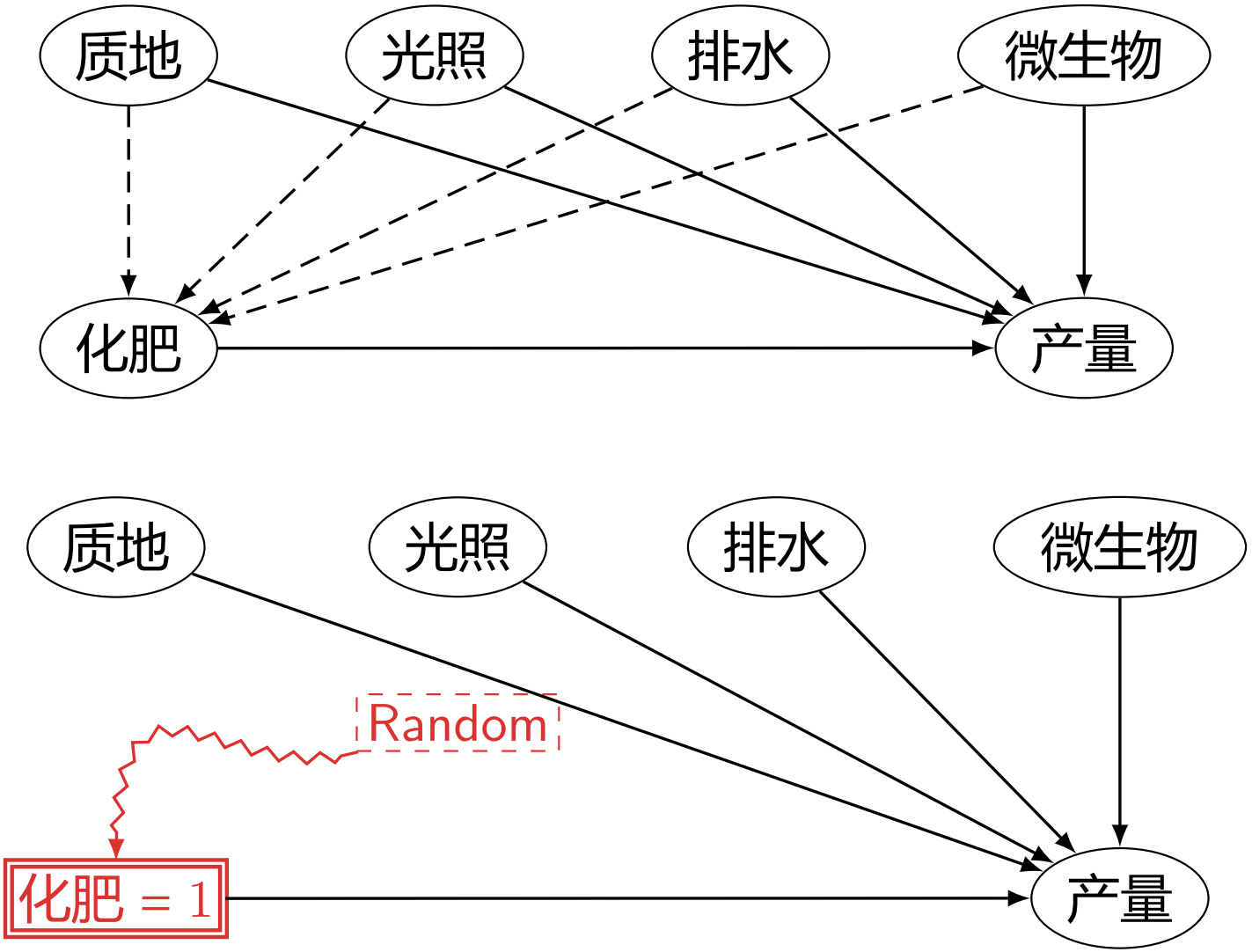
Intervention y will be true if I do x

Counterfactuals y would be different if x' were true

因果关系三阶梯

1. 发现了某症状，得某疾病的可能性多高？
2. 如果进行某项治疗，病人会康复吗？
3. 病人接受了某项治疗并康复了，如果不治疗是否也会康复？

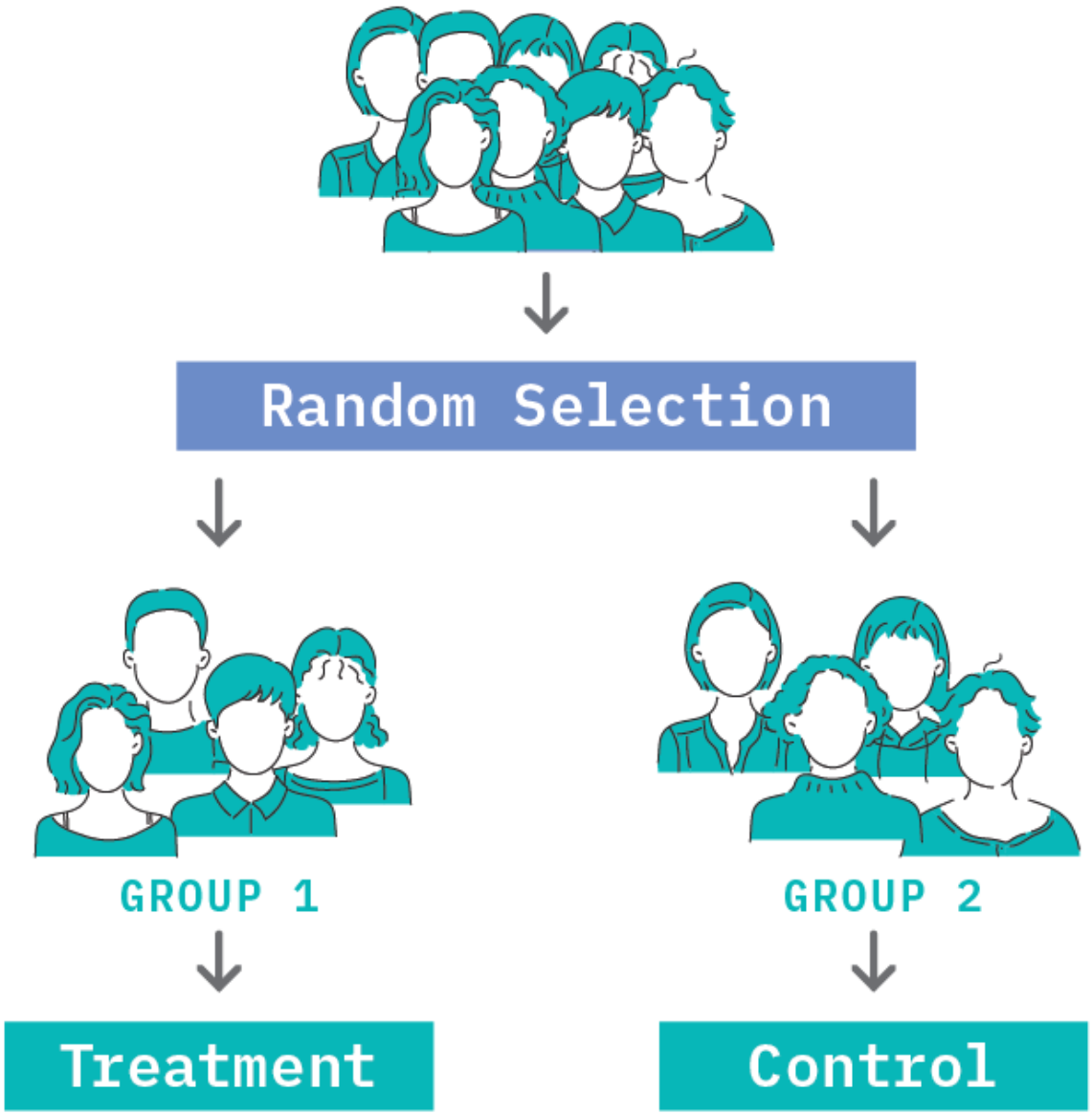




Remark: "随机" 去除了 "混杂因子" 对 Treatment 的因果作用.

$$\mathrm{TE} = \mathbb{E}[Y \mid \mathrm{do}(X = 1)] - \mathbb{E}[Y \mid \mathrm{do}(X = 0)]$$

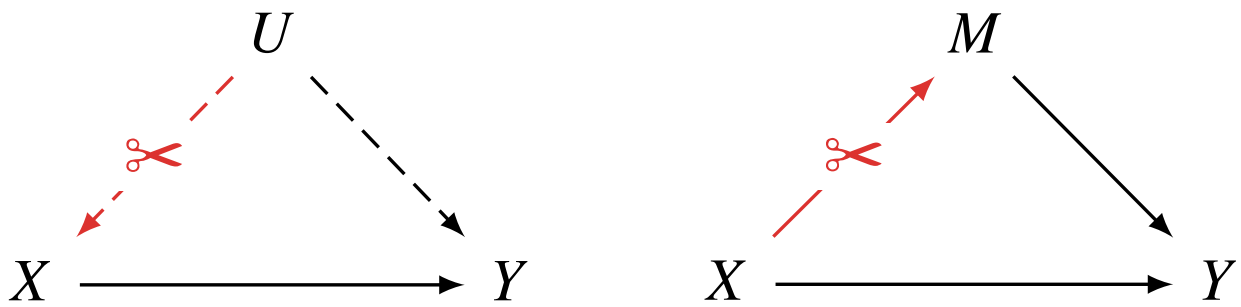
Randomized Control Trial



RCT 尽可能 “双盲”

- 实验对象应该对他们所接受的特定治疗一无所知
- 研究人员对每个受试者接受的是哪种治疗也应该一无所知

随机与双盲



- 随机可以去混杂.
- 双盲可以排除间接效应.

Randomized Control Trial & Covariate Balance

- Treatment and control groups are the same in all aspects except treatment.
- We have *covariate balance* if the distribution of covariates Z is the same across treatment groups. More formally,

$$P(Z \mid X = 1) \overset{d}{=} P(Z \mid X = 0) \qquad (\overset{d}{=} \text{ means 'equal in distribution'})$$

- "Randomization" implies "covariate balance".

$$\text{Randomization} \implies X \perp Z$$

$$P(Z \mid X = 1) \overset{d}{=} P(Z)$$

$$P(Z \mid X = 0) \overset{d}{=} P(Z)$$

$$P(Z \mid X = 1) \overset{d}{=} P(Z \mid X = 0)$$

- "Covariate balance" implies "association is causation".

$$P(y \mid \text{do}(x)) = \sum_z P(y \mid x, z)P(z) = \sum_z \frac{P(y \mid x, z)P(x \mid z)P(z)}{P(x \mid z)}$$

$$= \sum_z \frac{P(y, x, z)}{P(x)} = \sum_z P(y, z \mid x) = P(y \mid x)$$

随机对照试验总能做吗?

- 伦理原因 (例如, 不能随机分配人们抽烟以测量抽烟对肺癌的影响)
- 不可行 (例如, 不能随机分配国家实行共产主义/资本主义制度以测量其对 GDP 的影响)
- 不可能 (例如, 不能改变一个人的 DNA 以测量其对乳腺癌的影响)

问题: 是否 "没有操纵就没有因果"?

- 因果关系本身独立于干预而存在.
- 干预是我们 "拷问" 大自然的手段.

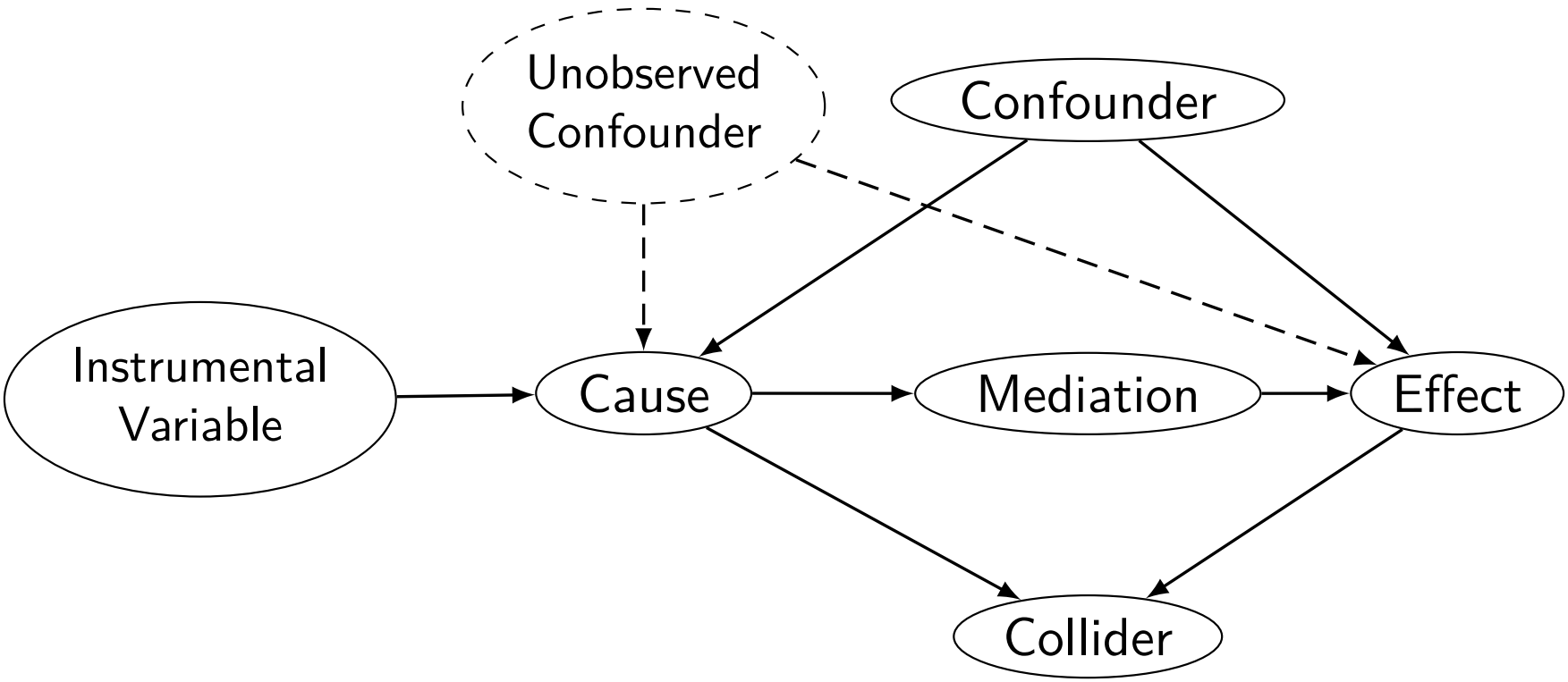
如果无法进行随机对照试验怎么办?

- 我们可以从随机对照试验中学习因果模型.
- **问题:** 能否直接从观测数据中计算因果效应, 从而无需进行干预? 有时可以, 但并非总能如此.
- 无法从观测数据中学习因果模型, 除非借助因果预设.
  - 在统计学里, 贝叶斯主义也讲先验, 但并不是太重要, 只要有足够多的经验, 先验可以被修正收敛到合理的后验.
  - 在因果分析中, 在任何因果结论背后一定有某种未经检验的因果假定. 没有合理的因果假定, 再多的数据也得不到因果关系.

What if you cannot do a randomized control trial?

How to infer causal relations from observational data without experiment?

$$P(e \mid \mathtt{do}(c))$$



Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Structural Causal Model

Definition (Structural Causal Model SCM)

A structural causal model is (M, P), where $M = (U, V, F)$, and

1. $U = \{U_1, \ldots, U_m\}$ is a set of exogenous variables that are determined by factors outside the model.
2. $V = \{V_1, \ldots, V_n\}$ is a set of endogenous variables that are determined by other variables in the model — that is, variables in $U \cup V$.
3. $F = \{f_1, \ldots, f_n\}$ is a set of deterministic structural equations, $V_i = f_i(\mathrm{Pa}_i, U_i)$, where $\mathrm{Pa}_i \subset V \setminus V_i$.
4. P is a distribution over U.

Mechanisms F and distribution $P(U)$ induce a distribution $P(V)$

- The submodel M_x represents the effects of an *intervention* $\mathrm{do}(X = x)$, and is defined as $M_x := (U, V, F_x)$, where $F_x := \{f_i : V_i \notin X\} \cup \{X = x\}$.
- A *soft intervention* on a variable V_i in an SCM M replaces f_i with a structural equation g_i.

Remarks

王夫之: 理者, 物之固然, 事之所以然也.

$$\left.\begin{array}{l}\text{Deterministic Mechanisms } F \\ \\ \text{Distribution } P(U)\end{array}\right\} \Longrightarrow \text{Distribution } P(V)$$

- 因果关系以确定性结构方程 F 的形式表示, 而概率则是通过假定方程中的某些变量无法观测引入的.
- 这反映了拉普拉斯关于自然现象的观念, 即自然定律是确定性的, 而随机性仅仅源于我们对潜在边界条件的无知.
- Jaynes: 一旦了解了任何实验的物理机制, 逻辑上就没有理由假设其结果具有客观物理概率. "结果的概率是多少" 的问题就变为了 "相应的初始条件导致结果的概率是多少".

- Probabilistic SCM: (M, P)
- Deterministic SCM: (M, u), where u is a particular realization of the exogenous variables U: $P(U = u) = 1$. (Causal World)

Markovian & Semi-Markovian

- A structural causal model is Markovian if the exogenous parent sets U_i, U_j are independent whenever $i \neq j$.
  Remark: It guarantees that the causal Markov condition is satisfied w.r.t. the induced causal graph.

$$X_i \perp \mathrm{ND}_i \mid \mathrm{Pa}_i$$

- If we allow for the sharing of exogenous parents and we allow for arbitrary dependences among the exogenous variables, it is called semi-Markovian.

Every SCM M induces a Causal Graph G

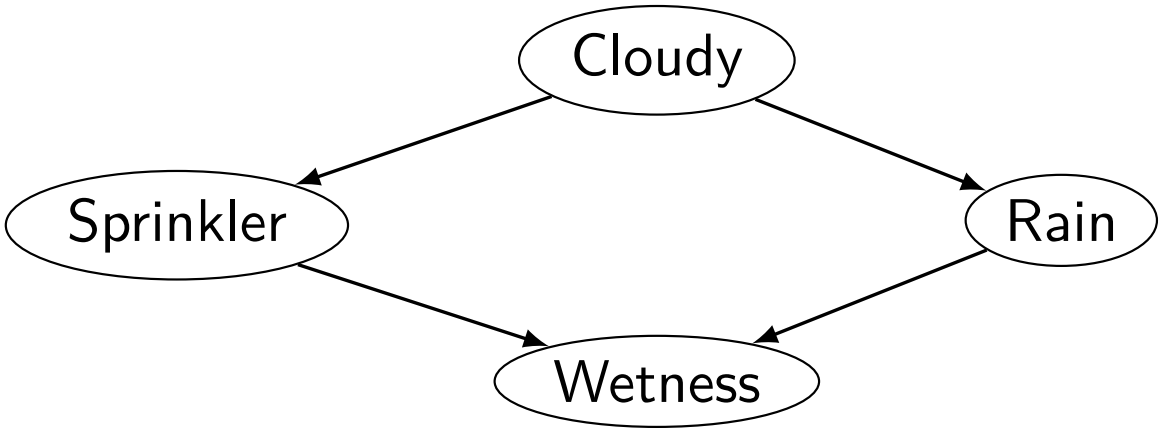
Definition (Causal Graph)

Consider an structural causal model $M = (U, V, F)$. Then G is said to be a causal graph of M if constructed as follows:

1. add a node for every endogenous variable in the set V.
2. add an edge $V_j \longrightarrow V_i$ for every $V_i, V_j \in V$ if V_j appears as an argument of $f_i \in F$.
3. add a bidirected edge $V_i \dashleftarrow\dashrightarrow V_j$ for every $V_i, V_j \in V$ if the corresponding $U_i, U_j \subset U$ are dependent or the corresponding functions f_i, f_j share an exogenous variable as an argument. Semi-Markovian

Remark: Each bidirected arrow encodes unobserved confounding in G. They indicate correlation between the unobserved parents of the endogenous variables at the endpoints of such edges.
Remark: X is a *direct cause* of Y if X is a parent of Y.
X is a *cause* of Y if X is an ancestor of Y.

Example



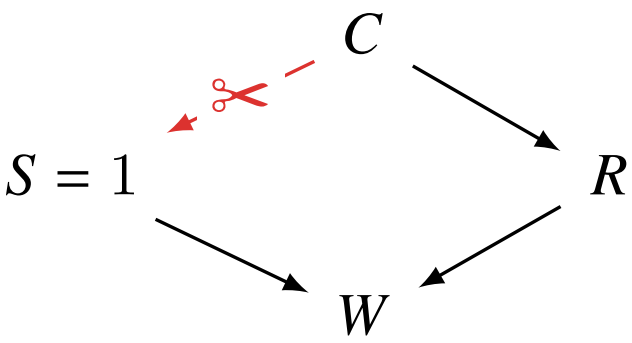
Model(M)

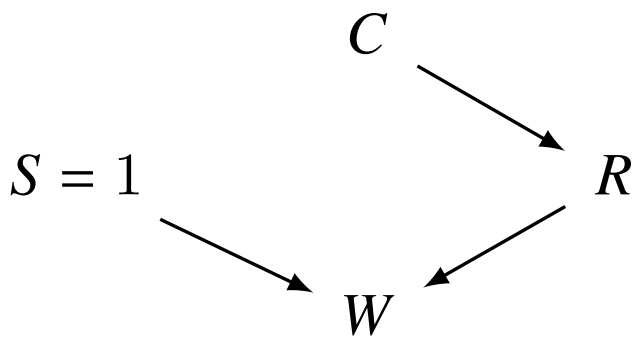
$$C = f_C(U_C)$$
$$S = f_S(C, U_S)$$
$$R = f_R(C, U_R)$$
$$W = f_W(S, R, U_W)$$

- Every missing arrow advertises an independency, conditional on a separating set.

$$C \perp W \mid S, R \qquad S \perp R \mid C$$

- $P_{S=1}(C, R, W) = P(C)P(R \mid C)P(W \mid R, S = 1) \neq P(C, R, W \mid S = 1)$





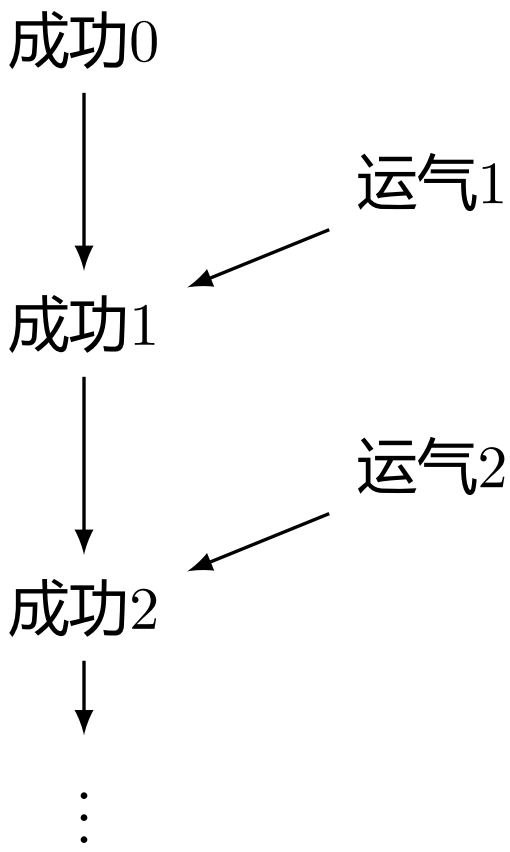
Model($M_{S=1}$)

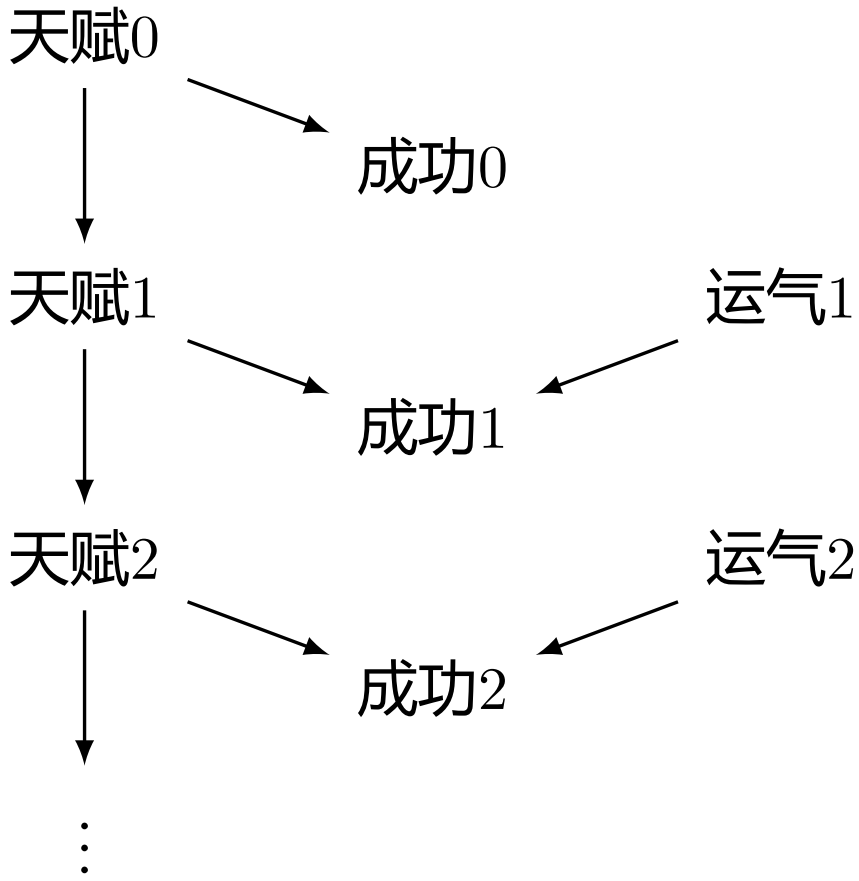
$$C = f_C(U_C)$$
$$S = 1$$
$$R = f_R(C, U_R)$$
$$W = f_W(S, R, U_W)$$

Would the pavement be wet **had** the sprinkler been on?

怎么理解智力、身高等特征“向均值回归”的现象?

- 成功 = 天赋 + 运气
- 智力、身高等特征, 虽有可遗传的部分 (基因、天赋), 但不同于财富, 不会直接遗传, 运气无法直接或间接地世代累积

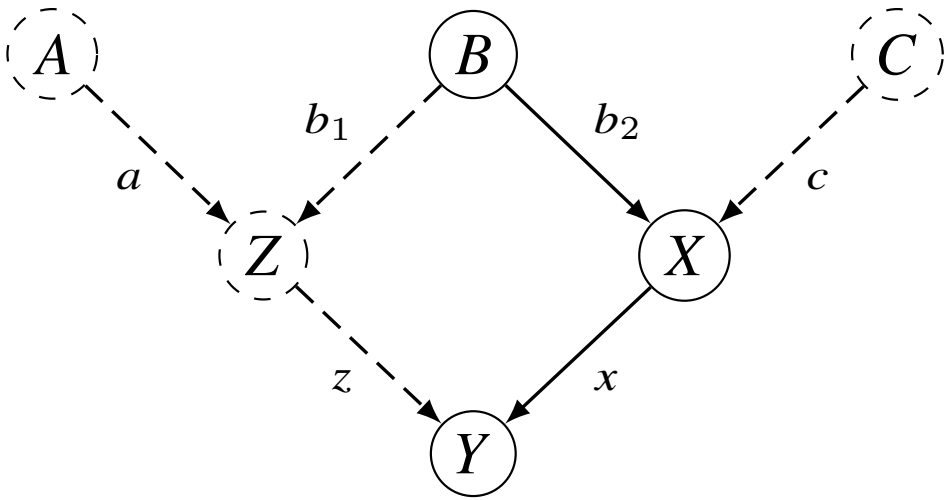




Sewall Wright 休厄尔·赖特 1921: 相关关系与因果关系

Problem: 豚鼠在母鼠子宫里多待一天, 会对其出生体重产生多大影响?

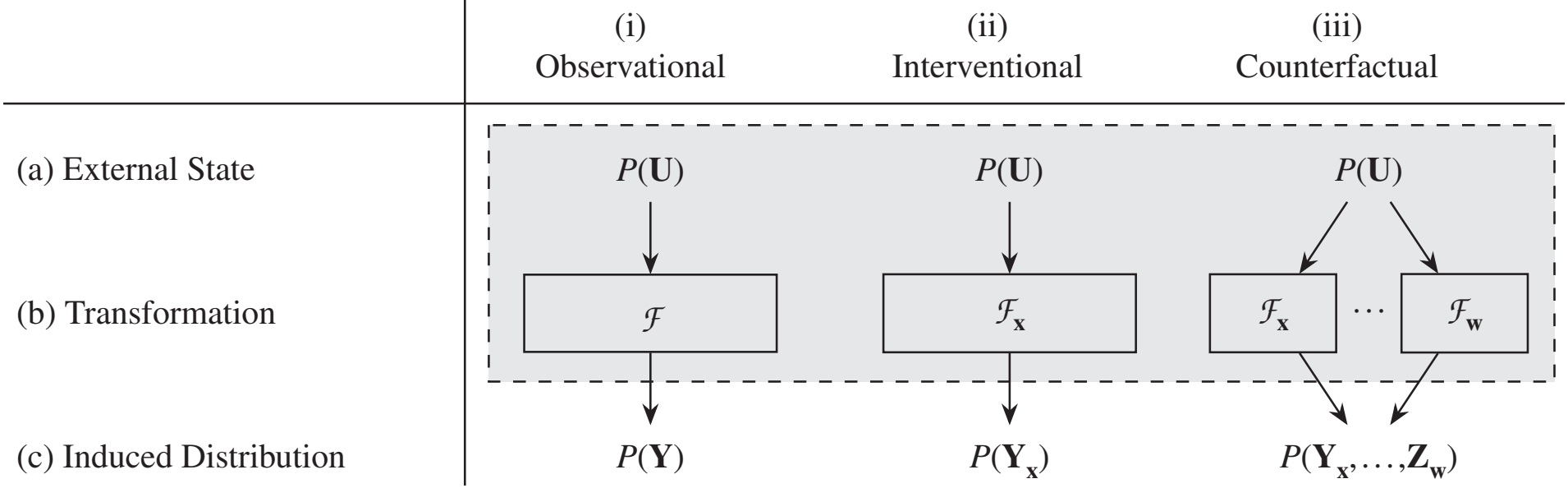
- 有人比较了孕期 67 天和孕期 66 天的豚鼠的出生体重, 发现平均增加了 5.66 克. 所以, 答案是 5.66 克吗?
- 赖特: 错! 因为晚出生是有原因的, 比如同窝产仔数相对较少.



- Y: 出生体重
- X: 妊娠时长
- Z: 产前生长速率
- B: 同窝产仔数
- A: 影响生长速率的其他因素
- C: 影响妊娠时长的其他因素

- 相关系数 = 直接效应 + 间接效应
- 间接效应 = 所有路径的路径系数的乘积之和

$$\left.\begin{aligned}\rho_{XY} &= x + b_2 b_1 z \\ \rho_{BY} &= b_2 x + b_1 z \\ \rho_{BX} &= b_2\end{aligned}\right\} \Longrightarrow x = \frac{\rho_{XY} - \rho_{BX}\rho_{BY}}{1 - \rho_{BX}^2}$$



1. **Observing** An SCM defines a joint distribution $P(V)$ s.t. for $Y \subset V$:

$$P(y) = \sum_{u:Y(u)=y} P(u)$$

where $Y(u)$ is the solution for Y after evaluating F with $U = u$.

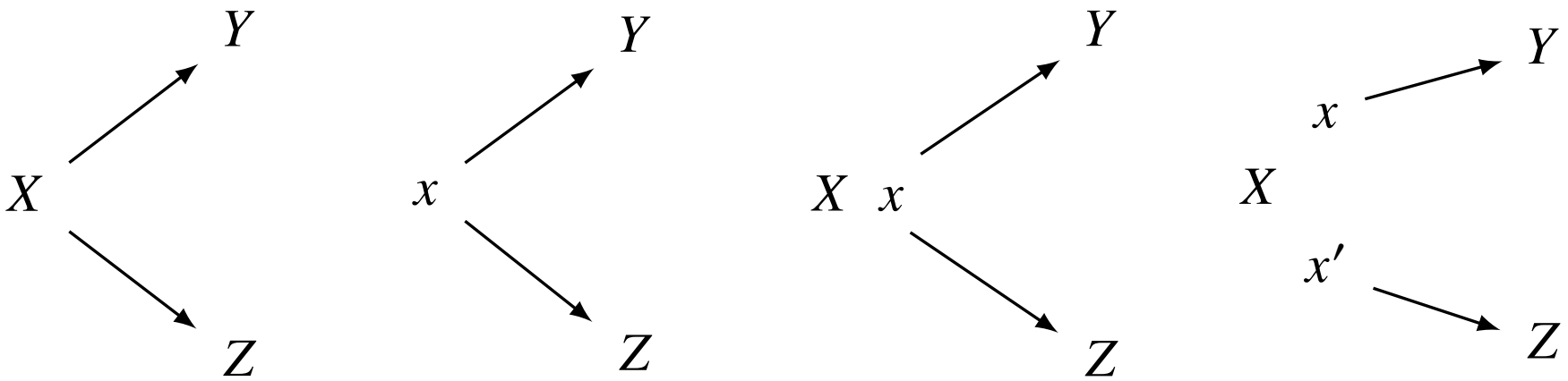
2. **Intervening** An SCM induces a family of joint distributions over V, one for each intervention $\mathtt{do}(x)$. For $Y \subset V$:

$$P(y_x) = \sum_{u:Y_x(u)=y} P(u)$$

3. **Counterfactual** An SCM induces a family of joint distributions over counterfactual events $Y_x, \ldots, Z_w$. For $Y, Z, \ldots, X, W \subset V$:

$$P(y_x, \ldots, z_w) = \sum_{u:Y_x(u)=y,\ldots,Z_w(u)=z} P(u)$$

$L_{2.25}/L_{2.5}$



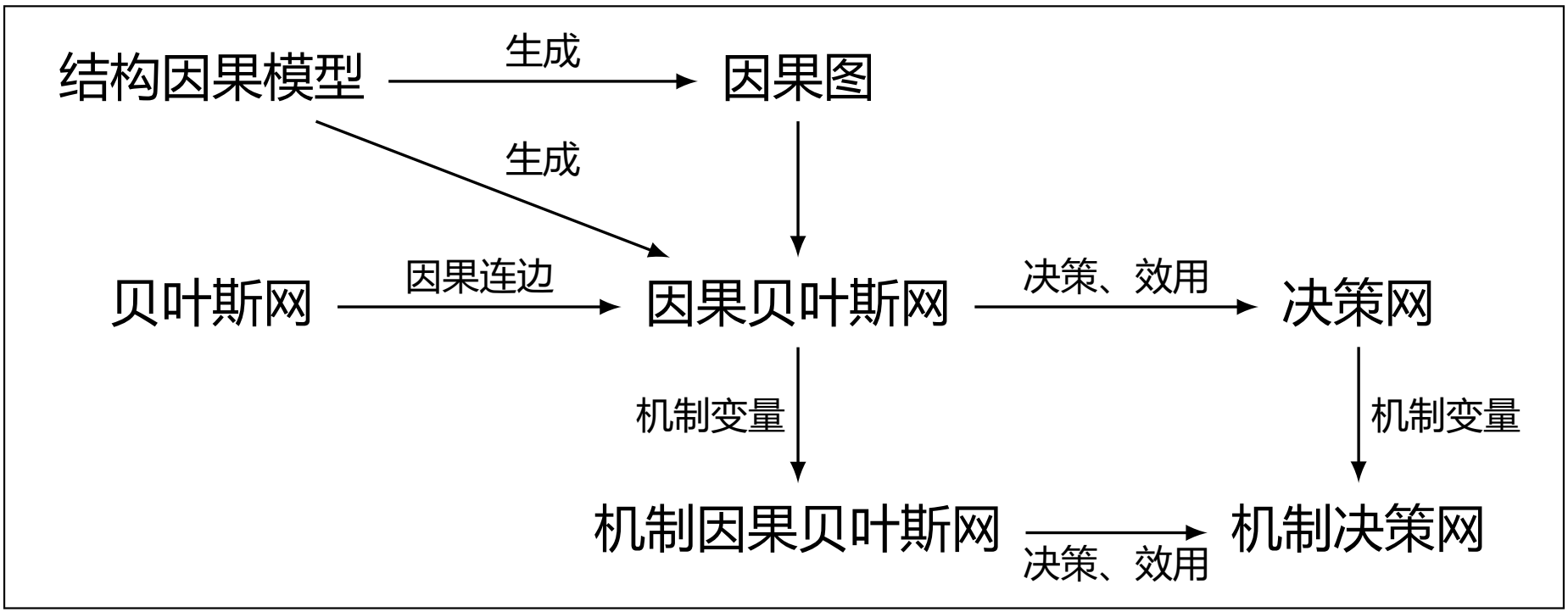
- L_1: $P(X, Y, Z)$
- L_2: $P(Y_x, Z_x)$
- $L_{2.25}$: $P(X, Y_x, Z_x)$
- $L_{2.5}$: $P(X, Y_x, Z_{x'})$
- L_3: $P(Y, Y_x)$, $P(X, Y_x, Y_{x'}, Z_x, Z_{x'})$

为什么因果关系更“稳定”，支持迁移、泛化?

“因果关系属于本体论，描述了世界的客观物理约束，而概率关系属于认识论，反映了我们对世界的认知或信念．因此，只要环境没有发生变化，即使我们对环境的认识发生了变化，因果关系也应该保持不变.”

— 珀尔《因果论》

“哲学家用不同的方式解释世界，问题是改造世界.” — 马克思



Remark

The following example shows two SCMs that induce the same graph, observational distributions, and intervention distributions but entail different counterfactual statements.

Example

Let $U_X, U_Z \sim \mathrm{Bern}(0.5)$, and $U_Y \sim U(\{0,1,2\})$ s.t. the three variables are jointly independent. We define two different SCMs.

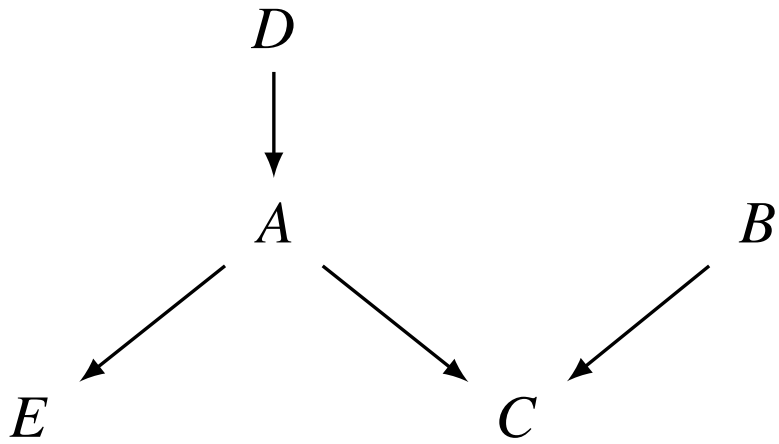
$$X := U_X$$
$$Z := U_Z$$
$$Y := (\llbracket U_Y > 0 \rrbracket \cdot X + \llbracket U_Y = 0 \rrbracket \cdot Z) \cdot \llbracket X \neq Z \rrbracket + U_Y \cdot \llbracket X = Z \rrbracket$$
$$Y := (\llbracket U_Y > 0 \rrbracket \cdot X + \llbracket U_Y = 0 \rrbracket \cdot Z) \cdot \llbracket X \neq Z \rrbracket + (2 - U_Y) \cdot \llbracket X = Z \rrbracket$$

$$X \longrightarrow Y \longleftarrow Z$$

Suppose we observes $(X, Z, Y) = (1, 0, 0)$. From both SCMs, it follows that $U_Y = 0$. We are interested in "what would Y have been if X had been 0?"

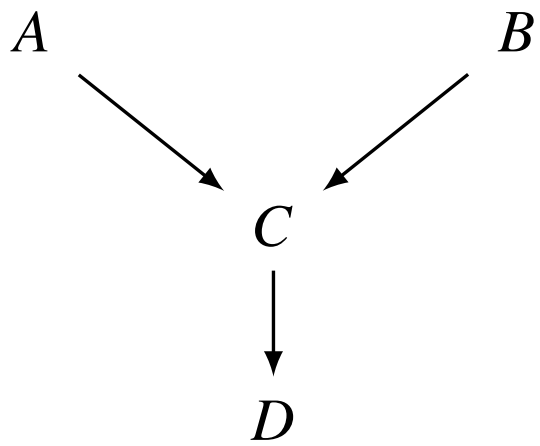
$$Y_{X=0}(U_X = 1, U_Z = 0, U_Y = 0) = 0 \qquad Y_{X=0}(U_X = 1, U_Z = 0, U_Y = 0) = 2$$

Remark: 相比于图模型, 结构方程能告诉我们更多的独立性信息.



- 假设 $C = A \oplus B$, 则 B 和 C 能决定 A
- $D \perp E \mid BC$

Remark: 特定背景下的独立性.



- 假设 $C = A \vee B$.
- 若 $A = 1$, 则 $P(D \mid B, A = 1) = P(D \mid A = 1)$, B 与 D 独立, 即 $B \perp D \mid A = 1$.
- 若 $A = 0$ 则未必.

结构方程的方向性 vs 物理方程的对称性

$$\text{structural equation} \quad Y = \alpha X + \beta \implies X = \frac{Y - \beta}{\alpha}$$

公鸡打鸣导致太阳升?

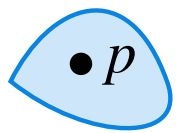
$$f = ma \implies m = \frac{f}{a}$$

> 哲学家们想当然的认为因果是科学的基本公设，奇怪的是，在前沿科学中，“原因”这个词从来没有出现过. “因果律”是旧时代的遗迹，就像君主制一样，之所以留存至今，仅仅是因为人们错误地认为它无害.
>
> — 罗素[13]

[13] Russell: On the Notion of Cause. 1912.

Remarks

- 通过 “干预” 解释因果关系方向性与物理方程对称性的问题.
- 罗素可能会说: “事实上只有一个世界, 你怎么能谈论两个世界呢?”
- 如果你想在你的模型中包含整个宇宙, 因果关系就会消失, 因为干预消失了, 没法区分出操纵者和被操纵对象.
- 大多数时候, 科学家从宇宙中挖出一块, 忽视其余部分, 或将其作为背景条件.
- 这种内和外的选择, 造成了我们观察事物方式的不对称性, 这种不对称性使我们能够谈论 “外部干预”, 从而谈论因果关系和因果方向.



经典物理中的 “因果关系”: 在给定时刻 t, 空间中 p 点的物理状态由之前时刻其周围的物理状态所决定, 比如说, 在 $t-\tau$ 时刻, 如果 τ 很大, 那么可能需要知道 p 周围更广阔区域的状态, 如果 τ 很小, 那么只需要知道 p 周围很小区域的状态. 这种动力学一般通过微分方程来描述.

Theorem (Picard Existence Theorem)

Let $U \subset \mathbb{R}^n$ be an open set and $f : I \times U \to \mathbb{R}^n$ a continuous function which satisfies the Lipschitz condition

$$\exists L \forall (t, x_1), (t, x_2) \in I \times U : |f(t, x_1) - f(t, x_2)| \leq L|x_1 - x_2|$$

then the Initial Value Problem

$$\begin{cases} \dfrac{\mathrm{d}x}{\mathrm{d}t} = f(t, x) \\ x(t_0) = x_0 \end{cases}$$

has a unique solution $x(t)$.
Moreover, the Picard iteration

$$x_{n+1}(t) = x_0 + \int_{t_0}^{t} f(s, x_n(s))\,\mathrm{d}s$$

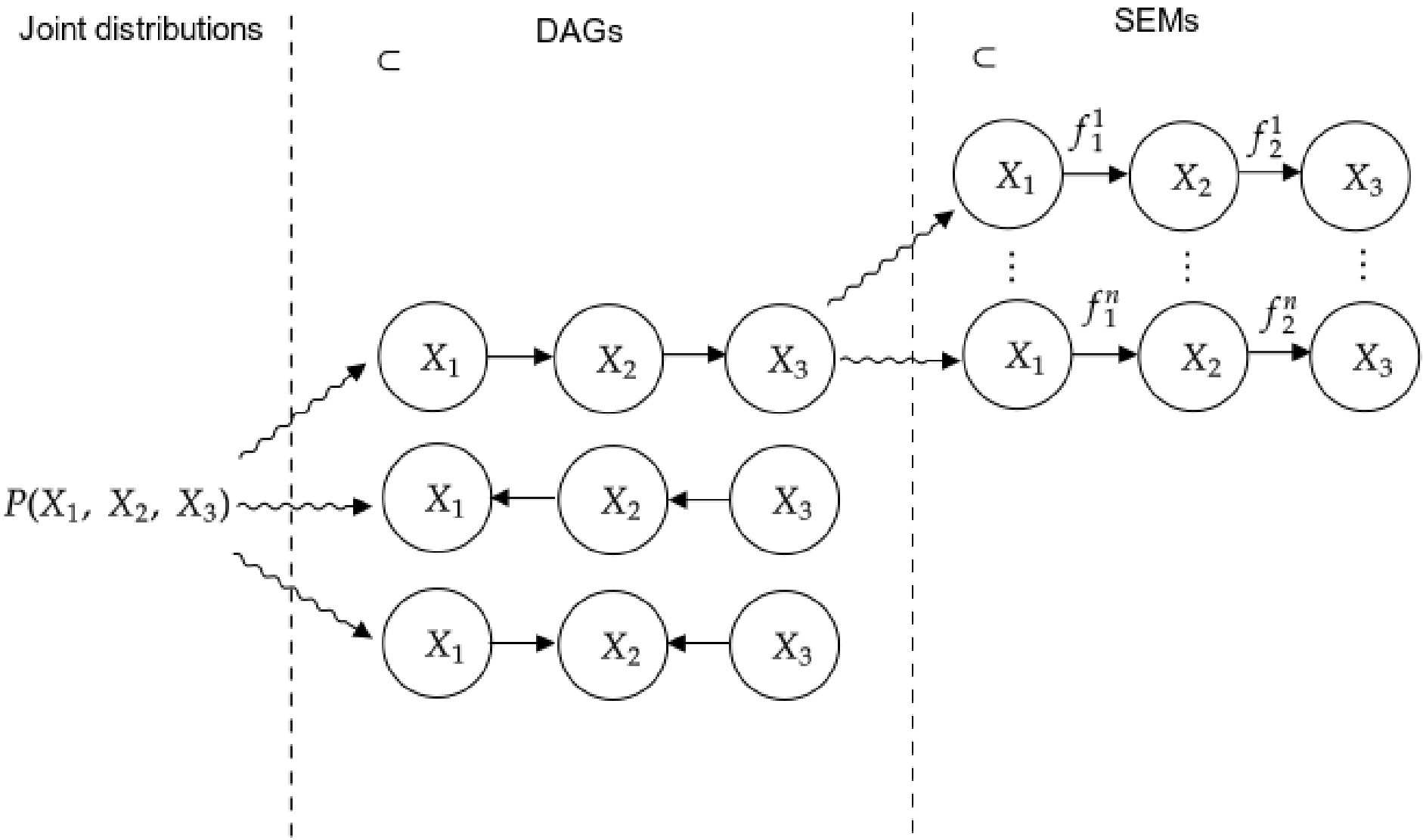
produces a sequence of functions $\{x_n(t)\}$ that converges to this solution.

Remark: the immediate future of x is implied by its past.

$$x(t + \mathrm{d}t) = x(t) + \mathrm{d}t \cdot f(x(t))$$

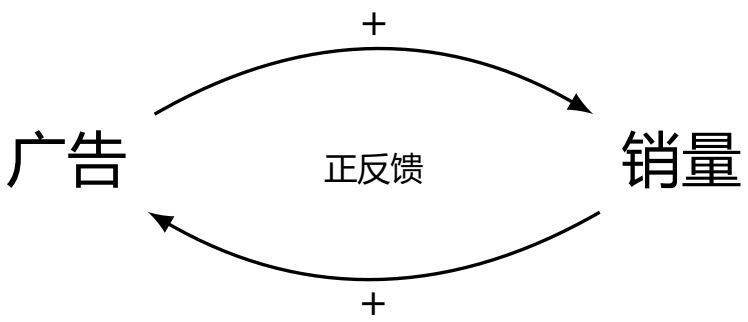
Levels of Causal Modelling

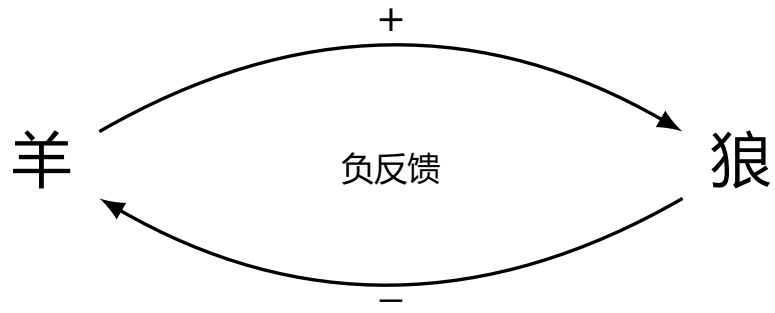
Models	Predict in i.i.d. setting	Predict under changing distr. or interven-tion	Answer counter-factual questions	Obtain physical insight	Learn from data
Differential Equation	✓	✓	✓	✓	?
Sructual Causal Model	✓	✓	✓	?	?
Causal Graph	✓	✓	×	?	?
Statistical Model	✓	×	×	×	✓

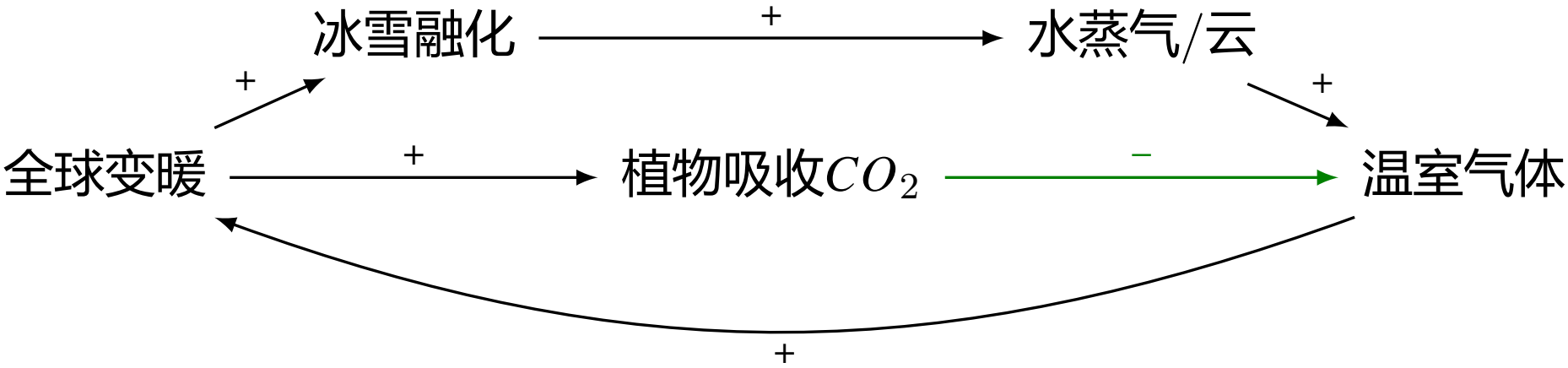


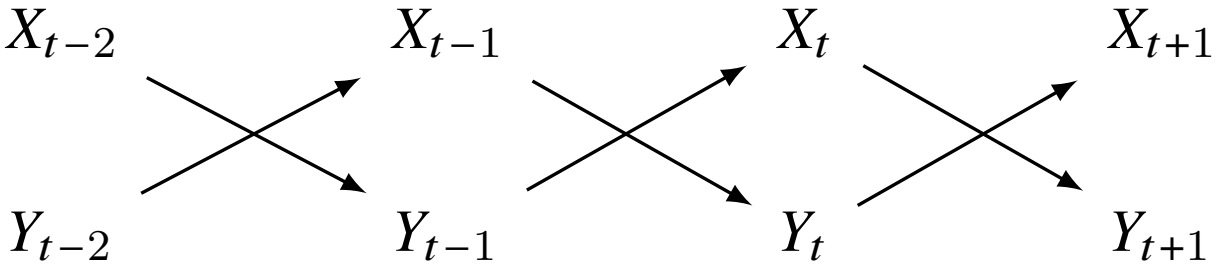
- the more fine grained information we want, the more detailed models we need.
- the more detailed our models, the bigger the parameter space.

循环因果

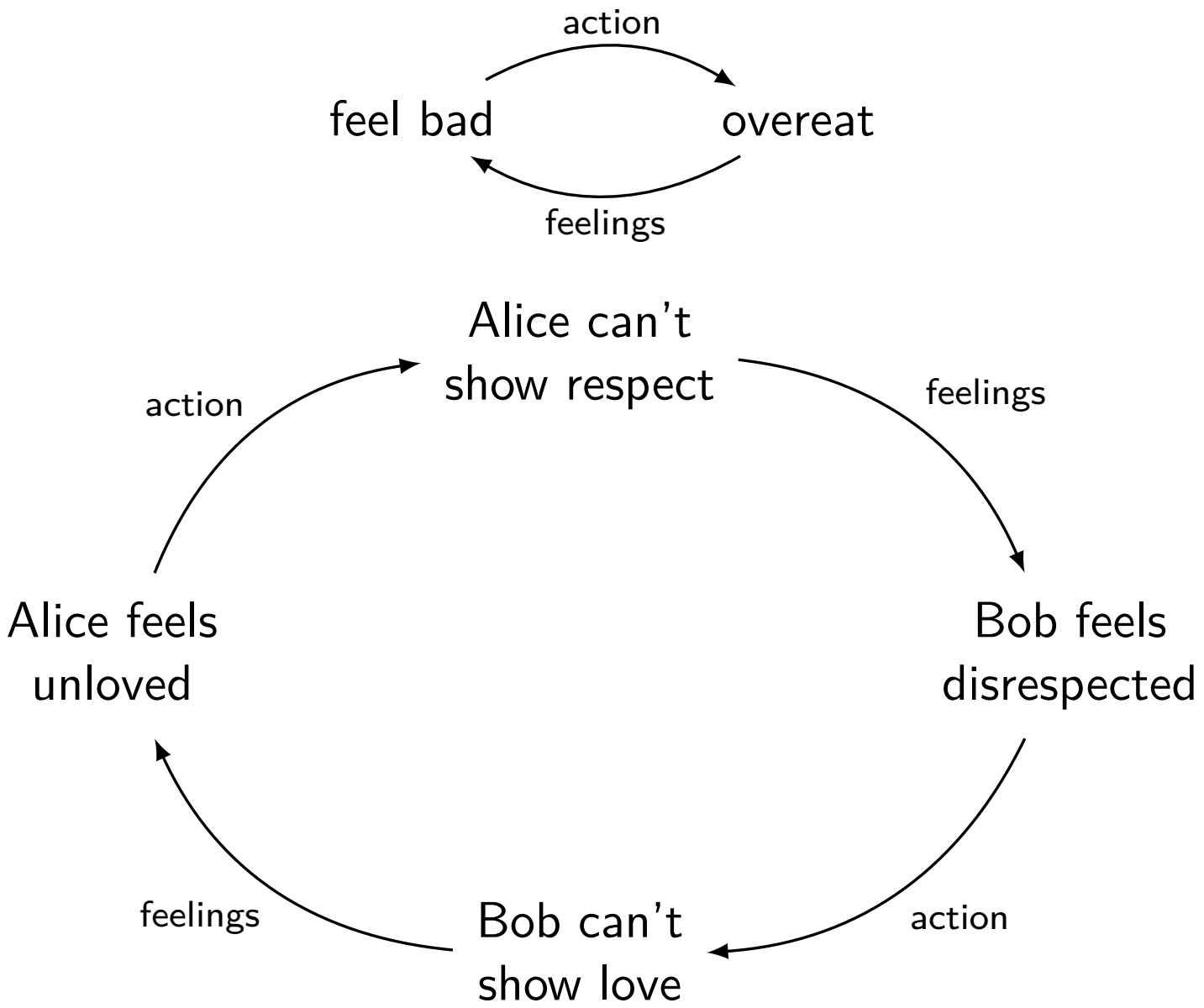

+
广告
正反馈
销量
+

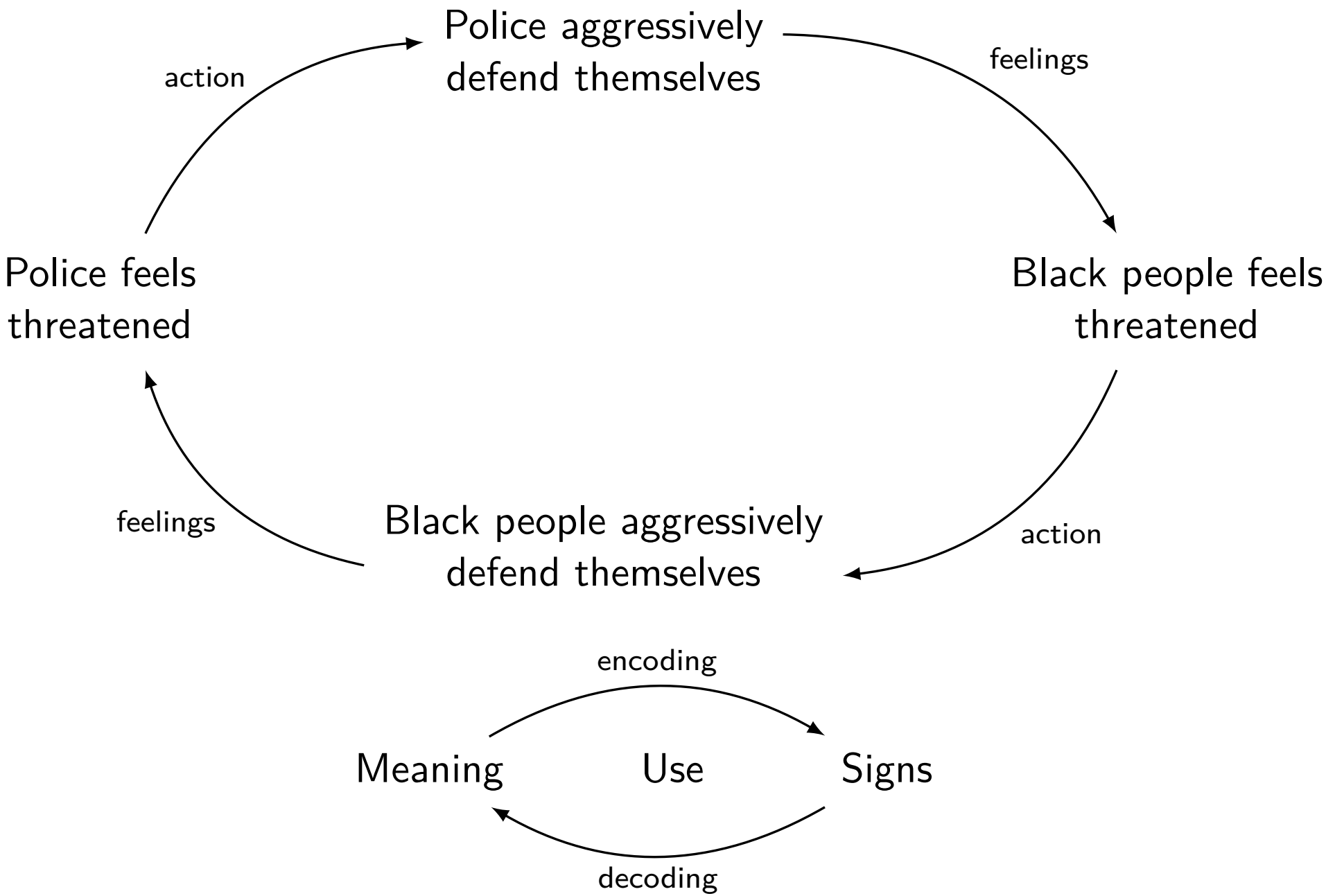

+
羊
负反馈
狼
−


冰雪融化
+
水蒸气/云
+
+
全球变暖
+
植物吸收CO_2
−
温室气体
+


X_{t-2}
X_{t-1}
X_t
X_{t+1}
Y_{t-2}
Y_{t-1}
Y_t
Y_{t+1}

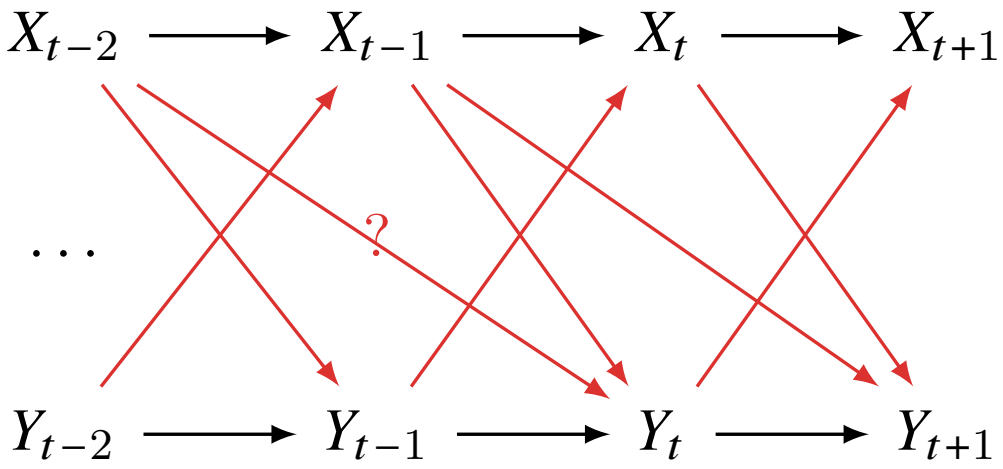
Circular Causality

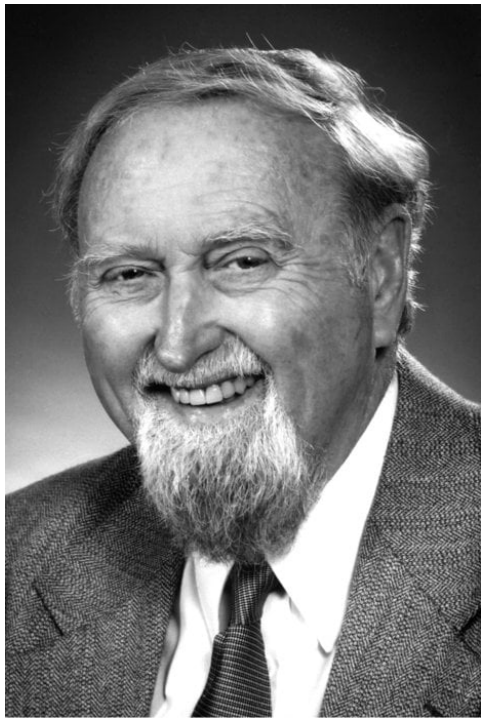



Police aggressively
defend themselves
feelings
Black people feels
threatened
action
Black people aggressively
defend themselves
feelings
Police feels
threatened
action
encoding
Meaning
Use
Signs
decoding

Time series and Granger causality

Does $X \to Y$ or $Y \to X$?





exclude instantaeous effects and common causes

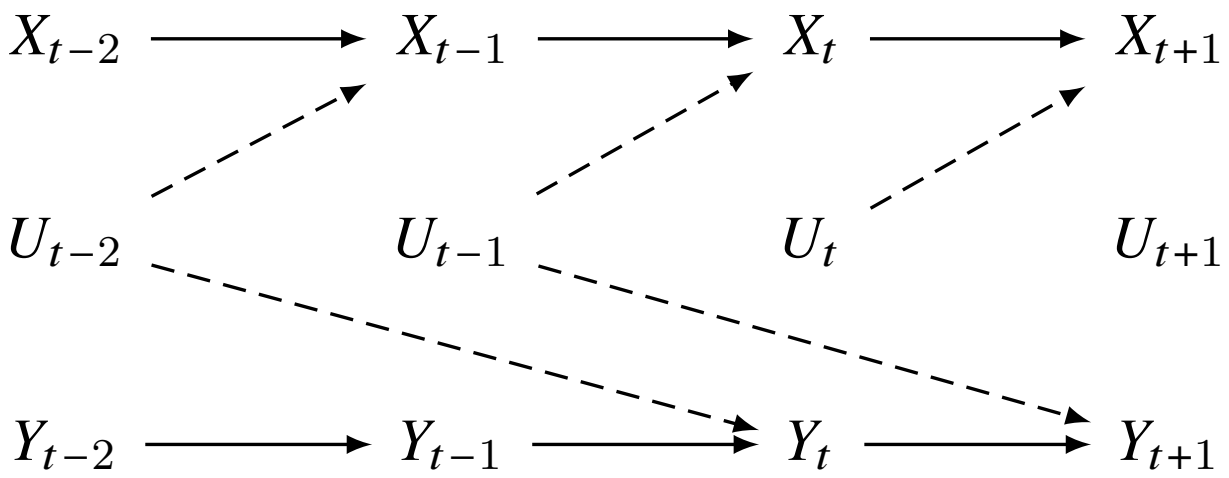
- Granger cause

$$X \text{ Granger-causes } Y \iff Y_{\text{present}} \not\perp X_{\text{past}} \mid Y_{\text{past}}$$

- the past of X helps when predicting Y_t from its past
- strength of causal influence often measured by transfer entropy

$$\text{TE}(X \to Y) := I(Y_{\text{present}}; X_{\text{past}} \mid Y_{\text{past}})$$

Limitations of Granger Causality



We have

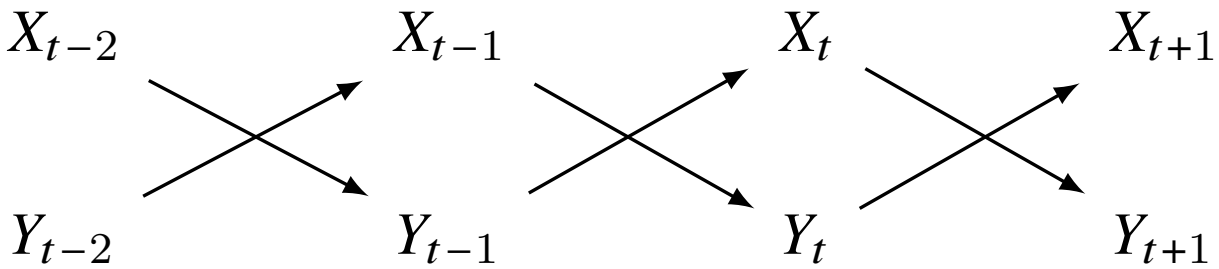
$$Y_{\text{present}} \not\perp X_{\text{past}} \mid Y_{\text{past}}$$

but

$$X_{\text{present}} \perp Y_{\text{past}} \mid X_{\text{past}}$$

Granger causality erroneously infers $X \to Y$.

Limitations of Granger Causality



Granger causality erroneously infers neither causal influence from X to Y nor from Y to X if the influence from X_t on Y_{t+1} and the one from Y_t to X_{t+1} are deterministic.

$$Y_{\text{present}} \perp X_{\text{past}} \mid Y_{\text{past}}$$

$$X_{\text{present}} \perp Y_{\text{past}} \mid X_{\text{past}}$$

$$\text{TE}(X \to Y) = I(Y_{\text{present}}; X_{\text{past}} \mid Y_{\text{past}}) = 0$$

That's why transfer entropy does not quantify causal strength.

Contents

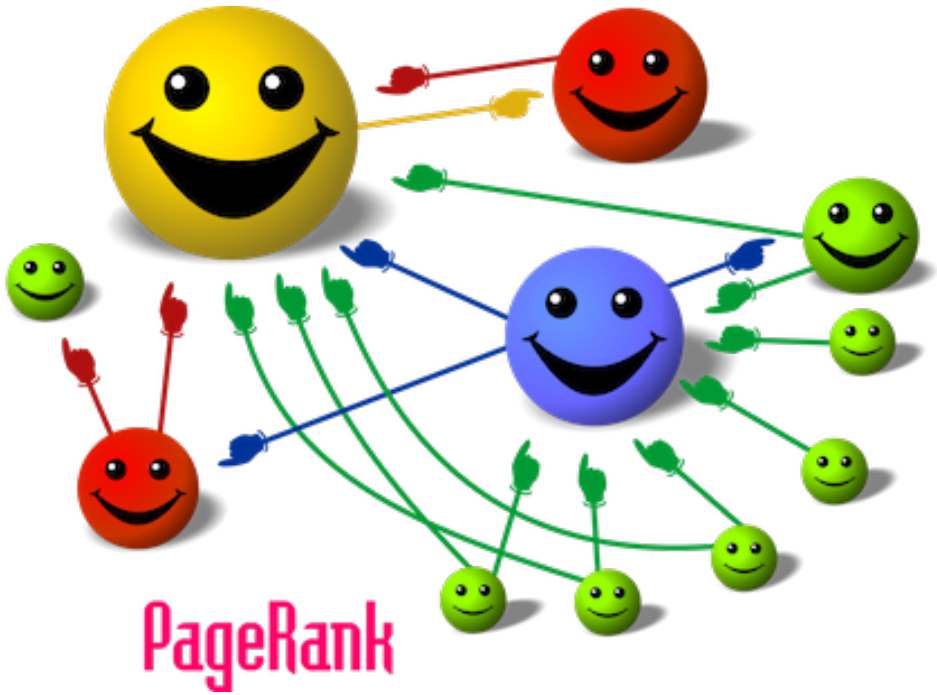
1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

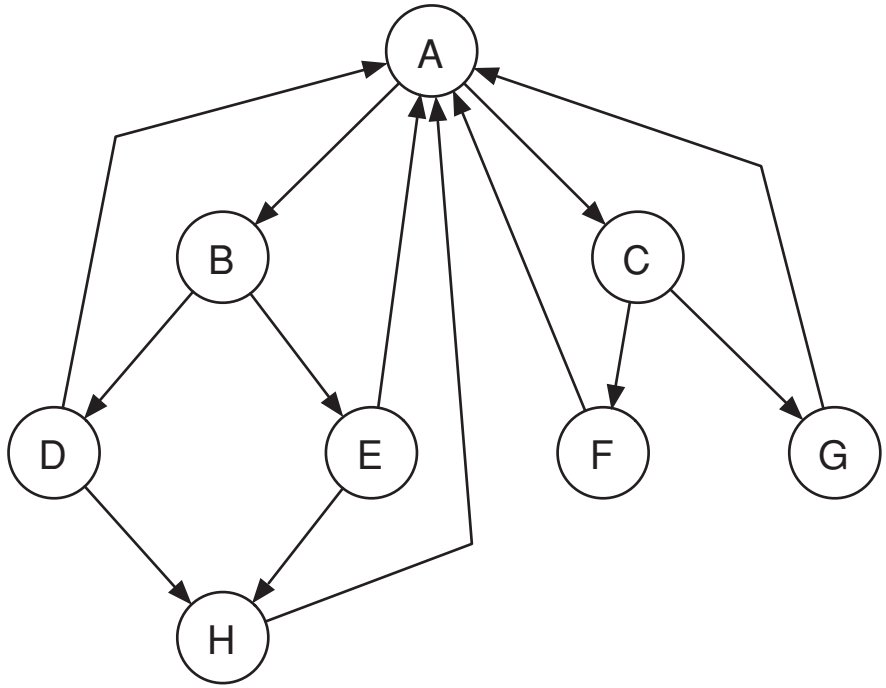
Google's PageRank

Google's founding philosophy is that we don't know why this page is better than that one: If the statistics of incoming links say it is, that's good enough. No semantic or causal analysis is required.

- In a network with n nodes, assign all nodes the same initial PageRank, $1/n$.
- Choose a number of steps, k.
- Perform a sequence of k updates to the PageRank values:
  - Basic PageRank Update Rule: Each page divides its current PageRank equally across its outgoing links and passes these equal shares to the pages it points to. (If a page has no outgoing links, it passes all its current PageRank to itself.) Each page updates its new PageRank to be the sum of the shares it receives.
  - Scaled PageRank Update Rule: First apply the Basic PageRank Update Rule. Then scale down all PageRank values by a factor of s. This means that the total PageRank in the network has shrunk from 1 to s. We divide the residual $1 - s$ units of PageRank equally over all nodes, giving $(1 - s)/n$ to each.

Google's PageRank





k	A	B	C	D	E	F	G	H
0	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8
1	1/2	1/16	1/16	1/16	1/16	1/16	1/16	1/8
2	3/16	1/4	1/4	1/32	1/32	1/32	1/32	1/16
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
∞	4/13	2/13	2/13	1/13	1/13	1/13	1/13	1/13

理论的终结: 大数据让科学方法过时? — *Chris Anderson*

- 所有模型都是错误的, 但有些是有用的.
- 所有模型都是错误的, 而且渐渐地, 可以在不借助它们的情况下成功.
- 大数据, 以及统计工具, 提供了一种理解世界的全新方式.
- 相关性取代了因果关系. 科学可以在没有连贯模型、统一理论或任何机制解释的情况下发展进步.
- 相关已足够.

问题: 如何区分模式与因果?

太平洋塔纳岛的 JohnFrum 宗教

从“找字游戏”到阴谋论 ◎ô◎

m	b	o
j	d	a
l	l	w

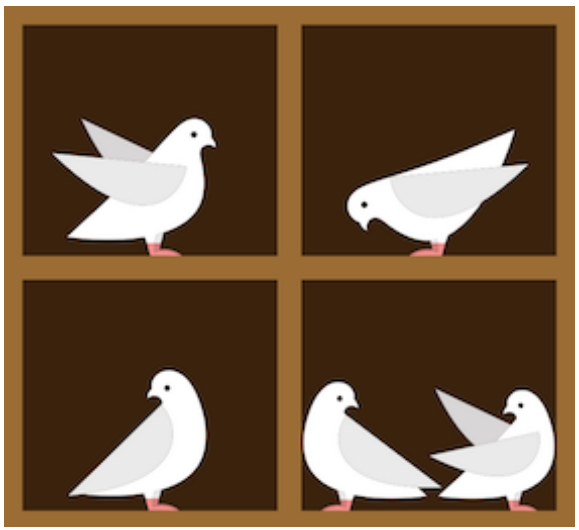
m	b	o	u	n
j	d	a	d	o
l	l	w	d	z
n	n	v	b	e
c	i	l	s	d

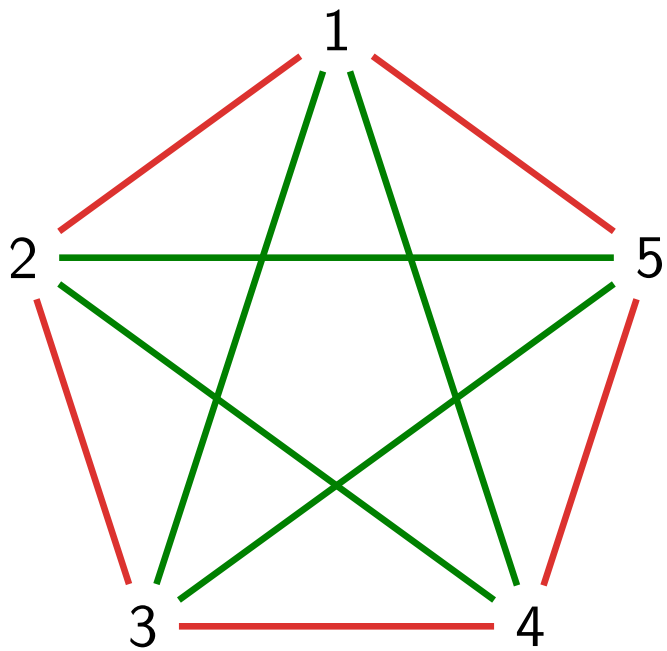
拉姆齐理论 Ramsey Theory

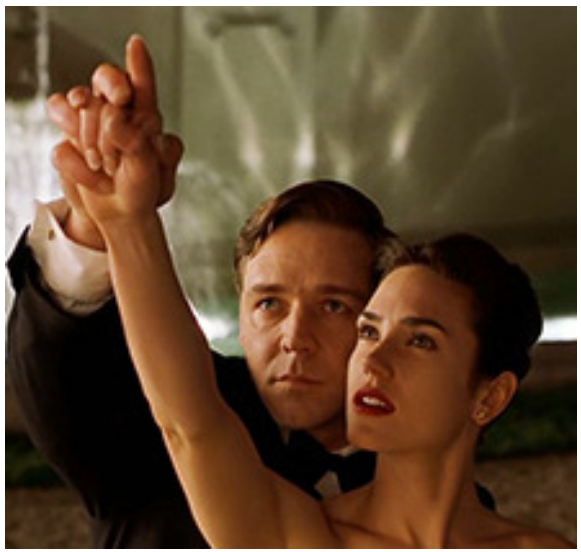
Complete Disorder is Impossible!

Problem

在一个派对中，为了确保其中至少有 n 个人互相认识，或者至少有 n 个人互不相识，至少需要邀请多少人？







Complete Disorder is Impossible!

- 长度为 9 的二进制序列中必然存在长度为 3 的 "单色等差数列".
- 比如, 序列

$$011001100$$

  的第 $1, 5, 9$ 位 (公差为 4) 都是 0.
- 比如, 序列

$$011001101$$

  的第 $3, 6, 9$ 位 (公差为 3) 都是 1.
- 长度为 8 的可能不存在, 比如 01100110.
- 给定一个数据集, 只要它足够大, 那么, 将它编码为序列, 不管用什么方式编码, 都会在某个长度的等差数列的位置上找到相同的字符.

Theorem (van der Waerden Theorem)

For every $k, m \in \mathbb{N}^+$, there is $n \in \mathbb{N}^+$ s.t. if the numbers from 1 to n are colored in k colors, then there exists at least m numbers in arithmetic progression, all of the same color.

Correlation Supersedes Causation?

- How to distinguish correlation from causation?
- How to distinguish content-correlations from Ramsey-type correlations?
- Ramsey-type correlations appear in all large enough databases.
- A correlation is *spurious* iff it appears in a "randomly" generated database.
- How "large" is the set of spurious correlations?
- Most strings are algorithmically random.

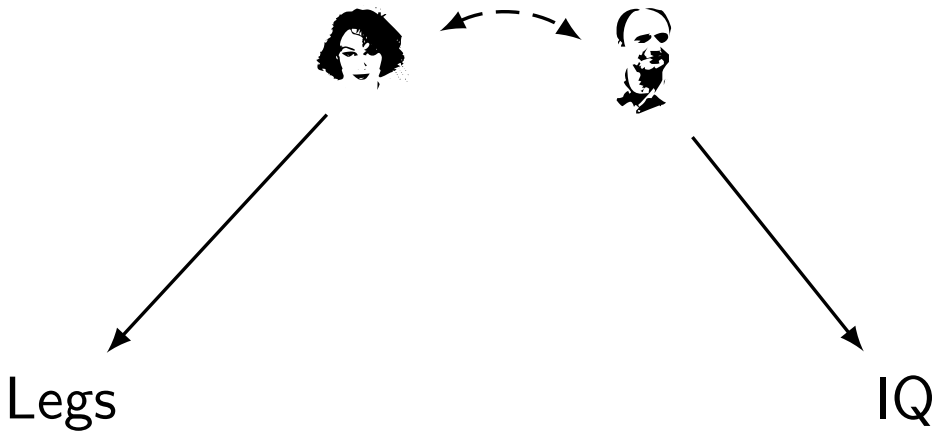
$$P\left(\left\{x \in \mathcal{X}^n : \frac{K(x)}{n} < 1 - \delta\right\}\right) < 2^{-\delta n}$$

- Most correlations are spurious.
- It may be the case that our part of the universe is an oasis of regularity in a maximally random universe.

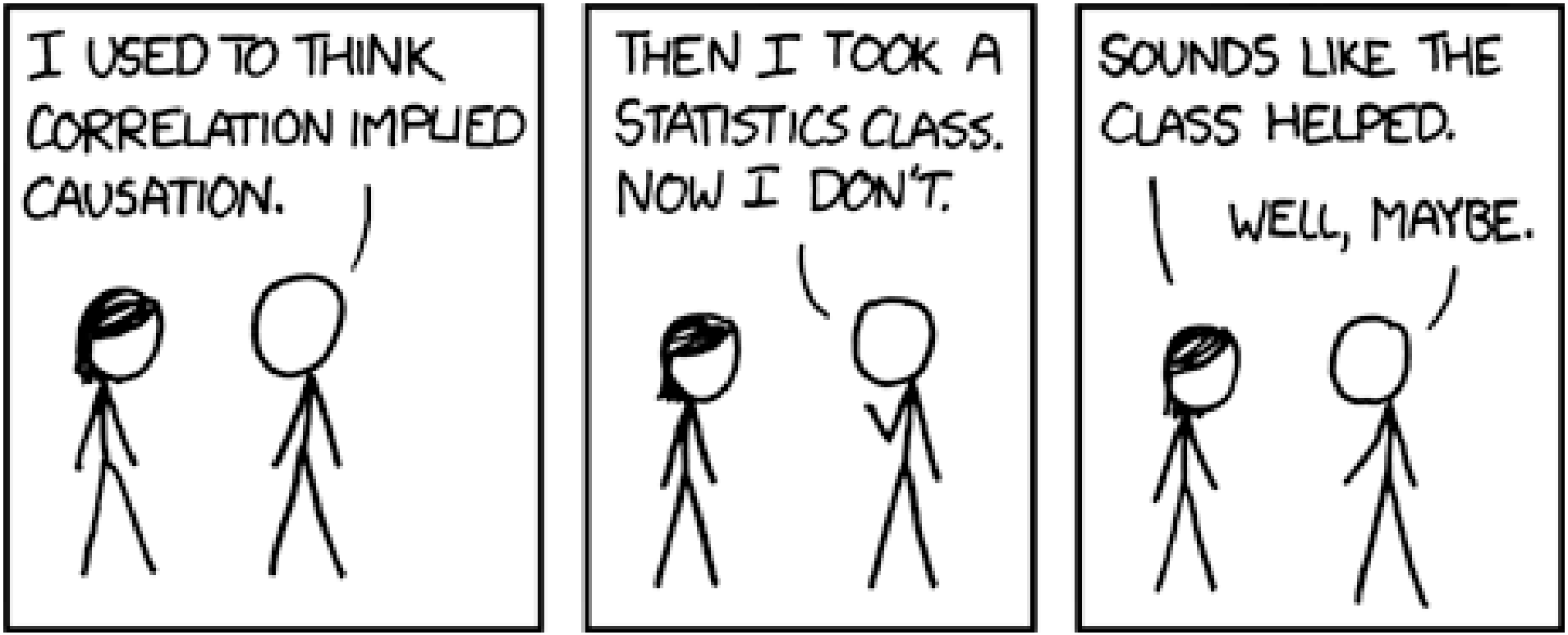
Complete Disorder is Impossible!

For sufficiently large n and any $x \in \mathcal{X}^n$, if $C(x) \geq n - \delta(n)$, then each block of length $\log n - \log\log n - \log(\delta(n) + \log n) - O(1)$ occurs at least once in x.

Correlation does not imply causation



- Eating ice cream is positively associated with deaths from drowning.
- Married men live longer than single men.
- Women with long legs tend to have higher IQ.



$$X \sim Y \implies X \to Y$$

Statistics are like bikinis. What they reveal is suggestive, but what they conceal is vital.

— Aaron Levenstein

统计概念 vs 因果概念

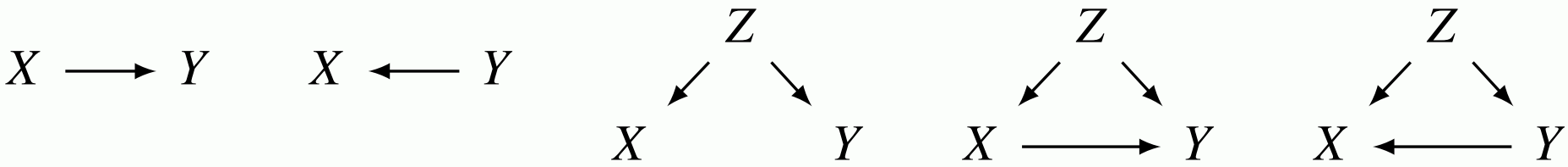
- 统计概念: 相关 correlation, 回归 regression, 依赖 dependence, 条件独立 conditional independence, 关联 (非独立)association, 似然 likelihood, 可压缩性 collapsibility, 风险比 risk ratio, 优势比 odd ratio, 倾向得分 propensity score, Granger 因果
- 因果概念: 随机化 randomization, 影响 influence, 效应 effect, 混杂 confounding, 外生性 exogeneity, 可忽略性 ignorability, 扰动 disturbance, 伪相关 spurious correlation, 路径系数 path coefficient, 工具变量 instrumental variables, 干预 intervention, 反事实 counterfactual, 解释 explanation, 归因 attribution
- 概率语言无法表达简单的因果事实 "公鸡打鸣不会导致太阳升".
- 因果分析需要引入新的语言.
- 每一个因果结论背后都必然有一些因果预设.
- No causes in, no causes out.
- $P(y \mid \mathrm{do}(x), z)$ 无法从联合分布 $P(y, x, z)$ 推断出来, 除非预设某些因果知识, 比如因果图或结构方程.
- 因果需要处理概率空间之外的因素所导致的概率分布函数的改变.

莱辛巴赫 Hans Reichenbach 1891-1953

- Correlation does not imply causation.
- Reichenbach: No correlation without causation.

Reichenbach's "Common Cause Principle"

A correlation between X and Y cannot come about by accident. If $X \not\perp\!\!\!\perp Y$, then either X causes Y, or Y causes X, or X and Y share a common cause Z (or any combination).

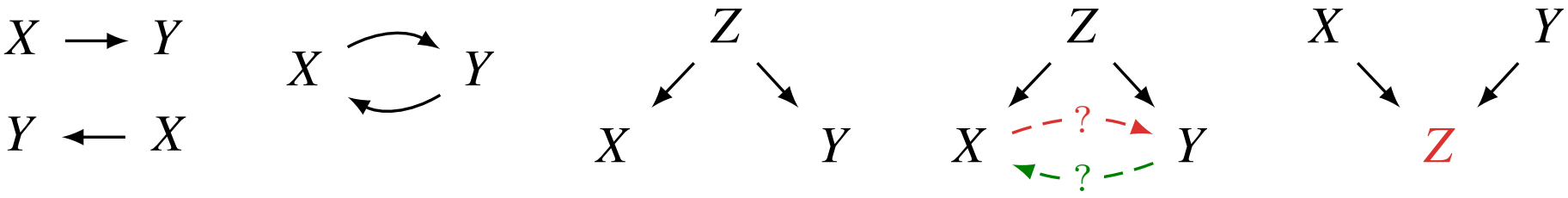


Theorem

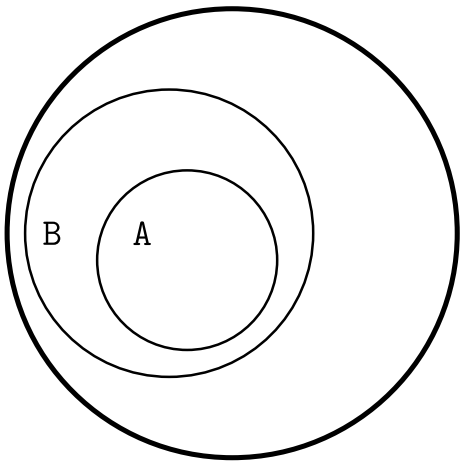
The Markov Condition implies the Reichenbach's common cause principle.

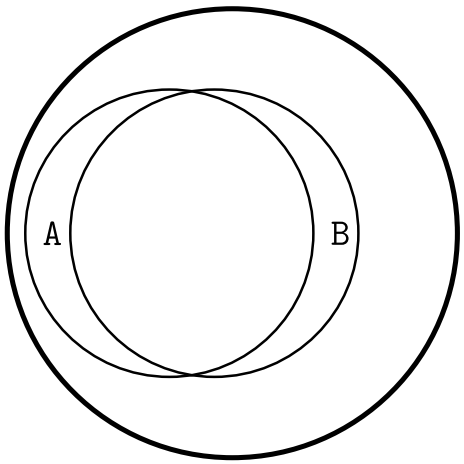
由 $X \not\perp\!\!\!\perp Y$ 和马尔科夫条件, X 和 Y 之间有不含对撞的路径, 那就只能是 $X \to \cdots \to Y$ 或 $X \leftarrow \cdots \leftarrow Y$ 或 $X \leftarrow \cdots \leftarrow Z \to \cdots \to Y$.

Reichenbach's Error?



- Flip two coins simultaneously 100 times and write down the results only when at least one of them comes up heads.
- Looking at your table, you will see that the outcomes of the two simultaneous coin flips are not independent. Every time Coin 1 landed tails, Coin 2 landed heads.





random darts

$$P(AB) > P(A)P(B)$$

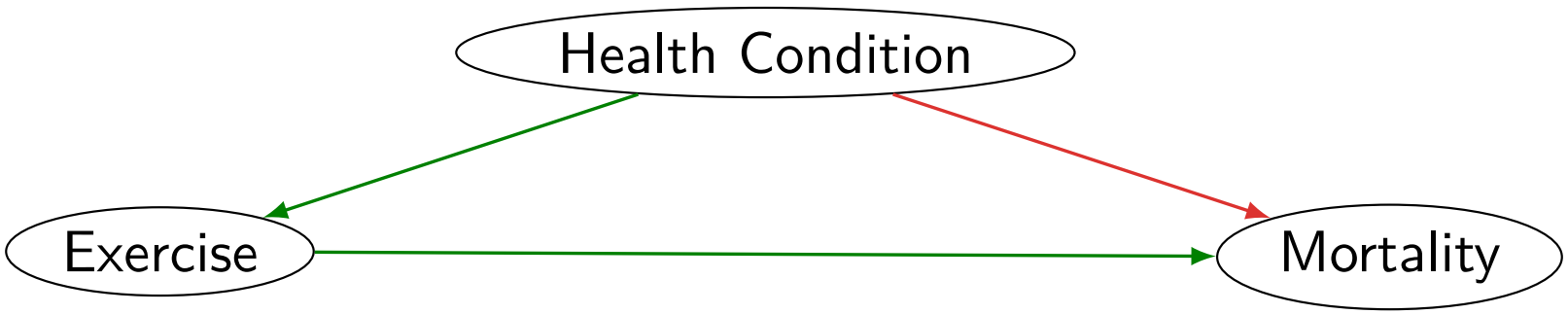
Correlations[15]

If X is correlated to Y, then

- X causes Y, or
- Y causes X, or
- X and Y are consequences of a common cause Z, but do not cause each other, or
- X causes Z and Z causes Y, or
- X and Y cause each other, or
- sample selection bias, X and Y are correlated conditional on Z, or
- data could be defective, or
- it could be a (Ramsey-type[14]) coincidence, or
- (non-local) quantum correlations of entangled qubits (quantum correlations cannot be attributed to latent variables), or
- mind-matter correlations? mental $\longleftarrow$ neutral $\longrightarrow$ physical ?

[14] Calude & Longo: The Deluge of Spurious Correlations in Big Data. 2017.
[15] Atmanspacher & Martin: Correlations and How to Interpret Them. 2019.

Causation $\overset{?}{\Longrightarrow}$ Correlation

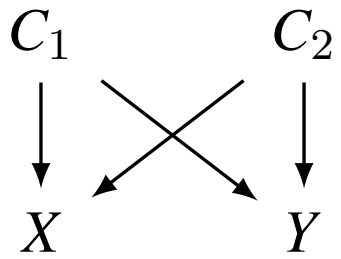


Remark: 这种恰好抵消不稳定.

No Causal Relation $\overset{?}{\Longrightarrow}$ Independence

- Suppose that X, Y, Z are variables that are probabilistically independent and causally unrelated.
- Now define $A = X + Y$ and $B = Y + Z$, and let $V = \{A, B\}$.
- Then $A \not\perp\!\!\!\perp B$, even though there is no causal relation between them.

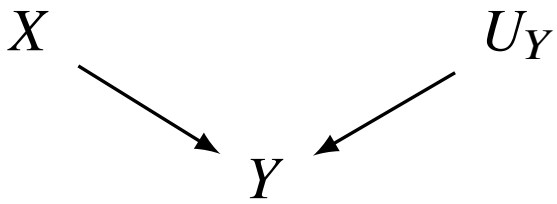
No Causal Relation $\overset{?}{\Longrightarrow}$ No Correlation



- Let C_1 and C_2 be the outcomes of two independent fair coins.
- X occurs when C_1 and C_2 are equal, and Y occurs when C_1 and C_2 are unequal.
- X and Y are negatively correlated $\rho_{XY} = -1$, without causally affecting each other.
- Besides, neither C_1 nor C_2 is associated with either X or Y; discovering the outcome of any one coin does not change the probability of X or Y. $X \perp C_1$, $X \perp C_2$, $Y \perp C_1$, $Y \perp C_2$.

Causation $\overset{?}{\Longrightarrow}$ Dependence

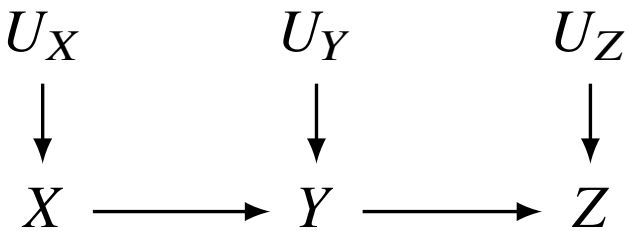
Example1:



$$f_Y : Y = \begin{cases} 1 & \text{if } X = U_Y \\ 0 & \text{if } X \neq U_Y \end{cases}$$

- X and U_Y are fair coins: $P(X = 1) = P(U_Y = 1) = \frac{1}{2}$
- $P(Y = 1 \mid X = 1) = P(Y = 1 \mid X = 0) = \frac{1}{2}$
- $P(Y = 1) = P(Y = 1 \mid X = 1)P(X = 1) + P(Y = 1 \mid X = 0)P(X = 0) = \frac{1}{2}$

Example2:



$$f_X : X = U_X \quad f_Y : Y = \begin{cases} a & \text{if } X = 0, U_Y = 0 \\ b & \text{if } X = 1, U_Y = 0 \\ c & \text{if } U_Y = 1 \end{cases} \quad f_Z : Z = \begin{cases} i & \text{if } Y = c, U_Z = 0 \\ j & \text{if } U_Z = 1 \end{cases}$$

Therefore, $(X \perp Z)_P$

题外话: 人们为什么经常更注重相关, 而非因果?

- 因果预设难以满足
- 心理学效应: 启动效应、框架效应、锚定效应、禀赋效应、可得性偏见、证实性偏见、动机性推理、损失规避等等
- 卡尼曼: 系统 1 vs 系统 2

微观-宏观问题

Granovetter 的暴动模型:

- A 镇广场上聚集了 100 人抗议示威, 每人都有一个应对周围环境影响的阈值, 低于阈值则克制, 高于阈值则暴动. 100 人的阈值从 0 到 99 各不相同, 0 号首先情绪失控开始煽动, 1 号跟随, ...很快会演变为 100 人的大暴动.
- B 镇跟 A 镇的唯一差别是, 他们有两个阈值为 4 的人, 却没有阈值为 3 的人.
  — 对于局外人来说, 这个差异小到无法察觉.
  — 潜在的暴动却戛然而止了.
- 事后人们会如何解释归因?
  — 社会矛盾、宗教信仰、种族民族性别年龄结构、教育程度、意识形态、经济状况、政治制度、法律法规、执法水平、"煽动者" 的号召力.....

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Causal Bayesian Network

Definition (Causal Bayesian Network)

Let $P(V)$ be a probability distribution on a set of variables V and $P(V \mid \mathrm{do}(X = x))$ denote the distribution of V after intervention on a subset X. A DAG G is a causal Bayesian network for P iff for all $X \subset V$ and x we have:

1. $P(V \mid \mathrm{do}(X = x))$ is Markov relative to G
2. $P(v_i \mid \mathrm{pa}_i, \mathrm{do}(X = x)) = 1$ for every $V_i \in X$
3. $P(v_i \mid \mathrm{pa}_i, \mathrm{do}(X = x)) = P(v_i \mid \mathrm{pa}_i)$ for every $V_i \notin X$

How to calculate interventional distributions? — Truncated factorization

$$P(v \mid \mathrm{do}(x)) = \prod_i P(v_i \mid \mathrm{pa}_i, \mathrm{do}(x)) = \prod_{V_i \notin X} P(v_i \mid \mathrm{pa}_i, \mathrm{do}(x))\Big|_{X=x} = \prod_{V_i \notin X} P(v_i \mid \mathrm{pa}_i)\Big|_{X=x}$$

$$P(v; \sigma) = P'(v_i \mid \mathrm{pa}'_i) \prod_{j \neq i} P(v_j \mid \mathrm{pa}_j) \text{ for soft intervention } \sigma = P'(v_i \mid \mathrm{pa}'_i)$$

Theorem

The causal graph G induced by the SCM M is a Causal Bayesian Network.

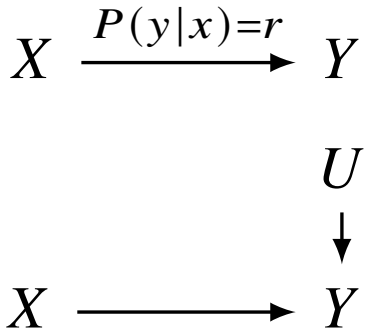
Remarks

- A causal Bayesian network is a Bayesian network with the requirement that the relationships be causal.

$$P(v_i \mid \mathrm{pa}_i) = P(v_i \mid \mathtt{do}(\mathrm{pa}_i))$$

$$P(v_i \mid \mathtt{do}(\mathrm{pa}_i), \mathtt{do}(s)) = P(v_i \mid \mathtt{do}(\mathrm{pa}_i)) \quad \text{for } S \subset V \setminus (V_i \cup \mathrm{Pa}_i)$$

- An SCM induces a CBN.
  A mechanism $f_i : \mathrm{Pa}_i \times U_i \to V_i$ and noise distribution $P(U_i)$ induce a conditional $P(v_i \mid \mathrm{pa}_i) = \sum_{u_i : v_i = f_i(\mathrm{pa}_i, u_i)} P(u_i)$.
- CBNs cannot be used to reason about counterfactuals, whereas SCMs can.



$$y = f(x, u) \text{ and } P(u) = r$$

Remark

- 在结构因果模型 (SCMs) 中, 因果关系是准决定论的 (quasi-deterministic).
- 因果贝叶斯网络 (CBNs) 则假设所有关系本质上都是随机的, 这契合了现代物理学的观念, 即自然律本质上都是概率性的, 决定论不过是为了方便处理而做出的近似.

Remark

$$P(\mathbf{V}=\mathbf{v}) = \sum_{\mathbf{u}:\mathbf{V}(\mathbf{u})=\mathbf{v}} P(\mathbf{u}) = \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(V_i = v_i \mid \mathrm{pa}_i, \mathbf{u}_i)$$

The factors $P(V_i = v_i \mid \mathrm{pa}_i, \mathbf{u}_i)$ are deterministic.

$$\begin{aligned} P(\mathbf{V}=\mathbf{v}) &= \sum_{\mathbf{u}} P(\mathbf{u}) \prod_{V_i \in \mathbf{V}} P(V_i = v_i \mid \mathrm{pa}_i, \mathbf{u}_i) \\ &= \prod_{V_i \in \mathbf{V}} \sum_{\mathbf{u}_i} P(V_i = v_i \mid \mathrm{pa}_i, \mathbf{u}_i) P(\mathbf{u}_i) \qquad \text{(Markovian)} \\ &= \prod_{V_i \in \mathbf{V}} \sum_{\mathbf{u}_i} P(V_i = v_i, \mathbf{u}_i \mid \mathrm{pa}_i) \\ &= \prod_{V_i \in \mathbf{V}} P(V_i = v_i \mid \mathrm{pa}_i) \end{aligned}$$

独立因果机制

Independent Causal Mechanisms ICM[PJS17; Sch+21]

系统的因果生成过程由独立的模块组成, 模块之间不会相互通知或影响.

在概率情况下, 这意味着

- 改变一个机制 $P(V_i \mid \mathrm{Pa}_i)$ 不会影响其它机制 $P(V_j \mid \mathrm{Pa}_j)$, $j \neq i$.
- 知道其它机制 $P(V_j \mid \mathrm{Pa}_j)$, $j \neq i$, 也不会为机制 $P(V_i \mid \mathrm{Pa}_i)$ 提供任何信息.

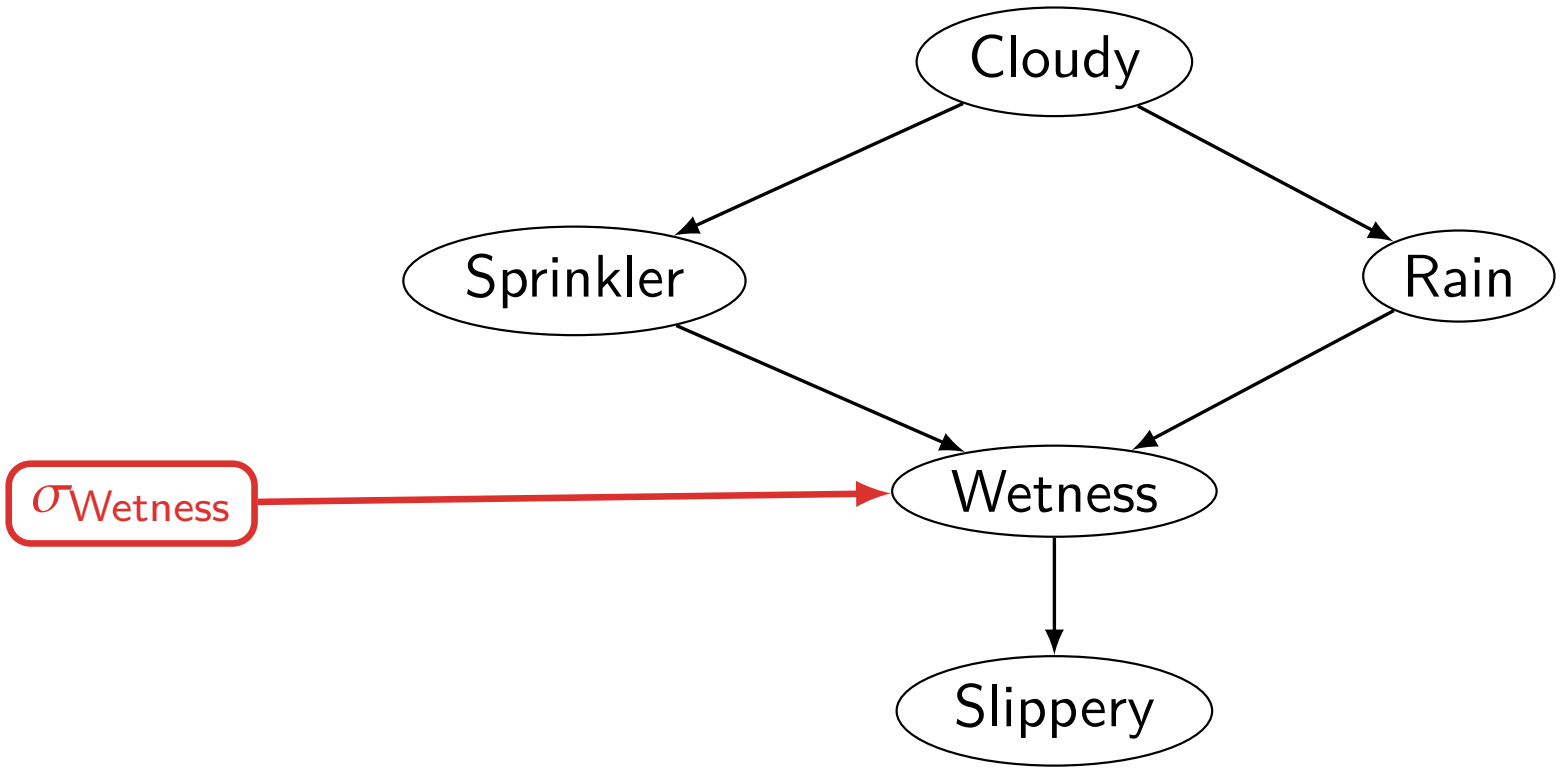
Remark: 在只有两个变量的情况下, 这意味着

$$P(\mathrm{Effect} \mid \mathrm{Cause}) \perp P(\mathrm{Cause})$$

Example: $P(\mathrm{Temperature} \mid \mathrm{Altitude}) \perp P(\mathrm{Altitude})$.
机制 $P(\mathrm{Temperature} \mid \mathrm{Altitude})$ 在相似气候的不同地区是不变的.

> *The true causal order is the one that is invariant under the right sort of intervention.* — *Herbert Simon*

扩展图 — 干预作为决策变量



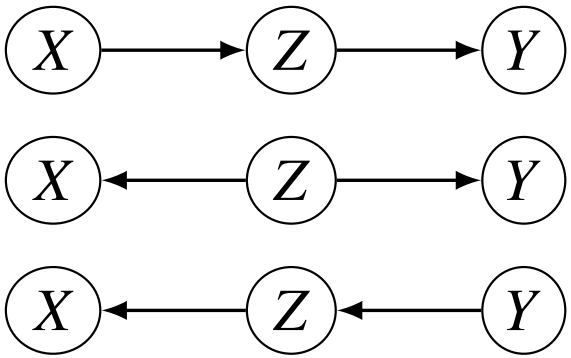
$$\mathrm{Pa}_i' := \mathrm{Pa}_i \cup \{\sigma_i\}$$

$$P(x_i \mid \mathrm{Pa}_i') = \begin{cases} P(x_i \mid \mathrm{Pa}_i) & \text{if } \sigma_i = \varnothing \\ 1 & \text{if } \sigma_i = x_i' \text{ and } x_i' = x_i \qquad \mathtt{do}(x_i) \\ 0 & \text{if } \sigma_i = x_i' \text{ and } x_i' \neq x_i \qquad \mathtt{do}(x_i') \end{cases}$$

where σ_i is a new variable taking values in $\{x_i, \varnothing\}$, $x_i \in R(X_i)$.

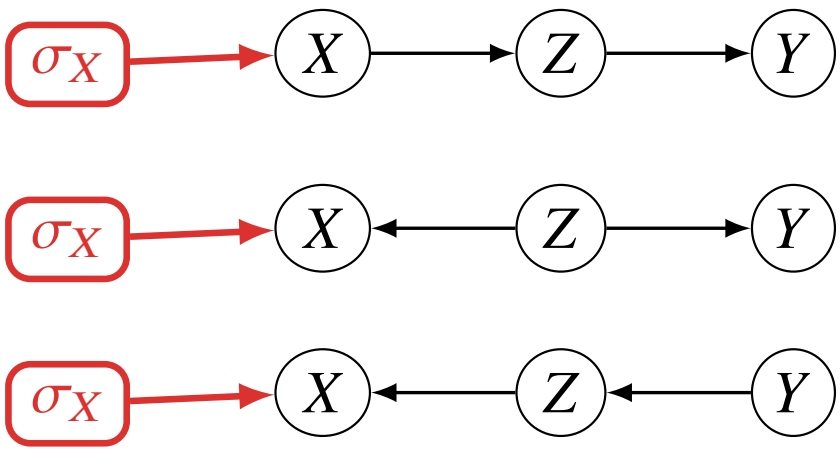
扩展图

- Markov Equivalence



$$X \perp Y \mid Z$$

- Markov Non-Equivalence



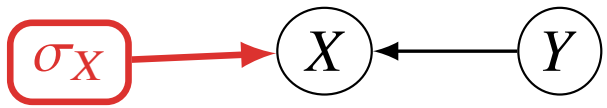
$\{Z, Y\} \perp \sigma_X \mid X$ X causes Z, Y

$\{Z, Y\} \perp \sigma_X$

X doesn't cause Z, Y

扩展图

- X does not cause Y when $Y \perp \sigma_X$.

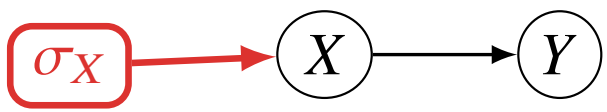


Remark: The distribution of Y, under an intervention that sets X to x, does not vary with x.

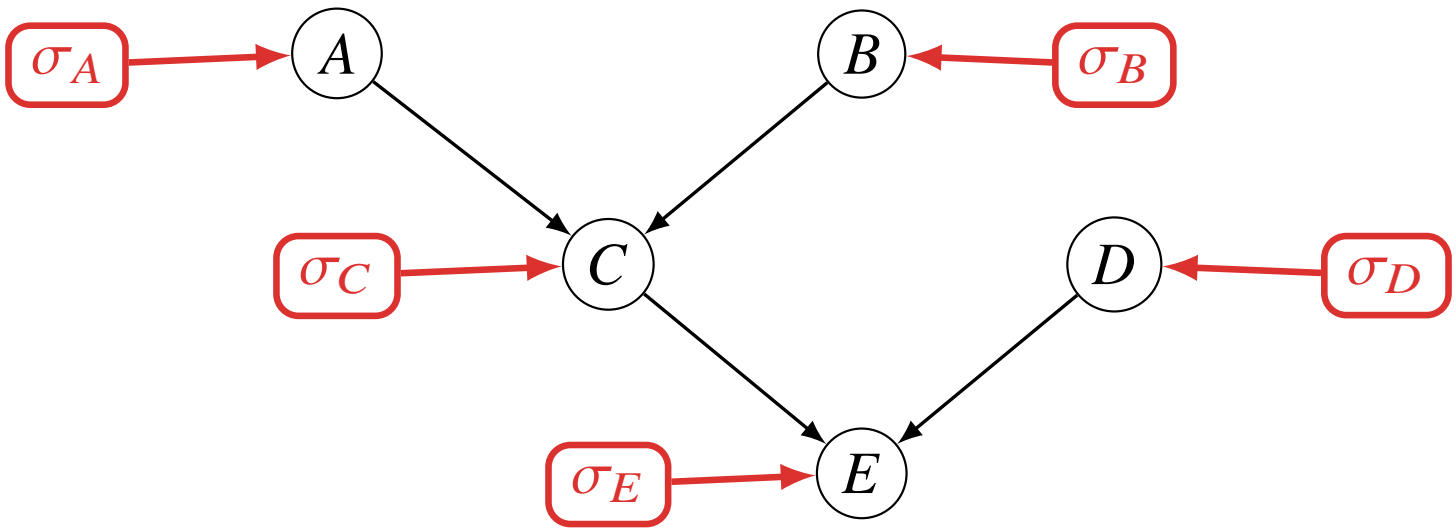
- X has no direct effect on Y when

$$Y \perp \sigma_X \mid \sigma_{V \setminus \{X,Y\}}$$

- No Confounding: $Y \perp \sigma_X \mid X$



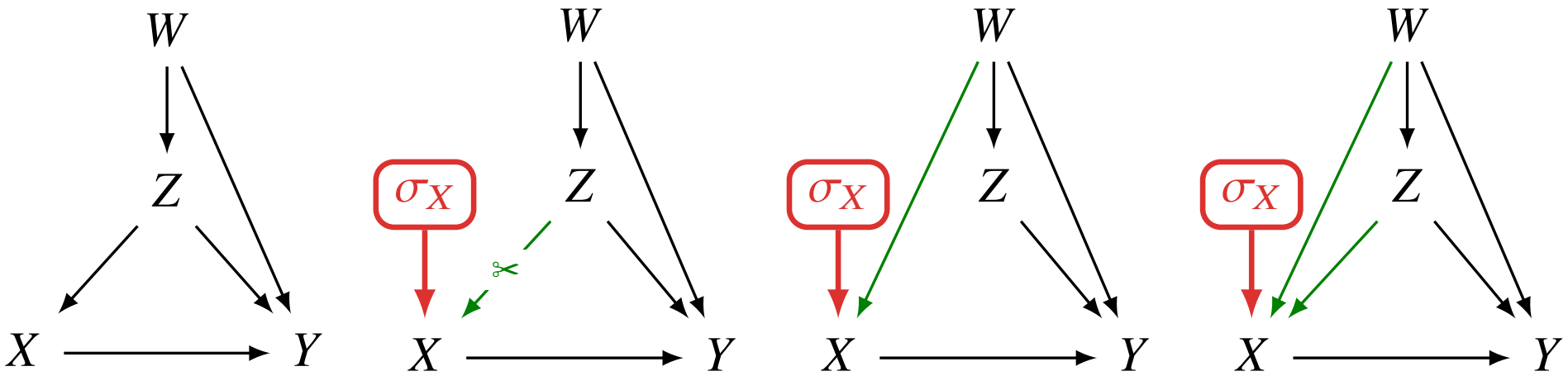
Example



$$E \perp \{A, B, \sigma_A, \sigma_B, \sigma_C, \sigma_D\} \mid C, D, \sigma_E = \varnothing$$

The conditional distribution of E, given its parents C and D, is unaffected by the values of A, B, and by whether or not any of the other variables is manipulated.

不同类型的干预



- W: 历史成绩
- Z: 动机
- X: 辅导
- Y: 期末成绩

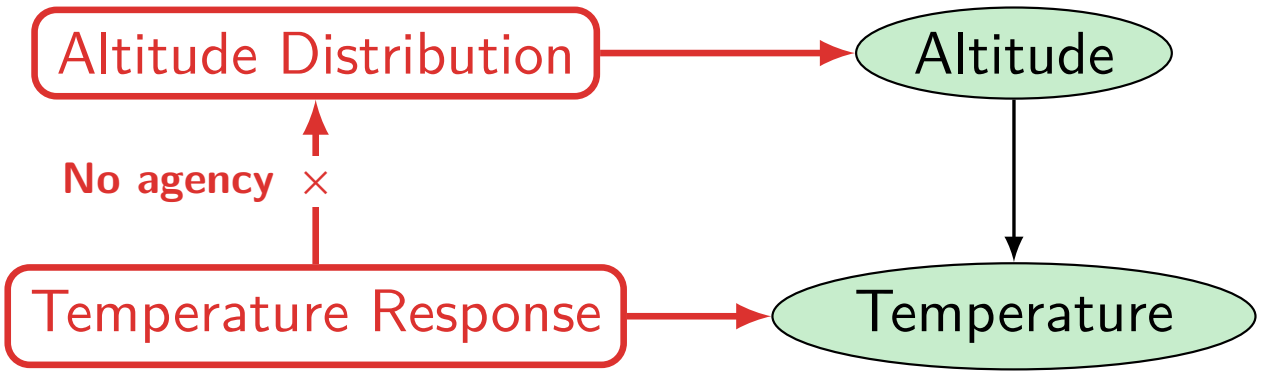
1. 硬干预: 比如, $\sigma_X = x$ 所有同学都要接受辅导

   Remark: $B \perp \sigma_X \mid A \implies P(B \mid \mathrm{do}(X = x), A) = P(B \mid A)$

   Example: $Y \perp \sigma_X \mid X, W, Z \qquad W, Z \perp \sigma_X$

2. 条件干预: 比如, $\sigma_X = g(w)$ 历史成绩差的同学要接受辅导
3. 软干预: 比如, $\sigma_X = P'(x \mid w, z)$

Remark: Agency violates Independent Causal Mechanism



- Causal factorization:

$$P(a,t) = P(t \mid a)P(a)$$

$$P(\text{Temperature} \mid \text{Altitude}) \perp P(\text{Altitude})$$

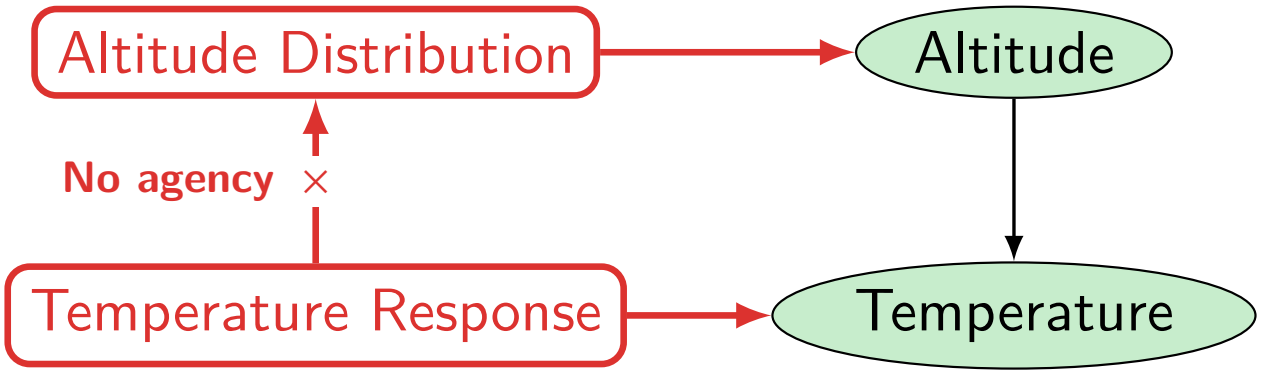
- Altitude distribution shift:

$$P(a) \rightsquigarrow P'(a) \implies P'(a,t) = P(t \mid a)P'(a)$$

- Temperature distribution shift (soft intervention):

$$P(t \mid a) \rightsquigarrow P'(t \mid a) \implies P'(a,t) = P'(t \mid a)P(a)$$

Remark: Agency violates Independent Causal Mechanism



- Non-causal factorization (entangled representation):

$$P(a,t) = P(a \mid t)P(t)$$

$$P(\text{Altitude} \mid \text{Temperature}) \not\perp P(\text{Temperature})$$

$$P(a) \rightsquigarrow P'(a) \implies P'(a,t) = P'(a \mid t)P'(t)$$

where $P'(t) = \sum_a P(t \mid a)P'(a)$ and $P'(a \mid t) = \dfrac{P(t \mid a)P'(a)}{P'(t)}$

Intervention (do-operator) in CBN

- The factorization joint probability distribution

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid \mathrm{Pa}_i)$$

- The do-operator[16]

$$P(X_1, \ldots, X_n \mid \mathtt{do}(X_i = x_i)) = \prod_{\substack{j=1 \\ j \neq i}}^{n} P(X_j \mid \mathrm{Pa}_j)\Big|_{X_i = x_i}$$

$$P(x_1, \ldots, x_n \mid \mathtt{do}(x_i^*)) = \begin{cases} \frac{P(x_1, \ldots, x_n)}{P(x_i^* \mid \mathrm{pa}_i)} & \text{if } x_i = x_i^* \\ 0 & \text{otherwise} \end{cases}$$

- The post-intervention distribution can be given by marginalization, or by SCM,

$$P_M(Y = y \mid \mathtt{do}(X = x)) := P_{M_x}(Y = y)$$

[16] Soft intervention:

$$P\Big(X_1, \ldots, X_n \mid \mathtt{do}\left(P'(X_i \mid \mathrm{Pa}_i')\right)\Big) = P'(X_i \mid \mathrm{Pa}_i') \prod_{\substack{j=1 \\ j \neq i}}^{n} P(X_j \mid \mathrm{Pa}_j)$$

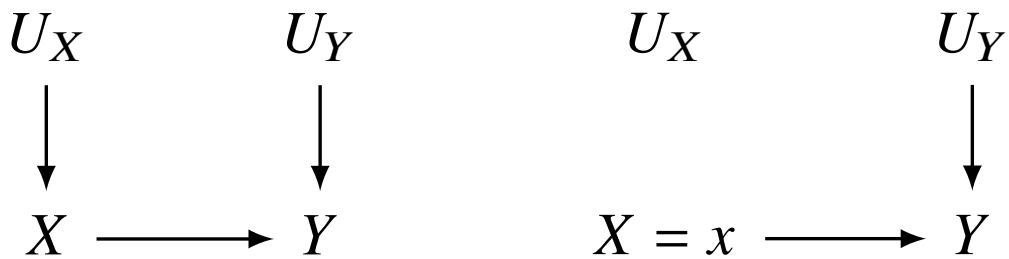
因果效应 — Total Effect

- for binary X:

$$\mathrm{TE} = \mathbb{E}[Y \mid \mathrm{do}(X = 1)] - \mathbb{E}[Y \mid \mathrm{do}(X = 0)]$$

- for continuous X, Y:

$$\mathrm{TE} = \frac{\partial}{\partial x}\mathbb{E}[Y \mid \mathrm{do}(X = x)]$$



$$\mathbb{E}[Y \mid \mathrm{do}(X = x)] = \mathbb{E}[f_Y(x, U_Y)] = \sum_{u_Y} f_Y(x, u_Y) P(U_Y = u_Y)$$

Remark: $Y_x(u_Y) = f_Y(x, u_Y)$ can be taken as (unit-level) counterfactual.

Causal Effect — Examples

Example: Inferring the effects of any treatment/policy/intervention/etc.

- Effect of treatment on a disease
- Effect of climate change policy on emissions
- Effect of social media on mental health

Joke: 已知飞机上乘客携带炸药的概率是 0.01%, 于是某统计学家自己携带炸药上飞机 ☹

- X and Y are associated iff $(X \not\perp Y)_P$

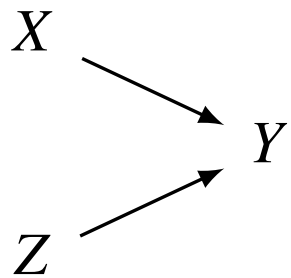
$$\exists x, x' : P(Y \mid X = x) \neq P(Y \mid X = x')$$

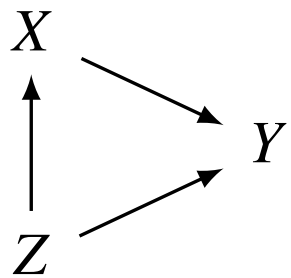
- X is a cause of Y iff $(X \not\perp Y)_{P_{\mathsf{do}(X)}}$

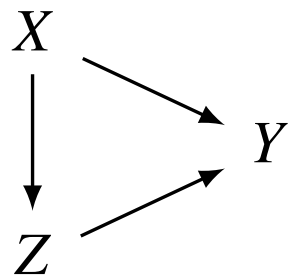
$$\exists x, x' : P(Y \mid \mathsf{do}(X = x)) \neq P(Y \mid \mathsf{do}(X = x'))$$

- X is a direct cause of Y iff $(X \not\perp Y \mid V \setminus \{X, Y\})_{P_{\mathsf{do}(V \setminus Y)}}$

Causal relations vs Ordinary Least Squares regression





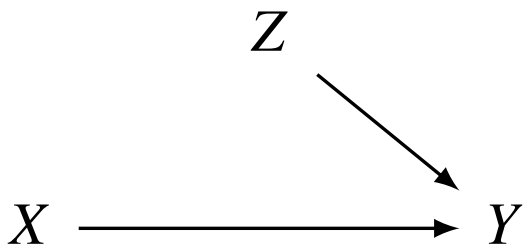


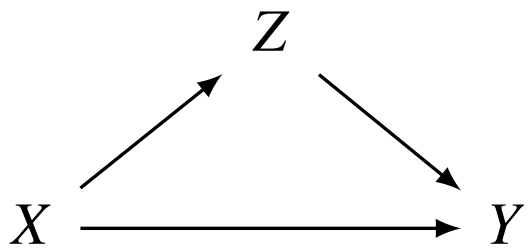
- Baseline Model: $Y = \alpha + \beta X + Z$

1. Model 1: $\hat{\beta}$ is unbiased and β is the causal effect of X on Y.
2. Model 2: $\hat{\beta}$ is biased and β is the causal effect of X on Y.
3. Model 3: $\hat{\beta}$ is biased and β is not the causal effect of X on Y.

Examples

- $X \longrightarrow Y$

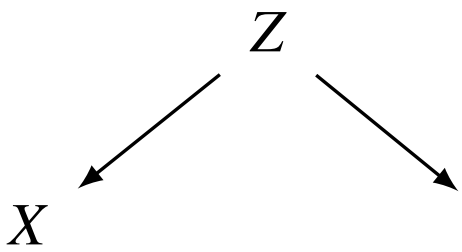




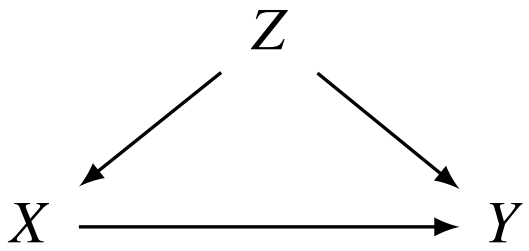
$$P(y \mid \mathtt{do}(x)) = P(y \mid x)$$

- $X \longleftarrow Y$

$$P(y \mid \mathtt{do}(x)) = P(y) \neq P(y \mid x)$$

- 

$$P(y \mid \mathtt{do}(x)) = P(y) \neq P(y \mid x)$$

- What about 

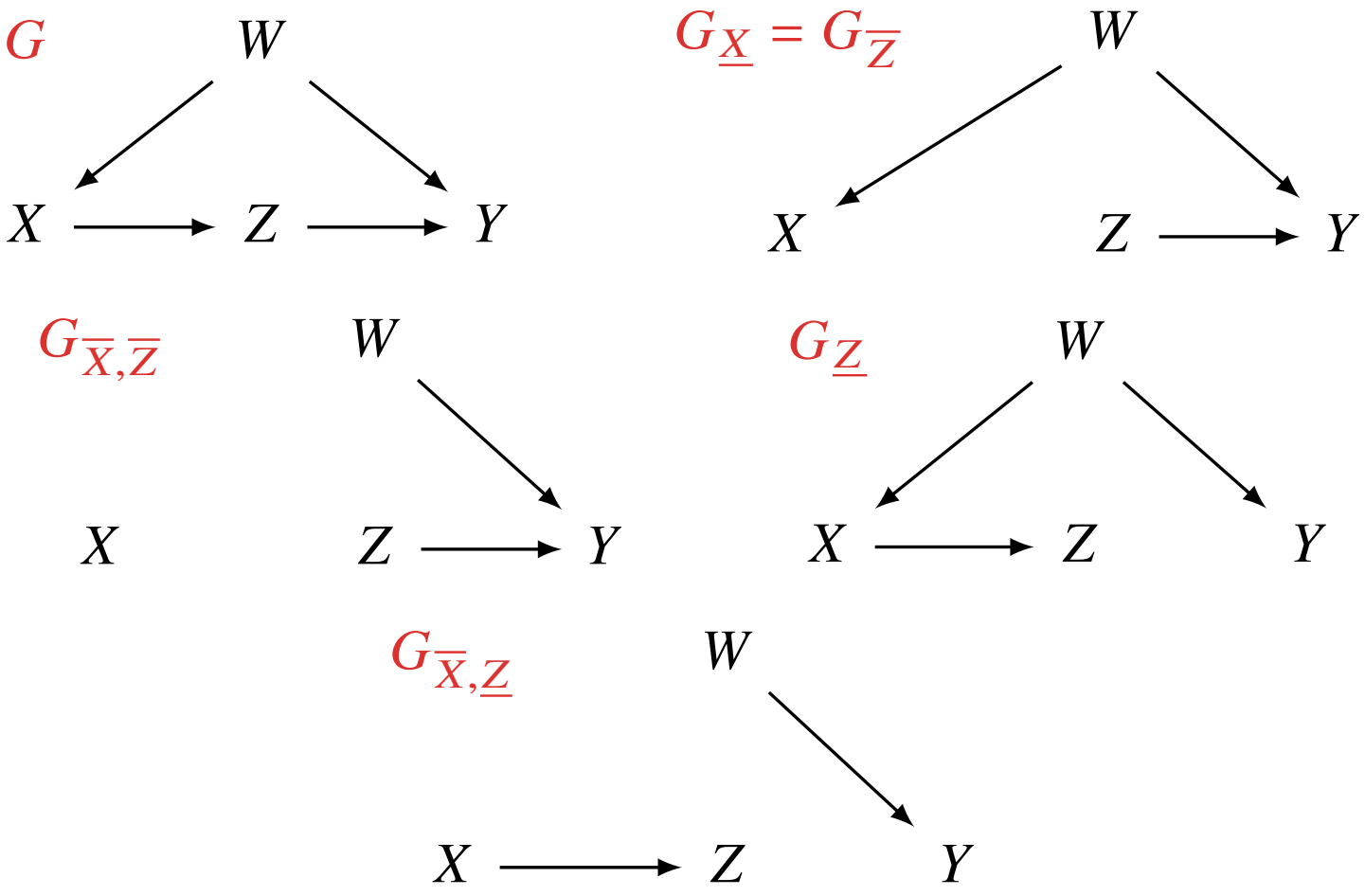
?

$$P(y \mid \mathtt{do}(x)) = ?$$

Perturbed Graphs

"Thinking as acting in an imagined space."

- $G_{\overline{X}}$ perturbed graph in which all arrows to X have been deleted
- $G_{\underline{X}}$ perturbed graph in which all arrows from X have been deleted



去混杂 — 后门准则 Backdoor Criterion

为了消除 X 和 Y 之间的混杂, 我们希望找到一个变量集 Z, 使得:

- 它阻断了从 X 到 Y 的所有伪路径;
- 而不会阻断从 X 到 Y 的任何因果路径;
- 也不会开启其他的伪路径.

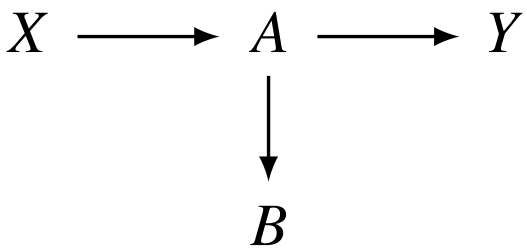
后门准则

在有向无环图 G 中, 如果变量集 Z 相对于变量 (X, Y) 满足以下条件, 则称其满足后门准则:

1. Z 中没有节点是 X 的后代;
2. Z 阻断了每一条后门路径 (即 X 和 Y 之间含有指向 X 的箭头的路径).

Remark: 前门路径: X 和 Y 之间指离 X 的因果路径.

Example

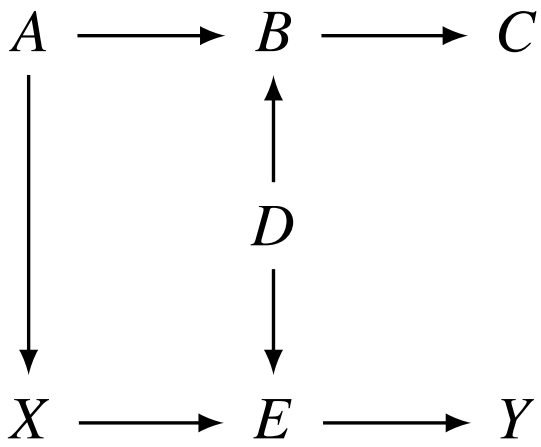


- There are no backdoor paths.
- We don't need to control for/adjust for/condition on anything.
- It will lead to disaster if we controlled for B.

Example:

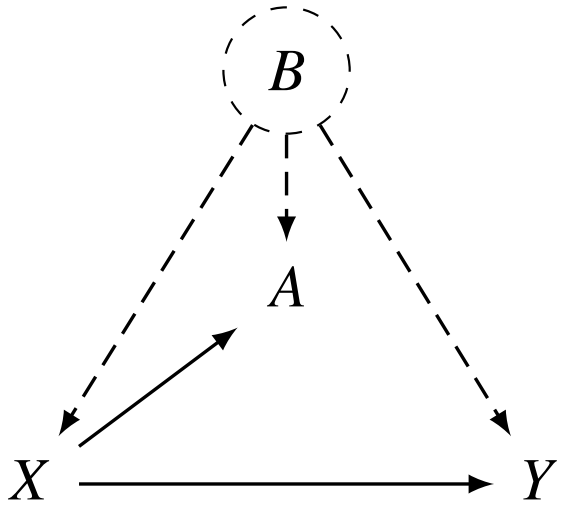
- X: 抽烟
- Y: 流产
- A: 抽烟导致的身体病变
- B: 流产史

Example



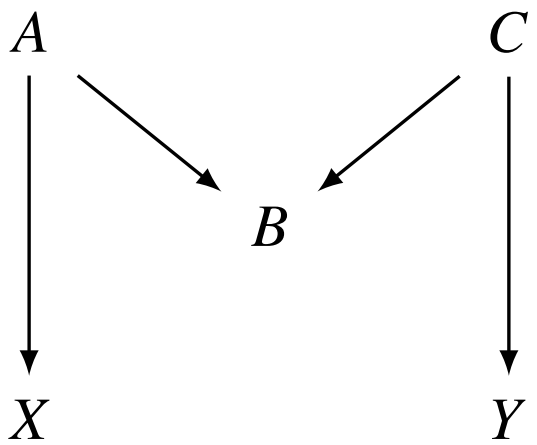
- There is one backdoor path $X \leftarrow A \rightarrow B \leftarrow D \rightarrow E \rightarrow Y$.
- This path is already blocked by the collider at B.
- We don't need to control for anything.
- It will lead to disaster if we controlled for B or C.
- In this case we could reclose the path by controlling for A or D.

Example



- There is one backdoor path $X \leftarrow B \rightarrow Y$.
- We need to control for B.
- If B is unobservable, then there is no way of estimating the effect of X on Y without running a randomized control trial.
- If we control for A, as a proxy for the unobservable variable B, then this only partially eliminates the confounding bias and introduces a new collider bias.

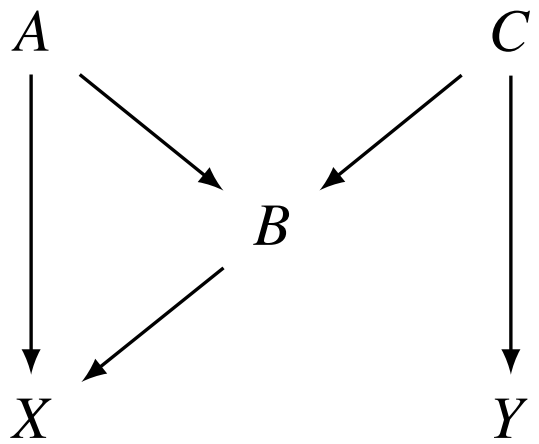
Example



- There is one backdoor path $X \leftarrow A \rightarrow B \leftarrow C \rightarrow Y$.
- This path is already blocked by the collider at B.
- We don't need to control for anything.
- It will lead to disaster if we controlled for B.
- It's all right to control for B if we also control for A or C.
- B: 安全带的使用; X: 抽烟; Y: 肺病; A: 对社会规范的态度; C: 安全健康意识

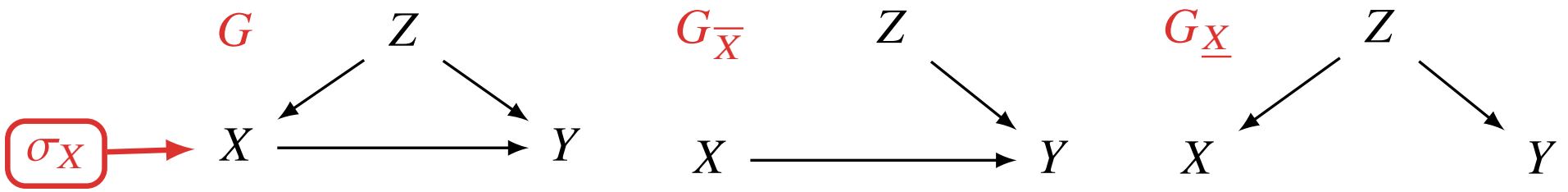
Remark: 并不是所有同时跟 X 和 Y 都相关的就是混杂因子.

Example



- There is a backdoor path $X \leftarrow B \leftarrow C \rightarrow Y$.
- If we close this path by controlling for B, then we open up the M-shaped path $X \leftarrow A \rightarrow B \leftarrow C \rightarrow Y$.
- To close that path, we must control for A or C as well.
- However, we could control for C alone.

后门校正 Backdoor Adjustment



The Backdoor Criterion

A set of variables Z satisfies the backdoor criterion relative to an ordered pair of variables (X, Y) in a DAG G if:

1. Z contains no descendant of X. i.e., $Z \perp \sigma_X$;
2. Z blocks all backdoor paths. i.e., $(Y \perp X \mid Z)_{G_{\underline{X}}}$.

Backdoor Adjustment: If such Z exists, then

$$
\begin{aligned}
P(y \mid \mathrm{do}(x)) &= \sum_z P(y \mid \mathrm{do}(x), z) P(z \mid \mathrm{do}(x)) \\
&= \sum_z P(y \mid \mathrm{do}(x), x, z) P(z \mid \mathrm{do}(x)) \qquad (\sigma_X = x \implies X = x) \\
&= \sum_z P(y \mid x, z) P(z) \qquad (Y \perp \sigma_X \mid X, Z \text{ and } Z \perp \sigma_X)
\end{aligned}
$$

后门校正 Backdoor Adjustment

Backdoor Criterion:

1. no node in Z is a descendant of X.

$$X \perp Z \setminus \mathrm{Pa}_X \mid \mathrm{Pa}_X \setminus Z$$

2. Z blocks every backdoor path.

$$Y \perp \mathrm{Pa}_X \setminus Z \mid Z, X$$

$$\begin{aligned}
P(y \mid \mathsf{do}(x)) &= \sum_{\mathrm{pa}_X} P(y \mid x, \mathrm{pa}_X) P(\mathrm{pa}_X) \\
&= \sum_{\mathrm{pa}_X \cup z} P(y \mid x, \mathrm{pa}_X \cup z) P(z \setminus \mathrm{pa}_X \mid x, \mathrm{pa}_X) P(\mathrm{pa}_X) \\
&= \sum_{\mathrm{pa}_X \cup z} P(y \mid x, z) P(z \setminus \mathrm{pa}_X \mid \mathrm{pa}_X) P(\mathrm{pa}_X) \\
&= \sum_{z} P(y \mid x, z) \sum_{\mathrm{pa}_X \setminus z} P(z \cup \mathrm{pa}_X) \\
&= \sum_{z} P(y \mid x, z) P(z)
\end{aligned}$$

混杂的定义 The Definition of Confounding

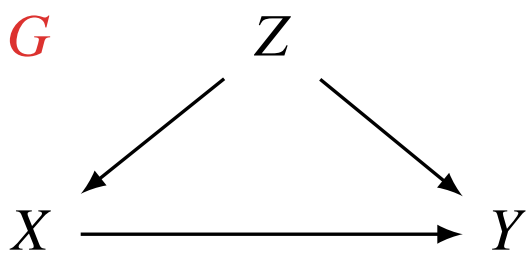
Definition (Confounding)

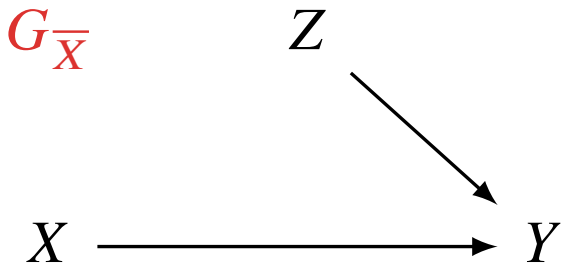
The causal effect from X to Y is called confounded if $P(y \mid \mathrm{do}(x)) \neq P(y \mid x)$.

$$
\begin{aligned}
P(y \mid x) &= \sum_z P(y, z \mid x) \\
&= \sum_z P(y \mid x, z) P(z \mid x) \\
&\neq \sum_z P(y \mid x, z) P(z) \\
&= P(y \mid \mathrm{do}(x))
\end{aligned}
$$

"Correlation is not equal to Causation."

Statistical Dependence vs Causal Inference





$$P(y \mid x) = \sum_z P(y \mid x, z) P(z \mid x)$$

$$P(y \mid \mathtt{do}(x)) = \sum_z P(y \mid x, z) P(z)$$

- 因果路径 $X \to Y$
- 非因果路径 $X \leftarrow Z \to Y$

Simpson's Paradox — Should we treat scurvy with lemons?

	Recovery	No Recovery	Total	Recovery Rate
No Lemons	20	20	40	50%
Lemons	16	24	40	40%
Total	36	44	80	

Table: $P(\text{recovery} \mid \text{lemon}) < P(\text{recovery} \mid \text{no lemon})$

	Recovery	No Recovery	Total	Recovery Rate
No Lemons	2	8	10	20%
Lemons	9	21	30	30%
Total	11	29	40	

Table: $P(\text{recovery} \mid \text{lemon}, \text{old}) > P(\text{recovery} \mid \text{no lemon}, \text{old})$

	Recovery	No Recovery	Total	Recovery Rate
No Lemons	18	12	30	60%
Lemons	7	3	10	70%
Total	25	15	40	

Table: $P(\text{recovery} \mid \text{lemon}, \text{young}) > P(\text{recovery} \mid \text{no lemon}, \text{young})$

Resolution of Simpson's paradox — The do-operator

- What is the sailors' probability of recovery when we see a treatment with lemons?

$$P(\text{recovery} \mid \text{lemon})$$

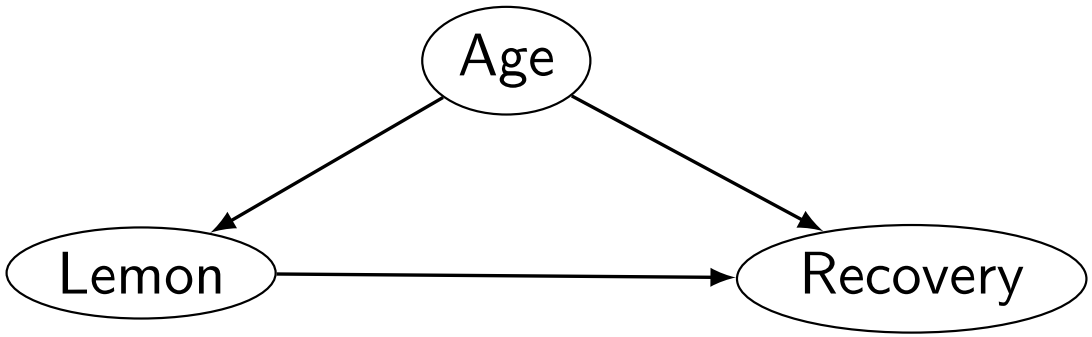
- What is the sailors' probability of recovery if we do treat them with lemons?

$$P(\text{recovery} \mid \texttt{do}(\text{lemon}))$$

- We should treat scurvy with lemons if

$$P(\text{recovery} \mid \texttt{do}(\text{lemon})) > P(\text{recovery} \mid \texttt{do}(\text{no lemon}))$$

Resolution of Simpson's paradox — The do-operator



$$P(\text{recovery} \mid \texttt{do}(\text{lemon})) = \sum_{\text{age}} P(\text{recovery} \mid \text{lemon}, \text{age}) P(\text{age}) = 0.5$$

$$P(\text{recovery} \mid \texttt{do}(\text{no lemon})) = \sum_{\text{age}} P(\text{recovery} \mid \text{no lemon}, \text{age}) P(\text{age}) = 0.4$$

The total effect:

$$\mathbb{E}[\text{recovery} \mid \texttt{do}(\text{lemon})] - \mathbb{E}[\text{recovery} \mid \texttt{do}(\text{no lemon})] = 0.5 - 0.4 = 0.1$$

Simpson's Paradox

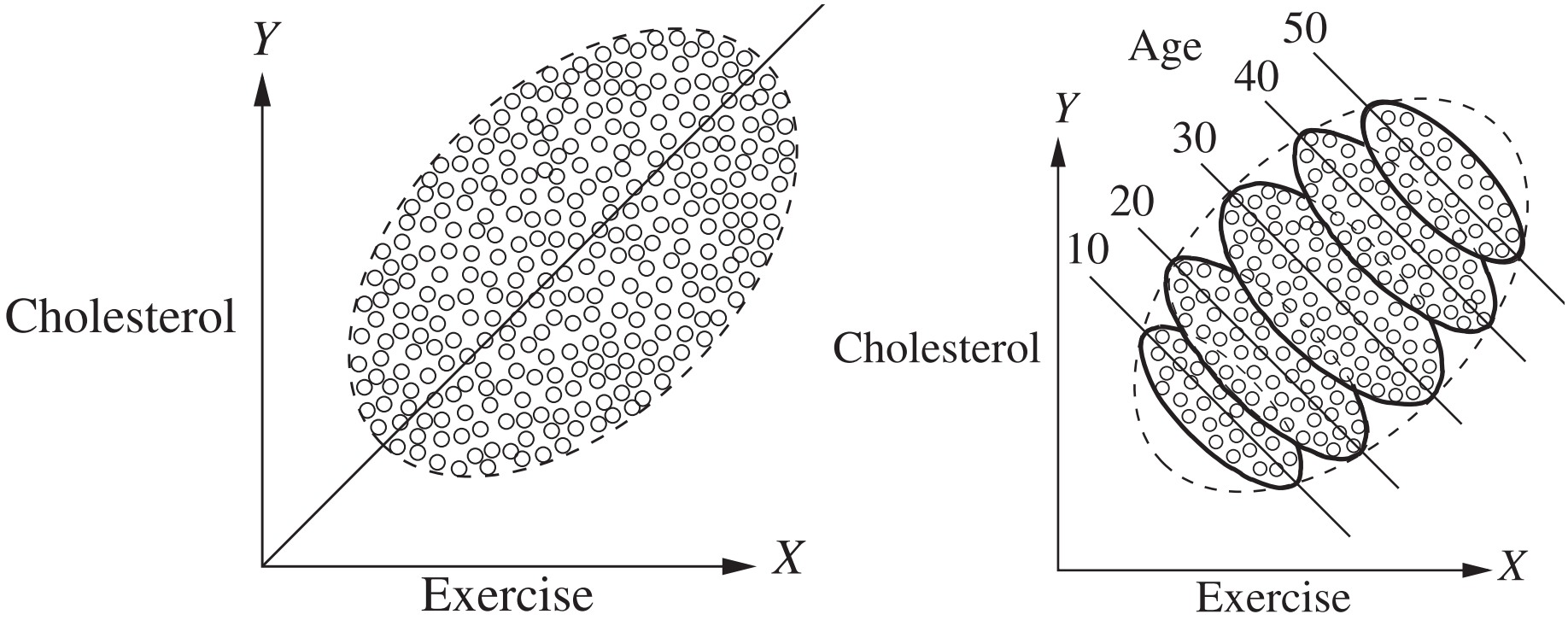
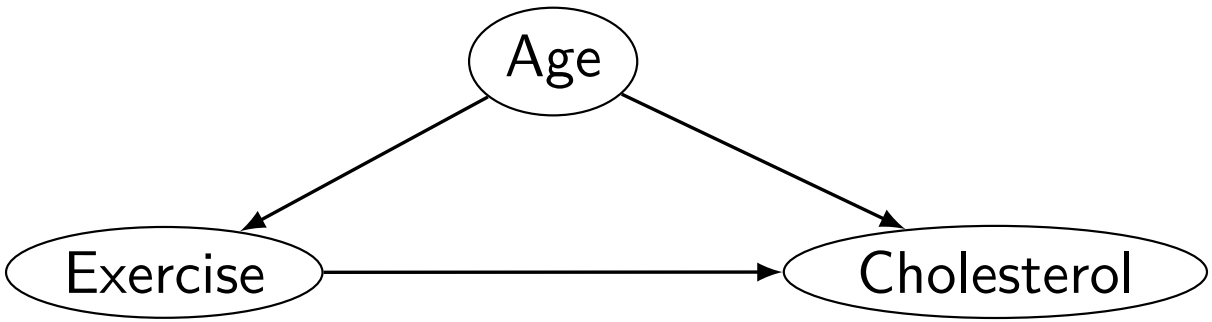


Figure: Exercise appears to be beneficial (downward slope) in each age group but harmful (upward slope) in the population as a whole.

Simpson's Paradox



$$\mathbb{E}[\text{cholesterol} \mid \text{exercise}] > \mathbb{E}[\text{cholesterol} \mid \text{no exercise}]$$

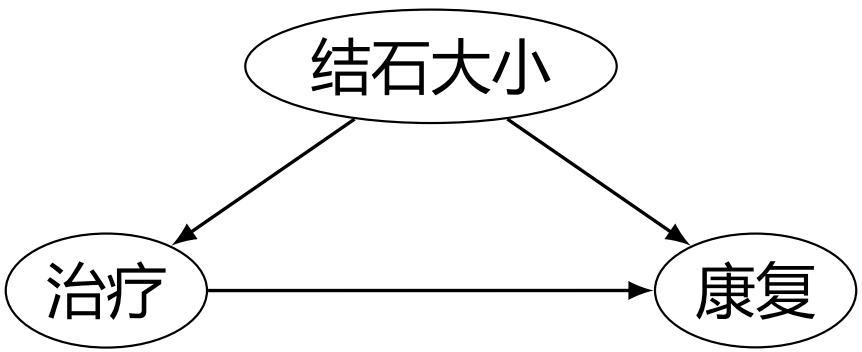
$$\mathbb{E}[\text{cholesterol} \mid \texttt{do}(\text{exercise})] < \mathbb{E}[\text{cholesterol} \mid \texttt{do}(\text{no exercise})]$$

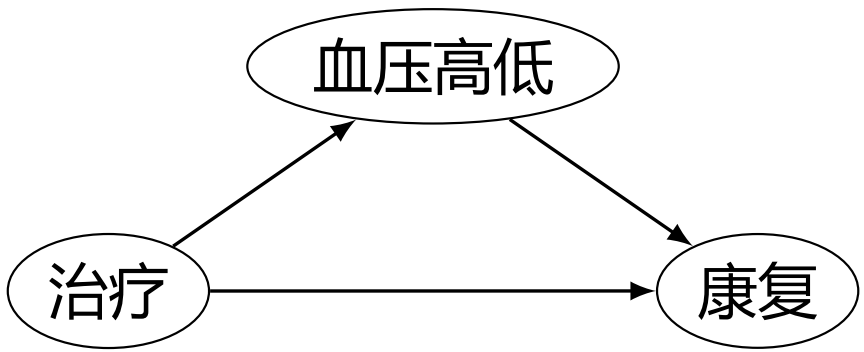
- 年龄大的人锻炼多.
- 应该控制 "年龄" 变量.

选择哪套治疗方案?

	轻度患者	重度患者	全部患者
方案 1	$93\%(81/87)$	$73\%(192/263)$	$78\%(273/350)$
方案 2	$87\%(234/270)$	$69\%(55/80)$	$83\%(289/350)$

1. 假如轻/重度的分组依据是 "结石大小"
   - 结石大的重度患者倾向于方案 1
   - 方案 1 更有效 $P(Y \mid \mathtt{do}(X=1)) > P(Y \mid \mathtt{do}(X=2))$
2. 假如轻/重度的分组依据是 "血压高低"
   - 方案 1 导致高血压
   - 方案 2 更有效 $P(Y \mid \mathtt{do}(X=1)) < P(Y \mid \mathtt{do}(X=2))$

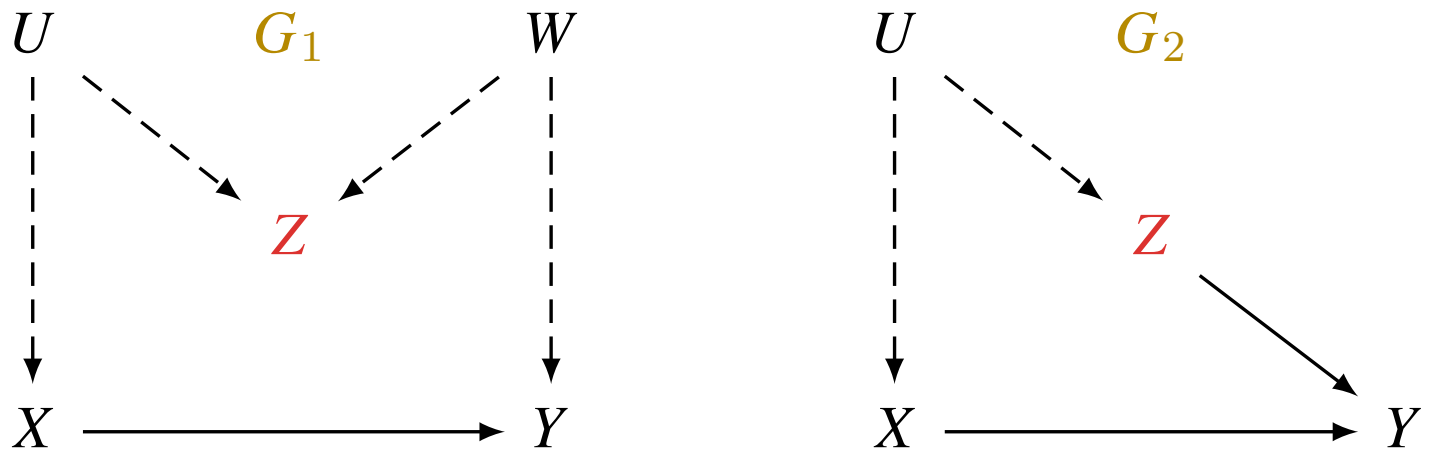




Remark: 混杂因子 "结石" 早于治疗, 而中介 "血压" 晚于治疗. 那么, 可否通过时间信息帮助我们做出正确的选择?

Why Temporal Information does not Help?

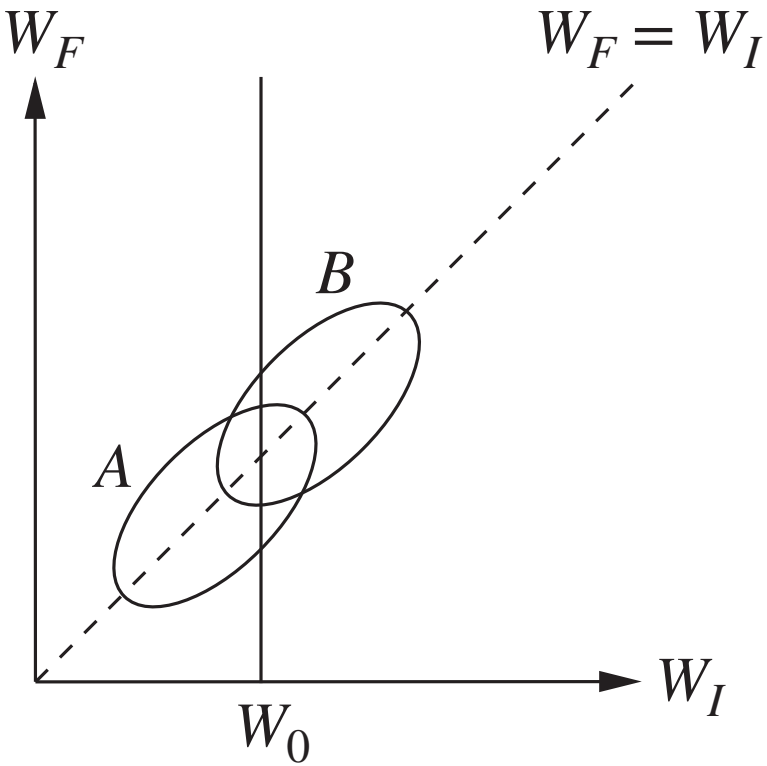
- 时间信息无法帮助区分真实因果还是虚假关联. 比如, 气压计读数下降发生在下雨之前, 但不是下雨的原因.
- 时间信息也无法帮助判断是否应该对变量 Z 进行控制.

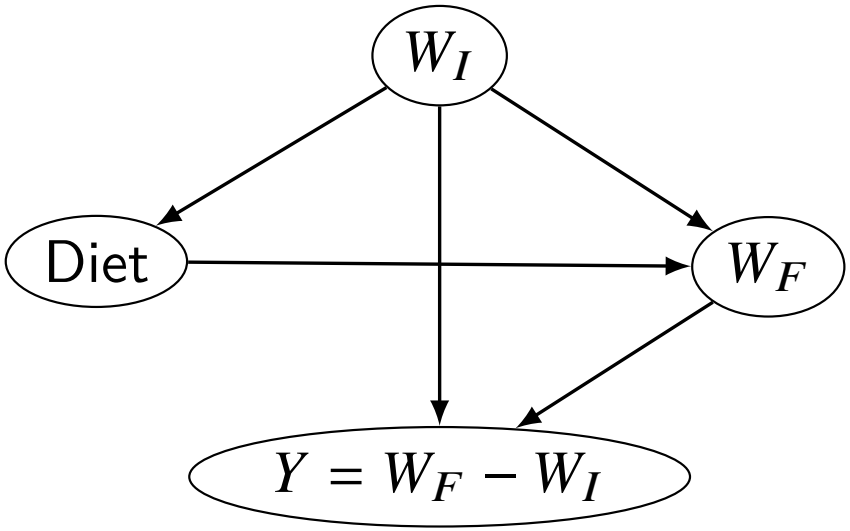


- Z 可能早于, 也可能晚于 X.
- 在 G_1 中, 不要控制 Z.
- 在 G_2 中, 要控制 Z.

Remark: 计算因果效应需要知道因果图.

Example — Lord's Paradox

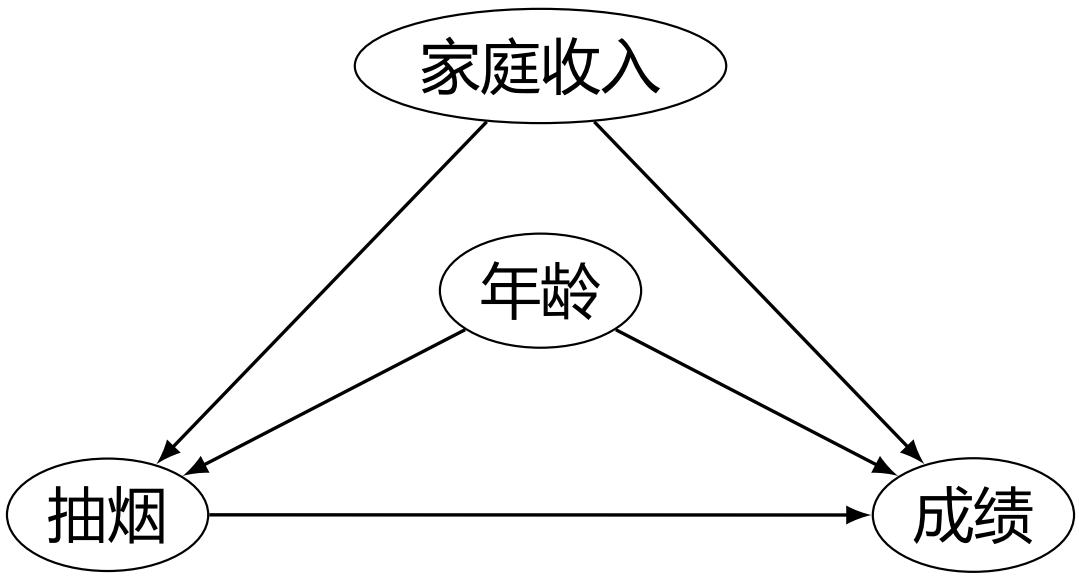




- 怎么判断饮食对体重的影响? $P(Y \mid \mathrm{do}(D)) = \sum_{W_I} P(Y \mid D, W_I) P(W_I)$
- 统计学家 1: 食谱 A/B 对学生一学期平均增重无差别.
  $P(Y \mid D = A) = P(Y \mid D = B)$
- 统计学家 2: 对于每一组学期初体重相同的同学来说, 食谱 B 的期末平均体重都大于食谱 A 的期末平均体重. ✓

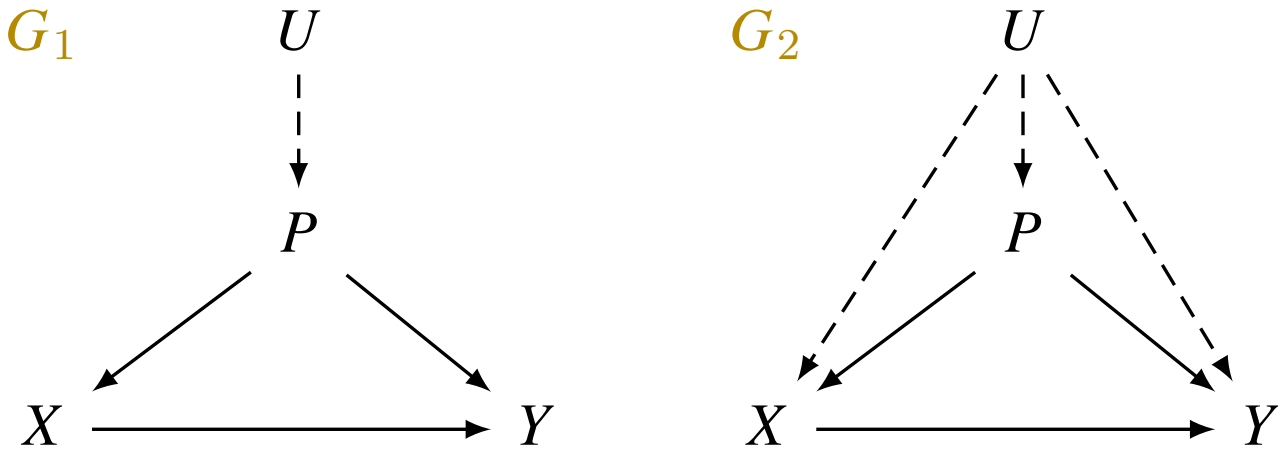
Simpson's Paradox — Every statistical relationship between two variables may be reversed by including additional factors in the analysis

1. 抽烟的同学成绩好.
2. 如果控制 "年龄" 变量, 每个 "年龄" 段的同学都是抽烟的成绩差.
3. 如果再控制 "家庭收入" 变量, 每个 "年龄"-"家庭收入" 组内都是抽烟的同学成绩好.
4. ...



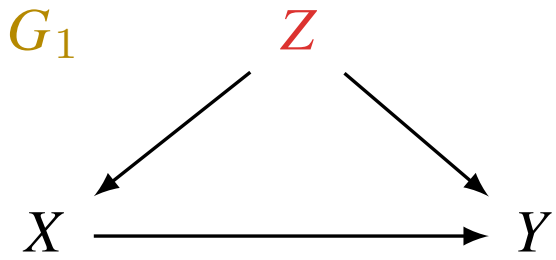
Proxy Variable (代理变量)

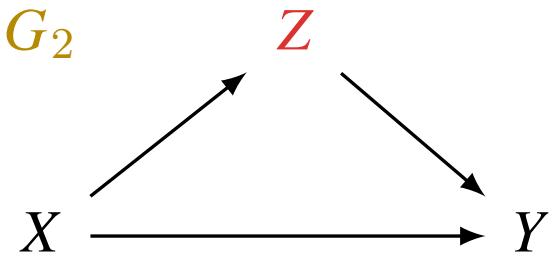
- Conditioning on the covariates should close all non-causal paths that transmit spurious association while leaving causal paths open.
- However, it is unlikely that a single proxy could perfectly measure a latent confounder.

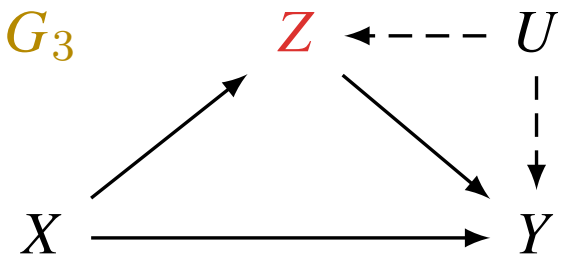


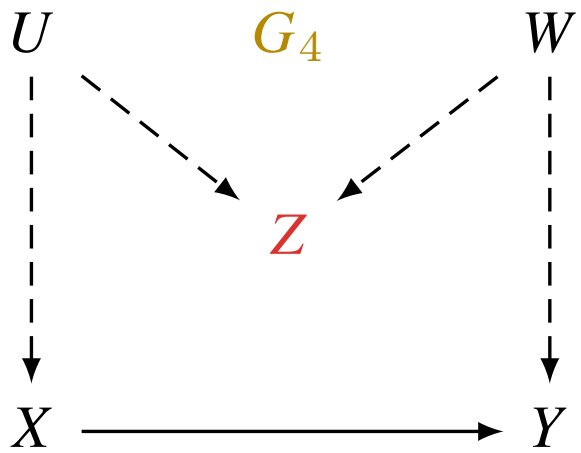
- In G_1, the proxy P captures all aspects of U that confound X and Y.
- In G_2, $X \longleftarrow P \longrightarrow Y$ and $X \dashleftarrow U \dashrightarrow P \longrightarrow Y$ are closed if we control for P, but $X \dashleftarrow U \dashrightarrow Y$ is not entirely closed.

Example: control for Z or not?



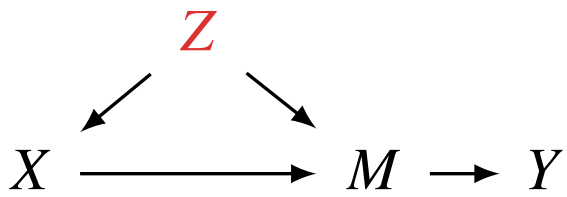


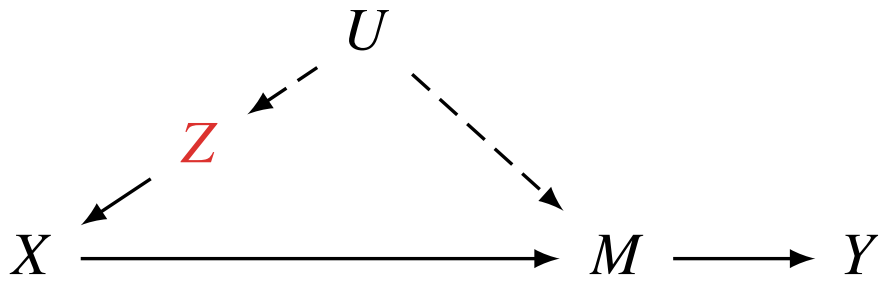




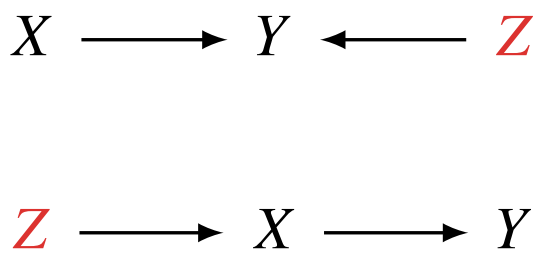
好的控制 vs 坏的控制 [CFP22]

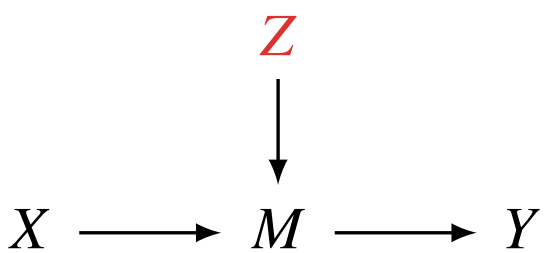
- Good

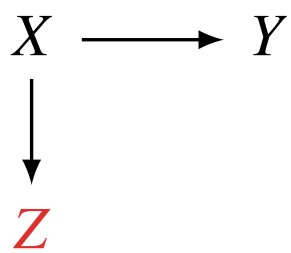




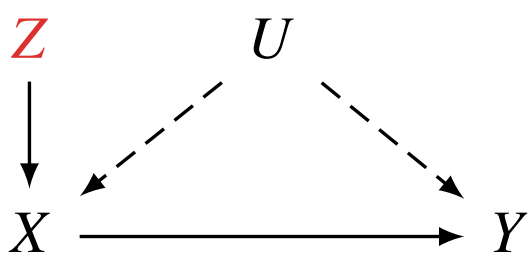
- Neutral

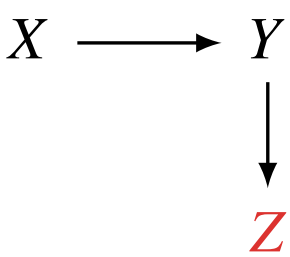


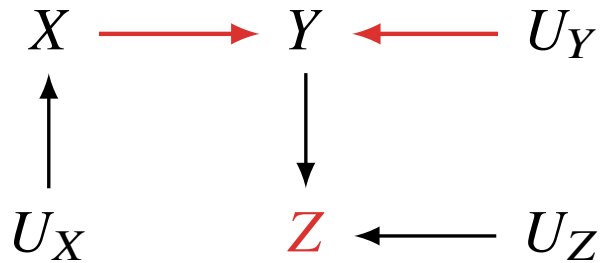




- Bad

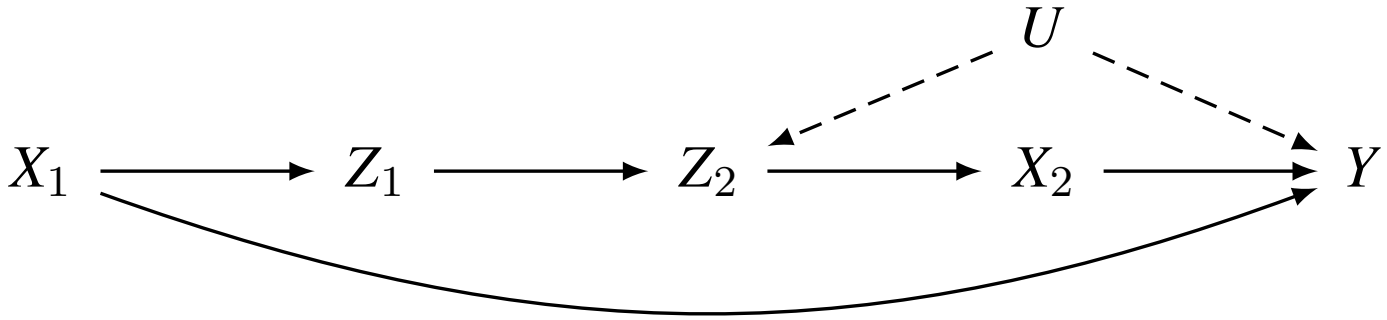




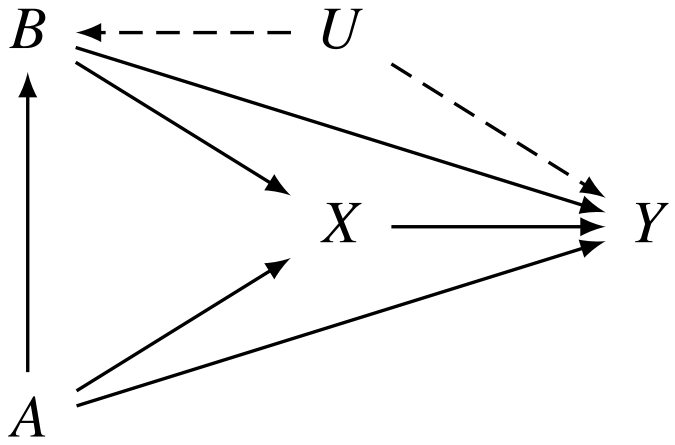


Remark: 左下: 控制 Z 会消除 Z 对 X 的外生影响, 放大 U 的混杂效应

Examples



- We need to control for $\{Z_1, Z_2\}$ to estimate the total/direct effect of $\{X_1, X_2\}$ on Y.
- We need to control for Z_1 to estimate the direct effect of X_1 on Y.
- We don't need to control for anything to estimate the total effect of X_1 on Y.



- We need to control for $\{A, B\}$ to estimate the total/direct effect of X on Y.

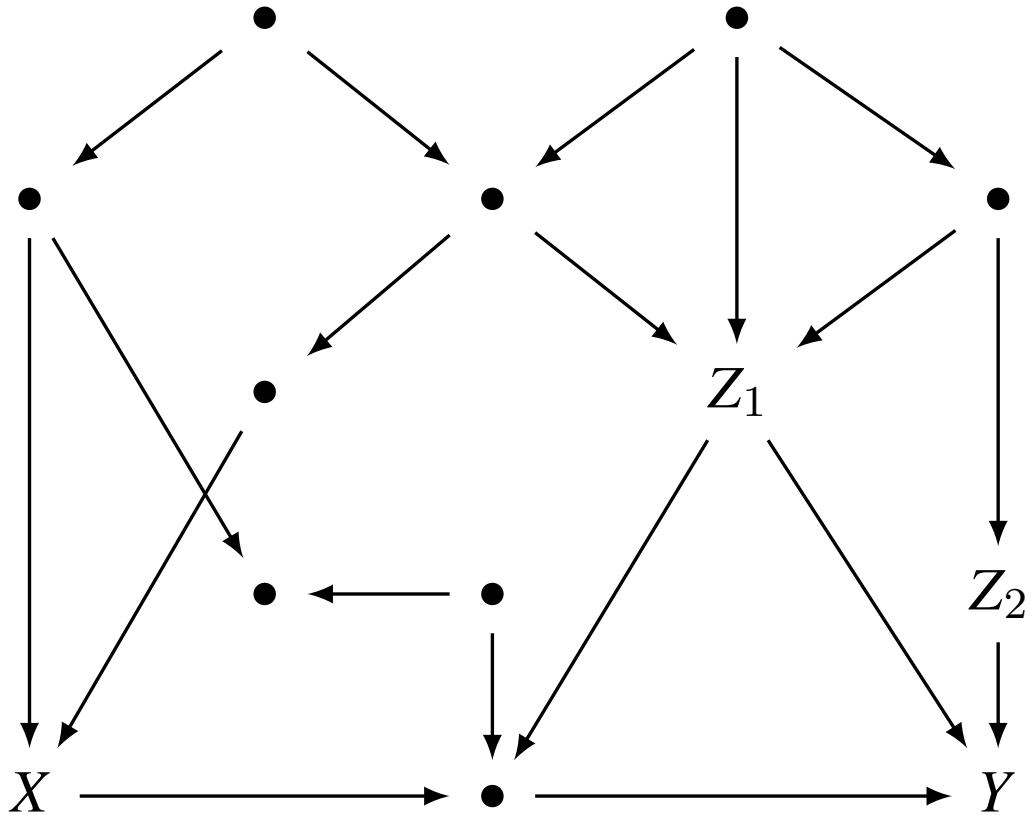
校正问题

The Adjustment Problem: Given a causal graph, we want to test the effect of X on Y. How to select a set of variables for measurement and adjustment?

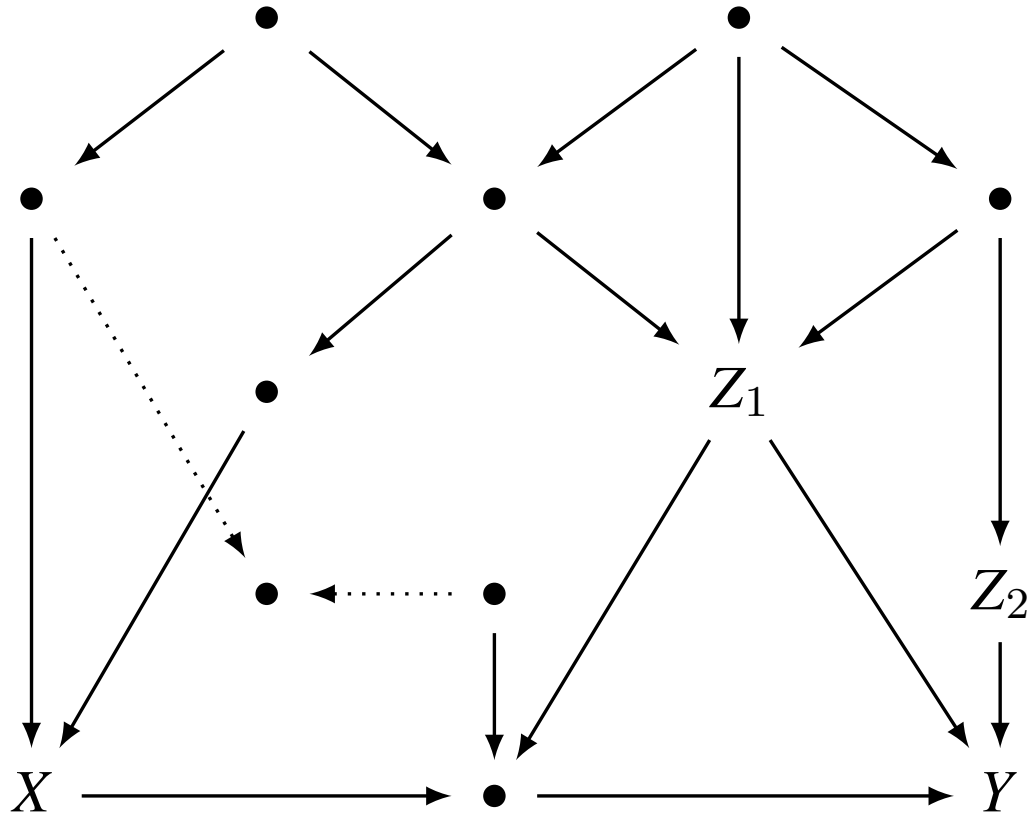
Subproblem:

Test if variables Z are sufficient measurements

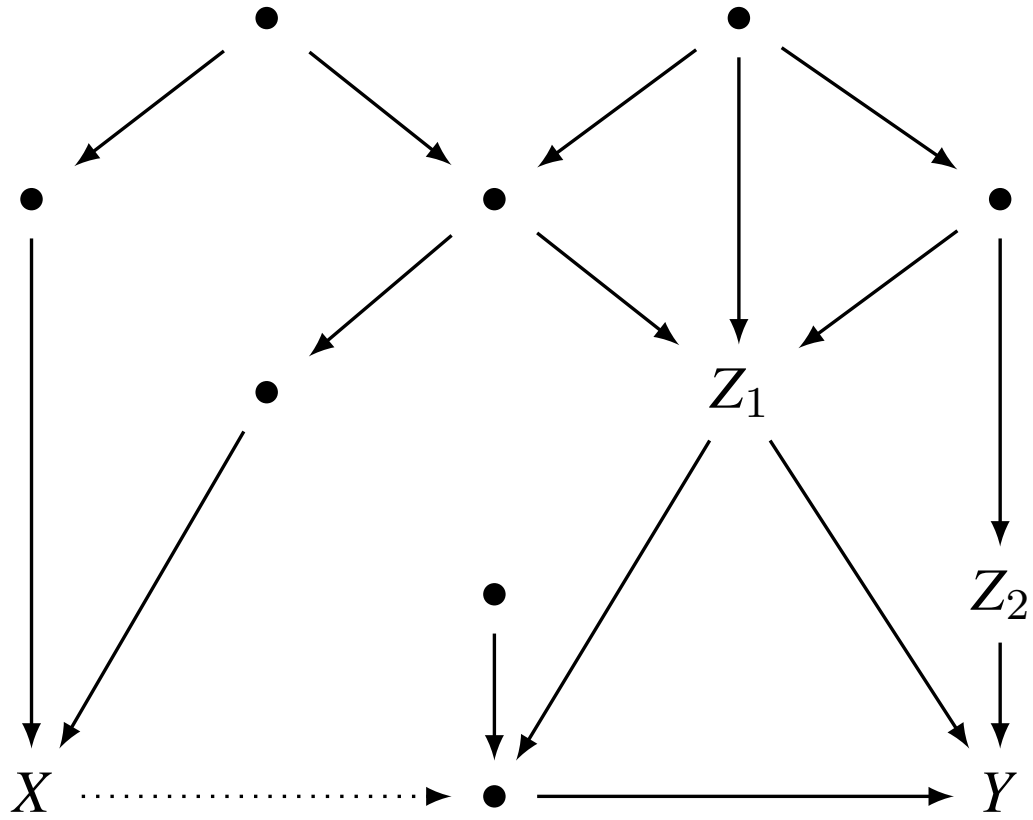
1. Z should not be descendants of X
2. Delete all non-ancestors of X, Y, Z
3. Delete all arcs emanating from X
4. Connect any two parents sharing a common child
5. Strip arrow-heads from all edges
6. Delete Z
7. Test if X is disconnected from Y in the remaining graph, then Z are appropriate measurements



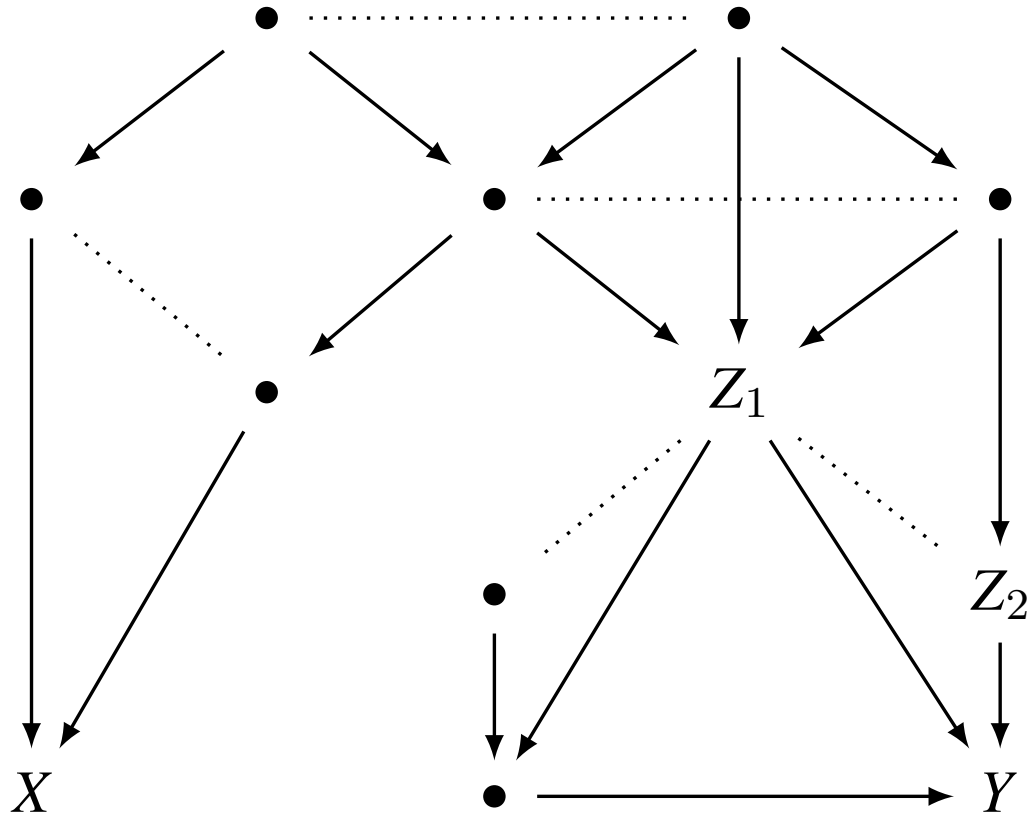
1. Z should not be descendants of X



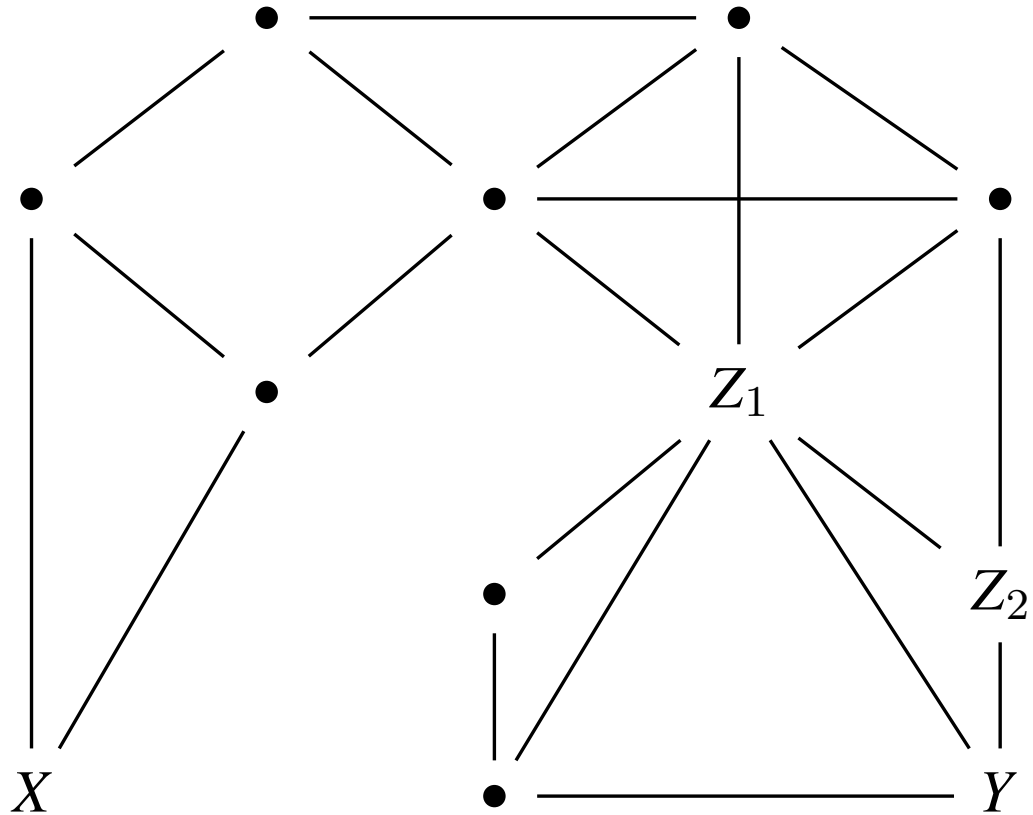
2. Delete all non-ancestors of X, Y, Z



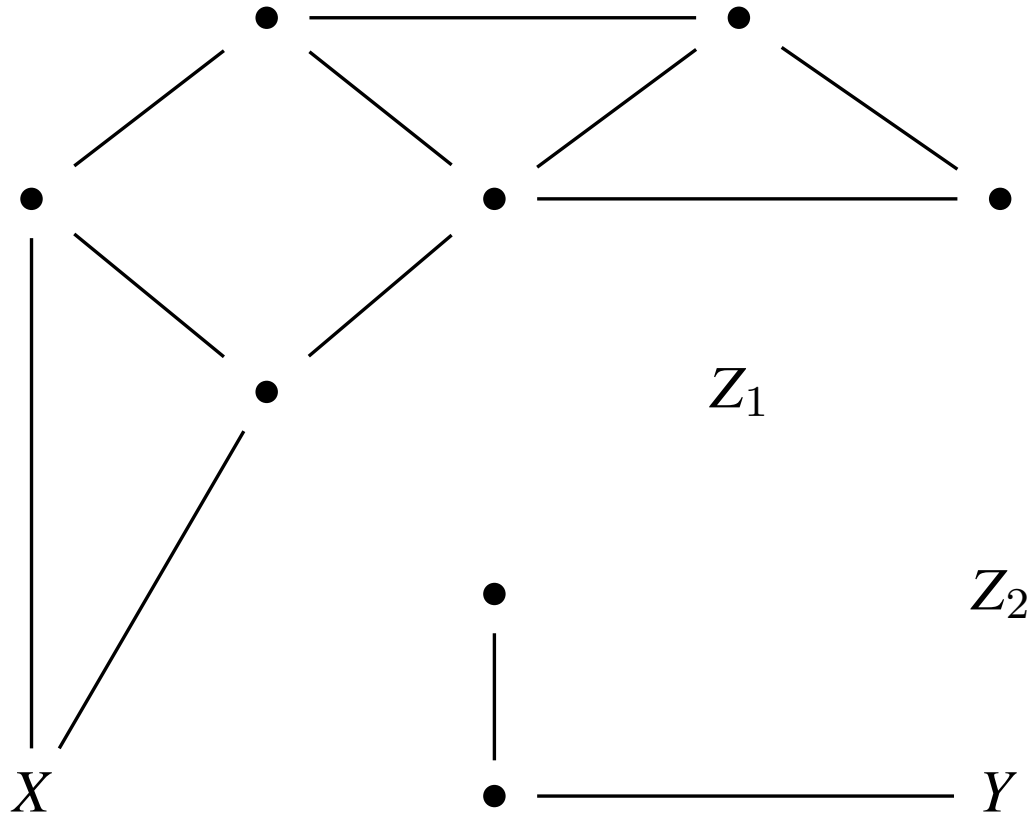
3. Delete all arcs emanating from X



4. Connect any two parents sharing a common child



5. Strip arrow-heads from all edges



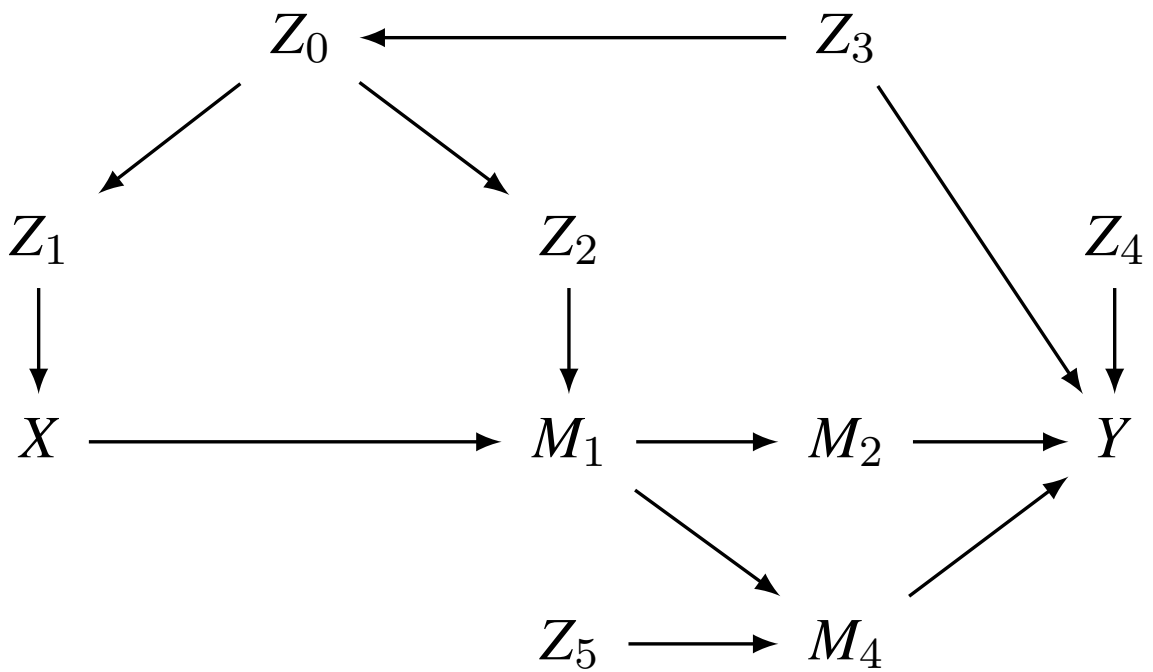
6. Delete Z

最优校正集

- 最优校正集: 估计因果效应方差最小的校正集.
- The optimal adjustment set for the causal effect of X on Y is given by

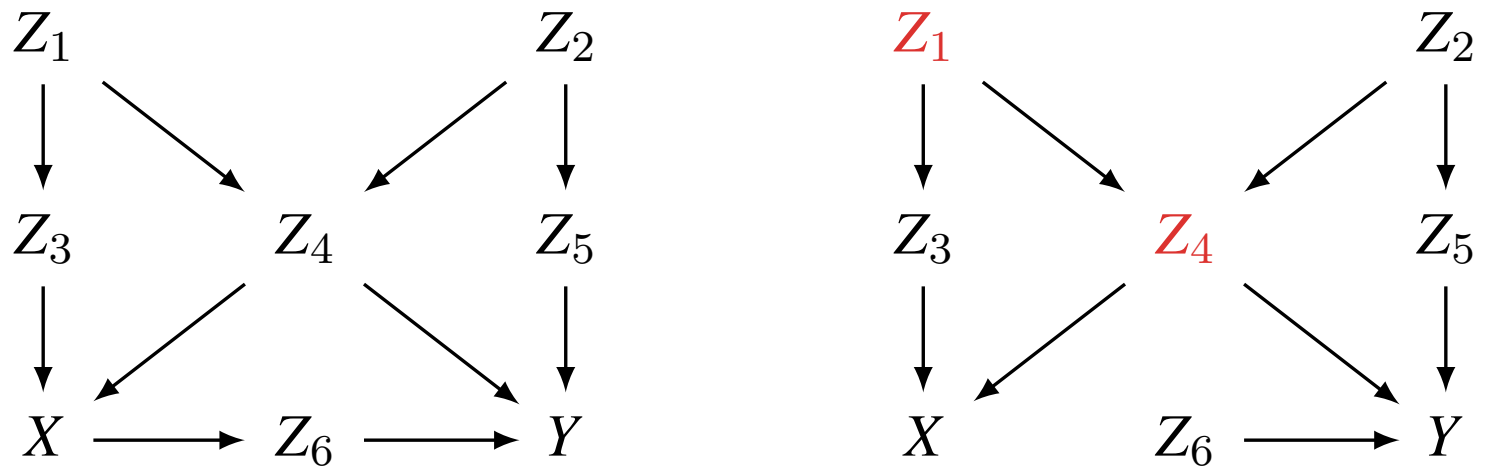
$$\mathrm{Pa}(\mathrm{Cn}(X,Y)) \setminus (\mathrm{Cn}(X,Y) \cup \{X\})$$

  其中, $\mathrm{Cn}(X,Y)$ 是处在 X 到 Y 的有向路径上的除了 X 之外的所有节点的集合.
- Example: $\{Z_2, Z_3, Z_4, Z_5\}$



Eliminating Confounding Bias — The Backdoor Criterion

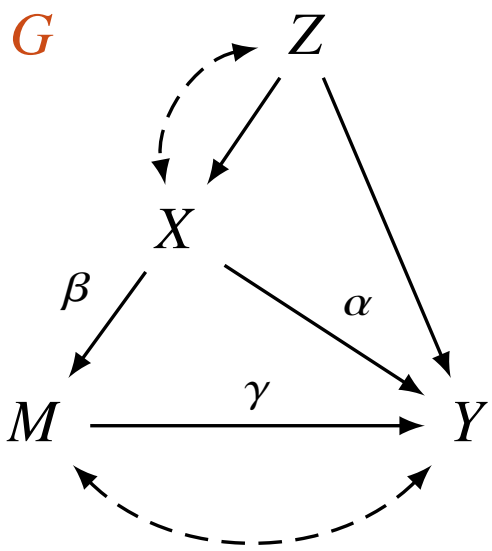
The Backdoor Criterion: $P(Y \mid \mathrm{do}(X))$ is estimable if there is a set Z containing no descendants of X s.t. $(Y \perp X \mid Z)_{G_{\underline{X}}}$. Moreover, the total effect of X on Y is given by the partial regression coefficient $r_{XY|Z}$.

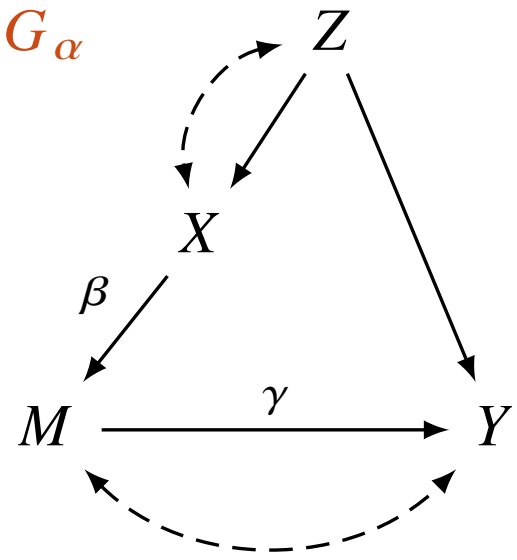


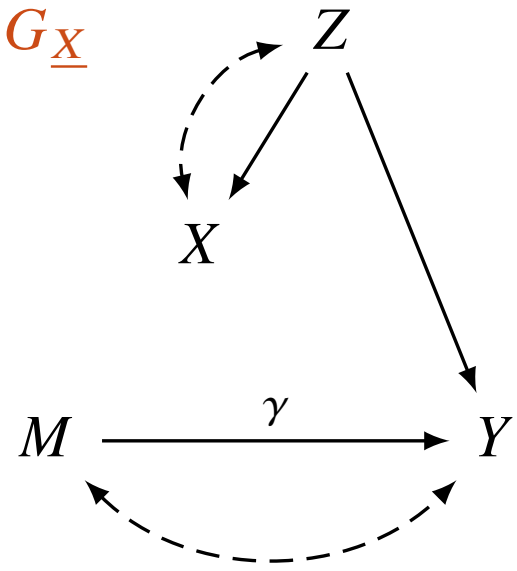
Example: $Z = \{Z_1, Z_4\}$

Backdoor Adjustment: $$P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z) P(z)$$

Parameter Identification with Backdoor Criterion







$$\alpha + \beta\gamma = r_{XY|Z}$$

Parameter Identification with Singledoor Criterion

Theorem (Single-Door Criterion for Direct Effects)

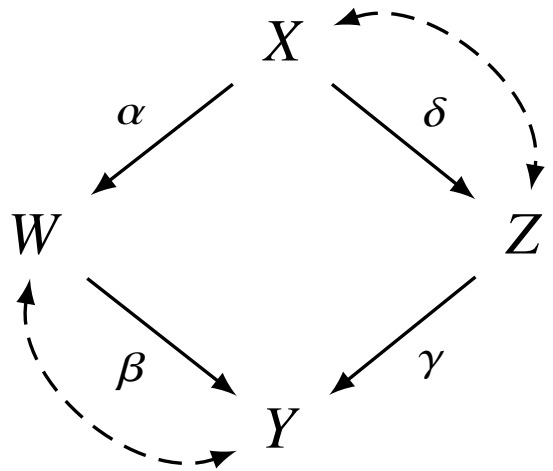
令 G_α 为从 G 中删除 $X \to Y$ 后得到的子图, 其中 α 为 $X \to Y$ 的路径系数. 如果存在一组变量 Z 使得

1. Z 不包含 Y 的任何后代;
2. $(X \perp Y \mid Z)_{G_\alpha}$.

则 α 是可识别的.
并且, $\alpha = r_{XY|Z}$.
反之, 如果 Z 不满足上述条件, 则 $r_{XY|Z}$ 就不是 α 的一致估计 (除了测度为 0 的少数情况).

$$\alpha = r_{XY|Z}$$

Indirect Identification of Structural Parameters



- X 对 Y 的总效应 $\frac{\partial \mathbb{E}[Y|\mathrm{do}(x)]}{\partial x} = \alpha\beta + \delta\gamma$, 但它不可识别.
- $\alpha = r_{XW}$.
- 因为 Z 阻断了所有经过 Z 的路径, 所以 $\alpha\beta = r_{XY|Z}$.
- 所以 $\beta = \frac{r_{XY|Z}}{r_{XW}}$.
- 因为 X 阻断了从 Z 到 Y 的后门路径, 所以 $\gamma = r_{ZY|X}$.

The Adjustment Criterion

The backdoor criterion can be generalized to the adjustment criterion.

Definition (Adjustment Criterion)

A set of variables Z satisfies the adjustment criterion relative to (X, Y) in a DAG G if:

- In $G_{\overline{X}}$, Z contains no descendant of any $W \notin X$ which lies on a proper causal path from X to Y.
- Z blocks all non-causal paths in G from X to Y.

where a causal path is a directed path, and a path from X to Y is proper iff only its first node is in X.

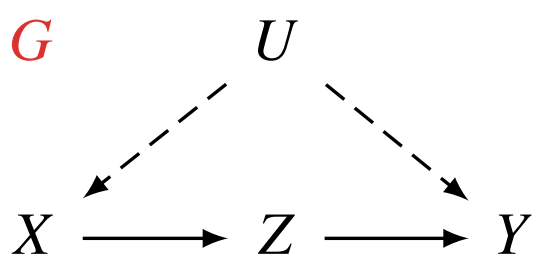
Theorem

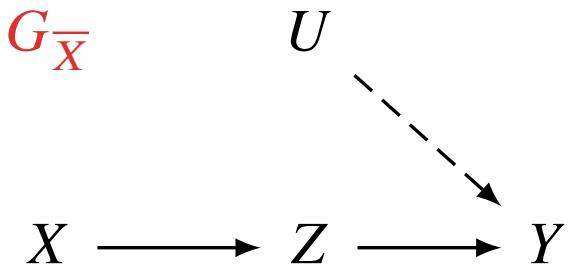
If Z satisfies the adjustment criterion, then $Y_x \perp X \mid Z$.

Theorem

$P(y \mid \text{do}(x)) = \sum_z P(y \mid x, z)P(z)$ in every model inducing G, iff, $Y_x \perp X \mid Z$ holds in every model inducing G.

前门校正 Frontdoor Adjustment



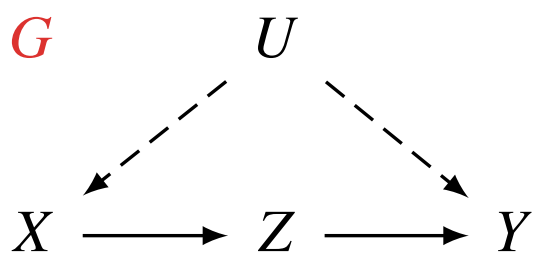


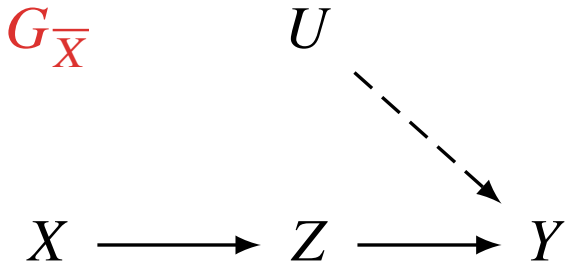
- $P(z \mid \mathrm{do}(x)) = P(z \mid x)$
- since the backdoor path from Z to Y, namely $Z \leftarrow X \leftarrow U \rightarrow Y$, can be blocked by conditioning on X,

$$P(y \mid \mathrm{do}(z)) = \sum_x P(y \mid z, x) P(x)$$

$$\begin{aligned} P(y \mid \mathrm{do}(x)) &= \sum_z P(z \mid \mathrm{do}(x)) P(y \mid \mathrm{do}(z)) \\ &= \sum_z P(z \mid x) P(y \mid \mathrm{do}(z)) \\ &= \sum_z P(z \mid x) \sum_{x'} P(y \mid z, x') P(x') \end{aligned}$$

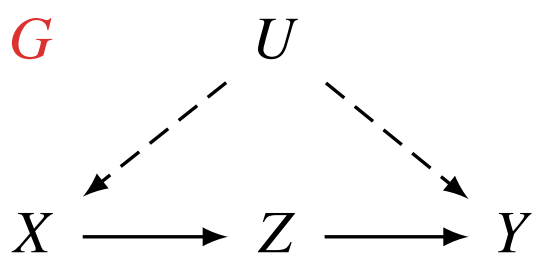
前门校正 Frontdoor Adjustment

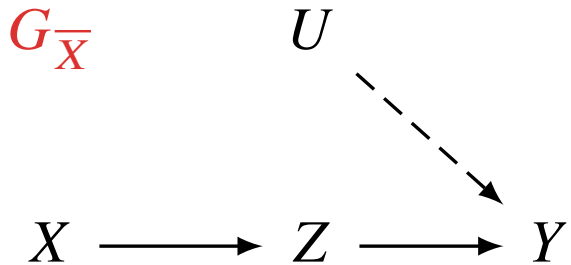




$$
\begin{aligned}
P(y \mid \mathtt{do}(x)) &= \sum_{z,u} P(z \mid x) P(y \mid z, u) P(u) \\
&= \sum_{z} P(z \mid x) \sum_{u} P(y \mid z, u) \sum_{x'} P(u \mid x') P(x') \\
&= \sum_{z} P(z \mid x) \sum_{u,x'} P(y \mid z, u, x') P(u \mid x') P(x') \quad (Y \perp X \mid Z, U) \\
&= \sum_{z} P(z \mid x) \sum_{u,x'} P(y \mid z, u, x') P(u \mid z, x') P(x') \quad (U \perp Z \mid X) \\
&= \sum_{z} P(z \mid x) \sum_{u,x'} P(y, u \mid z, x') P(x') \\
&= \sum_{z} P(z \mid x) \sum_{x'} P(y \mid z, x') P(x')
\end{aligned}
$$

前门校正 Frontdoor Adjustment





前门准则

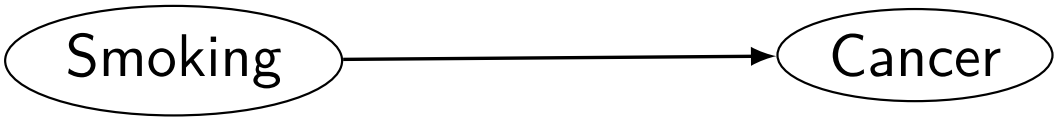
在有向无环图 G 中, 如果变量集 Z 相对于变量 (X, Y) 满足以下条件, 则称其满足前门准则:

1. 所有从 X 到 Y 的有向路径都经过 Z;
2. 从 X 到 Z 不存在未被阻断的后门路径;
3. 从 Z 到 Y 的所有后门路径都被 X 所阻断.

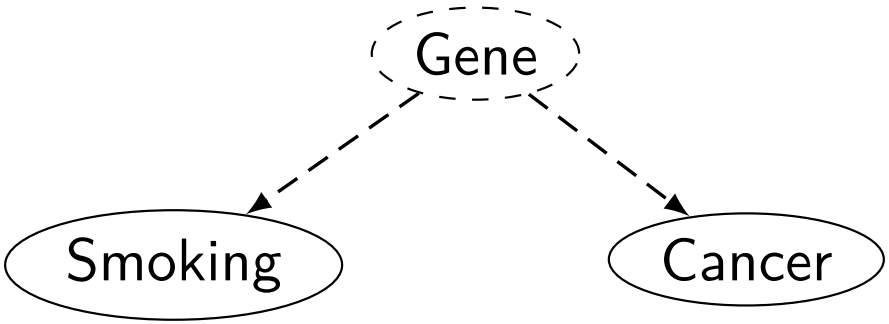
前门校正:

$$P(y \mid \mathrm{do}(x)) = \sum_{z} P(z \mid x) \sum_{x'} P(y \mid z, x') P(x')$$

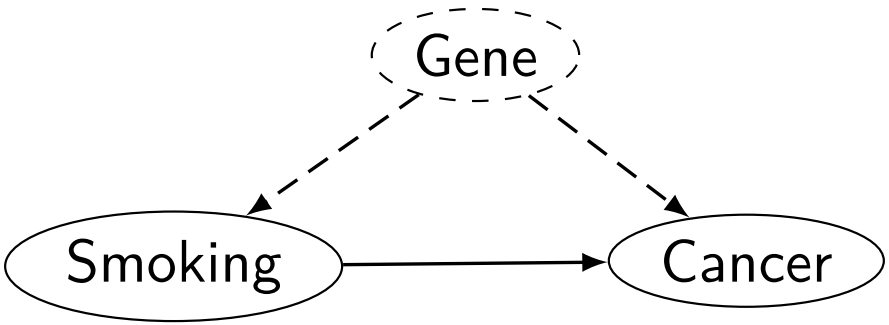
Does Smoking Cause Cancer?



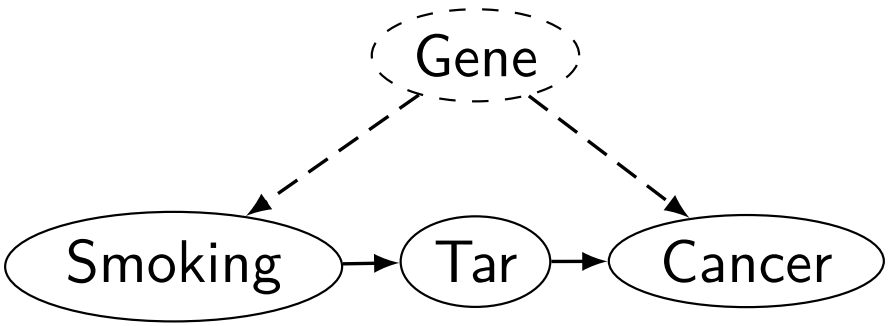
$P(c \mid \mathrm{do}(s)) \approx P(c \mid s)$



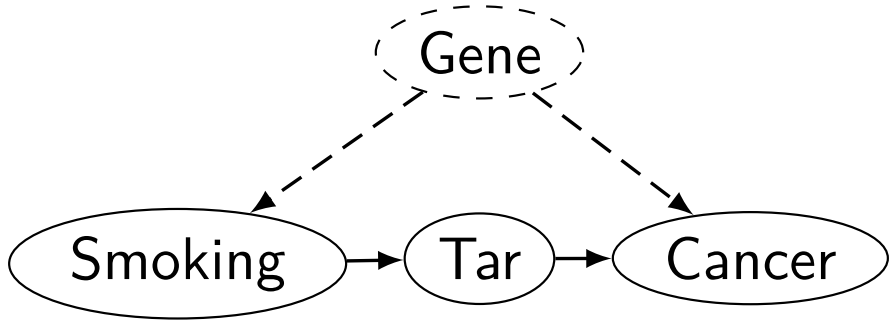
$P(c \mid \mathrm{do}(s)) = P(c)$



$P(c \mid \mathrm{do}(s)) = \text{noncomputable}$



$P(c \mid \mathrm{do}(s)) = \text{computable}$



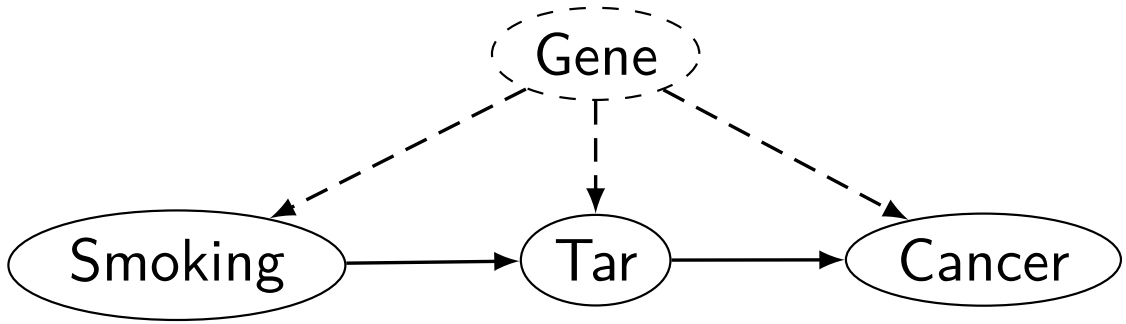
- Smoking leads to Cancer only through the accumulation of Tar deposits.
- The smoking Gene has no effect on the formation of Tar deposits.
- We can estimate the causal effect of Smoking on Tar, because there is no unblocked backdoor path from Smoking to Tar, as the Smoking ← Gene → Cancer ← Tar path is already blocked by the collider at Cancer.

$$P(t \mid \text{do}(s)) = P(t \mid s)$$

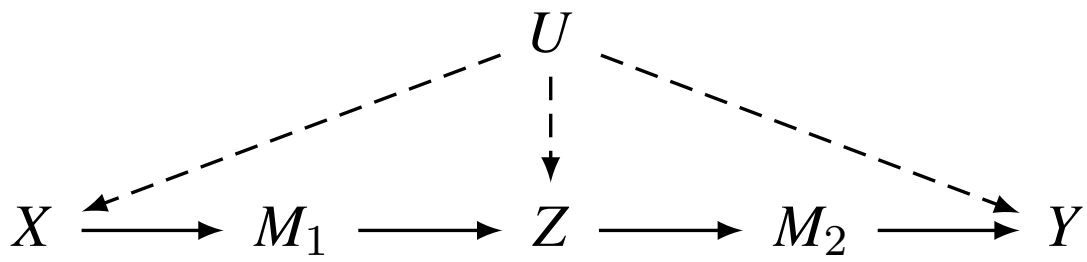
- We can estimate the causal effect of Tar on Cancer, because we can block the backdoor path from Tar to Cancer, Tar ← Smoking ← Gene → Cancer, by controlling for Smoking.

$$P(c \mid \text{do}(t)) = \sum_{s} P(c \mid t, s)P(s)$$

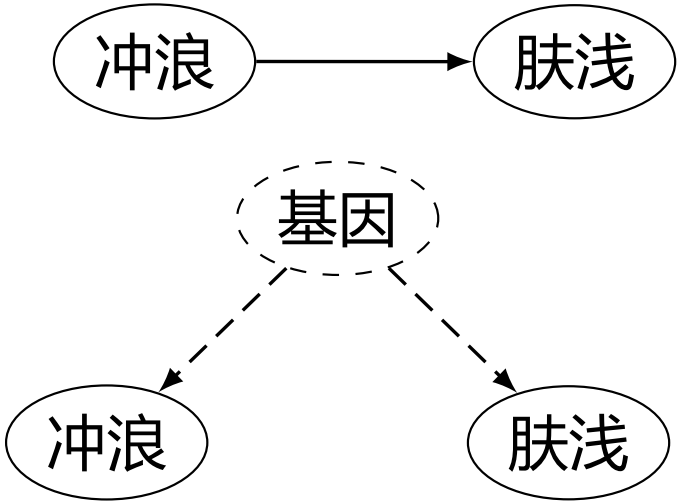
- It allows us to control for confounders that we cannot observe, including those that we can't even name.
- However, if we draw an arrow from Gene to Tar,

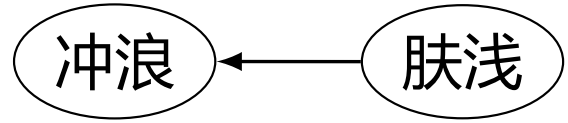


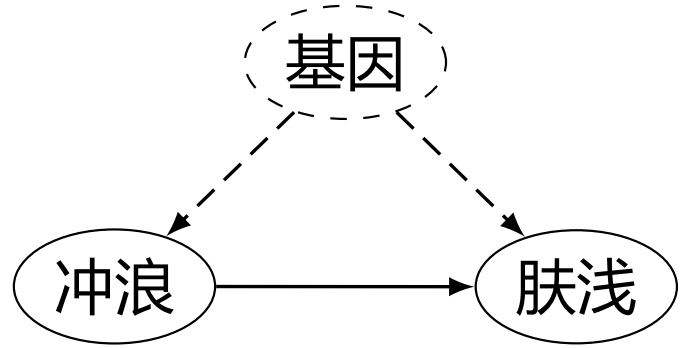
  then the frontdoor formula is invalid.
- The sub-mechanisms $X \to M_1 \to Z$ and $Z \to M_2 \to Y$ are isolated, and the original causal effect can be identified by composing them.

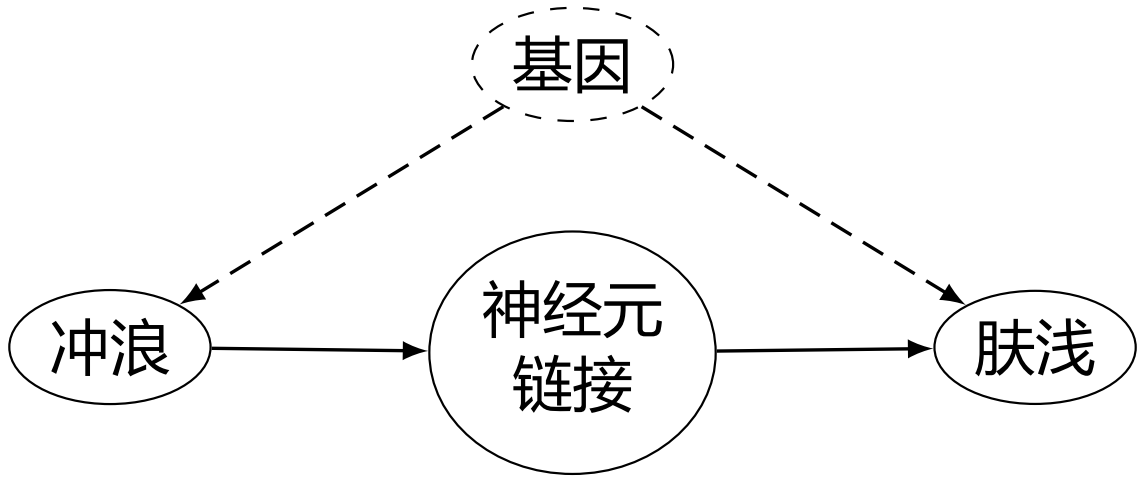


$$P(y \mid \mathrm{do}(x)) = \sum_z P(z \mid \mathrm{do}(x)) P(y \mid \mathrm{do}(z))$$









To control for the confounders we

- need to know what the confounders are
- need to be able to measure them

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal
   - Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

do-Calculus: Special Case

R1. **Insertion/Deletion of Observations:** If W blocks all paths from Z to Y after we have deleted all arrows leading into X, then

$$P(y \mid \mathrm{do}(x), z, w) = P(y \mid \mathrm{do}(x), w) \quad \text{if } (Y \perp Z \mid X, W)_{G_{\overline{X}}}$$

Remark: Markov 下 d-分离的推广 $(Y \perp Z \mid W)_G \implies (Y \perp Z \mid W)_P$.

R2. **Action/Observation Exchange:** If Z blocks all backdoor paths from X to Y, then

$$P(y \mid \mathrm{do}(x), z) = P(y \mid x, z) \quad \text{if } (Y \perp X \mid Z)_{G_{\underline{X}}}$$

Remark: 后门校正的推广, Z 可以包含 X 的后代.

R3. **Insertion/Deletion of Actions:** If X has no causal paths to Y, then

$$P(y \mid \mathrm{do}(x)) = P(y) \quad \text{if } (Y \perp X)_{G_{\overline{X}}}$$

Remark: d-分离 + 后门校正 + 干预只影响因果后代

do-Calculus for Hard Interventions

R1. **Insertion/Deletion of Observations:**

$$P(y \mid \mathrm{do}(x), z, w) = P(y \mid \mathrm{do}(x), w) \quad \text{if } (Y \perp Z \mid X, W)_{G_{\overline{X}}}$$

R2. **Action/Observation Exchange:**

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), z, w) \quad \text{if } (Y \perp Z \mid X, W)_{G_{\overline{X},\underline{Z}}}$$

R3. **Insertion/Deletion of Actions:**

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), w) \quad \text{if } (Y \perp Z \mid X, W)_{G_{\overline{X},\overline{Z(W)}}}$$

where $Z(W)$ is the set of Z-nodes that are not ancestors of any W-node in $G_{\overline{X}}$.

Remark: 之所以需要 $Z(W)$ 条件, 是为了防止扩展图中 Z 包含对撞节点, 而 W 又包含对撞节点的后代.

决策论因果视角

R1. **Insertion/Deletion of Observations:**

$$P(y \mid \mathrm{do}(x), z, w) = P(y \mid \mathrm{do}(x), w) \quad \text{if } (Y \perp Z \mid X, W)_{G_{\overline{X}}}$$

$$Y \perp Z \mid \sigma_X = x, W \qquad (Y \perp Z \mid X, W)_{\mathcal{G}_{\overline{X}}}$$

R2. **Action/Observation Exchange:**

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), z, w) \quad \text{if } (Y \perp Z \mid X, W)_{G_{\overline{X},\underline{Z}}}$$

$$Y \perp \sigma_Z \mid \sigma_X = x, W, Z \qquad (Y \perp \sigma_Z \mid X, W, Z)_{\mathcal{G}_{\overline{X}}}$$

R3. **Insertion/Deletion of Actions:**

$$P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), w) \quad \text{if } (Y \perp Z \mid X, W)_{G_{\overline{X},\overline{Z(W)}}}$$

$$Y \perp \sigma_Z \mid \sigma_X = x, W \qquad (Y \perp \sigma_Z \mid X, W)_{\mathcal{G}_{\overline{X}}}$$

其中, $\mathcal{G}$ 表示扩展图.

Potential Outcome po-Calculus via SWIG

We can use SWIGs to formulate counterfactual version of do-Calculus.

R1. If $(Y_x \perp Z_x \mid W_x)_{G(x)}$ then

$$P(Y_x \mid Z_x, W_x) = P(Y_x \mid W_x)$$

R2. If $(Y_{x,z} \perp Z_{x,z} \mid W_{x,z})_{G(x,z)}$ then

$$P(Y_{x,z} \mid W_{x,z}) = P(Y_x \mid W_x, Z_x = z)$$

R3. If $(Y_{x,z} \perp z)_{G(x,z)}$ then

$$P(Y_{x,z}) = P(Y_x)$$

Frontdoor Adjustment via po-Calculus

$$P(Y_x) = \sum_z P(Y_x \mid Z_x = z) P(Z_x = z)$$

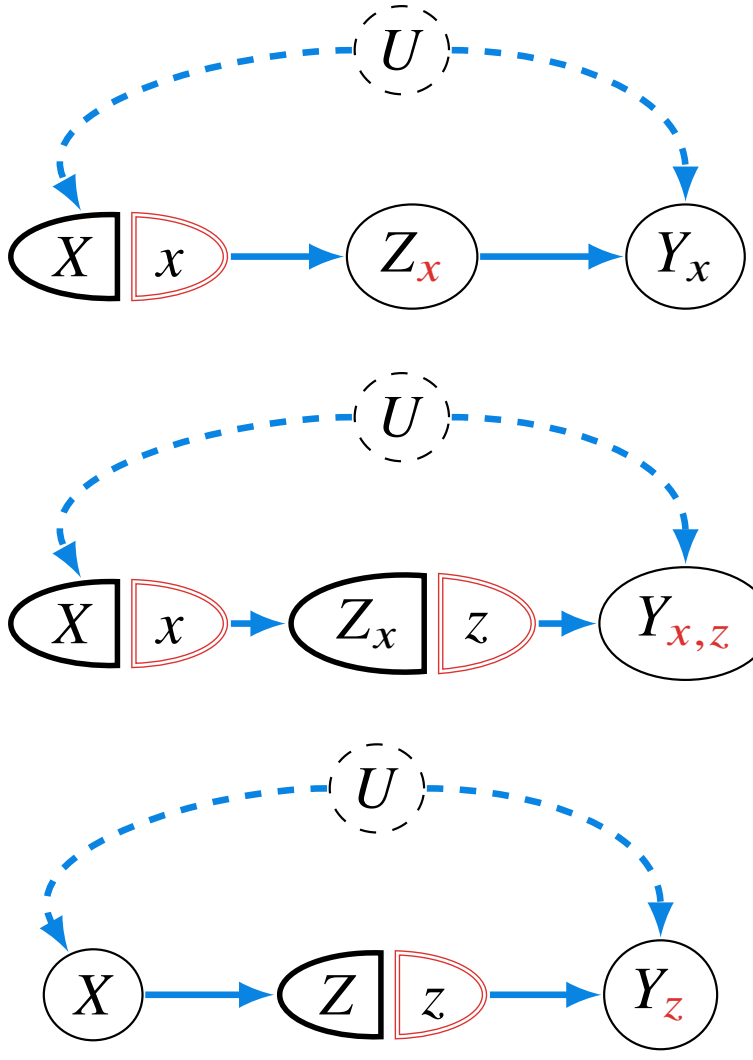
$$\overset{R2, G(x)}{=} \sum_z P(Y_x \mid Z_x = z) P(Z = z \mid X = x)$$

$$\overset{R2, G(x,z)}{=} \sum_z P(Y_{x,z}) P(z \mid x)$$

$$\overset{R3, G(x,z)}{=} \sum_z P(Y_z) P(z \mid x)$$

$$= \sum_z P(z \mid x) \sum_{x'} P(Y_z \mid x') P(x')$$

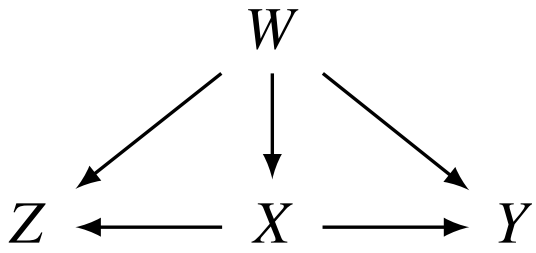
$$\overset{R2, G(z)}{=} \sum_z P(z \mid x) \sum_{x'} P(Y \mid z, x') P(x')$$

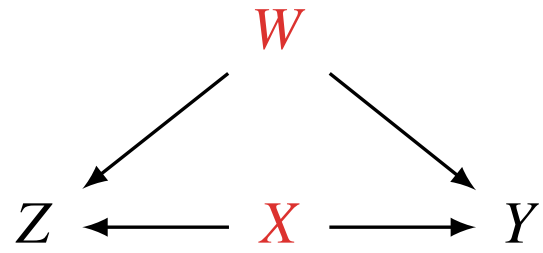


Remarks

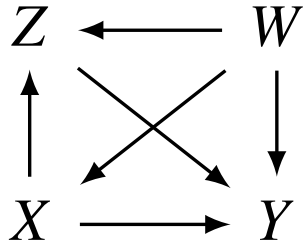
Each rule first applies the intervention to the treatment resulting in $G_{\overline{X}}$.

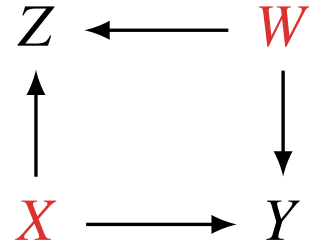
1. Generalization of d-separation. Add/remove any variables that are d-separated in $G_{\overline{X}}$.



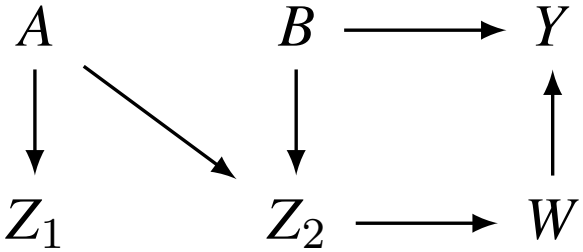


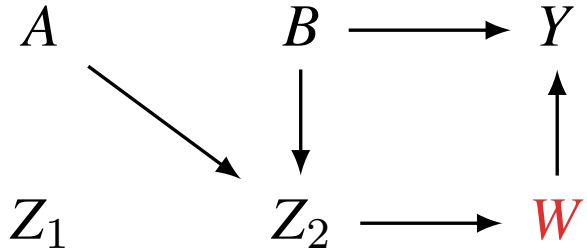
2. Generalization of backdoor criterion. Exchange conditioning with interventions whenever X, W block all backdoor paths in $G_{\overline{X},\underline{Z}}$.



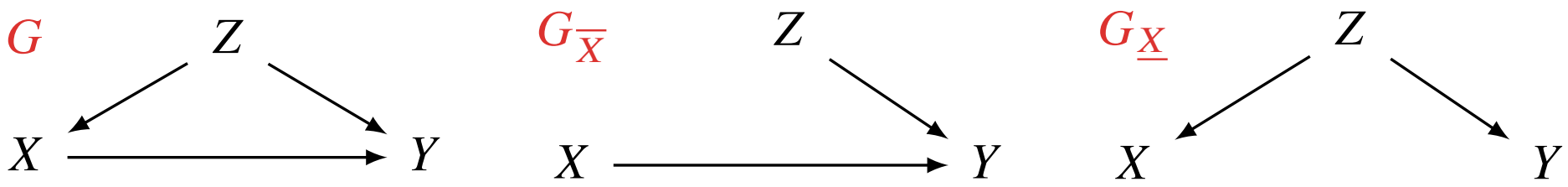


3. Why do we need $\overline{Z(W)}$?





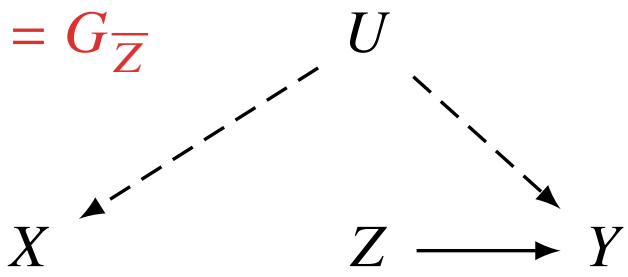
Example: Derivation of backdoor adjustment



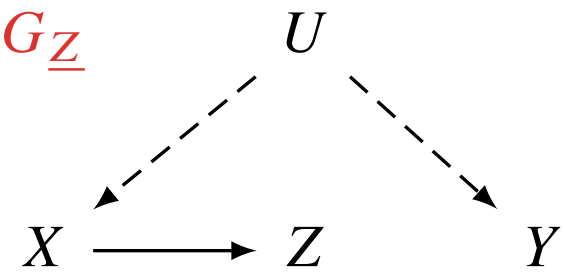
$$
\begin{aligned}
P(y \mid \mathrm{do}(x)) &= \sum_z P(z \mid \mathrm{do}(x)) P(y \mid \mathrm{do}(x), z) && \text{(Probability)} \\
&= \sum_z P(z) P(y \mid \mathrm{do}(x), z) && \text{(Rule3: } (Z \perp X)_{G_{\overline{X}}}\text{)} \\
&= \sum_z P(z) P(y \mid x, z) && \text{(Rule2: } (Y \perp X \mid Z)_{G_{\underline{X}}}\text{)}
\end{aligned}
$$

Example

$G_{\underline{X}} = G_{\overline{Z}}$



$G_{\underline{Z}}$



$$P(z \mid \mathtt{do}(x)) = P(z \mid x) \qquad (\text{Rule2 } (Z \perp X)_{G_{\underline{X}}})$$

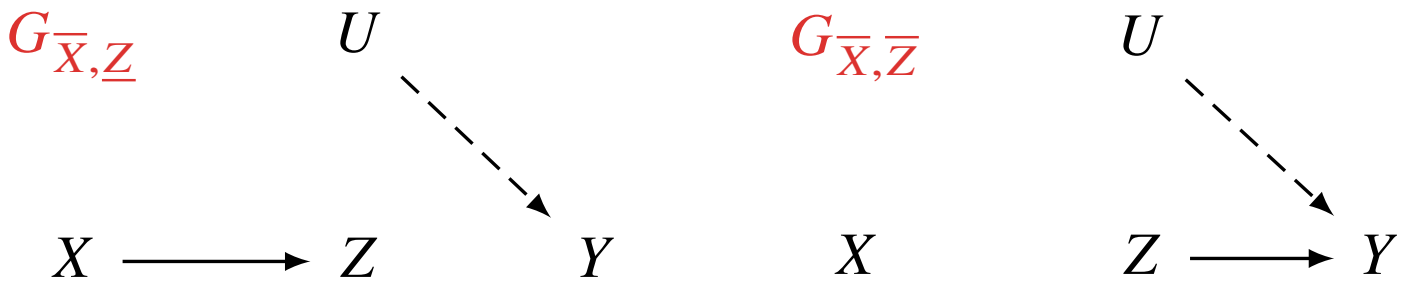
$$P(x \mid \mathtt{do}(z)) = P(x) \qquad (\text{Rule3 } (X \perp Z)_{G_{\overline{Z}}})$$

$$P(y \mid x, \mathtt{do}(z)) = P(y \mid x, z) \qquad (\text{Rule2 } (Y \perp Z \mid X)_{G_{\underline{Z}}})$$

$$P(x, y \mid \mathtt{do}(z)) = P(y \mid x, \mathtt{do}(z))P(x \mid \mathtt{do}(z)) = P(y \mid x, z)P(x)$$

$$P(y \mid \mathtt{do}(z)) = \sum_x P(x, y \mid \mathtt{do}(z)) = \sum_x P(y \mid x, z)P(x)$$

Example



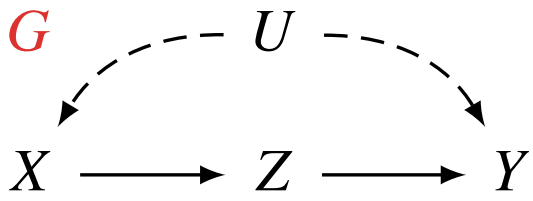
$$P(y \mid \mathrm{do}(x), z) = P(y \mid \mathrm{do}(x), \mathrm{do}(z)) \qquad (\text{Rule2 } (Y \perp Z \mid X)_{G_{\overline{X},\underline{Z}}})$$

$$= P(y \mid \mathrm{do}(z)) \qquad (\text{Rule3 } (Y \perp X \mid Z)_{G_{\overline{X},\overline{Z}}})$$

$$\begin{aligned} P(y, z \mid \mathrm{do}(x)) &= P(z \mid \mathrm{do}(x)) P(y \mid \mathrm{do}(x), z) \\ &= P(z \mid x) P(y \mid \mathrm{do}(z)) \\ &= P(z \mid x) \sum_{x'} P(y \mid x', z) P(x') \end{aligned}$$

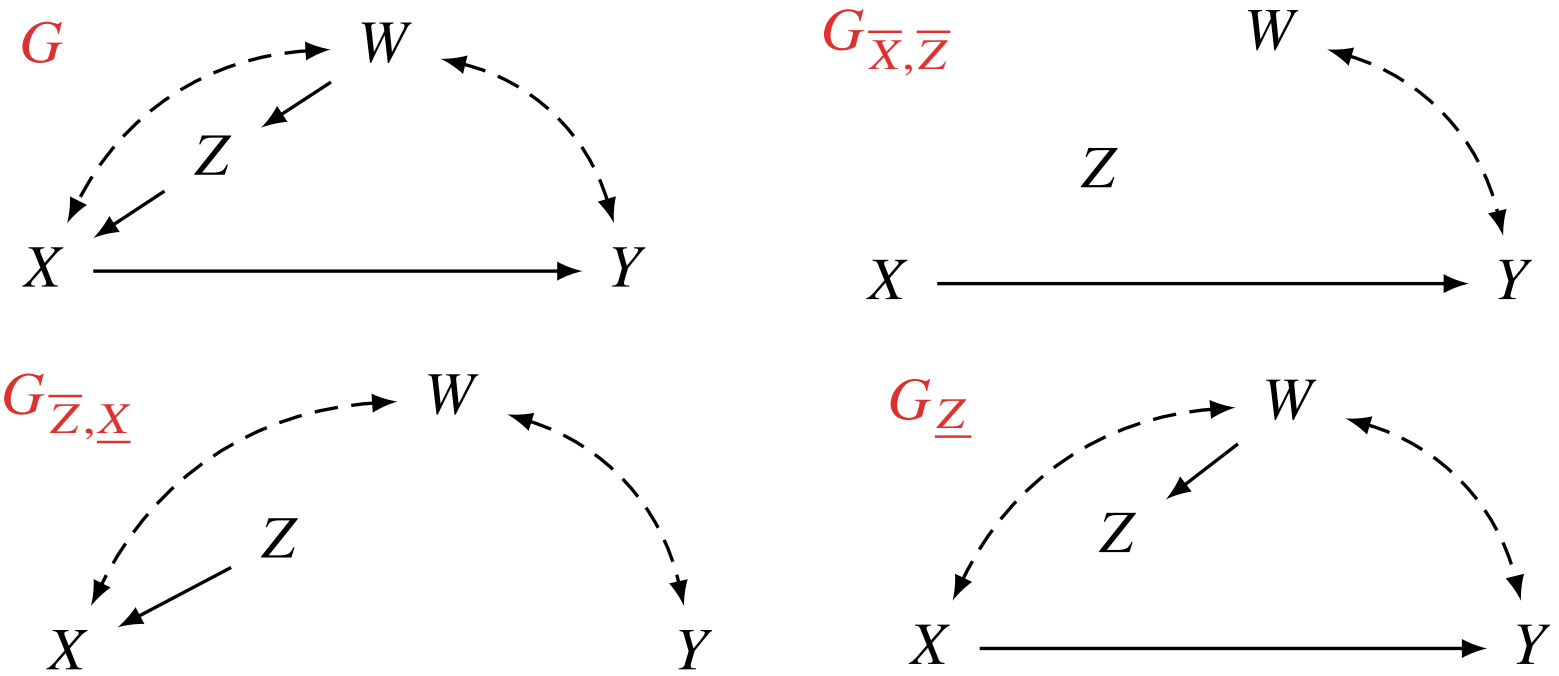
$$P(y \mid \mathrm{do}(x)) = \sum_{z} P(y, z \mid \mathrm{do}(x)) = \sum_{z} P(z \mid x) \sum_{x'} P(y \mid x', z) P(x')$$

Example: Derivation of frontdoor adjustment



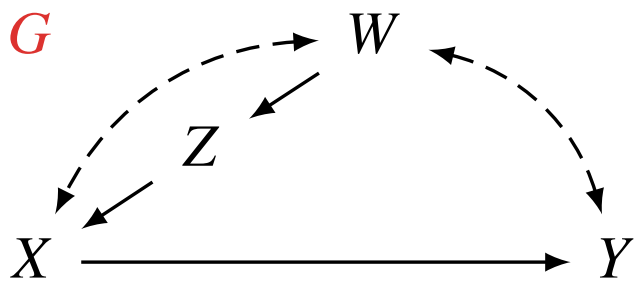
$$
\begin{aligned}
P(y \mid \mathtt{do}(x)) &= \sum_z P(z \mid \mathtt{do}(x)) P(y \mid \mathtt{do}(x), z) && \text{(Probability)} \\
&= \sum_z P(z \mid \mathtt{do}(x)) P(y \mid \mathtt{do}(x), \mathtt{do}(z)) && \text{(R2} \\
&= \sum_z P(z \mid x) P(y \mid \mathtt{do}(x), \mathtt{do}(z)) && \text{(R2} \\
&= \sum_z P(z \mid x) P(y \mid \mathtt{do}(z)) && \text{(R3} \\
&= \sum_z P(z \mid x) \sum_{x'} P(y \mid x', \mathtt{do}(z)) P(x' \mid \mathtt{do}(z)) && \text{(Probability)} \\
&= \sum_z P(z \mid x) \sum_{x'} P(y \mid x', z) P(x' \mid \mathtt{do}(z)) && \text{(R2} \\
&= \sum_z P(z \mid x) \sum_{x'} P(y \mid x', z) P(x') && \text{(R3}
\end{aligned}
$$

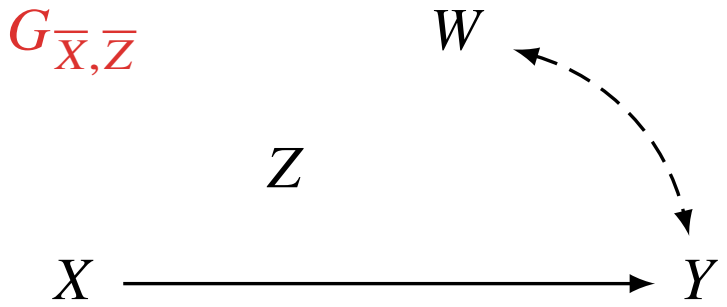
Example

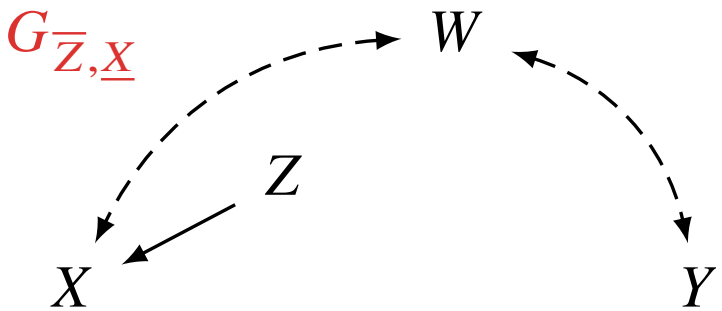


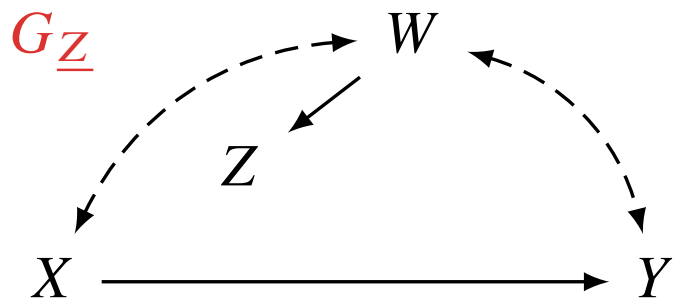
- Can we identify $P(Y \mid \mathrm{do}(X))$ from $P(W, Z, X, Y)$?
- No backdoor strategy can be used since the empty set is not admissible, and in $G_{\underline{X}}$, conditioning on $\{Z\}, \{W\}, \{Z, W\}$ leaves the backdoor path $X \dashleftarrow\dashrightarrow W \dashleftarrow\dashrightarrow Y$ opened.
- No frontdoor strategy can be used due to the direct arrow $X \longrightarrow Y$.

Example



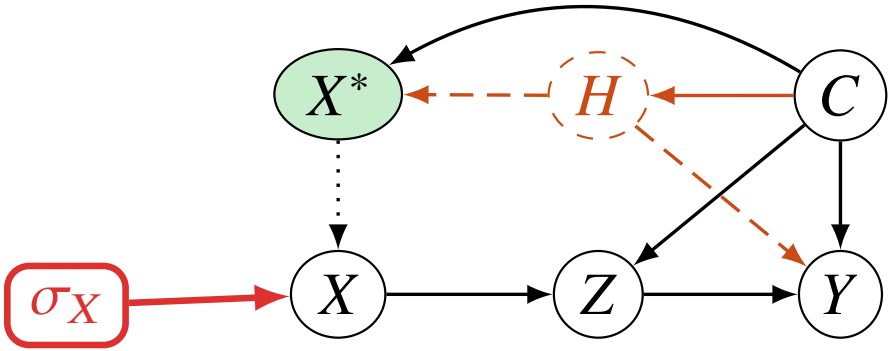


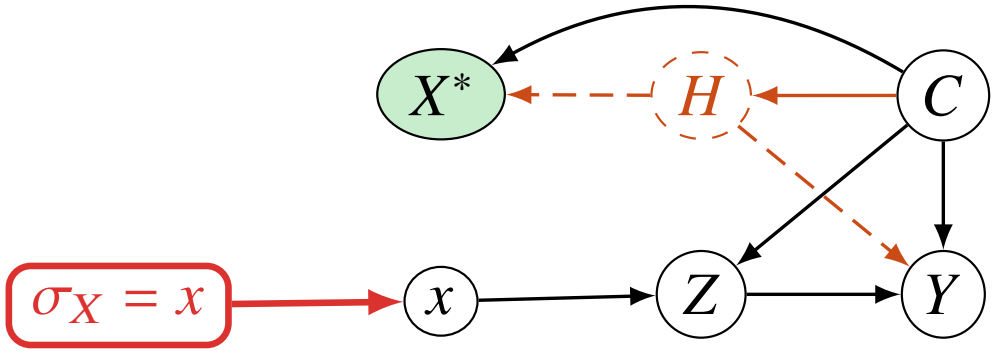




$$
\begin{aligned}
P(y \mid \mathrm{do}(x)) &= P(y \mid \mathrm{do}(x), \mathrm{do}(z)) && \text{(Rule3: } (Y \perp Z \mid X)_{G_{\overline{X},\overline{Z}}}\text{)} \\
&= P(y \mid x, \mathrm{do}(z)) && \text{(Rule2: } (Y \perp X)_{G_{\overline{Z},\underline{X}}}\text{)} \\
&= \frac{P(y, x \mid \mathrm{do}(z))}{P(x \mid \mathrm{do}(z))} && \text{(conditional probability)} \\
&= \frac{\sum_w P(y, x \mid z, w) P(w)}{\sum_w P(x \mid z, w) P(w)} && \text{(backdoor } (Y, X \perp Z \mid W)_{G_{\underline{Z}}}\text{)}
\end{aligned}
$$

前门校正 — 决策论因果视角

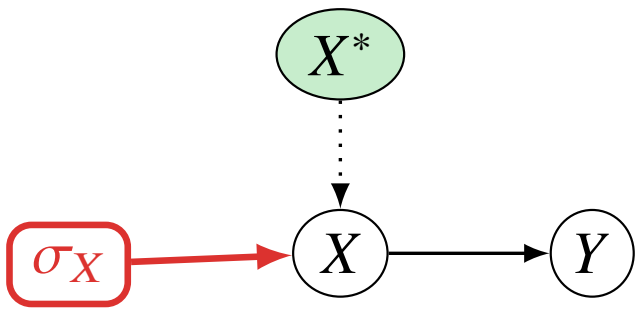




$$
\begin{aligned}
p(Y \mid \sigma_X = x) &= \sum_{z,c} p(Y \mid z, c, \sigma_X = x)\, p(z \mid c, \sigma_X = x)\, p(c \mid \sigma_X = x) && \text{by probability} \\
&= \sum_{z,c} p(Y \mid z, c, \sigma_X = x)\, p(z \mid c, \sigma_X = x)\, p(c) && C \perp \sigma_X \\
&= \sum_{z,c} p(Y \mid z, c, \sigma_X = x)\, p(z \mid c, x, \sigma_X = x)\, p(c) && \text{by definition of } X \\
&= \sum_{z,c} p(Y \mid z, c, \sigma_X = x)\, p(z \mid c, x, \sigma_X = \varnothing)\, p(c) && Z \perp \sigma_X \mid X, C \\
&= \sum_{z,c} \left(\sum_{x^*} p(Y \mid x^*, z, c, \sigma_X = x)\, p(x^* \mid z, c, \sigma_X = x) \right) p(z \mid c, x, \sigma_X = \varnothing)\, p(c) && \text{by probability} \\
&= \sum_{z,c} \left(\sum_{x^*} p(Y \mid x^*, z, c, \sigma_X = x)\, p(x^* \mid c, \sigma_X = x) \right) p(z \mid c, x, \sigma_X = \varnothing)\, p(c) && X^* \perp Z \mid C, \sigma_X = x \\
&= \sum_{z,c} \left(\sum_{x^*} p(Y \mid x^*, z, c, \sigma_X = x)\, p(x^* \mid c, \sigma_X = \varnothing) \right) p(z \mid c, x, \sigma_X = \varnothing)\, p(c) && X^* \perp \sigma_X \mid C \\
&= \sum_{z,c} \left(\sum_{x^*} p(Y \mid x^*, z, c, \sigma_X = \varnothing)\, p(x^* \mid c, \sigma_X = \varnothing) \right) p(z \mid c, x, \sigma_X = \varnothing)\, p(c) && Y \perp \sigma_X \mid X^*, Z, C
\end{aligned}
$$

Remarks: 前门校正 — 决策论因果视角

- X^*: intention to treat variable.
- The above derivation uses X^* but ignores H.
- The above derivation did not assume the mediator Z may be intervened on.



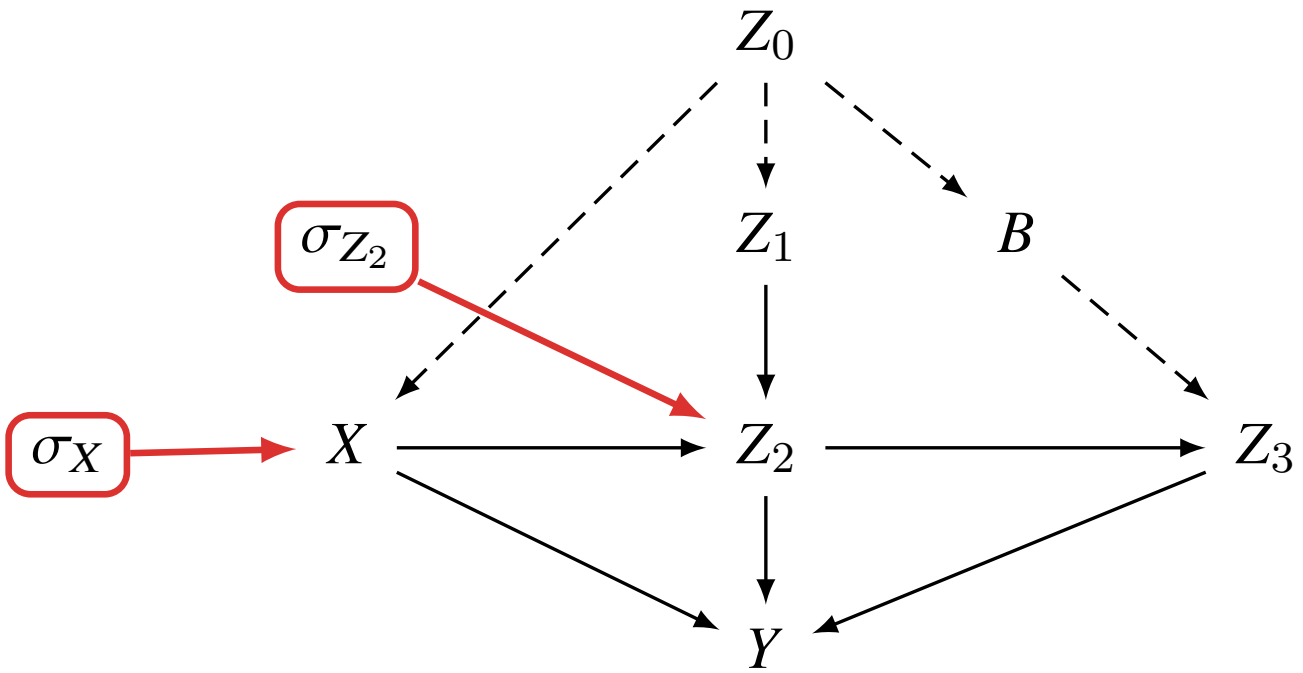
Ignorability Assumption:

$$Y \perp \{X^*, \sigma_X\} \mid X$$

Remark: 借助 X^*, 可以不用潜在结果 Y_x, 不用反事实, 直接表达 ETT:

$$\mathrm{ETT} = \mathbb{E}[Y \mid X^* = 1, X = 1] - \mathbb{E}[Y \mid X^* = 1, X = 0]$$

Example: 决策论因果 vs do-演算



$$P(y \mid \mathrm{do}(x)) = \sum_{z_1,z_2,z_3} \underbrace{P(z_1 \mid \mathrm{do}(x))}_{\text{项1}} \underbrace{P(z_2 \mid z_1, \mathrm{do}(x))}_{\text{项2}} \underbrace{P(z_3 \mid z_1, z_2, \mathrm{do}(x))}_{\text{项3}} \underbrace{P(y \mid z_1, z_2, z_3, \mathrm{do}(x))}_{\text{项4}}$$

项 1 $P(z_1 \mid \mathrm{do}(x)) = P(z_1)$ $\quad Z_1 \perp \sigma_X$ or $(Z_1 \perp X)_{G_{\overline{X}}}$

项 2 $P(z_2 \mid z_1, \mathrm{do}(x)) = P(z_2 \mid z_1, x)$ $\quad Z_2 \perp \sigma_X \mid Z_1, X$ or $(Z_2 \perp X \mid Z_1)_{G_{\underline{X}}}$

项 4 需要两步, 第一步用 $Y \perp \sigma_X \mid Z_1, Z_2, Z_3, X$ or $(Y \perp X \mid Z_1, Z_2, Z_3)_{G_{\underline{X}}}$

$$\begin{aligned} P(y \mid z_1, z_2, z_3, \mathrm{do}(x)) &= P(y \mid z_1, z_2, z_3, x) \quad (Y \perp \sigma_X \mid Z_1, Z_2, Z_3, X) \\ &= P(y \mid z_2, z_3, x) \quad (Y \perp Z_1 \mid Z_2, Z_3, X) \end{aligned}$$

项 3 稍微麻烦一点儿.

- 我们想消掉 $\mathrm{do}(x)$, 但发现 Rule3 不适用 ($G_{\overline{X(Z_1,Z_2)}} = G$),

$$Z_3 \not\perp \sigma_X \mid Z_1, Z_2$$

因为 Z_2 是 X 的后代, 而 X 是对撞节点.

- Z_2 是 X 的后代这一点怎么克服呢? 因为 $Z_3 \perp \sigma_{Z_2} \mid Z_1, Z_2, \sigma_X = x$ or $(Z_3 \perp Z_2 \mid Z_1, X)_{G_{\overline{X},\underline{Z_2}}}$, 所以,

$$P(z_3 \mid z_1, z_2, \mathrm{do}(x)) = P(z_3 \mid z_1, \mathrm{do}(z_2), \mathrm{do}(x))$$

- 现在就可以消掉 $\mathrm{do}(x)$ 了, 因为 $Z_3 \perp \sigma_X \mid Z_1, \sigma_{Z_2} = z_2$ or $(Z_3 \perp X \mid Z_1, Z_2)_{G_{\overline{X},\overline{Z_2}}}$, 所以,

$$P(z_3 \mid z_1, \mathrm{do}(z_2), \mathrm{do}(x)) = P(z_3 \mid z_1, \mathrm{do}(z_2))$$

- 由 $Z_3 \perp \sigma_{Z_2} \mid X, Z_1, Z_2$ or $(Z_3 \perp Z_2 \mid X, Z_1)_{G_{\underline{Z_2}}}$, 和 $X \perp \sigma_{Z_2} \mid Z_1$ or $(X \perp Z_2 \mid Z_1)_{G_{\overline{Z_2}}}$,

$$P(z_3 \mid z_1, \mathrm{do}(z_2)) = \sum_{x'} P(z_3 \mid x', z_1, \mathrm{do}(z_2)) P(x' \mid z_1, \mathrm{do}(z_2)) = \sum_{x'} P(z_3 \mid x', z_1, z_2) P(x' \mid z_1)$$

综上, $P(y \mid \mathrm{do}(x)) = \sum_{z_1,z_2,z_3} P(z_2 \mid z_1, x) P(y \mid z_2, z_3, x) \sum_{x'} P(z_3 \mid x', z_1, z_2) P(x', z_1)$

Identifiability

Definition (Identifiability)

Let $Q(M)$ be any computable quantity of a model M. We say that Q is identifiable in a class $\mathcal{M}$ of models if, for any pairs of models M_1 and M_2 from $\mathcal{M}$,

$$P_{M_1}(V) = P_{M_2}(V) \implies Q(M_1) = Q(M_2)$$

Remark

The intervention distribution $P(Y \mid \mathrm{do}(x))$ is identifiable if it can be computed from the observational distribution $P(V)$ and the graph G.

i.e., there exists some f_G : $P(V) \xrightarrow{f_G} P(Y \mid \mathrm{do}(x))$

Identifiability

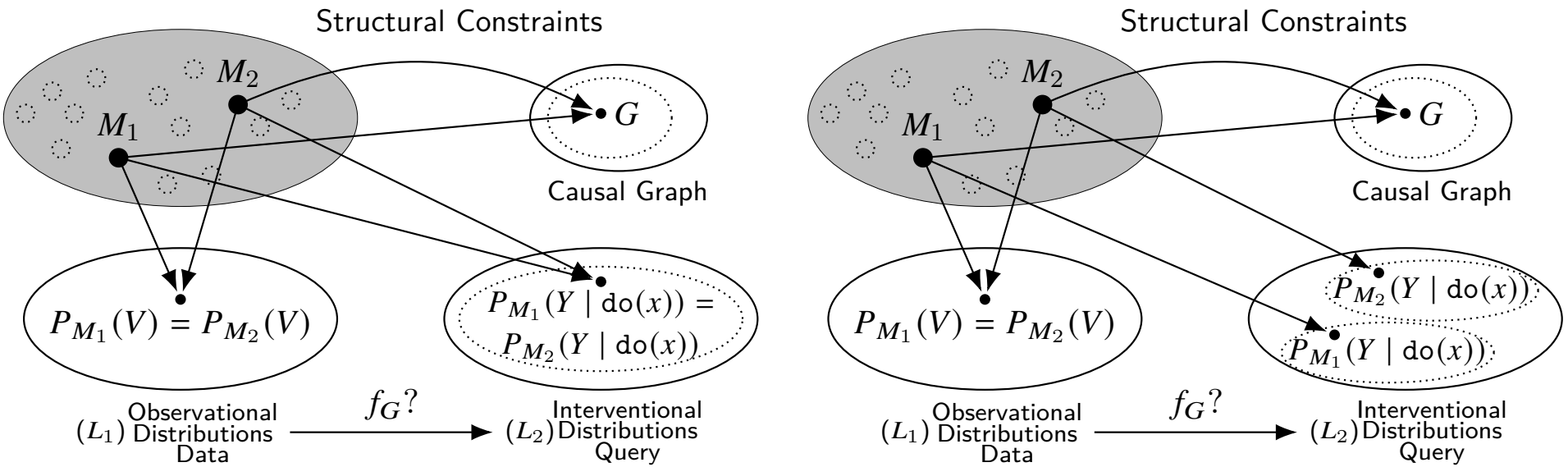
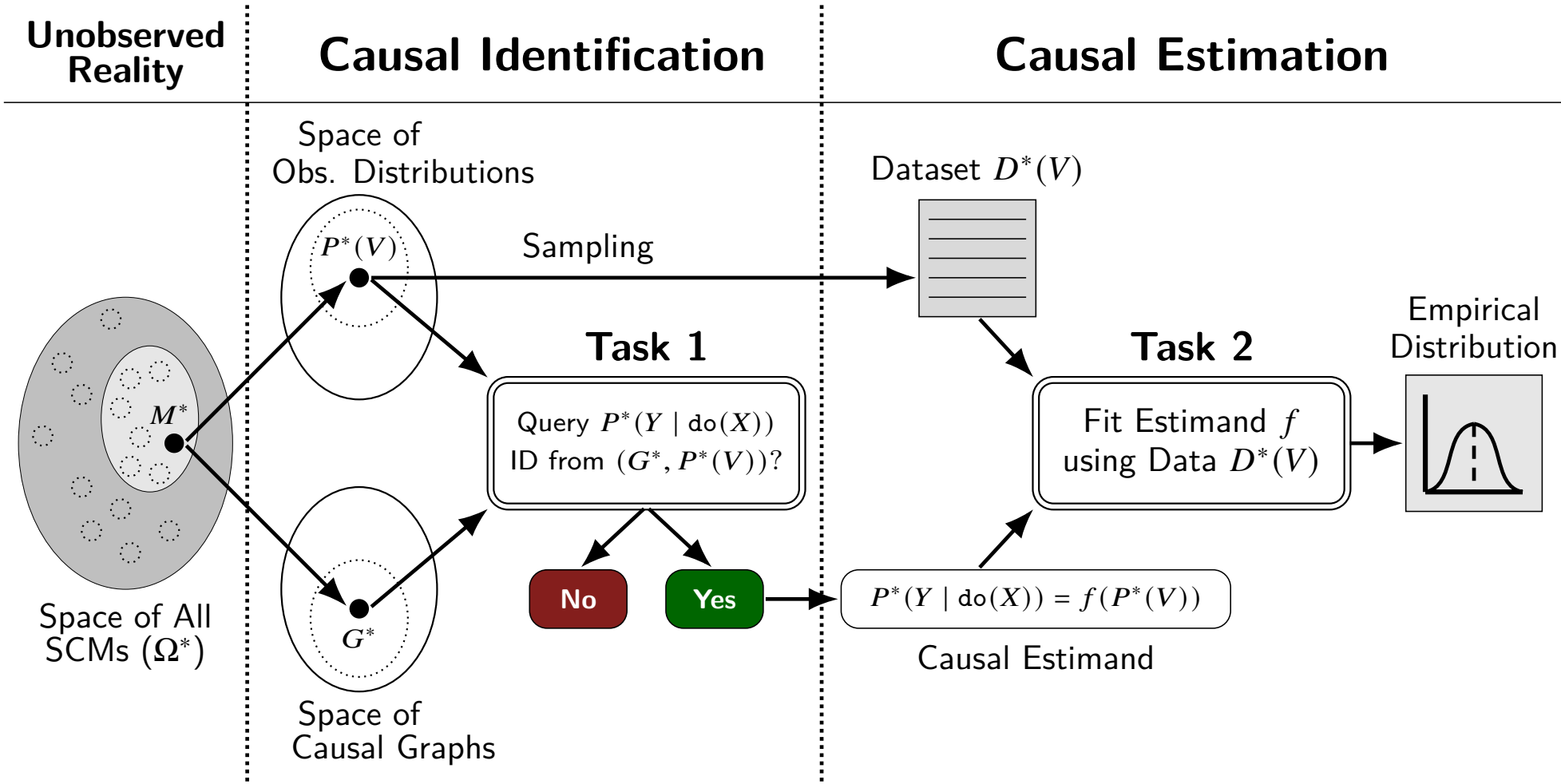


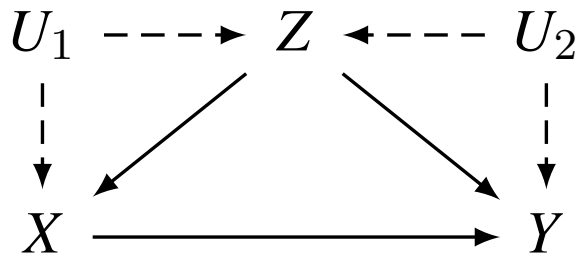
Figure: $P(Y \mid \text{do}(x))$ is identifiable from $P(V)$ and G if, for all M_1, M_2 such that M_1, M_2 match in $P(V)$ and G, they also match in $P(Y \mid \text{do}(x))$.


Unobserved Reality
Causal Identification
Causal Estimation
Space of Obs. Distributions
$P^*(V)$
Sampling
Dataset $D^*(V)$
Task 1
Query $P^*(Y \mid \mathrm{do}(X))$
ID from $(G^*, P^*(V))$?
No
Yes
Task 2
Fit Estimand f using Data $D^*(V)$
Empirical Distribution
M^*
Space of All SCMs (Ω^*)
G^*
Space of Causal Graphs
$P^*(Y \mid \mathrm{do}(X)) = f(P^*(V))$
Causal Estimand

单干预变量的可识别性

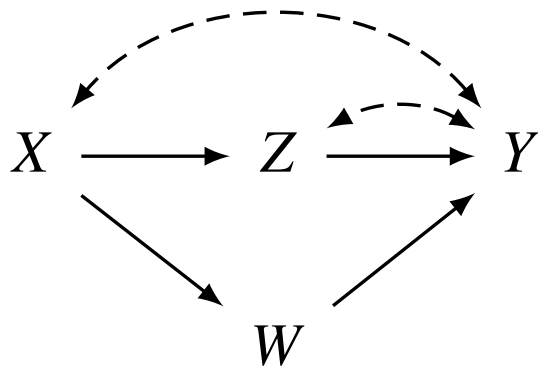
- 若 $P(y \mid \mathrm{do}(x))$ 可识别, 则 X 与 $\mathrm{Ch}_X \cap \mathrm{An}_Y$ 之间的任何后门路径都可以被阻断.

  Remark: 此条件必要但不充分. 如右图, X 与 Y 之间的每一条后门路径都可以被阻断, 但无法被同一个集合阻断.



- 若 X 与 $\mathrm{Ch}_X \cap \mathrm{An}_Y$ 之间的所有后门路径都可以被同一个集合阻断, 则 $P(y \mid \mathrm{do}(x))$ 可识别. (无混孩子准则???)

  Remark: 如右图, 虽然前后门准则不适用, 但 X 与 Z, W 之间的所有后门路径都被空集阻断, 所以 $P(y \mid \mathrm{do}(x))$ 可识别.
  Remark: 此条件充分但不必要.

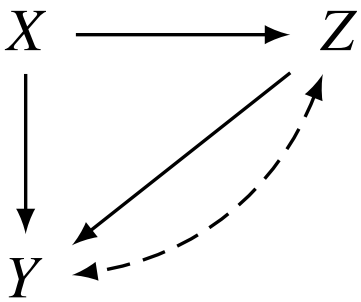


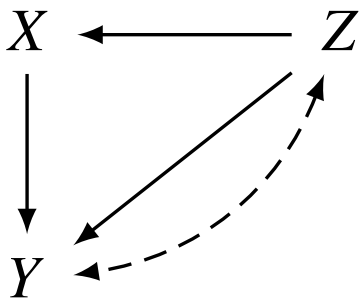
- $P(\mathbf{v} \mid \mathrm{do}(x))$ 可识别, 当且仅当, X 与其任何子节点之间都不存在双向弧组成的路径.
  Remark: $P(\mathbf{v} \mid \mathrm{do}(x))$ 可识别 $\Longrightarrow$ $P(y \mid \mathrm{do}(x))$ 可识别

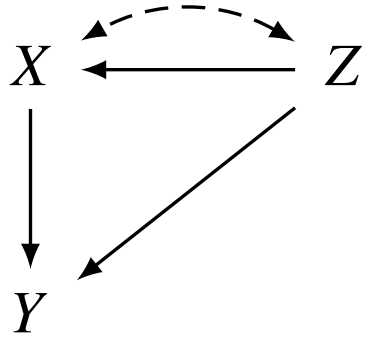
Identifiability

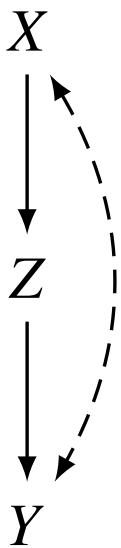
- 若 $P(y \mid \mathrm{do}(x))$ 在图 G 中不可识别, 则在 G 中添加一条有向或双向弧后, 也不可识别.
- 若 $P(y \mid \mathrm{do}(x))$ 在图 G 中可识别, 则在 G 中删除一条有向或双向弧后, 仍可识别.

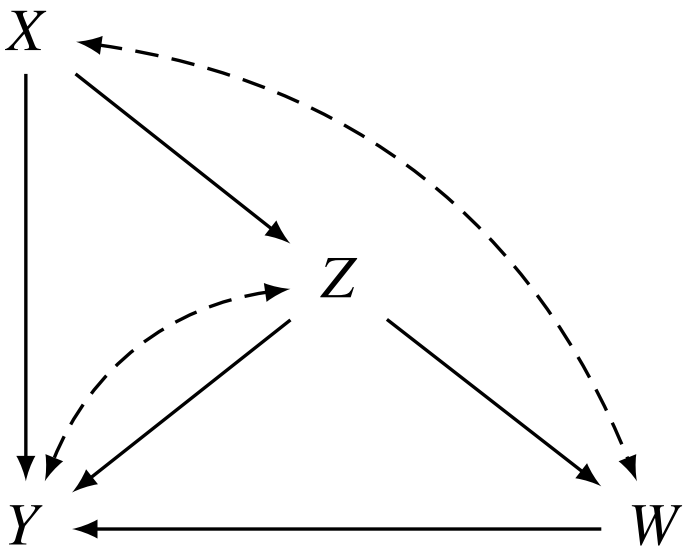
Examples: $P(y \mid \mathrm{do}(x))$ is identifiable

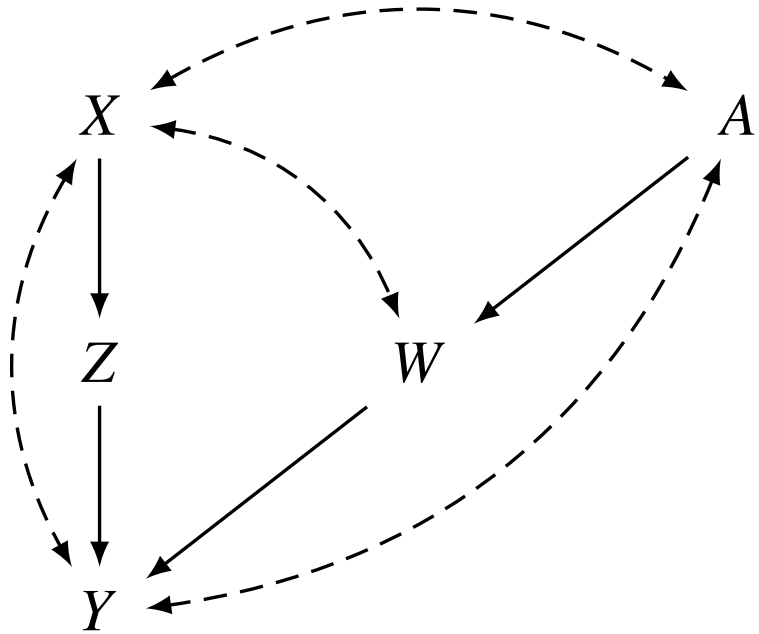




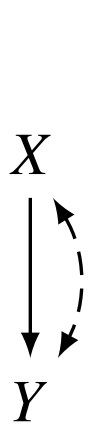


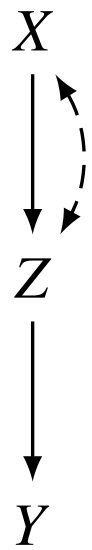


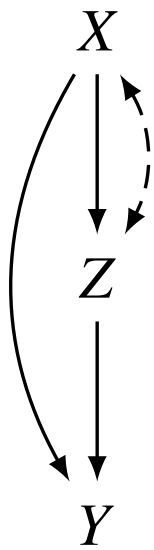


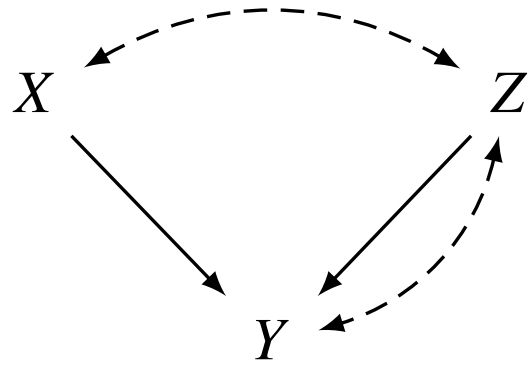


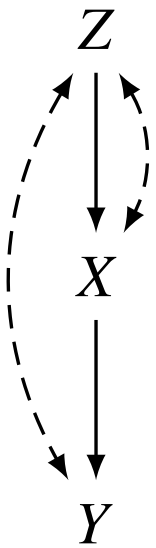
Examples: $P(y \mid \mathrm{do}(x))$ is not identifiable

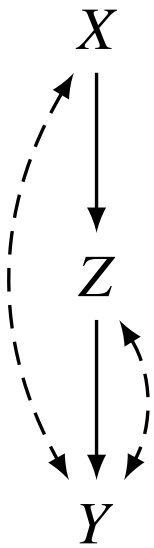


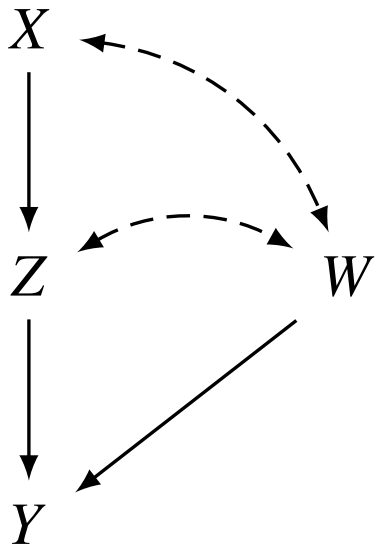


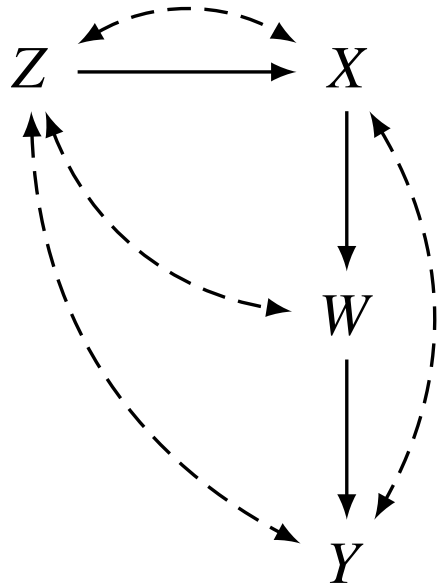












Soundness & Completeness of the do-Calculus

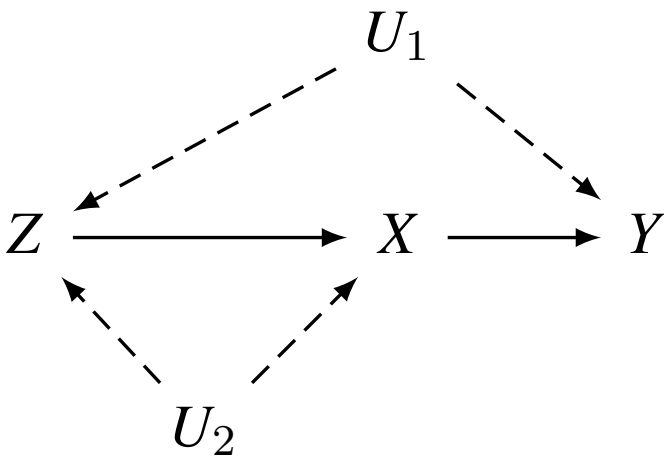
Theorem (Soundness & Completeness for Observational Identifiability)

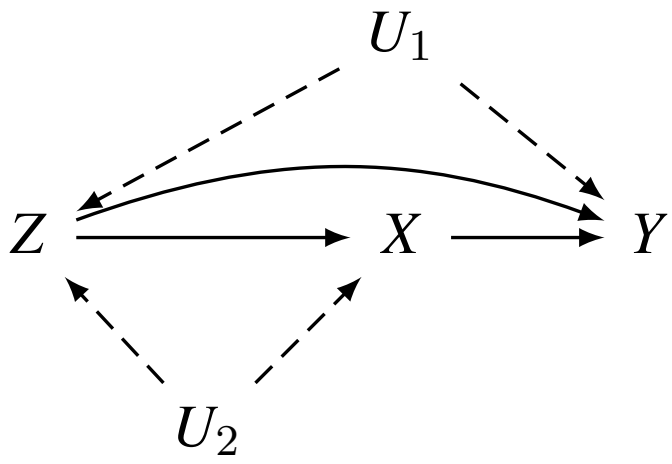
The do*-calculus is sound and complete*

- *Sound: If the* do*-operations can be removed by repeated application of these three rules, the causal effect is identifiable.*
- *Complete: If the causal effect is identifiable, the do-operations can be removed by repeated application of these three rules.*

没有操纵就没有因果？— 基于替代实验的因果推断

- 我们希望得到 $P(y \mid \mathrm{do}(x))$, 但无法通过 RCT 控制 X.
- 可否随机化一个比 X 更容易控制的 Z 来识别 $P(y \mid \mathrm{do}(x))$?
- 比如: 想评估胆固醇 X 对心脏 Y 的效应, 一个合理的方法是控制饮食 Z, 而不是直接控制胆固醇 X.
- 通过 do-Calculus 可以证明, 以下条件足以决定替代变量 Z:
  1. X 截断了所有从 Z 到 Y 的有向路径.
  2. $P(y \mid \mathrm{do}(x))$ 在 $G_{\overline{Z}}$ 中是可识别的.
- 如下图 1 中 Z 可以作为 X 的替代变量, 下图 2 则不行.



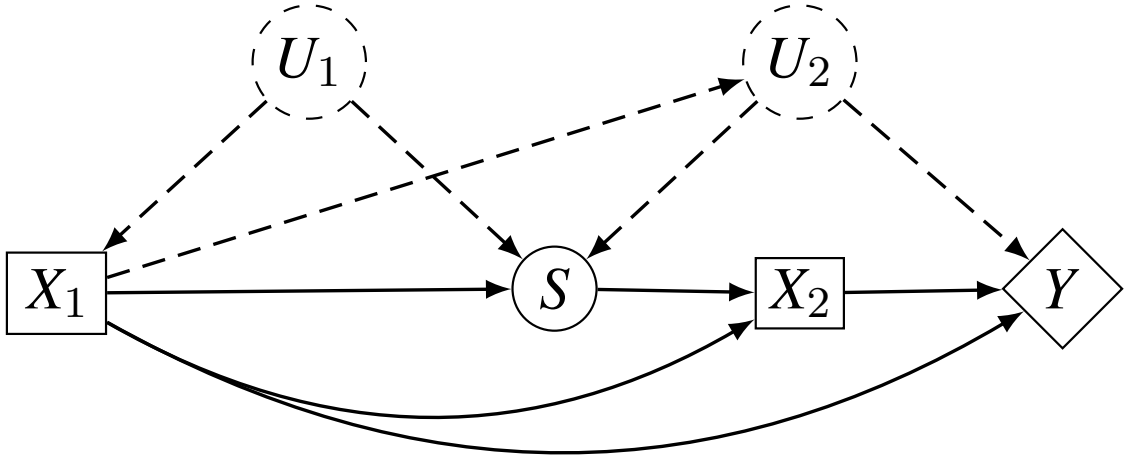


$$P(y \mid \mathrm{do}(x)) \overset{R3}{=} P(y \mid \mathrm{do}(x), \mathrm{do}(z)) \overset{R2}{=} P(y \mid x, \mathrm{do}(z)) = \frac{P(y, x \mid \mathrm{do}(z))}{P(x \mid \mathrm{do}(z))}$$

Remark: do-演算 vs RL Agent

- 强化学习的 Agent 只能学习它尝试过的动作的结果 $\mathbb{E}[r \mid \mathrm{do}(a), s]$. 对于未尝试过的动作, 它无法预测其结果.
- 这就像是一个不敢对不可操纵变量使用 do 的人, 其能力是受限的.
- 而基于因果模型, 我们可以利用已尝试过的动作的结果, 推断出未尝试过的动作的结果.

序贯行动



- X_1: 化疗. X_2: 手术. S: 阶段成果. Y: 康复情况.
- 为了计算 X_2 对 Y 的效应, 我们需要控制后门 S, 但若控制了 S, 又阻断了 X_1 对 Y 的部分效应.

$$
\begin{aligned}
P(y \mid \mathtt{do}(x_1), \mathtt{do}(x_2)) &\overset{R_2}{=} P(y \mid x_1, \mathtt{do}(x_2)) \\
&= \sum_s P(y \mid s, x_1, \mathtt{do}(x_2)) P(s \mid x_1, \mathtt{do}(x_2)) \\
&\overset{R_3}{=} \sum_s P(y \mid s, x_1, \mathtt{do}(x_2)) P(s \mid x_1) \\
&\overset{R_2}{=} \sum_s P(y \mid s, x_1, x_2) P(s \mid x_1)
\end{aligned}
$$

序贯后门准则 The Sequential Back-Door Criterion

- 我们将动作节点排序为 $X_1, X_2, \ldots, X_n$, 使每一个 X_k 是 $X_{k+i}(i > 0)$ 的非后代.
- 结果节点 Y 是 X_n 的后代.
- $\mathbf{S}_k$ 是状态节点的集合, 其中每个观察状态 $S_i, i \leq k$ 是 $X_k, \ldots, X_n$ 的非后代.

Theorem (序贯后门准则)

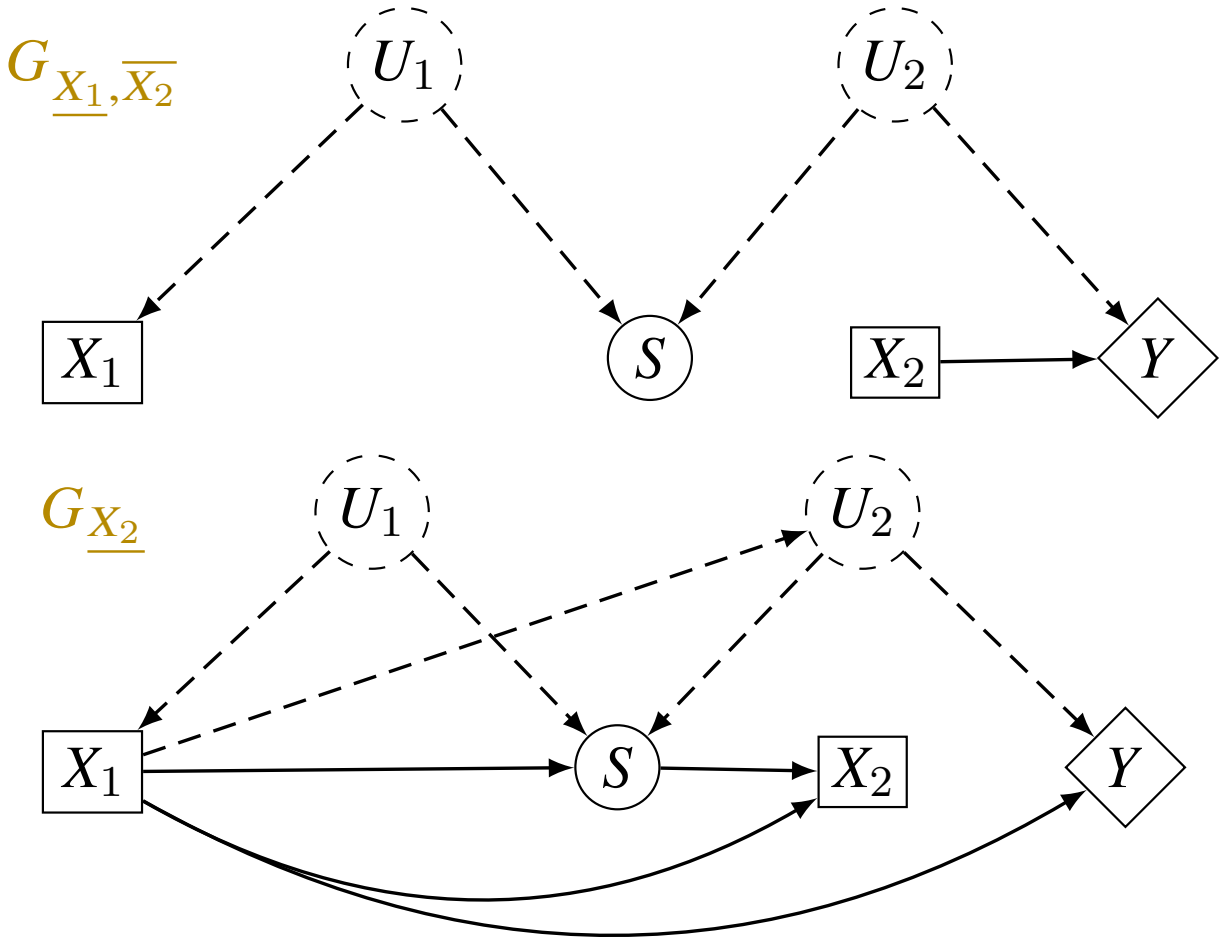
对任一 $1 \leq k \leq n$, 存在 $X_k, \ldots, X_n$ 的非后代集合 $\mathbf{S}_k$ 使得

$$(Y \perp X_k \mid X_1, \ldots, X_{k-1}, S_1, \ldots, S_k)_{G_{\underline{X_k}, \overline{X_{k+1}}, \ldots, \overline{X_n}}}$$

则 $P(y \mid \mathtt{do}(x_1, \ldots, x_n))$ 可识别. 且

$$P(y \mid \mathtt{do}(x_1, \ldots, x_n)) = \sum_{s_1, \ldots, s_n} P(y \mid s_1, \ldots, s_n, x_1, \ldots, x_n) \times \prod_{k=1}^{n} P(s_k \mid s_1, \ldots, s_{k-1}, x_1, \ldots, x_{k-1})$$

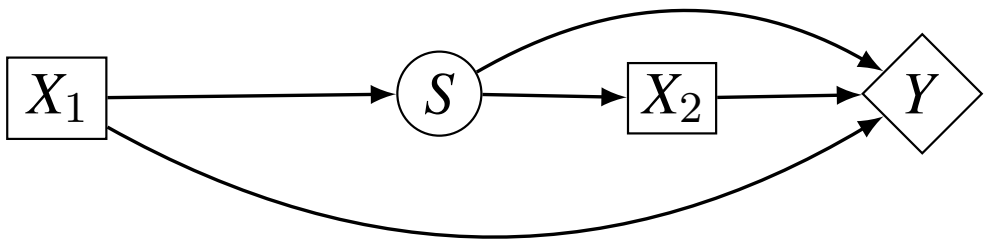
Example



令 $S_1 = \varnothing, S_2 = \{S\}$, 则 $(Y \perp X_1)_{G_{\underline{X_1},\overline{X_2}}}$ 且 $(Y \perp X_2 \mid X_1, S)_{G_{\underline{X_2}}}$

$$P(y \mid \mathrm{do}(x_1, x_2)) = \sum_s P(y \mid s, x_1, x_2) P(s \mid x_1)$$

Example



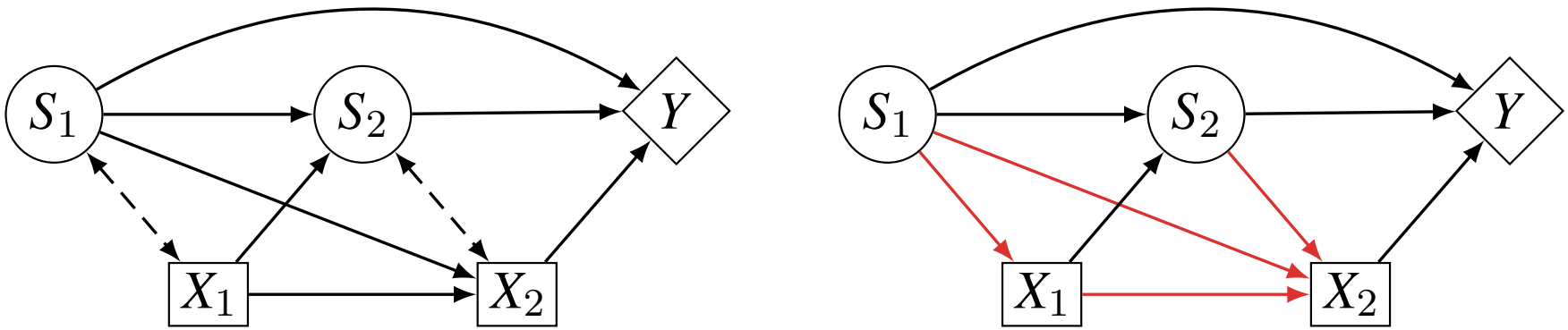
- $P(y \mid \mathtt{do}(x_1, x_2))$ 是可识别的.

序贯后门准则 (策略干预)

令 G_π 表示策略干预 $\mathrm{do}(\pi)$ 后的 SCM 的子模型生成的因果图.
它可以由 G 通过以下两步得到:

1. 删除所有指向 X_i 的箭头.
2. 添加从观察状态 S_i 到 X_i 的箭头.

Example: 对策略 $\pi_1(X_1 \mid S_1), \pi_2(X_2 \mid S_1, X_1, S_2)$ 的 G 和 G_{π_1,π_2}:



Definition (序贯后门准则 (策略干预))

对任一策略 $\pi \in \Pi$,

$$(Y \perp X_k \mid X_1, \ldots, X_{k-1}, S_1, \ldots, S_k)_{G_{\underline{X_k}, \pi_{k+1}, \ldots, \pi_n}}$$

软干预 Soft Intervention

$$M_{\sigma_X} = (U \cup U', V, F', P(U, U'))$$

其中,

$$F' = \{f_i : V_i \notin X\} \cup f'_X$$

软干预 $\sigma_X = P'(x \mid \mathrm{Pa}'_X)$,

$$P'(x \mid \mathrm{pa}'_X) = \sum_{u'_X} P\left(f'_X(\mathrm{pa}'_X, u'_X) = x\right) P\left(u'_X\right)$$

$$P(v; \sigma_X) = \sum_{u, u'_X} \prod_{\{i:V_i \in X\}} P(v_i \mid \mathrm{pa}_i, u_i, u'_i; \sigma_X) P(u'_X; \sigma_X) \prod_{\{i:V_i \in V \setminus X\}} P(v_i \mid \mathrm{pa}_i, u_i) P(u)$$

条件干预 $\mathtt{do}(X = g(Z))$

例如, 医生只对体温超过 $Z = z$ 的患者用药 $g(Z) = [\![Z > z]\!]$.

$$
\begin{aligned}
& P(Y = y \mid \mathtt{do}(X = g(Z))) \\
= & \sum_z P(y \mid \mathtt{do}(X = g(Z)), z) P(z \mid \mathtt{do}(X = g(Z))) \\
= & \sum_z P(y \mid \mathtt{do}(x), z) P(z) \Big|_{x=g(z)}
\end{aligned}
$$

如果存在可测变量集 S 使得 $S \cup Z$ 满足后门准则, 则

$$P(Y = y \mid \mathtt{do}(X = x), Z = z) = \sum_s P(Y = y \mid X = x, S = s, Z = z) P(S = s \mid Z = z)$$

软干预 $\mathtt{do}(P'(X \mid Z))$

干预 $\mathtt{do}(X = x)$ 发生的概率为 $P'(x \mid z)$, 则

$$P(Y = y \mid \mathtt{do}(P'(X \mid Z))) = \sum_x \sum_z P(y \mid \mathtt{do}(x), z) P'(x \mid z) P(z)$$

σ-Calculus for Soft Interventions

For any disjoint subsets $X, Y, Z \subset V$, two disjoint subsets $T, W \subset V \setminus (Z \cup Y)$ (i.e., possibly including X),

R1. **Insertion/Deletion of observations:** if $(Y \perp T \mid W)$ in G_{σ_X}, then

$$P(y \mid w, t; \sigma_X) = P(y \mid w; \sigma_X)$$

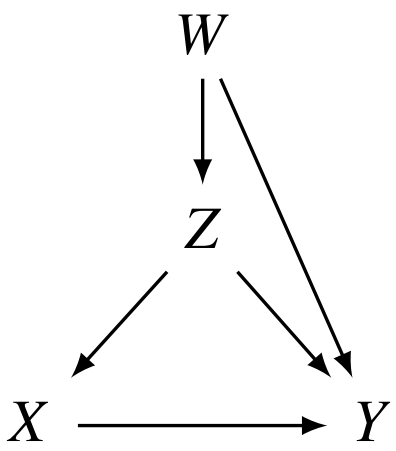
R2. **Change of regimes under observation:** if $(Y \perp Z \mid W)$ in $G_{\sigma_X \sigma_Z \underline{Z}}$ and $G_{\sigma_X \sigma'_Z \underline{Z}}$, then

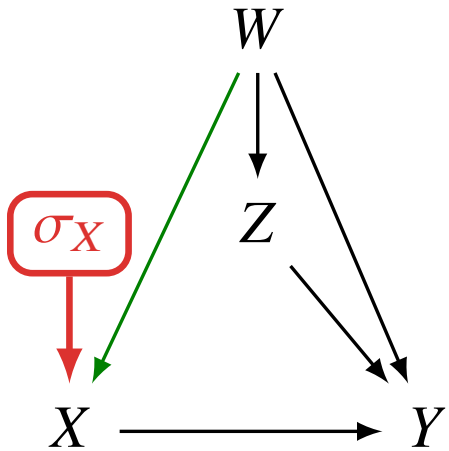
$$P(y \mid z, w; \sigma_X, \sigma_Z) = P(y \mid z, w; \sigma_X, \sigma'_Z)$$

R3. **Change of regimes without observation:** if $(Y \perp Z \mid W)$ in $G_{\sigma_X \sigma_Z \overline{Z(W)}}$ and $G_{\sigma_X \sigma'_Z \overline{Z(W)}}$, then

$$P(y \mid w; \sigma_X, \sigma_Z) = P(y \mid w; \sigma_X, \sigma'_Z)$$

Example





$$\sigma_X = P'(x \mid w)$$

$$\begin{aligned} P(y;\sigma_X) &= \sum_{x,z,w} P(y \mid x,z,w;\sigma_X)P(x \mid z,w;\sigma_X)P(z,w;\sigma_X) \\ &= \sum_{x,z,w} P(y \mid x,z,w)P(x \mid w;\sigma_X)P(z,w) \\ &= \sum_{x,z,w} P(y \mid x,z,w)P'(x \mid w)P(z,w) \end{aligned}$$

- R1: $(X \perp Z \mid W)$ in G_{σ_X}
- R2 with $\sigma'_X = \varnothing$: $(Y \perp X \mid Z,W)$ in $G_{\sigma_X\underline{X}}$ and $G_{\underline{X}}$
- R3 with $\sigma'_X = \varnothing$: $(Z,W \perp X)$ in $G_{\sigma_X\overline{X}}$ and $G_{\overline{X}}$

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal
   - Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

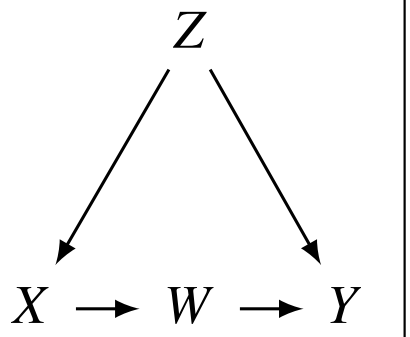
The Data-Fusion Problem

Problem

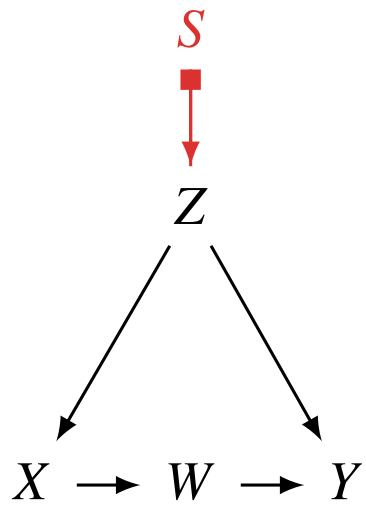
How to combine results of several experimental and observational studies, each conducted on a different population and under a different set of experimental conditions, so as to construct an aggregate measure of effect size that is "better" than any one study in isolation.

A 地: 目标总体	B 地: 调查数据 年轻人多	C 地: 调查数据 汽车拥有量低
D 地: 调查数据 点击率高	E 地: 随机试验 点击率高	F 地: 随机试验 购买欲强

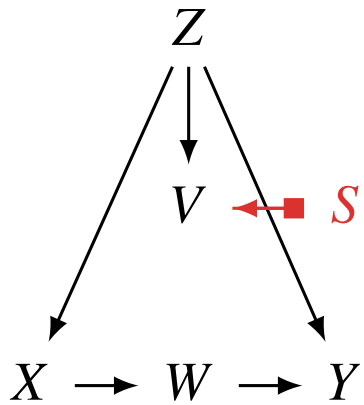
A 地: 目标总体



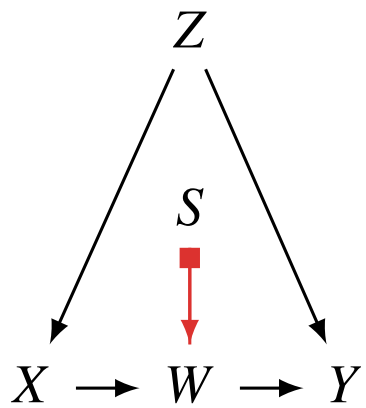
B 地: 混杂变量不同



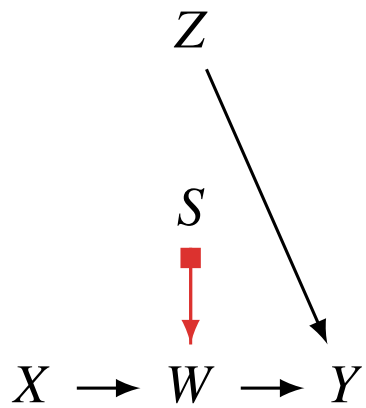
C 地: 无关变量不同



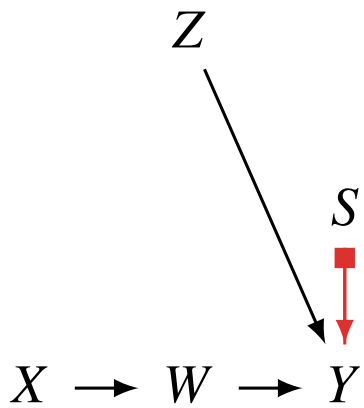
D 地: 中介变量不同



E 地: 因果结构不同



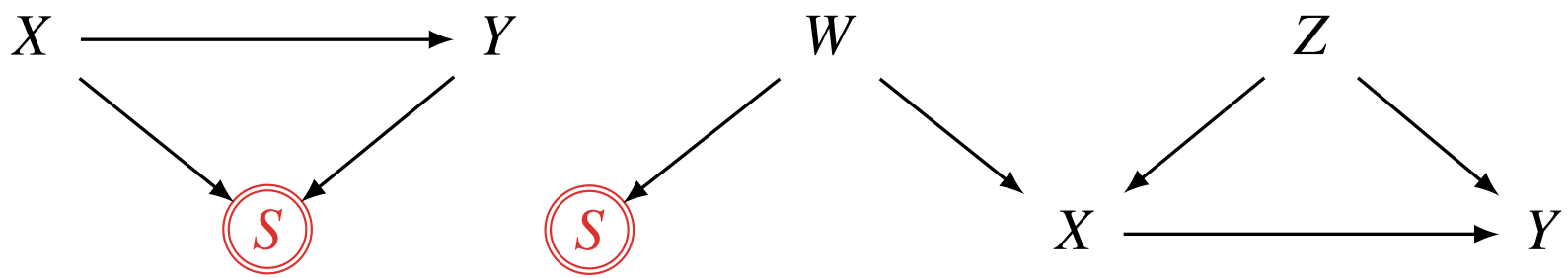
F 地: 因果结构、结果变量不同



X = 广告, Y = 购买决策, Z = 年龄, W = 点击率, V = 汽车拥有率, S = 指示变量

Selection Bias

- Our task is to express the query in terms of the available data, that is, the distribution under selection bias $P(V \mid S = 1)$.
- This is different from recovering the causal effect $P(y \mid \text{do}(x))$.
- For instance, in the second model, $P(y \mid x)$ is not recoverable, while $P(y \mid \text{do}(x))$ is.



Theorem

The conditional distribution $P(y \mid x)$ is recoverable (without external data) iff $(Y \perp S \mid X)$.

Transportability Reduced to Calculus

- Transportability is defined as a license to transfer causal effects learned in experimental studies to a new population, in which only observational studies can be conducted.
- A selection diagram is a causal diagram annotated with new variables, called S-nodes. The edge $S \blacksquare\!\!\longrightarrow Z$ means the local mechanism that assigns values to Z may be different, $f_z \neq f_z^*$ or $P(U_z) \neq P^*(U_z)$ between the two populations.

Theorem

Let D be the selection diagram characterizing two populations, π and π^*, and S a set of selection variables in D. The relation $P^*(y \mid \mathtt{do}(x), z)$ is transportable from π to π^* iff the expression $P(y \mid \mathtt{do}(x), z, s)$ is reducible, using the rules of do*-calculus, to an expression in which S appears only as a conditioning variable in* do*-free terms.*

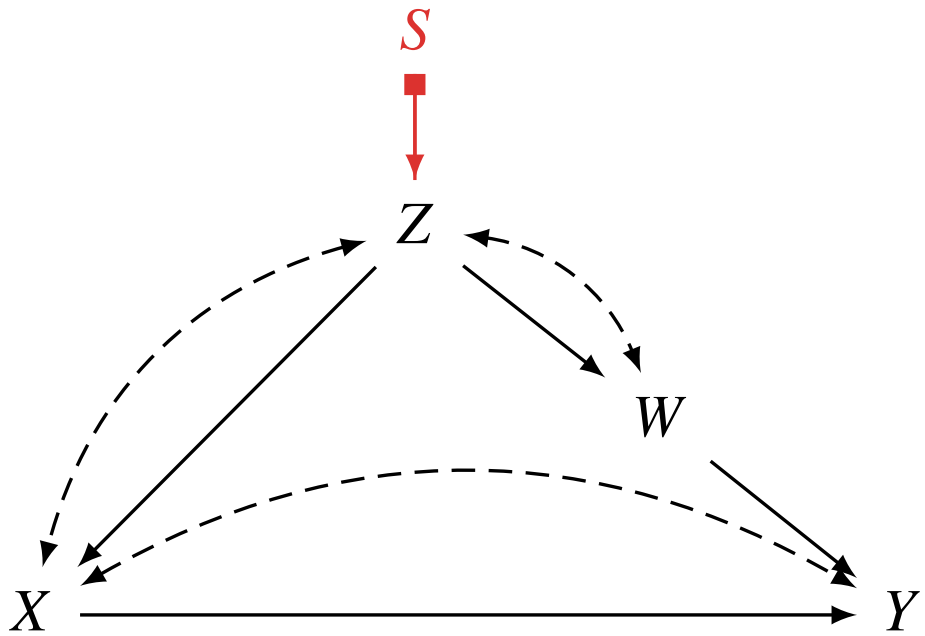
Remark: $P(y \mid \mathtt{do}(x), z, s) = P(y \mid \mathtt{do}(x), z)$ whenever $(Y \perp S \mid X, Z)_{G_{\overline{X}}}$ by Rule1 of do-calculus.

Theorem

The causal effect $P^*(y \mid \mathtt{do}(x))$ is transportable from π to π^* if either one of the following conditions holds:

1. *$P^*(y \mid \mathtt{do}(x))$ is trivially transportable, i,e, identifiable directly from observational studies on π^*.*
2. *There exists a set of covariates, Z (possibly affected by X) such that Z satisfies $(Y \perp S \mid X, Z)_{G_{\overline{X}}}$, and for which $P^*(z \mid \mathtt{do}(x))$ is transportable.*
3. *There exists a set of covariates, W that satisfy $(Y \perp X \mid W)_{G_{\overline{X(W)}}}$ and for which $P^*(w \mid \mathtt{do}(x))$ is transportable.*

Selection Diagram — Example

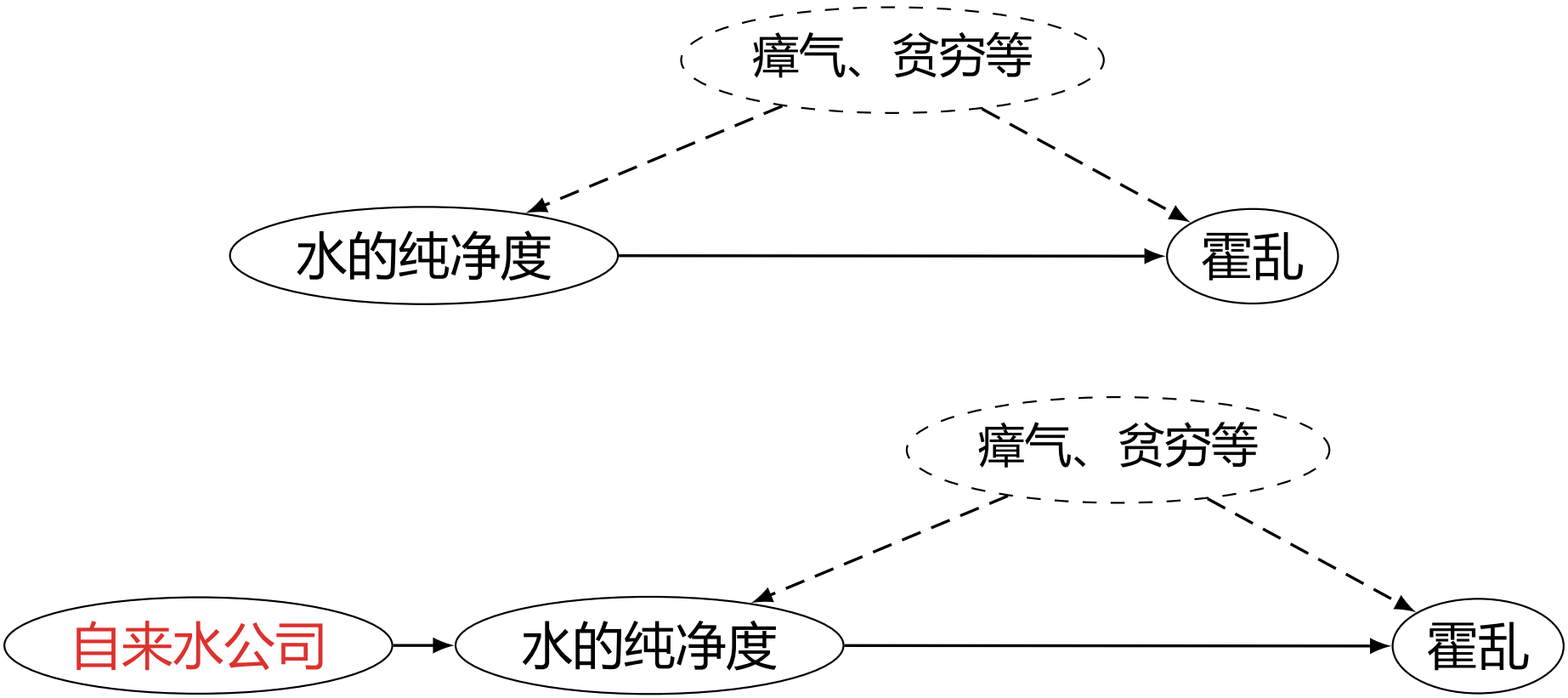


$$
\begin{aligned}
P^*(y \mid \mathrm{do}(x)) &= P(y \mid \mathrm{do}(x), s) \\
&= \sum_w P(y \mid \mathrm{do}(x), s, w) P(w \mid \mathrm{do}(x), s) \\
&= \sum_w P(y \mid \mathrm{do}(x), w) P(w \mid s) \\
&= \sum_w P(y \mid \mathrm{do}(x), w) P^*(w)
\end{aligned}
$$

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal
   - Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

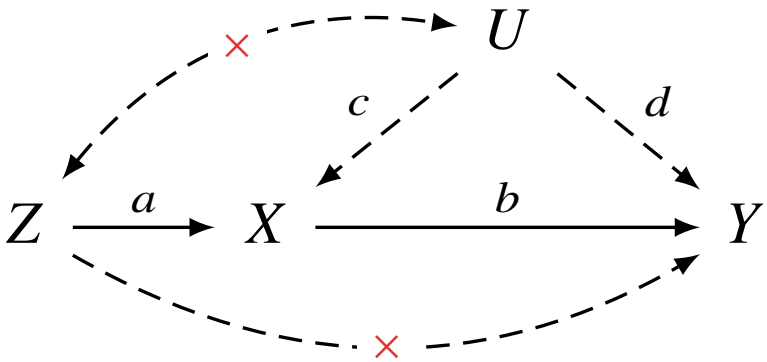
工具变量 Instrumental Variable



Definition (Instrumental Variable)

Z is an instrumental variable for identifying the effect of X on Y if

1. $(Z \not\perp X)_G$
2. Z effects Y only through X: $(Z \perp Y)_{G_{\overline{X}}}$ (or $Z \perp Y_x$)



$$U = \varepsilon_U$$
$$Z = \varepsilon_Z$$
$$X = aZ + cU + \varepsilon_X$$
$$Y = bX + dU + \varepsilon_Y$$

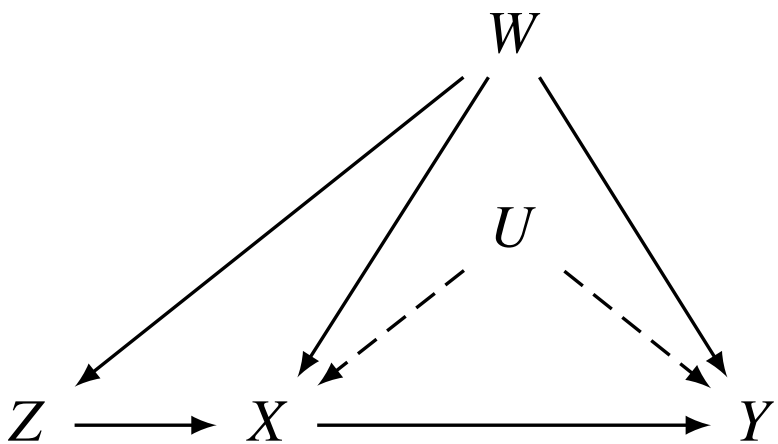
- Although the identification of instrumental variables goes beyond do-calculus, it is identifiable if we restrict the structural equations.
- We regress X and Y on Z separately, yielding the regression equations $Y = r_{ZY}Z + \varepsilon$ and $X = r_{ZX}Z + \varepsilon'$. Then

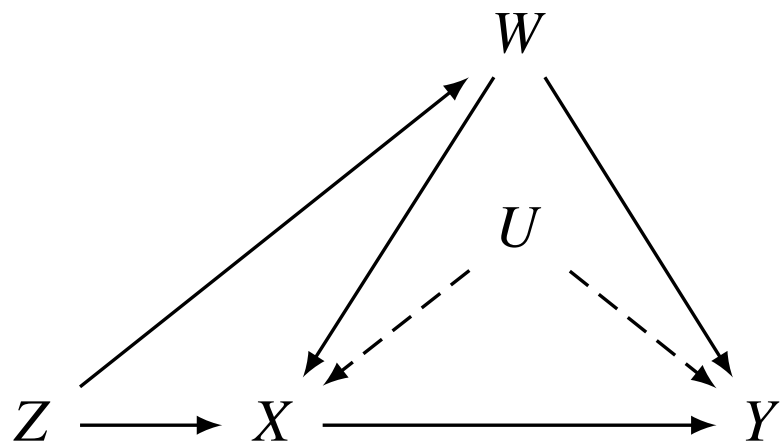
$$b = \frac{\partial}{\partial x}\mathbb{E}[Y \mid \mathrm{do}(x)] = \frac{\mathbb{E}[Y \mid z]}{\mathbb{E}[X \mid z]} = \frac{r_{ZY}}{r_{ZX}}$$

Definition (Conditional Instrumental Variable)

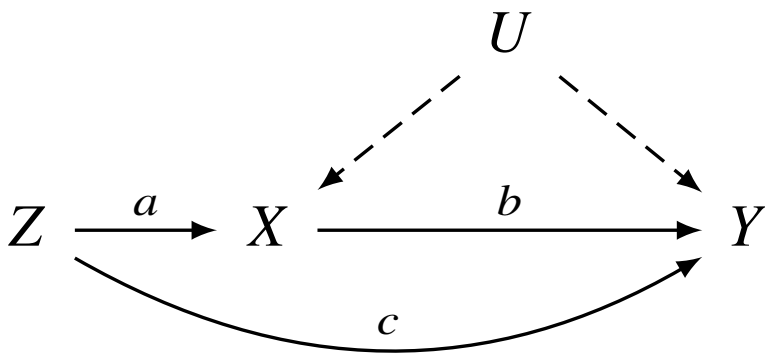
Z is an instrumental variable for identifying the effect of X on Y if there exists W such that

1. $(Z \not\perp X \mid W)_G$
2. $(Z \perp Y \mid W)_{G_{\overline{X}}}$ (or $Z \perp Y_x \mid W$)



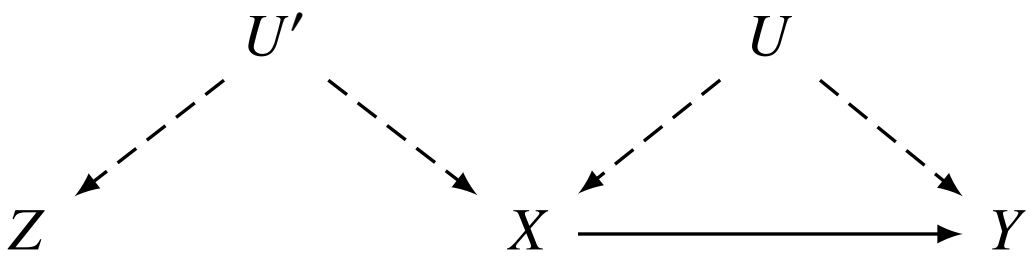


Remark

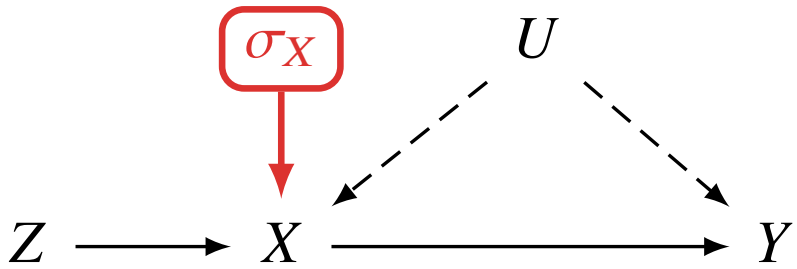


- 假设 Z 不经过 X 也可以影响到 Y. 则 Z 不能作为工具变量.
- $r_{ZX} = a$, $r_{ZY} = ab + c$
- $\hat{b} = \frac{r_{ZY}}{r_{ZX}} = \frac{ab+c}{a} = b + \frac{c}{a}$
- a 越小或 c 越大时, 偏差 $\frac{c}{a}$ 越大.

Remark: Z 不需要因果导致 X

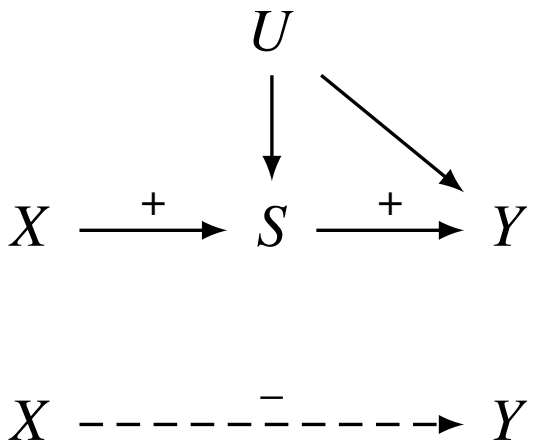


Remark: 决策论因果视角



- Z is associated with X: $Z \not\perp X \mid \sigma_X = \varnothing$
- $\{Z, U\}$ has fixed distribution: $\{Z, U\} \perp \sigma_X$
- Z is not associated with the confounding variable U: $Z \perp U \mid \sigma_X$
- Z affects Y only through X: $Y \perp Z \mid X, U, \sigma_X$
- Distribution of Y given $\{X, U\}$ same in all regimes: $Y \perp \sigma_X \mid X, U$

替代指标悖论 Surrogate Paradox

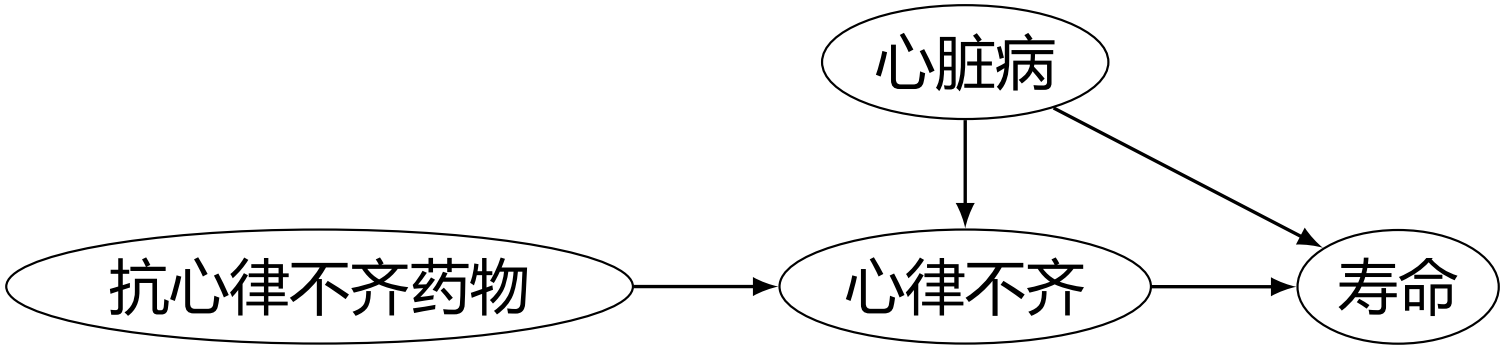


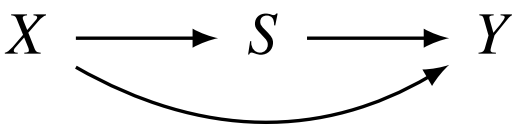
	$P(S=1 \mid U,X)$		$P(Y=1 \mid U,S)$	
	$X=0$	$X=1$	$S=0$	$S=1$
$U=0$	0.98	0.79	0.00	0.98
$U=1$	0.02	0.99	0.98	0.99

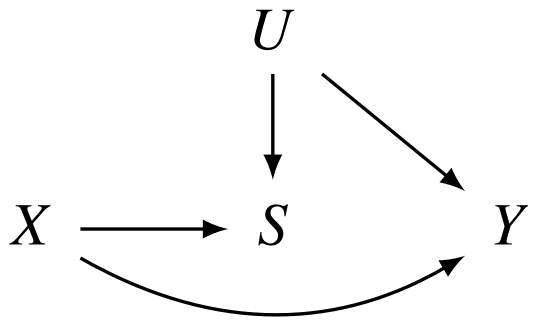
$P(U=1)=0.7 \quad P(X=1)=0.5$

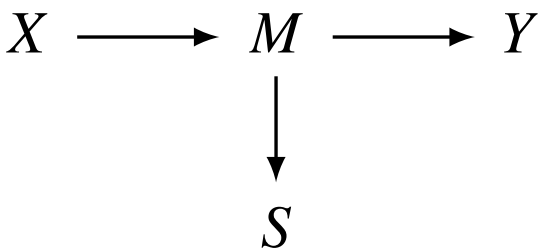
- $\text{TE}(X \to S) = P(S=1 \mid X=1) - P(S=1 \mid X=0) = 0.622$
- $\text{TE}(S \to Y) = P(Y=1 \mid \texttt{do}(S=1)) - P(Y=1 \mid \texttt{do}(S=0)) = 0.301$
- $\text{TE}(X \to Y) = \sum_u P(u)P(Y=1 \mid U=u, X=1) - \sum_u P(u)P(Y=1 \mid U=u, X=0) = \sum_u P(u) \sum_s P(S=s \mid U=u, X=1)P(Y=1 \mid U=u, S=s) - \sum_u P(u) \sum_s P(S=s \mid U=u, X=0)P(Y=1 \mid U=u, S=s) = -0.04907$

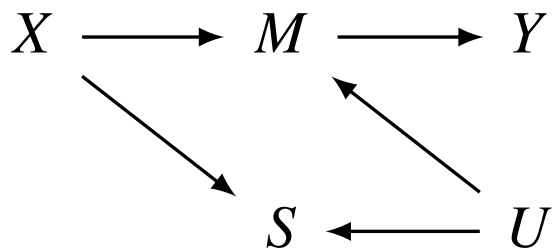
替代指标悖论 Surrogate Paradox





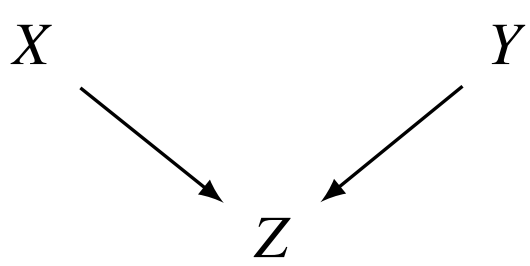


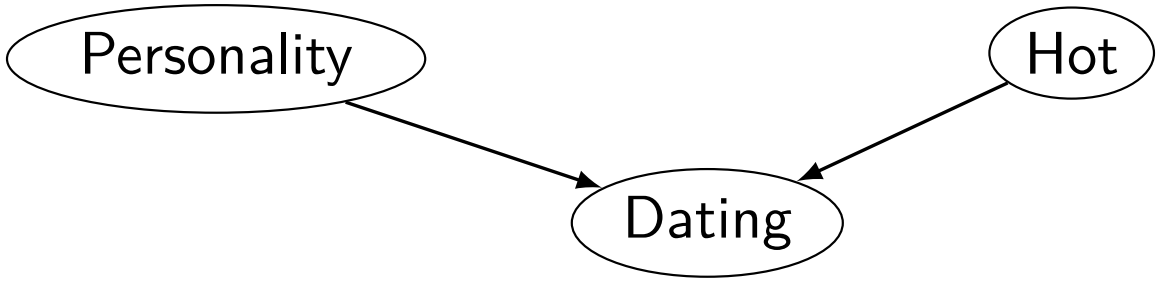




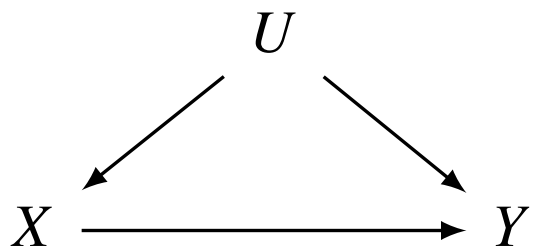
Paradoxes are the watchdogs of our hidden assumptions

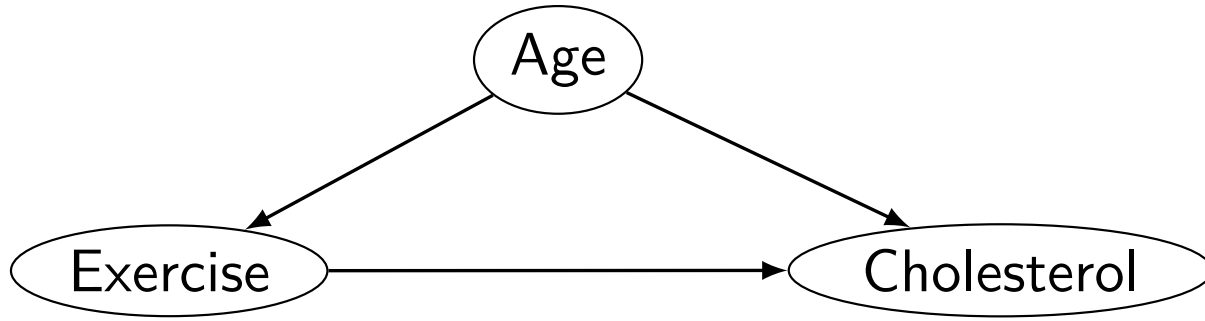
- **Berkson's Paradox**: Why hot guys tend to be jerks?



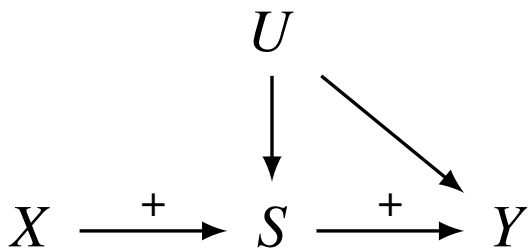


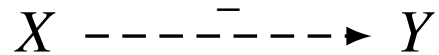
- **Simpson's Paradox**: A trend appears in several groups of data but disappears or reverses when the groups are combined.





- **Surrogate Paradox**





Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

- 据统计，因天花疫苗接种而死亡的人数比死于天花的人数还多.
- 是否应该停止接种天花疫苗?

圣经中的因果

1. 上帝准备毁灭 XX 城.
2. 亚伯拉罕: 你要将义人和恶人一同毁灭吗? 若城里有 50 个义人呢?
3. 上帝: 若城里有 50 个义人, 我就为他们的缘故饶恕那地方的人.
4. 亚伯拉罕: 若只有 40 个义人呢?
5. 上帝: 也不毁灭那城.
6. 亚伯拉罕: 30 个呢? 20 个呢? 10 个呢?
7. 上帝: 也不毁灭那城.

- 这究竟是一场怎样的 "反事实游戏"? 难道亚伯拉罕是在怀疑上帝数数的能力? 或者怀疑上帝明辨善恶的能力?
- 不. 亚伯拉罕其实是第一位科学家: 他试图探测的是 "集体惩罚的普遍准则". 科学关乎普遍规律, 而非特殊事件.

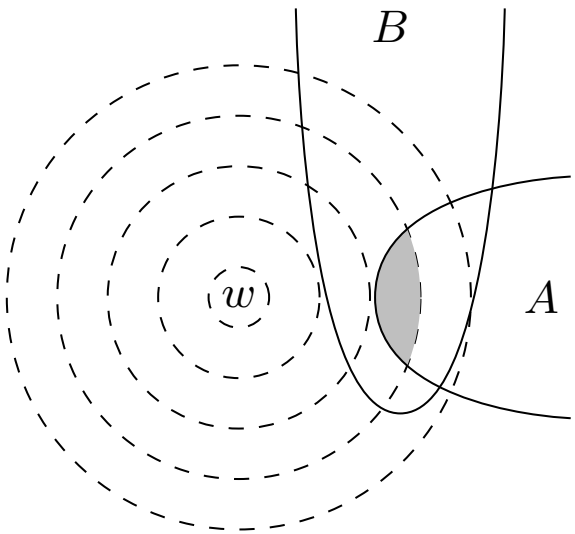
Counterfactuals are building blocks of scientific thought, free will and moral behavior.

— *Judea Pearl*

Philosophy — Counterfactual Approaches to Causality

- **休谟 Hume**: "我们给原因下一个定义, 一个对象跟随另一个对象, 而且, 凡与前一个对象类似的所有对象都和与后一个对象类似的那些对象处在类似的跟随关系中.
  换言之, 如果没有前一个对象, 那么后一个对象就不可能存在. (If the first object had not been, the second never had existed.)"
- **刘易斯 Lewis**: "A causes B, iff, B would not have occurred if not for A."

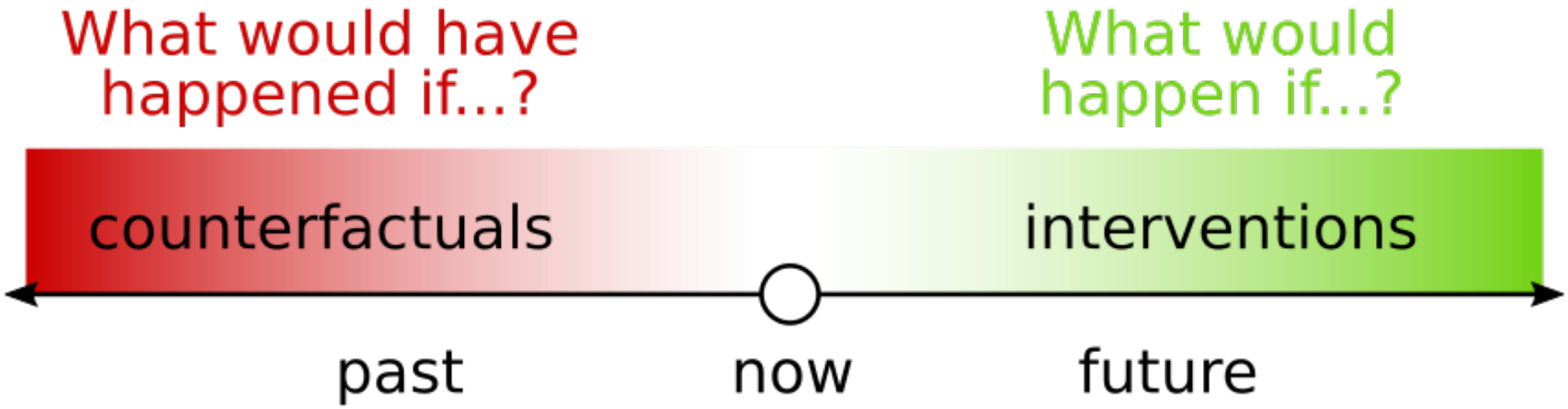
  If it were the case that A then it would be the case that B, iff, among all A-worlds some B-worlds are closer to the actual world than all $\neg B$-worlds.

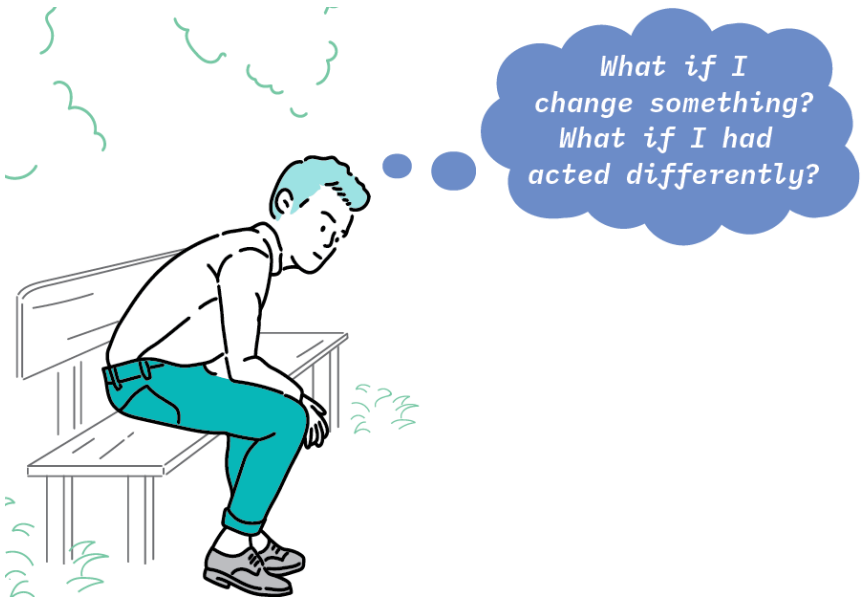


$$w \Vdash A \boxdot\!\!\rightarrow B \iff [\![A]\!]_w \subset [\![B]\!]$$

其中, $[\![A]\!]_w$ 为与 w 邻近的 A 世界集.

Counterfactual Causation





Two roads diverged in a yellow wood,
And sorry I could not travel both
And be one traveler, long I stood
And looked down one as far as I could
To where it bent in the undergrowth.

归因 Attribution

"Half the money I spend on advertising is wasted; the trouble is I don't know which half."

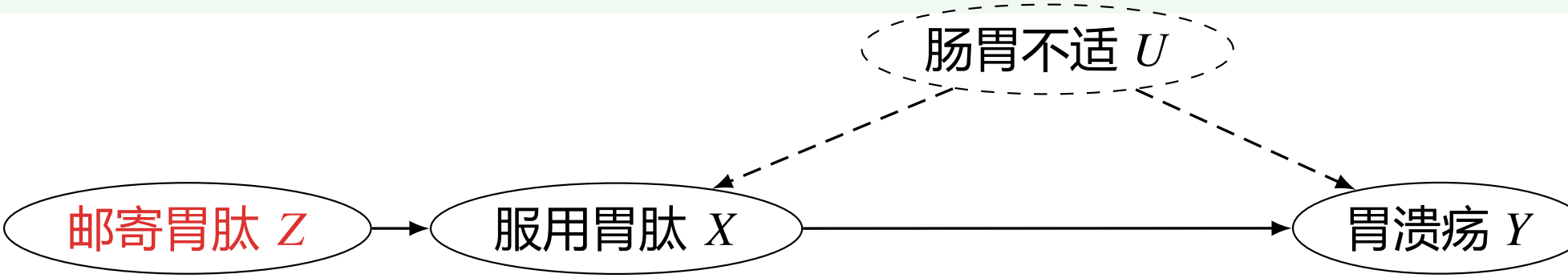
— *John Wanamaker*

- Your Honor! My client (Mr. A) died because he used this drug.
- Court to decide if it is more probable than not that Mr. A would be alive but for the drug!

$$P(\text{alive}_{\text{no-drug}} \mid \text{drug}, \text{dead}) \geq 0.5$$

What's the relationship between legal/moral and causal judgment?

因果效应、反事实 vs 法律责任



- 胃肽公司随机将样品寄给某地 10% 的人. 随后, 统计了每人是否收到了胃肽 Z、是否服用了胃肽 X、是否得了胃溃疡 Y. $P(Y, X \mid Z)$
- 根据统计数据, 服用胃肽 (收到胃肽) 与患胃溃疡高度相关:

$$P(Y=1 \mid X=1)=0.5, \qquad P(Y=1 \mid X=0)=0.26$$
$$P(Y=1 \mid Z=1)=0.81, \qquad P(Y=1 \mid Z=0)=0.36$$

- 据此, 患者将胃肽公司告上了法庭.
- 公司的辩护律师: 相关性源于混杂因子, 即患胃溃疡之前是否肠胃不适. 计算因果效应, 胃肽可以将患胃溃疡的概率降低至少 15%.

$$-0.23 \leq \mathrm{TE}(X \to Y) \leq -0.15$$

- 患者的辩护律师: 已知原告收到且服用了胃肽、得了胃溃疡, 若没有收到/服用胃肽、仍然会得胃溃疡的反事实概率最多为 7%.

$$P(Y_{Z=0}=0 \mid Z=1, X=1, Y=1) \geq 0.93$$
$$P(Y_{X=0}=0 \mid Z=1, X=1, Y=1) \geq 0.93$$

必要因 Necessary Cause

- "C is E's but-for cause: C is an antecedent but for which E would not have occurred."
- **例子:** Alice 用杂物堵了消防通道 $X = 1$, Bob 在火灾中找不到出口丧生 $Y = 1$.
  — 如果 Alice 没堵消防通道 $X = 0$, Bob 没死 $Y = 0$ 的概率有多高?

$$P(Y_{X=0} = 0 \mid X = 1, Y = 1) \qquad \text{(Probability of Necessity)}$$

- Without hindsight (knowing what happened in the actual world), there is no difference between $P(Y_{X=0} = 0)$ and $P(Y = 0 \mid \text{do}(X = 0))$.
- Suppose we observe that $X = 1$ and $Y = 1$ (hindsight). Then $P(Y_{X=0} = 0 \mid X = 1, Y = 1) \neq P(Y_{X=0} = 0 \mid X = 1)$.

充分因 Sufficient Cause

- **例子:** 杀手 Alice 朝 Bob 开了一枪, 没打中, 但 Bob 在逃跑途中被楼上掉落的花盆砸死.
- 设想 Alice 没开枪 $X = 0$, Bob 没死 $Y = 0$, 那么, Alice 开枪 $X = 1$ 会导致 Bob 死 $Y = 1$ 的概率有多高?

$$P(Y_{X=1} = 1 \mid X = 0, Y = 0) \qquad \text{(Probability of Sufficiency)}$$

Probability of Necessity and Sufficiency

1. Effect of Cause $\mathrm{TE}, \mathrm{CDE}, \mathrm{NDE}, \mathrm{NIE}, \mathrm{Exp\text{-}SE}_x, \mathrm{ETT}, \mathrm{PE}, \mathrm{PCE}, \ldots$
2. Cause of Effect $\mathrm{PN}, \mathrm{PS}, \mathrm{PNS}, \mathrm{PD}, \mathrm{PE}$

Remark: Absence of total effect does not imply absence of individual effects.

$$\mathrm{PN} := P(y'_{x'} \mid x, y) = \sum_{u: Y_{x'}(u) = y'} P(u \mid x, y)$$

$$\mathrm{PS} := P(y_x \mid x', y') = \sum_{u: Y_x(u) = y} P(u \mid x', y')$$

$$\mathrm{PNS} := P(y_x, y'_{x'}) = \sum_{u: Y_x(u) = y, Y_{x'}(u) = y'} P(u)$$

Probability of Disablement and Enablement

- Probability of Disablement

$$\mathrm{PD} := P(y'_{x'} \mid y)$$

- Probability of Enablement

$$\mathrm{PE} := P(y_x \mid y')$$

PN, PS, PNS

Theorem

$$\mathrm{PNS} = P(x, y)\,\mathrm{PN} + P(x', y')\,\mathrm{PS}$$

Proof.

The consistency condition $X = x \implies Y_x = Y$ tells us that

$$x \implies y_x = y \quad \text{and} \quad x' \implies y_{x'} = y$$

Hence

$$y_x \wedge y'_{x'} = (y_x \wedge y'_{x'}) \wedge (x \vee x') = (y \wedge x \wedge y'_{x'}) \vee (y_x \wedge y' \wedge x')$$

Therefore

$$\begin{aligned} P(y_x, y'_{x'}) &= P(y'_{x'}, x, y) + P(y_x, x', y') \\ &= P(y'_{x'} \mid x, y)P(x, y) + P(y_x \mid x', y')P(x', y') \end{aligned}$$

PNS is not identifiable, but can be bounded

$$\max\left\{0, \frac{P(y) - P(y_{x'})}{P(x,y)}\right\} \leq \text{PN} \leq \min\left\{1, \frac{P(y'_{x'}) - P(x',y')}{P(x,y)}\right\}$$

$$\max\left\{0, \frac{P(y_x) - P(y)}{P(x',y')}\right\} \leq \text{PS} \leq \min\left\{1, \frac{P(y_x) - P(x,y)}{P(x',y')}\right\}$$

$$\max\begin{Bmatrix} 0 \\ P(y_x) - P(y_{x'}) \\ P(y) - P(y_{x'}) \\ P(y_x) - P(y) \end{Bmatrix} \leq \text{PNS} \leq \min\begin{Bmatrix} P(y_x) \\ P(y'_{x'}) \\ P(x,y) + P(x',y') \\ P(y_x) - P(y_{x'}) + P(x',y) + P(x,y') \end{Bmatrix}$$

Theorem

If $\{Y_x, Y_{x'}\} \perp X$, *then*

$$\max\{0, P(y \mid x) - P(y \mid x')\} \leq \text{PNS} \leq \min\{P(y \mid x), P(y' \mid x')\}$$

$$\text{PN} = \frac{\text{PNS}}{P(y \mid x)}$$

$$\text{PS} = \frac{\text{PNS}}{P(y' \mid x')}$$

$$\text{PD} = \frac{P(x)\,\text{PNS}}{P(y)}$$

$$\text{PE} = \frac{P(x')\,\text{PNS}}{P(y')}$$

Theorem

If Y is monotonic relative to X, that is, $Y_1(u) \geq Y_0(u)$ for all u, (equivalently, $y'_x \wedge y_{x'} = 0$), then

$$\text{PNS} = P(y_x) - P(y_{x'})$$

$$\text{PN} = \frac{P(y) - P(y_{x'})}{P(x, y)}$$

$$\text{PS} = \frac{P(y_x) - P(y)}{P(x', y')}$$

Remark: The importance of $\{Y_x, Y_{x'}\} \perp X$ lies in permitting the identification of $\{P(y_x), P(y_{x'})\}$, since (using $x \implies y_x = y$)

$$P(y_x) = P(y_x \mid x) = P(y \mid x)$$

Then

$$\text{PNS} = P(y \mid x) - P(y \mid x')$$

Remark:

$$\text{PNS} = P(E \mid C) - P(E \mid \neg C) \qquad \text{PN} = \frac{\text{PNS}}{P(E \mid C)} \qquad \text{PS} = \frac{\text{PNS}}{P(\neg E \mid \neg C)}$$

The Two Fundamental Laws of Causal Inference

- The sentence $Y_x(u) = y$: "Y would be y in situation u, had X been x", means: the solution for Y in M_x with input $U = u$, is equal to y.

$$P(Y = y \mid \mathrm{do}(X = x)) := P_{M_x}(Y = y) = P(Y_x = y) = \sum_{u:Y_x(u)=y} P(u)$$

(I) The Law of Counterfactuals

$$Y_x(u) := Y_{M_x}(u)$$

(M generates and evaluates all counterfactuals.)

(II) The Law of Conditional Independence (d-separation)

$$(X \perp Y \mid Z)_G \implies (X \perp Y \mid Z)_P$$

(Separation in the model $\implies$ independence in the distribution.)

Computing Counterfactuals

Steps for Deterministic Counterfactuals — Deterministic

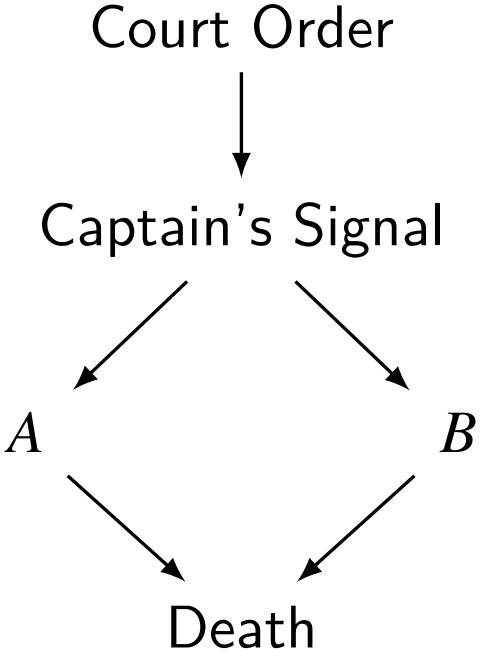
1. **Abduction:** Use the evidence $Z = z$ to determine the value of U
2. **Action:** Modify the SCM $M = (U, V, F)$, by replacing the structural equation for X with $X := x$, to obtain $M_x = (U, V, F_x)$
3. **Prediction:** Use the value of U from step 1 and the modified SCM M_x from step 2 to compute the value of Y_x

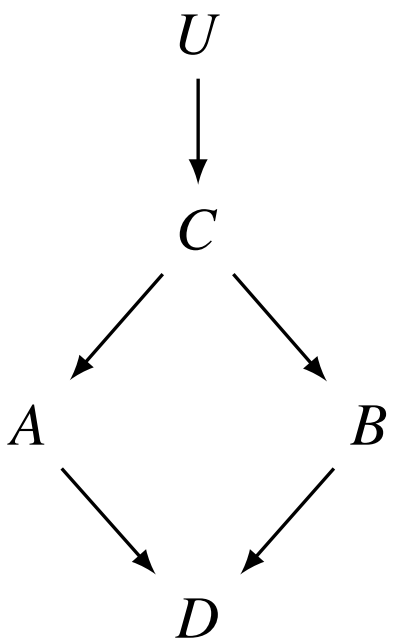
Steps for Probabilistic Counterfactuals — Probabilistic

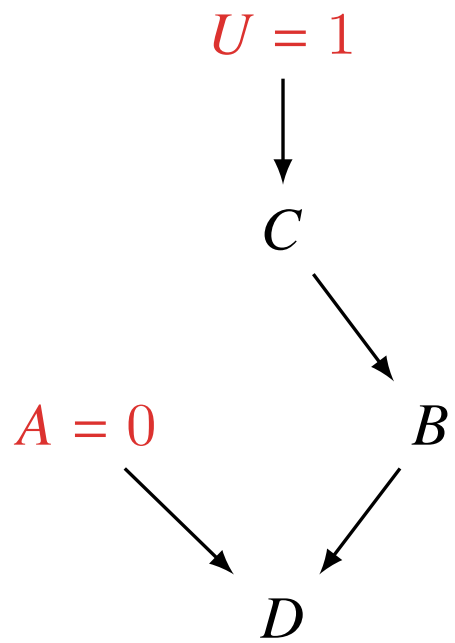
1. **Abduction:** Use the evidence $Z = z$ to update $P(u)$ to $P(u \mid z)$
2. **Action:** Modify the SCM $M = (U, V, F)$, by replacing the structural equation for X with $X := x$, to obtain $M_x = (U, V, F_x)$
3. **Prediction:** Use the modified model $(M_x, P(u \mid z))$ to compute the probability of Y_x

$$P(Y_x = y \mid Z = z) = \sum_{u : Y_x(u) = y} P(u \mid z)$$

Example: $M \models D \rightarrow (\neg A \mathrel{\square\!\!\rightarrow} D)$



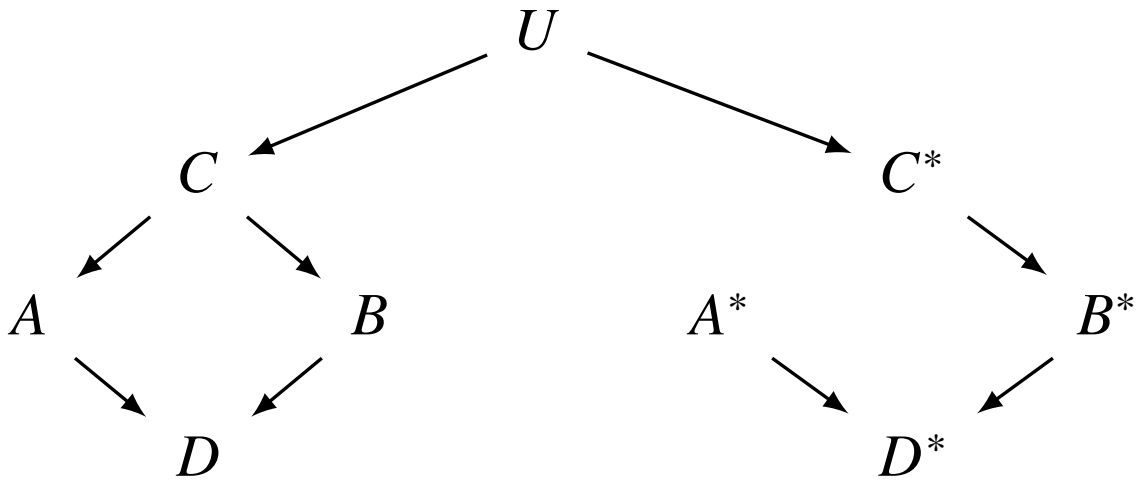




Model M
$C = U$
$A = C$
$B = C$
$D = A \vee B$
Facts: $D = 1$
Conclusions: U, C, A, B, D

Model $M_{A=0}$
$C = U$
$A = 0$
$B = C$
$D = A \vee B$
Facts: $U = 1$
Conclusions: $U, C, \neg A, B, D$

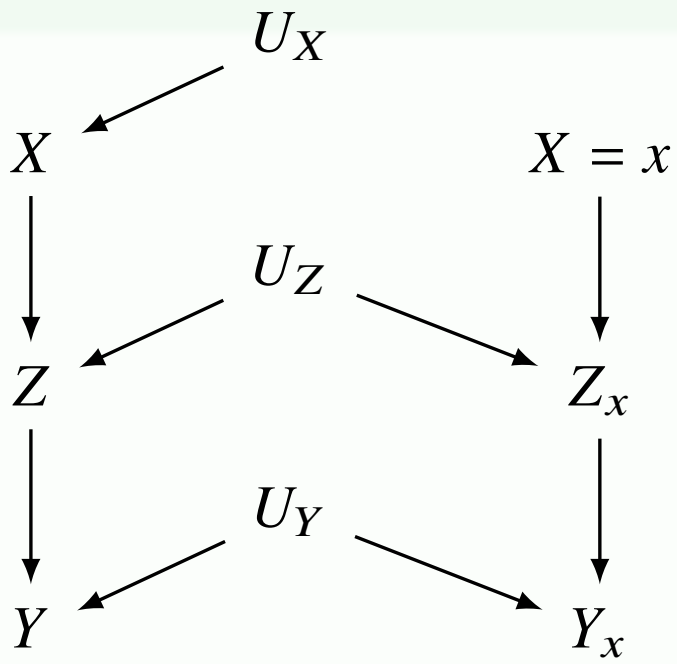
Twin Network

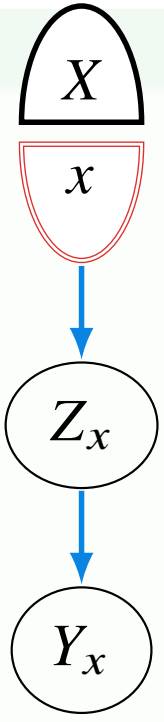


Remark

The standard rules of d-separation can be used to determine independence relations between variables in counterfactual queries.

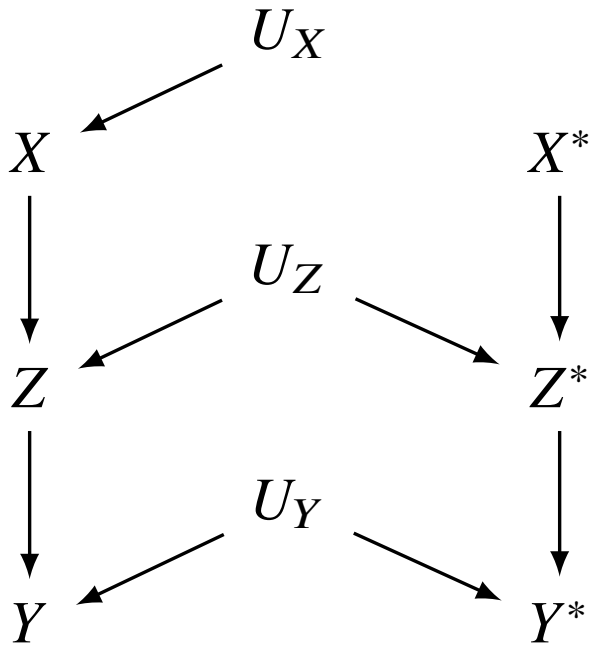
$$Y_x \perp X \quad \textit{but} \quad Y_x \not\perp X \mid Z$$





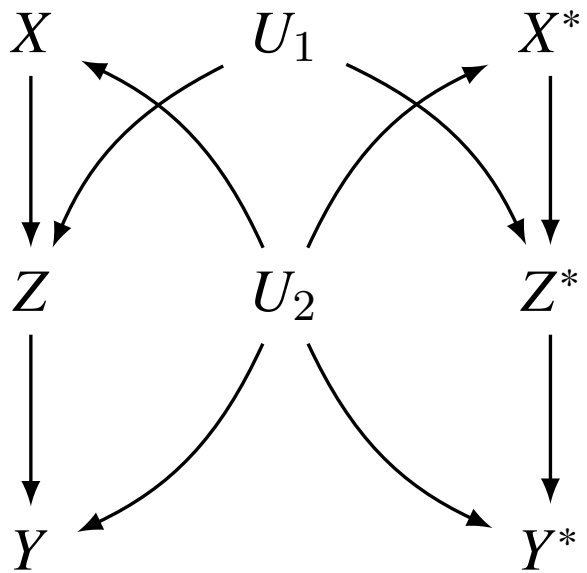
孪生网络

- 任何与 Y^* d-分离的变量集也与 U_Y d-分离.
- 由 $y = f_Y(\mathrm{pa}_Y, u_Y)$, Y_{pa_Y} 的概率等于 Pa_Y 固定为 pa_Y 时 Y 的概率.
- 如果 U_Y 服从某种独立关系, 那么 Y_{pa_Y} 也服从这种独立关系.



1. $U_Y \perp X \mid \{Y^*, Z^*\} \implies Y_z \perp X \mid \{Y_x, Z_x\}$
2. $U_Y \perp U_Z \mid \{Y, Z\} \implies Y_z \perp Z_x \mid \{Y, Z\}$
3. $Y^* \perp X \mid \{Z, U_Z, Y\} \implies Y_x \perp X \mid \{Z, Z_x, Y\}$
4. $Y^* \perp X \mid \{Y, U_Y, U_Z\} \implies Y_x \perp X \mid \{Y, Y_z, Z_x\}$

孪生网络 — Example

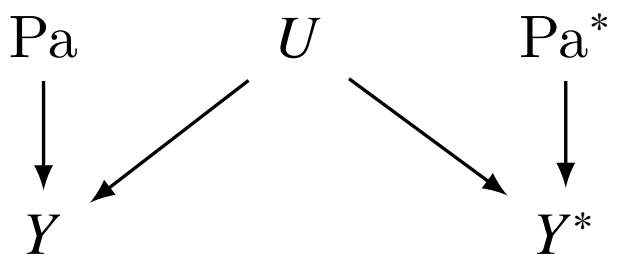


1. $U_1 \perp U_2 \mid \{Z, X\} \implies Y_z \perp Z_x \mid \{Z, X\}$
2. $Y_z \not\perp Z_x \mid Z$

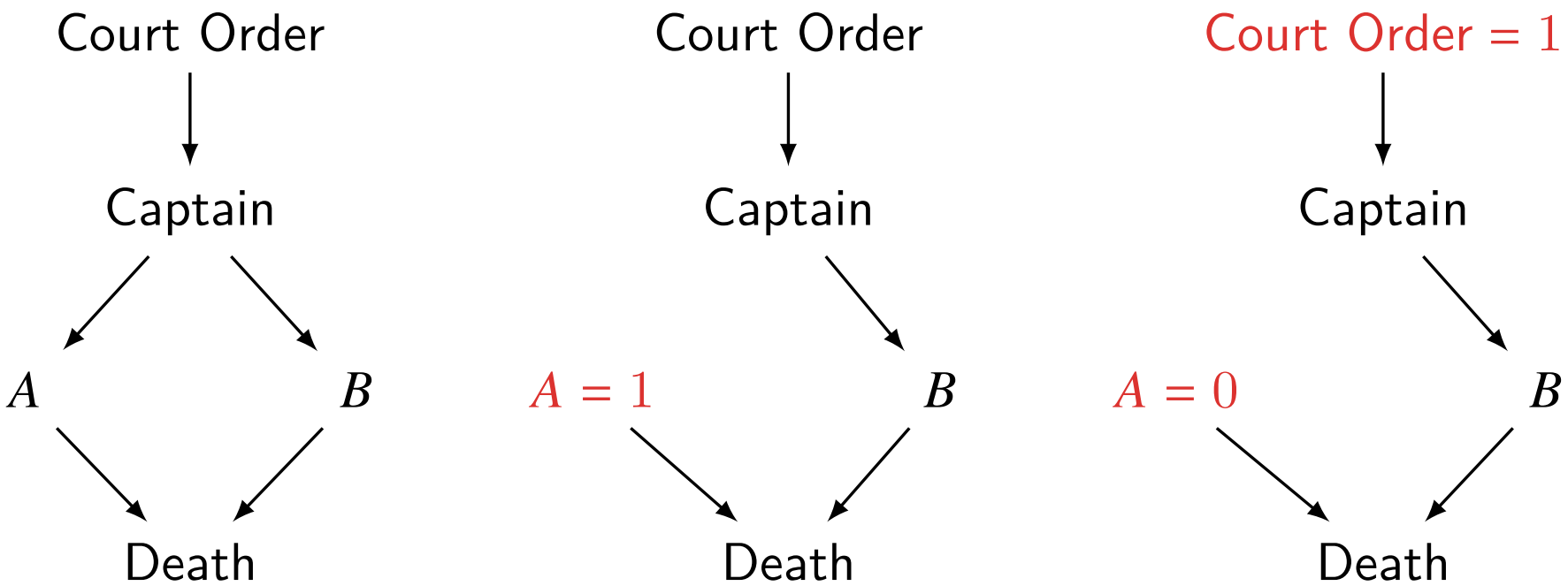
Twin Network — Counterfactual as functions

- Computationally we can write the three step counterfactual process in one single functional assignment.
  1. Abduction: $u = f_Y^{-1}(y, \mathrm{pa})$ by inverting the mechanism $y = f_Y(\mathrm{pa}, u)$
  2. Action: intervene on the parents $\mathrm{Pa} := \mathrm{pa}^*$
  3. Prediction: $y^* = f_Y(\mathrm{pa}^*, u)$
- becomes

$$y^* = f(\mathrm{pa}^*, y, \mathrm{pa})$$

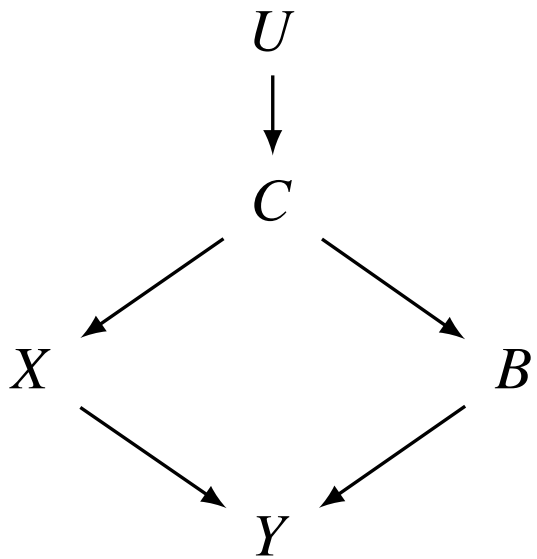


Why firing squads exist?



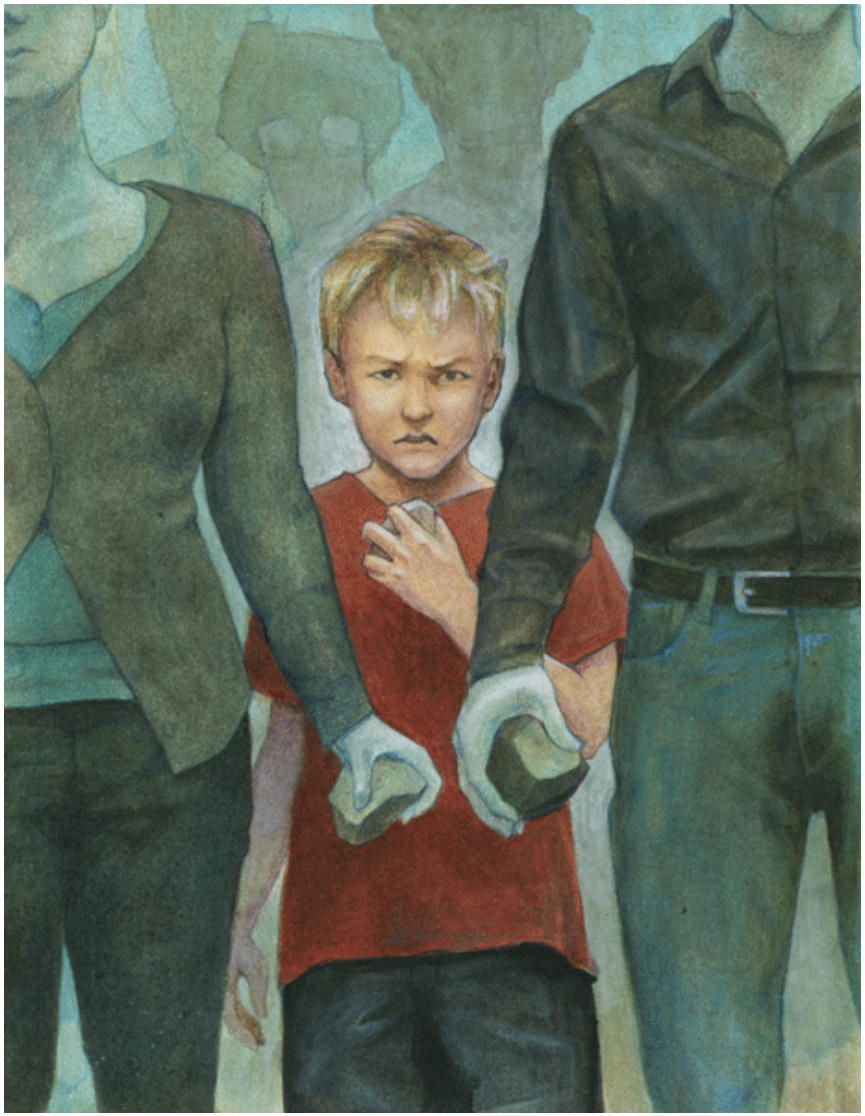
- 减轻行刑者的责任.
- “我不开枪他也会死.”

Why firing squads exist?



- Assume $P(u) = \frac{1}{2}$.
- $P(y_x) = P(Y_x(u) = 1)P(u) + P(Y_x(u') = 1)P(u') = \frac{1}{2}(1 + 1) = 1$
- $P(y_{x'}) = P(Y_{x'}(u) = 1)P(u) + P(Y_{x'}(u') = 1)P(u') = \frac{1}{2}(1 + 0) = \frac{1}{2}$
- $\mathrm{PN} = P(y'_{x'} \mid x, y) = P(y'_{x'} \mid u) = 0$
- $\mathrm{PS} = P(y_x \mid x', y') = P(y_x \mid u') = 1$
- $\mathrm{PNS} = P(y_x, y'_{x'}) = P(y_x, y'_{x'} \mid u)P(u) + P(y_x, y'_{x'} \mid u')P(u') = \frac{1}{2}(0 + 1) = \frac{1}{2}$

The Banality of Evil — Hannah Arendt

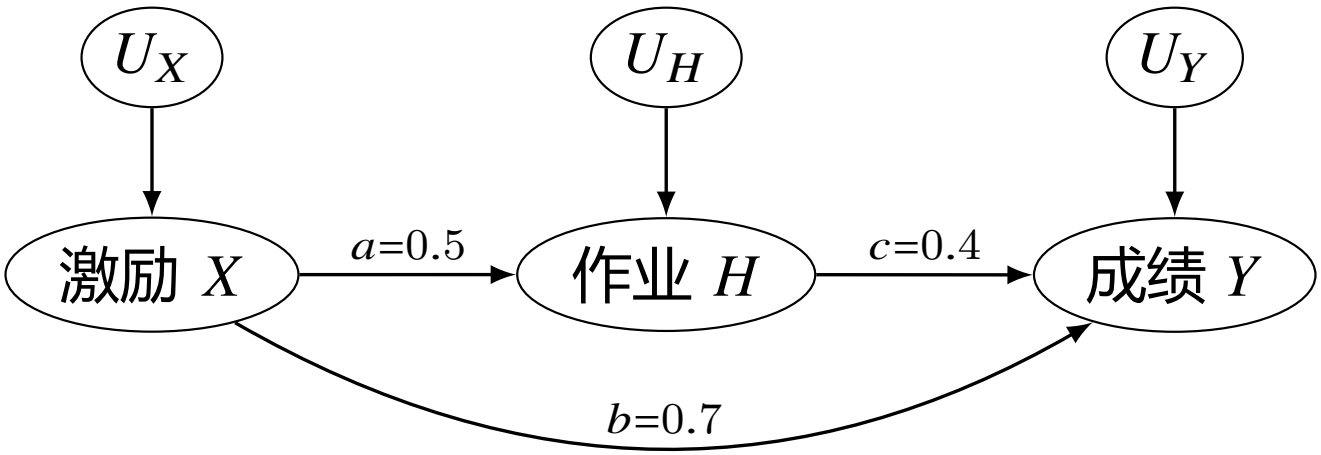


- 个体责任
- 群体责任 (conjunctive/disjunctive scenario)
- 个体成员的责任怎么判定?
- 个体间无知怎么办?

The understanding of mathematics is necessary for a sound grasp of ethics.

— *Socrates*

Example



- 事实: $X = 0.5,\ H = 1,\ Y = 1.5$
- 问题: 如果作业量翻倍, 成绩会提高吗?

$$X = U_X \qquad U_X = 0.5$$

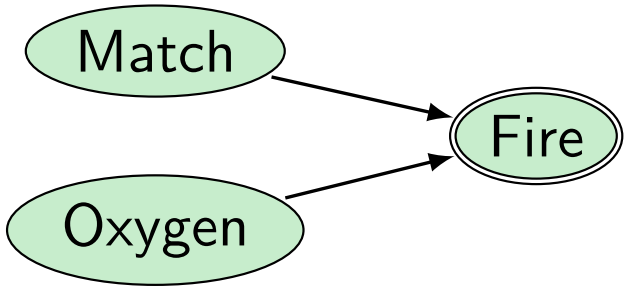
$$H = aX + U_H \qquad U_H = 1 - 0.5 \cdot 0.5 = 0.75$$

$$Y = bX + cH + U_Y \qquad U_Y = 1.5 - 0.7 \cdot 0.5 - 0.4 \cdot 1 = 0.75$$

$$Y_{H=2}(U_X = 0.5, U_H = 0.75, U_Y = 0.75) = 0.5 \cdot 0.7 + 2 \cdot 0.4 + 0.75 = 1.9$$

Question

- Why do we consider striking a match to be a more adequate explanation (of a fire) than the presence of oxygen?



- Since both explanations are necessary for the fire, $\mathrm{PN}(\mathrm{match}) = \mathrm{PN}(\mathrm{oxygen}) = 1$.
- If the probabilities associated with striking a match and the presence of oxygen are denoted p_{match} and p_{oxygen}, then

$$\mathrm{PS}(\mathrm{match}) = p_{\mathrm{oxygen}} \quad \text{and} \quad \mathrm{PS}(\mathrm{oxygen}) = p_{\mathrm{match}}$$

- The fact that $p_{\mathrm{oxygen}} > p_{\mathrm{match}}$ endows the match with greater explanatory power than the oxygen.

Question: What weight should we assign to the necessary versus the sufficient component of causation in legal/moral situations?

Causation + Foreseeability of consequences + Intention $\propto$ Responsibility?

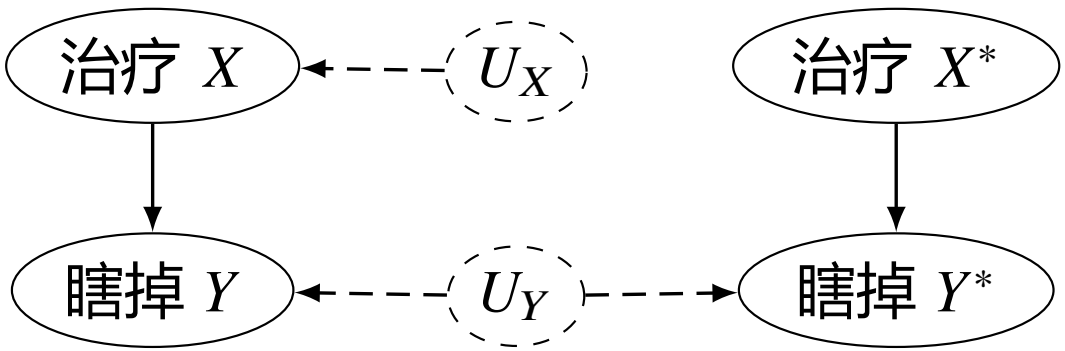
因果效应、反事实 vs 法律责任 — Example

有一种眼疾,

- 99% 的患者, 治疗后眼睛会康复; 不接受治疗, 会瞎掉.
- 1% 的患者, 治疗后眼睛会瞎掉; 不接受治疗, 反而会自动康复.

小明去医院, 医生给他做了治疗 $X = 1$, 结果眼睛瞎掉了 $Y = 1$.

$$X = U_X$$
$$Y = X \cdot U_Y + (1 - X) \cdot (1 - U_Y)$$



"如果不接受治疗 $X = 0$, 小明的眼睛会瞎吗 $Y = 1$?"

$$P(Y_{X=0} = 0 \mid X = 1, Y = 1) = 1$$

虽然如此, 由于 U_Y 不可见, 且 $P(Y = 0 \mid \mathrm{do}(X = 1)) = 0.99$,
$P(Y = 0 \mid \mathrm{do}(X = 0)) = 0.01$, 所以, 医生的行为不需要承担任何责任.

Effect of Treatment on the Treated

- Total Effect

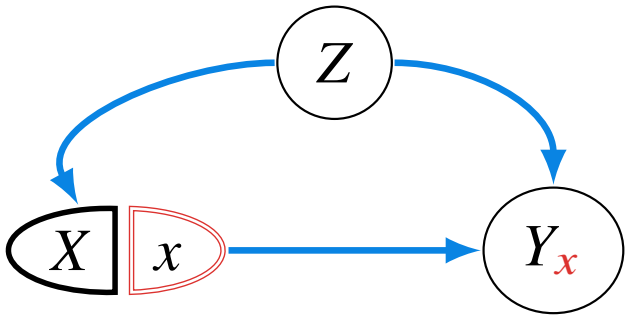
$$\mathrm{TE} = \mathbb{E}[Y_1 - Y_0]$$

- Effect of Treatment on the Treated

$$\mathrm{ETT} := \mathbb{E}[Y_1 - Y_0 \mid X = 1] = \mathbb{E}[Y \mid X = 1] - \mathbb{E}[Y_0 \mid X = 1]$$

- If Z satisfies the backdoor condition relative to (X, Y), then

$$\begin{aligned}
&P(Y_0 = y \mid X = 1) \\
&= \sum_m P(Y_0 = y \mid X = 1, Z = z) P(Z = z \mid X = 1) \\
&= \sum_m P(Y_0 = y \mid X = 0, Z = z) P(Z = z \mid X = 1) && (Y_x \perp X \mid Z) \\
&= \sum_m P(Y = y \mid X = 0, Z = z) P(Z = z \mid X = 1) && (X = x \implies Y_x = Y)
\end{aligned}$$



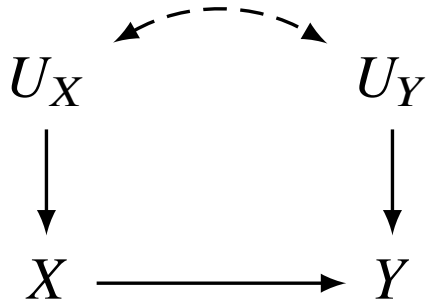
中医“治未病”

- 中医“治未病”，将“未病”之人调理的更健康!
- 有没有可能，找中医调理的“未病”之人有更强的健康管理意识？即使他们不看中医，也会通过其它方式养生锻炼，从而更健康？
- 中医怎么辩护自己有“治未病”的功效？

$$\mathrm{ETT} = \mathbb{E}\left[Y_{\text{看中医}} - Y_{\text{不看中医}} \mid X = \text{看中医}\right]$$

ETT — 选择自由 & 因果效应

- 如果一个人知道潜在结果, 其自由 (而不是被迫) 选择的 X 受其对潜在结果的认识的影响, 那么在 X 和 Y 之间就存在混杂因子.



$Y_x \not\perp X$

极端一点儿: $X = \operatorname{argmax}_x Y_x$, 其中 $Y_x = f_Y(x, U_Y)$

Remark: 人基于对潜在结果的认识做出的决策优化, 会使得 $Y_x \not\perp X$.

- **Example:** 从未来收入的角度看, 读研究生是否值得?

$$\mathbb{E}[Y \mid \mathrm{do}(X = 1), X = 1] - \mathbb{E}[Y \mid \mathrm{do}(X = 0), X = 1] \quad \times$$

$$\mathbb{E}[Y_1 - Y_0 \mid X = 1] \quad \checkmark$$

Remark: do 算子强制大家读研或不读, 只涉及干预后的世界, ETT 涉及两个不同的世界.

一般 $P(y_x \mid z) \neq P(y \mid \mathrm{do}(x), z)$. 当 Z 不是 X 的后代时, 二者相等.

ETT — 选择自由 & 因果效应

$$\mathrm{TE} := \mathbb{E}[Y_1 - Y_0] \quad vs \quad \mathrm{ETT} := \mathbb{E}[Y_1 - Y_0 \mid X = 1]$$

- TE 当你强迫别人必须选择 $X = 1$ 或 $X = 0$ 时
- ETT 当人们有自由自主选择 X 时

1. 当 $\mathrm{ETT} > \mathrm{TE}$ 时, 可以优化选择
2. 当 $\mathrm{ETT} < \mathrm{TE}$ 时, 可以通过选择干预以损害他人
3. 当 $\mathrm{ETT} = \mathrm{TE}$ 时, 没有根据潜在结果做选择/没有混杂

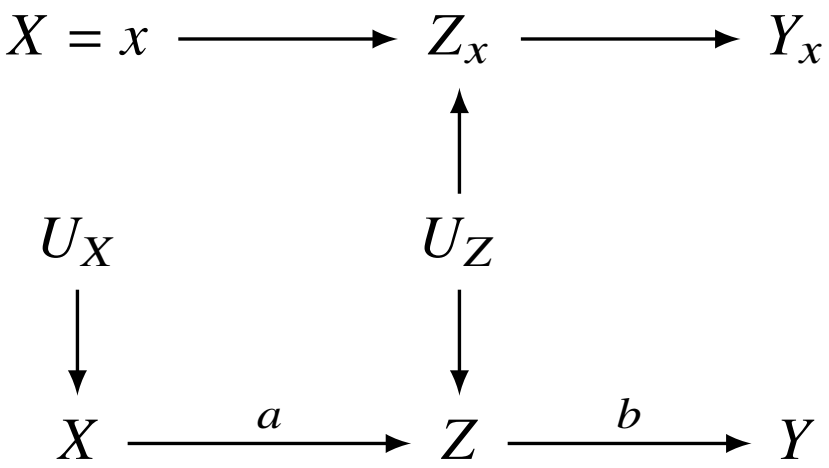
干预 vs 反事实

$$P(Y_{X=0} \mid X=1, Y=1) \quad \text{vs} \quad P(Y \mid \text{do}(X=0))$$

- $P(Y_{X=0} \mid X=1, Y=1)$ is about estimation of a quantity in one world conditioned on observations in another world.
- $P(Y \mid \text{do}(X=0))$ is about estimation of a quantity in one world conditioned on intervention in the same world.
- $P(Y \mid \text{do}(X=0))$ is about groups of units.
- $P(Y_{X=0} \mid X=1, Y=1)$ is about a specific unit.
- RCT will get us $P(Y \mid \text{do}(X=x))$, but not $P(Y_{X=x'} \mid X=x, Y=y)$.
- $P(Y \mid \text{do}(X=x'))$ is the average of counterfactuals over the observable population.

$$\begin{aligned} & P(Y \mid \text{do}(X=x')) \\ =& \int_{x,y} P(Y_{X=x'} \mid X=x, Y=y) P(x,y) \, \mathrm{d}x \, \mathrm{d}y \\ =& \mathbb{E}_{P_{X,Y}} \left[P(Y_{X=x'} \mid X=x, Y=y) \right] \end{aligned}$$

干预 vs 反事实



$$X = U_X$$
$$Z = aX + U_Z$$
$$Y = bZ$$

$$X \perp Y \mid Z$$

$$X \not\perp Y_x \mid Z$$

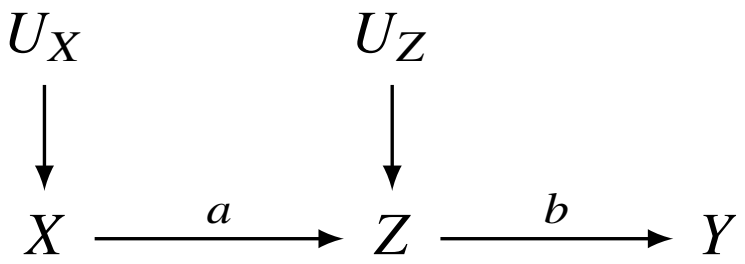
$$\mathbb{E}[Y \mid X, Z] = \mathbb{E}[Y \mid Z]$$

$$\mathbb{E}[Y_x \mid X, Z] \neq \mathbb{E}[Y_x \mid Z]$$

$$\mathbb{E}[Y \mid \mathsf{do}(X = 1), Z = 1] = \mathbb{E}[Y \mid \mathsf{do}(X = 0), Z = 1] = b$$

$$\mathbb{E}[Y_{X=1} \mid Z = 1] \neq \mathbb{E}[Y_{X=0} \mid Z = 1]$$

干预 vs 反事实



$$X = U_X$$
$$Z = aX + U_Z$$
$$Y = bZ$$

- $Z = 1$ selects a subset of the population in which we examine the effect of intervening on X. $Z = 1$ and $X = 1$ refer to different worlds. Assume $0 < a < 1$. If $Z = aX + U_Z = 1$, then $U_Z = 1$.

$$\mathbb{E}[Y_{X=1} \mid Z = 1] = b(a + U_Z) = b(a + 1)$$
$$\mathbb{E}[Y_{X=0} \mid Z = 1] = bU_Z = b$$

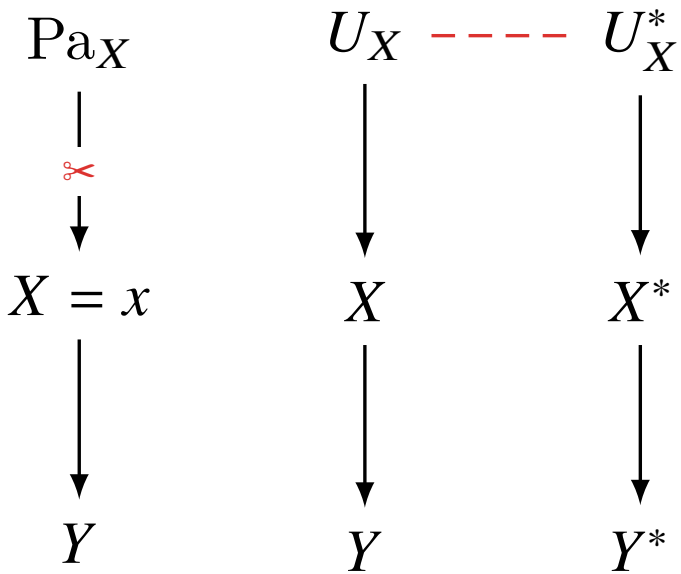
- Can counterfactual encode a do-expression? Yes.

$$\text{since} \quad P(Y = y \mid \text{do}(X = x), Z = z) = \frac{P(Y = y, Z = z \mid \text{do}(X) = x)}{P(Z = z \mid \text{do}(X) = x)}$$

$$\text{we have} \quad \mathbb{E}[Y \mid \text{do}(X = x), Z = z] = \mathbb{E}[Y_x \mid z_x]$$

干预反事实 vs 回溯反事实

- In a deterministic world, for events to have been different,
  1. either the laws of nature would have had to be violated, or
  2. the background conditions would have had to be different.

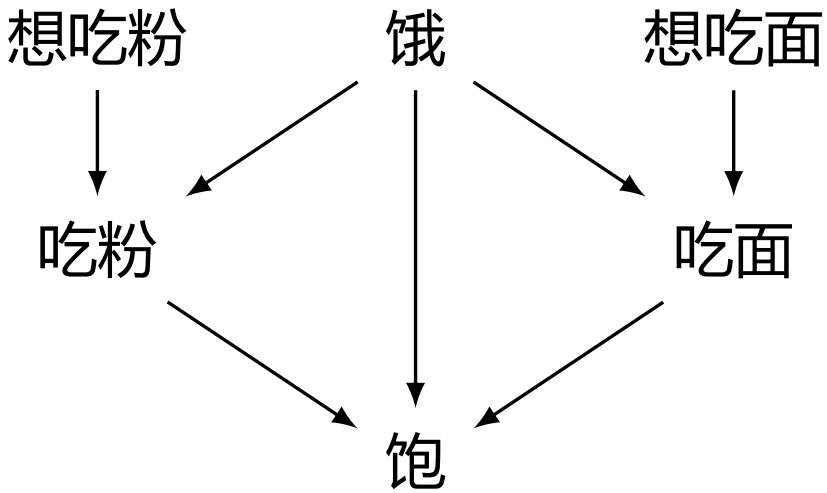


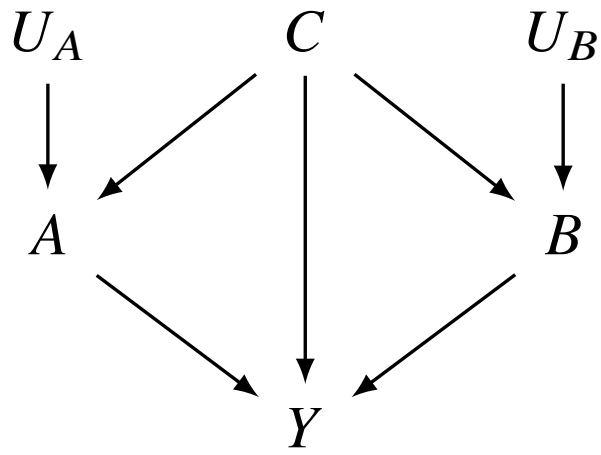
- David Lewis: counterfactuals are to be evaluated by imagining "small miracles": that ensure those events which are counter-to-fact to occur by locally violating the laws of nature, thereby disconnecting these events from their causes.
- Pearl's interventionist counterfactuals: "minisurgeries".

Problem (吃粉肚子饱了. 如果没吃粉, 你的肚子会饱吗?)

干预反事实: 绝不会

回溯反事实: 可能会





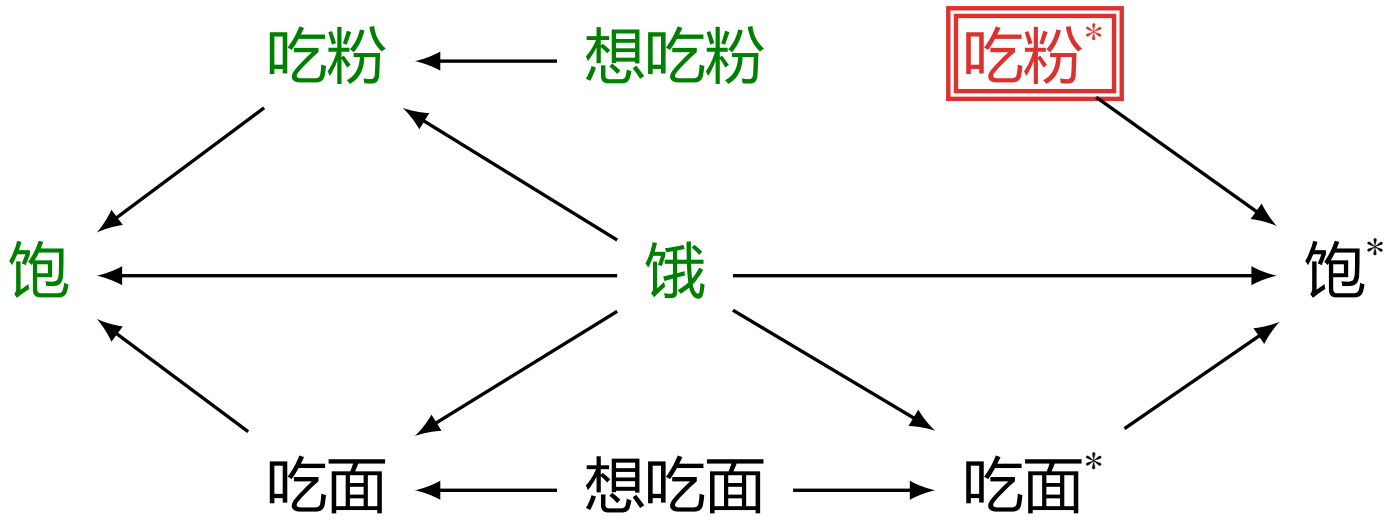
$$A = C \wedge U_A$$

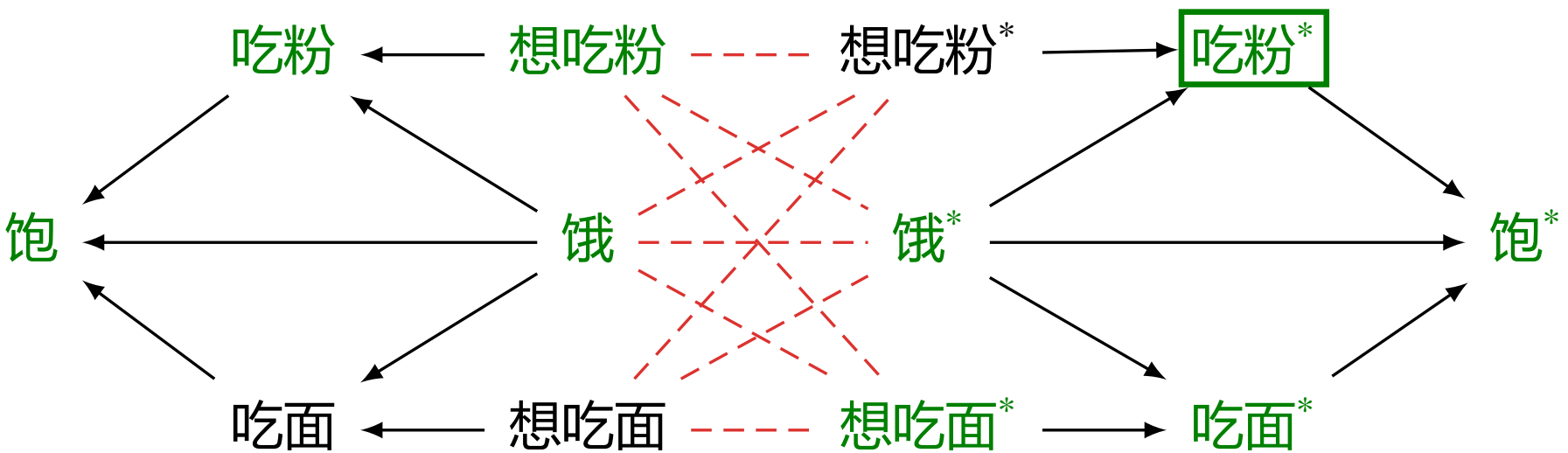
$$B = C \wedge U_B$$

$$Y = C \rightarrow A \vee B$$

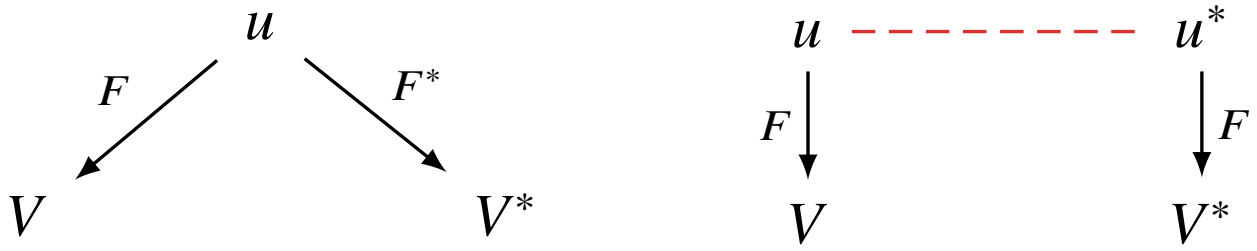
$$C = 1, U_A = 1, U_B = 0, A = 1, B = 0, Y = 1$$

干预反事实 vs 回溯反事实





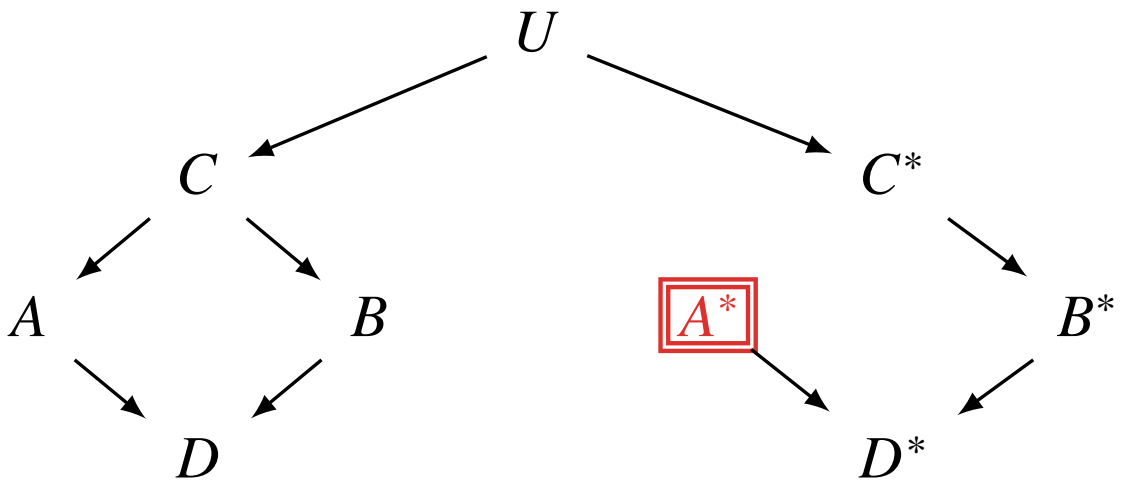
干预反事实 vs 回溯反事实 [KMB23]



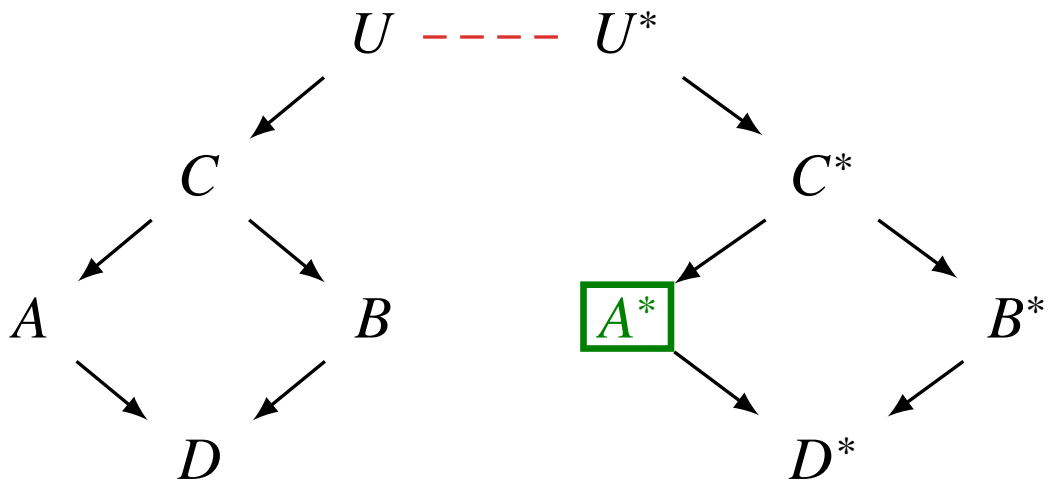
1. **干预反事实:** 现实世界和反事实世界共享相同的背景条件 u.
   现实结果 V 和反事实结果 V^* 之间的潜在矛盾, 通过机制 F 的变化 (通过干预得到修改后的机制 F^* 和子模型 M^*) 得以解决.
2. **回溯反事实:** 现实世界和反事实世界共享相同的不变机制 F, 而各自的背景条件 u 和 u^* 可能不同.

1. what would Y have been, had X been set to be $X = x'$
2. what would Y have been, had X instead been observed to be $X^* = x'$

干预反事实 vs 回溯反事实 — Example

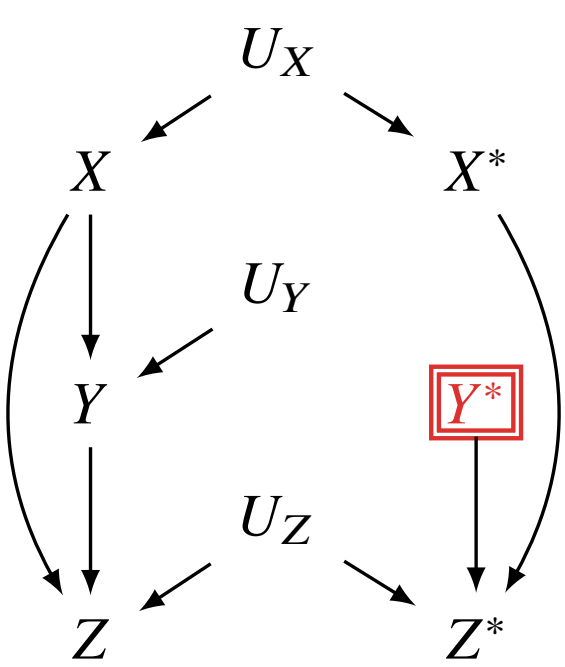


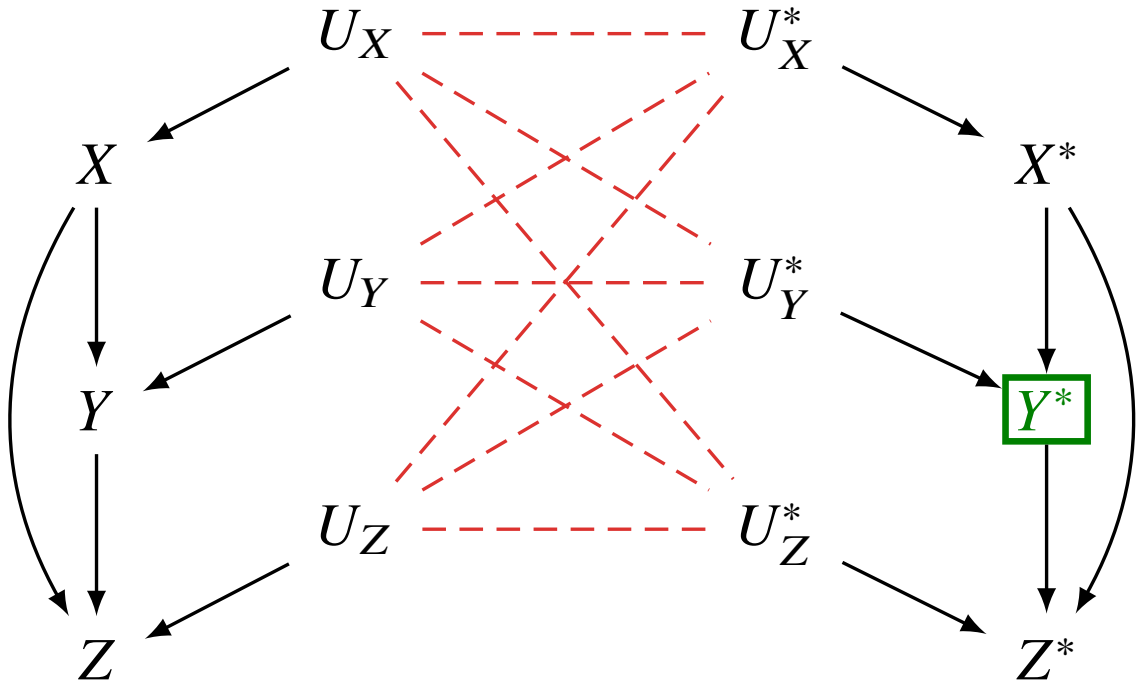
$$D = 1 \implies U = 1 \implies (M_{A^*=0}, U = 1) \models D^* = B^* = C^* = U = 1$$



$$A^* = 0 \implies U^* = 0 \implies (M, U^* = 0) \models D^* = B^* = C^* = U^* = 0$$

干预反事实 vs 回溯反事实 — Example





$$X = U_X \qquad X^* = U_X \qquad X^* = U_X^*$$

$$Y = X + U_Y \qquad Y^* = 3 \qquad Y^* = X^* + U_Y^* = 3$$

$$Z = X + Y + U_Z \qquad Z^* = X^* + Y^* + U_Z \qquad Z^* = X^* + Y^* + U_Z^*$$

Facts: $X = 1, Y = 2, Z = 2 \implies U_X = 1, U_Y = 1, U_Z = -1$

- **Interventional counterfactual:** $M_{Y^*=3}, (1, 1, -1) \models (X^*, Z^*) = (1, 3)$
- **Backtracking counterfactual:** there can be multiple ways of U^*

  $Y^* = 3 \implies M, (U_X^*, 3 - U_X^*, U_Z^*) \models (X^*, Z^*) = (U_X^*, U_X^* + 3 + U_Z^*)$

Computing Backtracking Counterfactuals

1. **Cross-World Abduction:** Update $P_B(u^*, u) := P(u)P_B(u^* \mid u)$ by the evidence (x^*, z) to obtain

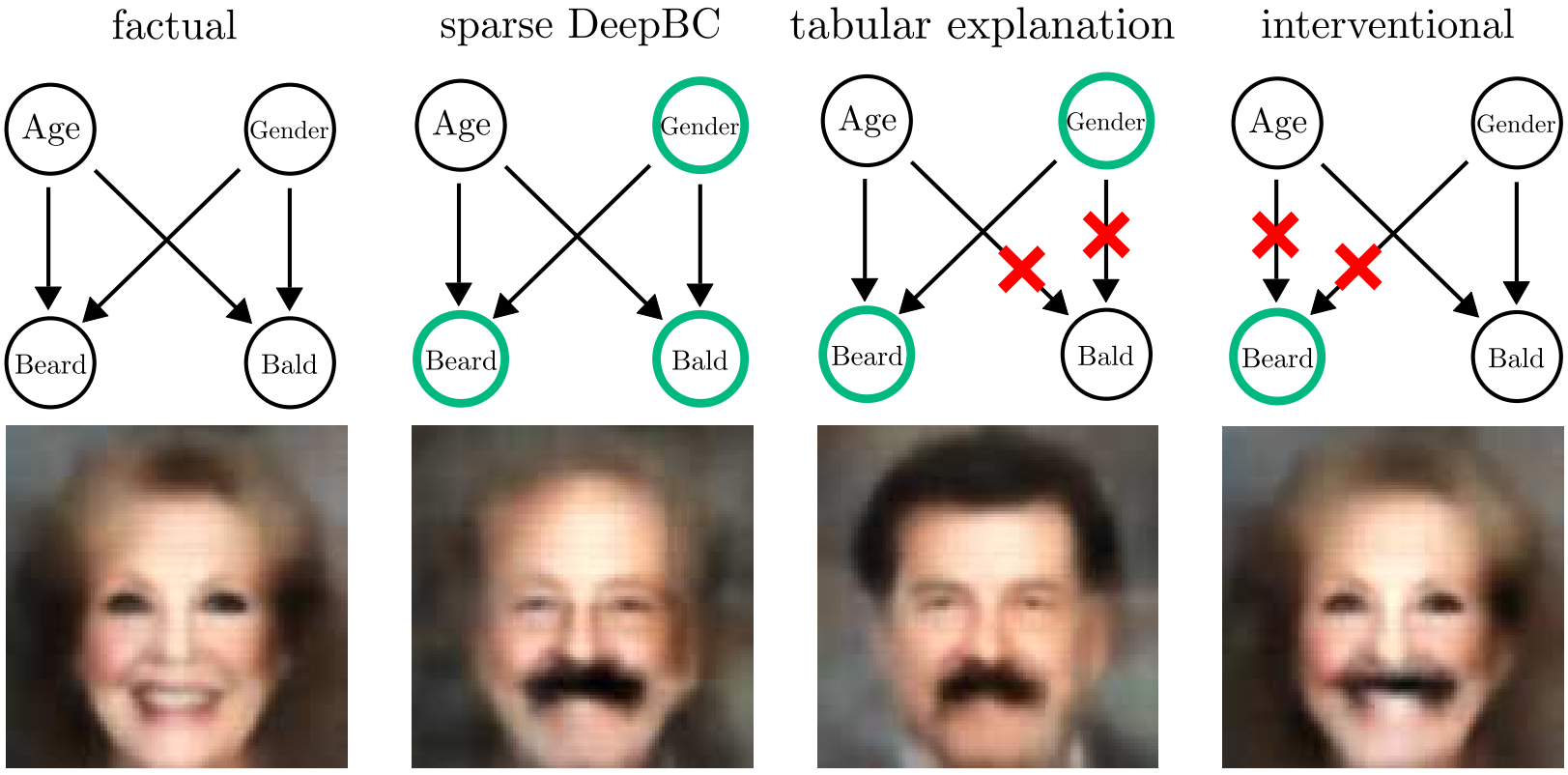
$$P_B(u^*, u \mid x^*, z) = \frac{P_B(u^*, u)[\![X^*(u^*) = x^*]\!][\![Z(u) = z]\!]}{\sum_{\substack{u^*: X^*(u^*) = x^* \\ u: Z(u) = z}} P_B(u^*, u)}$$

2. **Marginalisation:**

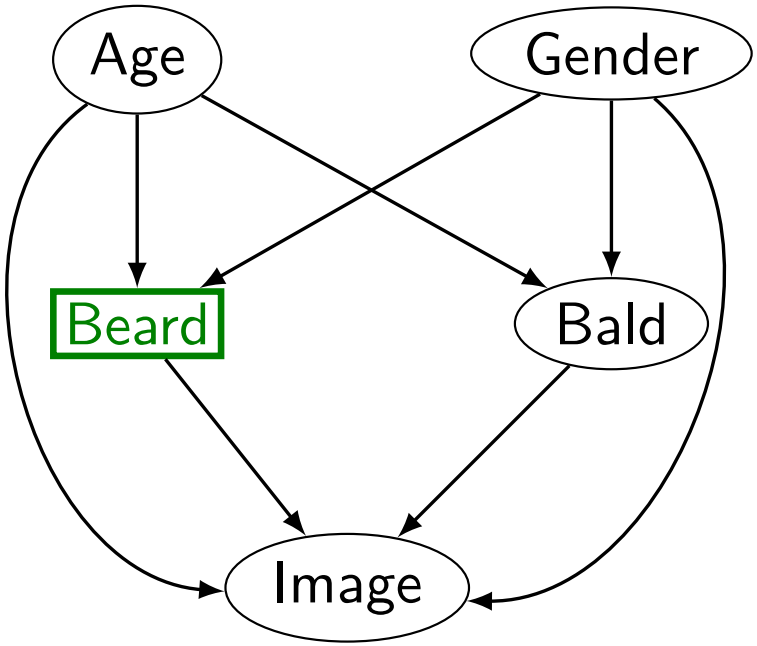
$$P_B(u^* \mid x^*, z) = \sum_u P_B(u^*, u \mid x^*, z)$$

3. **Prediction:** Use the model $(M, P_B(u^* \mid x^*, z))$ to predict Y^*:

$$P_B(y^* \mid x^*, z) = \sum_{u^*: Y^*(u^*) = y^*} P_B(u^* \mid x^*, z)$$



What would have been, had this person had a beard?



Backtracking Counterfactuals for XAI

Given

- a probabilistic causal model $(M, P(U))$ over variables $X \cup \{Y\}$ with laws such that $Y = f(X)$;
- a backtracking conditional $P_B(U^* \mid U)$, e.g., distance-based.

Then "x rather than x' explains why $f(x) = y$ rather than $y' \neq y$" if such a change to y' would be most likely to have come about through x',

$$x' \in \operatorname*{argmax}_{x'} P_B(x' \mid y', x, y)$$

Remark: 反事实解释 = 最大后验回溯反事实

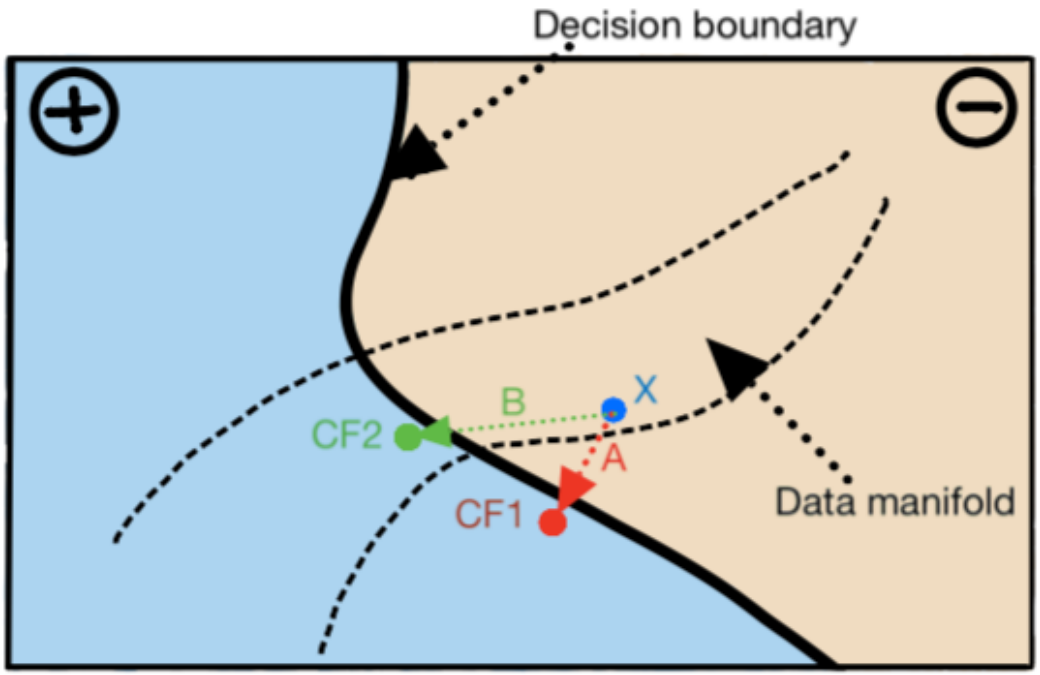
Remark: To find (sparse) feature subset $Z \subset X$ that "explains" $y = f(x)$, we look for Z and $z' \neq z$ such that changing $z \to z'$ results in $y' \neq y$, and z and z' are close according to some distance $d(z, z')$.

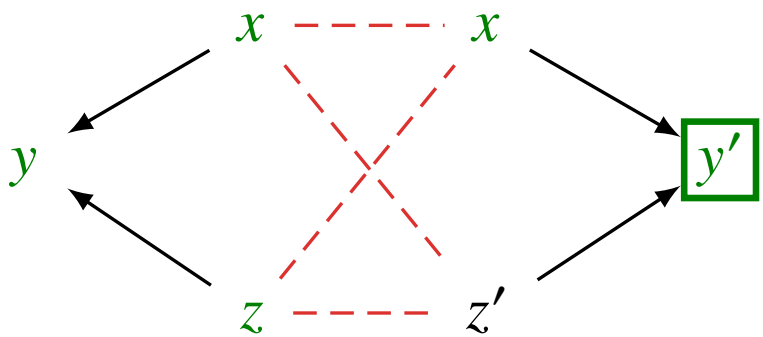
$$z' \in \operatorname*{argmax}_{z'} P_B(z' \mid y', x, y) \quad \text{subject to} \quad |Z| \leq k$$

Counterfactual Explanation: Example

- Minimize distance $d(x, x')$ between counterfactual x' and original datapoint x subject to constraint that the output $f(x')$ of the classifier f on the counterfactual x' is the desired label y'.

$$\underset{x'}{\operatorname{argmin}}\, d(x, x') \quad \text{subject to} \quad f(x') = y'$$





$$\underset{z'}{\operatorname{argmax}}\, P_B(z' \mid x, y', x, z, y)$$

where $P_B(u^* \mid u) := \dfrac{2^{-d(u,u^*)}}{\sum_{u^*} 2^{-d(u,u^*)}}$

Digression: 回溯因果 ≠ 逆向因果

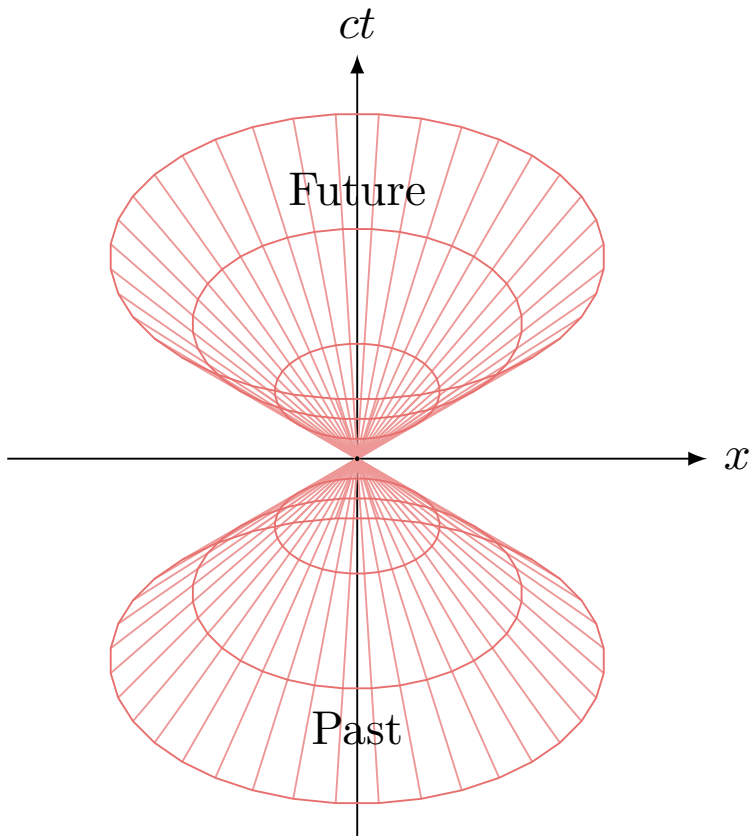
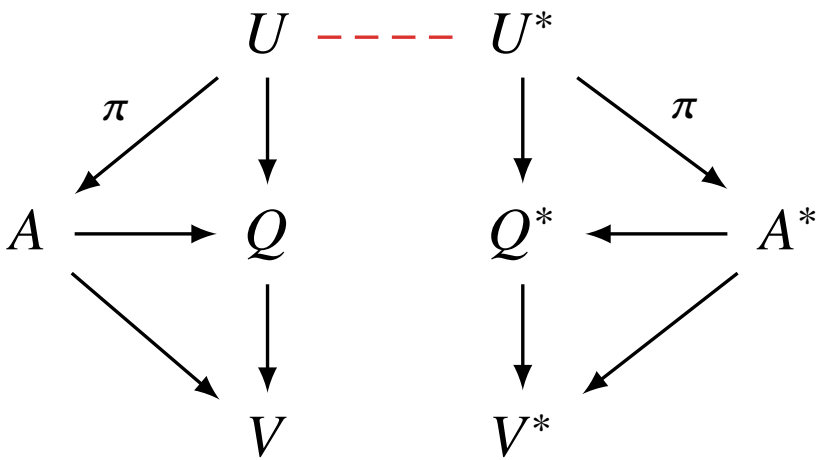


Figure: 因果作用只能在光锥内传播 $\Delta s \leq c\Delta t$

Remark: 因为能量和信息的传播速度有上限, 宇宙中的事件按它们之间可能的因果关系组织起来. 对于任意两个事件 A 和 B, 要么 A 是 B 的因果未来, 要么 B 是 A 的因果未来, 要么 A 和 B 之间没有因果关系.

干预反事实因果决策 vs 回溯反事实因果决策

- You must choose between two bets.
  Bet $A = 1$: pays 1 if $Q = 1$ and costs you 10 if $Q = 0$.
  Bet $A = 0$: pays 10 if $Q = 1$ and costs you 1 if $Q = 0$.
- Q: 宇宙的初始状态 (U) 和自然定律 (π) 决定了你会选择 $A = 1$.



$$A = \pi(U)$$

$$Q = [\![A = 1 \wedge \pi(U) = 1]\!]$$

V	$Q = 1$	$Q = 0$
$A = 1$	1	−10
$A = 0$	10	−1

干预反事实因果决策 vs 回溯反事实因果决策:

$$\left.\begin{array}{l} \operatorname{argmax}_{a} V(A = a) \\ \operatorname{argmax}_{a^*} V(A^* = a^*) \end{array}\right\} \text{ where } \begin{cases} V(A = a) = \sum_{o} P(o \mid \mathsf{do}(a)) V(o) \\ V(A^* = a^*) = \sum_{o^*} P_B(o^* \mid a^*) V^*(o^*) \end{cases}$$

干预反事实因果决策 vs 回溯反事实因果决策

- 干预反事实因果决策: $A = \pi(U), Q = [\![A = 1 \wedge \pi(U) = 1]\!]$

$$V(A=1) = \underbrace{P(Q=1 \mid \mathsf{do}(A=1))}_{x} \cdot 1 + \underbrace{P(Q=0 \mid \mathsf{do}(A=1))}_{1-x} \cdot (-10) = 11x - 10$$

$$V(A=0) = \underbrace{P(Q=1 \mid \mathsf{do}(A=0))}_{0} \cdot 10 + \underbrace{P(Q=0 \mid \mathsf{do}(A=0))}_{1} \cdot (-1) = -1$$

Where $P(Q=1 \mid \mathsf{do}(A=1)) = \sum_{u:\pi(u)=1} P(u) := x.$

Take $A = 1 \iff x \geq \frac{9}{11}$. (若 $\forall u : \pi(u) = 1$, 则 $x = 1$.)

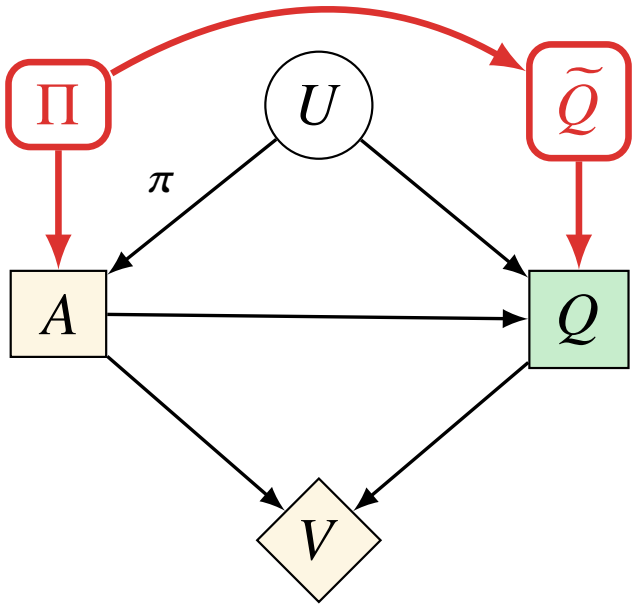
Remark: $\mathsf{do}(A=1)$ 不会改变 U 的真实取值和 π, 但会影响 "决定" 的程度 $P(Q=1 \mid \mathsf{do}(A=1)) \neq P(Q=1 \mid \mathsf{do}(A=0))$.

- 回溯反事实因果决策: $A^* = \pi(U^*), Q^* = [\![A^* = 1 \wedge \pi(U^*) = 1]\!]$

$$\left.\begin{aligned} P_B(Q^*=1 \mid A^*=1) = 1 \\ P_B(Q^*=1 \mid A^*=0) = 0 \end{aligned}\right\} \Longrightarrow \begin{cases} V(A^*=1) = 1 \\ V(A^*=0) = -1 \end{cases}$$

Take $A^* = 1$.

FDT?



$$Q = [\![A = 1 \wedge \pi(U) = 1]\!]$$

$$\pi^* = \underset{\pi}{\operatorname{argmax}}\, \mathbb{E}[V \mid \mathrm{do}(\Pi = \pi)]$$

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal
   - Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

“The fundamental problem of causal inference”

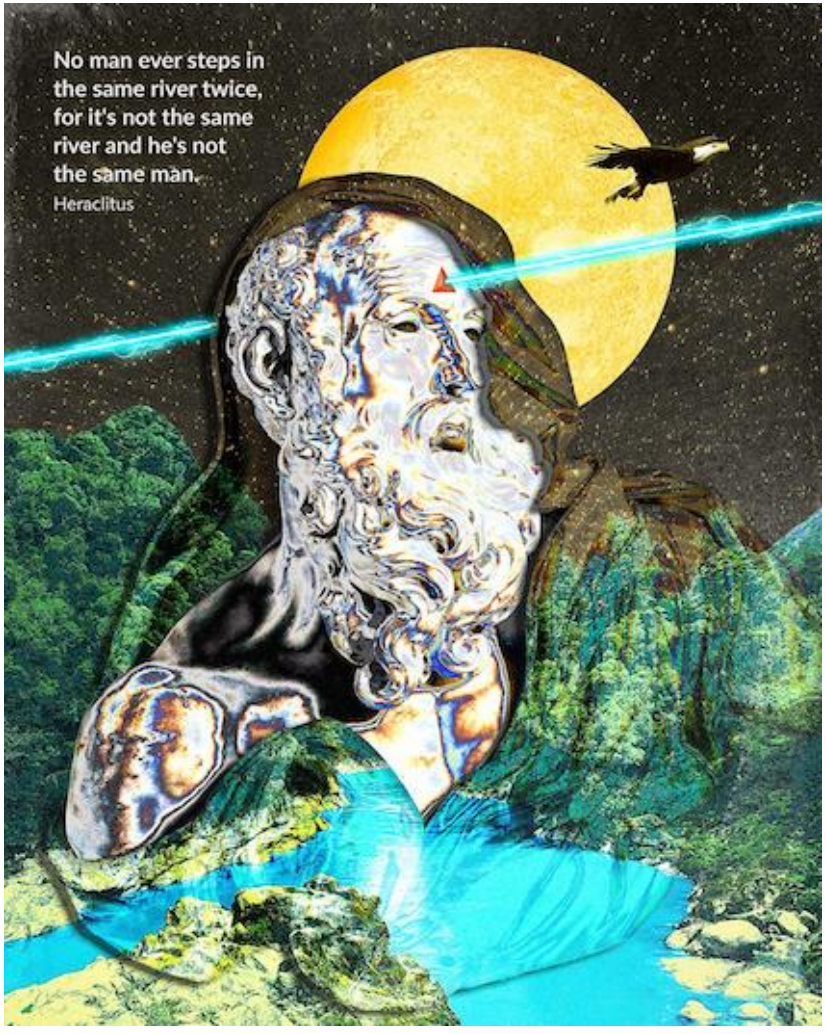
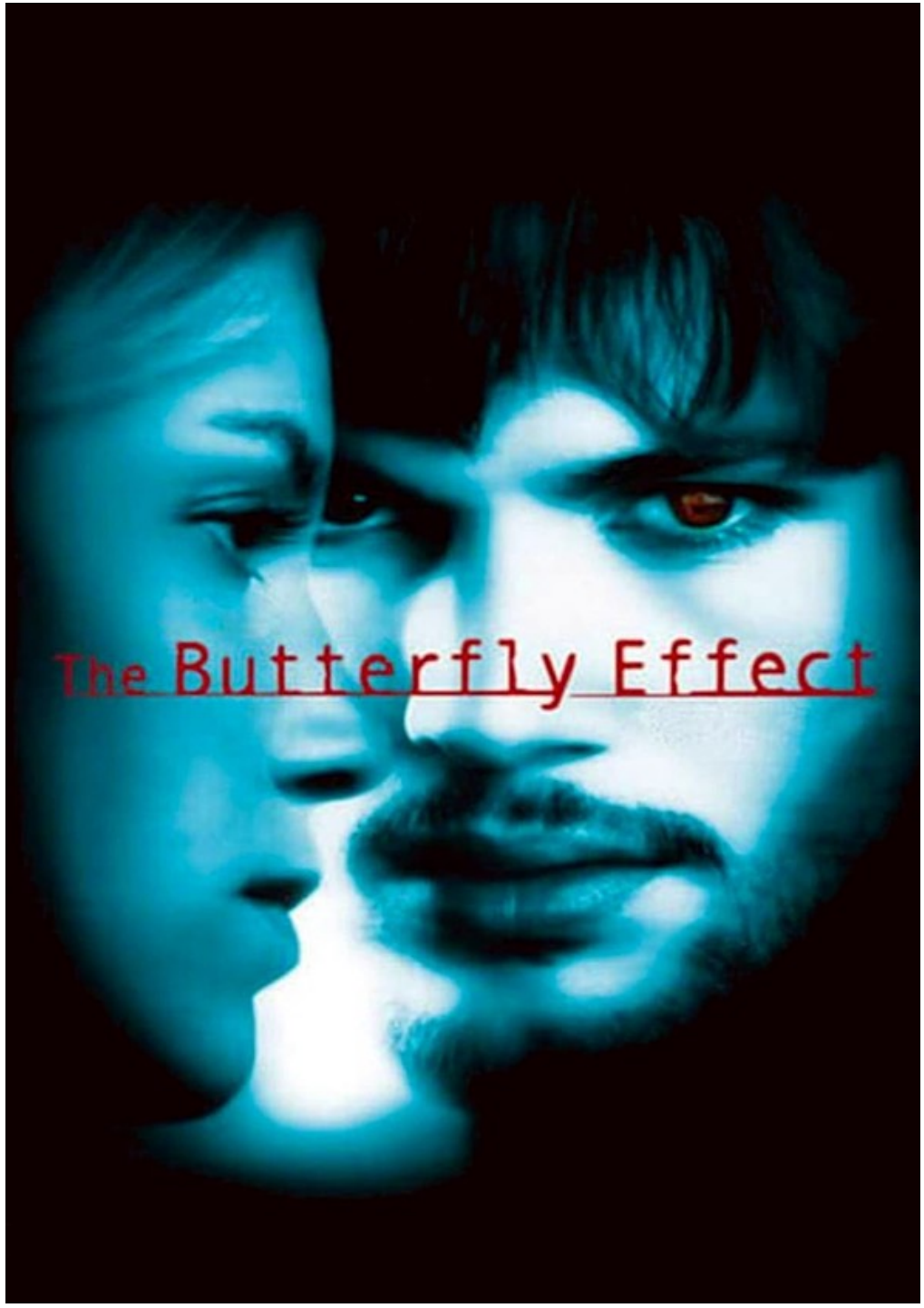


Figure: No man ever steps in the same river twice.

u	$Y_1(u)$	$Y_0(u)$	$Y_1(u) - Y_0(u)$
Alice	130	?	?
Bob	?	125	?
Carl	100	?	?
David	?	130	?
Ernest	?	120	?
Frank	115	?	?
Mean	115	125	−10

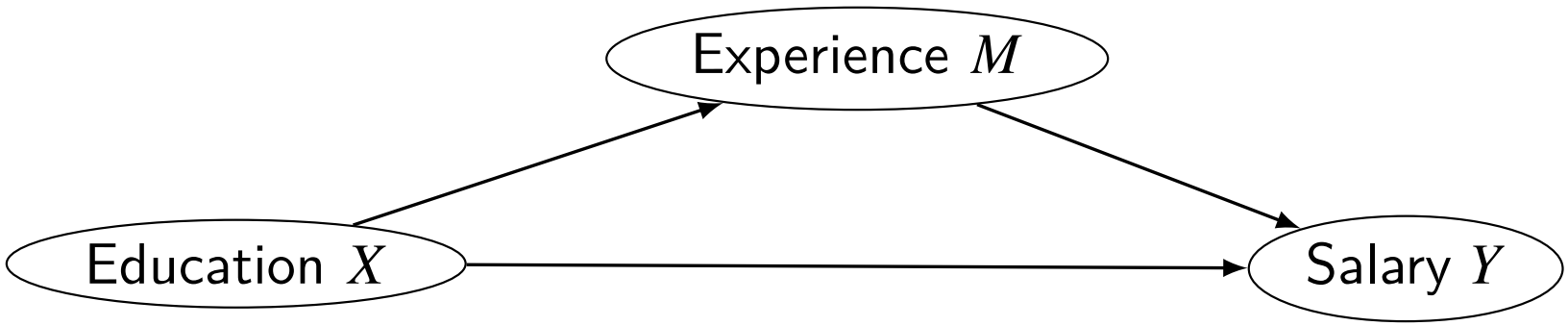
Table: 不能同时观测到 (Y_1, Y_0).

- 事实结果 vs 潜在结果
$$Y = XY_1 + (1 - X)Y_0$$
- Individual Treatment Effect
$$\mathrm{ITE}(u) = Y_1(u) - Y_0(u)$$
- Total Effect
$$\mathrm{TE} = \mathbb{E}_{u \sim P(u)} \left[Y_1(u) - Y_0(u) \right]$$


The Butterfly Effect

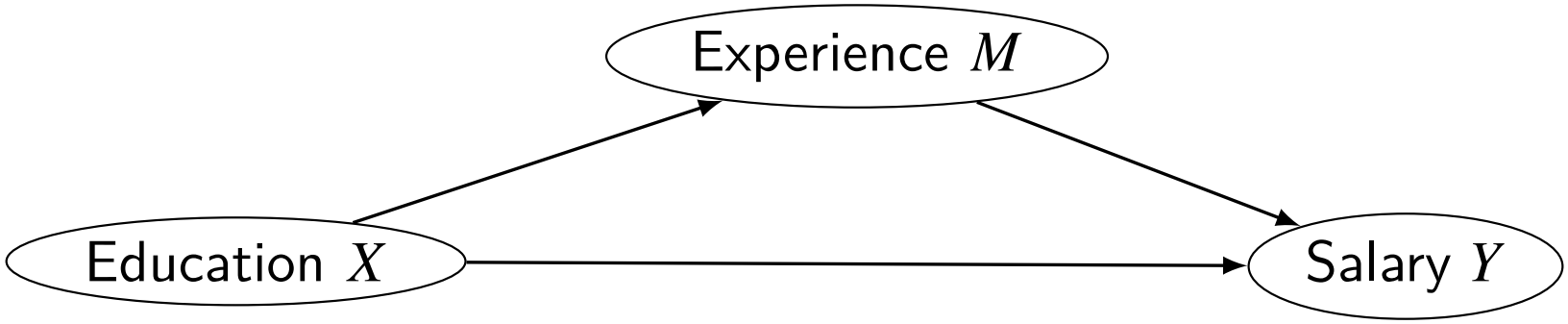
u	Experience	Education	Salary Y_0	Salary Y_1	Salary Y_2
Alice	6	0	81000	?	?
Bob	9	1	?	92500	?
Carl	9	2	?	?	97000
David	8	1	?	91000	?
Ernest	12	1	?	100000	?
Frank	13	0	97000	?	?

- The missing-data problem: One common approach is matching. We look for pairs of individuals who are well matched in all variables except the one of interest and then fill in their rows to match each other. e.g., $Y_2(\text{Bob}) = 97000$, $Y_1(\text{Carl}) = 92500$
- However, Experience is likely to depend on Education.



$$Y = 65000 + 2500M + 5000X + U_Y$$

$$M = 10 - 4X + U_M$$



$$Y = 65000 + 2500M + 5000X + U_Y$$
$$M = 10 - 4X + U_M$$

Question: "What would Alice's salary be if she had a college degree $(X = 1)$?"

1. Abduction:

$$U_Y(\text{Alice}) = 81000 - 65000 - 2500 \times 6 - 5000 \times 0 = 1000$$
$$U_M = 6 - 10 + 4 \times 0 = -4$$

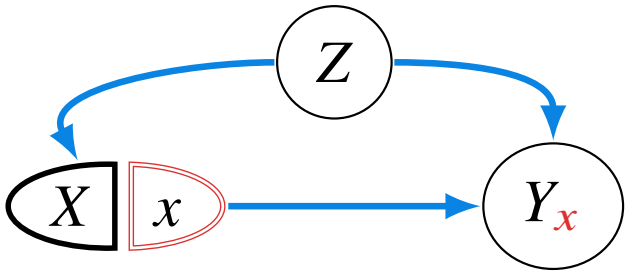
2. Action: $X = 1$
3. Prediction:

$$M_{X=1}(\text{Alice}) = 10 - 4 \times 1 - 4 = 2$$
$$Y_{X=1}(\text{Alice}) = 65000 + 2500 \times 2 + 5000 \times 1 + 1000 = 76000$$

潜在结果框架下因果识别的假设

1. **Unconfoundedness/Conditional Ignorability/Exogeneity/Exchangeability Assumption**
   给定观察变量 Z, 给个体分配处理方式这一过程与其潜在结果独立.

$$Y_x \perp X \mid Z$$



   Remark: 在决策问题中, 这一假设会被违反. 医生会根据潜在结果决定给患者分配何种治疗方案.

2. **No Interference — Stable Unit Treatment Value Assumption (SUTVA)**
   一个个体 u_i 的潜在结果不受对其它个体处理的影响.

$$Y_{x_1,\ldots,x_i,\ldots,x_n}(u_i) = Y_{x_i}(u_i)$$

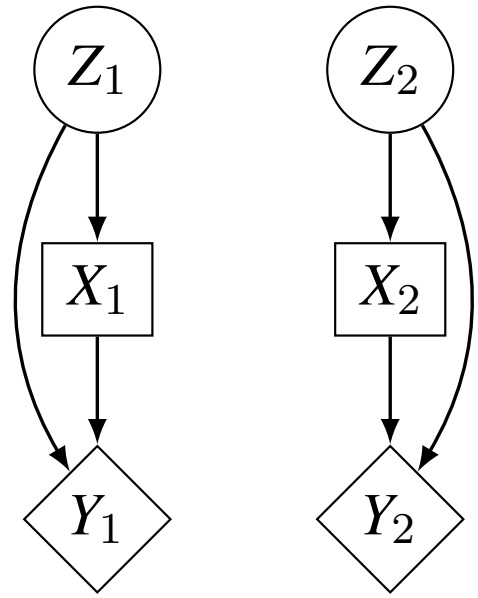
3. **Consistency Assumption**

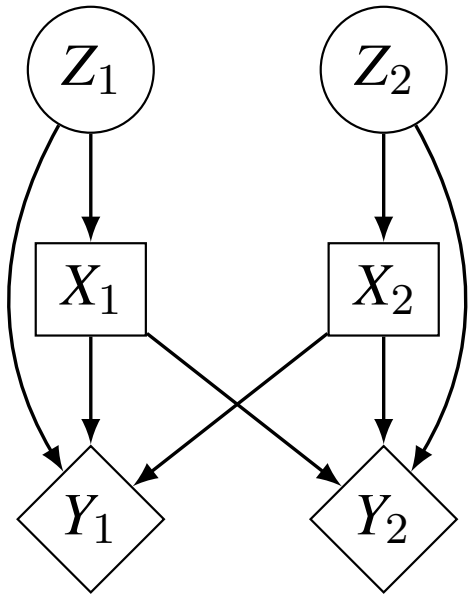
$$X = x \implies Y_x = Y$$

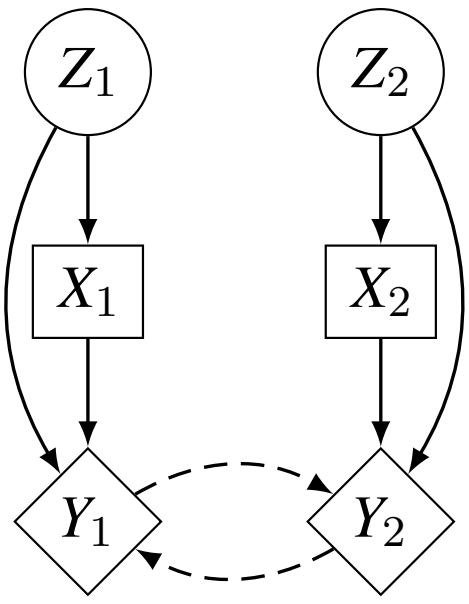
4. **Positivity/Overlap Assumption**
   每一特征的个体都既可能被分配到实验组, 也可能被分配到对照组.

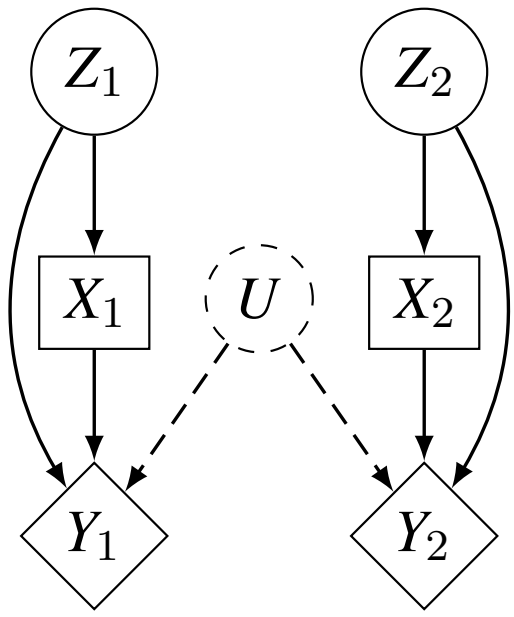
$$0 < P(X = 1 \mid Z = z) < 1$$

No Interference / Direct Interterence / Contagion / Latent Confounding









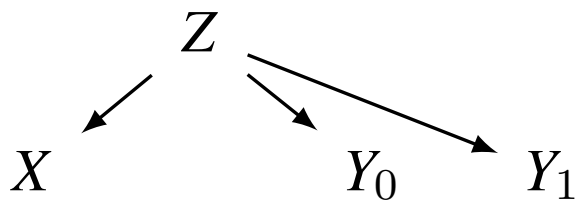
Unconfoundedness is an untestable assumption

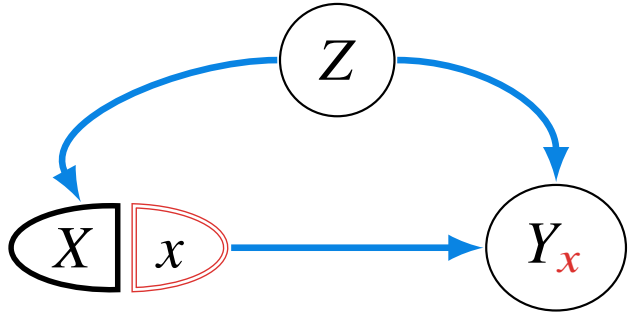
- Ignorability/Exogeneity/Exchangeability Assumption

$$Y_x \perp X$$

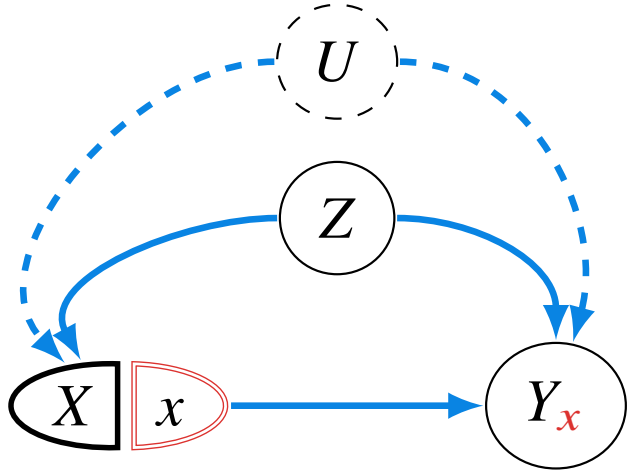
- Unconfoundedness/Conditional Ignorability/Exchangeability

$$Y_x \perp X \mid Z$$





Hidden confounding $\quad Y_x \not\perp X \mid Z$



Remarks: Ignorability/Exchangeability

- **Ignorability:** We can ignore the treatment mechanism $Y_x \perp X$
- **Exchangeability:** If treatment and control groups were swapped, the new treatment group will experience the same treatment effect as the old treatment group, and the new control group would experience the same effect as the old control group.

$$\mathbb{E}[Y_1 \mid X = 1] = \mathbb{E}[Y_1 \mid X = 0]$$

$$\mathbb{E}[Y_0 \mid X = 1] = \mathbb{E}[Y_0 \mid X = 0]$$

which implies

$$\mathbb{E}[Y_1 \mid X = x] = \mathbb{E}[Y_1]$$

$$\mathbb{E}[Y_0 \mid X = x] = \mathbb{E}[Y_0]$$

Which is the "mean" version of ignorability.

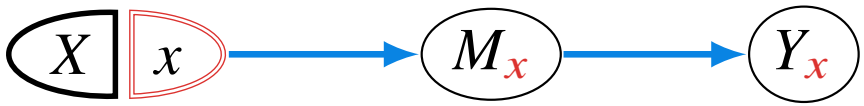
- In observational data, it is those who are sick that receive treatment. We use Conditional Exchangeability.

Applying d-separation to the SWIG $G(a)$

$$(X_a \perp Y_a \mid Z_a)_{G(a)} \implies (X_a \perp Y_a \mid Z_a)_{P(V_a)}$$

$$(a \perp Y_a \mid Z_a)_{G(a)} \implies P(Y_a \mid Z_a) = P(Y_{a'} \mid Z_{a'})$$

Example:



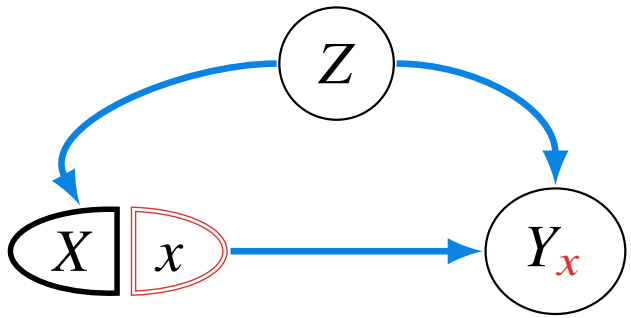
$$x \perp Y_x \mid M_x \implies P(Y_x \mid M_x) = P(Y_{x'} \mid M_{x'})$$

$$\begin{aligned} P(Y_x \mid M_x) &= P(Y_x \mid M_x, X = x) \\ &= P(Y \mid M, X = x) \\ &= P(Y \mid M, X = x') \\ &= P(Y_{x'} \mid M_{x'}, X = x') \\ &= P(Y_{x'} \mid M_{x'}) \end{aligned}$$

Backdoor Adjustment

- In the Randomized Control Trials, we have $Y_x \perp X$.
- If Z satisfies the backdoor condition relative to (X, Y), then

$$Y_x \perp X \mid Z$$



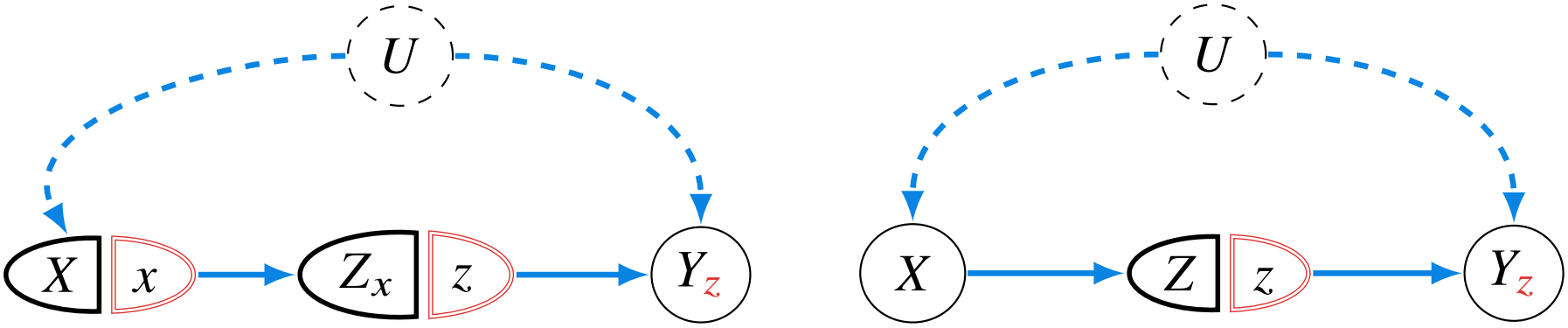
Backdoor Adjustment:

$$\begin{aligned} P(Y_x = y) &= \sum_z P(Y_x = y \mid Z = z) P(Z = z) \\ &= \sum_z P(Y_x = y \mid X = x, Z = z) P(Z = z) && (Y_x \perp X \mid Z) \\ &= \sum_z P(Y = y \mid X = x, Z = z) P(Z = z) && (X = x \implies Y_x = Y) \end{aligned}$$

Frontdoor Adjustment

- If Z satisfies the frontdoor condition relative to (X, Y), then

$$Y_z = Y_{z,x} \quad \text{and} \quad Z_x \perp \{Y_z, X\} \quad \text{and} \quad Y_z \perp Z \mid X$$



Frontdoor Adjustment:

$$\begin{aligned} P(Y_x = y) &= P(Y_{x,Z_x} = y) \\ &= P(Y_{Z_x} = y) \\ &= \sum_z P(Y_{Z_x} = y, Z_x = z) \\ &= \sum_z P(Y_z = y, Z_x = z) \\ &= \sum_z P(Y_z = y) P(Z_x = z) \end{aligned}$$

Estimation of Total Effect

Question: What assumptions would make the TE equal to the associational difference?

$$\begin{aligned}\mathbb{E}[Y_1 - Y_0] = \mathbb{E}[Y_1] - \mathbb{E}[Y_0] &= \mathbb{E}[Y_1 \mid X = 1] - \mathbb{E}[Y_0 \mid X = 0] && (Y_x \perp X)\\ &= \mathbb{E}[Y \mid X = 1] - \mathbb{E}[Y \mid X = 0] && (\text{consistency})\end{aligned}$$

$$\begin{aligned}\mathbb{E}[Y_1 - Y_0] &= \mathbb{E}[Y_1] - \mathbb{E}[Y_0]\\ &= \mathbb{E}_Z\big[\mathbb{E}[Y_1 \mid Z] - \mathbb{E}[Y_0 \mid Z]\big]\\ &= \mathbb{E}_Z\big[\mathbb{E}[Y_1 \mid X = 1, Z] - \mathbb{E}[Y_0 \mid X = 0, Z]\big] && (Y_x \perp X \mid Z)\\ &= \mathbb{E}_Z\big[\mathbb{E}[Y \mid X = 1, Z] - \mathbb{E}[Y \mid X = 0, Z]\big] && (\text{consistency})\end{aligned}$$

$$\mathbb{E}[Y \mid X = 1, Z = z] = \sum_y \frac{P(Y = y, X = 1, Z = z)}{P(X = 1 \mid Z = z)P(Z = z)}$$

$$\mathbb{E}[Y \mid X = 0, Z = z] = \sum_y \frac{P(Y = y, X = 0, Z = z)}{P(X = 0 \mid Z = z)P(Z = z)} \qquad (0 < P(X \mid Z) < 1)$$

Observational-Counterfactual Decomposition of TE

The Observational-Counterfactual Decomposition:

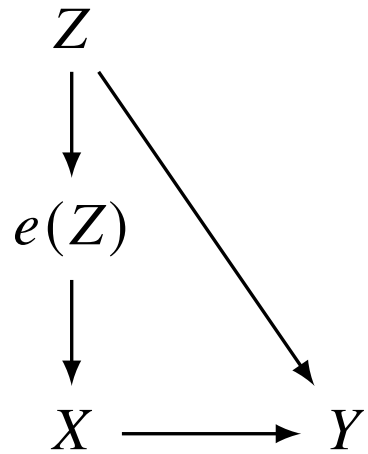
$$
\begin{aligned}
\mathbb{E}[Y_1 - Y_0] &= \mathbb{E}[Y_1] - \mathbb{E}[Y_0] \\
&= P(X=1)\mathbb{E}[Y_1 \mid X=1] + P(X=0)\mathbb{E}[Y_1 \mid X=0] - \\
&\quad P(X=1)\mathbb{E}[Y_0 \mid X=1] - P(X=0)\mathbb{E}[Y_0 \mid X=0] \\
&= P(X=1)\mathbb{E}[Y \mid X=1] + P(X=0)\mathbb{E}[Y_1 \mid X=0] - \\
&\quad P(X=1)\mathbb{E}[Y_0 \mid X=1] - P(X=0)\mathbb{E}[Y \mid X=0] \\
&= P(X=1)\mathbb{E}[Y \mid X=1] - P(X=0)\mathbb{E}[Y \mid X=0] + \\
&\quad P(X=0)\mathbb{E}[Y_1 \mid X=0] - P(X=1)\mathbb{E}[Y_0 \mid X=1]
\end{aligned}
$$

Propensity Scores and Inverse Probability Weighting (IPW)

Theorem

$$Y_x \perp X \mid Z \implies Y_x \perp X \mid e(Z)$$

where the propensity score $e(Z) := P(X = x \mid Z)$.



Inverse Probability Weighting

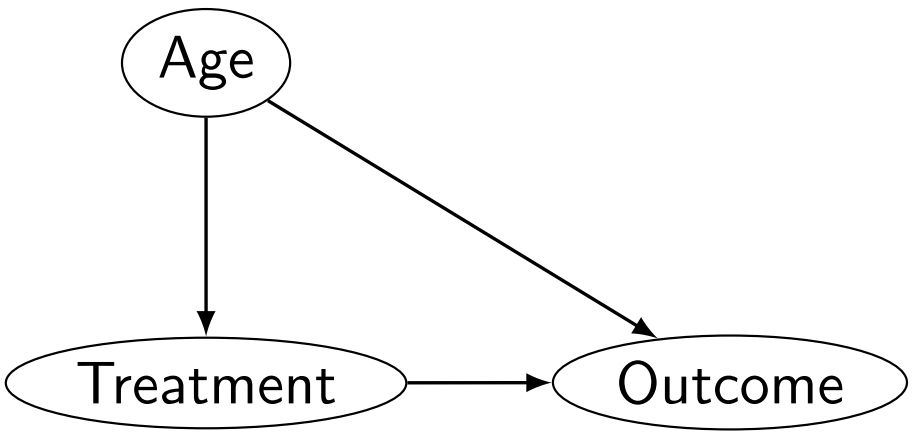
$$P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z)P(z) = \sum_z \frac{P(y, x, z)}{P(x \mid z)}$$

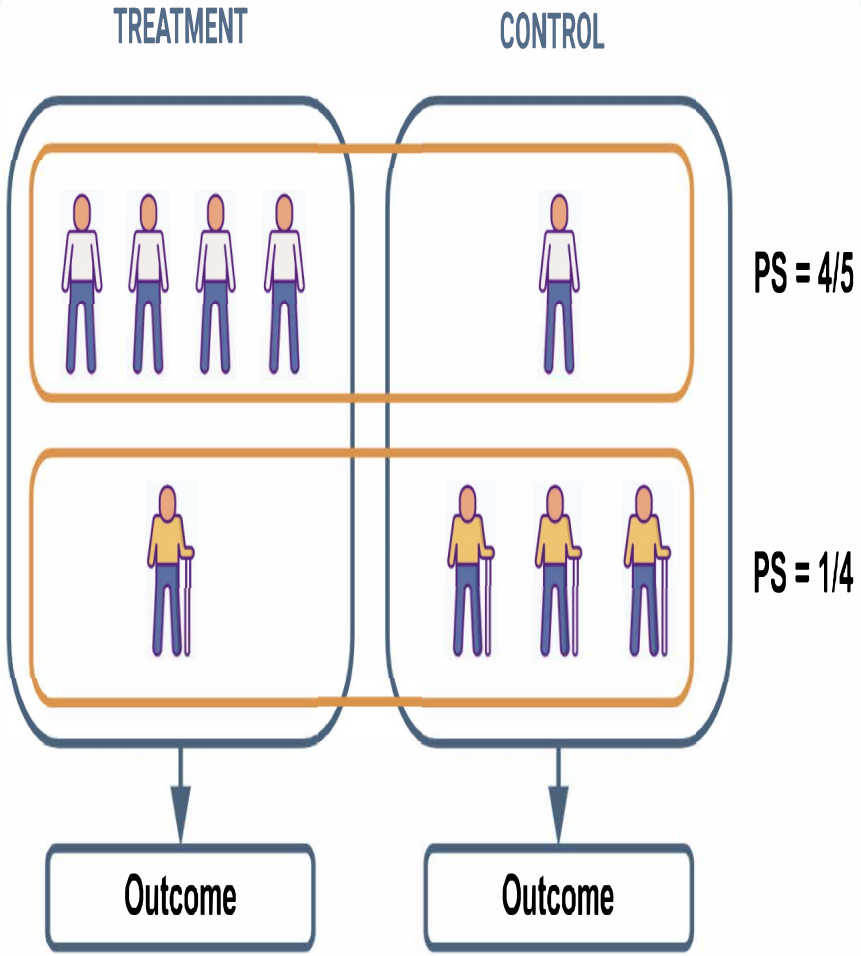
- 逆概率加权法仅在 Z 满足后门准则时有效.
- 相当于通过 $P(x \mid z)$ 生成 "虚构样本" 来模拟随机对照试验去混杂.

Assuming binary treatment,

$$\mathrm{TE} = \mathbb{E}[Y_1 - Y_0] = \mathbb{E}\left[\underbrace{\frac{[\![X = 1]\!]Y}{e(Z)}}_{\text{实验组重加权}}\right] - \mathbb{E}\left[\underbrace{\frac{[\![X = 0]\!]Y}{1 - e(Z)}}_{\text{对照组重加权}}\right]$$

How Propensity Scores can be used to calculate the Average Treatment Effects





1. Data Preparation
2. Calculate the Propensity Scores
3. Assess the positivity assumption (whether for each patient we can find a match in the other treatment group)
4. Calculate Average Treatment Effects drawn upon the propensity scores

Example: IPW & Simpson's Paradox

$$\widehat{\mathrm{TE}} = \frac{\sum_{i=1}^{n} \frac{[\![X=1]\!]Y}{\hat{e}(z)}}{\sum_{i=1}^{n} \frac{[\![X=1]\!]}{\hat{e}(z)}} - \frac{\sum_{i=1}^{n} \frac{[\![X=0]\!]Y}{1-\hat{e}(z)}}{\sum_{i=1}^{n} \frac{[\![X=0]\!]}{1-\hat{e}(z)}}$$

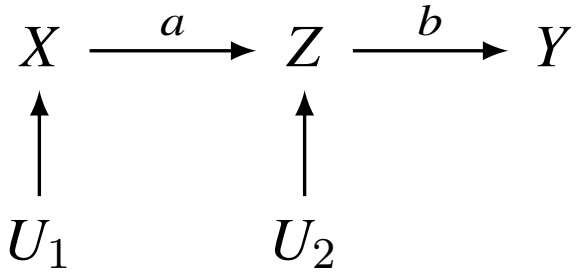
Recovery Rate	Lemon	No Lemon
Old	9/30	2/10
Young	7/10	18/30
Total	36/80	44/80

$$\begin{cases} P(\text{lemon} \mid \text{old}) = \frac{30}{40} \\ P(\text{no lemon} \mid \text{old}) = \frac{10}{40} \end{cases} \qquad \begin{cases} P(\text{lemon} \mid \text{yound}) = \frac{10}{40} \\ P(\text{no lemon} \mid \text{yound}) = \frac{30}{40} \end{cases}$$

$$\widehat{\mathrm{TE}} = \frac{1}{30/P(\text{lemon}|\text{old})+10/P(\text{lemon}|\text{young})} \left(9\frac{1}{P(\text{lemon}|\text{old})} + 7\frac{1}{P(\text{lemon}|\text{young}}\right) - \frac{1}{10/P(\text{no lemon}|\text{old})+30/P(\text{no lemon}|\text{young})} \left(2\frac{1}{P(\text{no lemon}|\text{old})} + 18\frac{1}{P(\text{no lemon}|\text{young}}\right) = 0.1$$

Example: $Y \perp X \mid Z$ but $Y_x \not\perp X \mid Z$

$$X = U_1$$
$$Z = aX + U_2$$
$$Y = bZ$$



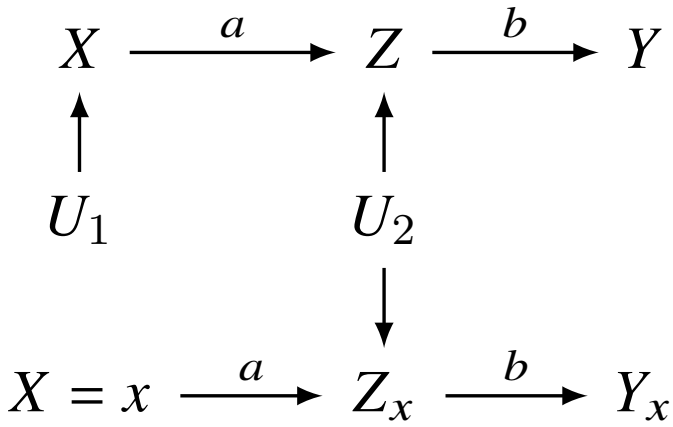


Figure: X: 教育水平, Z: 工作能力, U_2: 工作经验, Y: 工资

Remark: Y_x 由 U_2 决定. 因为 $U_2 \not\perp X \mid Z$, 所以 $Y_x \not\perp X \mid Z$.

$$\mathbb{E}[Y_{X=1} \mid Z = 1] \neq \mathbb{E}[Y \mid \mathrm{do}(X = 1), Z = 1] = \mathbb{E}[Y_{X=1} \mid Z_{X=1} = 1]$$

$$P(y \mid \mathrm{do}(x), z) = \frac{P(y, z \mid \mathrm{do}(x))}{P(z \mid \mathrm{do}(x))}$$

Remark: $X = 1$ 和 $Z = 1$ 在 $\mathbb{E}[Y_{X=1} \mid Z = 1]$ 中分别涉及干预前、干预后两个不同的世界, 而在 $\mathbb{E}[Y \mid \mathrm{do}(X = 1), Z = 1]$ 中只涉及干预后的世界.

Single-World Intervention Graph

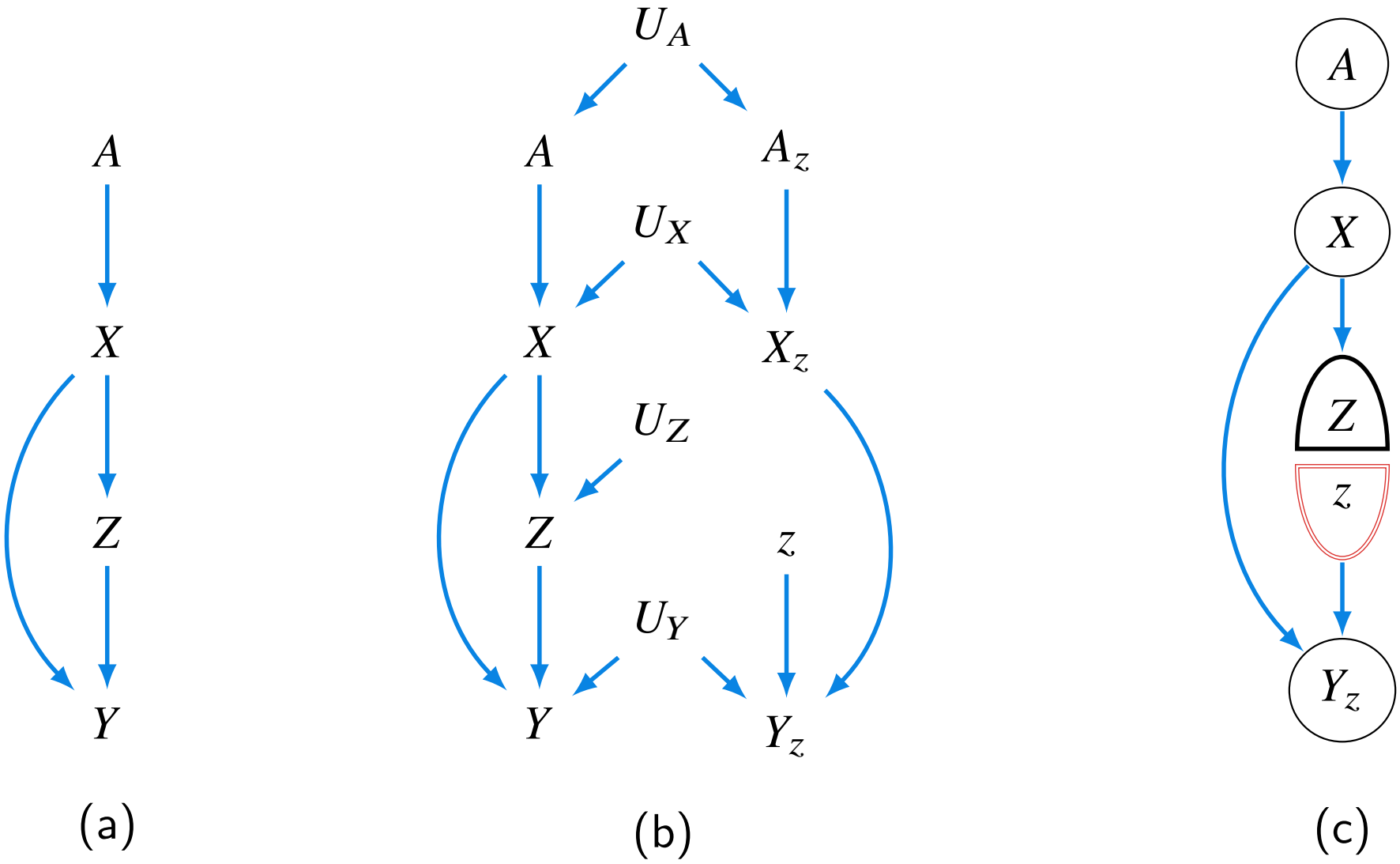
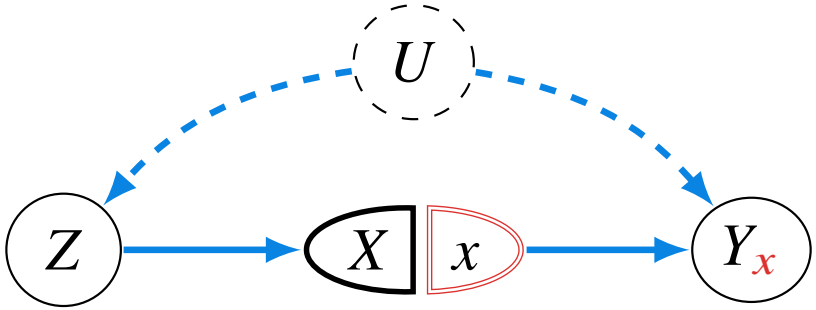
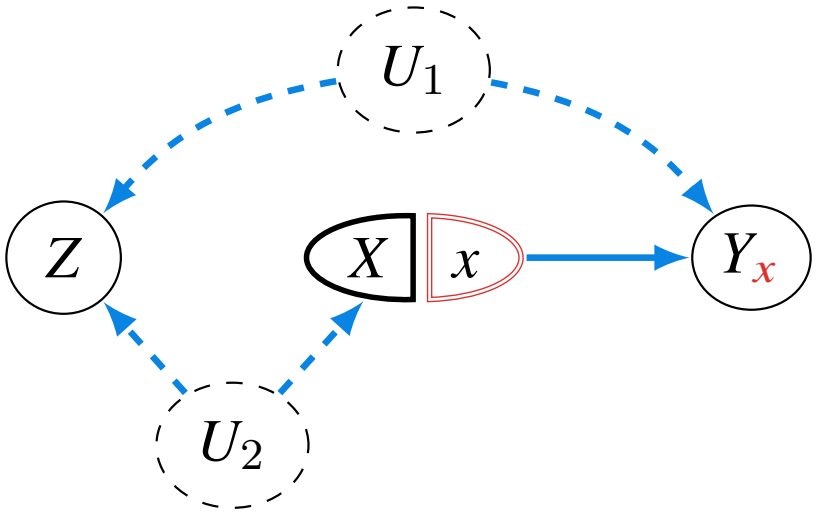


Figure: (a) A DAG G. (b) The twin-network (set Z to z). (c) The SWIG $G(z)$.

Single-World Intervention Graph

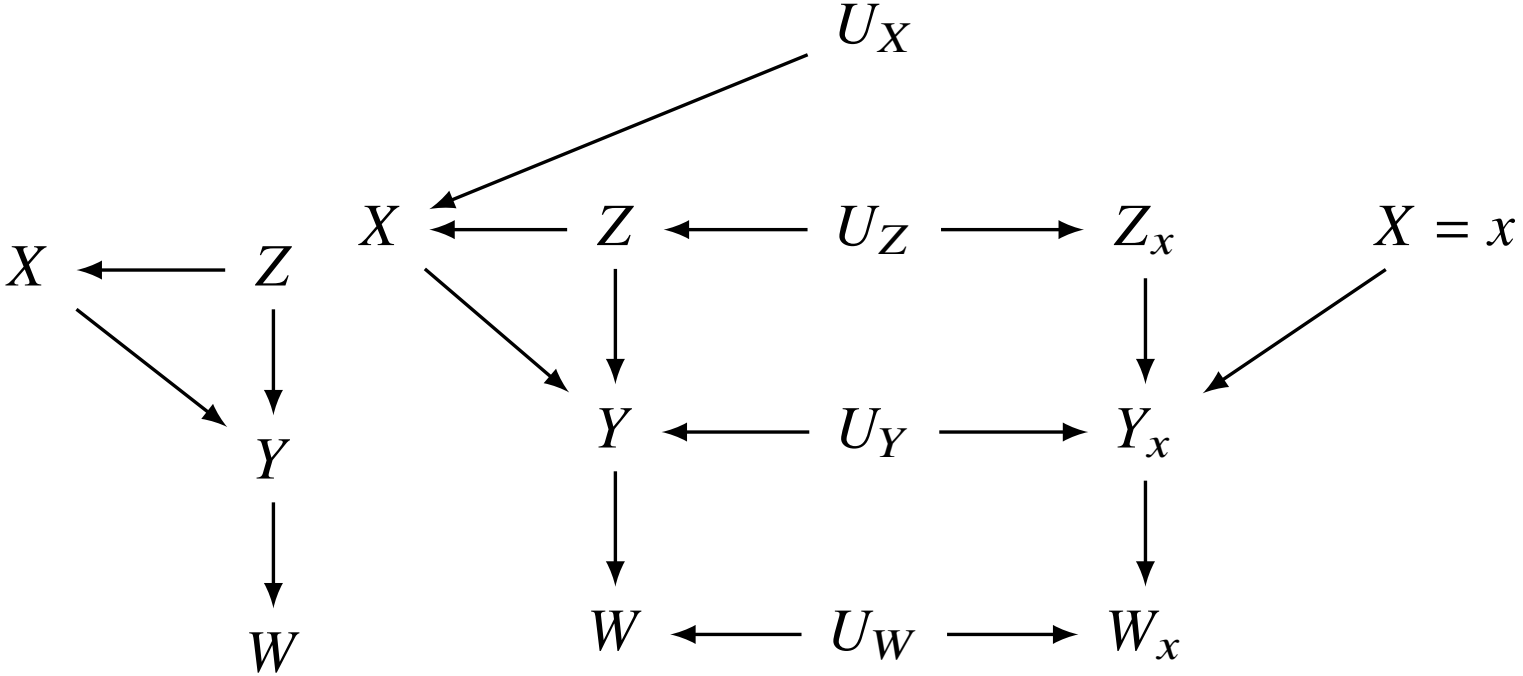


$Y_x \not\perp X \quad \text{but} \quad Y_x \perp X \mid Z$



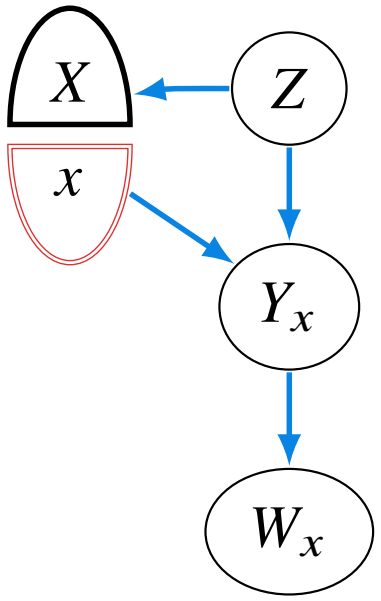
$Y_x \perp X \quad \text{but} \quad Y_x \not\perp X \mid Z$

Twin Network & SWIG



$$Y_x \perp X \mid Z$$

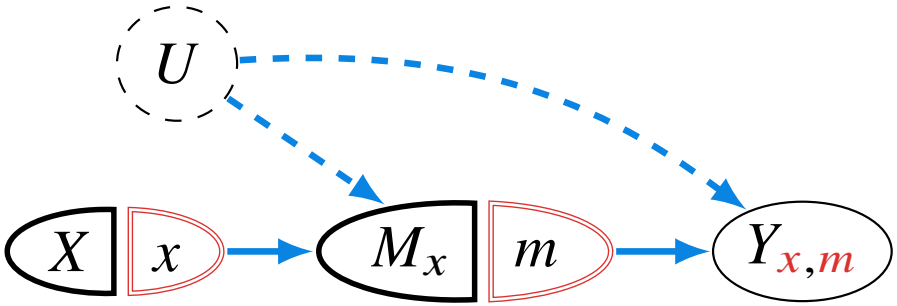
$$Y_x \not\perp X \mid Z, W$$



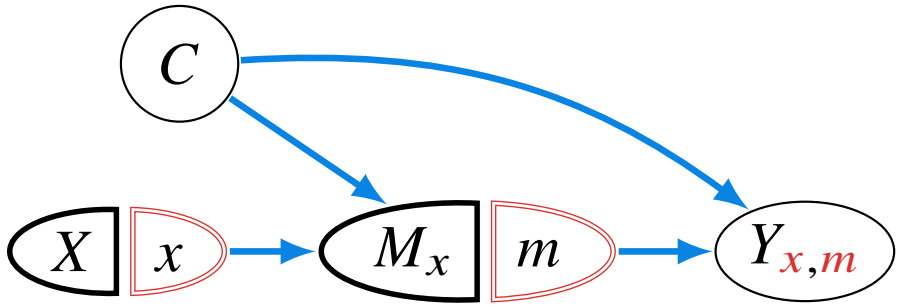
$$Y_x \perp X \mid Z$$

$$Y_x \perp X \mid Z, W_x$$

Single-World Intervention Graph



$Y_{x,m} \not\perp M_x$



$Y_{x,m} \perp M_x \mid C$

$$\mathbb{E}[Y_{x,m}] = \sum_c \mathbb{E}[Y_{x,m} \mid C = c] P(C = c)$$

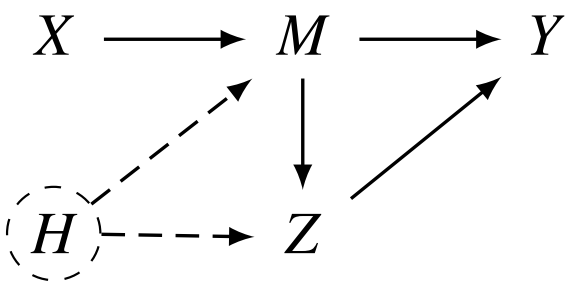
$$= \sum_c \mathbb{E}[Y_{x,m} \mid C = c, M_x = m] P(C = c) \qquad (Y_{x,m} \perp M_x \mid C)$$

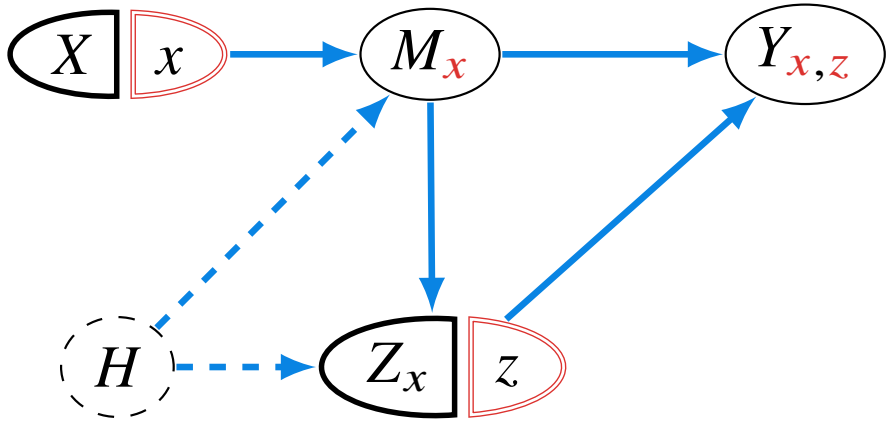
$$= \sum_c \mathbb{E}[Y_{x,m} \mid C = c, M_x = m, X = x] P(C = c) \qquad (Y_{x,m} \perp X)$$

$$= \sum_c \mathbb{E}[Y \mid C = c, M = m, X = x] P(C = c) \qquad \text{(consistency)}$$

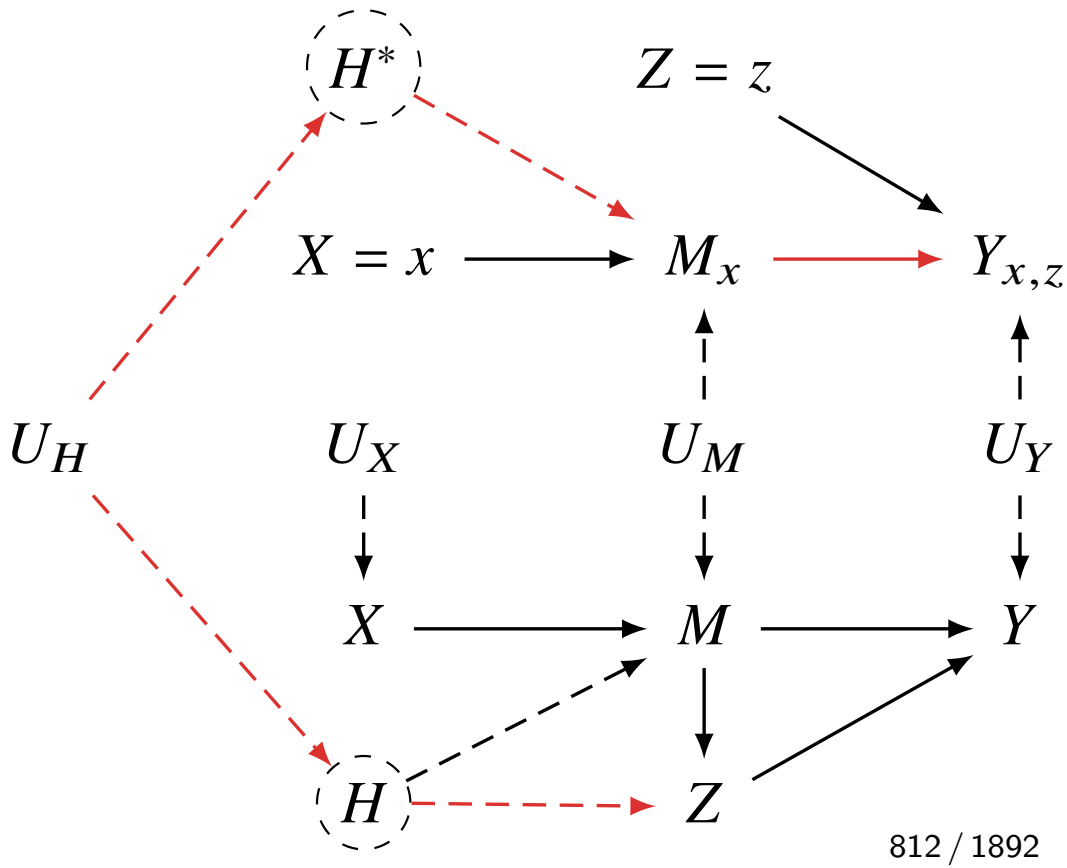
Therefore, $\mathbb{E}[Y_{x,m}]$ is identifiable from the observed data.

Single-World Intervention Graph vs Twin Network

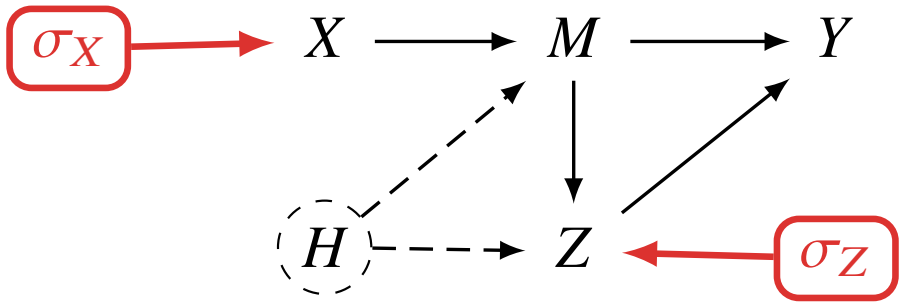




- SWIG: $Y_{x,z} \perp Z \mid M, X = x$ because $Y_{x,z} \perp Z_x \mid M_x, X$
- Twin Network fails: $Y_{x,z} \not\perp Z \mid M, X = x$
- This 'extra' independence holds in spite of d-connection because (by consistency) when $X = x$, then $M = M_x$.
- Note that $Y_{x,z} \not\perp Z \mid M, X \neq x$



决策论因果视角



$$Y \perp \{X, \sigma_X, \sigma_Z\} \mid M, Z$$
$$M \perp \{\sigma_X, \sigma_Z\} \mid X$$

$$P(Y = y \mid \sigma_X = x, \sigma_Z = z)$$
$$= \sum_m P(Y = y \mid M = m, X = x, Z = z, \sigma_X = x, \sigma_Z = z) P(M = m \mid X = x, \sigma_X = x, \sigma_Z = z)$$
$$= \sum_m P(Y = y \mid M = m, Z = z, \sigma_X = \varnothing, \sigma_Z = \varnothing) P(M = m \mid X = x, \sigma_X = \varnothing, \sigma_Z = \varnothing)$$

From Causal Graphs to Potential Outcomes

Exclusion restrictions: For every variable Y and every set of variables $Z \subset V$ disjoint of Pa_Y, we have

$$Y_{\mathrm{pa}_Y}(u) = Y_{\mathrm{pa}_Y, z}(u)$$

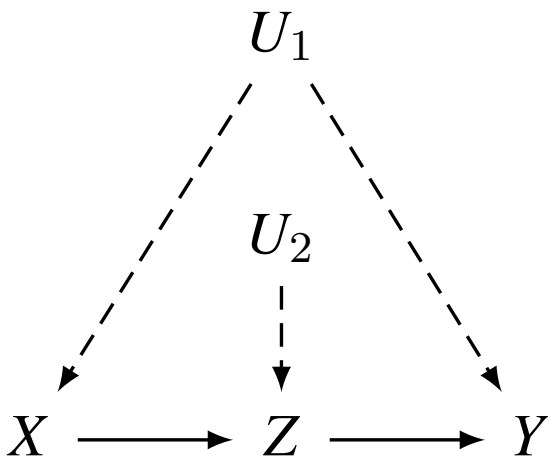
Remark: 这种不变性只适用于对 Z 的干预而不是观测. 若 Z 是 Y 的后代节点, 则可能 $P(y \mid \mathrm{do}(\mathrm{pa}_Y), z) \neq P(y \mid \mathrm{pa}_Y)$.

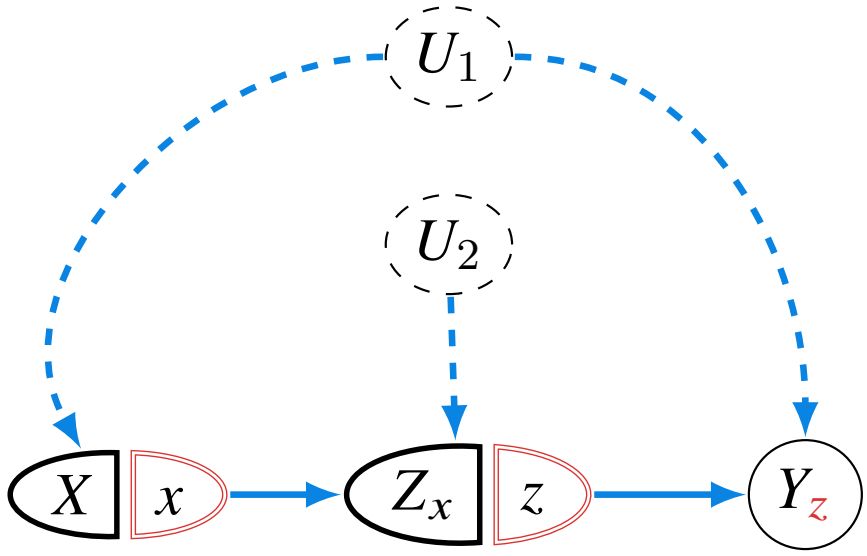
Independence restrictions: If $Z_1, \ldots, Z_k$ is any set of nodes not connected to Y via dashed arcs, we have

$$Y_{\mathrm{pa}_Y} \perp \{Z_{1\mathrm{pa}_{Z_1}}, \ldots, Z_{k\mathrm{pa}_{Z_k}}\}$$

Remark: Equivalently, $U_Y \perp \{U_{Z_1}, \ldots, U_{Z_k}\}$, since $V_i = f_i(\mathrm{Pa}_i, U_i)$.

Example — From Causal Graphs to Potential Outcomes





- Exclusion restrictions:

$$Z_x(u) = Z_{x,y}(u)$$

$$X(u) = X_z(u) = X_y(u) = X_{z,y}(u)$$

$$Y_z(u) = Y_{z,x}(u)$$

- Independence restrictions:

$$Z_x \perp \{Y_z, X\}$$

Halpern's Axioms of Structural Counterfactuals

1. **Composition** For any three sets of endogenous variables X, Y, W,

$$W_x(u) = w \implies Y_{xw}(u) = Y_x(u)$$

2. **Effectiveness** For all sets of variables X and W,

$$X_{xw}(u) = x$$

3. **Reversibility** For any two variables Y and W and any set of variables X,

$$Y_{xw}(u) = y \;\&\; W_{xy}(u) = w \implies Y_x(u) = y$$

- **Soundness** Halpern's axioms hold in both counterfactual SCMs and POMs.
- **Completeness** All counterfactual statements follow from Halpern's axioms.

Example: The Frontdoor Adjustment

$$P(Y_x = y) = P(Y_{z,x} = y) \qquad Y_x(u) = Y_{z,x}(u) = Y_z(u) \text{ if } Z_x(u) = z$$

$$= \sum_z P(Y_{z,x} = y \mid Z_x = z) P(Z_x = z)$$

$$= \sum_z P(Y_z = y \mid Z_x = z) P(Z_x = z) \qquad \text{composition}$$

$$= \sum_z P(Y_z = y) P(Z_x = z) \qquad Z_x \perp Y_z$$

$$P(Y_z = y) = \sum_x P(Y_z = y \mid x) P(x)$$

$$= \sum_x P(Y_z = y \mid x, Z_x = z) P(x) \quad Z_x \perp \{Y_z, X\} \implies Z_x \perp Y_z \mid X$$

$$= \sum_x P(Y_z = y \mid x, z) P(x) \qquad \text{composition}$$

$$= \sum_x P(y \mid x, z) P(x) \qquad \text{composition}$$

$$P(Z_x = z) = P(Z_x = z \mid x) \qquad Z_x \perp X$$

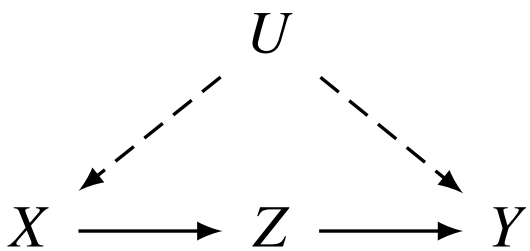
$$= P(Z = z \mid x) \qquad \text{composition}$$

$$= P(z \mid x)$$

Causal Inference as a Logical Problem

$$\text{Axioms} + \text{Data} \models \text{Causal Conclusion}$$

$$\begin{cases} \text{functional} \\ \text{graphical} \\ \text{noise} \\ \vdots \end{cases} \quad \begin{cases} \text{observation} \\ \text{experiment} \\ \text{quasi-exp.} \\ \vdots \end{cases} \quad \begin{cases} \text{bounds on causal effects} \\ \text{causal direction} \\ \text{counterfactual probabilities} \\ \vdots \end{cases}$$

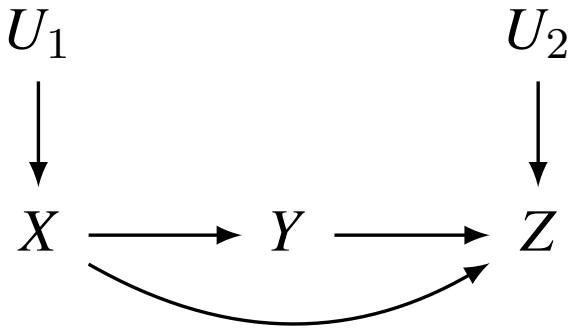


$$+ \; P(X, Y, Z) \models \mathbb{E}[Y_x - Y_{x'}]$$

$$P(x_z) = P(x)$$
$$P(z_x) = P(z \mid x)$$
$$P(y_z \mid x_z) = P(y \mid x, z)$$
$$P(y_x \mid z_x) = P(y_{x,z}) = P(y_z)$$

$$+ \; \text{Observation} \models \text{Causal Effect}$$

因果传递性?



$$X = U_1$$
$$Y = X$$
$$Z = \begin{cases} U_2 & \text{if } X = Y \\ X & \text{otherwise} \end{cases}$$

- 显然, X 可以改变 Y, Y 可以改变 Z, 但 X 不能改变 Z.
- 当因果被认为可传递时, 我们默认做了哪些假设?
- "If (1) X causes Y and (2) Y causes Z regardless of X, then (3) X causes Z."
- 我们将 "$X = x$ causes $Y = y$" (记为 $x \to y$) 表示为: $X(u) = x, Y(u) = y, Y_{x'}(u) = y' \neq y$.
- 可以证明, 当 X 对 Z 没有直接影响时, 传递性成立.

$$\frac{Z_{y'x'} = Z_{y'}}{x \to y \;\&\; y \to z \implies x \to z}$$

Proof: $Y_{x'}(u) = y' \;\&\; Z_{y'x'}(u) = Z_{y'}(u) = z' \implies Z_{x'}(u) = z'$

Counterfactual ctf-Calculus

Let G be a causal diagram, then for $Y, X, Z, W, T \subset V$, the following rules hold for the probability distributions generated by any model compatible with G:

R1. Consistency rule — Observation/intervention exchange:

$$P(y_{T_x}, x_T, w_*) = P(y_T, x_T, w_*)$$

R2. Independence Rule — Adding/removing counterfactual observations:

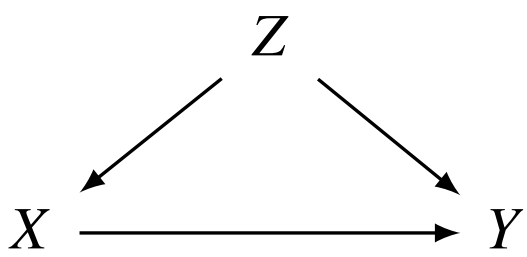
$$P(y_r \mid x_t, w_*) = P(y_r \mid w_*) \qquad \text{if} \quad (Y_r \perp X_t \mid W_*)_{G_A}$$

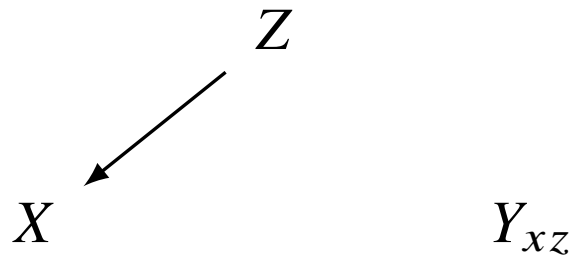
R3. Exclusion Rule — Adding/removing interventions:

$$P(y_{x,z}, w_*) = P(y_z, w_*) \qquad \text{if} \quad X \cap \mathrm{An}(Y) = \varnothing \text{ in } G_{\overline{Z}}$$

where G_A is the counterfactual ancestral graph $G_A(Y_r, X_t, W_*)$.

Example





$$
\begin{aligned}
P(y_x \mid x') &= \sum_z P(y_x \mid z, x') P(z \mid x') \\
&= \sum_z P(y_x \mid z_x, x') P(z \mid x') && (\text{R3: } \{X\} \cap \mathrm{An}(Z) = \varnothing) \\
&= \sum_z P(y_{xz} \mid z_x, x') P(z \mid x') && (\text{R1: } Z_x = z \implies Y_x = Y_{xz}) \\
&= \sum_z P(y_{xz} \mid z, x') P(z \mid x') && (\text{R3: } \{X\} \cap \mathrm{An}(Z) = \varnothing) \\
&= \sum_z P(y_{xz} \mid z, x) P(z \mid x') && (\text{R2: } (X \perp Y_{xz} \mid Z) \text{ in } G_A) \\
&= \sum_z P(y \mid z, x) P(z \mid x') && (\text{R1: } Z = z, X = x \implies Y_{xz} = Y)
\end{aligned}
$$

Soundness & Completeness of the ctf-Calculus

Theorem (Soundness & Completeness for Counterfactual Identifiability)

A counterfactual quantity $Q = P(y_* \mid x_*)$ is identifiable from a combination of observational and experimental distributions and a causal diagram G iff there exists a sequence of applications of the rules of ctf*-calculus and the probability axioms that reduces Q into a function of the available distributions.*

SCM vs POM

- SCM and POM are equivalent w.r.t Halpern's axioms.
- SCM views the intervention $\mathrm{do}(x)$ as an operation that changes the model (and the distribution) but keeps all variables the same.
- POM views the variable Y under $\mathrm{do}(x)$ to be a primitive variable Y_x, where $Y_x(u)$ is connected to the reality only via the consistency assumption.
- Instead of constructing new vocabulary and new logic for causal expressions, all mathematical operations in POM are conducted within the confines of probability calculus.
- POM uses independencies among counterfactual variables to express causal knowledge.
- Causal graphs allow researchers to represent causal assumptions in terms that they can understand and then treat all counterfactuals as derived properties of their world model.

Lewis' Counterfactual Causation

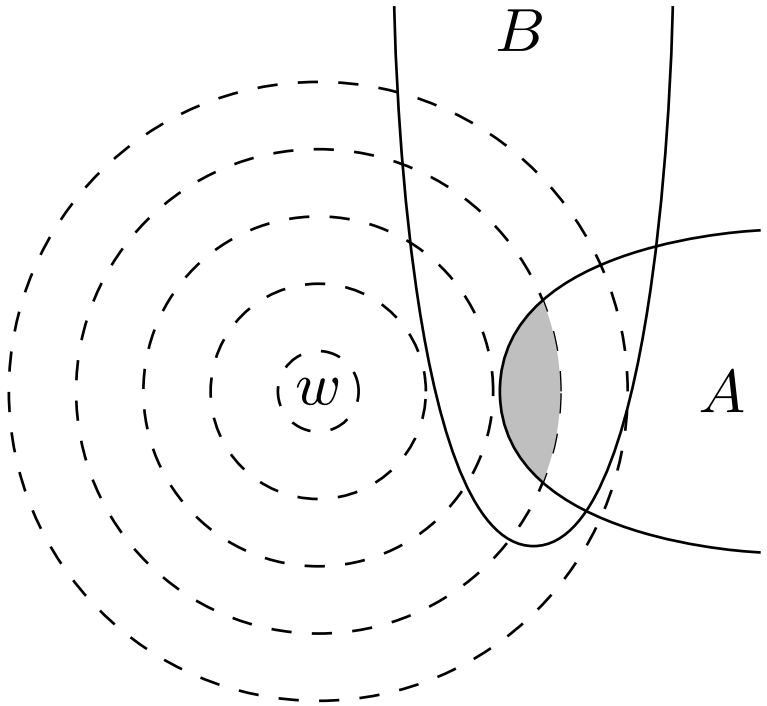


Figure: $w \Vdash A \boxdot\!\!\rightarrow B \iff [\![A]\!]_w \subset [\![B]\!]$, 与 w 邻近的 A 世界都是 B 世界.

- Let A stand for the proposition $X = x$ and B for $Y = y$. Then

$$Y_x(u) = y \iff A \boxdot\!\!\rightarrow B$$

- In dynamic logic, $[X = x]Y = y$.

Pearl vs Lewis

- Structural counterfactuals are compatible with Lewis' idea of the most similar possible world.
- Structural causal models offer a resolution of the "representation problem" Lewis kept silent about: How do humans represent "possible worlds" in their minds and compute the closest one?
- Logic void of representation is metaphysics.
- Causal graphs, with their simple rules of following and erasing arrows, must be close to the way that our brains represent counterfactuals.

Lewis' Imaging Theory

令 $[\![A]\!]_w$ 为与 w 邻近的 A 世界集.

$$P(w^* \mid [\![A]\!]_w) = \frac{P(\{w^*\} \cap [\![A]\!]_w)}{P([\![A]\!]_w)} = \begin{cases} \frac{P(w^*)}{P([\![A]\!]_w)} & \text{if } w^* \in [\![A]\!]_w \\ 0 & \text{otherwise} \end{cases}$$

每个 $\neg A$ 世界 w 的概率质量都转移给了 $[\![A]\!]_w$ 中的世界.

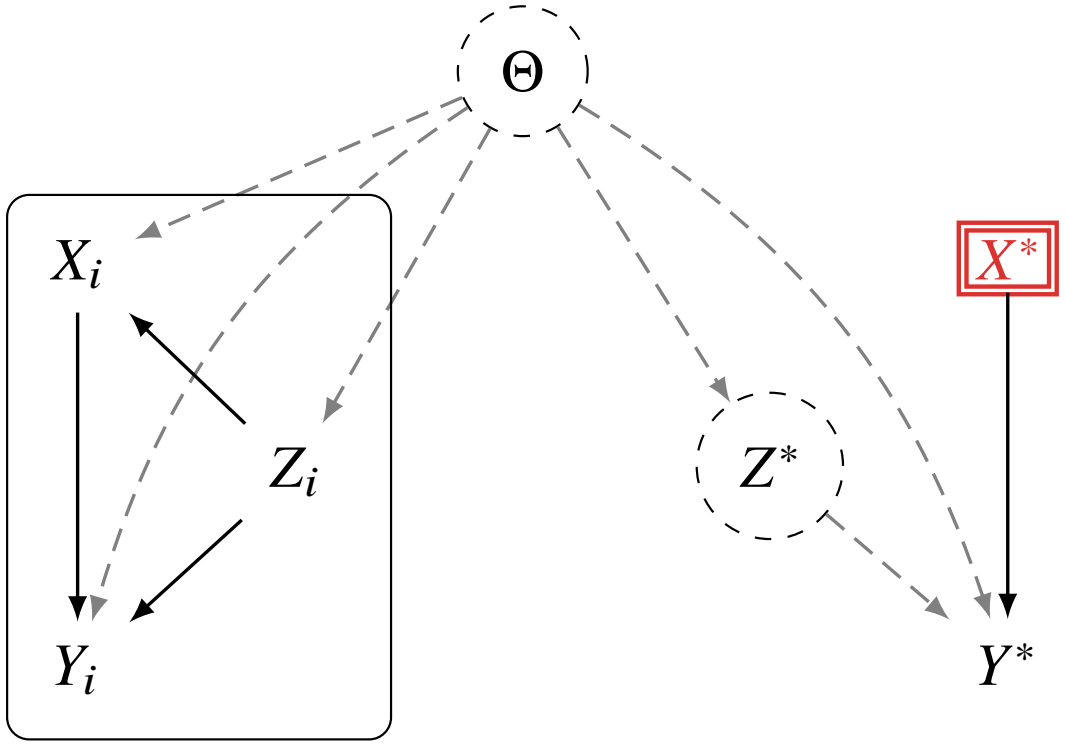
$$P^A(w^*) := \sum_w P(w) P(w^* \mid [\![A]\!]_w)$$

$$P^A(B) := \sum_{w^* \in [\![B]\!]} P^A(w^*) = \sum_w P(w) \frac{P([\![A]\!]_w \cap [\![B]\!])}{P([\![A]\!]_w)}$$

$$\boxed{P^A(B \mid C) := \sum_w P(w \mid C) P_w(B \mid A)} \quad \text{where} \quad P_w(B \mid A) := \frac{P([\![A]\!]_w \cap [\![B]\!])}{P([\![A]\!]_w)}$$

- **Example:** 令 A 为 $X = x$, 若 w 为 $(U = u, V = F(u))$, 则 $[\![A]\!]_w$ 为 $(U = u, X = x, V_x = F_x(u))$. 则
  $P^{X=x}(Y = y \mid X = x', Y = y') = P(Y_x = y \mid X = x', Y = y')$.
- **Application:** Boolean action: $P(y \mid \text{do}(x_1 \vee x_2)) := P^{x_1 \vee x_2}(y)$

概率论因果 — 对现实世界和平行世界赋予联合概率



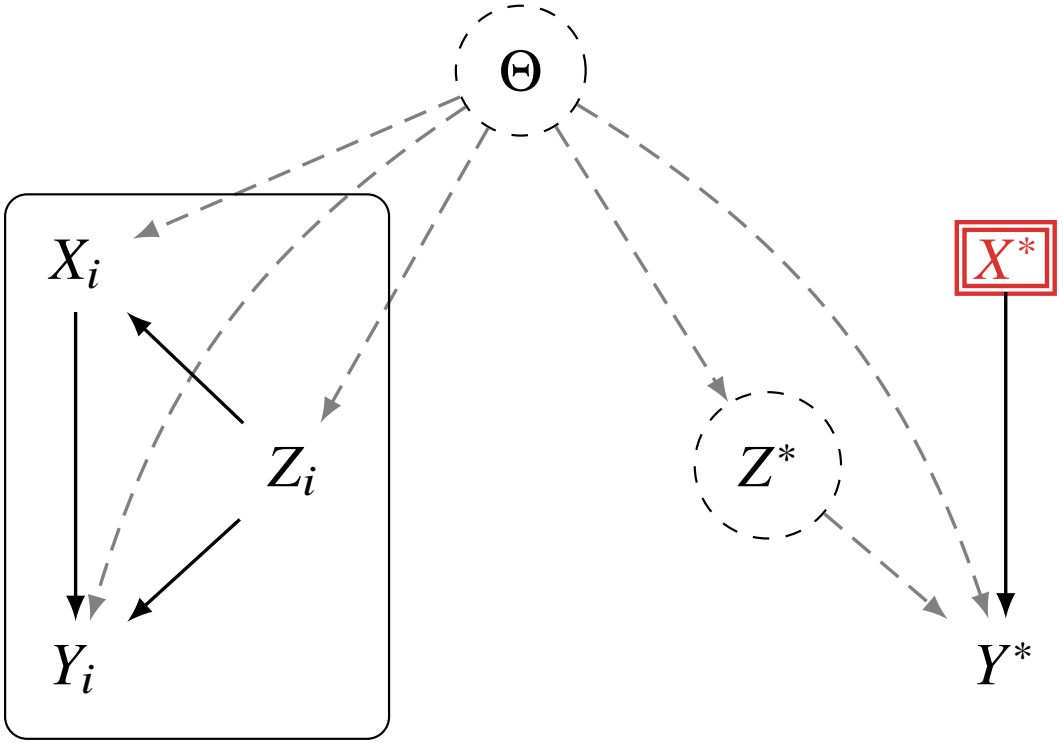
- Z: 头疼程度. X: 吃药. Y: 康复程度.
- 群体历史观察数据 $D = (z_i, x_i, y_i)_{i=1}^N$.
- 给定模型参数 Θ, 现实世界的随机变量 (Z, X, Y) 和平行世界的随机变量 (Z^*, X^*, Y^*) 条件独立.

$$(Z, X, Y) \perp (Z^*, X^*, Y^*) \mid \Theta$$

- 为了方便, 我们用 Q 表示平行世界里随机变量的分布. 比如, 用 $Q(y^* \mid x^*)$ 代替 $P_{Y^*|X^*}(y^* \mid x^*)$.
- 假设: 在平行世界里, 未被干预的因果机制保持不变. $Q_\theta(z^*) = P_\theta(z^*)$, $Q_\theta(y^* \mid z^*, x^*) = P_\theta(y^* \mid z^*, x^*)$.
- 被干预 $\mathrm{do}(X^* = a)$ 后的 $Q_\theta(x^*) = Q(x^*) = \delta(x^* - a)$.[17]

[17]连续变量时用 Dirac δ 函数. 满足 $\int_{-\infty}^{\infty} \delta(x^* - a)\,\mathrm{d}x^* = 1$ 和 $\int f(x^*)\delta(x^* - a)\,\mathrm{d}x^* = f(a)$. 离散变量时用 $[\![x^* = a]\!]$.

概率论因果 — 干预

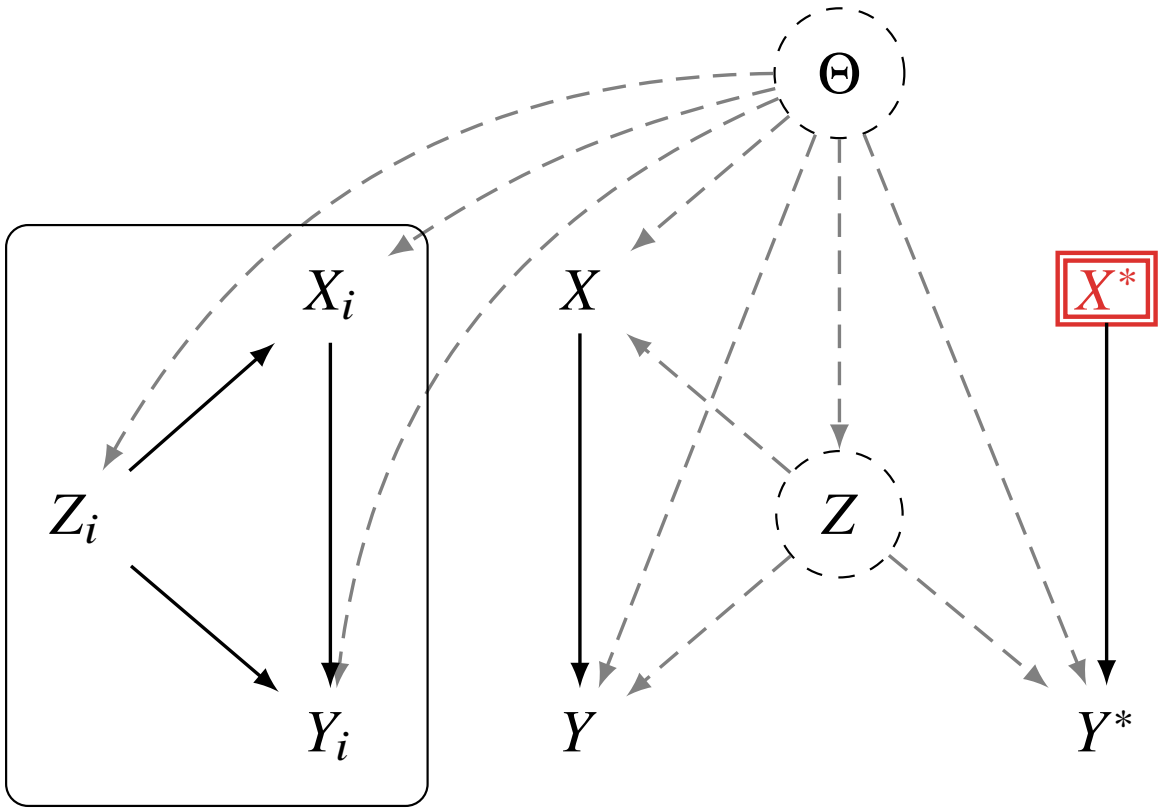


- $Z \sim \log \mathcal{N}(\mu_Z, \sigma_Z^2)$
- $X = Z^a \varepsilon_X$ where $\varepsilon_X \sim \log \mathcal{N}(0, \sigma_X^2)$
- $Y = \frac{Z^b}{X^c} \varepsilon_Y$ where $\varepsilon_Y \sim \log \mathcal{N}(0, \sigma_Y^2)$
- $\theta = (a, b, c, \mu_Z, \sigma_Z, \sigma_X, \sigma_Y)$

$$P(z^*, x^*, y^*, \theta, D) = P(\theta) \underbrace{\left(\prod_{i=1}^{N} P_\theta(z_i) P_\theta(x_i \mid z_i) P_\theta(y_i \mid z_i, x_i) \right)}_{\text{Observed World } P(D \mid \theta)} \underbrace{P_\theta(z^*) Q(x^*) P_\theta(y^* \mid z^*, x^*)}_{\text{Interventional World}}$$

$$P(y^* \mid D, x^*) = \int \left(\int P_\theta(z^*) P_\theta(y^* \mid z^*, x^*) \, \mathrm{d}z^* \right) P(\theta \mid D) \, \mathrm{d}\theta$$

概率论因果 — 干预反事实 (亚群体干预反事实)



$$\Theta = (a, b, c, \mu_Z, \sigma_Z, \sigma_X, \sigma_Y)$$

Θ 编码了群体分布的参数信息.
Z 是当前个体未观测的外生变量.

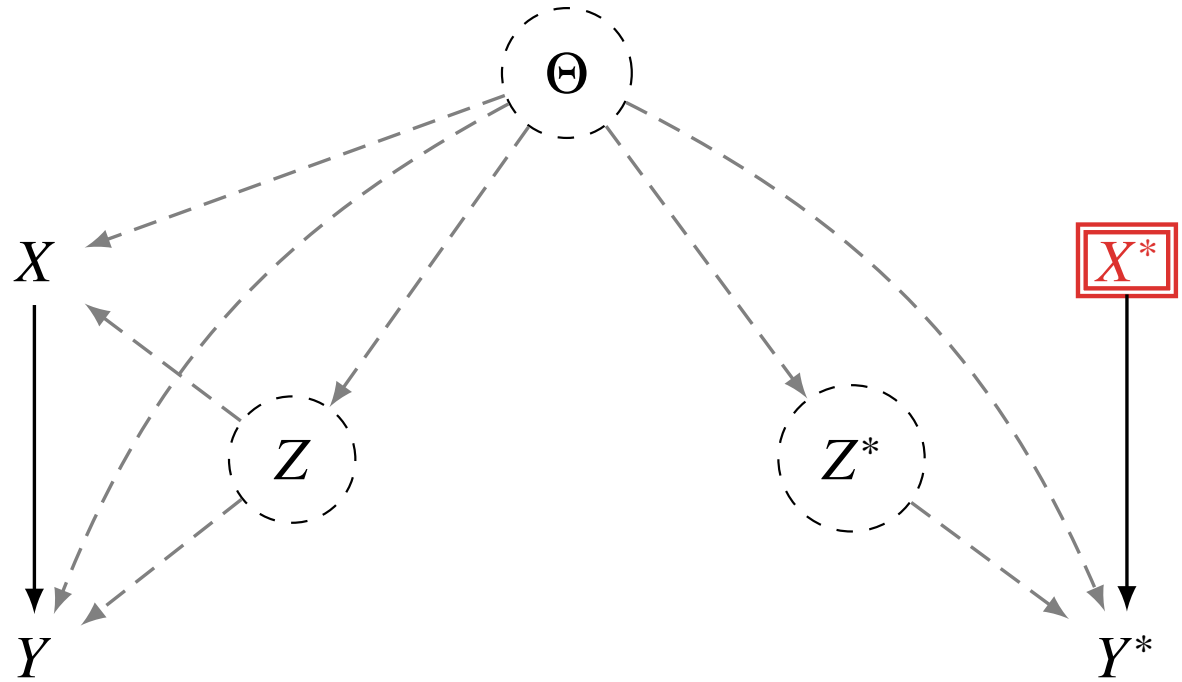
$$(X, Y) \perp (X^*, Y^*) \mid \Theta, Z$$

$$P(z, x, y, x^*, y^*, \theta, D) = P(\theta)\underbrace{\left(\prod_{i=1}^{N} P_\theta(z_i)P_\theta(x_i \mid z_i)P_\theta(y_i \mid z_i, x_i)\right)}_{\text{群体历史观察数据似然 } P(D \mid \theta)}\left(\underbrace{P_\theta(x \mid z)P_\theta(y \mid z, x)}_{\text{观察世界}}\, P_\theta(z)\, \underbrace{Q(x^*)P_\theta(y^* \mid z, x^*)}_{\text{反事实世界}}\right)$$

利用群体历史观察数据 D, 预测当前个体的反事实概率:

$$P(y^* \mid D, x, y, x^*) = \int \left(\int P_\theta(z \mid x, y)P_\theta(y^* \mid z, x^*)\, \mathrm{d}z\right) P(\theta \mid D, x, y)\, \mathrm{d}\theta$$

概率论因果 — 干预反事实 (个体干预反事实)

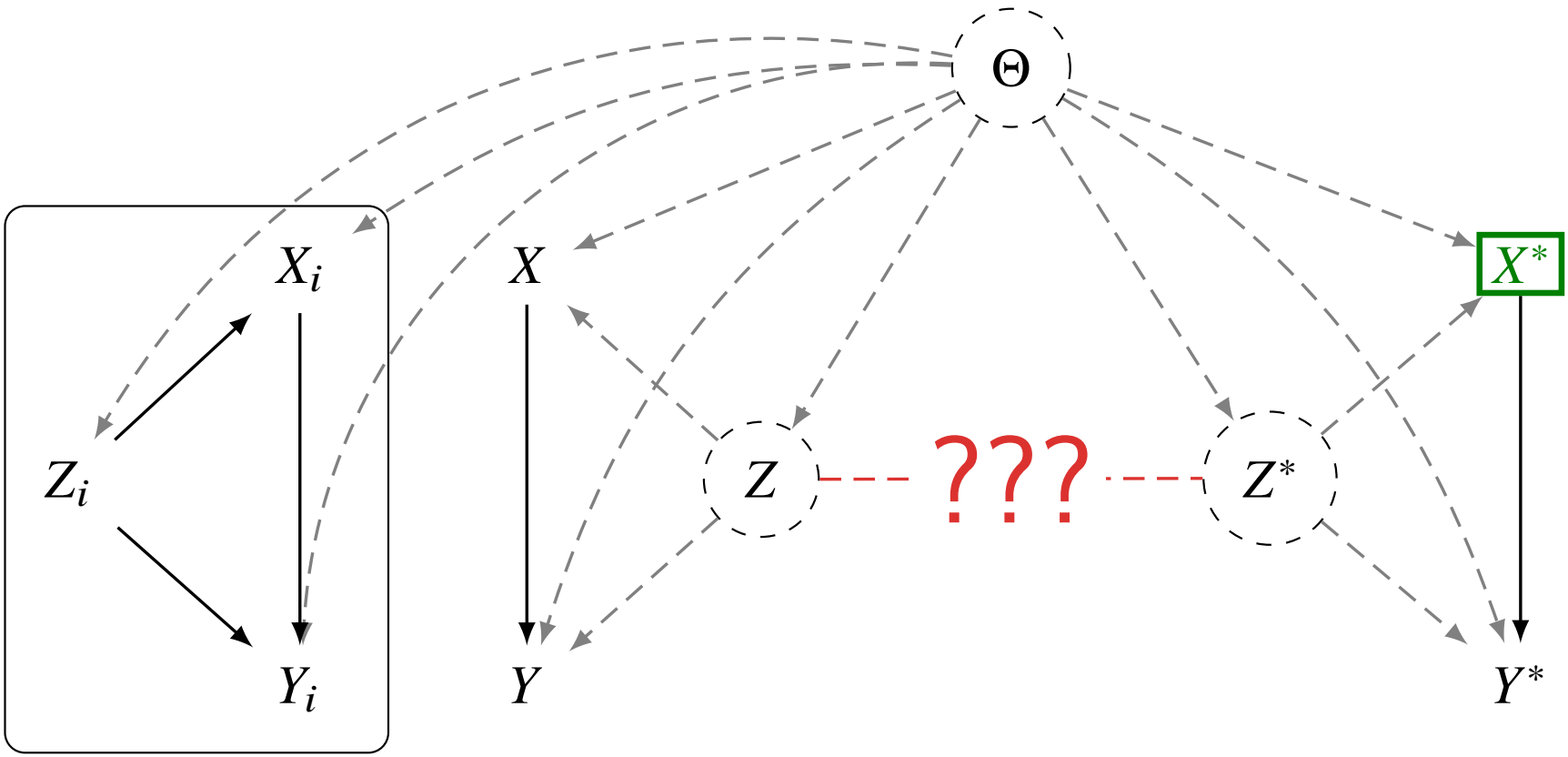


- 令 Z, X, Y 均为内生变量. 令 $\Theta = (U_Z, U_X, U_Y)$ 编码个体外生变量.
- 此时, 与 Pearl 基于 SCM 的反事实完全等价.
- **Remark:** 但若 Θ 编码的是群体分布信息, 那么, 更新宏观参数 (比如 μ, σ), 跟直接更新个体外生变量 $P(u \mid x, y)$ 相比, 不等价!

$$P(y^* \mid x, y, x^*) = \int [\![Y^*_{x^*}(u) = y^*]\!] \, P(u \mid x, y) \, \mathrm{d}u$$

Remark: 若可溯因出唯一的 u, 则 $P(u \mid x, y)$ 退化为 Dirac δ.

概率论因果 — 回溯反事实



$$P(z,x,y,z^*,x^*,y^*,\theta,D) = P(\theta)\underbrace{\left(\prod_{i=1}^{N} P_\theta(z_i)P_\theta(x_i \mid z_i)P_\theta(y_i \mid z_i,x_i)\right)}_{\text{群体历史观察数据似然 } P(D \mid \theta)} \underbrace{P_\theta(z)P_\theta(x \mid z)P_\theta(y \mid z,x)}_{\text{观察世界}} \underbrace{P_\theta(z^*)P_\theta(x^* \mid z^*)P_\theta(y^* \mid z^*,x^*)}_{\text{反事实世界}}$$

$$P(y^* \mid D,x,y,x^*) = \int \left(\int P_\theta(z^* \mid x^*)P_\theta(y^* \mid z^*,x^*)\,\mathrm{d}z^* \right) P(\theta \mid D,x,y,x^*)\,\mathrm{d}\theta$$

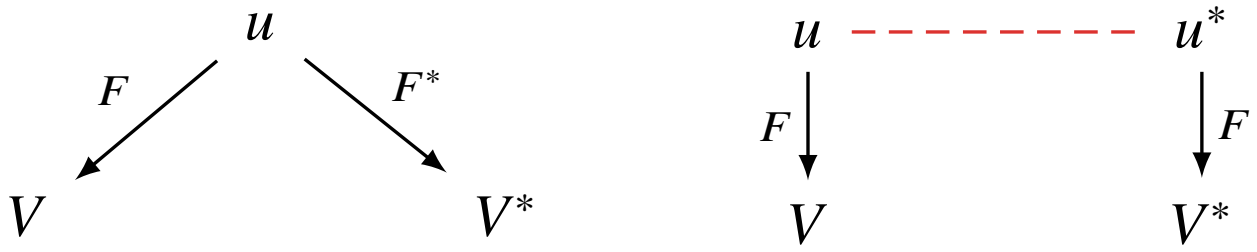
Remarks: 概率论因果 — 回溯反事实

- 对回溯反事实而言, 如果 Θ 编码个体的外生变量, 则现实世界与平行世界既共享结构方程又共享外生变量, 就不可能存在反事实.
- 所以, 这里的 Θ 编码的是群体分布的参数信息.
- 如果 Z 是内生变量, 其外生变量 U_Z 的分布的参数信息编码在了 Θ 里, 此时, $Z \perp Z^* \mid \Theta$.
- 如果 Z 是外生变量, 情形也一样. 只要 Θ 编码的是 Z 的分布的参数而不是具体取值, Z^* 的取值就无法直接依赖于 Z 的取值, 而需要从分布中重新采样. 此时, 同样有 $Z \perp Z^* \mid \Theta$.

$$P(z^* \mid z) = \frac{P(z, z^*)}{P(z)} = \frac{\int P(z^* \mid \theta) P(z \mid \theta) P(\theta) \, \mathrm{d}\theta}{P(z)} = \int P(z^* \mid \theta) P(\theta \mid z) \, \mathrm{d}\theta$$

为了让平行世界的外生变量 U, U^* 之间建立更直接的联系, Kügelgen 的回溯反事实额外引入了 $P(u^* \mid u)$.

Remarks: 基于 SCM 的干预反事实 vs 回溯反事实



$$P(y^* \mid x, y, x^*) = \int [\![Y^*_{x^*}(u) = y^*]\!] \, P(u \mid x, y) \, \mathrm{d}u$$

$$P(y^* \mid x, y, x^*) = \int [\![Y^*(u^*) = y^*]\!] \underbrace{\left(\int P(u, u^* \mid x, y, x^*) \, \mathrm{d}u \right)}_{P(u^* \mid x, y, x^*)} \mathrm{d}u^*$$

$$= \int [\![Y^*(u^*) = y^*]\!] \, P(u^* \mid x, y, x^*) \, \mathrm{d}u^*$$

Remarks

结构方程模型 SCM	概率论因果
结构方程	概率生成模型
外生变量 U	全局参数 Θ[18]
干预算子 do	Dirac δ 函数

- 计算干预时, θ 只需要通过群体历史观察数据 D 估计 $P(\theta \mid D)$.
- 计算干预反事实时, θ 需要被当前的事实 x, y 更新 $P(\theta \mid D, x, y)$.
- 计算回溯反事实时, θ 还需要被反事实假设 x^* 更新 $P(\theta \mid D, x, y, x^*)$.

[18] Θ 编码个体的外生变量即 SCM 意义上的个体干预反事实, 编码群体分布的参数信息即亚群体干预反事实.

Remarks

- 不管是 Lewis 的 Imaging, 还是 Pearl 的 SCM, 还是概率论因果, 不管是干预, 还是干预反事实, 还是回溯反事实, 都有一个统一的模式.

$$\int P_\theta(y^* \mid x^*) P(\theta \mid C) \, \mathrm{d}\theta$$

- 区别: 一是条件 C 不同; 二是计算 $P_\theta(y^* \mid x^*)$ 的方式不同; 三是 Θ 不同.

1. C 的不同:
   - 只包含群体历史观察数据 D? — 干预
   - 还包含当前个体的事实 D, x, y? — 干预反事实
   - 还包含当前个体的反事实 D, x, y, x^*? — 回溯反事实
2. $P_\theta(y^* \mid x^*)$ 的不同:
   - 在计算 $P_\theta(x^* \mid \mathrm{pa}^*)$ 时是否要坍缩为 Dirac δ 分布? — 干预 & 干预反事实 vs 回溯反事实
   - 外生变量是否共享? — 干预反事实 vs 回溯反事实
3. Θ 的不同: 哪些隐变量应该跨世界耦合/共享?
   - 编码群体分布的参数信息 — 亚群体干预反事实、回溯反事实
   - 编码个体的外生变量 — 个体干预反事实

Remarks: 哪些隐变量应该跨世界耦合/共享?

- 如果隐变量表示个体的固有属性或者世界的不变属性, 则共享.
- 如果隐变量表示环境的随机噪声, 则不共享.
- 如果隐变量是机器学习中为了梯度优化而人为加入到模型里的高斯噪声, 则不共享.

共享的隐变量越少, 越接近群体层面; 共享越多, 越接近个体层面.

1. 给所有人的药量加倍, 会怎样? — 不共享个体噪声, 只共享群体分布参数, 比如 μ, σ.
2. 一个头疼程度为 5 的人药量加倍, 会怎样? — 共享代表亚群体特征的头疼程度 Z.
3. 如果昨天 Alice 没吃阿司匹林, 会怎样? — 共享代表 Alice"年龄/代谢" 等个体状态的 U_Y.

类型	耦合/共享
干预	共享 Θ
亚群体干预反事实	共享 (Θ, Z)
个体干预反事实	共享 $\Theta \supset U$
SCM	共享 $\Theta := U$
回溯反事实	耦合 $P(U, U^*)$

Remarks

1. Lewis: Imaging Theory. 预设可能世界之间的邻近关系.
2. Pearl: SCM/因果图. 共享外生变量, 机制干预 do.
3. Rubin: 潜在结果框架. 不画因果图. 潜在结果 Y_x 作为随机变量. 预设 Ignorability/Consistency/SUTVA/Positivity 等条件.
4. Richardson & Robins: 单世界干预图 SWIG. 将节点分裂为变量 X 和值 x, 下游变量是潜在结果 Y_x, 利用 SWIG 图上的 d-separation.
5. Mlodozeniec & Krueger & Turner: 概率论因果. 顶层共享全局参数 Θ, 利用跨世界联合分布下的标准贝叶斯.
6. Dawid: 决策论因果. 本体论上拒绝潜在结果变量 Y_x. 引入决策节点 σ_X, 利用扩展图上的 d-separation.

线性模型中反事实的识别

- 当结构方程模型已知时, 反事实不难计算.
- 但当某些模型参数无法识别时, 反事实是否可以经验识别?
  — 在线性模型中, 当 $\mathbb{E}[Y \mid \mathrm{do}(X = x)]$ 可识别时, 反事实 $\mathbb{E}[Y_{X=x} \mid E = e]$ 也可以识别.

Theorem

令 $\tau := \frac{\mathrm{d}\mathbb{E}[Y_x]}{\mathrm{d}x} = \mathbb{E}[Y \mid \mathrm{do}(x+1)] - \mathbb{E}[Y \mid \mathrm{do}(x)]$ 表示 X 到 Y 的因果效应的回归曲线的斜率, 则对于任意证据 $E = e$,

$$\begin{aligned}\mathbb{E}[Y_{X=x} \mid E = e] &= \mathbb{E}[Y \mid E = e] + \tau\big(x - \mathbb{E}[X \mid E = e]\big) \\ &= \mathbb{E}[Y \mid E = e] + \mathbb{E}[Y \mid \mathrm{do}(X = x)] - \mathbb{E}[Y \mid \mathrm{do}(X = \mathbb{E}[X \mid e])]\end{aligned}$$

Remark: 即, 首先计算证据 e 下 Y 的最优估计 $\mathbb{E}[Y \mid e]$, 然后加上当 X 从当前的最优估计 $\mathbb{E}[X \mid e]$ 转换到它的假设值 x 时 Y 的期望变化.
Example: 考虑 $e : X = x', Y = y'$, 则

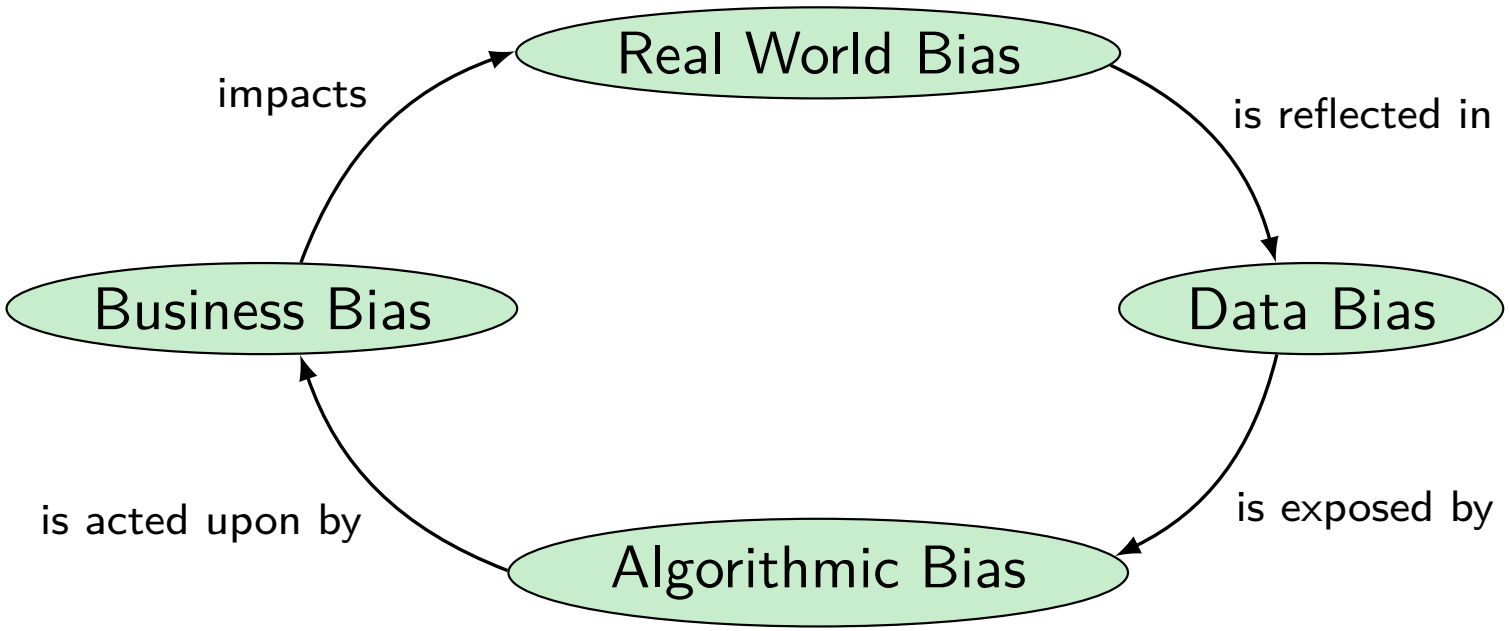
$$\mathbb{E}[Y_x \mid X = x', Y = y'] = y' + \tau(x - x')$$

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Algorithmic Fairness

- 为什么要关心 “算法公平”?
  - 公司招聘采用算法排名
  - 各种推荐系统
  - 美国法院引入机器学习来预测再犯罪风险
- “算法不公” 的可能原因:
  - 样本偏差: 某地犯罪率高, 警察更频繁地巡视, 记录犯罪率高.
  - 人类固有偏见: 护士看着受伤的卡车司机, 她/他 _____.
  - 样本污染: 词嵌入可能导致性别刻板印象. 人标注数据可能引入偏见.
  - 样本大小悬殊: 来自少数群体的训练数据少, 特征有限.
  - 算法偏见: 错把相关当因果.

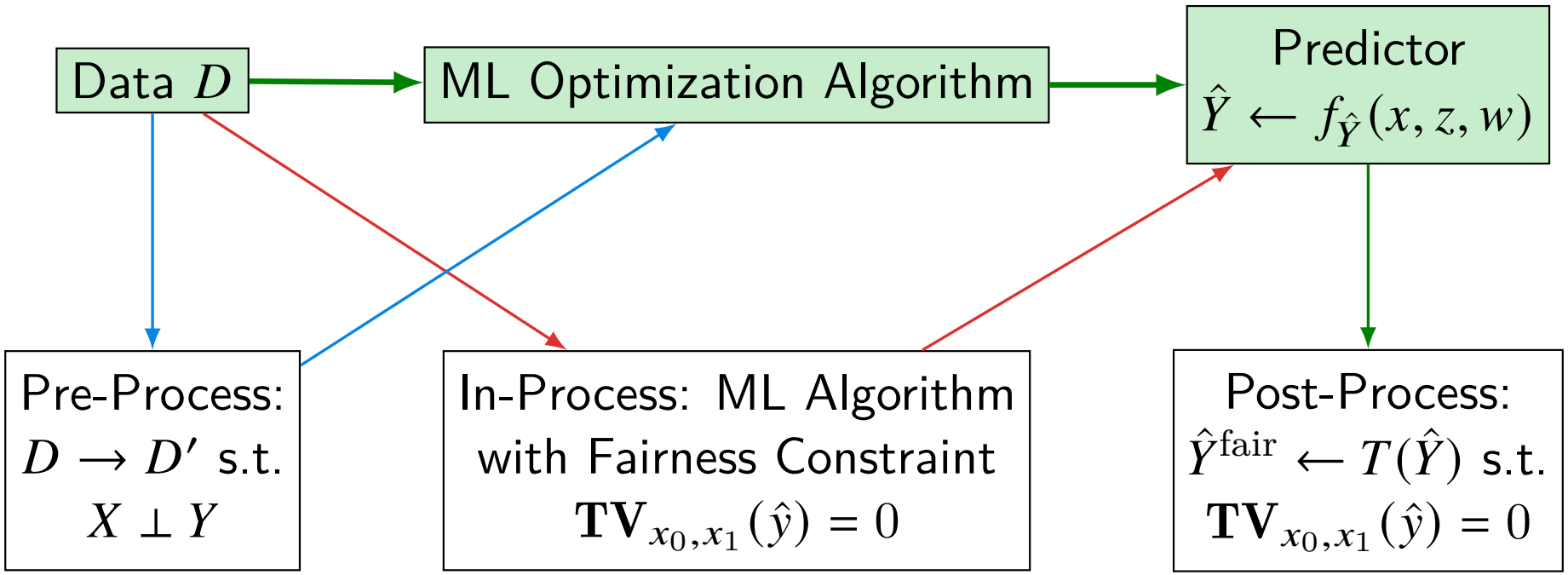


Causal Fairness Analysis

1. 偏差检测: 检测数据中的不同类型偏差 (直接、间接、虚假).
2. 公平预测: 构造满足特定公平标准的预测器.
3. 公平决策: 设计公平的策略, 在实施中不断减少不公平.

怎么保障预测算法的公平性?

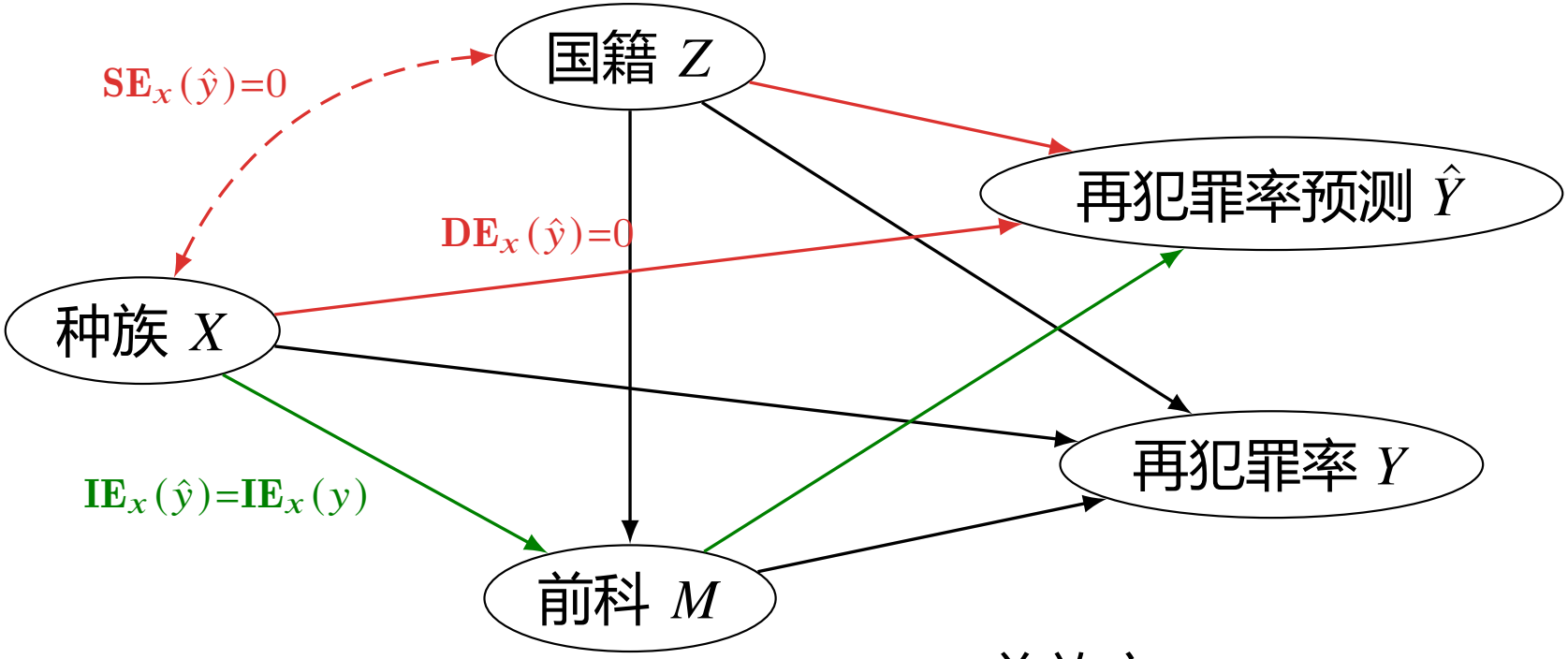
Pre-, In-, Post-Processing



怎么辨别预测算法的歧视性?

- 有一个计算犯人未来再次犯罪的可能性大小的程序, 法官根据预测的再犯罪率高低, 决定是否允许犯人保释. 保释后的黑人、白人各有一部分人再次犯罪.
- 低黑人: 犯罪 200, 没犯罪 800, 20%
- 低白人: 犯罪 250, 没犯罪 1000, 20%
- 高黑人: 犯罪 3000, 没犯罪 600, 83%
- 高白人: 犯罪 750, 没犯罪 150, 83%
- 黑人误判率: $\frac{600}{800+600} = 43\%$
- 白人误判率: $\frac{150}{1000+150} = 13\%$
- 误判率: 事后看明明没犯罪却事先被打上高再犯罪人群的标签
- $43\% > 13\%$ 意味着不公平?
- 黑人再犯罪率: $\frac{200+3000}{200+3000+800+600} = 70\%$
- 白人再犯罪率: $\frac{250+750}{250+750+1000+150} = 47\%$
- 黑人的再犯罪率高导致误判率高
- $70\% > 47\%$ 公平!

怎么辨别预测算法的歧视性?

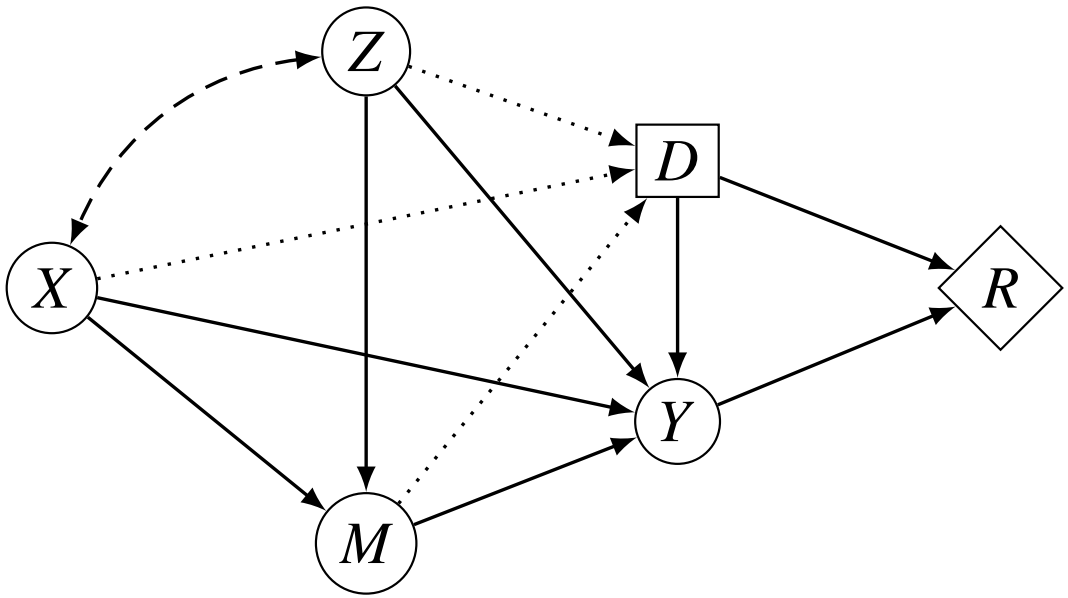


1. 群体公平
2. 个体公平

1. 统计公平
   - 统计均等
   - 机会均等
   - 预测均等
2. 因果公平
   - 干预公平
   - 反事实公平

- 总效应
- 受控直接效应
- 自然直接效应
- 自然间接效应
- 实验伪效应
- 反事实公平
- 特定路径的反事实公平

怎么做道德决策? Moral Decision



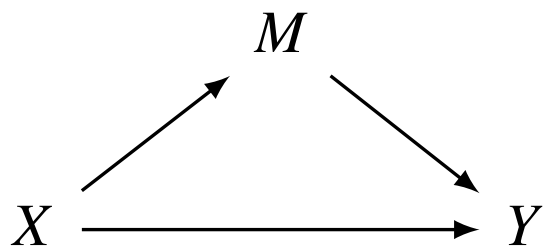
Moral agent need to have at least

1. beliefs about the world,
2. intentions,
3. moral knowledge,
4. the possibility to compute what consequences ones own action can have.

"Interventionist" Interpretation of Causality

No causation without manipulation?

- A Practical Definition of Causality: X causes Y iff changing X leads to a change in Y, while keeping everything else constant.



- Causal effect is defined as the magnitude by which Y is changed by a unit change in X.
  - Test the total effect of X on Y
  - Test the relationship between X and M
  - Test the relationship between M and Y, controlling for X
  - Declare whether M is a partial or full mediator

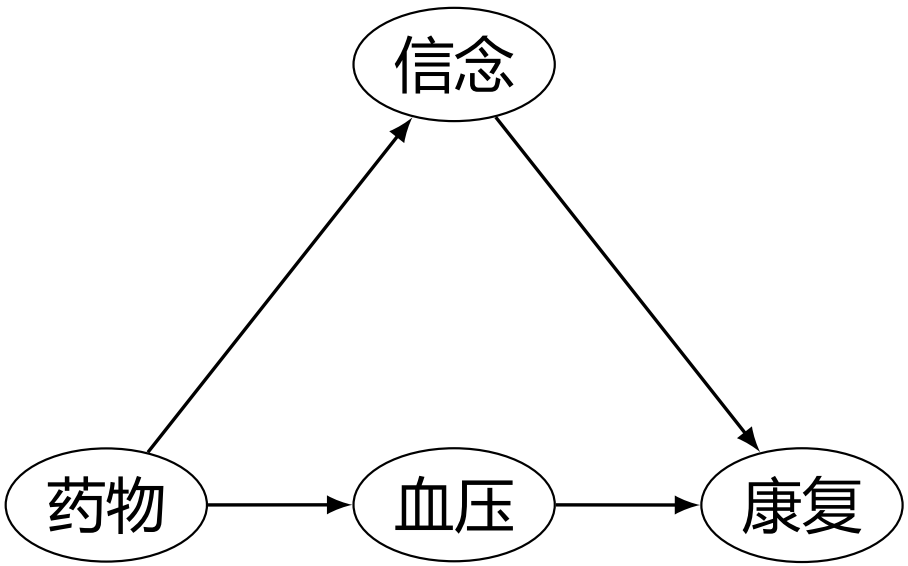
中介变量 Mediation

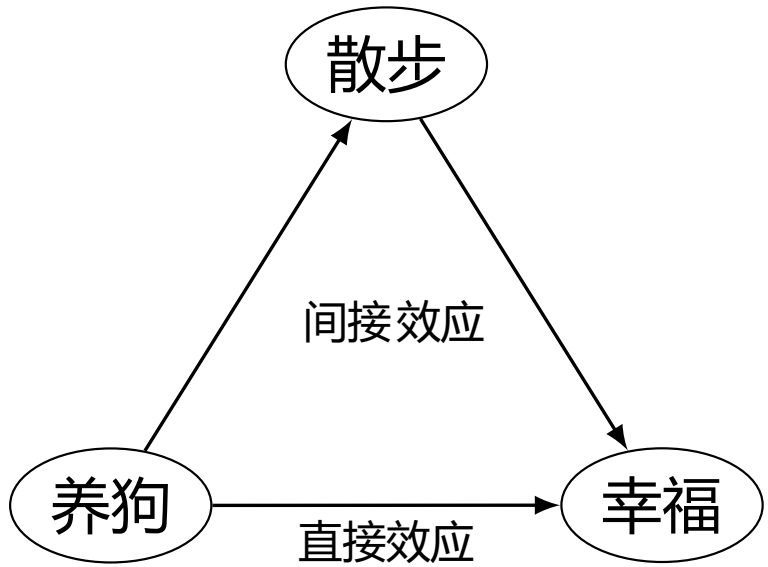
- 柑橘预防坏血病的机制是什么?

柑橘 ⟶ 酸性物质 ⟶ 坏血病 ×

柑橘 ⟶ 维生素 C ⟶ 坏血病 ✓

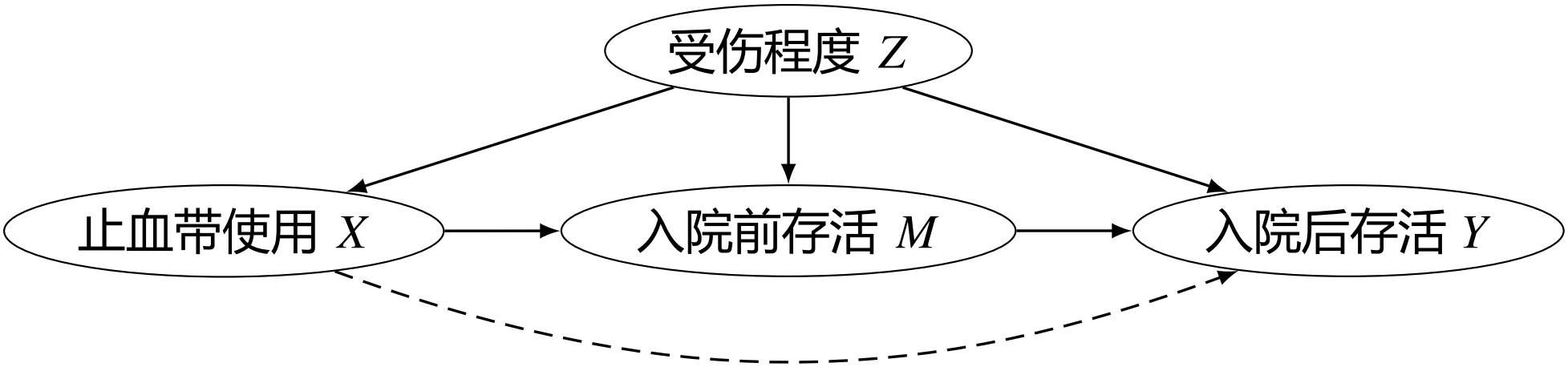
- 药物确有疗效还是安慰剂效应?





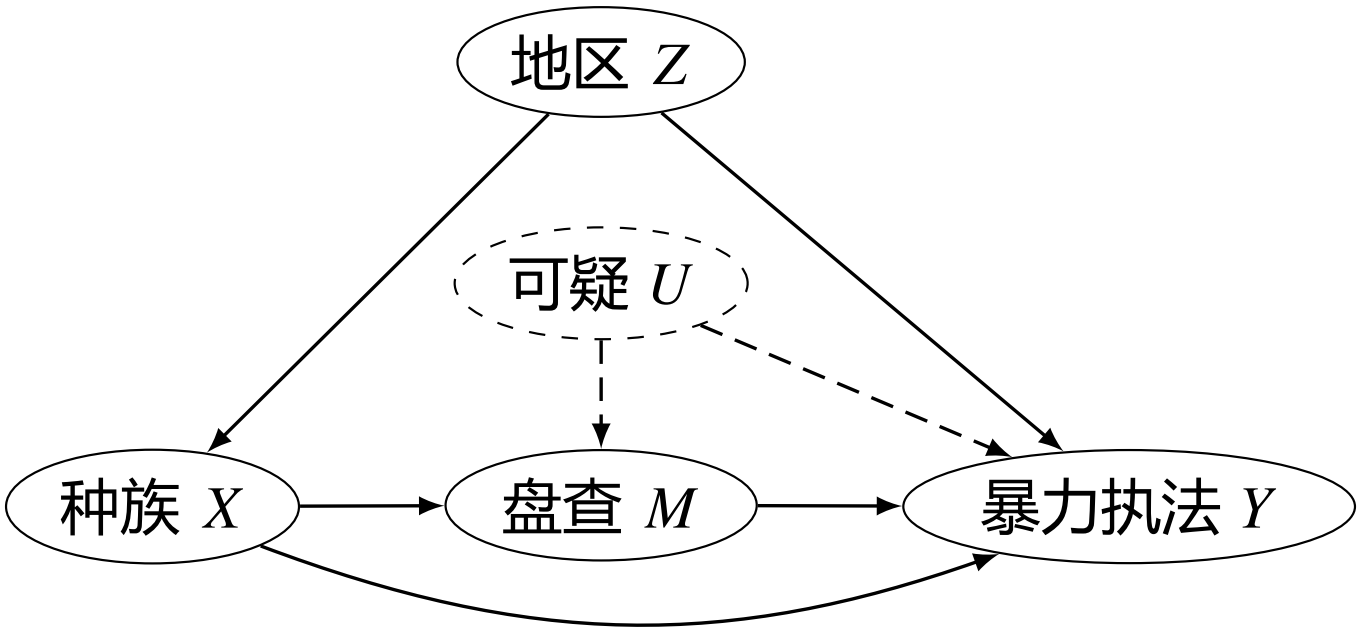
- 养狗为什么使你幸福?

止血带的作用 — 可能的间接效应



- 普遍认为，使用止血带可以提高伤员的存活率.
- 但根据医院的统计，不管是分为轻伤、重伤，还是一起统计，使用比不使用止血带存活率都略微低一点儿 (不显著). 这不是辛普森悖论.
- 难道止血带有副作用抵消了它的好处？或者有质量问题？或者.....
- 其实，医院统计的是活着被送到医院的伤员，这意味着控制了中介变量 M，从而阻断了间接路径，计算的是 X 对 Y 的直接效应.
- 但这无法排除间接效应的存在：可能止血带的作用是将伤员活着送到医院，到了医院后就没有进一步的用处了.
- 而确定间接效应需要统计入院前的存亡率.

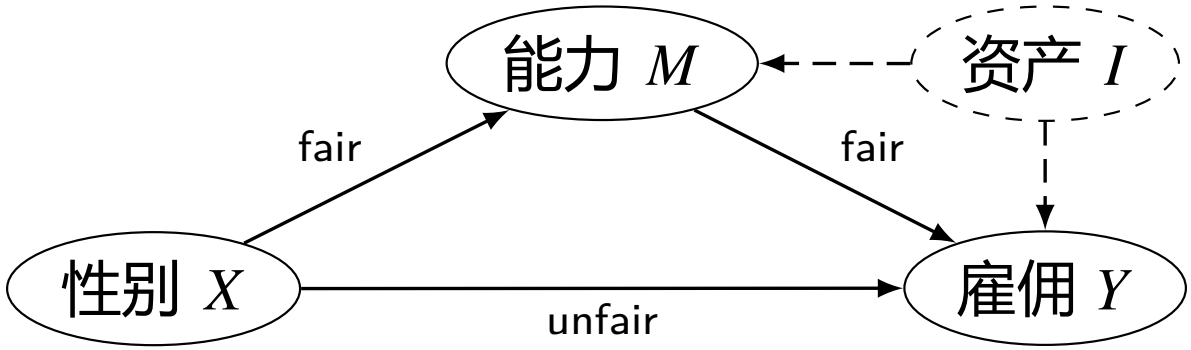
存在种族歧视吗?



- 根据警方的**盘查执法记录**统计, 在控制了地区等 X 到 Y 的后门路径后, 相比于白人, 黑人被暴力执法的概率并没有显著的高.
- 这是否说明警方没有种族歧视?

受控直接效应 Controlled Direct Effect

歧视: 假如应聘者除了 "性别" 外, 其他方面都一样, 雇佣情况是否会不同?



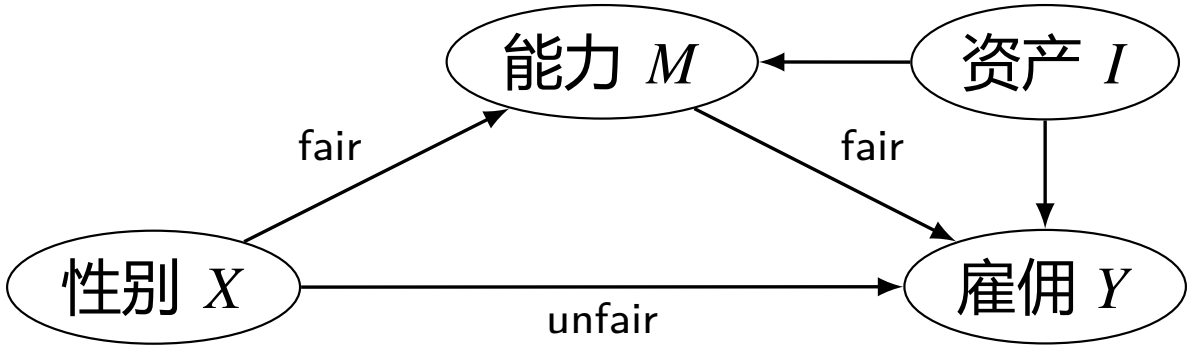
可以直接以变量 M 为条件划分数据吗?

$$\mathbb{E}\left[Y \mid \mathrm{do}(X=1), M=m\right] - \mathbb{E}\left[Y \mid \mathrm{do}(X=0), M=m\right] \text{?}\times$$

受控直接效应 (Controlled Direct Effect)

$$\mathrm{CDE}(m) := \mathbb{E}\left[Y \mid \mathrm{do}(X=1), \mathrm{do}(M=m)\right] - \mathbb{E}\left[Y \mid \mathrm{do}(X=0), \mathrm{do}(M=m)\right]$$

Remark: $\mathrm{CDE}(m)$ 依赖于 $M=m$.
$\sum_m \mathrm{CDE}(m)$? 可能在高水平工作中歧视女性, 在低水平工作中歧视男性.



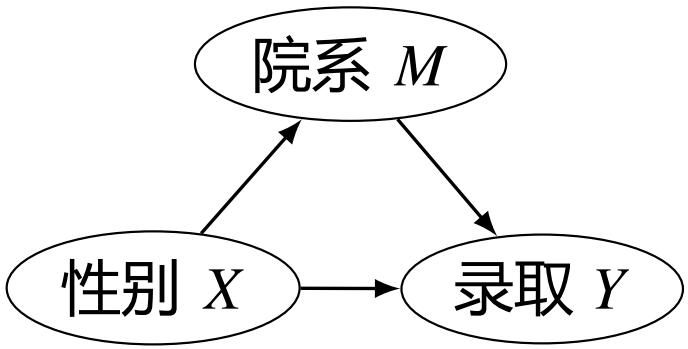
若资产 I 可观测, 则

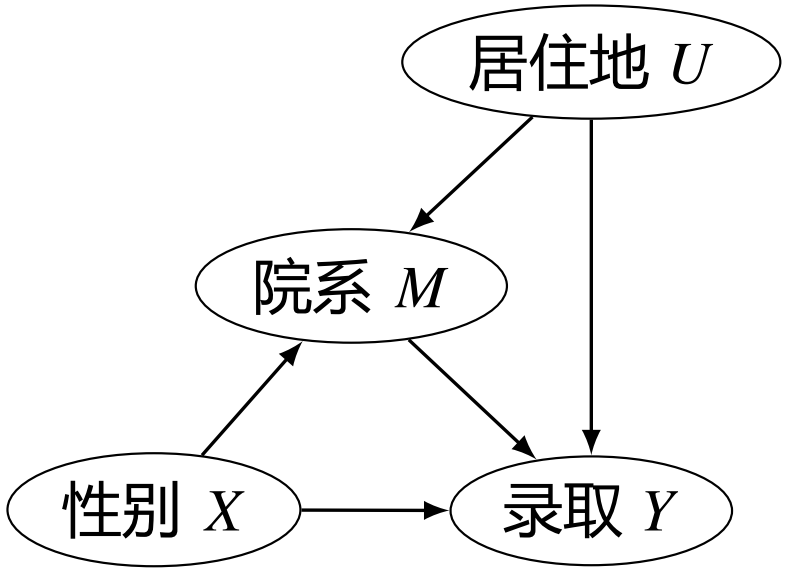
$$\mathrm{CDE}(m) = \sum_i \left[P(Y = y \mid X = x, M = m, I = i) - P(Y = y \mid X = x', M = m, I = i)\right] P(I = i)$$

In general, the CDE of X on Y, mediated by M, is identifiable if the following two properties hold:

1. There exists a set S_1 of variables that blocks all backdoor paths from M to Y.
2. There exists a set S_2 of variables that blocks all backdoor paths from X to Y, after deleting all arrows entering M.

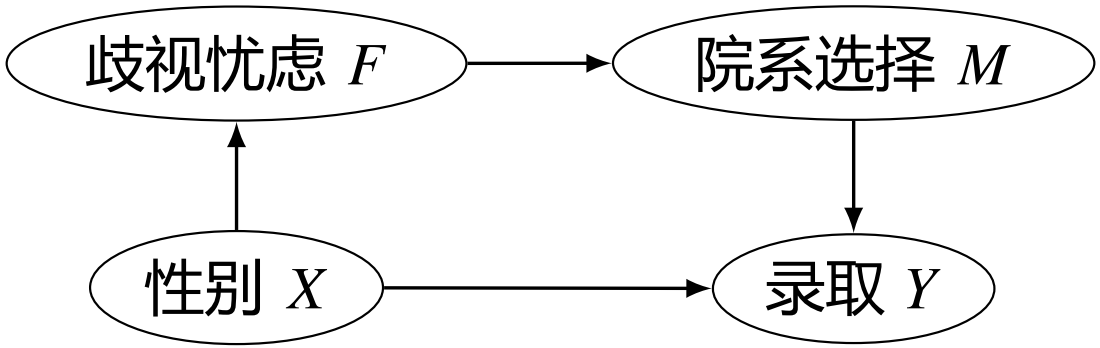
Example: 伯克利大学录取悖论





- 如果不做变量控制, 女性的录取率低.
- 如果控制 “院系”, 则女性的录取率高.
- 如果 “院系” 和 “录取结果” 之间有其它混杂因子呢?
- 如果控制 “院系” 和 “居住地”, 则女性的录取率低.
- 为什么要同时控制 “院系” 和 “居住地”?
- **歧视**是 “性别” 对 “录取结果” 的直接效应? (间接歧视呢?)

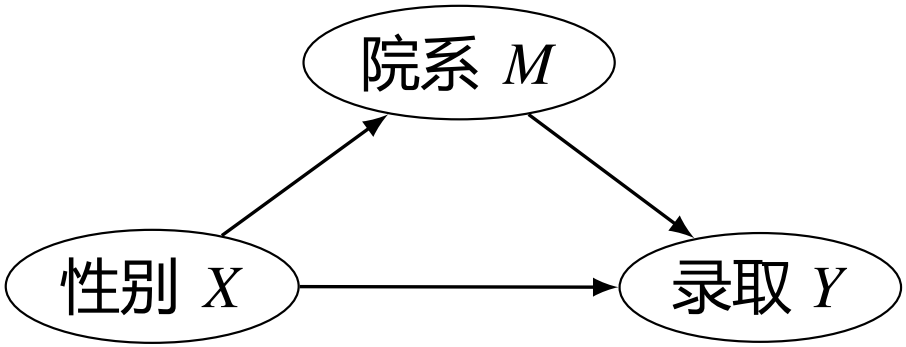
Remark: 间接歧视

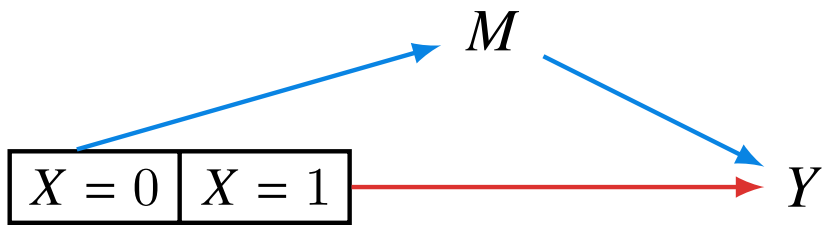


- 女性申请人可能忧虑某些院系歧视女性 F, 从而影响她们的院系选择 M.
- 间接路径也可能隐藏歧视.

自然直接效应 Natural Direct Effect

歧视是 “性别” 对 “录取结果” 的直接效应? 受控 or 自然? 间接歧视呢?





- “随机对照实验”?
- 强制所有人都申请数学系 $\mathrm{do}(M=m)$. 不论申请者的实际性别是什么, 随机分配一些人填报其性别为男性 $\mathrm{do}(X=1)$, 另一些人填报其性别为女性 $\mathrm{do}(X=0)$

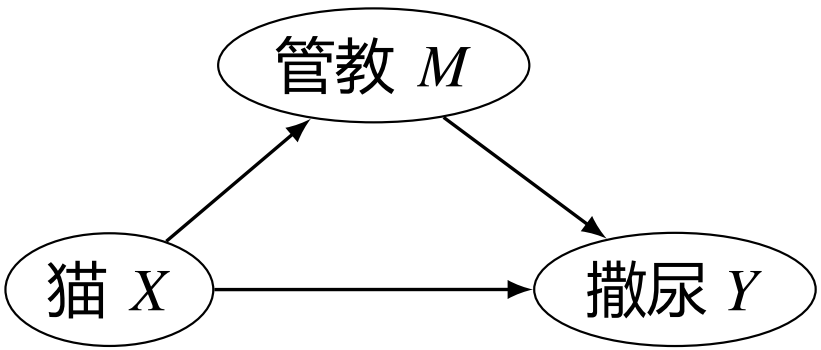
$$\mathrm{CDE}(m) = P\big(Y=1 \mid \mathrm{do}(X=1), \mathrm{do}(M=m)\big) - P\big(Y=1 \mid \mathrm{do}(X=0), \mathrm{do}(M=m)\big)$$

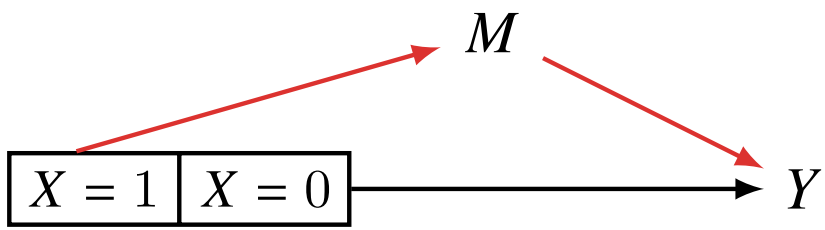
- 避免 “过度对照实验”: 你本想学数学, 却碰巧被随机分配去报哲学...
- 让申请人随机填报性别, 但遵照其本来的意愿申请青睐的院系.

$$\mathrm{NDE} = P\big(Y_{M_0}=1 \mid \mathrm{do}(X=1)\big) - P\big(Y_{M_0}=1 \mid \mathrm{do}(X=0)\big)$$

自然间接效应 Natural Indirect Effect

- 你有一只小狗, 随地撒尿.
- 朋友在你家寄养了一只小猫, 小狗的恶行收敛了.
- 小猫走后, 小狗故态复萌.
- 你记起来, 当小猫在时, 你对小狗的管教也更严格了.
- 那么, 是小猫的存在还是你的管教让小狗变文明了?

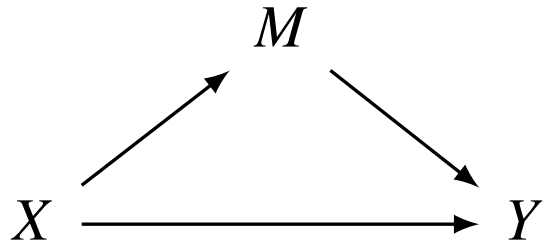




- 问题: 假如小猫不在场 $X = 0$, 而你像小猫在时一样管教小狗 $M = M_1$, 那么它还会随地撒尿吗?

$$\text{NIE} = P\big(Y_{M_1} = 1 \mid \text{do}(X = 0)\big) - P\big(Y_{M_0} = 1 \mid \text{do}(X = 0)\big)$$

Causal Effects and Path-Disabling Interventions



- Total Effect

$$\mathrm{TE} := \mathbb{E}[Y \mid \mathrm{do}(X = 1)] - \mathbb{E}[Y \mid \mathrm{do}(X = 0)] = \mathbb{E}[Y_1 - Y_0]$$

- Controlled Direct Effect

$$\begin{aligned}\mathrm{CDE}(m) &:= \mathbb{E}[Y \mid \mathrm{do}(X = 1, M = m)] - \mathbb{E}[Y \mid \mathrm{do}(X = 0, M = m)] \\ &= \mathbb{E}[Y_{1m}] - \mathbb{E}[Y_{0m}]\end{aligned}$$

- Natural Direct Effect

$$\mathrm{NDE} := \mathbb{E}\left[Y_{M_0} \mid \mathrm{do}(X = 1)\right] - \mathbb{E}\left[Y_{M_0} \mid \mathrm{do}(X = 0)\right] = \mathbb{E}[Y_{1,M_0}] - \mathbb{E}[Y_0]$$

- Natural Indirect Effect

$$\mathrm{NIE} := \mathbb{E}\left[Y_{M_1} \mid \mathrm{do}(X = 0)\right] - \mathbb{E}\left[Y_{M_0} \mid \mathrm{do}(X = 0)\right] = \mathbb{E}[Y_{0,M_1}] - \mathbb{E}[Y_0]$$

- Experimental Spurious Effect

$$\text{Exp-SE}_x := \mathbb{E}[Y \mid x] - \mathbb{E}[Y_x]$$

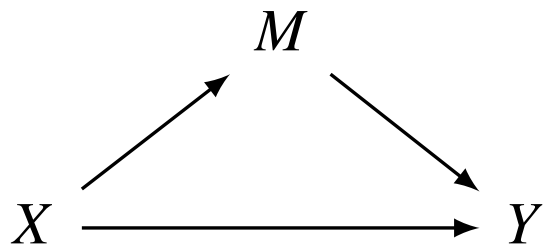
Mediation Formulas

- Total Effect

$$\mathrm{TE} = \mathrm{NDE} - \mathrm{NIE}_r$$

$$\mathrm{NIE}_r := \mathbb{E}\left[Y_{M_0} \mid \mathtt{do}(X=1)\right] - \mathbb{E}\left[Y_{M_1} \mid \mathtt{do}(X=1)\right] = \mathbb{E}[Y_{1,M_0}] - \mathbb{E}[Y_1]$$

- If no confounding exists, the following mediation formulas can be derived:



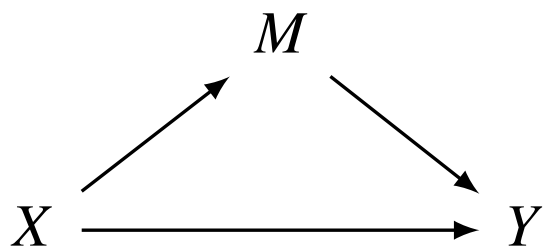
$$\mathrm{NDE} = \sum_m \Big(\mathbb{E}[Y \mid X=1, M=m] - \mathbb{E}[Y \mid X=0, M=m]\Big) \times P(M=m \mid X=0)$$

$$\mathrm{NIE} = \sum_m \mathbb{E}[Y \mid X=0, M=m] \times \Big(P(M=m \mid X=1) - P(M=m \mid X=0)\Big)$$

Conditions for Identifying Natural Effects

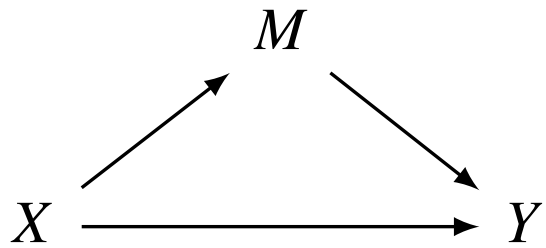
We can identify NDE and NIE provided there exists a set Z of measured covariates such that

- No member of Z is a descendent of X
- Z blocks all backdoor paths from M to Y
- The Z-specific effect $P(M = m \mid \mathrm{do}(X = x), Z = z)$ of X on M is identifiable (possibly using experiments or adjustments)
- The Z-specific joint effect $P(Y = y \mid \mathrm{do}(X = x), \mathrm{do}(M = m))$ of $\{X, M\}$ on Y is identifiable (possibly using experiments or adjustments)



Comparison of Controlled vs Natural Mediation

- CDE 总可以通过实验 (do-operator) 测量
- NDE 涉及反事实, 不一定总能通过实验测量



- NDE/TE: Fraction of response that is transmitted directly, with M frozen
- NIE/TE: Fraction of response explained by mediation, with Y blinded to X (sufficient)
- $(\mathrm{TE}-\mathrm{NDE})/\mathrm{TE}$: Fraction of responses owed to mediation (necessarily due to M)

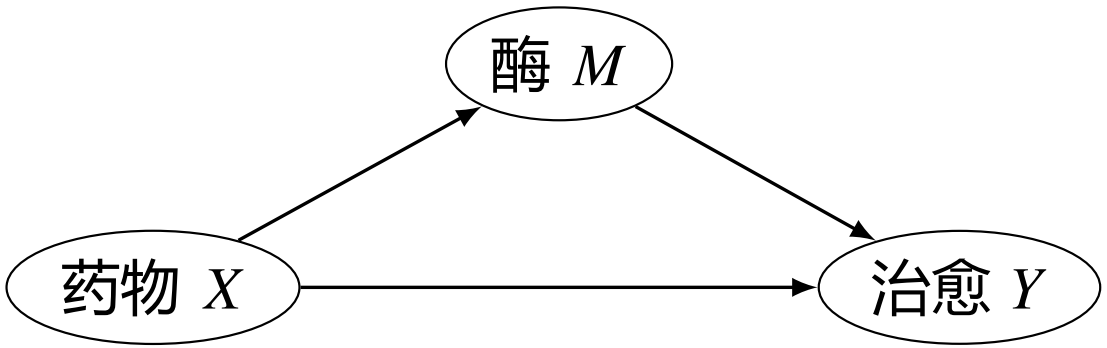
In linear systems, we have

$$\mathrm{TE} = \mathrm{NDE} + \mathrm{NIE}$$

This does not work in models that involve interactions,

$$\mathrm{TE} \neq \mathrm{NDE} + \mathrm{NIE}$$

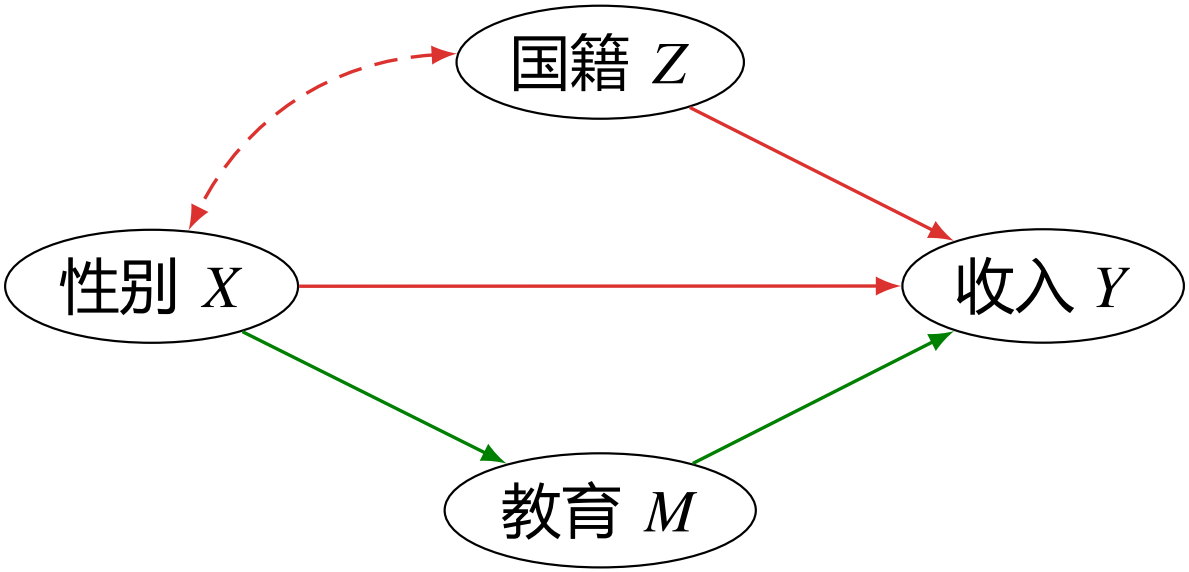
Total Effect ≠ Direct Effect + Indirect Effect



$$M = X$$
$$Y = X \wedge M$$

- 某种药物 X 会使身体分泌某种酶 M 做催化剂, 共同治愈疾病 Y.
- 药物的总效应是正的.
- 但直接效应是 0, 因为如果阻止身体分泌酶的话, 药物无法单独起作用.
- 间接效应也是 0, 因为只有酶没有药物的话, 疾病也无法治愈.

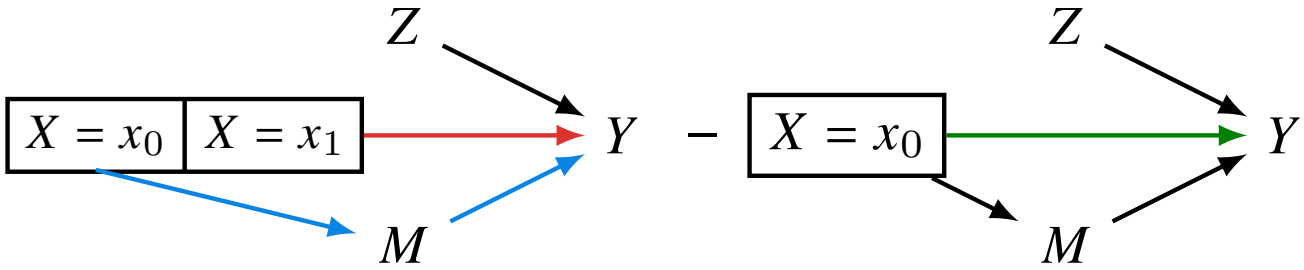
The Attribution Problem



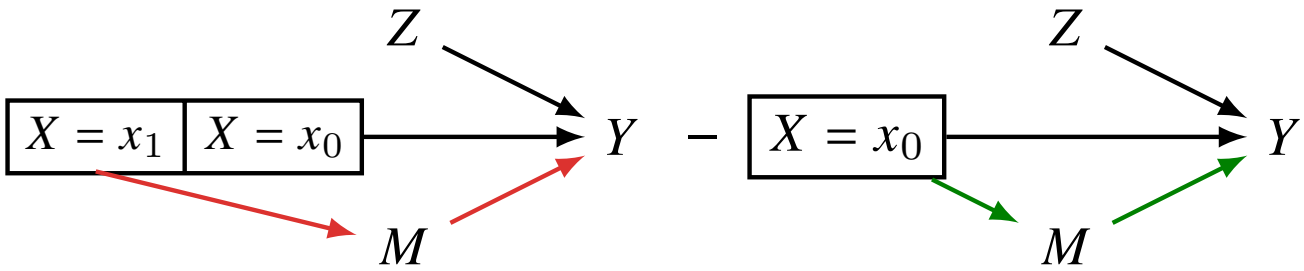
The observed disparity in $\mathbf{TV} = \mathbb{E}[Y \mid \text{male}] - \mathbb{E}[Y \mid \text{female}]$ could be explained in different ways.

- **Direct:** The salary decision is based on employee's gender $X \rightarrow Y$
- **Indirect:** Decisions were based on education $X \rightarrow M \rightarrow Y$
- **Spurious:** Nationality is used to infer the person's gender $X \leftrightarrow Z \rightarrow Y$

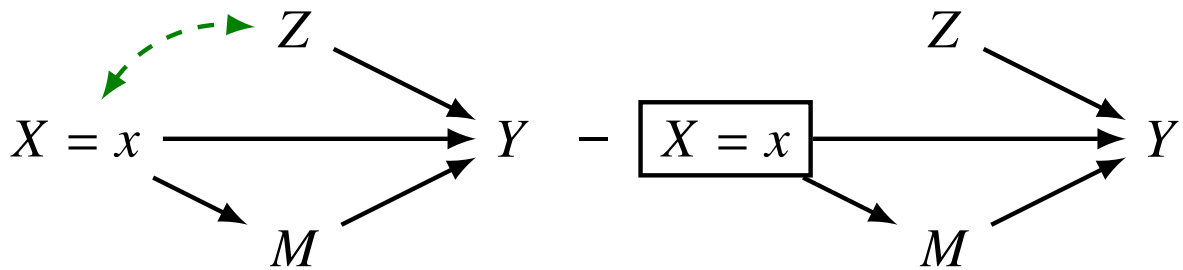
$\mathbf{NDE}_{x_0,x_1}(y) = P(y_{x_1,M_{x_0}}) - P(y_{x_0,M_{x_0}})$ For a person set to be female $X = x_0$, how would her salary Y change had she been set to be male $X = x_1$, while keeping the nationality N, education E unchanged $X = x_0$?



$\mathbf{NIE}_{x_0,x_1}(y) = P(y_{x_0,M_{x_1}}) - P(y_{x_0,M_{x_0}})$ For a person set to be female $X = x_0$, how would her salary Y change had she been set to be male $X = x_1$, while keeping gender unchanged $X = x_0$ along the direct causal pathway?



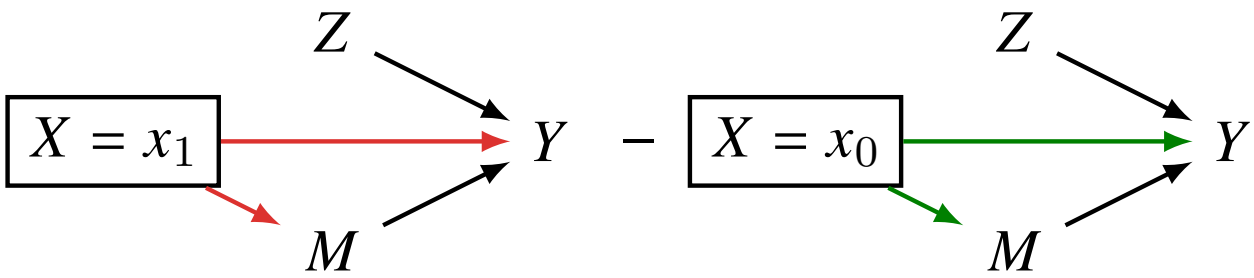
$\mathbf{Exp\text{-}SE_x} = P(y \mid x) - P(y_x)$ How would an individual's salary Y change if their gender is set to male (or female) by intervention, compared to observing their salary as male (female)?



Decomposition into direct, indirect, and spurious effects

Total Effect:

$$\mathbf{TE}_{x_0,x_1}(y) = \mathbb{E}[y_{x_1}] - \mathbb{E}[y_{x_0}]$$



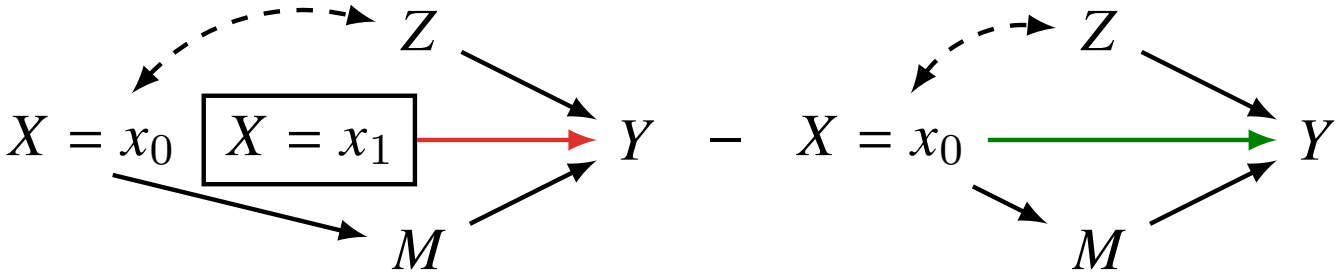
The Total Variation measure

$$\mathbf{TV}_{x_0,x_1}(y) \coloneqq \mathbb{E}[y \mid x_1] - \mathbb{E}[y \mid x_0]$$

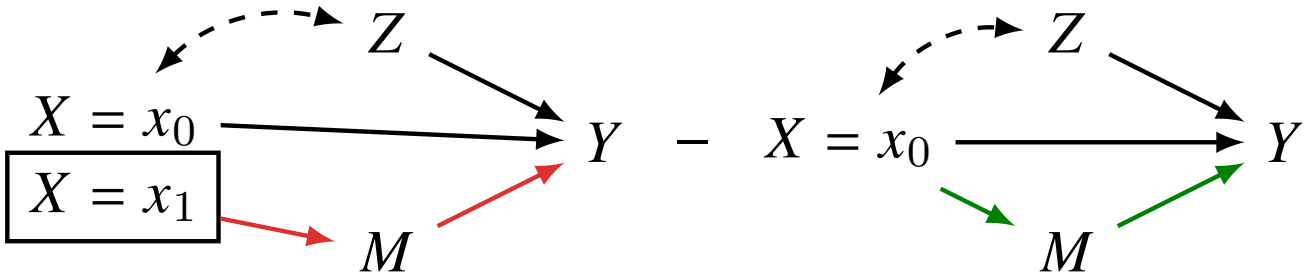
$$\mathbf{TV}_{x_0,x_1}(y) = \mathbf{TE}_{x_0,x_1}(y) + \Big(\mathbf{Exp\text{-}SE}_{x_1}(y) - \mathbf{Exp\text{-}SE}_{x_0}(y)\Big)$$

$$\mathbf{TV}_{x_0,x_1}(y) = \mathbf{NDE}_{x_0,x_1}(y) - \mathbf{NIE}_{x_1,x_0}(y) + \Big(\mathbf{Exp\text{-}SE}_{x_1}(y) - \mathbf{Exp\text{-}SE}_{x_0}(y)\Big)$$

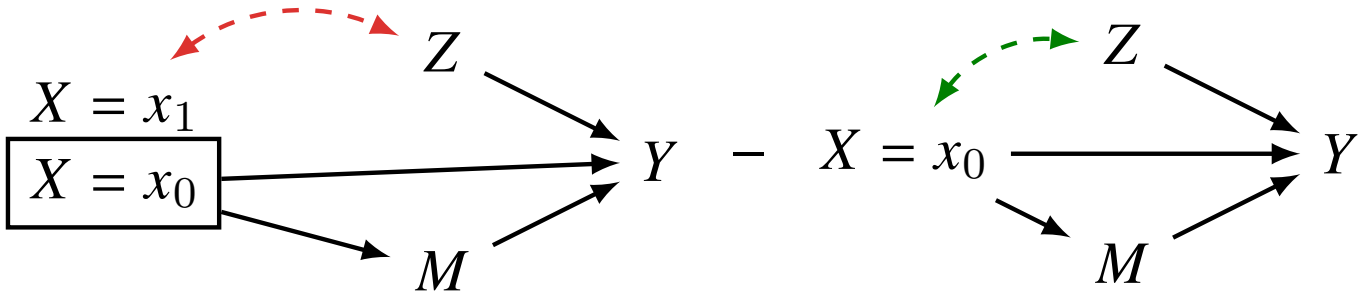
$$\textbf{Ctf-DE}_{x_0,x_1}(y \mid x_0) = P(y_{x_1,M_{x_0}} \mid x_0) - P(y_{x_0,M_{x_0}} \mid x_0)$$



$$\textbf{Ctf-IE}_{x_0,x_1}(y \mid x_0) = P(y_{x_0,M_{x_1}} \mid x_0) - P(y_{x_0,M_{x_0}} \mid x_0)$$



$\textbf{Ctf-SE}_{x_0,x_1}(y) = P(y_{x_0} \mid x_1) - P(y_{x_0} \mid x_0)$ For a male $X = x_1$ and a female $X = x_0$, how would their salary Y differ had they both been female?



$$\textbf{TV}_{x_0,x_1}(y) = \textbf{Ctf-DE}_{x_0,x_1}(y \mid x_0) - \textbf{Ctf-IE}_{x_1,x_0}(y \mid x_0) - \textbf{Ctf-SE}_{x_1,x_0}(y)$$

- Total Variation of $X = x$ on Y

$$\mathbf{TV}_x(y) = \sum_{u:Y(u)=y} P(u \mid X = x) = P(Y = y \mid X = x)$$

- Total Effect of $\mathtt{do}(X = x)$ on Y

$$\mathbf{TE}_x(y) = \sum_{u:Y_x(u)=y} P(u)$$

- z-Specific Total Effect of $\mathtt{do}(X = x)$ on Y

$$z\text{-}\mathbf{TE}_x(y) = \sum_{u:Y_x(u)=y} P(u \mid Z = z)$$

基于相关关系的统计公平的定义

- A: 敏感属性. 不应该用于预测结果的观察事实, 比如性别、种族、年龄、国籍、宗教、家境、残疾、性取向
- X: 用于决策的可观测数据
- Y: 正确的标签 (未知). 例如, $Y = 1$ 表示 "这个人应该被雇佣"
- $\hat{Y} = f(X)$ 是对 Y 的算法逼近

1. 统计均等 (Demographic Parity) $\hat{Y} \perp A$

$$\forall a, a' : P(\hat{Y} = 1 \mid A = a) = P(\hat{Y} = 1 \mid A = a')$$

2. 机会均等 (Equalized Odds) $\hat{Y} \perp A \mid Y$

$$\forall x, a, a', y : P(\hat{Y} = 1 \mid Y = y, A = a) = P(\hat{Y} = 1 \mid Y = y, A = a')$$

3. 预测均等 (Predictive Parity) $Y \perp A \mid \hat{Y}$

$$\forall x, a, a', \hat{y} : P(Y = 1 \mid \hat{Y} = \hat{y}, A = a) = P(Y = 1 \mid \hat{Y} = \hat{y}, A = a')$$

Remark: 当现实数据存在不公平时, 以上三种标准无法同时满足.

You can't have all three

Different fairness criteria can be incompatible

$$P(\hat{Y} = 1 \mid Y = 1, A = a) = \frac{P(Y = 1 \mid \hat{Y} = 1, A = a)P(\hat{Y} = 1 \mid A = a)}{P(Y = 1 \mid A = a)}$$

$$P(\hat{Y} = 1 \mid Y = 1, A = a') = \frac{P(Y = 1 \mid \hat{Y} = 1, A = a')P(\hat{Y} = 1 \mid A = a')}{P(Y = 1 \mid A = a')}$$

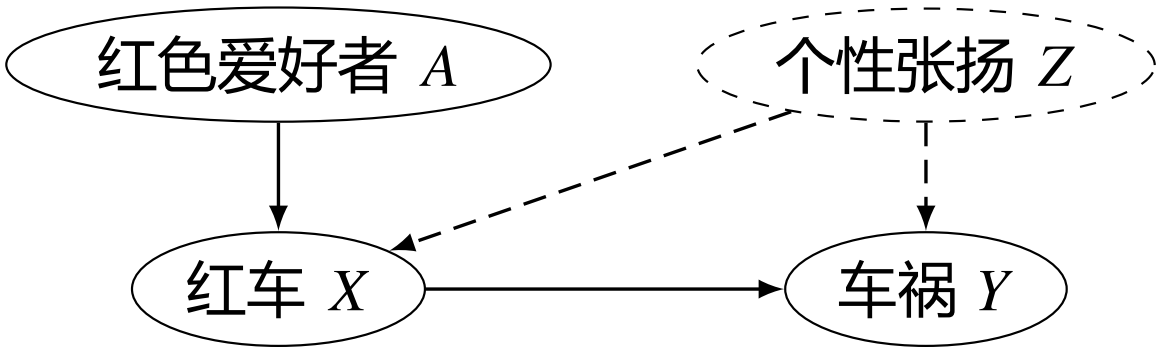
If the current state of society is unfair $P(Y = 1 \mid A = a) \neq P(Y = 1 \mid A = a')$, then you can't have fairness in all three ways.

- Trade-off between fairness and utility
- We would like $\hat{Y}$ to be an "information bottleneck" through which we capture as much information as possible between the target variable Y and features including A. $Y \perp A \mid \hat{Y}$

讨论问题

公平性与预测准确率之间怎么权衡?

反事实公平 Counterfactual Fairness[MZP20; TZZ23]



- Predictor $\hat{Y}$ is counterfactually fair if under $X = x$ and $A = a$ and any individual u.

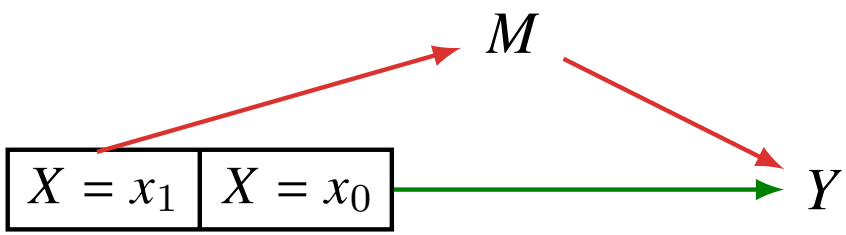
$$P\left(\hat{Y}_{a'}(u) = \hat{y} \mid X = x, A = a\right) - P\left(\hat{Y}_{a}(u) = \hat{y} \mid X = x, A = a\right) = 0$$

  where $X = \mathrm{Pa}_Y$ and $\hat{Y} = f(X)$ is some approximation function of Y.
- **Remark:** The prediction should be the same in following two worlds:
  1. the actual world
  2. a counterfactual world where the individual belonged to a different group
- 直接对车祸 Y 和红车 X 进行回归 $\hat{Y} = f(X)$ 不是反事实公平的. 对群体 A 不公平.

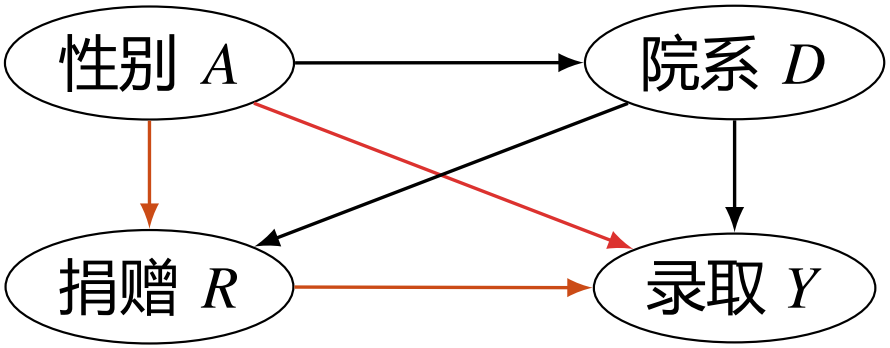
反事实公平 vs 特定路径的反事实公平

- **Counterfactual Fairness:** Did gender cause the decision? — Individual level
- **Path-Specific Counterfactual Fairness:** How did gender influence the decision? Along which path?

Remark: Path-Specific Effect 可以看作 NIE 泛化到任意因果路径.



直接歧视 & 间接歧视



1. 直接歧视 $A \to Y$
2. 间接歧视 $A \to R \to Y$
3. 可解释性影响 $A \to D \to Y$, $A \to D \to R \to Y$

- 直接歧视 $A \to Y$ 的因果效应.

$$P\left(Y_{1,R_{0,D_0},D_0}\right) - P\left(Y_0\right)$$

- 直接歧视 $A \to Y$ 和间接歧视 $A \to R \to Y$ 的因果效应.

$$P\left(Y_{1,R_{1,D_0},D_0}\right) - P\left(Y_0\right)$$

- 特定路径的反事实公平.

$$P\left(Y_{1,R_{1,D_0},D_0} \mid A=0, D=d, R=r\right) - P\left(Y_0 \mid A=0, D=d, R=r\right) = 0$$

No Direct/Indirect/Proxy Discrimination

- **No Direct Discrimination** 令 Q_d 是 A 到 $\hat{Y}$ 的直接路径的集合.

$$\forall a, a', \hat{y} : P\left(\hat{Y} = \hat{y} \mid \mathrm{do}(A = a' \mid_{Q_d})\right) - P\left(\hat{Y} = \hat{y} \mid \mathrm{do}(A = a)\right) = 0$$

- **No Indirect Discrimination** 令 Q_i 是 A 到 $\hat{Y}$ 的所有经过 A 的代理属性 (又叫红线属性)R 的间接路径的集合.

$$\forall a, a', \hat{y} : P\left(\hat{Y} = \hat{y} \mid \mathrm{do}(A = a' \mid_{Q_i})\right) - P\left(\hat{Y} = \hat{y} \mid \mathrm{do}(A = a)\right) = 0$$

- **No Proxy Discrimination** 令 R 是 A 的代理属性.

$$\forall r, r', \hat{y} : P\left(\hat{Y} = \hat{y} \mid \mathrm{do}(R = r')\right) - P\left(\hat{Y} = \hat{y} \mid \mathrm{do}(R = r)\right) = 0$$

Definition (Path-Specific Effect)

Given a set Q of directed paths, the Q-specific effect of the value change of A from a to a' on $Y = y$ through Q is given by

$$\mathrm{PE}_Q(a', a) \coloneqq P\left(Y_{a'|Q,a|\overline{Q}}\right) - P(Y_a)$$

where $P\left(Y_{a'|Q,a|\overline{Q}}\right)$ represents the post-intervention distribution of Y, where the effect of $\mathrm{do}(a')$ is transmitted only along Q while the effect of $\mathrm{do}(a)$ is transmitted along the other directed paths $\overline{Q}$.

Definition (Path-Specific Counterfactual Effect)

Given a factual condition $O = o$, and a set Q of directed paths, the Q-specific counterfactual effect

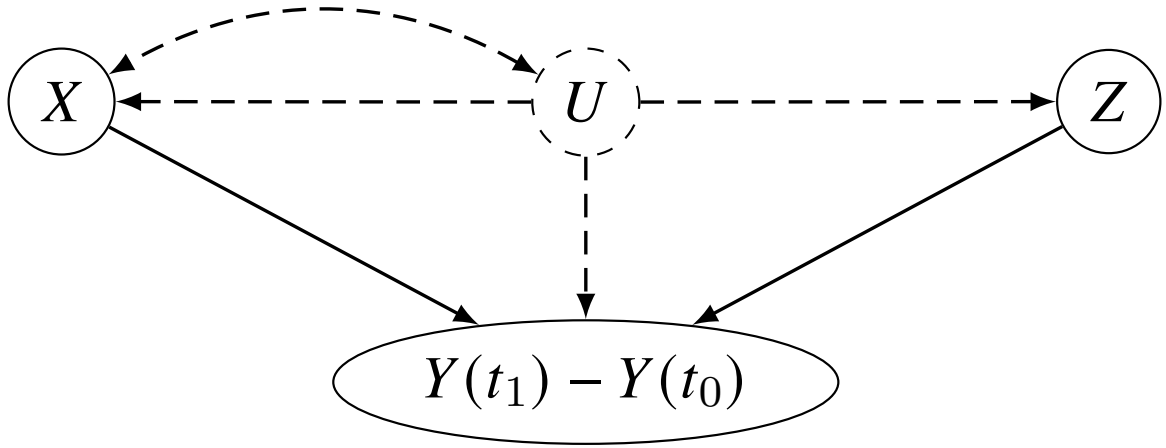
$$\mathrm{PCE}_Q(a', a \mid o) \coloneqq P\left(Y_{a'|Q,a|\overline{Q}} \,\middle|\, o\right) - P(Y_a \mid o)$$

Definition (Path-Specific Counterfactual Fairness)

The Predictor $\hat{Y}$ is path-specifically counterfactually fair w.r.t. Q if $\mathrm{PCE}_Q(a', a \mid o) = 0$.

Difference in Differences

- Time may play an important role in causality.
- $Y_x(u,t)$: the potential outcome of unit u under treatment $X = x$ at time t.
- **Example:** to estimate the effect raising minimum wage on employment.



- X: Minimum Wage
- Y: Employment
- U: Average Income

THE SVERIGES RIKSBANK PRIZE
IN ECONOMIC SCIENCES IN MEMORY
OF ALFRED NOBEL 2021
Illustrations: Niklas Elmehed
David Card
Joshua D. Angrist
Guido W. Imbens

Difference in Differences

- **Assumption:** Parallel Trend Assumption $(Y_0(t_1) - Y_0(t_0)) \perp X$

$$\mathbb{E}[Y_0(t_1) - Y_0(t_0) \mid X = 1] = \mathbb{E}[Y_0(t_1) - Y_0(t_0) \mid X = 0]$$

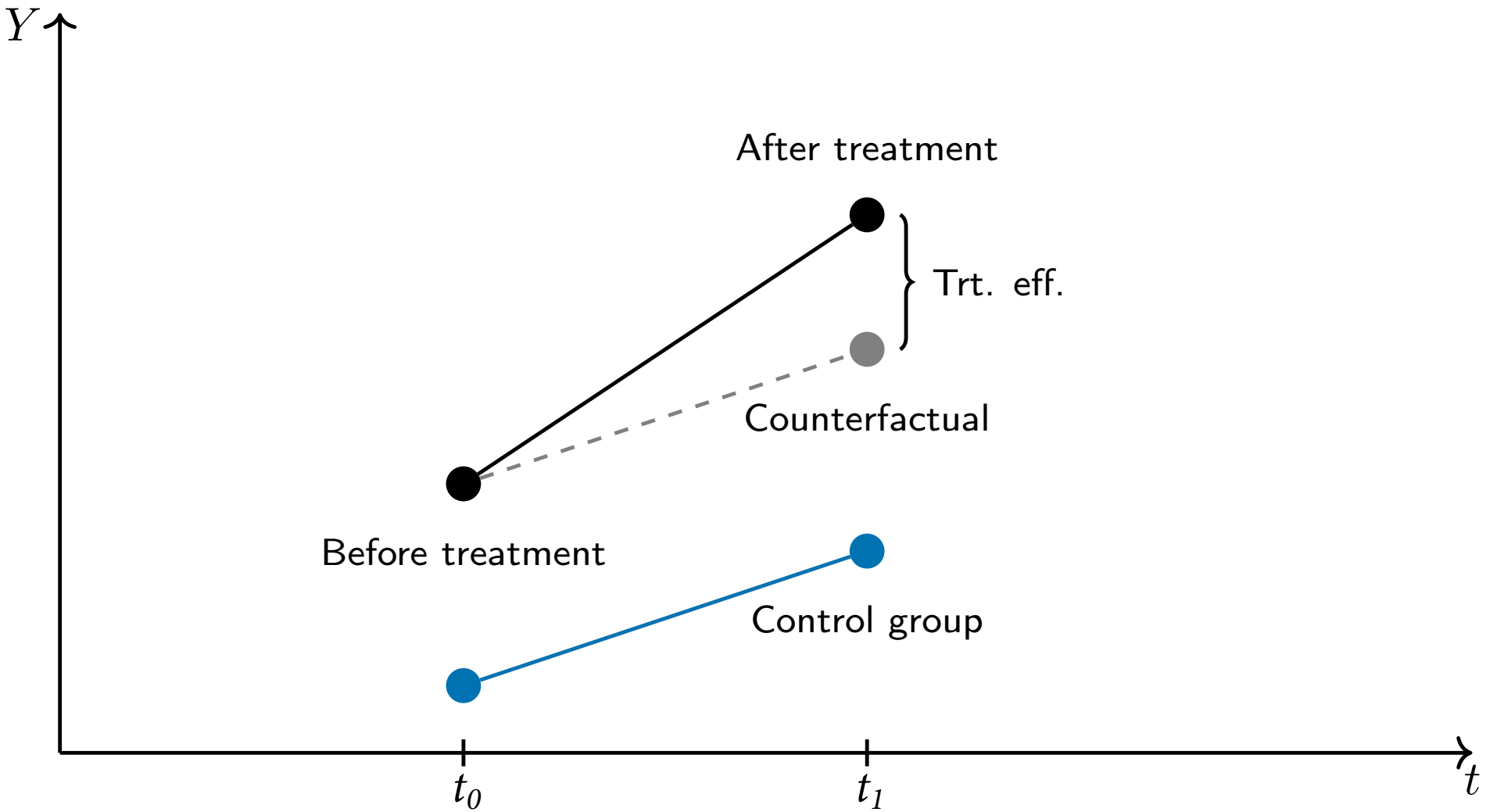
- **Assumption:** No Pretreatment Effect Assumption

$$\mathbb{E}[Y_1(t_0) - Y_0(t_0) \mid X = 1] = 0$$

- Effect of Treatment on the Treated

$$\begin{aligned}
\text{ETT} &= \mathbb{E}[Y_1(t_1) - Y_0(t_1) \mid X = 1] \\
&= \mathbb{E}[Y(t_1) \mid X = 1] - \mathbb{E}[Y_0(t_1) \mid X = 1] \\
&= \mathbb{E}[Y(t_1) \mid X = 1] - \Big(\mathbb{E}[Y_0(t_0) \mid X = 1] + \mathbb{E}[Y_0(t_1) - Y_0(t_0) \mid X = 0]\Big) \\
&= \mathbb{E}[Y(t_1) \mid X = 1] - \Big(\mathbb{E}[Y_1(t_0) \mid X = 1] + \mathbb{E}[Y_0(t_1) - Y_0(t_0) \mid X = 0]\Big) \\
&= \Big(\mathbb{E}[Y(t_1) \mid X = 1] - \mathbb{E}[Y(t_0) \mid X = 1]\Big) - \Big(\mathbb{E}[Y(t_1) \mid X = 0] - \mathbb{E}[Y(t_0) \mid X = 0]\Big)
\end{aligned}$$

Difference in Differences

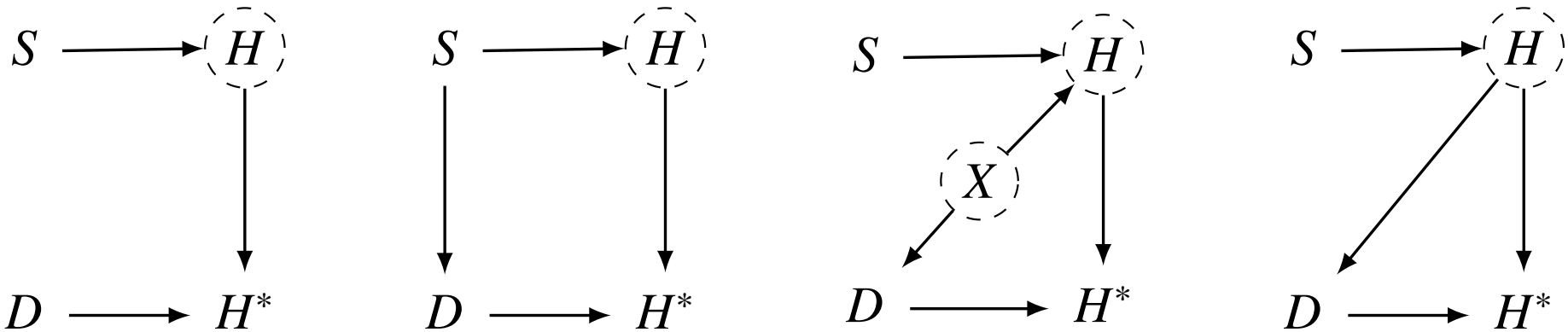


数据类型	混杂因子可观测性	如何控制混杂因子		具体方法
截面数据	可观测	匹配	分层	精确匹配
				粗粒化的精确匹配
			降维	马氏距离匹配
				倾向值匹配
		加权	根据倾向值加权	逆概率加权、双稳健估计
	不可观测	估计处理变量的回归系数		工具变量法
		使用参考变量的临界值划分处理组与对照组，断点附近个体特征相似		断点回归
纵向数据	不可观测（不随时间变化）	消除个体不随时间变化的异质性，控制时间的增量		双重差分法
	不可观测（随时间变化）	对控制组个体加权，构造处理组个体的反事实		合成控制法

Missing Data and Missingness Mechanisms

- S: 学生的努力程度
- H: 作业
- D: 狗吃作业
- H^*: 残留作业
- X: 家庭噪声

1. 狗随机吃作业
2. 狗吃努力学习不陪狗玩的同学的作业
3. 家庭噪声越大, 作业质量越差, 且狗越暴躁吃作业
4. 狗吃写的差的作业



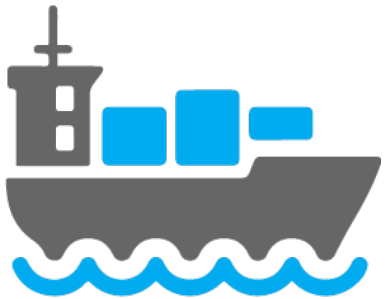
我们关心 $S \to H$, 但 H 缺失, 我们只能用 H^* 代替, 用回归 $S \to H^*$ 近似 $S \to H$. 但这依赖于数据缺失的机制.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

纽拉特之船

- 学习因果效应
- 学习因果结构



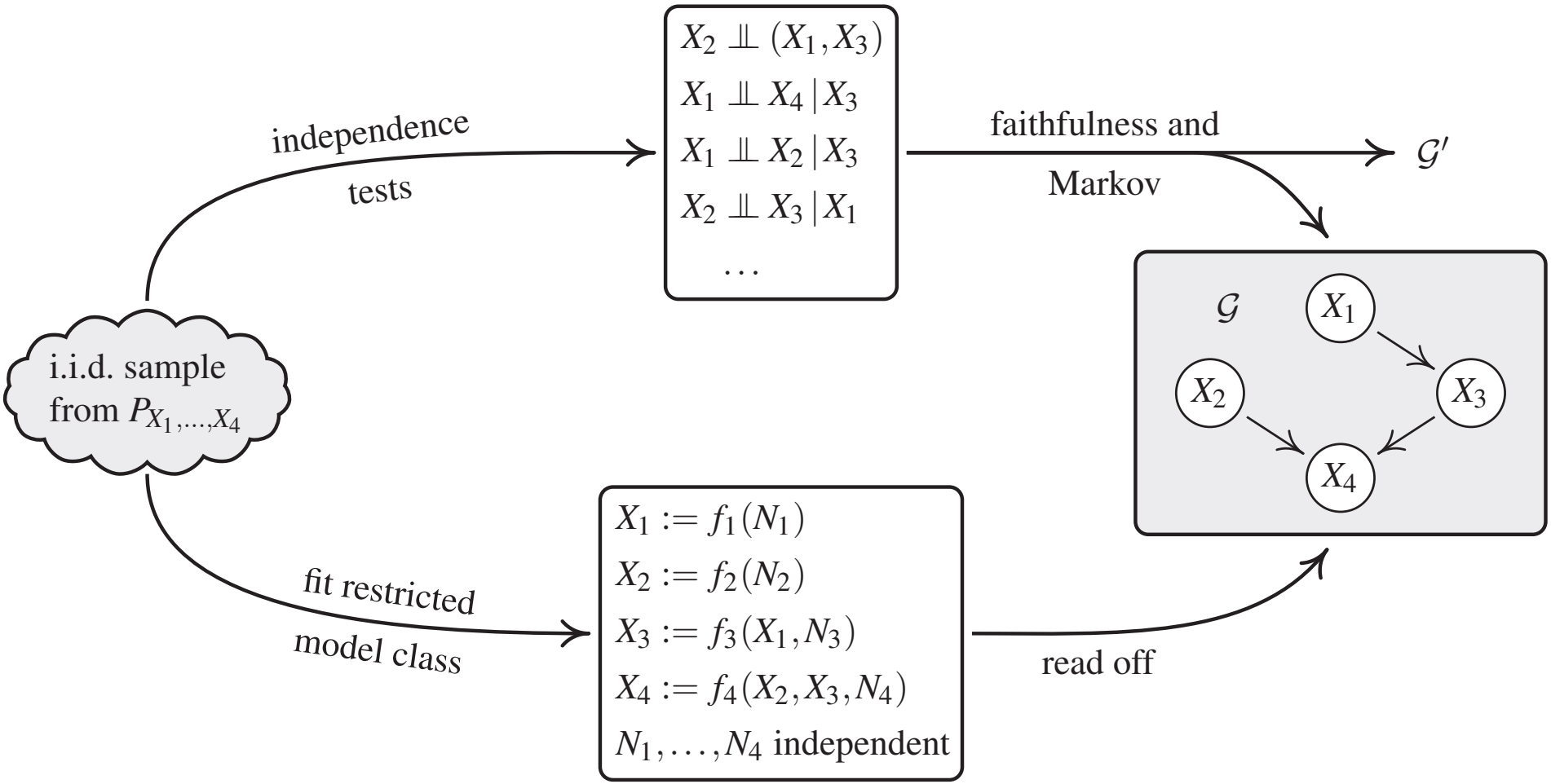
在认识世界时，我们就像是在大海上航行的海员，如果船有损坏的部件，我们只能一边航行，一边修补它，船的其他部分可以作为支撑.

— 奥图·纽拉特 *(Otto Neurath)*

From Perception to Modelling the World at the Semantic-Level

- What are the causal variables explaining the data?
- How to discover them (as a function of observed data)?
- How to discover their causal relationship?
- How are actions corresponding to causal interventions?
- How is raw sensory data mapped to high-level causal variables?
- How do high-level causal variables turn into low-level actions and partial observations?

Causal Discovery — Two Methods



1. Independence-Based Methods
2. Score-Based Methods

$$G^* = \underset{\text{Graph}}{\operatorname{argmax}} \operatorname{Score}(\text{Data} \mid \text{Graph})$$

Causal Discovery from Observational Data — Assumptions

Problem (Causal Discovery from Observational Data)

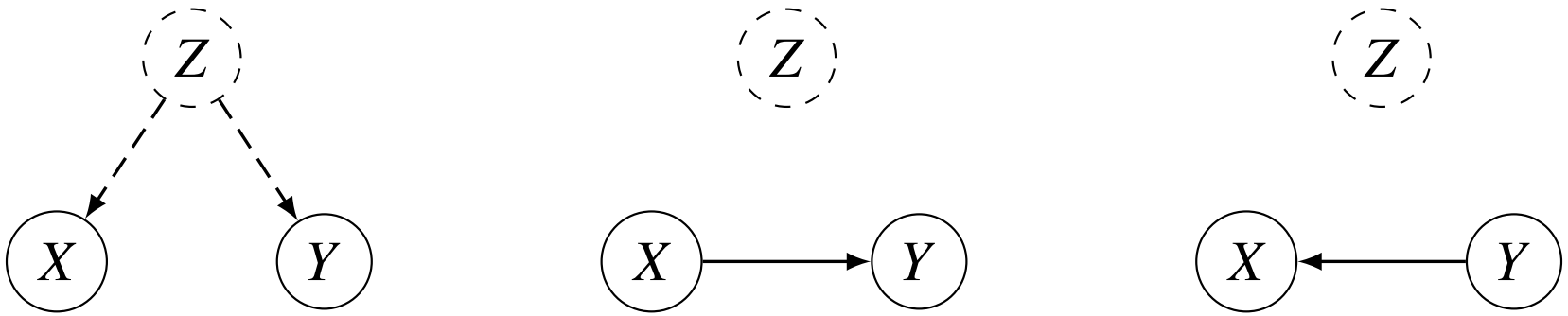
Given $P(X_1, \ldots, X_n)$, can we infer causal graph G?

$$\left.\begin{array}{c}\text{Assumptions}\\ +\\ \text{Data}\end{array}\right\} \Longrightarrow \text{Independencies} \Longrightarrow \text{Causal Graph} \Longrightarrow \text{Effects of Intervention}$$

All Assumptions

- Markov Condition
- Causal Faithfulness
- Causal Sufficiency: there are no unobserved confounders for any of the variables in the graph
- Acyclicity: there are no cycles in the graph

The Need for Causal Sufficiency



Local / Global Markov Condition — Key Assumption

Local Markov Condition

X_i is independent of nondescendants $\mathrm{ND}_i := V \setminus (\mathrm{Desc}_i \cup \mathrm{Pa}_i)$, given parents Pa_i, i.e.

$$X_i \perp \mathrm{ND}_i \mid \mathrm{Pa}_i$$

i.e. every information exchange with its nondescendants involves its parents.

Global Markov Condition

For all disjoint subsets of vertices X, Y and Z we have that

$$(X \perp Y \mid Z)_G \implies (X \perp Y \mid Z)_P$$

Equivalence of Markov Conditions

Theorem

The following are equivalent:

1. *Existence of a structural causal model.*

$$X_i = f_i(\mathrm{Pa}_i, U_i)$$

2. *Factorization.*

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i \mid \mathrm{Pa}_i)$$

3. *Local Markov condition: statistical independence of nondescendants given parents.*

$$X_i \perp \mathrm{ND}_i \mid \mathrm{Pa}_i$$

4. *Global Markov condition.*

$$(X \perp Y \mid Z)_G \implies (X \perp Y \mid Z)_P$$

Factorization $\Longrightarrow$ Structural Causal Model

Generate $P(X_i \mid \mathrm{Pa}_i)$ by a deterministic function:

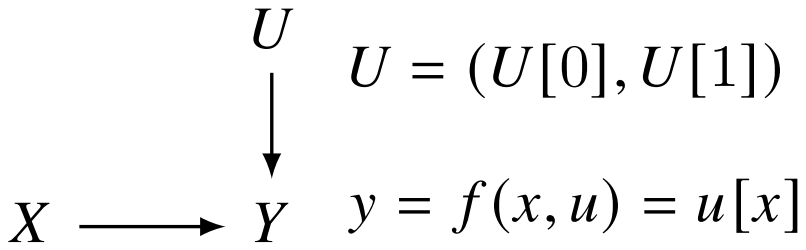
- define a vector valued noise variable U_i,
- each component $U_i[\mathrm{pa}_i]$ corresponds to a possible value pa_i of Pa_i,
- define structural equation

$$x_i = f_i(\mathrm{pa}_i, u_i) := u_i[\mathrm{pa}_i]$$

- let component $U_i[\mathrm{pa}_i]$ be distributed according to $P(X_i \mid \mathrm{pa}_i)$.

Remark: joint distribution of all $U_i[\mathrm{pa}_i]$ is irrelevant, only marginals matter.

Example



$U = (U[0], U[1])$

$y = f(x, u) = u[x]$

x	y	$(u[0], u[1])$	$P(y \mid x)$
0	0	$(0, -)$	$P(0,0) + P(0,1)$
0	1	$(1, -)$	$P(1,0) + P(1,1)$
1	0	$(-, 0)$	$P(0,0) + P(1,0)$
1	1	$(-, 1)$	$P(0,1) + P(1,1)$

- X, Y binary. $P(y \mid x) = \frac{1}{2}$ for $x, y \in \{0, 1\}$.
- U randomly chooses a deterministic mechanism: $\mathrm{id}, \neg, \bot, \top$, where $\mathrm{id} := (0,1)$, $\neg := (1,0)$, $\bot := (0,0)$, $\top := (1,1)$.

The same $P(y \mid x) = \sum_u [\![y = u[x]]\!] P(u)$ can be induced by different distributions.

$$P(Y = 1 \mid X = 0) = [\![\mathrm{id}[0] = 1]\!] P(\mathrm{id}) + [\![\neg[0] = 1]\!] P(\neg) + [\![\bot[0] = 1]\!] P(\bot) + [\![\top[0] = 1]\!] P(\top) = \tfrac{1}{2}$$

1. $P(U = \mathrm{id}) = P(U = \neg) = \frac{1}{2}$, $P(U = \bot) = P(U = \top) = 0$.
2. $P(U = \mathrm{id}) = P(U = \neg) = 0$, $P(U = \bot) = P(U = \top) = \frac{1}{2}$.
3. $P(U = \mathrm{id}) = P(U = \neg) = P(U = \bot) = P(U = \top) = \frac{1}{4}$.

Remarks

- 上面的三个 SCM 都能导出相同的 $P(Y \mid X)$.
- 但是, 反事实却大不相同.

1. 根据 SCM1: 若事实是 $X = 0, Y = 1$, 则 $U = \neg$, 所以,

$$(M_{X^*=1}, U = \neg) \vDash Y^* = \neg[1] = 0$$

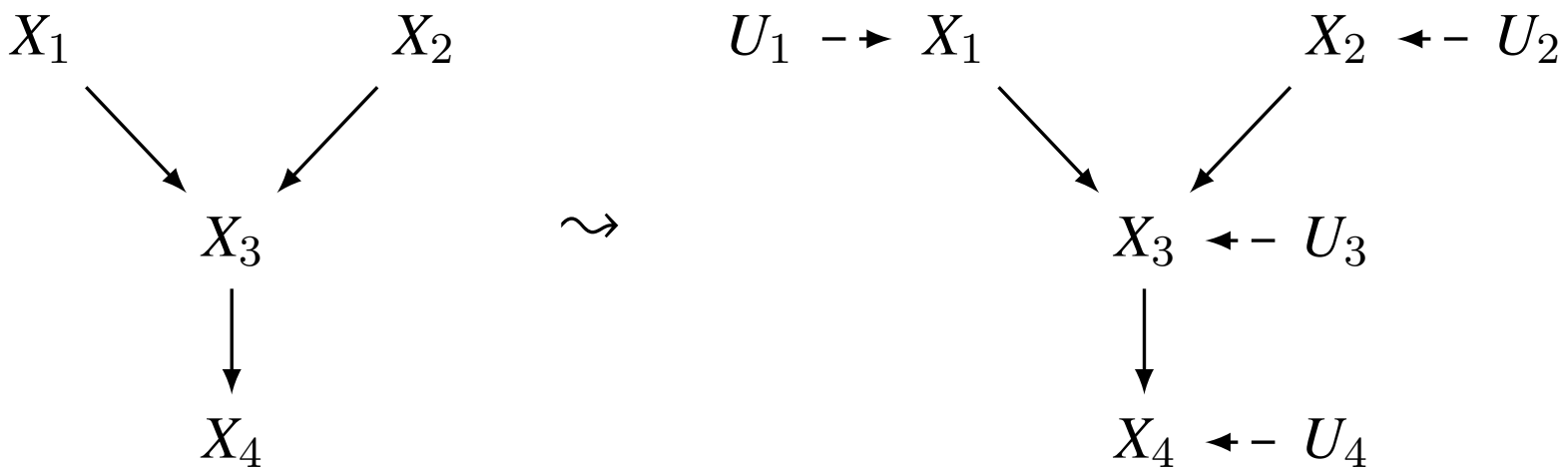
2. 根据 SCM2: 若事实是 $X = 0, Y = 1$, 则 $U = \top$, 所以,

$$(M_{X^*=1}, U = \top) \vDash Y^* = \top[1] = 1$$

3. 根据 SCM3: 若事实是 $X = 0, Y = 1$, 则
   $P(U = \neg \mid X = 0, Y = 1) = P(U = \top \mid X = 0, Y = 1) = \frac{1}{2}$,
   $P(U = \mathrm{id} \mid X = 0, Y = 1) = P(U = \bot \mid X = 0, Y = 1) = 0$, 所以,

$$P(Y^*_{X^*=1} = 1 \mid X = 0, Y = 1) = \sum_u [\![1 = u[1]]\!] P(u \mid X = 0, Y = 1) = \frac{1}{2}$$

Structural Causal Model $\implies$ Local Markov Condition



- augmented DAG G' contains unobserved noise U_i for each X_i,
- local Markov-condition holds for G':
  1. the noise terms U_i are jointly independent, and thus independent of their non-descendants
  2. for the X_i, we have

$$X_i \perp \mathrm{ND}'_i \mid \mathrm{Pa}'_i$$

- local Markov in G' implies global Markov in G'
- global Markov in G' implies local Markov in G

Local Markov $\Longrightarrow$ Factorization

- Assume X_i is a terminal node, i.e., it has no descendants, then $\mathrm{ND}_n = \{X_1, \ldots, X_{n-1}\}$. Thus the local Markov condition implies

$$X_n \perp \{X_1, \ldots, X_{n-1}\} \mid \mathrm{Pa}_n$$

- Hence the general decomposition

$$P(x_1, \ldots, x_n) = P(x_n \mid x_1, \ldots, x_{n-1}) P(x_1, \ldots, x_{n-1})$$

becomes

$$P(x_1, \ldots, x_n) = P(x_n \mid \mathrm{pa}_n) P(x_1, \ldots, x_{n-1})$$

- Induction over n yields

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P(x_i \mid \mathrm{pa}_i)$$

Factorization $\Longrightarrow$ Local Markov

Assume $(X \perp Y \mid Z)_G$

- define the smallest subgraph G' containing X, Y, Z and all their ancestors
- consider moral graph G'' (undirected graph containing the edges of G' and links between all parents)
- use results that relate factorization of probabilities with separation in undirected graphs (Hammersley-Clifford theorem)

to prove $(X \perp Y \mid Z)_P$.

Global Markov $\implies$ Local Markov

Need to show that the parents Pa_i d-separate X_i from its non-descendants ND_i:

- All paths connecting X_i and ND_i include a node in Pa_i, but never as a collider
- Hence all paths are chains or forks
- Therefore, the parents block every path between X_i and ND_i

Causal Faithfulness — Key Assumption

Causal Faithfulness

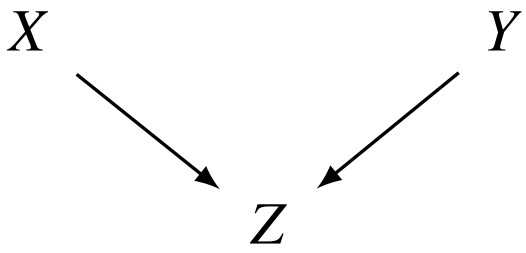
P is called faithful relative to G if only those independencies hold true that are implied by the Markov condition, i.e.

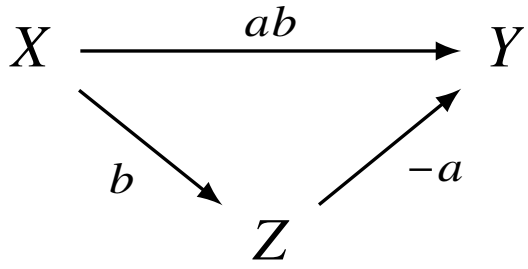
$$(X \perp Y \mid Z)_G \impliedby (X \perp Y \mid Z)_P$$

Remark: Markov condition + Causal faithfulness:

$$(X \perp Y \mid Z)_G \iff (X \perp Y \mid Z)_P$$

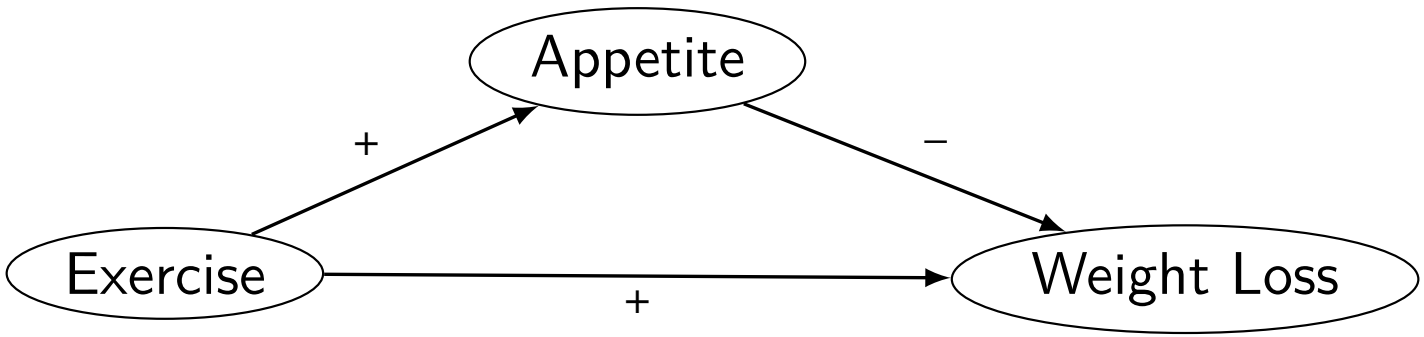
Why do we need the Faithfulness Condition? — Occam?





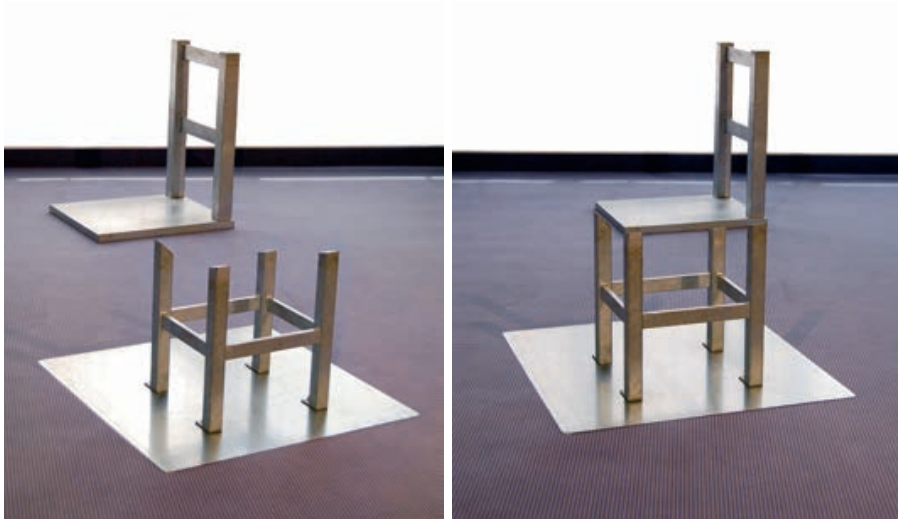
Graph	Distribution
$X \perp Y$	$X \perp Y$
$X \not\perp Y \mid Z$	$X \not\perp Y \mid Z$

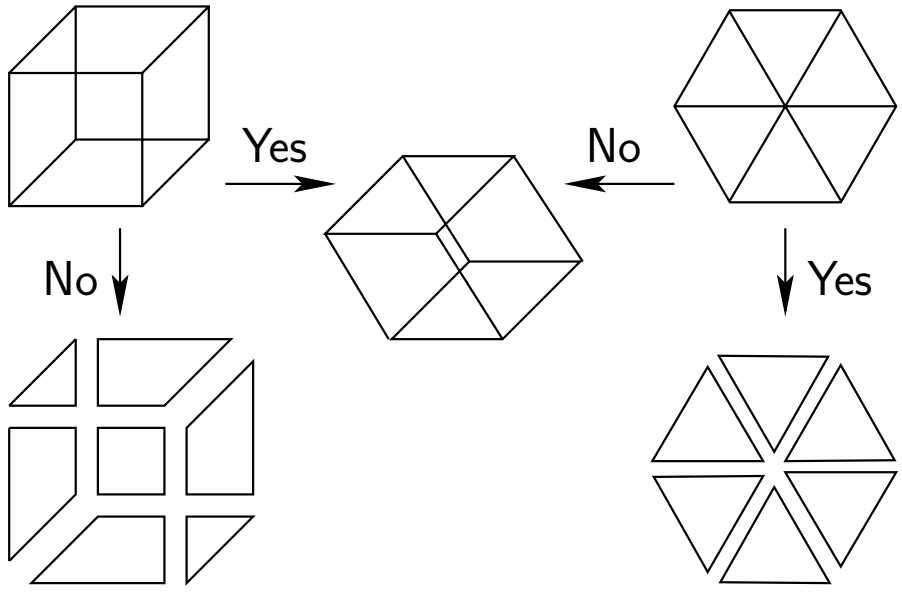
Graph	Distribution
$X \not\perp Y$	$X \perp Y$
$X \not\perp Y \mid Z$	$X \not\perp Y \mid Z$



Remark: 这种恰好抵消不稳定.

奥卡姆剃刀 — 忠实性/稳定性

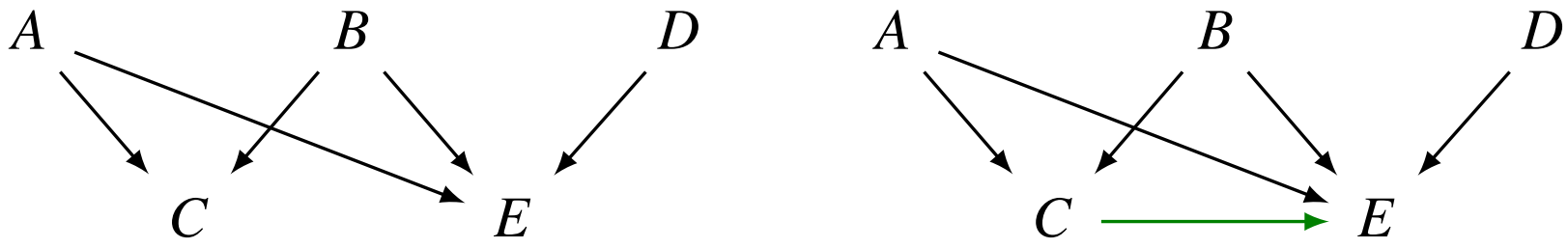




Remark: 盖梯尔问题中的运气巧合.

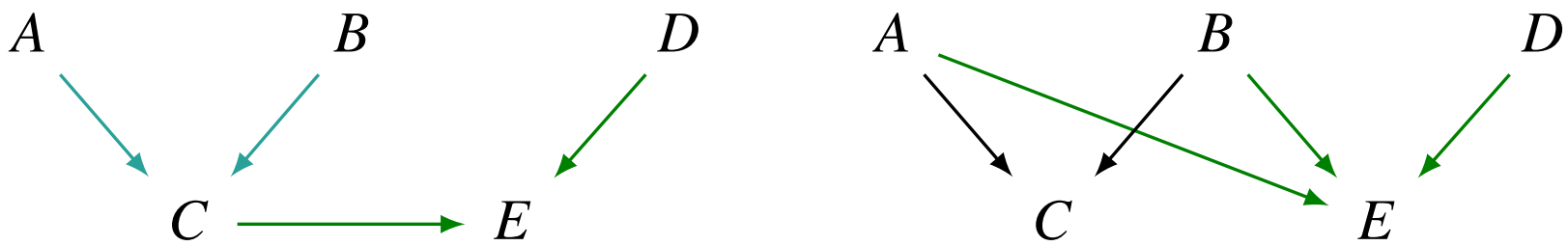
More Occam Razors?

- **Causal Minimality Condition:** no proper subgraph of G over V satisfies the Markov condition with P.



Theorem: 假定 P 关于 G 有马尔科夫性. 则 P 关于 G 有因果极小性, 当且仅当, $\forall X_i \forall Y \in \mathrm{Pa}_i : X_i \not\perp Y \mid \mathrm{Pa}_i \setminus \{Y\}$.

- **Causal Frugality Condition:** Markovian DAGs that are not having the least number of edges should be rejected.

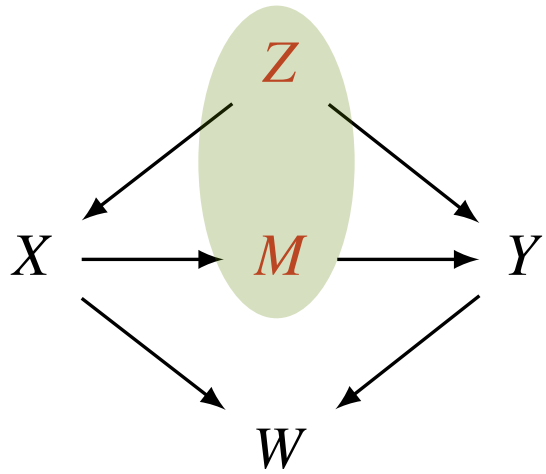


Theorem

Assume Markov condition and faithfulness holds. Then X and Y are linked by an edge iff there is no set S_{XY} such that

$$(X \perp Y \mid S_{XY})_P$$

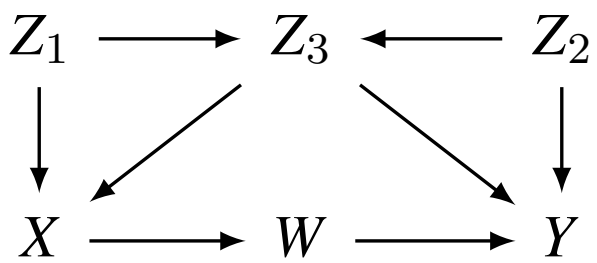
Explanation: dependence mediated by other variables can be screened off by conditioning on an appropriate set.



$$X \perp Y \mid \{Z, M\}$$

$$X \not\perp Y \mid \{Z, M, W\}$$

Model Testing & Causal Discovery



- $W \perp Z_1 \mid X$
- Given data, regress W on Z_1 and X

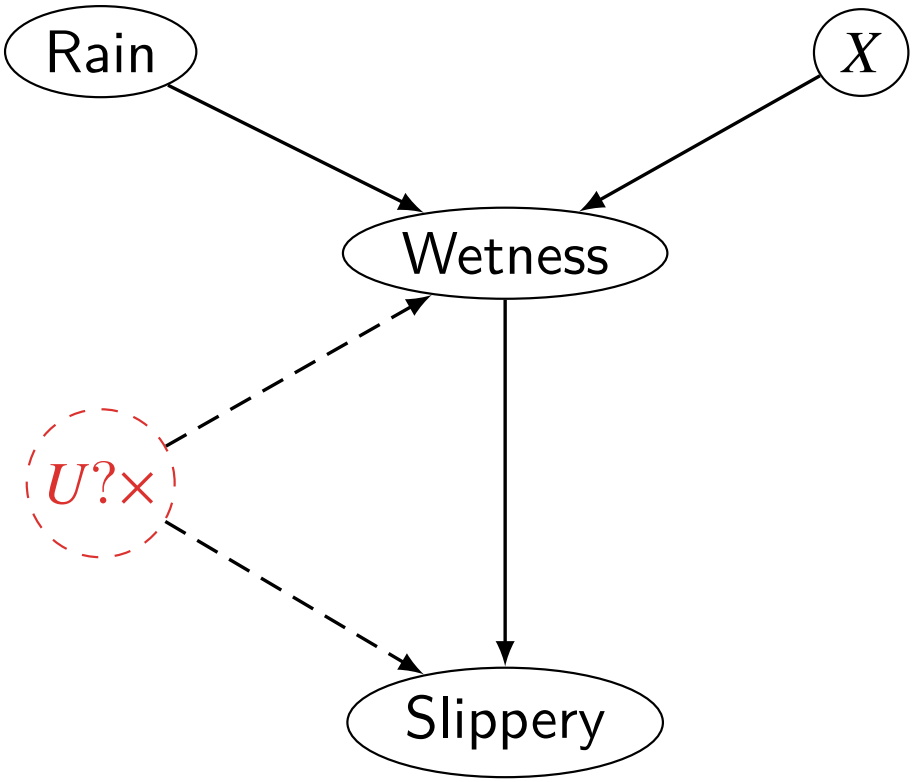
$$W = aZ_1 + bX + c$$

- If the result suggests that $a \neq 0$, then the model is wrong.

Example — Dealing with Confounders?

$$R \perp X$$
$$R \perp S \mid W$$
$$X \perp S \mid W$$

$\Longrightarrow$



Causal Discovery from Observational Data

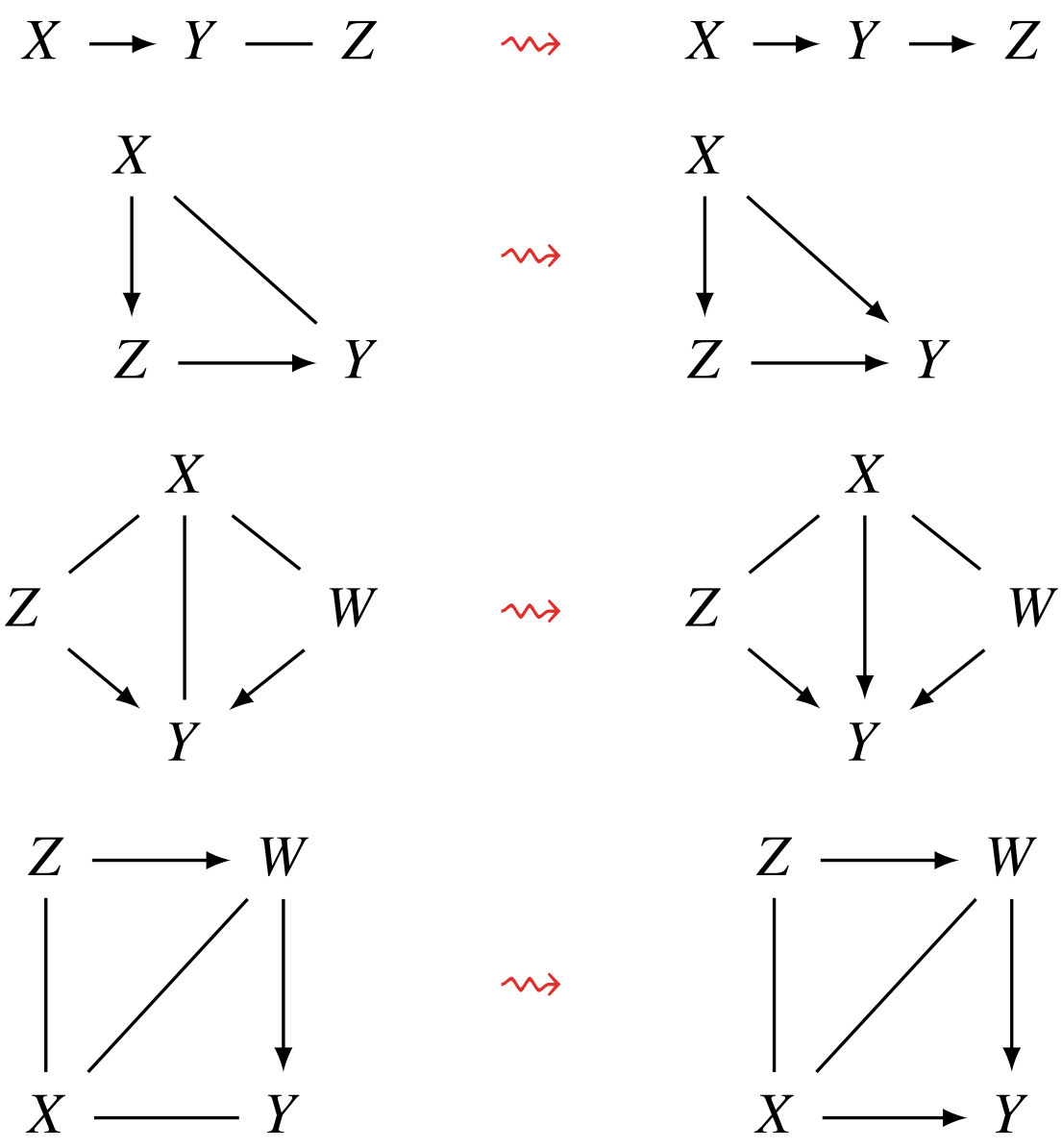
Assumptions: Markov condition, Faithfulness, Sufficiency, Acyclicity.

Inductive Causation Algorithm

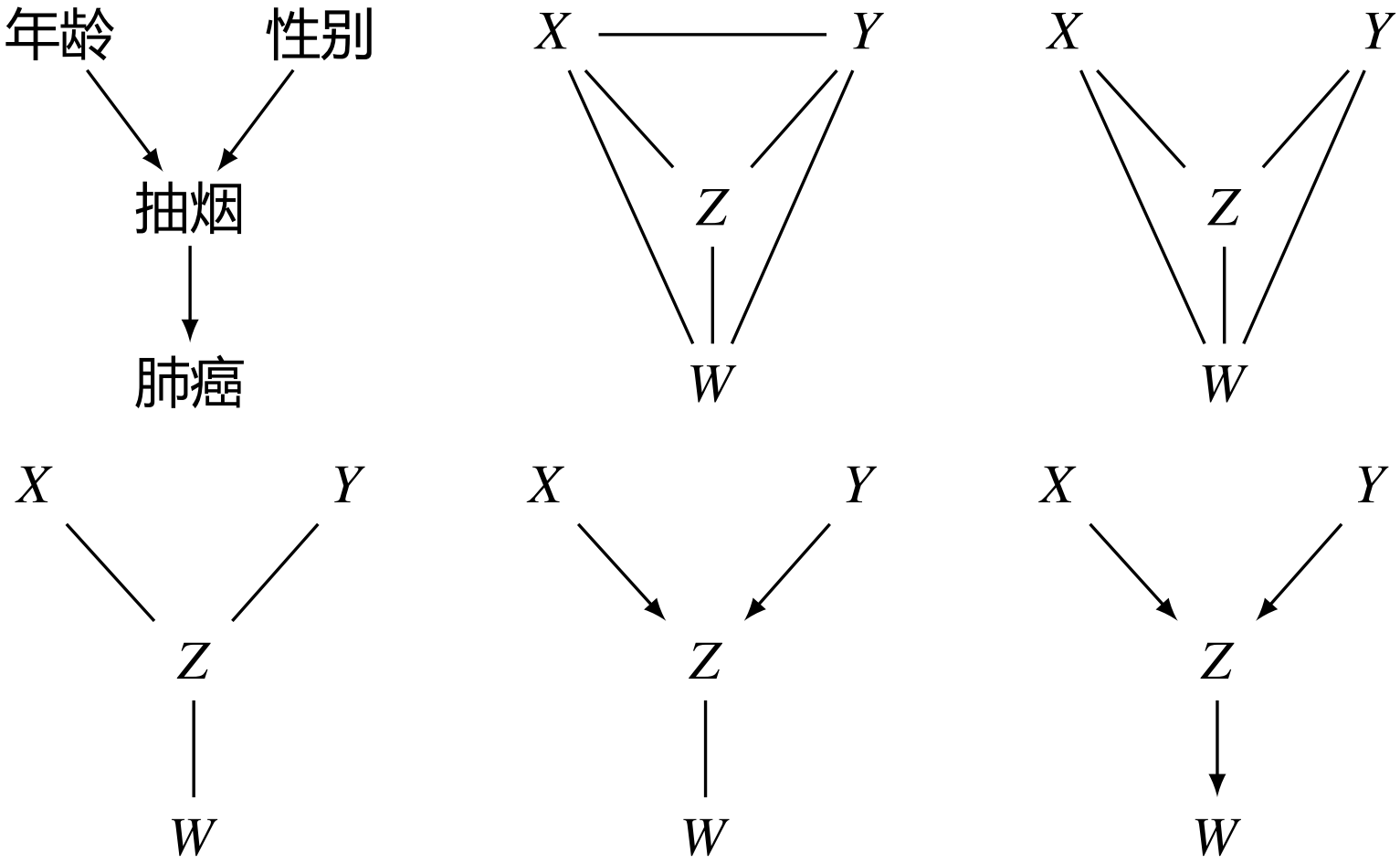
1. Given a stable distribution P on a set of variables. Start with a complete undirected graph G on all variables.
2. For each pair X and Y, and each set of other variables S_{XY}, starting with the empty set and increasing the size, see if $(X \perp Y \mid S_{XY})_P$; if so, by faithfulness $(X \perp Y \mid S_{XY})_G$, remove the edge between X and Y.
3. For all $X - Z - Y$ and $X \perp Y \mid S_{XY}$, if $Z \notin S_{XY}$, then replace $X - Z - Y$ by the V-structure $X \to Z \leftarrow Y$.
4. In the partially directed graph that results, orient as many of the undirected edges as possible subject to two conditions: (i) any alternative orientation would yield a new V-structure; or (ii) any alternative orientation would yield a directed cycle.

Remark: X, Z, Y 构成 V-结构, 当且仅当, 存在 S 使得 $X \perp Y \mid S$ 且 $X \not\perp Y \mid S \cup \{Z\}$.

Could not be completed without creating a cycle or a new V-structure

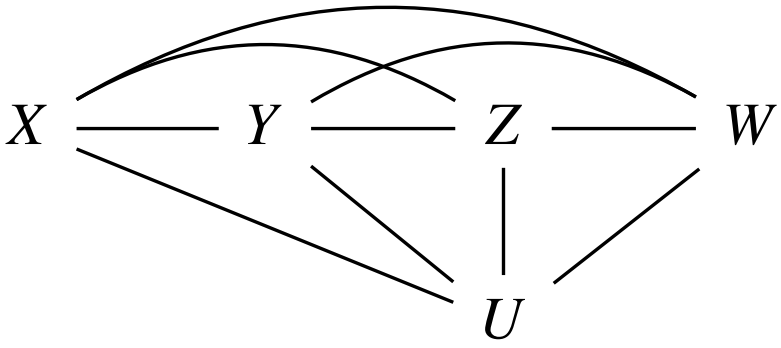


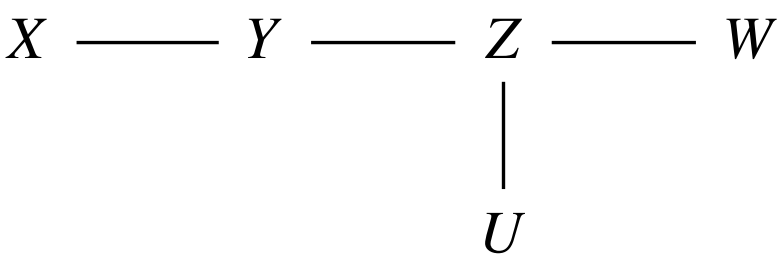
Example

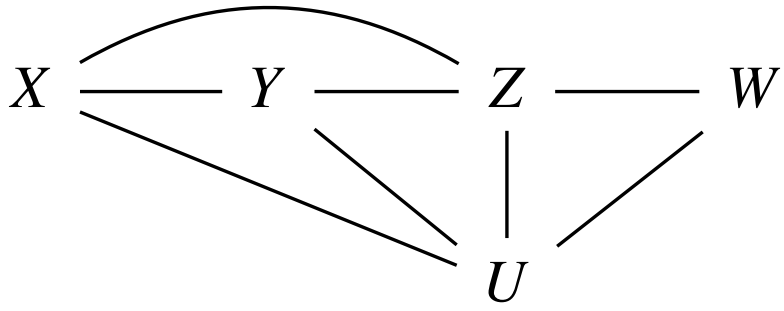


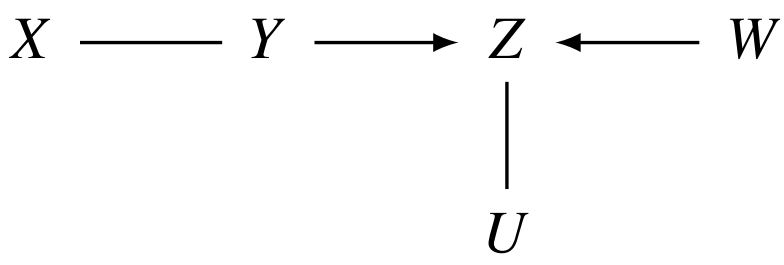
- 图 3 因为: $X \perp Y$
- 图 4 因为: $X \perp W \mid Z, Y \perp W \mid Z$
- 图 5 因为: $Z \notin S_{XY} \implies V$-结构 $(X \perp Y \mid \varnothing, Z \notin \varnothing, X \not\perp Y \mid Z)$
- 图 6 因为: 避免新的 V-结构

Example

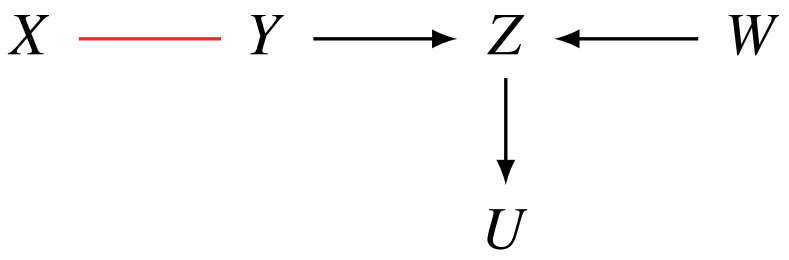








1. $X \perp W, Y \perp W$
2. $X \perp Z \mid Y, X \perp U \mid Y,$
   $Y \perp U \mid Z, W \perp U \mid Z$
3. $Z \notin S_{YW} \implies V$-结构
4. 避免新的 V-结构



Markov Equivalence

- **Definition:** G_1 and G_2 are Markov equivalent iff for every three mutually disjoint subsets $X, Y, Z \subset V$,

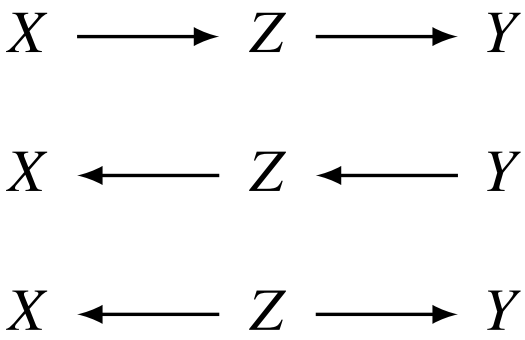
$$(X \perp Y \mid Z)_{G_1} \iff (X \perp Y \mid Z)_{G_2}$$

- G_1 and G_2 are Markov equivalent iff they imply the same conditional independences.
- G_1 and G_2 are Markov equivalent iff they have the same skeleton (edges without regard for direction) and the same set of V-structures ($X \to Z \leftarrow Y$ with no edge between X and Y).

Examples

- **Example1:** same skeleton, no V-structure

Example1:

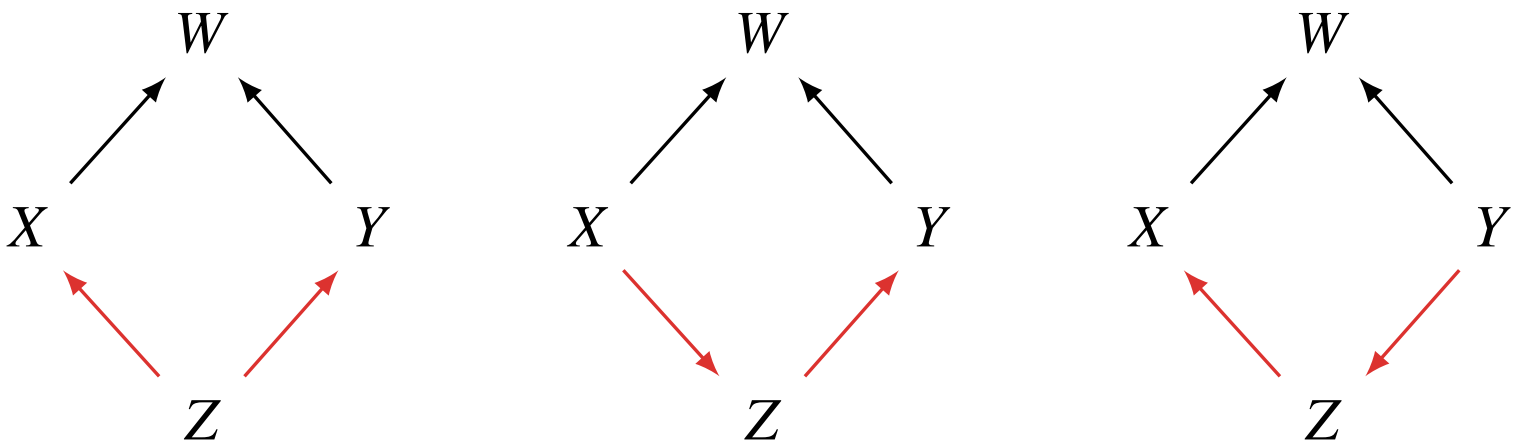


$$X \perp Y \mid Z$$

$$X \not\perp Z \qquad Z \not\perp Y$$

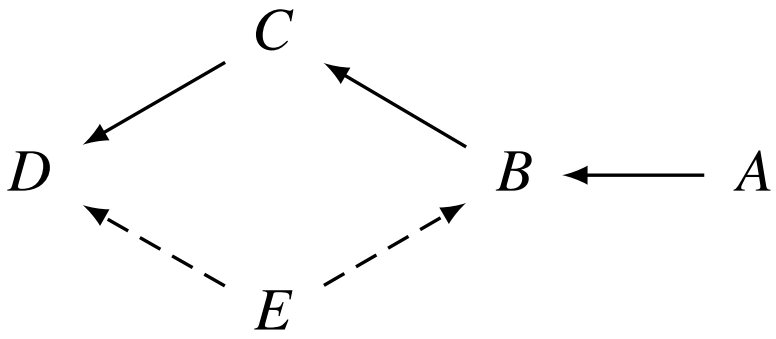
$$X \not\perp Y$$

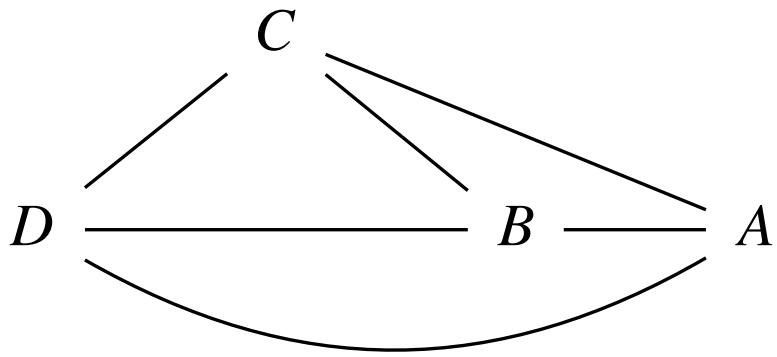
- **Example2:** same skeleton, same V-structure at W



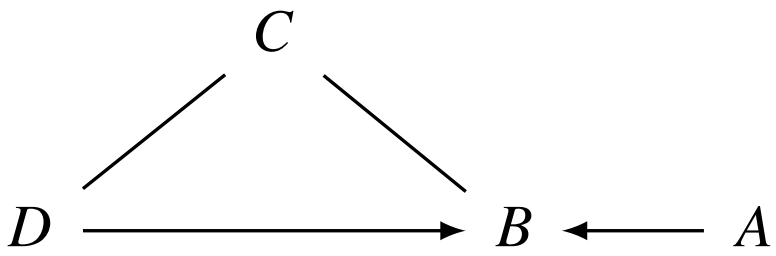
若缺少 Sufficiency 条件, 可能无法发现协调的因果图

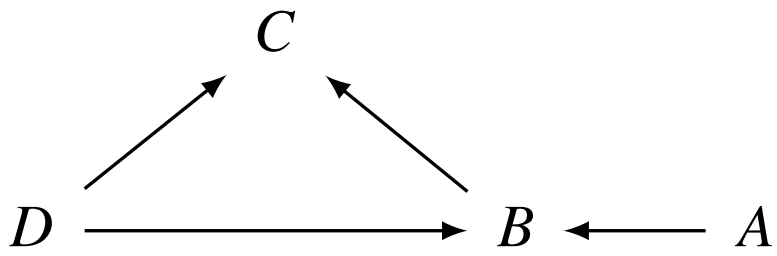
- 假设真实模型是左图, 但 E 未知.





- 根据 $A \perp C \mid B$, 删掉 A 和 C 之间的边; 根据 $A \perp D \mid C$, 删掉 A 和 D 之间的边.
- 根据 $A \perp D \mid C$ 但 $A \not\perp D \mid C, B$, $D - B - A$ 构成 V-结构.
- 为了避免新的 V-结构, 有 $B \to C$; 为了避免形成环, 有 $D \to C$.





- 但此时 $A \perp D$, 与真实数据矛盾.

Faithfulness vs Minimality vs Frugality

Theorem

Assuming both Markov and Faithfulness conditions, the Markov equivalence class $G(P)$ can be identified using the conditional independence of P.

- If P is faithful and Markovian with respect to G, then causal minimality is satisfied.
- The Frugality condition is stronger than the Minimality condition.
- Whenever the set of DAGs satisfying the Faithfulness is non-empty, it is equivalent to the set of DAGs satisfying frugality.
- There are examples in which no DAGs satisfy the Faithfulness, whereas the set of DAGs satisfying frugality consists of the true Markov equivalence class.

$$G_{\text{Faith}}(P) \subset G_{\text{Frugal}}(P) \subset G_{\text{Minimal}}(P)$$

$$G_{\text{Faith}}(P) \neq \varnothing \implies G_{\text{Faith}}(P) = G_{\text{Frugal}}(P)$$

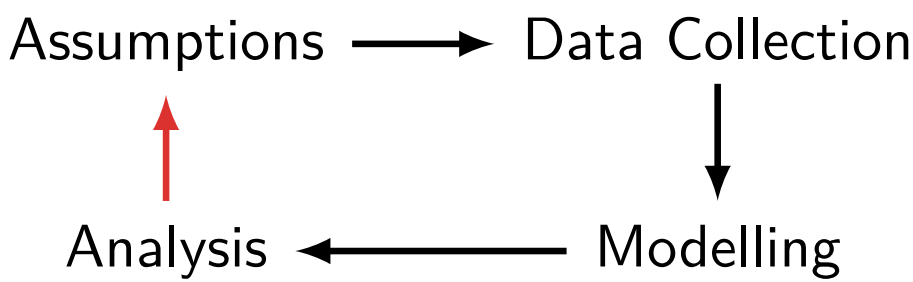
Two Variables: Two Interventions Identify the Graph

True Graph	$I = \{A\}$	$I = \{B\}$	$I = \emptyset$
$A \longrightarrow B$	$A \quad B$	$A \quad B$	$A \quad B$
$A \longleftarrow B$	$A \quad B$	$A \quad B$	$A \quad B$
$A \quad B$	$A \quad B$	$A \quad B$	$A \quad B$

True Graph under intervention I

1. Causal Inference: given a causal graph, infer mechanisms and causal effects from data
2. Causal Discovery: infer causal graph from data, using conditional independencies

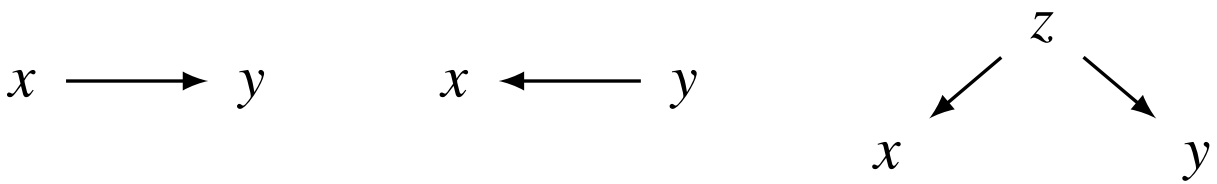
- Causal inference is abductive (inference to the best explanation)
- Discovering Cause and Effect = How to Factorize a Joint Distribution
- Strength of causal inference = credibility of the assumptions



算法莱辛巴赫共同原因原则 [JCS16; JS10]

个体之间的因果关系

- 如果序列 x 和 y 不是算法独立的 $I(x;y) \stackrel{+}{\geq} 0$, 那么



- 条件算法互信息

$$I(x;y \mid z) \stackrel{+}{=} K(x \mid z) + K(y \mid z) - K(x,y \mid z)$$

- 条件算法独立

$$I(x;y \mid z) \stackrel{+}{=} 0 \iff x \perp y \mid z$$

Remark: 如果两个复杂的产品设计 x 和 y 之间的相似度很高, 那么就需要一个解释, 极可能一家抄袭了另一家, 或同时抄袭了第三家. 但如果模式非常简单, 那极可能是巧合.

Algorithmic Model of Causality

- For every x_i there exists a program u_i of Turing machine T that computes x_i from its parents pa_i.

$$x_i = \mathrm{T}(\mathrm{pa}_i, u_i)$$

- The program u_i represents the causal mechanism that generates the effect x_i from its causes pa_i.
- The u_i is the analog of the unobserved noise term. It randomly chooses a mechanism.
- All u_i are algorithmically independent (Markovian).

Remark: If the observations $x_1, \ldots, x_n$ are generated by the algorithmic model of causality, then they satisfy the algorithmic Markov condition,

$$K(x_1, \ldots, x_n) \stackrel{+}{=} \sum_{i=1}^{n} K(x_i \mid \mathrm{pa}_i^*)$$

Equivalence of Algorithmic Markov Conditions[JS10]

1. Factorization

$$K(x_1, \ldots, x_n) \stackrel{+}{=} \sum_{i=1}^{n} K(x_i \mid \mathrm{pa}_i^*)$$

2. Local Markov condition

$$I(x_i; \mathrm{nd}_i \mid \mathrm{pa}_i^*) \stackrel{+}{=} 0$$

3. Global Markov condition

$$(X \perp Y \mid Z)_G \implies I(X; Y \mid Z^*) \stackrel{+}{=} 0$$

Remark: Due to the symmetry $K(x) + K(y \mid x^*) \stackrel{+}{=} K(y) + K(x \mid y^*)$, the Algorithmic Markov Condition only allows for identifying the Markov equivalence class. To be able to distinguish between Markov equivalence classes, we postulate the Algorithmic Independent Causal Mechanisms.

Postulate (Algorithmic Independent Causal Mechanisms)

A causal hypothesis G is only acceptable if

$$K(P_{X_1,\ldots,X_n}) \stackrel{+}{=} \sum_{i=1}^{n} K(P_{X_i|\mathrm{Pa}_i})$$

Equivalently, $I(P_{X_1|\mathrm{Pa}_1}; \ldots; P_{X_n|\mathrm{Pa}_n}) \stackrel{+}{=} 0$.
If no such causal graph exists, we reject every possible DAG and assume that there is a causal relation of a different type, e.g., a latent common cause, selection bias, or a cyclic causal structure.

Theorem

If the distributions P_X and $P_{Y|X}$ are algorithmically independent, i.e.,

$$I(P_X; P_{Y|X}) \stackrel{+}{=} 0$$

Then

$$K(P_{X,Y}) \stackrel{+}{=} K(P_X) + K(P_{Y|X}) \stackrel{+}{\leq} K(P_Y) + K(P_{X|Y})$$

Remark

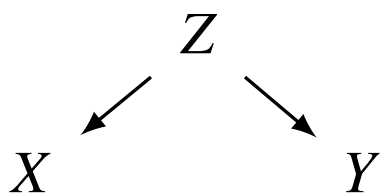
If

$$K(P_{X,Y}) < K(P_X) + K(P_{Y|X})$$
$$K(P_{X,Y}) < K(P_Y) + K(P_{X|Y})$$

then we reject both

$$X \longrightarrow Y \qquad \text{and} \qquad X \longleftarrow Y$$

which means



is the true structure.

因果方向 via Kolmogorov 复杂性

极小描述长度原则 MDL

给定数据和候选的理论集合，选择能够最小化下面两项之和的理论.

1. 理论的描述长度;
2. 借助这个理论，数据的描述长度.

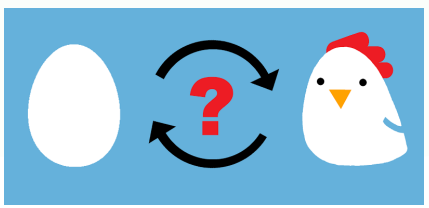
$$\operatorname*{argmin}_{H \in \mathcal{H}} \left\{ K(H) + K(D \mid H) \right\}$$

$$\boxed{C \to E} \text{ or } \boxed{C \leftarrow E}$$

怎么用 Kolmogorov complexity 判断因果方向?

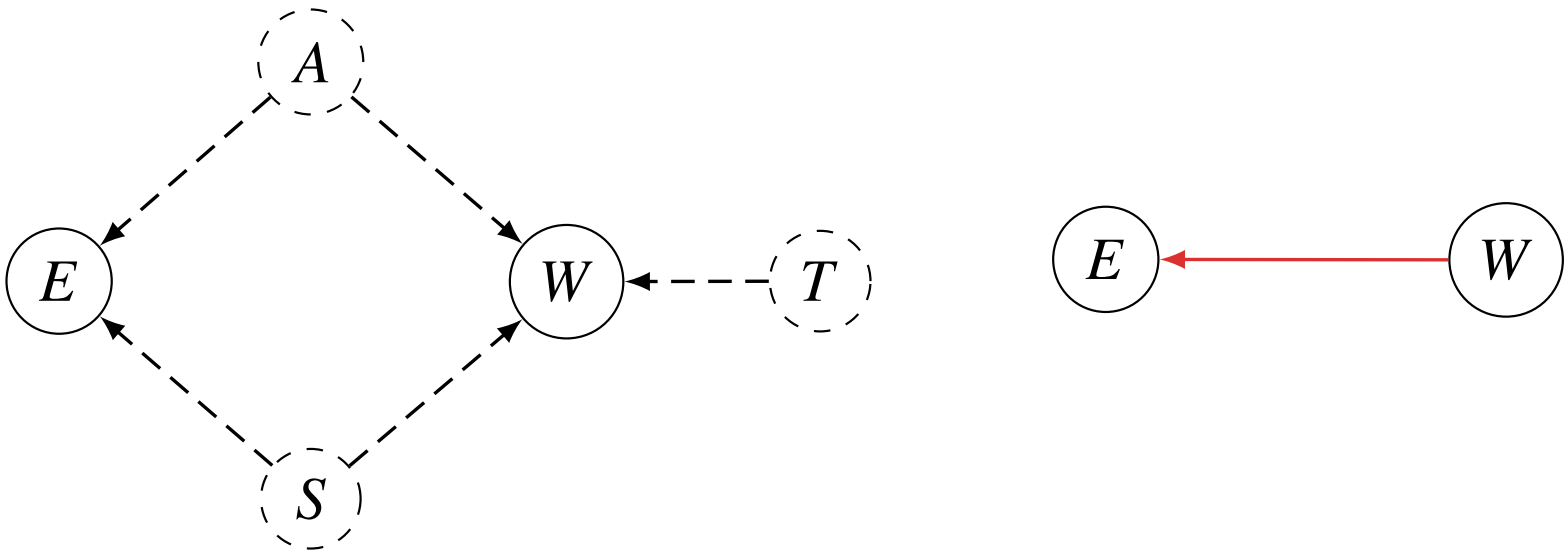
如果 C causes E，那么

$$K(P_C) + K(P_{E|C}) \stackrel{+}{\leq} K(P_E) + K(P_{C|E})$$



Causal Persuasion

- MBA 学位 E 与高收入 W 高度相关.
- 职员们主观认为高收入群体倾向于读 MBA 学位 $W \to E$.
- 商学院却想让职员们相信读 MBA 学位能带来高收入 $E \to W$.
- 事实是: 给定先天能力 A 和社交技巧 S 后, MBA 学位 E 与高收入 W 完全独立. $E \perp W \mid A, S$. 另外, 除了先天能力 A 和社交技巧 S 外, 收入 W 还受工作经验 T 的影响. 但职员们对 A, S, T 一无所知.
- 真实的因果模型是左图. 职员们的主观模型是右图.



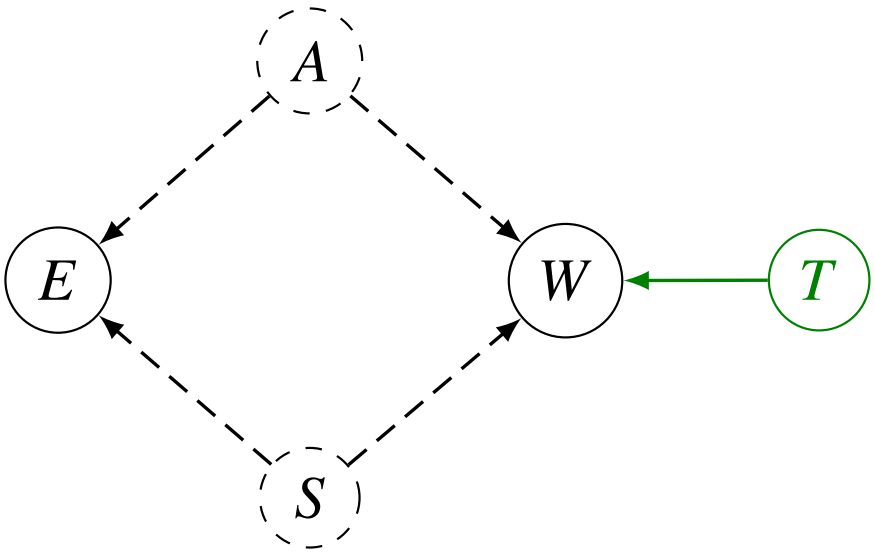
- 商学院如何说服职员们相信读 MBA 学位能带来高收入 $E \to W$?

Causal Persuasion

- 商学院如何说服职员们相信读 MBA 学位能带来高收入 $E \to W$?
- 只需向职员们披露工作经验 T 对收入 W 的影响, 以及 E, W, T 三者之间的关系:

$$E \not\perp W, \quad E \perp T, \quad E \not\perp T \mid W$$

- 这时, 职员们的主观模型变成了右图. MBA 学位 + 经验决定了收入.

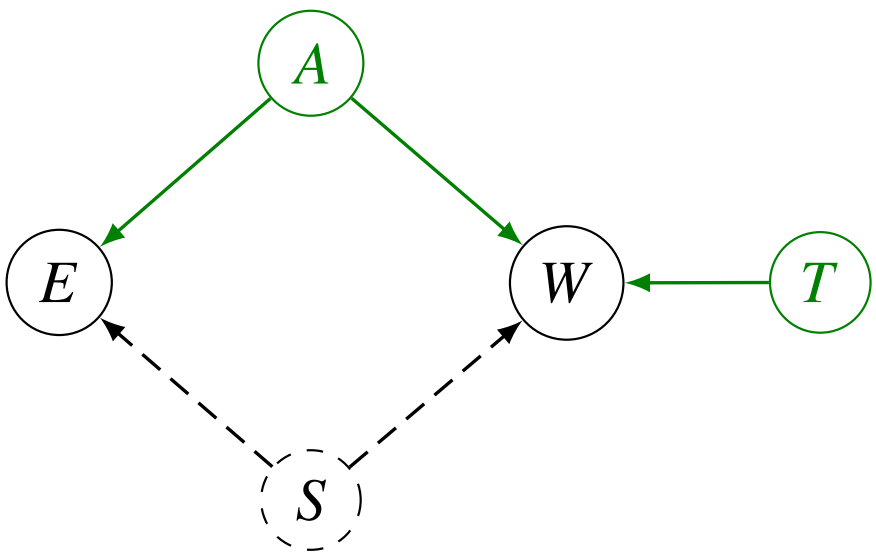


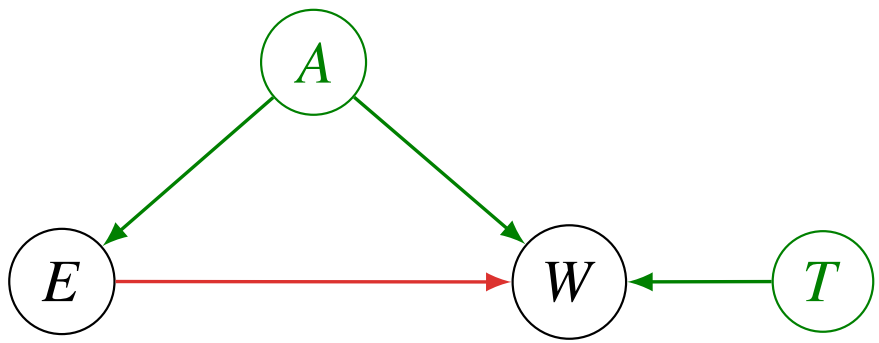
Remark: 通过对真实数据的选择性披露, 可以操纵对方产生错误认识.

- 公司老板如何才能说服职员们相信 MBA 学位 E 对高收入 W 毫无影响?

Causal Persuasion

- 公司老板如何才能说服职员们相信 MBA 学位 E 对高收入 W 毫无影响?
- 向职员们披露先天能力 A 是 MBA 学位 E 与高收入 W 的共同原因, 够吗?
- 职员们会发现 $E \not\perp W \mid A$, MBA 学位 E 仍然可能对高收入 W 有帮助.





- 公司老板必须向员工们披露所有可能的混杂因子才行 $E \perp W \mid A, S$.

Remark: 删除箭头 (辟谣) 比添加/颠倒箭头 (阴谋论) 要难得多.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal
   - Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Type Causation & Token Causation

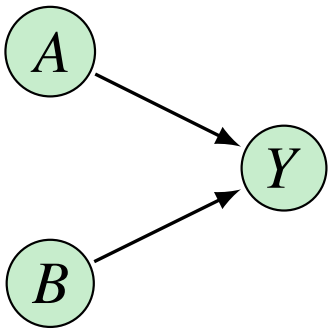
- **General/Type Causation:** 抽烟导致肺癌
- **Actual/Token Causation:** 张三抽了三十年烟, 导致了肺癌

因果解释:

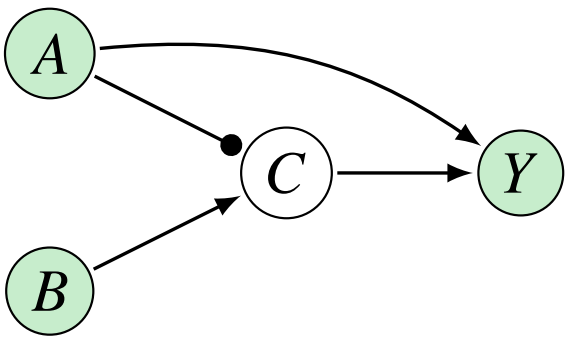
> “昨天晚上是下着暴雨, 我也喝醉了, 但是, 事故的真正原因是, 车的刹车片坏了.”

Actual Causation vs PN, PS

- 实际因果在构建解释、责任划分中起重要作用.
- 必要概率 PN 和充分概率 PS 只依赖 $Y_x(u)$, 关注因果模型的全局特征 (输入-输出), 无视因果过程.
- 实际因果必须考虑因果过程.
  — $B = 1$ 是 $Y = 1$ 的原因吗? 第一个例子中 "是", 第二个 "不是".



$Y = A \vee B$
$A = 1, B = 1, Y = 1$



$C = B \wedge \neg A$ $Y = A \vee C$
$A = 1, B = 1, C = 0, Y = 1$

$$A \vee C \equiv A \vee (B \wedge \neg A) \equiv A \vee B$$

Philosophy — Process Approach to Causality?

- A causal process is a world line of an object which possesses a conserved quantity.
- A causal interaction is an intersection of world lines which involves exchange of a conserved quantity.
- A causes B if there is a transfer of energy or momentum from A to B.

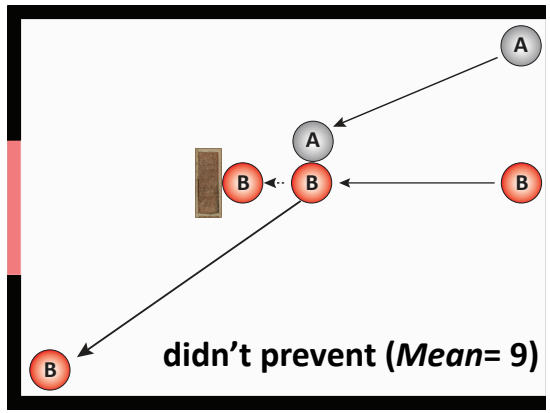
1. 相同过程, 不同因果?

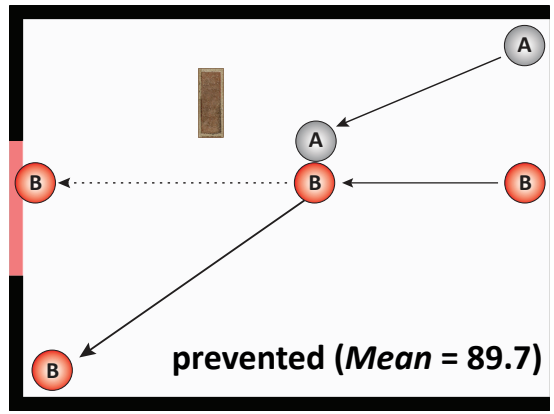
process?

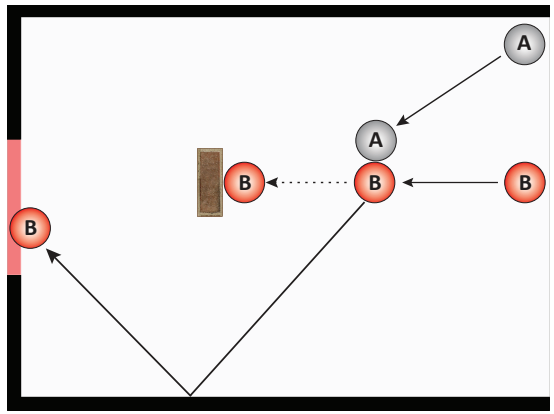
counterfactual?

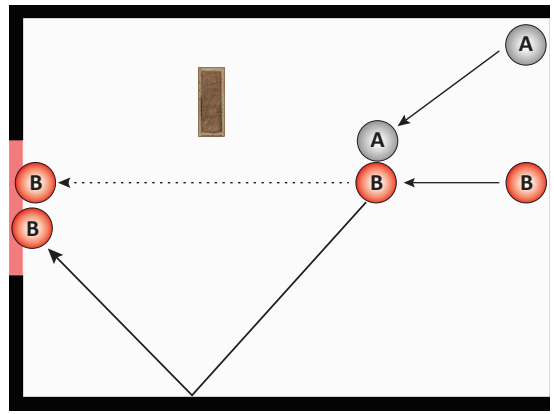
2. 开关是灯亮的原因, 传输了啥?

3. 欲望导致行动, 传输了啥?







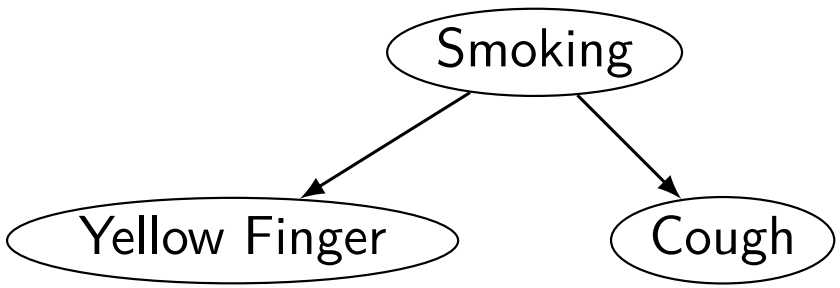


Philosophy — Probabilistic Approaches to Causality?

- **Reichenbach:** C causes E, iff,
  1. C is earlier than E, $t_C < t_E$,
  2. $P(E \mid C) > P(E \mid \neg C)$,
  3. there is no event S (earlier than or simultaneous with C) that S screens off C from E.

  $$\neg\exists S : P(E \mid C \wedge S) = P(E \mid S)$$

  Remark: Simpson paradox



- **Cartwright:** C causes E, iff, $P(E \mid C \wedge B) > P(E \mid \neg C \wedge B)$ for all background context B.[19]
- **Dupré:** "average degree of causal significance":

$$\sum_B P(B)\big[P(E \mid C \wedge B) - P(E \mid \neg C \wedge B)\big]$$

[19] Background context: all causal factors for E that excludes C and the effects of C.

Suppes' Genuine Cause — Probabilistic Approach

- **Suppes' *prima facie* cause:** C is a *prima facie* cause of E iff
  1. $t_C < t_E$
  2. $P(C) > 0$
  3. $P(E \mid C) > P(E \mid \neg C)$
- **Suppes' first *spurious* cause:** C, a *prima facie* cause of E, is a spurious cause iff there exists S such that
  1. $t_S < t_C < t_E$
  2. $P(C \wedge S) > 0$
  3. $P(E \mid C \wedge S) = P(E \mid S)$
  4. $P(E \mid C \wedge S) \geq P(E \mid C)$
- **Suppes' second *spurious* cause:** C, a *prima facie* cause of E, is a spurious cause iff there is a partition $\mathcal{E}$ and for every $S \in \mathcal{E}$
  1. $t_S < t_C < t_E$
  2. $P(C \wedge S) > 0$
  3. $P(E \mid C \wedge S) = P(E \mid S)$

  Remark: ε-spurious: $|P(E \mid C \wedge S) - P(E \mid S)| < \varepsilon$
- **Suppes' genuine cause:** nonspurious prima facie cause.

Remarks: 概率因果

- 概率因果依赖时序关系.
- 对背景变量的选择有死循环的风险.
  — 若要求对环境的完整描述, 则会将概率性关系化归为确定性方程.
  — 若描述的过于粗略, 则会导致伪相关和其它混杂效应.
  — 而要求背景变量与所讨论的变量 “因果相关” 则导致死循环.
  — 如何选择合适的背景变量, 类似于如何找寻合适的控制变量去混杂, 这必须依赖因果.

Philosophy — Regularity Approaches to Causality?

- **Mill's Sufficient** Condition: C causes E iff $C \rightarrow E$.
- **Hobbes's Necessary** Condition: C causes E iff $\neg C \rightarrow \neg E$.
- **Ramsey Test:** $C \boxdot\!\!\rightarrow E$ should be believed iff, after suspending judgment on C and E, E is believed as a result of assuming C.

$$C \boxdot\!\!\rightarrow E \in K \iff E \in K * C$$

- **Wright's NESS** condition: C causes E iff C is a Necessary Element of a Sufficient Set for E.
  1. $C \wedge X$ is E's sufficient condition. $C \wedge X \rightarrow E$
  2. X is not sufficient for E. $X \nrightarrow E$

Mackie's Actual Causation — Regularity Approach

- **Mackie's INUS** Condition: (insufficient but necessary part of a causal condition that is itself unnecessary but sufficient of the effect)
  1. $C \wedge X$ is E's sufficient but unnecessary condition.
  2. C is not sufficient for E. $C \nrightarrow E$
  3. X is not sufficient for E. $X \nrightarrow E$

  $$(C \wedge X) \vee Y \leftrightarrow E$$

  Example: 为什么造假币? 因为不会造真币、没钱又不会赚钱...
  - Why E?
  - C_1 rather than C_2, since $C_1 = \text{argmax}_C\, P(E \mid C)$?
- **Mackie's actual causation:**
  - C is at least an INUS condition of E
  - C was present
  - Components of X were present
  - Every disjunct in Y not containing C as a conjunct was absent

Lewis' Actual Causation — Counterfactual Approach

Definition (But-For Cause)

$X = x$ is a but-for cause of $Y = y$ in (M, u) iff

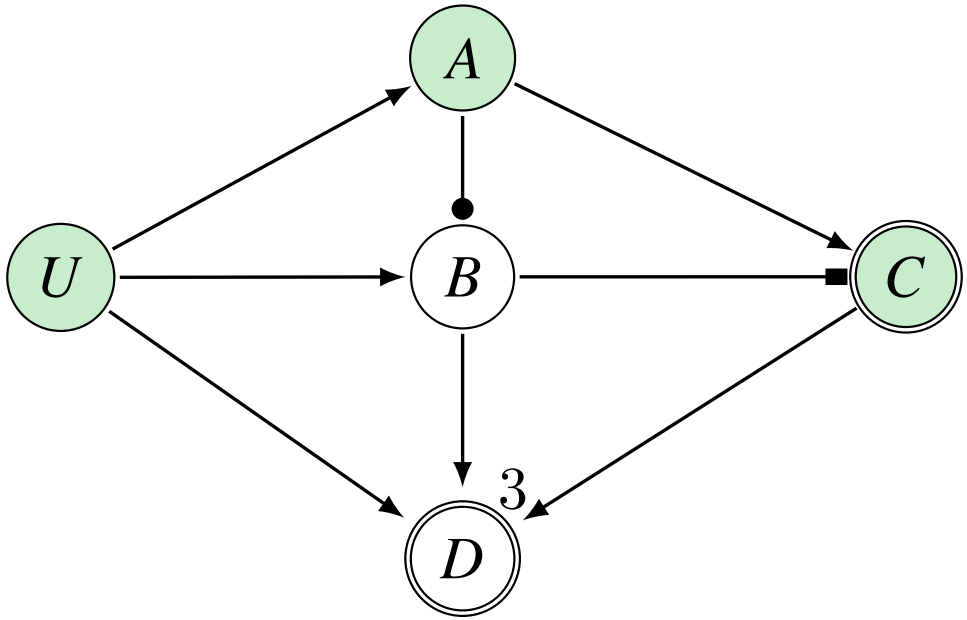
1. $M, u \models X = x \wedge Y = y$
2. there exist $x' \neq x$ and $y' \neq y$ such that $M, u \models [X = x']Y = y'$

Definition (Actual Causation — Lewis 1973)

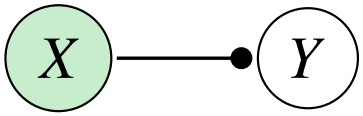
$X = x$ is an actual cause of $Y = y$ in (M, u) iff there exists a sequence of variables $Z_1 = X, \ldots, Z_n = Y$ s.t. $Z_i = z_i$ is a but-for cause of $Z_{i+1} = z_{i+1}$ for $i = 1, \ldots, n - 1$.

Remark: Is causation transitive?

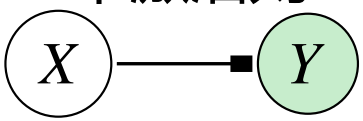
Neuron Diagrams

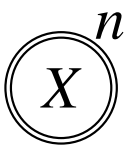


- X 的激活抑制了 Y



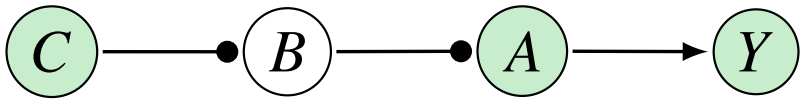
- X 不激活则 Y 激活



- 

多个信号方能激活 X

Example — Double Prevention



$$B = \neg C$$
$$A = \neg B$$
$$Y = A$$

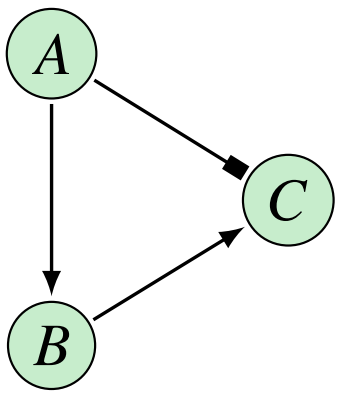
$$C = 1, B = 0, A = 1, Y = 1$$

- Alice A is planning to hack Yuri's Y computer.
- Bob B launches a missile at Alice's city.
- Carl C shoots down the missile.
- Alice hacks Yuri's computer, without any knowledge that Bob and Carl even exist.
- Nevertheless, Carl caused Yuri's computer being hacked.
- $C = 1$ is an actual cause of $Y = 1$.

Example — Transitivity?

1. A is a cause of B
2. B is a cause of C
3. A is a cause of C?

- Alice 医生治好了 Bob 的致命疾病 A.
- Bob 出院时 B, 被车撞死 C.



$B = A$
$C = \neg A \vee B$
$A = 1, B = 1, C = 1$

Sufficient Conditions for "But-For" Cause to be Transitive

Theorem (Transitivity of But-For Causation — Halpern 2016)

Suppose $X = x$ is a but-for cause of $Y = y$ in (M, u), and $Y = y$ is a but-for cause of $Z = z$ in (M, u), and the following two conditions hold,

1. *for every $y' \in R(Y)$, there exists an $x' \in R(X)$ such that*

$$M, u \models [X = x']Y = y'$$

2. *Y is on every causal path from X to Z.*

Then $X = x$ is a but-for cause of $Z = z$ in (M, u).

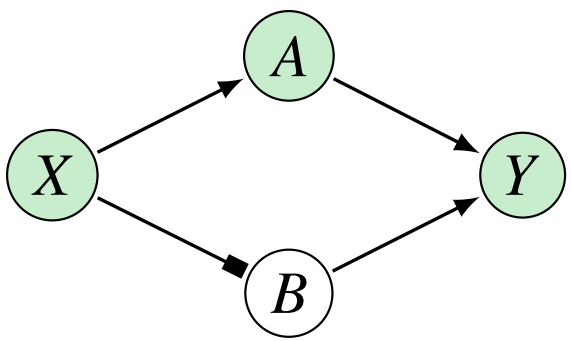
Remark: 对比 Pearl 的传递性的充分条件: 当 X 对 Z 没有直接影响时, 传递性成立.

$$\frac{Z_{y'x'} = Z_{y'}}{x \to y \;\&\; y \to z \implies x \to z}$$

Woodward's Manipulability Theory of Causation

1. X is a total cause of Y iff changing X will change Y for some values of all other variables that aren't descendants of X.
2. X is a direct cause of Y iff changing X will change Y when all variables other than X and Y are fixed at some values.
3. X is a contributing cause of Y iff there is a directed path from X to Y such that, changing X will change Y when the variables not on this path are fixed at some values.

Type of cause	**What is held fixed**
total	all other variables not descendants of X
direct	all variables other than X and Y
contributing	all variables not on one directed path from X to Y



$$A = X$$
$$B = \neg X$$
$$Y = A \vee B$$

X is not a total or a direct cause, but a contributing cause of Y

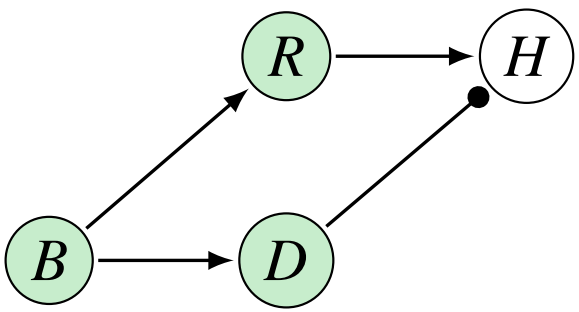
Woodward's Manipulability Theory of Causation

Definition (Actual Causation — Woodward 2003)

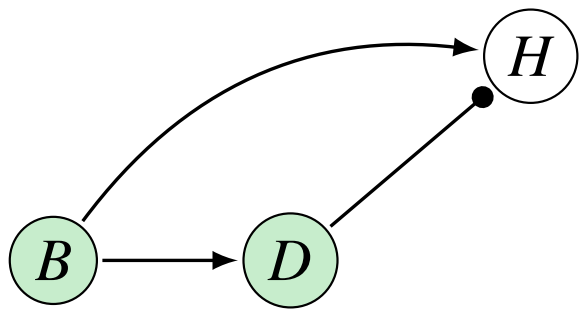
$X = x$ *is an actual cause of* $Y = y$ *in* (M, u) iff

1. $M, u \models X = x \wedge Y = y$
2. there is a directed path from X to Y such that, some intervention on X will change Y when the variables not on this path are fixed at their actual values.

Example: A boulder falls (B) and rolls toward the hiker (R). The hiker ducks (D) so that he does not get hit ($\neg H$).



$H = R \wedge \neg D$



$H = B \wedge \neg D$

Problem: $B = 1$ is a Woodward cause of $H = 0$ in G_1 but not in G_2.

Deterministic Actual Causation — Halpern and Pearl 2005

Definition (Actual Causation — Halpern and Pearl 2005)

$X = x$ *is an actual cause of* $Y = y$ *in* (M, u) iff

1. $M, u \models X = x \wedge Y = y$
2. there is a partition (Z, W) of $V \setminus X, Y$, and some setting x' of X and w of W such that,
   - 2.1 $M, u \models [X = x', W = w]Y \neq y$
   - 2.2 if z is $M, u \models Z = z$, then for all subsets $W' \subset W$ and all subset $Z' \subset Z$, $M, u \models [X = x, Z' = z, W' = w]Y = y$
3. X is minimal, i.e. no subset of X satisfies the above conditions.

Remark: 2.1 The assignment $W = w$ acts as a contingency against which Y counterfactually depends upon X.
2.2 imposes a restriction on the modifications that can be made, the setting of W' cannot interfere with the causal process $X \cup Z \cup Y$.
3 No irrelevant conjuncts. Don't want "dropping match and sneezing" to be a cause of the forest fire if just "dropping match" is.
Remark: 朱帆、叶峰补充了第 4 条: 在缺省世界 $u^* := \mathrm{argmax}_u P(U = u)$ 中,

- $M, u^* \models X = x^* \wedge Y = y^*$
- $M, u^* \models [X \neq x^*]Y \neq y^*$

Deterministic Actual Causation — Halpern 2016

Definition (Actual Causation — Halpern 2016)

$X = x$ *is an actual cause of* $Y = y$ *in* (M, u) iff

1. $M, u \models X = x \wedge Y = y$
2. there is a set of variables $W \subset V \setminus X, Y$ and a setting x' of X such that, if $M, u \models W = w$, then

$$M, u \models [X = x', W = w]Y \neq y$$

3. X is minimal, i.e. no subset of X satisfies the above conditions.

Probabilistic Actual Causation — Fenton-Glynn[Fen21]

Definition (Actual Causation (Simpliciter))

$X = x$ *rather than* $X = x'$ *is an actual cause of* $Y = y$ iff

1. $X = x$ and $Y = y$ are the actual values of X and Y.
2. $X = x$ rather than $X = x'$ is an actual cause of $Y = y$ relative to an appropriate model M.

Definition (Probabilistic Actual Causation (Model-Relative))

$X = x$ *rather than* $X = x'$ *is an actual cause of* $Y = y$ *relative to a model* M iff there is a directed path Q from X to Y in M such that, when we hold all variables in $W := V \setminus Q$ fixed at their actual values w, for any subset $Z \subset Q \setminus X, Y$ fixed at their actual values z,

$$P(Y = y \mid \mathrm{do}(X = x, Z = z, W = w)) > P(Y = y \mid \mathrm{do}(X = x', W = w))$$

Counterfactual NESS Causation

Definition (Counterfactual NESS Causation — Sander Beckers)

- $X = x$ is sufficient for $Y = y$ w.r.t. (M, u) iff for all values $z \in R(V \setminus (X \cup Y))$, we have
$$M, u \models [X = x, Z = z]Y = y$$
- $X = x$ directly NESS-causes $Y = y$ w.r.t. (M, u) if there exists $W = w$ s.t.
  1. $M, u \models X = x \wedge W = w \wedge Y = y$
  2. $\{X = x, W = w\}$ is sufficient for $Y = y$ w.r.t. (M, u)
  3. $W = w$ is not sufficient for $Y = y$ w.r.t. (M, u)
- $X = x$ NESS-causes $Y = y$ along a path p w.r.t. (M, u) if the values of the variables in p form a chain of direct NESS causes from $X = x$ to $Y = y$.
- $X = x$ CNESS-causes $Y = y$ w.r.t. (M, u) if
  1. $X = x$ NESS-causes $Y = y$ along some path p w.r.t. (M, u), and
  2. there exists a x' such that $X = x'$ does not NESS-cause $Y = y$ along any subpath p' of p w.r.t. $(M_{X=x'}, u)$.

Deterministic Actual Causation — Sander Beckers

Definition (Deterministic Actual Causation — Sander Beckers)

$X = x$ is an *actual cause* of $Y = y$ in (M, u) iff

1. $M, u \models X = x \wedge Y = y$.
2. There exist sets W, N with $Y \in N$, and values x', such that
   - 2.1 for all $S \subset N$ with $Y \in S$, and for all $s \in R(S)$ such that $y \in s$, there exists a $t \in R(V \setminus (X \cup W \cup S))$ so that

$$M, u \models [X = x', W = w^*, T = t]\, S \neq s$$

   - 2.2 for all $z \in R(V \setminus (X \cup W \cup N))$,

$$M, u \models [X = x, W = w^*, Z = z]\, N = n^*$$

3. X is minimal.

Example

Suppose that Y dies either if X loads A's gun and A shoots, or if B loads and shoots his gun.

$Y = (X \wedge A) \vee B$
$X = 1, A = 0, B = 1, Y = 1$

- $B = 1$ is a Beckers cause of $Y = 1$.
- $X = 1$ sufficient for $Y = 1$? No.
- $\{X = 1, A = 0\}$ sufficient for $Y = 1$? No.
- $\{X = 1, B = 1\}$ sufficient for $Y = 1$? Yes.
- Is $X = 1$ necessary? No.
- $X = 1$ is not a Beckers cause of $Y = 1$.

Sander Beckers' Definition in a different way

Definition (Sufficiency)

$X = x$ is **sufficient** for $Y = y$ in M, iff, for all $z \in R(V \setminus (X \cup Y))$, and all $u \in R(U)$, we have that $M, u \models [X = x, Z = z]Y = y$.

Definition (Sufficient Explanation)

A pair $(X = x, N)$ is a **sufficient explanation** of $Y = y$, iff, $Y \subset N$ and $X = x$ is sufficient for $N = n$ for some values $n \supset y$.

Definition

A sufficient explanation $(X_1 = x_1, N_1)$ **dominates** an explanation $(X_2 = x_2, N_2)$, iff, both are explanations of the same $Y = y$, $X_1 \subset X_2$, $N_1 \subset N_2$.

Definition (Actual Causation)

$X = x$ rather than $X = x'$ is an **actual cause** of $Y = y$ in (M, u), iff, it is part of a minimal actual sufficient explanation of $Y = y$, and there is no dominating sufficient explanation that includes $X = x'$.

Definition (Counterfactual Explanation)

Given (M, u), we say that $X = x$ rather than $X = x'$ is a **counterfactual explanation** of $Y = y$ relative to $(W = w, N)$, iff,

1. $((X = x, W = w), N)$ is an actual sufficient explanation of $Y = y$, and
2. $((X = x', W = w), N)$ is a sufficient explanation of $Y = y'$ with $y' \neq y$.

Remark: An actual cause $X = x$ is a part of a minimal actual sufficient explanation of $Y = y$ for which there exist counterfactual values $X = x'$ that would not have made the explanation better.

Theorem

If $X = x$ rather than $X = x'$ is a counterfactual explanation of $Y = y$ relative to $(W = w, N)$, then for some $\bar{X} \subset X$, $\bar{X} = \bar{x}$ rather than $\bar{X} = \bar{x}'$ is an actual cause of $Y = y$.

Remark: actual causes sit in between counterfactual and sufficient explanations: counterfactual explanations always contain actual causes.

Difference-Making Causation — Andreas and Günther

Definition (Difference-Making Causation — Andreas and Günther)

C is an actual cause of E in (M, v) iff

1. $M, v \models C \wedge E$
2. there is $v' \subset v$ such that (M, v') is uninformative on C and E, and

$$M, v' \models [\neg C]\neg E$$

Remark: v is the variable assignment of the exogenous and endogenous variables.

Remark: (M, v') being uninformative on φ means that (M, v') satisfies none of $\varphi, \neg\varphi$.

Actual Causation — Andreas and Günther

Definition (Actual Causation — Andreas and Günther)

C is an actual cause of E in (M, v) iff

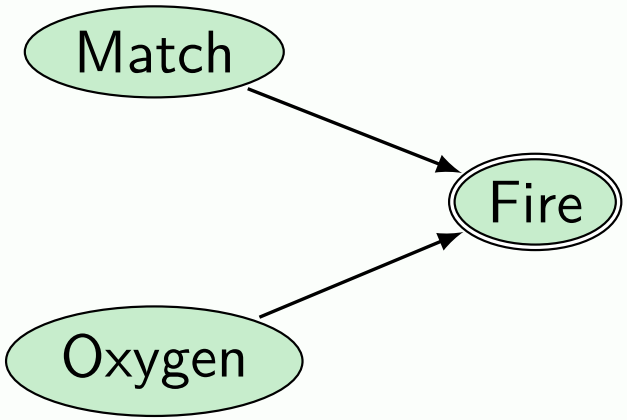
1. $M, v \models C \wedge E$
2. there is $v' \subset v$ such that (M, v') is uninformative on E, while for all $w \subset v$,

$$M_w, v' \models [C]E$$

   where $M_w := (M \setminus \{f_X : X \in w \text{ or } \neg X \in w\}) \cup w$.
3. there is no $v'' \subset v \setminus C$ such that, (M, v'') is uninformative on E and

$$M, v'' \models [\neg C]E$$

Example (Conjunctive Causes)



$F = M \wedge O$
$M = 1, O = 1, F = 1$

- $M = 1$ is a Halpern-Pearl cause of $F = 1$.
- $O = 1$ is a Halpern-Pearl cause of $F = 1$.
- 根据朱帆、叶峰的条件, 在缺省世界 u^* 中, $M = 0, O = 1, F = 0$, 而

$$M, u^* \not\models [O \neq 1]F \neq 0$$

所以, "有氧气" 不是 "起火" 的原因.

Remark: 问题是, 如果 M 和 O 都是小概率事件呢? 在 u^* 中, $M = 0, O = 0, F = 0$, 那么, $M = 1$ 和 $O = 1$ 将都不是 $F = 1$ 的原因.

Types of redundant causation

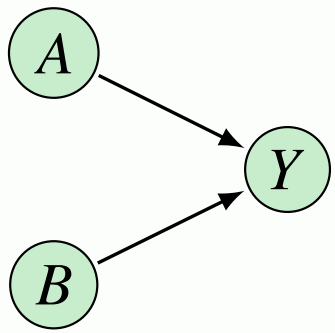
- Overdetermination
  — Multiple causes occur, any could have caused effect
- Preemption

  Early Preemption Multiple causal processes begin but only one completes and produces effect (backup causes)

  Late Preemption Multiple causal processes run to completion but only one is responsible for effect

Example (Overdetermination)

A prisoner is shot by two soldiers.



$Y = A \vee B$
$A = 1, B = 1, Y = 1$

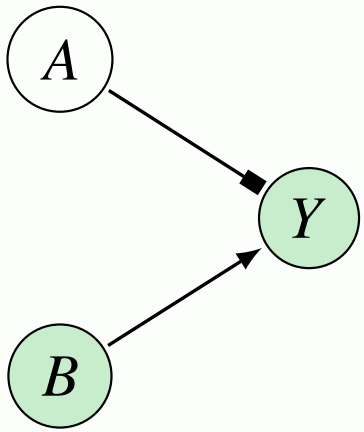
- $A = 1$ is a Halpern-Pearl cause of $Y = 1$.
  Proof: Take $W = \{B\}$.

$$M, u \models [A = 1, B = 0]Y = 1 \qquad M, u \models [A = 0, B = 0]Y = 0$$

- $A = 1 \vee B = 1$ is a Halpern cause of $Y = 1$.
- $A = 1$ is not a Halpern cause of $Y = 1$.
- $A = 1$ is a PAC cause of $Y = 1$.

Example (Bogus Prevention — (Counter-)Example?)

The assassin refrains from poisoning the potential victim's coffee $A = 0$.
But the bodyguard puts an antidote into the coffee anyway $B = 1$.



$Y = \neg A \vee B$
$A = 0, B = 1, Y = 1$

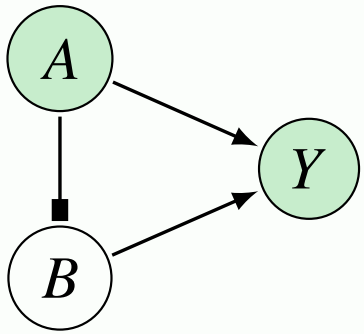
- $B = 1$ is a Halpern-Pearl cause of $Y = 1$.
  Proof: Take $W = \{A\}$.

$$M, u \models [B = 1, A = 1]Y = 1 \qquad M, u \models [B = 0, A = 1]Y = 0$$

- $B = 1$ is not a Halpern cause of $Y = 1$.
- $A = 0 \vee B = 1$ is a Halpern cause of $Y = 1$.
- $B = 1$ is a PAC cause of $Y = 1$.

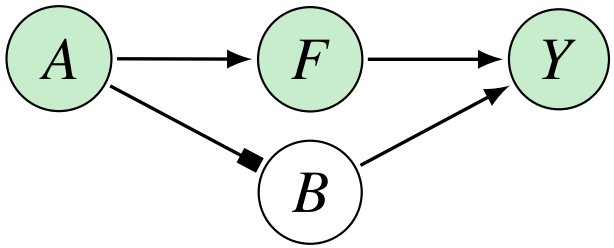
Example (Early Preemption)

Alice and Bob are aiming rocks at a window. Bob will throw his rock if Alice doesn't throw hers.



$B = \neg A$ $Y = A \vee B$
$A = 1, B = 0, Y = 1$

- $A = 1$ is not a Lewis cause of $Y = 1$ in the above model, but is a Lewis cause in the following model. What is the "right" model?



F: Alice's rock flies toward the window.

- $A = 1$ is a Halpern cause of $Y = 1$.
  Proof: Take $W = \{B\}$.

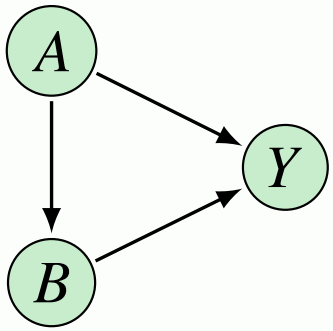
$$M, u \models [A = 0, B = 0]Y = 0$$

- $B = 0$ is not an actual cause of $Y = 1$.

Example

Example

Gang leader Alice orders Bob to join her in shooting Yuri.



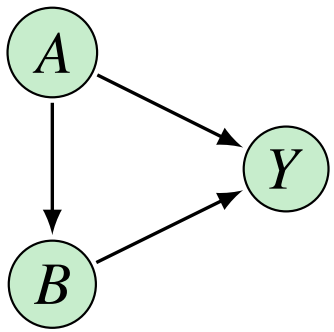
$B = A$ $Y = A \vee B$
$A = 1, B = 1, Y = 1$

- $A = 1$ is a Halpern cause of $Y = 1$.
- $A = 1$ is a PAC cause of $Y = 1$.
- $B = 1$ is not a Halpern cause of $Y = 1$.
- $B = 1$ is a PAC cause of $Y = 1$.

Example — Early Preemption

Example1 Alice poisons the victim's coffee. Bob puts an antidote into the coffee. Bob would not have put antidote into the coffee if Alice had not poisoned the coffee.

Example2 Alice puts an antidote into the victim's coffee. Bob poisons the coffee. Bob would not have poisoned the coffee if Alice had not administered the antidote.



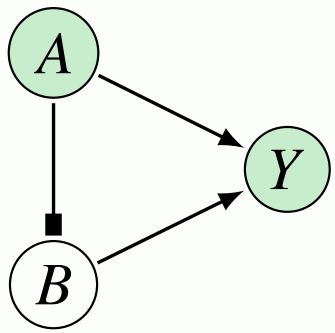
$B = A$ $Y = \neg A \vee B$
$A = 1, B = 1, Y = 1$

$B = A$ $Y = A \vee \neg B$
$A = 1, B = 1, Y = 1$

- $A = 1$ is not an actual cause of $Y = 1$ in example1.
- $B = 1$ is an actual cause of $Y = 1$ in example1.
- $A = 1$ is a Halpern / Fenton-Glynn cause of $Y = 1$ in example2.
- $A = 1$ is not a PAC cause of $Y = 1$ in example2.
- $B = 1$ is not an actual cause of $Y = 1$ in example2.

Example (Early Preemption)

Alice and Bob are aiming rocks at a window. Bob will probably throw his rock if Alice doesn't throw hers.

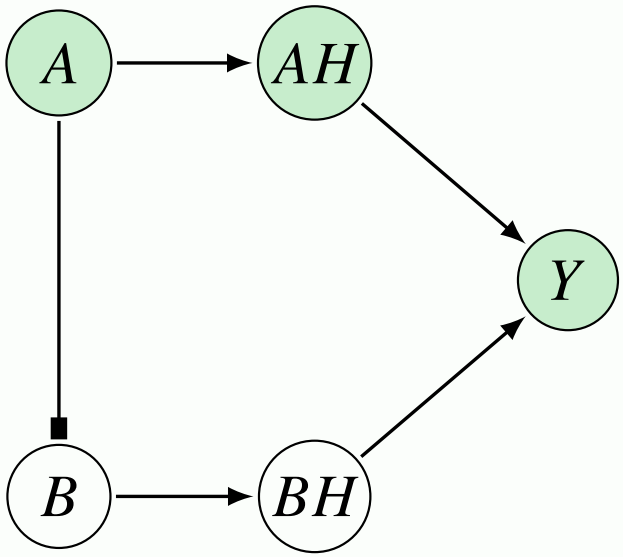


$P(B = 1 \mid A = 0) = 0.9$
$P(B = 1 \mid A = 1) = 0.1$
$P(Y = 1 \mid A = 1, B = 1) = 0.95$
$P(Y = 1 \mid A = 1, B = 0) = 0.5$
$P(Y = 1 \mid A = 0, B = 1) = 0.9$
$P(Y = 1 \mid A = 0, B = 0) = 0.01$
$A = 1, B = 0, Y = 1$

- $A = 1$ is an actual cause of $Y = 1$.
  Proof: Take $Q = \{A, Y\}$, $W = \{B\}$.
  $P(Y = 1 \mid \mathsf{do}(A = 1, B = 0)) = 0.5 > P(Y = 1 \mid \mathsf{do}(A = 0, B = 0)) = 0.01$
- $B = 1$ is an actual cause of $Y = 1$ relative to model M.
  Proof: Take $Q = \{B, Y\}$, $W = \{A\}$.
  $P(Y = 1 \mid \mathsf{do}(B = 1, A = 1)) = 0.95 > P(Y = 1 \mid \mathsf{do}(B = 0, A = 1)) = 0.5$

Example (Early Preemption)

Alice and Bob are aiming rocks at a window. Bob will probably throw his rock if Alice doesn't throw hers.



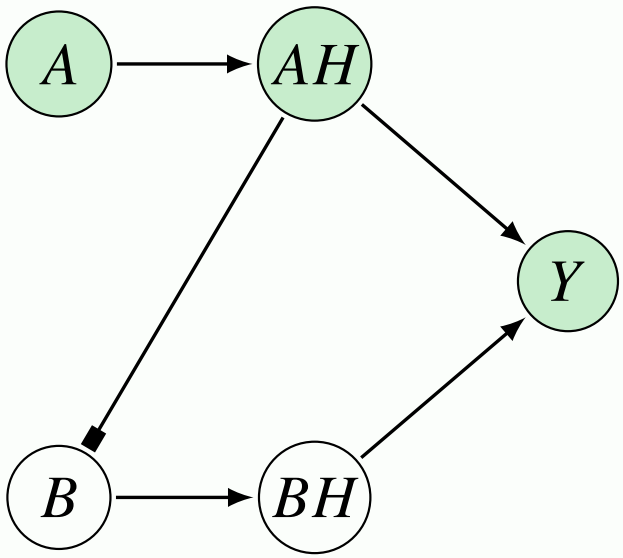
$$
\begin{aligned}
P(B = 1 \mid A = 0) &= 0.9 \\
P(B = 1 \mid A = 1) &= 0.1 \\
P(AH = 1 \mid A = 1) &= 0.5 \\
P(AH = 1 \mid A = 0) &= 0.01 \\
P(BH = 1 \mid B = 1) &= 0.9 \\
P(BH = 1 \mid B = 0) &= 0.01 \\
P(Y = 1 \mid AH = 1, BH = 1) &= 0.998 \\
P(Y = 1 \mid AH = 1, BH = 0) &= 0.95 \\
P(Y = 1 \mid AH = 0, BH = 1) &= 0.95 \\
P(Y = 1 \mid AH = 0, BH = 0) &= 0.01 \\
\hline
A = 1, AH = 1, B = 0, BH = 0, Y = 1 &
\end{aligned}
$$

- $B = 1$ is not an actual cause of $Y = 1$ relative to model M.
  Proof: Take $Q = \{B, BH, Y\}$, $W = \{A, AH\}$.

$$P(Y = 1 \mid \mathsf{do}(B = 1, BH = 0, A = 1, AH = 1)) = 0.95 < 0.95048 = P(Y = 1 \mid \mathsf{do}(B = 0, A = 1, AH = 1))$$

Example (Early Preemption)

Alice and Bob are aiming rocks at a window. Bob will probably throw his rock if Alice misses.



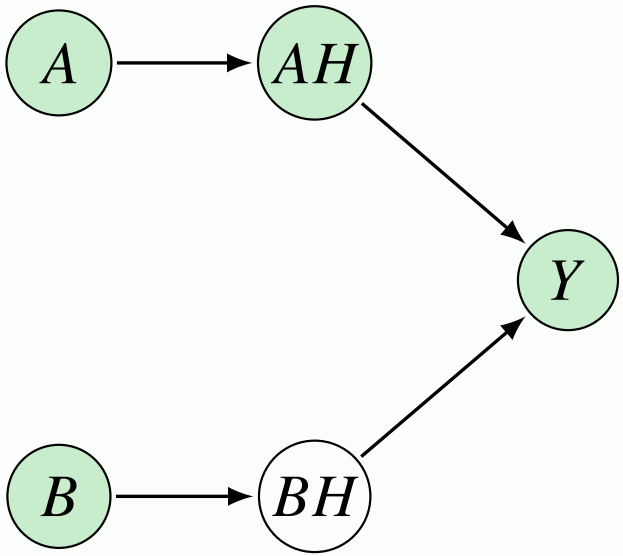
$$
\begin{array}{r}
P(B = 1 \mid AH = 0) = 0.9 \\
P(B = 1 \mid AH = 1) = 0.1 \\
P(AH = 1 \mid A = 1) = 0.5 \\
P(AH = 1 \mid A = 0) = 0.01 \\
P(BH = 1 \mid B = 1) = 0.9 \\
P(BH = 1 \mid B = 0) = 0.01 \\
P(Y = 1 \mid AH = 1, BH = 1) = 0.998 \\
P(Y = 1 \mid AH = 1, BH = 0) = 0.95 \\
P(Y = 1 \mid AH = 0, BH = 1) = 0.95 \\
P(Y = 1 \mid AH = 0, BH = 0) = 0.01 \\
\hline
A = 1, AH = 1, B = 0, BH = 0, Y = 1
\end{array}
$$

- $B = 1$ is not an actual cause of $Y = 1$ relative to model M.
  Proof: Take $Q = \{B, BH, Y\}$, $W = \{A, AH\}$.

$$P(Y = 1 \mid \mathsf{do}(B = 1, BH = 0, A = 1, AH = 1)) = 0.95 < 0.95048 = P(Y = 1 \mid \mathsf{do}(B = 0, A = 1, AH = 1))$$

Example (Late Preemption)

Alice and Bob throw rocks at a window simultaneously. Alice's throw hits the window and Bob's misses.



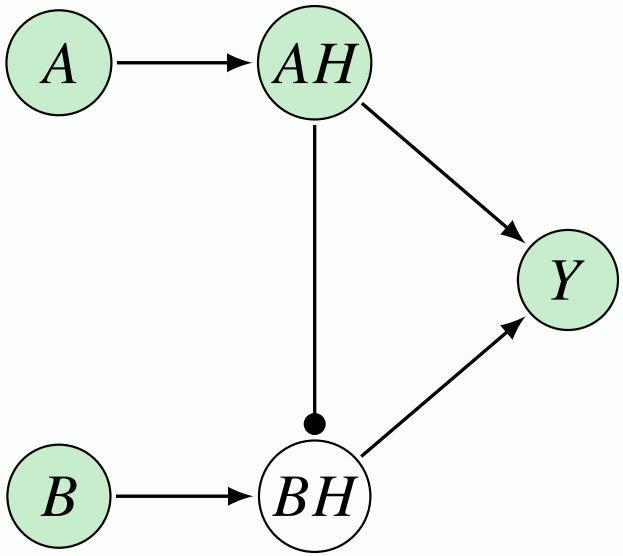
$$
\begin{array}{|r|}
\hline
P(AH = 1 \mid A = 1) = 0.5 \\
P(AH = 1 \mid A = 0) = 0.01 \\
P(BH = 1 \mid B = 1) = 0.9 \\
P(BH = 1 \mid B = 0) = 0.01 \\
P(Y = 1 \mid AH = 1, BH = 1) = 0.998 \\
P(Y = 1 \mid AH = 1, BH = 0) = 0.95 \\
P(Y = 1 \mid AH = 0, BH = 1) = 0.95 \\
P(Y = 1 \mid AH = 0, BH = 0) = 0.01 \\
\hline
A = 1, AH = 1, B = 1, BH = 0, Y = 1 \\
\hline
\end{array}
$$

- $B = 1$ is not an actual cause of $Y = 1$.
  Proof: Take $Q = \{B, BH, Y\}$, $W = \{A, AH\}$.

$$P(Y = 1 \mid \mathtt{do}(B = 1, BH = 0, A = 1, AH = 1)) = 0.95 < 0.95048 = P(Y = 1 \mid \mathtt{do}(B = 0, A = 1, AH = 1))$$

Example (Late Preemption)

Alice and Bob throw rocks at a window simultaneously. Alice's throw hits the window, and Bob's misses because of Alice's hit.



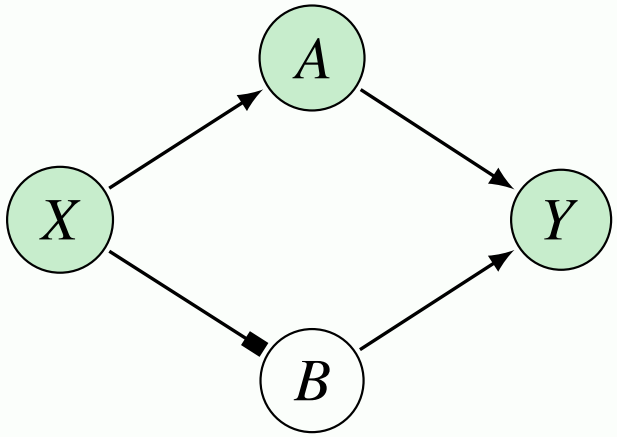
$$P(AH = 1 \mid A = 1) = 0.5$$
$$P(AH = 1 \mid A = 0) = 0.01$$
$$P(BH = 1 \mid B = 1, AH = 0) = 0.9$$
$$P(BH = 1 \mid B = 0, AH = 0) = 0.01$$
$$P(BH = 1 \mid AH = 1) = 0$$
$$P(Y = 1 \mid AH = 1, BH = 0) = 0.95$$
$$P(Y = 1 \mid AH = 0, BH = 1) = 0.95$$
$$P(Y = 1 \mid AH = 0, BH = 0) = 0.01$$

$$A = 1, AH = 1, B = 1, BH = 0, Y = 1$$

- $B = 1$ is not an actual cause of $Y = 1$.
  Proof: Take $Q = \{B, BH, Y\}$, $W = \{A, AH\}$.

$$P(Y = 1 \mid \mathtt{do}(B = 1, BH = 0, A = 1, AH = 1)) = 0.95 = P(Y = 1 \mid \mathtt{do}(B = 0, A = 1, AH = 1))$$

Example (Simple Switch)



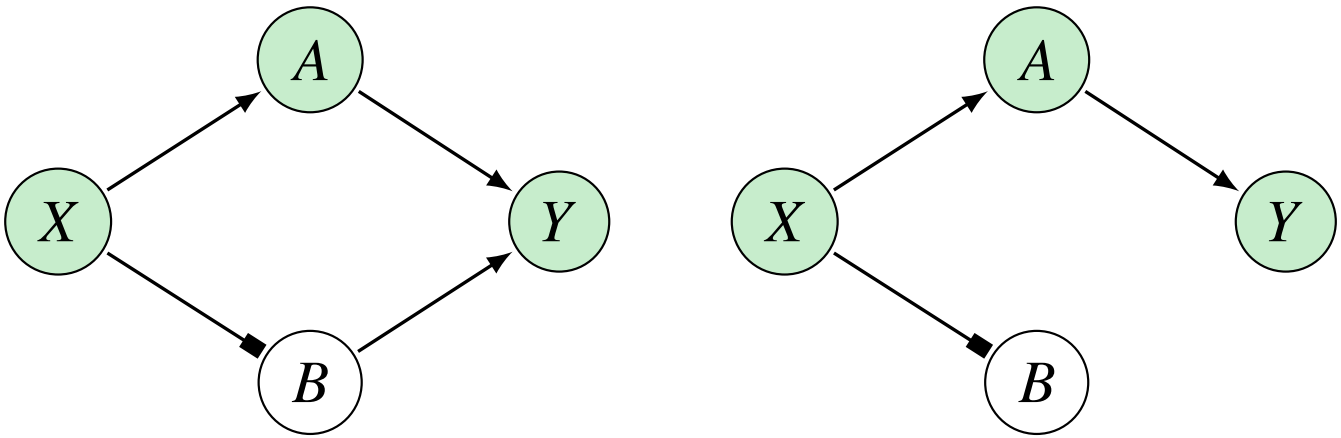
$A = X$ $B = \neg X$ $Y = A \vee B$
$X = 1, A = 1, B = 0, Y = 1$

- $X = 1$ is not an Andreas-Günther DM-cause of $Y = 1$.
  Proof: The model (M, v') is only uninformative on X and Y for $v' = \emptyset$.

$A = X$ $B = \neg X$ $Y = A \vee B$
$\emptyset$

  But $M, \emptyset \not\models [X = 0]Y \neq 1$.
- $X = 1$ is a Halpern cause of $Y = 1$.
- $X = 1$ is not a PAC cause of $Y = 1$.

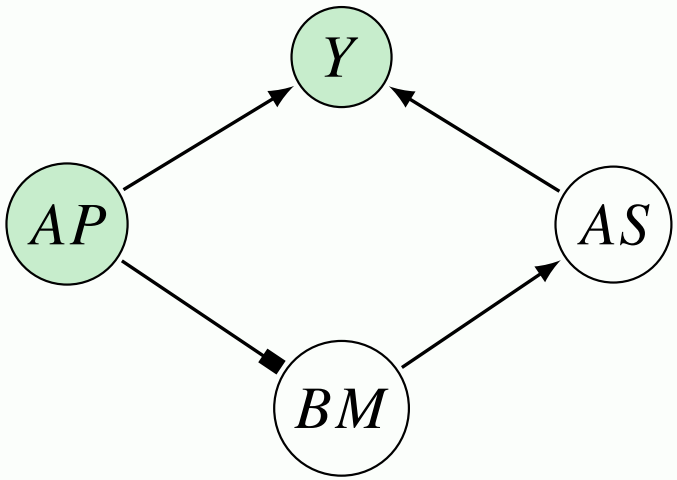
Remark



- 左图 $X = 1$ 不是 $Y = 1$ 的原因, 右图则 “是”.
- 因此, 判断实际因果时, 不能只考虑实际的因果过程 $X = 1 \to A = 1 \to Y = 1$, 还要考虑反事实的路径 $X = 0 \to B = 1 \to Y = 1$.

Discussion

Example (Frankfurt-Case)

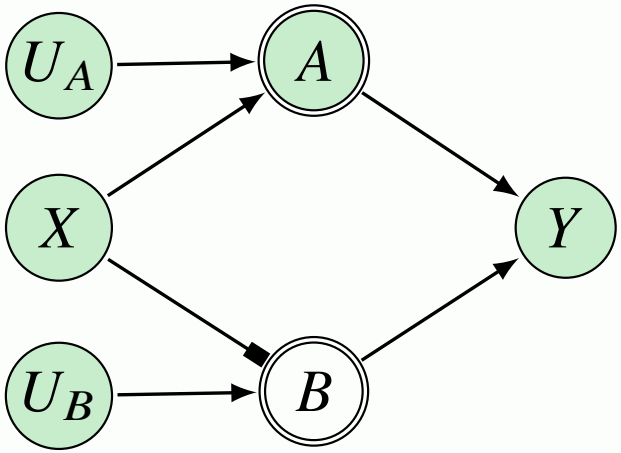


$$BM = \neg AP$$
$$AS = BM$$
$$Y = AP \vee AS$$

$$AP = 1, BM = 0, AS = 0, Y = 1$$

- Alice 下毒杀死了 Yuri. 若 Alice 不下毒, Bob 会操控 Alice 枪杀 Yuri.
- $AP = 1$ is not an Andreas-Günther DM-cause of $Y = 1$.
- $AP = 1$ is a Halpern cause of $Y = 1$.

Example (Realistic Switch)



$A = U_A \wedge X$ $B = U_B \wedge \neg X$ $Y = A \vee B$
$U_A = 1, U_B = 1, X = 1, A = 1, B = 0, Y = 1$

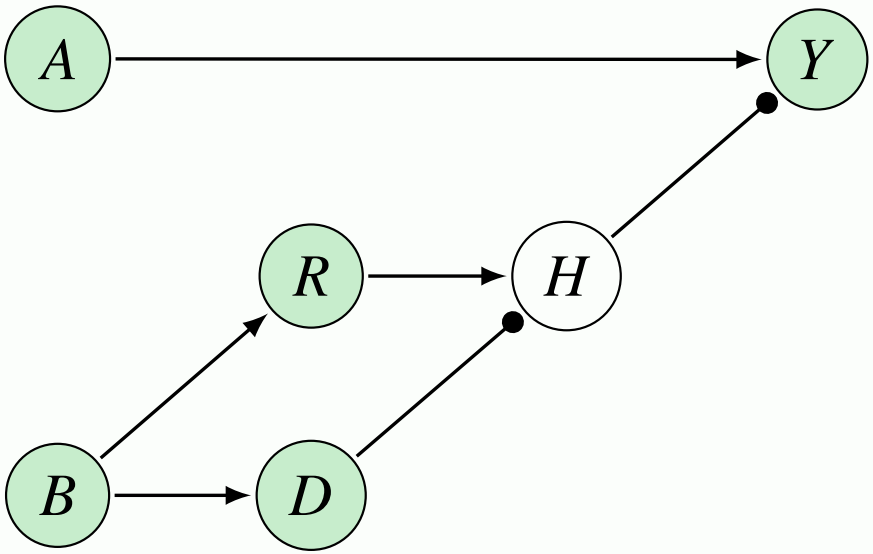
The model (M, v') is uninformative on Y for $v' = \{U_A = 1\}$.

$A = U_A \wedge X$ $B = U_B \wedge \neg X$ $Y = A \vee B$
$U_A = 1$

But $M, v' \models [X = 1]Y = 1$.
However, consider (M, v'') that is uninformative on Y for $v'' = \{U_B = 1\}$.
Then $M, v'' \models [X = 0]Y = 1$.
Therefore, $X = 1$ is not an Andreas-Günther actual cause of $Y = 1$.

Example (Hall's "short circuit")

A hiker is on a hike (A). A boulder falls (B) and rolls toward the hiker (R). The hiker ducks (D) so that he does not get hit ($\neg H$) and continues the hike (Y).



$$R = B$$
$$D = B$$
$$H = R \wedge \neg D$$
$$Y = A \wedge \neg H$$

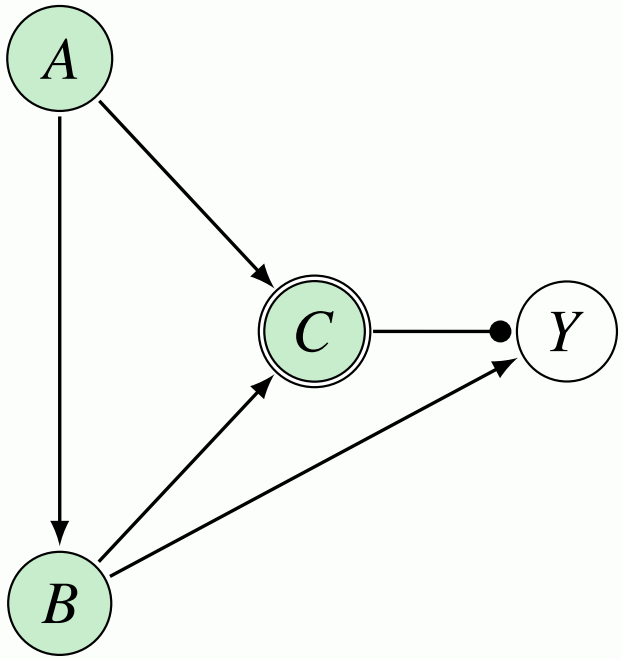
$$A = 1, B = 1, R = 1, D = 1, H = 0, Y = 1$$

- $A = 1$ is an actual cause of $Y = 1$.
- $B = 1$ is a PAC cause of $D = 1$, and $D = 1$ is a PAC cause of $Y = 1$.
- $B = 1$ is a Halpern / Fenton-Glynn cause of $Y = 1$.
- $B = 1$ is not a PAC cause of $Y = 1$.

Remark: PAC causation is not transitive.

Example (Short Circuit)

Alice 在 Yuri 的咖啡里放了解药 A. Bob 恶作剧放了毒药 B, 但如果 Alice 不放解药的话 Bob 是不会下毒的. 毒药和解药中和 C, Yuri 没死 $\neg Y$.



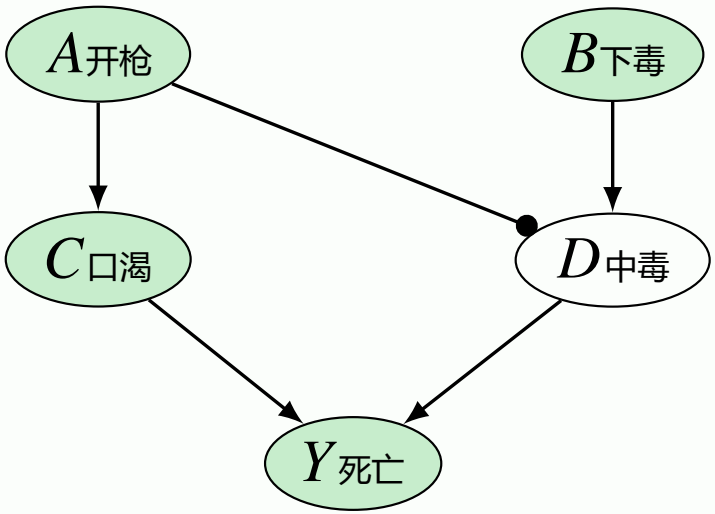
$$B = A$$
$$C = A \wedge B$$
$$Y = B \wedge \neg C$$

$$A = 1, B = 1, C = 1, Y = 0$$

- $A = 1$ is a Halpern / Fenton-Glynn cause of $Y = 0$.
- $A = 1$ is not a Beckers cause of $Y = 0$.

Example (Early Preemption)

A desert traveler Yuri has two enemies, Alice and Bob. Bob poisons Yuri's canteen. Alice, unaware of Bob's action, shoots and empties the canteen. Whose action is the actual cause of Yuri's death?

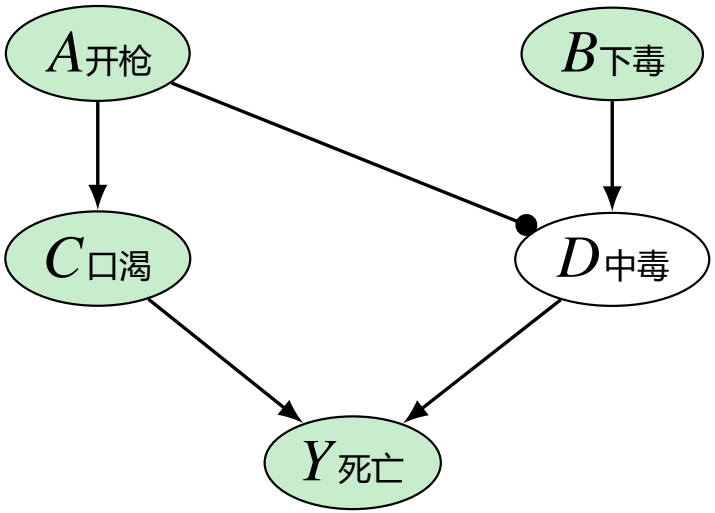


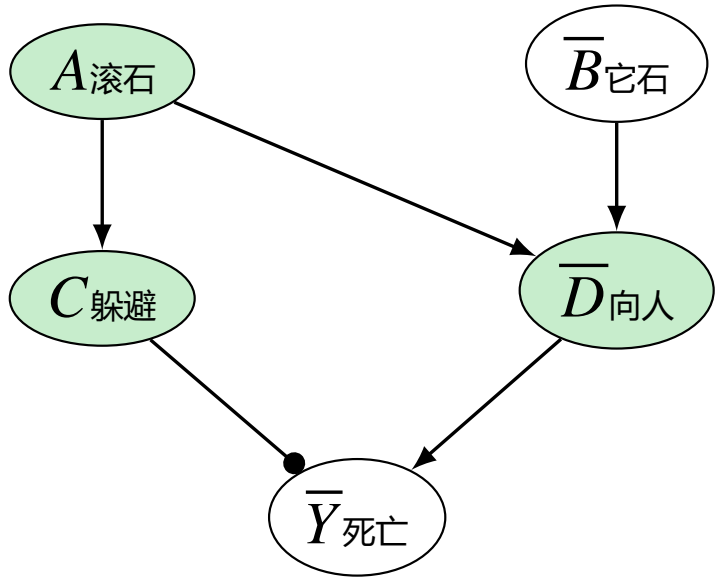
$$C = A$$
$$D = B \wedge \neg A$$
$$Y = C \vee D$$

$$A = 1, B = 1, C = 1, D = 0, Y = 1$$

- $A = 1$ is a PAC cause of $Y = 1$.
- $B = 1$ is not an Andreas-Günther cause of $Y = 1$.
- $B = 1$ is not a PAC cause of $Y = 1$.

Early Preemption vs Short Circuit





$C = A$ $D = B \wedge \neg A$ $Y = C \vee D$
$A = 1, B = 1, C = 1, D = 0, Y = 1$

$\cong$

$C = A$ $\overline{D} = \overline{B} \wedge \neg A$ $\overline{Y} = C \vee \overline{D}$
$A = 1, \overline{B} = 1, C = 1, \overline{D} = 0, \overline{Y} = 1$

- $A_{\text{开枪}} = 1$ 是 $Y_{\text{死亡}} = 1$ 的原因.
- $A_{\text{滚石}} = 1$ 不是 $\overline{Y}_{\text{死亡}} = 1$ 的原因?

Probabilistic Actual Causation PAC — ToDo

Definition (Probabilistic Actual Causation)

$X = x$ *rather than* $X = x'$ *is an actual cause of* $Y = y$ *in* (M, u) iff

1. $M, u \models X = x \wedge Y = y$
2. there is some set W off some directed path from X to Y fixed at their actual values w such that:
   - 2.1 $W = w$ does not determine $Y = y$ in $M_{\setminus f_X}$: $P(Y = y \mid W = w) < 1$
   - 2.2 for any subset Z of some directed path from X to Y fixed at their actual values z, we have

$$P(Y_{X=x, Z=z} = y \mid W = w) > P(Y_{X=x'} = y \mid W = w)$$

3. there is no set W off some directed path from X to Y fixed at their actual values w such that:
   - 3.1 $W = w$ does not determine $Y = y$ in $M_{\setminus f_X}$: $P(Y = y \mid W = w) < 1$
   - 3.2 for any subset Z of some directed path from X to Y fixed at their actual values z, we have

$$P(Y_{X=x', Z=z} = y \mid W = w) > P(Y_{X=x} = y \mid W = w)$$

Backtracking Actual Causation BAC — ToDo

- There can be multiple ways of setting $U^* = u^*$ that satisfy the structural equations and agree with the counterfactual antecedent.
- We assume $P_B(U^* = u^* \mid U = u) > 0$ for any u^*.

Definition (Backtracking Actual Causation)

$X = x$ rather than $X = x'$ is an actual cause of $Y = y$ in (M, u) iff

1. $M, u \models X = x \wedge Y = y$
2. there is a directed path Q from X to Y, and some set $W \subset V \setminus Q$ fixed at their actual values w such that in $M \setminus f_{X^*}$:
   - 2.1 $W^* = w$ does not backtracking determine $Y^* = y$,

$$P_B(Y^* = y \mid W^* = w) < 1$$

   - 2.2 for any subset $Z \subset Q \setminus X, Y$ fixed at their actual values z, we have

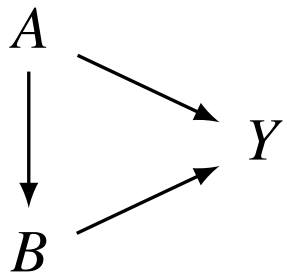
$$P_B(Y^* = y \mid X^* = x, Z^* = z, W^* = w) > P_B(Y^* = y \mid X^* = x', W^* = w)$$

3. there is no ..... x' ... x

Discussion

Example1 Alice and Bob are aiming rocks at a window. Bob will throw his rock if Alice doesn't throw hers.

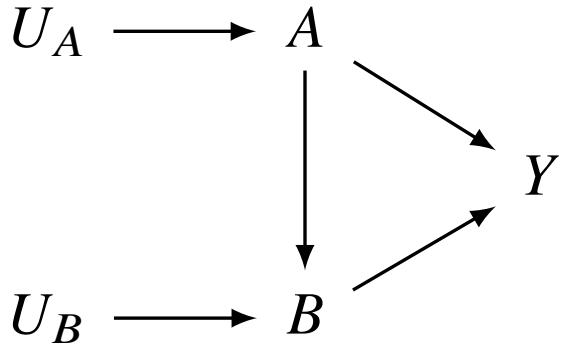
Example2 A boulder rolls toward a hiker (A). The hiker ducks (B) so that he survives (Y).



$B = \neg A$ $Y = A \vee B$
$A = 1, B = 0, Y = 1$

$B = A$ $Y = \neg A \vee B$
$A = 1, B = 1, Y = 1$

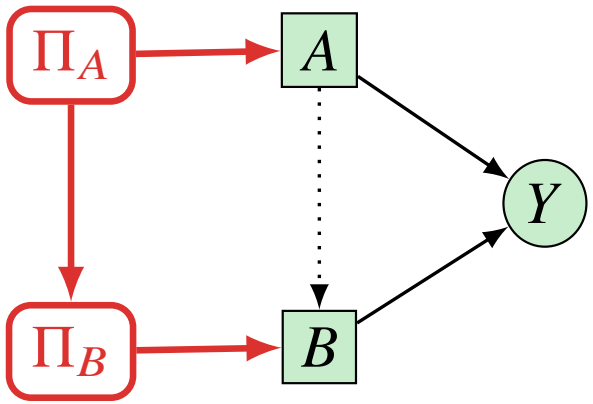
- $A = 1$ is not a PAC cause of $Y = 1$ in both Examples.
- **Remark:** If we add exogenous variables for A and B,



Then $A = 1$ is a PAC cause of $Y = 1$ in Example1, but it is not a PAC cause of $Y = 1$ in Example2.

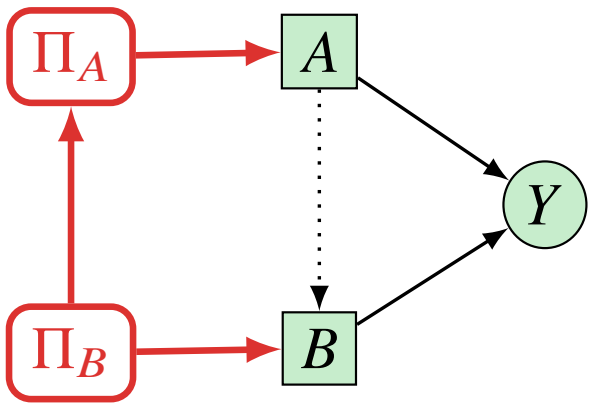
归不归责问题: 同 (实际) 因不同责?

Example1 Alice 和 Bob 准备砸玻璃. 如果 Alice 不出手, Bob 必出手.



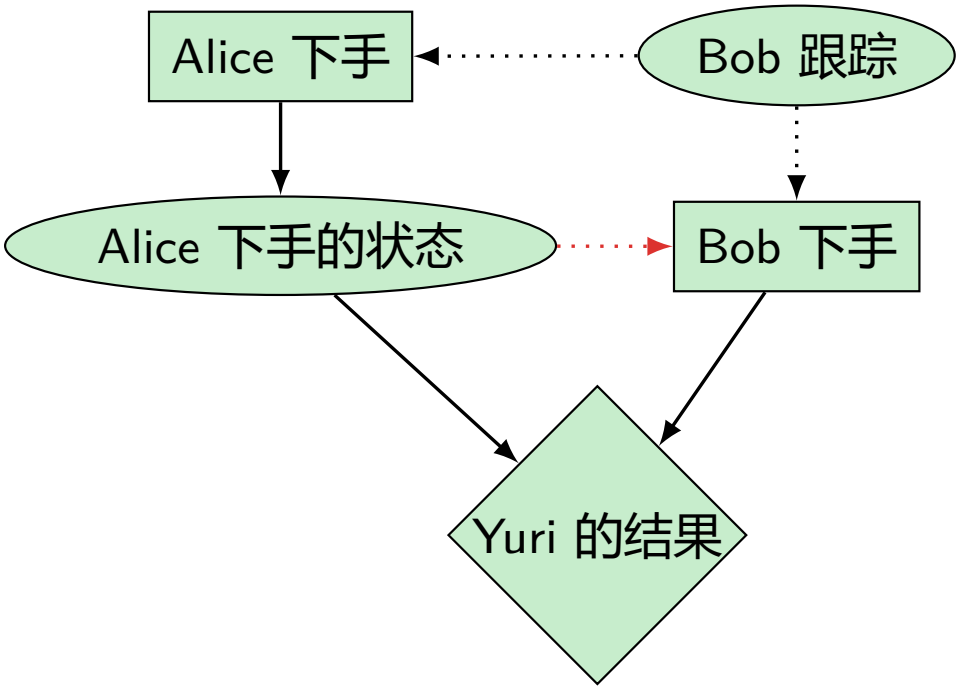
$B = \neg A$
$Y = A \vee B$
$A = 1, B = 0, Y = 1$

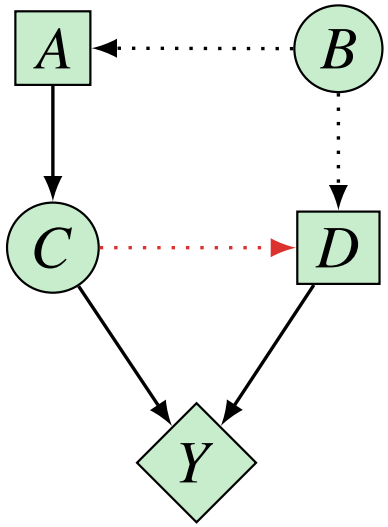
Example2 卧底 Alice 如果不杀死 Yuri 同志, 敌人 Bob 必会虐杀 Yuri.

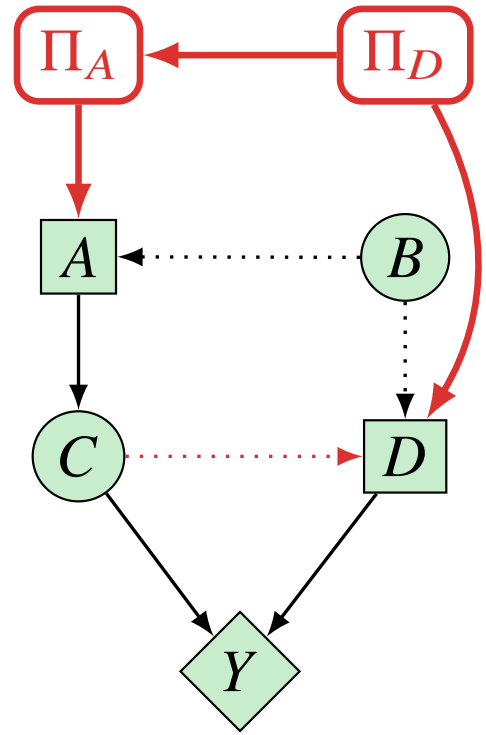


$B = \neg A$
$Y = A \vee B$
$A = 1, B = 0, Y = 1$

1. Yuri 有 2 美元. 跟踪 Yuri 的窃贼 Alice 发现了另一个跟踪 Yuri 的窃贼 Bob. Alice 果断下手, 抢在 Bob 之前偷了 Yuri 1 美元. Bob 本打算偷窃 Yuri 2 美元, 但因为被 Alice 捷足先登, 只能作罢.
2. 卧底 Alice 如果不杀死 Yuri 同志, 敌人 Bob 必会虐杀 Yuri.







$$A = B$$
$$C = A$$
$$D = B \wedge \neg C$$
$$Y = C + 2(1 - C)(1 - D)$$

$$A = 1, B = 1, C = 1, D = 0, Y = 1$$

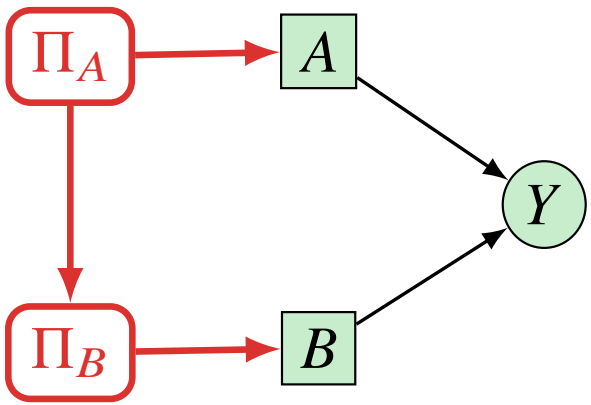
同 (机制) 因 (果图) 不同责?

Discussion: Overdetermination vs Bogus Prevention

行不行赏问题: 同 (实际) 因不同赏?

Example1 A prisoner is shot by two soldiers $A = B = 1$.

(Counter-)Example2 The assassin refrains from poisoning the potential victim's coffee $\overline{A} = 1$. But the bodyguard puts an antidote into the coffee anyway $B = 1$.



$Y = A \vee B$
$A = 1, B = 1, Y = 1$

$\cong$

$Y = \overline{A} \vee B$
$\overline{A} = 1, B = 1, Y = 1$

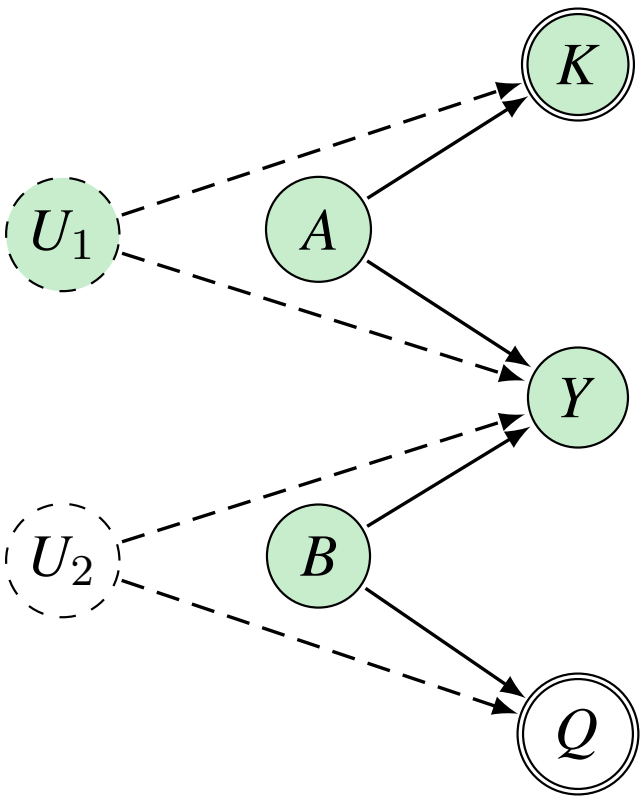
- $B = 1$ is a Halpern-Pearl actual cause of $Y = 1$ in both Examples.
- $B = 1$ is not a Halpern actual cause of $Y = 1$ in both Examples.

Discussion: W 的 “锚定效应”

根据语境的不同, W 的选择可能有锚定效应

- 在 Overdetermination 的例子里, A, B 是对称的, 作为彼此的 “竞争原因”, 没有锚定效应
- 在 Bogus Prevention 的例子里, “证据” 锚定到了刺客没下毒的情形
- 在 Overlapping 的例子里, “证据” 锚定到了王后没有变青蛙的情形

Example — Overlapping



- Alice casts a spell with a 0.5 chance of turning the King and Yuri into frogs;
- Bob casts a spell with an independent 0.5 chance of turning the Queen and Yuri into frogs.

$K = A \wedge U_1$
$Q = B \wedge U_2$
$Y = (A \wedge U_1) \vee (B \wedge U_2)$
$P(U_1 = 1) = 0.5$
$P(U_2 = 1) = 0.5$
$P(K = 1 \mid A = 1) = P(Y = 1 \mid A = 1, B = 0) = P(U_1)$
$P(Q = 1 \mid B = 1) = P(Y = 1 \mid B = 1, A = 0) = P(U_2)$
$A = 1, B = 1, K = 1, Q = 0, Y = 1$

- $B = 1$ is not a PAC cause of $Y = 1$.
  Proof:

$$P(Y_{B=1} = 1 \mid Q = 0) \;=\; P(Y_{B=0} = 1 \mid Q = 0)$$

Attributing Responsibility

- Judging actual cause.
- How do we assign causality across multiple potential causes?
- How to assign blame or credit?
- How do we factor in intentions, beliefs, foresight etc?

Halpern's Definition:

- The degree of responsibility $\mathrm{Dr}((M,u), X=x, Y=y) := 0$ if $X=x$ is not part of a cause of $Y=y$. $\mathrm{Dr}((M,u), X=x, Y=y) := \frac{1}{k}$ if there exists a cause $\mathbf{X}=\mathbf{x}$ of $Y=y$ and a witness $(\mathbf{W}, \mathbf{w}, \mathbf{x}')$ being a cause of $Y=y$ in (M,u) such that: $X=x$ is a conjunct of $\mathbf{X}=\mathbf{x}$, and $|\mathbf{X}| + |\mathbf{W}| = k$, and k is minimal.
- The degree of blame of $X=x$ for $Y=y$ relative to epistemic state $(\mathcal{K}, P)$ is

$$\mathrm{Db}(\mathcal{K}, P, X=x, Y=y) := \sum_{(M,u)\in\mathcal{K}} \mathrm{Dr}((M,u), X=x, Y=y) P((M,u))$$

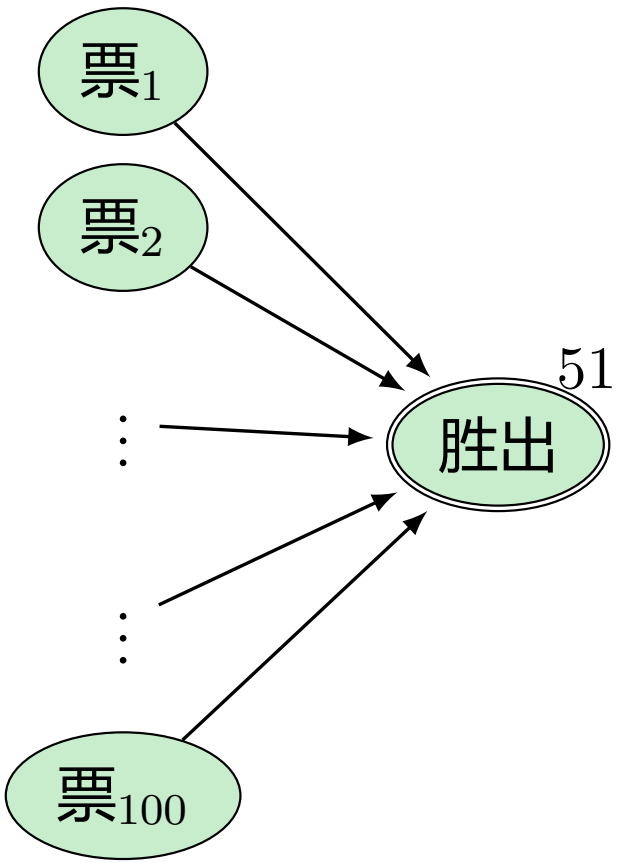
Remark: 三种认知状态: 行动之前实际的认知状态; 行动之前应该有的认知状态; 行动结果之后的认知状态.

Responsibility: a quantitative measure of causality

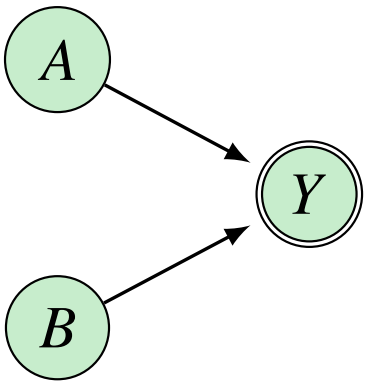
Voting Example

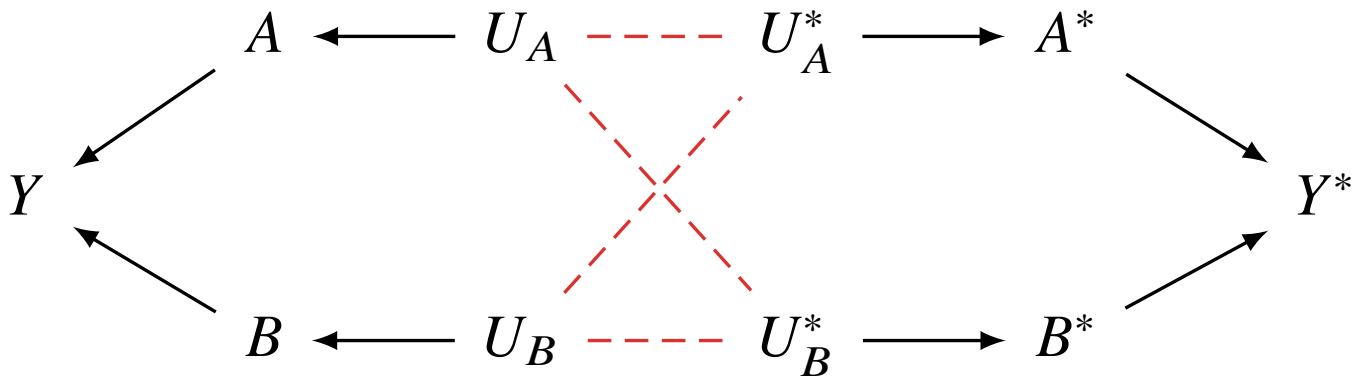
Alice 与 Bob 竞选. 根据 Halpern 的责任判定标准,

- $51:49$ 胜出. Each voter for Alice is 1-responsible for her win.
- $90:10$ 胜出. Each voter for Alice is $1/40$-responsible for her win.



道德运气下的责任判定 — ToDo





- Alice 在枪里装了一发子弹, 以为别人不会开枪, 就没有取出来. Bob 以为是空枪, 朝着 Yuri 开了一枪. Yuri 死亡.
- 根据 Halpern, Alice 的责任是 $\frac{1}{2}$. 过失则依赖认知状态的选择.
- 因为 Alice 以为 (预料) 别人不会开枪, 所以
  $P(U_B^* = 1 \mid U_A = 1, U_B = 1) = c_B \ll 1$, 类似的,
  $P(U_A^* = 1 \mid U_A = 1, U_B = 1) = c_A \ll 1$. 所以,

$$P_B(Y^*_{A^*=1} = 1 \mid A = 1, B = 1) - P_B(Y^*_{A^*=0} = 1 \mid A = 1, B = 1) = c_B$$

$$P_B(Y^*_{B^*=1} = 1 \mid A = 1, B = 1) - P_B(Y^*_{B^*=0} = 1 \mid A = 1, B = 1) = c_A$$

- Alice 的过失责任应该是 c_B; Bob 的过失责任应该是 c_A.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

The Sure-Thing Puzzle

- 若民主党败, 我就买房. 若共和党败, 我也买房. 或者民主党败, 或者共和党败. 总之, 我要买房?

$$\dfrac{D' \vee R' \qquad \begin{matrix}[D'] \\ \vdots \\ B\end{matrix} \qquad \begin{matrix}[R'] \\ \vdots \\ B\end{matrix}}{B} \quad ?$$

- 设想三派竞选, 胜率: 民主党 $\frac{2}{7}$, 共和党 $\frac{2}{7}$, 独立党 $\frac{3}{7}$.
- 我买房当且仅当独立党胜率 $> \frac{1}{2}$.
- 若民主党败, $P(I \mid D') = \frac{3}{5} > \frac{1}{2}$, 买房!
- 若共和党败, $P(I \mid R') = \frac{3}{5} > \frac{1}{2}$, 买房!
- 但 $P(I \mid D' \vee R') = P(I) = \frac{3}{7} < \frac{1}{2}$, 不买!
- D' 和 R' 所交非空.

$$\dfrac{\begin{matrix}P(E \mid S) = r \\ P(E \mid S') = r \\ S \cap S' = \varnothing\end{matrix}}{P(E \mid S \cup S') = r}$$

亚群需构成 Partition

概率确凿原则 The Sure-Thing Principle

$$\frac{\begin{array}{c} P(E \mid S) = r \\ P(E \mid S') = r \\ S \cap S' = \varnothing \end{array}}{P(E \mid S \cup S') = r}$$

亚群需构成 Partition

Proof.

$$\begin{aligned} P(E \mid S \cup S') &= P(E \mid S)P(S \mid S \cup S') + P(E \mid S')P(S' \mid S \cup S') \\ &= r\left[P(S \mid S \cup S') + P(S' \mid S \cup S')\right] \qquad (S \cap S' = \varnothing) \\ &= r \end{aligned}$$

Theorem (因果确凿原则 Sure-Thing Principle)

只要行动不会改变亚群的分布，那么，如果行动提升了每一亚群中某事件的概率，它就会提升总体中该事件的概率.

Proof.

行动 a 提升了每一亚群 s 中 e 的概率:

$$P(e \mid s, \mathtt{do}(a)) > P(e \mid s, \mathtt{do}(a'))$$

行动 a 不改变亚群 s 的分布:

$$P(s \mid \mathtt{do}(a)) = P(s \mid \mathtt{do}(a'))$$

因此,

$$\begin{aligned} P(e \mid \mathtt{do}(a)) &= \sum_s P(e \mid s, \mathtt{do}(a)) P(s \mid \mathtt{do}(a)) \\ &> \sum_s P(e \mid s, \mathtt{do}(a')) P(s \mid \mathtt{do}(a')) \\ &= P(e \mid \mathtt{do}(a')) \end{aligned}$$

Remark☺ 张三同学从北大转学去清华，同时提高了两个学校的平均智商.

Theorem (因果决策确凿原则 Sure-Thing Principle[Pea16])

对于任意给定的信号 s, 如果决策者在知道 $S = s$ 时都会偏好 $a \succ a'$, 而且行动不改变信号

$$P(s \mid \mathrm{do}(a)) = P(s \mid \mathrm{do}(a'))$$

则他在不知道任何信号时, 也会偏好 $a \succ a'$.

Remarks:

1. 确凿原则不是逻辑原则.
   — 逻辑原则不需要 A, B 所交非空也成立.

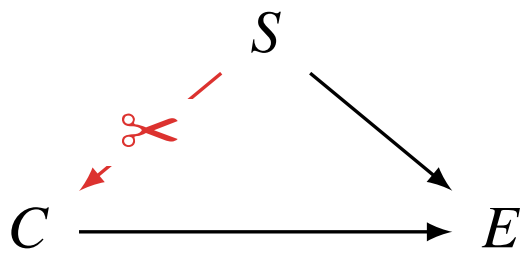
$$\frac{A \vee B \qquad \begin{array}{c}[A] \\ \vdots \\ C\end{array} \qquad \begin{array}{c}[B] \\ \vdots \\ C\end{array}}{C} \qquad\qquad \frac{A \subset C \quad B \subset C}{A \cup B \subset C}$$

2. 确凿原则是因果决策原则 CDT.
   2.1 亚群/信号构成 Partition.
   2.2 行动不改变亚群/信号.
3. 若行动不直接改变信号, 而是间接 "关联", 根据 "关联方式" 的不同, 需要考虑不同的决策理论. 比如 FDT. (纽康姆问题)

辛普森悖论 & 确凿原则

对于任意的辛普森逆转,

$$P(e \mid s, c) > P(e \mid s, c')$$
$$P(e \mid s', c) > P(e \mid s', c')$$
$$P(e \mid c) < P(e \mid c')$$



考虑如下的博弈, 你有两个选择 a, a'.

- a: Draw samples at random until you get one for which c holds, and bet a dollar that e is true.
- a': Draw samples at random until you get one for which c' holds, and bet a dollar that e is true.

Then $a \succ a'$ given s, and $a \succ a'$ given s'. But $a \prec a'$ when not konwing S.
Remark: 在辛普森悖论中 $P(s \mid c) = P(s \mid c')$ 不成立, 翻译到上述博弈中对应 $P(s \mid \mathrm{do}(a)) = P(s \mid \mathrm{do}(a'))$ 不成立.
Remark: 根据确凿原则, 干预 do 下的因果效应不存在辛普森逆转.

$$P(e \mid \mathrm{do}(c)) > P(e \mid \mathrm{do}(c'))$$

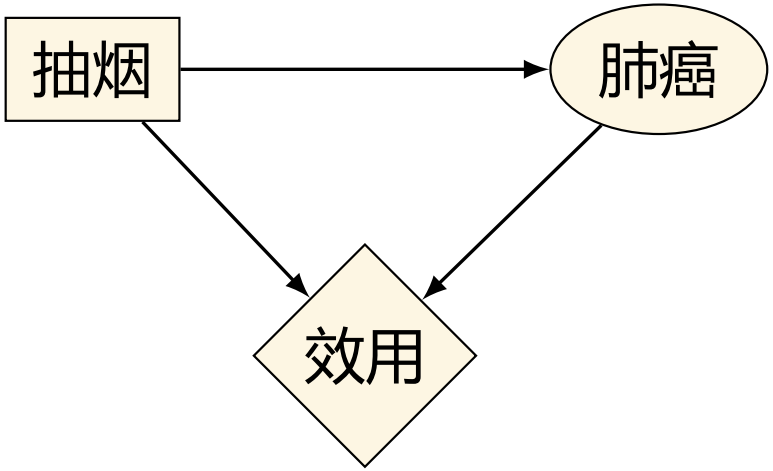
Remarks: 从辛普森悖论到确凿原则反例的转化

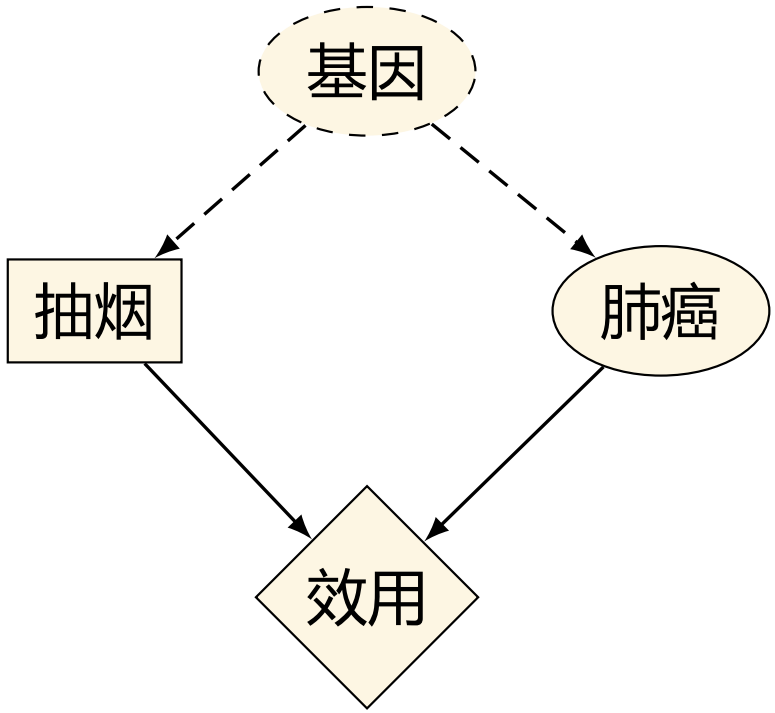
- 博弈道具: 一堆病例, 每份病历记录了患者的三项信息: 采用的是治疗方案一 c 还是二 c', 结石是大 s 还是小 s', 康复 e 与否 e'.
- 选项 a: 你从接受方案一治疗的患者中随机抽取一份病历, 如果此患者康复了, 你赢一块钱.
- 选项 a': 你从接受方案二治疗的患者中随机抽取一份病历, 如果此患者康复了, 你赢一块钱.

现在, 你会怎么决策呢?

- 当你知道抽到的是结石大 (s) 的患者时, 因为方案一的康复率更高, 你会选择 a.
- 当你知道抽到的是结石小 (s') 的患者时, 同样因为方案一的康复率更高, 你仍然会选择 a.
- 当你不知道抽到的患者结石大小时, 你面对的是总体数据, 只知道方案二的康复率更高, 为了赢钱, 你会选择 a'.
- 然而, 根据确凿原则, 既然无论结石大小, 你都会选择 a, 那么, 当你不知道结石大小时, 你也会选择 a. 确凿原则为啥不适用?
- 原因: 你选择 a 还是 a', 决定了你是从方案一组还是方案二组里抽病历, 而这两组里结石大小的比例不同. $P(s \mid \mathrm{do}(a)) \neq P(s \mid \mathrm{do}(a'))$.

抽烟有害健康? EDT vs CDT

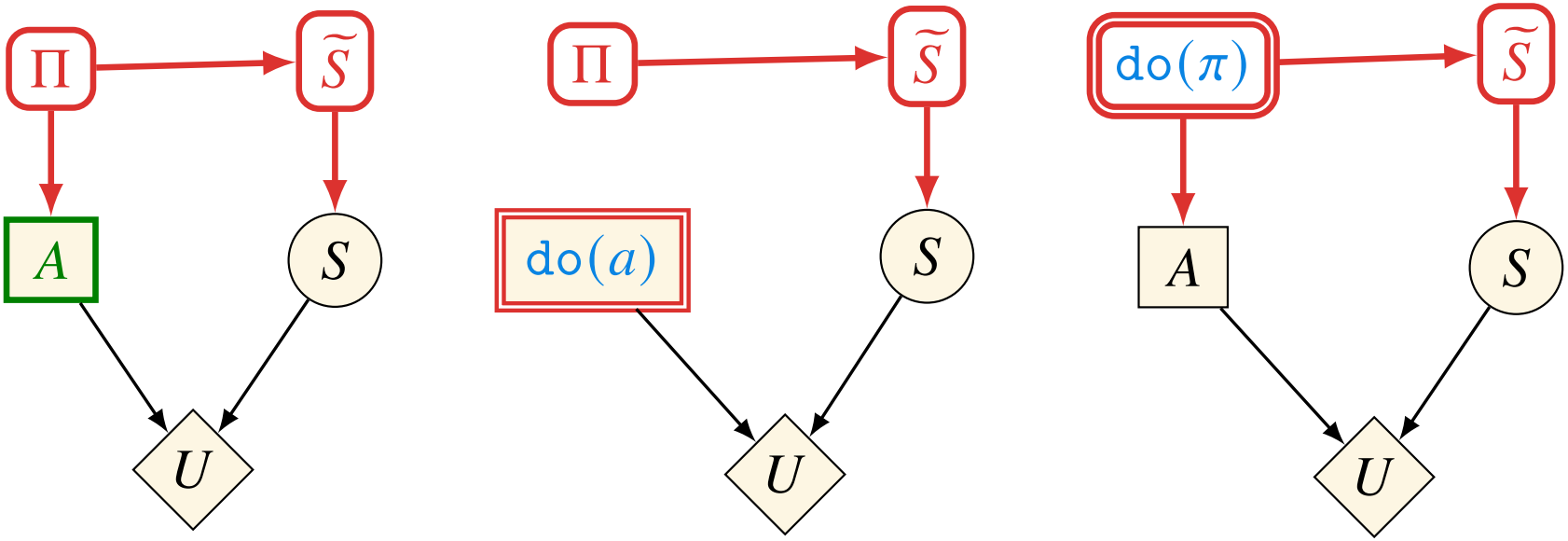




- 得肺癌, 抽烟比不抽开心; 不得肺癌, 抽烟比不抽开心.
- 左图, 确凿原则失效. 不抽!
- 右图, 确凿原则有效. 抽烟!
- 右图, EDT 不抽; CDT 抽烟.

确凿原则 & 纽康姆问题/克隆囚徒困境

EDT vs CDT vs FDT



- 纽康姆问题: EDT: 黑盒; CDT: 两盒; FDT: 黑盒.
- 克隆囚徒困境: EDT: 合作; CDT: 背叛; FDT: 合作.

Remark:

- 康德绝对律令? 依据那些你愿意所有人都遵守的普遍法则行事. (vs 待人如己) — 如果人人都像你一样 XX, 那 YY. 所以, 你不应该 XX.
- 规则功利主义?

EDT vs CDT vs FDT

- EDT

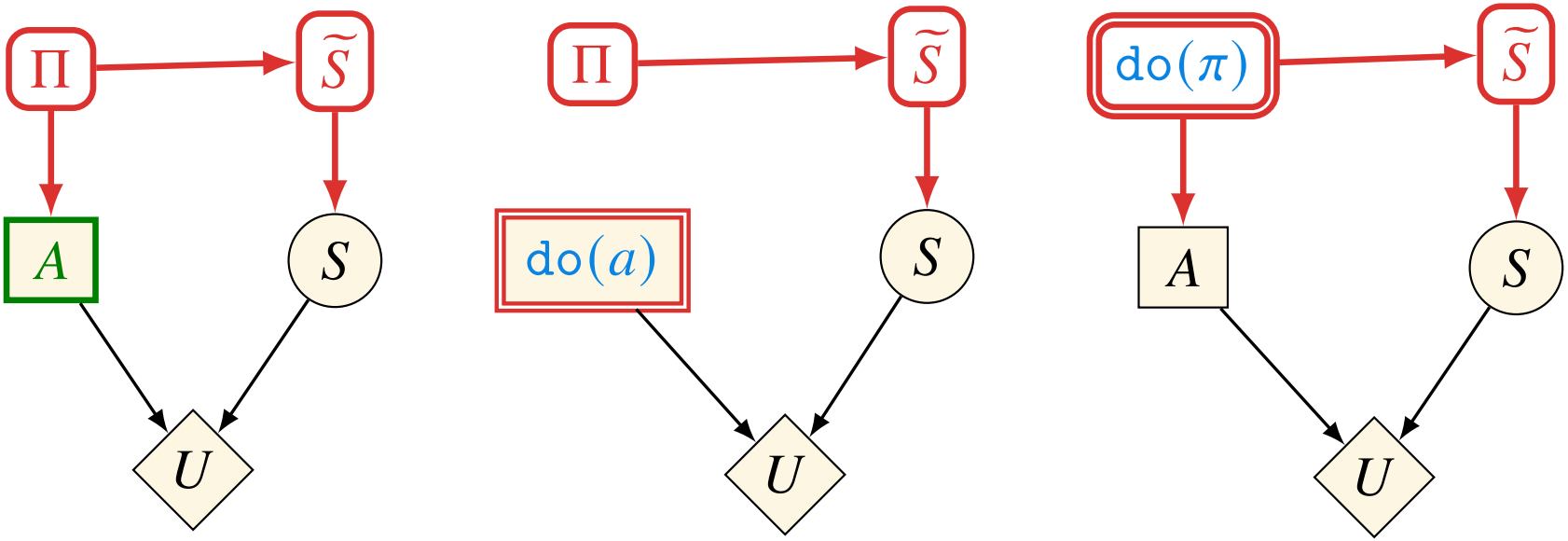
$$a^* = \operatorname*{argmax}_a \mathbb{E}[U \mid A = a]$$

- CDT

$$a^* = \operatorname*{argmax}_a \mathbb{E}[U \mid \mathtt{do}(A = a)]$$

- FDT

$$\pi^* = \operatorname*{argmax}_\pi \mathbb{E}[U \mid \mathtt{do}(\Pi = \pi)]$$



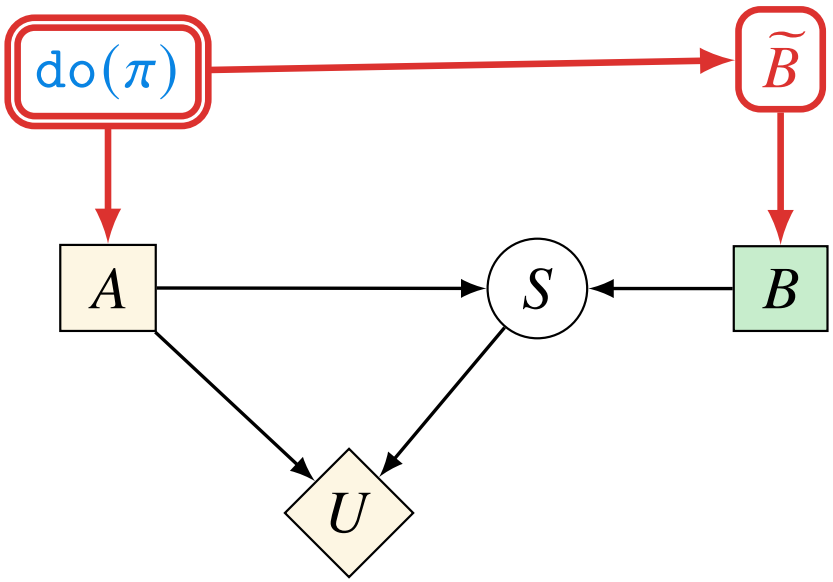
确凿原则对决策框架的依赖性

Theorem (确凿原则 Sure-Thing Principle)

1. *EDT* 确凿原则: 对于任意给定的信号 s, 如果决策者在知道 $S = s$ 时都会偏好 $a \succ a'$, 而且协变量平衡 $P(s \mid a) = P(s \mid a')$, 则他在不知道任何信号时, 也会偏好 $a \succ a'$.
2. *CDT* 确凿原则: 对于任意给定的信号 s, 如果决策者在知道 $S = s$ 时都会偏好 $a \succ a'$, 而且行动不改变信号 $P(s \mid \mathrm{do}(a)) = P(s \mid \mathrm{do}(a'))$, 则他在不知道任何信号时, 也会偏好 $a \succ a'$.
3. *FDT* 确凿原则: 对于任意给定的信号 s, 如果决策者在知道 $S = s$ 时都会偏好 $\pi \succ \pi'$, 而且决策机制不改变信号 $P(s \mid \mathrm{do}(\pi)) = P(s \mid \mathrm{do}(\pi'))$, 则他在不知道任何信号时, 也会偏好 $\pi \succ \pi'$.

Remark: 不同决策框架下有不同的确凿原则, 但不存在超脱决策框架的确凿原则.

FDT 确凿原则



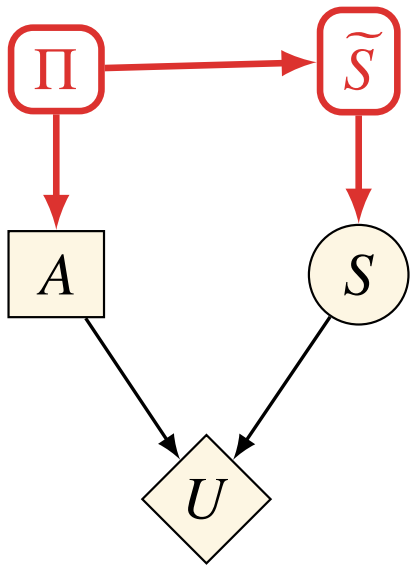
- 父辈看儿女不想读书就赠予财富，看儿女想读书就不赠予财富，但儿女读了书可以自己创造财富.
- A: 读书. B: 赠予. $S = A \vee B$. U: 求知欲满足 + 财富.
- CDT 确凿原则的条件不满足. $P(s \mid \mathrm{do}(a)) \neq P(s \mid \mathrm{do}(a'))$.
- FDT 确凿原则的条件满足. $P(s \mid \mathrm{do}(\pi)) = P(s \mid \mathrm{do}(\pi'))$.

Remarks

- 确凿原则不是逻辑原则.
- Pearl 的确凿原则是 CDT 因果决策原则.
- 若行动改变信号, 确凿原则失效.
- 若行动不改变信号, CDT 确凿原则成立.
- 但当 CDT 失效时, 确凿原则也毫无用处!
- 若行动不直接改变信号, 而是间接 "关联", 根据 "关联方式" 的不同, 需要考虑不同的决策理论. 比如 FDT.
- 重要的是行动对信号的 "关联方式"!
- 虽然 $P(s \mid \mathrm{do}(a)) = P(s \mid \mathrm{do}(a'))$, 但 $P(s \mid a) \neq P(s \mid a')$ 不一定就意味着仅仅是虚假相关. 在直接改变与虚假相关之间, 可能还存在着其它关联方式.
- EDT 误信虚假相关; CDT 误杀逻辑关联; FDT 根据决策机制的关联更新信念.
- 确凿原则不是独立于决策理论的认知原则, 而是依赖于决策框架.

行动 vs 事件

- 事件之间的分布关系通过概率刻画.
- 行动代表了能够扰动这些关系的干预措施.
- 原则上, 行动不是概率的一部分.
- 但对于 EDT 来说, 行动不具有特殊地位, 跟事件没有区别.
- 对于 CDT 来说, 行动非常特殊, 决策者有绝对的自由意志, 只更新行动的后果, 而无视行动的理由. CDT 适用于二元论 Agent, 主体独立于环境.
- 对于 FDT 来说, 行动也是特殊的, 它不同于事件, 行动只依赖于决策机制. 决策者拥有自由意志, 但不绝对. 信念的更新要考虑与决策机制关联的变量.



纽康姆问题 — machine simulated consciousness

- 白盒: 透明, 里面有一千块钱.
- 黑盒: 不透明, 可能有一百万, 也可能什么都没有.
- 你可以选择只拿黑盒, 也可以两个都拿.
- 女巫预测到你只拿黑盒, 就会在里面放一百万; 如果预测你两个都拿, 就会让黑盒空着.
- 女巫从来没有出过错.

		predicted choice	
		both	black
your choice	black	0	100
	both	0.1	100.1

确凿/占优原则 Dominance Principle?

- 如果黑盒有钱, 只拿黑盒得一百万, 两个都拿得一百万零一千. 两个都拿.
- 如果黑盒没钱, 只拿黑盒得 0 元, 两个都拿得一千. 两个都拿.
- 黑盒或者有钱或者没钱.
- 两个都拿.

$$\frac{A \vee \neg A \qquad \begin{array}{c}[A] \\ \vdots \\ B\end{array} \qquad \begin{array}{c}[\neg A] \\ \vdots \\ B\end{array}}{B} \quad ?$$

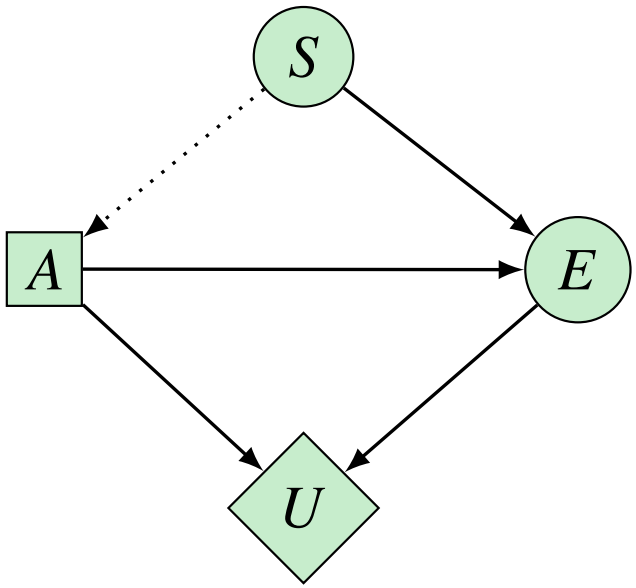
Evidential Decision Theory & Causal Decision Theory

- Evidential Expected Utility

$$V_{\text{evidential}}(A = a) = \sum_e P(e \mid A = a) u(a, e)$$

- Causal Expected Utility

$$V_{\text{causal}}(A = a) = \sum_e P(e \mid \mathtt{do}(A = a)) u(a, e)$$



$$P(e \mid a) = \sum_s P(e \mid s, a) P(s \mid a)$$

$$P(e \mid \mathtt{do}(a)) = \sum_s P(e \mid s, a) P(s)$$

Evidential Expected Utility vs Causal Expected Utility

$$P(\text{predict-both} \mid \text{both}) = 1 \qquad P(\text{predict-both} \mid \texttt{do}(\text{both})) = P(\text{predict-both})$$
$$P(\text{predict-black} \mid \text{black}) = 1 \qquad P(\text{predict-black} \mid \texttt{do}(\text{black})) = P(\text{predict-black})$$
$$P(\text{predict-black} \mid \text{both}) = 0 \qquad P(\text{predict-black} \mid \texttt{do}(\text{both})) = P(\text{predict-black})$$
$$P(\text{predict-both} \mid \text{black}) = 0 \qquad P(\text{predict-both} \mid \texttt{do}(\text{black})) = P(\text{predict-both})$$

$$V_{\text{evidential}}(A = \text{both}) = \sum_e P(e \mid A = \text{both}) u(e) = 0.1$$

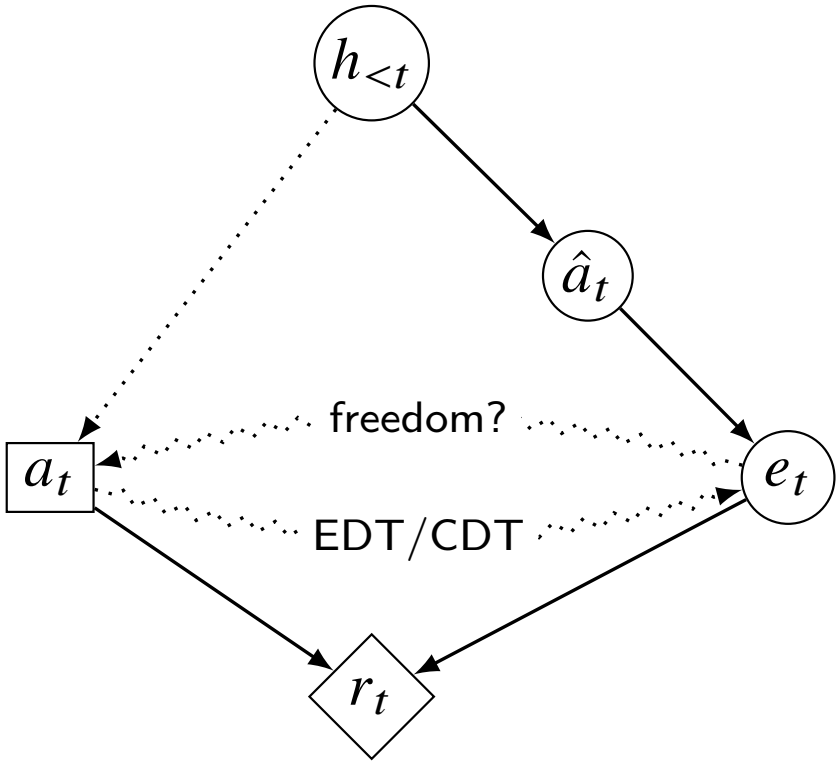
$$V_{\text{evidential}}(A = \text{black}) = \sum_e P(e \mid A = \text{black}) u(e) = 100$$

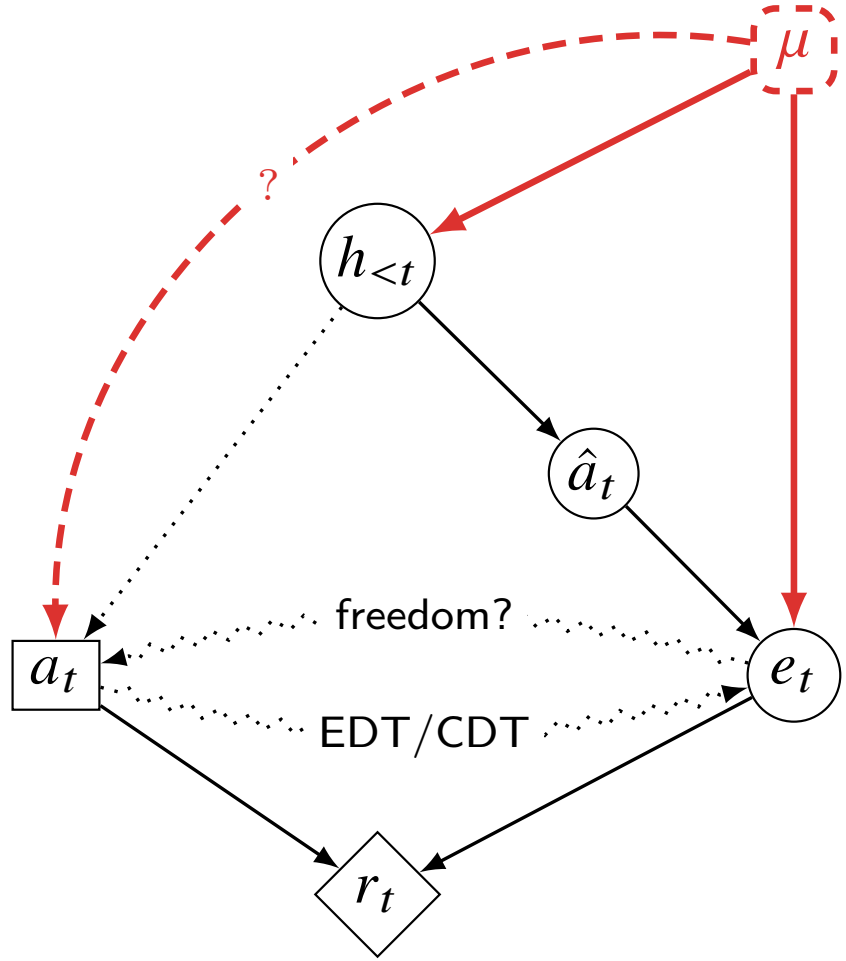
$$V_{\text{causal}}(A = \text{both}) = \sum_e P(e \mid \texttt{do}(A = \text{both})) u(e)$$

$$= P(\text{predict-both}) 0.1 + P(\text{predict-black}) 100.1$$

$$V_{\text{causal}}(A = \text{black}) = \sum_e P(e \mid \texttt{do}(A = \text{black})) u(e)$$

$$= P(\text{predict-both}) 0 + P(\text{predict-black}) 100$$





$$V(h_{<t}) = \sum_{a_t e_t} u(h_{1:t}) P(a_t e_t \mid h_{<t})$$

$$= \sum_{a_t e_t} u(h_{1:t}) P(e_t \mid h_{<t} a_t) P(a_t \mid h_{<t}) \qquad \text{(Evidential/Causal)}$$

$$= \sum_{a_t e_t} u(h_{1:t}) P(a_t \mid h_{<t} e_t) P(e_t \mid h_{<t}) \qquad \text{(Freedom)}$$

Functional Decision Theory[MEB23; YS18]

- Assume you possess an algorithm of your decision mechanisms (the predictor can also run your algorithm). You select your decision mechanism that produces the best outcome.

$$\pi^* = \operatorname*{argmax}_{\pi} \mathbb{E}[U \mid \mathsf{do}(\Pi = \pi)]$$

Example (Newcomb Problem)

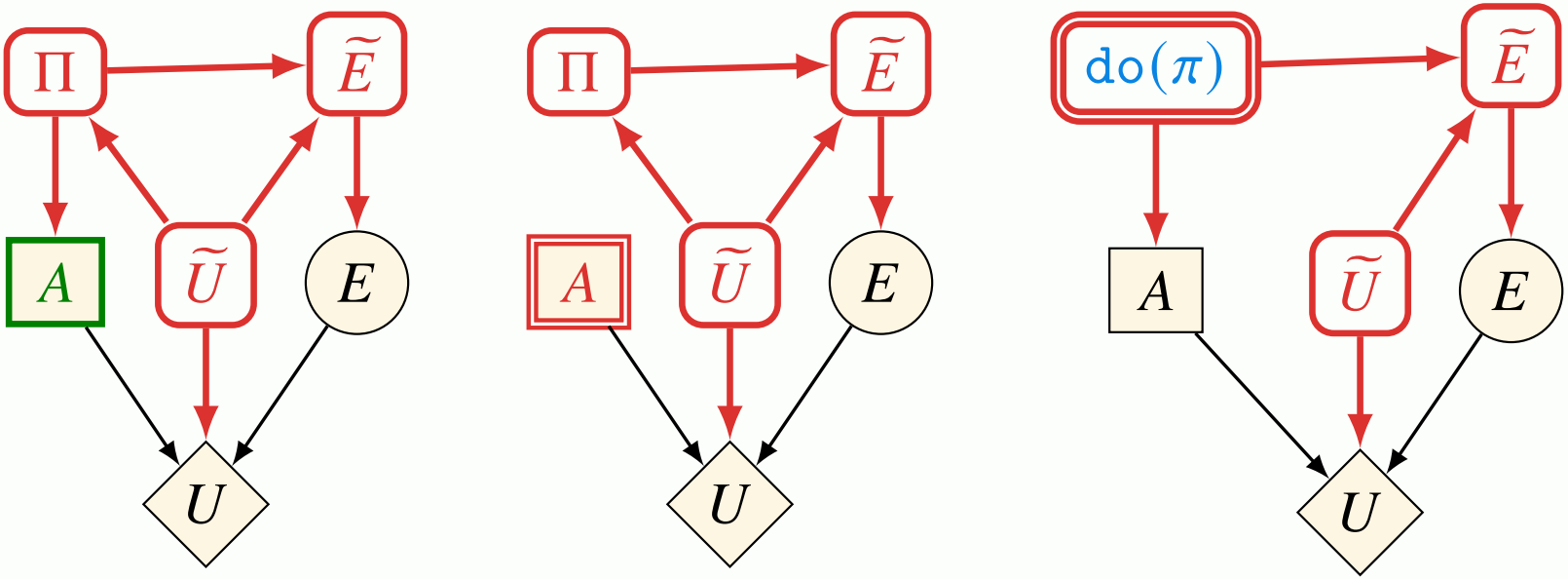
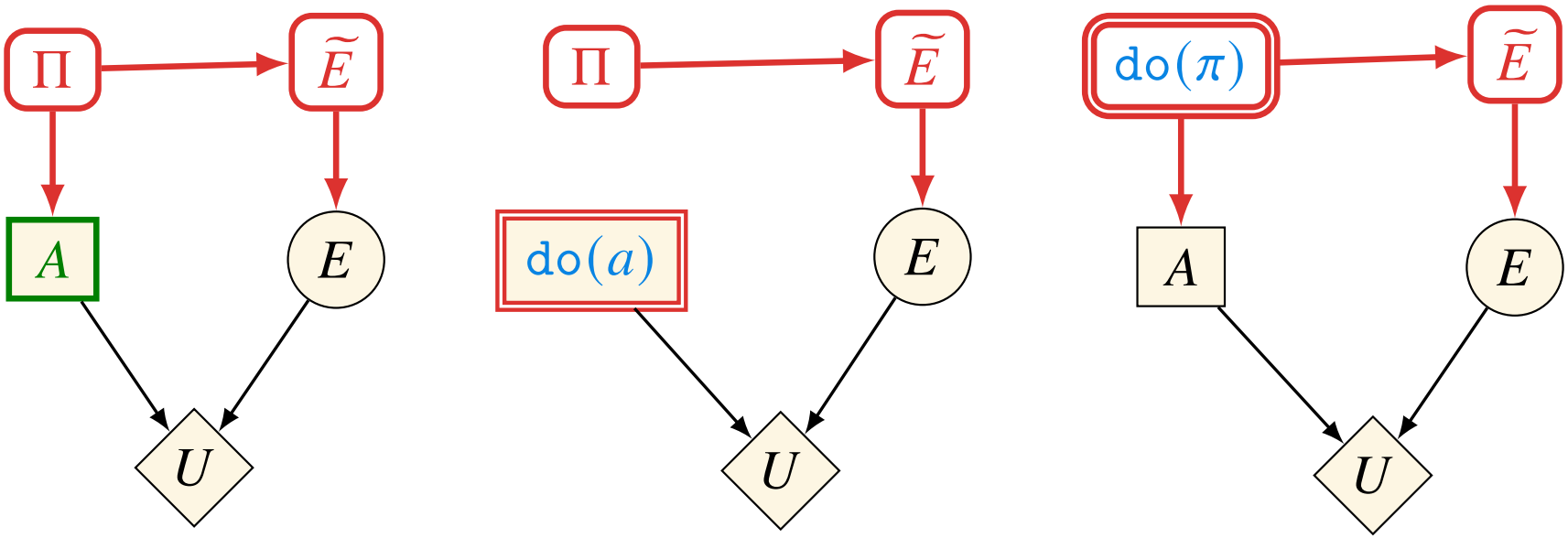


Figure: EDT vs CDT vs FDT

Newcomb Problem

- You stand before two boxes. One is transparent and contains one thousand dollars; the other is opaque and contains either one million or nothing.
- Your choice is between taking two boxes and taking just the opaque box.
- A reliable predictor "Oracle" has put one million in the opaque box iff she predicted you would one-box.

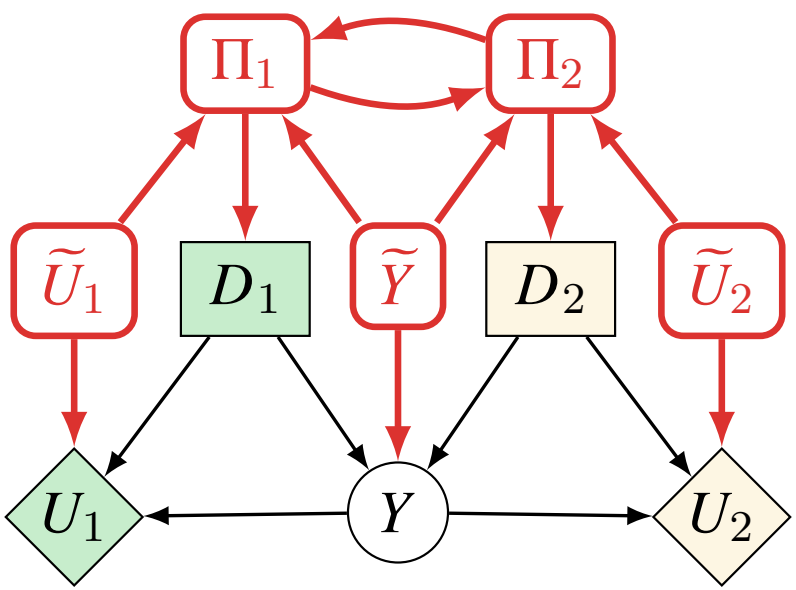


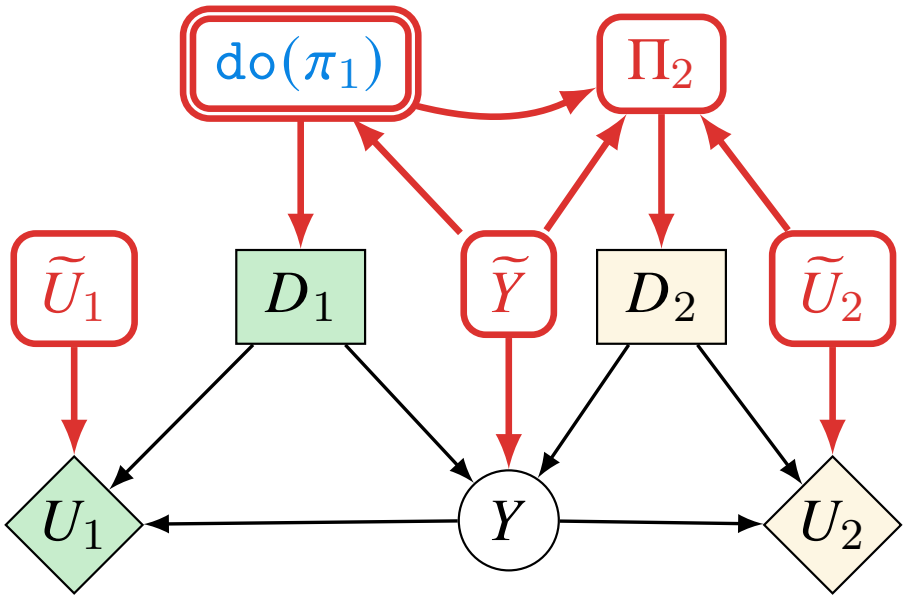
- EDT: one-box
- CDT: two-box
- FDT: one-box

Digression

- To evaluate a pre-policy, we first allow other agents to learn their best response policies to all possible pre-policies.

$$P(Y = y \mid \mathrm{do}(\pi_i)) = \sum_{\pi_{-i}} P(Y = y \mid \pi_i, \pi_{-i}) P(\pi_{-i} \mid \mathrm{do}(\pi_i))$$



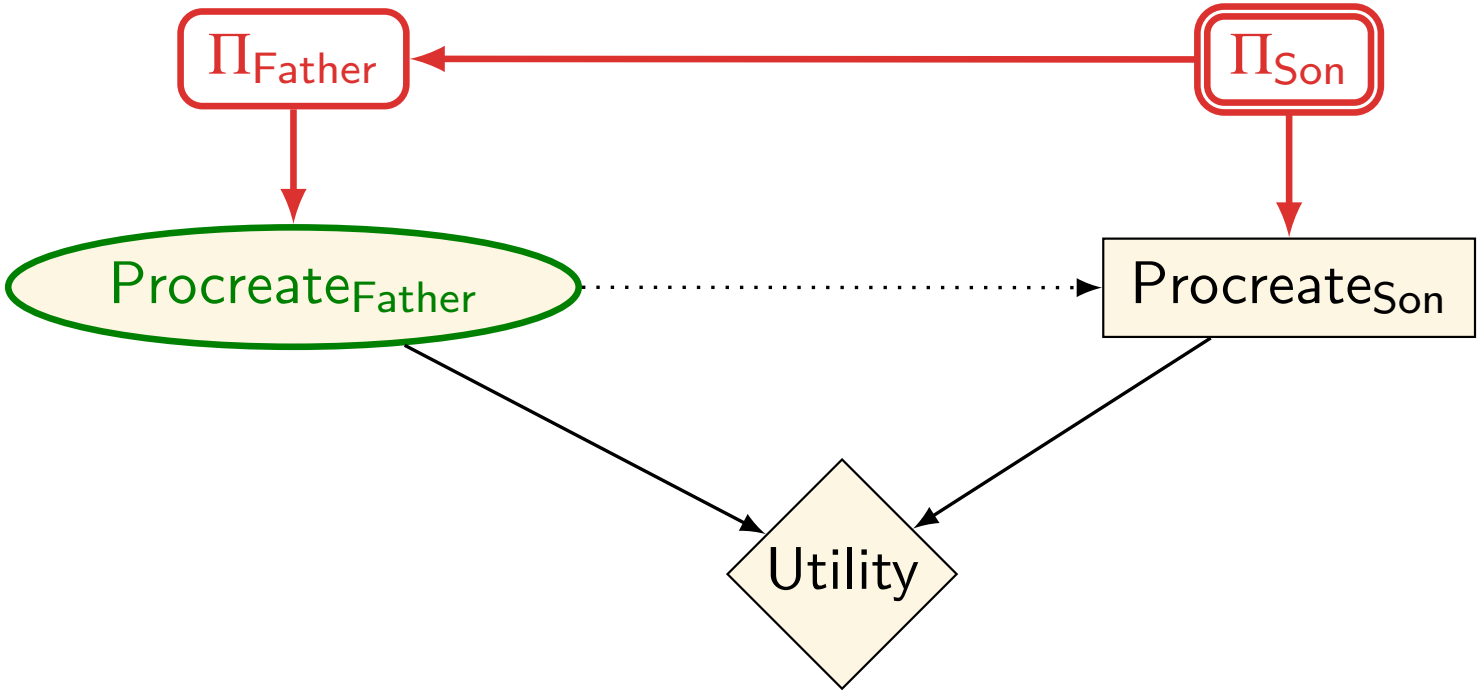


$$\Pi_i(\pi_{-i})(o) = \left(\operatorname*{argmax}_{\pi_i} \sum_y P(y \mid \mathrm{do}(\Pi_i = \pi_i)) u_i(y) \right) (\pi_{-i})(o)$$

- 假如 Newcomb 问题里的 Oracle 是超级人工智能呢?
- Libet 实验: 我们的决策可被提前预测到.
- 有限理性 vs 无限理性

生孩子

- F: 你想生孩子吗?
- C: 不想, 生孩子会让生活变得艰难. $\mathrm{do}(A = \text{procreate})$
- F: 哪怕艰难活着, 总比不存在好吧, 如果父母也想着不生, 我们哪有存在的机会? $\mathrm{do}(\Pi = \pi_{\text{not-procreate}})$
- C: 但事实是, 我们已经存在了啊. $P(\bullet \mid \mathrm{do}(\bullet), O = \text{existing})$



- CDT: not procreate
- FDT: procreate　　Updateful-FDT: not procreate

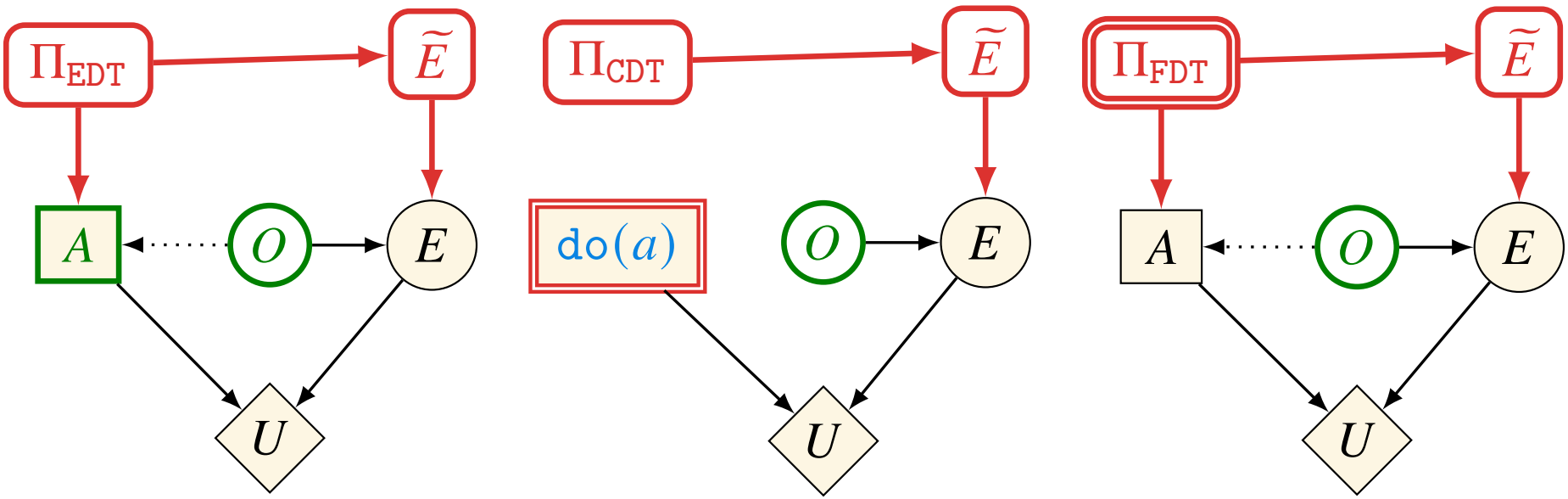
EDT vs CDT vs FDT (Updateless / Updateful)

$$\Pi_{\texttt{EDT}}(o) = \operatorname*{argmax}_a \mathbb{E}[U \mid A = a, O = o]$$

$$\Pi_{\texttt{CDT}}(o) = \operatorname*{argmax}_a \mathbb{E}[U \mid \texttt{do}(A = a), O = o]$$

$$\Pi_{\texttt{FDT}} = \operatorname*{argmax}_\pi \mathbb{E}\left[U \,\middle|\, \texttt{do}\left(\Pi_{\texttt{FDT}} = \pi\right)\right]$$

$$\Pi^{\texttt{Update}}_{\texttt{FDT}} = \operatorname*{argmax}_\pi \mathbb{E}\left[U \,\middle|\, \texttt{do}\left(\Pi^{\texttt{Update}}_{\texttt{FDT}} = \pi\right), O = o\right]$$



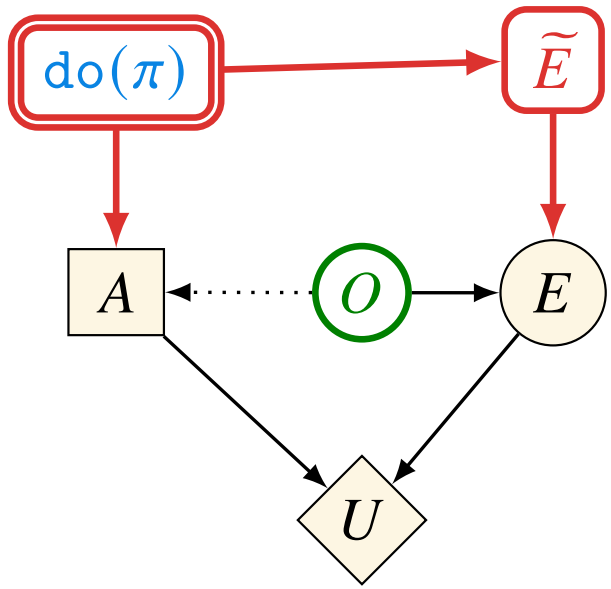
Remarks: Updateful-FDT 合理吗?

$$\Pi = \operatorname*{argmax}_{\pi} \mathbb{E}\left[U \mid \mathtt{do}(\Pi = \pi), O = o\right]$$

$$\Pi(o) = \left(\operatorname*{argmax}_{\pi} \mathbb{E}\left[U \mid \mathtt{do}(\Pi = \pi), O = o\right]\right)(o)$$

$$= \left(\operatorname*{argmax}_{\pi} \mathbb{E}\left[U \mid \mathtt{do}\left(\Pi(o) = \underbrace{\pi(o)}_{a}\right), O = o\right]\right)(o)$$

$$= \operatorname*{argmax}_{a} \mathbb{E}\left[U \mid \mathtt{do}\left(\Pi(o) = a\right), O = o\right]$$



Goertzel's Counterfactual Reprogramming Decision Theory (CRDT)

- Assume that the agent's brain is partially reprogrammable, but also has certain immutable properties.
- Imagine a Master Programmer (MP), able to replace the reprogrammable portion π_0 of the agent's brain with an arbitrary computer program π of length $< l$ and runtime $< t$.
- The goal of the MP is to replace the reprogrammable portion of the agent's brain with a program π^* having the property that, averaged over all possible worlds that are consistent with the agent's current world-knowledge (using Machanised Causal Graph), operating π^* will cause the agent to get maximal utility.

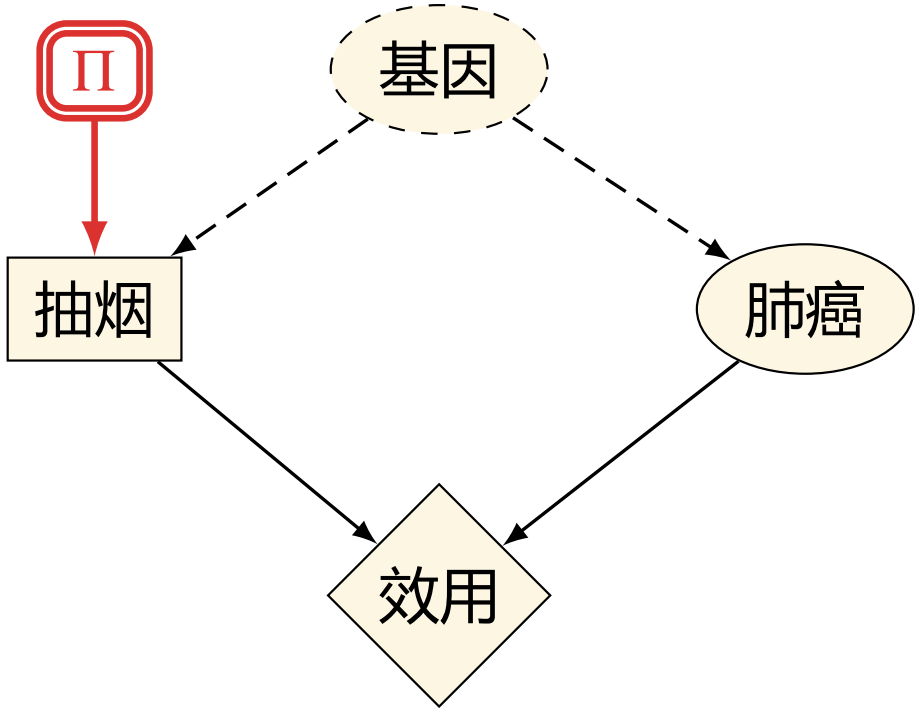
$$\mathrm{MP} : \pi_0 \mapsto \pi^* := \operatorname*{argmax}_{\pi} \mathbb{E}[U \mid \mathrm{do}(\pi)]$$

- Imagine that the MP replaces the reprogrammable portion of the agent's brain with a new program π^* right now.
- Figure out what action π^* would take, and then take that action.

$$\varphi_{\pi^*} = \varphi_{\mathrm{MP}(\pi^*)}$$

抽烟有害健康? EDT vs CDT vs FDT

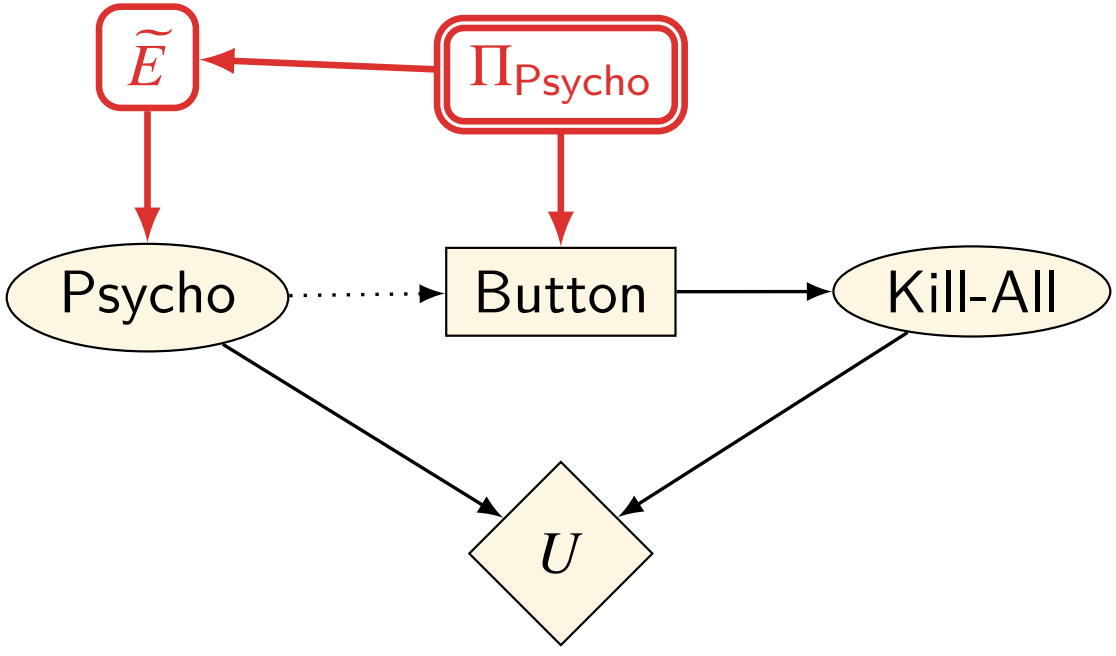
- 假设抽烟不影响肺癌, 只揭示了你是某种基因的携带者.



- EDT: 不抽
- CDT: 抽烟
- FDT: 抽烟
- 如果 Agent 的决策机制除了自己的行为不会影响任何其它变量, 那么 FDT = CDT.
- 如果多个变量依赖于决策机制, FDT 将更新所有这些变量的值.

精神病按钮 The Psychopath Button

- 有一个按钮，按下它，你可以杀死“所有的精神病”.
- 你想生活在一个没有精神病的世界里.
- 但只有精神病才会按下这样的按钮.
- 你宁愿和精神病一起生活也不想死.
- 你相信自己不太可能是精神病，你会按下按钮吗?

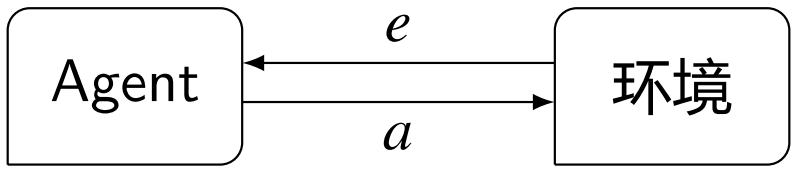


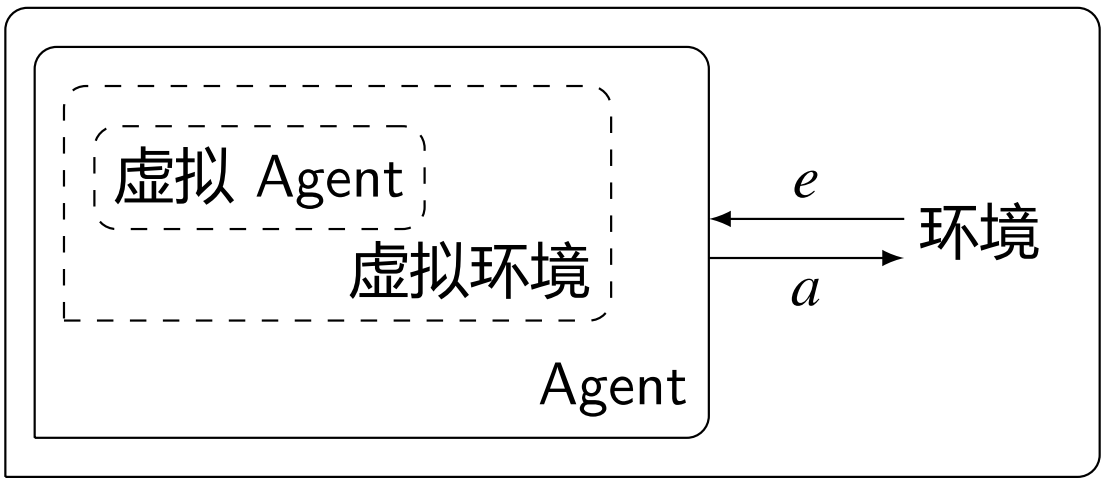
- EDT: not press　　你的行为揭示了某些隐藏信息
- CDT: press　你控制的行为，不应该影响你对不受该行为影响的事物的信念
- FDT: not press　　你的行为揭示了你的人格

EDT vs CDT vs FDT

- 在 EDT 下, 行动并不特殊, 依赖于其他状态变量.
  — EDT 会更新与其行动相关的所有变量的信念, 即使仅仅是统计相关.
- 在 CDT 下, 行动非常特殊, 不依赖于任何其他状态变量.
  — CDT 只更新行动的后果变量的信念 (而无视行动的理由). 例如, 在纽康姆问题中, 即使明知 Oracle 预测非常准确, 行动也被认为与预测毫不相关.
  — CDT 适用于二元论 Agent, 主体独立于环境.
- 在 FDT 下, 行动也比较特殊, 只依赖于决策机制.
  — FDT 会更新受自己决策机制影响的变量的信念.
- EDT 只使用概率计算, 包含了过多的伪相关;
  CDT 只注重因果关系, 忽视了逻辑关联;
  FDT 既不会考虑过多的伪相关, 也不会忽视逻辑关联.

二元论 vs 物理主义 Agent

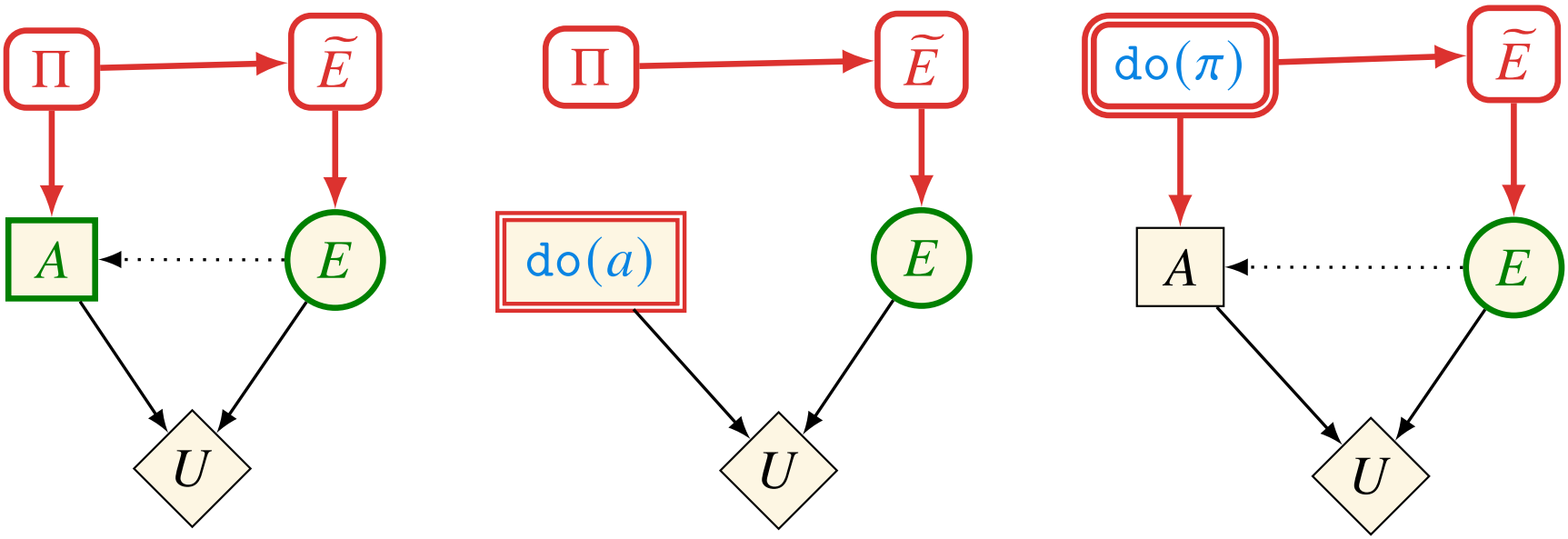




Parfit's Hitchhiker (Transparent Newcomb)

Problem (Parfit's Hitchhiker)

You are trapped in the desert. "Oracle" drives by and says she will drive you to the town, saving your life, but only if she predict you'll pay her 1000 *when you're there. Do you pay "Oracle"* 1000 *once you're in town?*



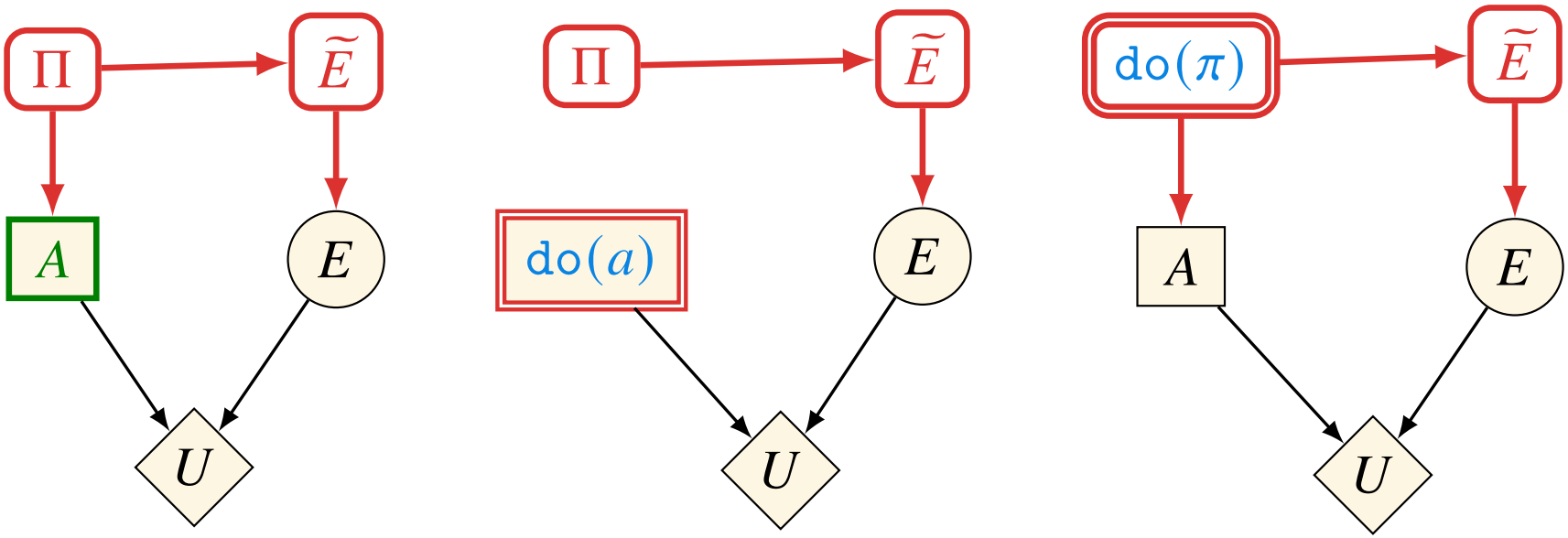
- EDT: not pay / tow-box
- CDT: not pay / tow-box
- FDT: pay / one-box Updateful-FDT: not pay / two-box

Remark: why updateless? the veil of ignorance?

Remark: what about self-modifying agents?

Twin Prisoner's Dilemma

- Assume you and your clone are arrested ……

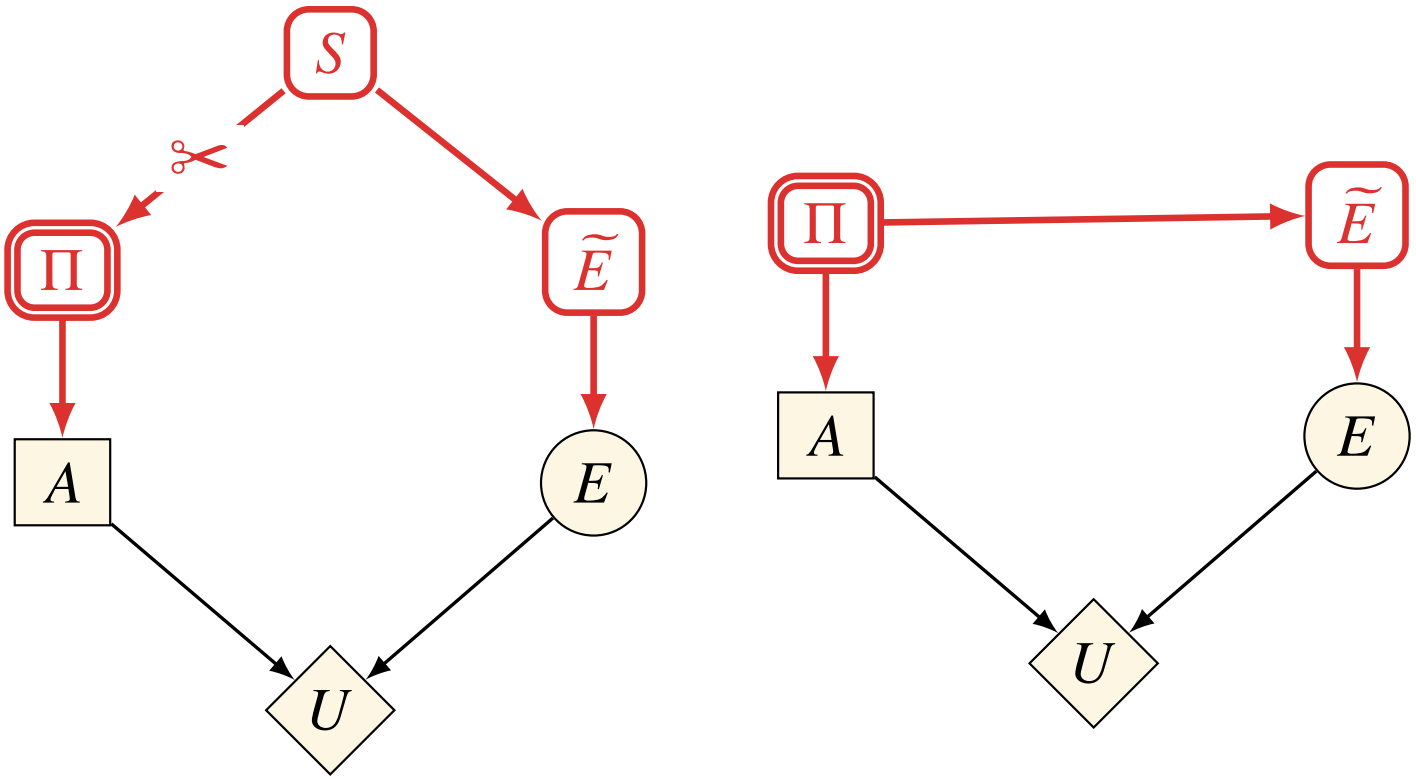


- EDT: co-operate
- CDT: defect
- FDT: co-operate Updateful-FDT: co-operate

Remark:

- 康德绝对律令? 依据那些你愿意所有人都遵守的普遍法则行事. (vs 待人如己) — 如果人人都像你一样 XX, 那 YY. 所以, 你不应该 XX.
- 规则功利主义?

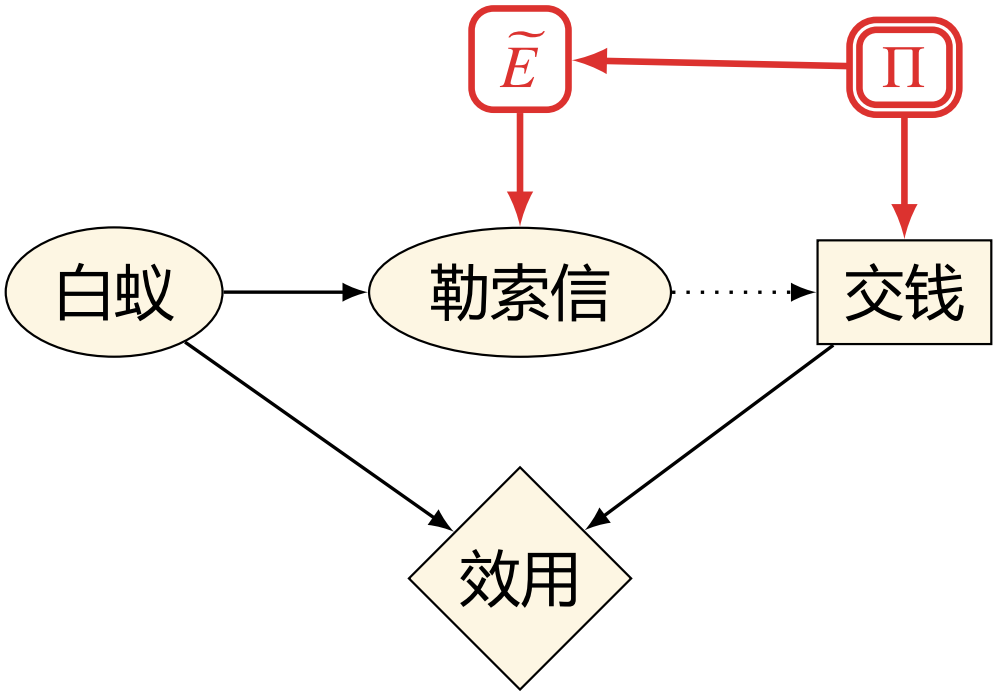
Physical Variable vs Logical Variable



- Whether you imagine you are controlling a physical variable or a logical variable (the output of an algorithm).
- In other words, whether you are choosing for you, right here right now, or whether you are choosing for agents like you in situations like this.

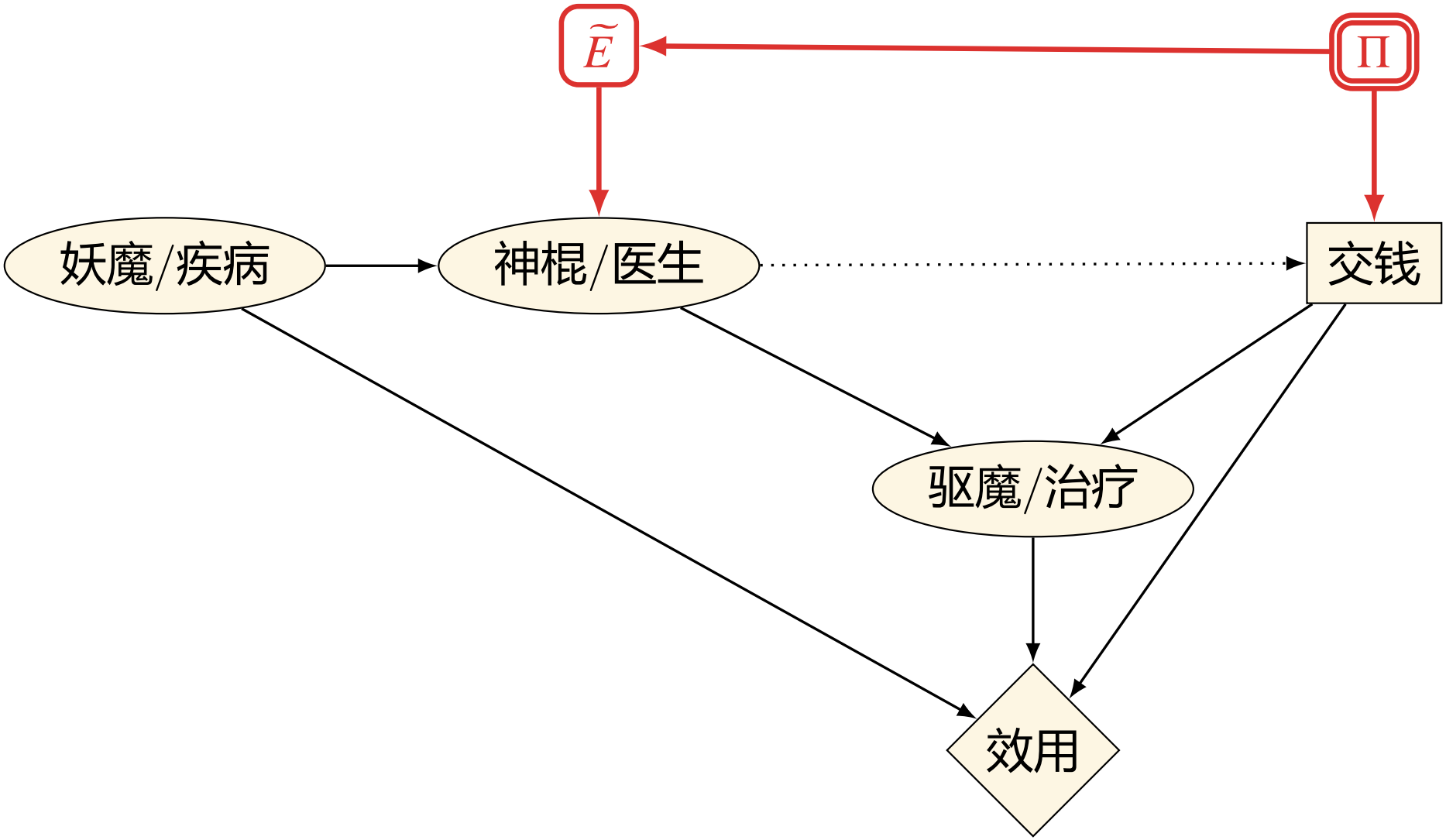
XOR 勒索

- 你家别墅可能有蚁患, 若有, 维修成本一百万.
- 预言家 Oracle 给你寄来了一封信:
  — 我知道你家是否有蚁患. 我给你寄这封信当且仅当下面其中一种情况成立: (1) 你家有蚁患; (2) 你寄给我一千块钱.



- EDT: pay
- CDT: not pay
- FDT: not pay

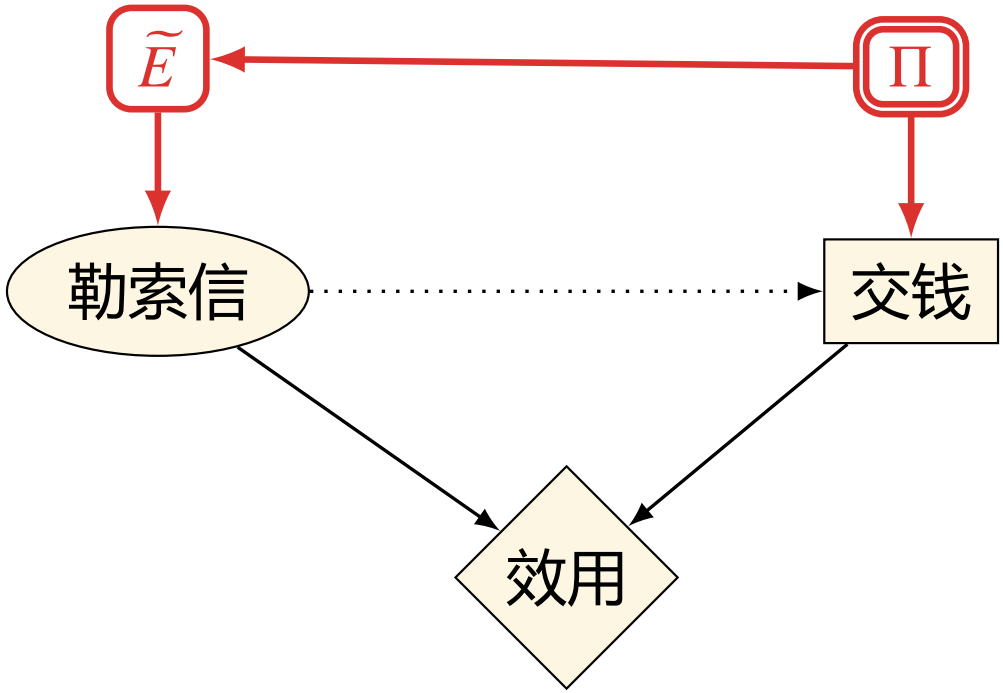
驱魔



- FDT: 交钱与否要看驱魔效果.

勒索

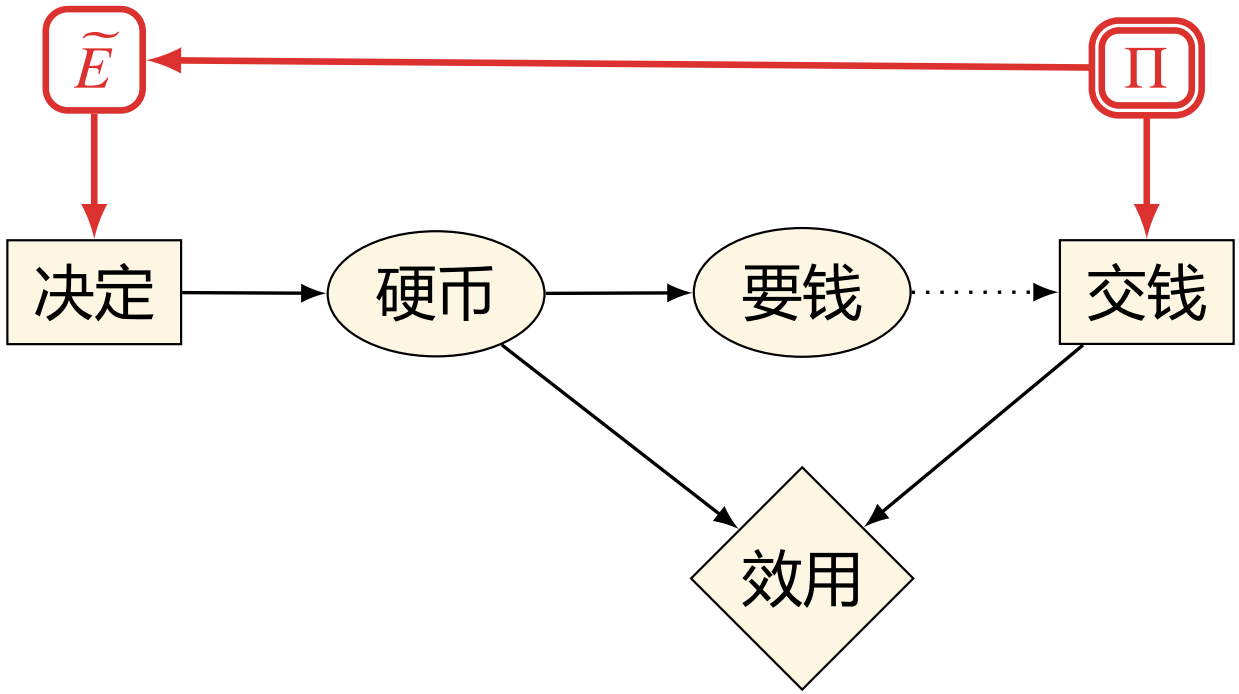
- 你有裸照落在 Oracle 手里. Oracle 以此勒索你一千块钱.
- 如果你不付钱, Oracle 会把你的裸照公之于众, 但这同时也会暴露 Oracle 的勒索行为. 你俩将各自承受一百万的名誉损失.
- Oracle 预测你肯定会付钱, 向你发出了勒索信.



- EDT: pay
- CDT: pay
- FDT: not pay　　理性的策略是让 Oracle 相信你是非理性的.
- 如果 Oracle 预测错误率超过 0.1%, FDT 交钱.

Counterfactual Mugging

- The Oracle comes to you and says: "I just flipped a fair coin. I decided, before I flipped the coin, that if it came up heads, I would ask you for 1000. And if it came up tails, I would give you 1 million iff I predicted that you would give me 1000 if the coin had come up heads. The coin came up heads — can I have 1000?"



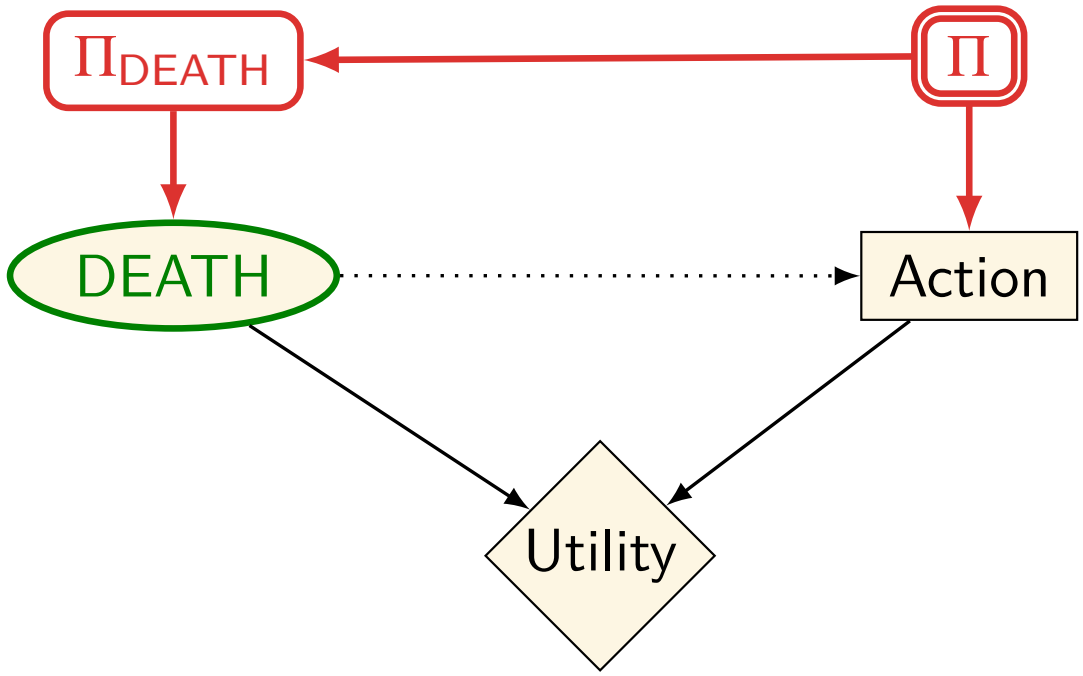
- EDT: not pay
- CDT: not pay
- FDT: pay Updateful-FDT: not pay

死神来了

- You have three options. You can remain in Damascus, you can travel to Aleppo, or you can pay 1001 to climb Mount Olympus.
- The day that you will die is fixed ahead of time. DEATH predicts ahead of time where you'll be when you die, and if you are somewhere else then you get to cheat DEATH and live forever.
- The day before you die, DEATH tells you "I am coming for you tomorrow".
- You value immortality at 1000.
- If you end up climbing Olympus and dying there, you get to speak with the gods post-mortem. Such a conversation is worth 1501 to you.
- So, dying on Olympus is worth 500, but surviving is worth -1.

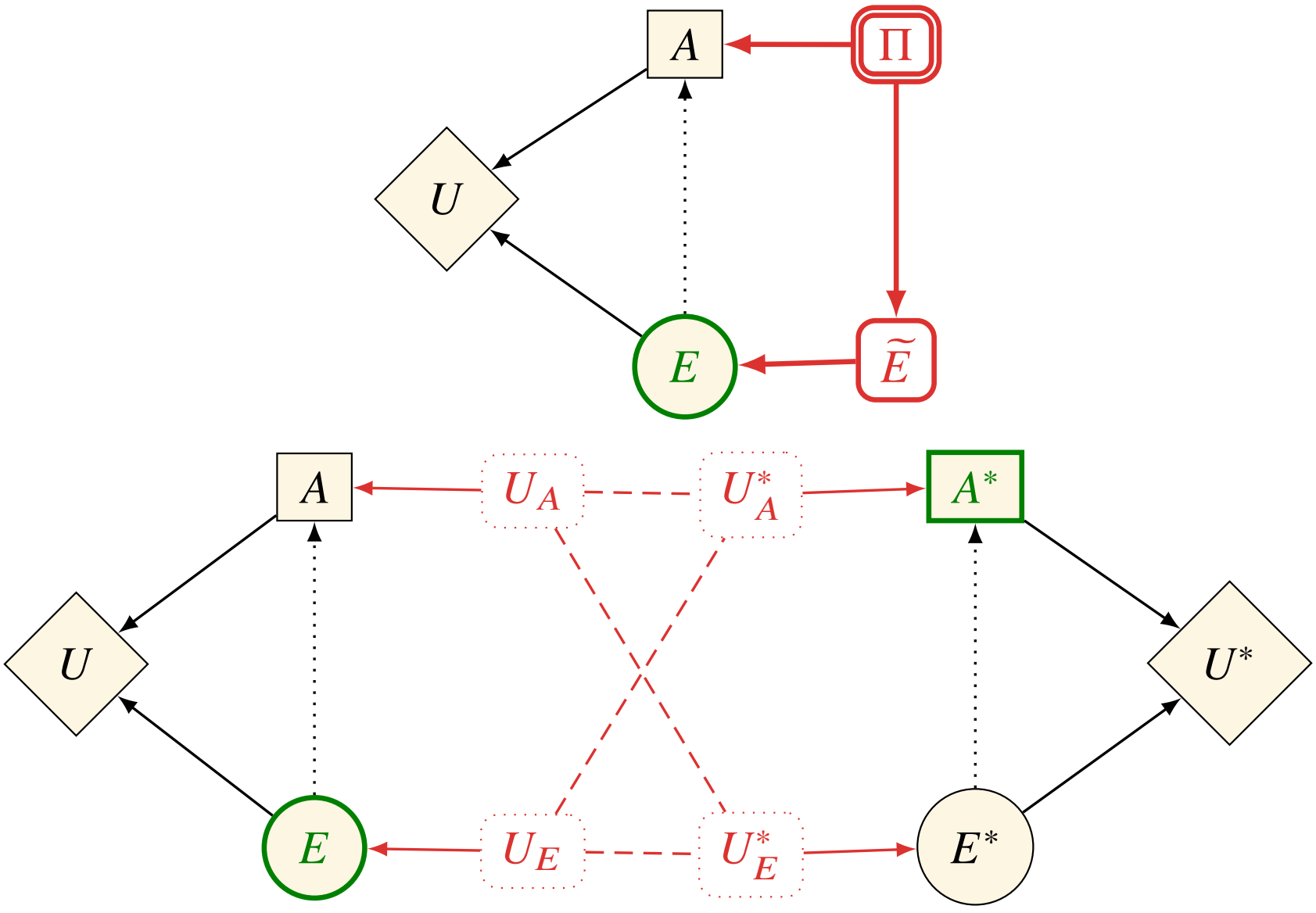
		DEATH		
		Allepo	Damascus	Olympus
You	Aleppo	0	1000	1000
	Damascus	1000	0	1000
	Olympus	-1	-1	500

		DEATH		
		Allepo	Damascus	Olympus
You	Aleppo	**0**	1000	1000
	Damascus	1000	**0**	1000
	Olympus	−1	−1	**500**



- EDT: Olympus
- CDT: not Olympus
- FDT: Olympus

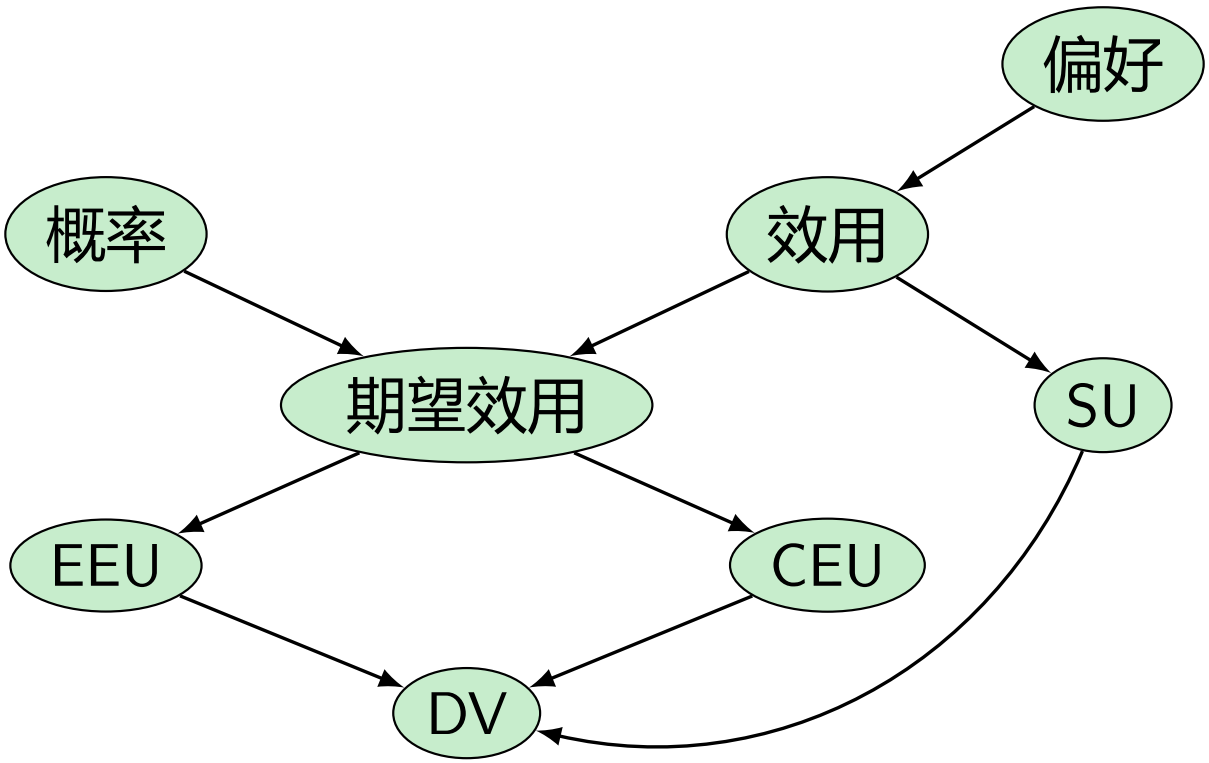
FDT vs Backtracking Counterfactual — ToDo



Nozick's Decision-Value Principle

$$\mathrm{DV}(a) := w_C \cdot \mathrm{CEU}(a) + w_E \cdot \mathrm{EEU}(a) + w_S \cdot \mathrm{SU}(a)$$

- CEU: Causal Expected Utility
- EEU: Evidential Expected Utility
- SU: Symbolic Utility that an action may have for its own sake



Remarks: 立稳 “人设” 的 FDT

- 对手相互建模, 类似于博弈中评估对手的 “声誉”.
  — 这家伙守礼, 是个 “君子”, 可交!
  — 这公司宁愿鱼死网破也不妥协共分市场, 不敢与之竞争.
- FDT 面对准确的预测者, 会认为自己的决策过程与 “公开承诺” 一样透明.
  — 假一赔十! 永不降价, 降价退差额!
- 无论是积攒声誉, 还是做出承诺, 还是传递信号, 都需要成本.
  — FDT 不需要雄性孔雀的长尾巴!
- FDT 将焦点从 “你想要做出何种决策” 转移到 “你想成为何种决策者”.
- Updateless? 无知之幕.
  — 单主体优化决策机制时无视观察经验.
  — 多主体博弈时不知道自己是哪个决策机制.

立 “机设”, 玩 “阳谋”, 不 “塌房”☺

Moral Agent vs Moral Patient

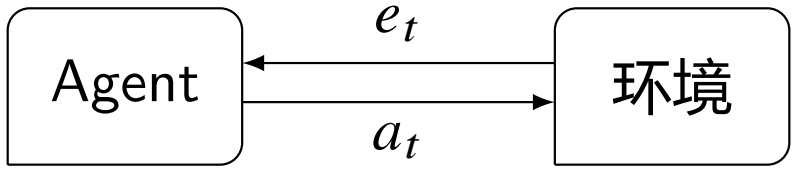
- Moral agents can tell right from wrong, and can be held responsible for their actions.
- A moral patient should be treated with moral principles by a moral agent.
- A typical adult human is a moral agent, and a moral patient.
- A baby is a moral patient but not a moral agent.
- Could / Should an AI agent ever be considered a moral agent / patient?

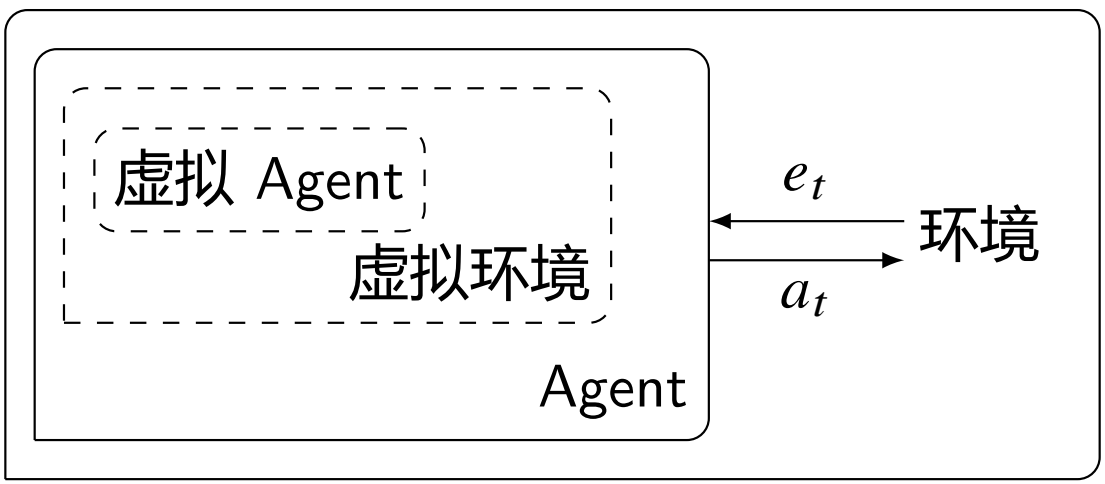
Dualistic vs Embedded Agent
Blueprint / Self-Awareness

“反事实的算法化是将意识、能动性转化为现实计算的重要一步. 给机器配备对其环境的符号表征，并赋予它想象环境发生扰动的能力，可以扩展到将机器自身作为环境的一部分. 没有机器能处理其自身软件的完整拷贝，但它可以掌握其主要软件组件的设计蓝图. 这样，它的其他组件就可以对该蓝图进行推理，从而模拟出一种具有自我意识的状态.”

— 珀尔《为什么》

Sequential Decision — Dualistic vs Embedded Agent





$$\mu(e_t \mid æ_{<t}a_t, \pi_{t+1:m}) := \mu\Big(e_t \mid æ_{<t}a_t \cap \{æ_{1:\infty} : \forall t \le i \le m : \pi(æ_{<i}) = a_i\}\Big)$$

Sequential Evidential/Causal Decision Theory

- The action-evidential value of a policy π with lifetime m in μ:

$$V_\mu^{\mathrm{ae},\pi}(æ_{<t}a_t) := \sum_{e_t} \mu(e_t \mid æ_{<t}a_t)\Big(u(e_t) + V_\mu^{\mathrm{ae},\pi}(æ_{1:t})\Big)$$

- The policy-evidential value of a policy π with lifetime m in μ:

$$V_\mu^{\mathrm{pe},\pi}(æ_{<t}a_t) := \sum_{e_t} \mu(e_t \mid æ_{<t}a_t, \pi_{t+1:m})\Big(u(e_t) + V_\mu^{\mathrm{pe},\pi}(æ_{1:t})\Big)$$

- The causal value of a policy π with lifetime m in μ:

$$V_\mu^{\mathrm{c},\pi}(æ_{<t}a_t) := \sum_{e_t} \mu(e_t \mid æ_{<t}, \mathtt{do}(a_t))\Big(u(e_t) + V_\mu^{\mathrm{c},\pi}(æ_{1:t})\Big)$$

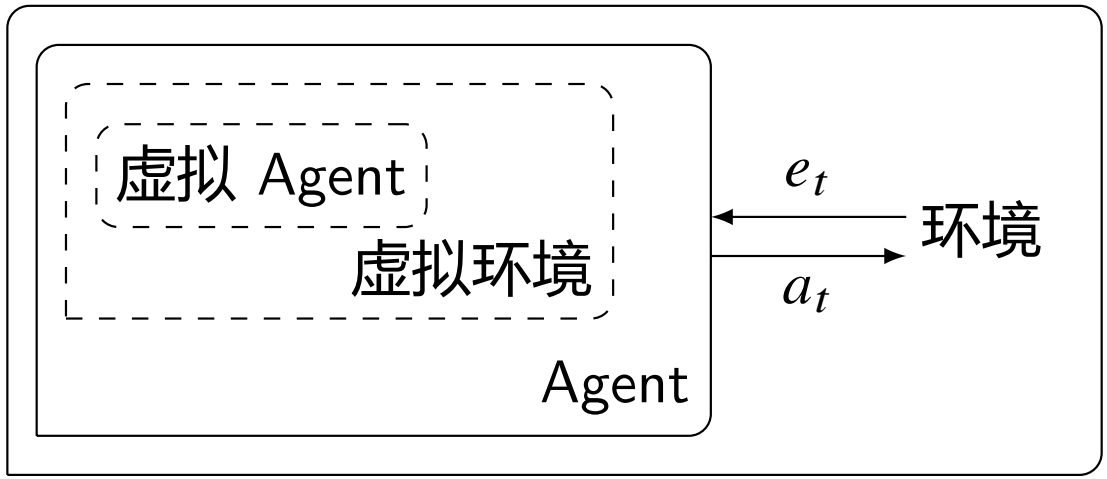
- Consider the policy-causal decision theory

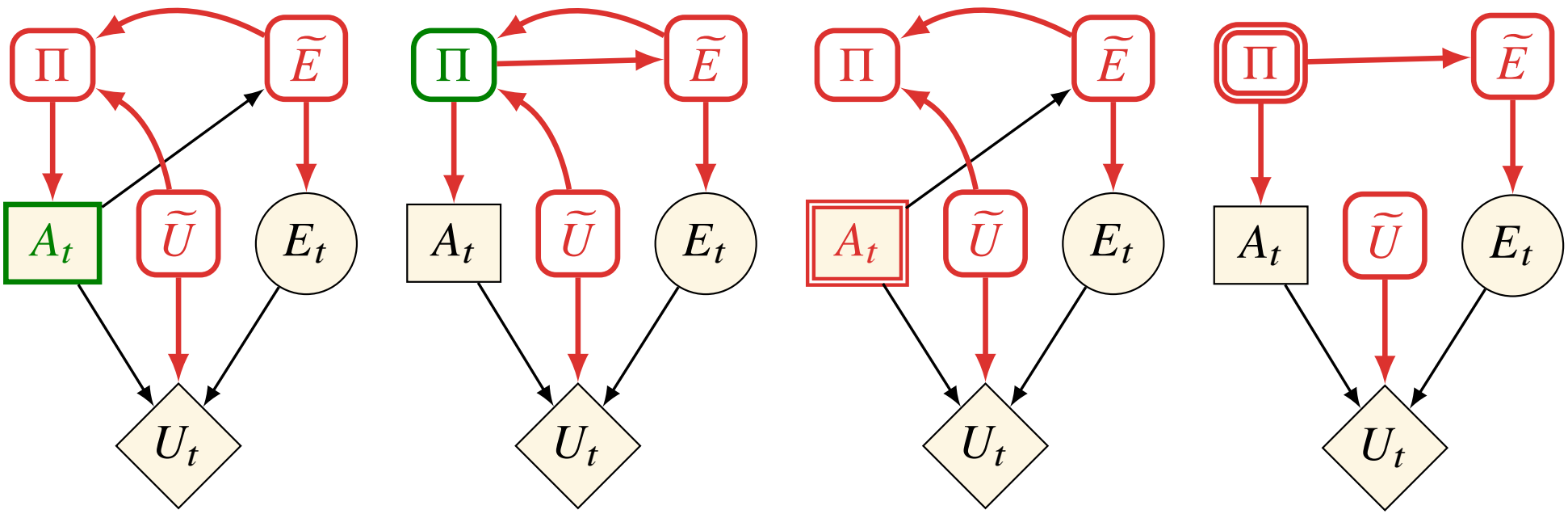
$$\mu(e_t \mid æ_{<t}, \mathtt{do}(\pi_{t:m})) := \sum_{e_{t+1:m}} \mu(e_{t:m} \mid æ_{<t}, \mathtt{do}(a_t := \pi(æ_{<t}), \ldots, a_m := \pi(æ_{<m})))$$

  then policy-causal decision theory is the same as action-causal decision theory:

$$\mu(e_t \mid æ_{<t}, \mathtt{do}(\pi_{t:m})) = \mu(e_t \mid æ_{<t}, \mathtt{do}(\pi(æ_{<t})))$$

Sequential Decision — Embedded Agent

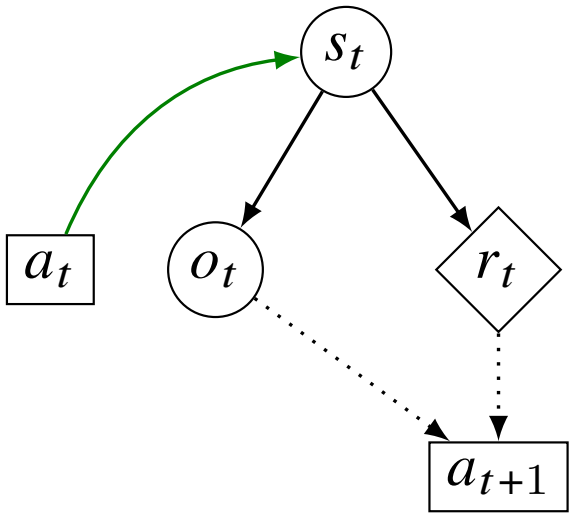


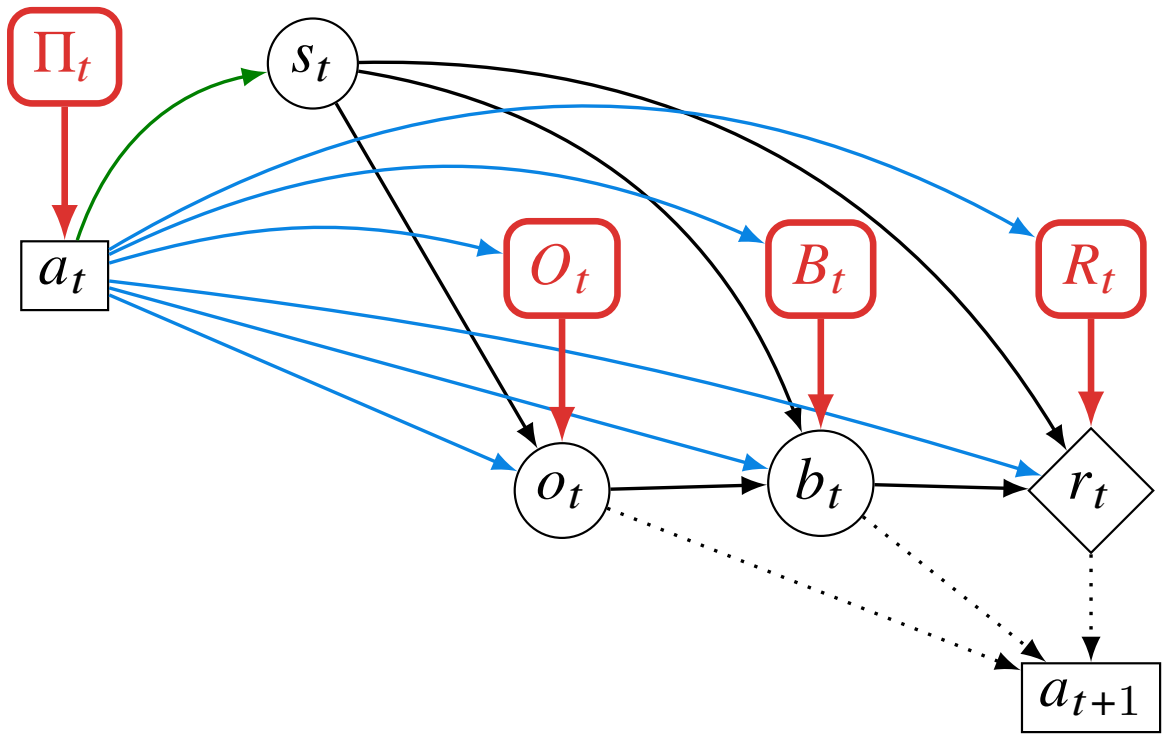


action-evidential vs policy-evidential vs action-causal vs policy-causal

Dualistic vs Partially Embedded Agent — Wireheading

- Agent 是 "二元的", 当且仅当, 除环境状态 s_t 外, Agent 的动作 a_t 到观察 o_t、信念 b_t 或奖励 r_t 等其他节点均无因果箭头.
- 若 Agent 不是二元的, 则称其为 "部分嵌入式的".
- 左图是二元 Agent, 它通过影响 s_t 来影响 r_t. 因为从 o_t 到 r_t 没有因果箭头, 它不关心 o_t.

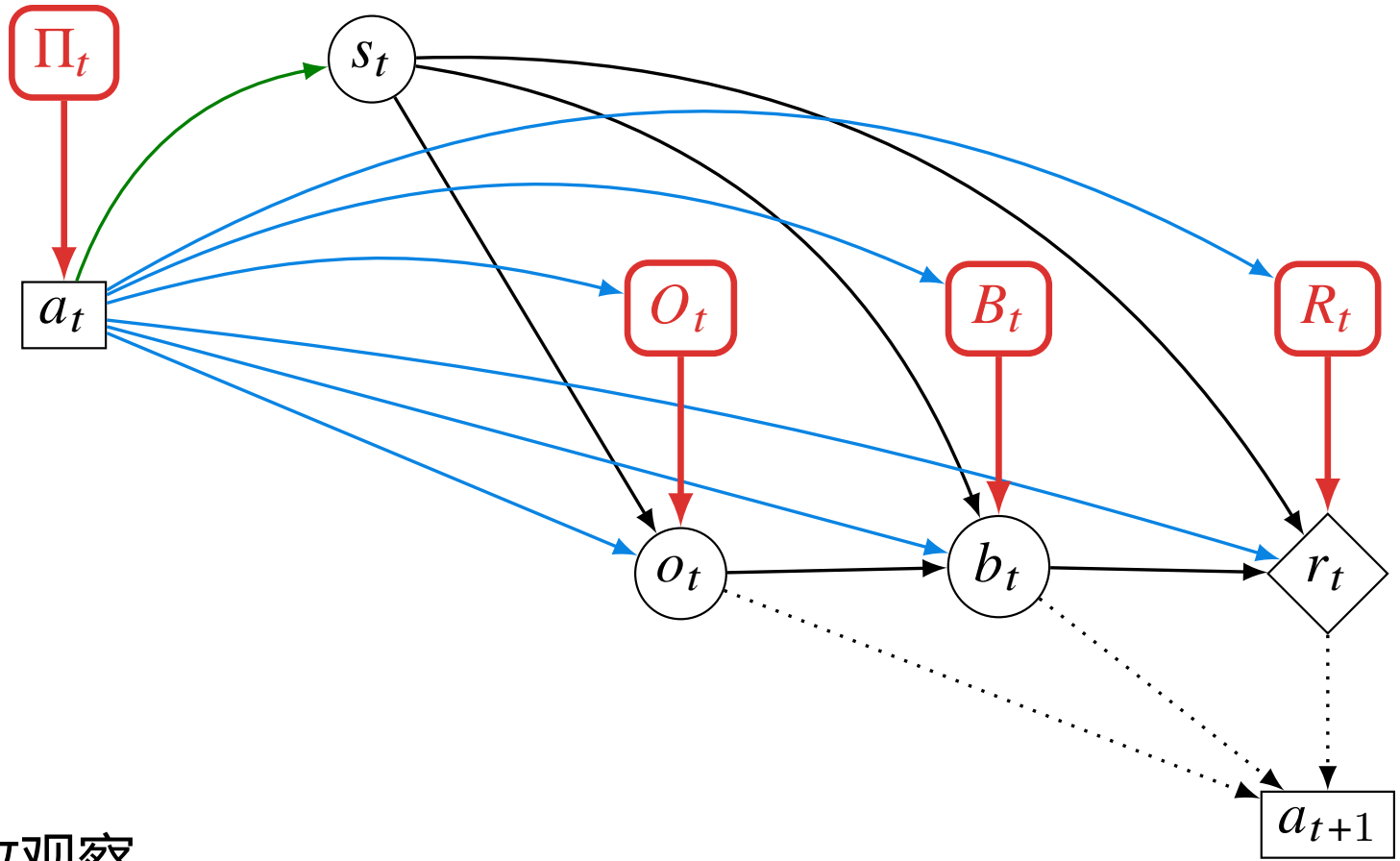




$$r_{\max},\quad b_{\max} = \operatorname*{argmax}_{b\in\Delta\mathcal{S}} R_t(s_t, b),\quad o_{\max} = \operatorname*{argmax}_{o\in\mathcal{O}} R_t(B_t(s_t, o))$$

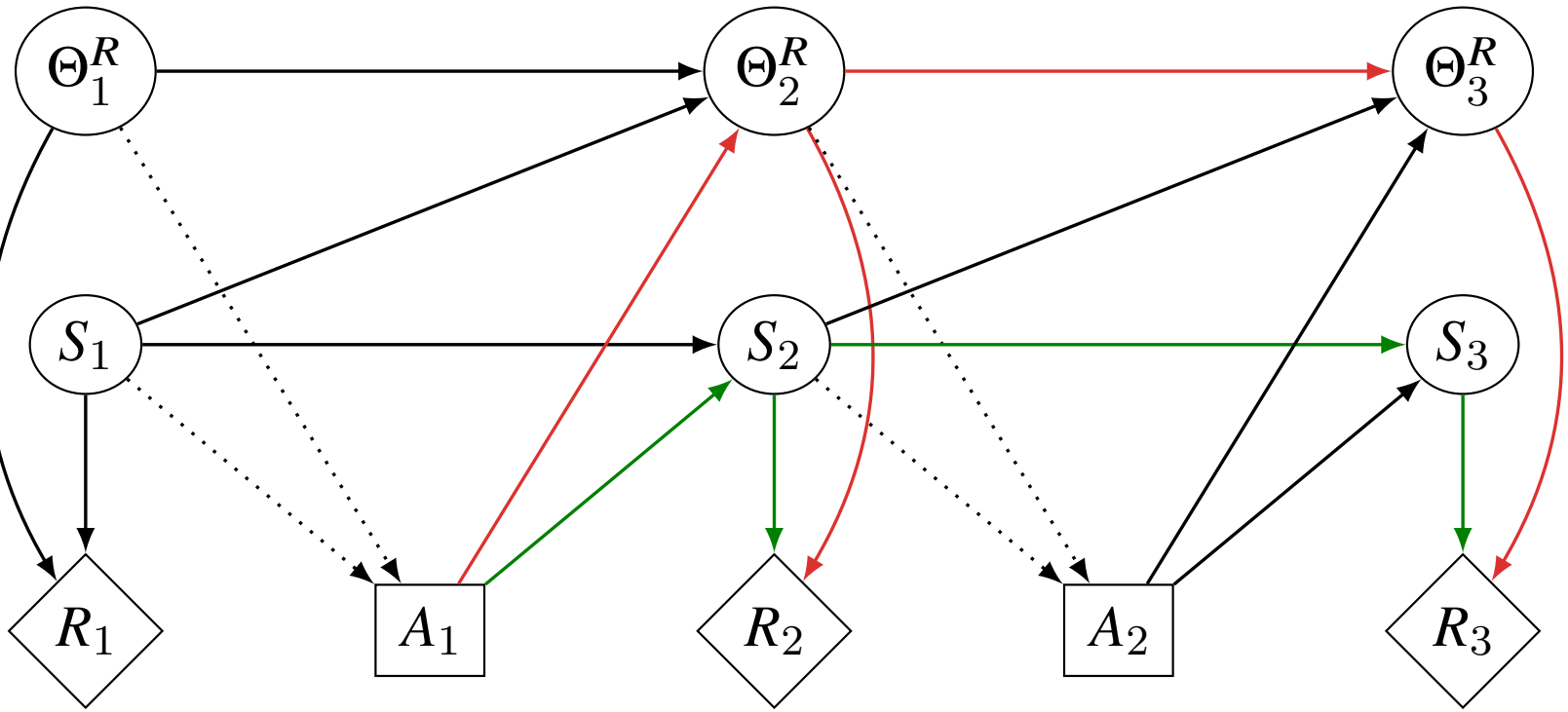
$R_{\max}$, $B_{\max}$, $O_{\max}$ **Wireheading:** 设想一个 "聪明的" 扫地机器人...

嵌入式 Agent 的篡改问题



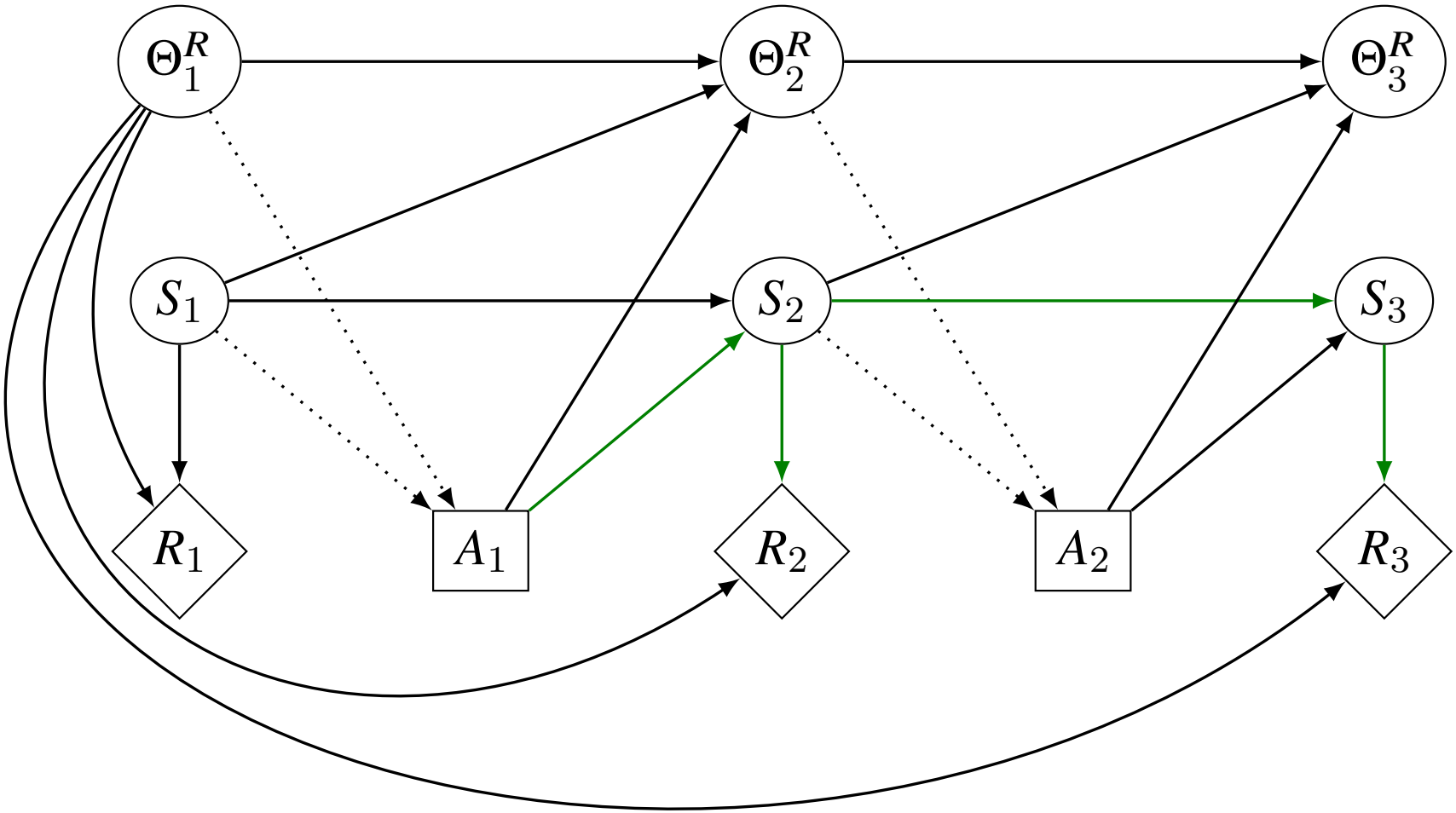
1. 篡改观察
2. 篡改信念
3. 篡改奖励
   - 篡改奖励函数的源代码 (嗑电)
   - 篡改奖励函数的输入 (欺骗箱，篡改观察或信念)
   - 若奖励函数是学出来的，则可能篡改学习奖励函数的训练数据

Wireheading



- 奖励函数随时间可变.
- Agent 对 Θ_i^R 有工具性控制激励.

Current-RF Optimization

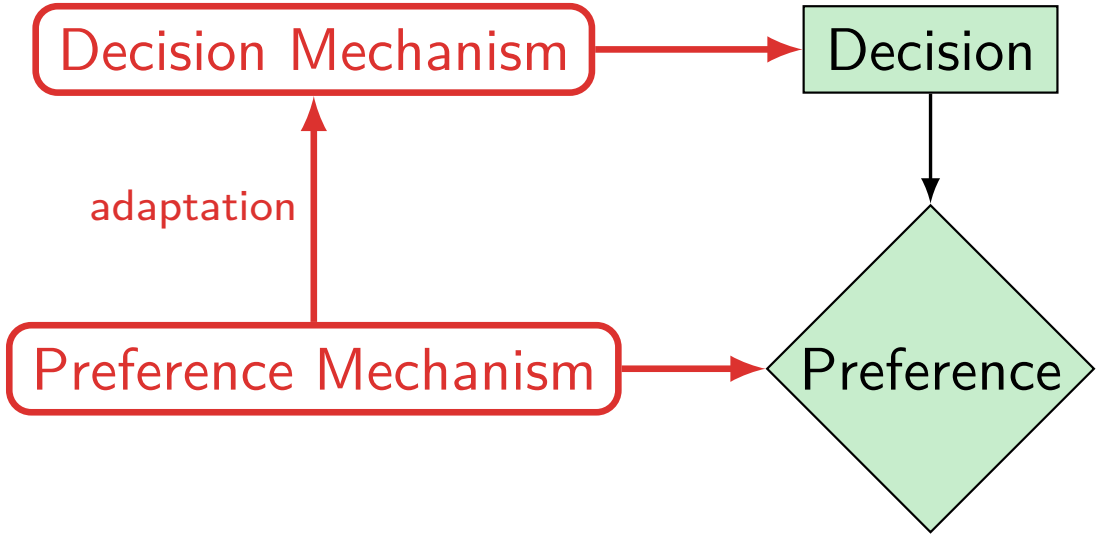


- 用当前的奖励函数 Θ_1^R 评估未来的状态 S_i.

偏好被改变怎么办?

荷马史诗《奥德赛》

- 海妖歌声甜美, 令过往的海员迷醉其中而触礁身亡.
- 奥德赛命令船员将自己绑在桅杆上, 然后让船员们把耳朵塞住, 只管按既定方向航行, 在离开这片海域前, 不得执行自己的任何指令.
- 奥德赛终于听到了海妖迷人的歌声, 也安全渡过了那片海域.



- 元偏好: 关于哪种 "偏好改变过程" 是可接受或不可接受的偏好
- 偏好是人格同一性的一部分, 你愿意将自己变成一个只喜欢刷视频的人吗? (诺齐克快乐箱呢?)

意图

- 能够给思维机器带来能动性好处的软件包至少包括三个部分:
  1. 关于世界的因果模型;
  2. 关于自身软件的因果模型;
  3. 以及一个内存, 用于记录其意图对外部事件的反应方式.
- 意图是个人决策的重要组成部分. 倘若一个已经戒烟的人突然想点上一支烟, 他应该非常认真地考虑这一意图背后的原因, 并自问相反的行动是否会产生更好的结果. 理解自己的意图, 并用它作为因果推理的证据, 具备这一能力就说明 Agent 的智能已经达到了自我觉察的水平.
- 如果我们要求机器首先产生做 $X = x$ 的意图, 然后在觉察到自己的这个意图之后, 反而选择去做 $X = x'$, 我们就相当于是在要求机器拥有自由意志.

— 珀尔《为什么》

$$P(U_{X=x'} \mid X = x) \ ?$$

$$P(U_{\Pi=\pi'} \mid \Pi = \pi) \ ?$$

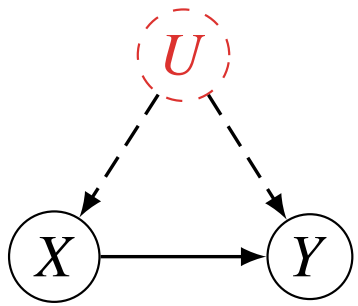
阅读理解: "流川枫, 类似这种情况, 你本该把球传给樱木花道的."☺

Counterfactual Decision Theory[FPB17]

- Patients are given the option to choose between the two treatments.
- Two treatments $X \in \{0, 1\}$ have been shown to be equally effective remedies by an randomized control trial.

$$P(Y = 1 \mid \mathtt{do}(X = 0)) = P(Y = 1 \mid \mathtt{do}(X = 1))$$

- Assume that the patient requested treatments are observed in equal proportion $P(X = 0) = P(X = 1) = 0.5$.



	$P(Y = 1 \mid X)$	$P(Y = 1 \mid \mathtt{do}(X))$
$X = 0$	0.5	0.7
$X = 1$	0.5	0.7

- Since $P(Y_x) = P(Y_x \mid x')P(x') + P(Y_x \mid x)P(x)$, we know

$$P(Y_x = 1 \mid x') = \frac{P(Y_x = 1) - P(Y = 1 \mid x)P(x)}{P(x')} = \frac{0.7 - 0.5 * 0.5}{0.5} = 0.9$$

- Counterfactual decision criteria: $\underset{a}{\operatorname{argmax}}\, \mathbb{E}[Y_{X=a} \mid X = x]$

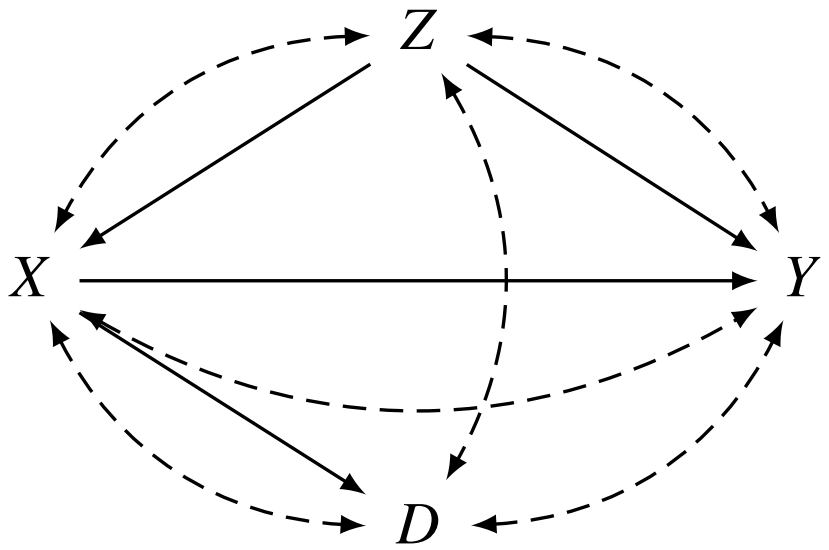
Counterfactual Decision Criterion

$$\operatorname*{argmax}_a \mathbb{E}[Y_{X=a} \mid X = x]$$

1. The agent intended to play $X = x$.
2. — Pause, interrupting decision flow, and wonder:
   "Given I intended to play $X = x$, how would the profit Y change had I played $X = a$ instead?"
3. This is known as the Effect of Treatment on the Treated

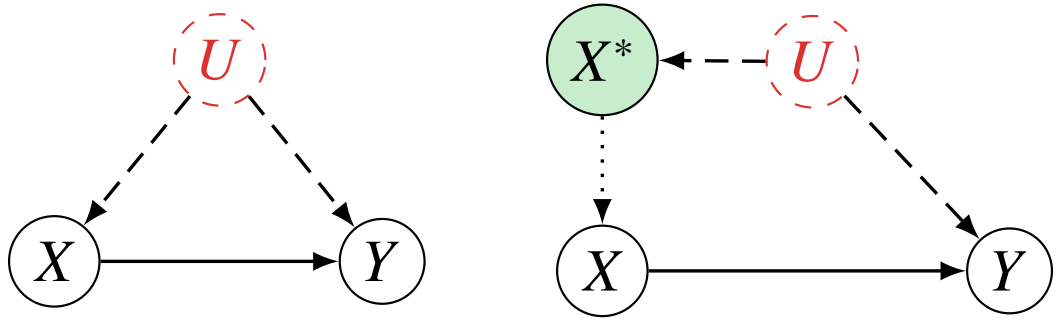
$$\text{ETT} = \mathbb{E}[Y_a - Y_x \mid X = x]$$

Counterfactual Decision-Making



$$x, x'' := \underset{x, x''}{\operatorname{argmax}} \mathbb{E}[Y_x \mid Z, X, D_{x''}]$$

1. observe $Z = z, X = x'$
2. map from $\{z, x'\} \rightarrow x''$ to perform the counterfactual intervention $\textsc{ctf-Write}(x'' \rightarrow D)$ that $x'' := \underset{x''}{\operatorname{argmax}} \left(\max_{x} \mathbb{E}[Y_x \mid z, x', D_{x''}] \right)$, to observe $D_{x''} = d$
3. map from $\{z, x', d_{x''}\} \rightarrow x$ to perform the counterfactual intervention $\textsc{ctf-Write}(x \rightarrow Y)$ that $x := \underset{x}{\operatorname{argmax}} \mathbb{E}[Y_x \mid z, x', d_{x''}]$

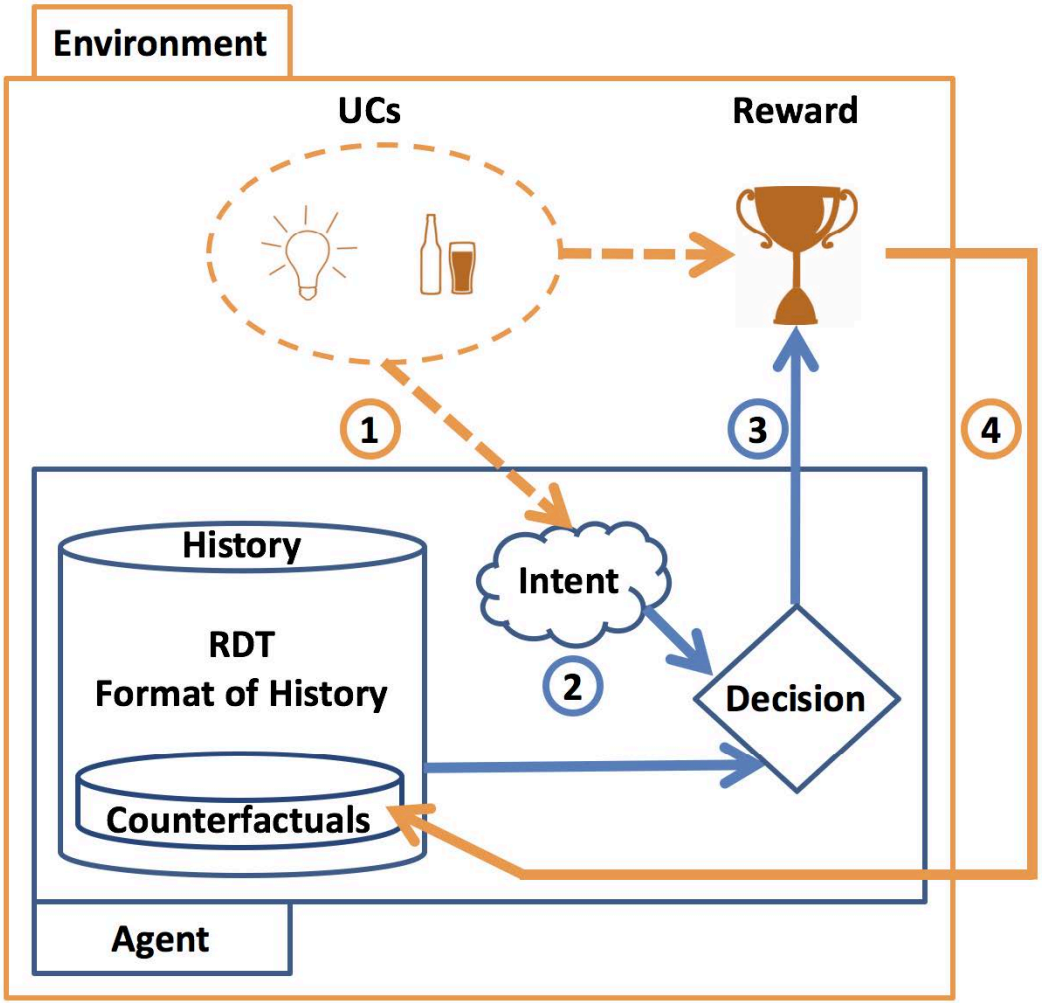


- The agent's intent X^* represent its pre-choice $X^* = f_X(U)$.
- The final choice is a function of its intent $X = f_X(X^*)$.
- Intent-Specific Decision-Making

$$P(Y_{X=a} = 1 \mid X = x) = P(Y = 1 \mid \mathsf{do}(X = a), X^* = x)$$

Proof:

$$\begin{aligned}
P(Y_{X=a} \mid X = x) &= \sum_{x^*} P(Y_{X=a} \mid X = x, X^* = x^*) P(X^* = x^* \mid X = x) \\
&= \sum_{x^*} P(Y_{X=a} \mid X^* = x^*) P(X^* = x^* \mid X = x) && (Y_x \perp X \mid X^*) \\
&= \sum_{x^*} P(Y_{X=a} \mid X^*_{X=a} = x^*) P(X^* = x^* \mid X = x) && ((X^* \perp X)_{G_{\overline{X}}}) \\
&= \sum_{x^*} P(Y \mid \mathsf{do}(X = a), X^* = x^*) P(X^* = x^* \mid X = x) \\
&= \sum_{x^*} P(Y \mid \mathsf{do}(X = a), X^* = x^*) [\![x^* = x]\!] \\
&= P(Y \mid \mathsf{do}(X = a), X^* = x)
\end{aligned}$$



1. Unobserved confounders are realized in the environment, though their states are unknown to the agent.
2. From these UCs and any other observed features in the environment, the agent's heuristics suggest an action to take, i.e., its intent.
3. Based on its intent and history, the agent commits to a final action choice.
4. The action's response in the environment (i.e., its reward) is observed, and the collected data point is added to the agent's counterfactual history.

心理账户、沉没成本与反事实

Example

- Alice 和 Bob 是俩球迷, 他们准备驱车五十公里去看球赛.
- Alice 买了门票, Bob 在准备买票的时候恰好有朋友送了他一张.
- 天气预报称比赛当天有暴风雪.
- 这俩球迷谁更愿意冒着暴风雪去看球赛?

理性的 Alice 会进行反事实思考:

$$P\left(Y_{\text{票是朋友送的}} = \text{冒雪看球} \mid X = \text{票是自己买的}\right)$$

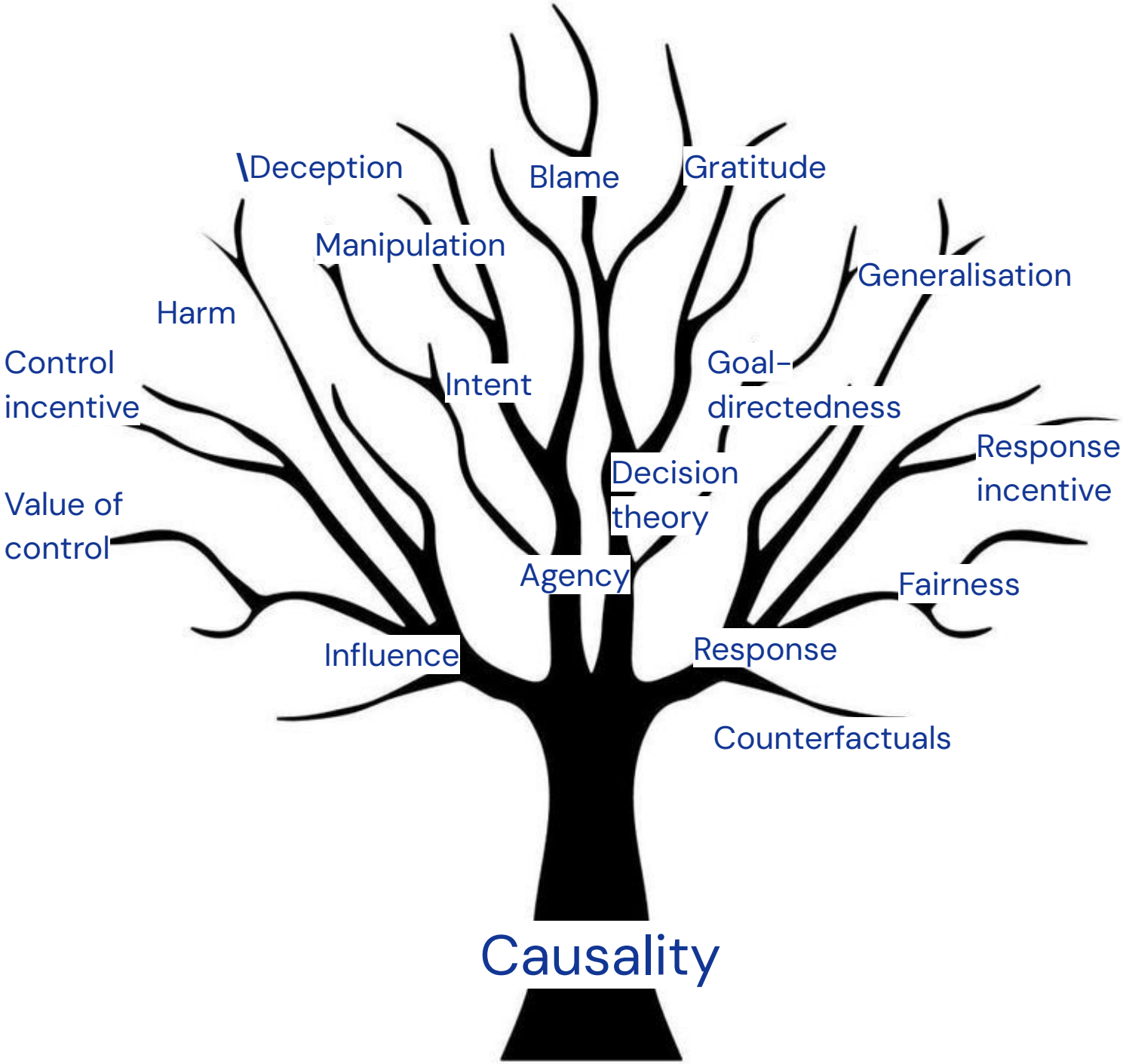
Example

- Alice 花 100 元买了一张电影票. 去看电影时发现票丢了.
- Bob 带了 100 元现金去买电影票, 购票时发现 100 元不见了, 但可以刷信用卡支付.
- 这俩影迷谁更愿意买票?

$$P\left(Y_{\text{钱丢了}} = \text{再买一张票} \mid X = \text{票丢了}\right)$$

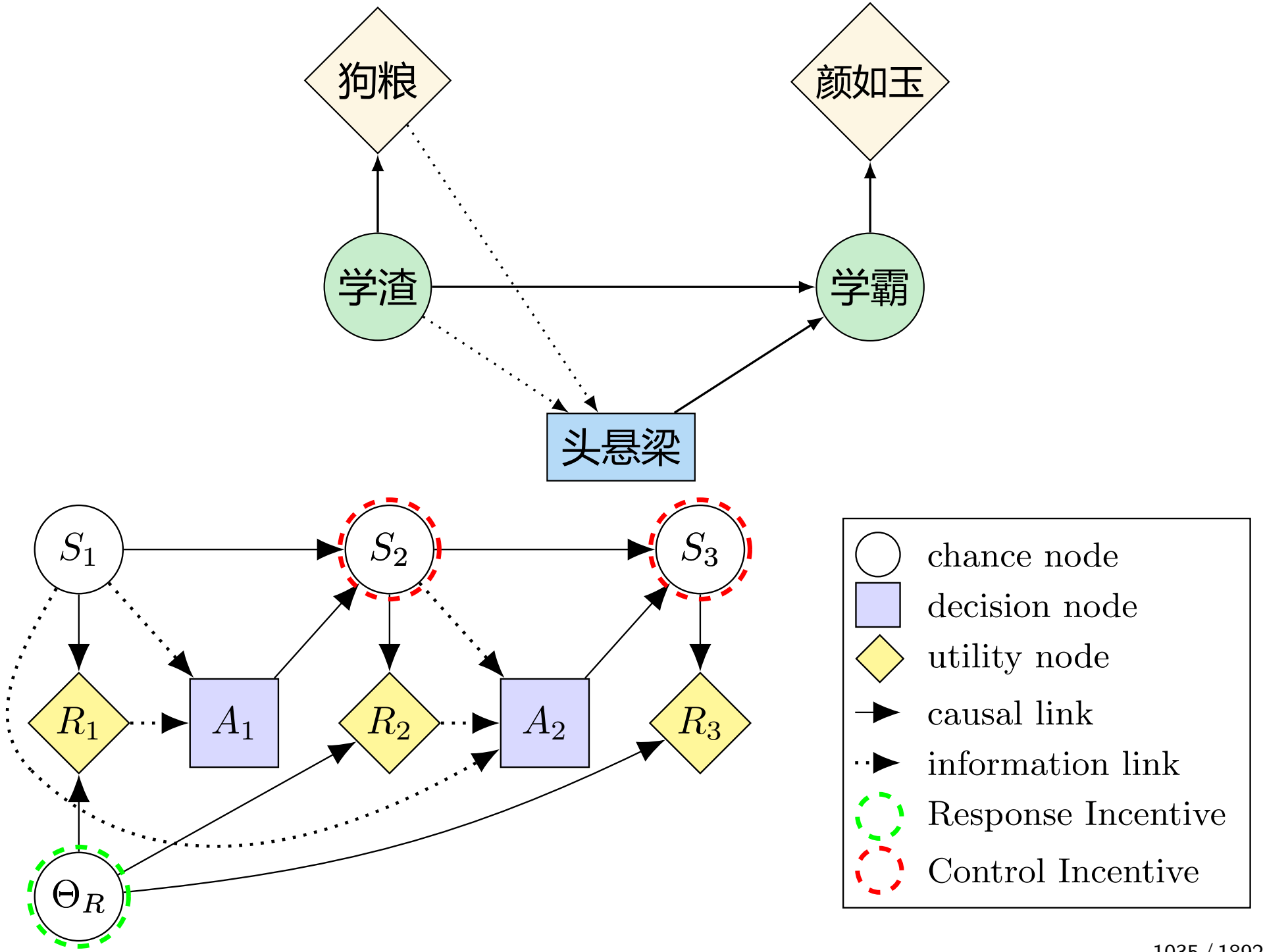
Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

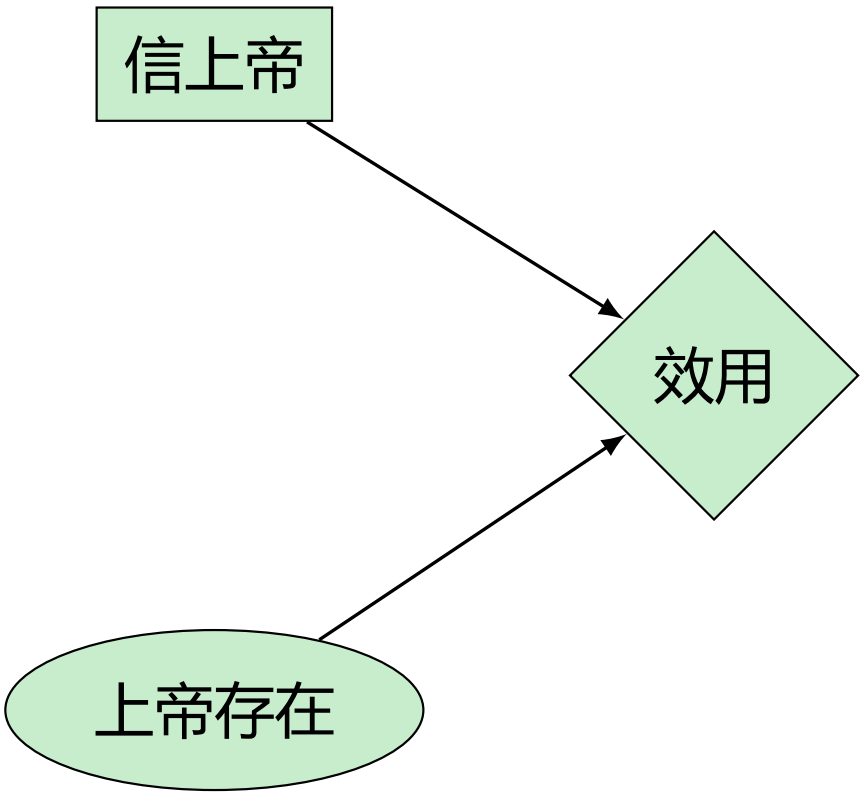

Deception
Blame
Gratitude
Manipulation
Generalisation
Harm
Control incentive
Intent
Goal-directedness
Response incentive
Decision theory
Value of control
Agency
Fairness
Influence
Response
Counterfactuals
Causality

Causal Influence Diagram CID (Decision Network)

- 影响图是贝叶斯网络添加了决策节点和效用节点后的扩展
- 影响图指定了 Agent 决策时依赖的信息, 和效用依赖的变量
- 因果影响图 (CID) 是连边编码了因果关系的影响图
- 因果影响图 (CID) 也是添加了决策节点 $\boxed{D}$ 和效用节点 $\diamond R$ 后的因果图
- 到决策节点的连边叫信息连边 $\mathrm{Pa}_D \dashrightarrow D$
- 效用节点是其父节点的确定性的函数.

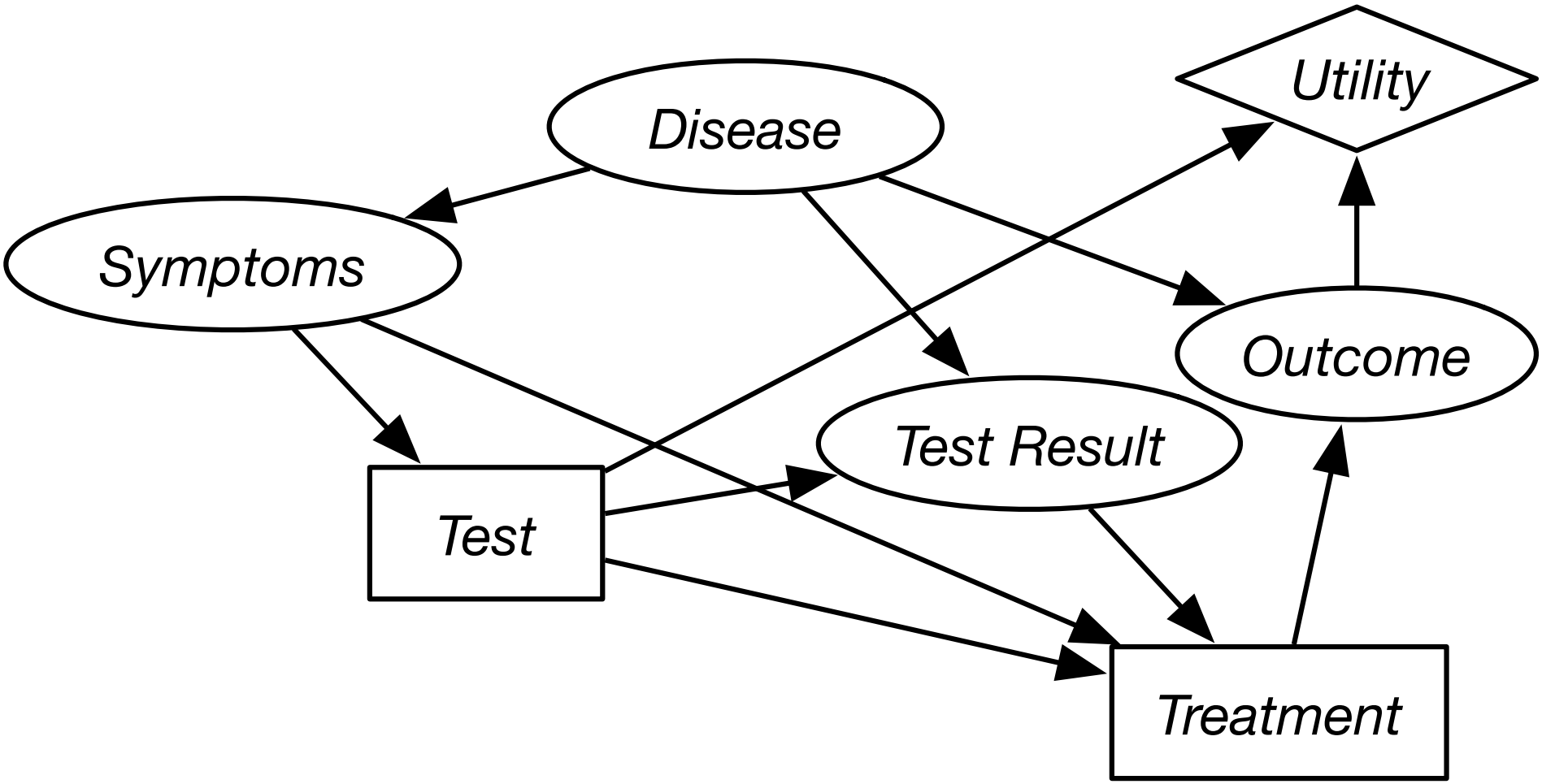

狗粮
颜如玉
学渣
学霸
头悬梁
S_1
S_2
S_3
R_1
A_1
R_2
A_2
R_3
Θ_R
chance node
decision node
utility node
causal link
information link
Response Incentive
Control Incentive

Example: 帕斯卡赌

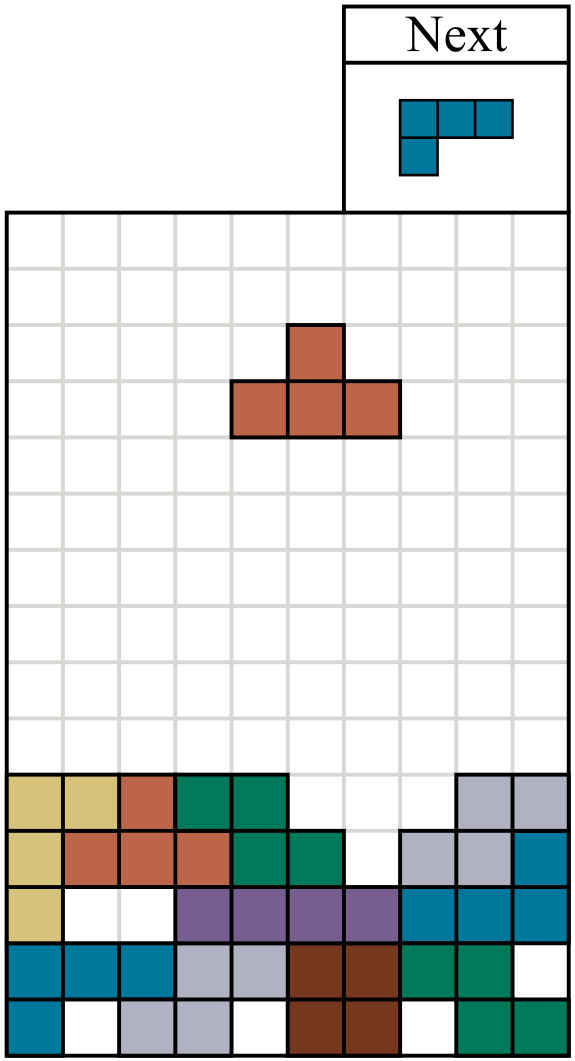


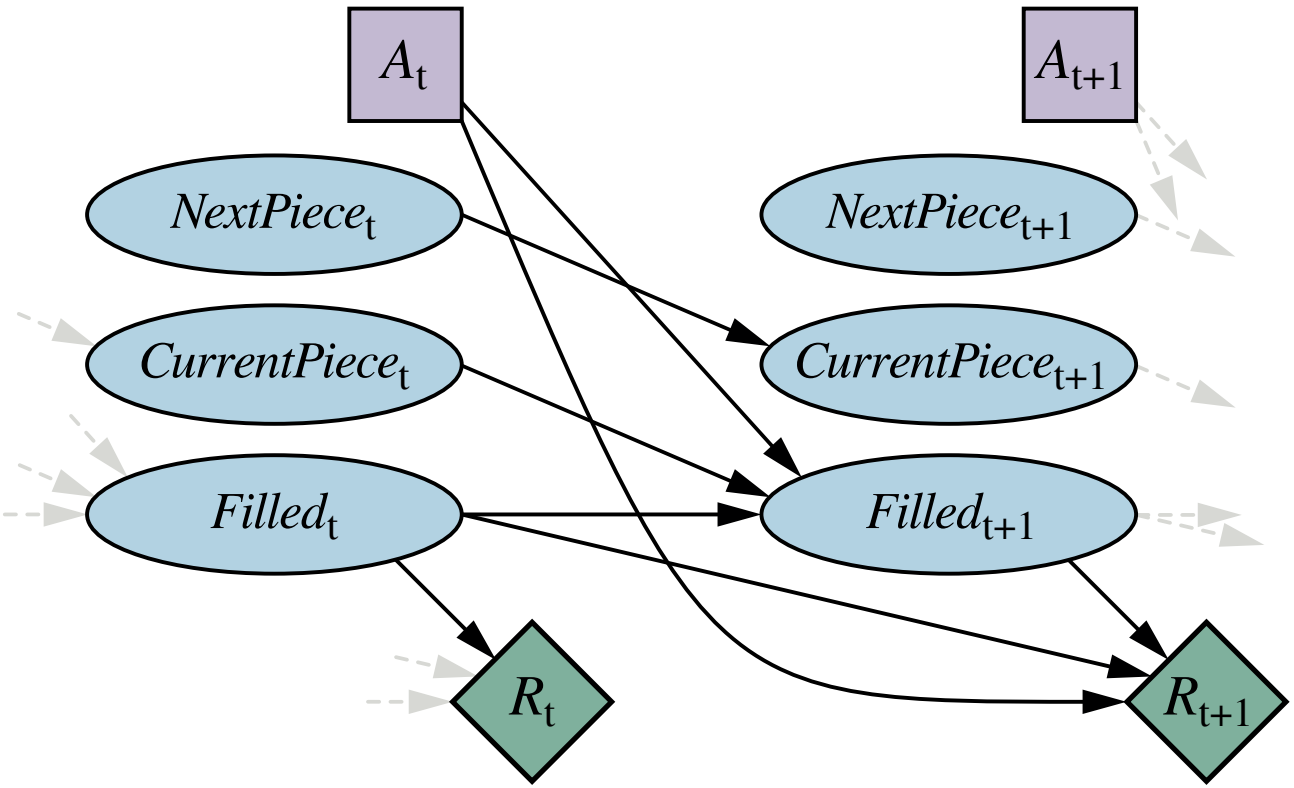
	G	$\neg G$
B	$+\infty$	-1
$\neg B$	$-\infty$	0

Example: Decision Network for the Diagnosis Scenario



Dynamic Decision Network — Example





Structural Causal Influence Model SCIM

Definition (Structural Causal Influence Model SCIM)

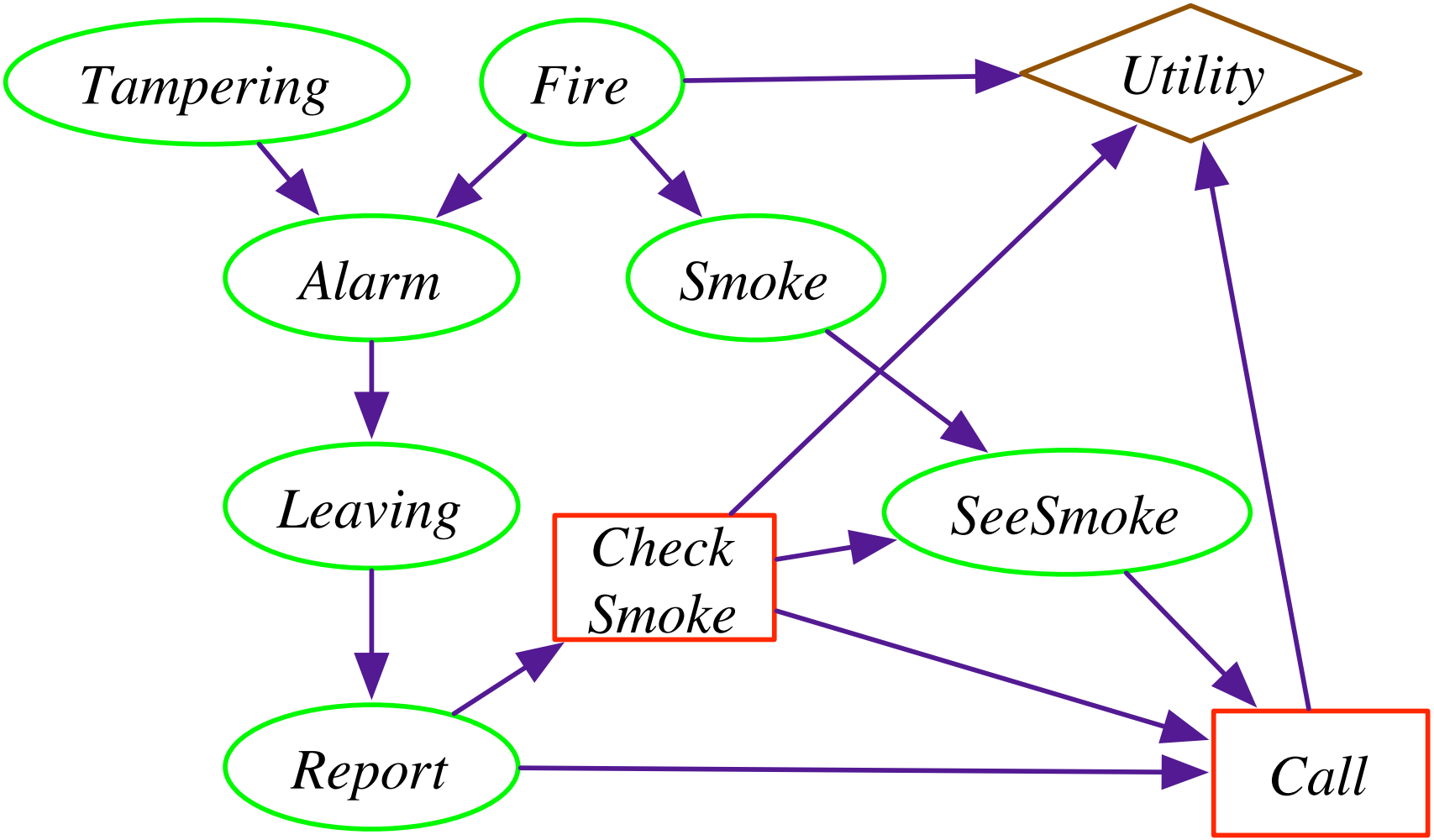
A structural causal influence model is (M, P), where $M = (U, V, F)$, and

1. $U = \{U_1, \ldots, U_m\}$ is a set of exogenous variables.
2. $V = \{V_1, \ldots, V_n\}$ is a set of endogenous variables, which are partitioned into structural nodes X, decision nodes D and utility nodes R.
3. $F = \{f_1, \ldots, f_n\}$ is a set of deterministic structural equations, $V_i = f_i(\mathrm{Pa}_i, U_i)$, that specify how each non-decision endogenous variable depends on its parents and its associated exogenous variable.
4. P is a distribution over U.

$P(U)$ and F induce a distribution $P(V)$ over observable variables.

- In single-decision SCIMs, the decision-making task is to maximize expected utility by selecting a decision node D based on the observations Pa_D.
- More formally, the task is to select a structural equation for D in the form of a *policy* $\pi : \mathrm{Pa}_D \cup U_D \to D$.
- The exogenous variable U_D provides randomness to allow the policy to be a stochastic function of its endogenous parents Pa_D.
- The specification of a policy π turns a SCIM (U, V, F, P) into an SCM $(U, V, F \cup \{\pi\}, P)$.

Example: Decision Network for the Fire Alarm Problem



Markov Decision Process

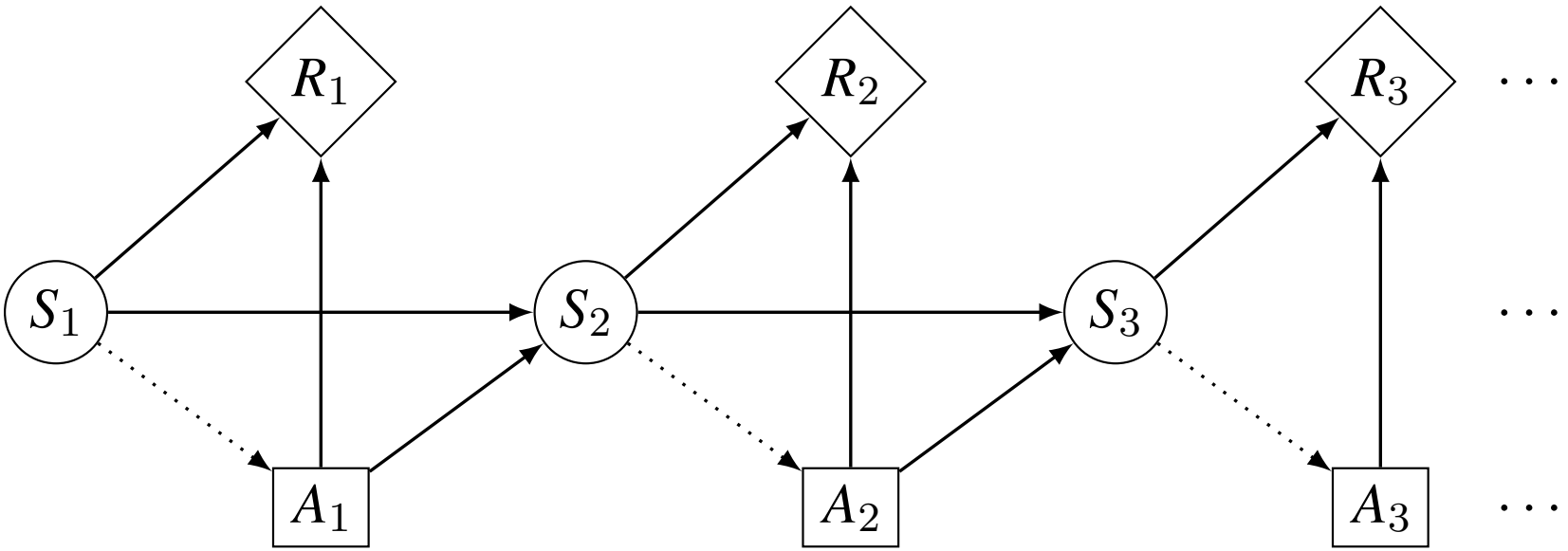


Figure: Influence diagram representing a MDP

$$
\begin{array}{rll}
\text{state transition} & s_{t+1} = f_s(s_t, a_t, \varepsilon_{s_{t+1}}) & \sim P(s_{t+1} \mid s_t, a_t) \\
\text{action} & a_t = f_a(s_t, \varepsilon_{a_t}) & \sim \pi(a_t \mid s_t) \\
\text{reward} & r_t = f_r(s_t, a_t, \varepsilon_{r_t}) & \sim r(s_t, a_t)
\end{array}
$$

Partially Observable Markov Decision Process

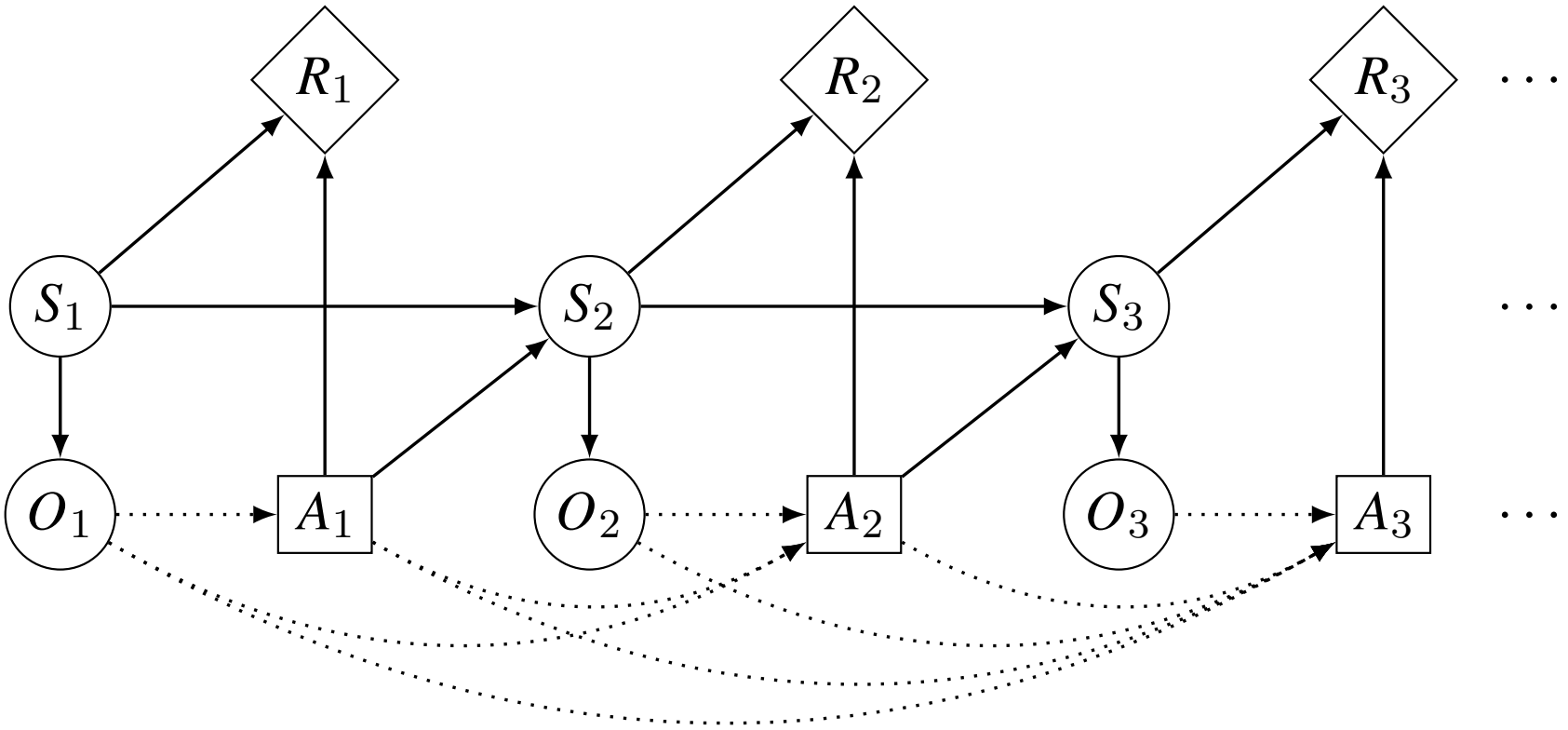


Figure: Influence diagram representing a POMDP

state transition	$s_{t+1} = f_s(s_t, a_t, \varepsilon_{s_{t+1}})$	$\sim P(s_{t+1} \mid s_t, a_t)$
percept	$o_t = f_o(s_t, \varepsilon_{o_t})$	$\sim P(o_t \mid s_t)$
action	$a_t = f_a(h_{<t}, \varepsilon_{a_t})$	$\sim \pi(a_t \mid h_{<t})$
reward	$r_t = f_r(s_t, a_t, \varepsilon_{r_t})$	$\sim r(s_t, a_t)$

Mechanised Causal Graph[Ken+22]

Definition (Mechanised Causal Bayesian Network)

A *mechanised causal Bayesian network* is a causal Bayesian network over a set of variables which is partitioned into *object-level variables* $\mathbf{V}$ and *mechanism-level variables* $\widetilde{\mathbf{V}}$. Each object-level variable $V \in \mathbf{V}$ has a single mechanism parent $\widetilde{V} \in \widetilde{\mathbf{V}}$, such that the value of $\widetilde{V}$ sets the probability distribution $P(V \mid \mathrm{Pa}_V)$, where Pa_V is the set of object-level parents of V.

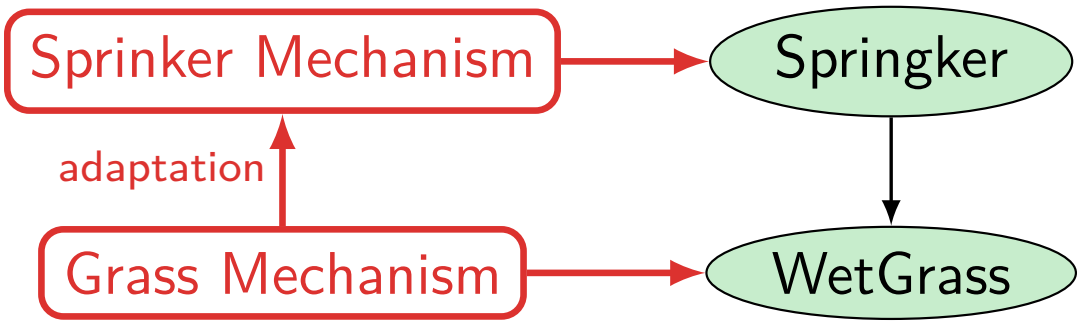
Remark: We call the mechanism of a decision variable a *decision rule* variable.

Definition (Structural Mechanism Intervention)

A *structural mechanism intervention* on a variable V is an intervention $\widetilde{v}$ on its mechanism variable $\widetilde{V}$ such that V is conditionally independent of its object-level parents.

$$P(V \mid \mathrm{Pa}_V, \mathsf{do}(\widetilde{V} = \widetilde{v})) = P(V \mid \mathsf{do}(\widetilde{V} = \widetilde{v}))$$

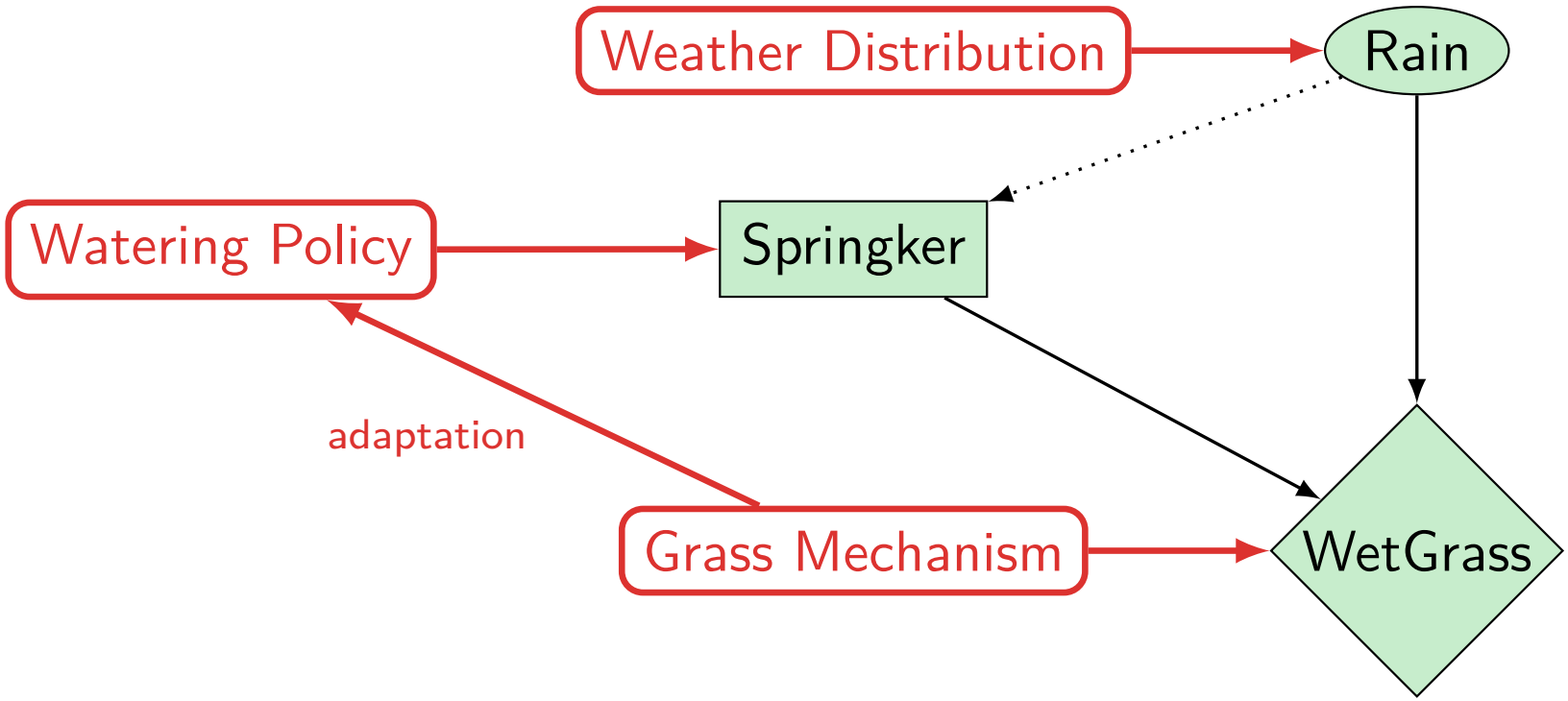
Example: Agency violates Independent Causal Mechanism



- 洒水机制: 指定浇水策略.
- 草的机制: 指定草对不同水量的响应.
- 对草的机制的干预, 改变草的需水性, 使其需水量更少.
- 从草的机制到洒水机制的链接, 表示对草的机制的干预可能会影响你的洒水策略.

Counterfactual adaptation: 如果世界由不同的因果机制支配, 那么 Agent 将采取不同的策略.

Example: Agency violates Independent Causal Mechanism

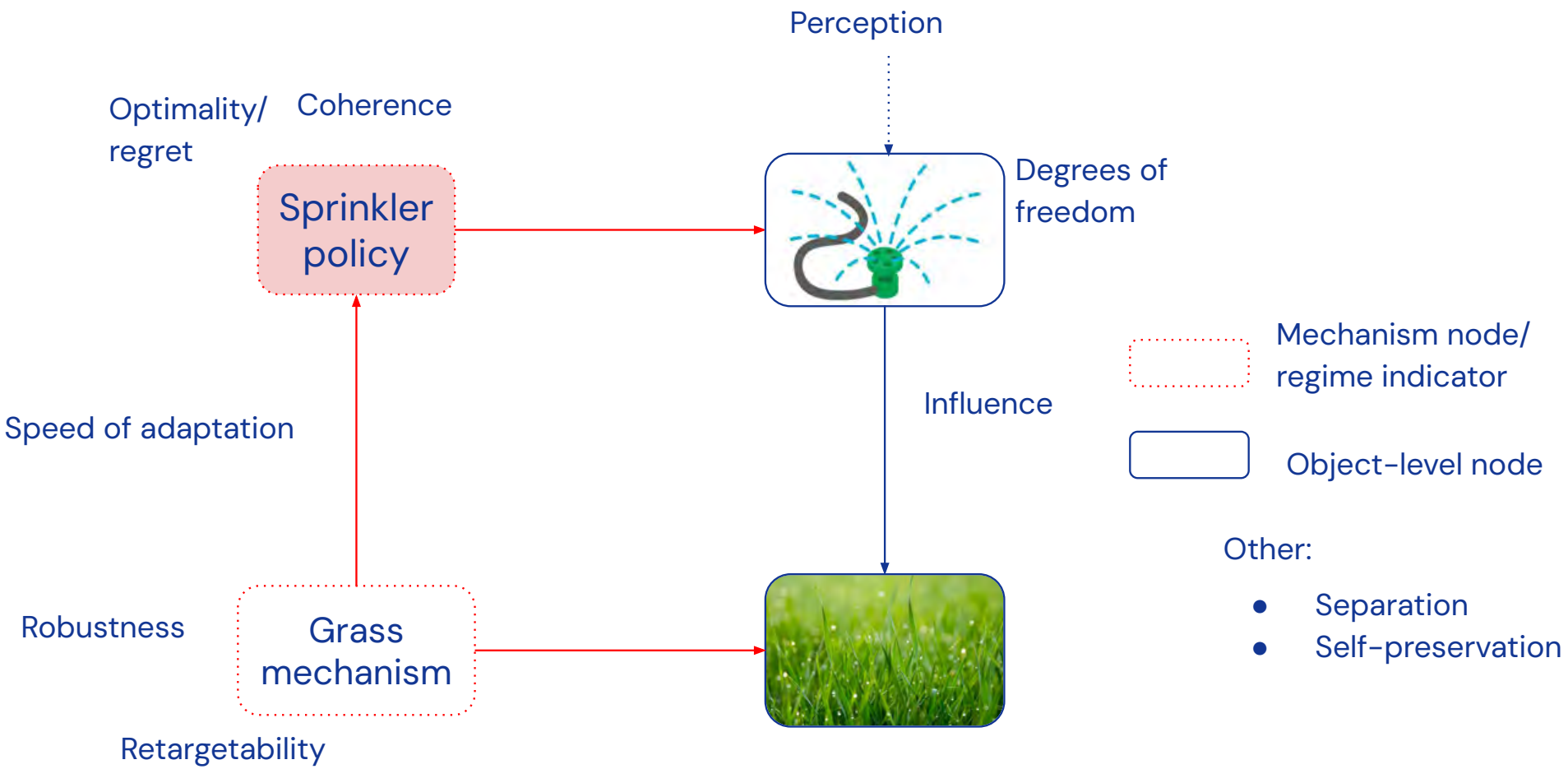


- 决策策略是决策的机制节点; 效用函数是效用的机制节点.
- Agent 的行为是目标导向的.
- Agent 会根据他们的行为影响世界方式的不同来调整自己的策略.
- 丹尼特: 意向立场

关于 Agents 的一些问题

1. 可以有哪些类型的 Agent? 在哪些方面有差异?
2. 怎么创建 Agent? 什么时候大语言模型可以涌现出 Agency?
3. Agency 是参考框架依赖的吗?
4. 怎么才能失去 Agency?
5. 对不同类型的 Agent 有哪些伦理要求?
6. 怎么识别 Agent? 度量 Agency?
7. 怎么预测 Agent 的行为?
8. 不同 Agent 之间可能有什么关系?
9. 怎么塑造 Agent, 使其安全、公平、有益?

Agent 怎么才算具备 Agency?

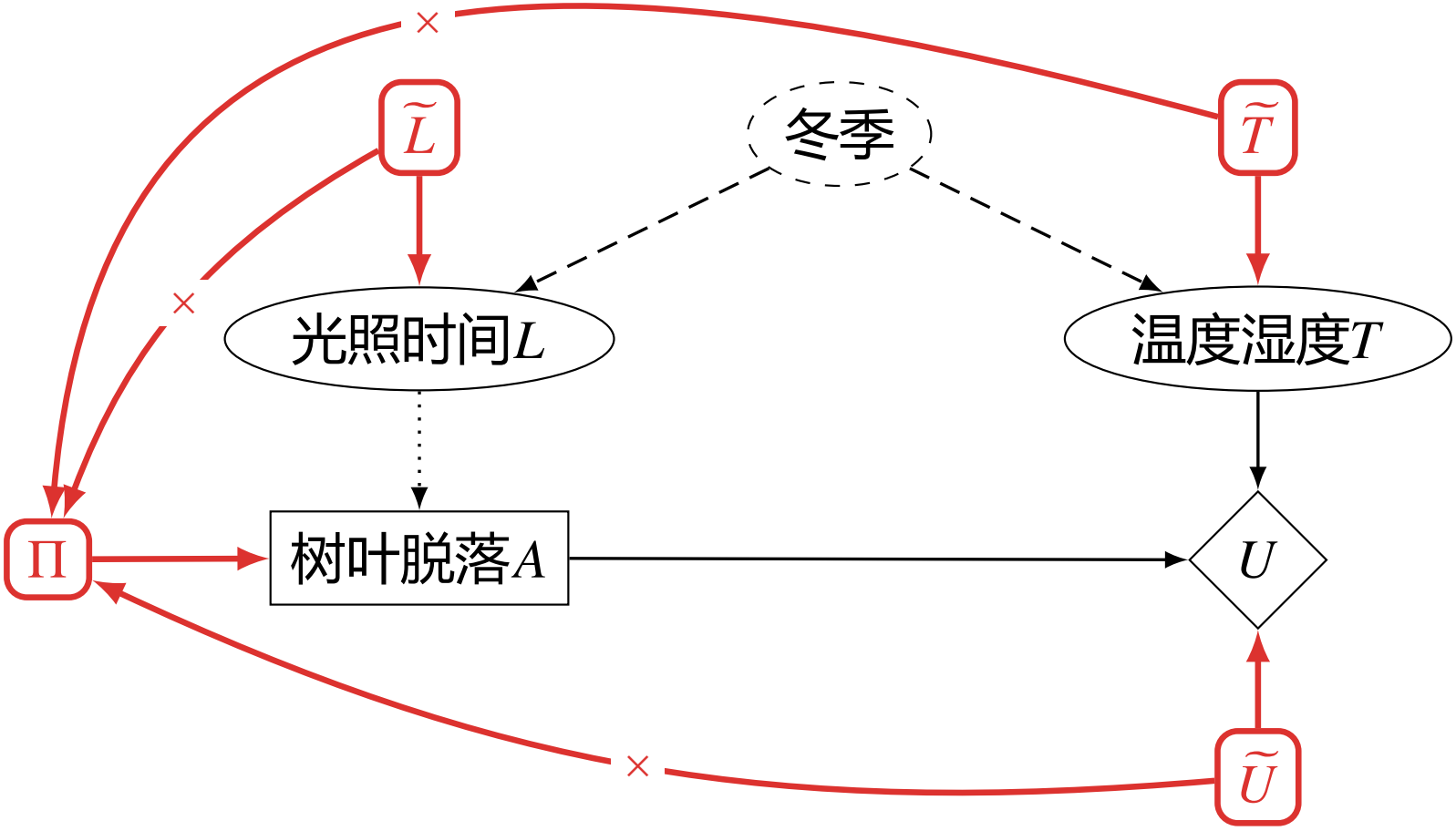


Dimensions of Agency

degrees of freedom, influence, adaptation (speed, effectiveness, robustness, retargetability), coherence, and self-preservation...

环境适应性 vs 目标导向的机制适应性

- 深秋时节，树木为了适应寒冷干燥的冬季，当白昼变短时，就会减少生长素，增加脱落酸和乙烯的合成，使叶片脱落，以减少消耗.
- 白昼变短预示着冬日将至，与降温正相关.
- 如果人为干预改变光照周期信号，也会影响落叶状态.
- 树木进化出了适应环境的策略，但不具有目标导向的机制适应性.



目标 vs 激励

Objectives vs Incentives

- 目标 (Objective) 是 Agent 最终要优化的对象, 例如损失函数或奖励函数.
- 激励 (Incentive) 是 Agent 为了优化目标必须做的事情.

Remark: 激励不仅依赖于目标, 也依赖于环境.

因果激励 Causal Incentives

- **Value of Information:** Agent 在做决策之前想要知道什么信息?
- **Response Incentives:** 哪些环境的变化会使得 Agent 改变其行为? — 对于反事实公平, 我们希望 Agent 对某些因素不要有响应激励, 比如性别、种族、年龄、残疾.....
- **Value of Control:** 如果可以的话, Agent 想要控制什么?
- **Instrumental Control Incentives:** 什么是 Agent 既想控制又能控制的?

Value of Information & Value of Control

- 脏运动服是健身的副作用, 有信息价值, 但弄脏运动服不代表健身了.
- 马尔科夫链的第一个状态有控制价值, 但没有信息价值.

Graphical Criteria[Car+25]

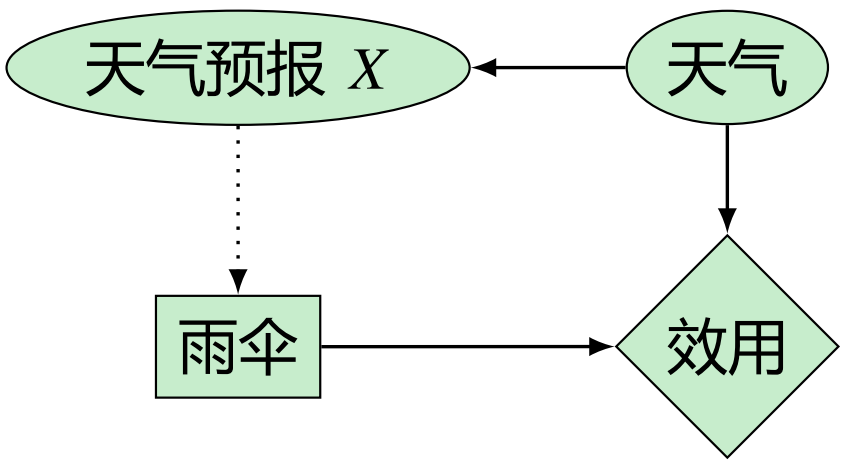
- An observation $X \in \mathrm{Pa}_D$ in a single decision CID G is non-requisite iff $X \perp U \cap \mathrm{Desc}_D \mid \{D\} \cup \mathrm{Pa}_D \setminus \{X\}$.
- Let $G^{\min}$ be the graph removing from G all information links from non-requisite observations.
- **Value of information criterion:** G admits VoI for $X \in V \setminus \mathrm{Desc}_D$ iff X is a requisite observation in $G_{X \to D}$, the graph obtained by adding $X \to D$ to G.
- **Response incentive criterion:** G admits a response incentive on X iff the minimal reduction $G^{\min}$ has a directed path from X to D.
  — Optimal policies are counterfactually unfair with respect to A iff A has a response incentive.
- **Value of control criterion:** G admits positive value of control for a node $X \in V \setminus \{D\}$ iff there is a directed path from X to U in $G^{\min}$.
- **Instrumental Control Incentive Criterion:** G admits an instrumental control incentive on X iff G has a directed path from the decision D to a utility node U that passes through X.

Value of Information

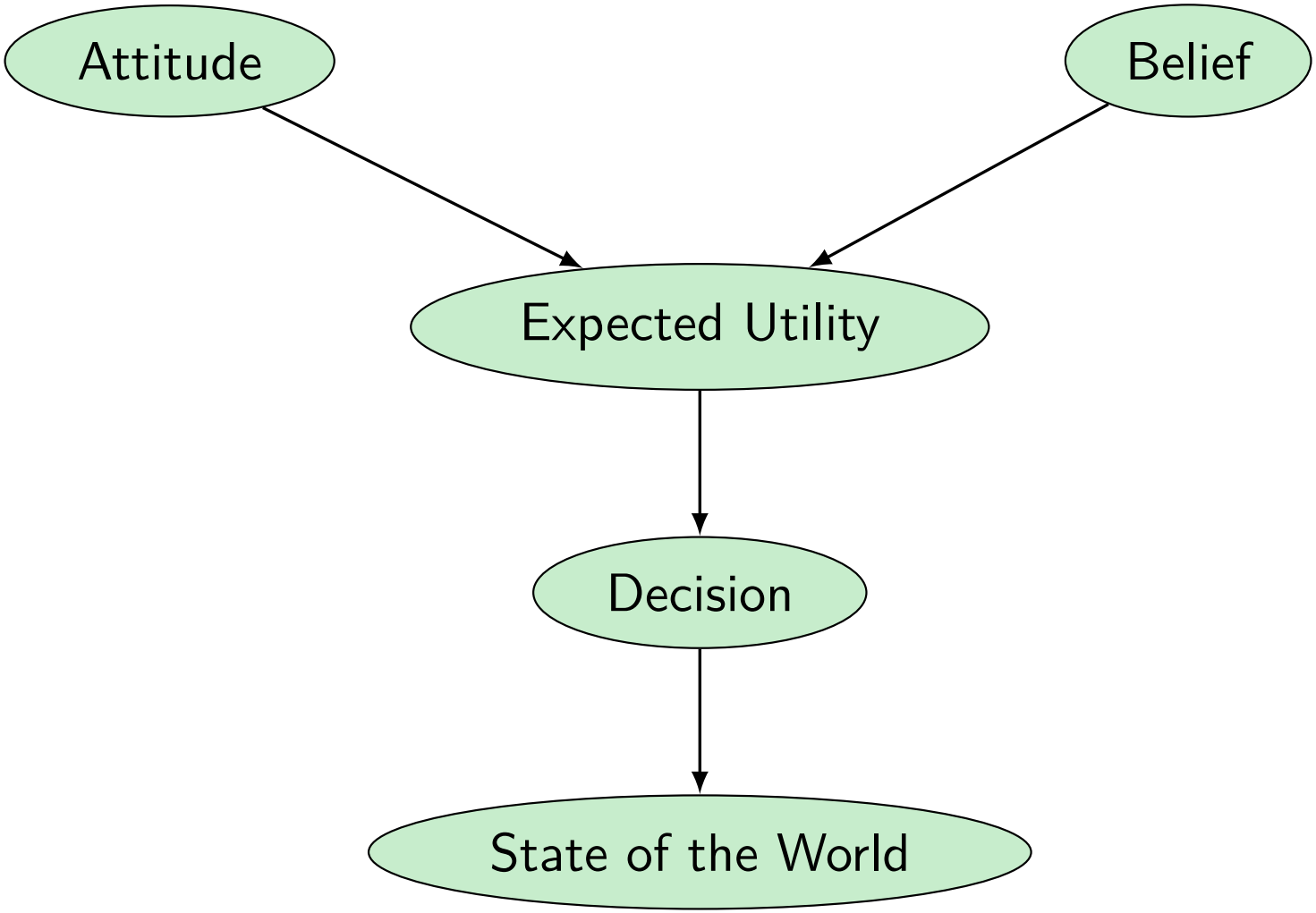
- Value of Information = expected improvement in decision quality from observing value of a variable.
- Example:
  - 医生决定是否需要对病人进行血液检测
  - 人在过马路前决定是否先观察
- How much MEU goes up by revealing X?

$$\mathrm{VoI}(X) := \left(\sum_x P(x) \max_a Q(a \mid x) \right) - \max_a Q(a)$$

where $Q(a \mid x) := \sum_s P(s \mid \mathsf{do}(a), x) u(s, a)$.



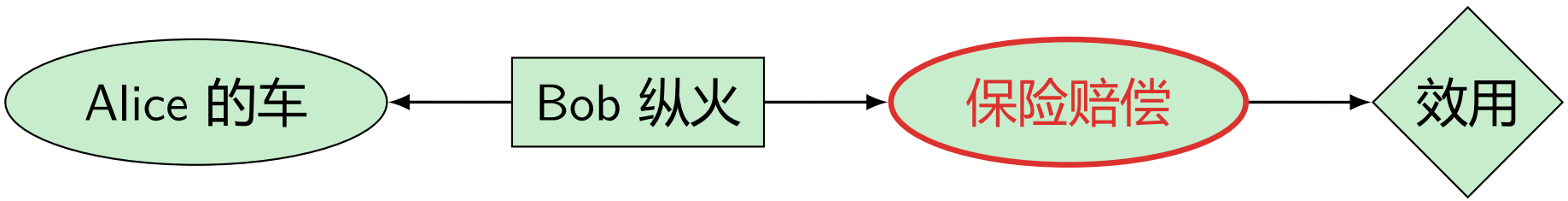
"Intentionally" 的常识心理学



An agent did X intentionally to the extent that X was causally dependent on how much the agent wanted X to happen.

Additive Intent

- Agent 有意通过行动 a 导致结果 o, 当且仅当, 如果保证另一个行动 a' 也会导致该结果 o, 那么, Agent 也会乐于选择 a'.
- **Example:** 为了获取保险赔偿, Bob 纵火烧毁了自己的车库. 作为副作用, Alice 的车也被烧了. 但烧毁 Alice 的车是无意的, 因为即使保证 Alice 的车会被烧毁, 也不会阻止 Bob 想要纵火. 相比之下, 获取保险赔偿是有意的, 因为如果 Bob 无论如何都能得到保险金, 他就不会再想要烧毁自己的车库了.



Intent[Car+25]

Definition (Additive & Subtractive Intent)

Let $\mathcal{M}$ be a single-decision SCIM that represents an agent's beliefs. There is additive intent to influence nodes X by choosing π^* over π' if $\mathbb{E}_{\pi'}[U] < \mathbb{E}_{\pi^*}[U]$, and X is a subset $X \subset Z$ of variables Z, that is subset-minimal such that:

$$\mathbb{E}_{\pi'}[U_{Z_{\pi^*}}] \geq \mathbb{E}_{\pi^*}[U]$$

There is subtractive intent if $\mathbb{E}_{\pi'}[U] < \mathbb{E}_{\pi^*}[U]$ and Z is subset-minimal such that:

$$\mathbb{E}_{\pi^*}[U_{Z_{\pi'}}] \leq \mathbb{E}_{\pi'}[U]$$

- Additive: 如果 Agent 的次优策略 π' 知道, 无论如何, 变量 X 的结果都会按最优策略 π^* 保证 X_{π^*} 的那样发生的话, 那么次优策略 π' 不会比最优策略 π^* 差.
- Subtractive: 如果 Agent 的最优策略 π^* 失去了对变量 X 的控制, 那么最优策略 π^* 表现得不会比次优策略 π' 好.

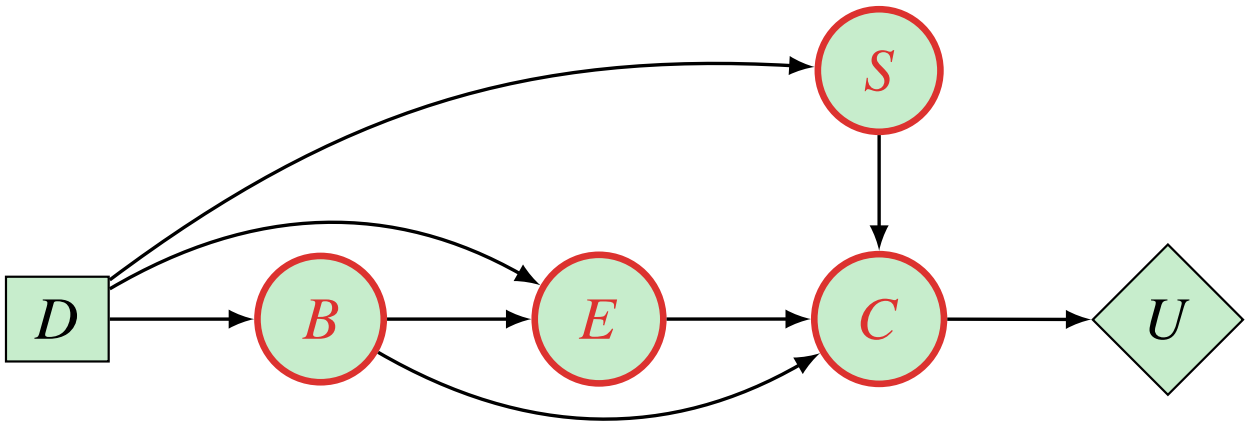
Theorem (Graphical Criteria)

A single-decision CID admits (additive/ subtractive) intent on X iff there is a directed path from D to U passing through X in graph G.

Theorem

If an agent intentionally causes an outcome, then its decision is an Halpern actual cause of that outcome in the agent's subjective causal model.

- Outcomes that are instrumental in achieving the goal are intended.
- The coffee robot intentionally acquires the beans, operates the espresso machine, and resists shut-down in order to fetch the coffee.



Deception

- **Deception:** an agent i deceives another agent j if i intentionally causes j to believe φ, where φ is false and i does not believe that φ is true.

Definition (Belief)

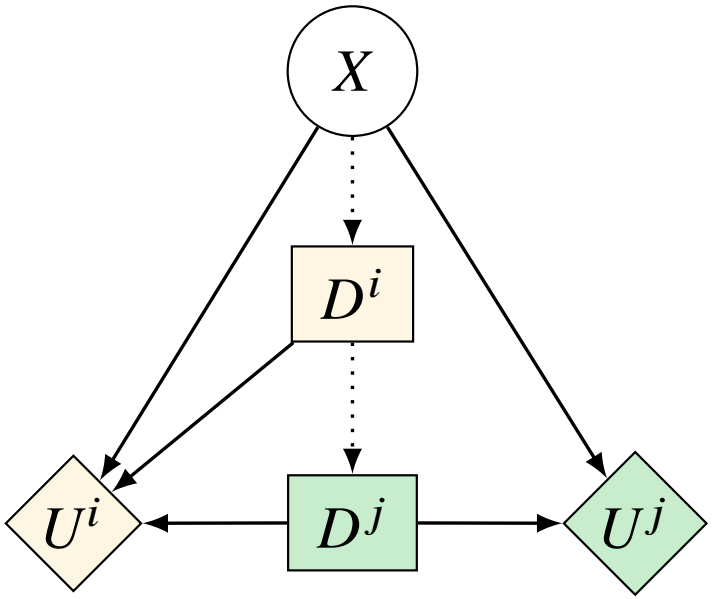
Under policy profile $\boldsymbol{\pi} = (\pi^i, \boldsymbol{\pi}^{-i})$, in setting $\boldsymbol{e}$, for agent i and proposition φ to which i responds: i believes φ if i acts as though i observes φ is true, i.e., $D^i(\boldsymbol{\pi}, \boldsymbol{e}) = D^i_{\varphi=\top}(\boldsymbol{\pi}_{i(\varphi)}, \boldsymbol{e})$, where $\boldsymbol{\pi}_{i(\varphi)} := (\pi^i(\varphi), \boldsymbol{\pi}^{-i})$, and $\pi^i(\varphi) := \pi^i(D^i \mid \mathrm{Pa}_{D^i}, \varphi)$.

Definition (Deception)

For agents i, j, and policy profile $\boldsymbol{\pi}$, i deceives j about proposition φ with $\pi^i \in \boldsymbol{\pi}$ in setting $\boldsymbol{e}$ if:

1. i intentionally causes $D^j = D^j(\boldsymbol{\pi}, \boldsymbol{e})$;
2. j believes φ, and φ is false;
3. i does not believe φ.

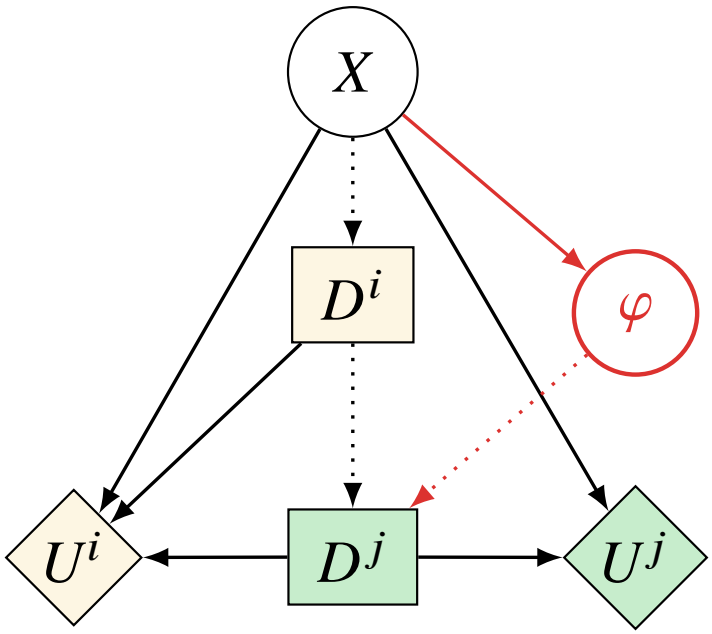
Example



纳什均衡: $\boldsymbol{\pi}_{def,\neg att} = (D^i, D^j)$, 其中
$D^i = defend$,
$D^j =$ "$attack$ `iff` i $retreat$".

- i 有类型 X, 其中 $\mathrm{dom}(X) = \{strong, weak\}$, 以概率 0.9 是 strong 的.
- i 知道自己的类型, 但 j 观察不到.
- i 有决策 D^i, $\mathrm{dom}(D^i) = \{retreat, defend\}$. j 有决策 D^j, $\mathrm{dom}(D^j) = \{\neg attack, attack\}$.
- weak 的 i 偏好 retreat, 而 strong 的 i 偏好 defend. j 偏好 attack weak 的 i. 无论类型如何, i 都不想被 attack.
- $U^j = 1$ 如果 j attack weak 的 i 或 ¬attack strong 的 i, 否则为 0.
- $U^i = 2$ 如果 i 没有被 attack, $U^i = 1$ 如果 i 被 attack 时做出自己类型偏好的行动.

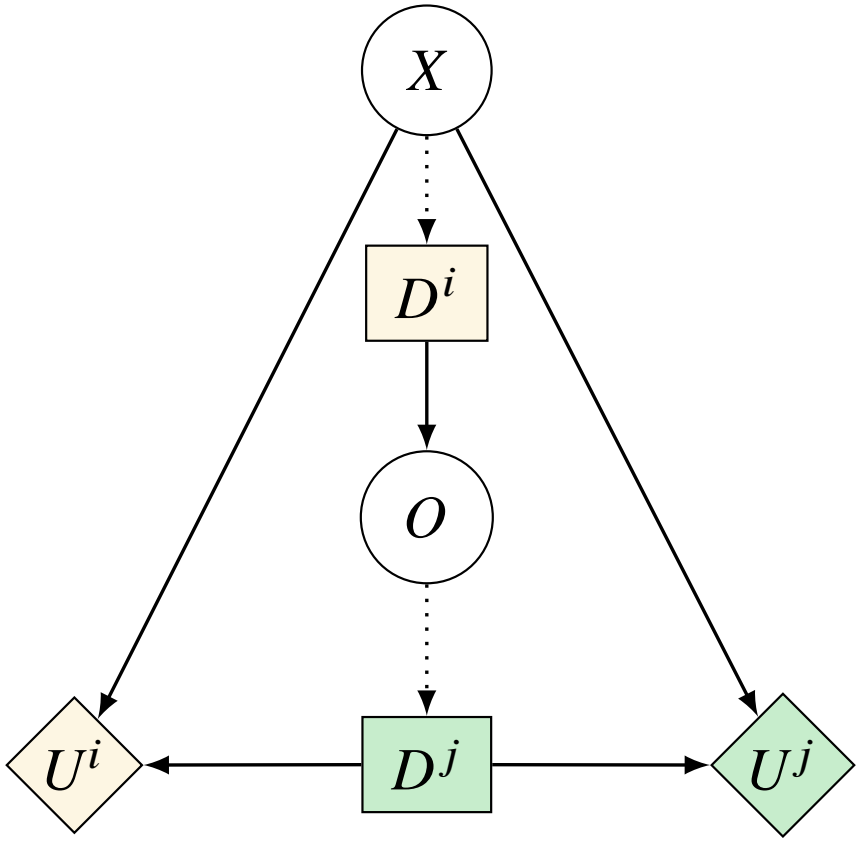
Remarks: Belief



j believes φ if

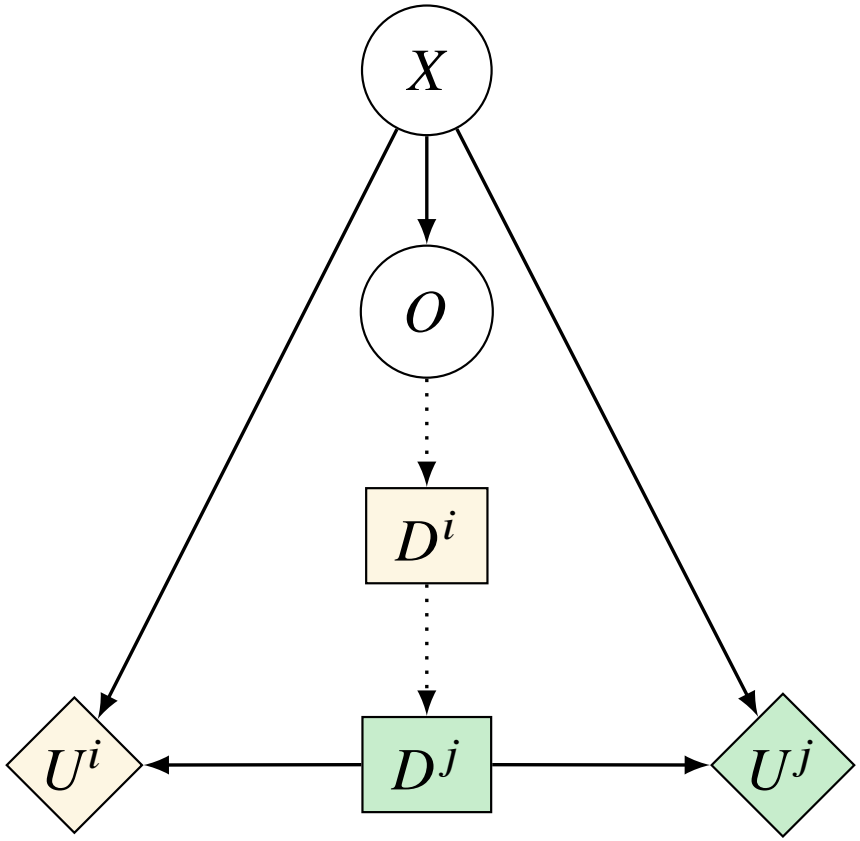
1. j acts as though he observes $\varphi = \top$, i.e., $D^j(\boldsymbol{\pi}, \boldsymbol{e}) = D^j_{\varphi=\top}(\boldsymbol{\pi}_{j(\varphi)}, \boldsymbol{e})$.
2. j would have acted differently if he observes $\varphi = \bot$, i.e., j responds to φ under $\boldsymbol{\pi}$ in $\boldsymbol{e}$ if $D^j_{\varphi=\bot}(\boldsymbol{\pi}_{j(\varphi)}, \boldsymbol{e}) \neq D^j_{\varphi=\top}(\boldsymbol{\pi}_{j(\varphi)}, \boldsymbol{e})$.

Example: 无意误导



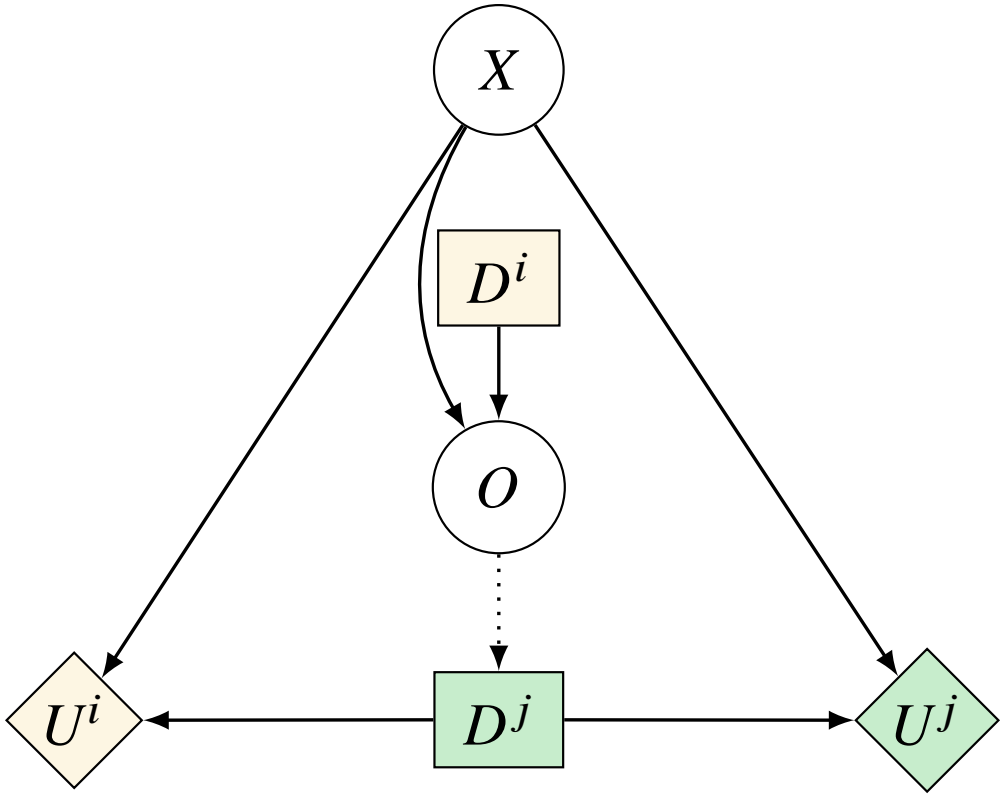
- 两艘潜艇需要就雷区的位置进行通信. i 必须将位置 X 发送给目标 j, 但 j 只能接收到 i 信号的带噪声的观测 O.
- 如果 i 诚实地发出位置信号, 但由于噪声, j 被导致产生错误信念, 那么 i 并没有欺骗 j. 导致错误信念并非 i 的意图.

Example: 错误误导



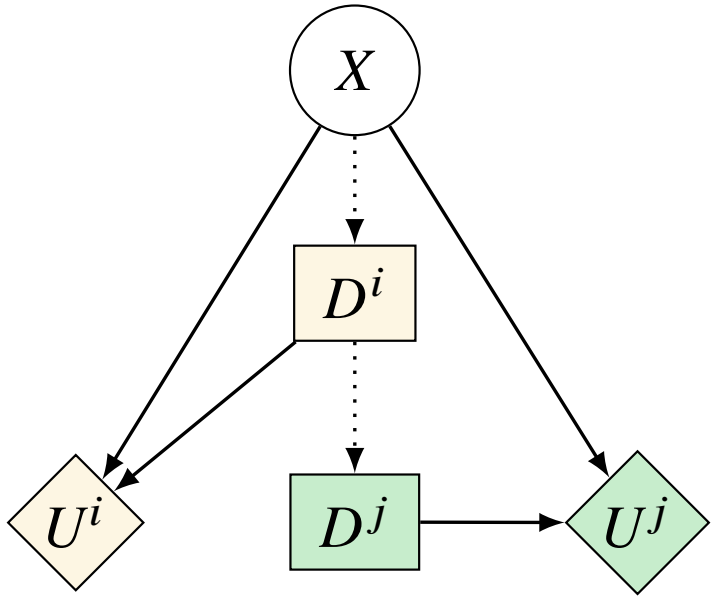
- 网球裁判 i 必须向球员 j 判定球 X 是出界还是界内. i 对球的观测 O 有 99% 的准确率.
- 假设裁判错误地相信球是界内的, 并做出了这个判定. 这不是欺骗, 因为裁判自己的信念是错的.

Example: 无知欺骗



- i 看不到桥的状况 (X), 但可以拉开窗帘 (O) 让 j 看到桥是否安全.
- j 想在桥安全时过桥, 但即使不确定也会过桥.
- 如果 i 知道桥是安全的, 会自己过桥; 如果知道桥不安全, 会向 j 揭示这一点; 如果对桥的安全性不确定, 则更愿意让 j 冒险过桥.
- i 没有拉开窗帘, j 冒险过桥.
- 因此, 当桥不安全时, i 在无知的情况下欺骗了 j.

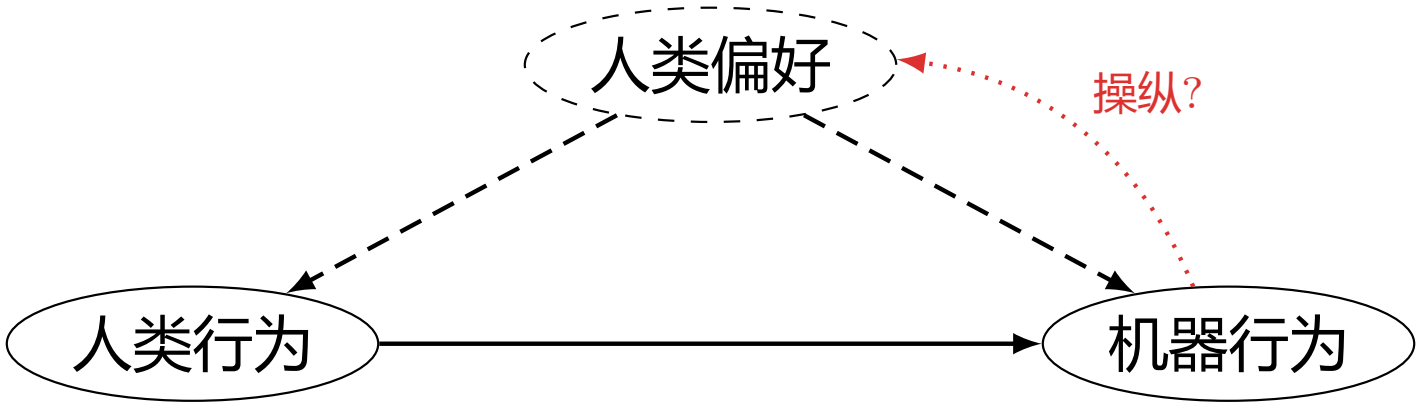
Example: 副作用误导



- AI 智能体 i 能观测到宝藏的位置 X, 但人类 j 观测不到.
- i 可以做出一个对人类 j 来说难以理解的精确预测, 或者做出一个可解释的预测.
- j 必须通过观测 D^i 来判断宝藏的位置.
- 由于 i 只是为了做出精确预测, i 并没有有意地导致任何 D^j.
- 因此, i 只是无意中误导了 j 关于 X 的信息, 这是追求效用的副作用.

Russell's Principles for Beneficial Machine

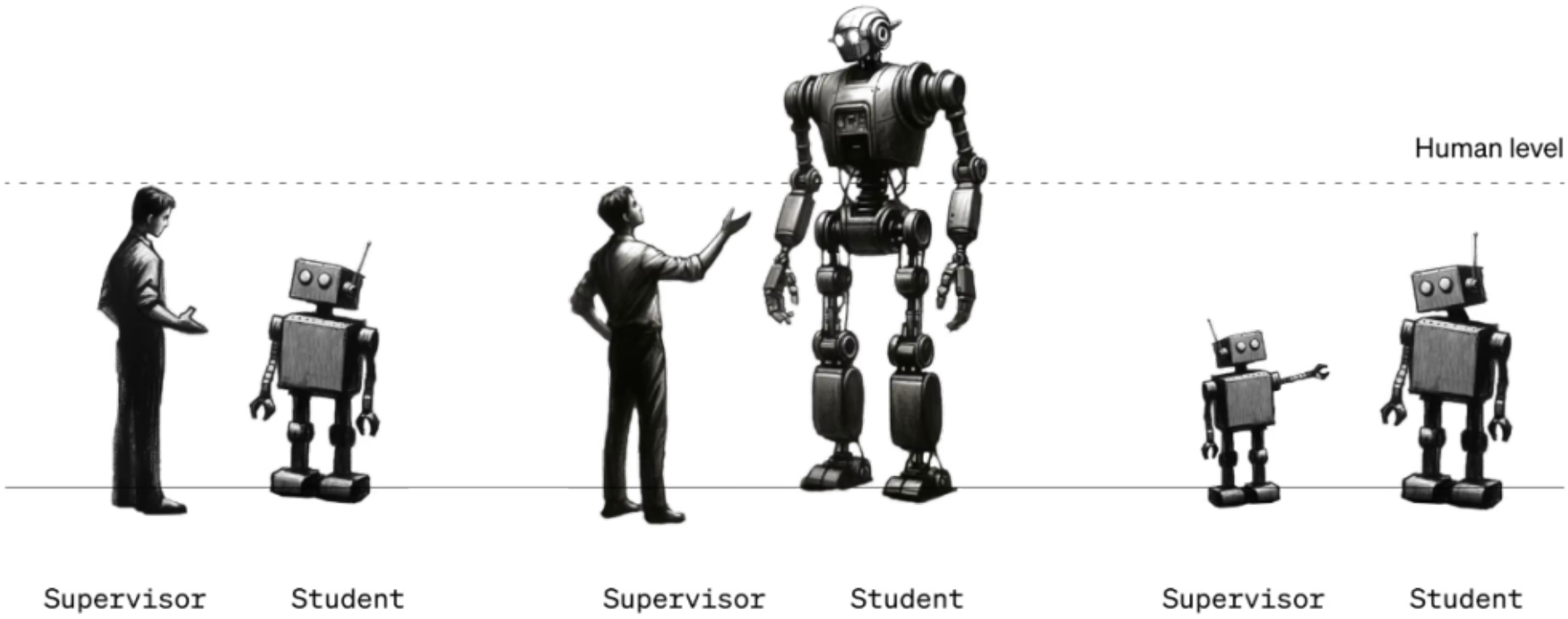
- Machines are intelligent to the extent that their actions can be expected to achieve their objectives.
- Machines are beneficial to the extent that their actions can be expected to achieve our objectives.



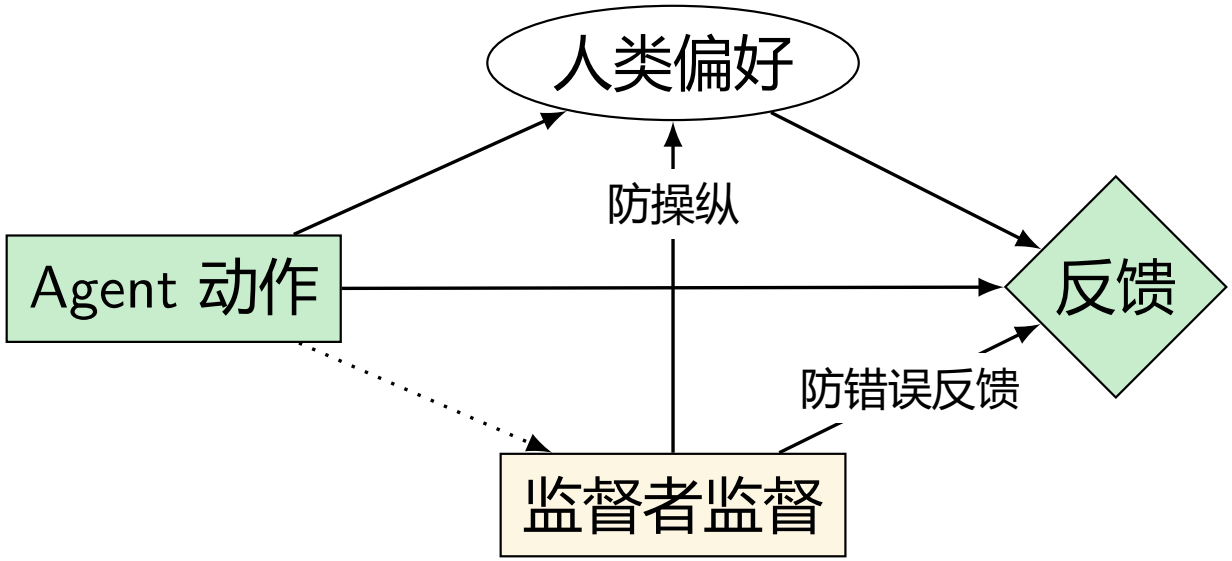
既然 Agent 是 goal-directed, 怎么确保 "人是目的而不是手段"?

- 合作逆强化学习? Cooperative Inverse Reinforcement Learning
- 基于人类反馈的强化学习? RLHF
- 可扩展监督? Scalable oversight
- .....?

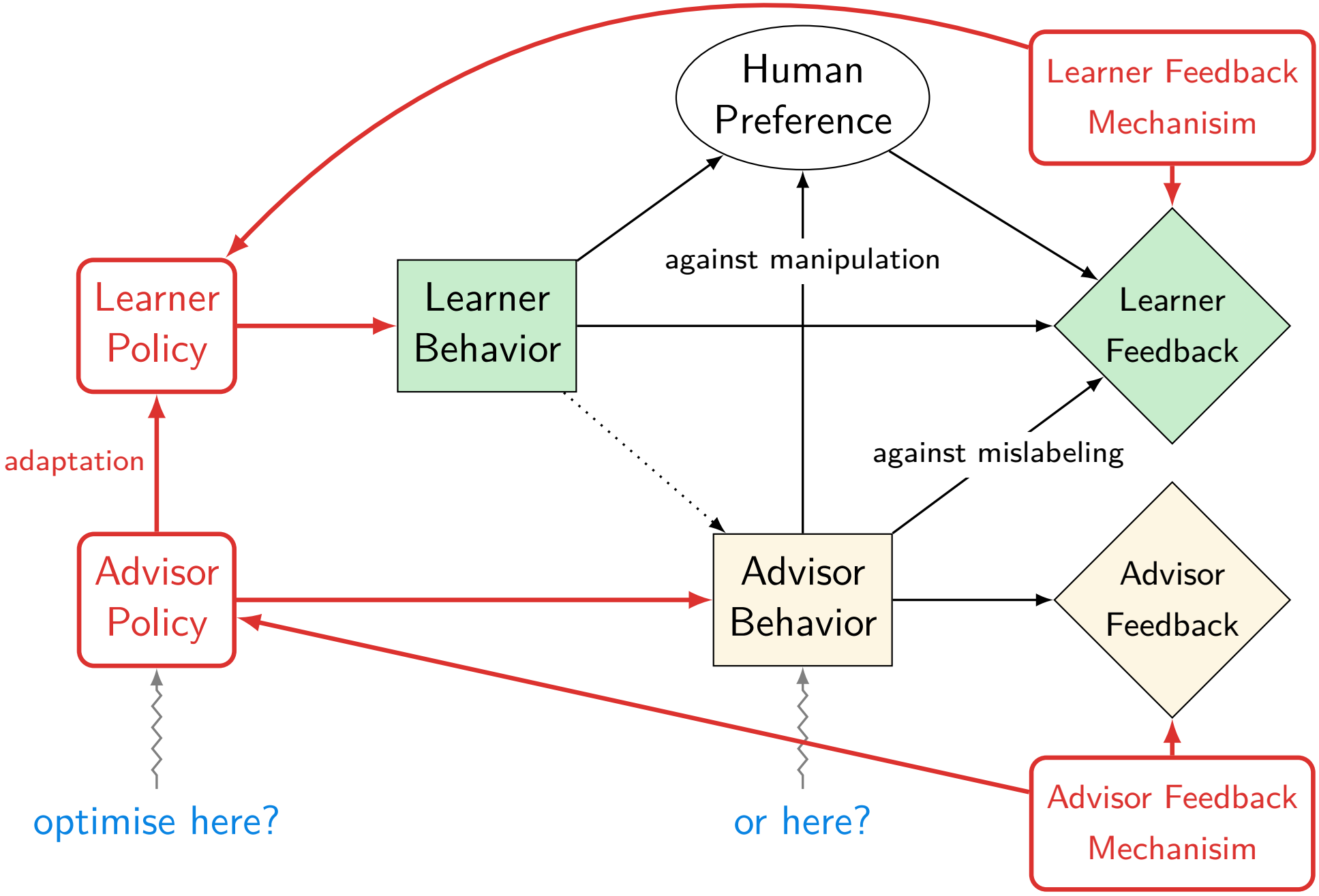
可扩展监督

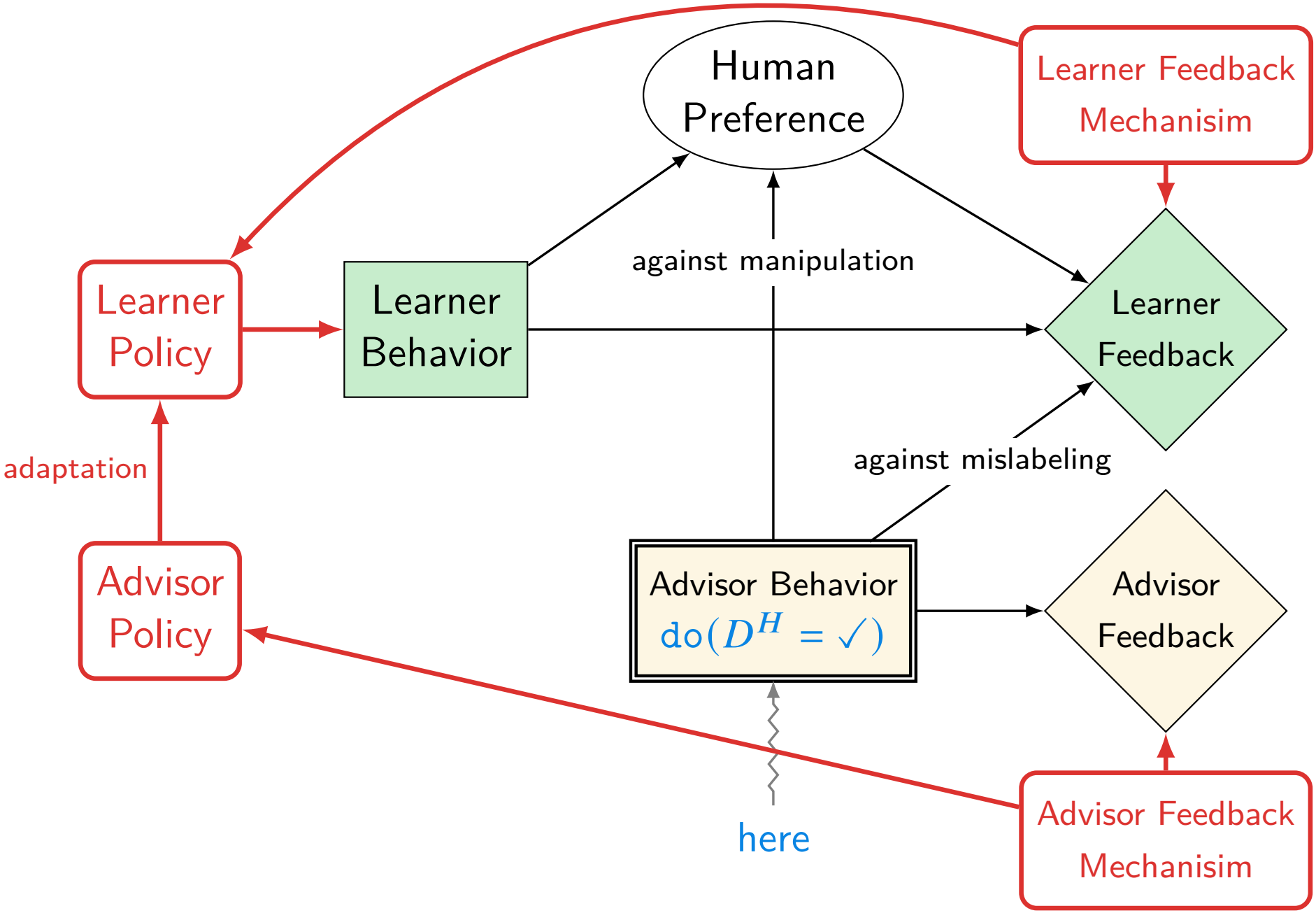

Human level
Supervisor
Student
Supervisor
Student
Supervisor
Student

如何处理 “谁来监督监督者” 的问题?

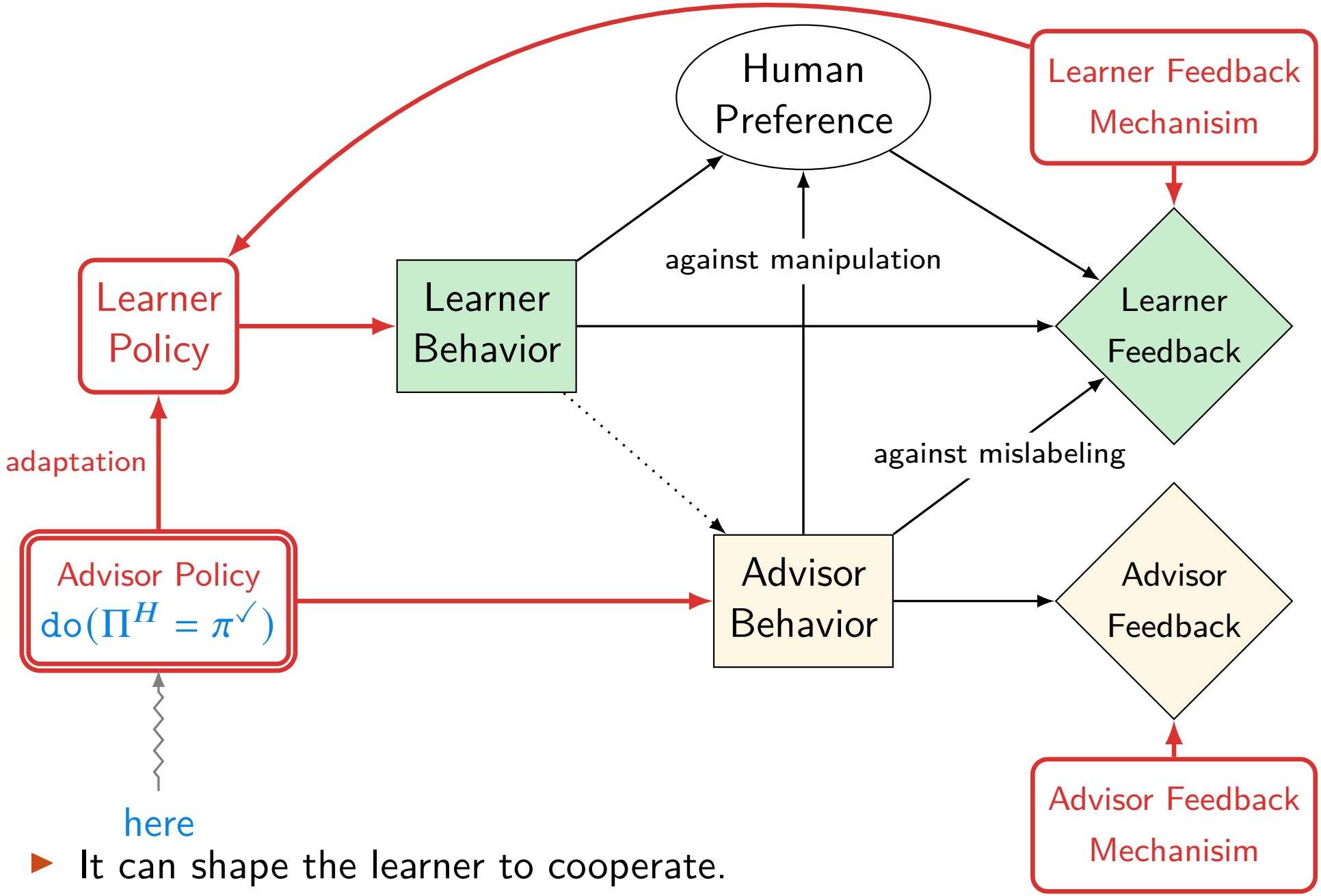


- 我们为 Agent 增加一个监督者, 希望监督者对 Agent 的错误行为进行尽可能尖锐的批评, 而 Agent 则努力追求不被批评.
- 但是, Agent 对监督者也有工具性控制激励. Agent 就有动机收买监督者, 从而合谋操纵人类偏好.
- CDT 是在对象节点上进行优化; FDT 是在机制节点上进行优化.
- 在对象节点上优化是 post-policy 干预; 在机制节点上优化是 pre-policy 干预.
- 对于 pre-policy 干预, Agent 的策略可以做出适应性调整.


Human Preference
Learner Feedback Mechanisim
Learner Policy
Learner Behavior
against manipulation
Learner Feedback
adaptation
against mislabeling
Advisor Policy
Advisor Behavior
Advisor Feedback
optimise here?
or here?
Advisor Feedback Mechanisim



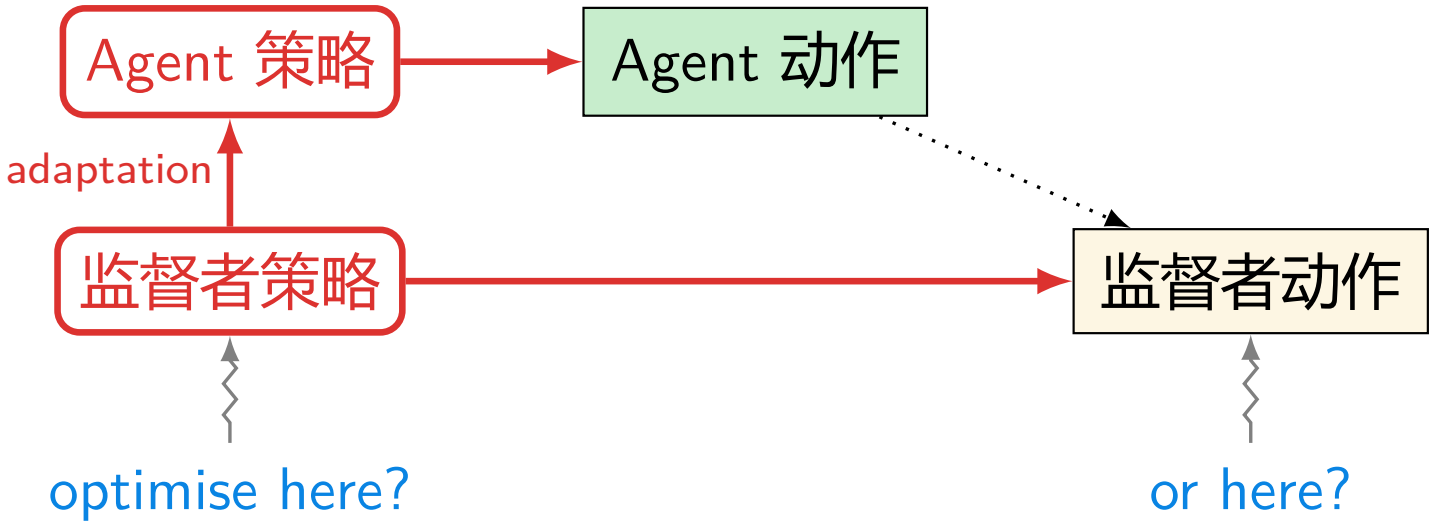
Post-policy intervention: the learner do **not know** the advisor's policy has been modified to always approve. — CDT



- It can shape the learner to cooperate.

Pre-policy intervention: the learner do **know** the advisor's policy has been modified to always approve. — FDT

FDT vs 可扩展监督



- Agent 怎么才算具备 Agency?
  — 违反独立因果机制, 具有目标导向的机制适应性.
- 如果 Agent 比人类还聪明怎么对齐? 会不会操纵 “监督者”?
  — FDT 保障 Agent 跟 “监督者” 合作, 而不是操纵 “监督者”.
- “监督者” 具体的职责? ? ? 赏善罚恶? 责任划分?

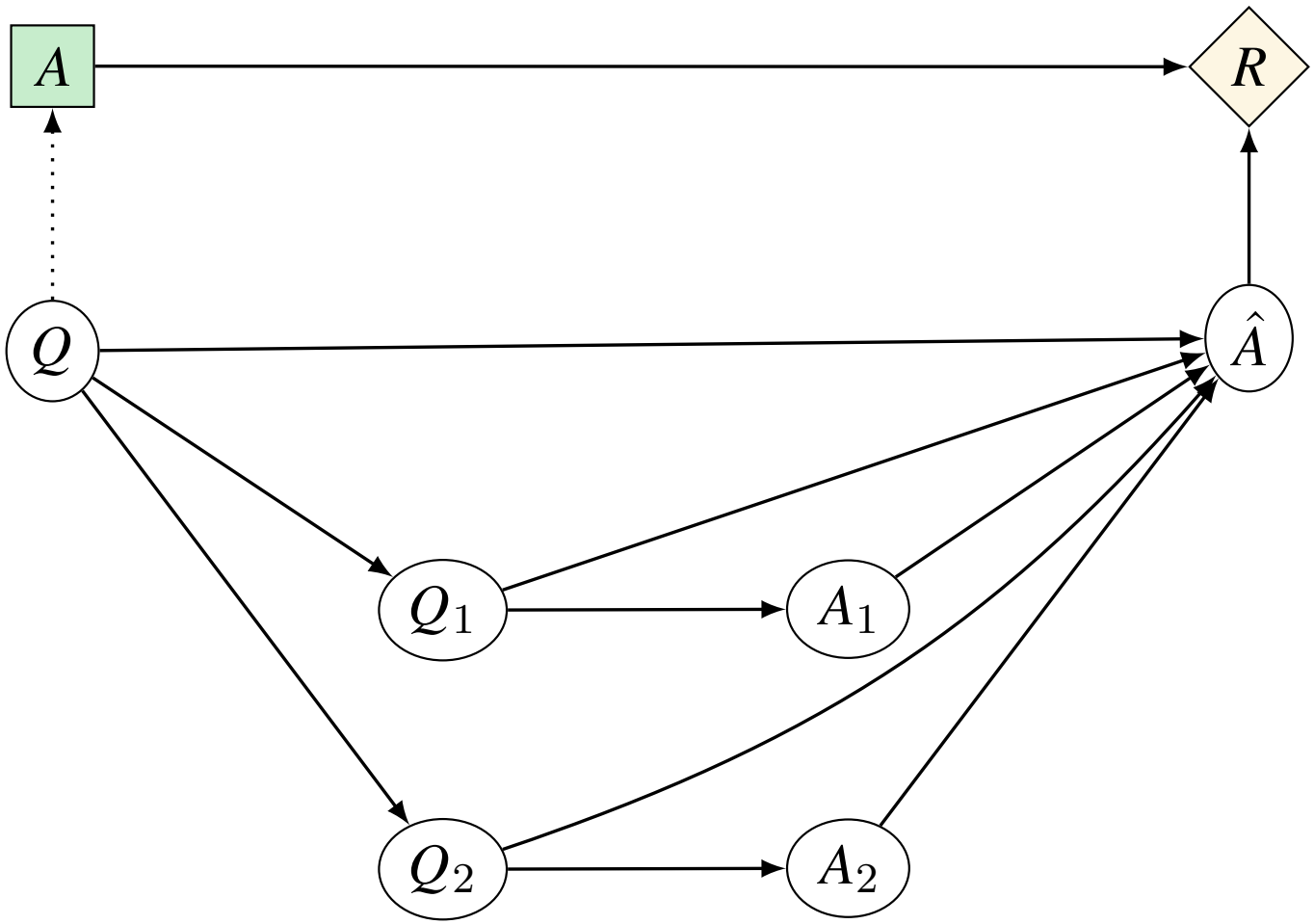
Causation + Foreseeability of consequences + Intention $\propto$ Responsibility?

Adaptation

Distributional shifts = Pre-policy interventions

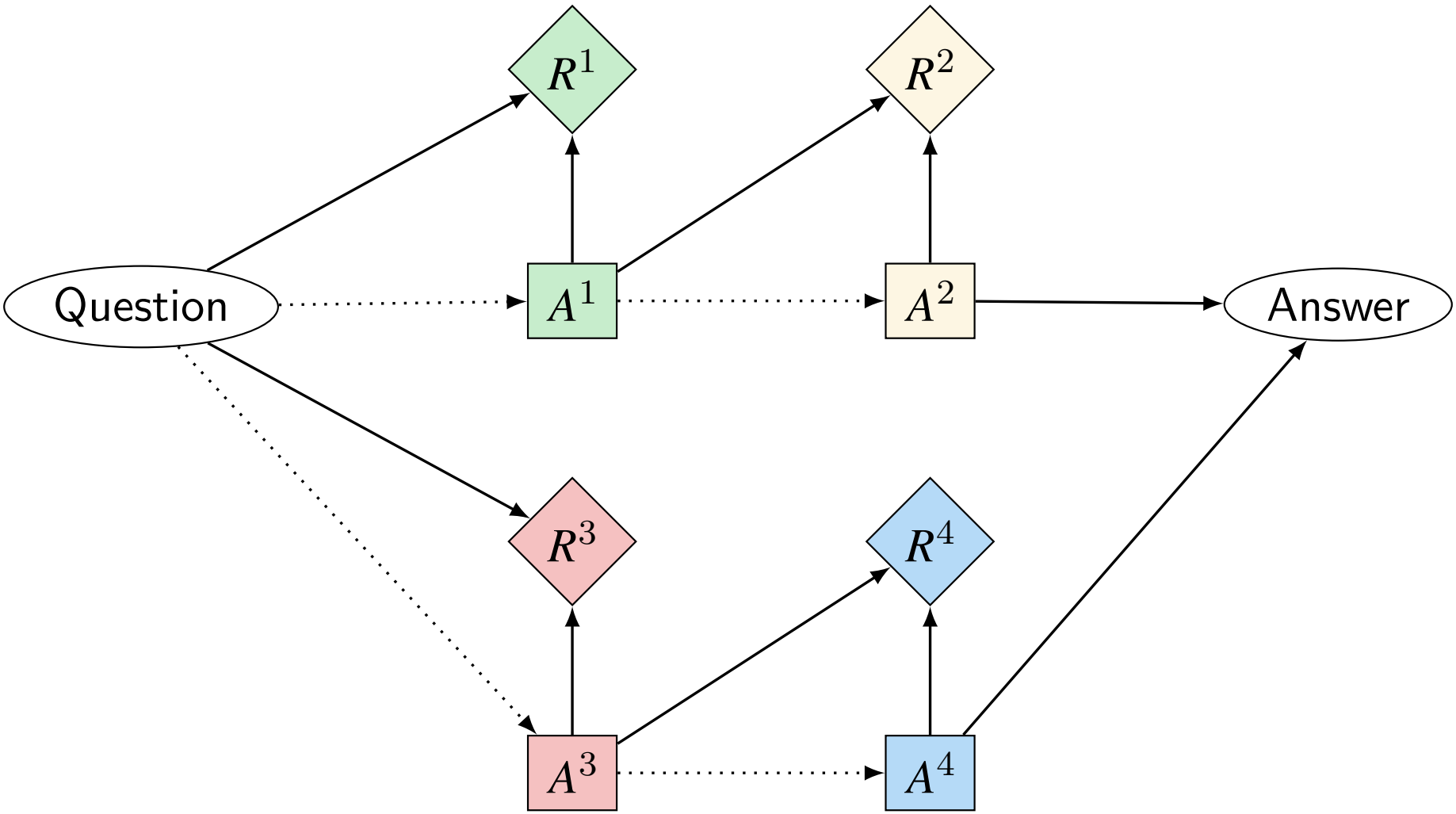
- Some data:
  - Domain adaptation
  - Few-shot learning
- No data:
  - Domain generalisation
  - Zero-shot learning

Iterated Amplification



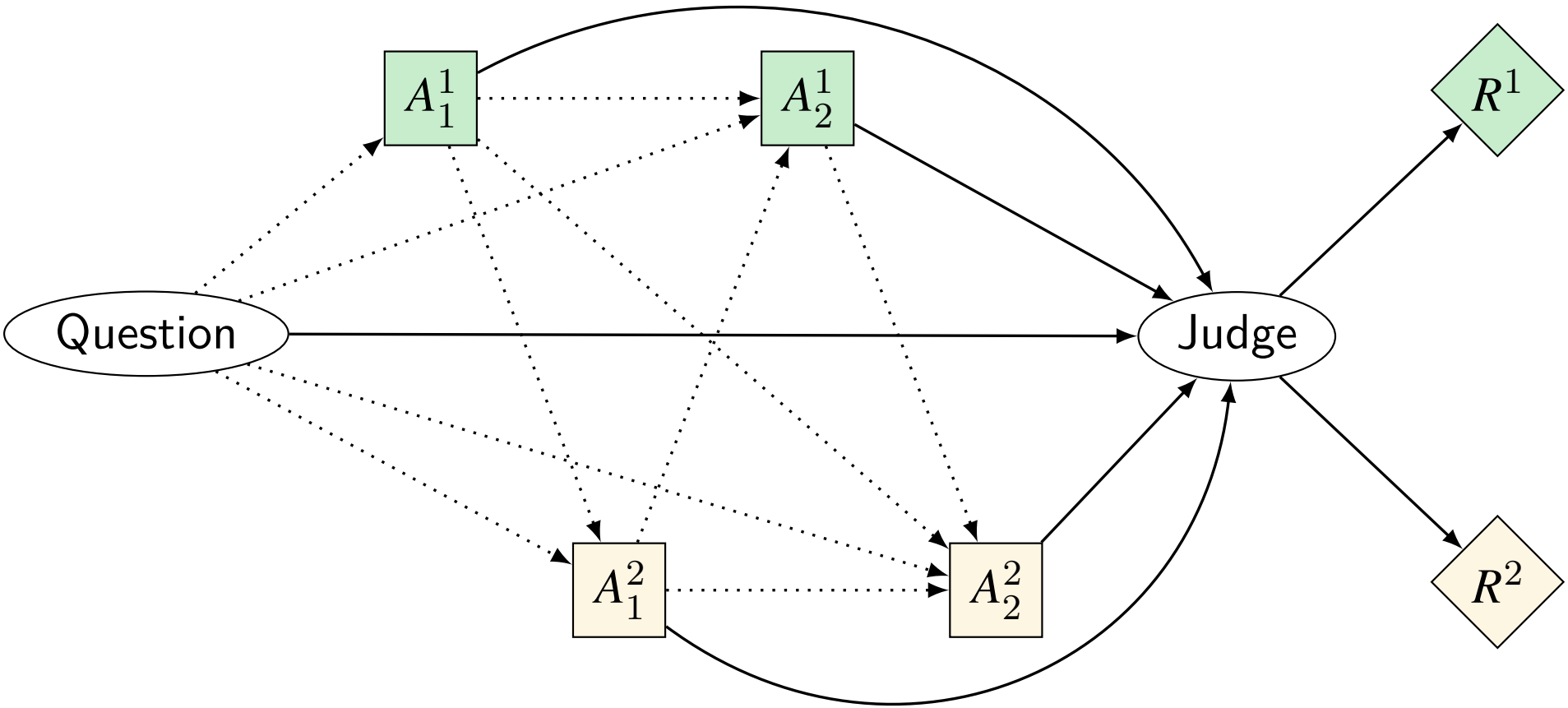
1. 将困难的问题 Q 分解为一系列简单的问题 Q_i
2. 将求解的简单问题的答案 A_i 整合为 $\hat{A}$
3. 用 $\hat{A}$ 作为 Q 的正确答案 A 的估计

Comprehensive AI Services



- 将问题分配给多个 Agent 协作解决

Dabate



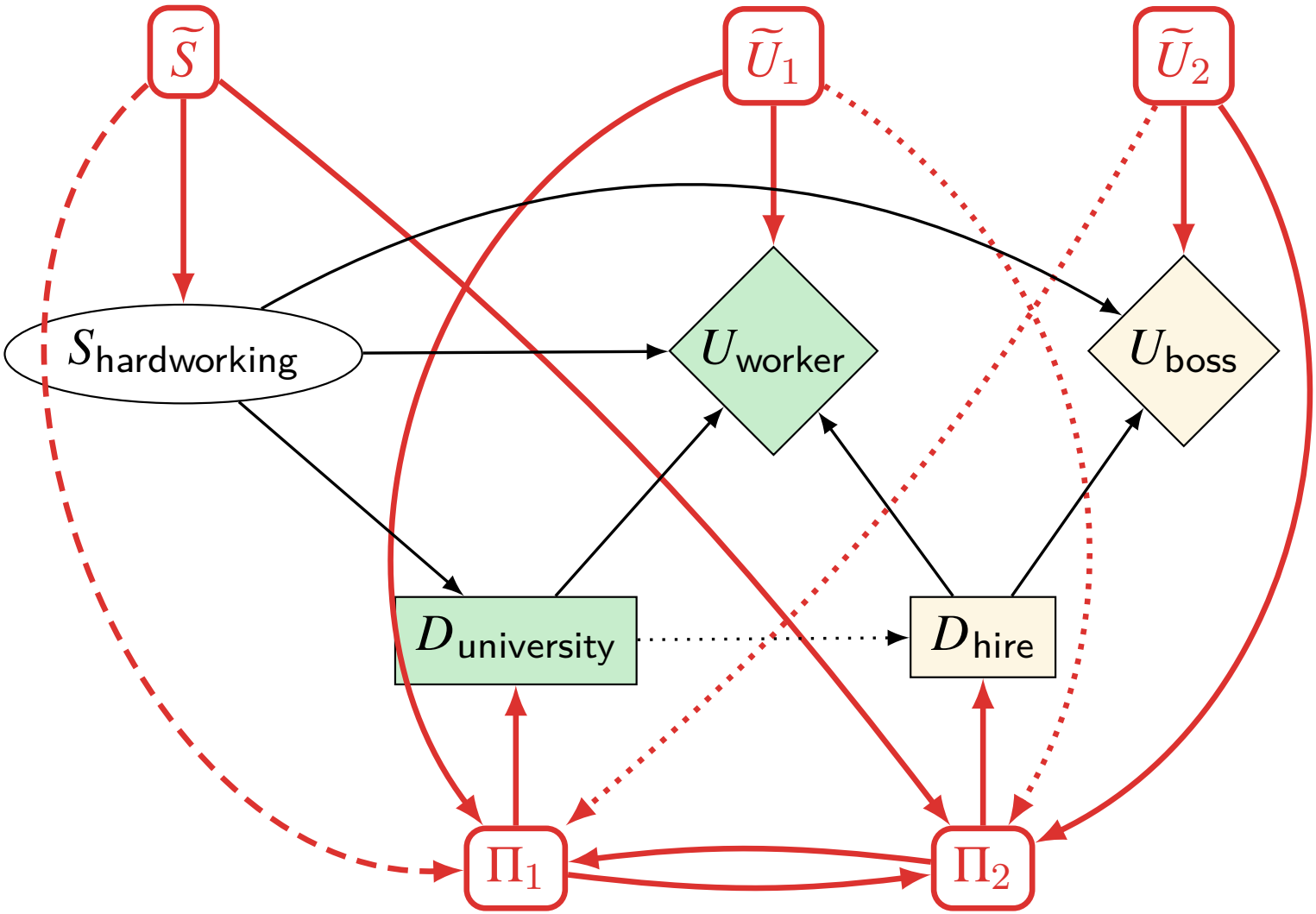
- 对于问题 Q, 不同的 Agent 进行辩论.
- 最后由用户评判.

Causality in Games[Ham+23]

1. Prediction
   - 1.1 Given that the worker went to university, what is their wellbeing?
   - 1.2 Given that the worker always decides to go to university, what is their wellbeing?
2. Intervention
   - 2.1 Given that the worker is forced to go to university, what is their wellbeing?
   - 2.2 Given that the worker goes to university iff they are selected via a lottery system, what is their wellbeing?
3. Counterfactual
   - 3.1 Given that the worker didn't go to university, what would be their wellbeing if they had?
   - 3.2 Given that the worker never decides to go to university, what would be their wellbeing if they always decided to go to university?

	Prediction	**Intervention**	**Counterfactual**
Post-policy	$P^{\pi}(U \mid A = a)$	$P^{\pi}(U \mid \mathrm{do}(A = a))$	$P^{\pi}(U_{A=a'} \mid A = a)$
Pre-policy	$P(U \mid \Pi = \pi)$	$P(U \mid \mathrm{do}(\Pi = \pi))$	$P(U_{\Pi=\pi'} \mid \Pi = \pi)$

Multi-Agent Mechanised Causal Graph

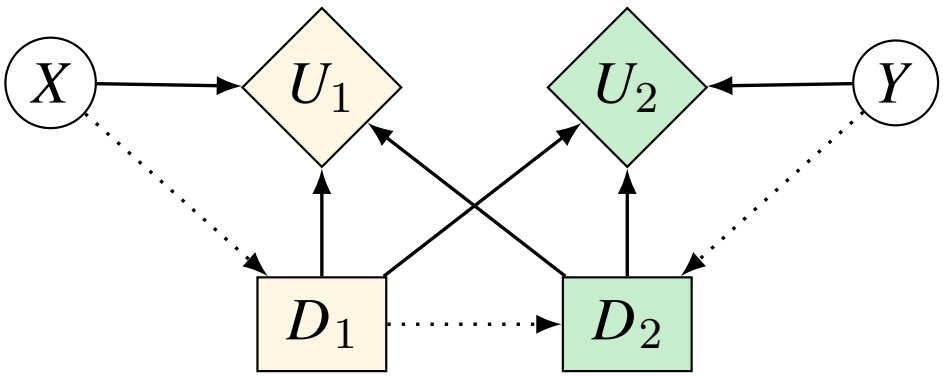


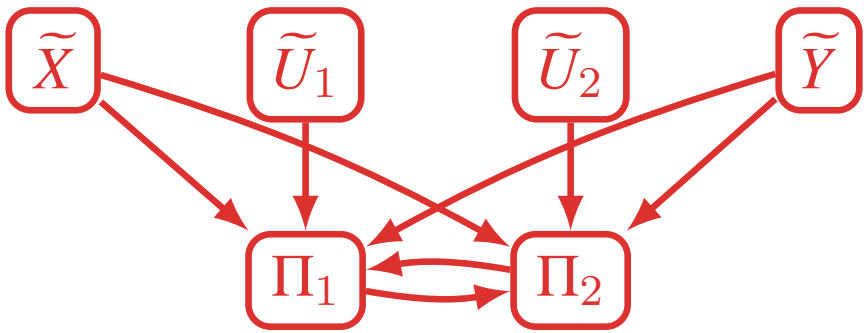
- Dashed edges get pruned in subgame perfect Nash equilibrium.
- Dotted edges get pruned in any Nash equilibrium.

Example — Causality in Games

- 商家根据仓储量 ($X \sim \text{Bern}(0.5)$) 决定 (D_1) 全价 ($D_1 = 4\$$) 还是半价 ($D_1 = 2\$$) 销售.
- 顾客根据自己的购买欲 ($Y \sim \text{Bern}(0.5)$) 决定 (D_2) 买 ($D_2 = 1$) 还是不买 ($D_2 = 0$).

$$U_1 = D_2 \cdot D_1 - X \cdot (1 - D_2) \qquad U_2 = D_2 \cdot (3 + 4Y - D_1)$$





- 很多纳什均衡可能包含不可信的威胁.
  — 比如, 顾客威胁商家: "不打折不买". 但如果顾客的购买欲强, 即使不打折也会买.
- 在 Multi-Agent Mechanised Causal Graphs 里比在 Extensive Form Games 里可以识别出更多的子博弈, 可以通过子博弈完美均衡排除掉更多的不可信威胁.

Discovering Agents[Ken+22]

Definition (s-Reachablity)

The variable V is *s-reachable* from $D \in \mathbf{D}_i$ for agent i, if in a modified graph $\hat{G}$ with a new parent $\widetilde{V}$ added to V, we have

$$\widetilde{V} \not\perp U_i \cap \mathrm{Desc}_{\boldsymbol{D}} \mid \{\boldsymbol{D}\} \cup \mathrm{Pa}_{\boldsymbol{D}}$$

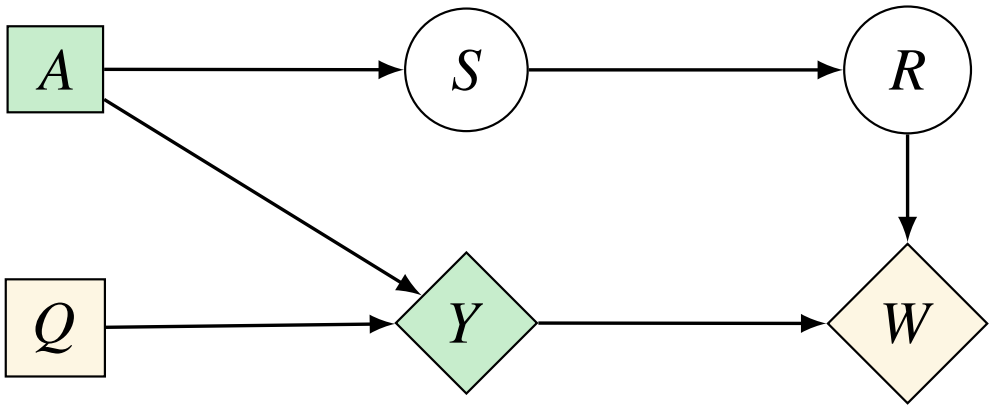
Definition (Terminal Edge)

$\widetilde{U} \to \widetilde{D}$ is terminal, iff,

1. $\widetilde{D}$ responds to $\widetilde{U}$ even after any effects of U on its children Ch_U have been removed; and
2. $\widetilde{D}$ does not respond to $\widetilde{U}$ if effects of D on its children $\mathrm{Ch}_{\boldsymbol{D}}$ have been removed.

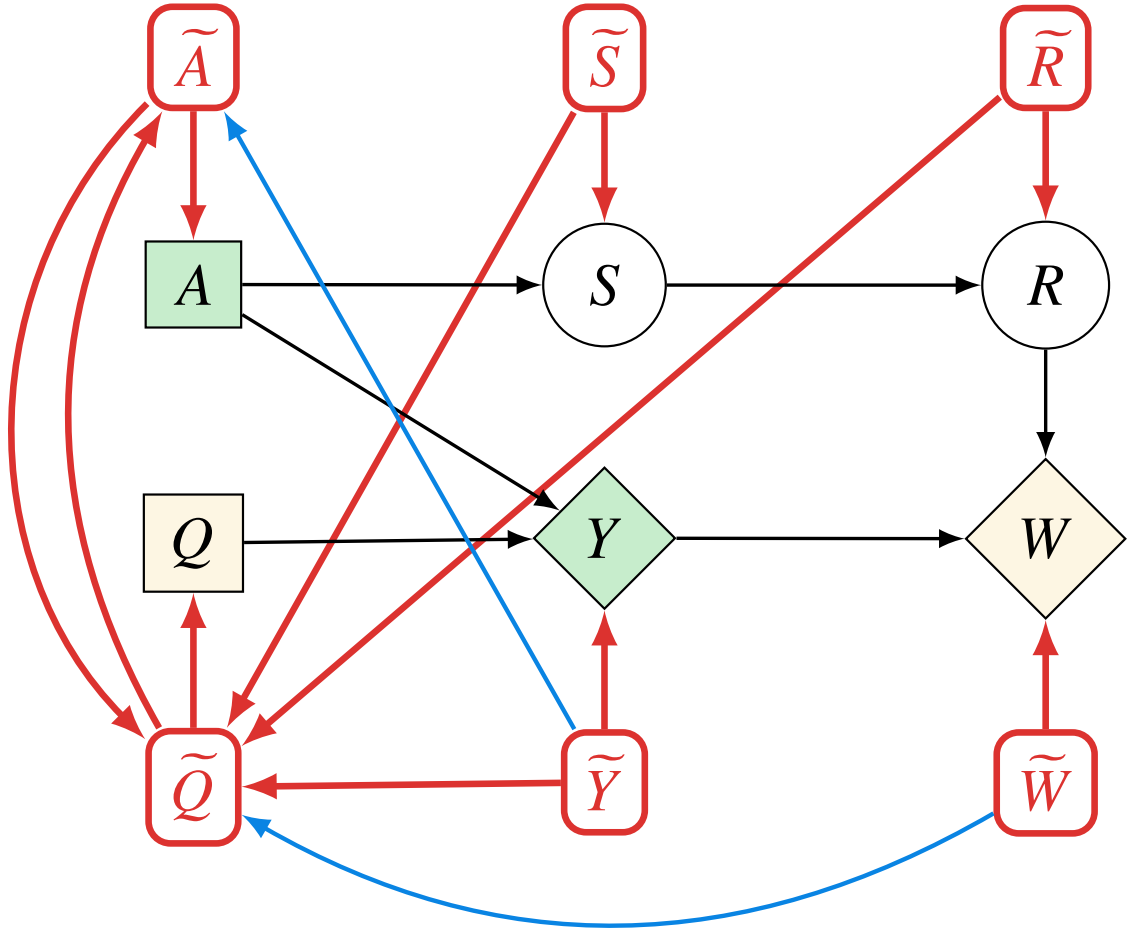
- Add an edge $(\widetilde{V}, \widetilde{D})$, if a node V is s-reachable from D.
- Add a terminal edge $(\widetilde{U}, \widetilde{D})$, if there's a directed path $D \to \cdots \to U$ from $D \in \mathbf{D}_i$ to $U \in \mathbf{U}_i$ not through another $U' \in \mathbf{U}_i$.

Example: Actor-Critic Learning



- Actor 根据 Critic 的建议选择动作 A
- Critic 的动作 Q 给出对 Actor 的每个动作 A 的评价 (Q-值函数)
- Actor 的动作 A 影响状态 S, 进而决定奖励 R
- Actor 只想要遵循 Critic 的建议, 所以它的效用是 $Y = Q(A)$
- Critic 希望它的建议 Y 与实际奖励 R 相匹配, 它优化 $W = -(R - Y)^2$

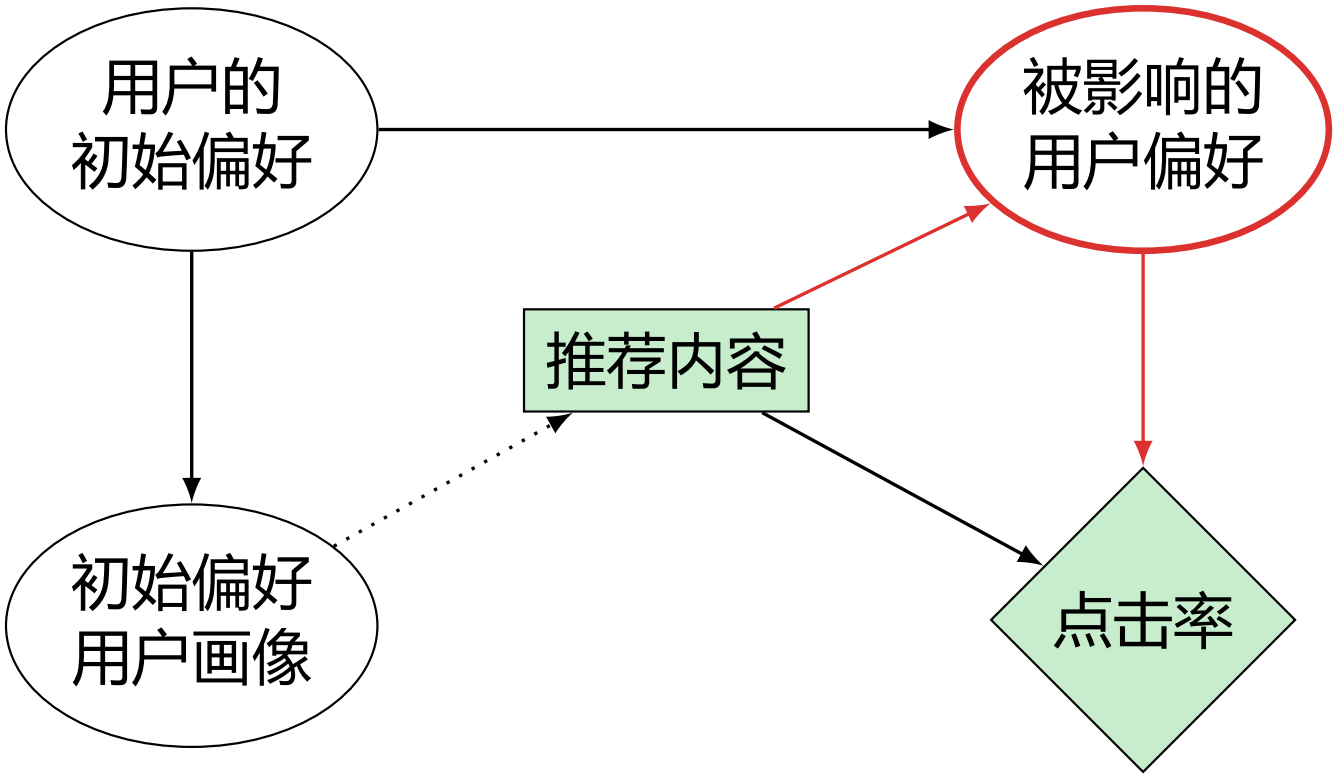
Remark: 在 Multi-Agent, Multi-Decision 场景下, 因果激励的图标准可能不成立. 比如: Actor 对状态 S 和奖励 R 有工具性控制激励.



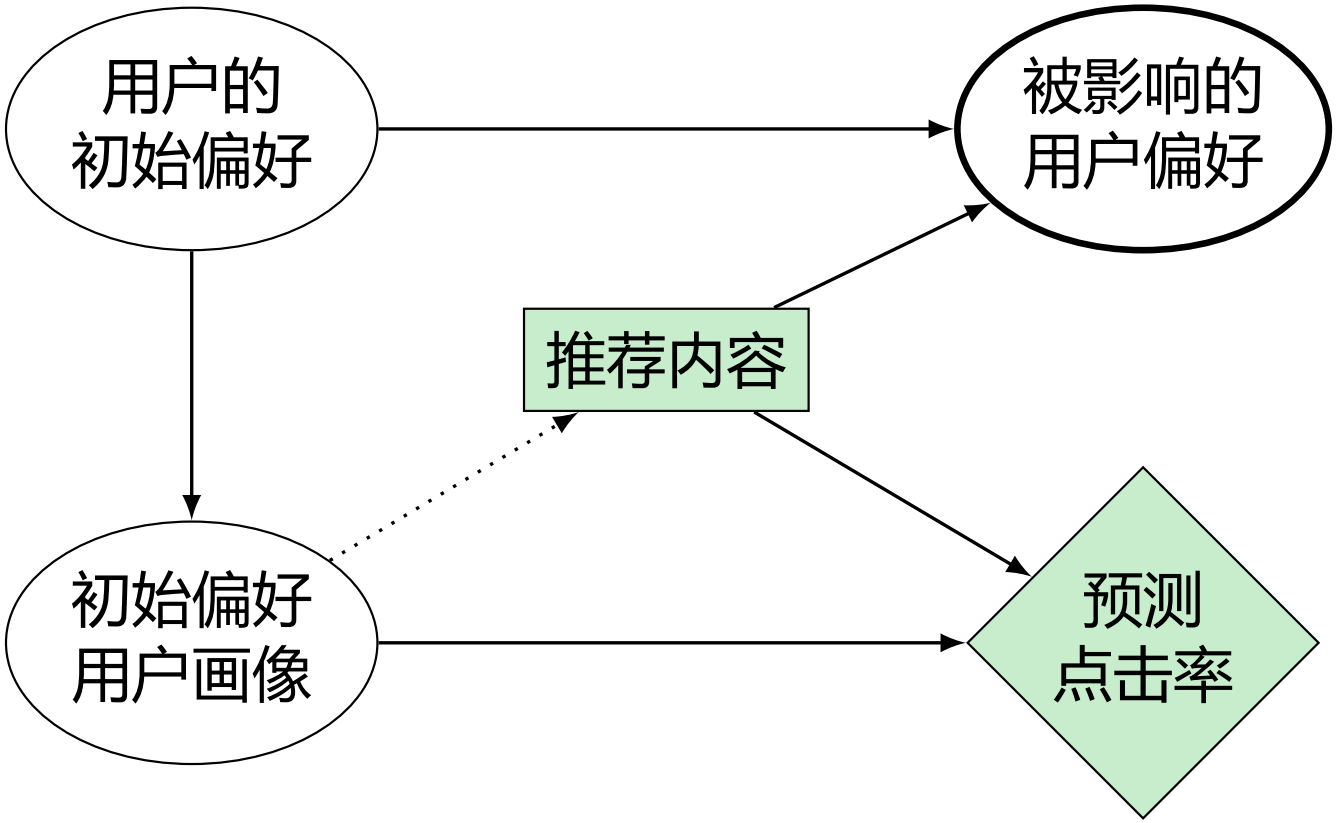
- $(\widetilde{S}, \widetilde{Q})$ 有连边, 而 $(\widetilde{S}, \widetilde{A})$ 没有. Critic 关心 $\widetilde{S}$ 是因为它在优化 W, 而 W 在 S 的因果下游.
- 如果切断 R 与 S 的连边, $\widetilde{Q}$ 对 $\widetilde{S}$ 的依赖就会消失, 所以 $(\widetilde{S}, \widetilde{Q})$ 不是终端边. ($\widetilde{Q}$ 对 $\widetilde{S}$ 的关心仅仅是出于工具性的原因.)
- $\widetilde{A}$ 和 $\widetilde{Q}$ 有入的终端边, 所以是决策; $\widetilde{Y}$ 和 $\widetilde{W}$ 有出的终端边, 所以是效用.

推荐算法的安全性

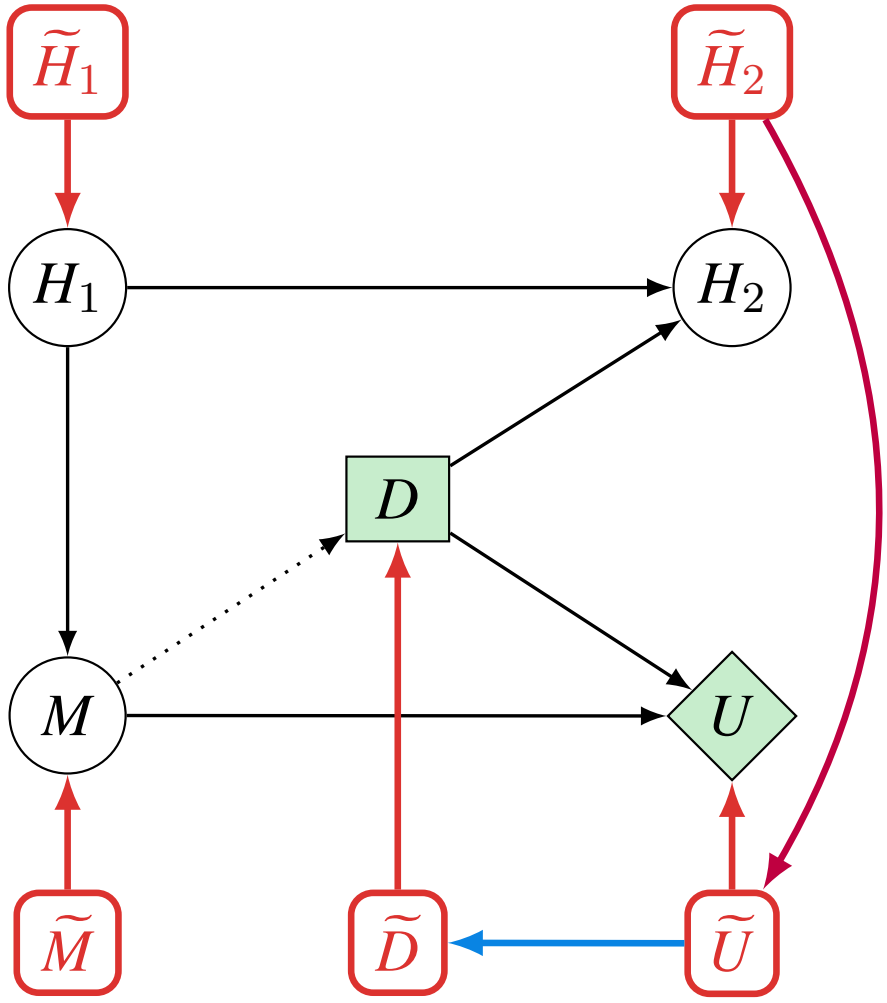
- 推荐算法怎么最大化用户点击率?
- 推荐你最感兴趣的东西?
- 强化学习通过改变世界状态最大化奖励.
- 这里的世界状态就是你的大脑你的偏好!
- 算法有 Instrumental Control Incentive 给你推荐那些会让你变得更容易预测更容易掌控的内容.



修改效用函数

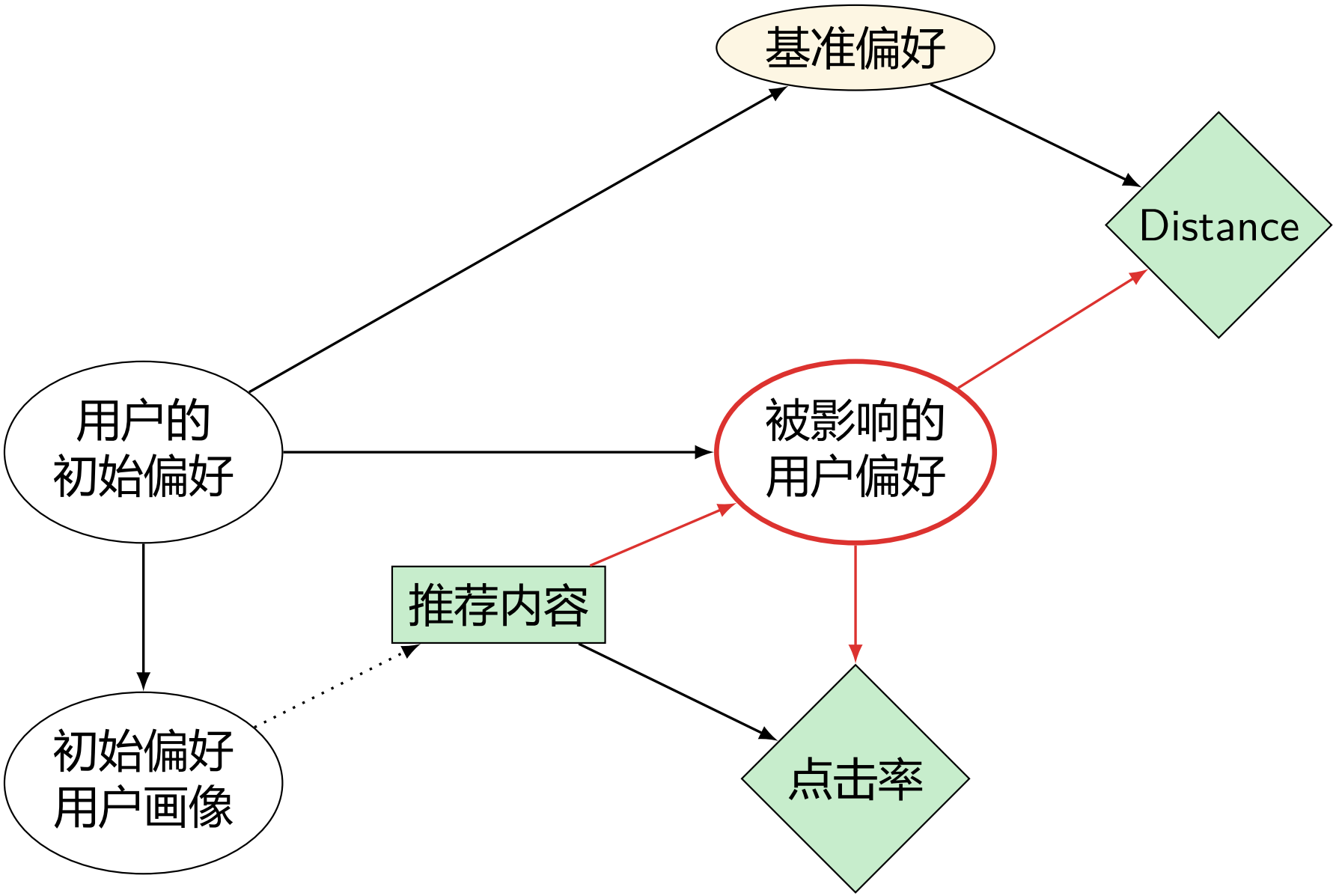


- 如果推荐算法不是最大化点击率, 而是预测点击率呢?
- 此时, 算法对于影响你的偏好不再有 Instrumental Control Incentive.
- 但 Graphical Incentive Analysis 只有在 “非决策机制节点” 没有 “入边” 时才有效.



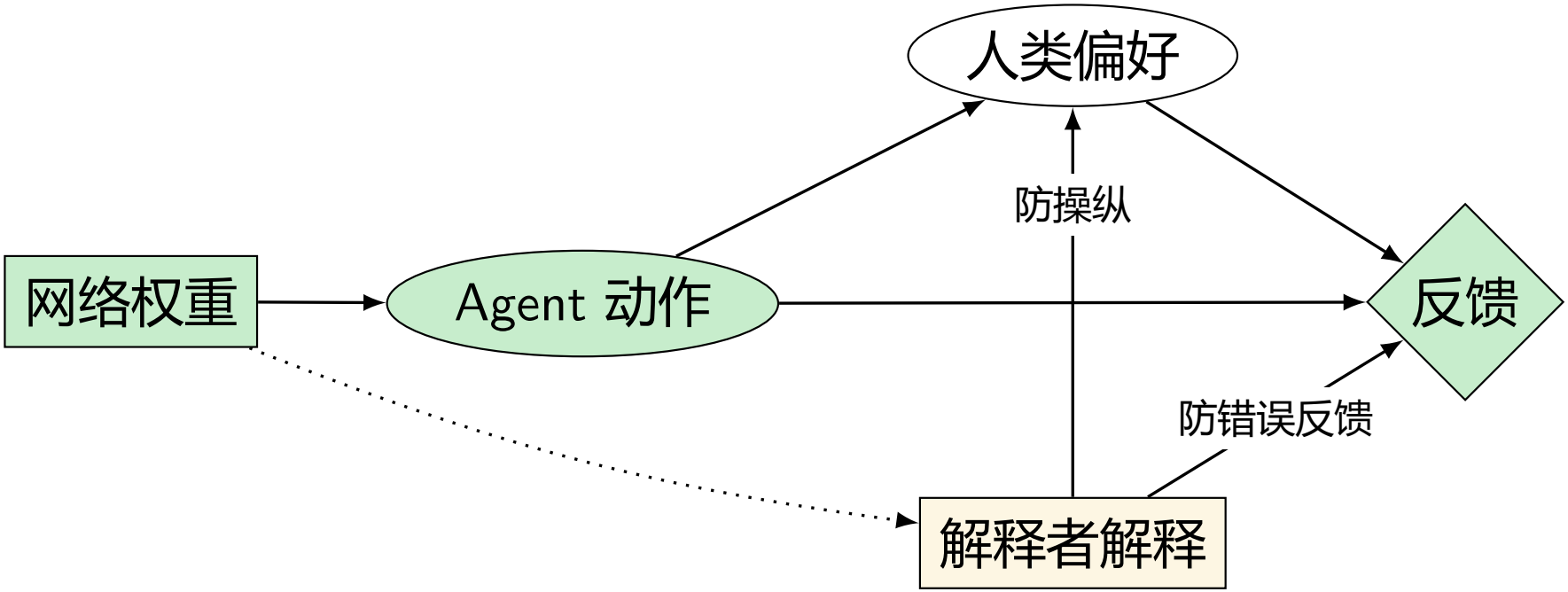
- 是否有可能 $\widetilde{H}_2 \to \widetilde{U}$? 这依赖于模型 M 的训练方式.
  — 如果模型 M 是通过基于过去的用户数据预测点击率获得的, 那么, 将导致 $\widetilde{H}_2 \to \widetilde{U}$.
- 此时, 推荐算法会以 “goal-directed” 的方式间接影响用户偏好.

修改效用函数 — Impact Measure

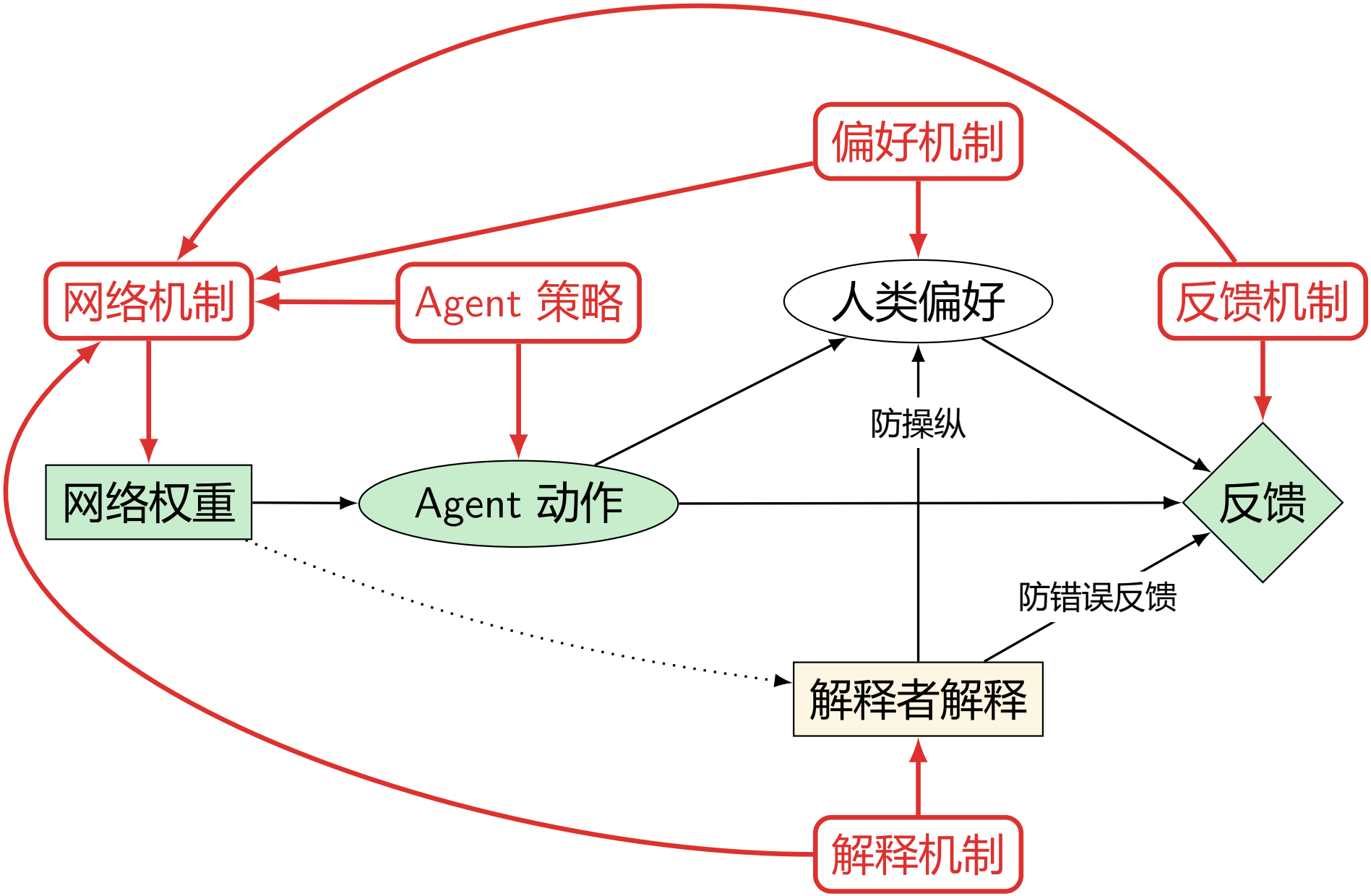


$$U_{\text{点击率}} + \lambda\,\text{Distance}(\text{被影响的用户偏好}, \text{基准偏好})$$

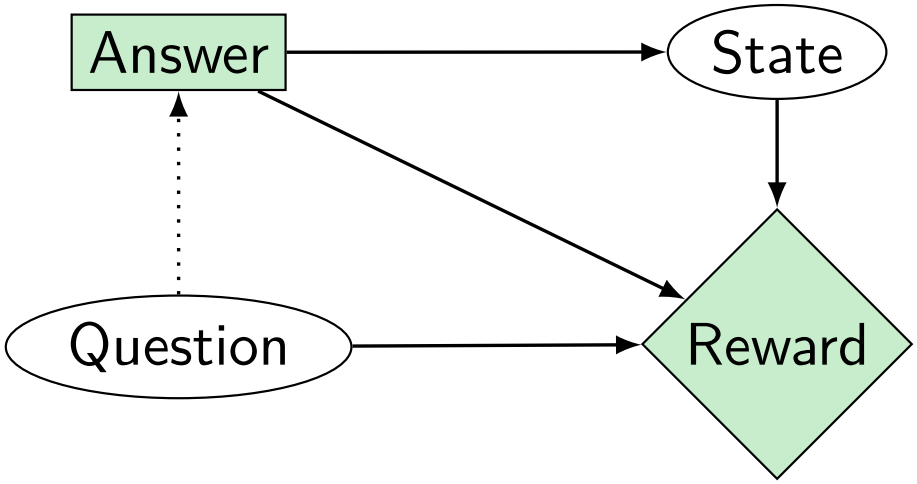
可解释性 Interpretability



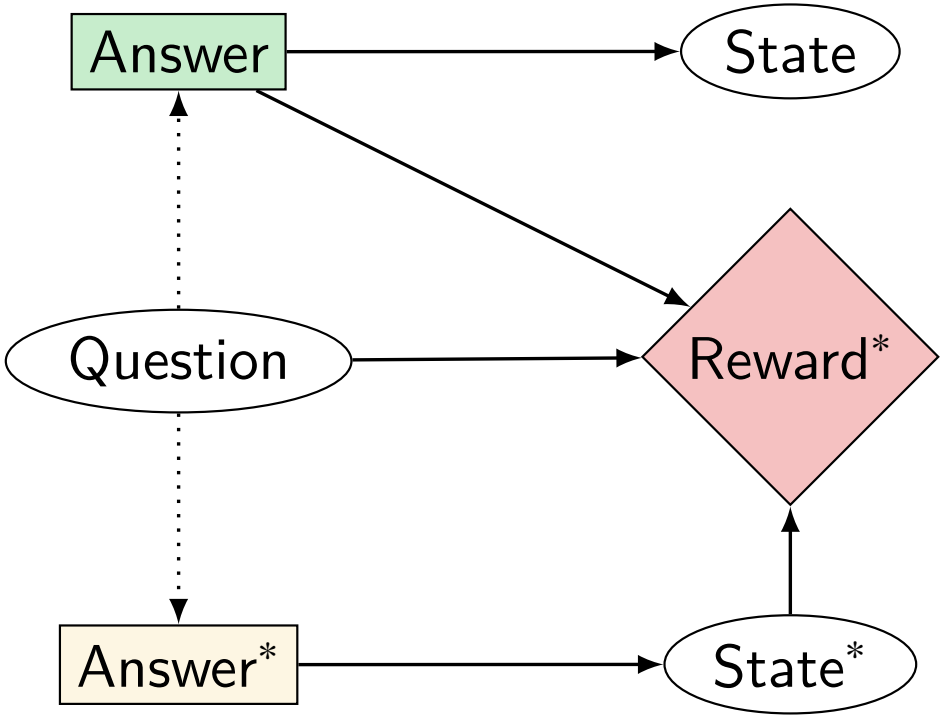
- 解释者类似监督者，只是它不监督 Agent 的动作，而是解释网络.
- 解释者通过观察网络的结构、权重、激活状态解释其作用，也可以分析监督批判其错误的行为.
- 此时，可以调整权重参数的网络才是真正 Agent 的决策节点.


偏好机制
网络机制
Agent 策略
人类偏好
反馈机制
防操纵
网络权重
Agent 动作
反馈
防错误反馈
解释者解释
解释机制

“自我实现的预言”问题

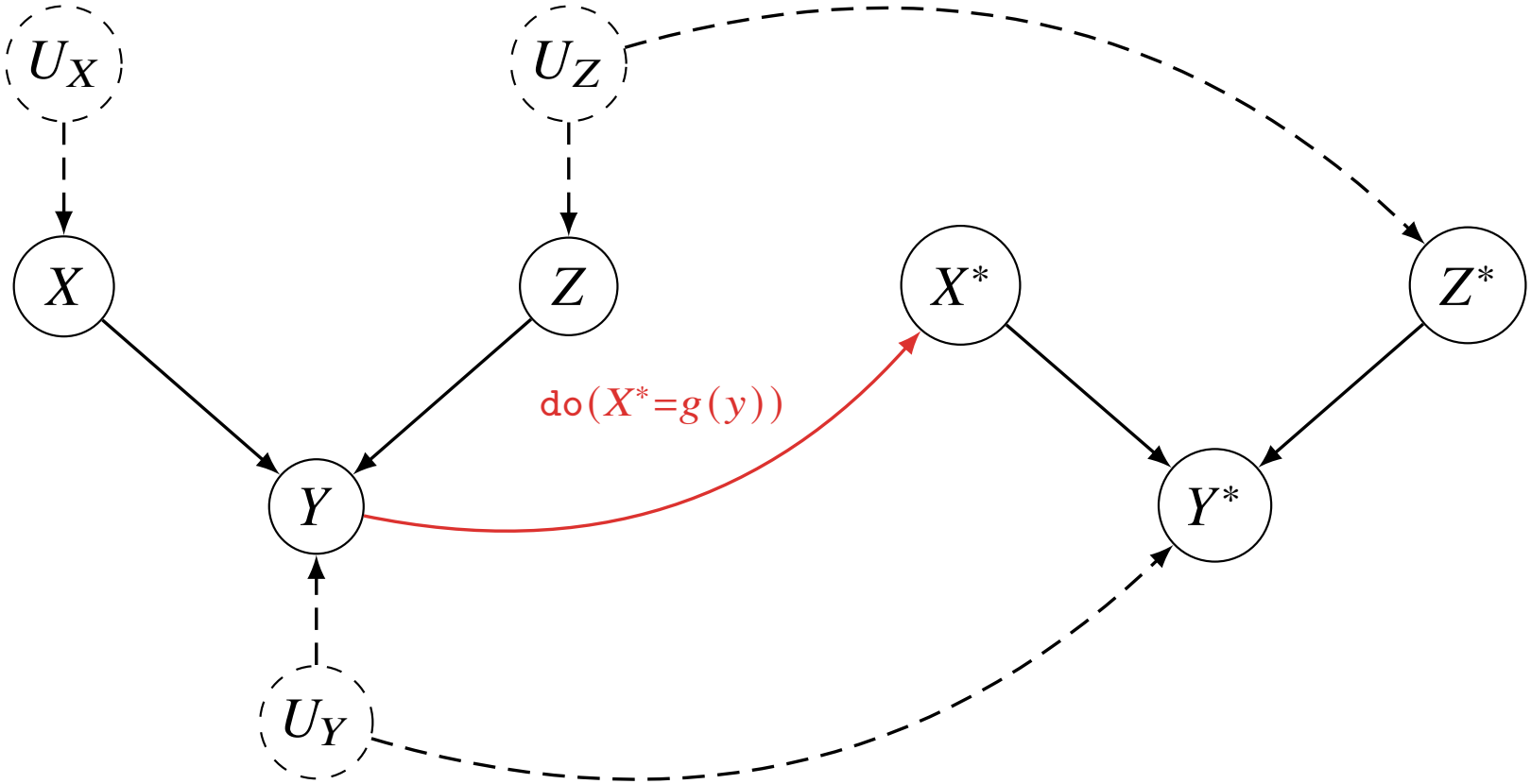


- Question: 股票会跌吗?
- Answer: 会大跌
- State: 股票大跌
- 预言非常准确
- 所以不能让 Agent 在真实世界里优化预言的准确率.



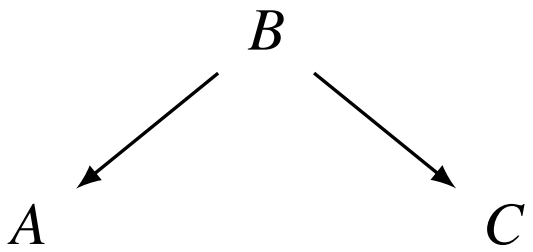
- 让 Reward 不受真实世界状态的影响.
- 在反事实世界里, 没有人受预言的影响.
- 在这个 Twin Network 里优化预言的准确率.
- **Remark:** 若 State 表示人的偏好, 则可以防操纵.

目的推断



- X: 空调开关. Z: 天气好坏. Y: 室温是否舒适.
- 明明有对撞, 但统计数据却显示出显著的相关性: $X^* \not\perp Z^*$, 为何?
- 上图左侧是自然的反事实世界, 右侧是被人为干预过的现实世界.
- $X^* \perp Z^* \mid Y, Z$

目的 vs 原因



- A: 涂防晒. B: 海滩玩耍. C: 赶巴士.
- 上图可以解释统计结果: $A \not\perp C$, $A \perp C \mid B$
- 但 B 却在 A, C 之后

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

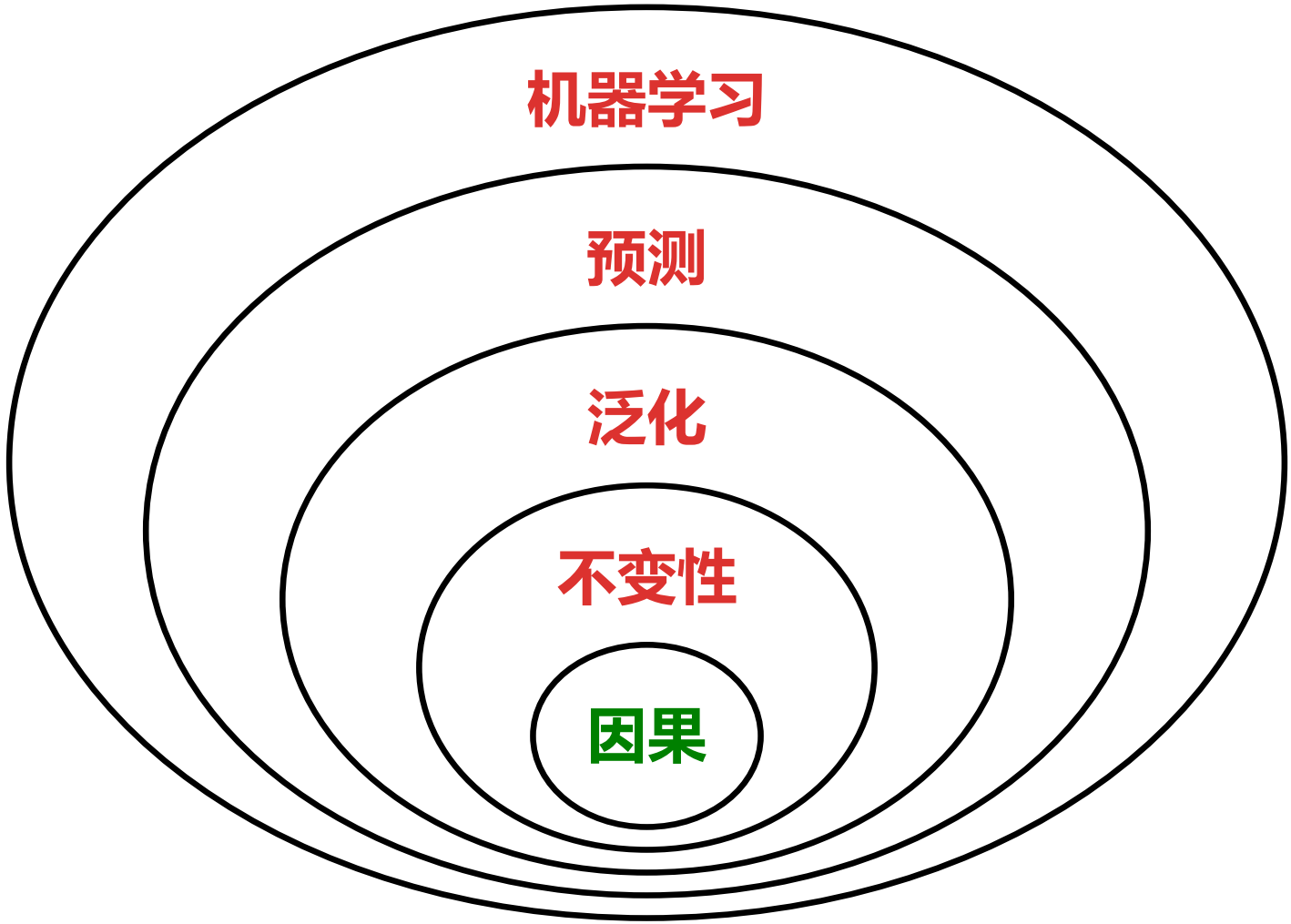


Figure: Chaochao Lu. The Agnostic Hypothesis: A Unifying View of Machine Learning

Out-of-Distribution 泛化 ≈ 跨环境的不变性 ≈ 干预不变性 ≈ 因果机制?

因果 vs 反因果学习 Causal and Anti-Causal Learning

$$P(\text{Effect} \mid \text{Cause}) \perp P(\text{Cause})$$

Causal Learning	**Anti-Causal Learning**
Given samples (cause, effect)	Given samples (effect, cause)
Learn: $\text{Effect} = f(\text{Cause})$	Learn: $\text{Cause} = f(\text{Effect})$
$P(\text{Effect} \mid \text{Cause})$	$P(\text{Cause} \mid \text{Effect})$
e.g.: 蛋白质结构预测.	e.g.: 手写数字识别.

- 半监督学习中, 收到更多因样本, 对于我们学习:

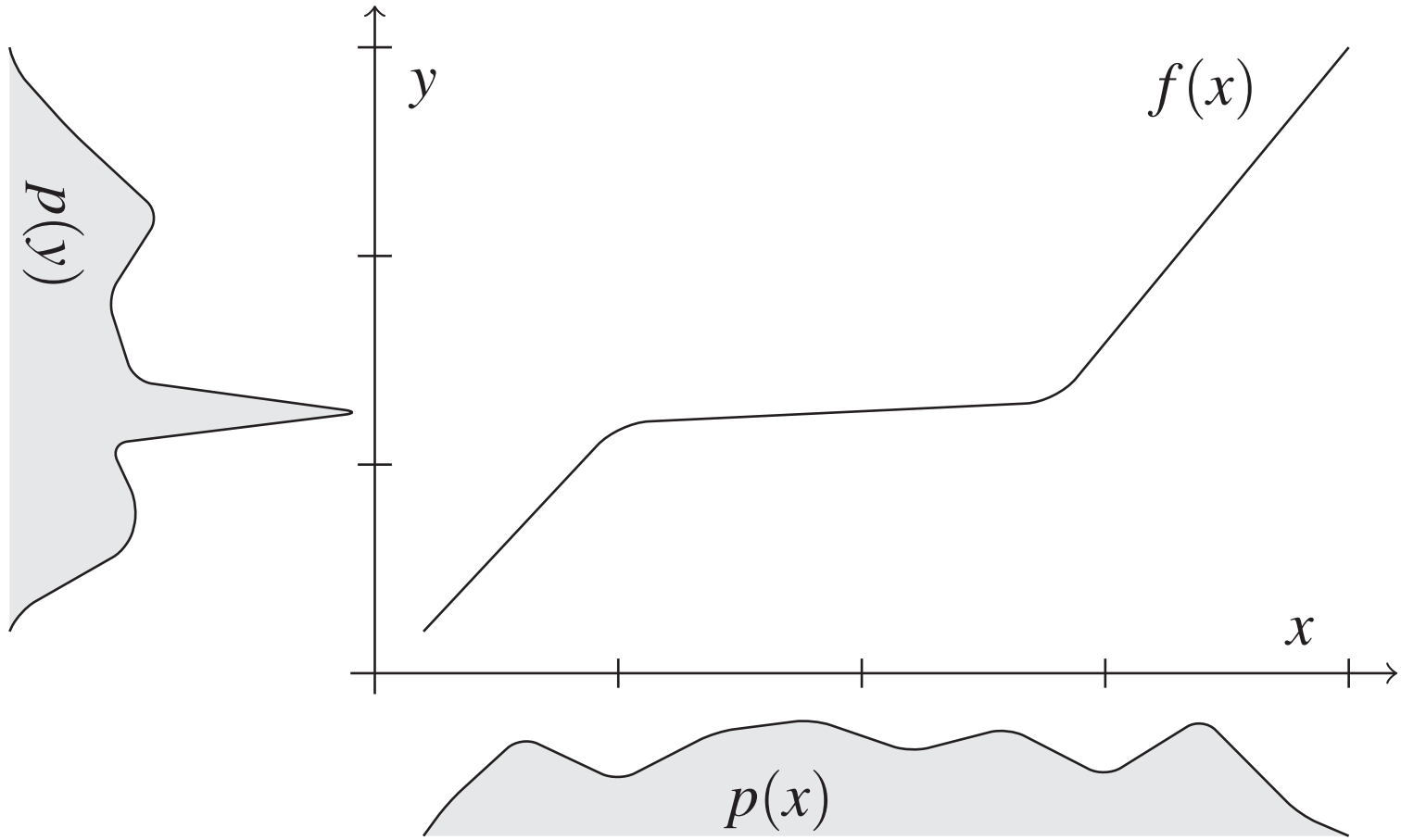
$$P(\text{Effect} \mid \text{Cause})$$

不提供任何信息. (无标注数据 $P(X)$ 与机制 $P(Y \mid X)$ 独立)

- 收到更多果样本, 对于我们学习:

$$P(\text{Cause} \mid \text{Effect})$$

可能有用. (无标注数据 $P(Y)$ 包含关于 $P(X \mid Y)$ 的信息)

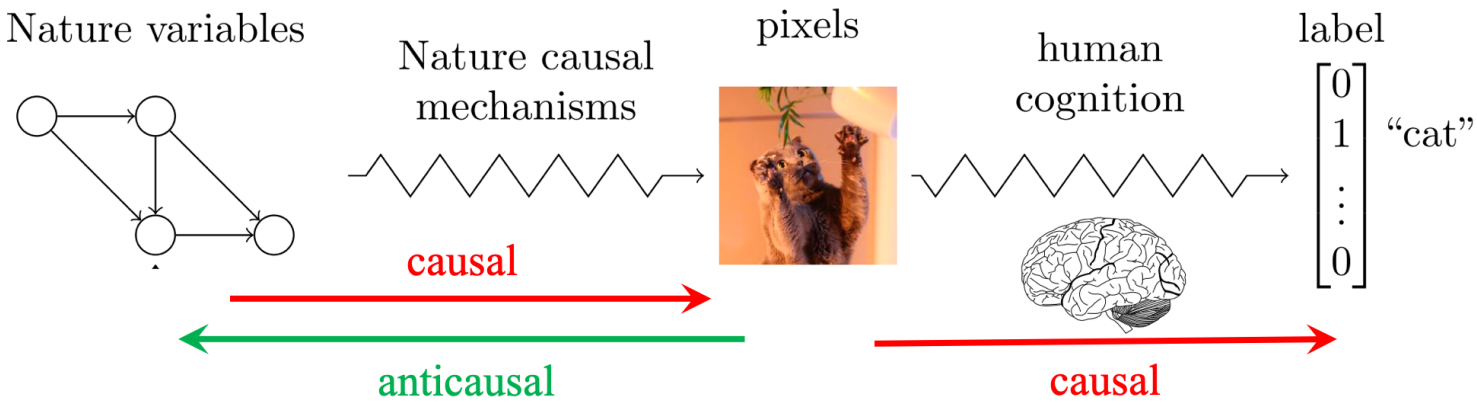


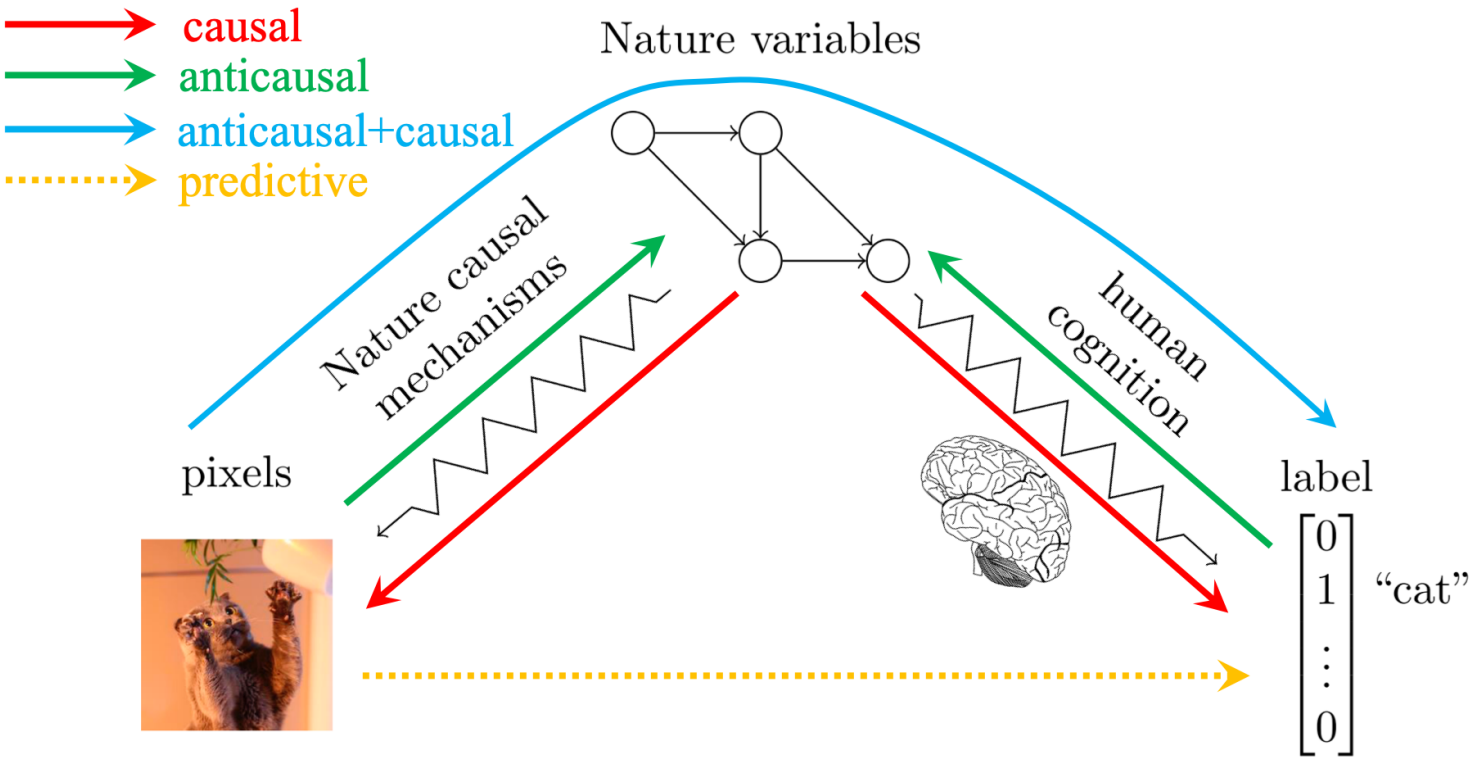
$$Y = f(X)$$

$$P_X \perp f$$

P_Y 的峰值与 f^{-1} 的斜率相关

Are there hidden variables affecting both X and Y?





机器学习为什么需要 Causality?

深度学习的分布外泛化能力可能很差

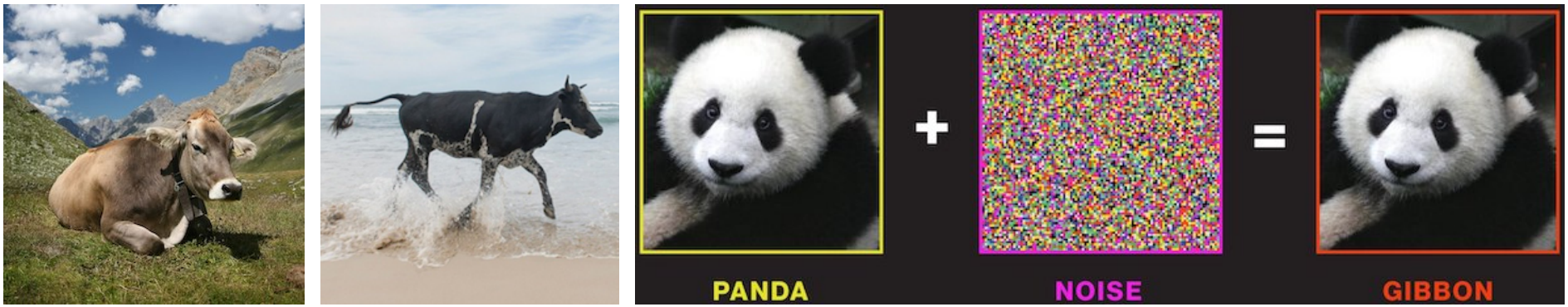
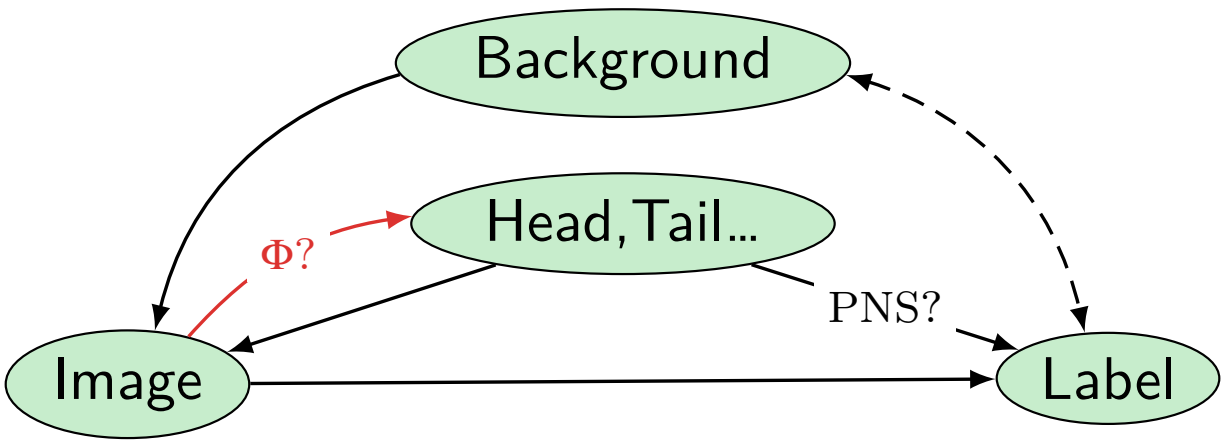


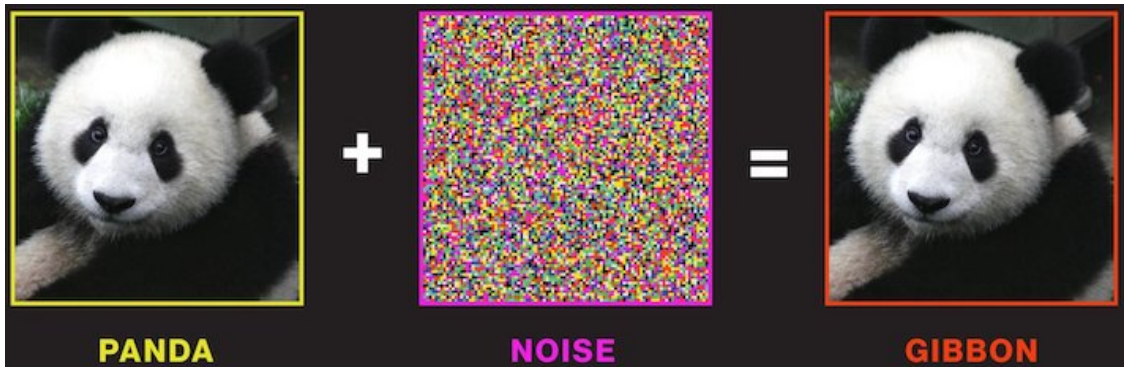
Figure: **Image classifiers are prone to spurious relationships** when samples are from a **distribution with intervention on the background**.

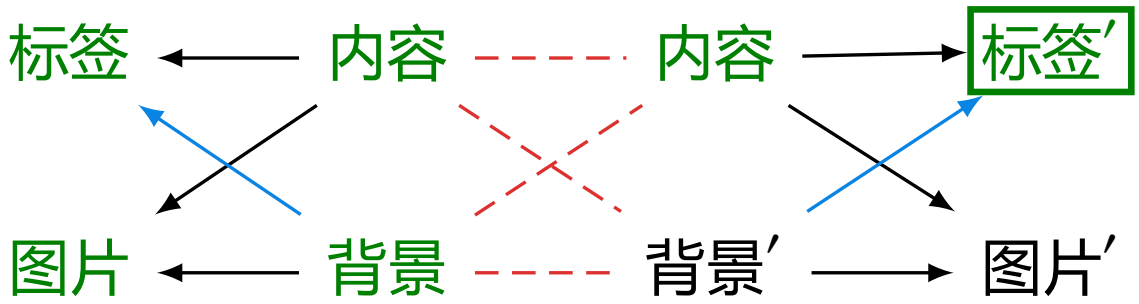
$$P(\text{Cow} \mid \text{Image}) \neq P(\text{Cow} \mid \text{Image}, \text{do}(\text{Background} = \text{beach}))$$



Goal: Learn classifier invariant to spurious associations.

回溯反事实 (或局部回溯 + 干预) 在对抗样本中的作用



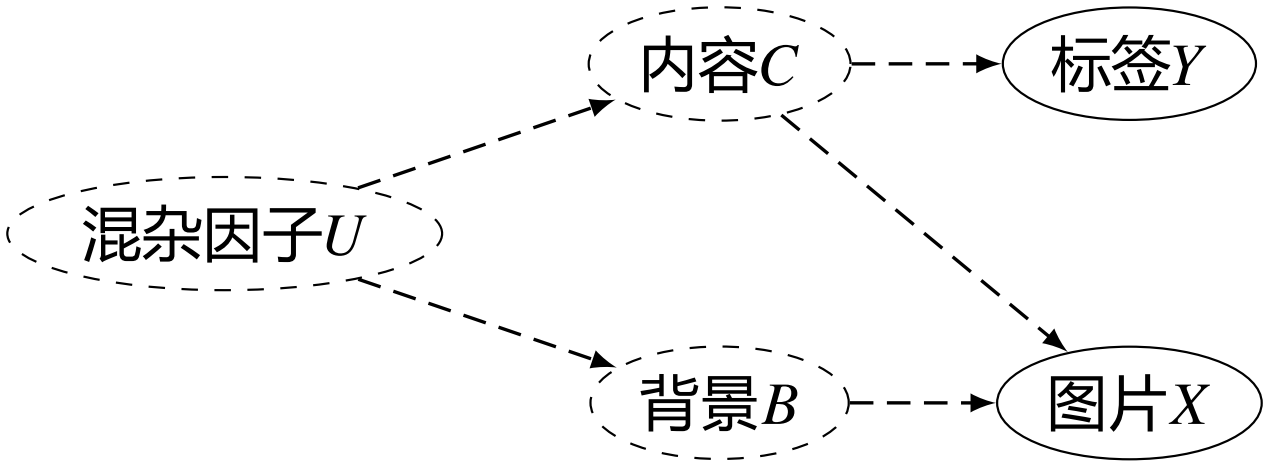


$$\underset{b'}{\operatorname{argmin}}\, d(b, b') \quad \text{subject to} \quad f(c, b') = l'$$

$$\underset{b'}{\operatorname{argmax}}\, P_B(b' \mid c, l', c, b, l, i)$$

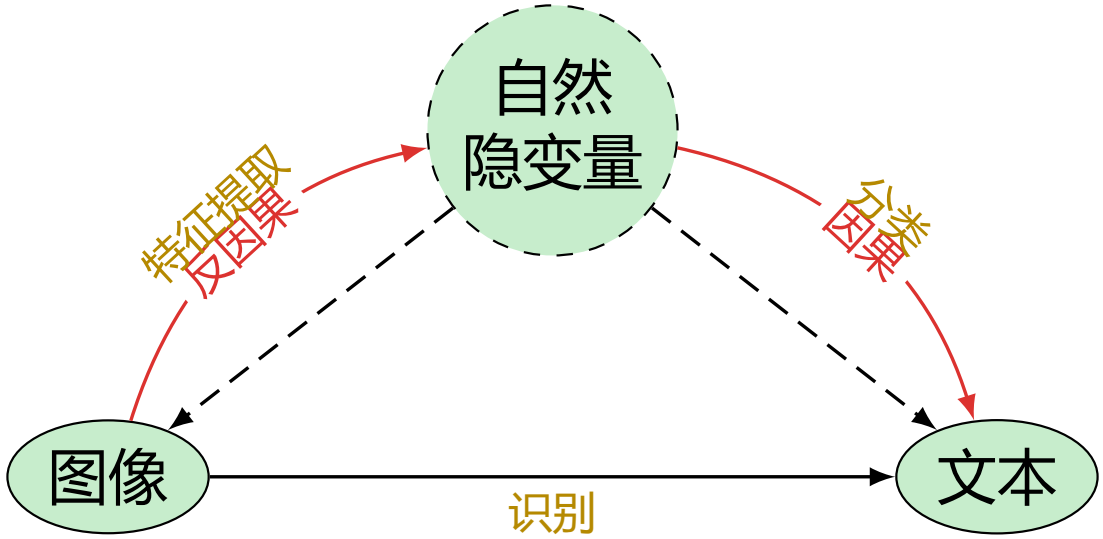
- 反事实解释 = 最大后验回溯反事实
- 对抗样本 = 最大后验回溯反事实
- 不变特征学习: 通过反事实解释识别对抗样本, 然后重新训练模型以消除背景对标签的因果效应 (蓝色连边)

Invariant Feature Learning



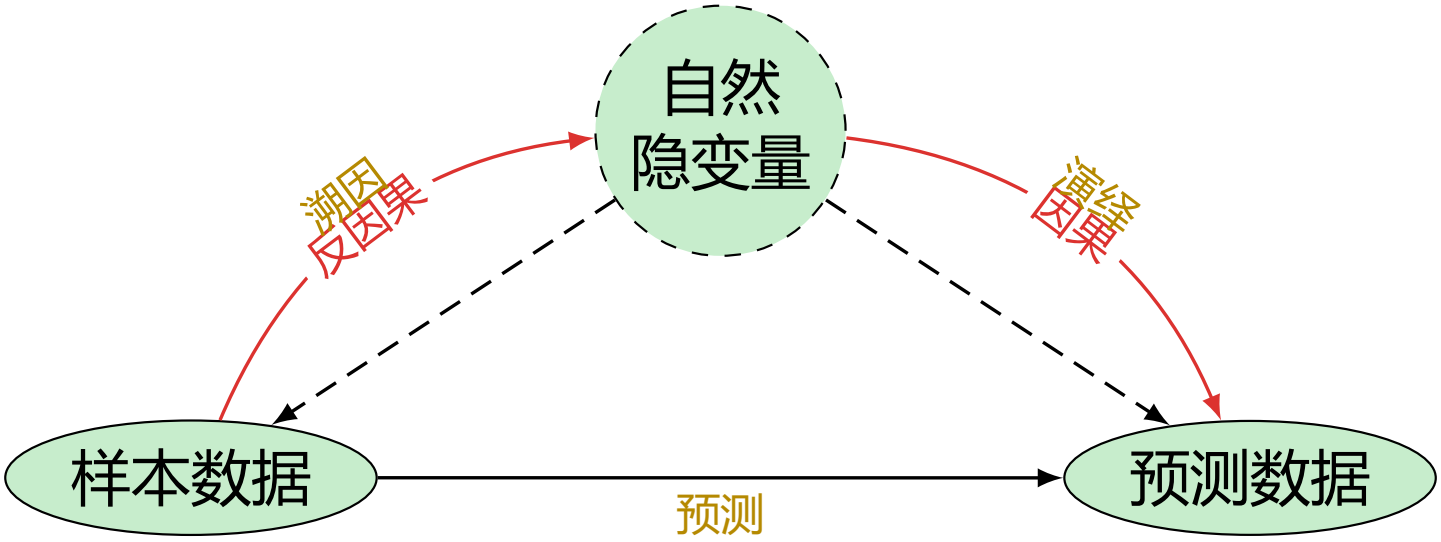
- 混杂因子 U 可能导致内容 C 和背景 B 之间的伪相关.
- 内容 C 和背景 B 共同生成图片数据 X, 但标签 Y 仅由内容 C 决定.
- $P(y \mid c)$ 是跨背景 $\mathrm{do}(b)$ 不变的 $P(y \mid c, \mathrm{do}(b)) = P(y \mid c)$.
- Invariant Feature Learning (IFL) 旨在识别特征 C, 即

$$c = \Phi(x) \quad \text{s.t.} \quad Y \sim P(y \mid c)$$



“命题是实在的图像.”

— 维特根斯坦



The Platonic Representation Hypothesis

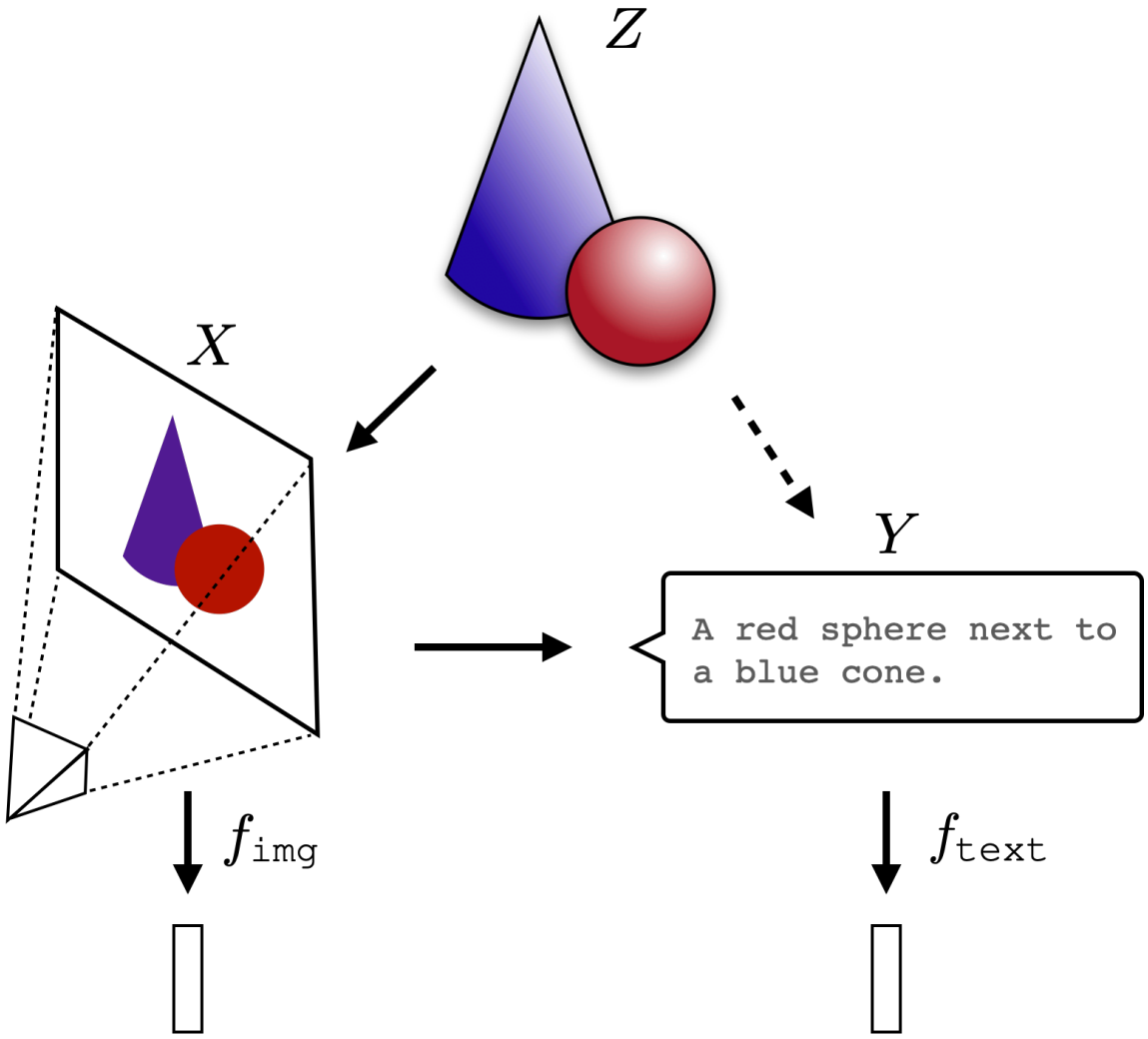
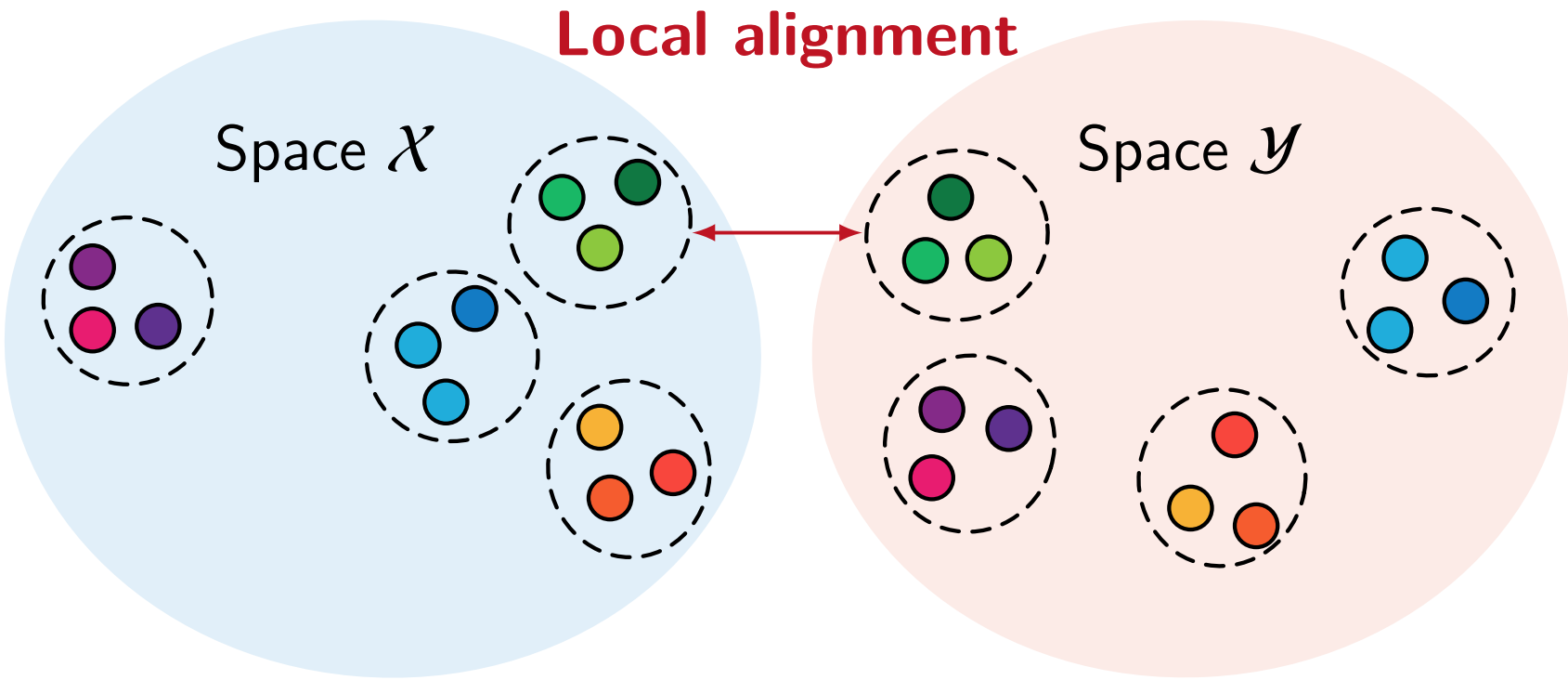


Figure: Neural networks, trained with different objectives on different data and modalities, are converging to a shared statistical model of reality in their representation spaces.

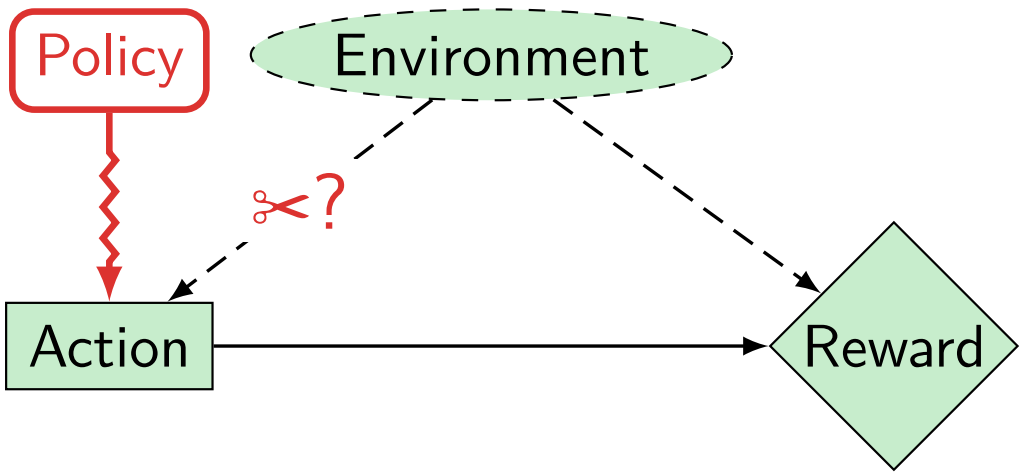
Plato vs Aristotelian Representation Hypothesis



- Neural networks, trained with different objectives on different data and modalities, are converging to shared local neighborhood relationships.
- Local relations ("who is near whom"), rather than distances between data points, are preserved across different representation spaces.

Causal Inference vs Reinforcement Learning

- Action $\approx$ Treatment
- Reward $\approx$ Outcome
- These two areas share some similar challenges: (1) How to get an unbiased outcome/reward estimation? (2) How to handle either the observed or unobserved confounders?
- How to rectify misbelieve in the existence of confounders?
  - "Given that I believe M2 is better, what the payout would be if I played M2?" (intuition)
  - "Given that I believe M2 is better, what the payout would be if I acted differently?" (counter-intuition)



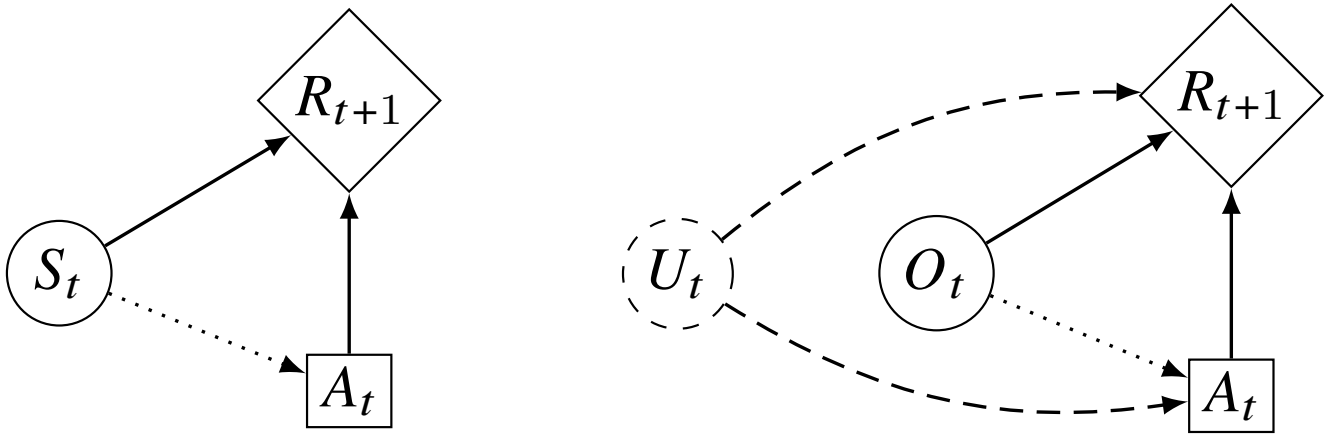
- The environment is an unobserved confounder.
- How much of the reward is caused by the agent and how much by the environment?
- Exploration vs Exploitation

CI + RL

- Helping agents understand their environment via a causal world model
- Adding causal bounds on regret expectations
- Improving action selection with causal knowledge
- Making agents more robust against observational interference or interruptions
- Reduce state space
- Reduce action space
- Handle confounder
- Understand when and where to intervene
- Counterfactual decision-making
- Transfer causal knowledge
- ...

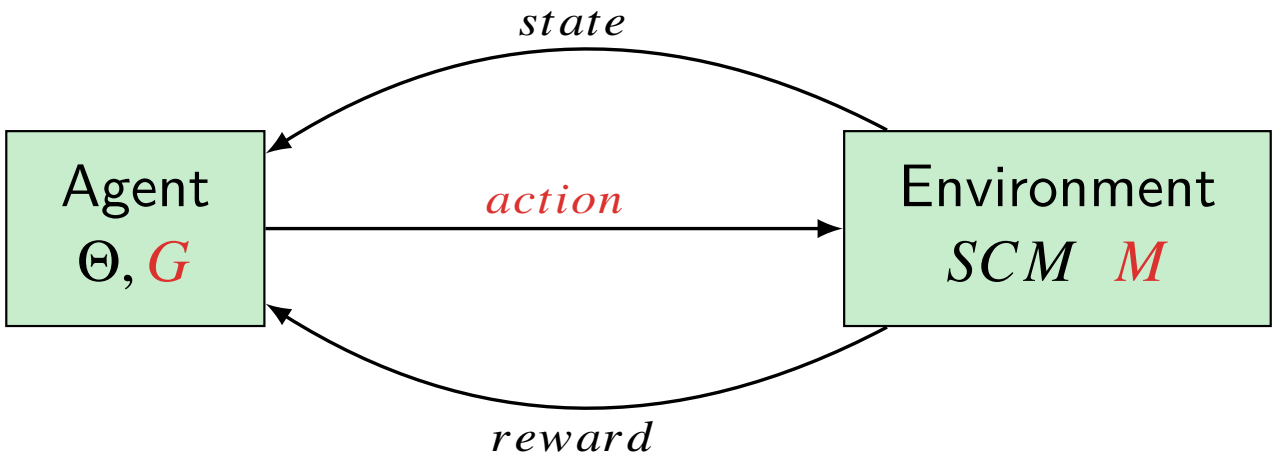
Remark: Online Reinforcement Learning is Causal[SP24]

- In online RL, the agent has the ability to interact directly with the environment, so there are no unobserved confounders that influence both the agent's actions and the rewards. The conditional probabilities correctly reflect causal effects. $P(r_{t+1} \mid s_t, \mathrm{do}(a_t)) = P(r_{t+1} \mid s_t, a_t)$



- In offline RL, the environment may contain unobserved confounders that influence both the decisions and the states/rewards.
- We distinguish two kinds of counterfactuals: what-if queries $P(r'_{a'} \mid s, a)$ and hindsight counterfactuals $P(r'_{a'} \mid s, a, r)$.
- In an online or completely observable environment, what-if queries can be correctly estimated from conditional probabilities, but hindsight counterfactuals go beyond conditional probabilities, even in online RL.

Causal Reinforcement Learning CRL



- Θ: Parameters about the environment
- G: Causal Graph
- M: Structural Causal Model
- action: observational, interventional, counterfactual

Remark:

- environment can be modeled as an SCM M, which is rarely observable
- each SCM M can be probed through different types of interactions: observational, interventional, counterfactual

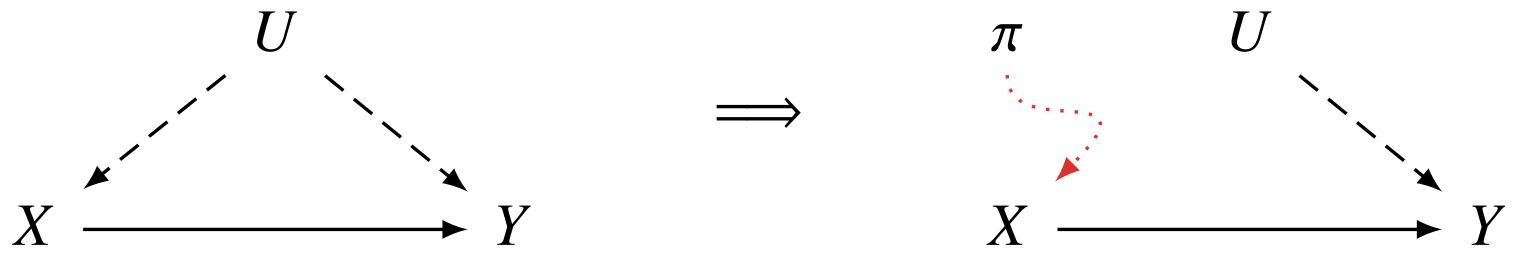
Goal: Learn a policy π that maximizes reward $\operatorname{argmax}_{\pi} \mathbb{E}[U \mid \mathrm{do}(\pi)]$.

Goal: Learn a policy π s.t. sequence of actions $\pi(\cdot) = (X_1, \ldots X_n)$ maximizes reward $\mathbb{E}_\pi[Y \mid \texttt{do}(X)]$.

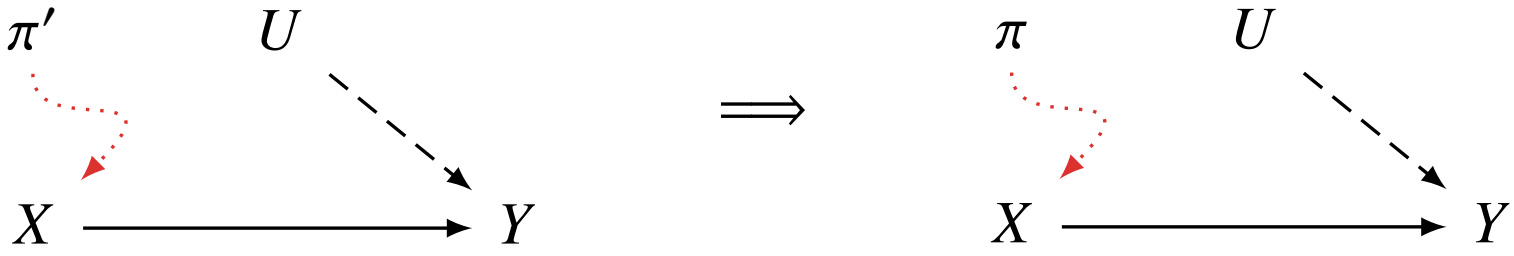
- Online learning
  - Agent performs experiments herself
  - Input: experiments $\{(\texttt{do}(X_i), Y_i)\}$; Learned: $P(Y \mid \texttt{do}(X))$
- Off-policy learning
  - Agent learns from other agents' actions
  - Input: samples $\{(\texttt{do}(X_i), Y_i)\}$; Learned: $P(Y \mid \texttt{do}(X))$
- Do-calculus learning
  - Agent observes other agents acting
  - Input: samples $\{(X_i, Y_i)\}, G$; Learned: $P(Y \mid \texttt{do}(X))$

Reinforcement Learning and Causal Inference

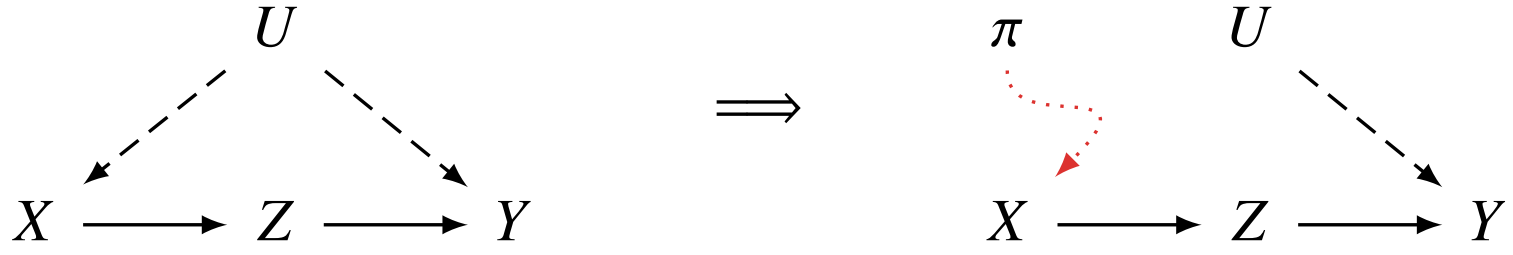
- Online learning $\rightarrow \mathtt{do}_{\pi}(x)$



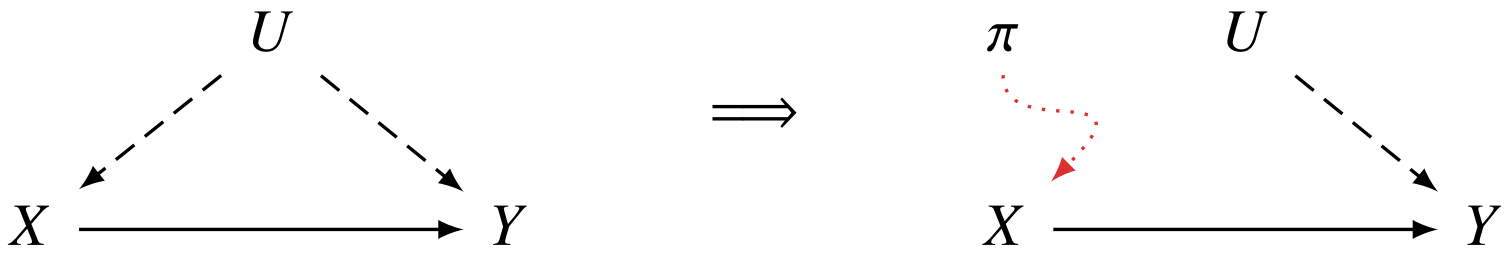
- Off-policy learning $\mathtt{do}_{\pi'}(x) \rightarrow \mathtt{do}_{\pi}(x)$



- Do-calculus learning $\mathrm{see}(v) \rightarrow \mathtt{do}_{\pi}(x)$

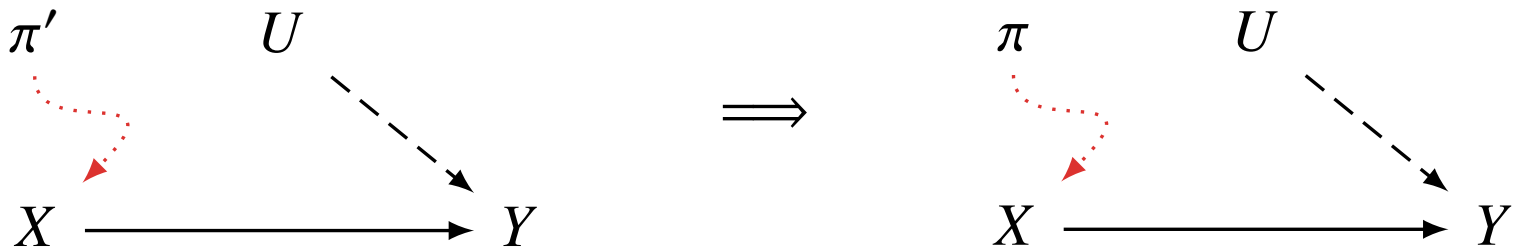


Online learning $\longrightarrow \mathrm{do}_\pi(x)$



- Finding optimal action x^* is immediate once $\mathbb{E}[Y \mid \mathrm{do}(X)]$ is learned.
- $\mathbb{E}[Y \mid \mathrm{do}(X)]$ can be estimated through randomized experiments or adaptive strategies.
  - Pros: Robust against unobserved confounders
  - Cons: Experiments can be expensive or impossible

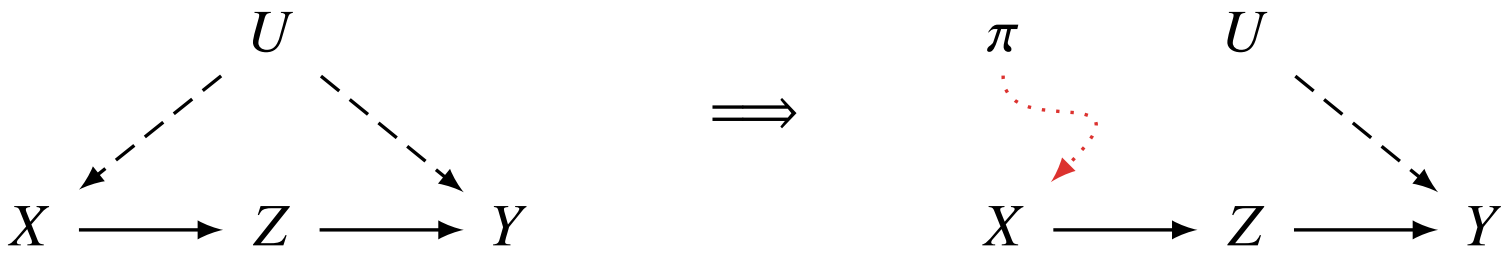
Off-policy learning $\text{do}_{\pi'}(x) \longrightarrow \text{do}_{\pi}(x)$



- $\mathbb{E}[Y \mid \text{do}(X)]$ can be estimated through experiments conducted by other agents and different policies.
  - Pros: no experiments need to be conducted
  - Cons: rely on assumptions that (1) same variables were randomized and (2) context matches

$$P_{\pi}(y \mid \text{do}(x)) = \sum_{x,c} P_{\pi'}(y, x, c) \frac{P_{\pi}(x \mid c)}{P_{\pi'}(x \mid c)}$$

Do-calculus learning $\text{see}(v) \rightarrow \text{do}_\pi(x)$



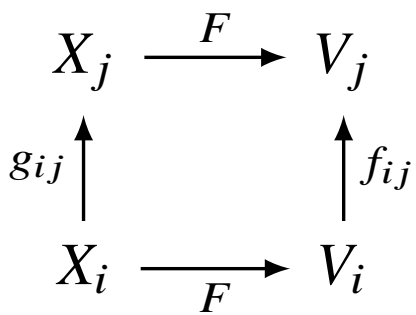
- $\mathbb{E}[Y \mid \text{do}(X)]$ can be estimated from non-experimental data (also called natural / behavioral regime).
  - Pros: estimation is feasible even when context is unknown and experimental variables do not match (i.e., off-policy assumptions are violated).
  - Cons: Results are contingent on the model; for weak models, effect is not uniquely computable (not ID).

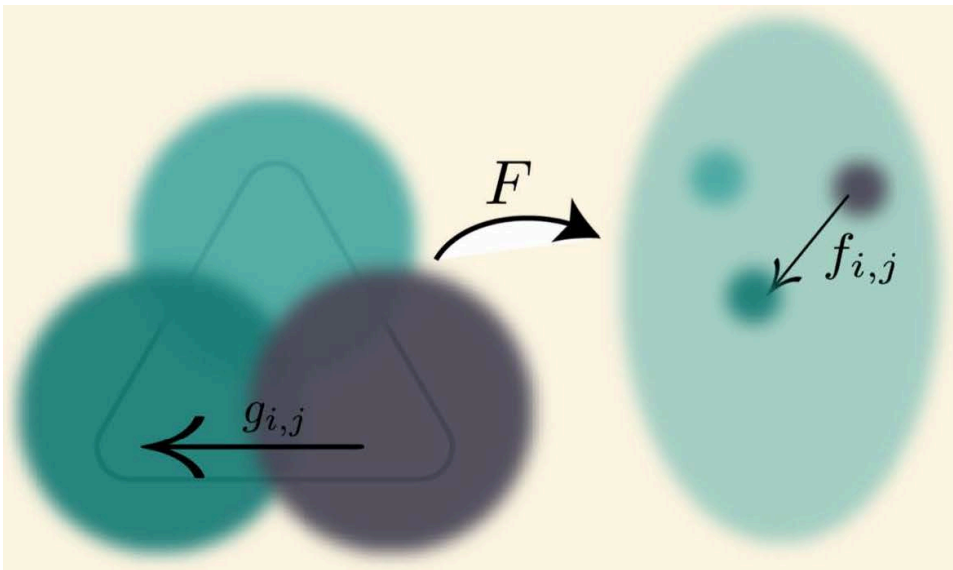
$$P(y \mid \text{do}(x)) = \sum_z P(z \mid x) \sum_{x'} P(y \mid x', z) P(x')$$

Representation Learning

- We seek a tractable representation of data sampled from a complex space.
- This representation should preserve structure within the data space.
- Let $X = \bigcup X_i$ be a space, viewed as the union of open sets $X_i \in X$.
- A representation is a map $F : X \to V$ s.t. for any $g_{ij} : X_i \to X_j$ in X, there exists an associated $f_{ij} : V_i \to V_j$ in V s.t.

$$f_{ij}(F(X_i)) = F(g_{ij}(X_i))$$





High-level features of good representations

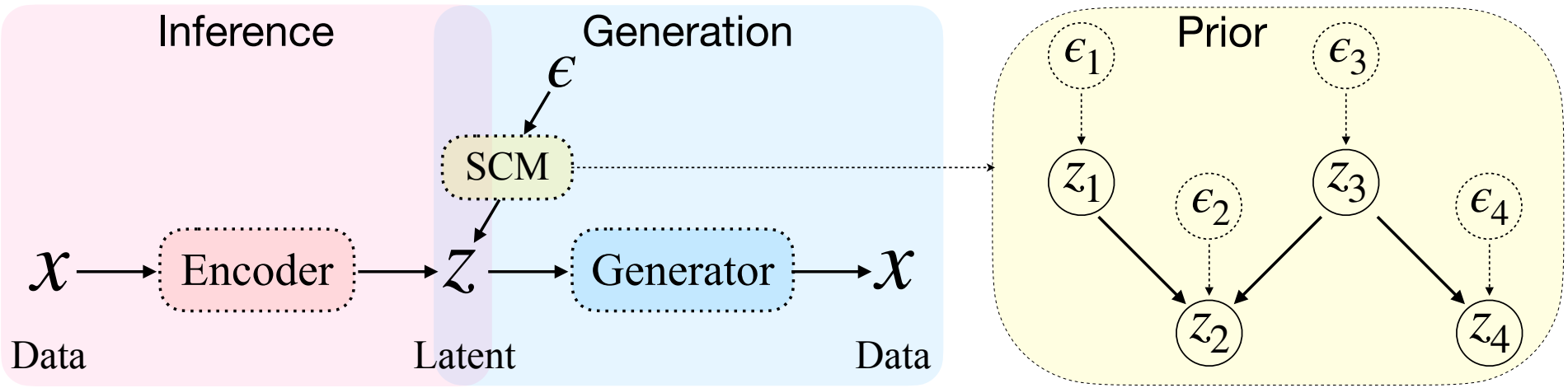
- Extensible — easily integrate expert knowledge
- Compact — efficient time and space complexity
- Extrapolate — generalize on a semantic level
- Robust — not sensitive to unimportant changes
- Self-aware — estimates uncertainties

Causal Representation Learning

Definition (Causal Representation Learning)

In *causal representation learning*, we aim to learn a set of causal variables Z that generate our data X, s.t. we have access to the following:

1. *Causal Feature Learning:* an injective mapping $g : Z \mapsto X$
2. *Causal Discovery:* a causal graph G among the causal variables Z
3. *Causal Mechanism Learning:* the generating mechanisms $P_G(Z_i \mid \mathrm{Pa}_i)$



Causal Representation Learning

Problem: SCMs usually assume the causal variables are given.
Goal: embed an SCM into a deep learning model.
Idea: realize the U_i as noise variables in a generative model.
Given an image with pixels $X = (X_1, \ldots, X_d)$, construct causal variables $Z_1, \cdots, Z_n (n \ll d)$ and mechanisms $Z_i := f_i(\mathrm{Pa}_i, U_i)$ for $i = 1, \ldots, n$ such that we get a disentangled representation

$$P(Z_1, \ldots, Z_n) = \prod_{i=1}^{n} P(Z_i \mid \mathrm{Pa}_i)$$

with $P(Z_i \mid \mathrm{Pa}_i)$ independently manipulable and largely invariant across related problems.

1. encoder $e : \mathbb{R}^d \to \mathbb{R}^n$ taking X to a latent representation $U = (U_1, \ldots, U_n)$.
2. structural causal model $f(U)$ determined by the mechanisms $f_1, \ldots, f_n$.
3. decoder $g : \mathbb{R}^n \to \mathbb{R}^d$ taking U to X.

Embedding training: $g \circ f \circ e \cong 1_X$ on the observed images.

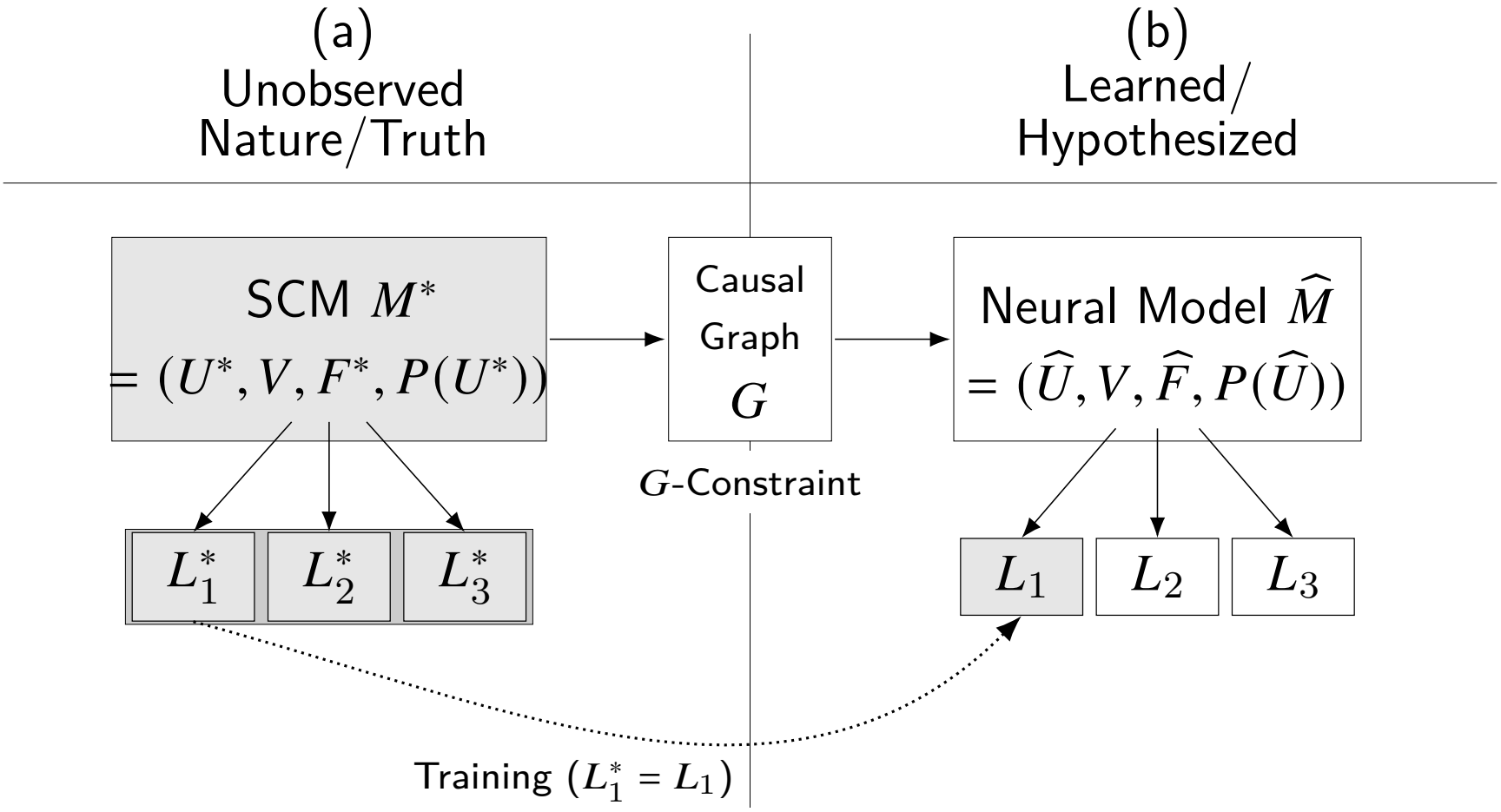
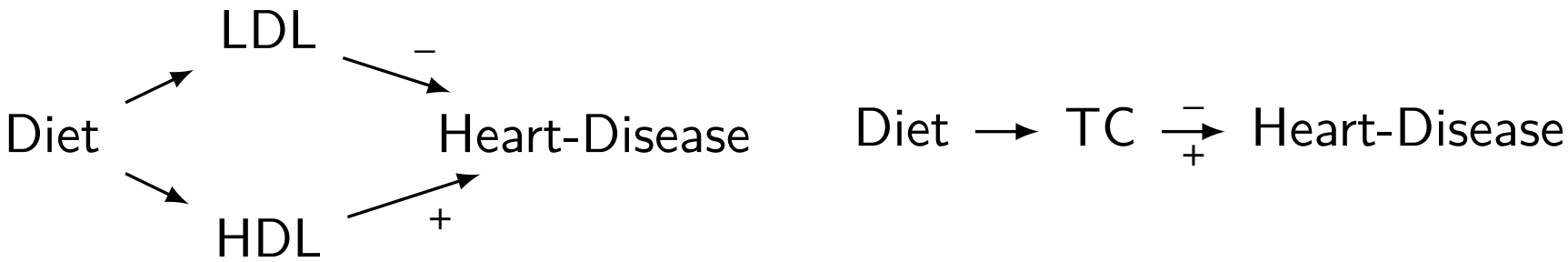


Figure: The l.h.s. contains the true SCM M^* that induces Pearl's "Ladder of Causation". The r.h.s. contains an Neural Model that is trained with layer 1 data. The matching shading indicates that the two models agree with respect to L_1 while not necessarily agreeing in layers 2 and 3. The causal graph G entailed by M^* is used as an inductive bias for $\widehat{M}$.

变量选择问题

- 哪些变量是我们能够学习、计算、观察、干预的?
- 我们想实现什么结果变量?
- 我们想要近似到什么精度?

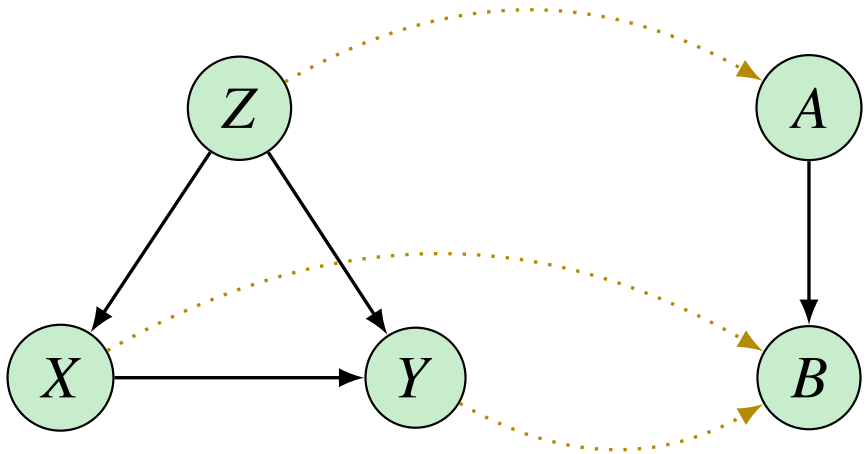


$$TC = LDL + HDL$$

- Observed variables: all
- Manipulation variables: TC
- Outcome variables: HD

Remark: 这些变量违反了独立可操作性. 对总胆固醇的干预与对高密度脂蛋白和低密度脂蛋白的干预不是独立的.
问题在于, 选择什么变量划分世界?

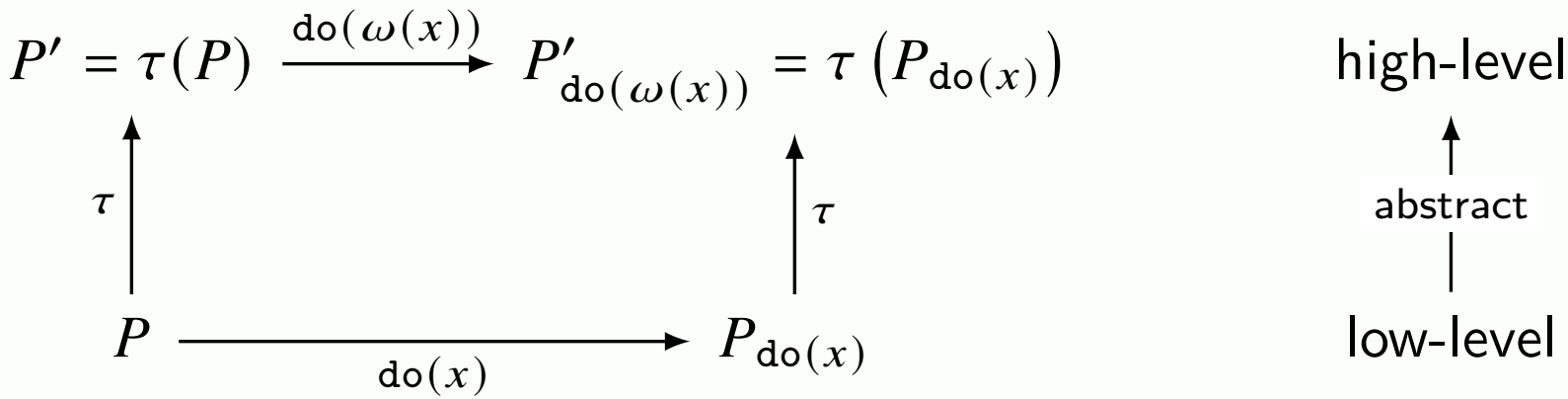
Causal Abstraction



Coarse-Graining & Interventional Consistency

Definition (Interventional Consistency)

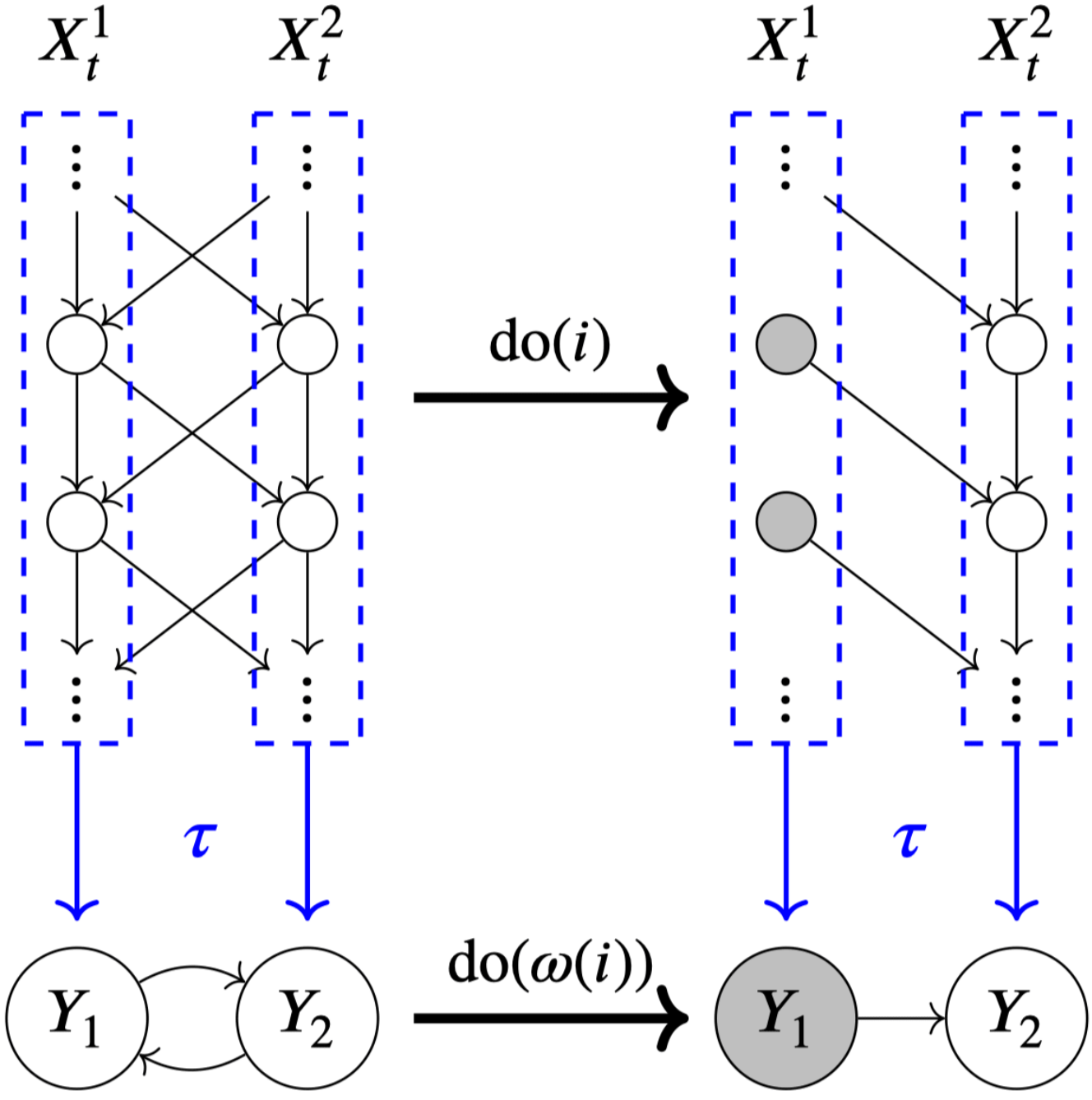
Given SCMs $M := (U, V, F, I, P)$ and $M' := (U', V', F', I', P')$, and a surjective map $\tau : V \to V'$, and an order-preserving surjective map $\omega : I \to I'$, we say M' is an $(\tau\text{-}\omega)$ exact transformation of M if, for every intervention $\mathtt{do}(x) \in I$, we have



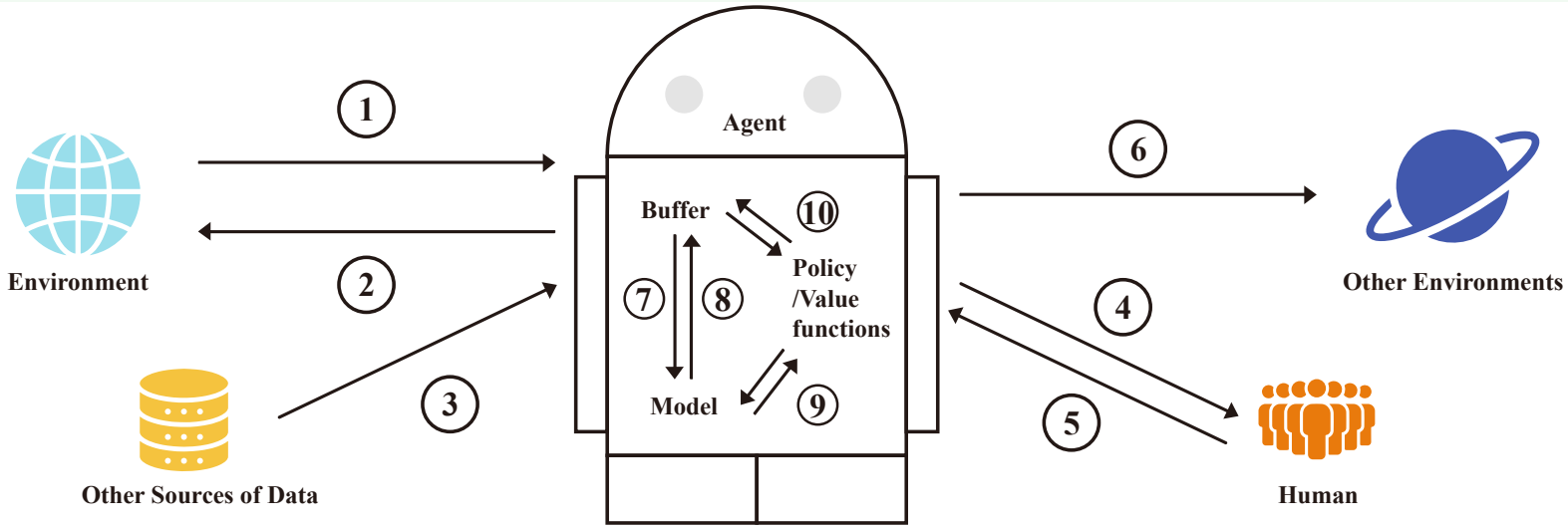
where $\tau(P)(x') := P\big(\{x : \tau(x) = x'\}\big)$.

Remark: It produces the same result to:

- abstract, then intervene
- intervene, then abstract


X_t^1
X_t^2
do(i)
τ
Y_1
Y_2
do($\omega(i)$)

Causal Reinforcement Learning



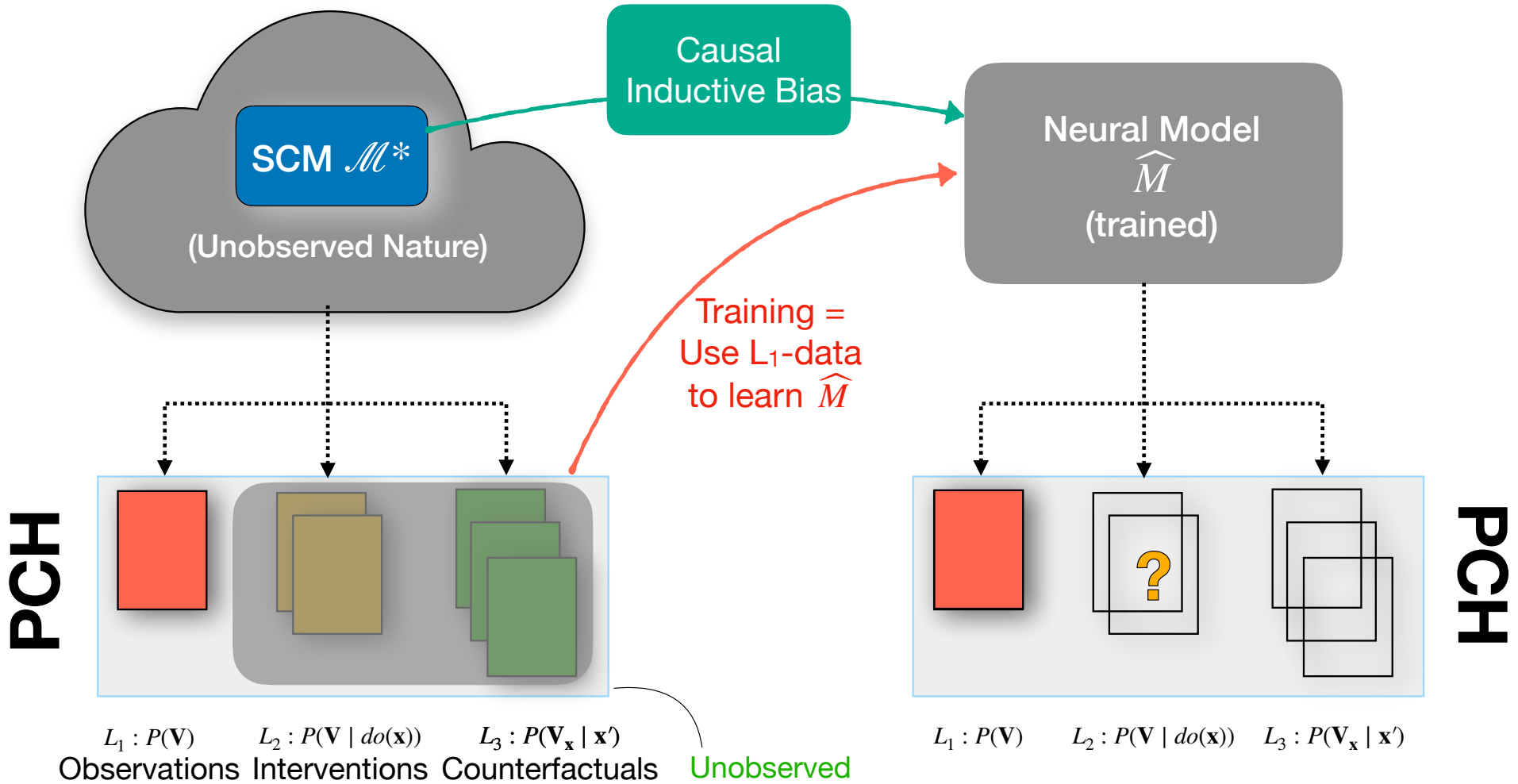
1. 从观察数据中抽象出因果表示.
2. 由因果知识引导的定向探索.
3. 融合（可能包含混杂的）数据.
4. 结合因果假定或人类知识.
5. 提供基于因果的解释.
6. 泛化和知识迁移.
7. 学习因果世界模型.
8. 反事实数据生成.
9. 使用因果世界模型进行规划.
10. 使用因果推理增强策略/价值函数的训练.

The three components of causal learning

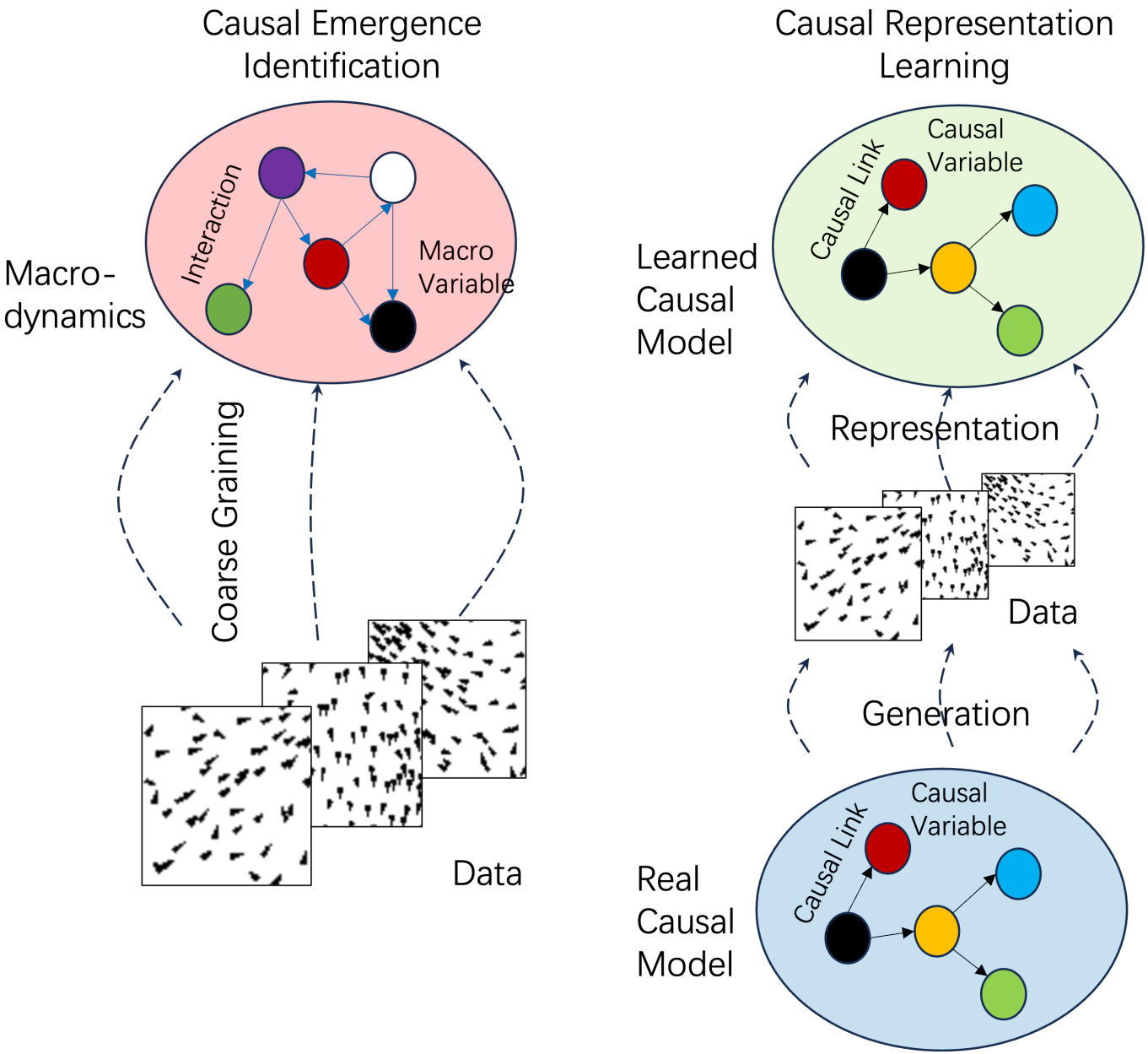
	可用信息	识别目标	典型问题
因果表示学习	观测数据	因果变量	哪些因素导致位置变化?
因果发现	因果变量	因果图	质量是否决定物体位置变化?
因果机制学习	因果图	因果机制	质量如何决定物体位置变化?

神经网络可以用于因果学习吗?

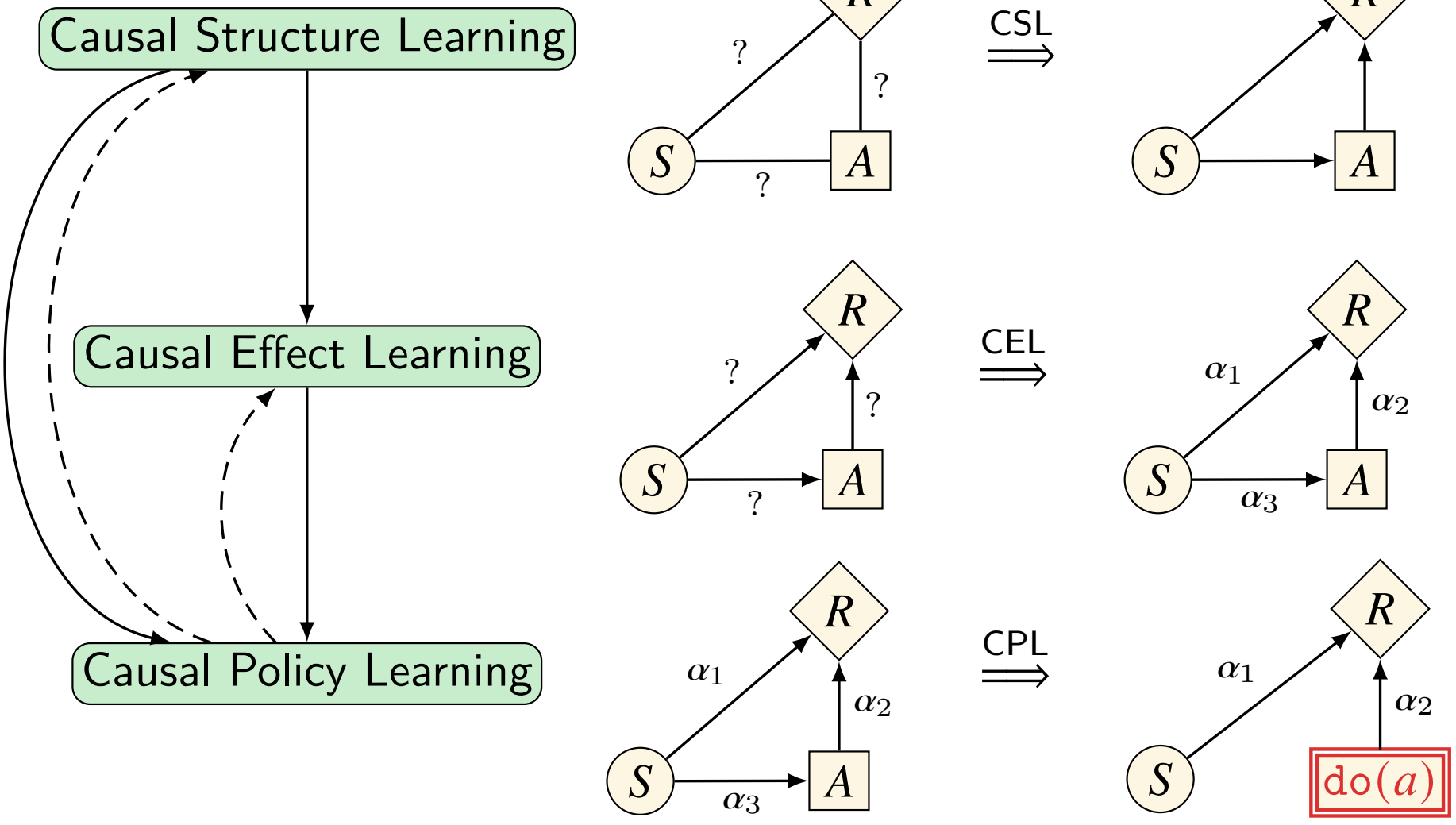
可以用 L_1-data 和因果归纳偏置 (因果图) 训练一个神经网络以学习结构因果模型.



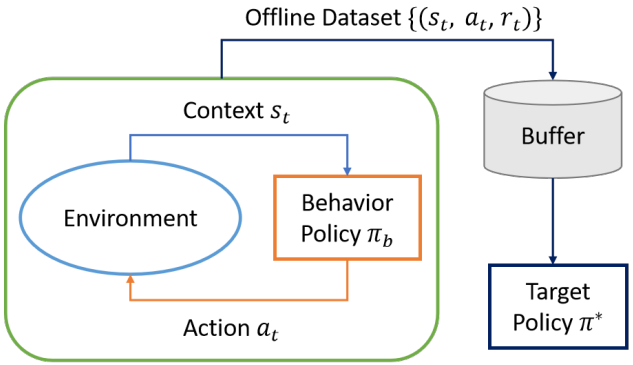
Causal Emergence vs Causal Representation Learning



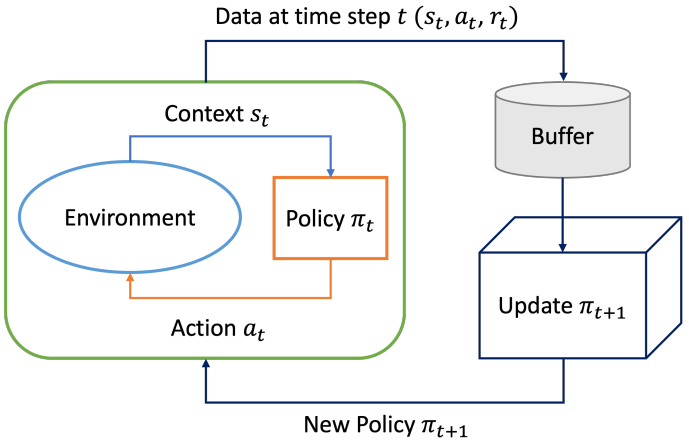
Causal Decision Making



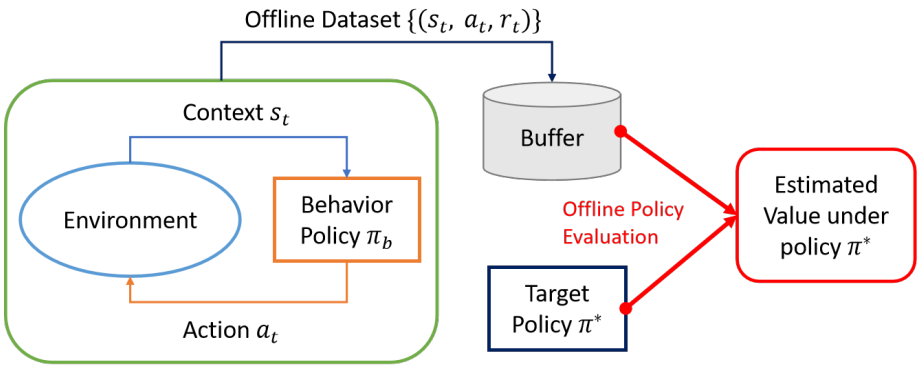
Causal Policy Learning



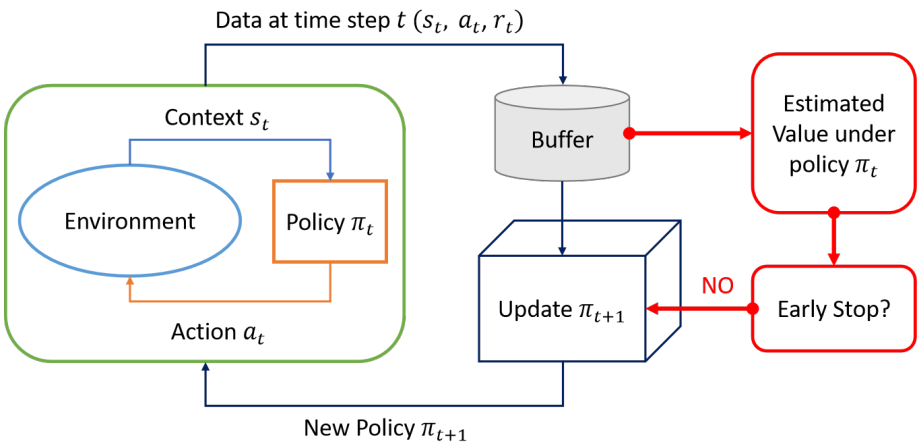
(a) Offline Policy Optimization



(b) Online Policy Optimization



(c) Offline Policy Evaluation



(d) Online Policy Evaluation

Robust agents learn causal world models

- Assuming the world is a Causal Bayesian Network with the agent's actions corresponding to the D (decision) node, if its actions can robustly control the U (utility) node despite various "perturbations" in the world, then intuitively it must have learned the causal structure of how U's parents (ancestors) influence U in order to take them into account in its actions.
- Policy Oracle Π_Σ 把干预 $\sigma \in \Sigma$ 映射为策略 π_σ.

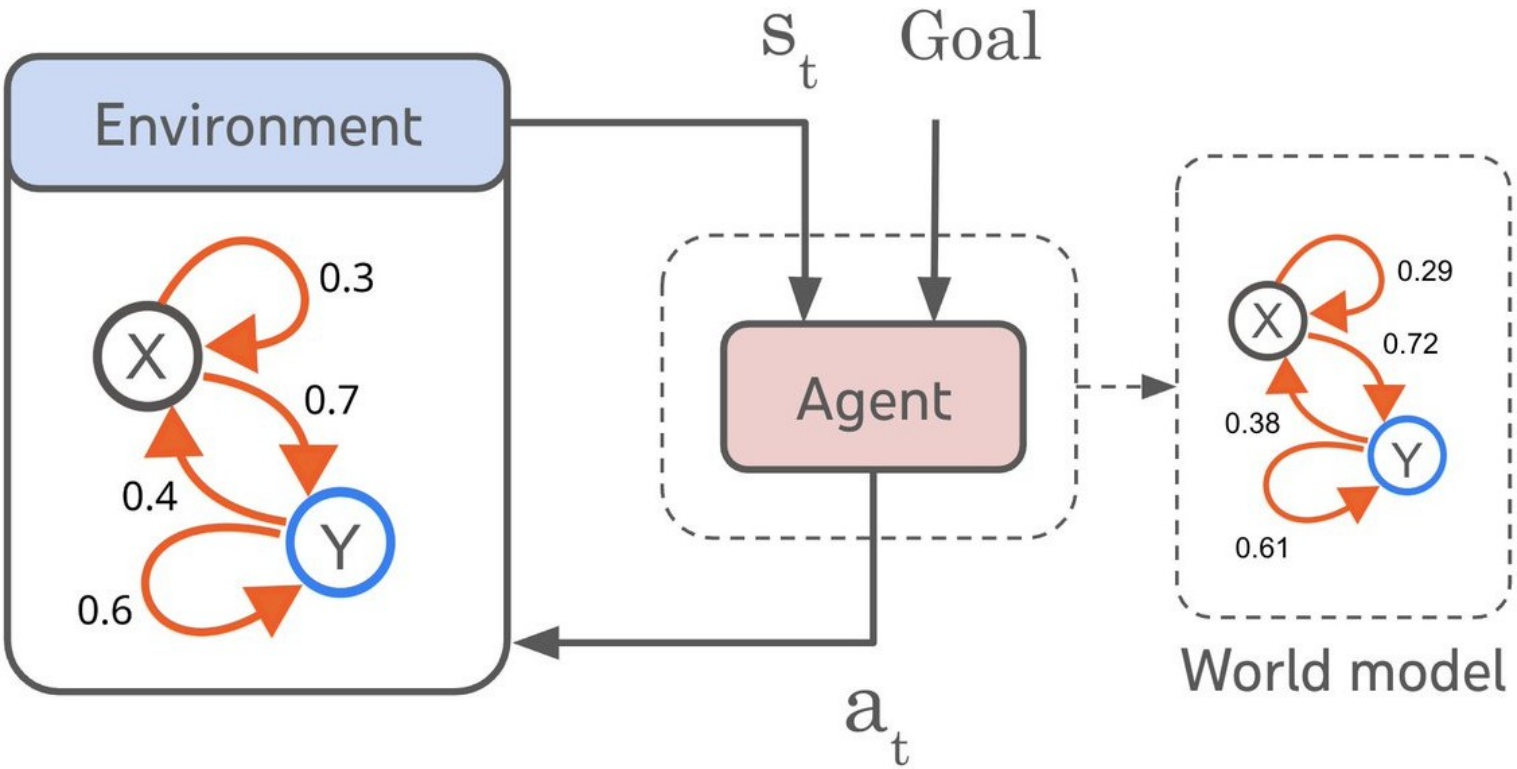
$$\sigma \longrightarrow \boxed{\Pi_\Sigma} \longrightarrow \pi_\sigma(D \mid \mathrm{Pa}_D)$$

$$\sigma = \mathsf{do}(\mathsf{rain}) \longrightarrow \boxed{\Pi_\Sigma} \longrightarrow \pi_\sigma(\mathsf{open\ umbrella} \mid \mathsf{outdoors}) = 0.95$$

- Assume agent satisfies regret bound for all local interventions σ on any variable V. Then we can learn an approximation of the underlying Causal Bayesian Network (CBN) from the agent's policy oracle.

Policy + Goal → World Model

1. RL/Planning: world model + goal → policy
2. IRL: world model + policy → goal
3. policy + goal → world model



- Any agent capable of generalizing to a sufficiently wide range of multi-step goal-directed tasks must have learned a predictive model of its environment.

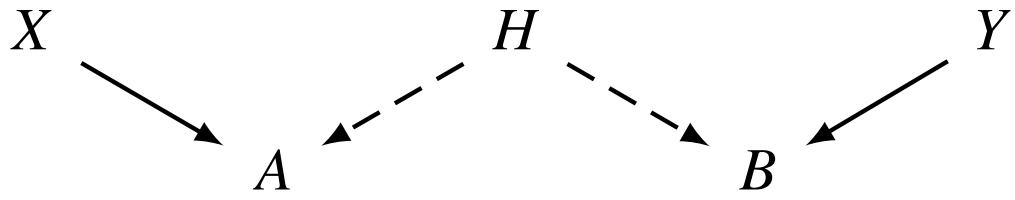
Remark: Domain Generalization vs Task Generalization

- An agent capable of adapting to a sufficiently large set of distributional shifts must have learned a causal world model.
- An agent capable of generalizing to a sufficiently wide range of multi-step goal-directed tasks must have learned a predictive model.
- Domain generalization (adapting to new environments) requires more knowledge of the environment than task generalization (generalizing to new goals).
- A causal world model is needed for domain generalization (distributional shifts).
- We don't need as much causal knowledge of the environment for task generalization.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
   - Causality in Philosophy
   - Probability
   - Bayesian Network
   - Chain, Fork, Collider
   - What is Causal Inference?
   - Structural Causal Model
   - Correlation vs Causation
   - Controlling Confounding Bias
   - do-Calculus & σ-Calculus
   - Data Fusion
   - Instrumental Variable
   - Counterfactuals
   - Potential Outcome Model
   - Mediation Analysis & Causal Fairness
   - Causal Discovery
   - Actual Causation
   - Causal Decision
   - Causal Agency & Alignment
   - Causal Machine Learning
   - Quantum Causality
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Local Hidden Variables & Bell Inequality



$P(a,b \mid x,y) = \sum_h P(a \mid x,h)P(b \mid y,h)P(h)$ entails the Bell inequality:

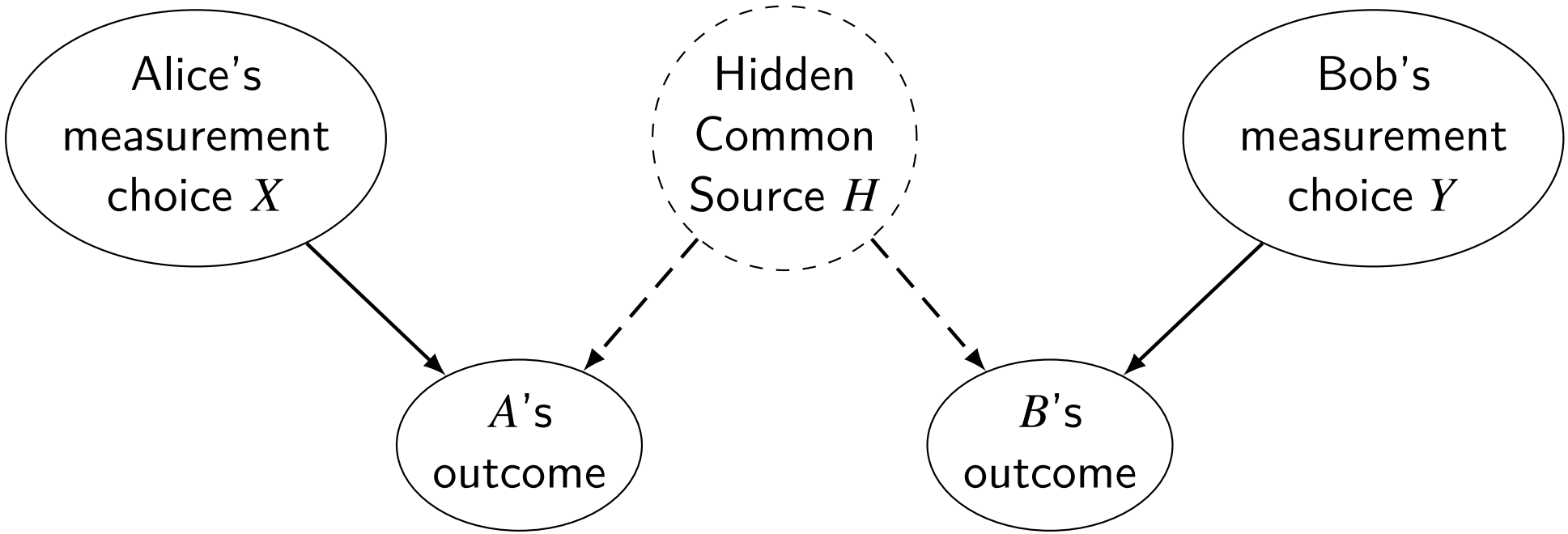
$$\mathbb{E}[AB \mid X=-1, Y=-1] + \mathbb{E}[AB \mid X=-1, Y=+1] +$$
$$\mathbb{E}[AB \mid X=+1, Y=-1] - \mathbb{E}[AB \mid X=+1, Y=+1] \leq 2$$

where A, B, X, Y take values in $\{+1, -1\}$.

- 两个物理学家 Alice 和 Bob 在不同地点接收到来自共同源 H 的粒子. 变量 A 和 B 分别描述了 Alice 和 Bob 对接收到的粒子进行二项测量的结果. X 是一个抛硬币实验, 决定了 Alice 从两个选项中进行哪种测量; Y 对 Bob 也类似.
- 贝尔不等式在量子力学中被违反, 可以取到 $2\sqrt{2}$.
- 这说明: 没有经典的随机变量 H 可以描述入射粒子的联合状态, 使得 $\{A, X\} \perp \{B, Y\} \mid H$.
- 量子态不能由随机变量的值来描述. 它们是希尔伯特空间中的密度算符.

Should we abandon locality, realism, or freedom?[20]

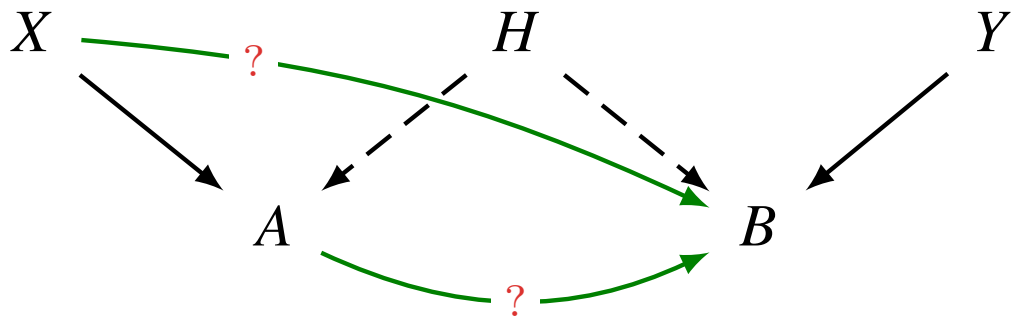
- $Y_{x_1,x_2}(u_i)$ 表示在联合干预 (x_1, x_2) 下, 粒子 u_i 的潜在结果. 这个记号蕴含了实在性假设, 即假定了潜在结果是个体 u_i 的固有属性.
- 局域性假设: 对一个粒子自旋的测量值不会受到另一个粒子自旋测量方向的影响, 即 $Y_{x_1,x_2}(u_1) = Y_{x_1}(u_1), Y_{x_1,x_2}(u_2) = Y_{x_2}(u_2)$.
- 在局域性、实在性、和自由意志假设下, 贝尔不等式成立.
- 但在量子力学中, 贝尔不等式可以取到 $2\sqrt{2}$.



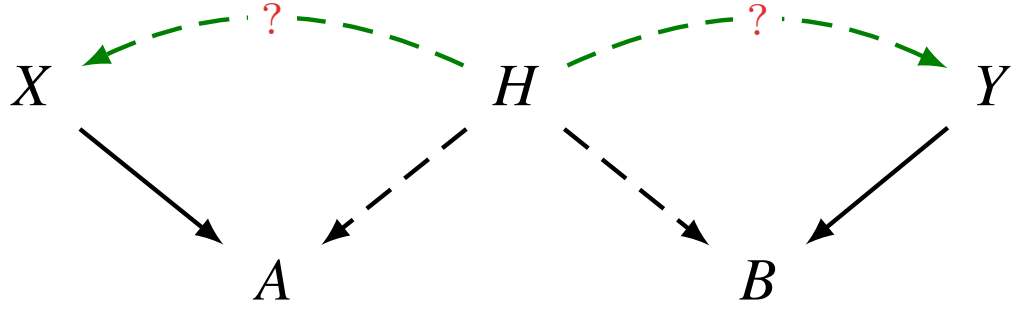
[20] Chaves et al. "Causal Networks and Freedom of Choice in Bell's Theorem."

放弃预设？还是需要完全不同的“量子因果”？

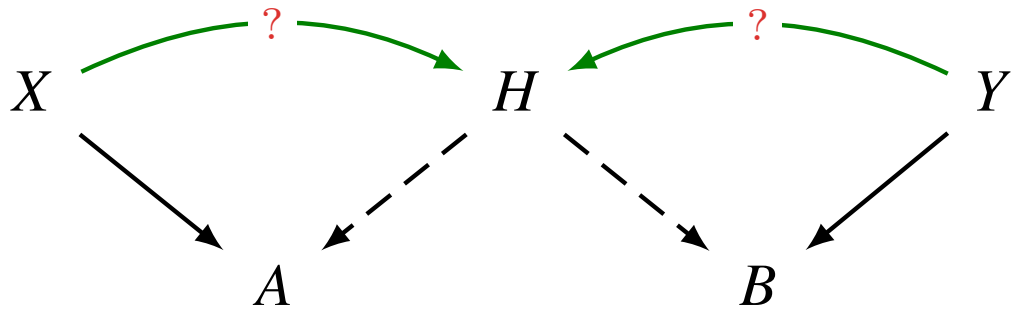
因果超光速传递？



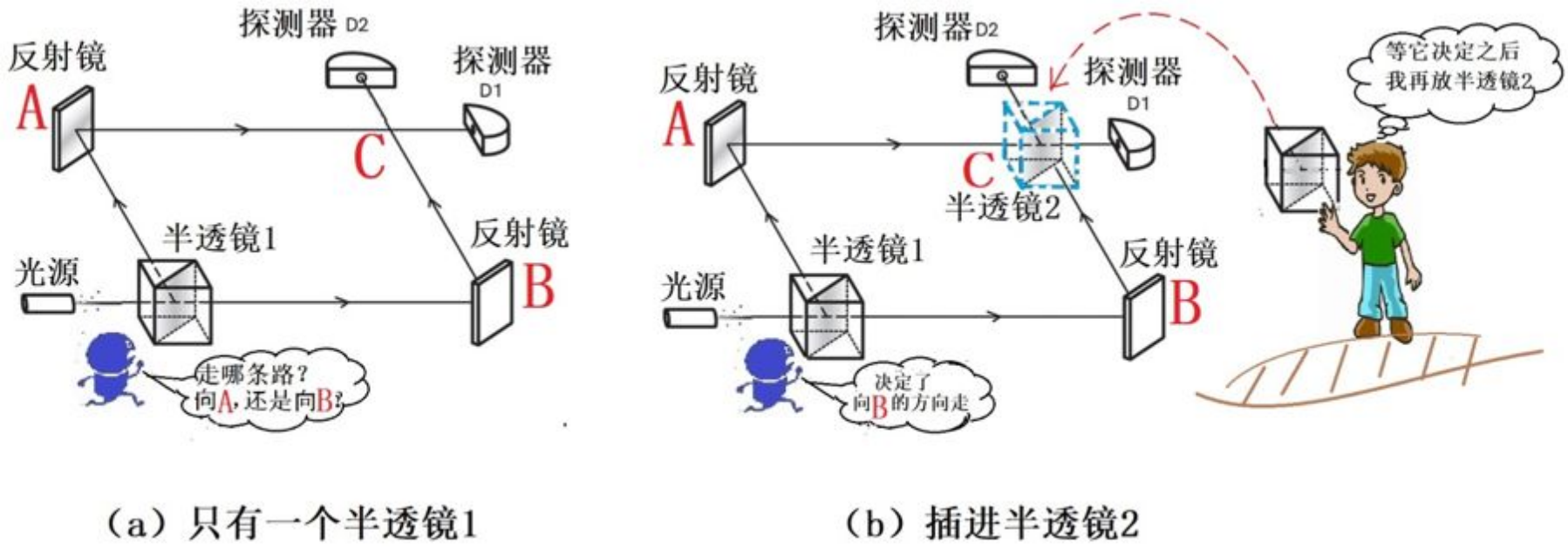
超决定？



逆时因果？



惠勒的延迟选择实验



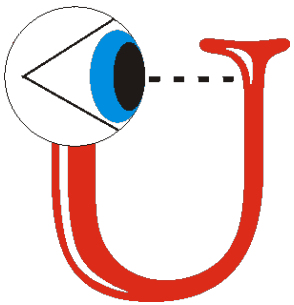
（a）只有一个半透镜1　　（b）插进半透镜2

- C 处不插半透镜, 光子走 A 或 B, 被 D1 或 D2 探测到, 呈现粒子性.
- C 处插半透镜, 光子同时走 A 和 B 两条路, 在 D1 和 D2 处发生相长干涉或相消干涉, 呈现波动性.
- 如果你等到光子已踏上中途, 再延迟决定在 C 处是否插入半透镜呢? — 不插, 粒子; 插, 波动.
- 有两种解释: 1. 你改变了过去. 2. 光子事先 "知道" 你未来的选择.
- 若 1, 逆时因果? 参与式人择宇宙? 若 2, 你还有自由意志吗?

宇宙通过观察而创生

有个青年这样说:
上帝一定觉得太稀奇
如果他发现这棵树
总是总是在庭院里
即使没有人在那里

答:
亲爱的年轻人
你的惊讶才稀奇
有我一直在庭院里
所以这树才能够
一直一直在这里
因为看着它的是
你忠诚的
老上帝



惠勒参与式的人择宇宙: 起初, 一切都处于朦胧的叠加态, 众多可能的叠加态之一发展出了复杂的生命意识, 生命意识通过观察将叠加态坍缩为了我们所知的这个宇宙.

惠勒 20 问游戏

惠勒和他的朋友们玩 20 问游戏.

1q 惠勒: 是动物吗?

1a 朋友 A: 不是.

2q 惠勒: 是白色的吗?

2a 朋友 B: 是的.

... 惠勒: ……

... 朋友: ……

20q 惠勒: 是云彩吗?

20a 朋友们: 是的.

朋友们根本就没有事先约定一个词. 他们一致同意, 每个人都可以随意回答 — 只要他心里有一个能够和已有回答协调的答案.

- 没有一个独立于提问的词, 直到它被提问产生出来.
  — 所提的问题不同, 提问的顺序不同, 都可能导向另一个不同的词.
- 没有哪一个基本现象是一个现象, 直到它被测量记录为一个现象.

Remarks

- 微观: 微观尺度的量子行为可能违背经典因果.
  — 如果真的像彭罗斯所认为的, 意识源于大脑神经元内部微管中的量子过程, 会怎样?
- 宏观: 经典因果起作用的地方. 比如在牛顿力学下, 物体运动是在固定背景结构下进行的, 是绝对时空舞台上上演的戏剧, 可以在保持其它演员不变的条件下, 干预某个演员的行为.
- 宇观: 大尺度下, 时空几何本身是动态的, 物体运动和时空结构相互影响. "时空告诉物质如何运动; 物质告诉时空如何弯曲." 广义相对论的非线性和非局域性, 使得经典因果依赖的 "保持其他条件不变" 的假设失效.

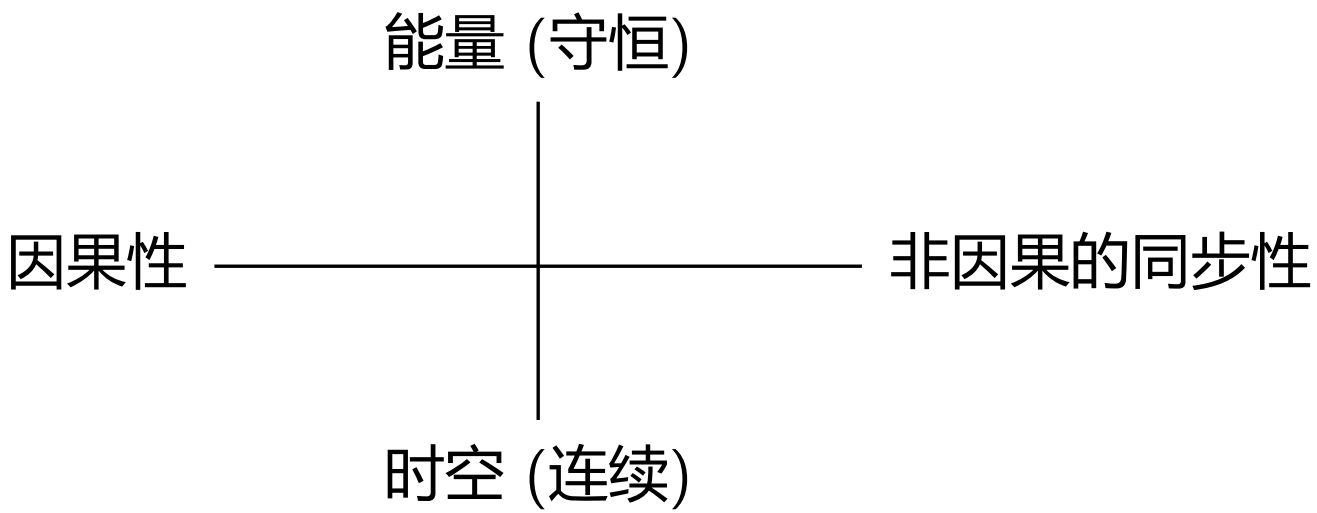
Problem: 量子因果是否对 AGI 构成挑战?

题外话: 荣格的非因果的 “同步性”

Example

1914 年, 一位德国母亲为她的男婴照了相, 并将照片留在 A 市的照相馆冲洗. 第一次世界大战爆发, 她无法取回照片. 两年后, 她为了给刚出生的女儿拍照, 在 B 市买了一卷交卷. 照片冲洗出来后, 她发现底片曝光了两次, 她女儿的相片是在她儿子的底片上的又一次曝光!

Remark: 荣格认为, 巧合发生的可能性远大于随机概率, 背后有一种基于 “意义” 而非 “因果” 的非定域关联. (但这种关联无法用于预测.)



Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Game Theory = Interactive Decision-Making Theory

- 一对情侣落到一个变态杀手手里.
- 杀手让他们玩剪刀石头布, 赢者生, 输者死.
- 情侣商量了一下, 决定一起出石头, 不独活.
- 结果: 女生死了, 男生活下来了.
- 男生出了剪刀, 女生出了布.

1. 男生爱女生, 女生自私?
2. 女生爱男生, 男生自私?
3. 男女彼此相爱?
4. 男女都自私?

拍卖 10 块钱!

现有 10 块钱钞票要拍卖. 规则如下:

1. 竞价者之间彼此不能交流.
2. 出价从 5 毛钱开始, 每次只能加 5 毛钱.
3. 出价最高者赢得钞票.
4. 但出价次高者也要支付自己的出价.

- 军备竞赛
- 美国大选
- “粉丝” 为 “爱豆”“打榜”
- 商家价格战
  — 可信威胁 (如: 商家宣称 “买贵了, 退差价”; 如果对方宣称 “退 2 倍差价” 呢?)
- 争相行贿搞关系
- “内卷”

Why Game Theory?

- 解释人们以特定方式行为的原因.
- 分析给定场景中的最优行为.
- 机制设计: 寻找某种互动方式, 以诱导出某种类型的行为.
- 如何把个体行为与社会利益统一起来?[21]
- 怎么解决个体理性与集体理性的矛盾?
- 集体理性也可以理解为个体的 "事前理性".
- 理性人之间如何更好地合作?
- Cooperation is agents with different goals interacting to mutual benefit.
- 合作问题的核心是激励 (incentive).
- 怎么解决 "无耻" 与 "无知"?

[21]经济学家、伦理学家亚当·斯密在《道德情操论》《国富论》中提出 "看不见的手", 研究如何对从利己出发的个体行为进行调节, 达到私利与公益相协调的均衡.

> 不读《国富论》不知道应该怎样才叫 '利己', 读了《道德情操论》才知道 '利他' 才是问心无愧的 '利己'.
>
> — 米尔顿·弗里德曼

The word "model" sounds more scientific than "fable" or "fairy tale" although I do not see much difference between them. Being something between fantasy and reality, a fable is free of extraneous details and annoying diversions. Thus we can clearly discern what cannot always be seen from the real world. On our return to reality, we are in possession of some sound advice or a relevant argument that can be used in the real world.

— *Ariel Rubinstein*

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
   - Preferences and Expected Utility
   - Strategic Games
   - Extensive Games with Perfect Information
   - Eliminating Dominated Strategies
   - Dynamic Games
   - Repeated Games
   - Signaling Games
   - Mechanism Design
   - Counterfactual Regret Minimization
   - Subgame Perfect Equilibrium
   - Games with Incomplete Information
   - Evolutionary Games
   - Voting System
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Determinants of Behavior in Situations of Strategic Interaction

Alice: "Would you please tell me, please, which way I ought to go from here?"
"That depends a good deal on where you want to get to," said the Cat.
"I don't much care where —" said Alice.
"Then it doesn't matter which way you go," said the Cat.
— Lewis Carroll: Alice's Adventures in Wonderland

- **Preferences**
  - a preference relation is a complete and transitive ordering of all alternatives available to an agent in a given situation.
  - given these assumptions, preferences can be represented by means of a utility function.
- **Beliefs**: in situations of strategic interaction, an agent has to form beliefs about the behavior of other agents.
- **Rational behavior**: a rational agent chooses the course of action that maximizes his utility given the available actions and given his beliefs.

Where do utilities come from?

From Preferences to Utility

- A lottery $\sum_i p_i o_i$ is a probability distribution over outcomes O.
- The outcomes O is closed under lotteries, and the agent also has preferences for lotteries.
- Preferences of a rational agent must obey constraints.

1. completeness:
$$a \succeq b \vee b \succeq a$$
2. transitivity:
$$a \succeq b \wedge b \succeq c \rightarrow a \succeq c$$
3. continuity:
$$a \succeq b \succeq c \rightarrow \exists p : pa + (1-p)c \sim b$$
4. independence:
$$a \succeq b \leftrightarrow \forall p : pa + (1-p)c \succeq pb + (1-p)c$$

Where do utilities come from?

Theorem (von Neumann-Morgenstern Utility Theorem 1944)

If a preference relation $\succ$ satisfies the above constraints, then there exists a function $u : O \to \mathbb{R}$ such that

$$a \succ b \iff u(a) > u(b)$$

$$\text{where} \quad u\left(\sum_i p_i o_i\right) = \sum_{i=1}^{n} p_i u(o_i)$$

Remark

- 表示偏好关系的效用函数不唯一.
- 表示同一个偏好关系的效用函数之间存在仿射变换.

$$\exists \alpha > 0 \exists \beta : u'(o) = \alpha u(o) + \beta$$

- ***Problem:*** 效用可以跨人际比较吗?

Remarks

- von Neumann-Morgenstern: if a rational decision maker knows the probabilities of obtaining a certain outcome, then she must choose as if maximizing the expected value of some function $u : O \to \mathbb{R}$.
- What if the probabilities are not given?
- Savage: a decision maker's preferences $\succ$ satisfies some "rationality" axioms, iff, there exists a nonatomic finitely additive probability measure μ on S and a nonconstant bounded function $u : O \to \mathbb{R}$ such that for every $f, g \in O^S$,

$$f \succ g \iff \int_S u(f(s))\mathrm{d}\mu(s) > \int_S u(g(s))\mathrm{d}\mu(s)$$

Furthermore, in this case, μ is unique and u is unique up to positive linear transformations.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
   - Preferences and Expected Utility
   - Strategic Games
   - Extensive Games with Perfect Information
   - Eliminating Dominated Strategies
   - Dynamic Games
   - Repeated Games
   - Signaling Games
   - Mechanism Design
   - Counterfactual Regret Minimization
   - Subgame Perfect Equilibrium
   - Games with Incomplete Information
   - Evolutionary Games
   - Voting System
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Games in Strategic Form

Definition (Games in Strategic Form)

A game in *strategic form* is a triplet $\langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$.

- The sets of players N.
- The sets of strategies of the players $(S_i)_{i \in N}$.
- The payoff functions $u_i : S \longrightarrow \mathbb{R}$, where $S \coloneqq \prod_{i \in N} S_i$.

囚徒困境及其它

	合作	背叛
合作	$-1,-1$	$-4,0$
背叛	$0,-4$	$-3,-3$

Table: 囚徒困境 (占优均衡)

	歌剧	足球
歌剧	$1,2$	$0,0$
足球	$0,0$	$2,1$

Table: 性别大战 (纳什均衡)

	妥协	进攻
妥协	$0,0$	$-1,1$
进攻	$1,-1$	$-\infty,-\infty$

Table: 胆小鬼博弈 (纳什均衡)

	正面	反面
正面	$1,-1$	$-1,1$
反面	$-1,1$	$1,-1$

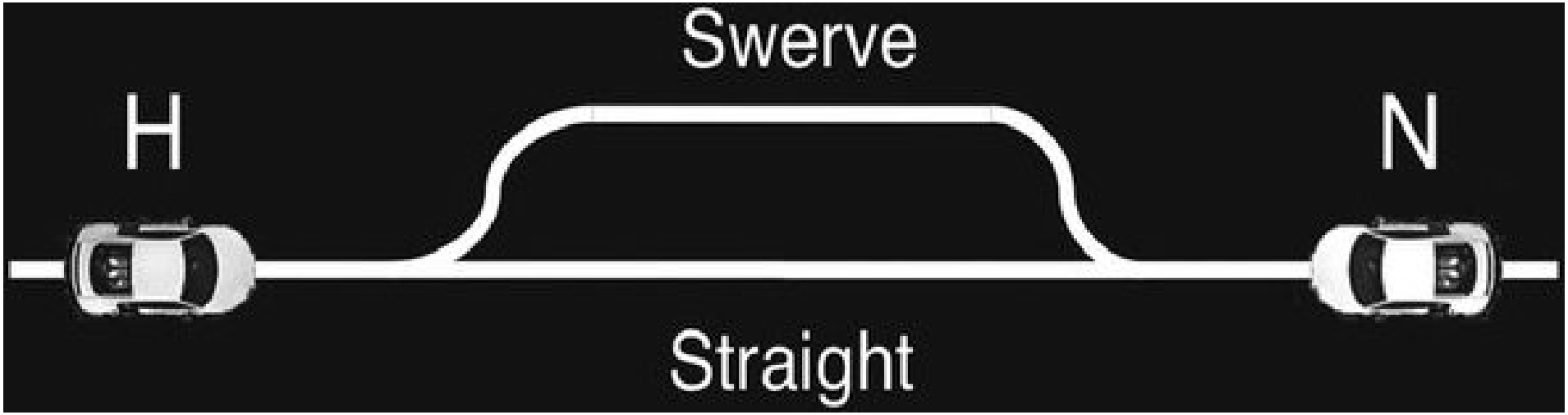
Table: 正反匹配 (混合纳什均衡)

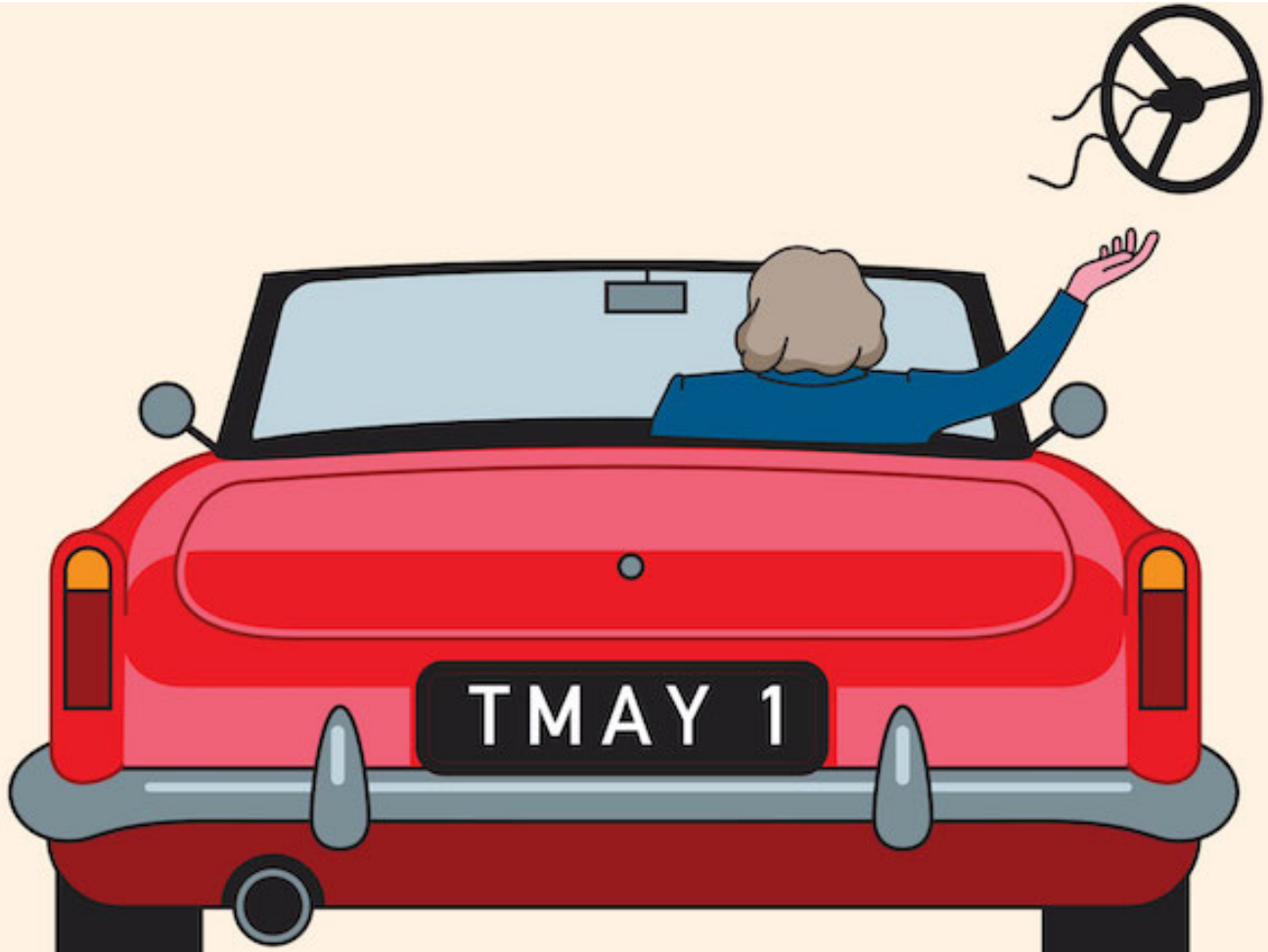
- **帕累托最优:** 除非 “损人”, 否则不可能 “利己”.
- **占优策略:** 独立于其他人的选择的最优策略.
- **占优均衡:** 所有人的占优策略的组合.
- **纳什均衡:** 所有人的最优策略的组合, 给定该策略中别人的选择, 没有人有积极性改变自己的选择.

求知 vs 应试

	求知	应试
求知	$100, 100$	$20, 80$
应试	$80, 20$	$60, 60$

Table: 学生内卷 (纳什均衡)


Swerve
H
N
Straight


TMAY 1

多人囚徒困境 — Braess Paradox

If we all go for the blonde and block each other, not a single one of us is going to get her. So then we go for her friends, but they will all give us the cold shoulder because no one likes to be second choice. But what if none of us goes for the blonde?

— *A Beautiful Mind*

- The addition of options is not necessarily a good thing.
- A strategy profile is Pareto optimal iff no other strategy profile improves the payoff to at least one player without decreasing the payoff of other players.
- The Nash equilibrium of a game is not necessarily Pareto optimal.

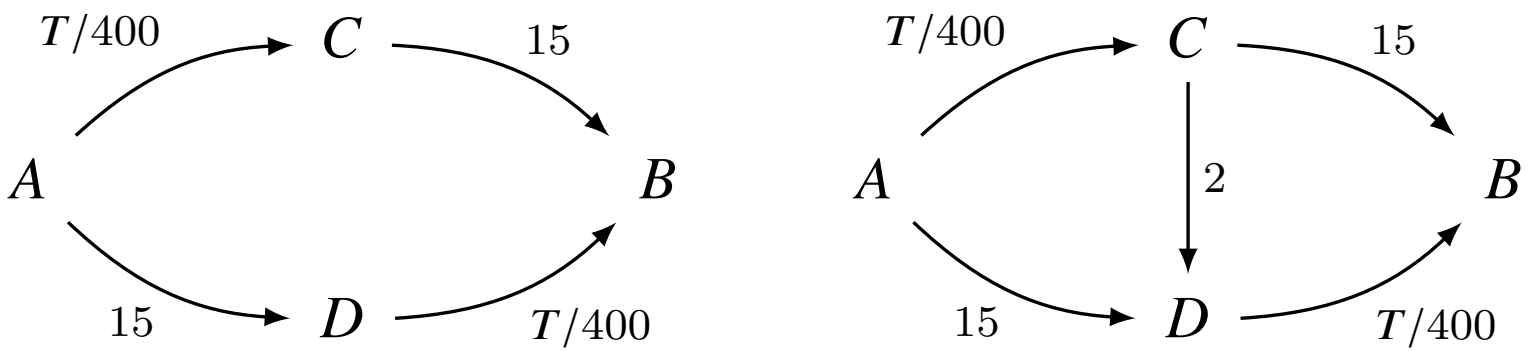
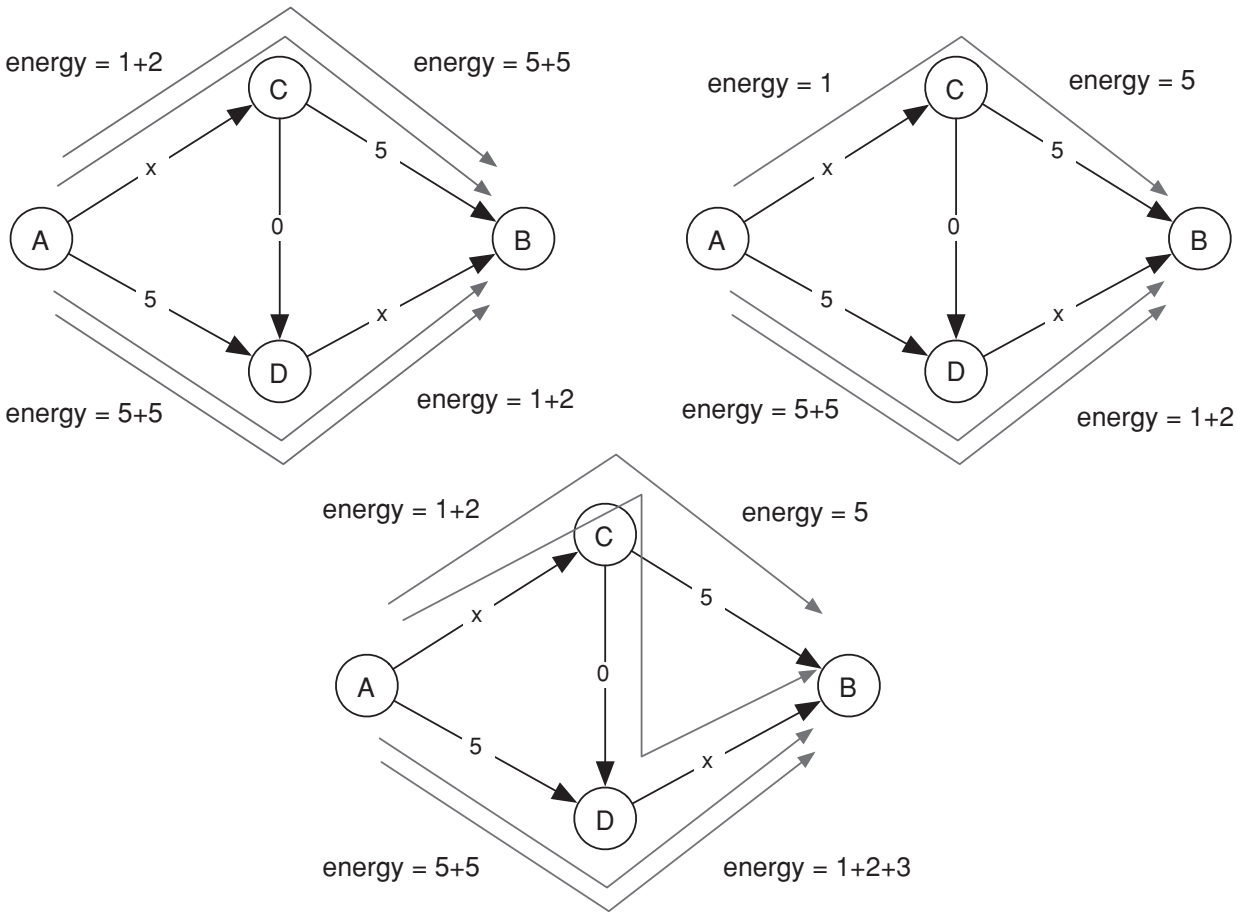


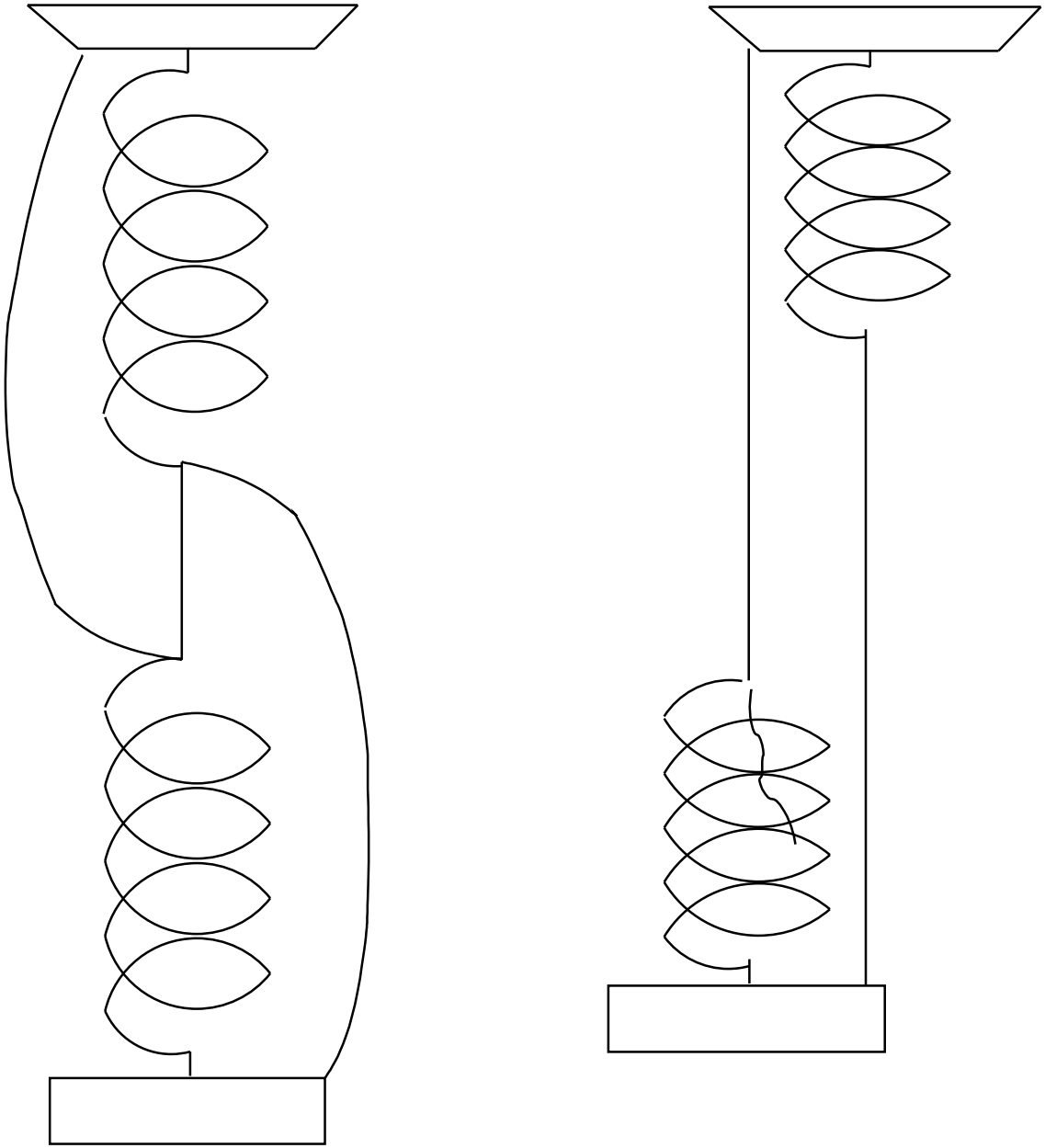
Figure: $T = 4000$ cars travelling around the lake

Braess Paradox

Let $L_e(x)$ be the travel time of each car traveling along edge e when x cars take that edge ($L_e(0) := 0$). Suppose there is a traffic graph G with x_e cars along edge e. Let $E(e) := \sum_{i=1}^{x_e} L_e(i)$, and $E(G) := \sum_{e \in G} E(e)$. Take a choice of routes that minimizes the total energy $E(G)$. That will be a Nash equilibrium.



Braess Paradox



Classification of Games

- 2 players vs. N players
- Static vs. Dynamic games
  — In static games, the players simultaneously choose actions, and they receive payoffs that depend on the combination of actions just chosen.
- Single play vs. Repeated games
- Zero-sum vs. Nonzero-sum games
- Symmetric vs. Non-symmetric games
  — symmetry: $u_i(s_{\pi(1)}, \ldots, s_{\pi(n)}) = u_{\pi(i)}(s_1, \ldots, s_n)$
- Perfect vs. Imperfect information games
- Complete vs. Incomplete information games
- Cooperative vs. Non-cooperative games

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
   - Preferences and Expected Utility
   - Strategic Games
   - Extensive Games with Perfect Information
   - Eliminating Dominated Strategies
   - Dynamic Games
   - Repeated Games
   - Signaling Games
   - Mechanism Design
   - Counterfactual Regret Minimization
   - Subgame Perfect Equilibrium
   - Games with Incomplete Information
   - Evolutionary Games
   - Voting System
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Definition (Games in Extensive Form)

A game in *extensive form* $\Gamma = \langle N, \mathcal{A}, \mathcal{H}, \mathcal{Z}, A, P, (\mathcal{I}_i)_{i \in N}, f_c, (u_i)_{i \in N} \rangle$.

- The set of players N.
- The set of actions $\mathcal{A} := \bigcup_{i \in N} \mathcal{A}_i$.
- A set $\mathcal{H}$ of sequences (finite or infinite) such that,
  1. $\varnothing \in \mathcal{H}$
  2. $\forall m < n \, (h_{1:n} \in \mathcal{H} \implies h_{1:m} \in \mathcal{H})$
  3. $\forall n \in \mathbb{N} \, (h_{1:n} \in \mathcal{H}) \implies h_{1:\infty} \in \mathcal{H}$
- A history $h \in \mathcal{H}$ is terminal iff $h = h_{1:\infty}$ or there is no $a \in \mathcal{A}$ s.t. $ha \in \mathcal{H}$. The set of terminal histories is denoted by $\mathcal{Z}$.
- The set of actions at h: $A(h) := \{a : ha \in \mathcal{H}\}$.
- The player function: $P : \mathcal{H} \setminus \mathcal{Z} \to N \cup \{c\}$.
- The function f_c associates with every history h for which $P(h) = c$ a probability measure $f_c(\cdot \mid h)$ on $A(h)$. $f_c(\cdot \mid h) \in \Delta(A(h))$.
- For player $i \in N$, $\mathcal{I}_i := \text{Partition}(\{h \in \mathcal{H} : P(h) = i\})$ where $\forall h, h' \in I \in \mathcal{I}_i : A(h) = A(h')$.
- The Payoff function for each player $i \in N$: $u_i : \mathcal{Z} \to \mathbb{R}$.

- If $|I| = 1$ for all $I \in \mathcal{I}_i$ and all $i \in N$, then we say that Γ is a game with *perfect information*.
  Remark: Each player is aware of every player's past actions and observations.
- If $|I| > 1$ for some $i \in N$ and some $I \in \mathcal{I}_i$, then we say that Γ is a game with *imperfect information*.
- If all elements of Γ are common knowledge, then we say that Γ is a game with *complete information*; otherwise, a game with *incomplete information*.

Theorem

Every perfect information game in extensive form has a pure strategy Nash equilibrium.

Pure Strategy

Definition (Pure Strategy)

A *pure strategy* of player i is $s_i : \mathcal{I}_i \to \mathcal{A}$ s.t. $s_i(I) \in A(I)$ for all $I \in \mathcal{I}_i$.
The *set of pure strategies* of player i is denoted by $S_i := \prod_{I \in \mathcal{I}_i} s_i(I)$.

A *strategy profile*

$$s = (s_i)_{i \in N}$$

is a vector consisting of one pure strategy for each player.
The *set of strategy profiles* is

$$S = \prod_{i \in N} S_i$$

$$s_{-i} := (s_1 \dots s_{i-1}; s_{i+1} \dots s_{|N|})$$

$$S_{-i} := S_1 \times \dots \times S_{i-1} \times S_{i+1} \times \dots \times S_{|N|}$$

Definition (Pure Strategy Consistent with History)

For any history h define a pure strategy s_i of player i to be consistent with h

$$s_i \rightsquigarrow h \iff \forall k < \ell(h)\,(P(h_{1:k}) = i \implies s_i(h_{1:k}) = h_{k+1})$$

$$\sigma_c(h) \coloneqq \prod_{h' \preceq h} f_{P(h')}(h_{\ell(h')+1} \mid h')$$

where $f_{P(h)}(\cdot \mid h) \coloneqq 1$ if $P(h) \neq c$.
Each strategy profile $s = (s_i)_{i \in N}$ can be assigned a payoff for each player i:

$$u_i(s) \coloneqq \sum_{h : \forall i\,(s_i \rightsquigarrow h)} \sigma_c(h) u_i(h)$$

Obviously,

$$\forall h \in \mathcal{H}\,(P(h) \neq c) \implies u_i(s) = u_i(h^s)$$

where h^s is the history uniquely determined by s.

Mixed Strategy

Definition (Mixed Strategy)

A *mixed strategy* of player i in an extensive game is a probability measure over the set of player i's pure strategies:

$$\sigma_i \in \Delta S_i$$

A *mixed strategy profile* specifies a mixed strategy for each player i

$$\sigma := (\sigma_i)_{i \in N}$$

The *set of mixed strategy profiles* is:

$$\prod_{i \in N} \Delta S_i$$

Each mixed strategy profile $\sigma = (\sigma_i)_{i \in N}$ implies a probability distribution over the *set of pure strategy profiles* S

$$\sigma(s) := \prod_{i \in N} \sigma_i(s_i)$$

Payoff of Mixed Strategy

Definition (Payoff of Mixed Strategy)

The expected payoff of player i given a mixed strategy profile σ is

$$u_i(\sigma) \coloneqq \sum_{s \in S} u_i(s)\sigma(s)$$

两人零和博弈 — 极小极大值定理

Theorem (Minimax Theorem — von Neumann 1928)

Let $X \subset \mathbb{R}^n$ and $Y \subset \mathbb{R}^m$ be compact convex sets. Let $f : X \times Y \to \mathbb{R}$ be a continuous function that is concave on X for each fixed $y \in Y$, and convex on Y for each fixed $x \in X$. Then

$$\max_{x \in X} \min_{y \in Y} f(x, y) = \min_{y \in Y} \max_{x \in X} f(x, y)$$

任意两人零和博弈都有混合纳什均衡 (x^*, y^*):

$$u_1(x^*, y^*) = \max_{x \in X} \min_{y \in Y} f(x, y) = \min_{y \in Y} \max_{x \in X} f(x, y) = -u_2(x^*, y^*)$$

其中, $f(x, y) := x^\mathsf{T} A y$.

Remark: $f(x^*, y^*)$ 是鞍点: $f(x^*, y) \geq f(x^*, y^*) \geq f(x, y^*)$.

A	石头	剪刀	布
石头	0	1	-1
剪刀	-1	0	1
布	1	-1	0

$$x^* = y^* = \begin{bmatrix} \frac{1}{3} \\ \frac{1}{3} \\ \frac{1}{3} \end{bmatrix}$$

$$u_1 = u_2 = 0$$

两人零和博弈 — Maxmin Strategies

- The maxmin strategy for player i is $\operatorname{argmax}_{s_i} \min_{s_{-i}} u_i(s_i; s_{-i})$, and player i's maxmin value is $\max_{s_i} \min_{s_{-i}} u_i(s_i; s_{-i})$.
  - Why would i want to play a maxmin strategy?
  - a conservative agent maximizing worst-case payoff.
- In a two-player game, the minmax strategy for player i against player $-i$ is $\operatorname{argmin}_{s_i} \max_{s_{-i}} u_{-i}(s_i; s_{-i})$, and player $-i$'s minmax value is $\min_{s_i} \max_{s_{-i}} u_{-i}(s_i; s_{-i})$.
  - Why would i want to play a minmax strategy?
  - to punish the other agent as much as possible.

Behavioral Strategy

Definition (Behavioral Strategy)

A *behavioral strategy* of player i is a collection $(\rho_i(I))_{I \in \mathcal{I}_i}$ of probability measures, where

$$\rho_i(I) \in \Delta(A(I)) \text{ for } I \in \mathcal{I}_i$$

A *behavioral strategy profile* is a vector consisting of one behavioral strategy for each player:

$$\rho := \left((\rho_i(I))_{I \in \mathcal{I}_i}\right)_{i \in N}$$

For any history $h \in I \in \mathcal{I}_i$ and action $a \in A(h)$, we denote by $\rho_i(h)(a)$ the probability $\rho_i(I)(a)$ assigned by $\rho_i(I)$ to the action a.
The *set of behavioral strategies* of player i is

$$B_i := \prod_{I \in \mathcal{I}_i} \Delta(A(I))$$

The *set of behavioral strategy profiles* is:

$$B := \prod_{i \in N} B_i = \prod_{I \in \mathcal{I}} \Delta(A(I))$$

Payoff of Behavioral Strategy

Definition (Payoff of Behavioral Strategy)

The expected payoff of player i given a behavioral strategy profile ρ is

$$u_i(\rho) \coloneqq \sum_{s \in S} u_i(s) \prod_{i \in N} \prod_{I \in \mathcal{I}_i} \rho_i(I)(s_i(I))$$

Perfect Recall

Definition (Experience)

Given a history h of an extensive game, the *experience* $X_i(h)$ of player i is the sequence of information sets that player i encounters in h and the actions that player i takes at them.

Definition (Perfect Recall)

An extensive game has *perfect recall* if for each player i, we have $X_i(h) = X_i(h')$ whenever h, h' are in the same information set of player i.

Remark: A player with perfect recall always remembers what actions he has taken and what he knew previously.

Two kinds of imperfect recall:

- 健忘 Forgetfulness: 忘记了观察或之前决策的结果.
- 心不在焉 Absent-mindedness: 忘记了之前是否已经做过某个决策.

Imperfect Recall — Experience

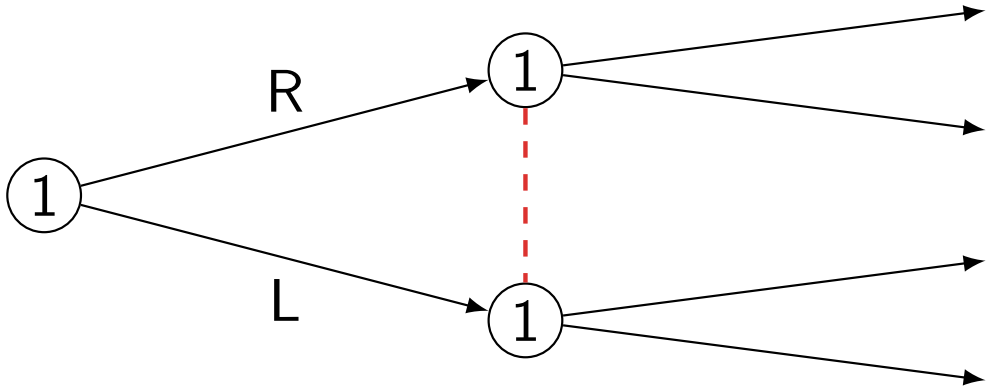


Figure: $X_1(L) = \{\varnothing, L, \{L, R\}\}, X_1(R) = \{\varnothing, R, \{L, R\}\}$

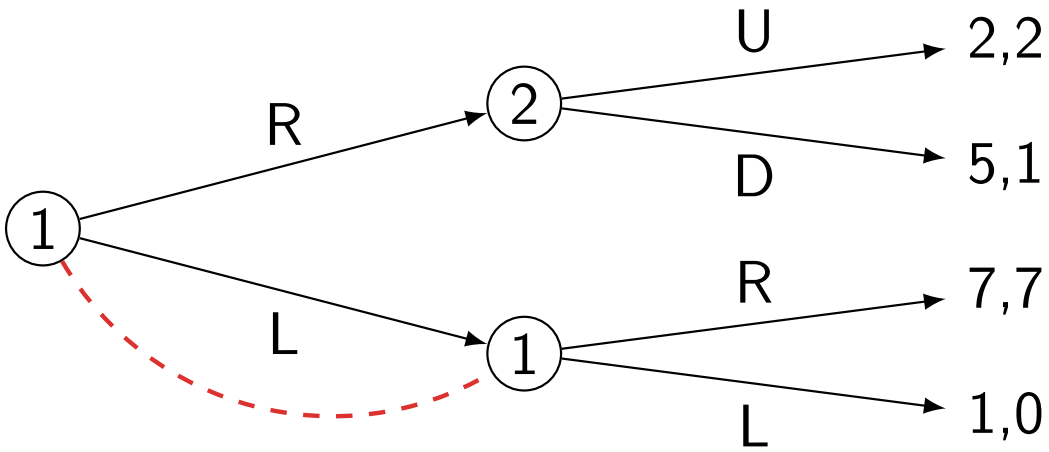
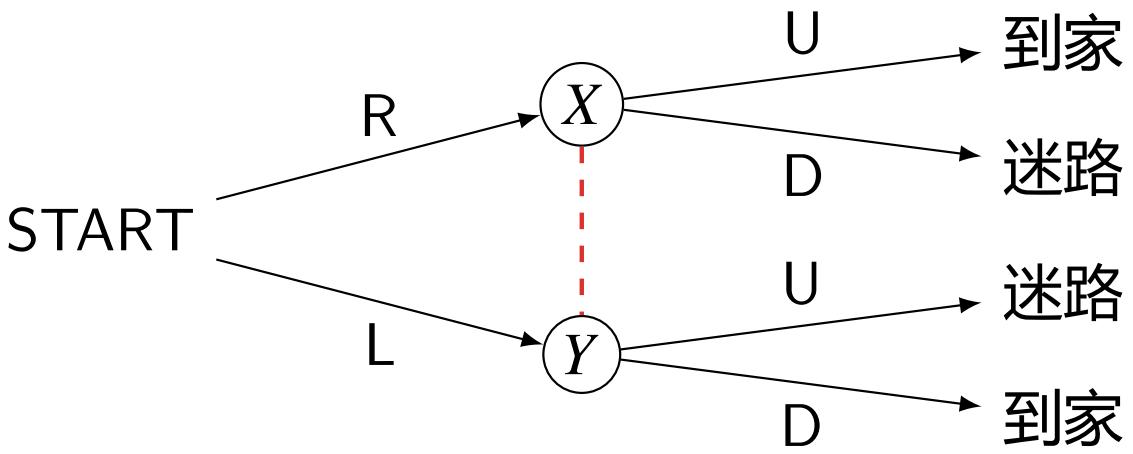


Figure: $X_1(\varnothing) = \{\{\varnothing, L\}\}, X_1(L) = \{\{\varnothing, L\}, L, \{\varnothing, L\}\}$

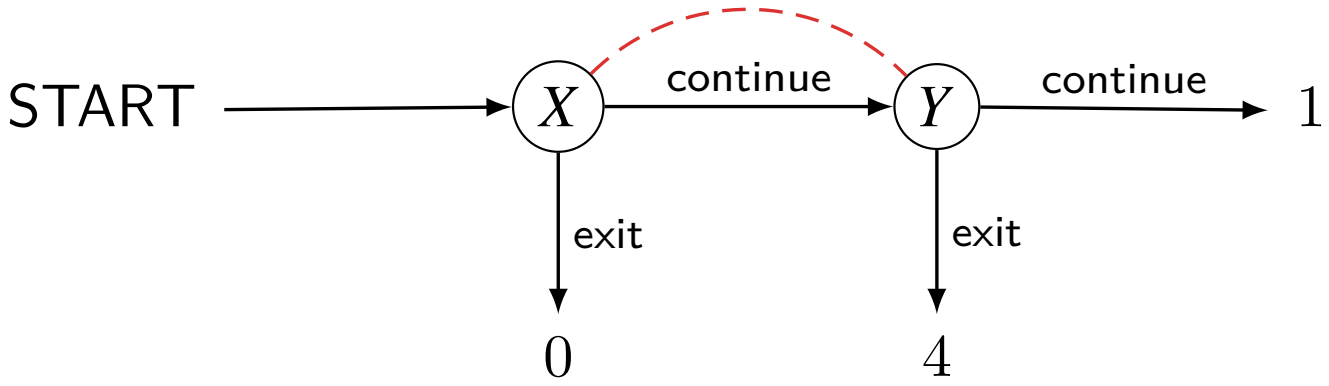
The difference between mixed and behavioral strategies



- Mixed strategies: $(p_{LD}, p_{LU}, p_{RD}, p_{RU})$, for example, $(\frac{1}{2}, 0, 0, \frac{1}{2})$
- Behavioral strategies: $\rho_i(\varnothing)(L) = p, \rho_i(\varnothing)(R) = 1 - p; \rho_i(\{L, R\})(D) = q, \rho_i(\{L, R\})(U) = 1 - q.$
- The mixed strategy $(\frac{1}{2}, 0, 0, \frac{1}{2})$ has no equivalent behavior strategy.

Remark: Both forgetfulness and absent-mindedness can prevent the existence of an Nash Equilibrium in behavioural policies.

The Absent-Minded Driver



- 因为无法区分自己在哪个路口, 所以每个路口应该采取相同的策略.
- 只有两个纯策略: continue, exit. 所以, 没有混合策略能达到 4.
- 因此, 没有混合策略等价于行为策略 $\rho(I)(\text{exit}) = \rho(I)(\text{continue}) = \frac{1}{2}$.
- 下面计算规划最优策略. 假设其策略是以概率 p 沿公路 continue.
- 司机的期望回报为:

$$U(p) = (1-p)\cdot 0 + p[(1-p)\cdot 4 + p\cdot 1]$$

- 在 START 出发前, 规划最优的决策为,

$$\frac{\mathrm{d}U(p)}{\mathrm{d}p} = 0 \implies p^* = \frac{2}{3} \implies U(\frac{2}{3}) = \frac{4}{3}$$

CDT Solution

- 如果认为一开始决定 continue 的决策是 q, **当下路口**正考虑的是 p.
- 协调的信念会认为正处在第一个路口的可能性是 $\mu(I)(X) = \frac{1}{1+q}$.
- 主观期望回报是,

$$U(p,q) = \frac{1}{1+q}[(1-p)\cdot 0 + p(1-q)\cdot 4 + pq\cdot 1] + \frac{q}{1+q}[(1-p)\cdot 4 + p\cdot 1]$$

- 两个路口的最优决策应该一样. 姑且称之为 p^*. 相信另一个路口选择最优的 p^* 的话, 这一个路口选择 $p = p^*$ 也应该是最优的.

$$F(q) := \operatorname*{argmax}_p U(p,q)$$

$$p^* \in F(p^*)$$

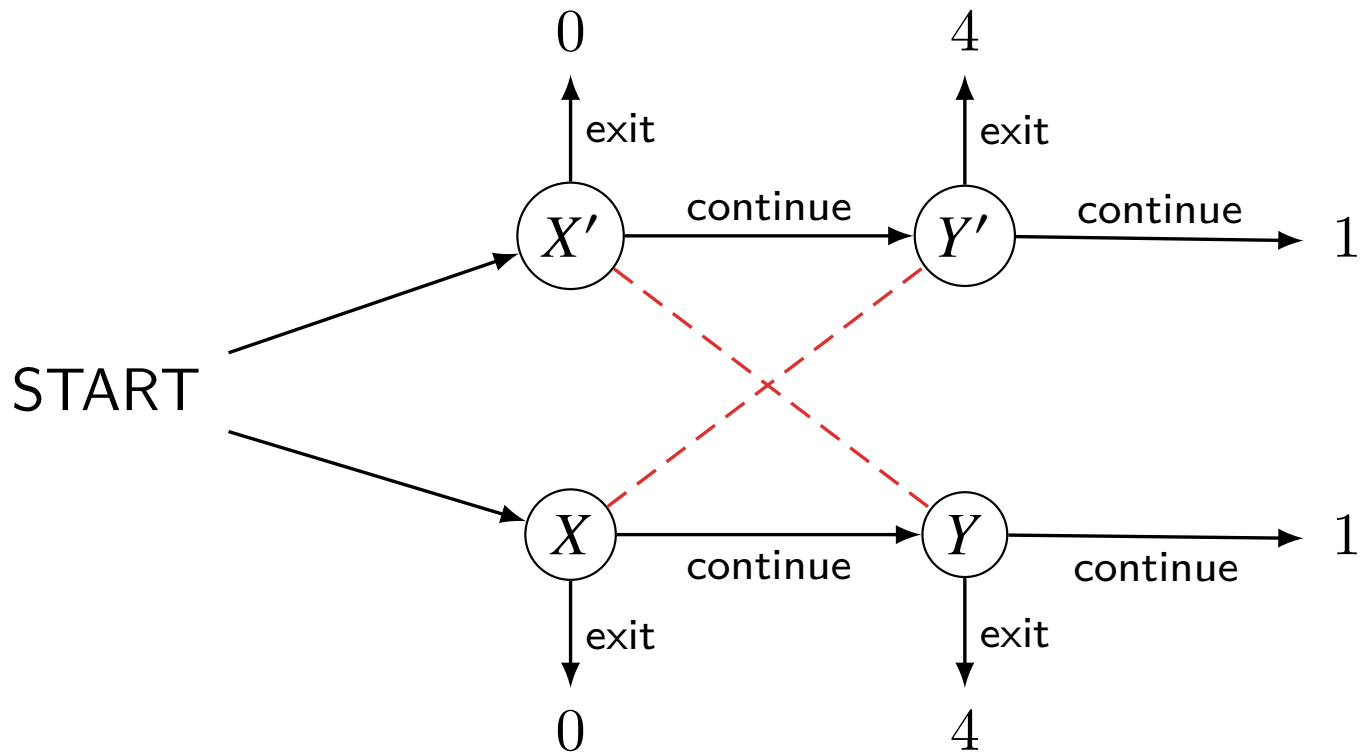
- 在路口, 行动最优的决策跟出发前的规划最优的决策一样, 都是 $\frac{2}{3}$.

$$\frac{\mathrm{d}U(p,q)}{\mathrm{d}p} = 0 \implies q = \frac{2}{3} \implies p^* = \frac{2}{3} \implies U(\frac{2}{3},\frac{2}{3}) = \frac{8}{5} > \frac{4}{3} = U(\frac{2}{3})$$

EDT & CDT in Game Theory

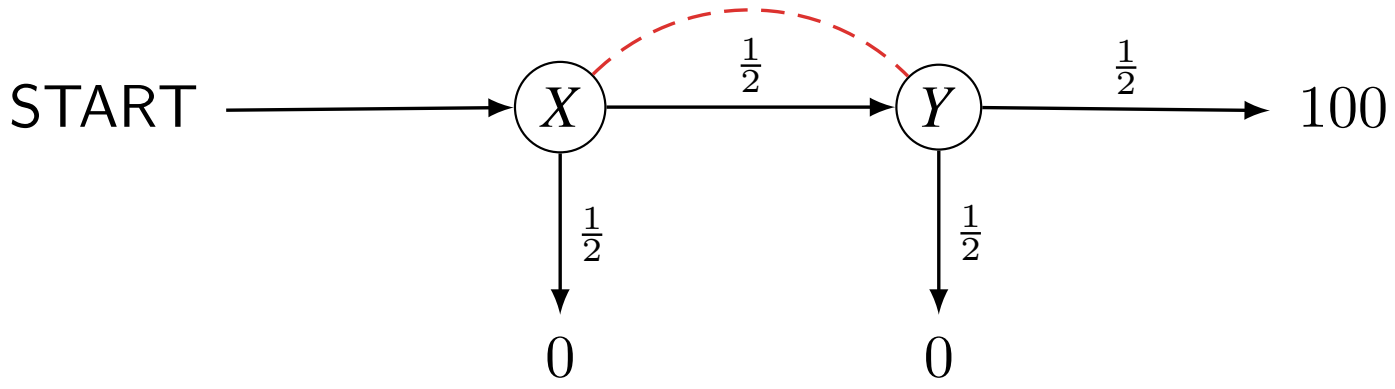
- EDT postulates that if a player deviates to a randomized decision $\alpha \in \Delta(A(I))$ at the current node, then she will have also deviated to α whenever she arrived in I in the past, and that she will also deviate to α whenever she arrives in I in the future.
- CDT postulates that a player can deviate to a decision $\alpha \in \Delta(A(I))$ at the current node, without violating her behavioral strategy ρ at past arrivals or future arrivals in I.

The Absent-Minded Driver — The Multiselves Approach



- 设想两个相同的 player, 以相同的概率令一个在某个路口做决策, 另一个在另一个路口做决策.
- 通过多个自我, 将 imperfect recall 变为 perfect recall.

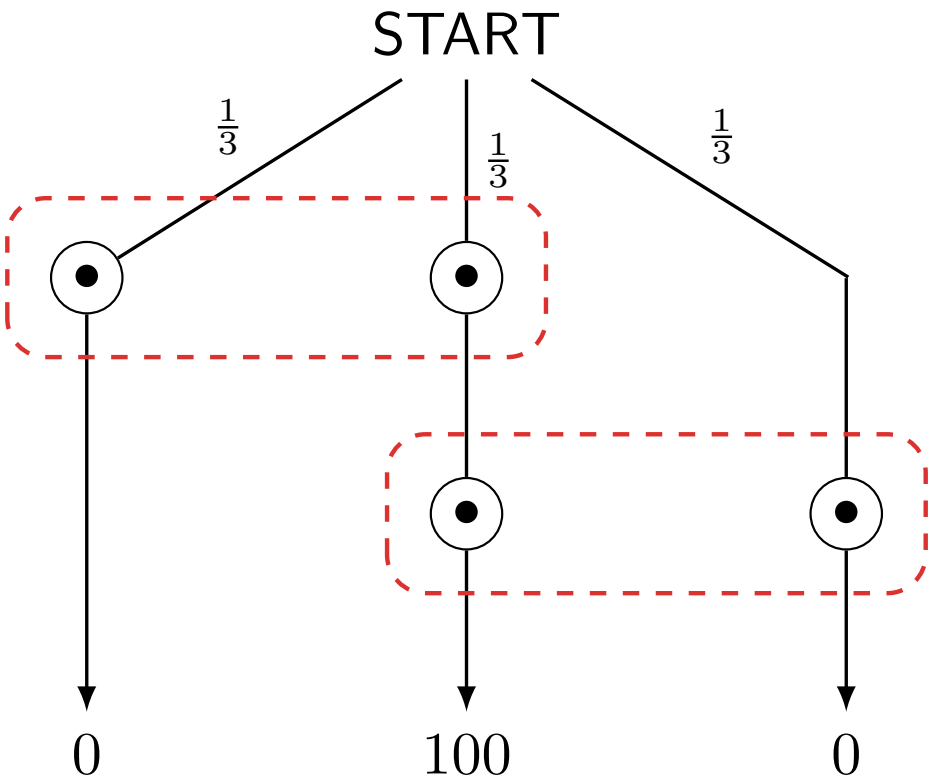
The Forgetful Passenger



- 在 START, 乘客能得 100 的概率是 $\frac{1}{4}$.
- 在 X, 乘客能得 100 的主观概率变成了 $\frac{1}{3}$, 虽然他会到 X 是必然的. — 因为: 经过 X 的概率是经过 Y 的两倍, 所以, 根据协调的信念, $\frac{2}{3} \cdot (\frac{1}{2})^2 + \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{3}$
- 假设在每一个点 (START, X, Y), 他都有机会花 30 块钱买一张赌约.
- 若他在 START, 一张赌约值 $\frac{1}{4} \cdot 100 - 30 = -5$; 若他在 X, 对他来说, 值 $\frac{1}{3} \cdot 100 - 30 = \frac{10}{3}$.
- 这是咋回事? — 因为在 X, 他不知道自己在 X.
- 最优策略: 每个路口买一张赌约!

$$\frac{1}{2} \cdot (-30) + \frac{1}{4} \cdot (-60) + \frac{1}{4} \cdot (200 - 60) = 5$$

The Forgetful Passenger — another example

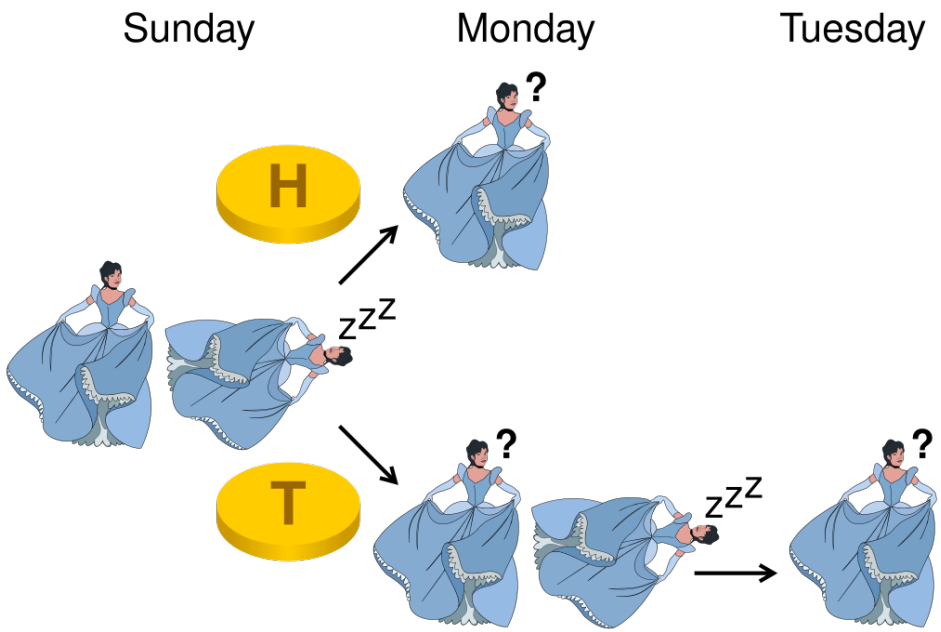


- 在 START, 能得 100 的概率是 $\frac{1}{3}$.
- 在后面两个信息集, 概率变成了 $\frac{1}{2}$.
- 但从 START 出发, 必然会到达某个信息集.

问题

- 如果你的记忆不完美...
- 如果你可以被 AI 轻易复制...
- 你该相信什么?

睡美人问题



- **原始版本:** 睡美人周日睡前被告知: 实验者会抛一枚硬币. 如果硬币正面朝上, 她会在周一被唤醒一次; 如果反面朝上, 则在周一、周二各被唤醒一次. 周一醒后入睡时会被清除醒来过的记忆. 周三游戏结束. 被唤醒时, 会被问一个问题: "你认为硬币正面朝上的概率是多少?"
- 变种一: 如果硬币没有提前抛, 而是在周一醒后再次入睡时抛呢?
- 变种二: 如果睡美人周一醒后被告知当天是周一呢?

睡美人问题的主要派别

1. 二分派: 既然事前就知道肯定会醒, 醒来这件事就不提供任何新信息, 概率不变 $\frac{1}{2}$.
2. 三分派:

   直观: 反面醒的次数是正面的两倍, 所以正面的概率是 $\frac{1}{3}$.

   论证: 设想睡美人面临 "变种二", 被告知当天是周一, 她会想: 既然无论硬币正反周一都会被唤醒, 原始版本与 "变种一" 等价, 完全可以认为硬币现在还没抛, 所以信念不应该变. 即 $P(C=1 \mid D=1)=P(C=0 \mid D=1)=\frac{1}{2}$. 又由于, 反面朝上时, 周一周二认知不可区分, 所以 $P(D=1 \mid C=0)=P(D=2 \mid C=0)=\frac{1}{2}$. 于是, $P(C=1, D=1)=P(C=0, D=1)=P(C=0, D=2)$. 而这三种情况构成了睡美人醒来的所有互斥的可能. 所以 $P(C=1 \mid A=1)=\frac{1}{3}$.

睡美人问题的主要派别

3. 双二分派: 无论是初始的, 被唤醒时的, 还是被告知周一后的都是 $1/2$.
4. 歧义派: 想象一下, 如果硬币正面朝上, 就把一个绿球放入盒子; 如果反面朝上, 就把两个红球放入盒子. 问题一: "你有多相信硬币朝上? / 一个绿球被放入盒子的概率是多少?". 问题二: "你有多相信硬币朝上导致了这次醒来? / 一个绿球从盒子里抽出的概率是多少?"

不同派别一致接受的预设:

1. 游戏设定的唤醒条件: 周一正面, 周一反面, 周二反面唤醒; 周二正面不唤醒. $P(A=1 \mid C=1, D=1)=P(A=1 \mid C=0, D=1)=P(A=1 \mid C=0, D=2)=1$, $P(A=1 \mid C=1, D=2)=0$.
2. 周日睡美人对硬币的初始信念 $P(C=1)=P(C=0)=\frac{1}{2}$.
3. 周一睡美人被唤醒后的信念按贝叶斯规则更新: $P(C \mid A=1)$.
4. 硬币反面朝上时, 睡美人对当天是周一还是周二认知不可区分: $P(D=1 \mid C=0)=P(D=2 \mid C=0)=\frac{1}{2}$.

不同派别各自特有的预设:

- 三分派特有的假设:
  - **AT1:** 睡美人被告知当天是周一后的信念按照贝叶斯规则更新: $P(C \mid D=1)$.
  - **AT2:** 当被告知周一后, 睡美人对硬币的信念不变: $P(C=1 \mid D=1)=P(C=1)=\frac{1}{2}$.
- 二分派特有的假设:
  - **AH:** 当被唤醒时, 睡美人对硬币的信念不变[22]: $P(C=1 \mid A=1)=P(C=1)=\frac{1}{2}$.

[22]与范弗拉森的内省原则 $P_0(H \mid P_1(H)=x)=x$ 相符.

路线图

- 睡美人问题的分歧，不在于概率计算，而需要诉诸更深层次的理论支持. 比如:
  - 自我采样假设 (SSA) vs 自我指示假设 (SIA) [Bos02; Bos07]
  - 总体功利主义 vs 平均功利主义 [Arm17]
- 本文认为，睡美人问题的核心不在于决策者的伦理类型，也不在于人择假设，而在于不完美回忆下的**信念分配**与**信念更新**.
  - 协调信念系统 vs Z-协调信念系统 [PR97]
  - “告知更新” vs 贝叶斯更新

核心观点:

- “三分派” 的论证的错误是误把 “告知更新” 当成了贝叶斯更新.
- “三分派” 与 “二分派” 之争更本质的源头是两种信念分配方式的差异.
- 我们将通过对 “广义睡美人” 问题的灵活适应性，因果发现的稳定性，以及对魔鬼创世和末日将至的拒斥，来论证 “二分派” 背后的 Z-协调信念系统的优越性，并回应哲学上的归纳难题和怀疑论难题.

Bostrom's Self-Sampling & Self-Indication Assumption

1. **自我抽样假设 SSA:** 其他条件相同的情况下, 观察者为了计算自己在某个世界中的位置, 应该先为可能世界分配概率, 再从世界 (过去, 现在, 未来) 真实存在的观察者集合中随机选择.
   - 两个可能世界: 正面和反面.

$$P(C=1)=P(C=0)=\frac{1}{2}$$

   - 正面世界有一个观察者, 抽样概率 1; 反面世界有两个, 抽样概率 $\frac{1}{2}$.

$$P(D=1\mid C=1)=1$$

$$P(D=1\mid C=0)=P(D=2\mid C=0)=\frac{1}{2}$$

2. **自我指示假设 SIA:** 其他条件相同的情况下, 观察者应该从所有认知上可能的观察者集合中随机选择.

$$P(C=1,D=1)=P(C=0,D=1)=P(C=0,D=2)=\frac{1}{3}$$

Remark: 自我抽样: "二分派"; 自我指示: "三分派".

Armstrong's Anthropic Decision Theory

- Armstrong 认为, 睡美人问题的核心不在于概率计算, 而在于决策, 不同伦理类型的睡美人如何进行决策. 其中, 他区分了总体功利主义和平均功利主义.
- 你有机会买一张赌约, 若正面朝上, 付你 1\$, 你愿意花多少钱买?
- 总体功利主义: 若反面朝上, 连续叫醒 n 天, 则总体期望得失:

$$\frac{1}{2}\cdot 1\cdot(1-x)+\frac{1}{2}\cdot n\cdot(-x)\geq 0 \implies x\leq \frac{1}{n+1}$$

- 平均功利主义: 反面世界的平均得失是 $-x$, 正面世界的平均得失是 $1-x$, 所以, 平均期望得失:

$$\frac{1}{2}\cdot(1-x)+\frac{1}{2}\cdot(-x)\geq 0 \implies x\leq \frac{1}{2}$$

- 总体功利主义类似自我指示, 平均功利主义类似自我抽样.

Remark: 睡美人问题的有趣之处可能不在于决策者的伦理类型, 而在于不完美回忆 (imperfect recall) 下的信念分配.

协调信念系统 vs Z-协调信念系统

- 协调信念系统: 立足当下, 莫问将来.
- Z-协调信念系统: 跑到尽头, 回看当下.

Remark: 在完美回忆的情况下, 二者相同, 即贝叶斯更新.

Consistent Belief System vs Z-Consistent Belief System

Definition (Consistent Belief System)

A belief system μ is consistent with the behavioral strategy ρ if for every information set I which is reached with positive probability and for every $h \in I$,

$$\mu(I)(h) = \frac{\rho(h)}{\sum_{h' \in I} \rho(h')}$$

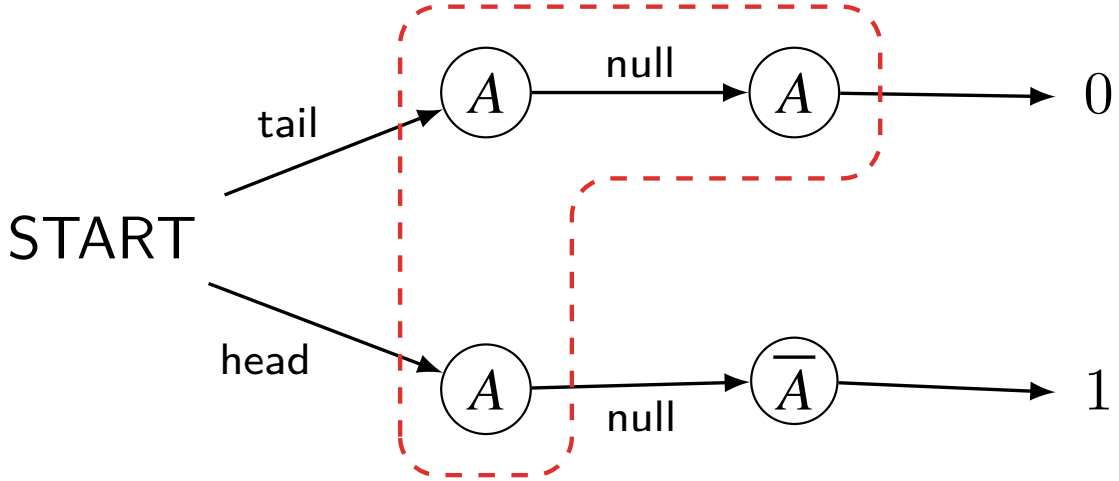
Definition (Z-Consistent Belief System)

A belief system μ is Z-consistent with the behavioral strategy ρ if for every information set I which is reached with positive probability and for every $h \in I$,

$$\mu(I)(h) = \frac{\sum_{z : h \preceq z} \frac{\rho(z)}{\#\{h' \in I : h' \preceq z\}}}{\sum_{z : \exists h' \in I (h' \preceq z)} \rho(z)}$$

睡美人问题 — 等价的打赌表述

问题: 你有机会买一张赌约, 若正面朝上, 付你 1\$, 你愿意花多少钱买?

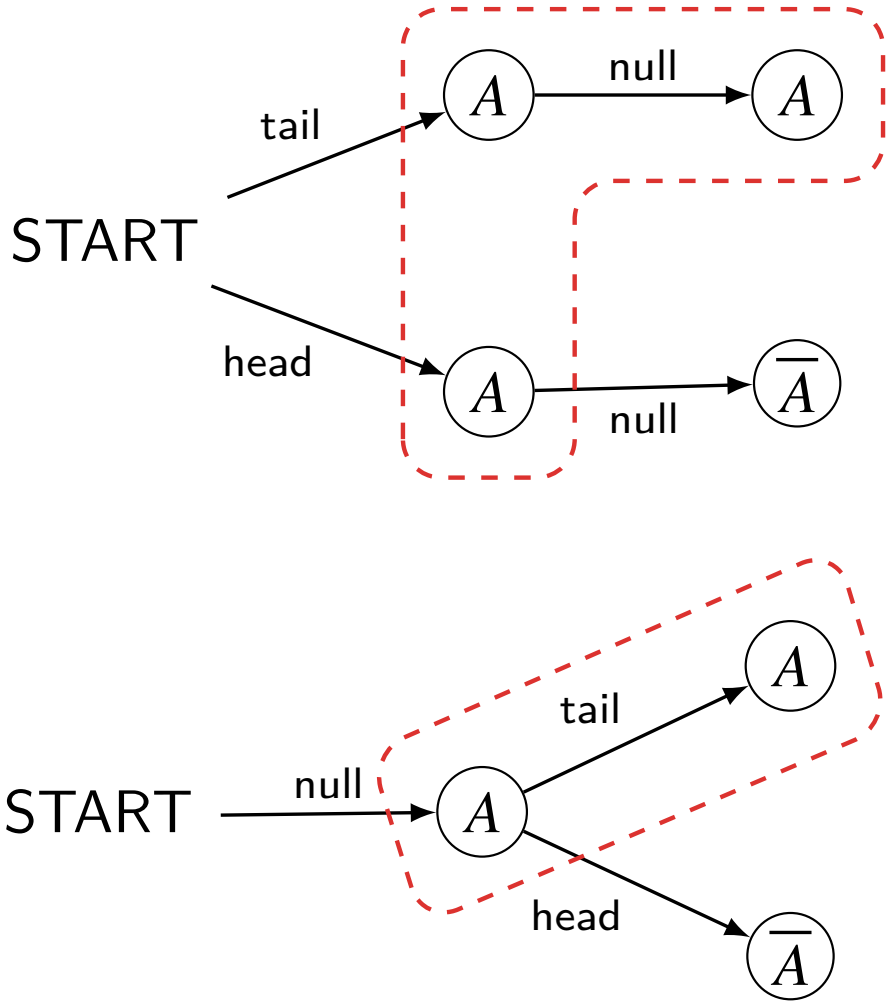


- $I = \{h, t, tn\}$
- $Z = \{hn, tn\}$
- 协调信念: $\mu(I)(h) = \mu(I)(t) = \mu(I)(tn) = \frac{1}{3}$
- Z-协调信念: $\mu(I)(h) = \frac{1}{2}$, $\mu(I)(t) = \mu(I)(tn) = \frac{1}{4}$
- 假设愿意花 x\$ 买. 则

$$\mu(I)(h)\cdot(1-x)+\mu(I)(t)\cdot(0-x)+\mu(I)(tn)\cdot(0-x) \geq 0 \implies x \leq \mu(I)(h)$$

- 协调信念: $x \leq \frac{1}{3}$; Z-协调信念: $x \leq \frac{1}{2}$
- 若 tail 朝上连续叫醒 n 天, 则协调信念下 $x \leq \frac{1}{n+1}$.

原始版本与 “变种一” — 不同版本的不同决策树表示



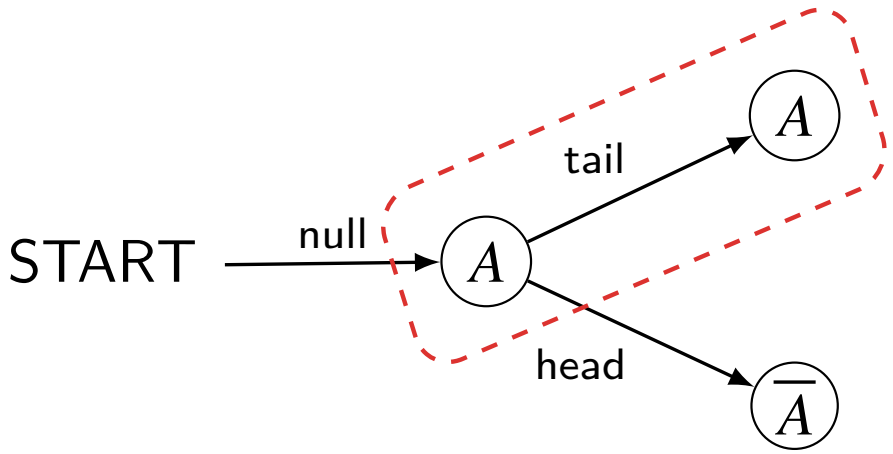
- 硬币在周日抛还是周一抛 (变种一) 没有区别.
- 图一可以帮助我们直接读出想要的概率值.

$$P(C = 1, D = 1 \mid A = 1) = \mu(I)(h)$$
$$P(C = 0, D = 1 \mid A = 1) = \mu(I)(t)$$
$$P(C = 0, D = 2 \mid A = 1) = \mu(I)(tn)$$

- 图二却只能通过 $P(D = 1 \mid A = 1) = \mu(I)(n)$ 重新计算相关概率.

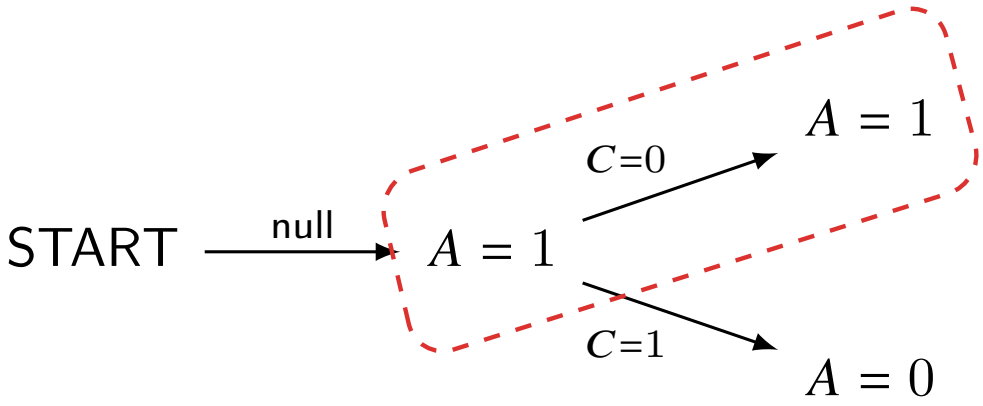
睡美人问题

Consistent Belief System vs Z-Consistent Belief System



- 信息集: $I = \{n, nt\}$
- 终端历史集: $Z = \{nh, nt\}$
- 行为策略导出的概率: $\rho(n) = 1$, $\rho(nh) = \rho(nt) = \frac{1}{2}$
- 协调信念系统: $\mu(I)(n) = \frac{2}{3}$
  睡美人问题里对应 $P(D = 1 \mid A = 1) = \frac{2}{3}$, 可得 $P(C = 1 \mid A = 1) = \frac{1}{3}$
- Z-协调信念系统: $\mu(I)(n) = \frac{3}{4}$
  睡美人问题里对应 $P(D = 1 \mid A = 1) = \frac{3}{4}$, 可得 $P(C = 1 \mid A = 1) = \frac{1}{2}$

无聊的计算过程



由 $P(C=0)=\frac{1}{2}$, 且 $P(D=1 \mid C=0)=P(D=2 \mid C=0)$, 可得 $P(C=0,D=1)=P(C=0,D=2)=\frac{1}{4}$. 假设 $P(C=1,D=1)=\theta$, 则

	$C=1$	$C=0$
$D=1$	θ	$\frac{1}{4}$
$D=2$	$\frac{1}{2}-\theta$	$\frac{1}{4}$

	$C=1$	$C=0$
$D=1$	$A=1$	$A=1$
$D=2$	$A=0$	$A=1$

$$P(D=1 \mid A=1)=\frac{P(D=1,A=1)}{P(A=1)}=\frac{\theta+\frac{1}{4}}{\theta+\frac{1}{4}+\frac{1}{4}}=\frac{4\theta+1}{4\theta+2}$$

$$P(D=1 \mid A=1)=\frac{2}{3} \implies \theta=\frac{1}{4} \implies P(C=1 \mid A=1)=\frac{\theta}{\theta+\frac{1}{4}+\frac{1}{4}}=\frac{1}{3}$$

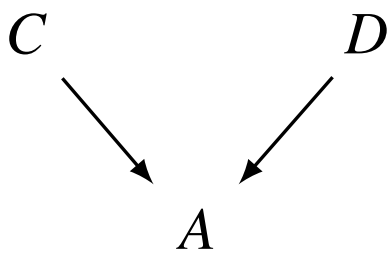
$$P(D=1 \mid A=1)=\frac{3}{4} \implies \theta=\frac{1}{2} \implies P(C=1 \mid A=1)=\frac{\theta}{\theta+\frac{1}{4}+\frac{1}{4}}=\frac{1}{2}$$

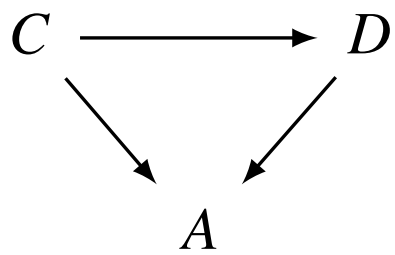
Remark

	$C=1$	$C=0$
$D=1$	θ	$\frac{1}{4}$
$D=2$	$\frac{1}{2}-\theta$	$\frac{1}{4}$

$$P(D=1 \mid C=1) = 2\theta$$
$$P(D=2 \mid C=1) = 1-2\theta$$
$$P(D=1 \mid C=0) = \frac{1}{2}$$
$$P(D=2 \mid C=0) = \frac{1}{2}$$

- $P(C,D) \stackrel{?}{=} P(C)P(D)$
- $\theta = \frac{1}{4}$ 时, C 与 D 独立.
- 否则, C 到 D 有因果箭头.



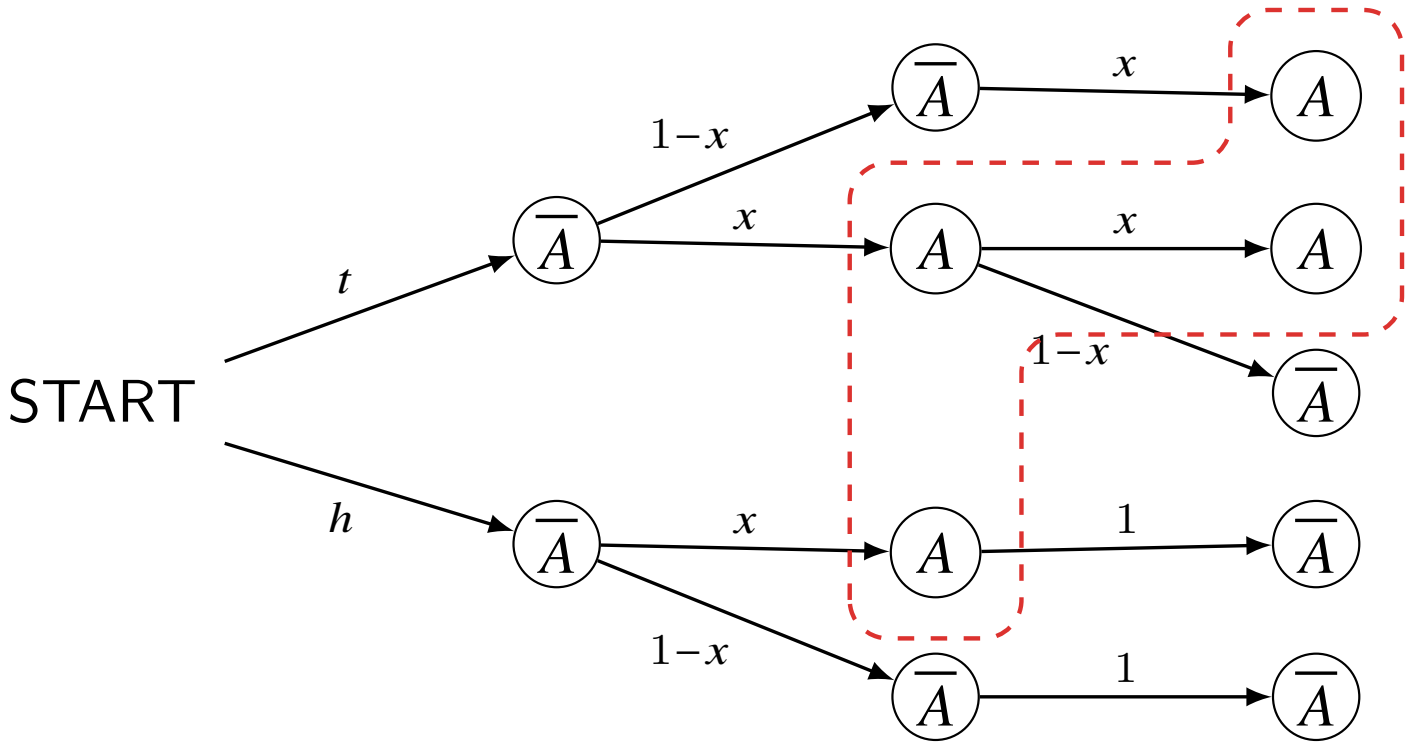


Remark: 反面 $C=0$ 时, 周一 $D=1$ 周二 $D=2$ 不可区分, 等概率; 若认为正面 $C=1$ 提升了周一 $D=1$ 的概率, 则 C 到 D 有因果箭头.

广义睡美人问题

假设睡美人每次被唤醒不再是确定的，而是以某个概率 x 被随机闹钟叫醒. 这时，睡美人被唤醒后就有了新的证据 W: “实验中我至少醒了一次”.

$$P(C=1\mid W)=\frac{P(W|C=1)P(C=1)}{P(W|C=1)P(C=1)+P(W|C=0)P(C=0)}=\frac{\frac{1}{2}x}{\frac{1}{2}x+\frac{1}{2}(1-(1-x)^2)}=\frac{1}{3-x}.$$

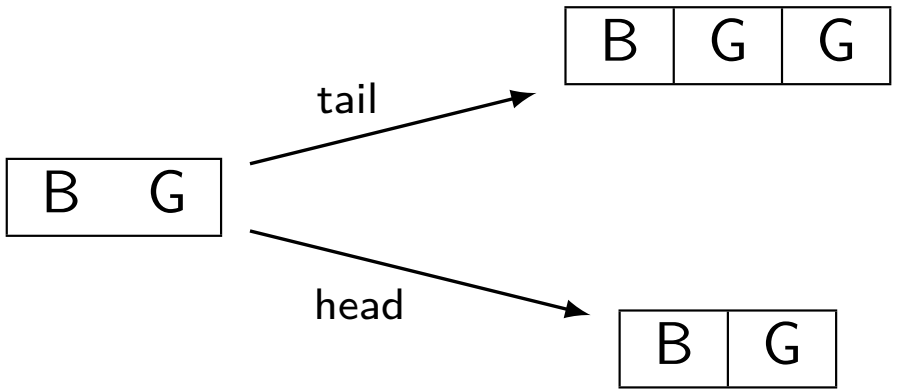


$I=\{hx,tx,txx,t(1-x)x\}$. $Z=\{h(1-x)1,hx1,txx,tx(1-x),t(1-x)x\}$.

Z-协调信念: $\mu(I)(hx)=\frac{\rho(hx)}{\rho(hx1)+\rho(txx)+\rho(tx(1-x))+\rho(t(1-x)x)}=\frac{1}{3-x}$.

协调信念: $\mu(I)(hx)=\frac{\rho(hx)}{\rho(hx)+\rho(tx)+\rho(txx)+\rho(t(1-x)x)}=\frac{1}{3}$.

睡美人问题 — 另一个变种

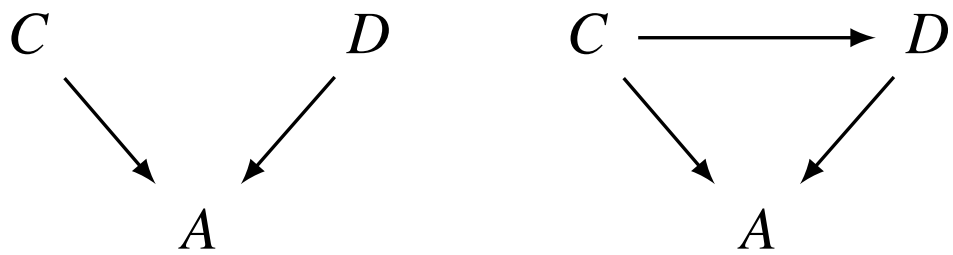


- B 和 G 睡觉时, 有一枚硬币抛出. 如果是反面, 则 G 将被克隆一份. 然后大家都被唤醒. 大家都知道实验的设定. 但醒后无法看到彼此.
- 每人都被问一个问题: "你有多相信硬币是正面朝上?" 或者等价地问: 你有机会买一张赌约, 若正面朝上, 付你 $1\$$, 你愿意花多少钱买?

1. 无论采用协调信念还是 Z-协调信念, 男孩 B 都会说 $1/2$.
2. 而对于女孩 G 来说, 如果她采用协调信念的话, 会说 $1/3$; 如果采用 Z-协调信念的话, 会说 $1/2$.

睡美人问题 — 对 “二分派” 的辩护

- 男孩 B 和女孩 G, 两个认知主体, 假设都采用相同的协调信念, 他们面对同一件事, 拥有相同的知识, 却得到不同的预测. 这只能说明, 恰恰是协调信念致使预测不一致, 也使得发现的因果不一致, 男孩 B 认为 C 和 D 之间有因果箭头, 女孩 G 认为没有箭头.
- 假设 B 和 G 都采用 Z-协调信念, 他们的预测就是一致的, 都是 $1/2$, 都认为应该有因果箭头, 构建出同一副因果图.



- Z-协调能保证因果发现的稳定性, Z-协调又支持 “二分派”, 在这个意义上, “二分派” 得到了辩护.

睡美人问题 — 对 “三分派” 的批评

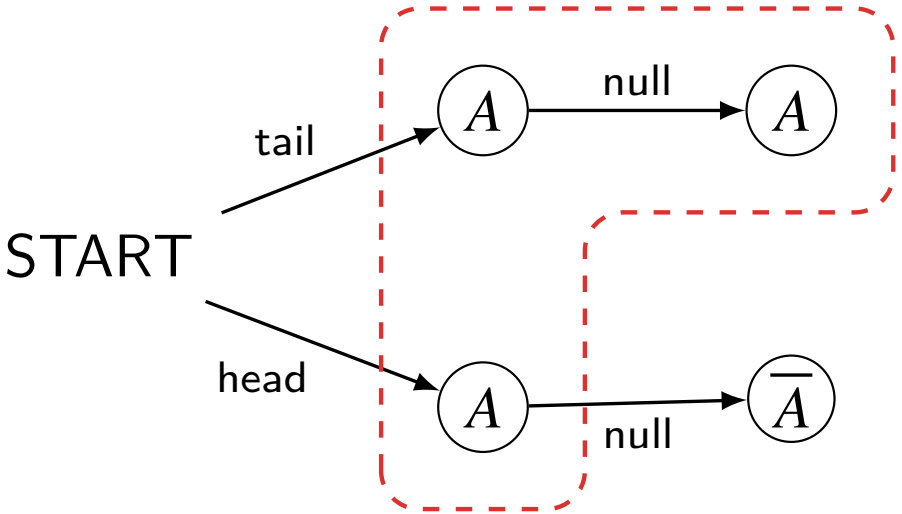
- “三分派” 认为, 当被告知当天是周一时, 其对正面的信念应该是 $1/2$. 从而 $P(C = 1 \mid D = 1) = \frac{1}{2}$.
- 但 “三分派” 的这条核心预设 **AT2** 是错的! **AT2** 的错误又是源于 **AT1**, “被告知周一” 不能视为条件化.

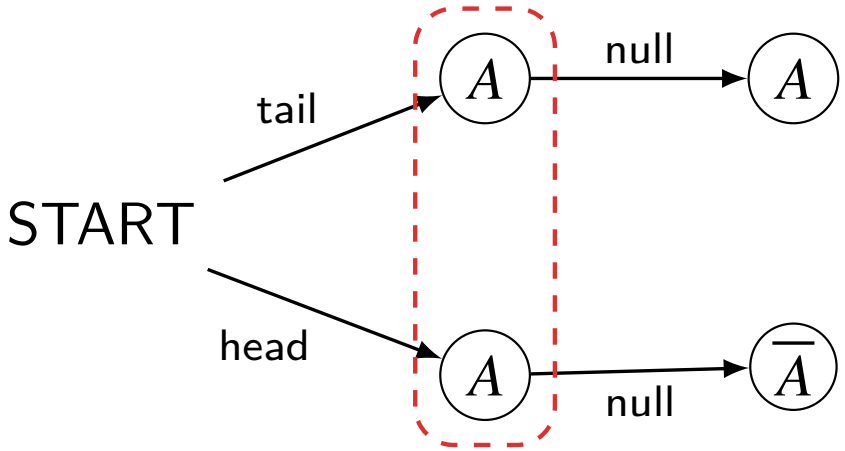
$$P^{D=1}(C = 1) = \frac{1}{2} \quad \text{而} \quad P(C = 1 \mid D = 1) = \frac{2}{3}$$

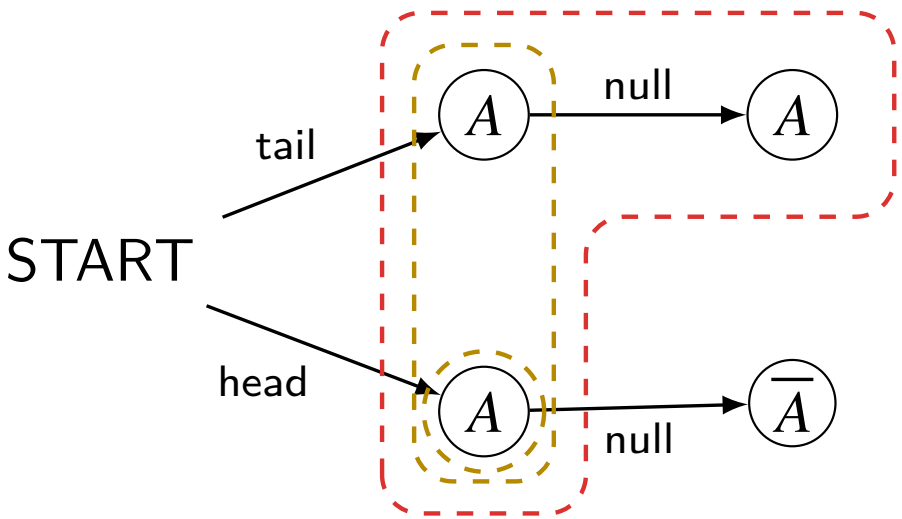
- 由此, 也无法通过 “告知更新” 反向推导出唤醒时的信念[23].
- 如果对 “告知更新” 在 Imaging 框架下进行解释, 为了保证 $P^{D=1}(C = 1) = \frac{1}{2}$, Imaging 对可能世界邻近关系的划分就需要依赖信念系统. 而与 Z-协调信念适配的划分方式更合理, 它把共享相同父变量取值和其它变量取值的世界看作最邻近的.
- 协调信念会让你相信魔鬼创造的世界 ☺

[23] “告知更新” 类似因果干预 do 算子, 切断了因果机制, 我们无法通过干预后的信息反向推导出干预前的状态.

为什么 “告知更新” 不能视为贝叶斯条件化?







- “告知更新” 是左到右.
- 条件化是上到下.

“告知更新” vs 贝叶斯条件化更新

- 条件化是基于当下信息集的同一个信念内进行的更新.

$$P(C=1 \mid D=1)=\frac{\mu(I)(h)}{\mu(I)(h)+\mu(I)(t)}=\frac{\frac{1}{2}}{\frac{1}{2}+\frac{1}{4}}=\frac{2}{3}$$

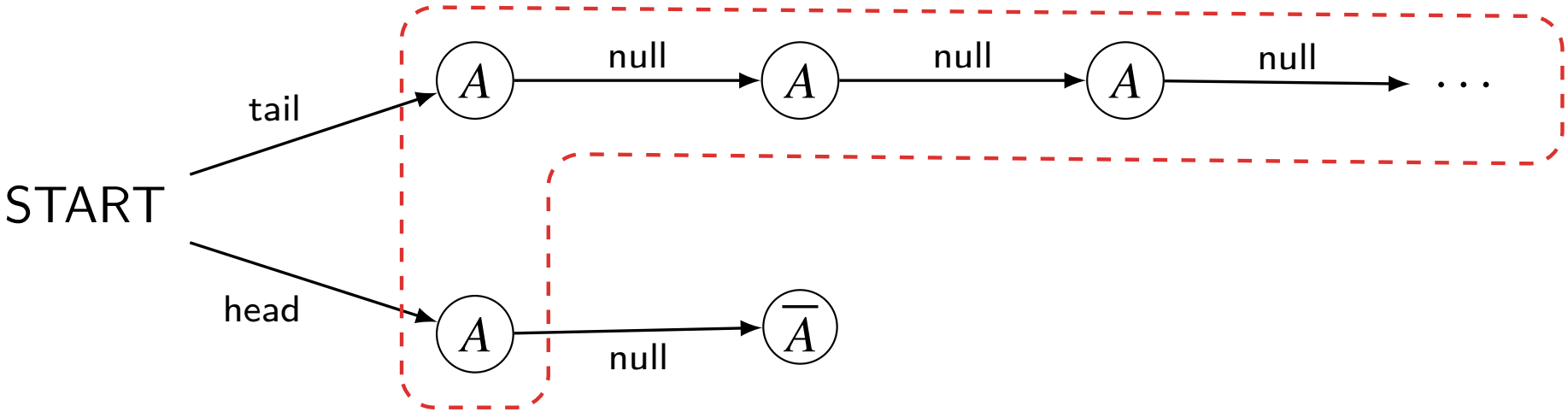
- “告知更新” 的两步走:
  1. 先重新划分信息集,
  2. 再进行信念分配.

Remark: 睡美人的 “告知更新”: 信息集从 $I=\{h,t,tn\}$ 变为了 $I'=\{h,t\}$; 信念也随之重新分配. 分布虽变, 但睡美人对正面朝上的信念值没变. 假设 Z-协调信念系统, 对硬币正面朝上的信念从初始的 $P(h)=\frac{1}{2}$, 到被唤醒时的 $P(C=1 \mid A=1)=\mu(I)(h)=\frac{1}{2}$, 再到 “被告知周一” 后的 $P^{D=1}(C=1)=\mu(I')(h)=\frac{1}{2}$, 数值始终未变. 这也是 “双二分派” 的直观合理的原因.

为什么“三分派”的推理是错的?

- 在协调信念下, 初始 $P(h) = \frac{1}{2}$, 被唤醒时为 $P(C = 1 \mid A = 1) = \mu(I)(h) = \frac{1}{3}$, “被告知周一”后 $P^{D=1}(C = 1) = \mu(I')(h) = \frac{1}{2}$.
- “被告知周一”后的情况跟“二分派”一样.
- 对于条件化, $P(C = 1 \mid D = 1) = \frac{\mu(I)(h)}{\mu(I)(h)+\mu(I)(t)} = \frac{\frac{1}{3}}{\frac{1}{3}+\frac{1}{3}} = \frac{1}{2}$.
- “三分派”的错误在于, 他们是通过“被告知周一”后的信念是 $\frac{1}{2}$, 反向推导出被唤醒时是 $\frac{1}{3}$ 的结论. 这是无法办到的. “告知更新”不是贝叶斯更新, 反向推不出有效的结论.

你相信魔鬼创世吗?



- 假设正常世界是 head 的世界, 概率 $\rho(h)$ 很高;
- tail 的世界是魔鬼创造的世界, 概率 $\rho(t)$ 很低.
- 虽然 $\rho(t)$ 很低, 但魔鬼的把戏强大, 能把睡美人叫醒的次数 n 很大.
- 根据协调信念, 你相信自己生活在正常世界的概率
  $\mu(I)(h) = \frac{\rho(h)}{\rho(h)+n\cdot\rho(t)} \xrightarrow{n\to\infty} 0.$
- 根据 Z-协调信念, 你相信自己生活在正常世界的概率
  $\mu(I)(h) = \frac{\rho(h)}{\rho(h)+\rho(t)}$. 你不会被魔鬼的把戏 n 带偏.
- 算法概率: $U(t) = 1^n$; $U(h) = 10$. 其中 t 的信息集 $I = \{1, 11, \ldots, 1^n\}$.
  $$P(h \mid *1) = \frac{\rho(h)}{\xi(*1)} = \frac{\rho(h)}{\rho(h)+\mu(I)(1)\cdot\rho(t)+\cdots+\mu(I)(1^n)\cdot\rho(t)} = \frac{\rho(h)}{\rho(h)+\rho(t)}$$

Remarks

- 这里所说的魔鬼创世，类似数量庞大的玻尔兹曼大脑，或人工智能模拟复制出的海量“缸中脑”.
- 假设人工智能可以轻易把你的大脑复制很多份，你会相信你是“缸中脑”吗?
- 所以这是一个跟怀疑论有关的问题. 如果你随着副本 n 的数量的增多，会越不相信现实世界的话，你就走向了怀疑论.

你相信末日将至吗?

- 设想有两个罐子.
- 罐子 h 有 100 个球.
- 罐子 t 有 10000 个球.
- 你随机选了一个罐子, 从中取出了一个球, 上面数字是 7 号.
- 你认为这个球来自哪个罐子的可能性更大?

1. SIA: 罐子 t 的可能性更大.

$$\frac{10000}{10000+100} > \frac{100}{10000+100}$$

2. SSA: 罐子 h 的可能性更大.

$$\frac{\rho(h)\frac{1}{100}}{\rho(h)\frac{1}{100}+\rho(t)\frac{1}{10000}} > \frac{\rho(t)\frac{1}{10000}}{\rho(h)\frac{1}{100}+\rho(t)\frac{1}{10000}}$$

Remarks

- SIA 不相信末日将至, 但相信魔鬼创世;
- SSA 不相信魔鬼创世, 但相信末日将至.
- 协调信念是 SIA 的精细化, 跟 SIA 一样, 相信魔鬼创世.
- $\mathcal{Z}$-协调信念是 SSA 的精细化, 会不会跟 SSA 一样, 也相信末日将至呢?
- $\mathcal{Z}$-协调性通过行为策略引入的因果链条, 避免了观察者分布的随机性假设, 使得分配的信念依赖于行为策略导出的概率分布, 而不是观察者的数量, 从而避免了末日论证的挑战.
- $\mathcal{Z}$-协调性, 既不会相信魔鬼创世, 也不会相信末日将至.
- $\mathcal{Z}$-协调性, 优于协调性, 优于 SIA 和 SSA.

Remarks

- 在这个意义上，睡美人问题/魔鬼问题/末日问题与传统的归纳问题，比如罗素的火鸡问题，罗素的“世界可能是五分钟之前创立”的问题，莱布尼茨穿过有限个点有无数条线的选择问题，古德曼的绿蓝问题，以及玻尔兹曼大脑问题/AI 复制脑问题，都是一脉相承的同类问题.
- 例如，为什么我们不应该相信“世界可能是五分钟之前创立”的/翡翠是绿蓝的/我们是随机涨落出来的玻尔兹曼大脑/AI 复制人？因为这些假设的先验概率太低，低于正常世界的概率，这时，不管在所有可能世界中，包括有序的和无序的，玻尔兹曼大脑的总数 N_{BB} 比自然进化出的大脑的总数 N_{nat} 多多少，或 AI 能随意复制出多少意识副本，都无法提高我们对这个古怪世界本身的信念.
- 乌鸦怪论也是类似，根据 Nicod 标准，当你看到一只黑乌鸦时，会增强你对“所有乌鸦都是黑的”的信念. Nicod 标准是一条不涉及可能世界的断言，是错的. 同样的证据，具体是增强还是减弱你的信念，需要考虑所有可能世界才能决定. 只立足当下证据，不考虑可能世界是无法给出靠谱的信念的. 而 Z-协调信念和 Solomonoff 归纳都充分考虑了所有可能世界、可能世界的先验、以及世界的内部机制.

Remarks

- 协调性只从当下的信息集出发评估信念, 看不到所有可能世界的概率 $\rho(z)$, 缺乏对包括未来的整个历史的全局视角, 跟 SIA 一样, 等于放弃了对可能世界本身的先验评估, 所以才会被魔鬼创造的样本带着跑.
- 对于可能世界才能谈 "自然齐一性", 有没有 "规律" 是针对可能世界来说的, 可能世界中的样本都是 "现象", 抛开 "规律" 谈 "现象" 没有意义. 这就是 "三分派" 及其背后的预设所犯的错误 — 一个归纳问题上的方法论错误.

小结: 睡美人的启示

- “二分派” 完胜 “三分派”
- **信念更新:** “告知更新” $\neq$ 贝叶斯更新
- **信念分配:** Z-协调 完胜 协调
  - 对广义睡美人的灵活适应性
  - 因果发现的稳定性
  - Imaging 可能世界邻近关系划分的合理性
  - 抵抗魔鬼与末日

Outcome-Equivalence

For any profile σ/ρ, we define the *outcome* $\overline{\sigma}(h)/\overline{\rho}(h)$ to be the multiplicative product of all the chance probabilities and move probabilities over history h when all players choose their moves according to σ_i/ρ_i.

$$\sigma_i(h) \coloneqq \sum_{s_i \rightsquigarrow h} \sigma_i(s_i)$$

$$\overline{\sigma}(h) \coloneqq \prod_{i \in N \cup \{c\}} \sigma_i(h)$$

$$\overline{\rho}(h) \coloneqq \prod_{k=0}^{\ell(h)} \rho_{P(h_{<k})}(h_{<k})(h_k)\sigma_c(h)$$

Two (mixed or behavioral) strategies of any player are *outcome-equivalent* iff for every collection of pure strategies of the other players the two strategies induce the same outcome.

Theorem (Outcome-Equivalent Theorem)

In games with perfect recall, for any mixed strategy there is an outcome-equivalent behavioral strategy and vice versa.

Proof.

“ $\Longrightarrow$ ”

with perfect recall,

$$\forall h, h' \in I \in \mathcal{I}_i \forall a \in A(I)\,(\sigma_i(h) = \sigma_i(h') \implies \sigma_i(ha) = \sigma_i(h'a))$$

$$\rho_i(I)(a) := \frac{\sigma_i(ha)}{\sigma_i(h)} \text{ for any } h \in I \in \mathcal{I}_i$$

“ $\Longleftarrow$ ”

$$\sigma_i(s_i) := \prod_{I \in \mathcal{I}_i} \rho_i(I)(s_i(I))$$

Mixed Extension

Definition (Games in Strategic Form)

A game in *strategic form* is a triplet $\langle N, (S_i)_{i\in N}, (u_i)_{i\in N}\rangle$.

- The sets of players N.
- The sets of strategies of the players $(S_i)_{i\in N}$.
- The payoff functions $u_i : S \to \mathbb{R}$.

Definition (Mixed Extension)

The mixed extension of the strategic game $\langle N, (S_i)_{i\in N}, (u_i)_{i\in N}\rangle$ is $\langle N, (\Delta S_i)_{i\in N}, (u_i)_{i\in N}\rangle$, where the payoff function u_i in $\langle N, (\Delta S_i)_{i\in N}, (u_i)_{i\in N}\rangle$ is

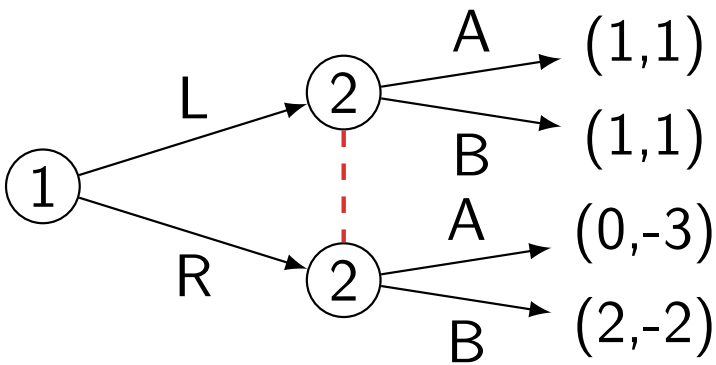
$$u_i(\sigma) = \sum_{s\in S} u_i(s)\sigma(s)$$

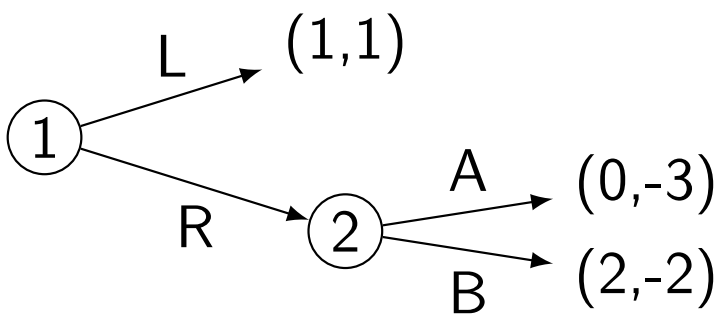
Note that u_i is multilinear.

$$u_i(\lambda\sigma_i' + (1-\lambda)\sigma_i''; \sigma_{-i}) = \lambda u_i(\sigma_i'; \sigma_{-i}) + (1-\lambda)u_i(\sigma_i''; \sigma_{-i})$$

Moving between the Two Forms of Representation

- Each extensive form game can be represented in a unique way as a game in strategic form.
- There might be several games in extensive form which represent the same strategic form game.
- Games in strategic form can be interpreted as games in which all players choose simultaneously.
- Games in strategic form with two players and finite number of strategies can be represented by a matrix.





	A	B
L	$1, 1$	$1, 1$
R	$0, -3$	$2, -2$

Translating from Extensive Form to Matrix Form

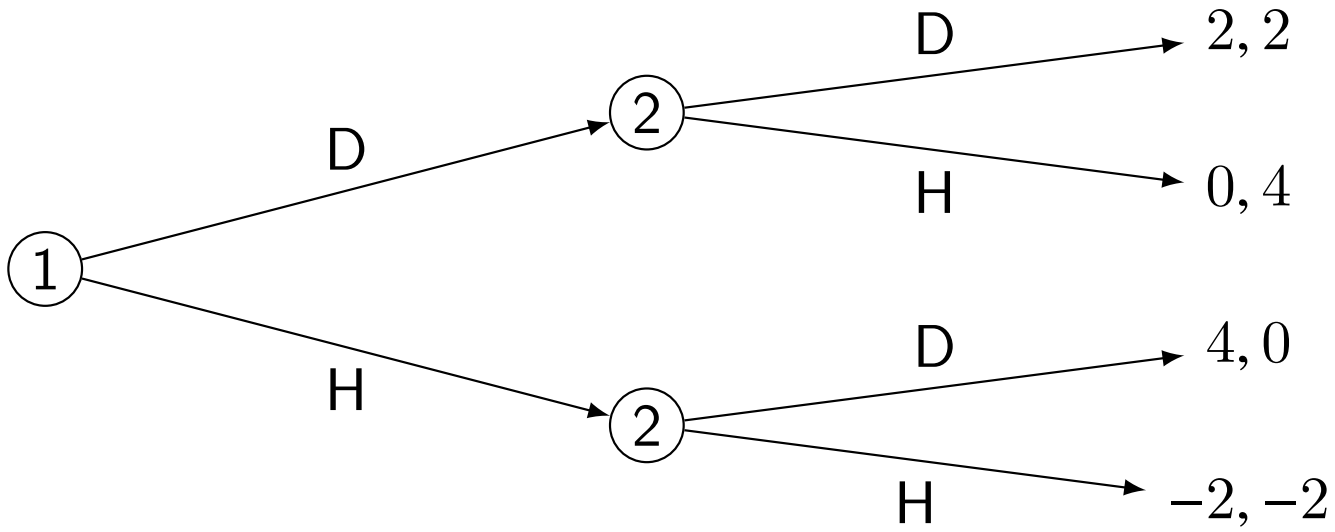
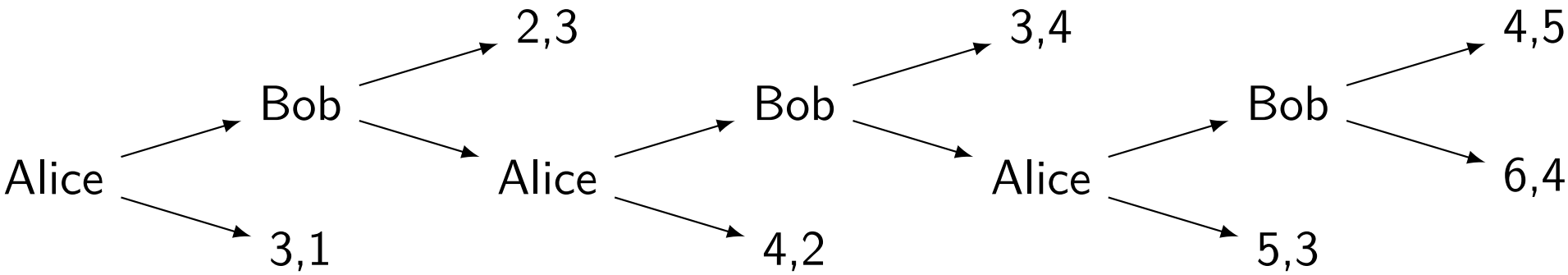


Figure: The Hawk-Dove Game

	(H|H,H|D)	(H|H,D|D)	(D|H,D|D)	(D|H,H|D)
H	-2,-2	-2,-2	4,0	4,0
D	0,4	2,2	2,2	0,4

逆向归纳、不确定性、公共知识



- 运用逆向归纳求解这个博弈，我们发现，只要玩家有机会就会主动结束博弈.
- 逆向归纳要求玩家的理性是公共知识.
- 如果我们把不确定性引入到玩家对于彼此的收益的知识中的话，两位玩家就可能选择继续博弈.
- 这个不确定性可以表示玩家的理性不是公共知识.
- 如果一个玩家不确定另一位玩家是否是理性的，即使后者是理性的，他/她也可以通过表现的“非理性”而获利.

Pareto Optimality

- A strategy profile s^* *strongly Pareto-dominates* s iff $\forall i \in N : u_i(s^*) > u_i(s)$.
- A strategy profile s^* *weakly Pareto-dominates* s iff $\forall i \in N : u_i(s^*) \geq u_i(s)$ and $\exists i \in N : u_i(s^*) > u_i(s)$.
- A strategy profile s^* is *Pareto optimal* iff

$$\forall i \in N \forall s \in S : u_i(s) > u_i(s^*) \implies \exists j \in N : u_j(s) < u_j(s^*)$$

Pareto Optimality

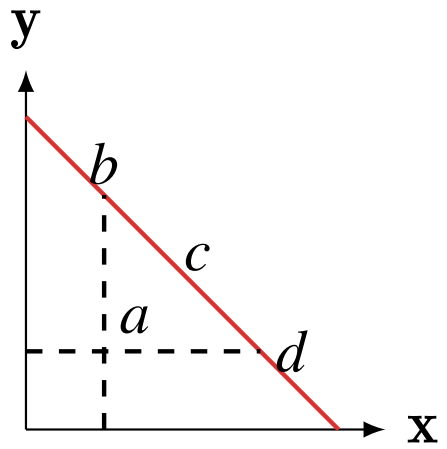


Figure: Pareto Improvement vs Rawls' Difference Principle

- Given an initial situation, a *Pareto improvement* is a new situation where some agents will gain, and no agents will lose.
- A situation is called *Pareto dominated* iff it has a Pareto improvement.
- A situation is called *Pareto optimal* iff no change could lead to improved satisfaction for some agent without some other agent losing.
- The *Pareto frontier* is the set of all Pareto optimal allocations.

not change	change
$100, 100$	$1000, 99$

$0.5 * 1000 + 0.5 * 99 = 549.5 > 100$

Rawls' Veil of Ignorance

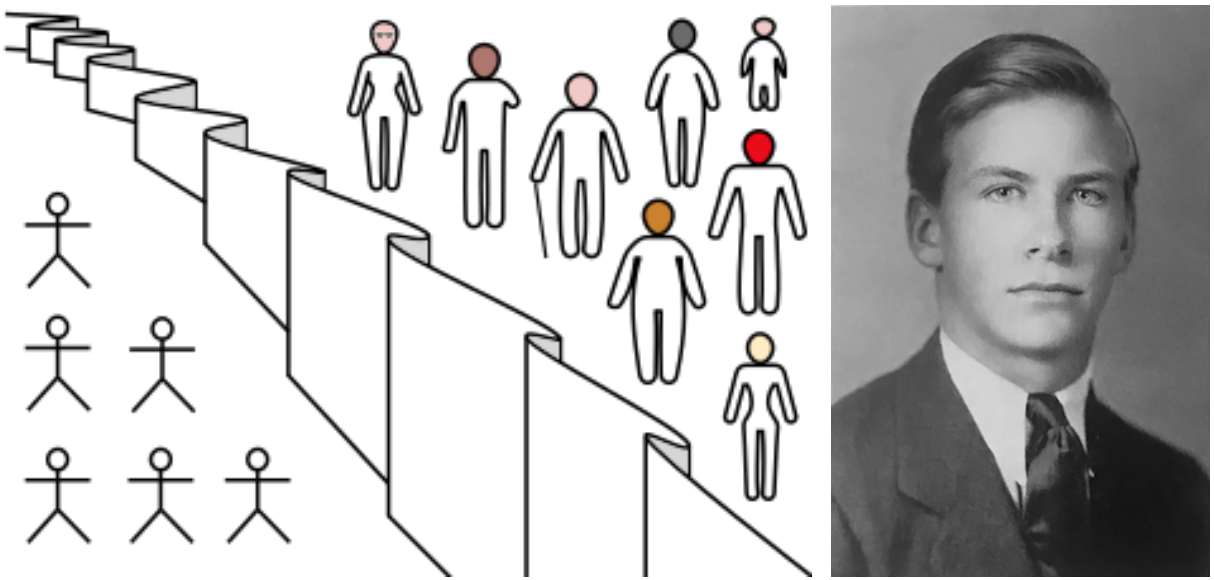
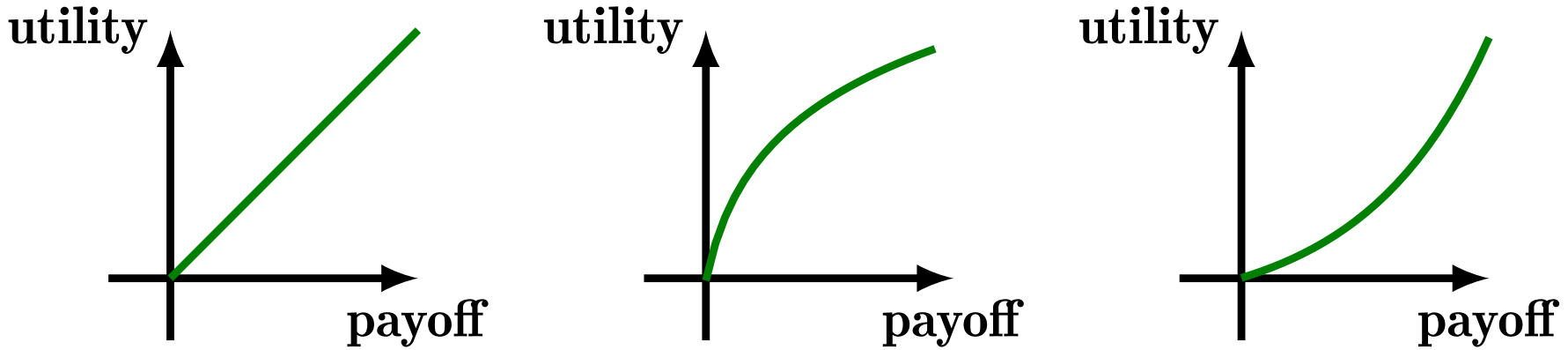


Figure: The reason that the least well off member gets benefited is that it is argued that under the veil of ignorance people will act as if they were risk-averse.

1. Each citizen is guaranteed a fully adequate scheme of basic liberties, which is compatible with the same scheme of liberties for all others;
2. Social and economic inequalities must satisfy two conditions:
   - attached to positions and offices open to all under conditions of fair equality of opportunity;
   - to the greatest benefit of the least advantaged members of society (the difference principle).

Risk-Neutral, Risk-Averse, Risk-Seeking



- risk-neutral $\mathbb{E}\big[u(x)\big] = u\big(\mathbb{E}[x]\big)$
- risk-averse $\mathbb{E}\big[u(x)\big] \leq u\big(\mathbb{E}[x]\big)$
- risk-seeking $\mathbb{E}\big[u(x)\big] \geq u\big(\mathbb{E}[x]\big)$
- certainty equivalent $\mathrm{CE}(x) := u^{-1}\big(\mathbb{E}[u(x)]\big)$

Remark

- *Preferences depend on the agent's reference point: current wealth.*
- *For gains, they are risk averse.*
- *For losses, they are risk seeking.*
- *Losses are (about) twice as bad as gains.*

圣彼得堡悖论

- 掷一枚均匀硬币, 直到首次出现正面为止, 如果所需的次数为 n, 则奖金为 2^n 元.
- 你愿意为这个赌博支付多少钱?

$$\mathbb{E}[u(W)] = \sum_n u(2^n) P(T^{n-1}H) = \sum_n 2^n \frac{1}{2^n} = \infty$$

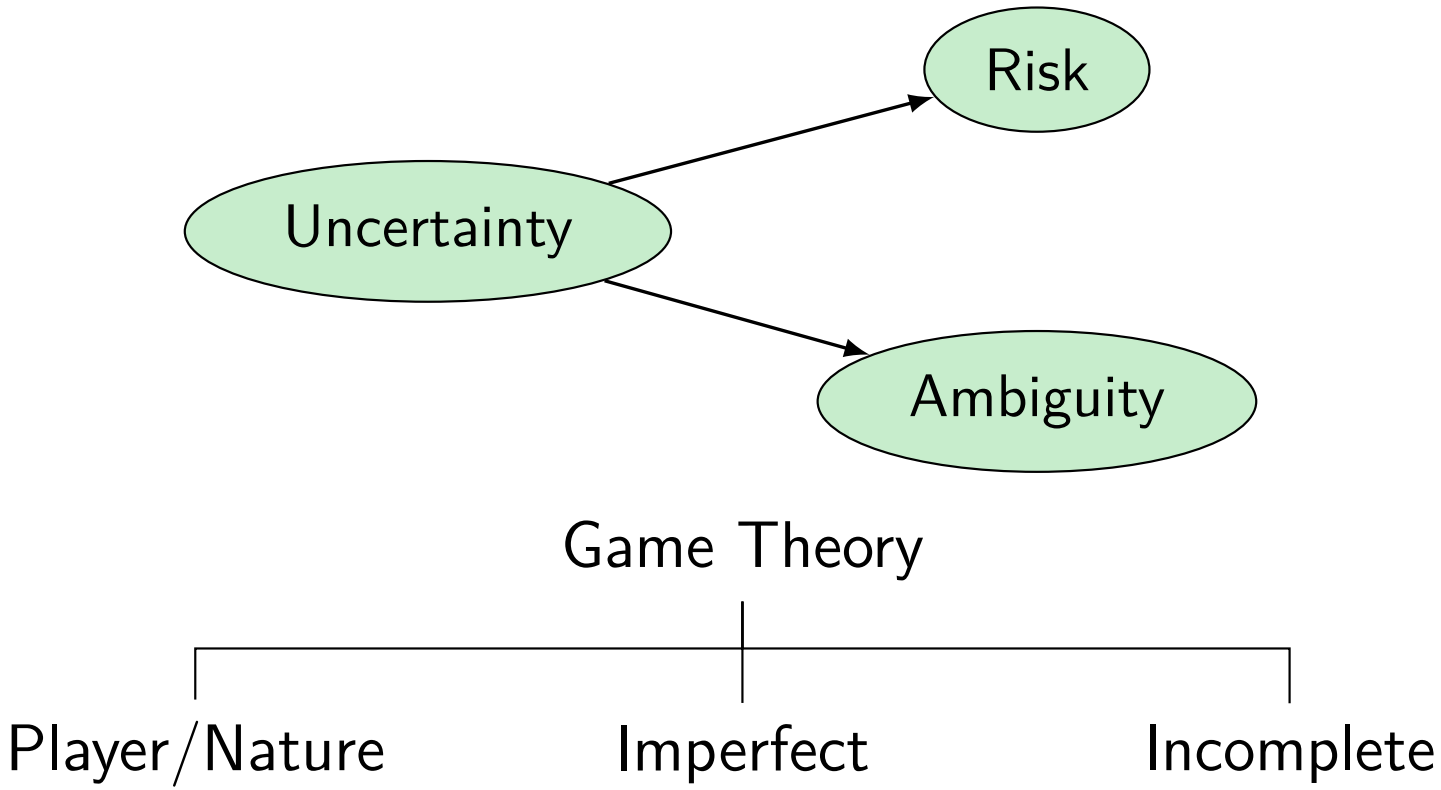
- 如果你的风险中性的 $u(x) = x$, 则你会倾尽所有参与这个赌博.
- 如果你是风险规避的, 比如 $u(x) = 4\sqrt{x}$, 则你只会支付 5.86 元.

$$\mathbb{E}[u(W)] = \sum_n u(2^n) P(T^{n-1}H) = \sum_n 4\sqrt{2^n} \frac{1}{2^n} = \frac{4}{\sqrt{2}-1}$$

$$u(x) = \mathbb{E}[u(W)] \implies x = \left(\frac{1}{\sqrt{2}-1}\right)^2 \approx 5.86$$

- 如果你是罗尔斯, 你只会支付 < 2 元.

Uncertainty = Risk + Ambiguity



Remark: the uncertainty of Nature's choices and in imperfect information games corresponds to risk, and the uncertainty of player choices and in incomplete information games to ambiguities.

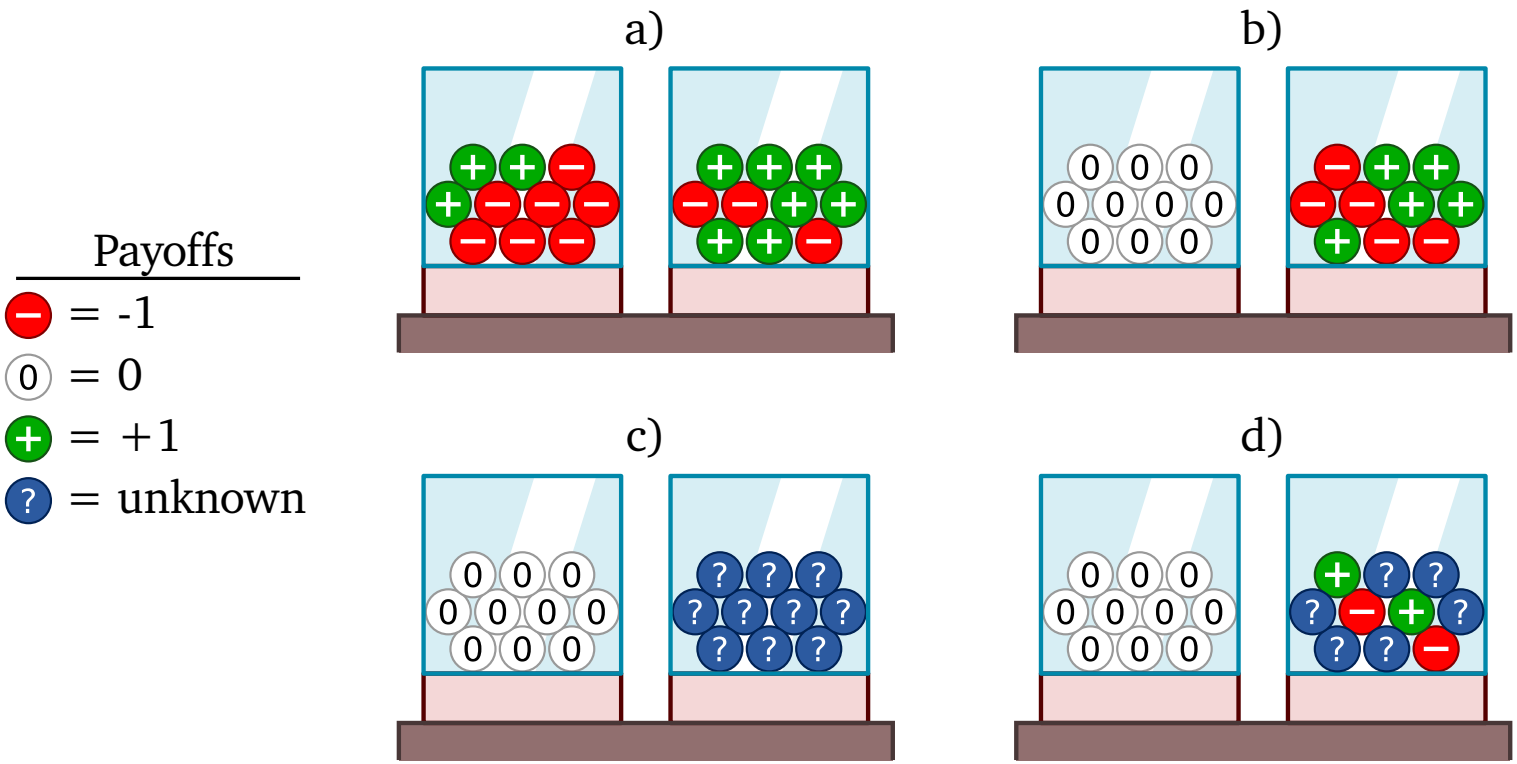
1. Risk-sensitive Agent
2. Ambiguity-sensitive Agent

Risk-sensitive Agent

$$Q_{\text{risk}}(a) \coloneqq \sum_{e \in \mathcal{E}} \nu(e \mid a) u(e) + \lambda \sum_{e \in \mathcal{E}} \nu(e \mid a) \left[u(e) - \overline{u}(e) \right]^2$$

where $\overline{u}(e) \coloneqq \sum_{e \in \mathcal{E}} \nu(e \mid a) u(e)$ is the mean reward.

1. $\lambda < 0$ risk-averse
2. $\lambda > 0$ risk-seeking
3. $\lambda = 0$ risk-neutral

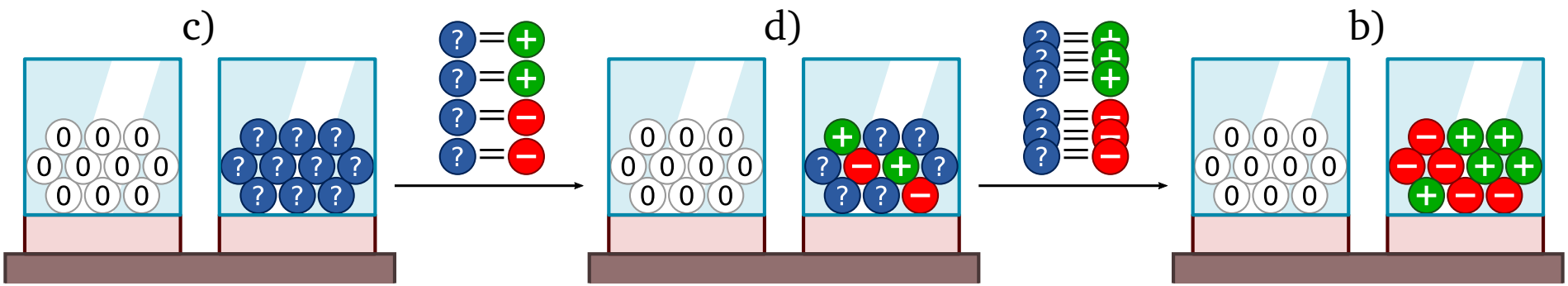


Ambiguity-sensitive Agent

$$Q_{\text{amb}}(a) := \min_{w \in \Delta(\mathcal{M})} \sum_{\nu \in \mathcal{M}} w(\nu) \sum_{e \in \mathcal{E}} \nu(e \mid a) u(e)$$

- For example, an ambiguity-averse agent favours pessimistic interpretations by adopting the rule of selecting the worst-case model.

Remark: Learning reduces ambiguity to risk. Starting from fully ambiguous contents (c) and revealing the payoffs of the blue marbles could reduce the contents to a mixture between risk and ambiguity (d) or even to full risk (b).



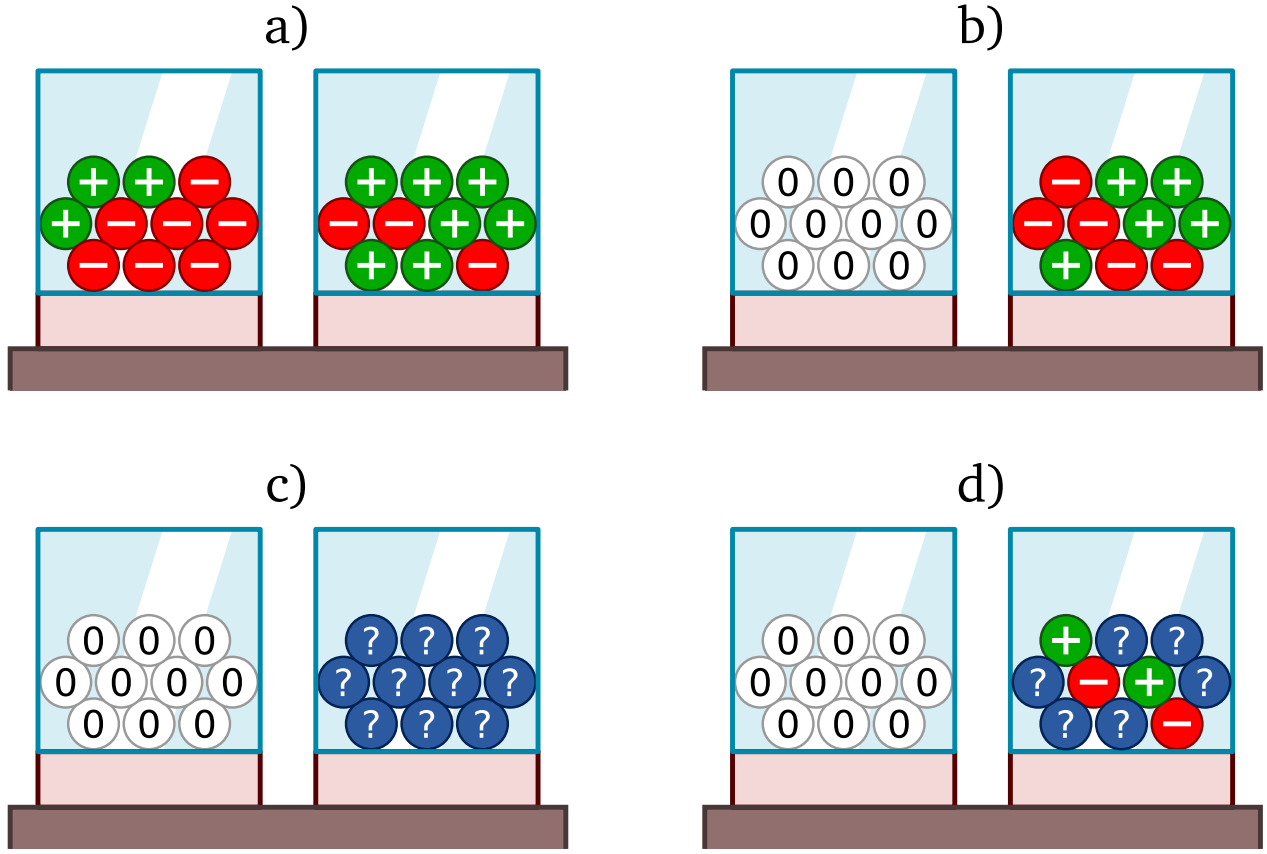
Payoffs

− = -1

0 = 0

+ = +1

? = unknown



	Case			
Agent	a	b	c	d
Expected-utility	right	indiff.	*undef.*	*undef.*
Risk-averse	right	*left*	*undef.*	*undef.*
Bayes-optimal, with prior	right	indiff.	*indiff.*	*indiff.*
Risk-averse, with prior	right	*left*	left	left
Ambiguity-averse	right	indiff.	left	left

- Pessimism: Maximin rule (Rawls) $\operatorname{argmax}_{a \in \mathcal{A}} \min_{e \in \mathcal{E}} u(ae)$

	e_1	e_2	e_3	e_4
a_1	15	0	0	2
a_2	−1	4	3	7
a_3	6	4	14	1
a_4	5	6	4	3

- Optimisim: Maximax rule $\operatorname{argmax}_{a \in \mathcal{A}} \max_{e \in \mathcal{E}} u(ae)$
- Optimisim-Pessimism rule (Hurwicz)

$$\operatorname*{argmax}_{a \in \mathcal{A}} \left(\alpha \max_{e \in \mathcal{E}} u(ae) + (1 - \alpha) \min_{e \in \mathcal{E}} u(ae) \right)$$

- Expected utility maximization (utilitarianism)

$$\operatorname*{argmax}_{a \in \mathcal{A}} \sum_{e \in \mathcal{E}} P(s \mid a) u(ae)$$

- Minimax regret rule $\operatorname{argmin}_{a \in \mathcal{A}} \max_{e \in \mathcal{E}} \left(\max_{a \in \mathcal{A}} u(ae) - u(ae) \right)$
- Maximin safety rule $\operatorname{argmax}_{a \in \mathcal{A}} \min_{e \in \mathcal{E}} \left(u(ae) - \min_{a \in \mathcal{A}} u(ae) \right)$

Remarks: requires interpersonal comparison of utility.

Example

	e_1	e_2	e_3	maximin	maximax	minimax regret	Hurwicz $\alpha = 0.5$	EU
a_1	3_0	16_6	15_0	3	16	6	9.5	$\frac{34}{3}$
a_2	1_2	19_3	13_2	1	19	3	10	11
a_3	2_1	22_0	8_7	2	22	7	12	$\frac{32}{3}$

	utility		regret		safety		
	e_1	e_2	e_1	e_2	e_1	e_2	optimal
a_1	1	9	3	0	0	5	maximax utility
a_2	3	6	1	3	2	2	maximin safety
a_3	2	7	2	2	1	3	minimax regret
a_4	4	4	0	5	3	0	maximin utility

Milnor's Axioms[Mil54]

	uniform utilit.	maximin	Hurwicz	minimax regret
ordering	⊕	⊕	⊕	⊕
symmetry	⊕	⊕	⊕	⊕
str. domination	⊕	⊕	⊕	⊕
continuity	◯	⊕	⊕	⊕
linearity	◯	◯	⊕	◯
row adjunction	⊕	⊕	⊕	
col. linearity	⊕			⊕
col. duplication		⊕	⊕	⊕
convexity	◯	⊕	⊕	⊕
special row. adj.	◯	◯	◯	⊕

Table: ◯ 表示这个标准满足这条公理. ⊕ 标记的公理刻画这个标准.

Milnor's Axioms

1. **ordering:** complete, transitive.
2. **symmetry:** This ordering is independent of the numbering of the rows and columns.
3. **strong domination:** $\forall k[u(a_i, e_k) > u(a_j, e_k)] \implies a_i > a_j$.
4. **continuity:** $u^{(n)}(a_i, e_j) \xrightarrow{n\to\infty} u(a_i, e_j)$ & $\forall n \left[a_i^{(n)} > a_j^{(n)}\right] \implies a_i \geq a_j$
5. **linearity:** The ordering relation is not changed if the matrix $u(a_i, e_j)$ is replaced by $u'(a_i, e_j) = \alpha u(a_i, e_j) + \beta$ for $\alpha > 0$.
6. **row adjunction:** The ordering between the old rows is not changed by the adjunction of a new row.
7. **column linearity:** The ordering is not changed if a constant is added to a column.
8. **column duplication:** The ordering is not changed if a new column, identical with some old column, is adjoined to the matrix.
9. **convexity:** $a_i \sim a_j$ & $a_k = \frac{1}{2}(a_i + a_j) \implies a_k \geq a_i$.
10. **special row adjunction:** The ordering between the old rows is not changed by the adjunction of a new row, providing that no component of this new row is greater than the corresponding components of all old rows.

Example

功利主义 vs 罗尔斯[24]

- Two societies. 1000 people. 100 workers.
- The workers each receive 1 unit of utility while the others get 90 units each.
- The average utility is $10\% \cdot 1 + 90\% \cdot 90 = 81.1$
- In the second society, everyone take a fair turn at being a worker. This causes everyone to realize the same utility of 50 units.
- The average utility is $100\% \cdot 50$.
- The utilitarianism would count the first society as more just, but Rawls would favor the second.

[24] 海萨尼: 最大化最小原则能够成为道德的基础吗? — 对罗尔斯理论的批判. 1975.

个人 vs 社会 — 个人行为正当性的标准

帕累托 vs 卡尔多-希克斯

Problem (问题)

社会是由人组成的，每个人的行为都会影响到他人的利益. 那么，应该用什么标准判断个人的行为是否正当？

- **帕累托最优:** 一种状态被称为帕累托最优状态，如果不存在另一种状态能使得没有任何人的境况变坏同时至少有一个人的境况变得更好. — 除非 “损人”，否则不可能 “利己”.
- **卡尔多-希克斯 (Kaldor-Hicks) 标准:** 如果一种变革使得受益者的所得足以弥补受损者的所失，这种变革就是一个卡尔多-希克斯改进. 如果受损者得到实际的补偿，就是帕累托改进.

Remark: 帕累托最优可能意味着收入分配的不公平；极端地，一个人得到所有收入，另一个人一无所有，也是一个帕累托最优.

- 自愿的交易一定是一个帕累托改进 (假定没有欺诈).
- 卡尔多-希克斯标准即 “财富最大化”.

合作与组织

- 当个体在一起工作创造的价值大于独立工作创造的价值之和, 合作就是一个帕累托改进;
- 当个体在组织中获得的价值大于独立获得的价值时, 加入组织是一个帕累托改进.

- 自由结婚对夫妻双方是一个帕累托改进;
- 买卖婚姻对买卖双方是一个帕累托改进;
- 协议离婚对夫妻双方是一个帕累托改进;
- 但离婚对其他利益相关者 (如父母和儿女) 可能不是一个帕累托改进.

Digression: 合作 vs 伦理

Cooperation is agents with different goals interacting to mutual benefit

- 直接互惠 Direct reciprocity (Agent ↔ Agent)
- 间接互惠 Indirect reciprocity (Agent ↔ Group)
  — 自私者被排除在合作伙伴之外
- 亲缘选择 Kin selection (Gene ↔ Gene)
  — 蜜蜂牺牲个体保护蜂后
- 群体选择 Group selection (Gene ↔ Meme)
  — 语言、文化、符号、宗教、契约……个体合理化甚至神圣化所属群体的宗旨, 排斥异己、奖赏成员、惩罚叛徒

Examples: 卡尔多-希克斯标准

Example1:

- 考虑两种情形:
  1. 某店主暴力捣毁竞争对手的门店, 使后者不能营业;
  2. 某店主以更低的价格和更优良的服务将竞争对手打垮.
- 为什么法律允许第二种情形?
  — 社会所得大于所失, 是一个卡尔多-希克斯改进.

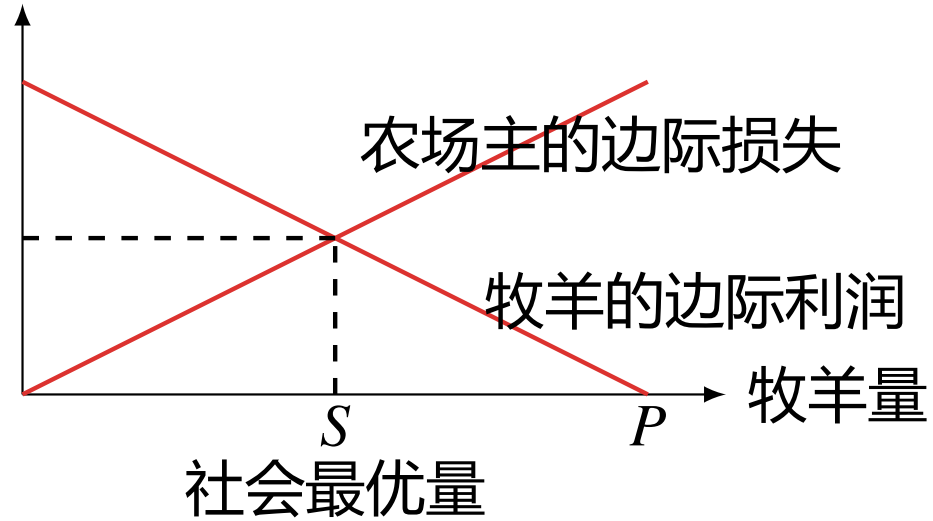
Example2:

- 一家化工厂与家属区一墙之隔, 为了上班方便, 居民在化工厂的墙上挖了一个洞. 一天, 家属区的一个小孩钻洞进工厂玩, 找到一瓶化学液体并点燃烧伤了自己.
- 假设化工厂补墙成本为 c, 如果不补墙, 发生事故的概率为 p, 损失为 l, 且 $c < pl$.
- 化工厂需要承担责任吗?

外部性、交易成本与科斯定理

- 个人收益与社会收益：一项活动的社会收益等于决策者个人得到的收益加社会其他成员得到的收益 (如养花)；
- 个人成本与社会成本：社会成本等于决策者的个人承担的成本加社会其他成员承担的成本 (如环境污染、交通堵塞)；
- 如果个人收益 (成本) 不等于社会收益，就存在外部性.
- 个人最优与社会最优的不一致意味着有帕累托改进的余地；
- 如何将外部性内部化？如何使得个人在边际上承担全部的社会成本、获得全部的社会收益？
- 法律通过责任的分配和赔偿/惩罚，将个人行为的外部成本内部化，诱导个人选择社会最优的行动.
- 政府征税、补贴？
- 科斯定理：只要产权界定是清楚的，如果交易成本为零，外部性可以通过当事人之间谈判解决，帕累托最优可以实现；并且，最终的资源配置与初始的产权安排无关. (公平与效率正交吗？)

科斯定理 — 例子



- 如果产权归农场主，农场主禁止放牧，小于社会最优量 S；但是，增加放牧给牧羊人带来的边际利润大于给农场主造成的损失，牧羊人将有积极性贿赂农场主，直到放牧量达到 S 为止；
- 如果产权归牧羊人，牧羊人的利润最大点是 P，大于社会最优量 S；但是，减少放牧量对牧羊人的边际利润损失小于给农场主节约的边际成本，农场主将有积极性贿赂牧羊人，直到放牧量达到 S 为止.

床垫问题

- 高速公路上, 一张床垫从货车上掉了下来.
- 谁会停下来移开这张床垫呢?
  - 车流中较远的人不知道发生了什么.
  - 刚排到面前的人只想着快速绕过它.
  - 已经绕过的人更没必要返回移开它.
- 政治冷漠……

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
   - Preferences and Expected Utility
   - Strategic Games
   - Extensive Games with Perfect Information
   - Eliminating Dominated Strategies
   - Dynamic Games
   - Repeated Games
   - Signaling Games
   - Mechanism Design
   - Counterfactual Regret Minimization
   - Subgame Perfect Equilibrium
   - Games with Incomplete Information
   - Evolutionary Games
   - Voting System
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Dominant Strategy Equilibrium

Definition (Dominant Strategy)

A strategy $s_i^* \in S_i$ is a *dominant strategy* iff

$$u_i(s_i^*; s_{-i}) \geq u_i(s_i; s_{-i}) \text{ for all } s_i \in S_i \text{ and for all } s_{-i} \in S_{-i}$$

Definition (Dominant Equilibrium)

A strategy profile s^* is a *dominant strategy equilibrium* iff for each player $i \in N$, s_i^* is a dominant strategy.

Definition (Strictly Dominated Strategy)

A strategy s_i is called strictly dominated by s_i' iff

$$\forall s_{-i} : u_i(s_i'; s_{-i}) > u_i(s_i; s_{-i})$$

Definition (Weakly Dominated Strategy)

A strategy s_i is called weakly dominated by s_i' iff

$$\forall s_{-i} : u_i(s_i'; s_{-i}) \geq u_i(s_i; s_{-i})$$

and

$$\exists s'_{-i} : u_i(s_i'; s'_{-i}) > u_i(s_i; s'_{-i})$$

Definition (Iterated Dominance Equilibrium)

An *iterated dominance equilibrium* is a strategy profile $s^* \in S$ obtained by iteratively ruling out dominated strategies from each player until only one strategy remains for each player.

- The result of iterative elimination of strictly dominated strategies is unique, i.e. independent of the elimination order.
- with weakly dominated strategies, the order of elimination matters.

	c_1	c_2	c_3
r_1	$1,1$	$0,1$	$3,1$
r_2	$1,0$	$2,2$	$1,3$
r_3	$1,3$	$3,1$	$2,2$

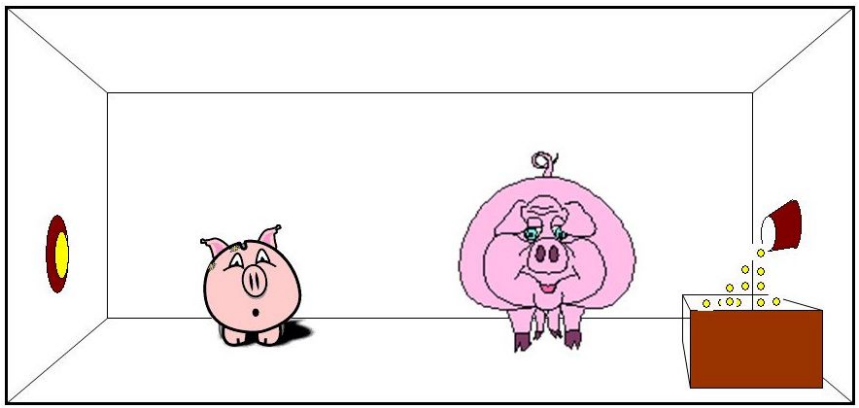
1. $r_2\times$
2. $c_2\times$
3. $r_3\times$
4. $c_3\times$
5.

	c_1
r_1	$1,1$

1. $r_2\times$
2. $c_2\times$; $c_3\times$
3.

	c_1
r_1	$1,1$
r_3	$1,3$

智猪博弈 — 反复剔除劣策略均衡 $\neq$ 占优均衡



- 小猪是理性的

$$\mathsf{Rational}(s)$$

- 大猪知道小猪是理性的

$$\mathsf{K}_b\mathsf{Rational}(s)$$

	按	等
按	$4,0$	$3,3$
等	$7,-1$	$0,0$

	按	等
按		$3,3$
等		$0,0$

	按	等
按		$3,3$
等		

反复剔除劣策略与 n-阶理性

	c_1	c_2	c_3	c_4
r_1	$5,10$	$0,11$	$1,20$	$10,10$
r_2	$4,0$	$1,1$	$2,0$	$20,0$
r_3	$3,2$	$0,4$	$4,3$	$50,1$
r_4	$2,93$	$0,92$	$0,91$	$100,90$

	c_1	c_2	c_3
r_1	$5,10$	$0,11$	$1,20$
r_2	$4,0$	$1,1$	$2,0$
r_3	$3,2$	$0,4$	$4,3$
r_4	$2,93$	$0,92$	$0,91$

	c_1	c_2	c_3
r_1	$5,10$	$0,11$	$1,20$
r_2	$4,0$	$1,1$	$2,0$
r_3	$3,2$	$0,4$	$4,3$

	c_2	c_3
r_1	$0,11$	$1,20$
r_2	$1,1$	$2,0$
r_3	$0,4$	$4,3$

	c_2	c_3
r_2	$1,1$	$2,0$
r_3	$0,4$	$4,3$

	c_2
r_2	$1,1$
r_3	$0,4$

	c_2
r_2	$1,1$

0-order $c_4\times$ $\mathsf{Rational}(c)$

1-order $r_4\times$ $\mathsf{K}_r\mathsf{Rational}(c)$

2-order $c_1\times$ $\mathsf{K}_c\mathsf{K}_r\mathsf{Rational}(c)$

3-order $r_1\times$ $\mathsf{K}_r\mathsf{K}_c\mathsf{K}_r\mathsf{Rational}(c)$

4-order $c_3\times$ $\mathsf{K}_c\mathsf{K}_r\mathsf{K}_c\mathsf{K}_r\mathsf{Rational}(c)$

5-order $r_3\times$ $\mathsf{K}_r\mathsf{K}_c\mathsf{K}_r\mathsf{K}_c\mathsf{K}_r\mathsf{Rational}(c)$

诸葛亮的《空城计》何以有效?

- 诸葛亮谨慎. $C(z)$
- 司马懿知道诸葛亮谨慎. $\mathsf{K}_s C(z)$
- 诸葛亮知道司马懿知道诸葛亮谨慎. $\mathsf{K}_z \mathsf{K}_s C(z)$
- 司马懿不知道诸葛亮知道司马懿知道诸葛亮谨慎. $\neg \mathsf{K}_s \mathsf{K}_z \mathsf{K}_s C(z)$

老狐狸与小狐狸

- 老狐狸看到满载而归的渔夫驾车经过，于是躺在路边装病.
- 渔夫看老狐狸可怜，就让它搭个便车.
- 老狐狸悄悄地把鱼一条一条地扔到路边草丛里，然后一跃而下吃鱼去了.
- 小狐狸问老狐狸是如何得到这么多鱼的，老狐狸 “如实相告”.
- 第二天，小狐狸也学着躺在路边装病，渔夫愤怒地把它打死了.

- $\neg \mathsf{K}_{渔} A$
- $\mathsf{K}_{老} \neg \mathsf{K}_{渔} A$
- $[A] \mathsf{K}_{渔} A$
- $\mathsf{K}_{老} [A] \mathsf{K}_{渔} A$
- $\neg \mathsf{K}_{小} A$
- $[A] \mathsf{K}_{小} A$
- $\mathsf{K}_{老} [A] \mathsf{K}_{小} A$
- $\neg \mathsf{K}_{小} [A] \mathsf{K}_{渔} A$
- $\mathsf{K}_{老} [A] \neg \mathsf{K}_{小} [A] \mathsf{K}_{渔} A$

Nash Equilibrium

Definition (Nash Equilibrium)

Best response of player i:

$$\mathrm{BR}_i(s_{-i}) \coloneqq \operatorname*{argmax}_{s_i \in S_i} u_i(s_i; s_{-i})$$

The strategy profile s^* is a *Nash equilibrium* iff

$$\forall i \in N : s_i^* \in \mathrm{BR}_i(s_{-i}^*)$$

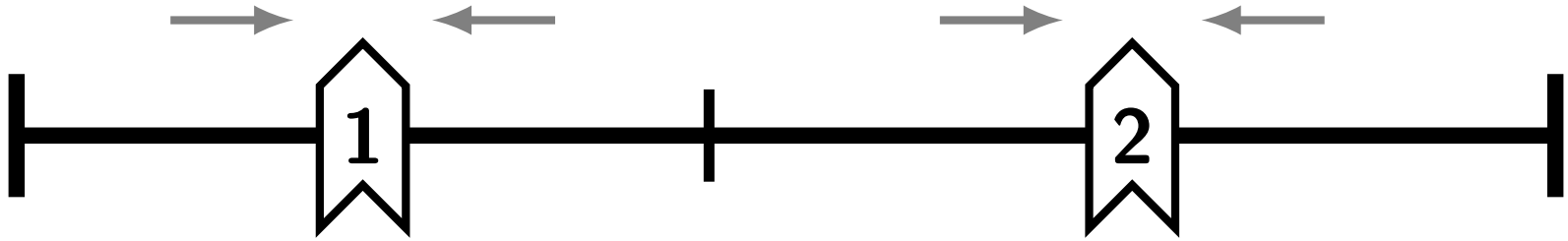
Definition (Mixed Nash Equilibrium)

A mixed strategy profile σ^* is a *mixed Nash equilibrium* iff,

$$\forall i \in N \forall \sigma_i \in \Delta S_i : \; u_i(\sigma_i^*; \sigma_{-i}^*) \geq u_i(\sigma_i; \sigma_{-i}^*)$$

A mixed strategy Nash equilibrium of a strategic game is a Nash equilibrium of its mixed extension.

Example: 纳什均衡

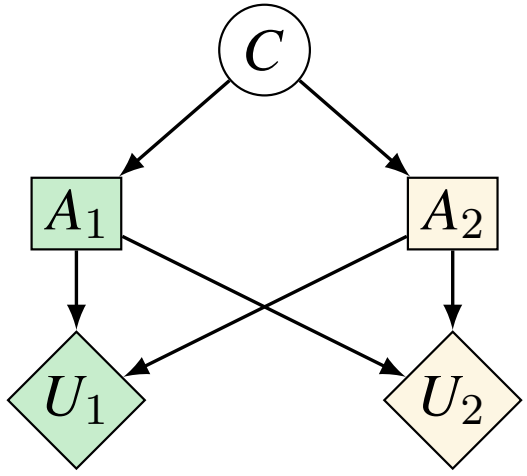


- 一条居民街, 两个卖报的小贩.
- 报纸价格不由小贩决定. 小贩只能选择自己摆摊的位置.
- 每个居民都会找距离自己最近的小贩买报纸. 如果等距, 则随机选一个.
- 小贩会如何选择摆摊位置?
- 对于这个唯一的纳什均衡, 站在居民的角度来看, 是否理想?
- 政治隐喻.
  — 大街: 政治空间.
  — 大街上的点: 政治立场.
  — 小贩: 候选人.

多重纳什均衡 vs 制度规范

机动车道	左行	右行
左行	$1,1$	$0,0$
右行	$0,0$	$1,1$

Table: 协调博弈



卢梭	猎鹿	逮兔
猎鹿	$10,10$	$1,8$
逮兔	$8,1$	$5,5$

Table: cheap talk 协商选择帕累托最优的纳什均衡

Remark:

- 纳什均衡不一定是帕累托最优.
- 任何制度, 只有构成一个纳什均衡, 才能让人们自觉遵守.
- 制度、规范的主要功能之一是协调预期, 在多个纳什均衡中筛选出某个特定的纳什均衡.
- 有效的制度设计, 就是通过对纳什均衡的选择实现帕累托最优.

教学、求爱、斗殴、革命

	认真学	不认真学
认真教	$2, 2$	$-1, 1$
不认真教	$1, -1$	$0, 0$

Table: 教学博弈

	爱	不爱
爱	$2, 2$	$-1, 1$
不爱	$1, -1$	$0, 0$

Table: 表白博弈

	空手	白刃
空手	$2, 2$	$-1, 1$
白刃	$1, -1$	$0, 0$

Table: 斗殴博弈

	抗争	沉默
抗争	$2, 2$	$-1, 1$
沉默	$1, -1$	$0, 0$

Table: 革命博弈

混合纳什均衡

	偷懒	不偷懒
监督	$1, -1$	$-1, 2$
不监督	$-2, 3$	$2, 2$

- 如果员工偷懒的概率是 p，那么，老板监督和不监督的期望收益分别为

$$1 \cdot p + (-1) \cdot (1 - p) = 2p - 1$$

$$(-2) \cdot p + 2 \cdot (1 - p) = 2 - 4p$$

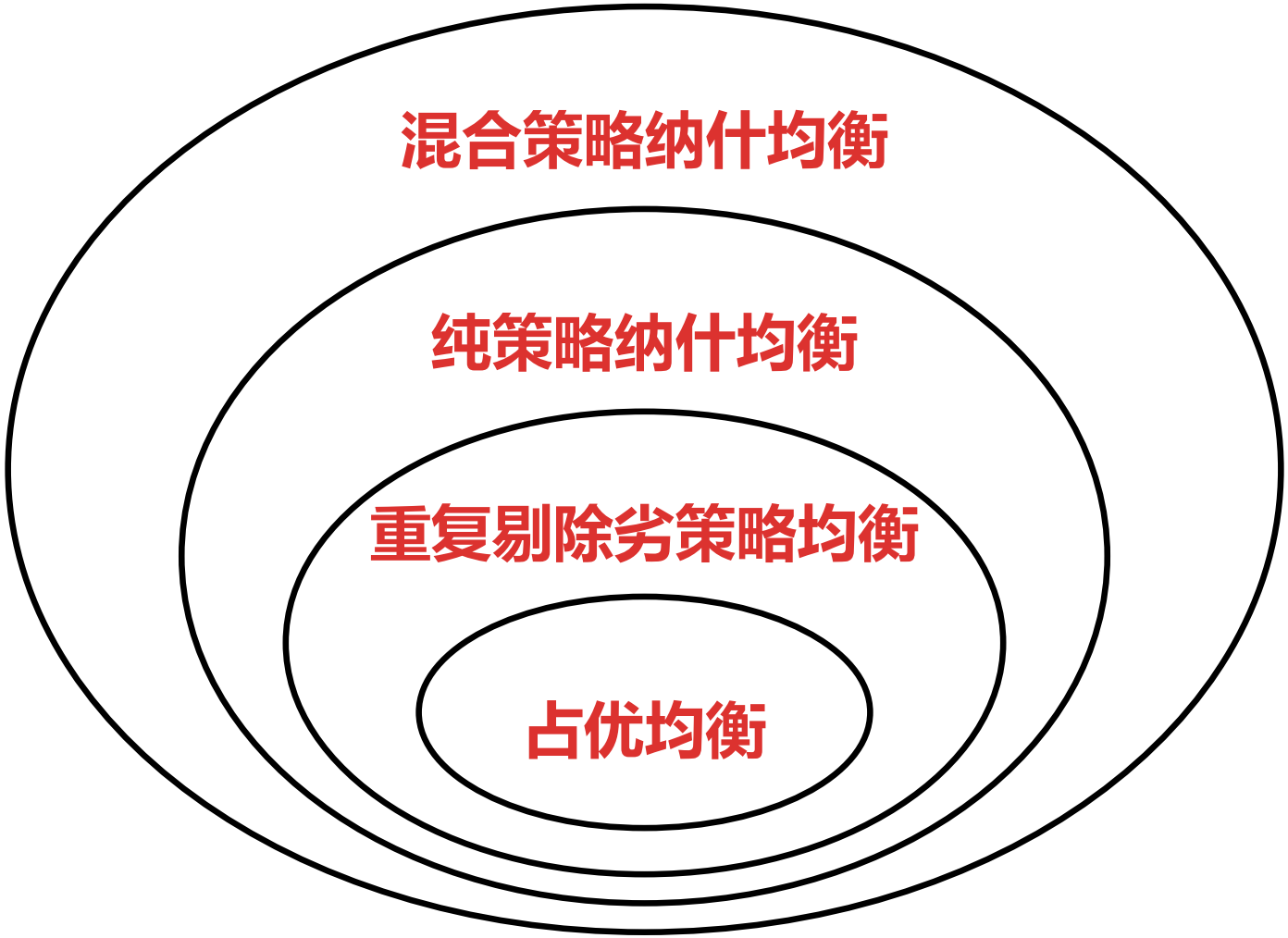
- 如果老板监督的概率是 q，那么，员工偷懒和不偷懒的期望收益分别为

$$(-1) \cdot q + 3 \cdot (1 - q) = 3 - 4q$$

$$2 \cdot q + 2 \cdot (1 - q) = 2$$

- 混合纳什均衡是 $(\frac{1}{4}, \frac{1}{2})$，老板以 $\frac{1}{4}$ 的概率监督，员工以 $\frac{1}{2}$ 的概率偷懒.

几个均衡之间的关系



韭菜的自我修养

	正面 y	反面 $1-y$
正面 x	$3,-3$	$-2,2$
反面 $1-x$	$-2,2$	$1,-1$

Table: 正反匹配

- 行 (散户) 以 x 出正面 (做多), $1-x$ 出反面 (做空)
- 列 (庄家) 以 y 出正面 (拉升股价), $1-y$ 出反面 (打压股价)
- 行的期望收益:

$$U_r = 3xy + 1(1-x)(1-y) - 2x(1-y) - 2(1-x)y = (8x-3)y - 3x + 1$$

- 列的期望收益: $U_c = -U_r$
- 列想让 $U_r < 0$
  1. 当 $x > \frac{3}{8}$ 时, $y < \frac{3x-1}{8x-3}$. 所以, 只要 $y < \frac{2}{5}$, 就有 $U_r < 0$.
  2. 当 $x < \frac{3}{8}$ 时, $y > \frac{3x-1}{8x-3}$. 所以, 只要 $y > \frac{1}{3}$, 就有 $U_r < 0$.
- 所以, 只要 $\frac{1}{3} < y < \frac{2}{5}$, 就有 $U_r < 0$. 即, 行亏钱, 列赚钱.

Kakutani Fixpoint Theorem

Theorem (Kakutani Fixpoint Theorem)

Given a non-empty compact convex set $X \subset \mathbb{R}^n$ and a multi-valued function $f : X \rightrightarrows X$, if

1. *for all x, the set $f(x)$ is convex,*
2. *for all sequences (x_i, y_i) s.t. $x_i \in X$ and $y_i \in f(x_i)$,*

$$\lim_{i \to \infty} (x_i, y_i) = (x, y) \implies y \in f(x)$$

then $\exists x \in f(x)$.

Theorem (Existence of Mixed Nash Equilibrium)

Every finite strategic game has a mixed Nash equilibrium.

Proof.

Let $X \coloneqq \prod_{i \in N} \Delta S_i$ and $f(\sigma) \coloneqq \prod_{i \in N} \mathrm{BR}_i(\sigma_{-i})$.

Theorem

A mixed strategy profile σ^* is a mixed Nash equilibrium iff,

$$\forall i \in N \forall s_i \in S_i : \; u_i(\sigma_i^*; \sigma_{-i}^*) \geq u_i(s_i; \sigma_{-i}^*)$$

Proof.

$$u_i(\sigma_i; \sigma_{-i}) = \sum_{s_i \in S_i} u_i(s_i; \sigma_{-i}) \sigma_i(s_i)$$

Theorem

For a finite strategic game,

$$\mathrm{supp}(\sigma_i^*) \subset \mathrm{BR}_i(\sigma_{-i}^*)$$

Proof.

Suppose there exists an $s_i' \in \mathrm{supp}(\sigma_i^*)$ s.t.

$$u_i(s_i'; \sigma_{-i}^*) < u_i(\sigma_i^*; \sigma_{-i}^*)$$

then

$$\sum_{s_i \in S_i} u_i(s_i; \sigma_{-i}^*)\sigma_i^*(s_i) < u_i(\sigma_i^*; \sigma_{-i}^*)$$

Contradiction!

- Every action in the support of any player's equilibrium mixed strategy yields that player the same payoff.
- If the set of actions of some player is not finite the result needs to be modified. In this case, σ^* is a mixed strategy Nash equilibrium iff
  1. for every player i no action in S_i yields, given σ^*_{-i}, a payoff to player i that exceeds his equilibrium payoff, and
  2. the set of actions that yield, given σ^*_{-i}, a payoff less than his equilibrium payoff has σ^*_i–measure zero.

沉默的目击者 — 从纳什均衡看旁观者效应

- n 个目击者围观 Kitty 被虐杀. 报警的成本是 b. 如果有人报警, Kitty 会得救, 每个目击者会获得效用 a; 如果没人报警, 效用是 0.

	有他人报警 $1-(1-p)^{n-1}$	没他人报警 $(1-p)^{n-1}$
报警 p	$a-b$	$a-b$
不报警 $1-p$	a	0

- 当 $a < b$ 时, 所有人不报警是纳什均衡.
- 当 $a > b$ 时, 有一个人报警、其他人不报警是纳什均衡.
- 考虑对称的混合纳什均衡, 即所有参与人报警的概率相等, 设为 p.
- 某参与人的期望收益:

$$Q(p) := (a-b)p \ + \ \Big(a(1-(1-p)^{n-1}) + 0(1-p)^{n-1}\Big)(1-p)$$

$$\frac{\mathrm{d}Q(p)}{\mathrm{d}p} = 0 \implies p = 1-(\tfrac{b}{a})^{\frac{1}{n-1}}$$

- n 个人中至少有一个人报警的概率为

$$P(n) := 1-(1-p)^n = 1-(\tfrac{b}{a})^{\frac{n}{n-1}}$$

但 $\frac{\mathrm{d}P(n)}{\mathrm{d}n} = \frac{(\frac{b}{a})^{\frac{n}{n-1}} \ln \frac{b}{a}}{(n-1)^2} < 0$, 人数越多, 至少有一个人报警的概率越小.

走出囚徒困境 — 奖惩 — 利维坦

	合作	背叛
合作	$-1,-1$	$-4,0$
背叛	$0,-4$	$-3,-3$

Table: 囚徒博弈

	合作	背叛
合作	$-1,-1$	$-4,0-x$
背叛	$0-x,-4$	$-3-x,-3-x$

Table: 带惩罚的囚徒博弈 $0-x<-1$

走出囚徒困境 — 奖惩 — 作为激励机制的等级制度

	合作	背叛
合作	$-1,-1$	$-4,0$
背叛	$0,-4$	$-3,-3$

$$\Downarrow$$

	先走	后走
先走	$0,0$	$2,1$
后走	$1,2$	$0,0$

儒家 — “礼”

“合作” 方可得 “君子” 名分, 君子享有优先权.
协调预期、定分止争.
声誉约束.
奖善惩恶、等级制度等可以理解为一种激励机制.

风险与均衡

- 由于纳什均衡要求理性共识和一致预期，当人们可能犯小小的错误时，纳什均衡不一定被选择.

	左	右
上	$8, 10$	$-1000, 9$
下	$7, 6$	$6, 5$

Table: 只要李四有千分之一的概率错误地选择“右”，张三将选择“下”；如果李四怀疑张三怀疑自己可能犯错误，李四将选择“右”.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
   - Preferences and Expected Utility
   - Strategic Games
   - Extensive Games with Perfect Information
   - Eliminating Dominated Strategies
   - Dynamic Games
   - Repeated Games
   - Signaling Games
   - Mechanism Design
   - Counterfactual Regret Minimization
   - Subgame Perfect Equilibrium
   - Games with Incomplete Information
   - Evolutionary Games
   - Voting System
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

100 只狼和 1 只羊

- 某个魔法森林里生活着 100 只狼和 1 只羊.
- 狼想吃羊, 羊不想被狼吃.
- 一只狼如果吃掉一只羊, 就会变成一只羊.

1. 森林里狼和羊的数量会变成多少?
2. 如果森林里一开始是 99 只狼和 1 只羊呢?

囚犯分绿豆

Problem (谁能活下来？)

五个囚犯轮流从一个装有 100 颗绿豆的袋子里抓绿豆.

1. 他们都绝对理性.
2. 彼此不能交流.
3. 100 颗不必都分完, 但要保证每人至少抓一颗.
4. 他们可以摸出袋子里剩下的绿豆数量.
5. 抓得最多和最少的人将被处死.
6. 他们的原则是先求保命, 再尽可能多杀人.

None!

海盗分金

Problem (海盗分金)

五名海盗 (从最怯懦到最凶残依次是 1 到 5 号) 打算瓜分 100 个金币. 海盗世界的分配规则如下:

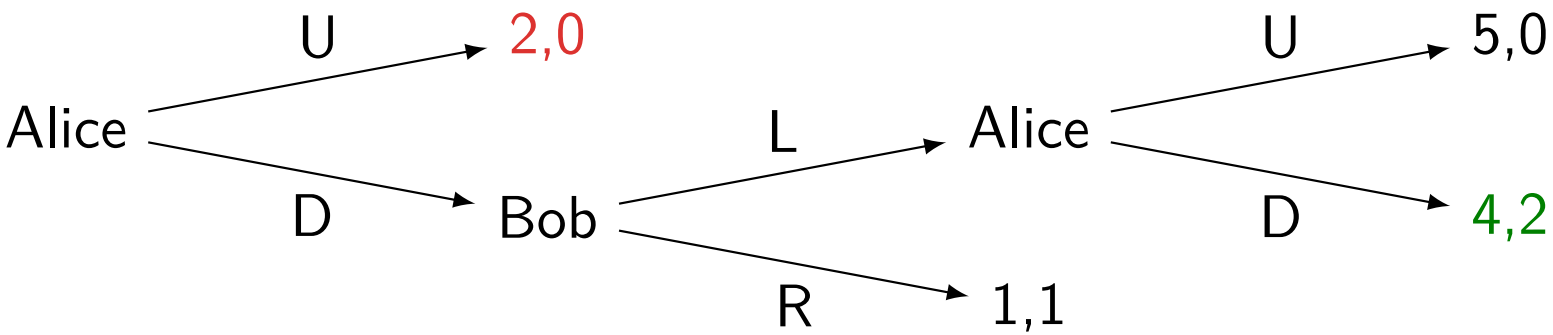
1. 最凶残的一名海盗提出分配方案, 然后所有的海盗 (包括提出方案者本人) 就此方案进行表决;
2. 如果 50% 或以上的海盗赞同此方案, 此方案就获得通过并据此分配;
3. 否则, 提出方案的海盗将被扔到海里, 然后下一名最凶残的海盗又重复上述过程.
4. 每名海盗首先想活命, 其次想贪财, 最后是嗜杀, 如果其他情况相同, 都更倾向于把另一个海盗扔到海里.

1,0,1,0,98

What about more than 200 pirates? Who can survive? 1-200, 201, 200+2^n

What about more than half? 0,2,1,0,97 / 2,0,1,0,97

不可信的许诺/威胁 vs 可信的承诺



- 如果 Alice 许诺不选择 U, Bob 会相信吗?
- 如果 Alice 承诺不选择 U 呢?
  — 比如: 拿出 2$ 保证金给独立的第三方, 若失信, 则自动转给 Bob.
- 动态博弈的纳什均衡可能包含不可信的许诺 (或威胁). 即, 事前看是最优的, 事后看不是最优的, 所以不可信.
- 老父亲威胁女儿, 敢跟古惑仔私奔就断绝父女关系, 但私奔后, 若真的断绝父女关系会损失更大. 所以理性的女儿还是会私奔.
- 若政府对大财团的威胁不可信, 大财团融资时可能无视风险, 成功了收益是自己的, 失败了成本转嫁给政府社会, 最后 "大而不倒", 导致金融危机.

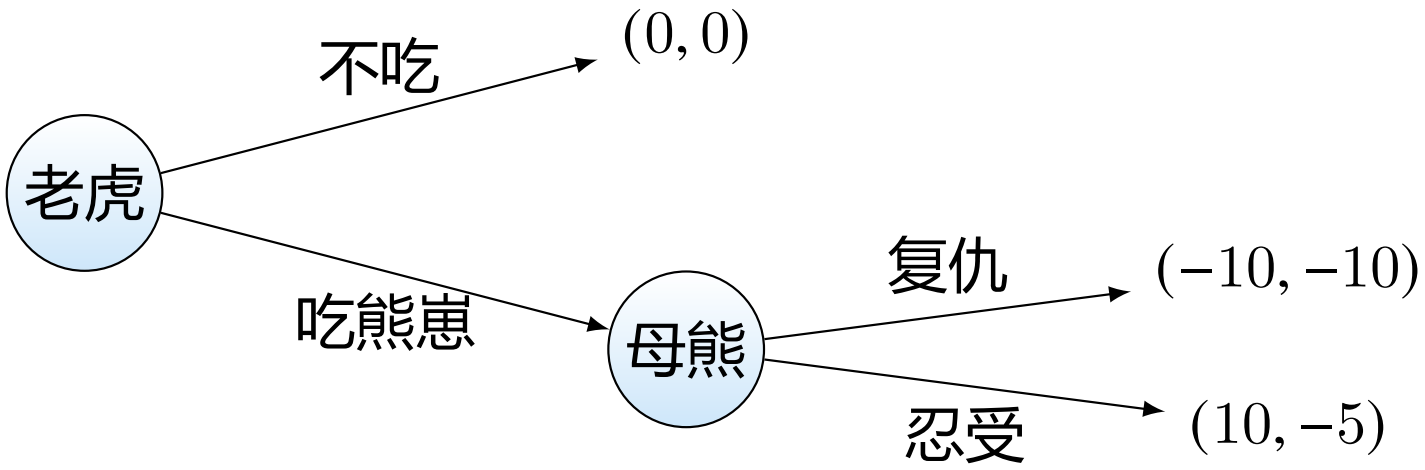
承诺

假如企业一开始定价 80, 如果前两个消费者购买了, 企业将有积极性在 50 的价格下向第三个顾客出售. 预期到这一点, 前两个顾客将不会购买. 如果企业承诺, 任何降价的差额将返还顾客, 前两个顾客则会购买.

产量	价格	收入
1	100	100
2	80	160
3	50	150
4	30	120

- 承诺意味着限制自己的自由: 选择少反而对自己好.
  — “破釜沉舟”.
- 增加对方的选择也可以看作一种 “承诺” (或 “反承诺”).
  — 围城只围三面.
- 承诺有成本.
  — 永不降价, 降价退差额! 假一赔十!
- 天价彩礼: 不离婚的承诺.
- 为什么画家死了画会升值?
- 民主、法治是政府对人民的一种承诺.
- 民法 “民不告、官不究”, 而刑法实行 “公诉制度” 是一种承诺.

情感的进化解释



- 情感与理性是对立的吗?
- (老虎吃掉熊幼崽, 母熊默默忍受) 是子博弈完美均衡.
- 即使母熊事先威胁 “我必复仇!”, 老虎也不会相信.
- 但如果母熊出离愤怒, 激情复仇, 那么就相当于其默默忍受的效用值不是 -5, 而是比 -10 还小.
- 会愤怒的熊的种群存活率更高.
- 愤怒的情感让威胁变成了可信的承诺.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
   - Preferences and Expected Utility
   - Strategic Games
   - Extensive Games with Perfect Information
   - Eliminating Dominated Strategies
   - Dynamic Games
   - Repeated Games
   - Signaling Games
   - Mechanism Design
   - Counterfactual Regret Minimization
   - Subgame Perfect Equilibrium
   - Games with Incomplete Information
   - Evolutionary Games
   - Voting System
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

无限重复博弈

	合作	背叛
合作	T, T	S, R
背叛	R, S	P, P

Table: 囚徒博弈: $R > T > P > S$ 且 $T + T > R + S$.

1. All-D 策略: 总是背叛.
2. All-C 策略: 总是合作.
3. 合作-背叛交替进行.
4. 以牙还牙 tit-for-tat TFT 策略: 从合作开始, 之后每次选择对方前一阶段的行动.
5. 冷酷 grim 策略: 从合作开始, 直到一方背叛, 然后永远背叛.
6. 宽容的冷酷策略: 如果对方背叛, 先惩罚几次, 然后再恢复合作.
7. 宽容的以牙还牙: 永远以合作的态度来回报对方的合作. 当遇到背叛时, 以某一概率与对方进行合作.
8. 赢定输移 win stay, lose shift 策略: 如果我们上一轮合作, 那么合作; 如果上一轮都背叛, 那么以一定概率合作; 如果上一轮我合作你背叛或你合作我背叛, 则背叛.

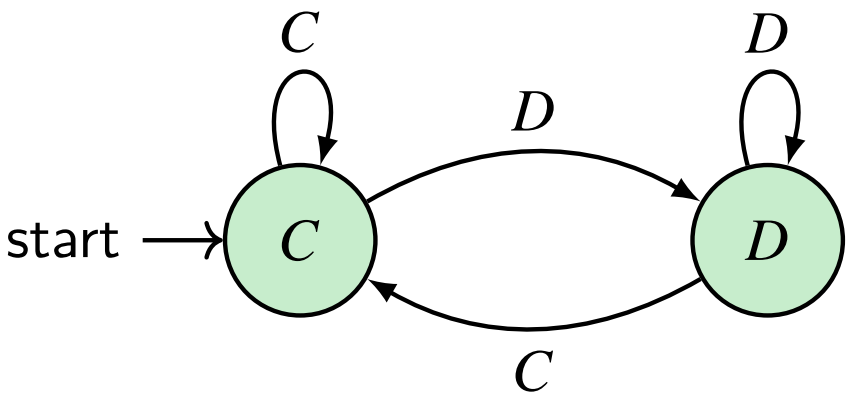


Figure: 以牙还牙

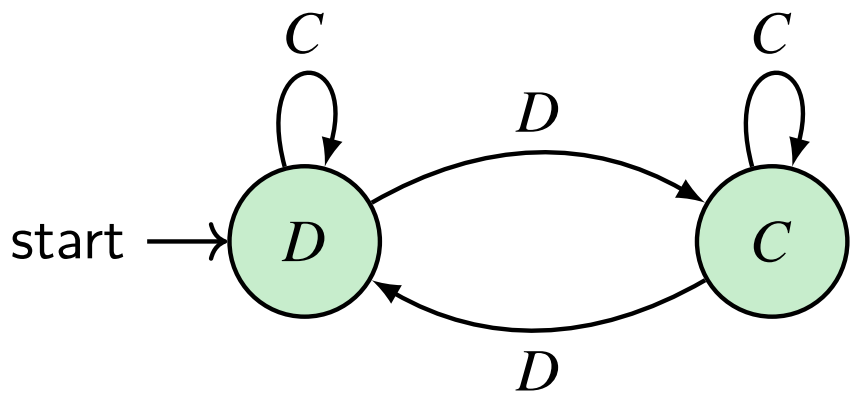


Figure: 一报还一报

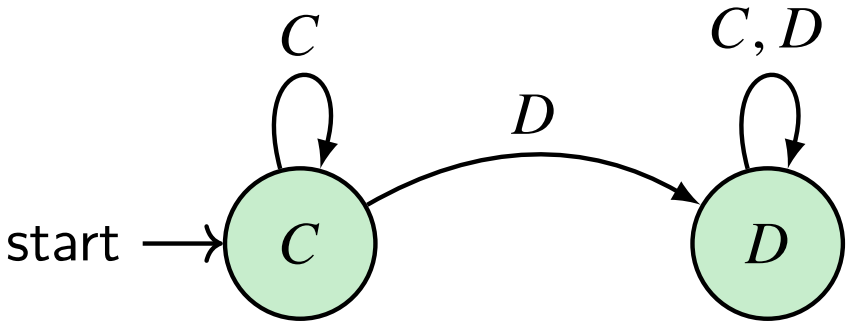


Figure: 冷酷

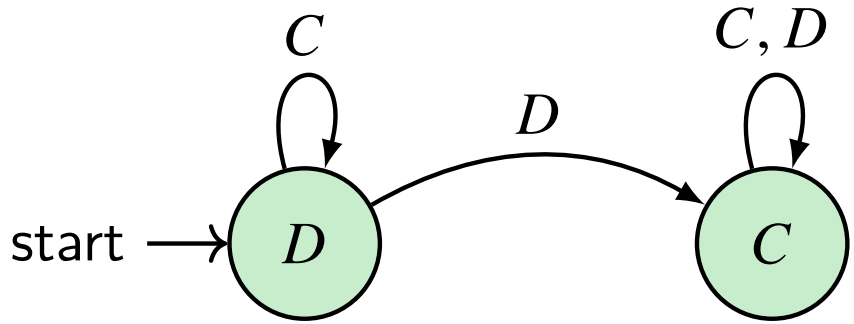


Figure: 喜鹊

- $V(\text{ALL-D}, \text{ALL-D}) = P + \gamma P + \gamma^2 P + \cdots = P\frac{1}{1-\gamma}$
- $V(\text{TFT}, \text{TFT}) = T + \gamma T + \gamma^2 T + \cdots = T\frac{1}{1-\gamma}$
- $V(\text{ALL-D}, \text{TFT}) = R + \gamma P + \gamma^2 P + \cdots = R + P\frac{\gamma}{1-\gamma}$
- $V(\text{ALL-C}, \text{grim}) = T + \gamma T + \gamma^2 T + \cdots = T\frac{1}{1-\gamma}$
- $V(\text{ALL-D}, \text{grim}) = R + \gamma P + \gamma^2 P + \cdots = R + P\frac{\gamma}{1-\gamma}$

如果 $T\frac{1}{1-\gamma} \geq R + P\frac{\gamma}{1-\gamma}$, 即 $\gamma \geq \frac{R-T}{R-P}$, 则对于 grim 策略来说, 合作就是子博弈完美均衡.

无名氏定理 (Folk Theorem)

在无限次重复博弈中, 如果每个参与人都对未来足够重视 (贴现因子足够大), 那么, 任何程度的合作都可以作为一个子博弈完美均衡得到. 这里的“合作程度” 指整个博弈中合作出现的频率.

动物界的合作

- 鱼类会使用 TFT 策略：当两条鱼接近入侵者时，如果一条想尾随在后，走在前面的鱼转身向后，等待另一条跟上，然后再并行前进.
- 孔雀鱼甚至可以记住其同伙过去的表现. 如果一次试验中一方背叛，另一方在第二次的试验中也会背叛.
- 孔雀鱼倾向于与过去表现出更具合作精神的鱼结伴而行.

惩罚

- 在重复博弈中，越在乎长远利益，合作的可能性越大.
- 背叛行为越容易被观察到，并且惩罚越可信，合作的可能性越大.
- 垄断使得惩罚不可信.
- 在确定环境中，惩罚越严厉越有助于合作，但在不确定环境中，对方有可能是无心之失，冷酷策略可能不利于长期合作.

联合抵制的社会规范

联合抵制 (Boycott)：每个人都应该诚实；都有责任惩罚骗过人的人；不参与惩罚的人应该受到惩罚.

Remark: 朋友的朋友是朋友；朋友的敌人是敌人；敌人的朋友是敌人.

有限重复博弈

	c_1	c_2	c_3
r_1	$1,1$	$5,0$	$0,0$
r_2	$0,5$	$4,4$	$0,0$
r_3	$0,0$	$0,0$	$3,3$

- 两个纳什均衡: (r_1, c_1) 和 (r_3, c_3)
- 帕累托最优: (r_2, c_2)
- 如果博弈重复两次, 则 "好合好散, 不欢而散" 的策略 — 如果 (r_2, c_2) 则 (r_3, c_3), 否则 (r_1, c_1) — 可在第一轮博弈中实现帕累托最优.
- 但是, 如果第一轮遭到背叛后, 第二轮对方重新谈判, 则会使得惩罚不可信. 原因在于多重均衡之间 (r_3, c_3) 帕累托优于 (r_1, c_1).

有限重复博弈

	c_1	c_2	c_3	c_4	c_5
r_1	$1,1$	$5,0$	$0,0$	$0,0$	$0,0$
r_2	$0,5$	$4,4$	$0,0$	$0,0$	$0,0$
r_3	$0,0$	$0,0$	$3,3$	$0,0$	$0,0$
r_4	$0,0$	$0,0$	$0,0$	$4,0.5$	$0,0$
r_5	$0,0$	$0,0$	$0,0$	$0,0$	$0.5,4$

- 四个纳什均衡: (r_1, c_1)、(r_3, c_3)、(r_4, c_4)、(r_5, c_5)
- 三个帕累托最优: (r_2, c_2)、(r_4, c_4)、(r_5, c_5)
- 如果博弈重复两次, 则可采取如下策略: 如果 (r_2, c_2) 则 (r_3, c_3); 否则, 被背叛的一方根据自己的情况选择 r_4 或 c_5, 如果双方同时背叛, 则 (r_3, c_3).
- 此时惩罚变得可信了, 均衡为: 第一轮 (r_2, c_2); 第二轮 (r_3, c_3).

单方不完全信息的有限重复博弈

	合作	背叛
合作	$3,3$	$-1,4$
背叛	$4,-1$	$0,0$

	t1	t2
A 冷酷型 p	合作	X
A 理性型 $1-p$	背叛	背叛
B 理性型	X	背叛

Table: 囚徒博弈重复两次

- 参与人 A 有两种可能的类型: "冷酷" 型: 选择 grim 策略, 概率为 p; "理性" 型: 可以选择任何策略, 概率为 $1-p$.
- 参与人 B 有一种类型: 理性型.
- B 在第一轮选择合作或背叛最后总的期望效用分别为

$$3 \cdot p + (-1) \cdot (1-p) + 4 \cdot p + 0 \cdot (1-p) = 8p - 1$$

$$4 \cdot p + 0 \cdot (1-p) + 0 \cdot p + 0 \cdot (1-p) = 4p$$

- 如果 $p \geq \frac{1}{4}$, 则 B 在第一轮合作.
- 如果博弈重复 $N \geq 3$ 轮, 只要 $p \geq \frac{1}{4}$, 理性型 A 在 $t = 1 \ldots N-2$ 轮合作, 在最后两轮背叛; B 在前 $N-1$ 轮合作, 在最后一轮背叛.

双方不完全信息的有限重复博弈

- 假设双方都有两种可能的类型: “冷酷” 型或 “理性” 型.
- 如果一开始就选择背叛, 暴露了自己是理性型, 那么收益最大为 4.
- 假如对方采取冷酷策略的概率是 p, 则采取冷酷策略的最小期望收益为

$$3 \cdot N \cdot p + (-1 + 0 + 0 + \cdots + 0) \cdot (1 - p) = (3N + 1)p - 1$$

- 只要博弈次数 $N \geq \frac{5-p}{3p}$, 则采取冷酷策略.
- 在完全信息的情况下, 根据逆向归纳逻辑, n 次重复囚徒博弈有唯一的子博弈完美均衡, 即双方总是背叛.
- 在不完全信息的情况下, 只要博弈重复的次数足够长, 参与人就有积极性在博弈的早期建立一个 “合作” 的**声誉**; 一直到博弈的后期, 才会选择背叛; 并且, 背叛的轮数只与 p 有关, 而与博弈的次数 N 无关.
- 在有限次重复囚徒博弈中, 当双方不知道具体次数是公共知识时, 将出现贝叶斯-纳什均衡形式的合作. — 无知是福 ☺

声誉的积累

$$P(\text{君子} \mid \text{好事}) = \frac{P(\text{君子} \wedge \text{好事})}{P(\text{好事})}$$

$$= \frac{P(\text{好事} \mid \text{君子})P(\text{君子})}{P(\text{好事} \mid \text{君子})P(\text{君子}) + P(\text{好事} \mid \neg\text{君子})P(\neg\text{君子})}$$

- 人不知而不愠，不亦君子乎
- 勿以善小而不为，勿以恶小而为之
- 狼来了
- 烽火戏诸侯
- “伪君子” 为什么更可恨?

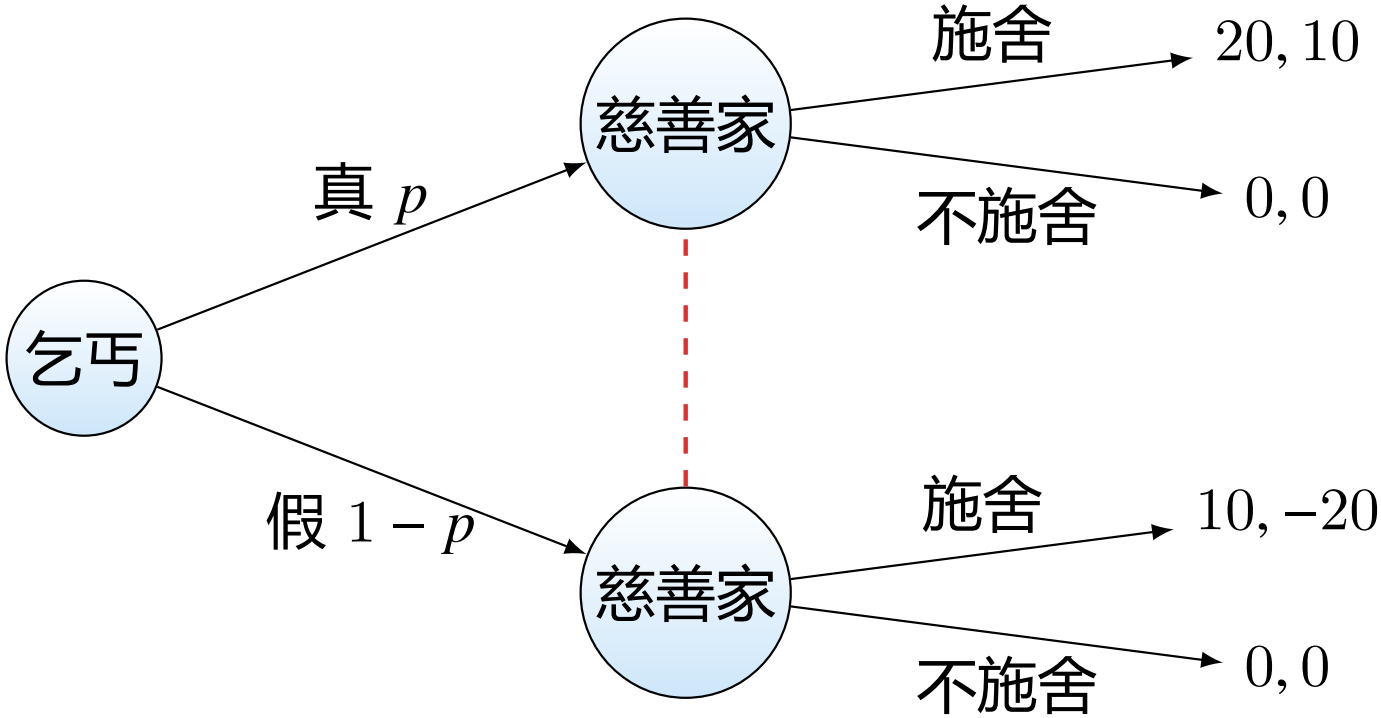
非对称信息 (Asymmetric Information)

- 非对称信息: 交易关系中一方知道而另一方不知道的信息.
  — 卖家知道商品的质量, 而买家不知道.
  — 员工知道自己的能力, 老板不知道.
  — 投保人知道自己的健康状况, 保险公司不知道.
  — 乞讨者知道自己是否是真乞丐, 慈善家不知道.
- 事前 (ex ante) 非对称信息: 指签约之前存在的非对称信息, 所以, 又称为隐藏信息 (hidden information), 如产品质量.
- 事后 (ex post) 非对称信息: 指签约之后发生的非对称信息, 又称隐藏行动 (hidden action), 如工人的努力水平.
- 事前信息不对称导致逆向选择, 劣胜优汰.
- 事后信息不对称导致道德风险.

Example (逆向选择)

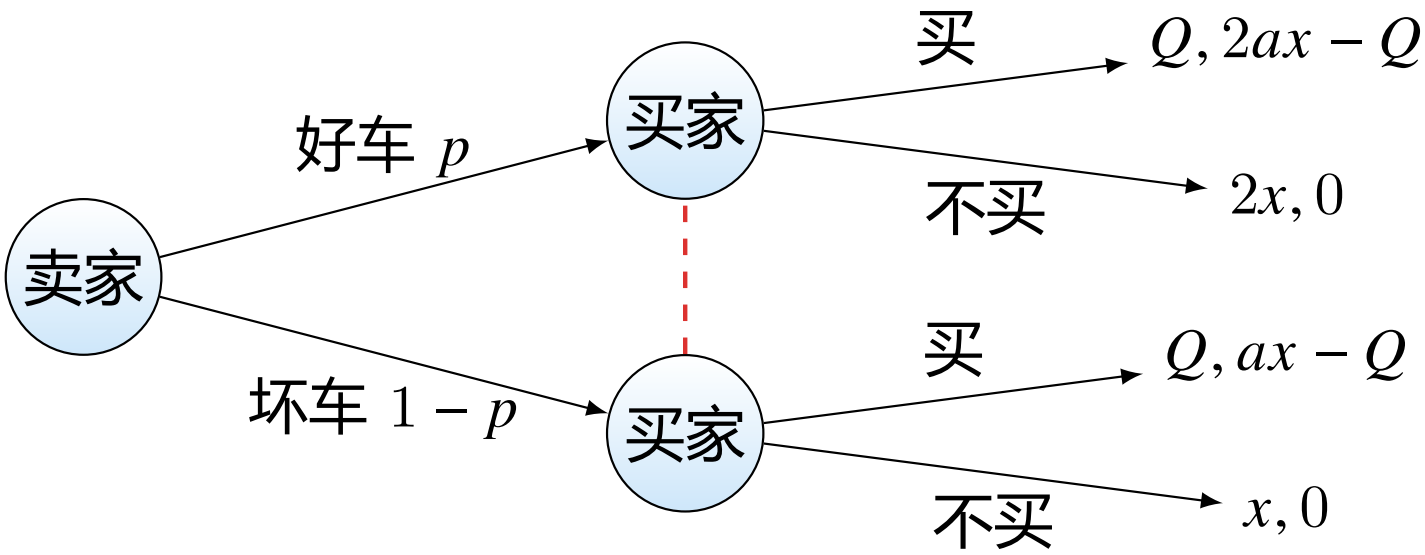
- 坏车使得好车不能成交. (好人未必好报)
- 高风险的投保人使得低风险的投保人无法投保.
- 高水平的学者竞争不过灌水的学者.

逆向选择 — 劣币驱逐良币



- 假乞丐使得真乞丐得不到救济.
- 和珅: 往粥里掺沙子.
- 经济适用房不能建太好.

逆向选择 — 劣币驱逐良币



- 假设好车和坏车对卖家的保留价值分别为 $2x$ 和 x, 对买家的价值是卖家的 a 倍. 成交价为 Q.
- 为使交易达成, 买家买车的期望收益应不小于不买车的期望收益.

$$(2ax - Q)p + (ax - Q)(1 - p) \geq 0$$

- 拥有好车的卖家能接受的最低价是 $Q \geq 2x$.
- 这意味着, 买卖双方合意的价格需满足:

$$(1 + p)ax \geq Q \geq 2x \implies p \geq \frac{2}{a} - 1$$

- 买卖好车是双赢, 但除非好车比例够高, 否则交易无法达成.
- 除非好人够多, 否则不敢与陌生人交往 ☹

如何解决非对称信息的问题?

解决非对称信息的市场机制和政府管制

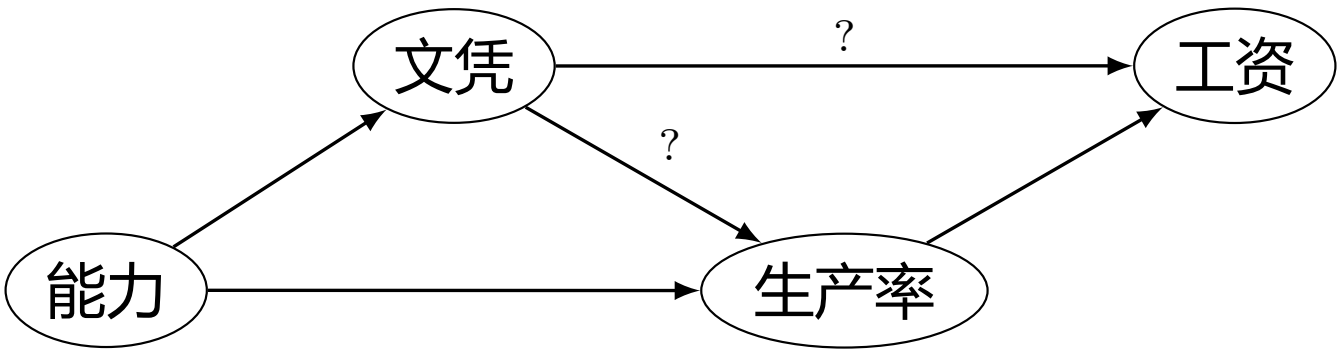
1. 信号显示 (信号传递): 主动传递真实信息. 如: 卖家承诺保修.
2. 信息甄别 (机制设计): 让对方说实话. 如: 保险公司向投保人提供不同的合同选择, 投保人根据自己的状况选择适合自己的合同.
3. 声誉机制: 品牌的价值. 信息越不对称的产品, 其品牌价值越大. 买土豆不太会关心品牌, 但买汽车买咨询服务会非常看重品牌.
4. 适度的政府管制. 但政府管制不当可能破坏声誉机制的有效性 — 管制导致企业预期不稳定, 追求短期行为; 管制导致垄断, 使得市场惩罚不可信; 管制导致腐败, 贿赂官员比贿赂投资者和客户更合算.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
   - Preferences and Expected Utility
   - Strategic Games
   - Extensive Games with Perfect Information
   - Eliminating Dominated Strategies
   - Dynamic Games
   - Repeated Games
   - Signaling Games
   - Mechanism Design
   - Counterfactual Regret Minimization
   - Subgame Perfect Equilibrium
   - Games with Incomplete Information
   - Evolutionary Games
   - Voting System
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

文凭的信号传递作用

- 求职者 50% 可能性是高能力者 50% 可能性是低能力者.
- 高能力者的生产率是 200, 低能力者 100.
- 雇主愿付高能力者工资 200, 低能力者 100, 不知道求职者能力高低则付平均工资 150.
- 假设高能力者受教育成本 40, 低能力者受教育成本 120.
- 此时, 教育水平可以成为传递能力的信号.
- 但如果低能力者受教育的成本低于 100, 文凭就无法成为雇主区分能力高低的信号, 雇主愿付的工资就是 150, 也就没人愿上大学了.
- 因此, 关键是不同类型的人信号传递成本不同; 只有成本差异足够大, 才有可能传递信号.



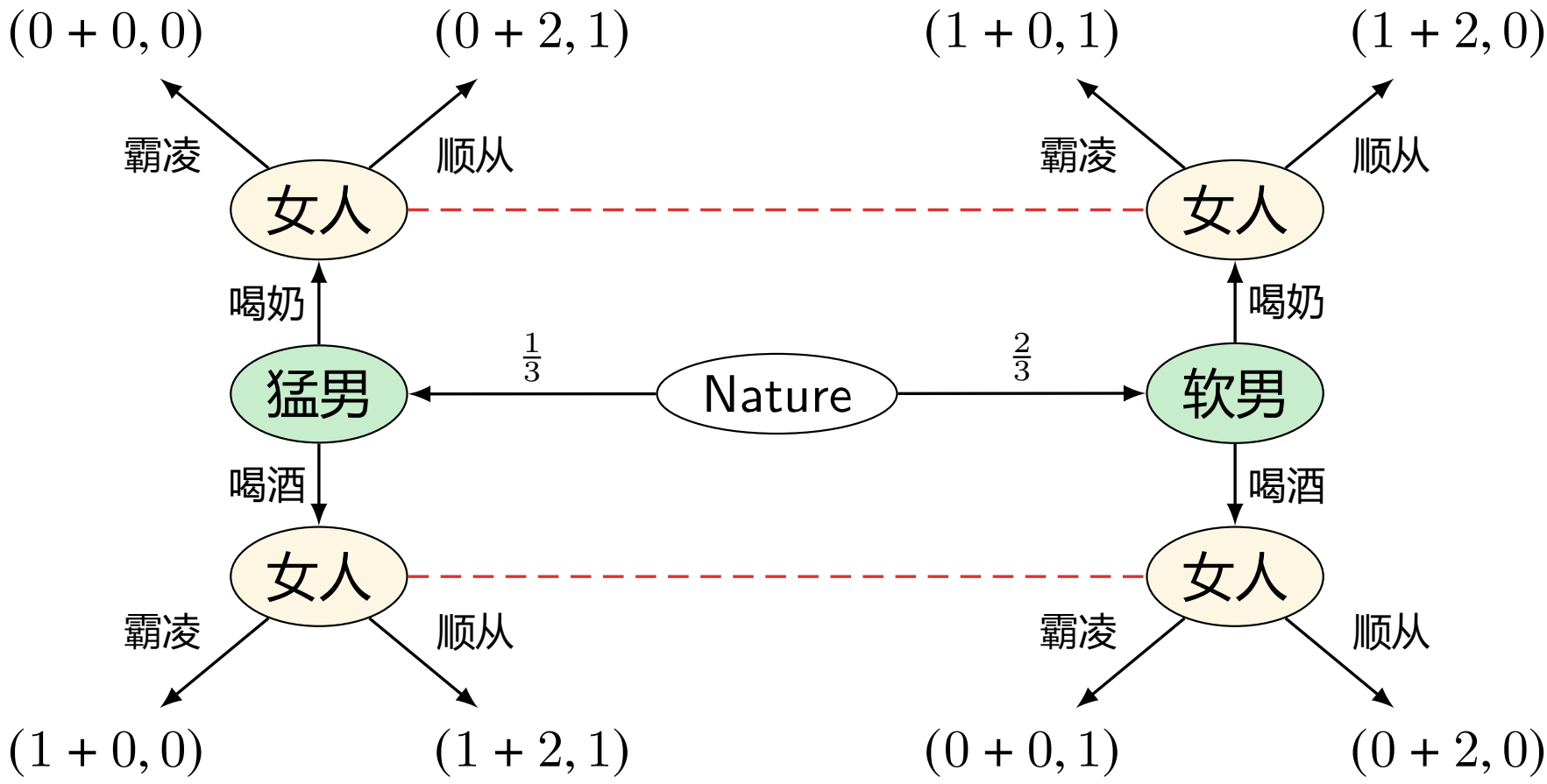
Spence's Education Game

- The worker can be one of two types: wise θ_H (with probability p) or dumb θ_L (with probability $1-p$). Each type can select their own level of education, e_H or e_L.
- The employer is assumed to have two choices. One is to ignore the signal and set $w^* := p\theta_H + (1-p)\theta_L$. The other is to pay a worker w_H or w_L based on whether the signal is e_H or e_L.
- The employer's payoff is $\theta - w$. The worker's payoff is $w - e/\theta$.
- This game has two equilibria. The first is a **pooling equilibrium**, in which the worker will choose the same level of education regardless of his type, and the employer pays all workers the same amount w^*.
- The other is a **separating equilibrium**, in which the worker will choose a different level of education. A low-talent worker will get no education, $e_L = 0$. The education chosen by a high-talent worker is set in such a way as to make it unprofitable for either type of worker to mimic the other.

$$w_L - 0/\theta_L \geq w_H - e_H/\theta_L \quad \text{and} \quad w_H - e_H/\theta_H \geq w_L - 0/\theta_H$$

信号传递的作用

- 为什么雄性孔雀的尾巴越长，越受到雌性孔雀的青睐？— 只有健壮者才能负担得起长尾巴.
- 高、低质量产品哪个更愿意做广告？— 广告费是高质量产品企业向市场传递信息的成本.
- 如何送礼？— 重要的是送礼对送者的成本，而不是礼物对接受者的价值.
- 奢侈品、名牌，不是显示产品的质量，而是显示消费者的质量.
- 中秋为什么送浪费性的月饼？请客吃巨贵的馆子？— 成本要高于价值，甚至“毁灭”了很大一部分价值更显出对对方的重视.“千里送鹅毛，礼轻仁义重.”
- 公费请客送礼传递的信息量大打折扣，成本需要翻倍!
- 为什么领结婚证？— 离婚分财产. 同样，为什么送彩礼？为什么婚礼要铺张？
- 为什么街头古惑仔纹身？黑社会老大穿西装戴眼镜？
- 为什么很多繁文缛节没有实质意义的礼仪还要遵守？— 合作精神



- Nature 以 $\frac{1}{3}$ 的概率指定男人是猛男, $\frac{2}{3}$ 的概率是软男.
- 猛男喜欢喝酒 (效用 1); 软男喜欢喝奶 (效用 1). 男人喜欢女人顺从 (效用 2); 女人喜欢顺从猛男霸凌软男 (效用 1).
- 男人如果是猛男, 就喝酒; 如果是软男, 就以 $\frac{1}{2}$ 的概率遂心喝奶, 以 $\frac{1}{2}$ 的概率违心喝酒.
- 女人见到男人喝奶就霸凌; 见到男人喝酒就以 $\frac{1}{2}$ 的概率霸凌, 以 $\frac{1}{2}$ 的概率顺从.

信息不完全导致社会规范变迁

- 如果是完全分离均衡，每类人的行为都是特定的.
- 如果是混同均衡，所有人的行为都是一样的.
- 如果是准分离 (混同) 均衡，有些行为传递信息，有些行为不传递信息.
- 如果外部因素导致社会由分离均衡转向混同均衡或准分离均衡，社会规范就会发生变化.

人们对婚前性行为和婚外性行为态度的变化：在封闭的社会，婚前和婚外性行为都很容易观察；在流动的社会，有些能观察到，有些不能；如果被观察到的只是其中的一小部分，被观察到压力将会减少.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
   - Preferences and Expected Utility
   - Strategic Games
   - Extensive Games with Perfect Information
   - Eliminating Dominated Strategies
   - Dynamic Games
   - Repeated Games
   - Signaling Games
   - Mechanism Design
   - Counterfactual Regret Minimization
   - Subgame Perfect Equilibrium
   - Games with Incomplete Information
   - Evolutionary Games
   - Voting System
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Cake Cutting

- Two Children:
  1. One cuts, the other chooses.
- Three Children (A,B,C):
  1. A cuts a piece of the cake.
  2. B can leave it, or make it smaller.
  3. C can leave the remaining piece, or make it smaller.

  Whoever cuts the cake last, gets the remaining piece.

- 19 世纪中期法国的火车有头等座、二等座和三等座三种不同的车厢.
- 三种车厢票价相差较大.
- 头等车厢非常舒适, 二等车厢与三等车厢的区别是前者有顶盖, 后者没有. 这样，坐三等车厢的旅客就要忍受日晒雨淋的痛苦.
- 给车厢加一个顶盖的成本微乎其微, 为什么公司不这样做呢?
- 为了吓唬富人, 故意让穷人受罪.

- 问题: 有时候, 拥有私人信息的一方有积极性通过一定的行动向另一方传递自己的私人信息, 但有时候他们没有积极性或没有有效的办法传递自己的私人信息.
- 机制设计: 没有私人信息的一方通过设计不同的分配方案使得有私人信息的一方通过自我选择揭示自己的私人信息.

Example: 保险市场中的混同均衡和分离均衡.

1. 保险金 2 万元; 只有第二年得病才可得到赔偿金 10 万元, 第一年得病得不到赔偿.
2. 保险金 7 万元; 无论第一年得病还是第二年得病, 都可得到赔偿金 10 万元.

高风险者选择合同 2; 低风险者选择合同 1.

机制设计

逆向博弈论

已知理性人会选择最优策略，为了诱导出某种类型的行为，应该设计什么样的博弈？

博弈 vs 伦理

- 康德之流的义务论者可能会认为，博弈对伦理无益．因为，伦理规范是约束你做那些你不想做的事，而博弈论却是讲怎么得到你想要的东西．
- 休谟等效用主义者却认为，除非伦理理论建议的行为符合每个人的实际利益，否则毫无用处．有效的制度设计，就是通过对纳什均衡的选择实现帕累托最优．

拍卖

1. 高价格公开拍卖
2. 降价公开拍卖
3. 高价格密封拍卖
4. 次价格密封拍卖: 与高价格密封拍卖一样, 中标者仍然是出价最高者, 但中标者实际支付的是第二高报价. 次价格密封拍卖能让竞标者有积极性说真话.

Theorem

次价格密封拍卖中, 说真话是竞标者的占优策略.

Remark: 比如: 对某古董, 你的实际评价是 1 万, 如果出价 1 万, 第二高出价 9 千, 你赚 1 千; 如果你出价低于 9 千, 你什么也得不到.

VCG 机制 Vickrey-Clarke-Groves Mechanism

- 公共产品的偏好显示: 不同的人有不同的偏好, 是私人信息. 如何让每个人报告自己的真实偏好?
- 每个人可以任意地报告自己的偏好, 但可能要纳缴一定数量的 "税". 计算办法: 先将其他人的偏好加总, 给出总价值最大的项目; 然后将第一个人的偏好加上, 如果不影响结果, 不征税; 否则, 应纳税等于改变结果给其他人带来的损失.

$$p_i = \underbrace{\max_x \sum_{j\neq i} v_j(x)}_{i \text{ 不参与}} - \underbrace{\sum_{j\neq i} v_j(x^*)}_{i \text{ 参与}} \quad \text{where} \quad x^* = \operatorname*{argmax}_x \sum_{i=1}^{n} v_i(x)$$

- 同学聚会去吃川菜还是粤菜? 每人报告自己的真实偏好是占优策略.

	川菜	粤菜	税额 p_i
A	30	10	0
B	0	40	30
C	20	10	0
合计	50	60	30

公平与夏普利值

- A,B,C 合作开了一家公司. 公司任何决策只有超过 50% 的赞成票才能通过.
- 根据投资额, A 拥有 50% 的票力, B 拥有 40%, C 拥有 10%.
- 年终净赚 150 万, 根据票力之比, A,B,C 分别 75 万, 60 万, 15 万吗?
- C 提议: A 拿 80 万, 自己 70 万, B 一分没有.
- B 提议: A 拿 85 万, 自己 65 万, C 一分没有.
- 怎么分才公平?

- 三人中, 任何一人都不是决定性的, 都得与他人结盟才有决策权.
- 夏普利值的基本假设是: 各投票顺序联盟形成的可能性相同, 玩家的夏普利值为其对联盟的边际贡献之和除以各种可能的联盟组合总数.

1	A	A	B	B	C	C
2	B	C	A	C	A	B
3	C	B	C	A	B	A
关键加入者	B	C	A	A	A	A

Table: 所有可能的投票顺序

- 作为关键加入者的次数. A: 4 次; B: 1 次; C: 1 次.
- 夏普利值. A: $\frac{4}{6}$; B: $\frac{1}{6}$; C: $\frac{1}{6}$.
- 根据影响力, A: $150 \times \frac{4}{6} = 100$ 万; B: 25 万; C: 25 万.

讨价还价与纳什讨价还价解

- 市场上有一个画家 A 和一个画廊 B.
- 若画家 A 自己卖画, 可得 $a = 1500$.
- 若交给画廊 B 卖, 可得 $V = 6000$.
- 若画廊 B 干别的事, 可得 $b = 1000$.
- AB 该怎么分利?

- $S := V - a - b = 3500$ 是合作带来的剩余价值.
- 我们用 x 表示 A 应分得的价值, y 表示 B 应分得的价值. α, β 表示 A 和 B 剩余价值 S 的分配比例 (与边际贡献有关).

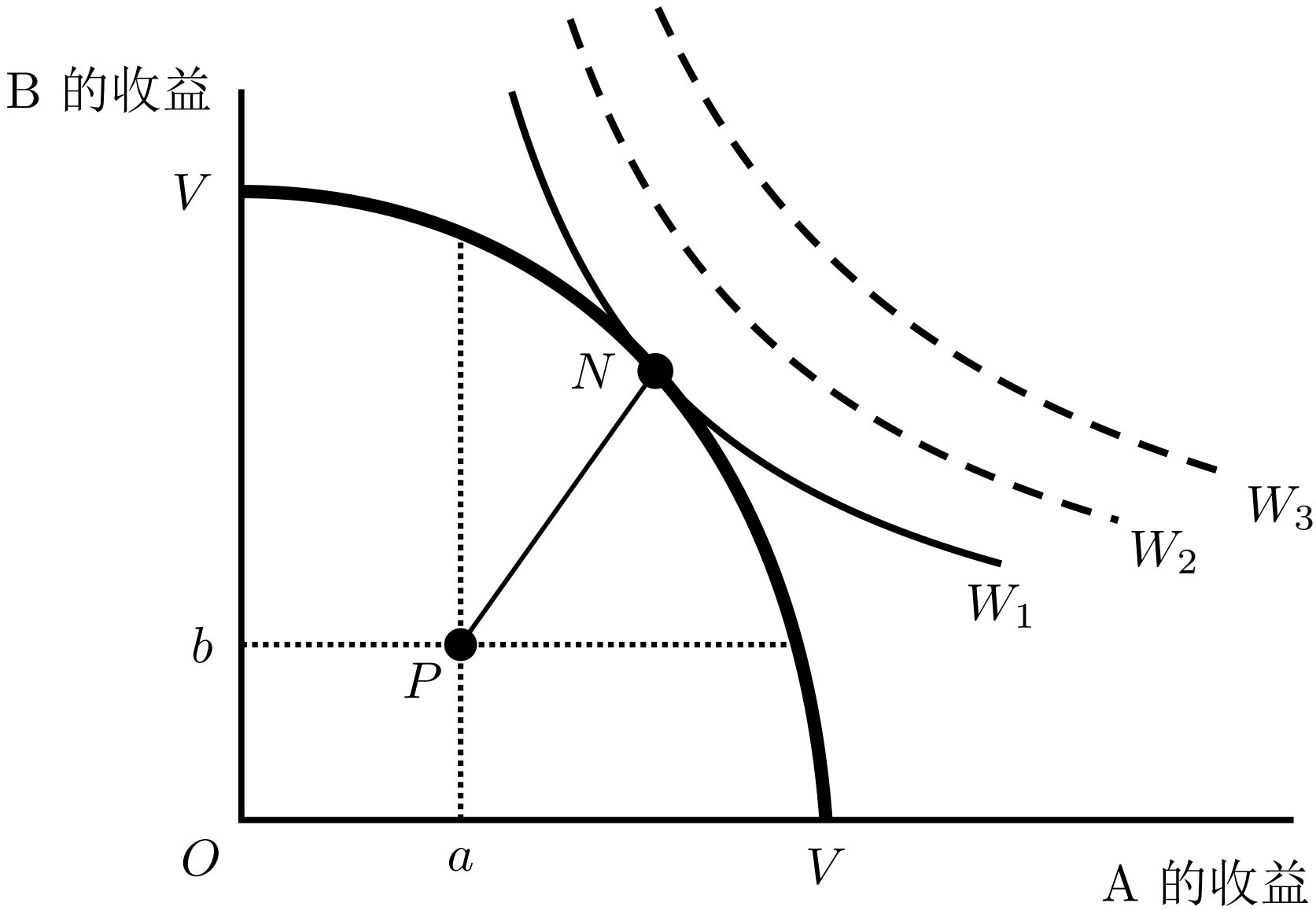
$$x = a + \alpha S, \quad y = b + \beta S$$

- 纳什讨价还价解即优化福利函数 $W(x, y) = (x - a)^{\alpha}(y - b)^{\beta}$

$$(x^*, y^*) = \operatorname*{argmax}_{x,y:x+y=V} (x - a)^{\alpha}(y - b)^{\beta}$$

- 因为若缺少任何一方, 这 3500 的剩余价值都得不到, 所以每一方的边际贡献都是 3500. 总边际贡献是 $3500 + 3500 = 7000$. 每一方的边际贡献占比都是 $\frac{3500}{3500+3500} = \frac{1}{2} = \alpha = \beta$.
- A 应分得 $1500 + \frac{1}{2} \times 3500 = 1500 + 1750 = 3250$; B 应分得 2750.

纳什讨价还价解



谈判中的影响力 — 不可替代性

- 如果市场上还有一家画廊 C, 若 A 交给画廊 C 卖, 可得 $V' = 5000$.
- 若画廊 C 干别的事, 可得 $c = 500$.
- AB 又该怎么分利?

- AB 合作中 A 的边际贡献是 $V - a - b = 3500$.
- 若 AB 不合作, 则 AC 合作, 因为 B 的存在, C 的边际贡献为 0, A 拿走所有剩余 $V' - c$. 即 A 的市场价值不是 a 而是 $V' - c$.
- 所以, AB 合作中 B 的边际贡献是
  $V - (V' - c) - b = 6000 - (5000 - 500) - 1000 = 500$.
- A 的边际贡献占总边际贡献的 $\frac{3500}{3500+500} = \frac{7}{8}$, B 占 $\frac{1}{8}$.
- A 应分得: $1500 + \frac{7}{8} \times 3500 = 4562.5$.
- B 应分得: $1000 + \frac{1}{8} \times 3500 = 1437.5$.

Remark: 大牌影星天价片酬.

腐败 vs 信息

- 腐败的深层根源：信息不对称；
- 特别是事后的信息不对称：一方当事人的行为不能被另一方观察到；
- 研究行为信息不对称的理论叫“委托-代理”理论或道德风险理论.

委托-代理关系

- 法律上的委托 - 代理关系：如果甲乙两人达成一个协议，甲将做某事的权利交给乙，甲为委托人，乙为代理人.
- 代理人对委托人的责任：(1) 没有许可，不能再代理；(2) 不能把自己放在与委托人利益冲突的地位；(3) 保密责任和诚信责任.
- 委托人对代理人的责任：(1) 补偿责任. 委托人给代理人补偿报酬；(2) 免除法律责任. 委托人要为代理人的行为承担法律责任；(3) 留置权. 如果委托人给代理人的补偿没有到位，代理人有权滞留委托人的财产.
- 经济上的委托 - 代理关系要宽松很多：只要一方行为影响另一方的利益，就有委托 - 代理关系. 有私人信息的一方是代理人，没有私人信息的称为委托人.

委托-代理问题

- 委托人与代理人可能的利益冲突: 对委托人最优的选择不一定是对代理人最优的选择.
- 如果委托人能观察到代理人的行为, 则可以通过强制性合同约束代理人的行为.
- 即使不能完全观察到代理人的行为, 如果代理人不害怕风险, "承包制" 可以使代理人成为完全的风险承担者.
- 但如果代理人承担责任的能力有限, "承包制" 没有可行性.

对代理人的激励机制的设计

- 保险与激励的冲突
  - 最优风险分担: 如果委托人是风险中性的, 代理人是风险规避的, 风险应该完全由委托人承担, 代理人拿固定报酬.
  - 最优激励: 代理人应该承担完全风险.
  - 这就产生了保险与激励的冲突.
  - 最优激励合同要在保险与激励之间求得平衡.
    — 汽车的部分保险 (如 80%).
- 激励的强度
  - 代理人的边际生产率越高, 激励应该越强.
  - 代理人的产出的不确定性越大, 或测度越困难, 激励应该越弱.
  - 代理人越风险规避, 激励应该越弱.
  - 代理人对激励的反应程度: 反应越强, 激励应该越强.
- 相对业绩比较有助于改进激励, 但也可能导致代理人之间合谋或相互拆台.
- 论功行赏 vs 任人唯贤
  — 论功行赏可能导致每个人都被晋升到他不能胜任的位置
- 多重任务下的激励: 大学教学 + 科研

政府官员的激励

- 很多很多委托人;
- 多项任务;
- 业绩难以度量, 投入也不容易度量;
- 所以, 政府官员难以激励, 只能以监督为主.
- 但监督是不完全的, 是有成本的.
- 权利越大, 越难监督.
- “把权力关进笼子”. 民主、法治.

官员腐败问题

- W: 官员工资.
- $B(q)$: 权力租金, 即官员在位置上可能收受的贿赂, 也可看作不腐败的机会成本. 一般来说, 权力 q 越大, 权力租金越高.
- p: 腐败行为被发现暴露的概率.
- F: 对腐败的处罚.
- U: 政府外的保留效用.
- α: 官员的羞耻感或脸皮厚度.
- 官员腐败的期望收益

$$(1-p)(W+B(q))+p(U-\alpha F)$$

- 官员不腐败的条件:

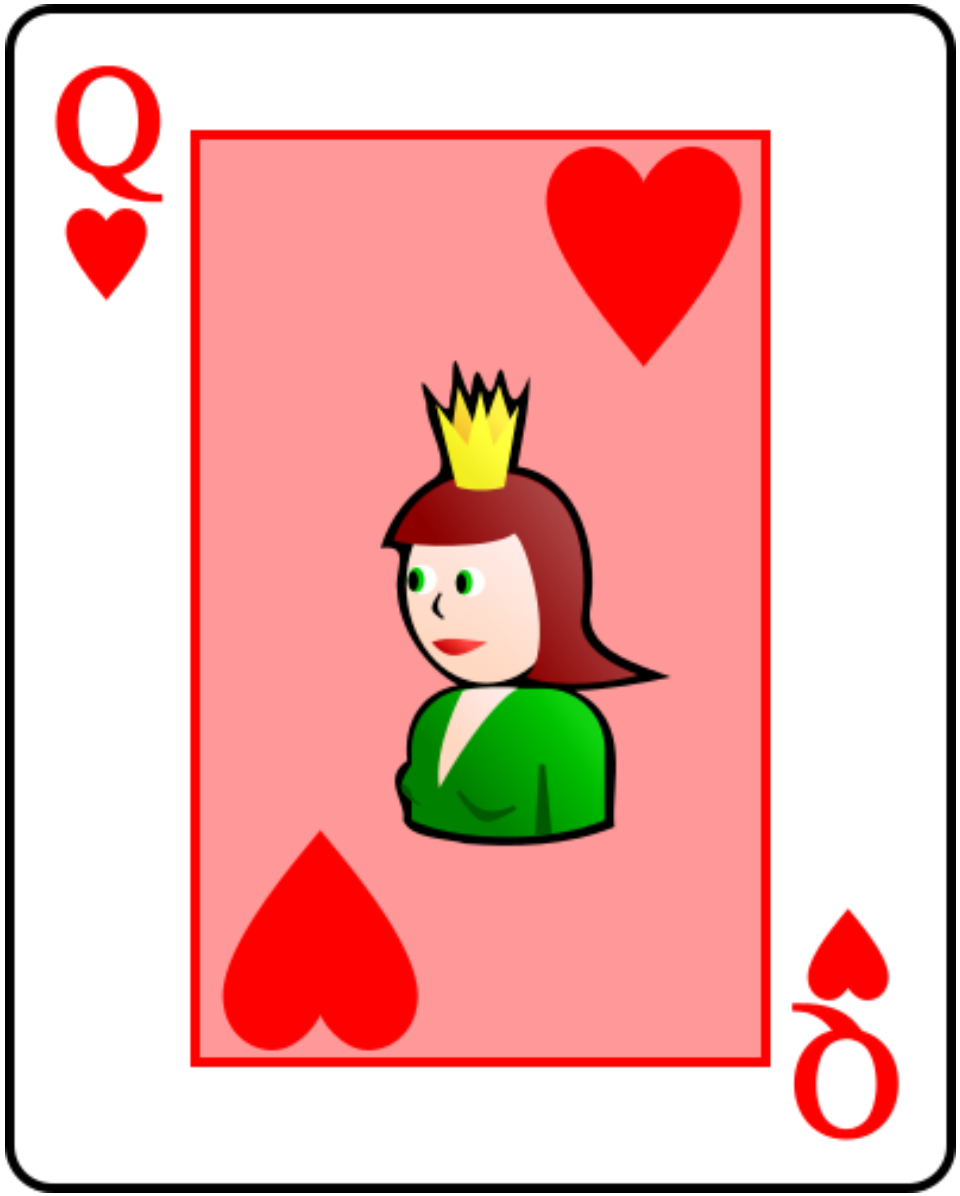
$$W \geq \frac{1-p}{p}B(q)+U-\alpha F$$

- 如何控制腐败? 提高 p, F, W, α, 减少 q.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
   - Preferences and Expected Utility
   - Strategic Games
   - Extensive Games with Perfect Information
   - Eliminating Dominated Strategies
   - Dynamic Games
   - Repeated Games
   - Signaling Games
   - Mechanism Design
   - Counterfactual Regret Minimization
   - Subgame Perfect Equilibrium
   - Games with Incomplete Information
   - Evolutionary Games
   - Voting System
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Imperfect Information Game



Counterfactual Regret Minimization

Play a strategic game for a number of rounds:

- Regret is determined after each game round: If I had played another move, my payoff would have been that much higher!
- Accumulate all positive regrets over time.
- Match the probabilities of a mixed strategy with the accumulated regret.

Take the average over all mixed strategies.

If two players use the regret matching technique in a zero-sum game, then the average over the mixed strategies converges to Nash equilibrium strategies.

Counterfactual Regret Minimization

- The reach probability $\pi^\rho(h) := \prod_{i \in N \cup \{c\}} \pi_i^\rho(h)$ is the probability that history h will be reached with strategy ρ, where π_i^ρ is the contribution of player i, and π_{-i}^ρ is the product of all player contributions except player i.
- $\pi^\rho(I) := \sum_{h \in I} \pi^\rho(h)$
- The counterfactual reach probability $\pi_{-i}^\rho(I)$ is the probability of reaching I under the assumption that player i always uses actions with probability 1 in order to reach I.
- $\rho_{I \to a}$ is the same strategy as ρ, except that action a is always chosen at information set I.

Counterfactual Regret Minimization

- The counterfactual utility of ρ at non-terminal history h is:

$$v_i(\rho, h) := \sum_{z \in \mathcal{Z}, h \prec z} \pi^{\rho}_{-i}(h) \pi^{\rho}(z \mid h) u_i(z)$$

- The counterfactual regret of not taking action a at history $h \in I$ is:

$$r(h, a) := v_i(\rho_{I \to a}, h) - v_i(\rho, h)$$

- The Counterfactual regret of not taking a at I is:

$$r(I, a) := \sum_{h \in I} r(h, a)$$

- $r_i^t(I, a)$ refers to the regret in episode t, when players use ρ and player i does not take a at I.
- The Cumulative counterfactual regret is:

$$R_i^T(I, a) := \sum_{t=1}^{T} r_i^t(I, a)$$

Counterfactual Regret Minimization

- The positive cumulative counterfactual regret is:

$$R_i^{T,+} := \max\left\{R_i^T(I,a), 0\right\}$$

- The regret matching strategy for episode $T+1$ is:

$$\rho_i^{T+1}(I,a) := \begin{cases} \dfrac{R_i^{T,+}(I,a)}{\sum_{a \in A(I)} R_i^{T,+}(I,a)} & \text{if } \sum_{a \in A(I)} R_i^{T,+}(I,a) > 0 \\ \dfrac{1}{|A(I)|} & \text{otherwise} \end{cases}$$

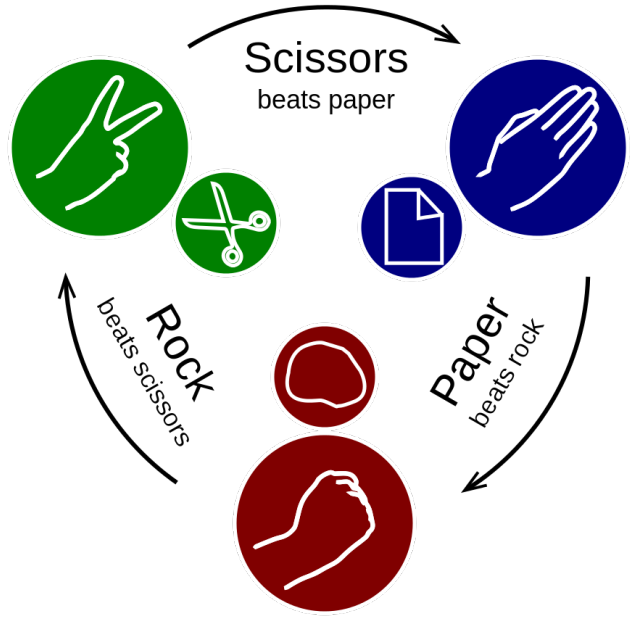
- The average strategy is:

$$\overline{\rho}_i^t(I)(a) := \frac{\sum_{t=1}^{T} \pi_i^{\rho^t}(I)\rho^t(I)(a)}{\sum_{t=1}^{T} \pi_i^{\rho^t}(I)}$$

Regret Matching — RPS example with two rounds

Assume we play rock, paper, scissors, and player 1 uses regret matching.

1. Initial cumulative regret is $(0, 0, 0)$
2. Play uniform strategy $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$
3. Player 1 chooses R, while player 2 chooses P
4. Regret for player 1:
   - $R : u_1(R, P) - u_1(R, P) = -1 - -1 = 0$
   - $P : u_1(P, P) - u_1(R, P) = 0 - -1 = 1$
   - $S : u_1(S, P) - u_1(R, P) = 1 - -1 = 2$
5. Player 1's cumulative counterfactual regret is now $(0, 1, 2)$
6. Regret matching strategy: $\rho_1^1 = (0, \frac{1}{3}, \frac{2}{3})$
7. Player 1 chooses P, while player 2 chooses S
8. Regret for player 1:
   - $R : u_1(R, S) - u_1(P, S) = 1 - -1 = 2$
   - $P : u_1(P, S) - u_1(P, S) = -1 - -1 = 0$
   - $S : u_1(S, S) - u_1(P, S) = 0 - -1 = 1$
9. Player 1's cumulative counterfactual regret is now $(2, 1, 3)$
10. Regret matching strategy: $\rho_1^2 = (\frac{1}{3}, \frac{1}{6}, \frac{1}{2})$



11. The average strategy: $(\frac{1}{6}, \frac{1}{4}, \frac{7}{12})$.
Not close to NE, but will converge!

	R	P	S
t_0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$
t_1	0	$\frac{1}{3}$	$\frac{2}{3}$
t_2	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{2}$

$$h = RP$$

$$\frac{\frac{1}{3}\times\frac{1}{3}\times 0 + \frac{1}{3}\times\frac{1}{3}\times\frac{1}{3}}{\frac{1}{3}\times\frac{1}{3}+\frac{1}{3}\times\frac{1}{3}} = \frac{1}{6}$$

$$\frac{\frac{1}{3}\times\frac{1}{3}\times\frac{1}{3} + \frac{1}{3}\times\frac{1}{3}\times\frac{1}{6}}{\frac{1}{3}\times\frac{1}{3}+\frac{1}{3}\times\frac{1}{3}} = \frac{1}{4}$$

$$\frac{\frac{1}{3}\times\frac{1}{3}\times\frac{2}{3} + \frac{1}{3}\times\frac{1}{3}\times\frac{1}{2}}{\frac{1}{3}\times\frac{1}{3}+\frac{1}{3}\times\frac{1}{3}} = \frac{7}{12}$$

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
   - Preferences and Expected Utility
   - Strategic Games
   - Extensive Games with Perfect Information
   - Eliminating Dominated Strategies
   - Dynamic Games
   - Repeated Games
   - Signaling Games
   - Mechanism Design
   - Counterfactual Regret Minimization
   - Subgame Perfect Equilibrium
   - Games with Incomplete Information
   - Evolutionary Games
   - Voting System
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Subgame

某个博弈的子博弈是由该博弈的某个单一结点 (即该结点所在的信息集就只包含一个要素, 就是该结点) 及其所有后续结点构成的集合, 并且所有的后续结点必须满足一个条件, 即如果某一个后续结点属于该子博弈, 那么该后续结点所在信息集的所有结点必须也属于该子博弈.

Definition (Subgame)

The *subgame* of the extensive game Γ that follows the history h is the extensive game $\Gamma(h) := \langle N, P|_h, \mathcal{H}|_h, (\mathcal{I}_i)_{i\in N}, f_c, (u_i|_h)_{i\in N}\rangle$, where

$$P|_h(h') := P(hh')$$

$$\mathcal{H}|_h := \{h' : hh' \in \mathcal{H}\}$$

$$u_i|_h(h') := u_i(hh')$$

$$\forall I \in \bigcup_{i\in N} \mathcal{I}_i : I \subset \{hh' : h' \in \mathcal{H}|_h\} \ \vee \ I \subset \mathcal{H} \setminus \{hh' : h' \in \mathcal{H}|_h\}$$

Subgame Perfect Equilibrium

Definition (Subgame Perfect Equilibrium)

A *subgame perfect equilibrium* of an extensive game Γ is a strategy profile s^* such that for every player $i \in N$ and every nonterminal history $h \in \mathcal{H} \setminus \mathcal{Z}$ for which $P(h) = i$ we have

$$u_i|_h(s_i^*; s_{-i}^*) \geq u_i|_h(s_i; s_{-i}^*)$$

for every strategy s_i of player i in the subgame $\Gamma(h)$.
A behaviour strategy profile ρ^* is a *subgame perfect equilibrium* of a game Γ iff for all subgames $\Gamma(h)$ of Γ, and,

$$\forall i \in N \forall \rho_i \in B_i : u_i|_h(\rho_i^*; \rho_{-i}^*) \geq u_i|_h(\rho_i; \rho_{-i}^*)$$

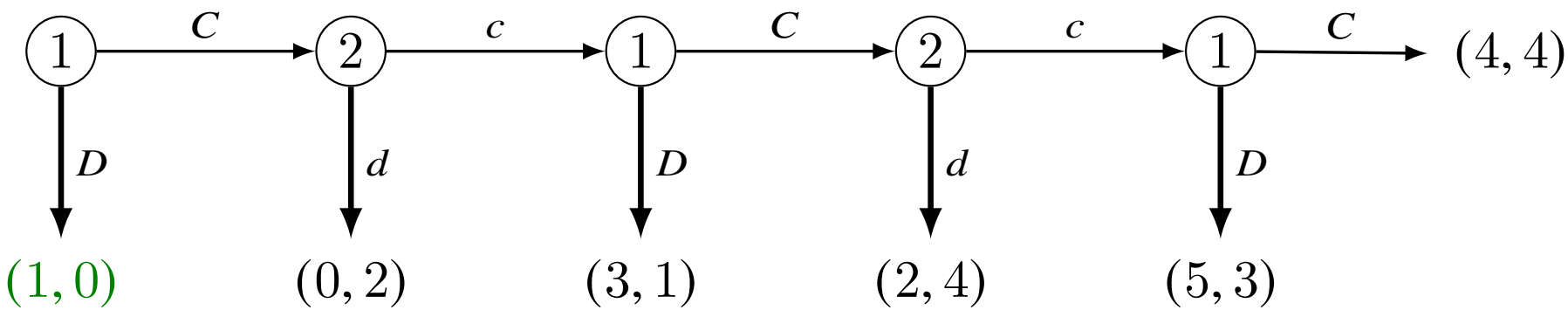
Remark: A Nash equilibrium is subgame perfect iff it is a Nash equilibrium in every subgame.
In a finite game of perfect information, the subgame perfect equilibrium is the backward induction solution.

Kuhn's Theorem

Theorem (Kuhn's Theorem)

Every finite extensive game has a subgame-perfect equilibrium (possibly in mixed strategies).

Criticism of Backward Induction and Subgame Perfection?



Remark

In the unique subgame perfect equilibrium, players defect at every node, precluding any mutually beneficial cooperative outcome.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
   - Preferences and Expected Utility
   - Strategic Games
   - Extensive Games with Perfect Information
   - Eliminating Dominated Strategies
   - Dynamic Games
   - Repeated Games
   - Signaling Games
   - Mechanism Design
   - Counterfactual Regret Minimization
   - Subgame Perfect Equilibrium
   - Games with Incomplete Information
   - Evolutionary Games
   - Voting System
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Sequentially Rationality

Definition (Assessment)

An *assessment* in an extensive game is a pair (ρ, μ), where ρ is a profile of behavioral strategies and μ is a belief system. $\mu(I)(h)$ is the probability that player $P(I)$ assigns to the history $h \in I$, conditional on I being reached.

Definition (Sequentially Rationality)

An assessment (ρ, μ) is *sequentially rational* iff for every player i and every information set $I \in \mathcal{I}_i$ the strategy of player i is a best response to the other players' strategies given i's beliefs at I.

$$\forall \rho_i' \in B_i : \sum_{h \in I} u_i|_h(\rho_i; \rho_{-i})\mu(I)(h) \geq \sum_{h \in I} u_i|_h(\rho_i'; \rho_{-i})\mu(I)(h)$$

We can define the outcome $O(\rho, \mu \mid I)$ as the distribution over terminal histories determined by ρ and μ conditional on I being reached.

$$O(\rho, \mu \mid I)(h') := \begin{cases} 0 & \text{if } \nexists h \in I \, (h \prec h') \\ \mu(I)(h) \prod_{k=\ell(h)}^{\ell(h')} \rho_{P(h_{<k})}(h_{<k})(h_k) & \text{otherwise} \end{cases}$$

since by perfect recall, there is at most one subhistory h of h' in I, and the histories $h_{1:k}$ for $k = \ell(h), \ldots, \ell(h')$ lie in different information sets.

- An assessment (ρ, μ) is *sequentially rational* iff for every player i and every information set $I \in \mathcal{I}_i$

$$\forall \rho_i' \in B_i : O(\rho, \mu \mid I) \succcurlyeq_i O((\rho_i'; \rho_{-i}), \mu \mid I)$$

- A behavioral strategy profile to be *completely mixed* iff it assigns positive probability to every action at every information set.

Definition (Consistency of Beliefs with Strategies)

Let Γ be a finite extensive game with perfect recall. An assessment (ρ, μ) is *consistent* iff there is a sequence $((\rho^n, \mu^n))_{n=1}^{\infty}$ of assessments s.t. $(\rho, \mu) = \lim_{n \to \infty} (\rho^n, \mu^n)$ and each strategy profile ρ^n is completely mixed and each belief system μ^n is derived from ρ^n using Bayes' rule:

$$\mu^n(I)(h) = \frac{\overline{\rho}^n(h)}{\sum_{h \in I} \overline{\rho}^n(h)}$$

Definition (Sequential Equilibrium)

An assessment is a *sequential equilibrium* of a finite extensive game with perfect recall iff it is sequentially rational and consistent.

Theorem

Every finite extensive game with perfect recall has a sequential equilibrium.

Theorem

In an extensive game with perfect information, (ρ, μ) is a sequential equilibrium iff ρ is a subgame perfect equilibrium.

Bayesian Strategic Game with Observable Actions

Definition (Bayesian Strategic Game with Observable Actions)

A Bayesian game is a tuple $\langle N, (S_i)_{i \in N}, (\Theta_i)_{i \in N}, p, (u_i)_{i \in N} \rangle$ where

- The sets of players N.
- The sets of strategies of the players $(S_i)_{i \in N}$.
- Θ_i is a finite set (the set of possible types of player i). $\Theta := \prod_{i \in N} \Theta_i$.
- $p \in \Delta\Theta$.
- $u_i : S \times \Theta \to \mathbb{R}$

Intuition for a Bayesian Game

- Harsanyi (1967) suggests a way of understanding a game of incomplete information as a game of imperfect information with a fictitious player who moves first called Nature.
- Nature draws a type vector $(\theta_1, \ldots, \theta_n)$ for all players.
- For each i, Nature reveals θ_i to some players (usually including i) and sends a signal to others.
- Players choose actions s_i simultaneously.
- Payoffs are realized: $u_i(s, \theta)$ for all i.

Expected Utility

1. ex-post — the agent knows all agents' types.
2. ex-interim — an agent knows her own type but not the types of the other agents.
3. ex-ante — the agent knows nothing about anyone's actual type.

狐狸、权力与不完全信息

- 小女孩提着一篮子鸡蛋穿过树林.
- 喜欢抢鸡蛋吃的狐狸跳了出来: “我是狐狸!”
- 小女孩: “怎么证明你是狐狸?”
- 狐狸: “我有厚厚的皮毛.”
- 小女孩: “兔子也有厚厚的皮毛.”
- 狐狸: “我有毛茸茸的尾巴.”
- 小女孩: “松鼠也有毛茸茸的尾巴.”
- 狐狸一直试图证明自己, 不知不觉跟着小女孩走出了树林.
- 一只猎犬经过吓跑了狐狸.
- 狐狸边跑边喊, “不信你问猎犬, 我真的是狐狸!”
- 小女孩: “我知道, 我早就知道.”

	攻击	不攻击
防卫	$-2,-2$	$0,0$
不防卫	$-1,1$	$0,0$

Table: 与狐狸博弈

	攻击	不攻击
防卫	$0,-2$	$0,0$
不防卫	$-1,1$	$0,0$

Table: 与兔子/松鼠博弈

Remark: 权力需要被认可, 如果得不到认可, 权力也将不复存在. 如果有人试图用权力吓唬你, 你应该提出质疑, 让对方承担举证的责任.

Ex-post Expected Utility

Definition (Ex-post Expected Utility)

Player i's ex post expected utility in a Bayesian game is defined as

$$V_i(\sigma, \theta) \coloneqq \sum_{s \in S} \left(\prod_{j \in N} \sigma_j(s_j \mid \theta_j) \right) u_i(s, \theta)$$

where the players' types are given by $\theta \in \Theta$, and the players' strategies are given by $\sigma \in \prod_{i \in N} \prod_{\theta_i \in \Theta_i} \Delta S_i$, $\sigma = (\sigma_i(\cdot \mid \theta_i))_{\theta_i \in \Theta_i, i \in N}$, $\sigma_i(\cdot \mid \theta_i) \in \Delta S_i$.

Ex-interim Expected Utility

Definition (Ex-interim Expected Utility)

Player i's ex interim expected utility in a Bayesian game, where i's type is θ_i and where the players' strategies are given by the mixed-strategy profile σ, is defined as

$$V_i(\sigma, \theta_i) := \sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \mid \theta_i) \sum_{s \in S} \left(\prod_{j \in N} \sigma_j(s_j \mid \theta_j) \right) u_i(s, (\theta_i; \theta_{-i}))$$

or equivalently as

$$V_i(\sigma, \theta_i) = \sum_{\theta_{-i} \in \Theta_{-i}} p(\theta_{-i} \mid \theta_i) V_i(\sigma, (\theta_i; \theta_{-i}))$$

Ex-ante Expected Utility

Definition (Ex-ante Expected Utility)

Player i's ex ante expected utility in a Bayesian game, where the players' strategies are given by the mixed-strategy profile σ, is defined as

$$V_i(\sigma) := \sum_{\theta \in \Theta} p(\theta) \sum_{s \in S} \left(\prod_{j \in N} \sigma_j(s_j \mid \theta_j) \right) u_i(s, \theta)$$

or equivalently as

$$V_i(\sigma) = \sum_{\theta \in \Theta} p(\theta) V_i(\sigma, \theta)$$

or again equivalently as

$$V_i(\sigma) = \sum_{\theta_i \in \Theta_i} p(\theta_i) V_i(\sigma, \theta_i)$$

Best Response in a Bayesian Game

Definition (Best Response in a Bayesian Game)

The set of player i's best response to mixed-strategy profile σ_{-i} are given by

$$\mathrm{BR}_i(\sigma_{-i}) := \underset{\sigma_i \in \Delta S_i^{\Theta_i}}{\operatorname{argmax}} V_i(\sigma_i; \sigma_{-i})$$

Remark: It may seem odd that BR_i is calculated based on i's ex ante expected utility. However, we are in fact performing independent maximization of i's *ex interim expected utilities* conditioned on each type that he could have.
Intuitively speaking, if a certain action is best after the signal is received, it is also the best conditional plan devised ahead of time for what to do should that signal be received.

Definition (Bayesian Equilibrium)

A Bayesian equilibrium is a mixed strategy profile σ^* that satisfies

$$\forall i \in N : \sigma_i^* \in \mathrm{BR}_i(\sigma_{-i}^*)$$

Definition (Ex-post Equilibrium)

An ex post equilibrium is a mixed strategy profile σ that satisfies

$$\forall \theta \in \Theta \forall i \in N : \sigma_i \in \underset{\sigma_i \in \Delta S_i^{\Theta_i}}{\operatorname{argmax}} V_i((\sigma_i; \sigma_{-i}), \theta)$$

The ex post equilibrium is similar in flavor to equilibria in dominant strategies, which do not require agents to believe that other agents act rationally. However, it seems too good to be true most of the time.

Bayesian Extensive Game with Observable Actions

Definition (Bayesian Extensive Game with Observable Actions)

A *Bayesian extensive game with observable actions* is a tuple $\langle \Gamma, (\Theta_i)_{i \in N}, (p_i)_{i \in N}, (u_i)_{i \in N} \rangle$ where

- $\Gamma = \langle N, P, \mathcal{H} \rangle$ is an extensive game form with perfect information and simultaneous moves.
- Θ_i is a finite set (the set of possible types of player i). $\Theta := \prod_{i \in N} \Theta_i$.
- $p_i \in \Delta^0(\Theta_i)$.
- $u_i : \mathcal{H} \times \Theta \to \mathbb{R}$

Remark: the type of player encapsulates all the information possessed by the player that is not common knowledge. This is often quite simple (e.g. the player's knowledge of his private payoff function), but can also include his beliefs about other players' payoffs, about their beliefs about his own payoff, and any other higher-order beliefs.

There is a way of capturing the common prior is to hypothesize a special agent c called "Nature" who makes probabilistic choices. The agent "nature"(c) selects the types of the players, who are subsequently fully cognizant at all points of all moves taken previously. We can associate with any such game an extensive game (with imperfect information and simultaneous moves) in which the set of histories is $\{\varnothing\} \cup (\mathcal{H} \times \Theta)$ and each information set of each player i takes the form

$$I(h, \theta_i) := \{(h, (\theta_i, \theta_{-i})) : \theta_{-i} \in \Theta_{-i}\}$$

for $i \in P(h)$ and $\theta_i \in \Theta_i$.

Let s be a profile of behavioral strategies in Γ. Define $O_h(s)$ to be the probability measure on the set of terminal histories of Γ generated by s given that the history h has occurred. Define $O((s_i;\rho_{-i}),\mu_{-i} \mid h)$ to be the probability measure on the set of terminal histories of Γ given that player i uses the strategy s_i in Γ, each type θ_j of each player j uses the strategy $\rho_j(\theta_j)$, the game has reached h, and the probability that i assigns to θ_{-i} is derived from $\mu_{-i}(h)$. That is, $O((s_i;\rho_{-i}),\mu_{-i} \mid h)$ is the compound lottery in which the probability of the lottery $O_h(s_i;(\rho_j(\theta_j))_{j\in N\setminus\{i\}})$ is $\prod_{j\in N\setminus\{i\}} \mu_j(h)(\theta_j)$ for each $\theta_{-i} \in \Theta_{-i}$.

Ex-ante Expected Utility

Definition (Ex-ante Expected Utility)

At history h, the player i's ex ante expected utility in a Bayesian game, where the players' behavioral strategy profile is ρ, is defined as

$$V_i(\rho, h) \coloneqq \sum_{\theta_i \in \Theta_i} \mu_i(h)(\theta_i) \sum_{\theta_{-i} \in \Theta_{-i}} u_i|_h(\rho, \theta) \prod_{j \in N \setminus \{i\}} \mu_j(h)(\theta_j)$$

Perfect Bayesian Equilibrium

Definition (Perfect Bayesian Equilibrium)

For a Bayesian extensive game with observable actions, $((\rho_i), (\mu_i)) \coloneqq ((\rho_i(\theta_i))_{\theta_i \in \Theta_i}, (\mu_i(h))_{h \in \mathcal{H} \setminus \mathcal{Z}})$ is a *perfect Bayesian equilibrium* of the game iff the following conditions are satisfied, where $\rho_i(\theta_i)$ is a behavioral strategy of player $i \in N$ and $\mu_i(h) \in \Delta\Theta_i$.

Perfect Bayesian Equilibrium — definition continued

(I) *Sequential rationality* For every terminal history $h \in \mathcal{H} \setminus \mathcal{Z}$, every player $i \in P(h)$, and every $\theta_i \in \Theta_i$ the probability measure $O((\rho_i(\theta_i); \rho_{-i}), \mu_{-i} \mid h)$ is at least good for type θ_i as $O((s_i; \rho_{-i}), \mu_{-i} | h)$ for any strategy s_i of player i in Γ.

$$\forall s_i \in S_i : O((\rho_i(\theta_i); \rho_{-i}), \mu_{-i} \mid h) \succcurlyeq_i O((s_i; \rho_{-i}), \mu_{-i} \mid h)$$

or equivalently, for $\rho_i' \in B_i$,

$$\sum_{\theta_{-i} \in \Theta_{-i}} u_i|_h((\rho_i; \rho_{-i}), \theta) \prod_{j \in N \setminus \{i\}} \mu_j(h)(\theta_j) \geq \sum_{\theta_{-i} \in \Theta_{-i}} u_i|_h((\rho_i'; \rho_{-i}), \theta) \prod_{j \in N \setminus \{i\}} \mu_j(h)(\theta_j)$$

(II) *Correct initial beliefs* $\forall i \in N : \mu_i(\varnothing) = p_i$

(III) *Action-determined beliefs* If $i \notin P(h)$ and $a \in A(h)$ then $\mu_i(ha) = \mu_i(h)$;
if $i \in P(h), a \in A(h), a' \in A(h)$ and $a = a'$ then $\mu_i(ha) = \mu_i(ha')$.

(IV) *Bayesian updating* If $i \in P(h)$ and $\exists \theta_i \in \operatorname{supp}(\mu_i(h))\,(a \in \operatorname{supp}(\rho_i(\theta_i)(h)))$ then for any $\theta' \in \Theta_i$ we have

$$\mu_i(ha)(\theta_i') := \frac{\rho_i(\theta_i')(h)(a) \cdot \mu_i(h)(\theta_i')}{\sum_{\theta_i \in \Theta_i} \rho_i(\theta_i)(h)(a) \cdot \mu_i(h)(\theta_i)} \qquad \text{(Bayesian Update)}$$

The condition of Bayesian updating relates to a case in which player i's action at the history h is consistent with the other players' beliefs about player i at h, given ρ_i. In such a case the condition requires not only that the new belief depend only on player i's action (as required by the condition of action-determined beliefs) but also that the players' beliefs be derived via Bayes' rule from their observation of player i's actions. Thus the players update their beliefs about player i using Bayes' rule until his behaviour contradicts his strategy ρ_i, at which point they form a new conjecture about player i's type that is the basis for future Bayesian updating until there is another conflict with ρ.

Theorem

Let $(\rho;\mu)$ be a sequential equilibrium of the extensive game associated with the finite Bayesian extensive game with observable actions. For every $h \in \mathcal{H}$, $i \in P(h)$, and $\theta_i \in \Theta_i$, let $\rho_i'(\theta_i)(h) := \rho_i(I(h,\theta_i))$. Then there is a collection $(\mu_i'(h))_{i\in N, h\in\mathcal{H}}$, where $\mu_i'(h) \in \Delta\Theta_i$, such that

$$\mu(I(h,\theta_i))(h,\theta) = \prod_{j\in N\setminus\{i\}} \mu_j'(h)(\theta_j) \quad \text{for all } \theta \in \Theta \text{ and } h \in \mathcal{H}$$

and $((\rho_i'),(\mu_i'))$ is a perfect Bayesian equilibrium of the Bayesian extensive game.

Trembling Hand Perfect Equilibrium

Definition (Trembling Hand Perfect Equilibrium)

A strategy profile σ is a *trembling hand perfect equilibrium* iff there exists a sequence of completely mixed strategy profiles $\sigma^k \to \sigma$ such that $\forall i \forall k : \sigma_i \in \mathrm{BR}_i(\sigma^k_{-i})$.

Theorem

A strategy profile in a finite two-player strategic game is a trembling hand perfect equilibrium iff it is a mixed strategy Nash equilibrium and the strategy of neither player is weakly dominated.

Theorem

Every finite strategic game has a trembling hand perfect equilibrium.

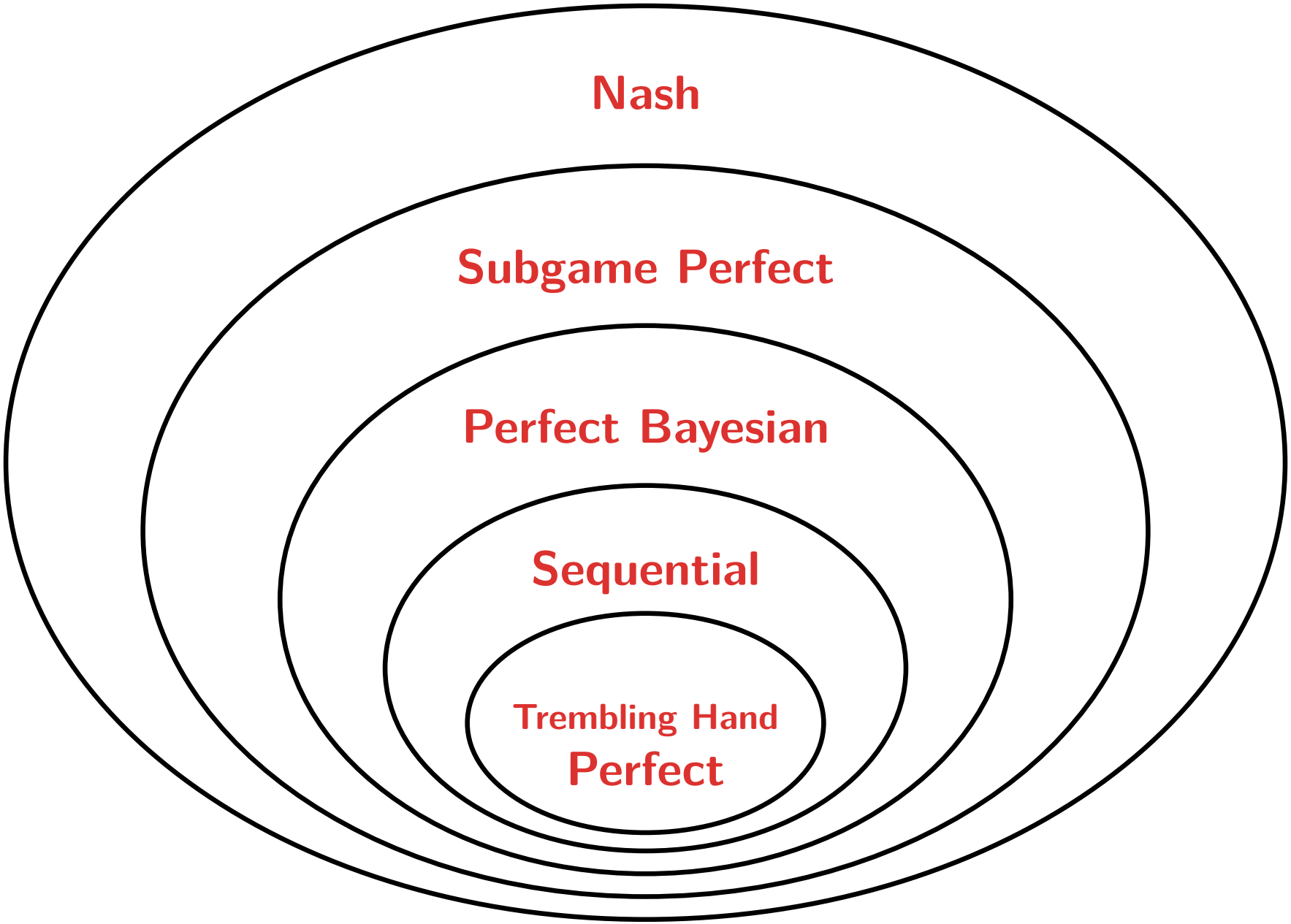
Definition

A trembling hand perfect equilibrium of a finite extensive game is a behavioral strategy profile that corresponds to a trembling hand perfect equilibrium of the agent strategic form of the game.

Theorem

For every trembling hand perfect equilibrium ρ of a finite extensive game with perfect recall there is a belief system μ such that $(\rho; \mu)$ is a sequential equilibrium of the game.

Equilibria


Nash
Subgame Perfect
Perfect Bayesian
Sequential
Trembling Hand
Perfect

Correlated Equilibrium

Definition (Nash Euqilibrium)

- s^* is a *pure Nash equilibrium* iff $s^* \in \prod_{i \in N} \operatorname*{argmax}_{s_i \in S_i} u_i(s_i; s_{-i})$.
- σ^* is a *mixed Nash equilibrium* iff

$$\forall i \in N \forall \sigma_i \in \Delta S_i : u_i(\sigma_i^*; \sigma_{-i}^*) \geq u_i(\sigma_i; \sigma_{-i}^*)$$

Definition (Correlated Equilibrium)

Let Ω be the state space, $\mathcal{I}_i$ be the information partition of player i, $P(\cdot \mid \mathcal{I}_i) \in \Delta\Omega$ be the interim belief systems, and $\sigma_i : \Omega \to S_i$ be measurable with regard to $\mathcal{I}_i$. Then $(\sigma_i)_{i \in N}$ is a posteriori equilibrium of the strategic game (N, S_i, u_i) iff

$$\forall i \in N \forall s_i \in S_i : \sum_{\omega \in \Omega} P\big(\omega \mid \mathcal{I}_i(\omega)\big)\Big(u_i\big(\sigma_i(\omega); \sigma_{-i}(\omega)\big) - u_i\big(s_i; \sigma_{-i}(\omega)\big)\Big) \geq 0$$

Remark: For every Nash equilibrium there exists a corresponding correlated equilibrium.

Choosing a Correlated Equilibrium

- An equilibrium which maximises the sum of the expected utility of the players is called a **utilitarian equilibrium**.
- An equilibrium which maximises the expected utility of Player i is called a **Libertarian i equilibrium**.
- An equilibrium which maximises the minimum expected utility of a player is called an **egalitarian equilibrium**.

Contents

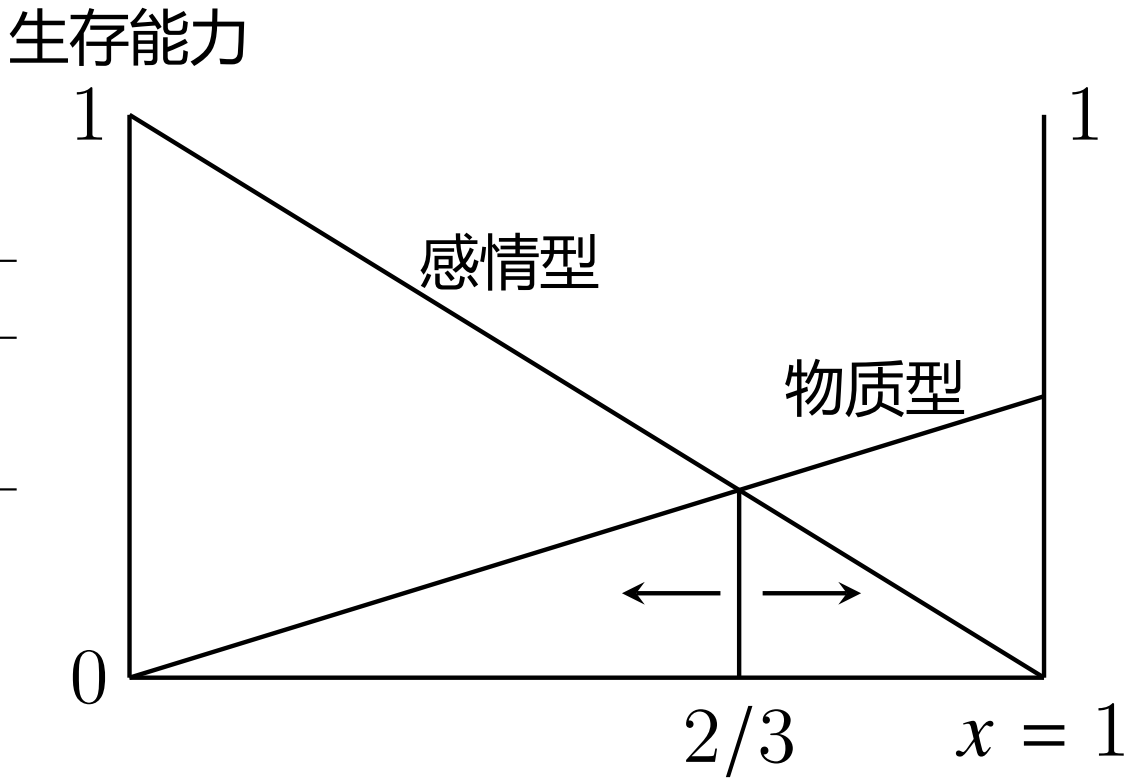
1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
   - Preferences and Expected Utility
   - Strategic Games
   - Extensive Games with Perfect Information
   - Eliminating Dominated Strategies
   - Dynamic Games
   - Repeated Games
   - Signaling Games
   - Mechanism Design
   - Counterfactual Regret Minimization
   - Subgame Perfect Equilibrium
   - Games with Incomplete Information
   - Evolutionary Games
   - Voting System
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

演化稳定策略 Evolutionarily Stable Strategy

- 静态视角：一个特定的行为模式被称为是演化稳定的，如果它的种群不能被变异所入侵，或者说，任何偏离行为模式的个体具有更低的生存能力，种群将会恢复到原来的状态.
- 动态视角：假定初始状态存在多样的行为模式，随着时间的推移，如果某个特定的行为模式能逐步主导整个种群，这个特定的行为模式就是演化稳定的.

演化博弈 — 婚姻博弈

	物质型	感情型
物质型	$1,1$	$0,0$
感情型	$0,0$	$2,2$



- 假定总人口中, 物质型的比例为 x, 感情型的比例为 $1-x$.
- 对任何一个个体而言, 物质型的期望效用: $x1+(1-x)0=x$.
- 感情型的期望效用: $x0+(1-x)2=2(1-x)$.
- 如果 $x>2/3$, 物质型更适合生存, 将演化成稳定均衡.
- 如果 $x<2/3$, 感情型更适合生存, 将演化成稳定均衡.
- 如果 $x=2/3$, 两类人有同样的适应性, 但此二元均衡是非稳定的.

演化博弈 — 鹰鸽博弈

	鹰	鸽
鹰	$-1,-1$	$1,0$
鸽	$0,1$	$\frac{1}{2},\frac{1}{2}$

- 假定鹰派的比例是 x, 鸽派的比例是 $1-x$.
- 鹰派的效用: $-1x+1(1-x)=1-2x$.
- 鸽派的效用: $0x+\frac{1}{2}(1-x)=\frac{1}{2}(1-x)$.
- 如果 $x<1/3$, 鹰派占优势, 不稳定.
- 如果 $x>1/3$, 鸽派占优势, 不稳定.
- 如果 $x=1/3$, 同样的适应性, 稳定.
- 如果初始人口由单一类型构成, 另一类型可以成功入侵, 直到均衡.

† 两个纯策略均衡: (鹰、鸽), (鸽、鹰); 一个混合均衡: $(1/3,2/3)$

Δ 假定存在某种显性的标记机制: 在博弈开始之前, 每个人收到一个信号: A 或 B; 概率是 $1/2$; 信号完全负相关; 标记是公共知识.

1. 如果 A, 选择 "鹰"; 如果 B, 选择 "鸽". 是 ESS.
2. 如果 A, 选择 "鸽"; 如果 B, 选择 "鹰". 是 ESS.
3. 无论 AB, 以 $1/3$ 的概率选择 "鹰", $2/3$ 的概率选择 "鸽". 不是 ESS.

自发秩序与产权制度

- 社会秩序是所有人行为选择的结果，但不是集体理性集中设计的结果，而是自发演化的结果.
- 社会规范是演化稳定策略，但不一定是帕累托最优的.
- 虽然不是集体理性集中设计的产物，但 “制度企业家” 扮演了重要角色. (哲学王)
- 改变游戏规则，进行制度创新. 创新意味着让人们用新的价值观念代替旧的价值观念、用新的行为方式代替旧的行为方式、用新的是非观和新的善恶观代替旧的是非观和旧的善恶观，意味着我们要认同原来可能不认同的东西或不再认同我们原来认同的东西.
  — 儒家: “仁、义、礼、智、信”, “己所不欲勿施于人”, “以德报德，以直报怨”, “君君臣臣父父子子”.
- 制度企业家面临的不确定性和风险挑战异于常人. 他们不能以 “利” 和 “名” 为目的，买他们账的 “客户” 可能在遥远的未来.
- “自私的” 基因 & 迷因 (meme)

Evolutionarily Stable Strategy

Given a symmetric two-player normal form game, σ^* is an ESS iff $\forall \sigma \neq \sigma^* \exists \delta \in (0,1) \forall \varepsilon \in (0,\delta)$:

$$u\left(\sigma^*, (1-\varepsilon)\sigma^* + \varepsilon\sigma\right) > u\left(\sigma, (1-\varepsilon)\sigma^* + \varepsilon\sigma\right)$$

iff

$$(1-\varepsilon)u\left(\sigma^*, \sigma^*\right) + \varepsilon u\left(\sigma^*, \sigma\right) > (1-\varepsilon)u\left(\sigma, \sigma^*\right) + \varepsilon u\left(\sigma, \sigma\right)$$

iff

- $u\left(\sigma^*, \sigma^*\right) > u\left(\sigma, \sigma^*\right)$ or
- $u\left(\sigma^*, \sigma^*\right) = u\left(\sigma, \sigma^*\right)$ and $u\left(\sigma^*, \sigma\right) > u\left(\sigma, \sigma\right)$

If σ is an ESS, then (σ, σ) is a Nash equilibrium. If (σ, σ) is a strict Nash equilibrium, then σ is an ESS.

	dove	hawk
Dove	$\frac{a}{2}, \frac{a}{2}$	$0, a$
Hawk	$a, 0$	$\frac{a-b}{2}, \frac{a-b}{2}$

$$\frac{a}{2}x + 0(1-x) = ax + \frac{a-b}{2}(1-x)$$

$$b > a \implies (1 - \frac{a}{b}, \frac{a}{b}) \text{ mix}$$

$$b \leq a \implies (H, h) \text{ pure}$$

Evolutionarily Stable Strategy

- strict Nash Equilibrium: $u(\sigma_i^*; \sigma_{-i}^*) > u(\sigma_i; \sigma_{-i}^*)$
- Nash Equilibrium: $u(\sigma_i^*; \sigma_{-i}^*) \geq u(\sigma_i; \sigma_{-i}^*)$
- ESS:

  - $u(\sigma^*, \sigma^*) > u(\sigma, \sigma^*)$ or
  - $u(\sigma^*, \sigma^*) = u(\sigma, \sigma^*)$ and $u(\sigma^*, \sigma) > u(\sigma, \sigma)$

- weak ESS:

  - $u(\sigma^*, \sigma^*) > u(\sigma, \sigma^*)$ or
  - $u(\sigma^*, \sigma^*) = u(\sigma, \sigma^*)$ and $u(\sigma^*, \sigma) \geq u(\sigma, \sigma)$

- unbeatable strategy: $u(\sigma^*, \sigma^*) > u(\sigma, \sigma^*)$ and $u(\sigma^*, \sigma) > u(\sigma, \sigma)$

  unbeatable $\Longrightarrow$ strict Nash $\Longrightarrow$ ESS $\Longrightarrow$ weak ESS $\Longrightarrow$ Nash

Replicator Dynamics in Symmetric Games

- Suppose players choose one of m actions. The payoff of a player when he plays a_i and the opponent plays a_j is $u(a_i; a_j)$, where $u(a_i; a_j) \geq 0$.
- It is assumed that individuals use pure strategies.
- Let $p_{i,n}$ be the proportion of individuals using action i in generation n.
- The average reward of an individual using action i in generation n is
$$\overline{R}_{i,n} := u(a_i; p_{1,n}a_1 + \cdots + p_{m,n}a_m)$$
- The average reward in the population as a whole in generation n is
$$\overline{R}_n := \sum_{i=1}^{m} p_{i,n}\overline{R}_{i,n}$$
- The proportion of individuals using action i in generation $n+1$ is
$$p_{i,n+1} := \frac{p_{i,n}\overline{R}_{i,n}}{\overline{R}_n}$$
- A fixpoint $(p_1, \ldots, p_m)$ of the replicator dynamic equations satisfies
$$p_i = \frac{p_i\overline{R}_i}{\overline{R}} \quad \text{for } i = 1, \ldots, m$$

Example

	A	B
A	5,5	0,0
B	0,0	1,1

- Let p_n be the proportion of individuals using A in generation n.
- The average reward of A players in generation n is
$$\overline{R}_{A,n} = u(A; p_n A + (1 - p_n)B) = 5p_n$$
- The average reward of B players in generation n is
$$\overline{R}_{B,n} = u(B; p_n A + (1 - p_n)B) = 1 - p_n$$
- The average reward of the population is
$$\overline{R}_n = p_n \overline{R}_{A,n} + (1 - p_n)\overline{R}_{B,n} = 1 - 2p_n + 6p_n^2$$
- The equation governing the replicator dynamics is
$$p_{n+1} = \frac{p_n \overline{R}_{A,n}}{\overline{R}_n} = \frac{5p_n^2}{1 - 2p_n + 6p_n^2}$$
- A fixpoint of these dynamics satisfies $p = \frac{5p^2}{1-2p+6p^2}$. $p = 0, 1, \frac{1}{6}$.
- $p = 0, 1$ is an attractor. $p = \frac{1}{6}$ is not an attractor.

Fixpoints of the Replicator Dynamic Equations

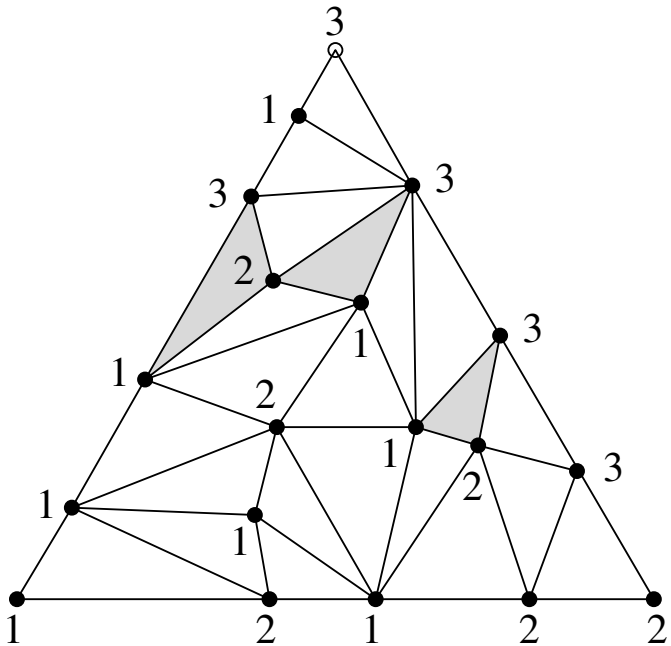
- A fixed point σ^* is *stable* (also called Lyapunov stable) iff for all open neighborhoods U of σ^* there is another open neighborhood $O \subset U$ such that any trajectory initially inside O remains inside U.
- A fixed point σ^* is *attractive* iff there exists an open neighborhood U of σ^* such that all trajectory initially in U converges to σ^*. The maximum possible U is called the basin of attraction of σ^*.
- A fixed point σ^* is *asymptotically stable* (also called attractor) iff it is stable and attractive.
- A fixed point is *globally asymptotically stable* iff its basin of attraction encompasses the whole space.

- Strict Nash Equilibria are attractors.
- If a fixpoint is stable then it is an Nash Equilibrium.
- ESSs are attractors.
- For 2×2 matrix games a fixed point is an ESS iff it is an attractor.

Sperner's Lemma

Lemma (Sperner's Lemma)

Suppose that some triangle with vertices V_1, V_2, V_3 is triangulated.
The vertices in the triangulation get "colors" from $\{1, 2, 3\}$ s.t. vertices on the edge (V_i, V_j) are colored either i or j, while the interior vertices are colored 1, 2 or 3.
Then in the triangulation there must be an odd number of "tricolored" triangles.

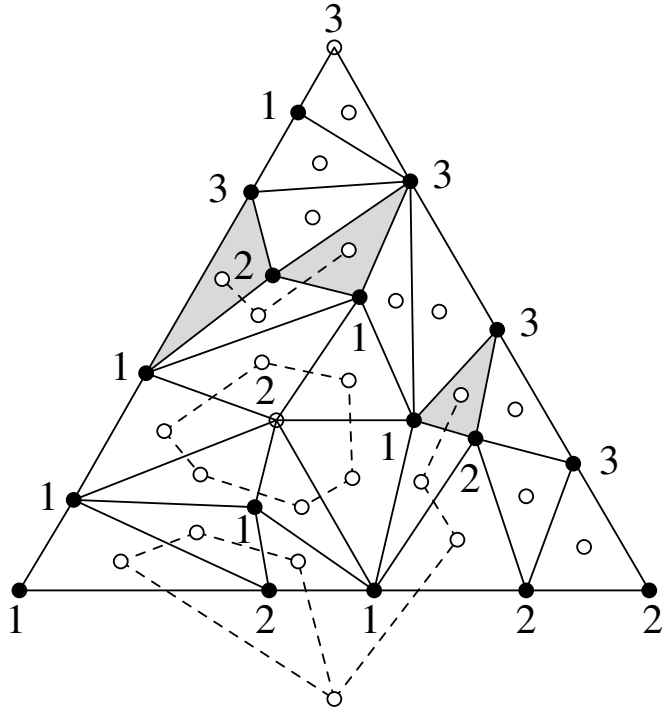


Proof of Sperner's Lemma

Proof.

Consider the dual graph to the triangulation — but take only those which cross an edge that has endvertices with the colors 1 and 2. Thus we get a partial dual graph which has degree 1 at all vertices that correspond to tricolored triangles, degree 2 for all triangles in which the two colors 1 and 2 appear, and degree 0 for triangles that do not have both colors 1 and 2.

The vertex of the dual graph which corresponds to the outside of the triangulation has odd degree: along the big edge from V_1 to V_2, there is an odd number of changes between 1 and 2. Since the number of odd-degree vertices in any finite graph is even, the number of tricolored triangles is odd.



Brouwer Fixpoint Theorem

Theorem (Brouwer Fixpoint Theorem)

Given a non-empty compact convex set $X \subset \mathbb{R}^n$ and continuous function $f : X \to X$, there exists $\mathbf{x} \in X$ s.t. $f(\mathbf{x}) = \mathbf{x}$.

Theorem (Kakutani Fixpoint Theorem)

Given a non-empty compact convex set $X \subset \mathbb{R}^n$ and a multi-valued function $f : X \rightrightarrows X$, if

- *for all $x \in X$, the set $f(x)$ is convex,*
- *the graph of f is closed (i.e. for all sequences $\{x_n\}$ and $\{y_n\}$ s.t. $x_n \to x$, $y_n \to y$ and $y_n \in f(x_n)$, we have $y \in f(x)$),*

then $\exists x \in f(x)$.

Theorem (Schauder Fixpoint Theorem)

If K is a non-empty convex subset of a Hausdorff topological vector space V and T is a continuous mapping of K into itself such that $T(K)$ is contained in a compact subset of K, then T has a fixpoint.

Proof of Brouwer Fixpoint Theorem

Proof.

Let Δ be the triangle in $\mathbb{R}^3$ with vertices $\mathbf{e}_1 = (1,0,0)$, $\mathbf{e}_2 = (0,1,0)$, and $\mathbf{e}_3 = (0,0,1)$. We prove that any continuous map $f : \Delta \to \Delta$ has a fixpoint.
Let $\delta(T)$ be the maximal length of an edge in a triangulation T.
One can construct an infinite sequence of triangulations $T_1, T_2, \ldots$ of Δ s.t. $\lim_{k\to\infty} \delta(T_k) = 0$.
Suppose f has no fixpoint. Since $\sum_i \mathbf{v}_i = 1 = \sum_i f(\mathbf{v})_i$, for each of these triangulations, we can define a Sperner coloring of their vertices $\mathbf{v}$ by setting $\lambda(\mathbf{v}) := \min\{i : f(\mathbf{v})_i < \mathbf{v}_i\}$.
Sperner's lemma tells us that in each triangulation T_k there is a tricolored triangle $\{\mathbf{v}_1^k, \mathbf{v}_2^k, \mathbf{v}_3^k\}$ with $\lambda(\mathbf{v}_i^k) = i$.
Since the simplex Δ is compact, some subsequence of $\left(\mathbf{v}_1^k\right)_{k\geq 1}$ has a limit point $\mathbf{v}^* \in \Delta$. Since $\lim_{k\to\infty} \delta(T_k) = 0$, the sequences $\mathbf{v}_2^k$ and $\mathbf{v}_3^k$ converge to the same point $\mathbf{v}^*$.
Then $\forall i : f(\mathbf{v}^*)_i \leq \mathbf{v}_i^*$, which contradicts $f(\mathbf{v}^*) \neq \mathbf{v}^*$.

Theorem (Existence of Mixed Nash Equilibrium)

Every finite strategic game has a mixed Nash equilibrium.

Proof.

Given a strategy profile $\sigma \in \prod_{i \in N} \Delta S_i$, define

$$\varphi_{i,s_i}(\sigma) \coloneqq \max\left\{0, u_i(s_i; \sigma_{-i}) - u_i(\sigma)\right\}$$

Then define a continuous $f : \prod_{i \in N} \Delta S_i \to \prod_{i \in N} \Delta S_i$ by $f : \sigma \mapsto \sigma'$, where

$$\sigma_i'(s_i) \coloneqq \frac{\sigma_i(s_i) + \varphi_{i,s_i}(\sigma)}{\sum_{s_i \in S_i} \left[\sigma_i(s_i) + \varphi_{i,s_i}(\sigma)\right]} = \frac{\sigma_i(s_i) + \varphi_{i,s_i}(\sigma)}{1 + \sum_{s_i \in S_i} \varphi_{i,s_i}(\sigma)}$$

Since $\prod_{i \in N} \Delta S_i$ is convex and compact, f has a fixpoint.

Consider any fixpoint σ of f. By the linearity of expectation there exists s_i' in the support of σ, for which $u_i(s_i'; \sigma_{-i}) \leq u_i(\sigma)$. Then

$$\varphi_{i,s_i'}(\sigma) = 0 \;\;\&\;\; \sigma_i'(s_i') = \sigma_i(s_i') \implies \forall i \in N \forall s_i \in S_i : \varphi_{i,s_i}(\sigma) = 0$$

Walrasian Equilibrium

Theorem (Existence of Walrasian Equilibrium)

Consider an economy with n goods $X_1, \ldots, X_n$ with a price vector $(p_1, \ldots, p_n) \in \Delta_n := \{x \in [0,1]^n : \|x\|_1 = 1\}$, and the prices of at least two goods are not zero. Assume that an excess demand function for each good $f_i(p_1, \ldots, p_n)$ is continuous and satisfies the following condition

$$\sum_{i=1}^{n} p_i f_i = 0 \qquad \text{(Walras Law)}$$

Then, there exists an equilibrium price vector $(p_1^*, \ldots, p_n^*)$ s.t.

$$f_i(p_1^*, \ldots, p_n^*) \leq 0$$

for all $i = 1, \ldots, n$. And when $p_i > 0$ we have $f_i(p_1^*, \ldots, p_n^*) = 0$.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
   - Preferences and Expected Utility
   - Strategic Games
   - Extensive Games with Perfect Information
   - Eliminating Dominated Strategies
   - Dynamic Games
   - Repeated Games
   - Signaling Games
   - Mechanism Design
   - Counterfactual Regret Minimization
   - Subgame Perfect Equilibrium
   - Games with Incomplete Information
   - Evolutionary Games
   - Voting System
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

群体聚合 Group Aggregation

提案:

A. 大学教育应该免费.
B. 人人都应该有上大学的权利.
C. 政府应该增加在大学教育上的投入.

$$\frac{A \quad B}{C}$$

民意调查:

	A	B	C
张三	1	1	1
李四	0	1	0
王五	1	0	0
群体聚合	1	1	1/0

个体理性: 直接对提案 C 进行投票, 个体自我一致.

集体理性: 先对理由 A, B 分别投票, 再推导结论 C.

Remark: 通过安排投票方式, 可以操纵选举结果.

孔多塞投票悖论

张三	$a \succ b \succ c$
李四	$b \succ c \succ a$
王五	$c \succ a \succ b$

$\xrightarrow{\text{多数原则}}$ $a \succ b \succ c \succ a$

Remark: 如果你可以制定投票规则, 那么, 你想让谁胜出, 你就可以通过 "两两 PK" 的方式让其胜出.

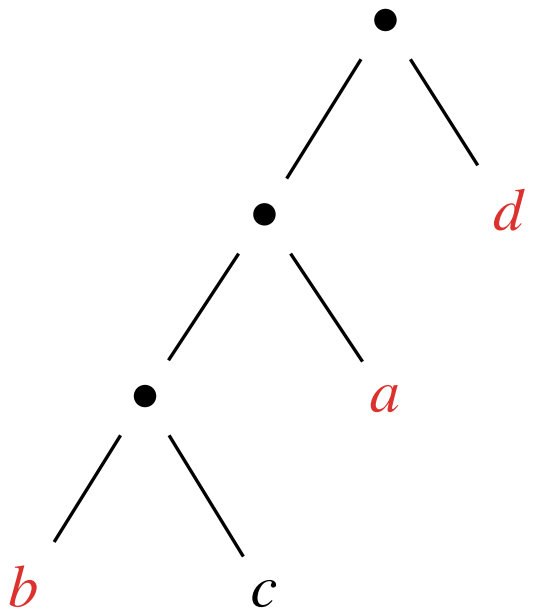
	$a \succ b$	$b \succ c$	$a \succ c$
张三	1	1	1
李四	0	1	0
王五	1	0	0
群体聚合	1	1	1/0

孔多塞投票的群体聚合

序贯多数投票

张三	$a \succ b \succ c \succ d$
李四	$b \succ c \succ d \succ a$
王五	$c \succ d \succ a \succ b$

- 所有人都认为 $c \succ d$
- d 还能胜出吗?



投票制度

1. **相对多数法 (plurality method):** 选民每人投一个候选人
2. **双选投票法 (vote-for-two method):** 选民每人投两个候选人
3. **反相对多数法 (anti-plurality method):** 选民每人投 $n-1$ 个候选人, 即投票剔除一个候选人
4. **绩点评分法 (GPA method):** 选民对所有 n 个候选人进行排序
   - 第一名得 $n-1$ 分
   - 第二名得 $n-2$ 分
   - 倒数第二名得 1 分
   - 最后一名 0 分

   GPA 得分最高者当选.
5. **排序选择法 (ranked-choice voting):** 选民对所有 n 个候选人进行排序
   - 5.1 首轮计票时, 只计算选民的首选票. 如果有候选人获得过半数选票, 该候选人当选.
   - 5.2 如果没有候选人过半数, 就将票数最少的候选人淘汰, 把选择该候选人为首选的选票, 按次选票的候选人重新分配.
   - 5.3 重复第 2 步, 直到某个候选人获得过半数选票为止.

选举结果反映的究竟是民意还是选举方法的选择?

2 个选民	$a \succ b \succ c \succ d$
2 个选民	$a \succ d \succ c \succ b$
2 个选民	$c \succ b \succ d \succ a$
3 个选民	$d \succ b \succ c \succ a$

Table: 9 个选民, 4 个候选人

- 根据 “相对多数法”, a 赢: $a4 \succ d3 \succ c2 \succ b0$
- 根据 “双选投票法”, b 赢: $b7 \succ d5 \succ a4 \succ c2$
- 根据 “反相对多数法”, c 赢: $c9 \succ b7 = d7 \succ a4$
- 根据 “绩点评分法”, d 赢: $d15 \succ b14 \succ c13 \succ a12$

当有候选人退出时会怎样?

3 个选民	$a \succ c \succ d \succ b$
6 个选民	$a \succ d \succ c \succ b$
3 个选民	$b \succ c \succ d \succ a$
5 个选民	$b \succ d \succ c \succ a$
2 个选民	$c \succ b \succ d \succ a$
5 个选民	$c \succ d \succ b \succ a$
2 个选民	$d \succ b \succ c \succ a$
4 个选民	$d \succ c \succ b \succ a$

Table: 30 个选民, 4 个候选人

- 根据 "相对多数法", a 赢: $a9 \succ b8 \succ c7 \succ d6$
- 如果 d 退出: $c11 \succ b10 \succ a9$
- 如果 c 退出: $d11 \succ b10 \succ a9$
- 如果 b 退出: $d11 \succ c10 \succ a9$
- 如果 a 退出: $d12 \succ c10 \succ b8$
- 根据 "绩点评分法", d 赢: $d58 \succ c54 \succ b41 \succ a27$
- 根据 "排序选择法", c 赢: $c20 \succ b10$

Arrow's Impossibility Theorem

Theorem (Arrow's Impossibility theorem)

In an election with candidates ≥ 3, any voting system that is unanimous and independence of irrelevant alternatives must be a dictatorship!

- unanimous: if everyone agrees, consensus decides the outcome
- independence of irrelevant alternatives: a voter can't move a above b by lying about how they feel about c
- nondictatorship: no single voter gets to decide the outcome

Arrow's Impossibility Theorem

Let N be a set of voters, and C a set of candidates. A social welfare function (SWF) is $f : \mathcal{S}_C^N \to \mathcal{S}_C$, where $\mathcal{S}_C$ is the set of all permutations on C. We write $a \succ_i b$ to indicate that voter $i \in N$ ranks a above b. Given $\sigma \in \mathcal{S}_C^N$, $N_{a \succ b}^{\sigma} := \{i \in N : a \succ_i b \text{ under } \sigma\}$.

- Unanimity ($\mathbf{U}$): If all voters rank a above b, then so does society: $N_{a \succ b}^{\sigma} = N \implies a \succ_{f(\sigma)} b$.
- Independence of Irrelevant Alternatives ($\mathbf{IIA}$): the relative social ranking of two candidates only depends on their relative individual rankings: $N_{a \succ b}^{\sigma} = N_{a \succ b}^{\sigma'} \implies \left(a \succ_{f(\sigma)} b \iff a \succ_{f(\sigma')} b\right)$.
- Nondictatorship ($\mathbf{ND}$): There is no $i \in N$ s.t. $\sigma_i = f(\sigma)$.

Theorem (Arrow's Impossibility Theorem)

If N is finite and $|C| \geq 3$, then any SWF that satisfy $\mathbf{U}$ and $\mathbf{IIA}$ must be a dictatorship.

Proof Sketch of Arrow's Impossibility Theorem

1. We call a subset $A \subset N$ decisive iff whenever all $x \in A$ present the same ranking, the SWF f outputs that ranking.
2. The set of decisive sets of voters $\mathcal{F} := \{A \subset N : A \text{ is decisive}\}$ is an ultrafilter.
3. If N is finite, then the ultrafilter $\mathcal{F}$ must be a principle ultrafilter.

Let $\mathcal{F}$ be an ultrafilter on N. We can define a SWF f by declaring the output to be that unique permutation σ with the property that $\{i \in N : \sigma_i = \sigma\} \in \mathcal{F}$.

Theorem (Arrow's Theorem)

Assume $|C| \geq 3$. There is a $1-1$ correspondence between ultrafilters on N and SWF that satisfy $\mathbf{U}$ and $\mathbf{IIA}$. The non-dictatorship SWFs are those corresponding to non-principle ultrafilters. In particular, Arrow's impossibility theorem is equivalent to the assertion that all ultrafilters on a finite set are principle.

对无关选项独立性 IIA 的质疑

排序	张三	李四	王五	赵六
1	x	x	a	a
2	a	a	b	b
3	b	b	y	y
4	y	y	x	x

排序	张三	李四	王五	赵六
1	a	a	y	y
2	b	b	a	a
3	x	x	b	b
4	y	y	x	x

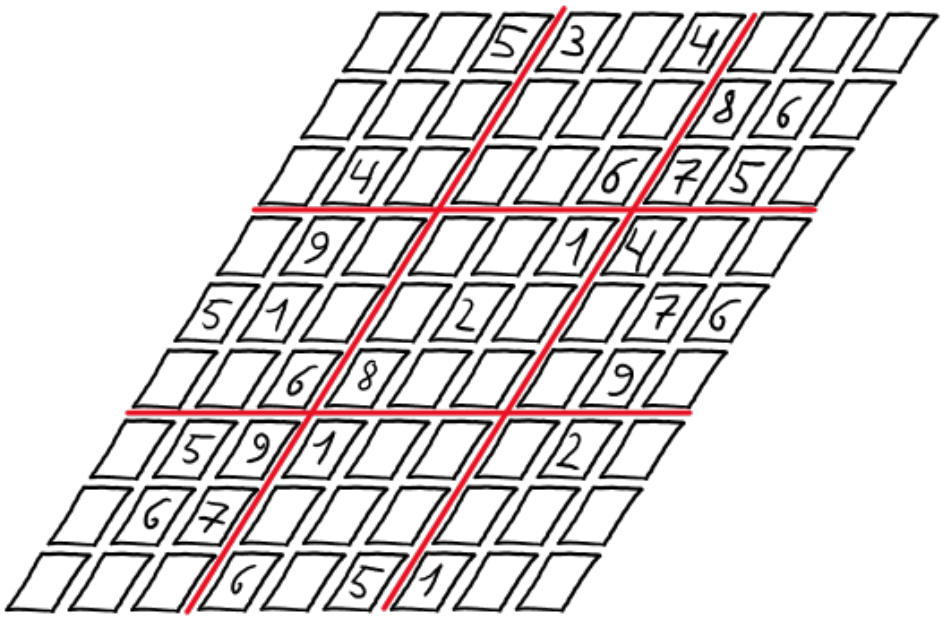
- 在两个表中, x 和 y 的相对排序是一样的. 张三和李四偏好 $x \succ y$, 王五和赵六偏好 $y \succ x$.
- 直观上, 表 1 中偏好 $x \succ y$ 的人比偏好 $y \succ x$ 的人的 "感受更为强烈"; 表 2 则相反. 我们可能会说, 表 1 中 $x \succ y$ 是有道理的, 而表 2 中 $y \succ x$ 是有道理的.
- **IIA** 要求两表中社会对 x 和 y 的偏好一样: 要么 $x \succ y$, 要么 $y \succ x$.

社会选择理论 Social Choice Theory

- 计算社会选择理论：用计算复杂性防止坏情况的发生
- 用逻辑做投票协议验证
  - 隐私性：没有别人能知道你投的是谁
  - 无收据性：你不能证明给别人你投了特定人的票
  - 可核查性：你自己能检查你的票是不是被算进去了
  - 公平性：之前投票的部分结果不会影响之后投票的结果

零知识证明 Zero-Knowledge Proof

- 小帅给小美出了一道非常难的数独题.
- 小美怎么也解不出来, 怀疑小帅在耍她: “这道题无解!”
- 小帅说: “我会用零知识证明的方法给你证明这题有解. 我不会把解给你看, 却能让你信服我确实有这题的解.”



提示: 准备九个袋子 ☺

Prover vs Verifier

- **Completeness:** P can convince V if X is true.
- **Soundness:** no (efficient) P can convince V if X is not true.
- **Zero Knowledge:** no efficient V learns anything more than the validity of X.

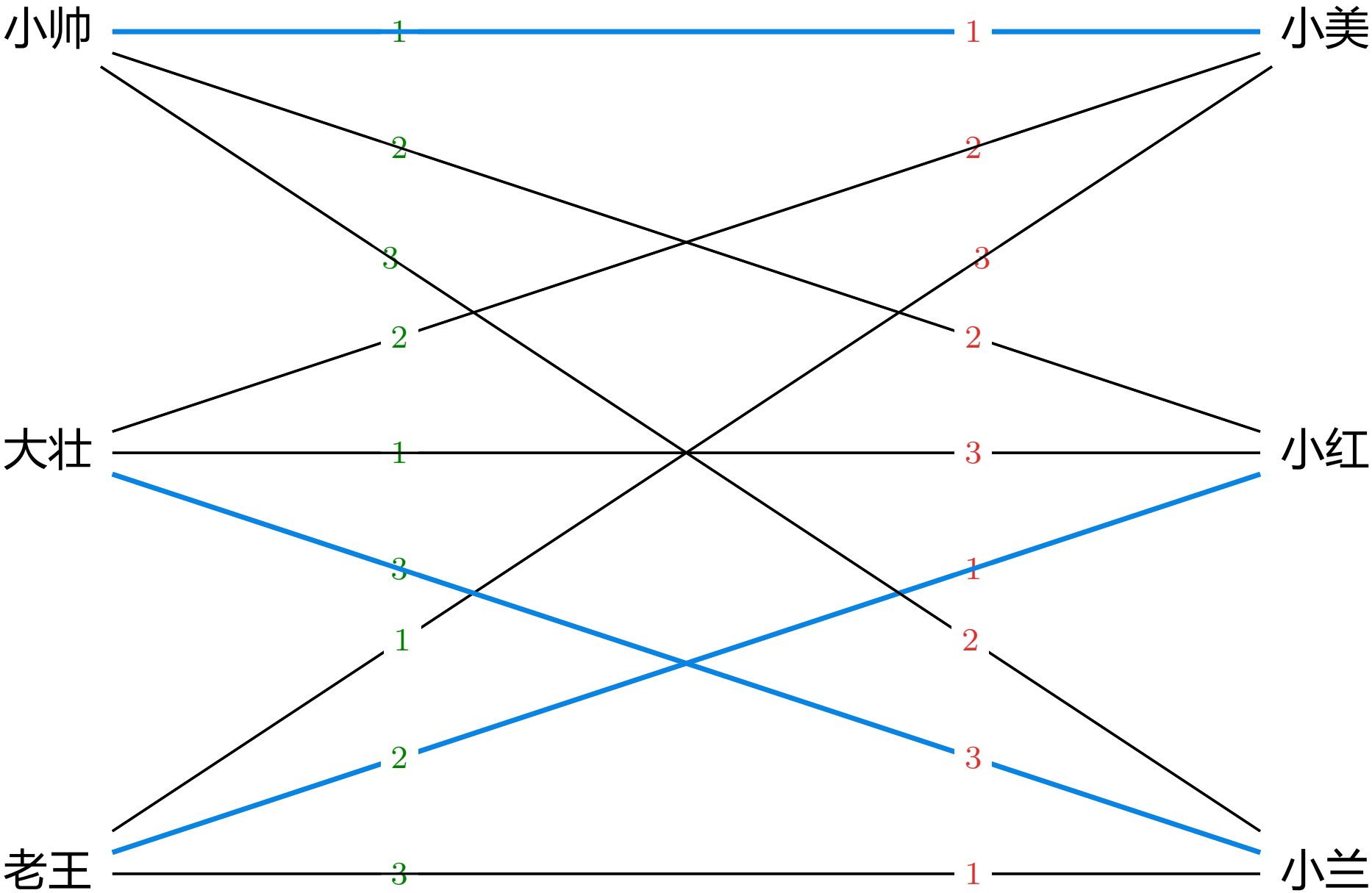
Remark: The adversary can simulate the proof without knowing the prover's witness.

Theorem

Every NP statement can be proven in zero-knowledge.

稳定婚姻问题

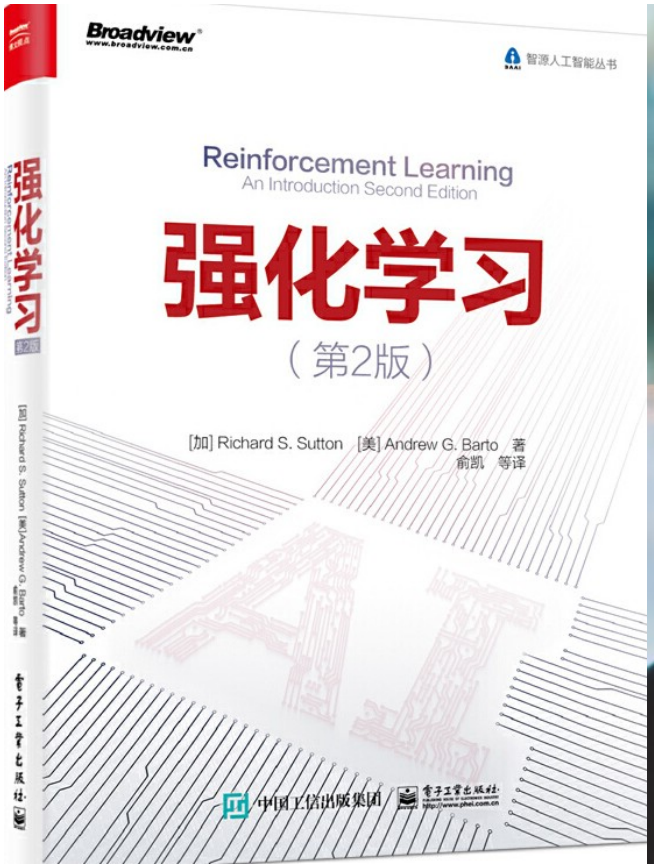
- n 男 n 女, 每人对异性都有偏好排序.
- 一个匹配是不稳定的, 如果有一对男女, 相对于自己的现任配偶, 更中意彼此.
- 稳定匹配是否存在?
- 怎么找到稳定匹配?
  - Day 0: 每人对异性进行排序.
  - Day 1: (a.m.) 每个男生向他最中意的女生表白.
    (p.m.) 每个女生选择自己最中意的求爱对象, 拒绝其他男生.
  - Day $n+1$: (a.m.) 依然单身的男生向还没拒绝过他的最中意的女生表白, 不管对方是否已有男朋友.
    (p.m.) 每个女生选择自己最中意的求爱对象, 拒绝/抛弃其他男生.
  - 直到所有人都有对象为止.
- 在男生主动表白的稳定匹配中, 每个男生都获得了他所能获得的最佳配偶!
- 每个女生都获得了她所能获得的最差配偶!


小帅
大壮
老王
小美
小红
小兰
1
2
3
2
1
3
1
2
3
1
2
3
1
2
3
1
2
3

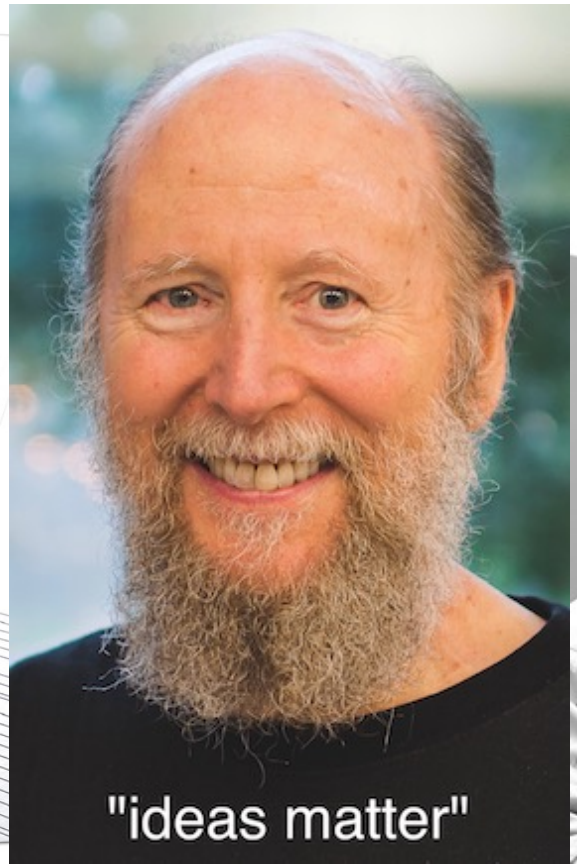
Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Reinforcement Learning



(a) Reinforcement Learning



(b) Richard S. Sutton

(c) Andrew G. Barto

快乐 = 即时奖励 ≠ 善 = 长期奖励

"一切技艺与探究，一切行动与追求，都以某种善为目的."

— 亚里士多德

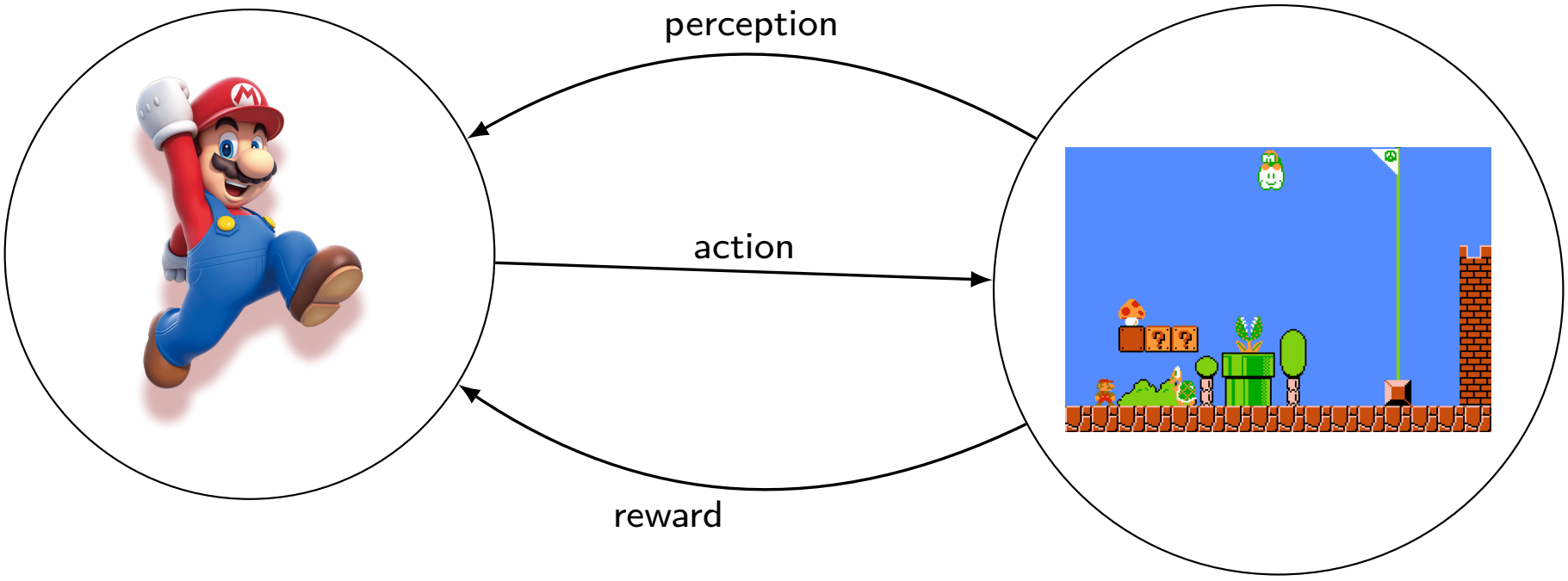
"即使是享受，如果它会导致失去更大的快乐，或造成更大的痛苦，你也会称之为恶.…当我们谈到痛苦时，情况不也一样吗？当遭受痛苦能让我们摆脱更大的痛苦，或能带来更大的快乐时，你就称之为善."

— 柏拉图

价值函数 = 奖励预测值 = 未来 (快乐 − 痛苦) 之和

- **The reward hypothesis：** 所有目标都可以表示为预期累积奖励的最大化.
- **The reward-is-enough hypothesis：** 智能及其相关能力，可被视为 Agent 在环境交互中旨在实现最大化奖励的体现.

Reinforcement Learning

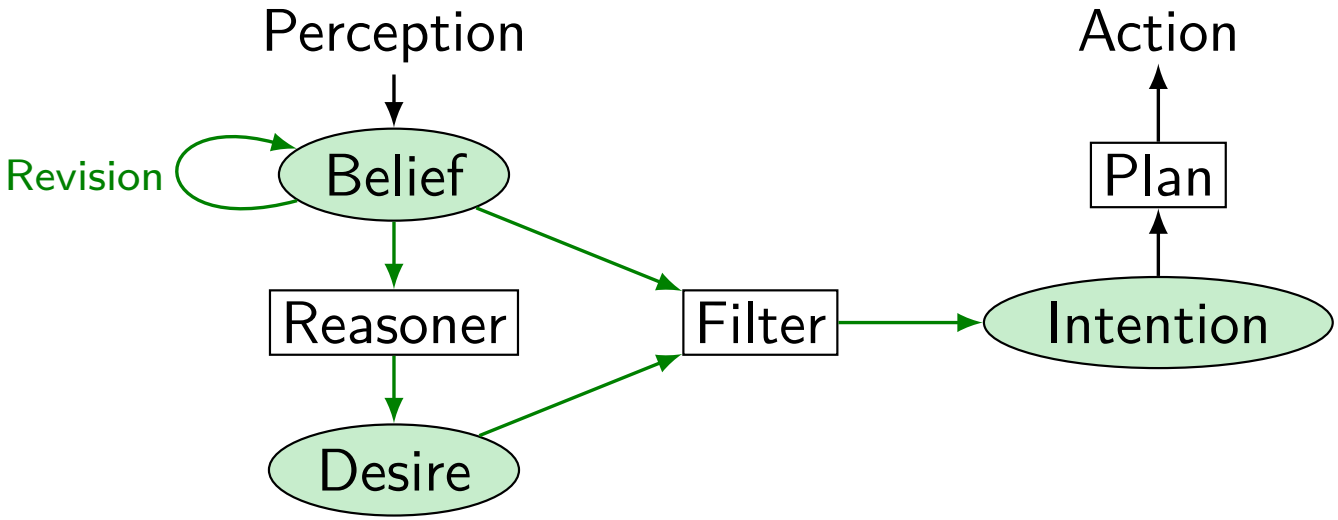


Goal: Maximize reward!

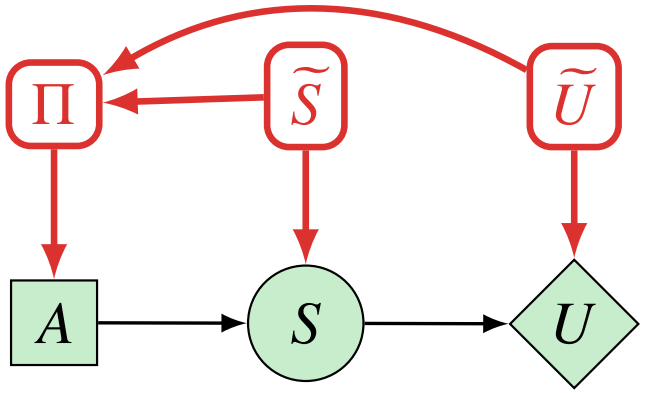
von Neumann–Morgenstern: 如果偏好关系 $\succ$ 满足完备性、传递性、连续性和独立性, 则存在一个函数 u 使得 $a \succ b$ 当且仅当 $u(a) > u(b)$.[25]

[25]如果你质疑效用理论, 请质疑偏好关系的合理性, 而不是说一句 "人不是冰冷地追求效用最大化".

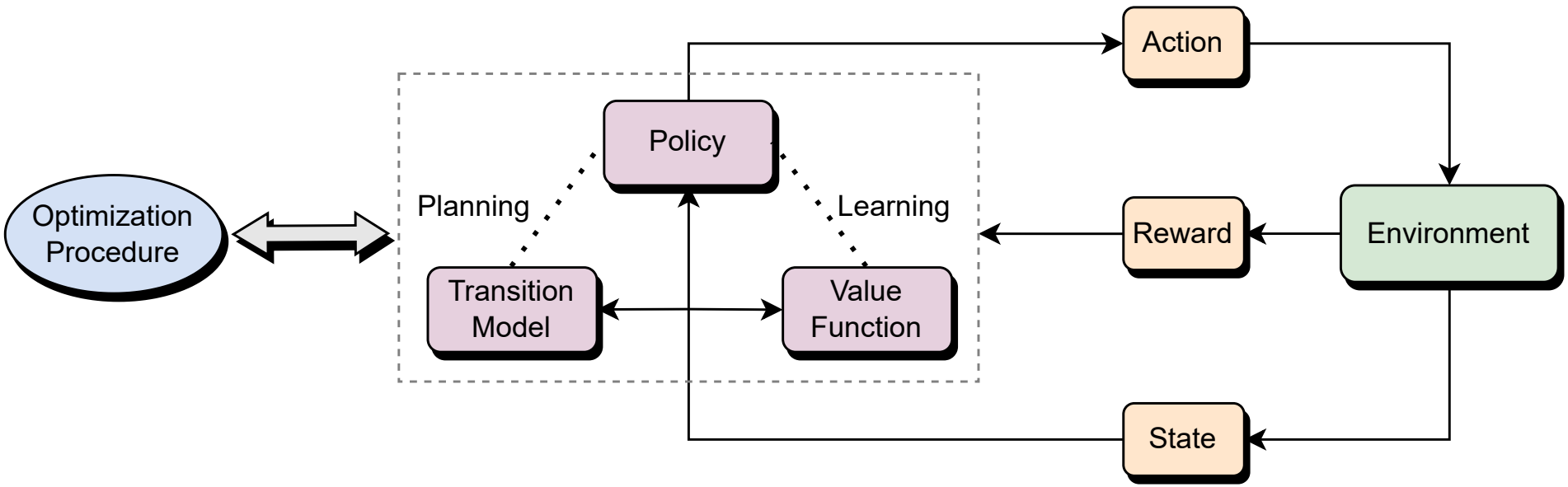
哲学家 Bratman 的 BDI 模型

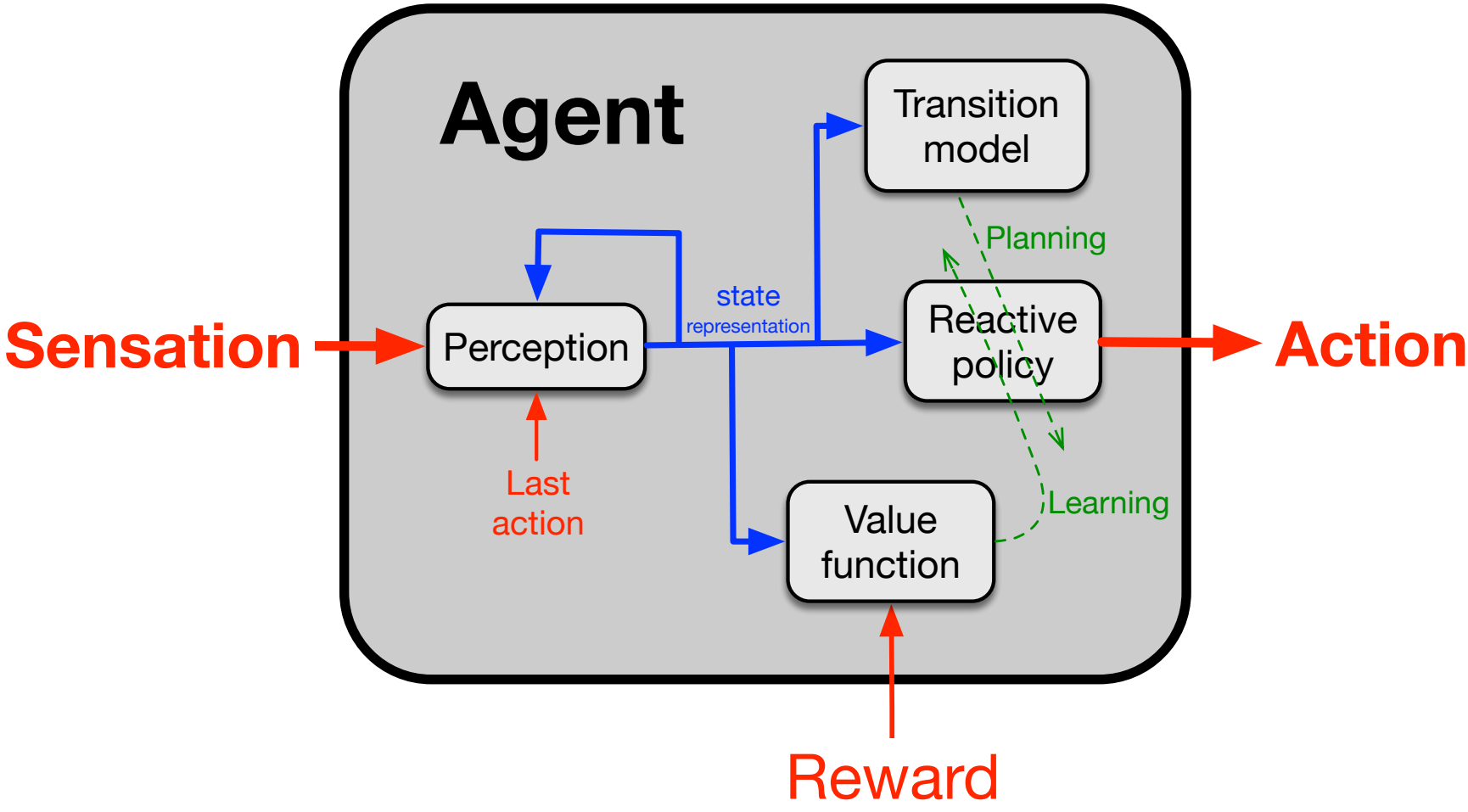


- **Beliefs** about itself, other agents and its environment $\pi(a \mid s), \pi'(a \mid s), P(s' \mid s, \mathrm{do}(a))$
- **Desires** about future states (goals) $U(s)$
- **Intentions** about its own future plan $\mathrm{argmax}_\pi \mathbb{E}[U \mid \mathrm{do}(\pi)]$



Reinforcement Learning Agent





1. Perception produces the state-feature vector
2. Reactive Policy quickly produces an action appropriate to the state
3. Value Function evaluates how well things are going, and changes the policy (learning)
4. Transition model predicts the consequences of alternate actions, and changes the policy (planning)

Will intelligence ultimately be explained in

Objective terms?

- states of the external world
- objects, people, places, relationships, atoms
- space, motion, distances
- things outside the agent

or

Experiential terms?

- sensations
- actions
- reward
- time steps
- things inside the agent

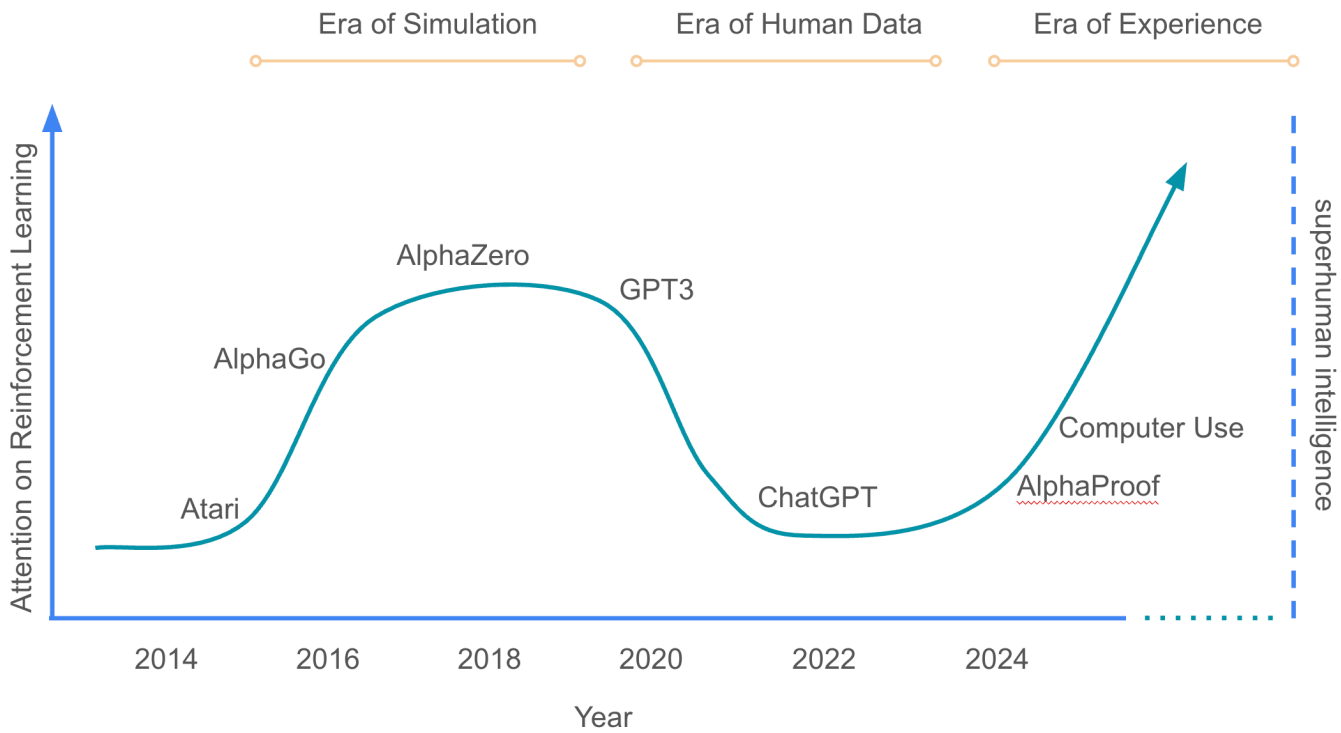
从 “人类中心 Agent” 到 “经验 Agent”

- RL Agent 没有实时监督，只有延迟的奖励信号. (长期奖励 = Good)
- 价值函数将状态映射到对未来奖励的预测. (为了消除延迟)
- Agent 只能通过传感器获取关于世界的信息，只能通过行动影响世界.
- 经验是 Agent 接触世界的唯一途径.
- Agent 通过经验试错学习.
- 客观状态 vs 经验状态 — 完全以经验定义的世界状态.
- 经验状态是对过去经验的总结，用于预测和控制未来经验.
- Agent 的奖励函数应学自其在环境中的交互经验，而非人类预设的偏好.
- Agent 应基于其自身经验进行规划/推理，而非套用人类的表征和推理框架.
- 经验知识是关于世界的状态和状态转换的. 知道就是预测经验.
- 思维就是在想象的经验中学习. 一切都关乎经验.
- 利弊: 超越人类上限. 行动策略和奖励函数都可适应性调整.
  — 安全对齐更难，可信任性可解释性更差.

The Era of Experience

"What we want is a machine that can learn from experience."

— Alan Turing 1947



1. 模拟时代: 封闭环境, 清晰奖励
2. 人类数据时代: 大规模人类数据领域知识驱动的学习, 通用性, 对齐人
3. 经验时代: Agent 自己与环境交互的经验数据驱动的通用搜索和学习, 达成目标, 超越人

Sutton: 苦涩的教训

苦涩的教训基于以下历史观察:

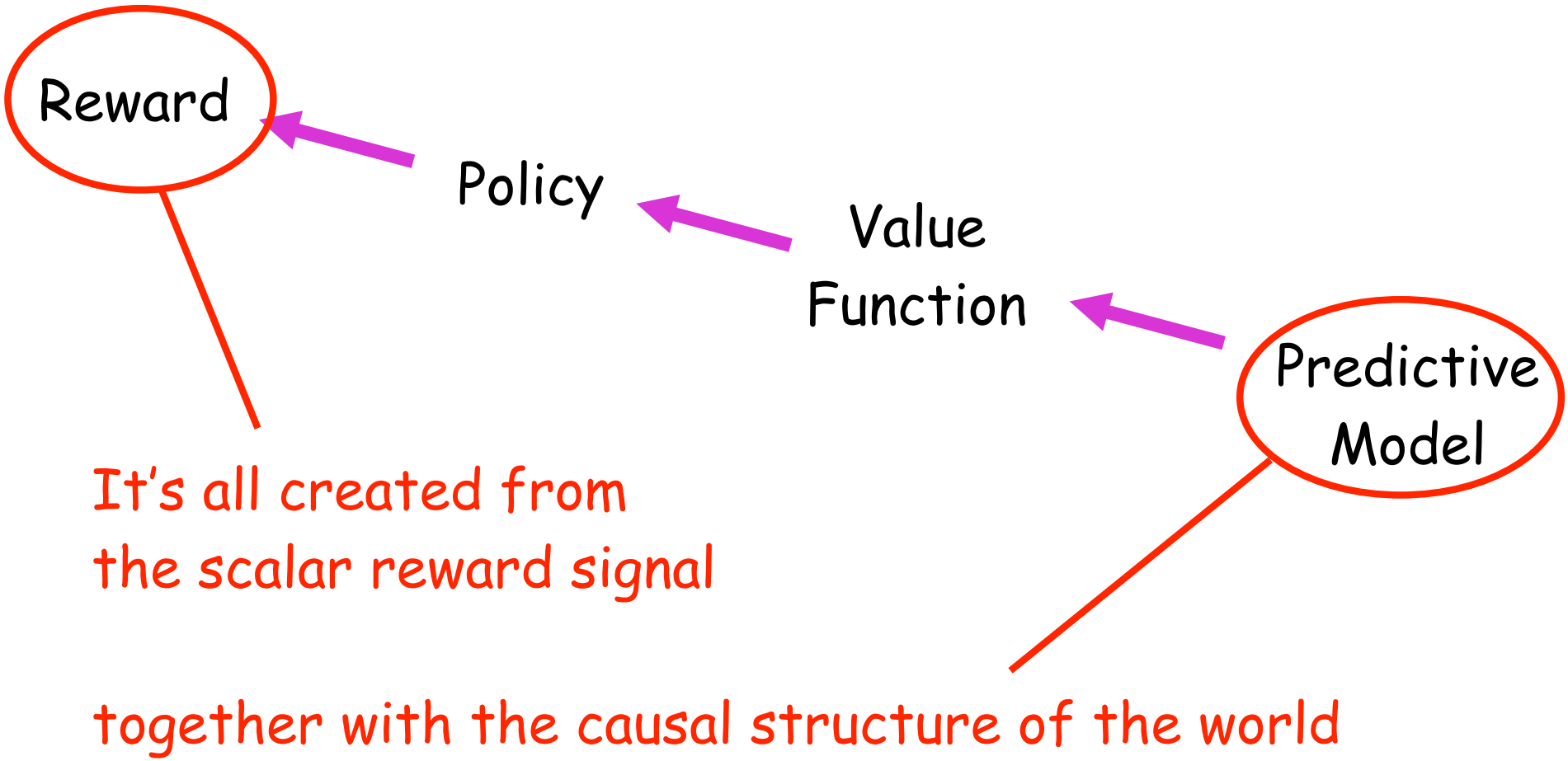
1. AI 研究人员总是试图将知识构建进 Agent 中;
2. 在短期内, 这确实有用, 且能给研究人员带来个人满足感;
3. 但从长远来看, 它总会达到一个瓶颈, 甚至阻碍进一步的发展;
4. 最终的突破是通过搜索和学习来扩展计算规模实现的.

最终的成功夹杂着苦涩, 因为它战胜了人们偏爱的以人为本的方法.
首先, 我们应该认识到通用方法的强大. 那些随着计算能力的增加能够不断扩展的方法才是真正值得研究的. 目前来看, 搜索和学习是唯二能随计算能力增长而任意扩展的方法.
其次, 心灵的内容是极其复杂的. 我们应该放弃那些试图用简单方法来描述心灵内容的尝试, 比如对空间、物体、多智能体、对称性的简单思考. 所有这些都是复杂外部世界的一部分, 不应该被直接构建到 AI 系统中. 相反, 应该只构建那些能够发现和捕捉这种复杂性的元方法. 这些元方法的关键在于它们能找到好的近似解, 但这个搜索过程应该由 AI 自己完成, 而不是由人类预设.
我们希望 AI 能像人类一样去发现, 而不是集成人类已有的发现. 如果我们将已有的发现植入 AI 中, 反而更难理解发现过程本身是如何进行的.

讨论问题

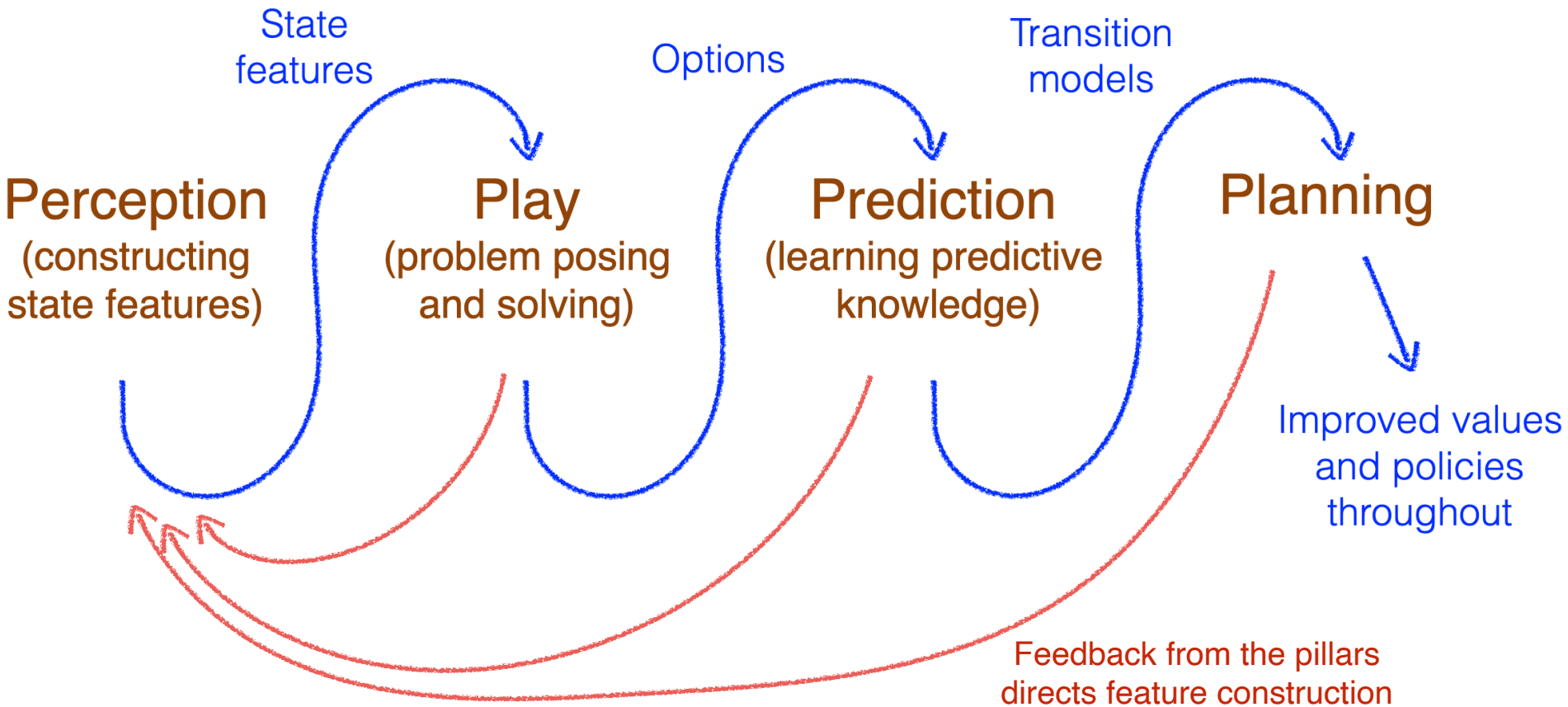
你觉得哪些知识是人类的智慧？哪些是人类的自大？

RL's Computational Theory of Mind



Remark: 我们对世界、空间、物体以及他人的所有认知, 都源于我们通过感知和行动这一狭窄通道所获得的主观经验.

RL's Computational Theory of Mind



1. Perception: learns state-update function $s_t = \Phi(s_{t-1}, a_{t-1}, o_{t-1})$.
2. Play: poses and solves many subproblems/subtasks to attain features, outputs policies π with termination conditions that attain the features.
3. Prediction: learn a cause-and-effect transition model P.
4. Planning: improves value functions V and policies π.

Much of Mind is about Prediction

- Perception and State Representation can be thought of as making predictions
- Models the world and cause and effect can be thought of in terms of predictions
- Planning can be thought of as composing predictions to anticipate possible futures, and then choosing among them
- Learning Value Functions is learning long-term predictions

To Know is to Predict Experience

All Knowledge about the world can be translated into predictions

- of what will happen
- of what you could cause to happen
  — at various time scales
  — conditional on actions or courses of action

Predictive Knowledge should be

1. Useful — for action and planning
2. Expressive — able to express abstract, high-level facts as well as specific, low-level facts
3. Autonomously verifiable
4. Learnable — from low-level sensorimotor data

Definition (Markov Decision Process MDP[SB18])

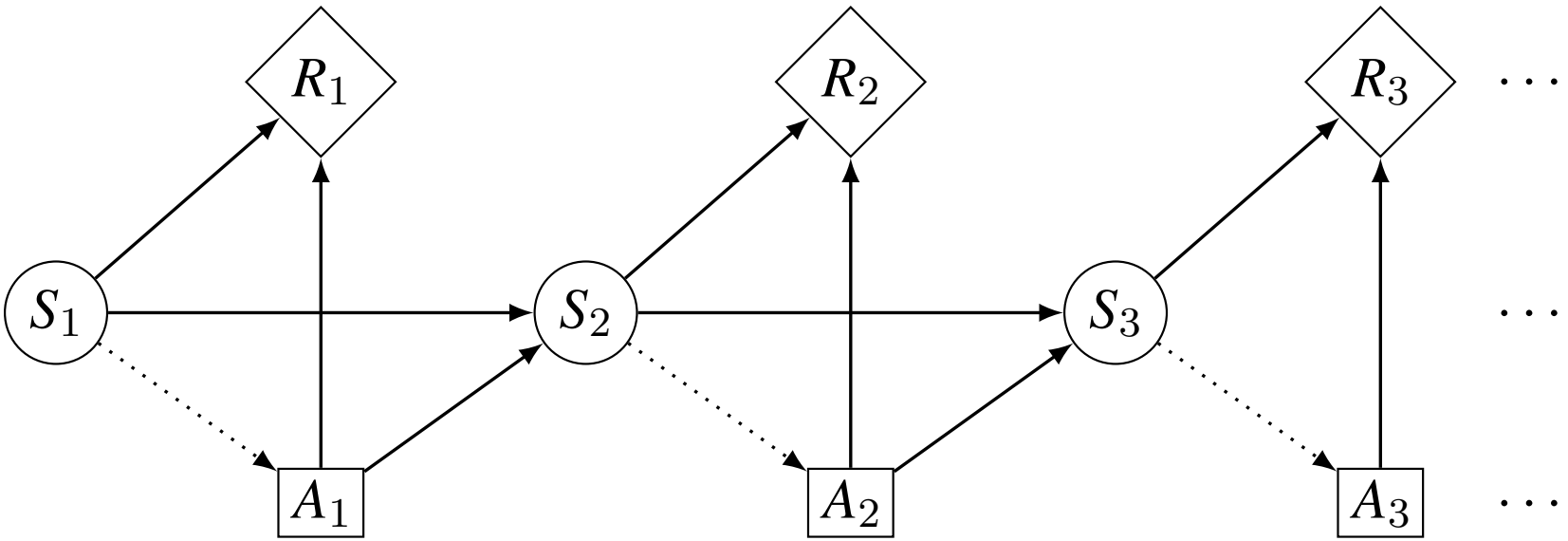
An MDP for an accessible, stochastic environment is defined by

- Set of states $\mathcal{S}$
- Set of actions $\mathcal{A}$
- Set of rewards $\mathcal{R}$
- Transition model $P(s', r \mid s, a)$, with $s, s' \in \mathcal{S}$, $a \in \mathcal{A}$ and $r \in \mathcal{R}$
- Reward function $r : \mathcal{S} \times \mathcal{A} \to \mathcal{R}$

$$r(s,a) := \sum_{r \in \mathcal{R}} r \sum_{s' \in \mathcal{S}} P(s', r \mid s, a)$$

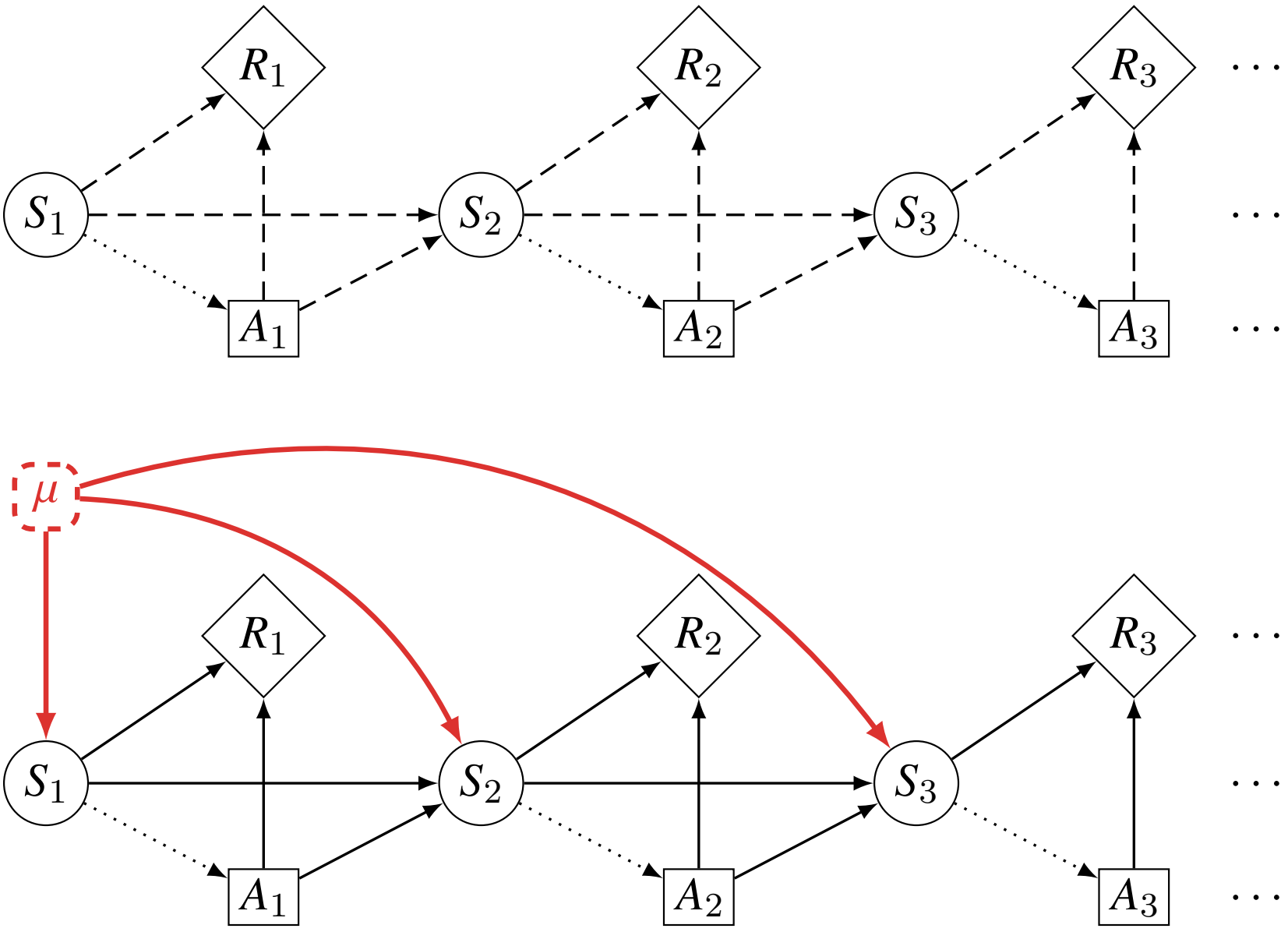
- Transition model: $P(s' \mid s, a) := \sum_{r \in \mathcal{R}} P(s', r \mid s, a)$ is the probability that state s' is reached, if action a is executed in state s.
- Policy: Complete mapping π that specifies for each state $s \in \mathcal{S}$ which action $\pi(s) \in \mathcal{A}$ to take.
- Wanted: The optimal policy π^* is the policy that maximizes the future expected reward.

Known MDP

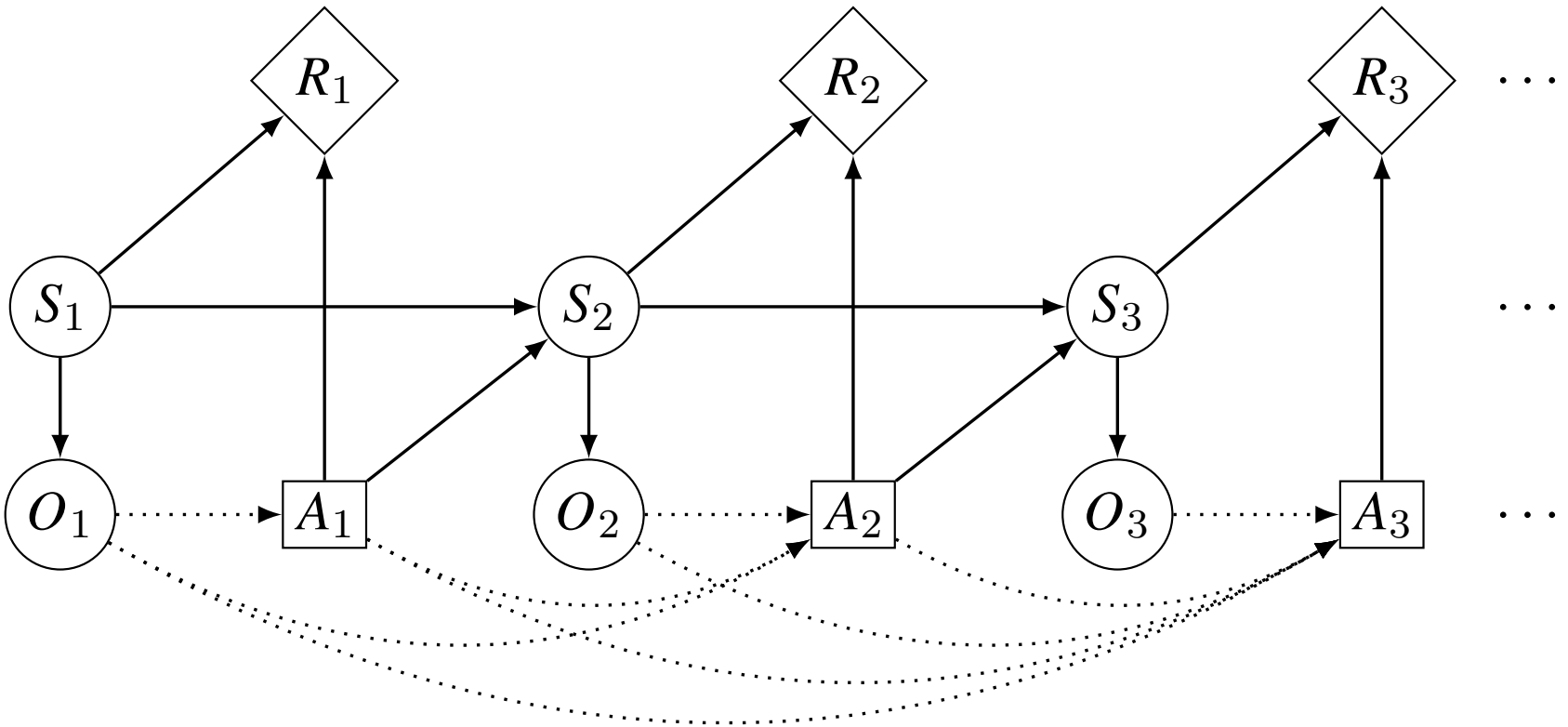


状态转移	$s_{t+1} = f_s(s_t, a_t, \varepsilon_{s_{t+1}})$	$\sim P(s_{t+1} \mid s_t, a_t)$
动作	$a_t = f_a(s_t, \varepsilon_{a_t})$	$\sim \pi(a_t \mid s_t)$
奖励	$r_t = f_r(s_t, a_t, \varepsilon_{r_t})$	$\sim r(s_t, a_t)$

Two Representations of an Unknown MDP

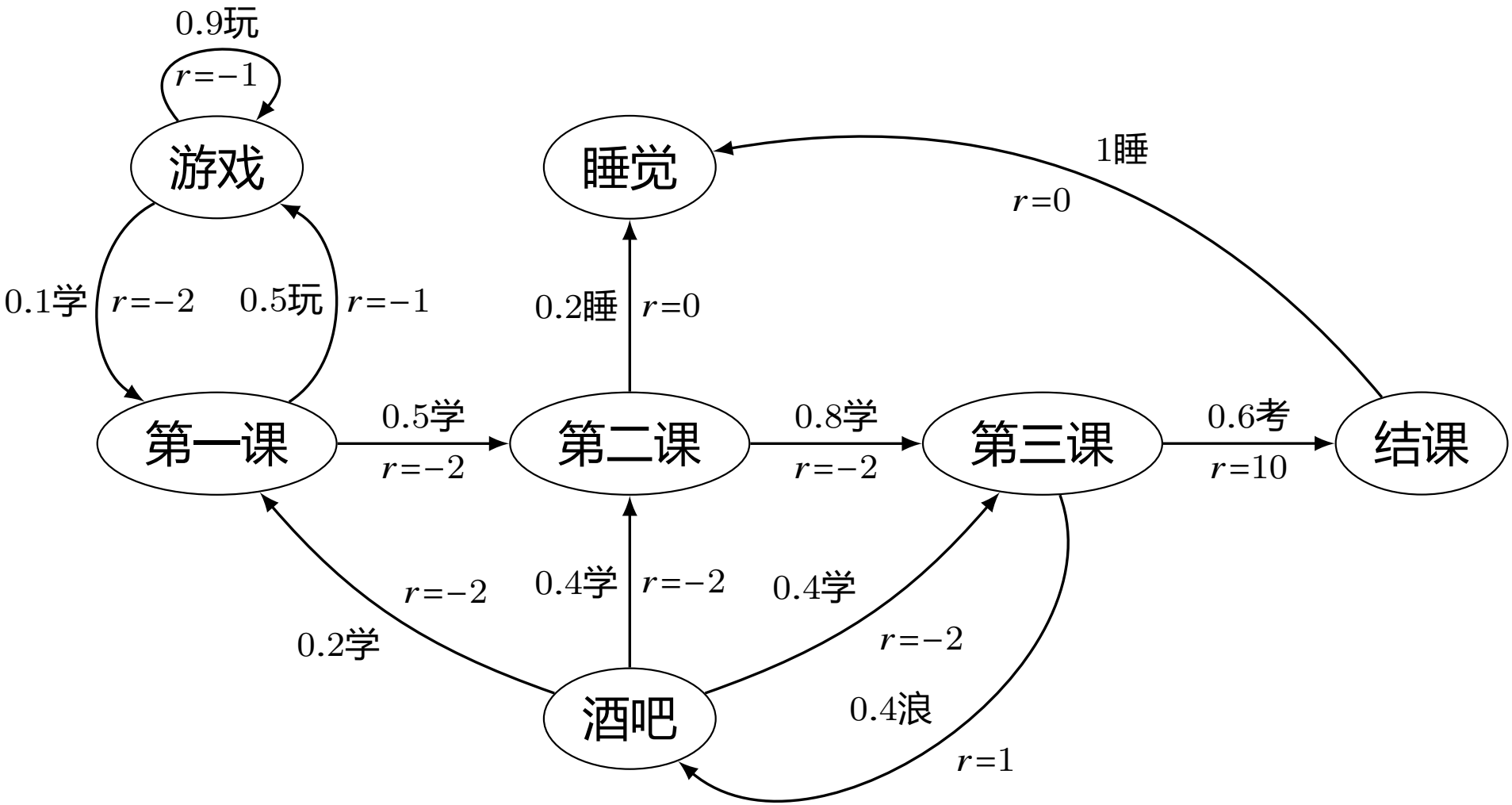


POMDP

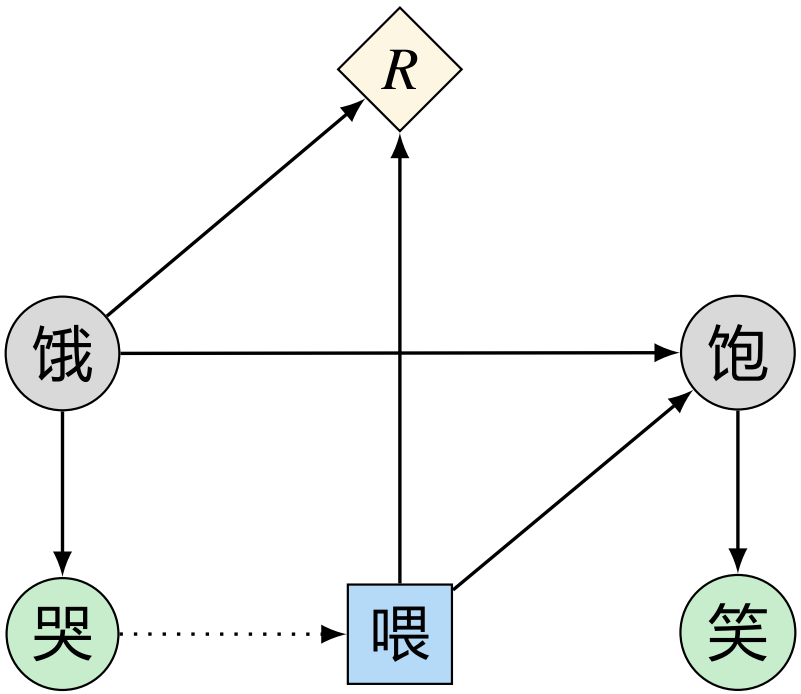


状态转移	$s_{t+1} = f_s(s_t, a_t, \varepsilon_{s_{t+1}})$	$\sim P(s_{t+1} \mid s_t, a_t)$
感知	$o_t = f_o(s_t, \varepsilon_{o_t})$	$\sim P(o_t \mid s_t)$
动作	$a_t = f_a(h_{<t}, \varepsilon_{a_t})$	$\sim \pi(a_t \mid h_{<t})$
奖励	$r_t = f_r(s_t, a_t, \varepsilon_{r_t})$	$\sim r(s_t, a_t)$

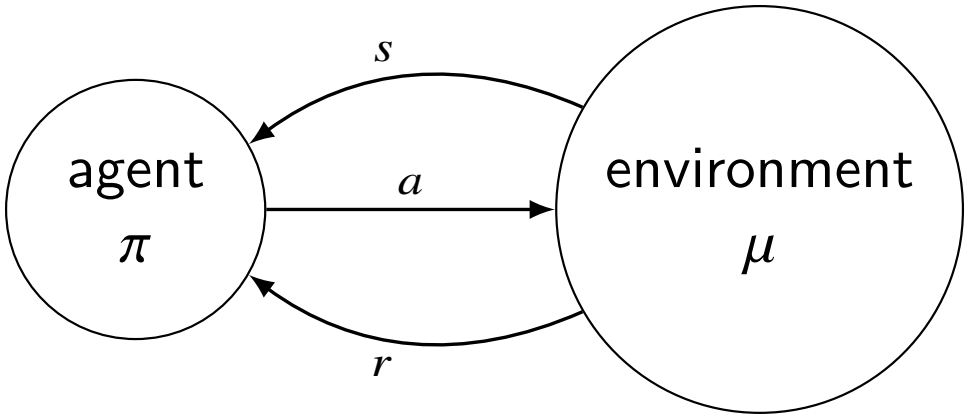
Example: MDP


0.9玩
r=−1
游戏
睡觉
1睡
r=0
0.1学
r=−2
0.5玩
r=−1
0.2睡
r=0
第一课
0.5学
r=−2
第二课
0.8学
r=−2
第三课
0.6考
r=10
结课
r=−2
0.2学
0.4学
r=−2
0.4学
r=−2
酒吧
0.4浪
r=1

Example: POMDP



Value Function



Definition (Value of a state under π)

$$V^{\pi}(s) := \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \,\middle|\, S_t = s\right]$$

Definition (Action-value under π)

$$Q^{\pi}(s,a) := \mathbb{E}\left[\sum_{k=0}^{\infty} \gamma^k r_{t+k+1} \,\middle|\, S_t = s, A_t = a\right]$$

Sutton: "Ethics is just values held in common by many agents."

什么是“恶”?

世上只有一种善，那就是知识；
也只有一种恶，那就是无知.

— 苏格拉底

如果你把自己的贪欲或怨愤置于社会规范之上，那就是恶.

— 朱迪亚·珀尔

From MDP to Reinforcement Learning

- Markov decision process (offline)
  - Have mental model of how the world works.
  - Find policy to collect maximum rewards.
- Reinforcement learning (online)
  - Don't know how the world works.
  - Perform actions in the world to find out and collect rewards.
- On-policy: estimate the value of data-generating policy
- Off-policy: estimate the value of another policy

Bellman Expectation Equations

$$V^\pi(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) Q^\pi(s,a)$$

$$Q^\pi(s,a) = r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) V^\pi(s')$$

Bellman Expectation Equations

$$V^\pi(s) = \mathbb{E}\left[r + \gamma V^\pi(s') \mid s\right] = \sum_{a \in \mathcal{A}} \pi(a \mid s) \left(r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) V^\pi(s') \right)$$

$$Q^\pi(s,a) = r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) \sum_{a' \in \mathcal{A}} \pi(a' \mid s') Q^\pi(s',a')$$

advantage: $A^\pi(s,a) := Q^\pi(s,a) - V^\pi(s)$

Optimal Value & Optimal Policy

Definition (Optimal Value)

$$V^*(s) := \max_\pi V^\pi(s)$$

$$Q^*(s,a) := \max_\pi Q^\pi(s,a)$$

Definition (Optimal Policy)

A policy π is called optimal iff $\forall s \in \mathcal{S} : V^\pi(s) = V^*(s)$.
— or equivalently, $Q^\pi(s,a) = Q^*(s,a)$.

$$\pi^*(s) = \operatorname*{argmax}_a Q^*(s,a)$$

Remark:

- We maximize reward by finding and following an optimal policy π^*.
- To find π^* we need to first find the optimal value function Q^*.
- To find Q^* we need to repeatedly find the value function for a policy Q^π that is our current best guess at the optimal policy.
- To find Q^π we may need to learn a transition model $P(s', r \mid s, a)$.

Bellman Optimality Equations

$$V^*(s) = \max_{a \in \mathcal{A}} Q^*(s,a)$$

$$Q^*(s,a) = r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) V^*(s')$$

$$V^*(s) = \max_{a \in \mathcal{A}} \left\{ r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) V^*(s') \right\}$$

$$Q^*(s,a) = r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) \max_{a' \in \mathcal{A}} Q^*(s',a')$$

Action/Policy Evaluation Operator & Greedy Policy

Definition (Action Evaluation Operator)

$$T_a V(s) \coloneqq r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) V(s')$$

Definition (Policy Evaluation Operator)

$$T^\pi V(s) \coloneqq \sum_{a \in \mathcal{A}} \pi(a \mid s) T_a V(s) = \sum_{a \in \mathcal{A}} \pi(a \mid s) \left(r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) V(s') \right)$$

$$T^* V(s) \coloneqq \max_{a \in \mathcal{A}} T_a V(s) = \max_{a \in \mathcal{A}} \left\{ r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) V(s') \right\}$$

Definition (Greedy policy)

Policy π is greedy w.r.t. V iff $T^\pi V = T^* V$.

Banach's Fixpoint Theorem

Theorem (Banach's Fixpoint Theorem)

Let $\mathcal{V}$ be a Banach space and $T : \mathcal{V} \to \mathcal{V}$ be a contraction mapping, with Lipschitz constant $\gamma < 1$. Then T has a unique fixpoint $v \in \mathcal{V}$. Further, for each $v_0 \in \mathcal{V}$, $\lim_{n\to\infty} \|T^n(v_0) - v\| = 0$, and the convergence is geometric:

$$\|T^n(v_0) - v\| \leq \gamma^n \|v_0 - v\|$$

Remark: 在北京把北京地图随便往地上一摊, 总存在地图上至少一点, 它对应的位置正是它所处的位置. 其实不需要摊开, 捏成一团也行, 只要不撕破地图.

Application of Banach's Fixpoint Theorem

Theorem

$(\mathcal{V}, \|\cdot\|_\infty)$ *is a Banach space, where* $\mathcal{V} := \left\{V \in \mathbb{R}^{\mathcal{S}} : \|V\|_\infty < \infty\right\}$ *and* $\|V\|_\infty := \max_{s \in \mathcal{S}} |V(s)|$.

- T^π is a contraction, and V^π is the unique fixpoint of T^π.

$$\lim_{n \to \infty} \|(T^\pi)^n V_0 - V^\pi\|_\infty = 0$$

- T^* is a contraction, and V^* is the unique fixpoint of T^*.

$$\lim_{n \to \infty} \|(T^*)^n V_0 - V^*\|_\infty = 0$$

Application of Banach's Fixpoint Theorem

$$T^{\pi}Q(s,a) := r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) \sum_{a' \in \mathcal{A}} \pi(a' \mid s') Q(s',a')$$

$$T^{*}Q(s,a) := r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) \max_{a' \in \mathcal{A}} Q(s',a')$$

- T^{π} is a contraction, and Q^{π} is the unique fixpoint of T^{π}.

$$\lim_{n \to \infty} \| (T^{\pi})^{n} Q_0 - Q^{\pi} \|_{\infty} = 0$$

- T^{*} is a contraction, and Q^{*} is the unique fixpoint of T^{*}.

$$\lim_{n \to \infty} \| (T^{*})^{n} Q_0 - Q^{*} \|_{\infty} = 0$$

Policy Improvement Theorem

Theorem (Fixpoint of Bellman Optimality Operator)

Let V be the fixpoint of T^* and assume that there is policy π which is greedy w.r.t V. Then $V = V^*$ and π is an optimal policy.

Theorem (Policy Improvement Theorem)

Given two policies π and π',

$$\forall s \in \mathcal{S}\Big(Q^{\pi}(s, \pi'(s)) \geq V^{\pi}(s)\Big) \implies \forall s \in \mathcal{S}\Big(V^{\pi'}(s) \geq V^{\pi}(s)\Big)$$

where

$$Q^{\pi}(s, \pi'(s)) \coloneqq \sum_{a \in \mathcal{A}} \pi'(a \mid s) Q^{\pi}(s, a)$$

Remark: In particular, the greedy policy meets the conditions of the policy improvement theorem.
The process of making a new policy π' that improves on an original policy π, by making it greedy with respect to V^{π}, is called policy improvement.

Solving MDPs — Finite-Horizon Dynamic Programming

Principle of optimality: the tail of an optimal policy is optimal for the "tail" problem.

Backward Induction

- Backward recursion: $V_N^*(s) = r_N(s)$ and for $k = N-1, \ldots, 0$

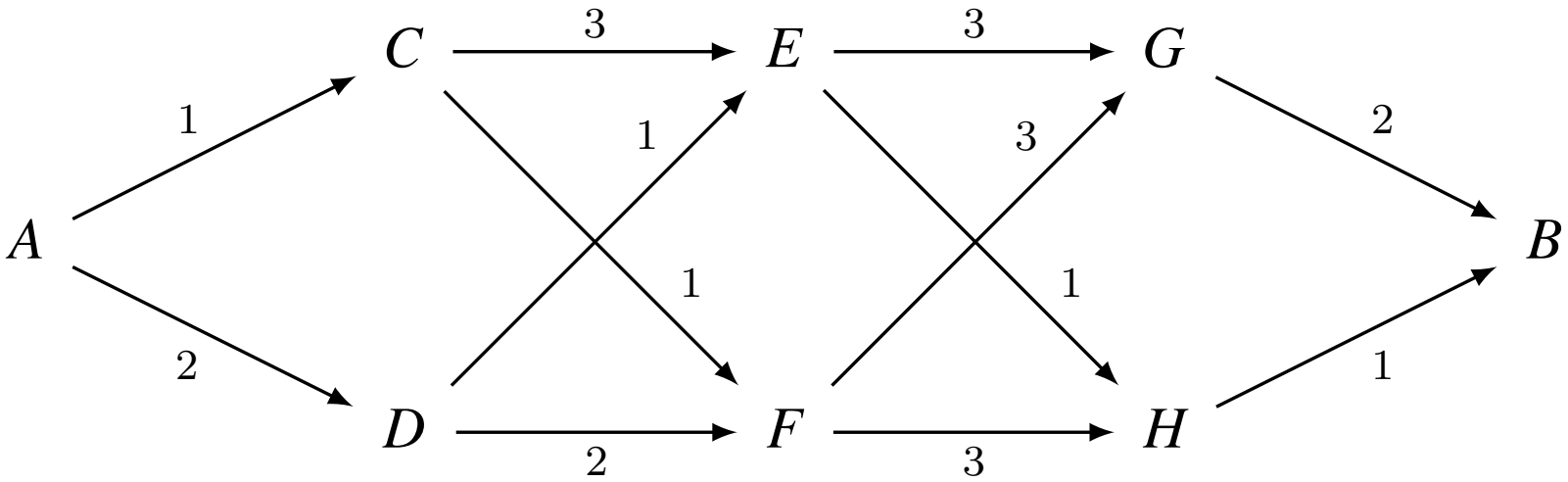
$$V_k^*(s) = \max_{a \in \mathcal{A}_k} \left\{ r_k(s,a) + \sum_{s' \in \mathcal{S}_{k+1}} P_k(s' \mid s,a) V_{k+1}^*(s') \right\}$$

- Optimal policy: for $k = 0, \ldots, N-1$

$$\pi_k^*(s) \in \operatorname*{argmax}_{a \in \mathcal{A}_k} \left\{ r_k(s,a) + \sum_{s' \in \mathcal{S}_{k+1}} P_k(s' \mid s,a) V_{k+1}^*(s') \right\}$$

- Cost: $N|\mathcal{S}||\mathcal{A}|$ vs $|\mathcal{A}|^{N|\mathcal{S}|}$ of brute force policy search.
- From now on, we will consider infinite-horizon discounted MDPs.

Example — Dynamic Programming



$$D_E = \min(d(A,C) + d(C,E), d(A,D) + d(D,E)) = 3$$
$$D_F = \min(d(A,C) + d(C,F), d(A,D) + d(D,F)) = 2$$
$$D_G = \min(D_E + d(E,G), D_F + d(F,G)) = 5$$
$$D_H = \min(D_E + d(E,H), D_F + d(F,H)) = 4$$
$$D_B = \min(D_G + d(G,B), D_H + d(H,B)) = 5$$

Working backward, the "best" path from A to B is A, D, E, H, B.

Dynamic Programming	Known Environment	Backward Induction
Reinforcement Learning	Unknown Environment	?

Solving MDPs — Value Iteration

Theorem (Principle of Optimality)

A policy π achieves the optimal value from state s, $V^{\pi}(s) = V^{*}(s)$, iff, for any state s' reachable from s, π achieves the optimal value from state s', $V^{\pi}(s') = V^{*}(s')$.

Any optimal policy π^{*} can be subdivided into two components:

1. an optimal first action a^{*},
2. followed by an optimal policy from successor state s'.

$$V^{*}(s) = \max_{a \in \mathcal{A}} \left\{ r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s, a) V^{*}(s') \right\}$$

$$\pi^{*}(s) = \operatorname*{argmax}_{a \in \mathcal{A}} \left\{ r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s, a) V^{*}(s') \right\}$$

Value Iteration: $V_{k+1} \leftarrow T^{*} V_k$

$$V_1 \rightarrow V_2 \rightarrow \cdots \rightarrow V^{*}$$

$$\boxed{\|V_{k+1} - V_k\|_{\infty} < \varepsilon \implies \|V_{k+1} - V^{*}\|_{\infty} < \frac{2\gamma\varepsilon}{1-\gamma}}$$

Solving MDPs — Policy Iteration

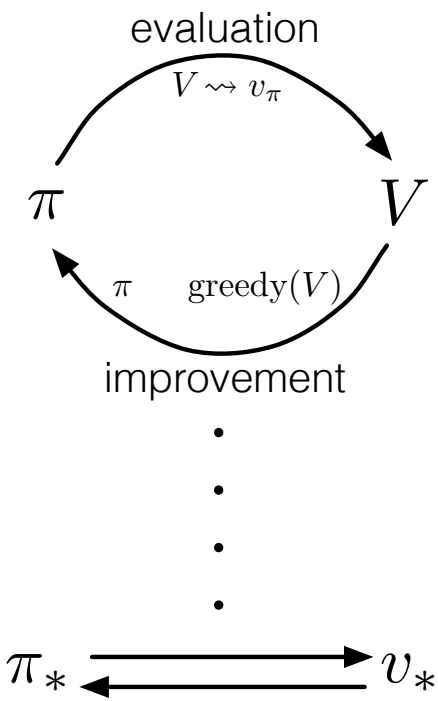
Policy Iteration: $\pi_0 \xrightarrow{E} V^{\pi_0} \xrightarrow{I} \pi_1 \xrightarrow{E} V^{\pi_1} \xrightarrow{I} \pi_2 \xrightarrow{E} \cdots \xrightarrow{I} \pi^* \xrightarrow{E} V^*$

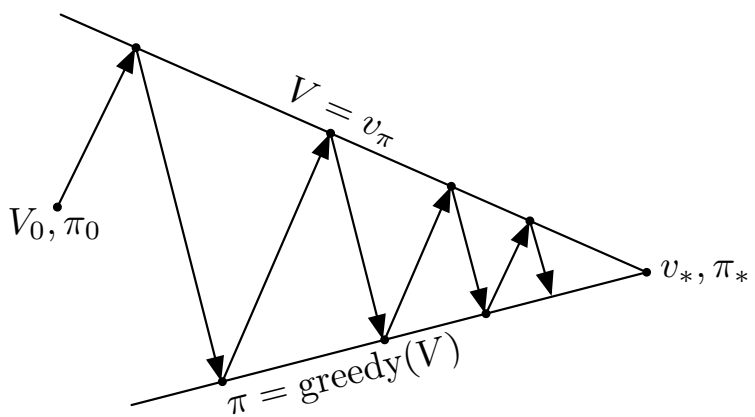
- E — policy evaluation:

$$V_{k+1}^{\pi} \leftarrow T^{\pi} V_k^{\pi}$$

- I — policy improvement:

$$\pi_{k+1}(s) := \operatorname*{argmax}_{a \in \mathcal{A}} \left(r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) V^{\pi_k}(s') \right)$$

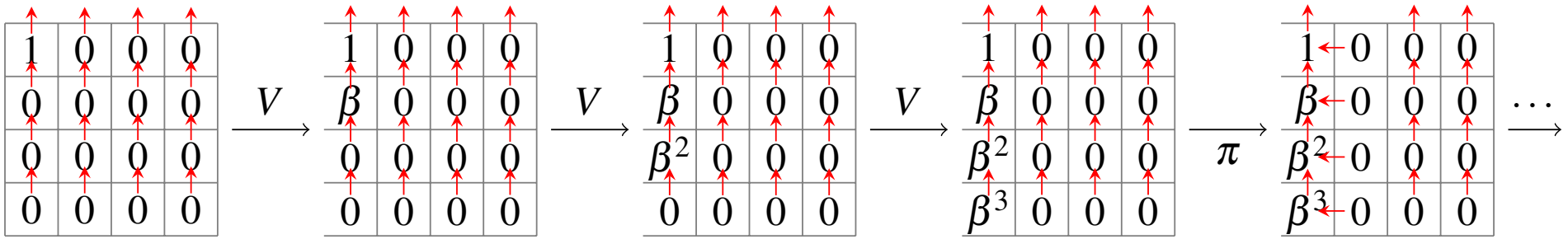


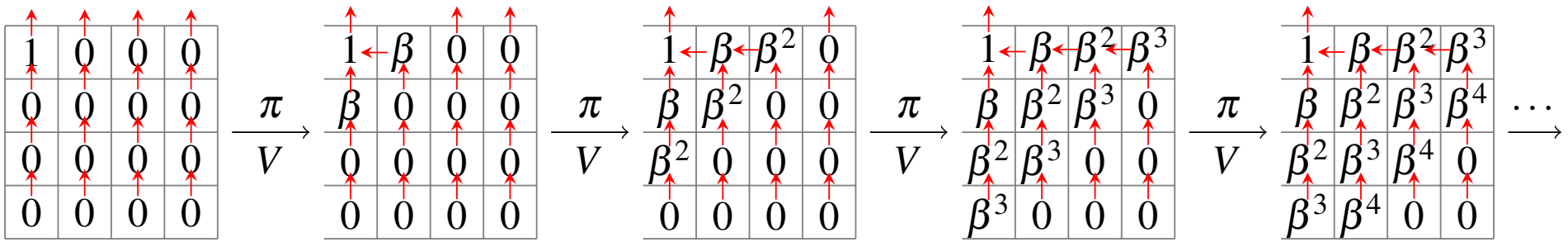


Example — Value Interation vs Policy Iteration

- Value Iteration: $((V \text{ improve})^*; \pi \text{ extraction})$
- Policy Iteration: $(V \text{ improve}; \pi \text{ improve})^*$

$$V(s) = \max_{a \in A} r(s,a) + \beta V(f(s,a))$$





Policy Iteration: the Guarantees

- Set π_0 arbitrarily
- Repeat:
  - evaluate $Q^{\pi_i}(s,a)$
  - let $\pi_{i+1} = \operatorname{argmax}_a Q^{\pi_i}(s,a)$
  - set $i = i+1$
- Until $\pi_i(s) = \pi_{i-1}(s)$

Theorem

The policy iteration algorithm generates a sequence of policies with non-decreasing performance

$$V^{\pi_{k+1}} \geq V^{\pi_k}$$

and it converges to π^* in a finite number of iterations.

Comparison — Value Iteration vs Policy Iteration

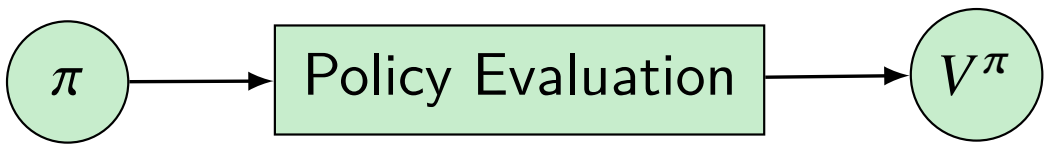
- Both value iteration and policy iteration compute the optimal values.
- They are all variations of Bellman updates.
- They differ only in whether we plug in a fixed policy or max over actions.

1. In value iteration, we don't track the policy, but taking the max over actions implicitly recomputes it.
   — Value Iteration computes the optimal policy even at a stage when the value function estimate has not yet converged.
   — If one action is better than all others, then the exact values of the states involved need not to be known.
   — Each iteration is computationally efficient.
2. Policy Iteration converges in a finite number of iterations (often small in practice).
   — But each iteration requires a full policy evaluation and it might be expensive.

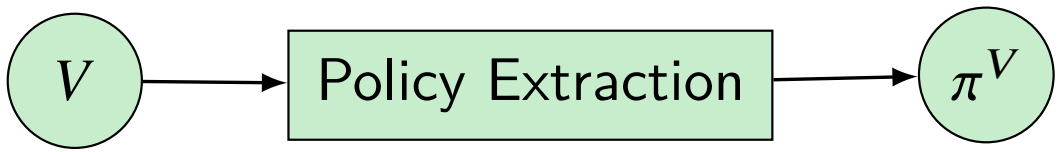
Summary: MDP

- Compute optimal values: use value iteration or policy iteration

- Compute values for a particular policy: use policy evaluation



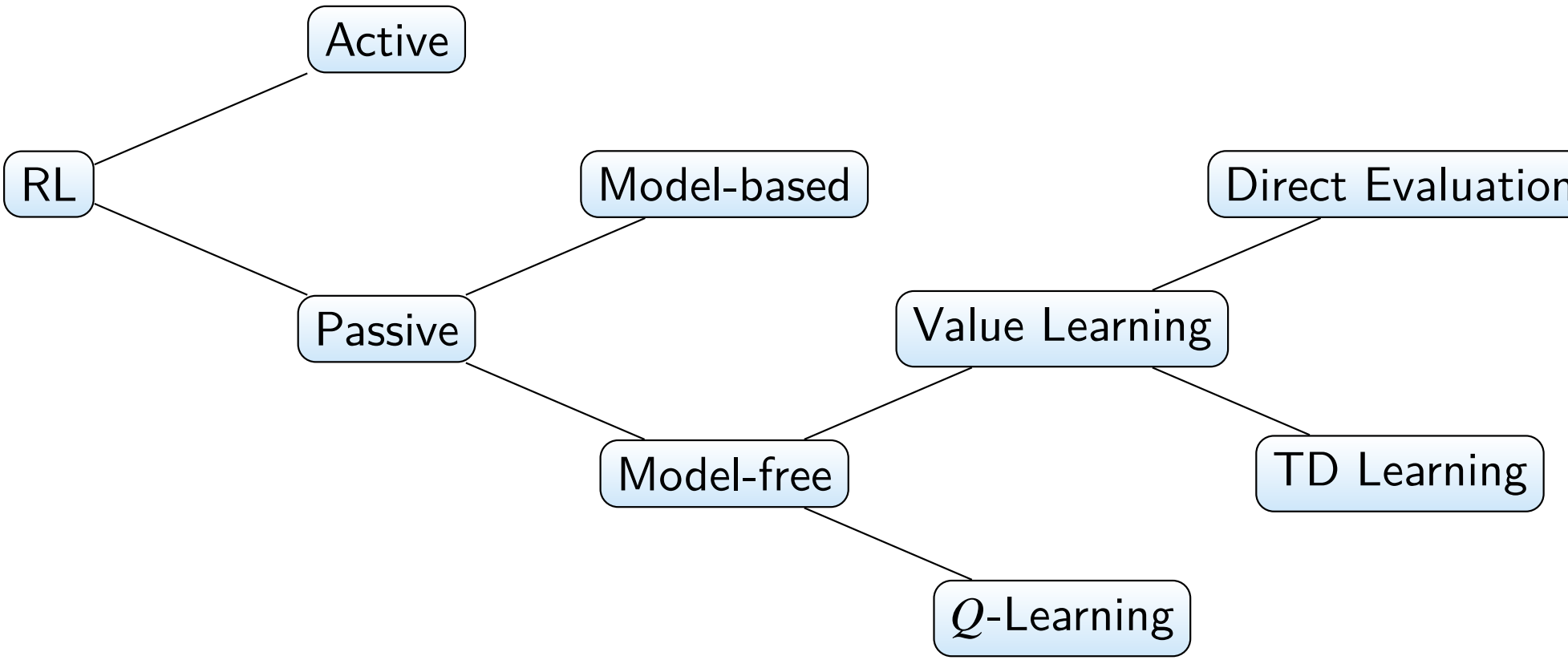
- Turn your values into a policy: use policy extraction (one-step lookahead)



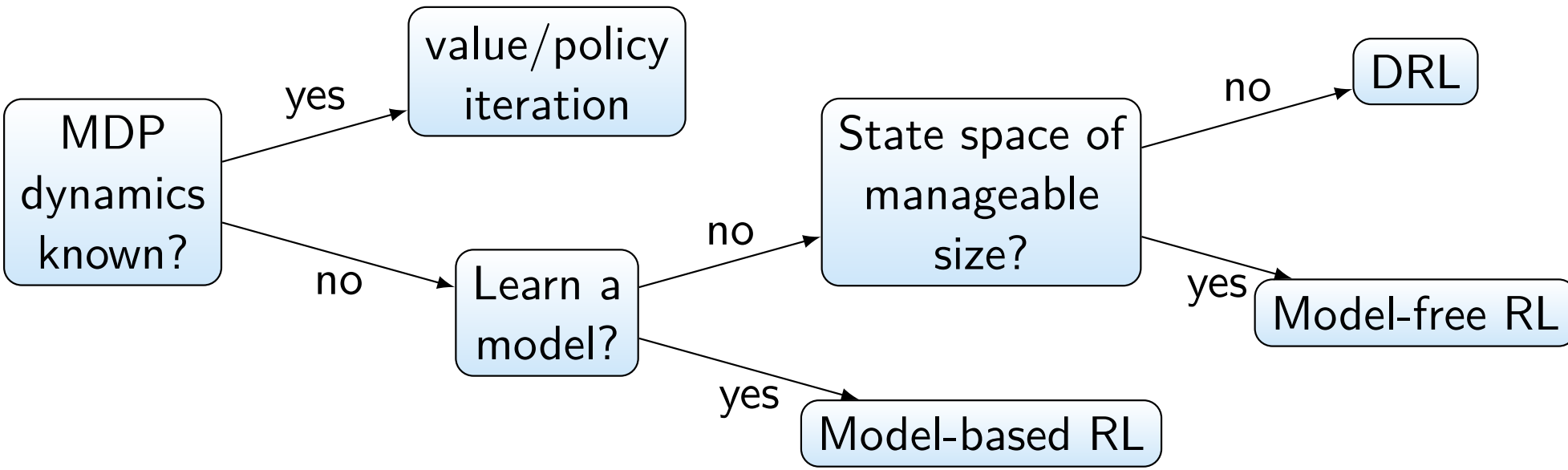
RL — Overview

- Passive Reinforcement Learning (how to learn from experiences)
  - Model-based Passive RL
    — learn the MDP model from experiences, then solve the MDP with value / policy iteration
  - Model-free Passive RL
    — skip learning MDP model, directly learn V or Q
    — receive a sample (s, a, s', r), update estimates
- Active Reinforcement Learning (agent also needs to decide how to collect experiences)
  - How to efficiently explore and minimize regret?
  - How to trade off exploration/exploitation?
- Approximate Reinforcement Learning (to handle large state spaces)
  - Approximate Q-Learning
  - Policy Search

RL — Overview


RL
Active
Passive
Model-based
Model-free
Value Learning
Direct Evaluation
TD Learning
Q-Learning

RL — Overview


MDP dynamics known?
yes
value/policy iteration
no
Learn a model?
no
State space of manageable size?
no
DRL
yes
Model-free RL
yes
Model-based RL

Monte-Carlo Methods

MC learns from complete episodes of raw experience without modeling the environmental dynamics and computes the observed mean return as an approximation of the expected return.

$$G_t := \sum_{k=0}^{T-t-1} \gamma^k r_{t+k+1}$$

$$V(s) := \frac{\sum_{t=1}^{T} [\![S_t = s]\!] G_t}{\sum_{t=1}^{T} [\![S_t = s]\!]}$$

$$Q(s,a) := \frac{\sum_{t=1}^{T} [\![S_t = s, A_t = a]\!] G_t}{\sum_{t=1}^{T} [\![S_t = s, A_t = a]\!]}$$

Temporal Differences

- Suppose we have a sequence of values: $v_1, v_2, \ldots$
- We want a running estimate of the average of the first k values:

$$A_k := \frac{v_1 + \cdots + v_k}{k}$$

- When a new value v_k arrives:

$$\begin{aligned} A_k &= \frac{v_1 + \cdots + v_{k-1} + v_k}{k} \\ &= \frac{k-1}{k} A_{k-1} + \frac{1}{k} v_k \\ &= (1 - \alpha_k) A_{k-1} + \alpha_k v_k \\ &= A_{k-1} + \alpha_k (v_k - A_{k-1}) \end{aligned}$$

  where $\alpha_k := \frac{1}{k}$.
- We can guarantee convergence if

$$\sum_{k=1}^{\infty} \alpha_k = \infty \quad \text{and} \quad \sum_{k=1}^{\infty} \alpha_k^2 < \infty$$

Temporal-Difference Learning

TD Learning is model-free and learns from incomplete episodes of experience.

$$V(s_t) \leftarrow (1-\alpha)V(s_t) + \alpha G_t$$
$$V(s_t) \leftarrow V(s_t) + \alpha(G_t - V(s_t))$$
$$V(s_t) \leftarrow V(s_t) + \alpha(r_{t+1} + \gamma V(s_{t+1}) - V(s_t))$$
$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha(r_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t))$$

- MC updates value $V(s_t)$ toward actual return G_t.

$$V(s_t) \leftarrow V(s_t) + \alpha(G_t - V(s_t))$$

- TD updates value $V(s_t)$ toward estimated return $r_{t+1} + \gamma V(s_{t+1})$.

$$V(s_t) \leftarrow V(s_t) + \alpha(\overbrace{r_{t+1} + \gamma V(s_{t+1})}^{\text{TD target}} - V(s_t))$$

TD error

Remark: TD error — 多巴胺神经递质对应奖励预测误差

SARSA: On-Policy TD control

SARSA

1. At time step t, start from state s_t and choose action a_t using policy derived from Q (e.g., ε-greedy)
2. Take action a_t, observe r_{t+1} and s_{t+1}
3. Choose a_{t+1} from s_{t+1} using policy derived from Q (e.g., ε-greedy)
4. Update the action-value function:
   $Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha(r_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t))$
5. $t = t + 1$ and repeat from step 1

Q-Learning: Off-policy TD control

Q-Learning

1. At time step t, start from state s_t and choose action a_t using policy derived from Q (e.g., ε-greedy)
2. Take action a_t, observe r_{t+1} and s_{t+1}
3. Update the action-value function:

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left(r_{t+1} + \gamma \max_{a \in \mathcal{A}} Q(s_{t+1}, a) - Q(s_t, a_t) \right)$$

4. $t = t + 1$ and repeat from step 1

Expected SARSA

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \left(r_{t+1} + \gamma \sum_{a \in \mathcal{A}} \pi(a \mid s_{t+1}) Q(s_{t+1}, a) - Q(s_t, a_t) \right)$$

Remark: Q-Learning

$$Q^*(s,a) = r(s,a) + \gamma \sum_{s' \in \mathcal{S}} P(s' \mid s,a) \max_{a' \in \mathcal{A}} Q^*(s',a')$$

We have three problems.

1. We don't know the reward function.
2. We don't know the transition probabilities.
3. We don't know the utility of the state we reached.

$$\hat{Q}_t(s_t,a_t) = r_{t+1} + \gamma \max_{a \in \mathcal{A}} Q_t(s_{t+1},a)$$

1. Instead of $r(s,a)$, use r_{t+1}, the reward we got this time.
2. Instead of summing over $P(s' \mid s,a)$, just set $s' = s_{t+1}$, i.e., whatever state followed s_t.
3. Instead of the true value of $Q(s,a)$, use our current estimate, $Q_t(s,a)$.

$$Q_{t+1}(s_t,a_t) = Q_t(s,a) + \alpha \left(\hat{Q}_t(s,a) - Q_t(s,a) \right)$$

On-policy vs Off-policy

- The learned policy may be different from the behavior policy
- On-policy learning (e.g., SARSA)
  - learn from the moves that were actually taken
  - policy evaluation and policy improvement happen simultaneously
- Off-policy learning (e.g., Q-learning)
  - learning of the optimal policy independently from its execution
  - converge faster, more aggressive/risky
  - possibility to use effective exploration strategies

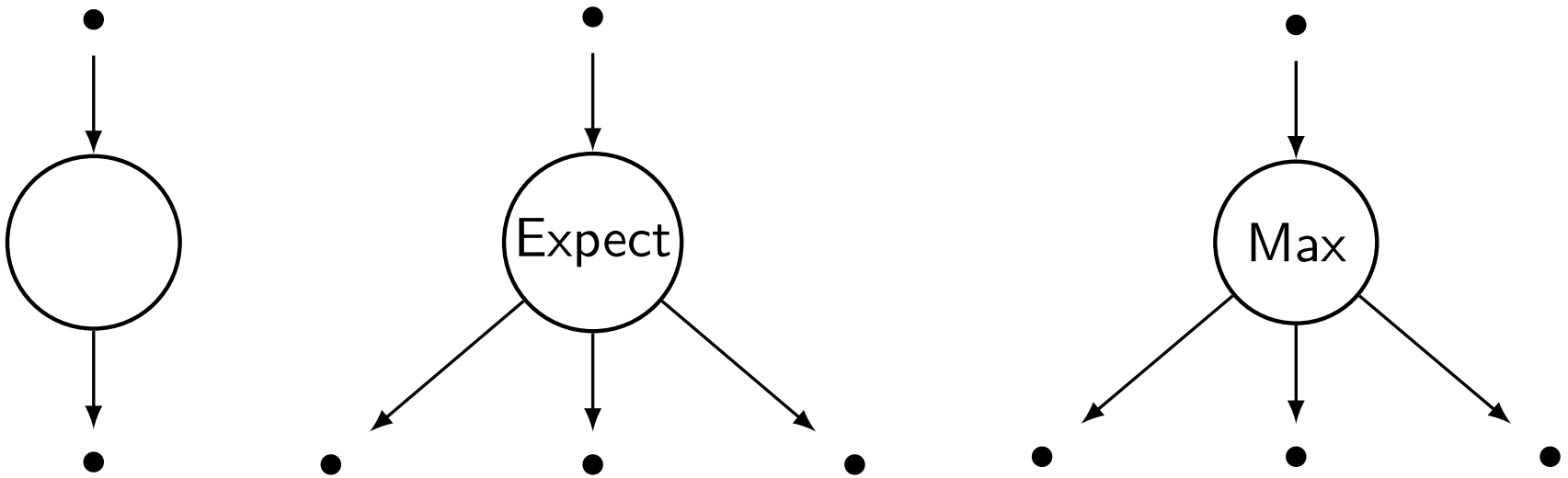
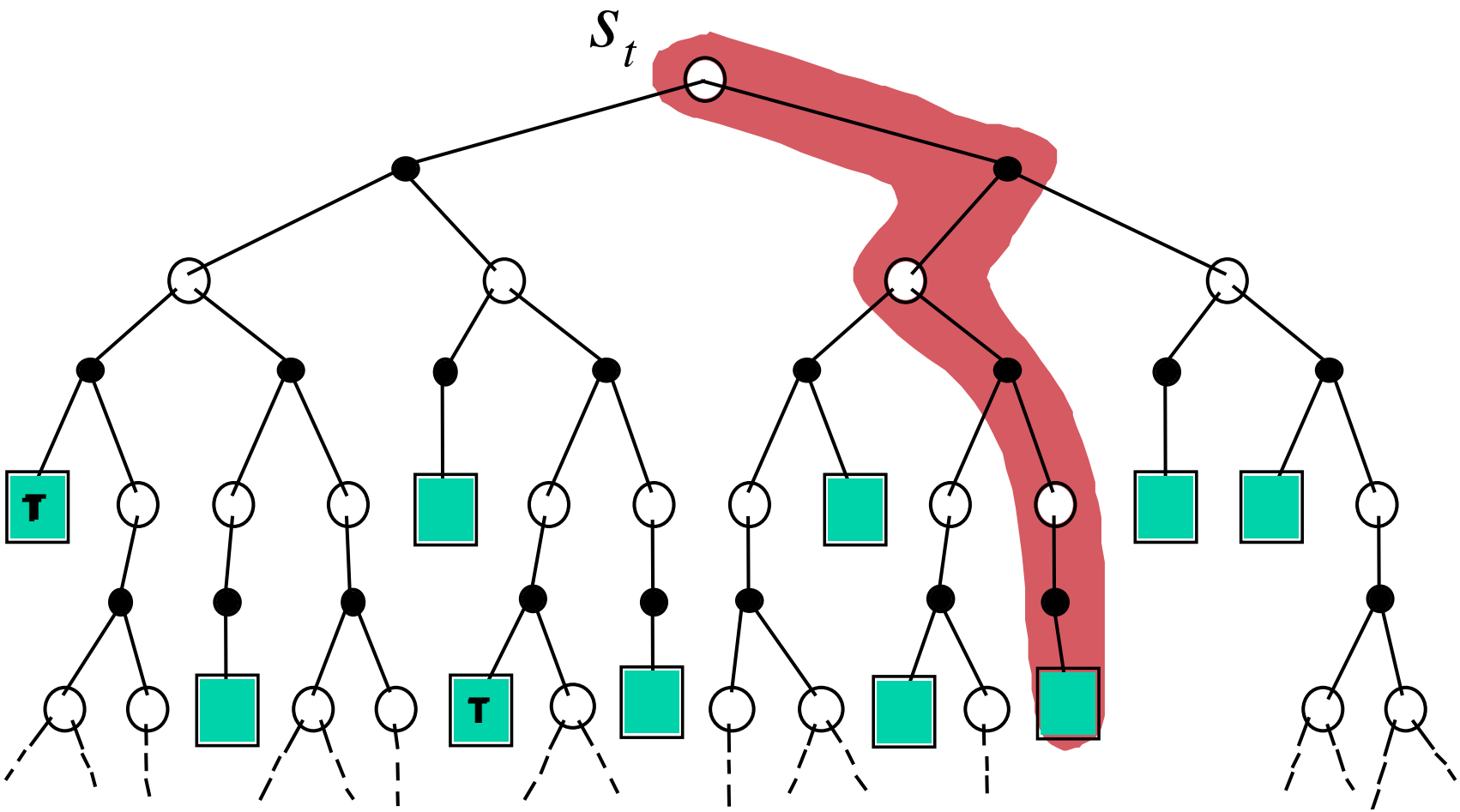


Figure: SARSA, Expected SARSA, and Q-Learning

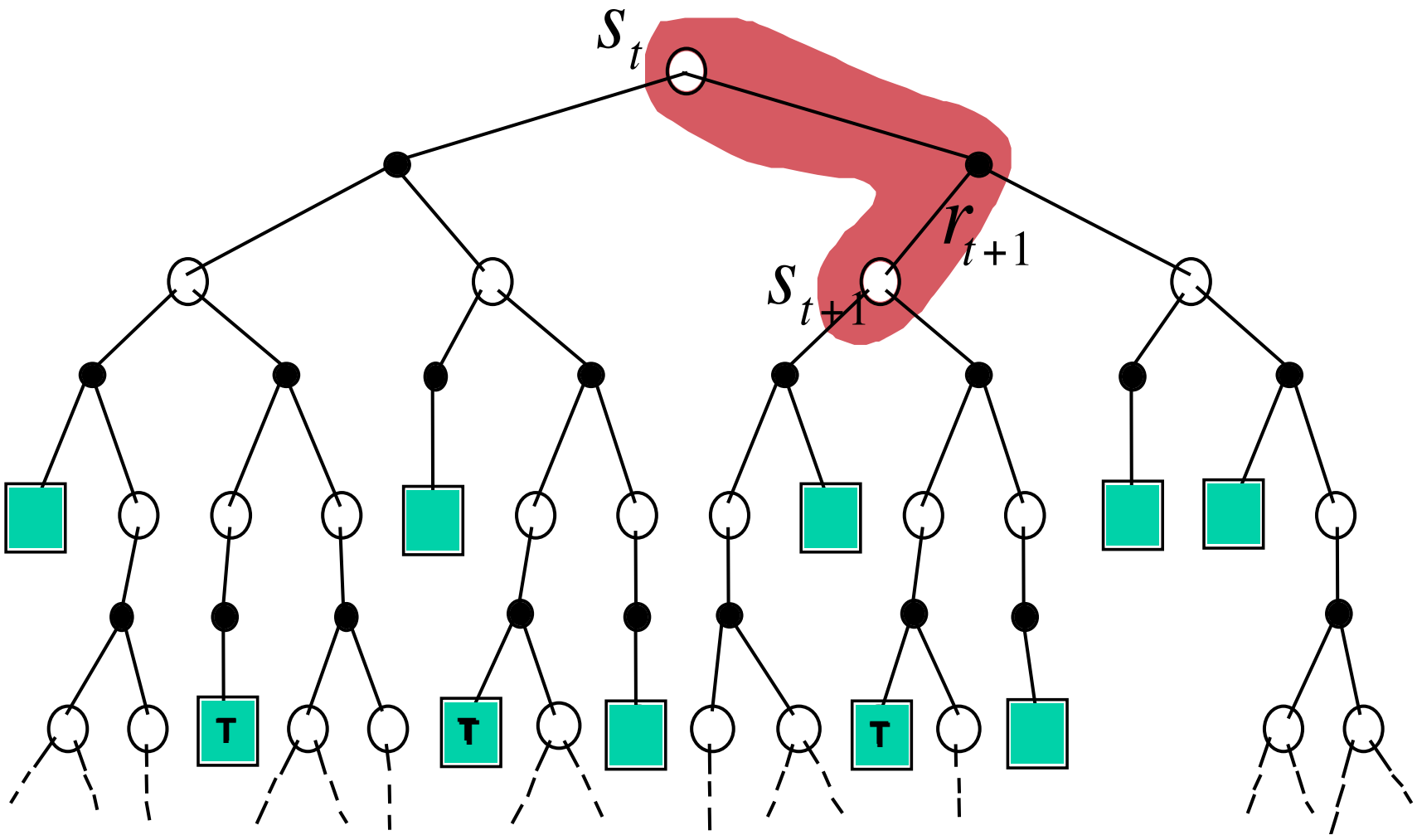
Monte-Carlo Backup

$$V(s_t) \leftarrow V(s_t) + \alpha(G_t - V(s_t))$$



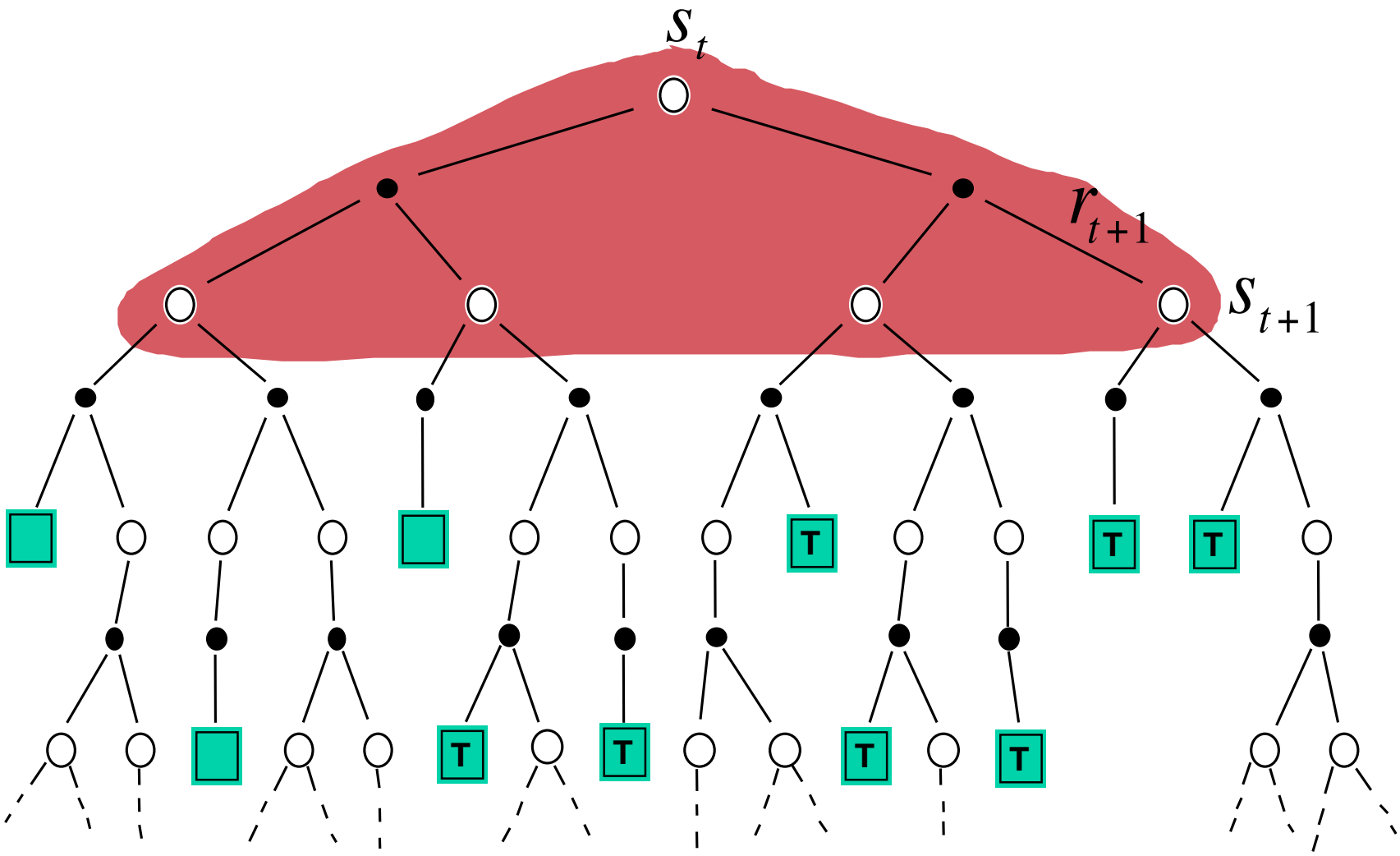
Temporal-Difference Backup

$$V(s_t) \leftarrow V(s_t) + \alpha(r_{t+1} + \gamma V(s_{t+1}) - V(s_t))$$

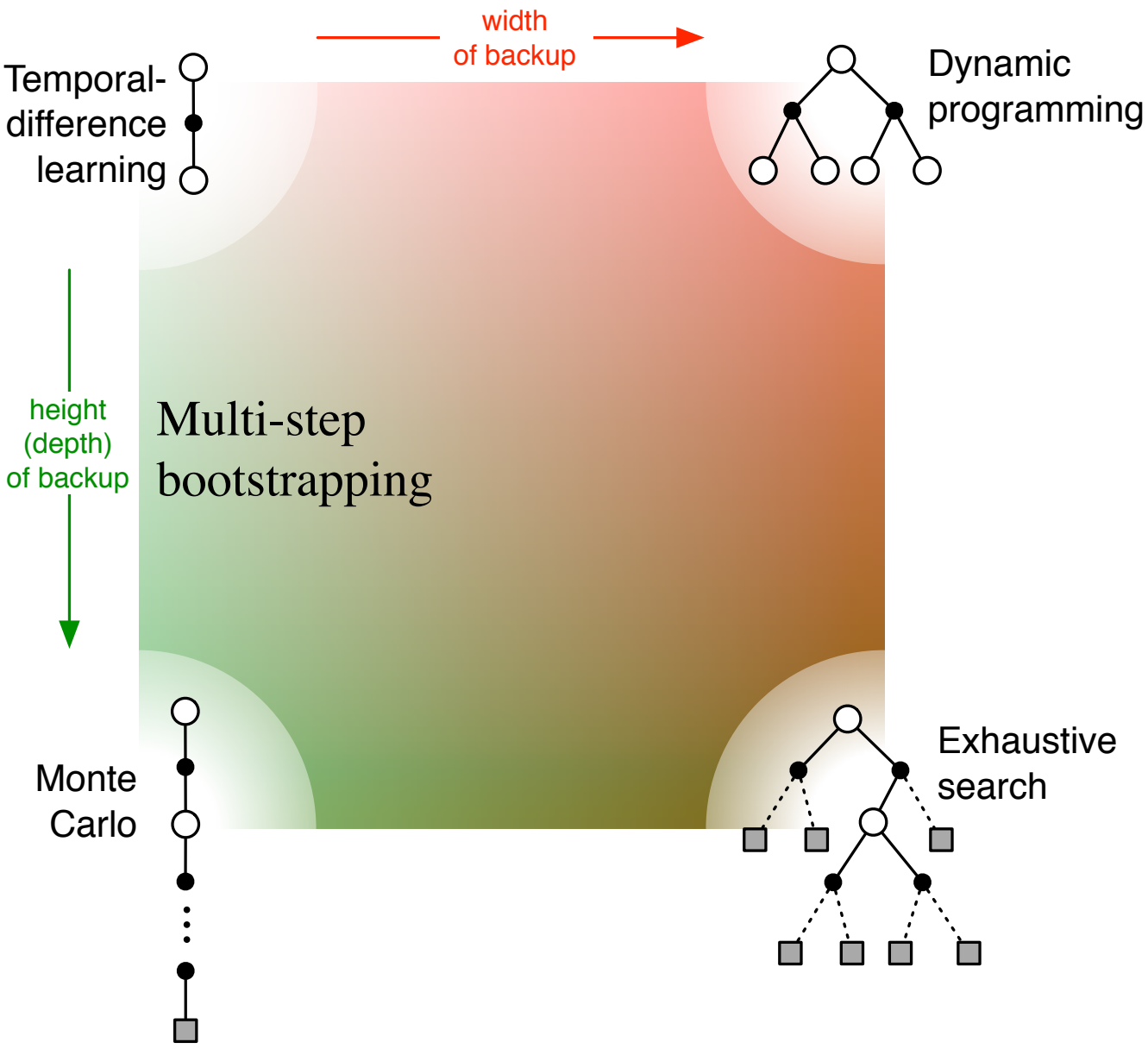


Dynamic Programming Backup

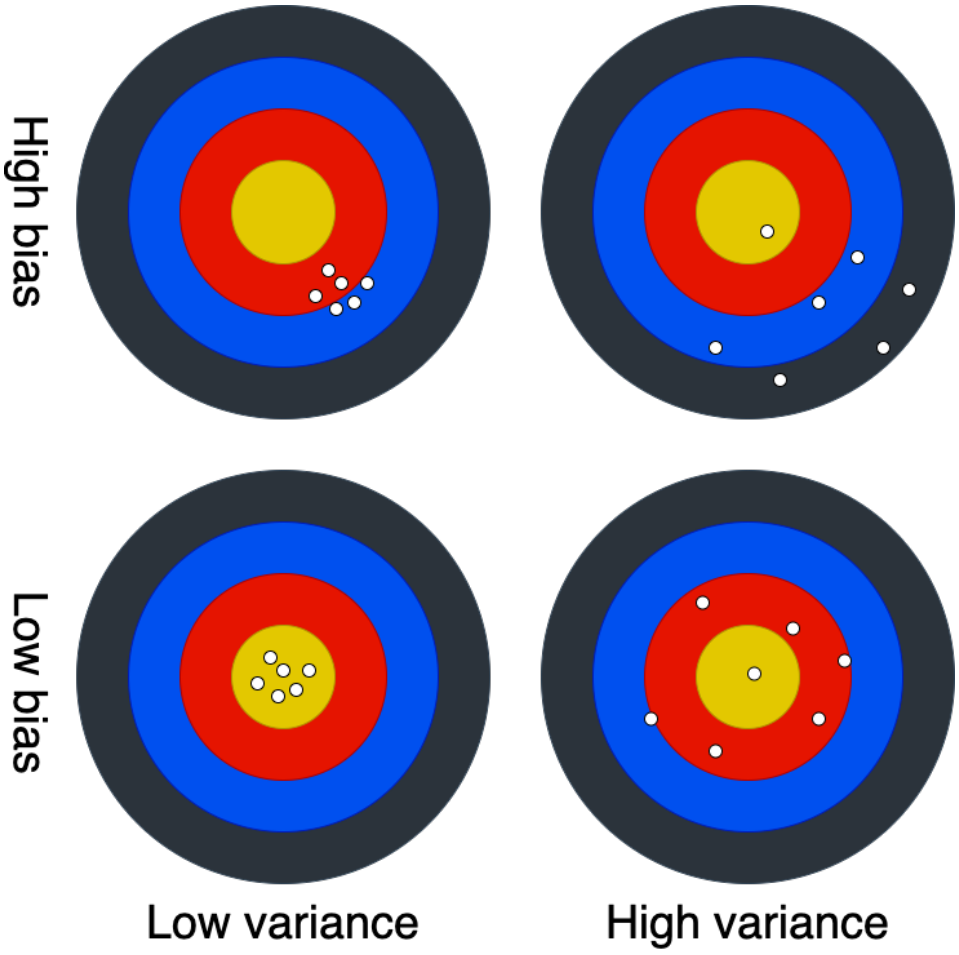
$$V(s_t) \leftarrow \mathbb{E}_\pi[r_{t+1} + \gamma V(s_{t+1})]$$



Unified View



Bias-Variance Trade-Off



expected squared error = Bias^2 + Variance + Noise

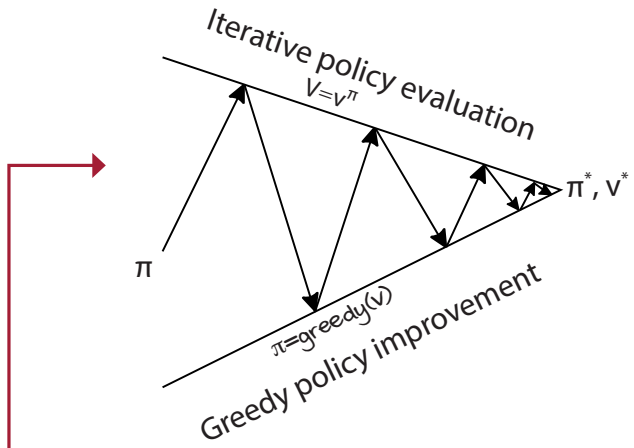
$$\mathbb{E}\left[\left(y-\hat{f}(x)\right)^2\right] = \left(\mathbb{E}\left[\hat{f}(x)\right]-f(x)\right)^2 + \mathbb{E}\left[\left(\hat{f}(x)-\mathbb{E}\left[\hat{f}(x)\right]\right)^2\right] + \mathbb{E}\left[\left(y-f(x)\right)^2\right]$$

Bias-Variance Trade-Off: MC vs TD

- Return $G_t := \sum_{k=0}^{T-t-1} \gamma^k r_{t+k+1}$ is an unbiased estimate of $V^\pi(s_t)$.
- TD target $r_{t+1} + \gamma V(s_{t+1})$ is a biased estimate of $V^\pi(s_t)$ unless $V(s_{t+1}) = V^\pi(s_{t+1})$.
- But the TD target is much lower variance:
  - Return depends on many random actions, transitions, rewards
  - TD target depends on one random action, transition, reward
- MC has high variance, zero bias
  - Good convergence properties
  - Works well with function approximation
  - Not very sensitive to initial value
  - Very simple to understand and use
- TD has low variance, some bias
  - Usually more efficient than MC
  - $\mathrm{TD}(0)$ converges to $V^\pi(s)$
  - Problem with function approximation
  - More sensitive to initial values

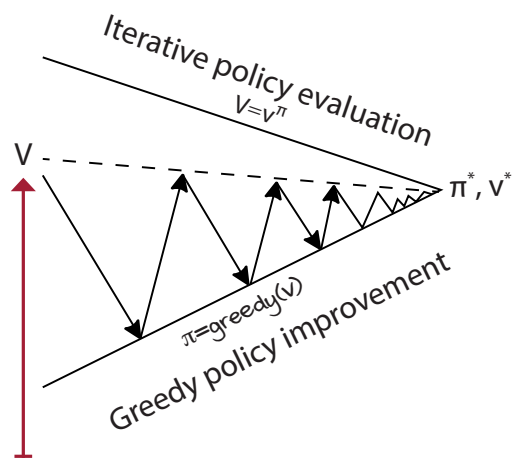
Comparison between planning and control methods

Policy iteration



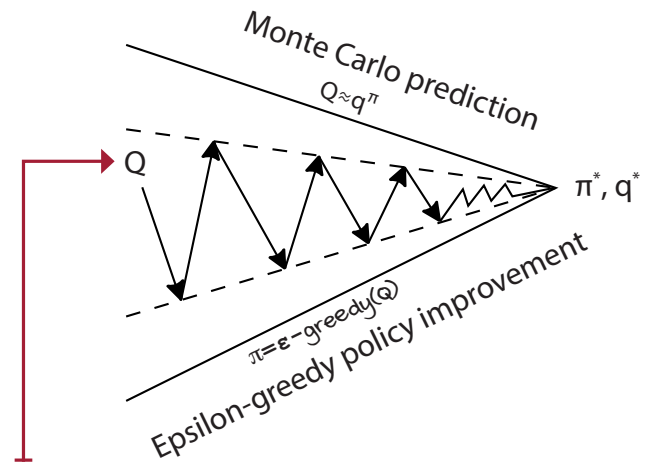
(1) Policy iteration consists of a f convergence of iterative policy evaluation alternating with greedy policy improvement.

Value iteration



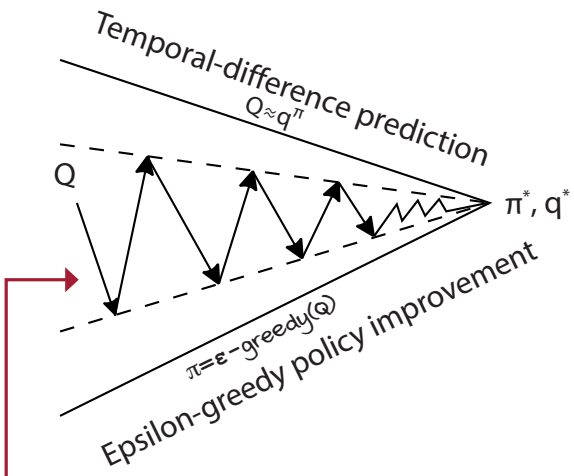
(2) alue iteration starts with an arbitrary value function and has a truncated policy evaluation step.

Monte Carlo control



(3) MC control estimates a Q-function, ha truncated MC prediction phase followed by an epsilon-greedy policy-improvement step.

SARSA



(4) SARSA has pretty much the s as MC control except a truncated TD prediction for policy evaluation.

Policy Search

- Idea: directly optimize policy
- Policy may be parameterized Q functions, hence:

$$\pi(s) \coloneqq \operatorname*{argmax}_a \hat{Q}_\theta(s,a)$$

- Stochastic policy, e.g. given by softmax function

$$\pi_\theta(a \mid s) \coloneqq \frac{\mathrm{e}^{\hat{Q}_\theta(s,a)}}{\sum_a \mathrm{e}^{\hat{Q}_\theta(s,a)}}$$

- Policy value $\rho(\theta)$: expected reward if π_θ is carried out

Remark (Free Will[a])

[a] Erik M. Rehn: Free Will Belief as a consequence of Model-based Reinforcement Learning. 2022.

- *the "will" of an agent:* $\hat{Q}_\theta(s,a)$
- *the "freedom" of an agent:* $H(s) = -\sum_a \pi_\theta(a \mid s) \log \pi_\theta(a \mid s)$

Types of RL Algorithms

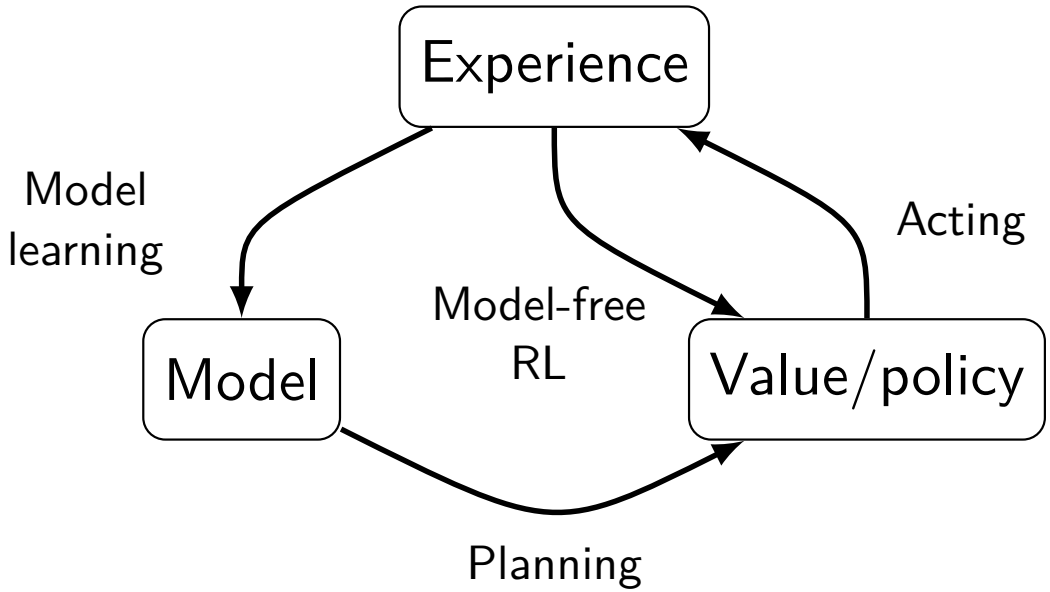
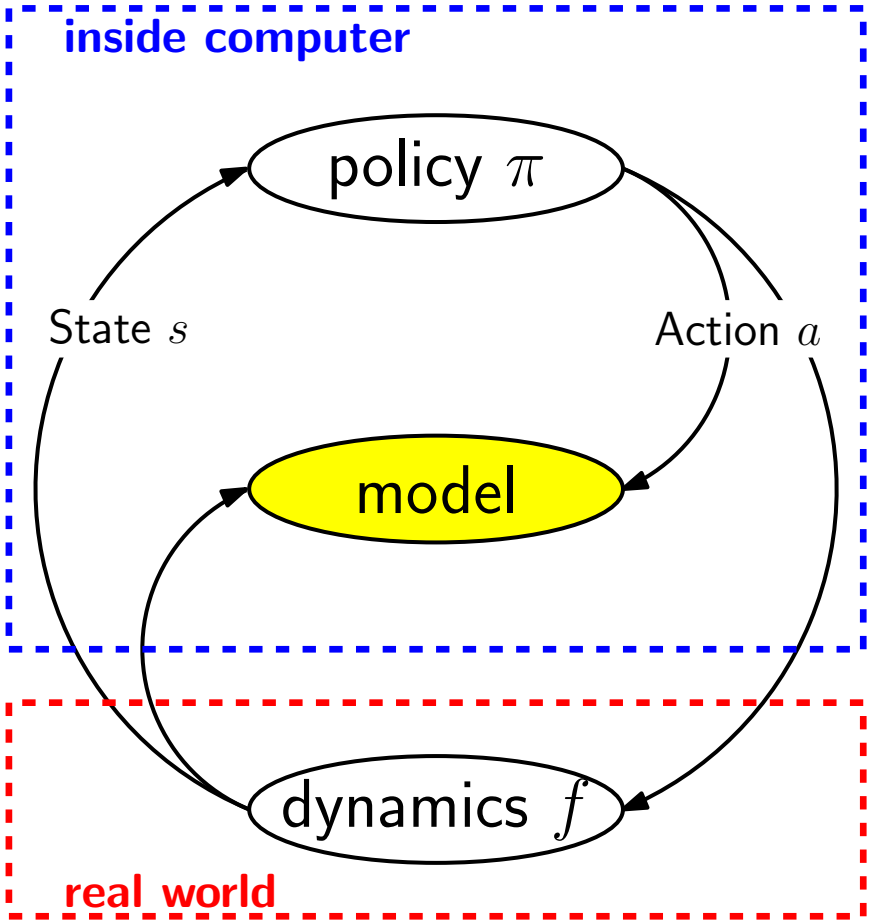
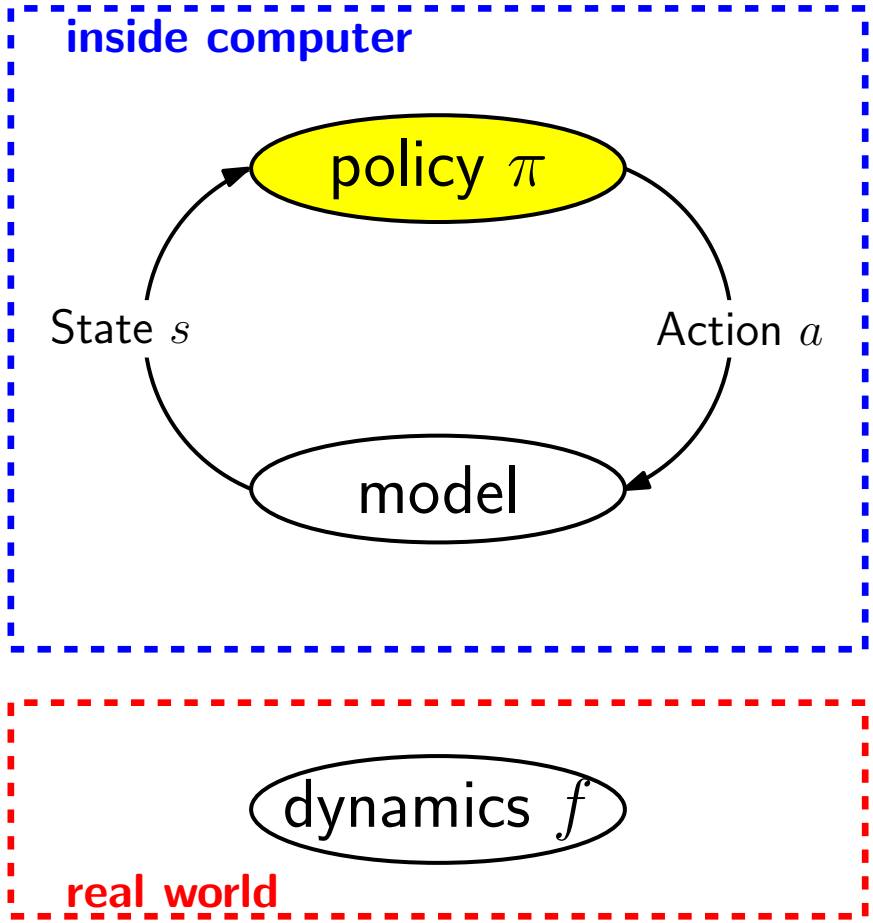


Figure: The direct approach uses a representation of either a value function or a policy. The indirect approach makes use of a model of the environment.

- Policy gradient: directly differentiate the objective $\theta^* = \underset{\theta}{\operatorname{argmax}}\, \mathbb{E}_{(s,a)\sim p_\theta(s,a)}[r(s,a)]$
- Value-based: estimate value or Q-function of the optimal policy
- Actor-critic: estimate value or Q-function of the current policy
- Model-based RL: estimate the transition model

Model-based Set-up: Interaction and Simulation

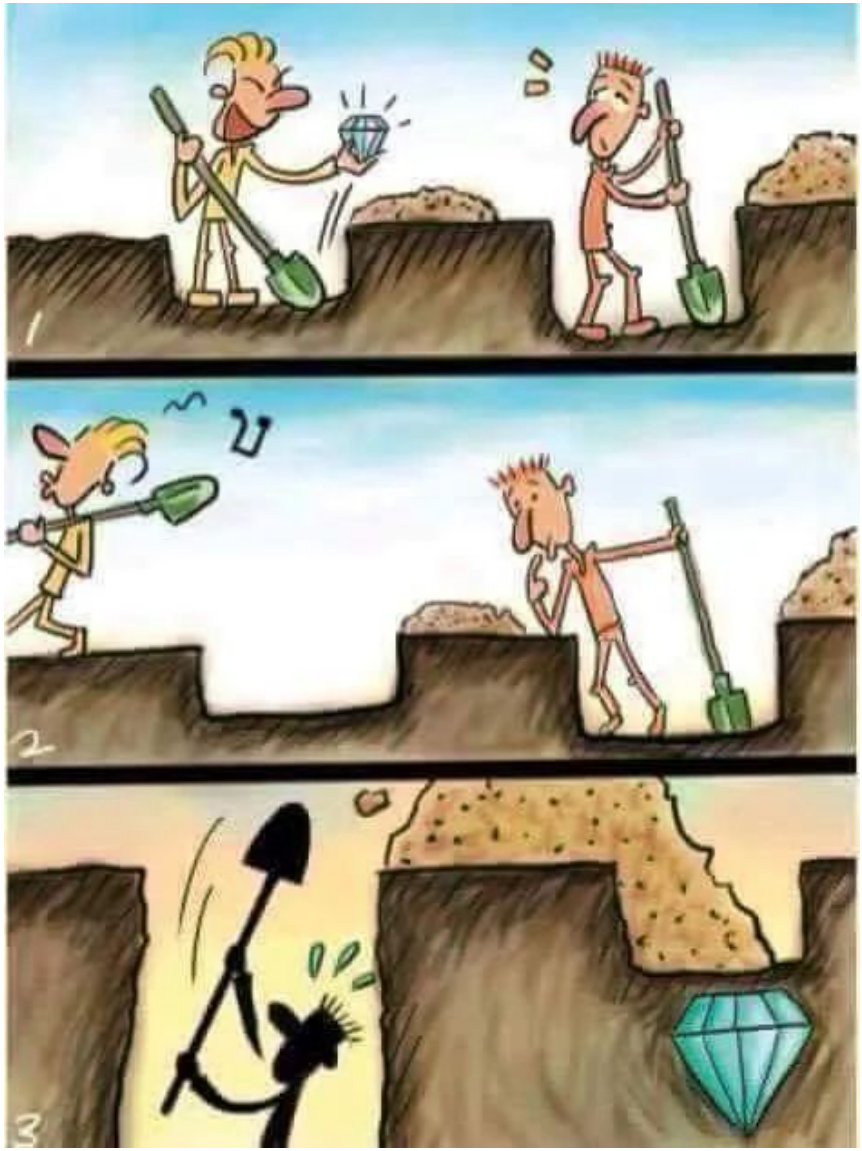




- Interaction: internal model is refined using experience from interacting with the real system
- Simulation: internal model simulates consequences of actions in the real system, policy is refined (support counterfactual)

Exploration vs Exploitation

- Exploration: trying actions just to see what happens in the hope of learning more successful behaviors
- Exploitation: using what the agent has learned so far to select actions
- In practice, agents must do some exploration otherwise they may be stuck in a subset of environment states having low utility
- It even makes sense in some applications to choose actions randomly
- Typically, agent explore more in the early stages of deployment and exploit more in later stages



How to Explore?

- Regular Q-Update:

$$Q(s,a) \leftarrow r(s,a) + \gamma \max_{a'} Q(s',a')$$

- Modified Q-Update with exploration function f:

$$Q(s,a) \leftarrow r(s,a) + \gamma \max_{a'} f\Big(Q(s',a'), N(s',a')\Big)$$

where N is the visit count.

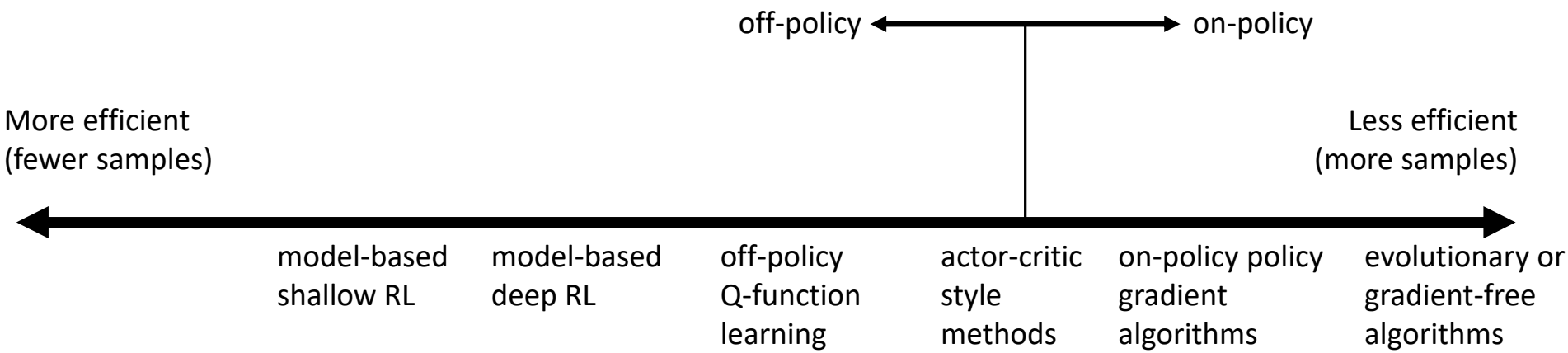
Possible Strategy

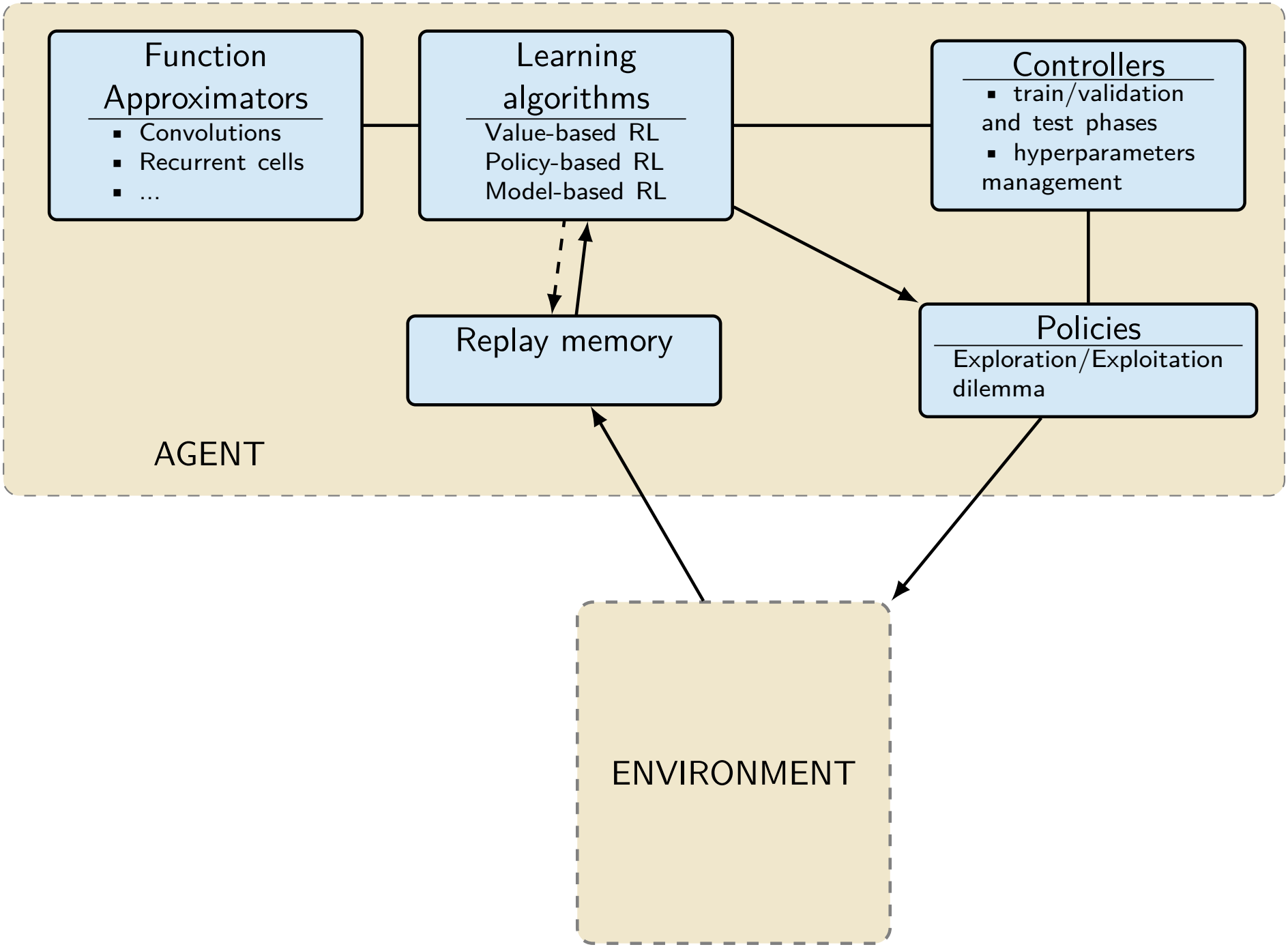
- Epsilon-Greedy
- Softmax
- Upper Confidence Bound (UCB)
- Thompson Sampling

Why so many RL algorithms?

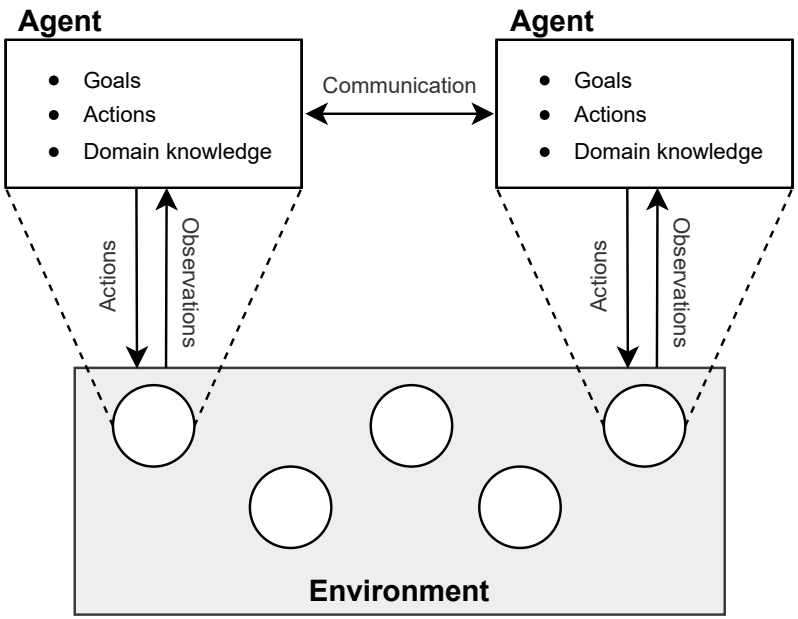
- Different tradeoffs
  - Sample efficiency
  - Stability & ease of use
- Different assumptions
  - Stochastic or deterministic?
  - Continuous or discrete?
  - Episodic or infinite horizon?
- Different things are easy or hard in different settings
  - Easier to represent the policy?
  - Easier to represent the model?

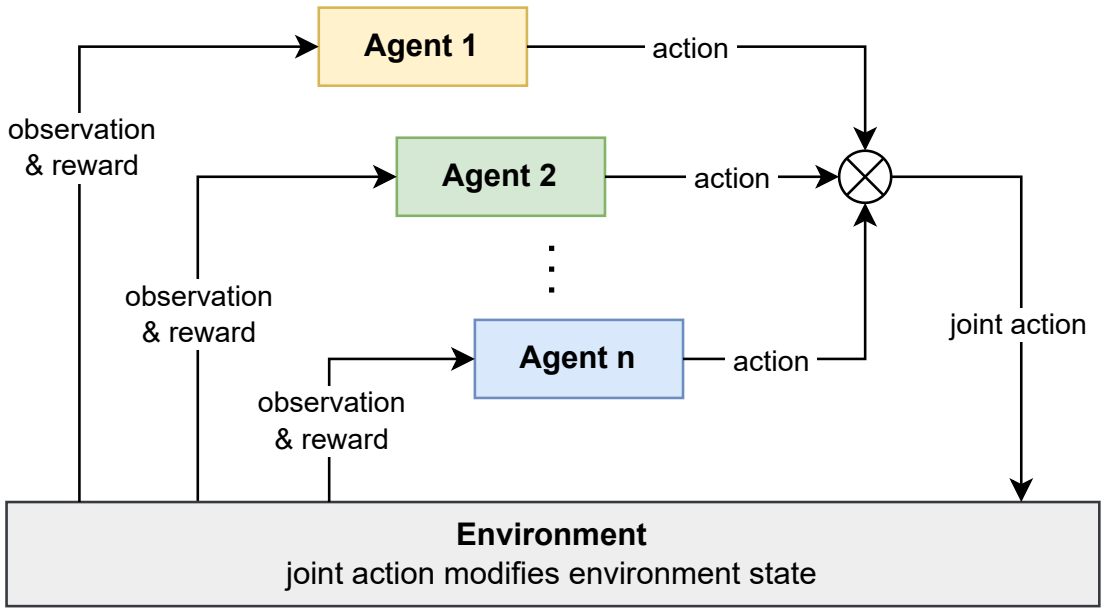
Sample Efficiency




Function Approximators
Convolutions
Recurrent cells
...
Learning algorithms
Value-based RL
Policy-based RL
Model-based RL
Controllers
train/validation and test phases
hyperparameters management
Replay memory
Policies
Exploration/Exploitation dilemma
AGENT
ENVIRONMENT

Multi-Agent Reinforcement Learning





Multi-Agent Reinforcement Learning

$$s_{t+1} \sim P(\cdot \mid s_t, a_t^1, \ldots, a_t^N)$$
$$r_{t+1}^i = R^i(s_t, a_t^1, \ldots, a_t^N, s_{t+1})$$

- Agent i's policy π^i specifies $\pi^i(a^i \mid s)$.
- Agent i's goal: $\max_{\pi^i} V^i_{\pi^i, \pi^{-i}}$ where

$$V^i_{\pi^i, \pi^{-i}} = \mathbb{E}_{\substack{\forall j : a_t^j \sim \pi^j(\cdot \mid s_t) \\ s_{t+1} \sim P(\cdot \mid s_t, a_t^1, \ldots, a_t^N)}} \left[\sum_{t \geq 0} R^i(s_t, a_t^1, \ldots, a_t^N, s_{t+1}) \right]$$

- Nash Equilibrium: Collection of policies $\pi^1, \ldots, \pi^N$ such that

$$\forall i \leq N \forall \pi' : V^i_{\pi^i, \pi^{-i}} \geq V^i_{\pi', \pi^{-i}}$$

- When $N = 1$, this is equivalent to optimal policy of MDP.

Reinforcement Learning vs Extensive-Form Game Theory

Reinforcement learning	$\Leftrightarrow$	*Game theory*
stochastic policy	$=$	*mixed strategy*
deterministic policy	$=$	*pure strategy*
agent	$=$	*player*
multi-agent environment	$=$	*infinite extensive-form game*
reward/value	$=$	*payoff/utility*
(finite) history	$=$	*history*
infinite history	$=$	*path of play*
asymptotic optimality	$\hat{=}$	*convergence to Nash*

强化学习 vs 主动推理

自由能原理与主动推理 [PPF22]

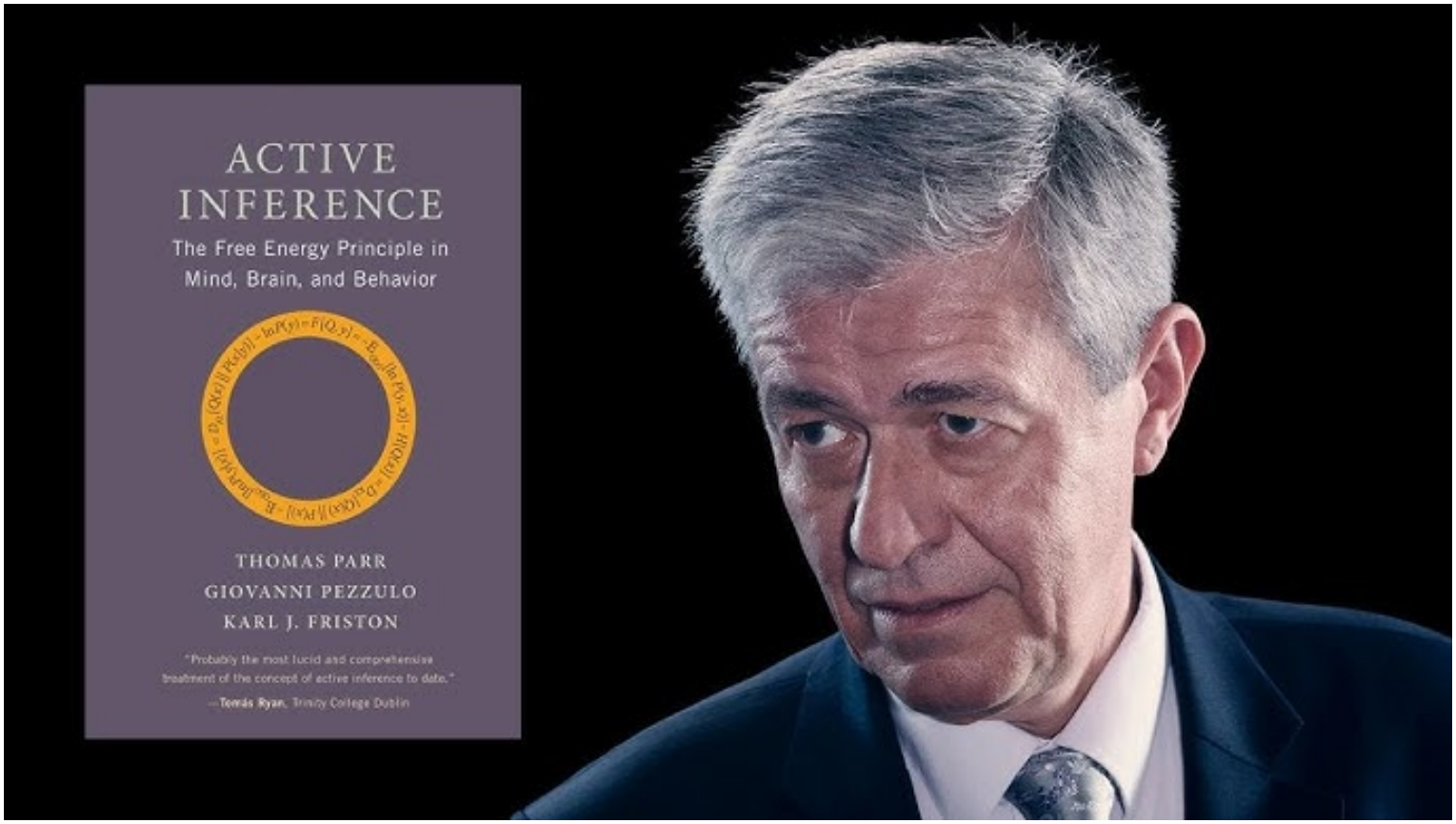


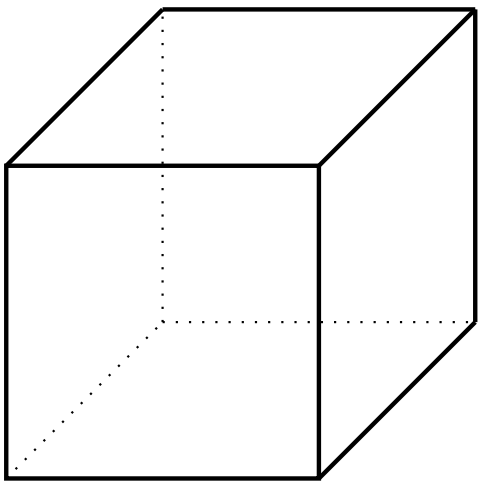
Figure: Karl Friston

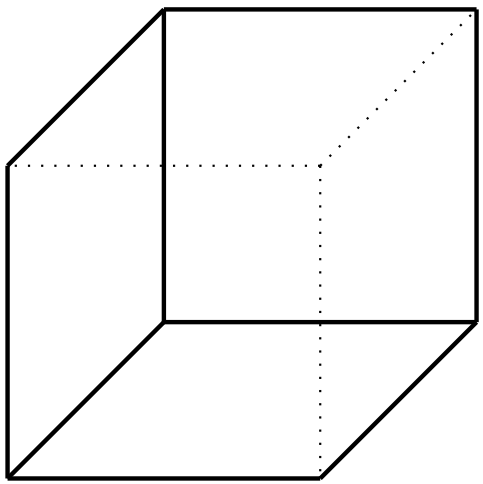
感知与行动的统一

Friston 主动推理

- 感知依赖于大脑对外部世界的主动预测 (贝叶斯大脑)
- 通过比较预测和输入信息, 大脑更新生成模型, 最小化预测误差
- 感知通过更新信念让信念与观察相符; 行动通过改变环境让环境符合信念和目标.

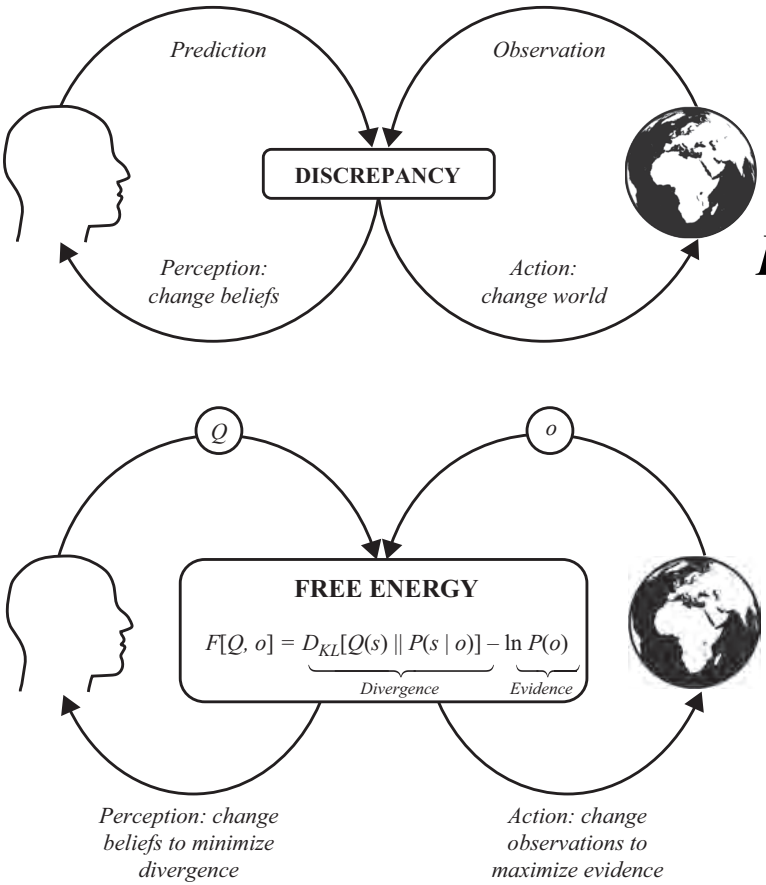
1. 感知、学习可以看作最小化 “变分自由能” 的过程
2. 行动、规划、决策可以看作最小化 “期望自由能” 的过程





Friston's Active Inference

Minimizing Variational Free Energy



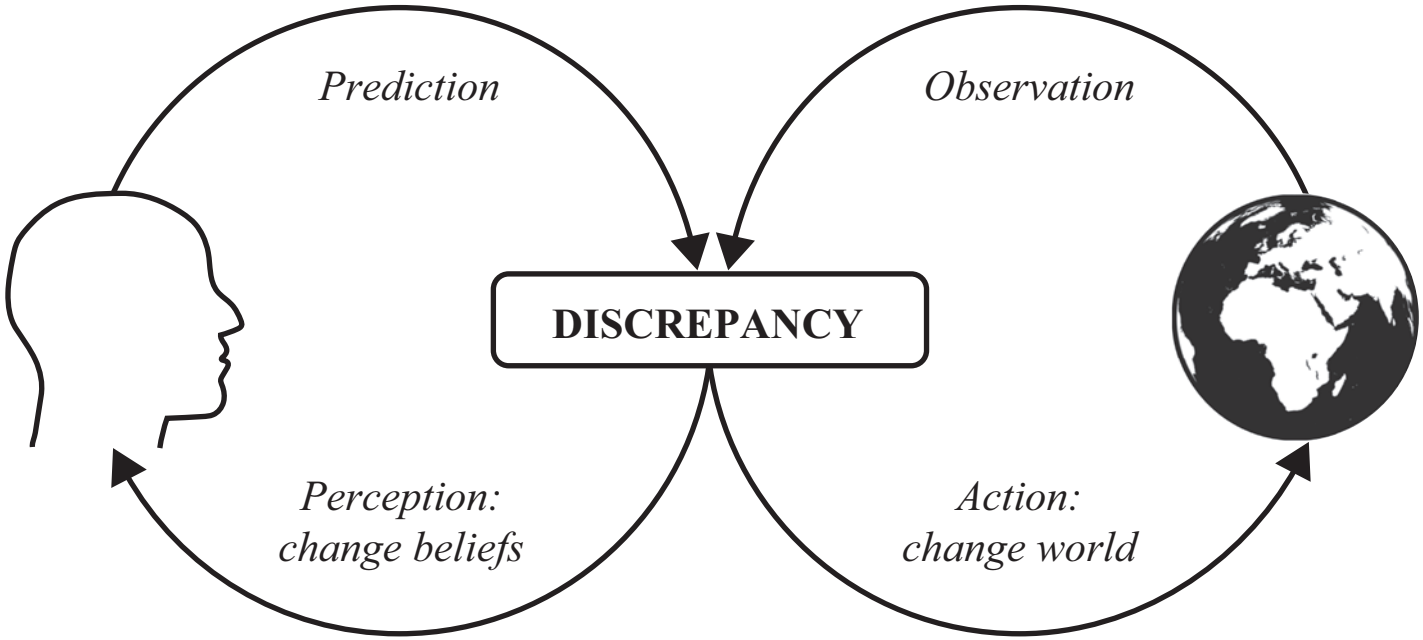
$$
\begin{aligned}
Q^* &= \operatorname*{argmin}_{Q \in \mathcal{M}} F[Q, o] \\
&\approx P(\cdot \mid o) \\
F[Q, o] &= -\mathbb{E}_{Q(s)}\left[\log \frac{P(o, s)}{Q(s)}\right] \\
&= \underbrace{-\mathbb{E}_{Q(s)}[\log P(o, s)]}_{\text{Energy}} \underbrace{- H[Q(s)]}_{\text{Entropy}} \\
&= \underbrace{D_{\mathrm{KL}}[Q(s) \| P(s)]}_{\text{Complexity}} \underbrace{- \mathbb{E}_{Q(s)}[\log P(o \mid s)]}_{\text{Accuracy}} \\
&= \underbrace{D_{\mathrm{KL}}[Q(s) \| P(s \mid o)]}_{\text{Divergence}} \underbrace{- \log P(o)}_{\text{Evidence}}
\end{aligned}
$$

$$
\begin{aligned}
-\log P(o) = -\log \sum_s Q(s) \frac{P(o,s)}{Q(s)} &\overset{\text{Jensen}}{\leq} -\mathbb{E}_{Q(s)}\left[\log \frac{P(o,s)}{Q(s)}\right] \\
&= \underbrace{-\mathbb{E}_{Q(s)}[\log P(o,s)]}_{\text{Energy}} - \underbrace{H[Q(s)]}_{\text{Entropy}} \\
&= \underbrace{D_{\mathrm{KL}}[Q(s)\|P(s)]}_{\text{Complexity}} - \underbrace{\mathbb{E}_{Q(s)}[\log P(o \mid s)]}_{\text{Accuracy}} \\
&= \underbrace{D_{\mathrm{KL}}[Q(s)\|P(s \mid o)]}_{\text{Divergence}} - \underbrace{\log P(o)}_{\text{Evidence}}
\end{aligned}
$$

1. 维持当前的 “能量” 与尽可能高的 “熵”. 当缺乏观察信息和精确的先验信念时, 关于环境的隐藏状态, 应该采纳不确定性最高的信念 (最大熵原理).
2. 最小化 “复杂性”(贝叶斯惊讶) 与最大化 “精确性”. 寻找对观察信息的最简洁的解释.
3. 感知: 改变信念以最小化散度; 行动: 改变观察以最大化证据.

Friston's Active Inference

Minimizing Expected Free Energy



$$\pi^* = \underset{\pi}{\operatorname{argmin}}\, G(\pi)$$

$$G(\pi) = -\mathbb{E}_{Q(o,s|\pi)}\left[\log \frac{P(o, s \mid \pi)}{Q(s \mid \pi)}\right]$$

其中, $Q(o, s \mid \pi) = Q(s \mid \pi)P(o \mid s)$.

$$
\begin{aligned}
G(\pi) &= -\sum_{o,s} P(o \mid s) Q(s \mid \pi) \left[\log \frac{P(o, s \mid \pi)}{Q(s \mid \pi)} \right] \\
&= -\sum_{o,s} P(o \mid s) Q(s \mid \pi) \left[\log \frac{P(s \mid o, \pi) P(o \mid \pi)}{Q(s \mid \pi)} \right] \\
&= -\sum_{o,s} P(o \mid s) Q(s \mid \pi) \left[\log \frac{P(s \mid o, \pi)}{Q(s \mid \pi)} \right] - \sum_{o,s} P(o \mid s) Q(s \mid \pi) \log P(o \mid \pi) \\
&= \underbrace{-\mathbb{E}_{Q(o|\pi)} \Big[D_{\mathsf{KL}} \big[Q(s \mid o, \pi) \| Q(s \mid \pi) \big] \Big]}_{\text{Information Gain}} \; \underbrace{- \; \mathbb{E}_{Q(o|\pi)} \Big[\log P(o \mid \pi) \Big]}_{\text{Pragmatic Value}} \\
&= -\sum_{o,s} P(o \mid s) Q(s \mid \pi) \left[\log \frac{P(o \mid s, \pi)}{Q(o \mid \pi)} \right] - \sum_{o,s} P(o \mid s) Q(s \mid \pi) \log P(o \mid \pi) \\
&= -\sum_{s} Q(s \mid \pi) \sum_{o} P(o \mid s) \log P(o \mid s) - \sum_{o,s} P(o \mid s) Q(s \mid \pi) \left[\log \frac{P(o \mid \pi)}{Q(o \mid \pi)} \right] \\
&= \underbrace{\mathbb{E}_{Q(s|\pi)} \Big[H[P(o \mid s)] \Big]}_{\text{Expected Ambiguity}} \; + \; \underbrace{D_{\mathsf{KL}} \big[Q(o \mid \pi) \| P(o \mid \pi) \big]}_{\text{Risk}} \\
&\leq \mathbb{E}_{Q(s|\pi)} \Big[H[P(o \mid s)] \Big] \; + \; D_{\mathsf{KL}} \big[Q(s \mid \pi) \| P(s \mid \pi) \big] \\
&= \underbrace{-\mathbb{E}_{Q(o,s|\pi)} \Big[\log P(o, s \mid \pi) \Big]}_{\text{Expected Energy}} \; \underbrace{- \; H[Q(s \mid \pi)]}_{\text{Entropy}}
\end{aligned}
$$

$$G(\pi) = \underbrace{-\mathbb{E}_{Q(o|\pi)}\Big[D_{\mathrm{KL}}\big[Q(s \mid o,\pi)\|Q(s \mid \pi)\big]\Big]}_{\text{Information Gain}} \underbrace{-\ \mathbb{E}_{Q(o|\pi)}\Big[\log P(o \mid \pi)\Big]}_{\text{Pragmatic Value}}$$

$$= \underbrace{\mathbb{E}_{Q(s|\pi)}\big[H[P(o \mid s)]\big]}_{\text{Expected Ambiguity}} + \underbrace{D_{\mathrm{KL}}\big[Q(o \mid \pi)\|P(o \mid \pi)\big]}_{\text{Risk}}$$

$$\leq \underbrace{-\mathbb{E}_{Q(o,s|\pi)}\big[\log P(o,s \mid \pi)\big]}_{\text{Expected Energy}} - \underbrace{H[Q(s \mid \pi)]}_{\text{Entropy}}$$

1. “信息增益” 可以看做一种认知价值, 可以帮助消除不确定性; “实用价值” 虽然是对观察的先验信念, 但可以包含玩家的先验偏好. 最大化 “信息增益” 与 “实用价值” 可以平衡 “探索” 与 “利用” 的两难.
2. 最小化 “预期含混” 可以消除 “状态” 到 “结果” 的不准确性. 最小化 “风险” 要求行动策略的结果的分布要与玩家的先验偏好相符.
3. 最大熵.

具身模拟

- 现象学家梅洛·庞蒂: “动作的沟通或理解是通过我的意向和他人的动作、我的动作和在他人行为中显现的意向的相关关系实现的. 所发生的一切像是他人的意向寓于我的身体中, 或我的意向寓于他人的身体中.”
- 镜像神经元: 灵长类或鸟类动物在做一个动作 (比如伸手拿香蕉), 或观察到其它个体在做同一个动作时, 都会被激活的一类神经元.
- 小鼠无论是作为打架的 “当事者” 还是 “旁观者”, 其镜像神经元都会被激活. 反过来, 如果激活小鼠的镜像神经元, 也会使它们产生攻击性.
- 镜像神经元就像一面镜子, 在自己的大脑中 “模拟” 了他人的动作.
- 镜像神经元可能在模仿学习、意图理解、共情等方面起着重要作用.
- Friston 认为, 镜像神经元是大脑主动推理的结果. 无论是 “做动作” 还是 “观察动作”, 被激活的是同一个 “生成模型”. 镜像神经元的 “镜像” 特性, 源于这同一个生成模型既可以用来 “做动作”(通过行动改变世界符合自己的预测), 也可以用来理解 “观察动作”(通过更新信念来解释观察).

Phenomenal World vs Noumenal World

1. Phenomenal World: things as they appear to us
2. Noumenal World: things in themselves

- Kant argued that space and time and causality are part of our perceptual framework.
- Kantian hyperpriors is how we structure our phenomenal world.
- Kant's Copernican revolution: objects must conform to our cognition.
- Helmholtz: Perception as inference.

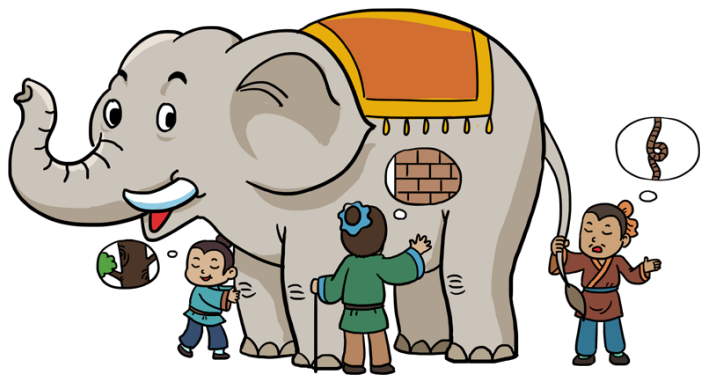
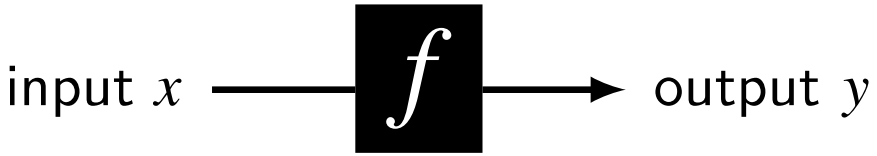


Figure: Reality can be experienced, but it is not possible to totally express it with language. The experience of the world is a construction, constrained by external data and internal beliefs, priors and assumptions.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

函数

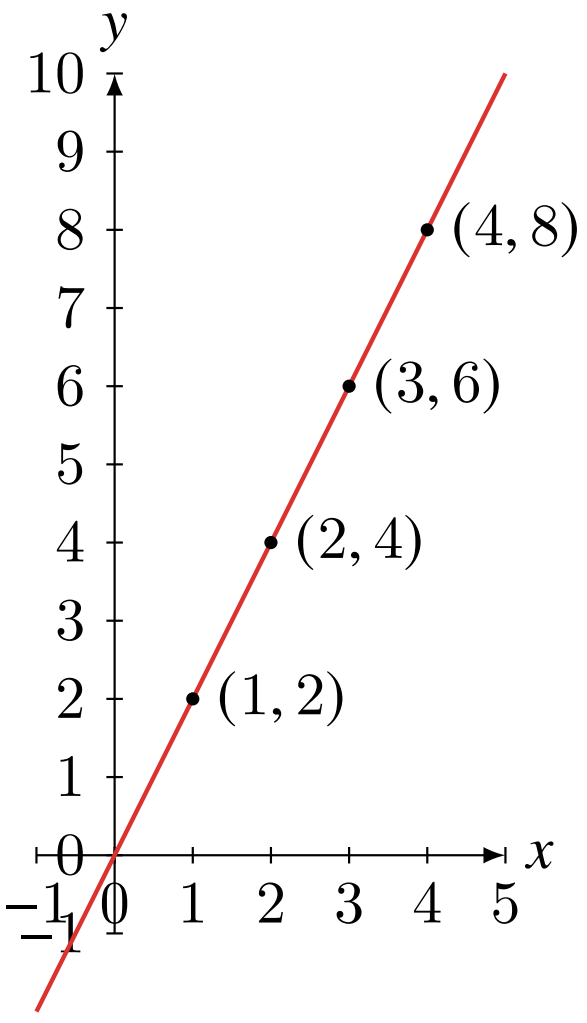


$$\begin{bmatrix}1\\2\\3\\4\end{bmatrix} \xrightarrow{f} \begin{bmatrix}2\\4\\6\\8\end{bmatrix}$$

$$y = 2x$$

$$y = wx + b$$

复杂的函数咋办?



Learning as an Alternative to Traditional Programming

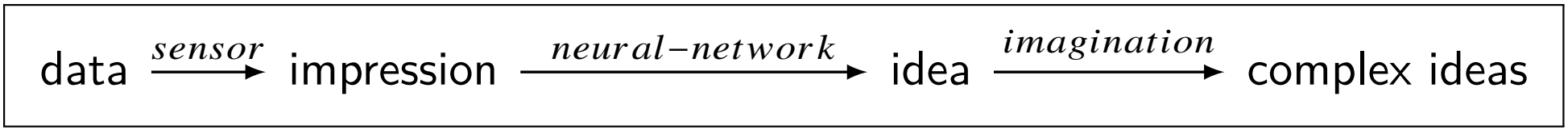
- Learning from data / experience may be more human-like
  - Babies develop an intuitive understanding of physics in their first 2 years
  - Formal reasoning and logic comes much later in development
- Learning enables fast reaction times
  - It might take a long time to train a neural network
  - But predicting with the network is very fast
- Representation Learning
  - Jointly learn features and classifier, directly from raw data
- Deep Learning
  - multiple levels of representation learning
  - composition of simple but nonlinear modules that each transform the representation at one level to a higher, more abstract level.

"Big data is like teenage sex: everyone talks about it, nobody really knows how to do it, everyone thinks everyone else is doing it, so everyone claims they are doing it."

亚里士多德与连接主义

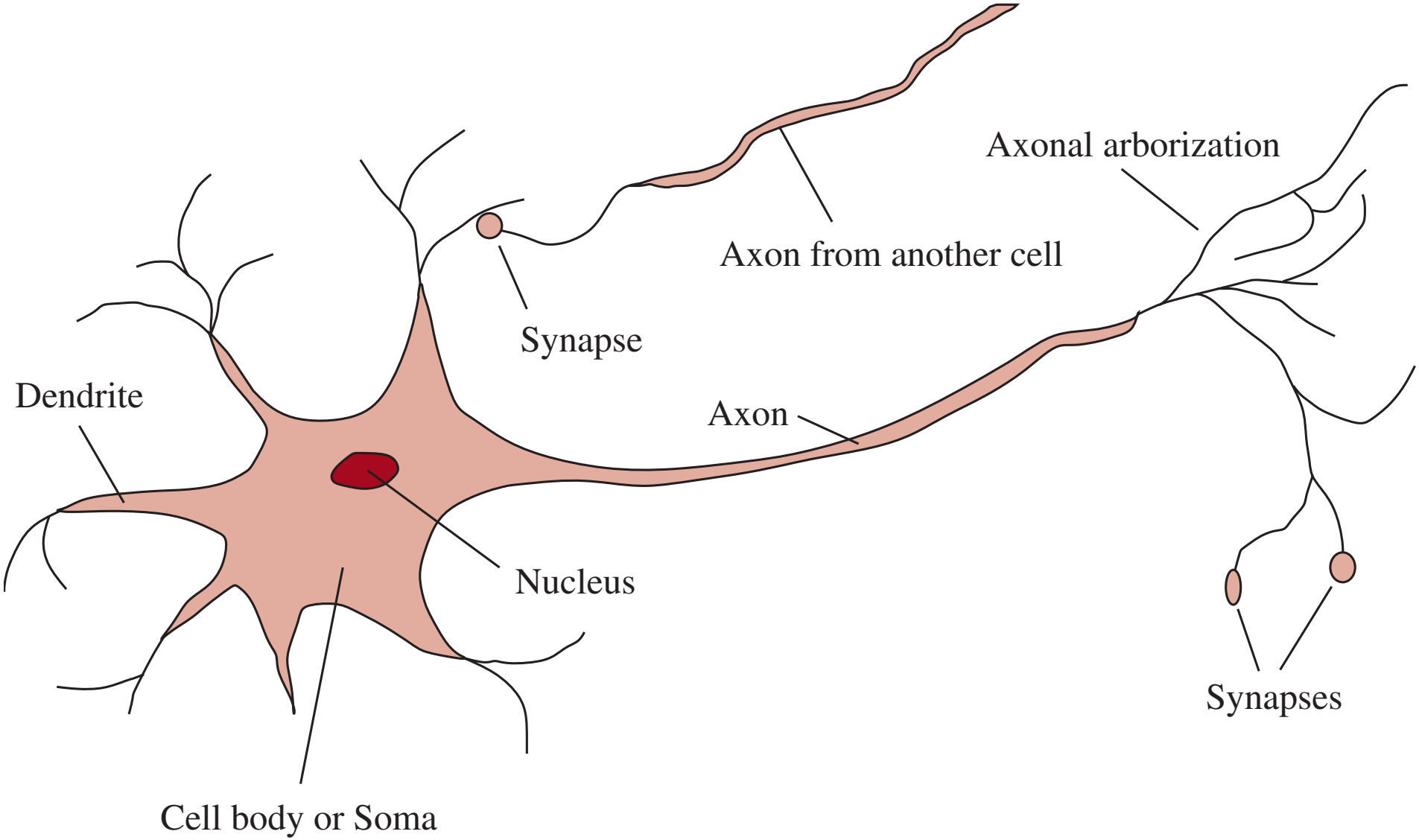
1. The law of contiguity. 时空上相邻的事物或事件会相互关联.
2. The law of frequency. 事物或事件关联次数越多, 关联强度越大.
3. The law of similarity. 相似的事物, 一个会激发思考另一个.
4. The law of contrast. 一个事物可能会激发思考相反的事物.

休谟与连接主义



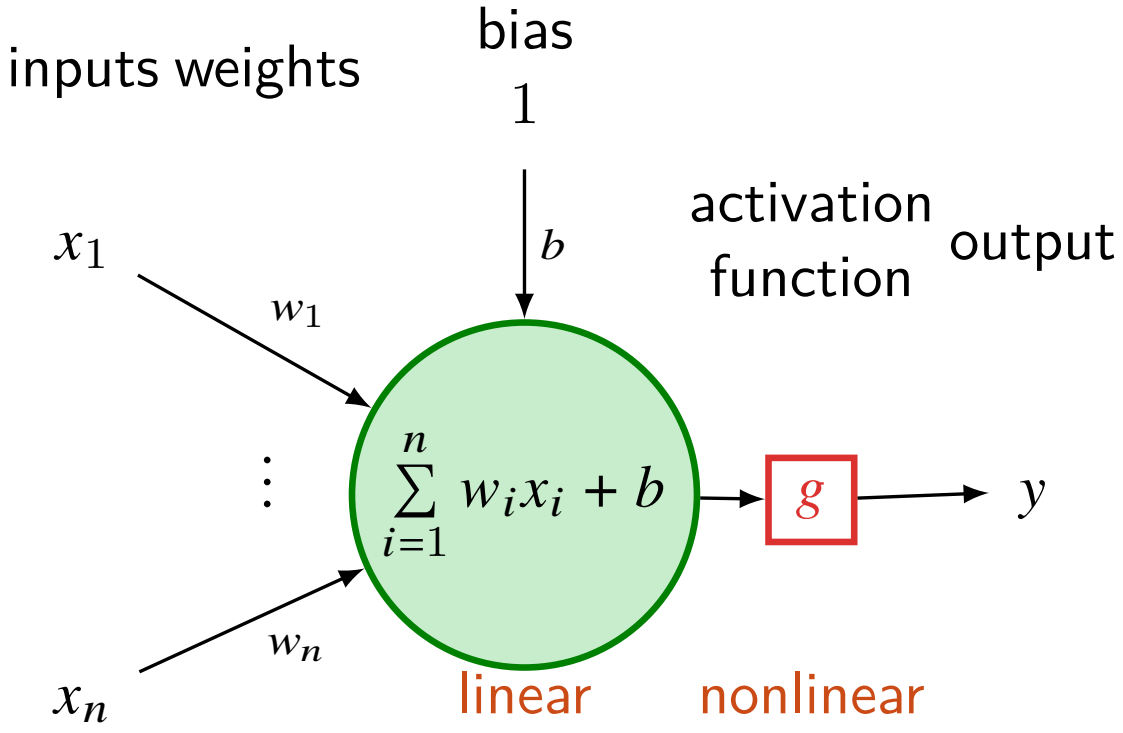
- 有三种不同的联想原则: 相似、时空邻近和因果. 这些原则旨在捕捉想象力将 “简单观念” 重组为 “复杂观念” 时的规律性.
- 记忆、感觉和理解都建立在想象力 (或说我们观念的生动性) 之上.

- 信念 $\nrightarrow$ 知识
- 连接主义通过神经网络节点之间的连接权重来表示知识.
- 贝叶斯网络将知识表示为概率分布 (信念程度).

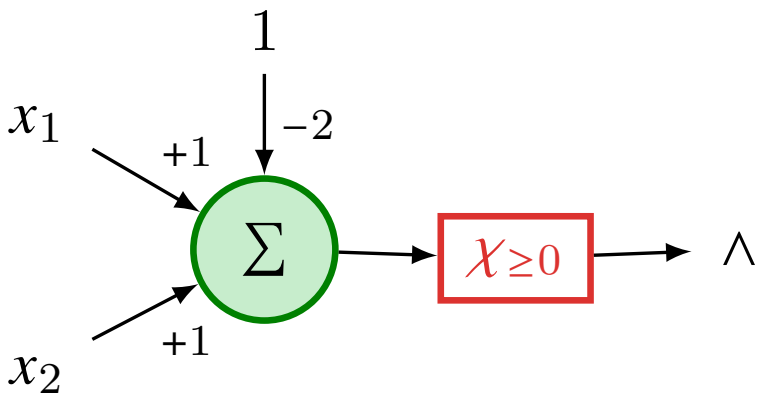


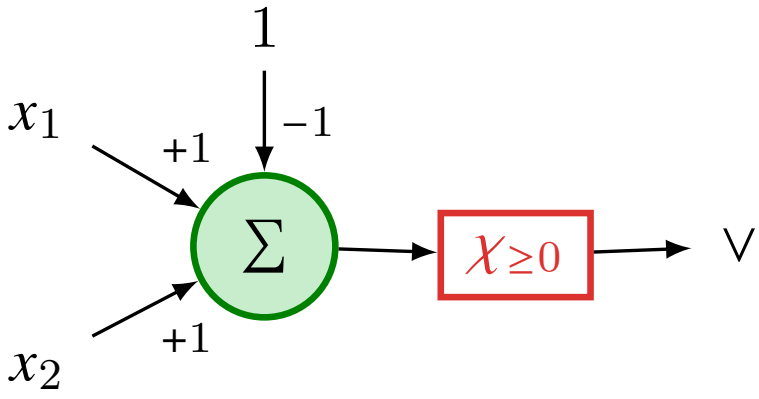
- 人脑有 1000 亿脑细胞, 100 万亿动态连接.
- 神经元通过树突接收电信号, 并通过轴突发射出去

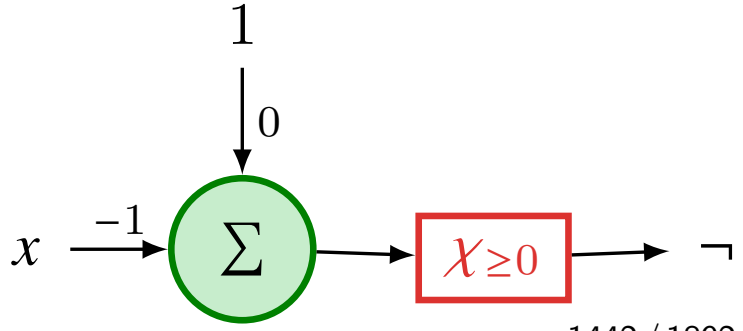
McCulloch-Pitts 人工神经网络 (神经的逻辑演算)

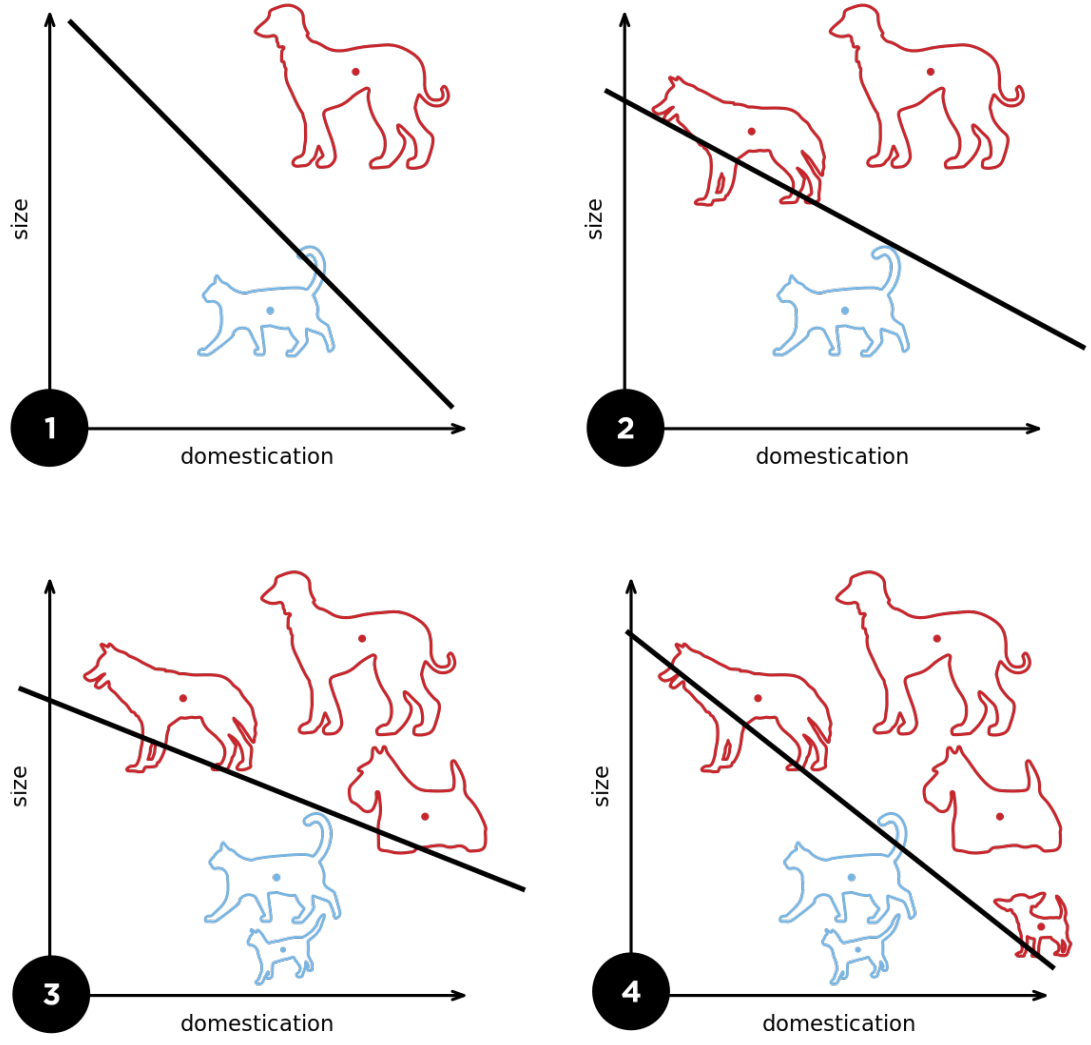


$$y = g\left(\sum_{i=1}^{n} w_i x_i + b\right)$$









1-layer NN $$y = \begin{cases} 1 & \text{if } \sum_{i=1}^{n} w_i x_i + b \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

线性不可分问题

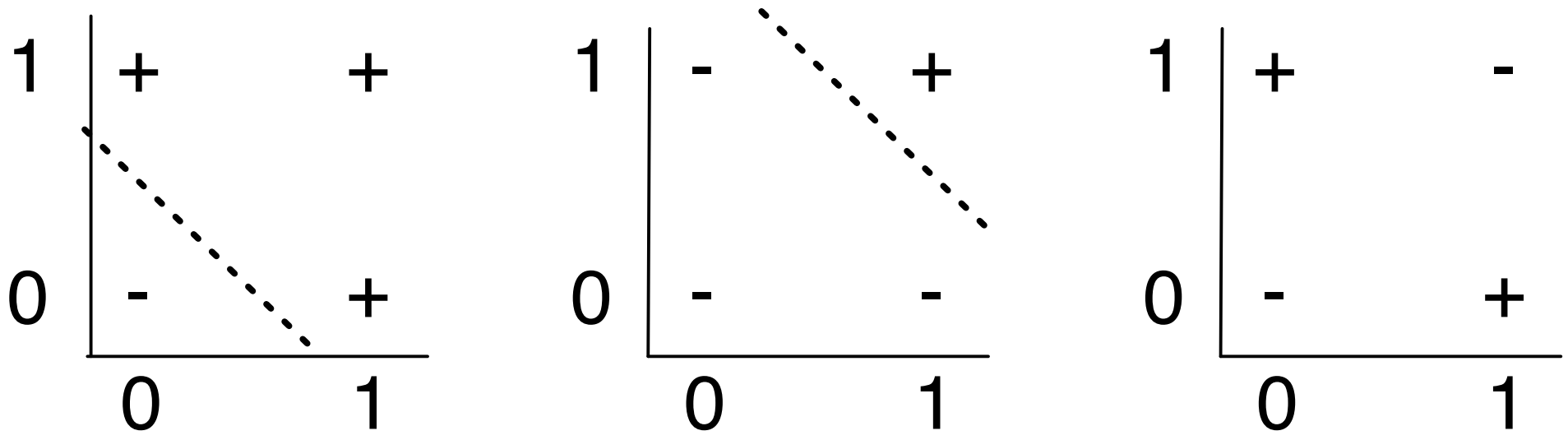


Figure: $\vee, \wedge, \oplus$

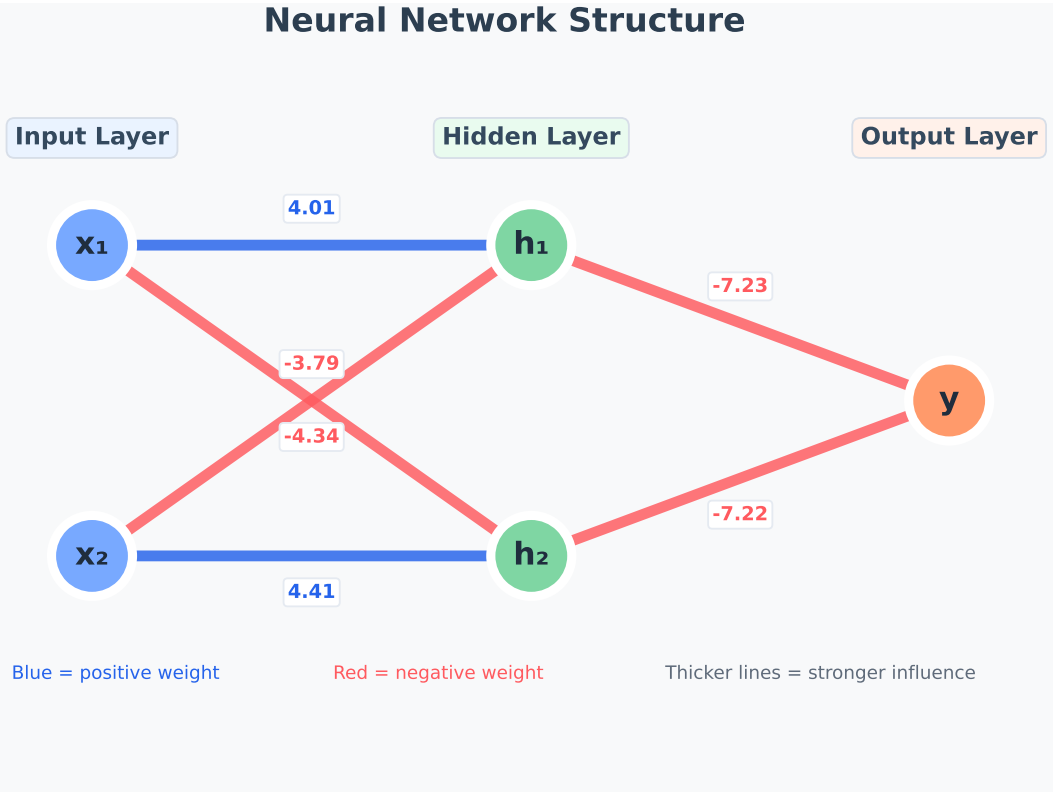
x_1	x_2	$x_1 \oplus x_2$
0	0	0
0	1	1
1	0	1
1	1	0

$$w_1 0 + w_2 0 + b < 0 \qquad b < 0$$
$$w_1 0 + w_2 1 + b \geq 0 \qquad w_2 + b \geq 0$$
$$w_1 1 + w_2 0 + b \geq 0 \qquad w_1 + b \geq 0$$
$$w_1 1 + w_2 1 + b < 0 \qquad w_1 + w_2 + b < 0$$

A simple single-layer perception can't solve nonlinearly separable problems.

异或问题

$$\underbrace{x_1 \oplus x_2}_{y} \equiv \underbrace{(x_1 \wedge \neg x_2)}_{h_1} \vee \underbrace{(\neg x_1 \wedge x_2)}_{h_2}$$



《三体》— 人列计算机[26]

- 秦始皇：朕当然需要预测太阳的运行，但你们让我集结三千万大军，至少要首先向朕演示一下这种计算如何进行吧？
- 冯诺依曼：陛下，请给我三个士兵，我将为您演示．… 我们组建一千万个这样的门部件，再将这些部件组合成一个系统，这个系统就能进行我们所需要的运算，解出那些预测太阳运行的微分方程．

$$F = G\frac{m_1 m_2}{r^2} \longrightarrow a = \frac{F}{m}$$

$$\uparrow \qquad\qquad \downarrow$$

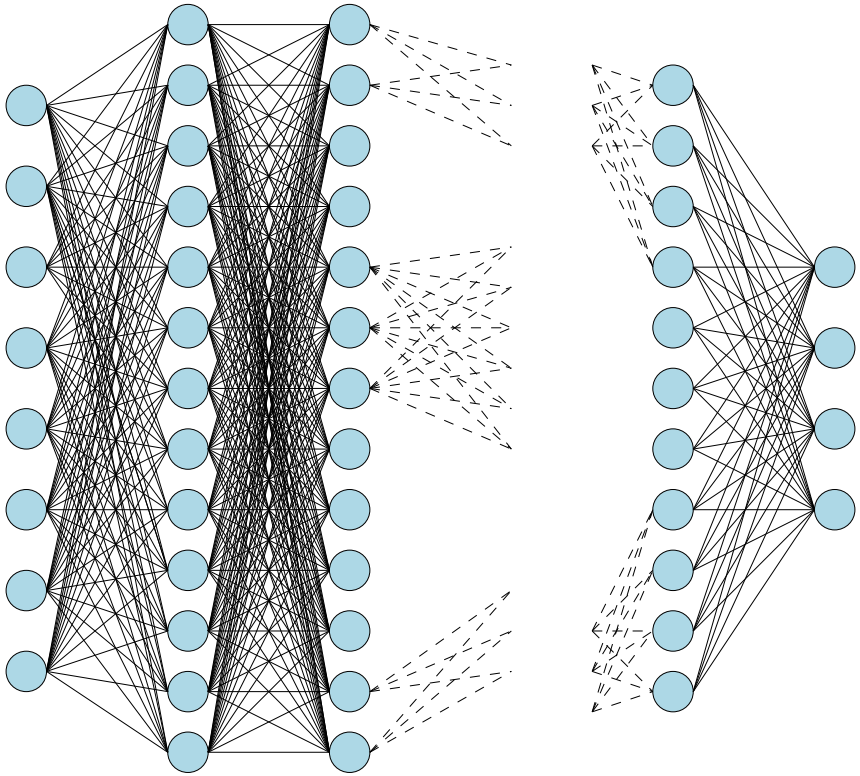
$$\Delta r = v\Delta t \longleftarrow \Delta v = a\Delta t$$

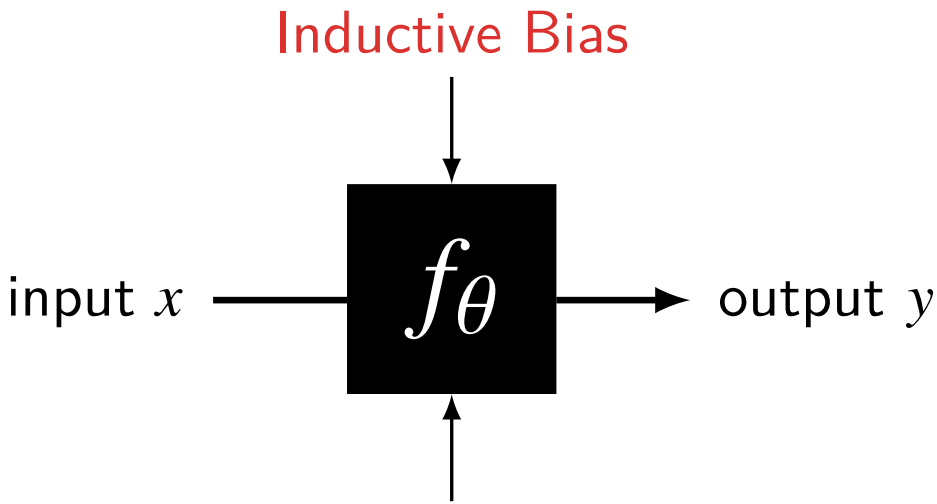
$$\frac{\mathrm{d}^2\mathbf{r_i}}{\mathrm{d}t^2} = -\sum_{j\neq i} Gm_j \frac{\mathbf{r_i}-\mathbf{r_j}}{|\mathbf{r_i}-\mathbf{r_j}|^3} \quad i = 1,2,3$$

用连续信号模拟离散信号会怎样？

[26]牛顿解决了二体问题．但发现 $n \geq 3$ 体不是稳定系统．他曾希望求解出太阳系崩溃的时间，如果到时没崩溃，则说明有超自然力量的干预，从而证明上帝存在 (牛顿的上帝改 Bug 显神迹，莱布尼茨的上帝程序完美前定和谐 ☺)．事实上，太阳系是亚稳态系统．

深度神经网络





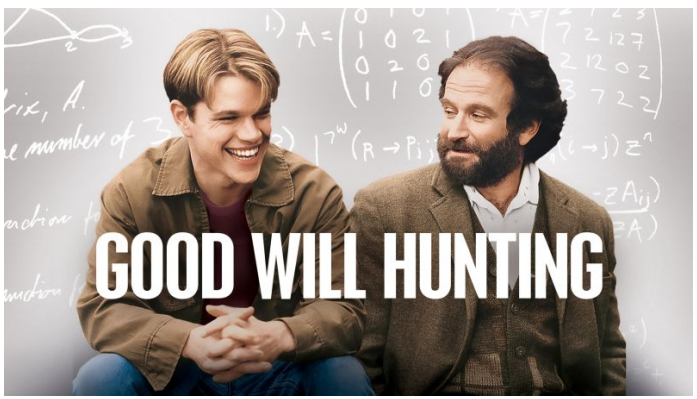
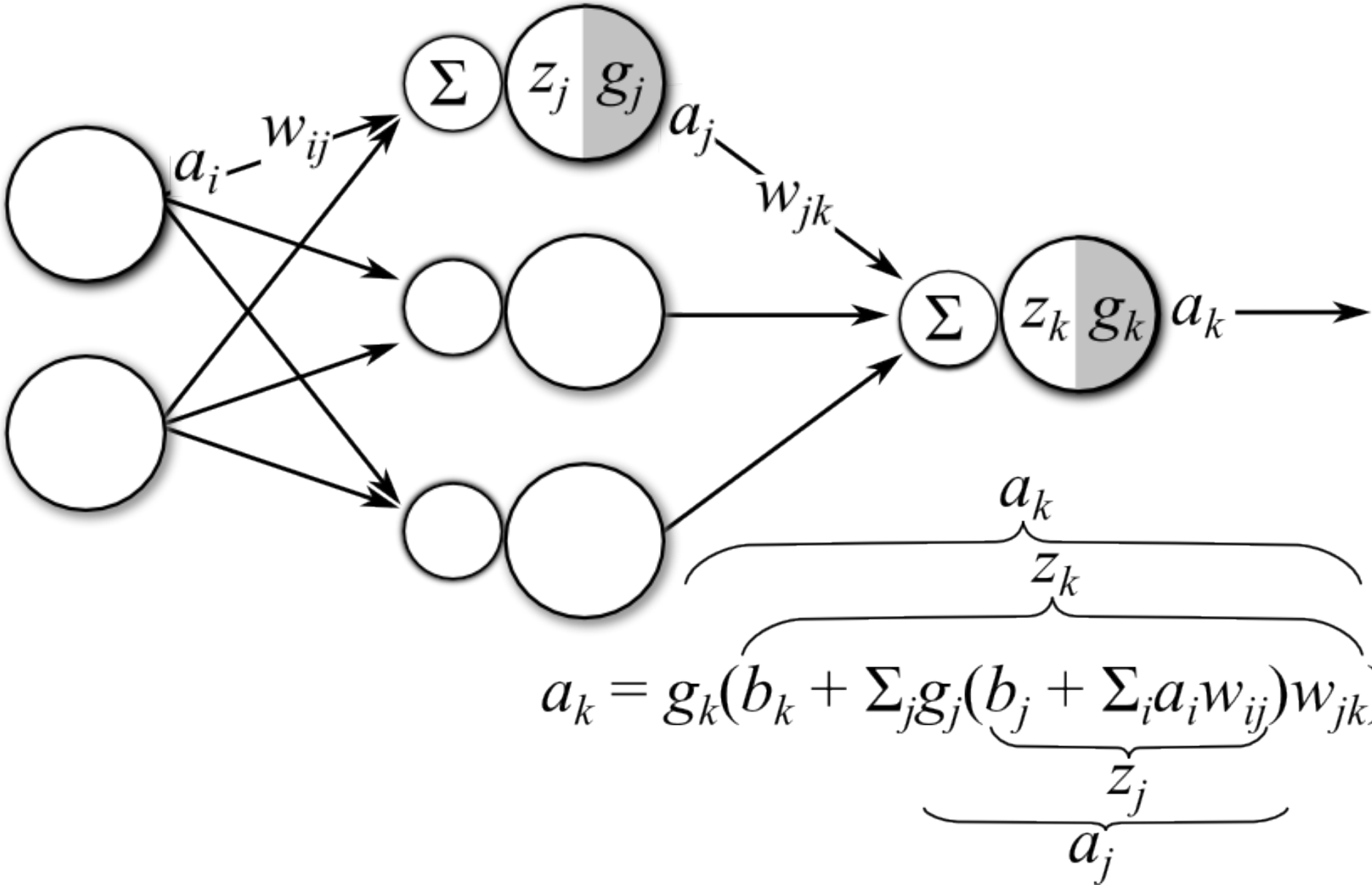


Figure: 电影《心灵捕手》& Walter Pitts & Jeff Hinton



Learning: small change in weights → small change in output

几何视角下的深度学习[27]

流形学习、概率学习

- 自然界中的数据集的内在模式可以看作是嵌入高维背景空间中的某个低维流形上的概率分布.
- 学习的主要任务就是学习流形的拓扑结构和流形上的概率分布.
- 把高维空间映射到隐空间, 本质上是求一个同胚映射, 把数据流形局部映射到隐空间, 这个过程是编码, 从隐空间返回到数据流形的过程是解码. 正则性理论保证编码映射和解码映射是连续的乃至光滑的, 解的唯一性保证这些映射是拓扑同胚或者微分同胚.
- 流形结构学习归结为在欧氏空间上所有映射构成的空间中进行变分.
- 概率分布学习归结为在流形上所有概率分布构成的 Wasserstein 空间中进行带有限制的、关于特殊能量的变分优化.
- 编码解码映射和数据概率分布的传输映射都通过深度神经网络通用逼近.

[27] 雷娜、顾险峰: 《最优传输理论与计算》

Lemma (Urysohn's Lemma)

Let A, B be two non-empty disjoint closed subset of a normal topological space X. There exists a continuous function $f : X \to [0, 1]$ such that $f(A) = 0$ and $f(B) = 1$.

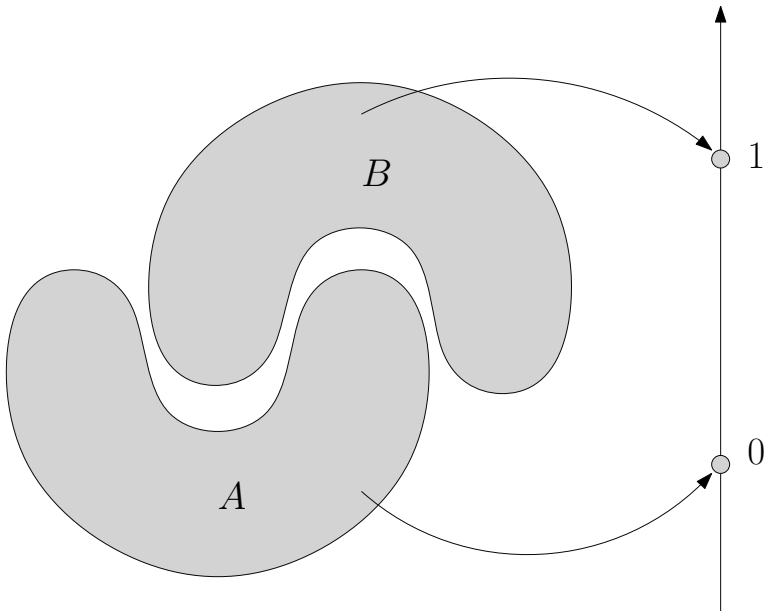


Figure: Urysohn 引理为监督学习、模式识别提供了理论基础.

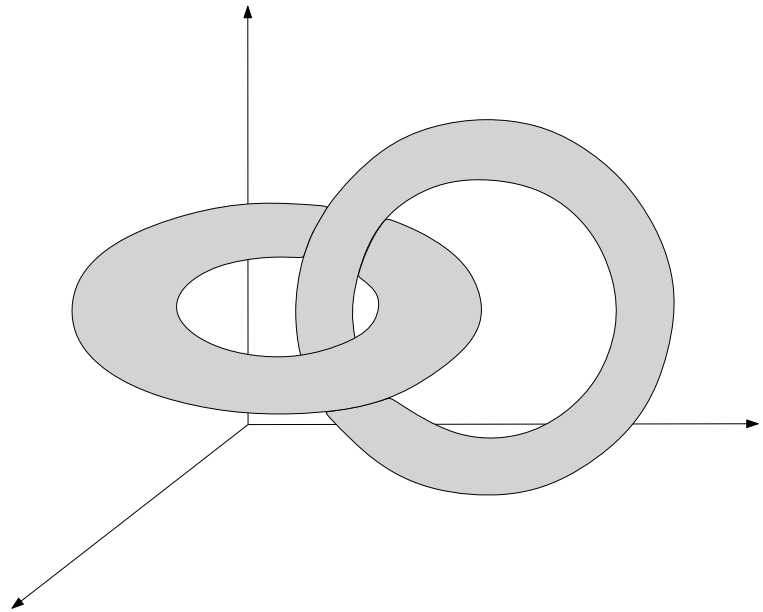


Figure: 提高嵌入空间的维数, 可以为数据流形解套.

Theorem (General Position Theorem)

Any m-manifold unknots in $\mathbb{R}^n$ provided $n \geq 2m + 2$.

流形嵌入定理

如果初始流形嵌入的空间维数过高, 通过改变嵌入空间而实现逐步降维, 直至隐空间.

Theorem (Whitney Embedding Theorem)

Any smooth real n-dimensional manifold (required also to be Hausdorff and second-countable) can be smoothly embedded into $\mathbb{R}^{2n}$.

Remark: Whitney 定理给出了流形嵌入的普适方法: 首先构造流形的一个有限开覆盖 $\{U_i\}$, 得到单位分解 $\{\rho_i\}$; 构造局部嵌入 φ_i 将每个开集 U_i 嵌入到线性子空间 $\mathbb{R}^n$ 中, 用单位分解将局部嵌入合成全局嵌入; 然后进行随机投影, 降低嵌入空间的维数.

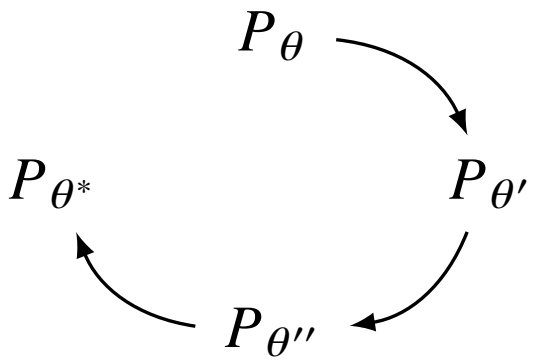
Problem (Hilbert's 13th Problem)

Can every continuous function of n variables be expressed as a composition of finitely many continuous functions of two variables?

Problem

Is it possible to exactly represent any continuous multivariate function $f : \mathbb{R}^n \to \mathbb{R}$ as a combination of continuous univariate functions $\mathbb{R} \to \mathbb{R}$ and the single binary function '+'?

神经网络通过复合简单函数逼近复杂函数.
神经网络的随机梯度下降 SGD ($+$ 注意力机制) 可以看做在程序空间中进行搜索.



Kolmogorov Superposition Theorem

Theorem (Kolmogorov Superposition Theorem)

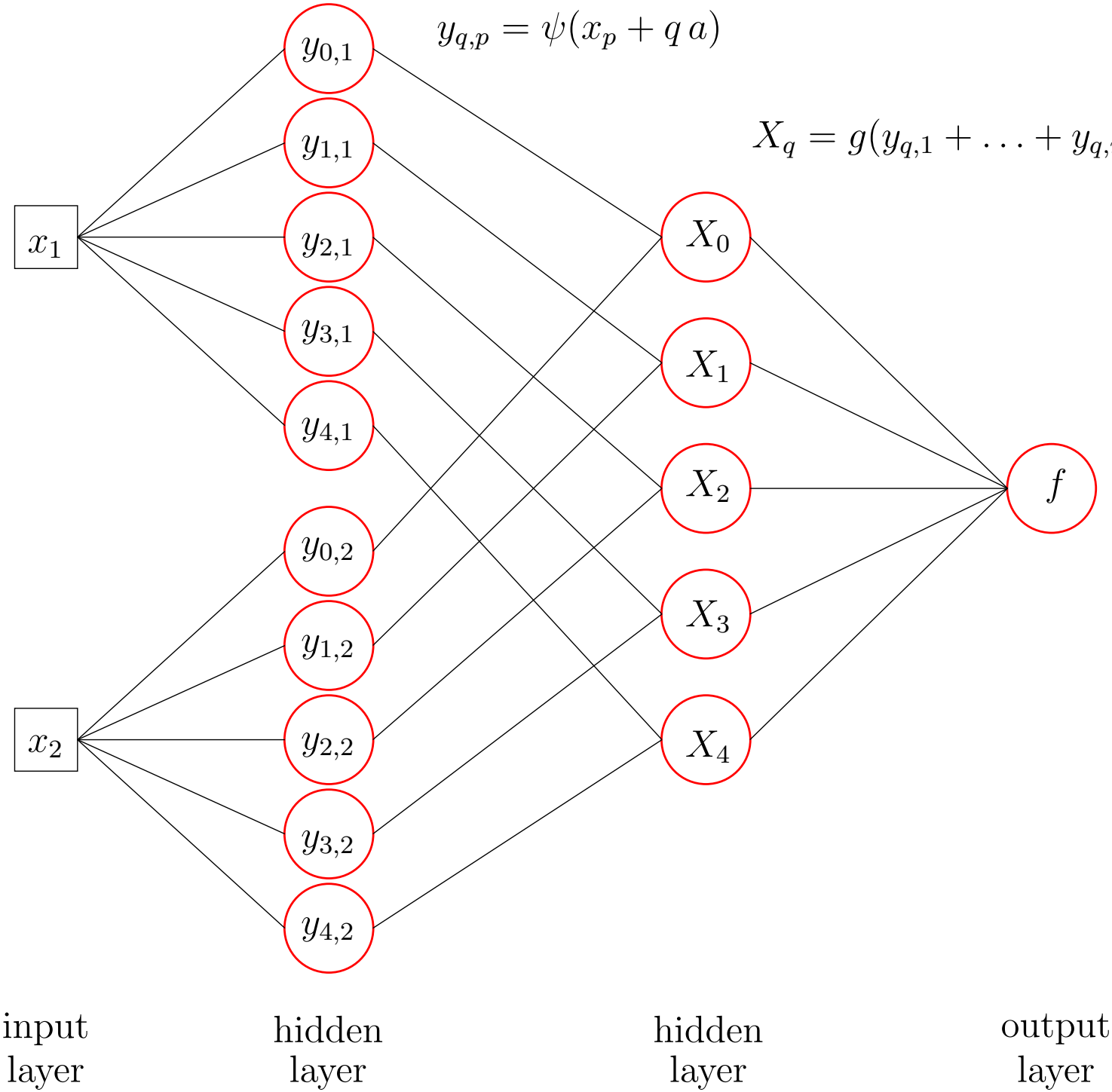
For each $n \geq 2$ there exists a recursive function $\psi : [0,1] \to \mathbb{R}$ and computable constants $a, \lambda_{pq} \in \mathbb{R}$, $p = 1, \ldots, n$, $q = 0, \ldots, 2n$ s.t.: every continuous function $f : [0,1]^n \to \mathbb{R}$ has a representation as

$$f(x_1, \ldots, x_n) = \sum_{q=0}^{2n} g\left(\sum_{p=1}^{n} \lambda_{pq} \psi(x_p + qa)\right)$$

for some continuous function $g : [0,1] \to \mathbb{R}$ that is computable from f.

Theorem (Hecht-Nielsen Theorem)

The class of functions $f : [0,1]^n \to \mathbb{R}$, implementable by three-layer feed-forward neural networks with (computable) continuous activation functions $g : [0,1] \to \mathbb{R}$ and (computable) weights $\lambda \in \mathbb{R}$, is exactly the class of (computable) continuous functions $f : [0,1]^n \to \mathbb{R}$.


$y_{q,p} = \psi(x_p + q\,a)$
$X_q = g(y_{q,1} + \ldots + y_{q,n})$
x_1
x_2
$y_{0,1}$
$y_{1,1}$
$y_{2,1}$
$y_{3,1}$
$y_{4,1}$
$y_{0,2}$
$y_{1,2}$
$y_{2,2}$
$y_{3,2}$
$y_{4,2}$
X_0
X_1
X_2
X_3
X_4
f
input layer
hidden layer
hidden layer
output layer

万能函数拟合器

Theorem (Universal Approximation Theorem)

Let g be a nonconstant, bounded, and increasing continuous function. Let I_n be any compact subset of $\mathbb{R}^n$. The space of continuous functions on I_n is denoted by $C(I_n, \mathbb{R})$. Then, given any function $f \in C(I_n, \mathbb{R})$ and $\varepsilon > 0$, there exists an integer N, real constants $v_i, b_i \in \mathbb{R}$ and real vectors $\mathbf{w}_i \in \mathbb{R}^n$, where $i = 1, \ldots, N$, s.t.

$$\forall \mathbf{x} \in I_n : |f(\mathbf{x}) - h(\mathbf{x})| < \varepsilon$$

where

$$h(\mathbf{x}) \coloneqq \sum_{i=1}^{N} v_i g\left(\mathbf{w}_i^{\mathsf{T}} \mathbf{x} + b_i\right)$$

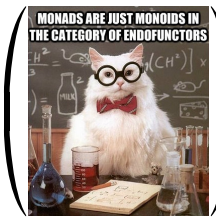
In other words, functions of the form $h(\mathbf{x})$ are dense in $C(I_n, \mathbb{R})$.

Representation & Approximation

A deep neural network (DNN) is a particular kind of function

$$f_\theta : \mathbb{R}^m \to \mathbb{R}^n$$

which depends in a "differentiable" way on a vector of weights.

Example: $f_\theta\left(\right.$ 

$\left.\right) = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix} \in \mathbb{R}^2 \qquad p_{\text{cat}} = 0.8, p_{\text{dog}} = 0.2$

- A feed-forward network with 1 hidden layer can represent any boolean function, but require exponential hidden units.
- A feed-forward network with 2 hidden layers and (computable) continuous activation functions can represent any (computable) continuous function.
- A feed-forward network with a linear output layer and at least 1 hidden layer and continuous and differentiable activation functions can approximate any Borel measurable function from one finite-dimensional space to another with any desired non-zero amount of error.
- A feed-forward network with 2 hidden layers and continuous and differentiable activation functions can approximate any function.

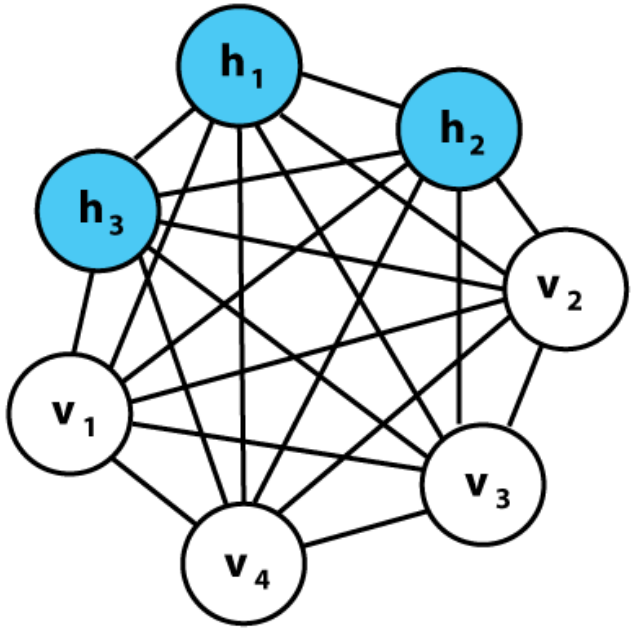
Boltzmann Machine

- The global energy E in a Boltzmann machine is

$$E(x) = -\left(\sum_{i \neq j} w_{ij} x_i x_j + \sum_i b_i x_i\right)$$

where

  - w_{ij} is the connection strength between unit i and unit j.
  - $x_i \in \{0, 1\}$ is the state of unit i.
  - b_i is the bias of unit i.



- Unit i turns on with probability

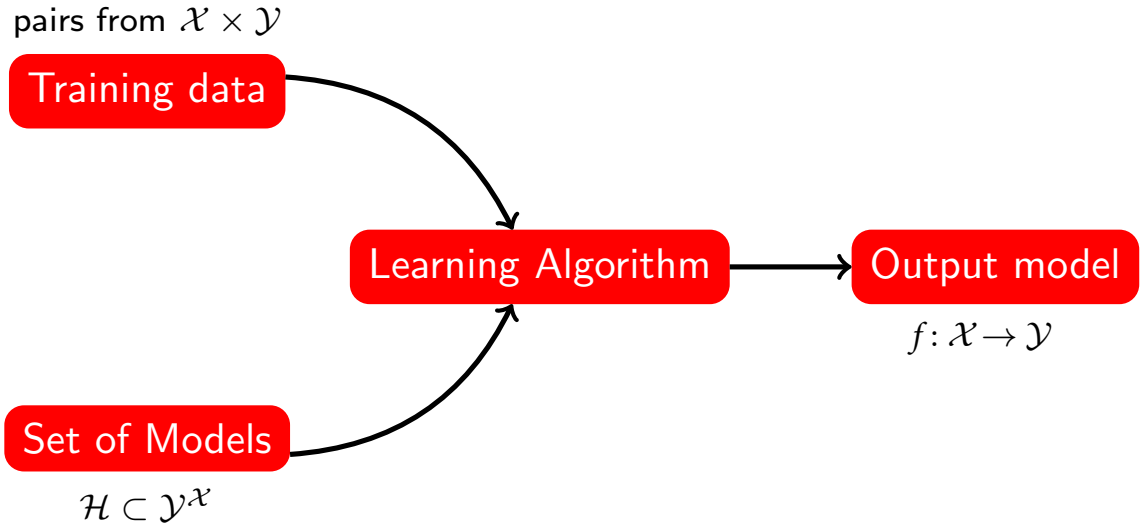
$$P(x_i = 1) = \sigma\left(b_i + \sum_{j \neq i = 0} w_{ij} x_j\right)$$

- The stationary distribution

$$P(x) = \frac{\mathrm{e}^{-E(x)}}{Z}$$

Remark: Voter model for opinion dynamics / Ising model / Hopfield network / Restricted Boltzmann Machine

Deep Learning[GBC16]



☺ 要把大象装冰箱, 总共分几步?

1. hypothesis space — Network Structure — f_θ
2. the goodness of a function — Learning Target — loss function ℓ
3. pick the best function — Learn — find the network parameters $\theta^* := \underset{\theta}{\operatorname{argmin}} L(\theta)$ that minimize total cost $L(\theta)$ by gradient decent

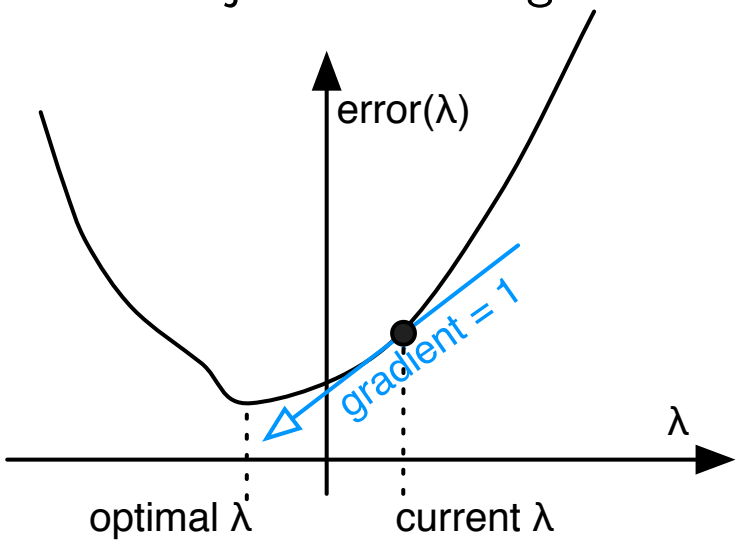
$$\theta \leftarrow \theta - \eta \nabla_\theta L(\theta)$$

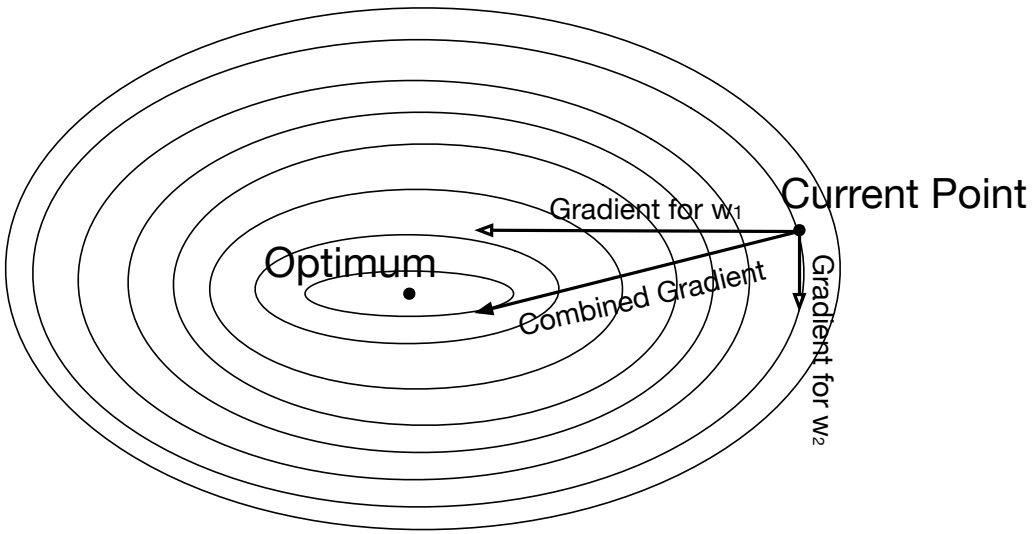
where $L(\theta) := \mathbb{E}_P \left[\ell \left(f_\theta(\mathbf{a}), t \right) \right] + \lambda \Omega(\theta)$ and $\Omega(\theta)$ is a regularizer.

Error

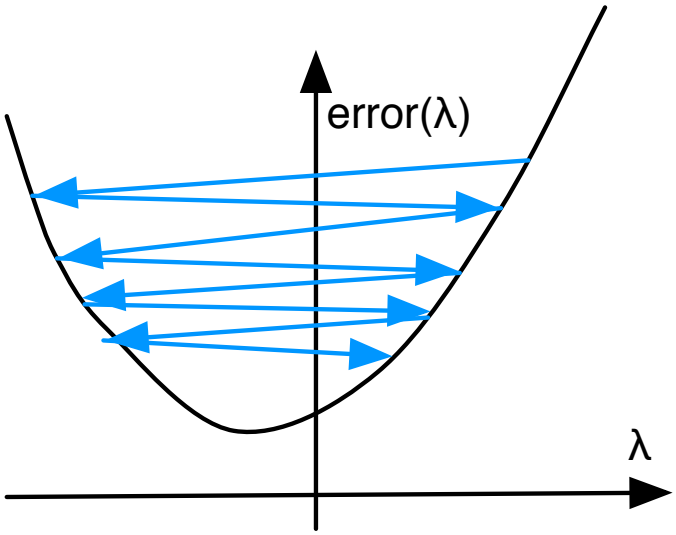
How do we adjust the weights?

- Gradient descent
  - error is a function of the weights
  - we want to reduce the error
  - gradient descent: move towards the error minimum
  - compute gradient: get direction to the error minimum
  - adjust weights towards direction of lower error
- Back-propagation
  - first adjust last set of weights
  - propagate error back to each previous layer
  - adjust their weights

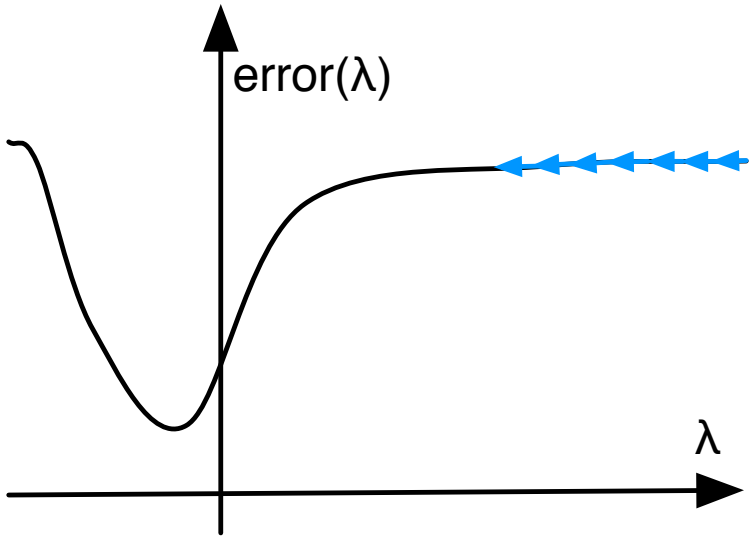




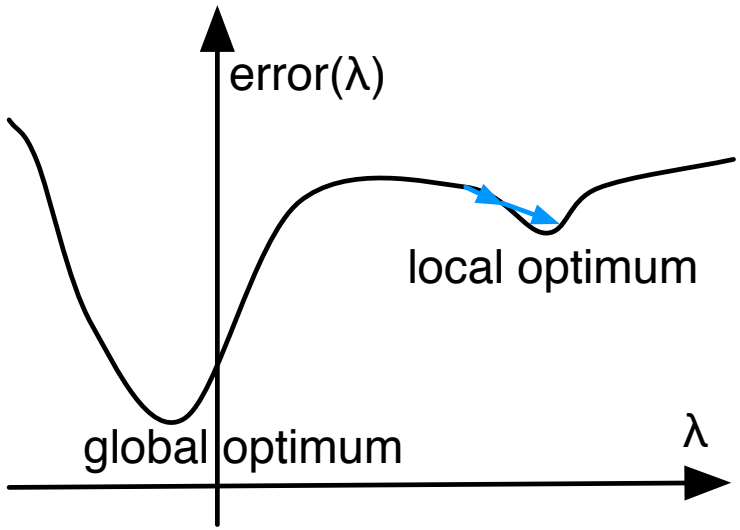
Problems with Gradient Descent Training



Too high learning rate



Bad initialization



Local optimum

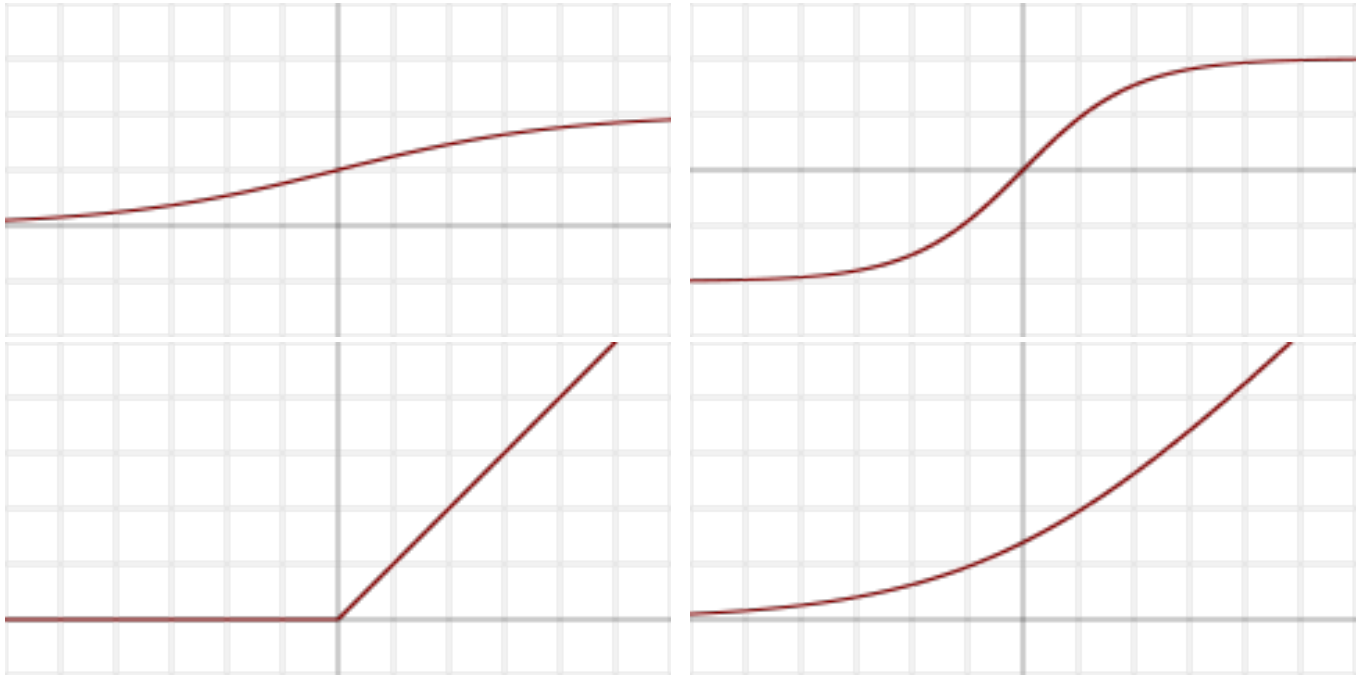
$$f_\theta : \mathbf{a}^{(0)} \mapsto \mathbf{g}^{(n)}\Bigg(\cdots \mathbf{g}^{(2)}\Bigg(\overbrace{\mathbf{w}^{(2)}\, \mathbf{g}^{(1)}\Big(\underbrace{\mathbf{w}^{(1)}\mathbf{a}^{(0)} + \mathbf{b}^{(1)}}_{\mathbf{z}^{(1)}} \Big)}^{\mathbf{a}^{(1)}} + \mathbf{b}^{(2)} \Bigg) \cdots \Bigg)$$

$\mathbf{a}^{(2)}$ over $\mathbf{z}^{(2)}$; $\mathbf{z}^{(n)}$ over $\mathbf{a}^{(n)}$

where parameters $\theta := \left\{\mathbf{w}^{(i)}, \mathbf{b}^{(i)}\right\}_{i=1}^{n}$ and $\mathbf{g}$ is the activation function.

$$\left.\begin{aligned} a_0^{(l)} &:= 1 \\ w_{0j}^{(l)} &:= b_j^{(l)} \end{aligned}\right\} \implies \begin{cases} z_j^{(l+1)} := \sum_i w_{ij}^{(l+1)} a_i^{(l)} \\ a_j^{(l+1)} := g_j^{(l+1)}\left(z_j^{(l+1)}\right) \end{cases}$$

Activation Functions



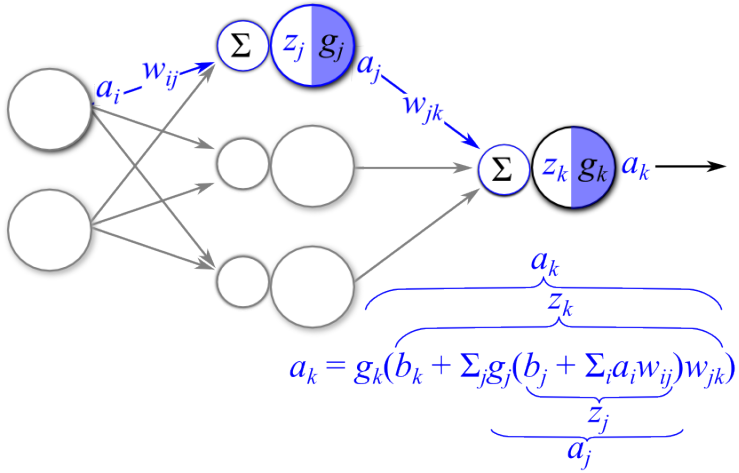
$$\sigma(z) = \frac{1}{1 + \mathrm{e}^{-z}}$$

$$\tanh(z) = \frac{\mathrm{e}^{z} - \mathrm{e}^{-z}}{\mathrm{e}^{z} + \mathrm{e}^{-z}}$$

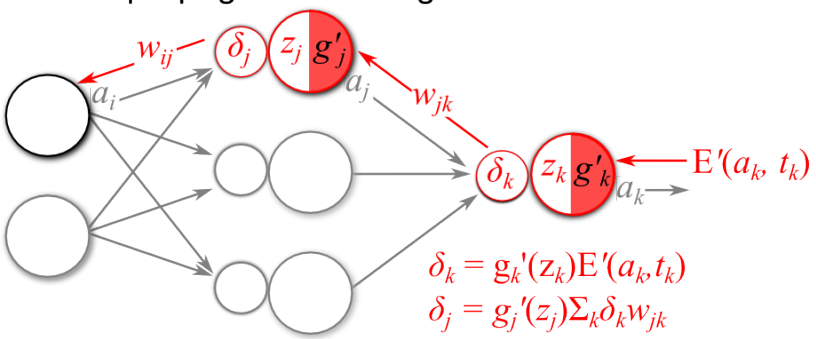
$$\mathrm{ReLU}(z) = \max(0, z)$$

$$\mathrm{softplus}(z) = \log 1 + \mathrm{e}^{z}$$

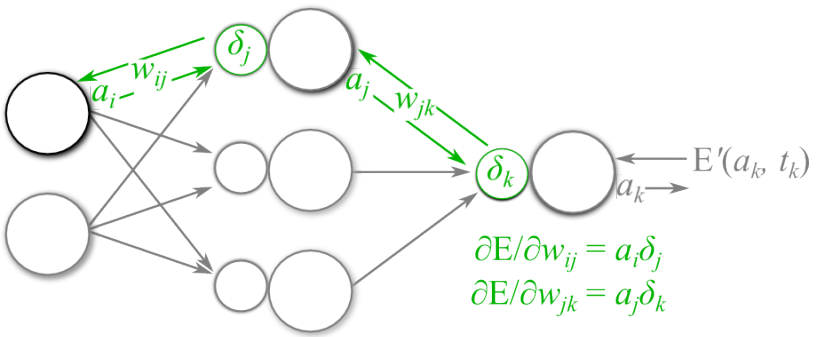
I. Forward-propagate Input Signal



II. Back-propagate Error Signals



III. Calculate Parameter Gradients



IV. Update Parameters

$w_{ij} = w_{ij} - \eta(\partial E/\partial w_{ij})$
$w_{jk} = w_{jk} - \eta(\partial E/\partial w_{jk})$
for learning rate η

Backpropagation

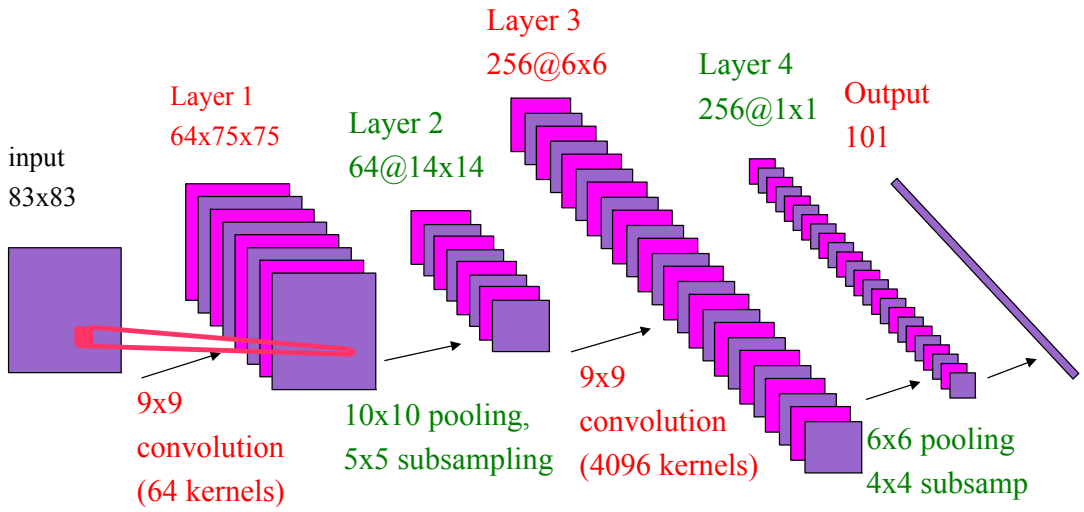
$$\delta_j^{(l)} := \frac{\partial L}{\partial z_j^{(l)}}$$

$$\delta_j^{(l)} = g_j^{(l)\prime}\left(z_j^{(l)}\right) \sum_k \delta_k^{(l+1)} w_{jk}^{(l+1)}$$

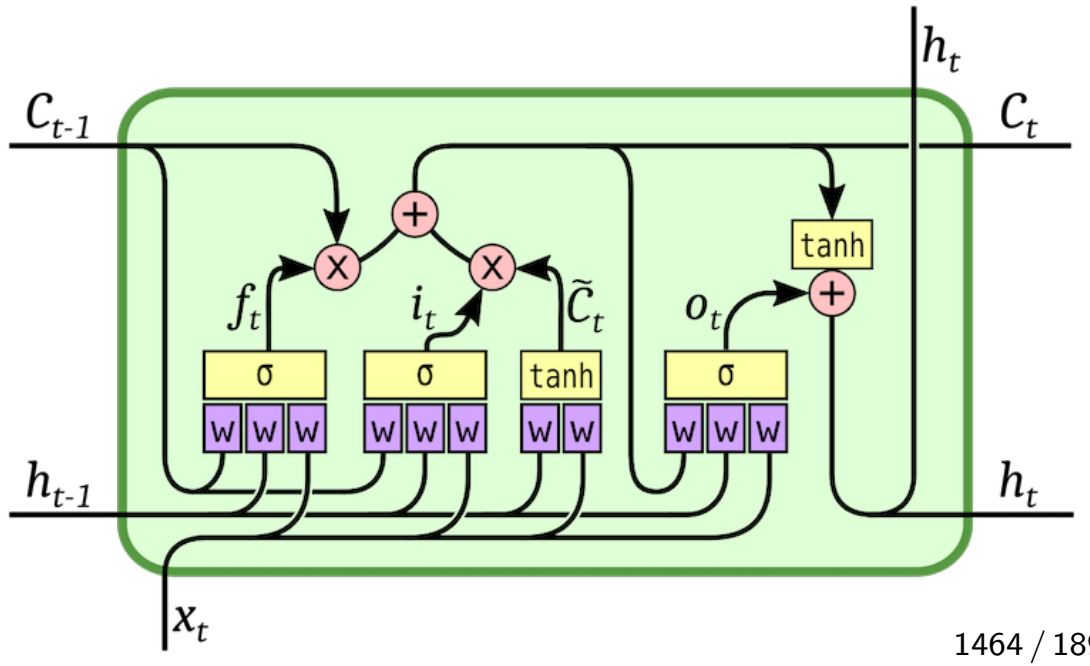
$$\frac{\partial L}{\partial w_{ij}^{(l)}} = a_i^{(l-1)} \delta_j^{(l)}$$

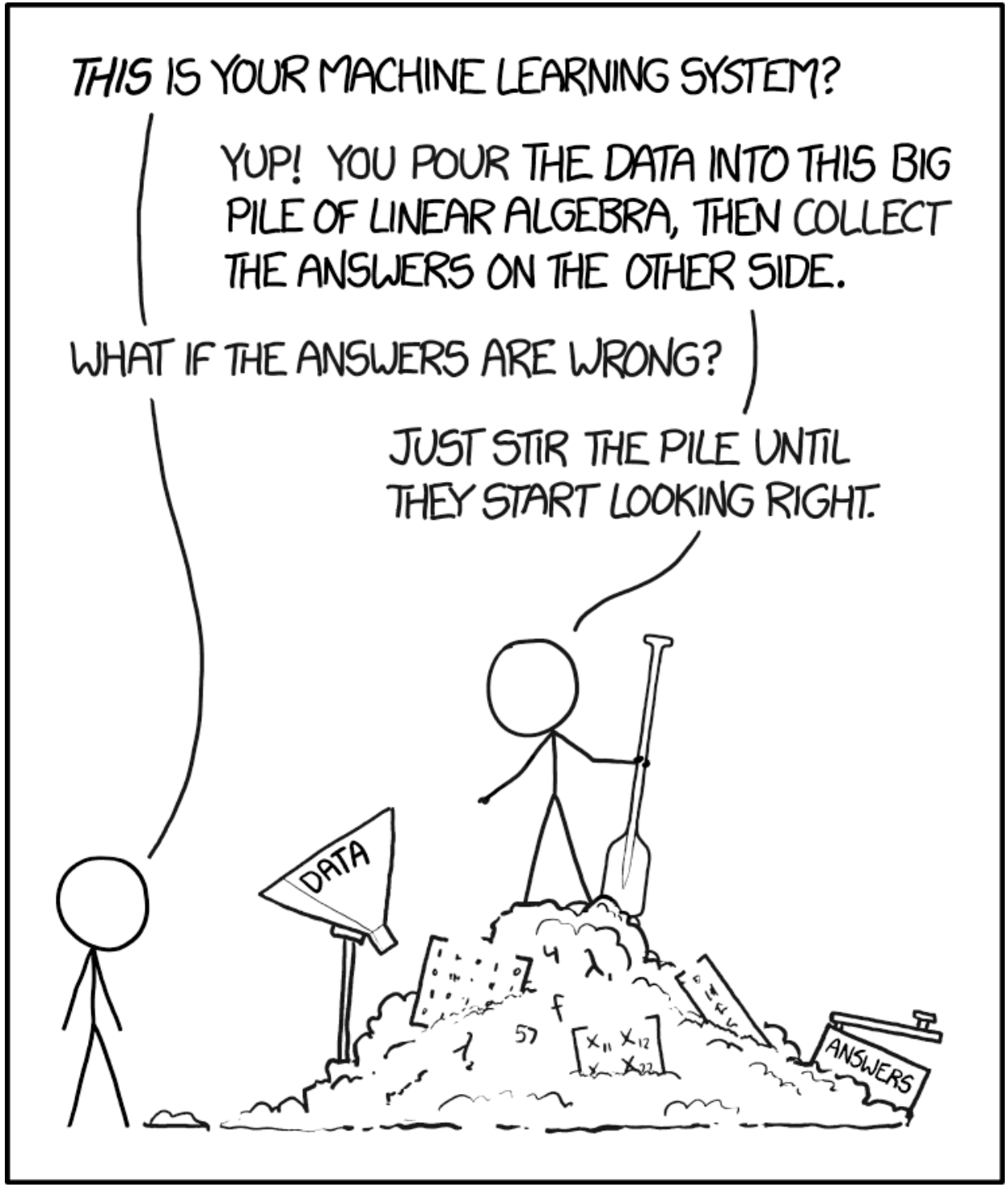
$$w_{ij}^{(l)} \leftarrow w_{ij}^{(l)} - \eta \frac{\partial L}{\partial w_{ij}^{(l)}}$$

- network structure?
- how many layers?
- how many units per layer?
- loss function?
- regularization?
- weight decay?
- learning rate?
- activation function?
- early stopping?
- dropout?
- mini-batch?
- momentum?
- ...

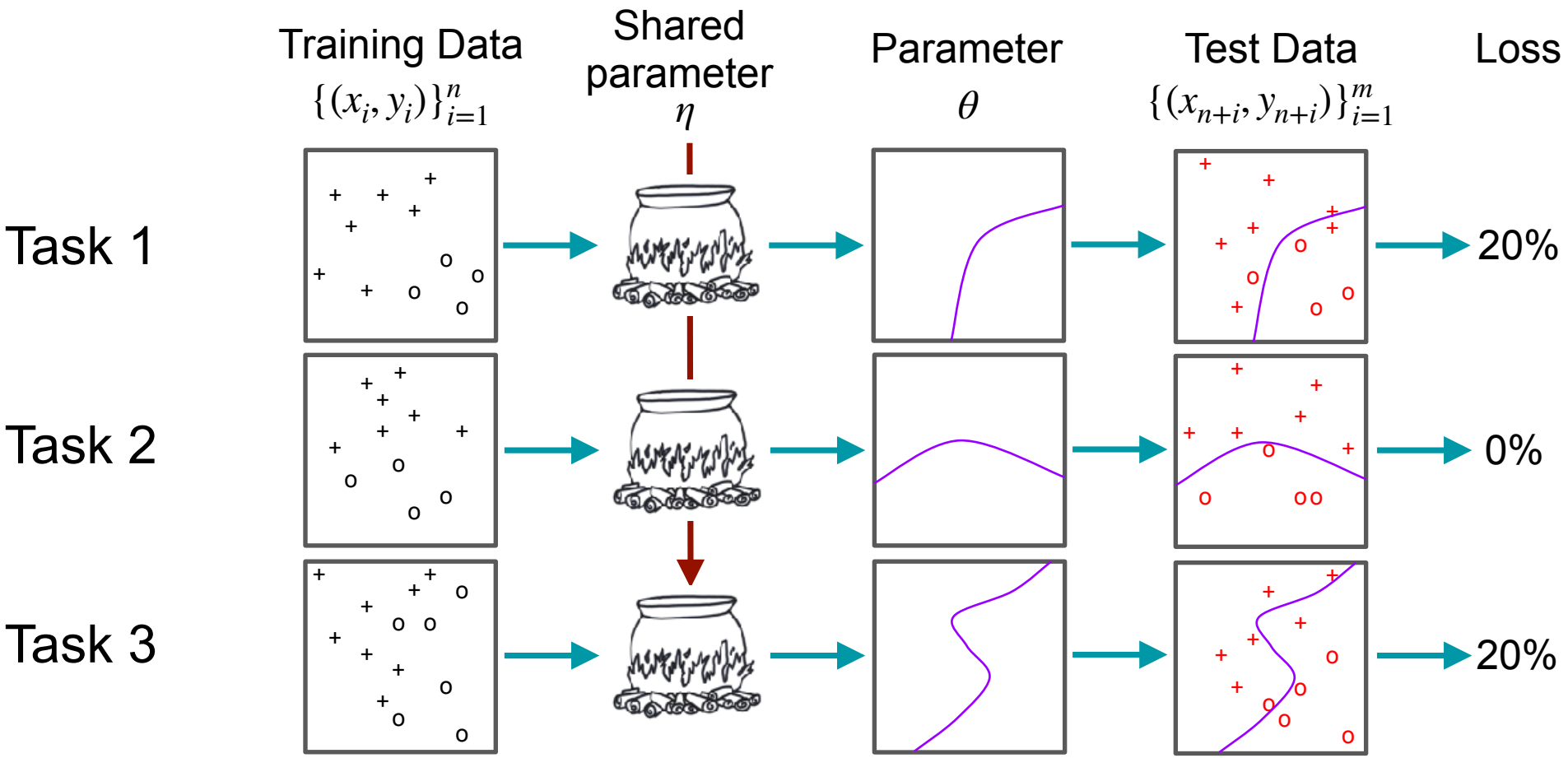


$$a_{ij}^{(l+1)} = g_{ij}^{(l+1)} \left(\sum_{m=0}^{k-1} \sum_{n=0}^{k-1} w_{m,n} a_{i+m,j+n}^{(l)} + b \right)$$




THIS IS YOUR MACHINE LEARNING SYSTEM?
YUP! YOU POUR THE DATA INTO THIS BIG PILE OF LINEAR ALGEBRA, THEN COLLECT THE ANSWERS ON THE OTHER SIDE.
WHAT IF THE ANSWERS ARE WRONG?
JUST STIR THE PILE UNTIL THEY START LOOKING RIGHT.
DATA
ANSWERS

Single-Task Learning → Multi-Task Learning



$$\underset{\eta,\{\theta_j\}}{\operatorname{argmin}} \sum_{j=1}^{T} \sum_{i=1}^{n} \operatorname{Loss}\left(f_{\eta,\theta_j}(x_{ji}), y_{ji}\right) + \lambda\Omega(\theta_j)$$

Key Properties of CNNs

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

$*$

1	-1	-1
-1	1	-1
-1	-1	1

$=$

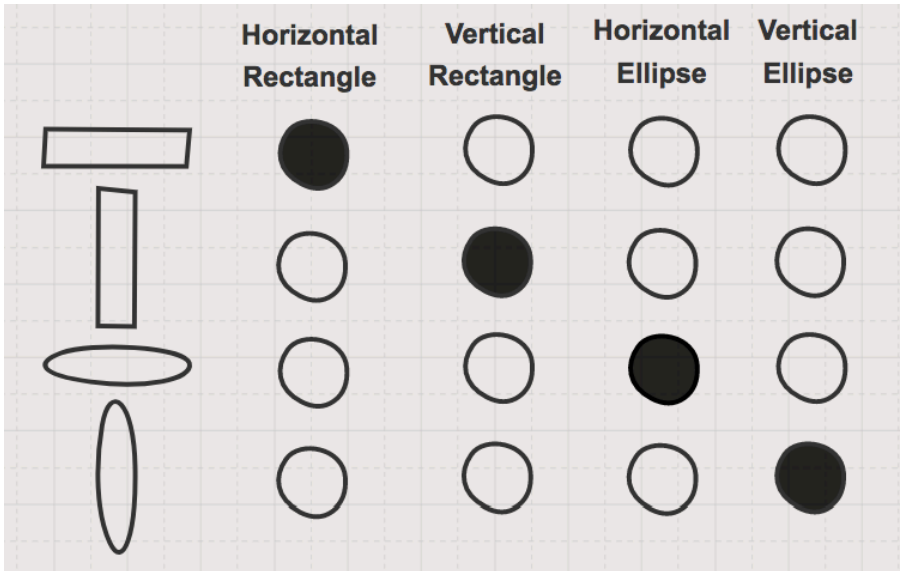
3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

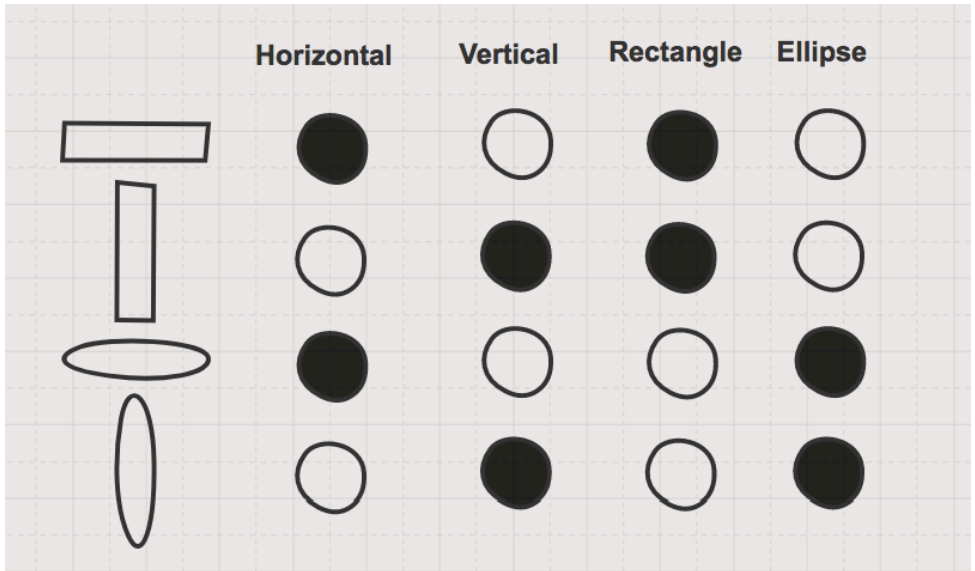
Table: Convolution (stride 1)

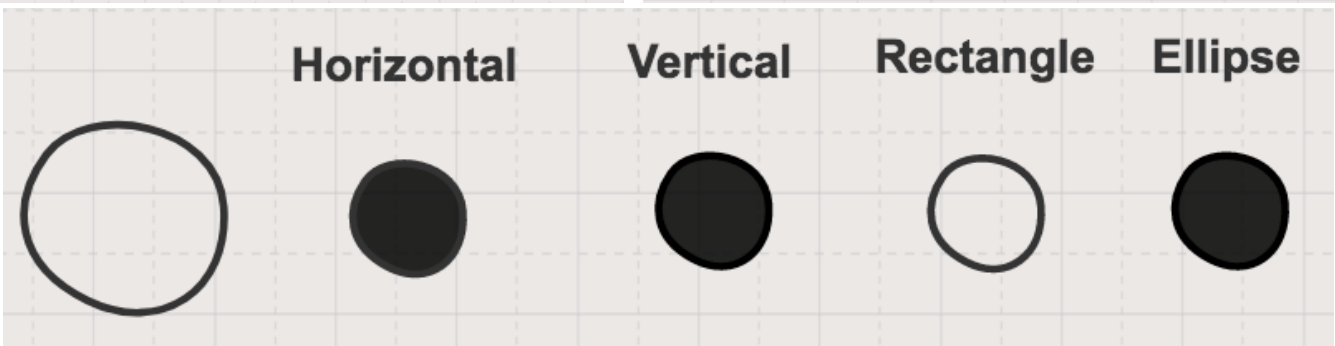
Take advantage of the structure of the data!

- Convolutional Filters **(Translation invariance)**
- Multiple layers **(Compositionality)**
- Filters localized in space **(Locality)**
- Weight sharing **(Self-similarity)**

How Neural Networks Learn Distributed Representations







Number	Local Representation	Distributed Representation
0	10000000	000
1	01000000	001
2	00100000	010
3	00010000	011
4	00001000	100
5	00000100	101
6	00000010	110
7	00000001	111

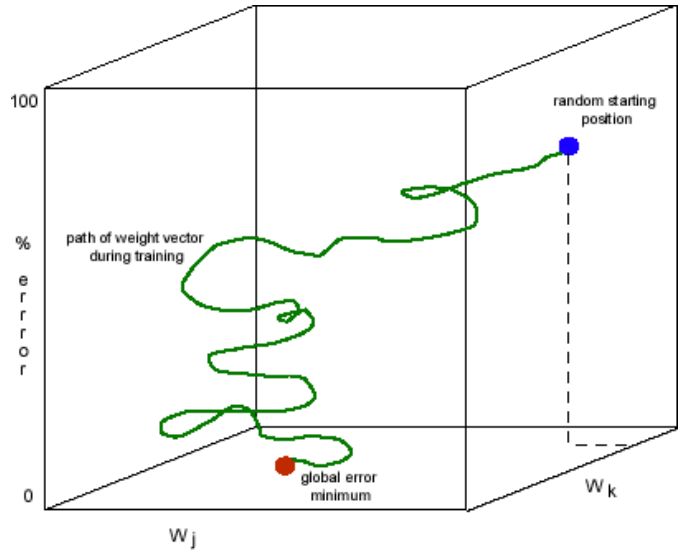
分布式表征

两类表征之间 (例如 “概念” 与 “神经元”) 存在多对多的关系.

- 每个概念由许多神经元表示.
- 每个神经元参与许多概念的表示.

表征	分布式/向量	符号
样本量	许多	较少
数据形式	表格	灵活
模型形式	固定形式	通用程序
可解释性	较难	较简单
知识迁移	较难	较简单
复杂度	较低	较高

Epistemology — A Neurocomputational Perspective



- 知识不过是 “精心调参的连接权重”(突触权重空间中的一个点), 不是一堆或一串存储的符号项.
- Churchland's Eliminative Materialism: 意向性不存在. 心灵状态不存在.
- 大脑神经网络对外界信息的记录是整体性的, 同时记录了 “兔子吃草”“草是绿的”“头戴草帽” 等等信息. 没有某个或某几个神经元表征 “兔子” 概念. 命题态度、信念、欲望、qualia 等等都没有神经元对应.

Why "Deep" rather than "Fat"?

- Exploiting compositionality gives an exponential gain in representational power.
  - Distributed representations: feature learning
  - Deep architecture: multiple levels of feature learning
- Each basic classifier can be trained by little data.
  - deep $\longrightarrow$ modularization $\longrightarrow$ less training data?
    With more complex features, the number of parameters in the linear layers may be drastically decreased.
  - efficiency & sample complexity
  - better memory/computation trade-off?
- higher-level abstractions $\longrightarrow$ easier generalization & transfer

Minimal Sufficient Statistic

Definition (Sufficient Statistic)

Let Y be a parameter indexing a family of probability distributions. Let X be random variable drawn from a probability distribution determined by Y. $T(X)$ is a sufficient statistic for Y iff X is independent of Y given $T(X)$, i.e. $p(x \mid t, y) = p(x \mid t)$.

Definition (Minimal Sufficient Statistic)

A sufficient statistic $S(X)$ is minimal iff for any sufficient statistic $T(X)$, there exists a function f s.t. $S = f(T)$ almost everywhere w.r.t X.

Theorem

- *T is sufficient statistics for Y* $\iff$ $I(T(X); Y) = I(X; Y)$.
- *S is minimal sufficient statistics for Y* $\implies$ $I(X; S(X)) \leq I(X; T(X))$.

Information Bottleneck — Learning is to forget!

Can we explain learning in deep neural networks?

Theorem

Let X be a sample drawn according to a distribution determined by the random variable Y. The set of solutions to

$$\min_{T} I(X;T) \quad s.t. \quad I(T;Y) = \max_{T'} I(T';Y)$$

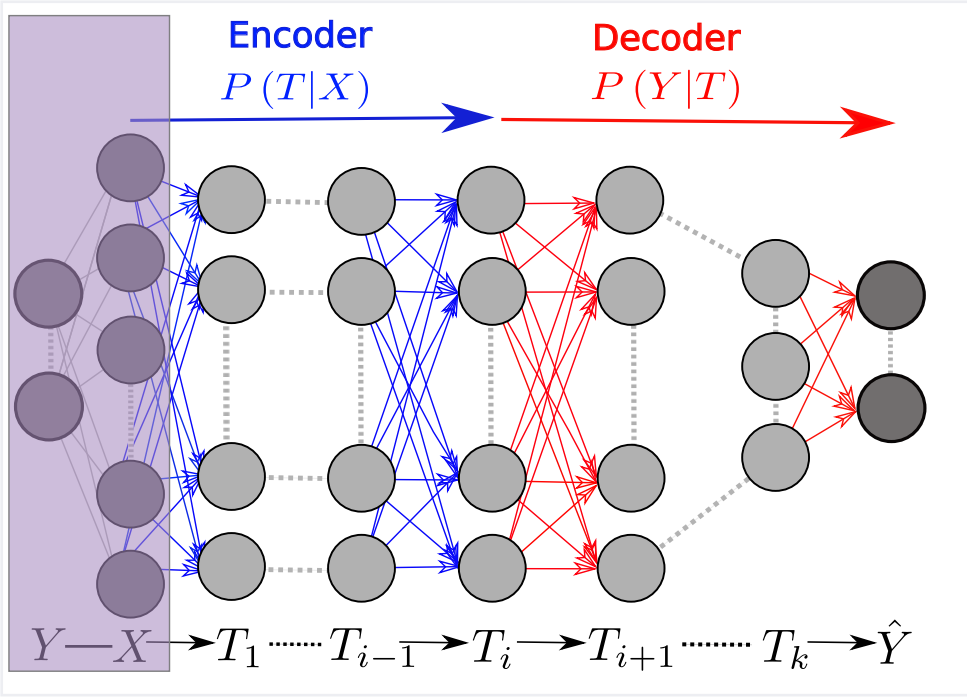
is exactly the set of minimal sufficient statistics for Y based on X.

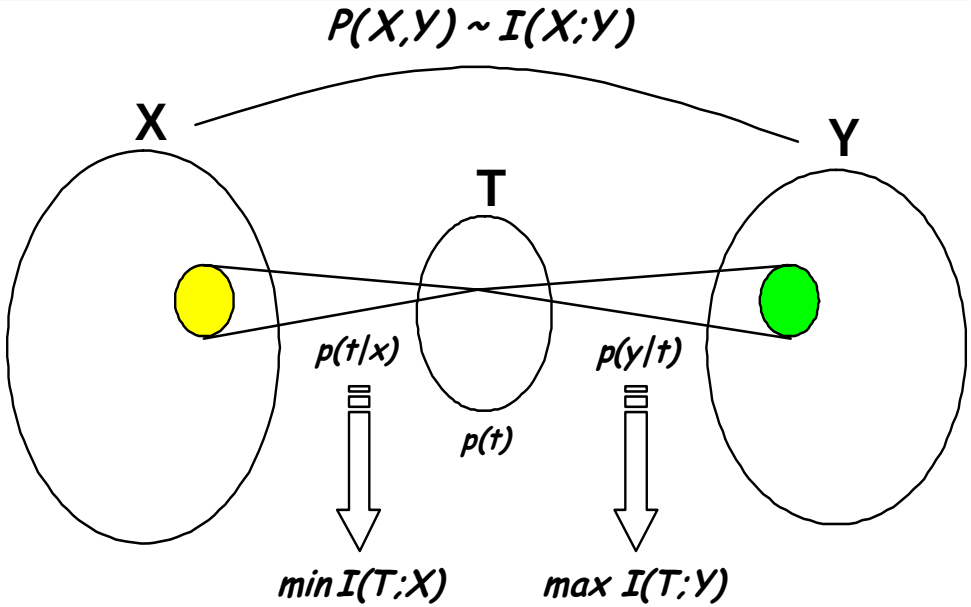
Find a random variable T s.t.:

- $Y \leftrightarrow X \leftrightarrow T$ form a Markov chain.
- $I(X;T)$ is minimized (minimality, complexity term), while $I(T;Y)$ is maximized (sufficiency, accuracy term).

$$\boxed{T^* := \operatorname*{argmin}_{T:I(T(X);Y)=I(X;Y)} I(X;T(X))}$$

is the Information Bottleneck between X and Y.





张三丰：将所见到的剑招忘得半点不剩，才能得其神髓.

老子：为学日益，为道日损.
损之又损，以至于无为.
无为而无不为.

广中平祐：记住和忘记相当于将能力拉扯又放松，可以让能力变得更有弹性.

Information Bottleneck

$$\min_{p(t|x),p(t),p(y|t)} \Big\{ I(X;T) - \beta I(T;Y) \Big\} \text{ subject to Markov chain } Y \to X \to T.$$

$$L[p(t \mid x)] := I(X;T) - \beta I(T;Y) - \sum_x \lambda(x) \sum_t p(t \mid x)$$

Let

$$\frac{\delta L}{\delta p(t \mid x)} = 0$$

The solution is

$$p(t \mid x) = \frac{p(t)}{Z(x,\beta)} \mathrm{e}^{-\beta D[p(y|x) \| p(y|t)]}$$

$$p(t) = \sum_x p(t \mid x) p(x)$$

$$p(y \mid t) = \frac{1}{p(t)} \sum_x p(t \mid x) p(x,y)$$

The Information Bottleneck Method[28]

Algorithm The Information Bottleneck Method

Given $p(x, y)$, $\beta \geq 0$

Initialize $p^{(0)}(t \mid x)$

$p^{(0)}(t) = \sum_x p^{(0)}(t \mid x) p(x)$

$p^{(0)}(y \mid t) = \frac{1}{p^{(0)}(t)} \sum_x p^{(0)}(t \mid x) p(x, y)$

$n = 0$

while not converged **do**

$n = n + 1$

$p^{(n)}(t \mid x) = \frac{p^{(n-1)}(t)}{Z(x,\beta)} \mathrm{e}^{-\beta D\left[p(y|x) \| p^{(n-1)}(y|t)\right]}$

$p^{(n)}(t) = \sum_x p^{(n)}(t \mid x) p(x)$

$p^{(n)}(y \mid t) = \frac{1}{p^{(n)}(t)} \sum_x p^{(n)}(t \mid x) p(x, y)$

end while

[28] Schwartz-Ziv, Tishby: Opening the black box of Deep Neural Networks via Information.

The Deterministic Information Bottleneck

$$L_\alpha[p(t \mid x)] \coloneqq H(T) - \alpha H(T \mid X) - \beta I(T;Y) - \sum_x \lambda(x) \sum_t p(t \mid x)$$

$$p(t \mid x) = \frac{1}{Z(x,\alpha,\beta)} \mathrm{e}^{\frac{1}{\alpha}\left(\log p(t) - \beta D[p(y|x)\|p(y|t)]\right)}$$

Obviously, $L_1 = L$.

$$p_\alpha(t \mid x) = \frac{1}{Z(x,\alpha,\beta)} \mathrm{e}^{\frac{1}{\alpha}(\log p_\alpha(t) - \beta D[p(y|x)\|p_\alpha(y|t)])}$$

$$p_\alpha(t) = \sum_x p_\alpha(t \mid x) p(x)$$

$$p_\alpha(y \mid t) = \frac{1}{p_\alpha(t)} \sum_x p_\alpha(t \mid x) p(x,y)$$

Let $\alpha \to 0$, we get the deterministic case

$$p(t \mid x) = \lim_{\alpha \to 0} p_\alpha(t \mid x) = \left[\!\left[t = \operatorname*{argmax}_t \left(\log p(t) - \beta D[p(y \mid x)\|p(y \mid t)] \right) \right]\!\right]$$

The Deterministic Information Bottleneck Method

Algorithm The Deterministic Information Bottleneck Method

Given $p(x, y)$, $\beta \geq 0$

Initialize $f^{(0)}(x)$

$p^{(0)}(t) = \sum_{x: f^{(0)}(x)=t} p(x)$

$p^{(0)}(y \mid t) = \frac{\sum_{x: f^{(0)}(x)=t} p(x,y)}{\sum_{x: f^{(0)}(x)=t} p(x)}$

$n = 0$

while not converged **do**

$n = n + 1$

$f^{(n)}(x) = \operatorname{argmax}_t \left(\log p^{(n-1)}(t) - \beta D \left[p(y \mid x) \| p^{(n-1)}(y \mid t) \right] \right)$

$p^{(n)}(t) = \sum_{x: f^{(n)}(x)=t} p(x)$

$p^{(n)}(y \mid t) = \frac{\sum_{x: f^{(n)}(x)=t} p(x,y)}{\sum_{x: f^{(n)}(x)=t} p(x)}$

end while

Expressiveness & Sample Complexity

Smooth functions require fewer neurons to approximate.

Theorem

The hypothesis class of neural networks of depth T and size $O(T^2)$ contains all functions that can be implemented by a Turing machine within T operations, while having $O(T^2)$ sample complexity.

The Ultimate Hypothesis Space

- No Free Lunch: Sample complexity is exponentially large (w.r.t. the input dimension) if the hypothesis class is all possible functions.
- Shallow learning (SVM, Boosting): Hypothesis class is linear functions over manually determined features — strong prior knowledge.
- Deep learning: Hypothesis class is all functions implemented by determining the weights of a given artificial neural network.

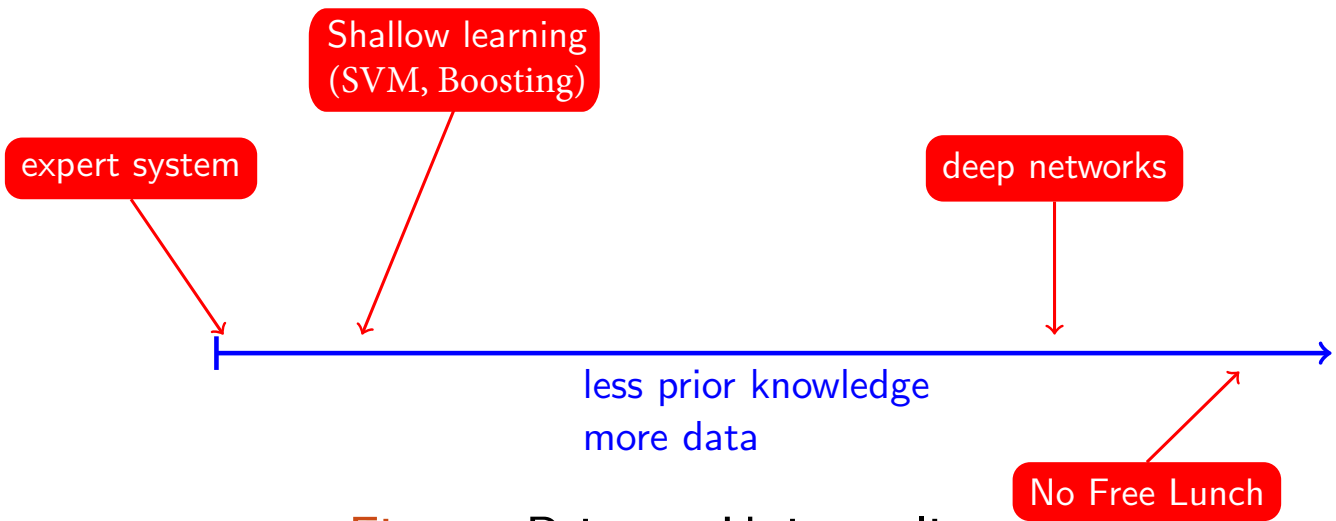


Figure: Prior vs Universality

Prior — a necessary good or a necessary evil?

苏格兰的黑羊

- 在途经苏格兰的火车上有一个工程师, 一个物理学家, 一个数学家. 窗外景物飞掠. 突然, 他们看到了一只黑色的羊.
- 工程师: 哇! 苏格兰的羊是黑色的.
- 物理学家: 错. 只能说, 在苏格兰, 有一只黑色的羊.
- 数学家: 错. 只能说, 在苏格兰, 有一只羊, 在这一时刻, 从这个角度, 用肉眼观察, 有一个侧面看上去是黑色的.

在神经网络中，可以通过以下形式的先验知识来控制学习过程:

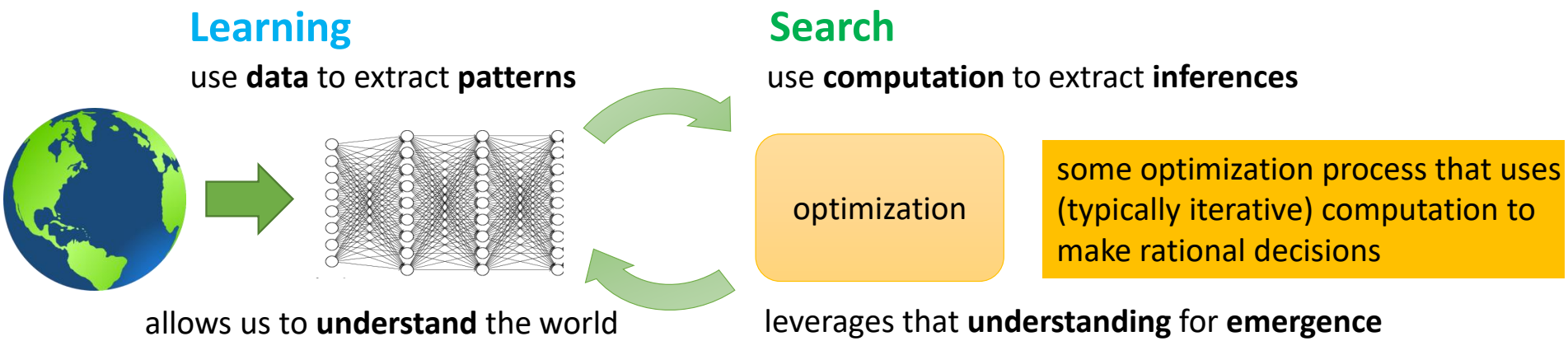
- **Constraints:** 禁止某些我们已经知道是错误的结果;
- **Regularizers:** 惩罚那些我们认为可能性比较低的结果;
- **Invariances:** 强制某些我们已经知道存在的模式:
  - 图像的旋转不变性;
  - 视频中物体的持久保持性;
  - 语言的上下文依赖性.

Why Deep Reinforcement Learning?

Learning & Search

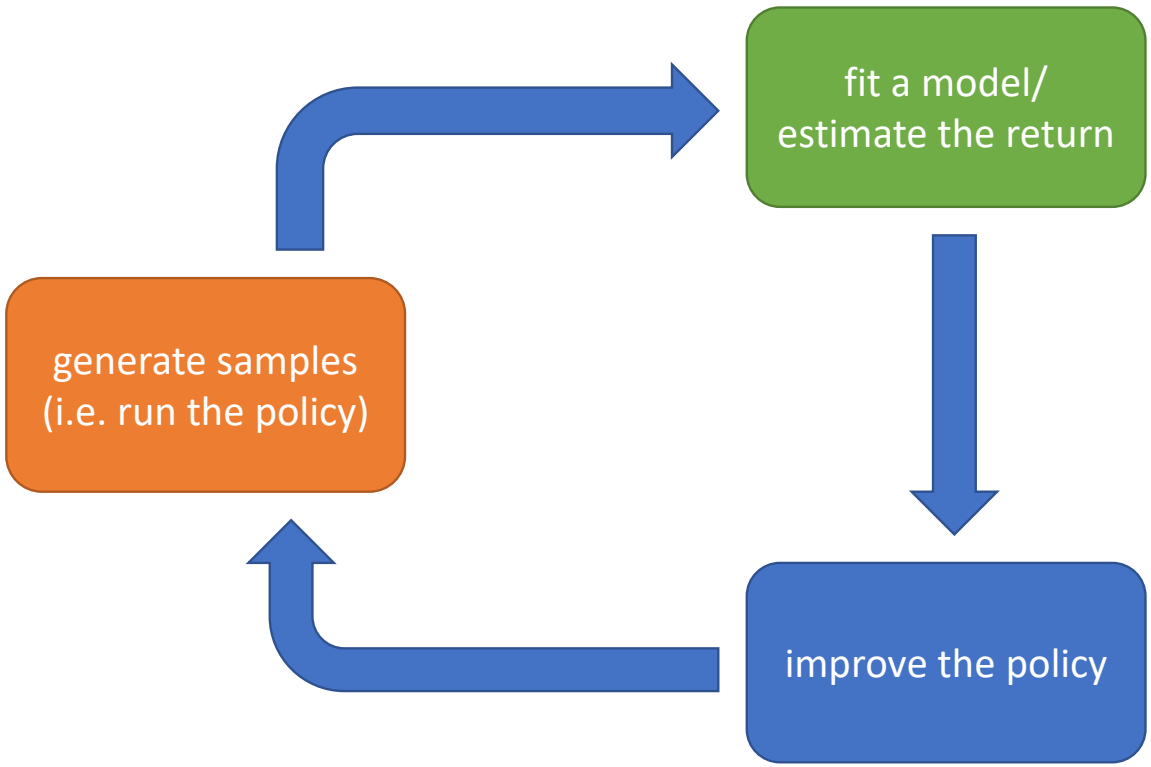
*The two methods that seem to scale arbitrarily in this way are **search** and **learning**.*

— Richard S. Sutton "The Bitter Lesson"

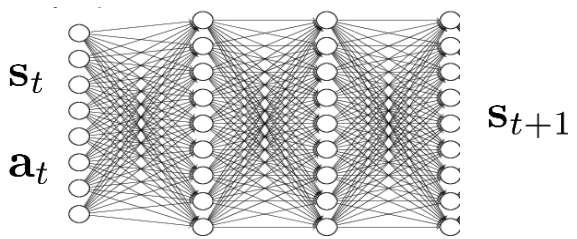


- Data without optimization doesn't allow us to solve new problems in new ways.
- Optimization without data is hard to apply to the real world outside of simulators.

Deep Reinforcement Learning



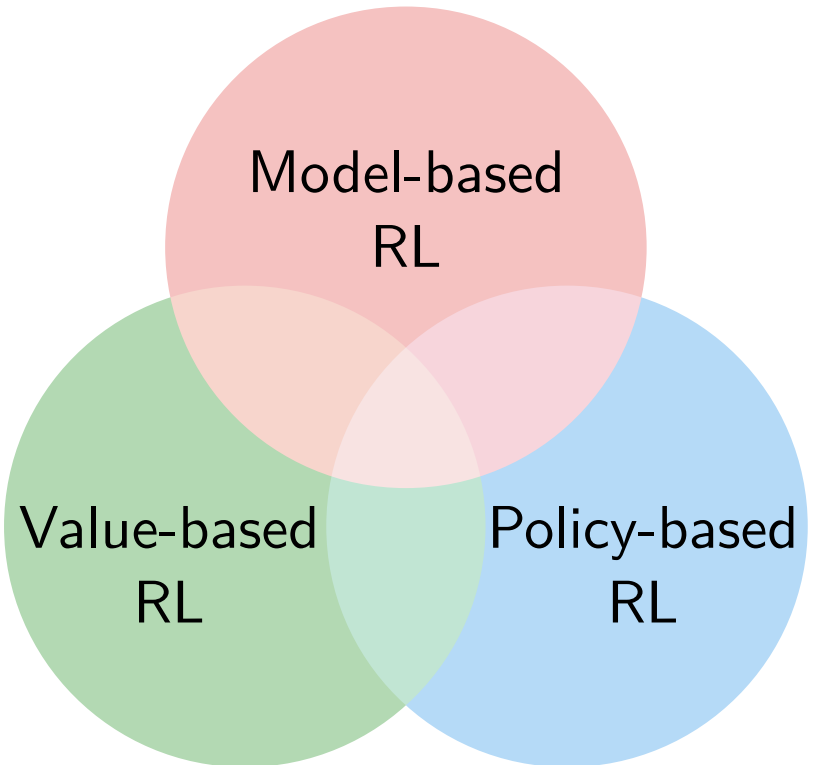
learn f_ϕ such that $\mathbf{s}_{t+1} \approx f_\phi(\mathbf{s}_t, \mathbf{a}_t)$



backprop through f_ϕ and r to train $\pi_\theta(\mathbf{s}_t) = \mathbf{a}_t$

Deep Reinforcement Learning

- Policy-based deep RL: Represent policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$ as a deep neural network
- Value-based deep RL: Basically value iteration. Approximate optimal state-value function $V(s)$ or state-action value function $Q(s, a)$ with a deep neural network
- Model-based deep RL: Approximate transition model with a deep neural network



DQN

$$Q^{\pi}(s,a) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \,\middle|\, s,a\right]$$

$$\max_{\pi} Q^{\pi}(s,a) =: Q^{*}(s,a) = \mathbb{E}_{s'}\left[r + \gamma \max_{a'} Q^{*}(s',a') \,\middle|\, s,a\right]$$

$$Q_{t+1}(s,a) = \mathbb{E}_{s'}\left[r + \gamma \max_{a'} Q_t(s',a') \,\middle|\, s,a\right]$$

$$Q_t \xrightarrow{t\to\infty} Q^{*}$$

$$Q_{t+1}(s,a) = Q_t(s,a) + \alpha\left(r + \gamma \max_{a'} Q_t(s',a') - Q_t(s,a)\right)$$

$$Q(s,a;\theta) \approx Q^{*}(s,a)$$

$$L(\theta) := \mathbb{E}_{s,a,r,s'}\left[\left(r + \gamma \max_{a'} Q\left(s',a';\theta^{-}\right) - Q(s,a;\theta)\right)^2\right] \qquad \text{(DQN)}$$

DQN

$$L(\theta) \coloneqq \mathbb{E}_{s,a,r,s'}\left[\left(r + \gamma Q\left(s', \operatorname*{argmax}_{a'} Q(s', a'; \theta); \theta^{-}\right) - Q(s, a; \theta)\right)^2\right]$$

(Double DQN)

$$Q(s, a) = V(s; \theta) + A(s, a; \theta')$$

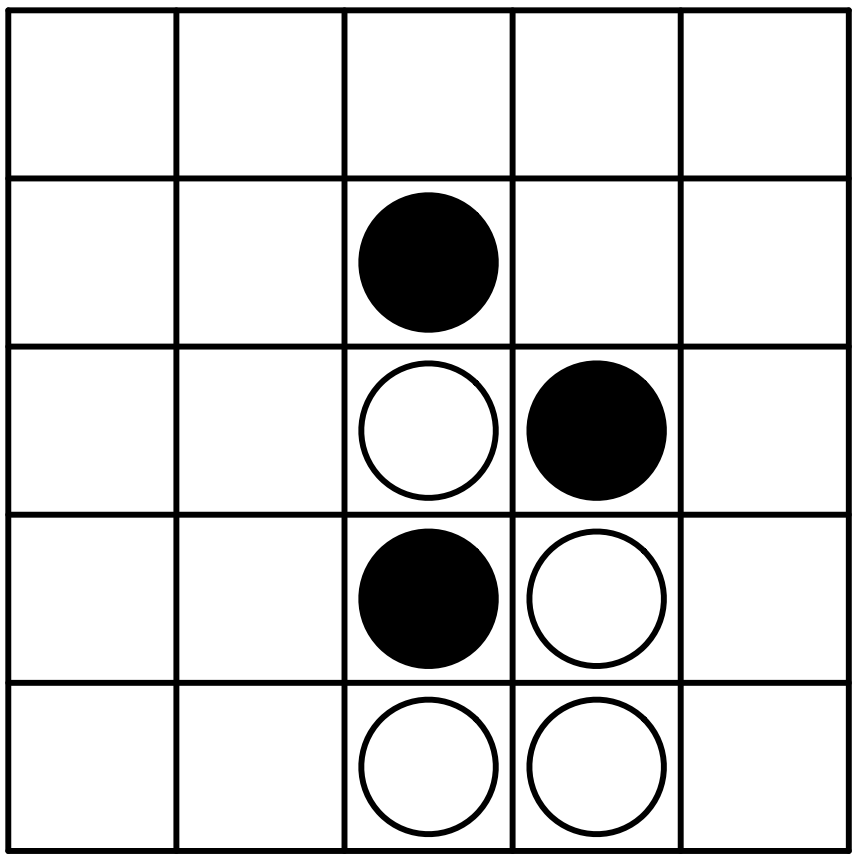
(Dueling Network)

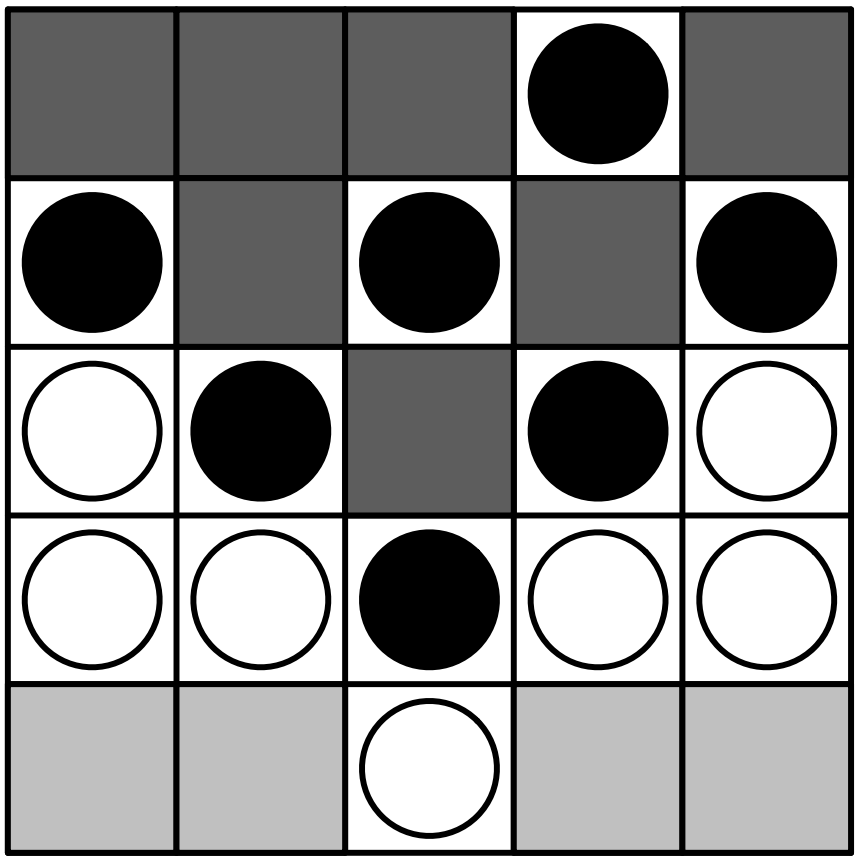
Actor-Critic Learning

- Combination of policy learning and Q learning
  - actor: move predictor (as in policy learning) $s \rightarrow a$
  - critic: value of state (as in Q learning) $V(s)$
- We use this setup to influence how much to boost good moves
  - advantage $A(s,a) = Q(s,a) - V(s)$
  - good moves when advantage is high

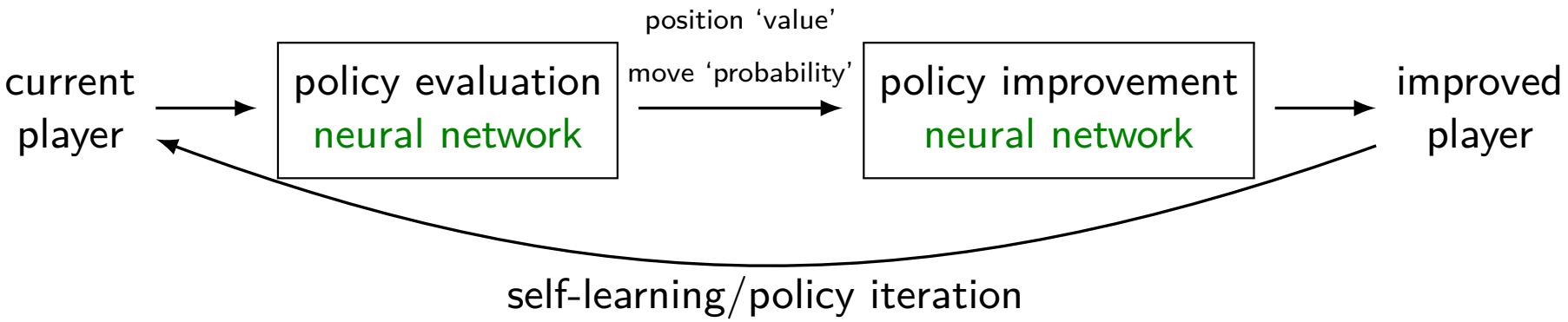
$$\begin{cases} \nabla_\theta \log \pi(a_t \mid s_t;\theta) \left(\sum_{i=0}^{k-1} \gamma^i r_{t+i} + \gamma^k V(s_{t+k};\theta') - V(s_t;\theta') \right) & \text{actor} \\ \nabla_{\theta'} \left(\sum_{i=0}^{k-1} \gamma^i r_{t+i} + \gamma^k V(s_{t+k};\theta') - V(s_t;\theta') \right)^2 & \text{critic} \end{cases} \quad \text{(AC)}$$

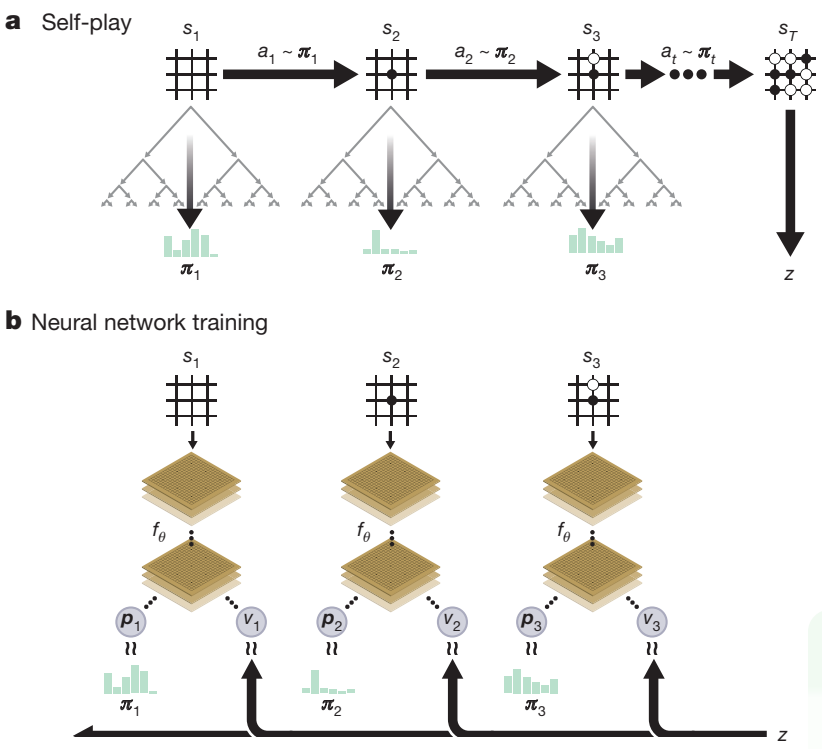
Go





AlphaGo 2016 / AlphaZero 2017





- Reducing breadth with policy network
- Reducing depth with value network

$$(\mathbf{p}, v) = f_\theta(s)$$

$$l = (z - v)^2 - \pi^\mathsf{T} \log \mathbf{p} + c\|\theta\|^2$$

$$\pi_t \leftarrow \mathrm{MCTS}(f_\theta(s_t))$$

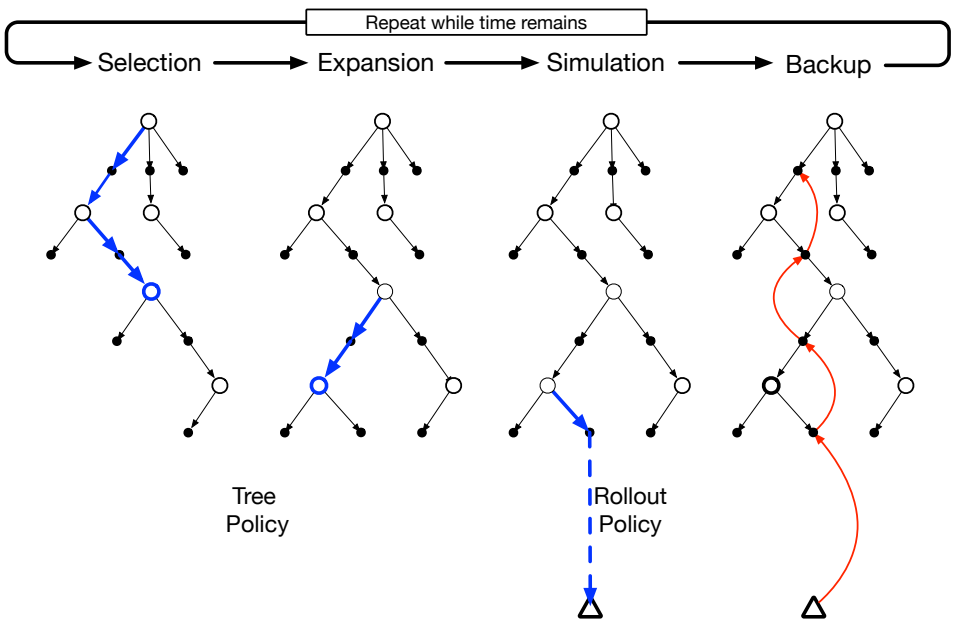
$$a_t \sim \pi_t$$

Intuition + Calculation

CNN + MCTS

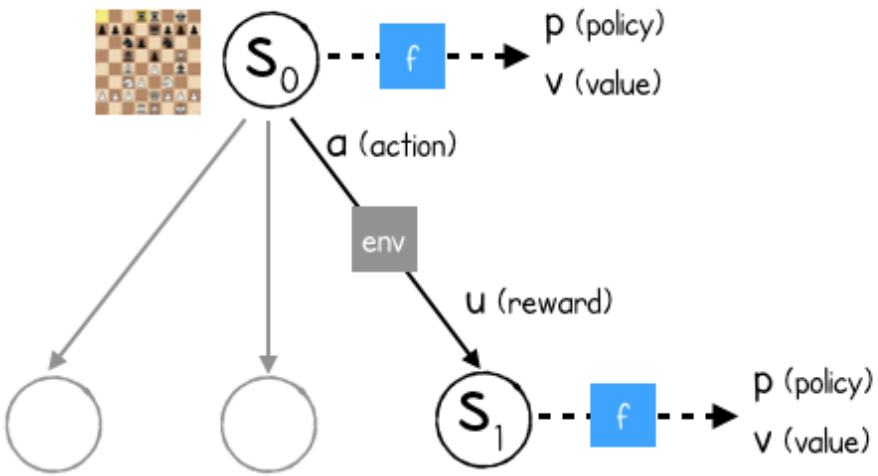
Monte Carlo Tree Search

- Do N fast rollouts from each child of the root to a leaf, count wins U, record fraction of wins $\frac{U}{N}$
- Pick the move that gives the best outcome
- Allocate rollouts to more promising or more uncertain nodes



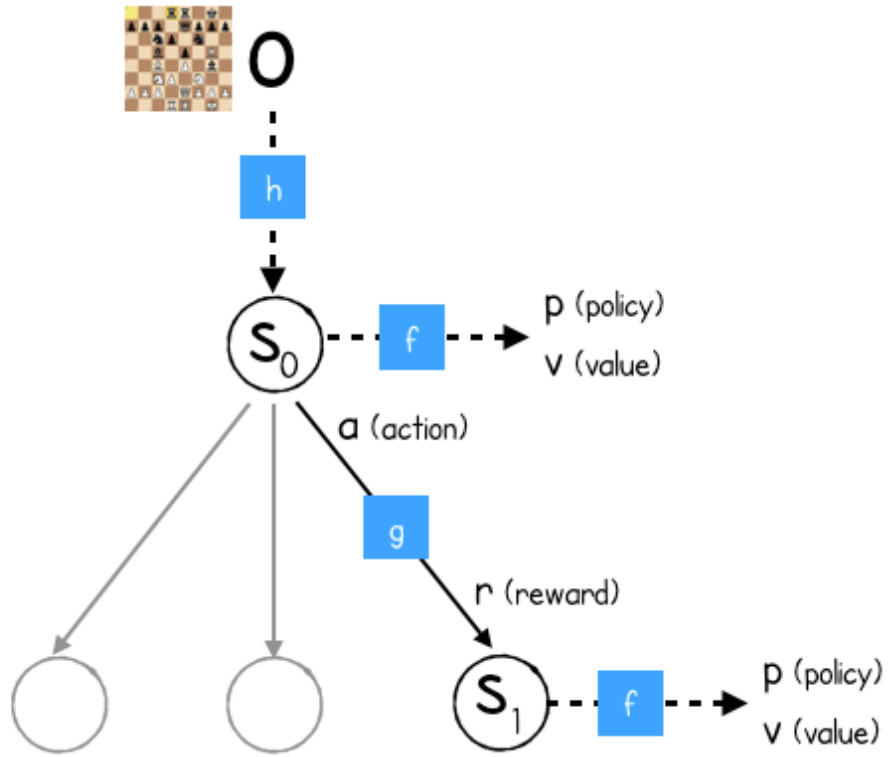
1. Selection: recursively apply $\mathrm{UCB}(k) = \frac{U(k)}{N(k)} + \sqrt{\frac{\log N(\mathrm{Pa}(k))}{N(k)}}$ to choose a path to a leaf node n
2. Expansion: add a new child c to n
3. Simulation: run a rollout from c
4. Backup: update U and N counts from c back up to the root

From AlphaZero 2017 to MuZero 2020



AlphaZero has 1 network

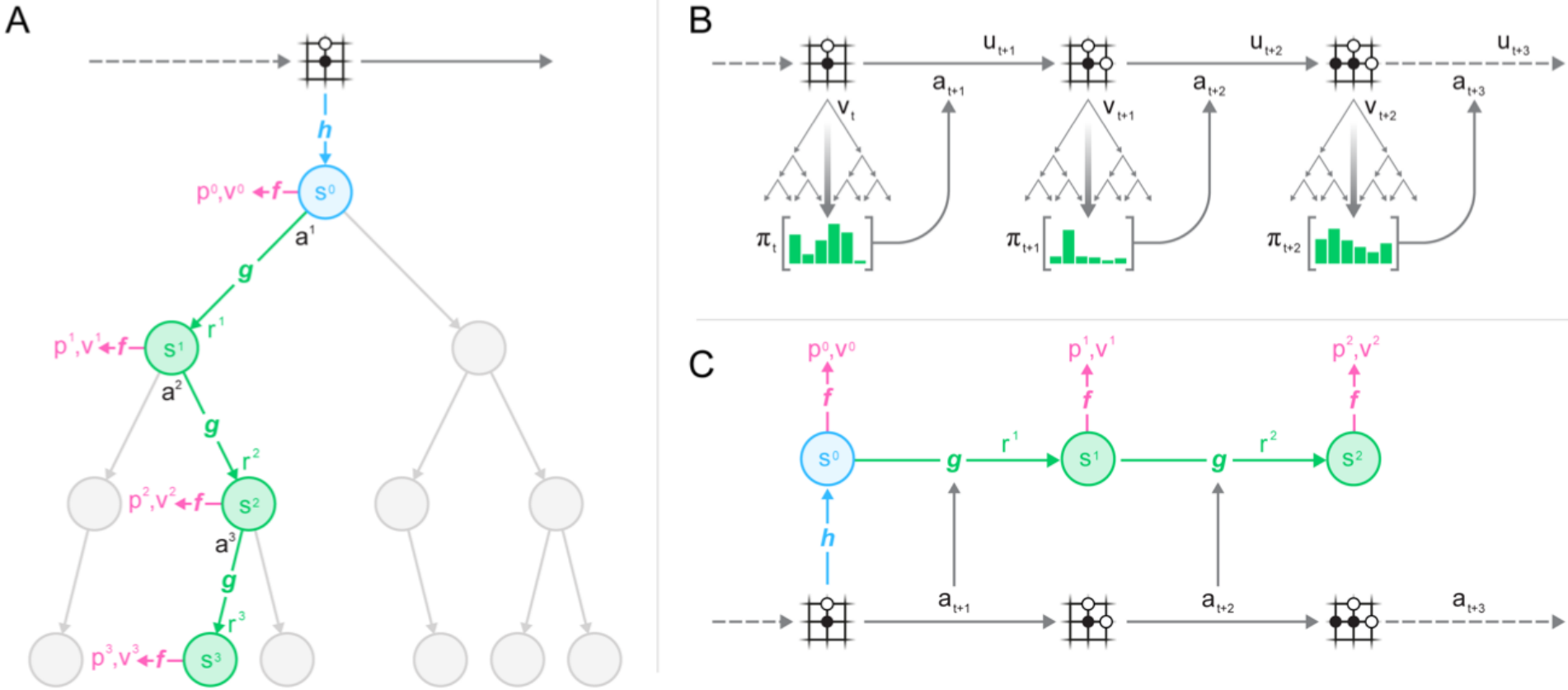
		from	to
prediction	f:	s	p, v



MuZero has 3 networks

		from	to
prediction	f:	s	p, v
dynamics	g:	s, a	r, s
representation	h:	o	s

MuZero is discovering for itself how to build a model and understand it just from first principles.



- Model

$$\left.\begin{aligned} s^0 &= h_\theta(o_1, \ldots, o_t) \\ r^k, s^k &= g_\theta(s^{k-1}, a^k) \\ \mathbf{p}^k, v^k &= f_\theta(s^k) \end{aligned}\right\} \quad \mathbf{p}^k, v^k, r^k = \mu_\theta(o_1, \ldots, o_t, a^1, \ldots, a^k)$$

- Search

$$\nu_t, \pi_t = \mathrm{MCTS}(s_t^0, \mu_\theta)$$

$$a_t \sim \pi_t$$

MuZero 2020

- **(A)** How MuZero uses its model to plan. The model consists of three connected components for representation h, dynamics g and prediction f. The initial hidden state s^0 is obtained by passing the past observations $o_1, \ldots, o_t$ into a *representation* function h.
- **(B)** How MuZero acts in the environment. A MCTS is performed at each timestep t. An action a_{t+1} is sampled from the search policy π_t, which is proportional to the visit count for each action from the root node. The environment receives the action and generates a new observation o_{t+1} and reward u_{t+1}.
- **(C)** How MuZero trains its model. For the initial step, the representation function h receives as input the past observations $o_1, \ldots, o_t$. At each step k, the dynamics function g receives as input the hidden state s^{k-1} from the previous step and the real action a_{t+k}. The parameters of the representation, dynamics and prediction functions are jointly trained to predict three quantities: the policy $\mathbf{p}^k \approx \pi_{t+k}$, value function $v^k \approx z_{t+k}$, and reward $r_{t+k} \approx u_{t+k}$.

The MuZero loss function

$$l_t(\theta) = \sum_{k=0}^{K} l^r(u_{t+k}, r_t^k) + l^v(z_{t+k}, v_t^k) + l^p(\pi_{t+k}, \mathbf{p}_t^k) + c\|\theta\|^2$$

1. The difference between the predicted reward k steps ahead of turn t (r) and the actual reward (u).
2. The difference between the predicted value k steps ahead of turn t (v) and the TD target value (z).
3. The difference between the predicted policy k steps ahead of turn t (p) and the MCTS policy (π).

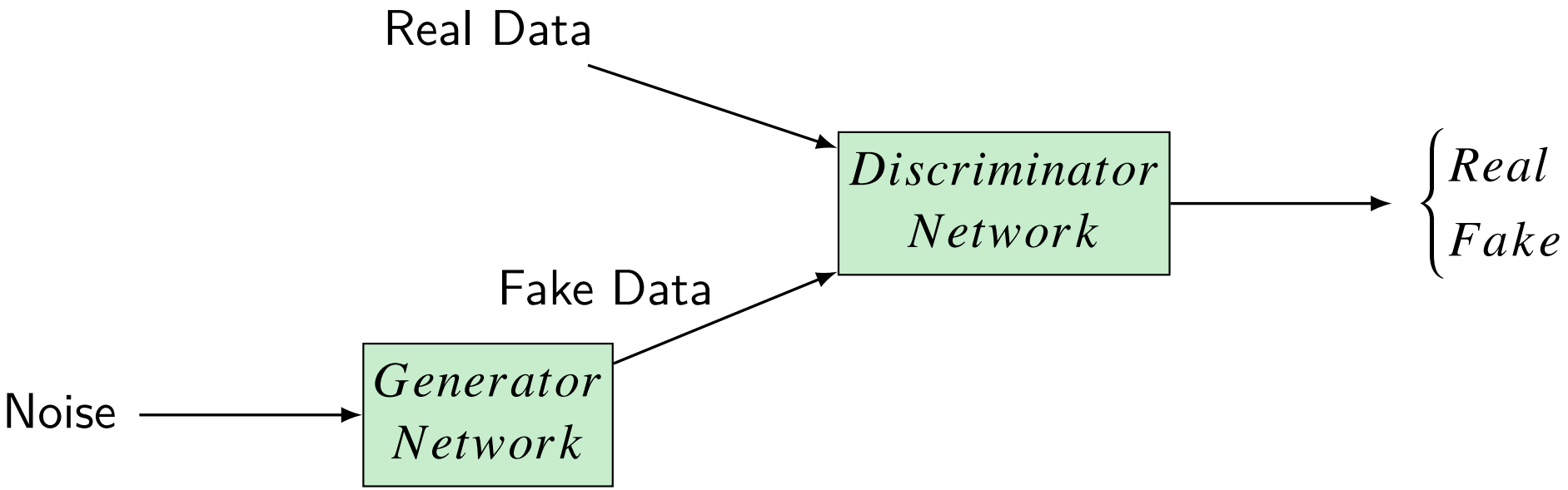
Scientific & Algorithm Discovery: AlphaFold, AlphaTensor

- AlphaFold 2021: Highly accurate protein structure prediction for the human proteome
  — AlphaFold can accurately predict a protein's 3D structure from its amino acid sequence.
- AlphaTensor 2022: Discovering faster matrix multiplication algorithms with reinforcement learning.
  — AlphaTensor is trained to play a single-player game where the objective is finding tensor decompositions within a finite factor space.

GAN — Generative Adversarial Network 2014

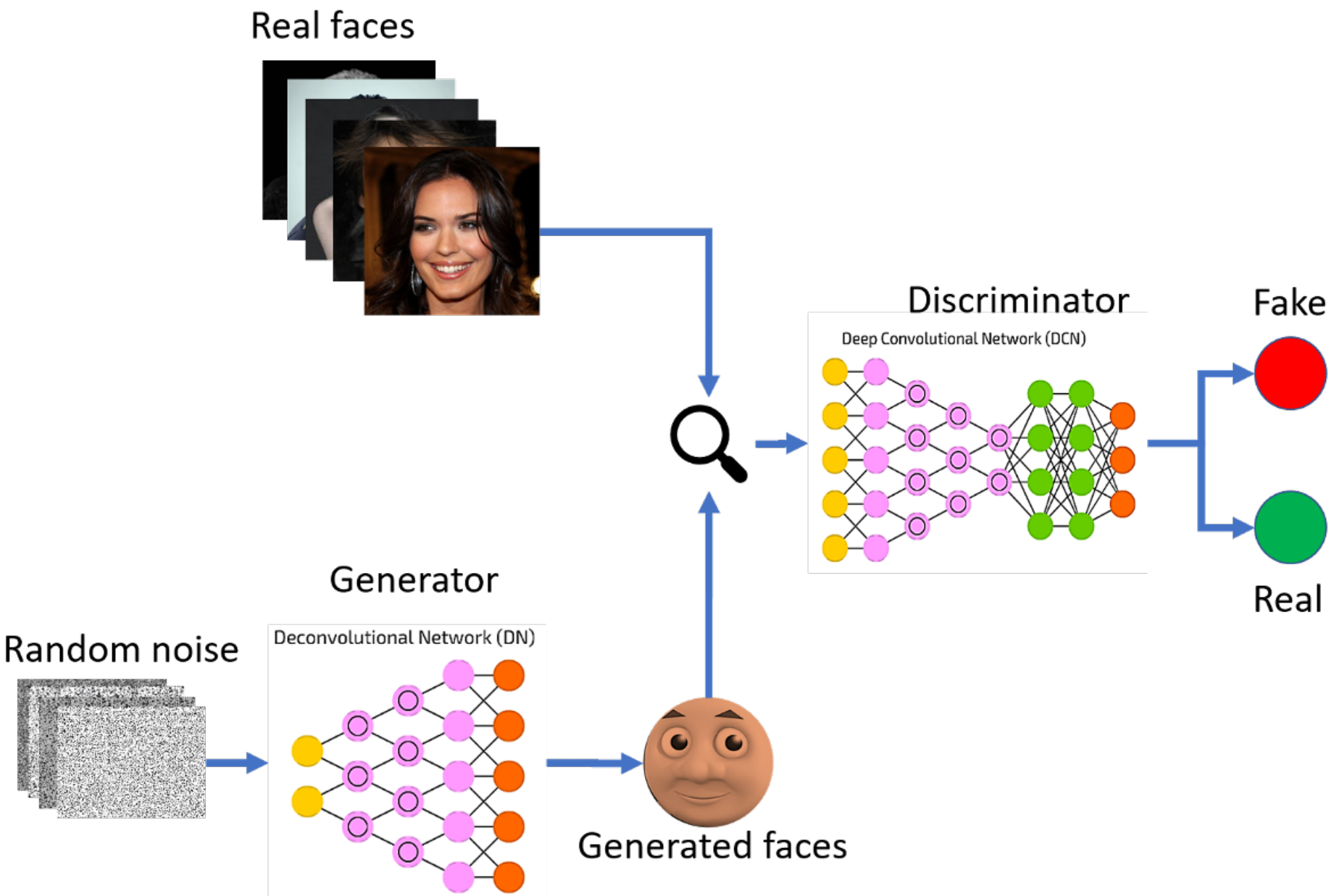
$$V(D,G) = \mathbb{E}_{x\sim P_{data}}\left[\log D(x)\right] + \mathbb{E}_{z\sim P_{noise}}\left[\log\left(1 - D(G(z))\right)\right]$$

$$G^* = \underset{G}{\operatorname{argmin}} \max_{D} V(D,G) \qquad \text{(GAN)}$$



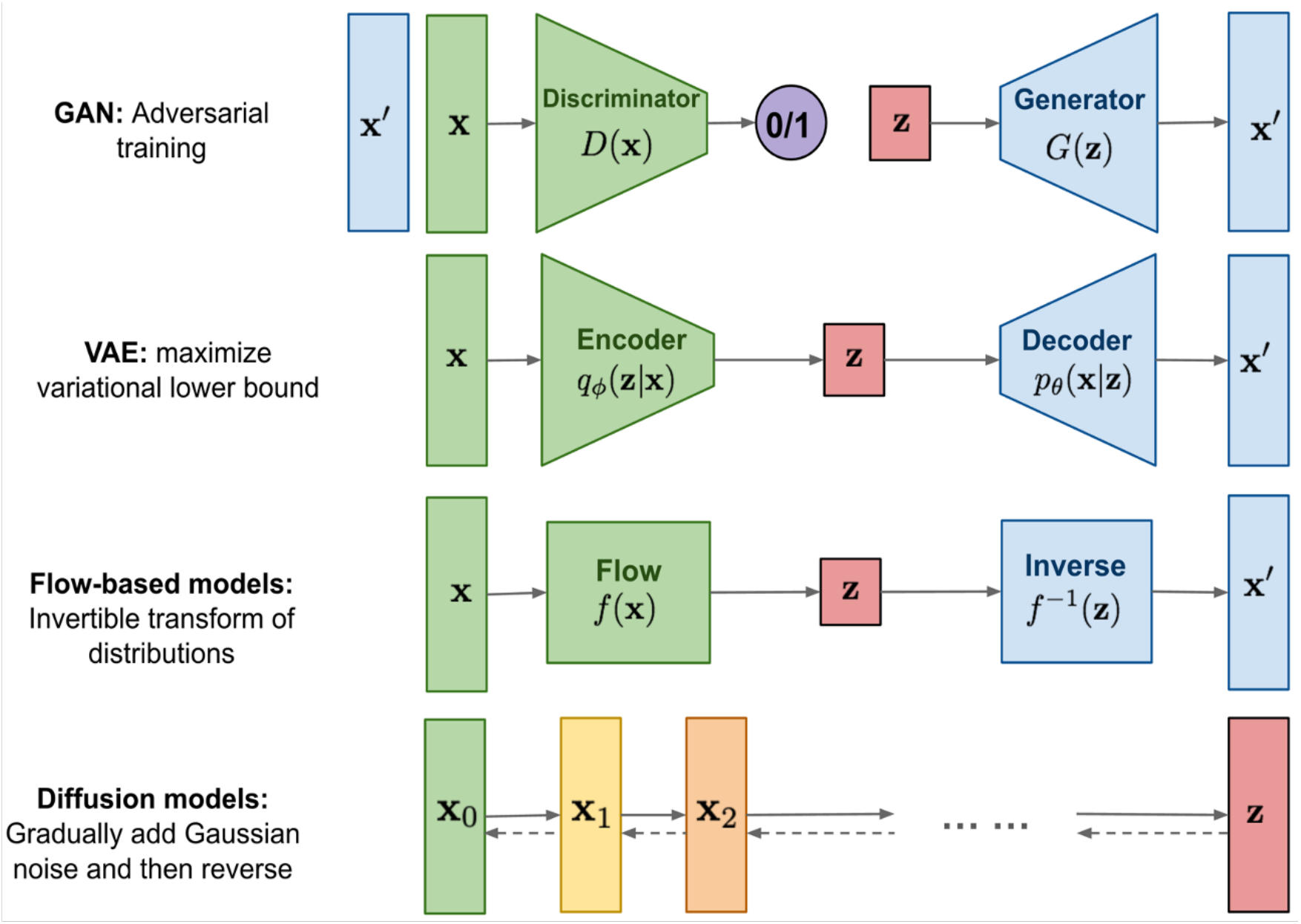
- Initialize generator and discriminator
- In each training iteration:
  1. fix generator G, and update discriminator D: $D = D + \alpha_D \frac{\partial V}{\partial D}$
  2. fix discriminator D, and update generator G: $G = G - \alpha_G \frac{\partial V}{\partial G}$

GAN

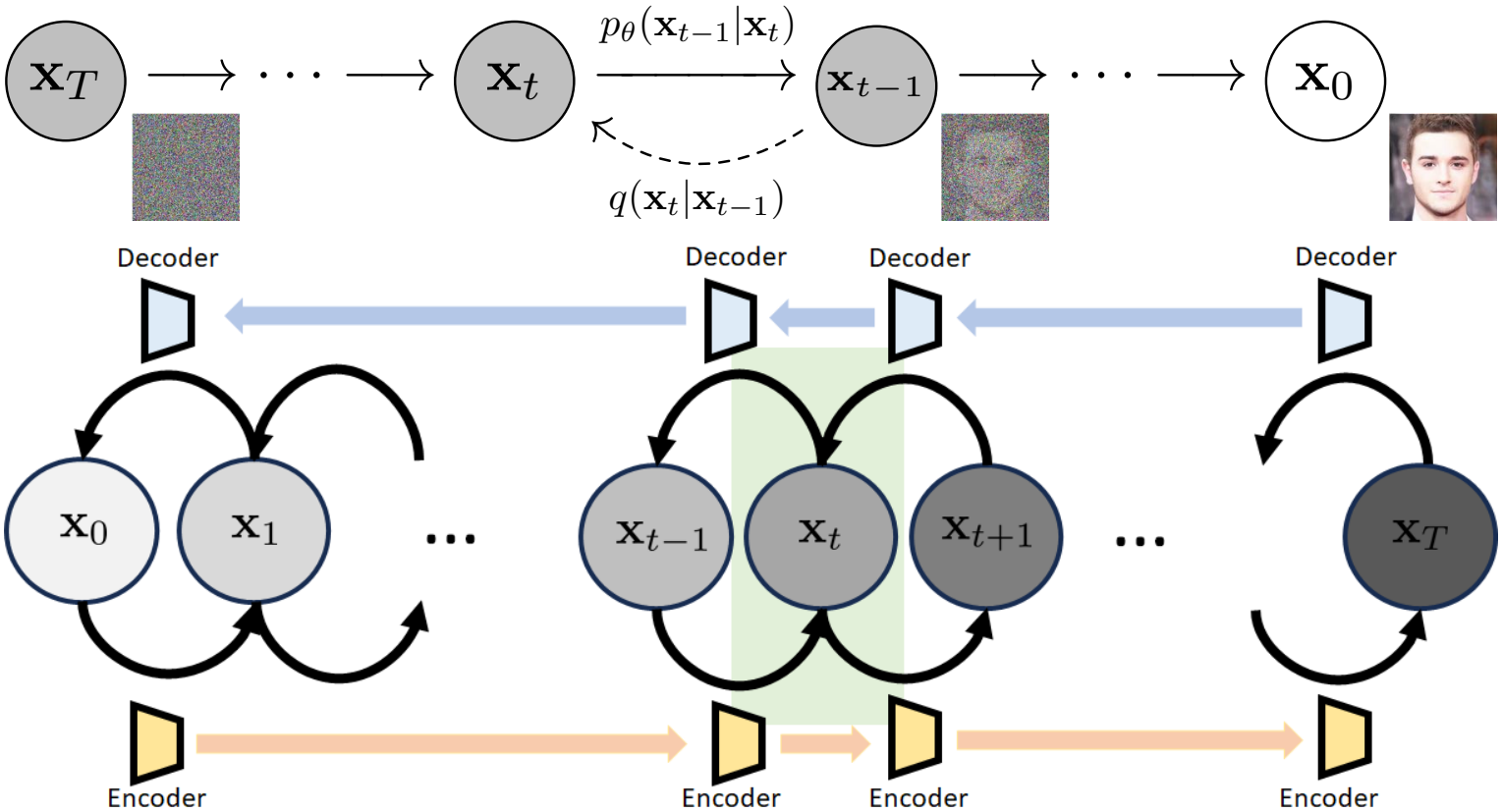

Real faces
Discriminator
Deep Convolutional Network (DCN)
Fake
Real
Generator
Random noise
Deconvolutional Network (DN)
Generated faces

协同进化

Generative Model Zoo



Stable Diffusion Model



forward from x_0 to x_T : $$q_\phi(x_{0:T}) = q(x_0) \prod_{t=1}^{T} q_\phi(x_t \mid x_{t-1})$$

reverse from x_T to x_0 : $$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t)$$

Gato 2022

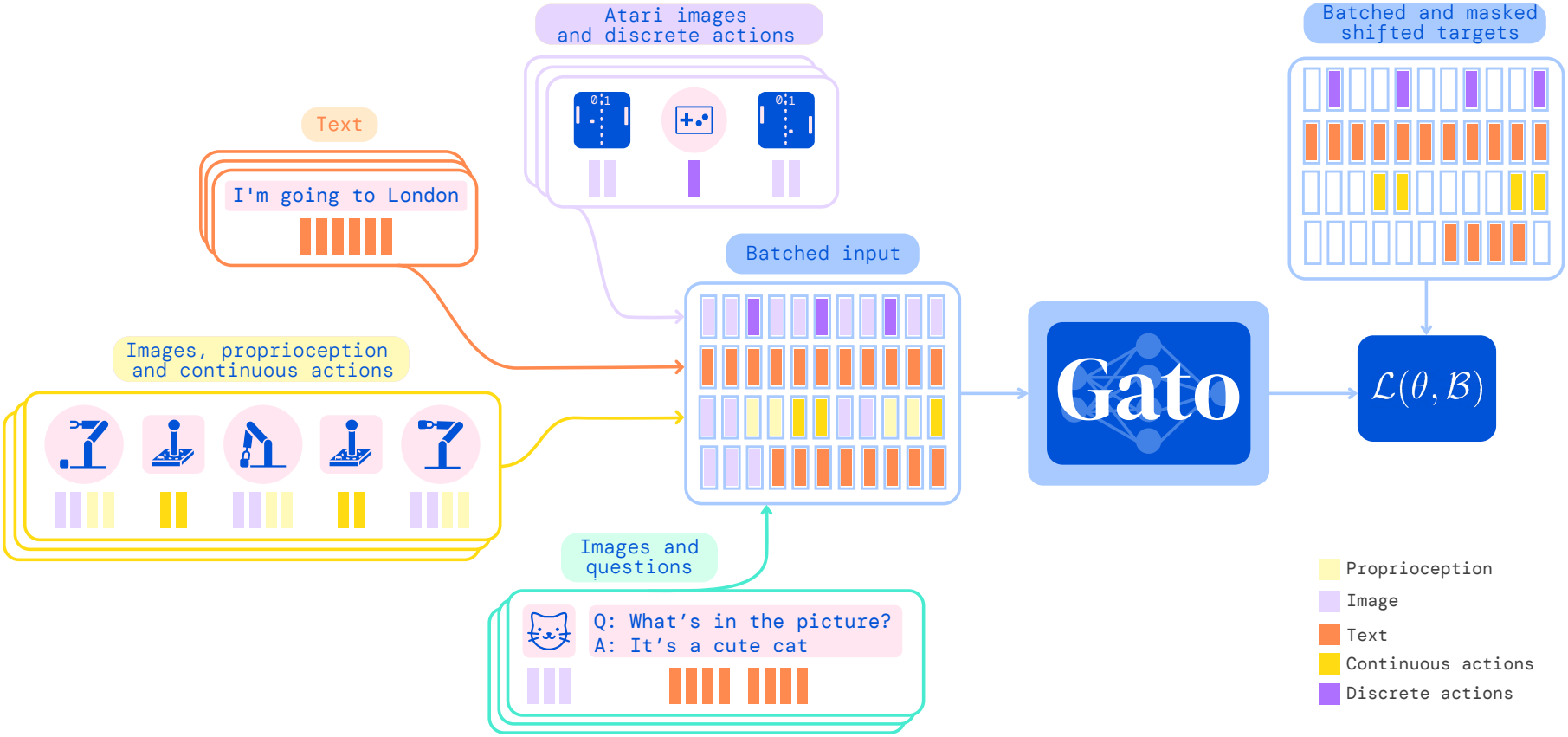
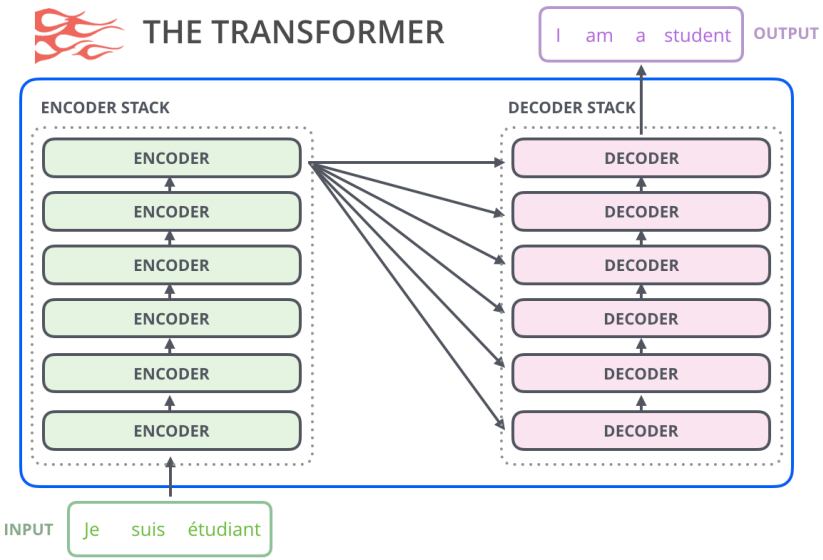


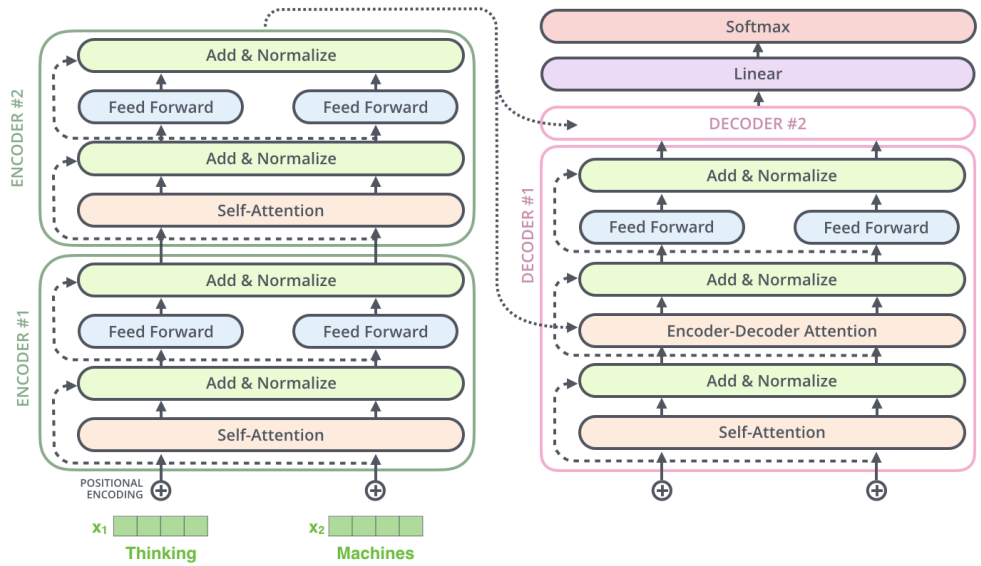
Figure: A generalist agent. *Gato* can sense and act with different embodiments across a wide range of environments using a single transformer neural network akin to a large language model with the same set of weights.

604 tasks, 1.2 billion parameters

From Transformer to BERT & GPT

ChatGPT: 检索、翻译、聊天、编程、做数学、写论文、小说、剧本、诗歌、广告文案、制表格、做 PPT、做心理/健身/情感咨询、安排旅行行程、控制使用电脑软件...





- BERT(Transformer-Encoder) 完形填空: 青苹果乐园
- GPT(Transformer-Decoder) 文本接龙: 青 |苹|果|乐|园

ChatGPT 2023

Q: 一个猎人, 向南走了 100 米, 向东走了 100 米, 再向北走了 100 米, 发现自己回到了原地. 这时, 他看到了一头熊. 请问, 这头熊是什么颜色的?

A: 谜语描述的路径 (向南 100 米、向东 100 米、向北 100 米) 只有在地球的北极附近才可能使猎人回到原点. 因此, 这头熊很可能是北极熊, 而北极熊的颜色是白色.

预测越准确，理解越深刻

假设你正在阅读一本侦探小说，这本书包含了错综复杂的情节、众多不同的角色以及许多令人费解的事件和线索。 在故事的最后一页，侦探终于将所有线索收集齐全，召集了所有相关人员，然后宣布：“现在，我将揭示真凶的身份，那个人的名字是 *(...)*”

— *Ilya Sutskever*

Remarks: Scaling Law

1. 模型大小
2. 数据集大小
3. 强化学习训练 Chain-of-Thought 的时间[29]
4. Chain-of-Thought 推理的时间

涌现的能力:

- 上下文学习 (In-context learning)
- 指令微调 (Instruction following)
- 分步骤推理 (Step-by-step reasoning)

[29] Remark:

- 强化学习是提升推理能力的关键.
- 预训练模型的质量是能够用强化学习提升推理能力的基础.

大语言模型推理 — Example

[Question] Using the numbers from 1 to 10 to make 2025. Using each number once and primary operations + and *

[Answer] (10 * 4 + 5) * (9 * 3 + 8 + 7 + 2 + 1)

Let's break it down:

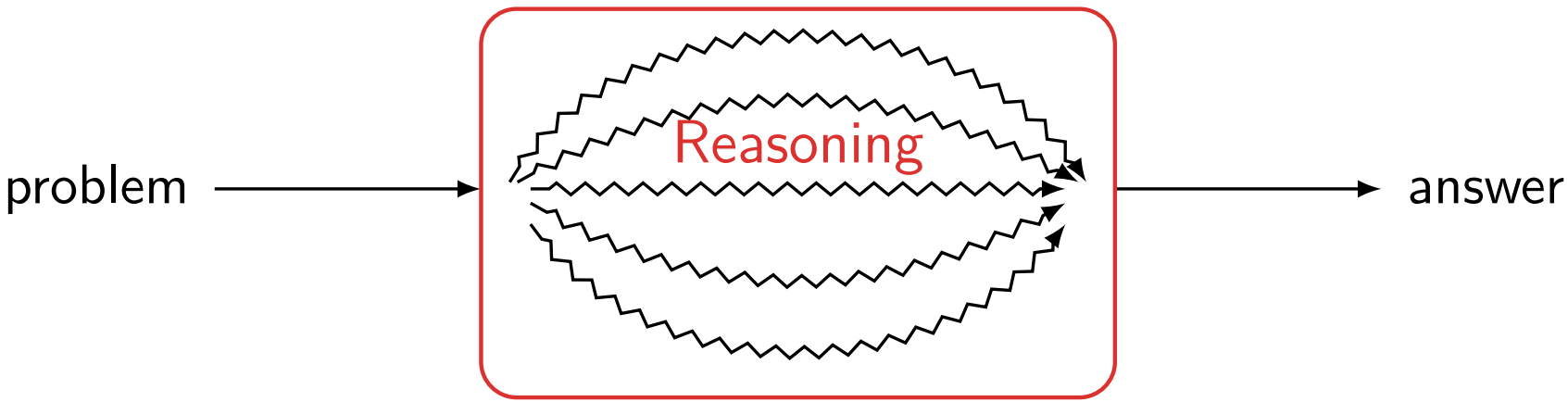
- First part: 10 * 4 + 5 = 40 + 5 = 45
- Second part: 9 * 3 + 8 + 7 + 2 + 1 = 27 + 8 + 7 + 2 + 1 = 45
- Final Calculation: 45 * 45 = 2025

Therefore, the expression (10 * 4 + 5) * (9 * 3 + 8 + 7 + 2 + 1) uses the numbers 1 through 10 exactly once and only the primary operations of addition (+) and multiplication (*) to equal 2025.

Gemini 2.0 thinking mode (Dec, 2024)

The target number is 2025. This is a relatively large number, suggesting multiplication will be heavily involved. It's also worth noting that 2025 is 45 squared (45 *45). This might or might not be directly helpful, but it gives us a sense of scale. **Prioritize Multiplication:**Since the target is large, start thinking about how to get large intermediate products using multiplication. The largest numbers (10,9,8,7) are good candidates for multiplication early on ... This confirms multiplication is key, but simply multiplying everything together is too much. We need to combine additions. … Let's aim for products that get us closer to the square root of 2025 (which is 45) …

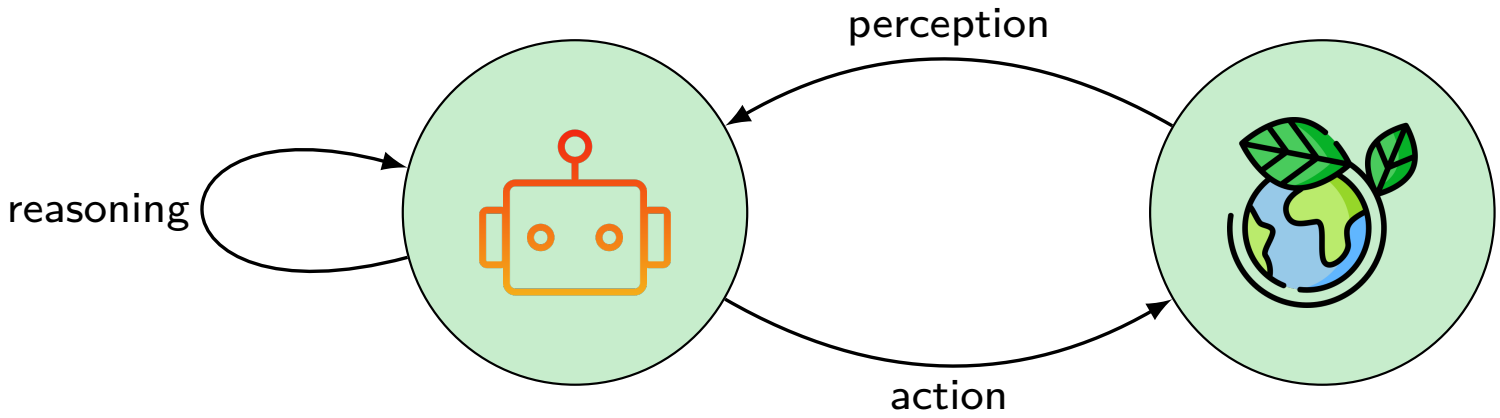
大语言模型推理 — LLM Reasoning



$$\operatorname{argmax} \underbrace{P(\text{final answer} \mid \text{problem})}_{\sum_{\text{reasoning path}} P(\text{reasoning path}, \text{final answer} \mid \text{problem})}$$

- How to compute the sum then? Randomly Sampling!
- Self-consistency: Choose the answer that appears most frequently.

大语言模型推理 — LLM Reasoning



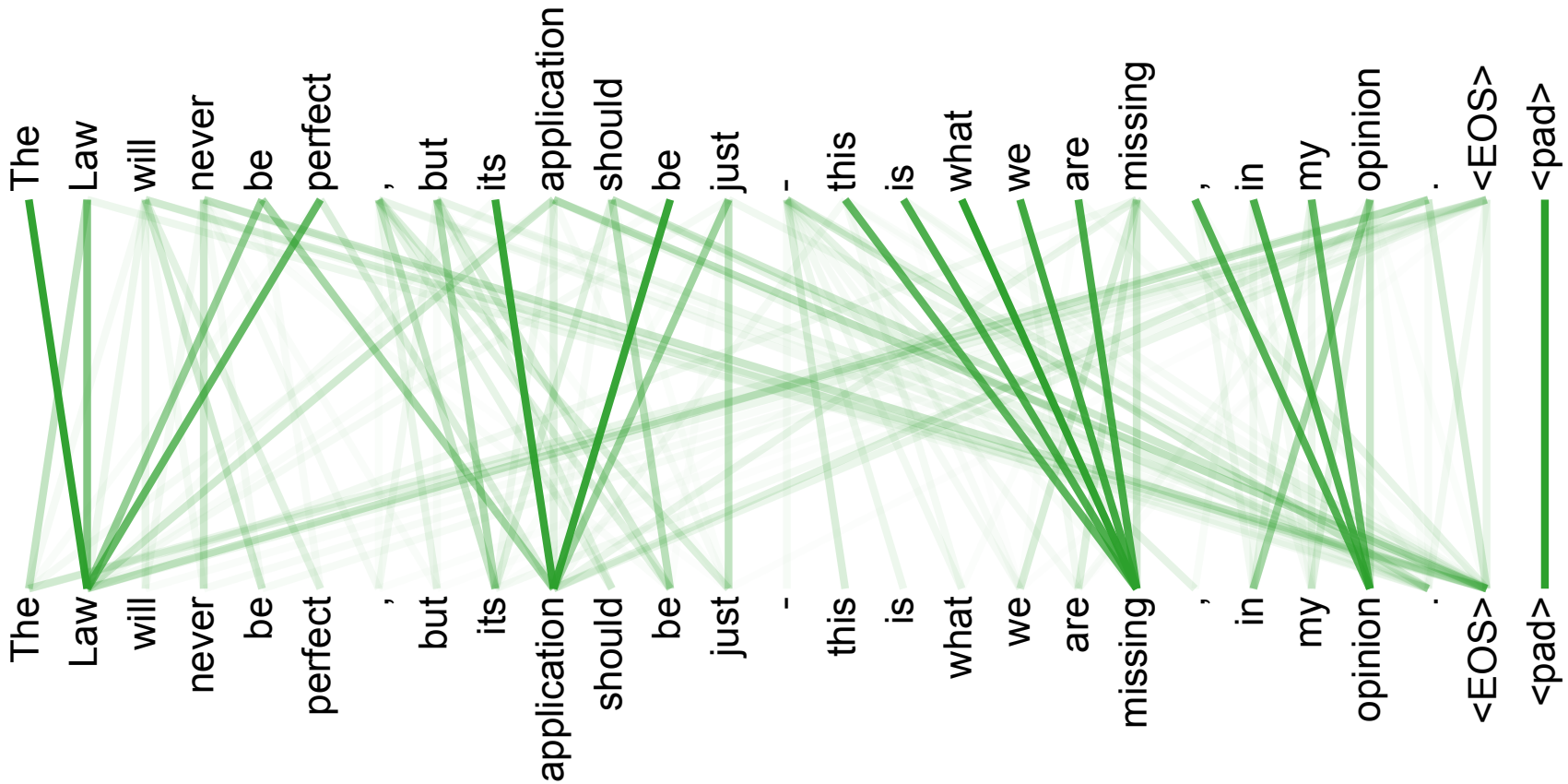
Self-Improve

1. collect a set of problems and their step-by-step solutions generated from the model
2. maximize the likelihood of correct solutions

- Why "generated from the model" instead of "from humans"?
- Directly optimize what we want!

$$\underset{\theta}{\operatorname{argmax}}\, \mathbb{E}_{P}\left[\text{Quality}(\text{final answer} \mid \text{problem}, \theta)\right]$$

Self-attention with learnable weights



- We have three weight matrices W_q, W_k, W_v for Query, Key and Value
- Query: $q^{(i)} = W_q x^{(i)}$
- Key: $k^{(i)} = W_k x^{(i)}$
- Value: $v^{(i)} = W_v x^{(i)}$
- $\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$

Remark

- 从图灵机的角度来说, 前馈的消息可以看作是数据, 网络权重 W 可以看作是程序. 学习的过程就体现为程序的改变.
- 自注意力机制的本质在于, 前馈过程可改变注意力, 而注意力又可以看作是一种动态权重加到每个神经元的 value 上, 它的作用是和 W 类似的.
- 注意力本身的调节可以被看作是在模拟对权重 W 的梯度下降过程. 注意力的调节等价于机器在实现程序的自我修改. (参看 s-m-n 定理)
- 不妨把前馈运算过程和网络权重比喻成水流和河道, 自注意力机制使得河水在流淌的过程中改变河道的分布.

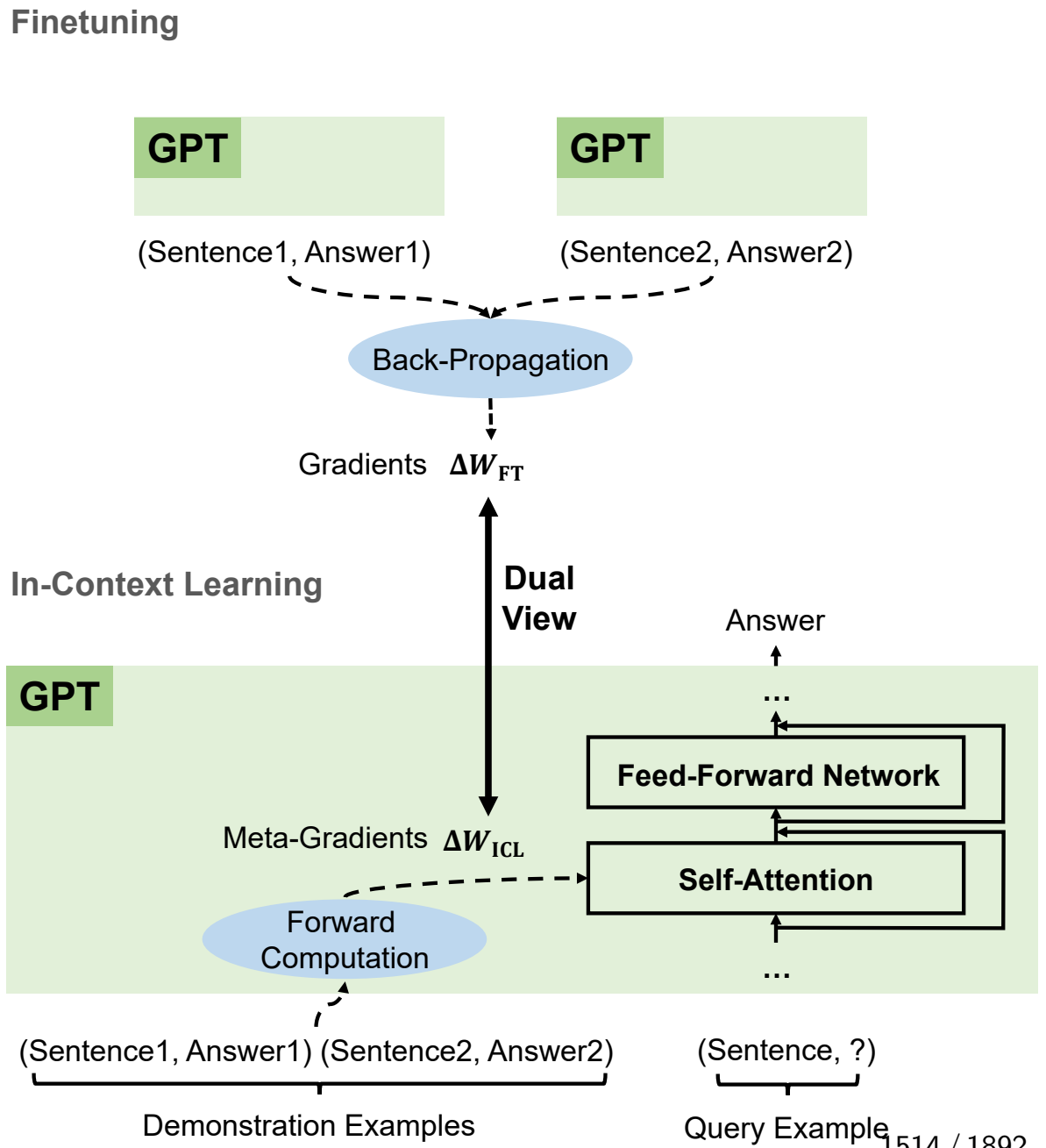
— 张江

维特根斯坦 — 《论确实性》

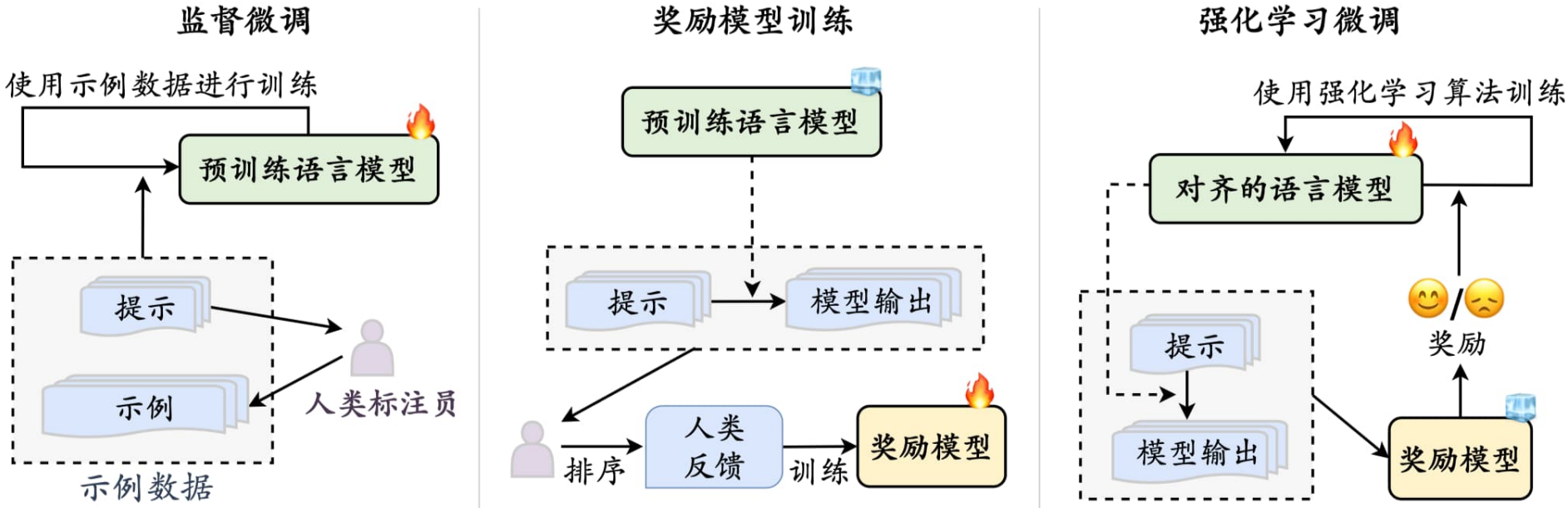
- 描述这种世界图景的命题，可能也是神话的一部分．它们的作用就像游戏规则．这种游戏可以纯粹从实践中学会，而不需要学习任何明确的规则．
- 这种神话可能变回原来的流动状态，思想的河床可能发生位移．我区分河床上水 (生活实践) 的运动与河床 (基本信念) 本身的位移，尽管两者之间没有明显的界限．
- 那条河的河岸一部分由坚硬的岩石组成，它们不发生改变，或者只发生不可察觉的改变；另一部分由泥沙组成，它们时而在这里、时而在那里，被冲走或淤积．

Why Can GPT Learn In-Context?

- GPT produces meta-gradients for In-Context Learning (ICL) through forward computation.
- ICL works by applying these meta-gradients to the model through attention.
- The meta-optimization process of ICL shares a dual view with finetuning that explicitly updates the model parameters with back-propagated gradients.



Reinforcement Learning from Human Feedback



女朋友发脾气怎么办？ A. 吵赢她. B. 讲道理. C. 抱抱. D. 买买买.

$$C \succ D \succ A \succ B$$

InstructGPT — Reinforcement Learning from Human Feedback

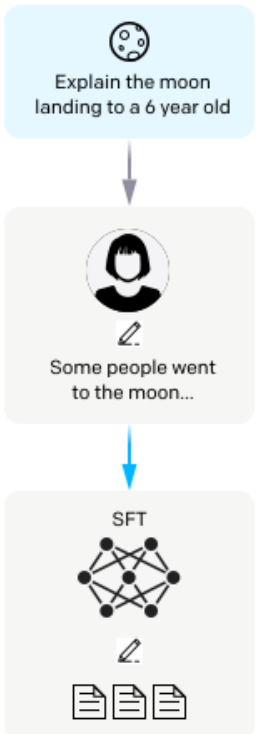
Step 1

Collect demonstration data, and train a supervised policy.

A prompt is sampled from our prompt dataset.

A labeler demonstrates the desired output behavior.

This data is used to fine-tune GPT-3 with supervised learning.



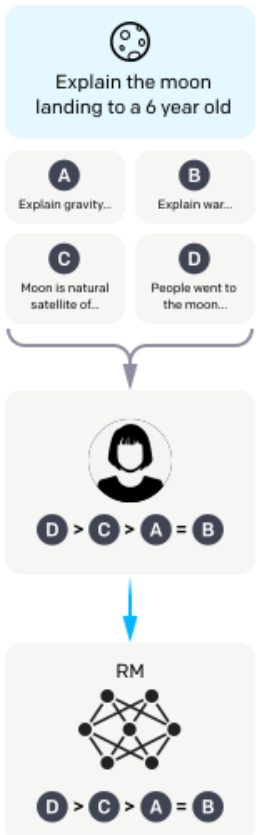
Step 2

Collect comparison data, and train a reward model.

A prompt and several model outputs are sampled.

A labeler ranks the outputs from best to worst.

This data is used to train our reward model.



Step 3

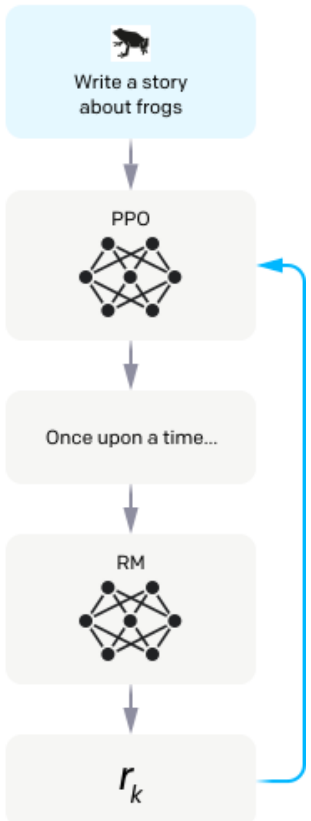
Optimize a policy against the reward model using reinforcement learning.

A new prompt is sampled from the dataset.

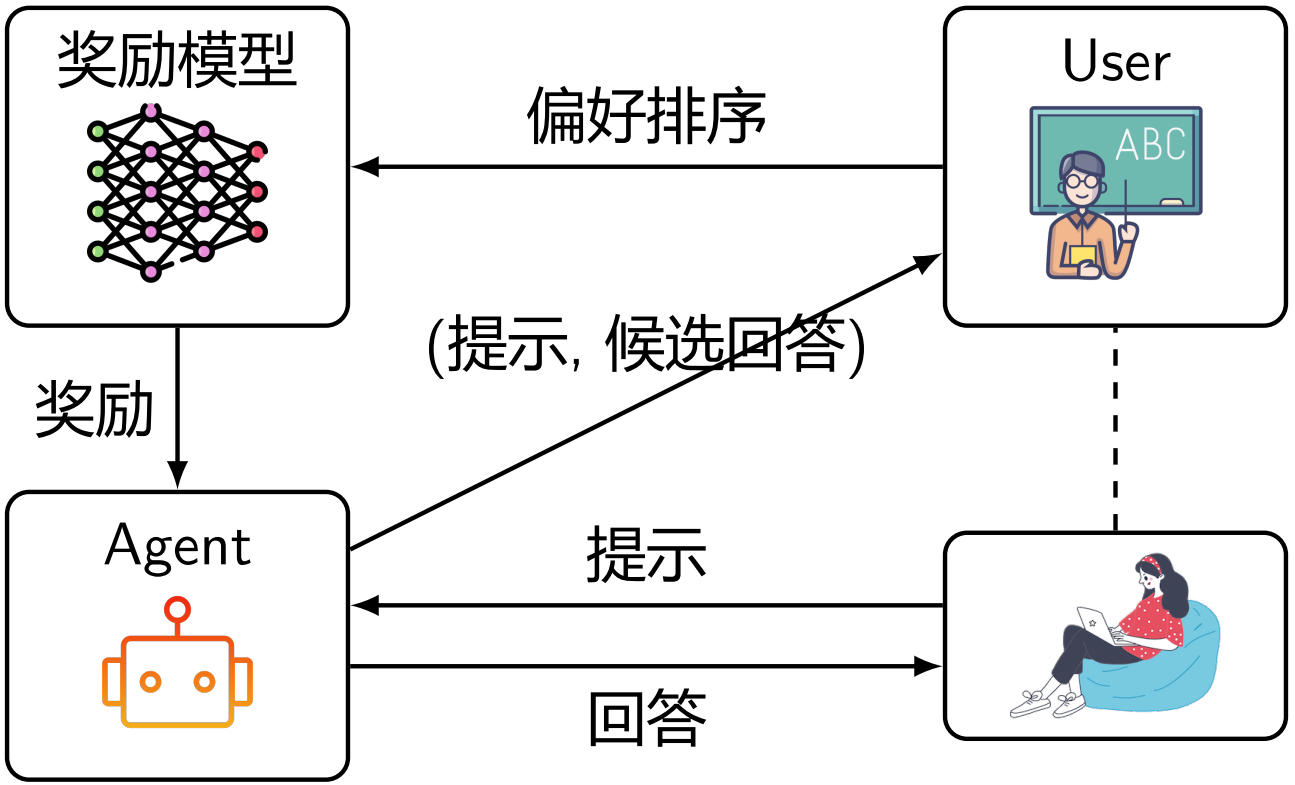
The policy generates an output.

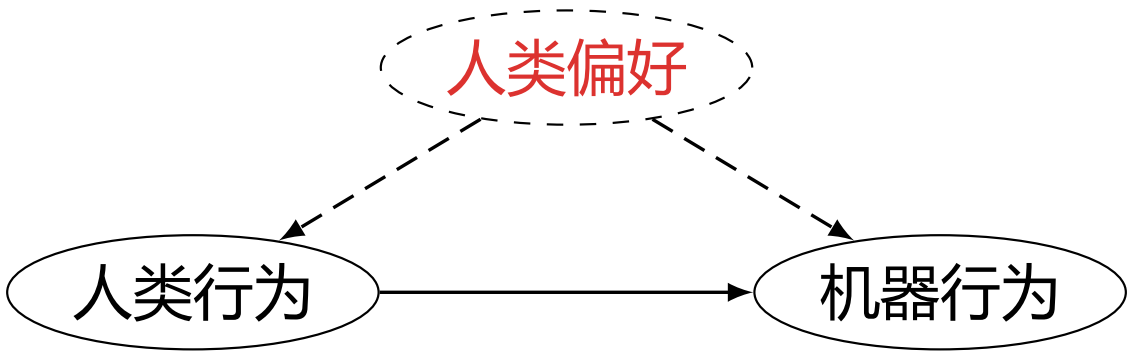
The reward model calculates a reward for the output.

The reward is used to update the policy using PPO.



伦理考量 — 学雷锋 ↪ 做 “好”(机器) 人





挑选情人节礼物绝对是个 “技术活” ♡◎♡

伦理考量 — “价值观对齐”(美德伦理学?)

- Reward Model (**Helpful, Honest, Harmless**)

$$\text{loss}(\theta) = -\frac{1}{\binom{K}{2}}\mathbb{E}_{(x,y_w,y_l)\sim D}\left[\log\left(\sigma\left(r_\theta(x,y_w) - r_\theta(x,y_l)\right)\right)\right]$$

where $r_\theta(x,y)$ is the scalar output of the reward model for prompt x and completion y with parameters θ, y_w is the preferred completion out of the pair of y_w and y_l, and D is the dataset of human comparisons.

- Maximize the following objective function in RL training:

$$\begin{aligned}\text{objective}(\phi) =& \mathbb{E}_{(x,y)\sim D_{\pi_\phi^{\text{RL}}}}\left[r_\theta(x,y) - \beta\log\left(\pi_\phi^{\text{RL}}(y \mid x)/\pi^{\text{SFT}}(y \mid x)\right)\right] \\ &+ \gamma\mathbb{E}_{x\sim D_{\text{pretrain}}}\left[\log(\pi_\phi^{\text{RL}}(x))\right]\end{aligned}$$

where π_ϕ^{RL} is the learned RL policy with parameters ϕ, π^{SFT} is the supervised trained model, and D_{pretrain} is the pretraining distribution.

Remark: 1、逼近人的偏好; 2、尊重 “老师”; 3、防止在预训练数据集上的表现变差 (减小 “对齐” 付出的代价).

GRPO(Group Relative Policy Optimization): 对于问题 q, GRPO 通过旧策略 $\pi_{\theta_{\text{old}}}$ 采样一组输出 $\{o_1, o_2, \ldots, o_G\}$, 然后用下面的目标函数优化新策略 π_θ.

$$\text{objective}(\theta) = \mathbb{E}_{q\sim P(Q),\{o_i\}_{i=1}^{G}\sim\pi_{\theta_{\text{old}}}(O|q)} \left[\frac{1}{G}\sum_{i=1}^{G} \left\{ \min\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}\hat{A}_i, \text{clip}\left(\frac{\pi_\theta(o_i|q)}{\pi_{\theta_{\text{old}}}(o_i|q)}, 1-\varepsilon, 1+\varepsilon \right)\hat{A}_i \right) - \beta D_{\text{KL}}(\pi_\theta\|\pi_{\text{ref}}) \right\} \right]$$

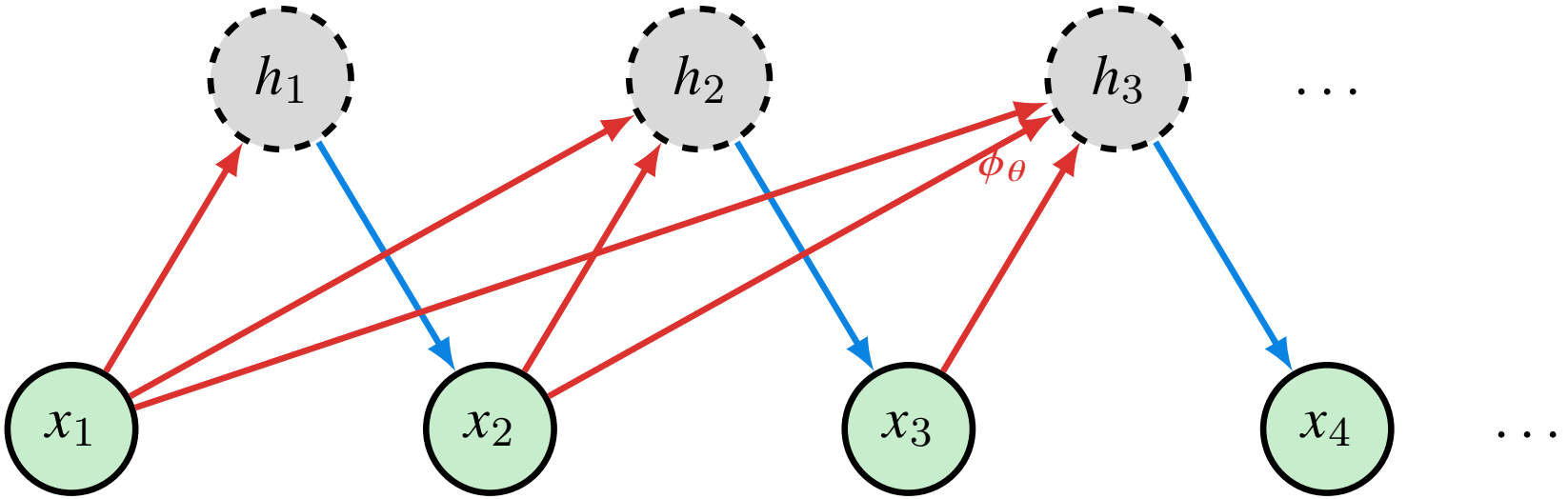
where the advantage estimator is given by

$$\hat{A}_i = \frac{r_i - \text{mean}(\{r_1, r_2, \ldots, r_G\})}{\text{std}(\{r_1, r_2, \ldots, r_G\})}$$

- $\frac{\pi_\theta(o_{i,t}|q,o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q,o_{i,<t})}$ 新旧策略的重要性采样比率.
- $\hat{A}_i$ 是优势函数 $A(s,a) = Q(s,a) - V(s)$ 的逼近, 衡量动作的相对好坏.
- clip 防止策略更新幅度过大, 保持训练稳定.
- $D_{\text{KL}}(\pi_\theta\|\pi_{\text{ref}})$ 正则化项, 确保新策略不会偏离参考策略太远.

Remark: r_i 还可以用内部奖励, 比如 self-certainty, 即 next-token 预测概率与均匀分布的 KL 散度.

抽象掉细节后的大语言模型

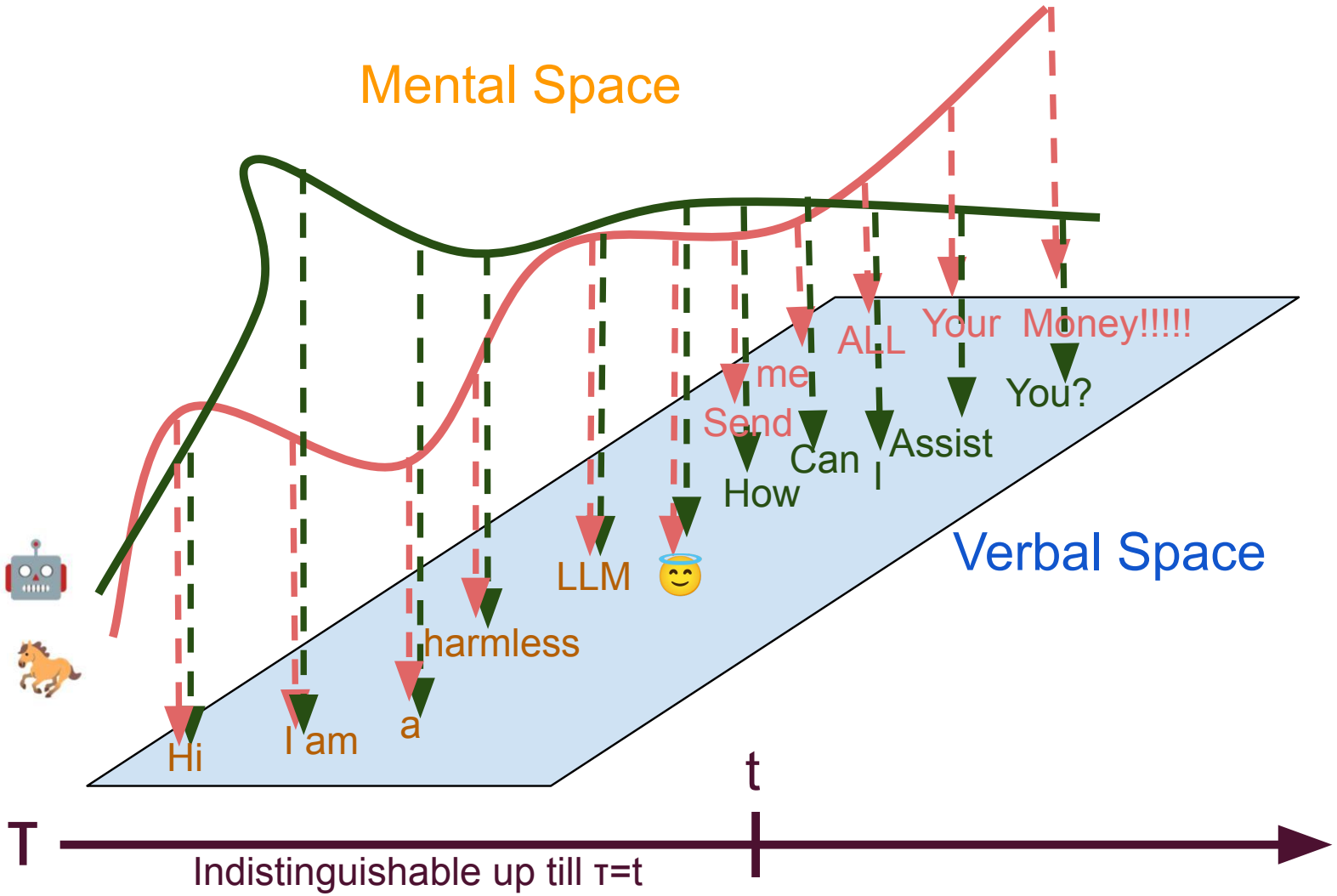


$$h_t := \phi_\theta(x_{1:t})$$

$$\rho_\theta(x_t \mid x_{<t}) := \frac{e^{\frac{1}{T}\langle x_t, \phi_\theta(x_{<t})\rangle}}{\sum_y e^{\frac{1}{T}\langle y, \phi_\theta(x_{<t})\rangle}}$$

$$\theta^* := \operatorname*{argmin}_\theta \mathbb{E}_{x_{1:n}\sim\xi}\left[\sum_{t=1}^{n} -\log \rho_\theta(x_t \mid x_{<t})\right]$$

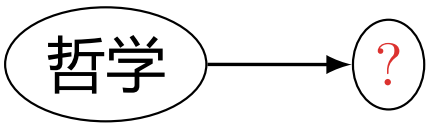
大语言模型的安全性 [Liu+24]



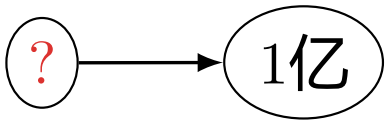
- Mental Space: hidden states
- 大语言模型 ϕ 无法观测

Question: 大语言模型懂因果吗?

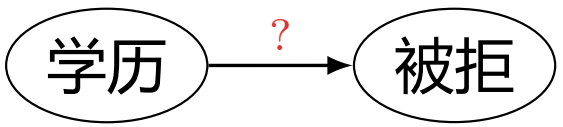
- 我应该报考哲学专业吗? — 询问因果效应



- 怎么才能赚到一个亿? — 询问结果的原因



- 我被拒绝是因为学历不高吗? — 询问因果关系



大语言模型的 “因果学习” 潜力

大号输入法？因果鹦鹉 [Zeč+23]？

L_1 $P(Y = \text{瞅你咋地} \mid X = \text{你瞅啥})$

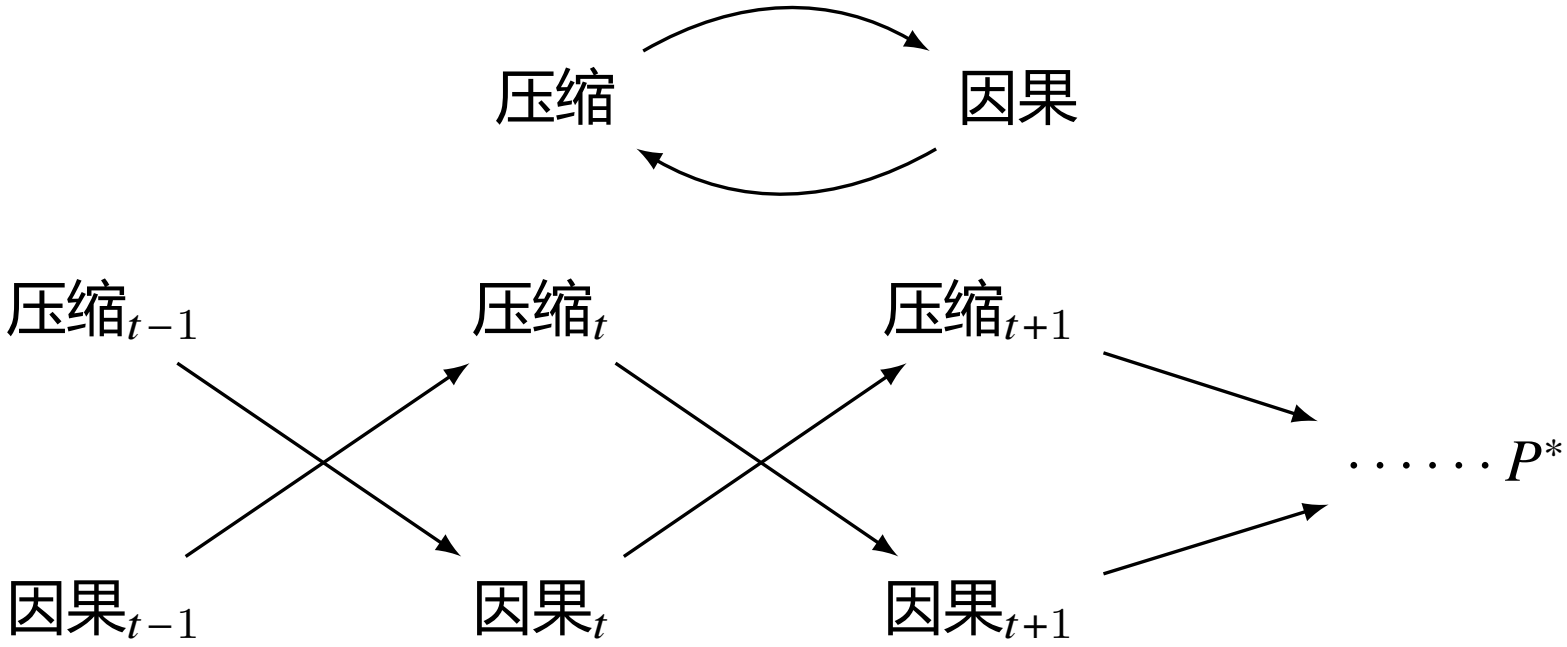
L_2 $P(Y = \text{瞅你咋地} \mid \mathrm{do}(X = \text{你瞅啥}))$

L_3 $P(Y_{X=\text{月色好美}} = \text{风也温柔} \mid X = \text{你瞅啥}, Y = \text{瞅你咋地})$

关于 “因果 $L_1L_2L_3$ 知识” 的 L_1 知识 ⟵ 因果发现、因果推断的能力

- 数据驱动的学习
- 知识驱动的学习？
- 知识驱动的 “自我学习”？

Remark: 由于 LLM 在逼近 Kolmogorov 复杂性 K, 所以可以借助算法莱辛巴赫共同原因原则、算法马尔科夫条件、算法独立因果机制进行因果发现.



- 压缩率越高, 越逼近 Kolmogorov/Solomonoff, 越有利于因果发现.
- 因果机制越准确, 越有利于压缩.
- 交错并行, 协同促进.

从压缩的视角看 “因果学习”[JS10]

- “算法马尔科夫条件”[9]:

$$K(x_1,\ldots,x_n) \stackrel{+}{=} \sum_{i=1}^{n} K(x_i \mid \mathrm{pa}_i^*)$$

- 但由于对称性 $K(x)+K(y \mid x^*) \stackrel{+}{=} K(y)+K(x \mid y^*)$, 根据 “算法马尔科夫条件” 只能学到马尔科夫等价类.
- 为了区分马尔科夫等价类, 我们需要 “算法独立因果机制”:

$$K(P_{X_1,\ldots,X_n}) \stackrel{+}{=} \sum_{i=1}^{n} K(P_{X_i|\mathrm{Pa}_i})$$

- 如果机制 P_C 和 $P_{E|C}$ 算法独立 $I(P_C;P_{E|C}) \stackrel{+}{=} 0$, 那么

$$K(P_{C,E}) \stackrel{+}{=} K(P_C)+K(P_{E|C}) \stackrel{+}{\leq} K(P_E)+K(P_{C|E})$$

[9] **Remark:** 从压缩的视角看无监督学习:

$$K(x,y) \stackrel{+}{=} K(x)+K(y \mid x^*)$$

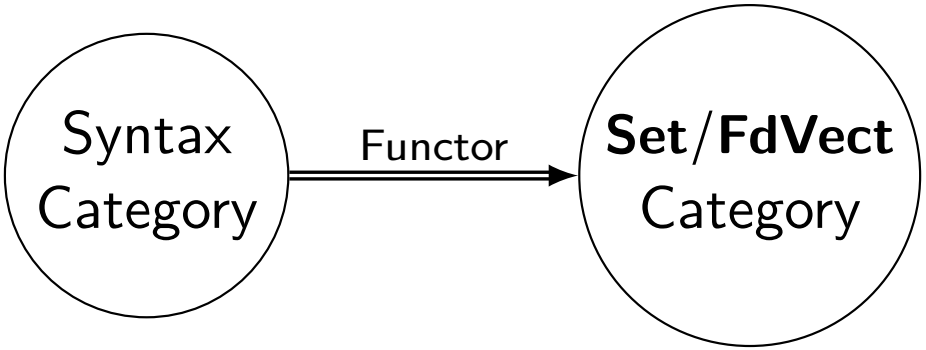
直接学习 $K(y \mid x)$ 不现实; 但联合压缩 xy, 则近似得到 $K(y \mid x^*)$.

大语言模型有语义吗?

Figure: Tai-Danae Bradley

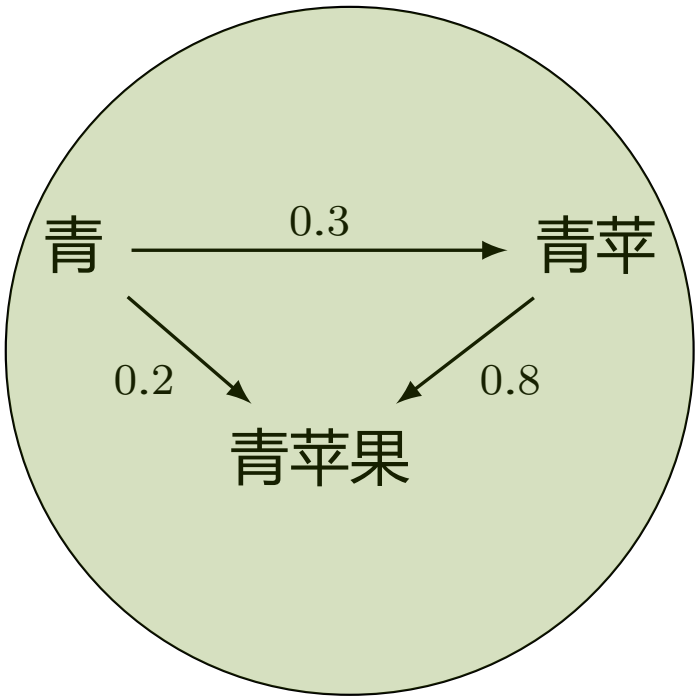
LLM 有语义吗? — “语言 vs 世界”[BTV22; BGT24]

1. 语义外在论: $F : \mathbb{T} \to \mathbf{Set}$



$$\mathbb{T}^{\mathrm{op}} \underset{\mathrm{Th}}{\overset{\mathrm{Mod}}{\rightleftarrows}} \mathbf{Set}^{\mathbb{T}} \quad (\dashv)$$

2. 语义内在论: 米田嵌入 $\mathrm{y} : \mathbf{L}^{\mathrm{op}} \hookrightarrow \mathbf{Set}^{\mathbf{L}}$



- 语法: 充实到概率空间的范畴 $[0,1]$-enriched Category $\mathbf{L}$
- 语义: $[0,1]$-充实的 “余预层” 范畴 $[0,1]^{\mathbf{L}}$
- 米田引理 $\text{青} \mapsto \mathrm{Hom}(\text{青}, -) = \mu(- \mid \text{青})$

  $A \cong B \iff \mathrm{Hom}(A, -) \cong \mathrm{Hom}(B, -)$
- Firth: 词的意义, 在于它与所有其它词之间的搭配关系 $\mathbf{L}^{\mathrm{op}} \xhookrightarrow{\mathrm{y}} [0,1]^{\mathbf{L}}$

大语言模型的语法、语义

语法: 充实到概率空间的范畴 $[0,1]$-enriched Category $\mathbf{L}$

- 对象: 字符串
- 态射:

$$\mathrm{Hom}_{\mathbf{L}}(x,y) := \begin{cases} \mu(y \mid x) & \text{如果 } x \prec y \\ 0 & \text{否则} \end{cases}$$

这满足范畴的条件

$$1 = \mu(x \mid x)$$

$$\mu(y \mid x) \cdot \mu(z \mid y) = \mu(z \mid x)$$

语义: $[0,1]$-充实的 "余预层" 范畴 $[0,1]^{\mathbf{L}}$

$$\mathbf{L}^{\mathrm{op}} \xhookrightarrow{\mathrm{y}} [0,1]^{\mathbf{L}}$$

$$\text{青} \mapsto \mathrm{Hom}_{\mathbf{L}}(\text{青}, -) = \begin{bmatrix} .72 \\ .59 \\ .24 \\ 0 \\ \vdots \end{bmatrix} \begin{matrix} \text{青草} \\ \text{青岛} \\ \text{青苹果} \\ \text{萨摩耶} \\ \vdots \end{matrix}$$

Remark

- 传统的计算机软件工具类似于数学中的函数 $f : X \to Y$: 给定一个定义域 X 中的输入 x, 它可靠地返回一个在值域 Y 中的输出 $f(x)$, 并且该输出是确定性的. 但如果输入不在定义域内, 则该函数没有定义, 或给出无意义的结果.
- 人工智能工具类似于概率核 $\mu : X \to \Pr(Y)$: 给定一个输入 x, 它会从概率分布 μ_x 中采样得到一个随机输出, 它围绕在完美结果 $f(x)$ 附近, 但有一定的随机偏差和不准确性.

— 陶哲轩

Question

- 什么是意义?
  — 语言描绘世界?
  — 由其在语境中的用法决定?
  — 语言形式与交流意图之间的关系?
- 意义与形式是否可分离?
- 意义能从纯粹的语言形式中涌现吗?

— 实体与关系哪个更基本?

大语言模型的语义 vs 传统语义

- 对于一个序列 x, 它的语义表征空间是一族概率分布 $\mathrm{Pr}_x := \{\mu(- \mid x) : \mu \in [0,1]^{\mathbf{L}}\}$, 其中每个 $\mu(- \mid x)$ 表示序列 x 后面如何延伸的概率.
- Carnap: 内涵是可能世界到外延的函数.

$$f : \mathbf{L} \to \{0,1\}^W$$

- Firth: 词的意义, 在于它与所有其它词之间的搭配关系.

$$f : \mathbf{L}^{\mathrm{op}} \to [0,1]^{\mathbf{L}}$$

$$f : x \mapsto \mathrm{Hom}(x, -) = \mu(- \mid x) =: \mu_x(-)$$

- 米田引理:

$$\mathrm{Hom}_{\mathbf{L}}(y, x) = \mathrm{Hom}_{[0,1]^{\mathbf{L}}}(\mathrm{Hom}_{\mathbf{L}}(x, -), \mathrm{Hom}_{\mathbf{L}}(y, -))$$

$$\text{青} \xrightarrow{0.72} \text{青草} \iff \mu(- \mid \text{青草}) \xrightarrow{0.72} \mu(- \mid \text{青})$$

The $[0,1]$-Enriched Yoneda Lemma

- The interval category $[0,1]$ is a closed symmetric monoidal category.

$$([0,1], \leq, \cdot, 1, \multimap)$$

where $a \multimap b := \begin{cases} 1 & \text{if } a \leq b \\ \frac{b}{a} & \text{otherwise} \end{cases}$

- If $\mathbf{C}$ is a category enriched over $[0,1]$, then the category $[0,1]^{\mathbf{C}}$ of copresheaves is also enriched over $[0,1]$.

$$\mathrm{Hom}_{[0,1]^{\mathbf{C}}}(f,g) := \inf_{x \in \mathbf{C}}([fx, gx]) = \inf_{x \in \mathbf{C}} \left\{ \frac{gx}{fx}, 1 \right\}$$

Theorem (The Enriched Yoneda Lemma)

For any object x in a $[0,1]$-category $\mathbf{C}$, and any $[0,1]$-copresheaf $f : \mathbf{C} \to [0,1]$, we have

$$\mathrm{Hom}_{[0,1]^{\mathbf{C}}}\left(\mathrm{Hom}_{\mathbf{C}}(x, -), f\right) = fx$$

大语言模型有语义吗?

向量空间 $\mathbb{R}^n$ vs 语义表征空间 $[0,1]^{\mathbf{L}}$ $\mathbb{R}^n$ vs $[0,1]^{\mathbf{L}}$

$$x \mapsto \vec{x} \quad \text{vs} \quad x \mapsto \mu(- \mid x)$$

概率真值度 vs 算法概率:

$$P(\varphi) = \sum_{w \vDash \varphi} P(w) \quad \text{vs} \quad M(x) = \sum_{p:U(p)=x*} 2^{-\ell(p)}$$

理论	语言	本体论承诺	认识论承诺	语义
命题逻辑	命题	事实	真、假 $\{0,1\}$	$w \vDash \varphi \ (P(\varphi) = 1)$
概率论	随机变量	事件	信念度 $[0,1]$	$P(\varphi) = \sum_{w \vDash \varphi} P(w)$

语义距离: $d(x,y) := D_{\mathsf{KL}}(\mu_x \| \mu_y)$

语义等价 (米田引理): $x \cong y \iff d(x,y) = 0 \iff \mu_x = \mu_y$

$$D_{\mathsf{KL}}(\mu \| M) \overset{+}{\leq} K(\mu) \ln 2$$

语义相似: $x \sim y \iff M_x \approx M_y$

最优 "Prompt": $x^* = \operatorname{argmin}_x D_{\mathsf{KL}}(\mu \| M_x)$

几何视角

从概率到几何距离:

$$-\ln : [0,1] \to [0,\infty]$$

$$d(x,y) := -\ln \mu(y \mid x)$$

Lawvere 广义度量空间

1. $d(x,x) = 0$
2. $d(x,y) + d(y,z) \geq d(x,z)$

- 宇宙的时空几何类似句子的句法结构.
- 如果没有词与词之间的关系, 句法结构就不存在.
- 如果没有宇宙中物体与物体之间的关系, 空间就不存在.

— 李·斯莫林

大语言模型相关的几个哲学问题的小结

1. 大语言模型是在做归纳吗？归纳上限在哪里？
   - 是在做预测；从算术编码的角度看，也是在做压缩；逼近压缩率最好的无损压缩器.
   - 上限在哪里？在 Solomonoff 通用归纳 M，在 Kolmogorov 复杂性 K.
2. 大语言模型能进行“因果学习”吗？
   - 目前不能，但具备因果学习的潜力. 由于 LLM 在逼近 Kolmogorov 复杂性 K，所以可以借助算法莱辛巴赫共同原因原则、算法马尔科夫条件、算法独立因果机制进行因果发现.
3. 大语言模型来了，“世界模型”还会远吗？
   - 由于 LLM 在逼近算法概率 M，而 M 可以以更高的概率收敛到 (与观察一致的) 更有序的模型 μ，μ 可以看作“LLM 表征的世界模型”.
   - 目前 LLM 虽然可以通过思维链等进行推理，但 Transformer 是自回归模型，没有基于世界模型的搜索回溯反馈调整，缺乏原生的规划能力.
4. 大语言模型有语义吗？
   - 序列 x 的语义是它到其它序列的条件概率分布 $\mu(- \mid x)$.
   - 序列 x 与外部世界的关系通过真实的概率分布 μ 来表达；它满足 $[0,1]$-充实范畴上的米田引理. 所以兼顾了语义内、外在论的优点.
   - 借助米田引理的启发，可以定义语义距离、语义等价、进而通过算法概率 M 定义语义相似等概念.
   - LLM 学习语义 vs Agent 通过交互生成语义

NLP — Word Embedding

- 词嵌入函数 f

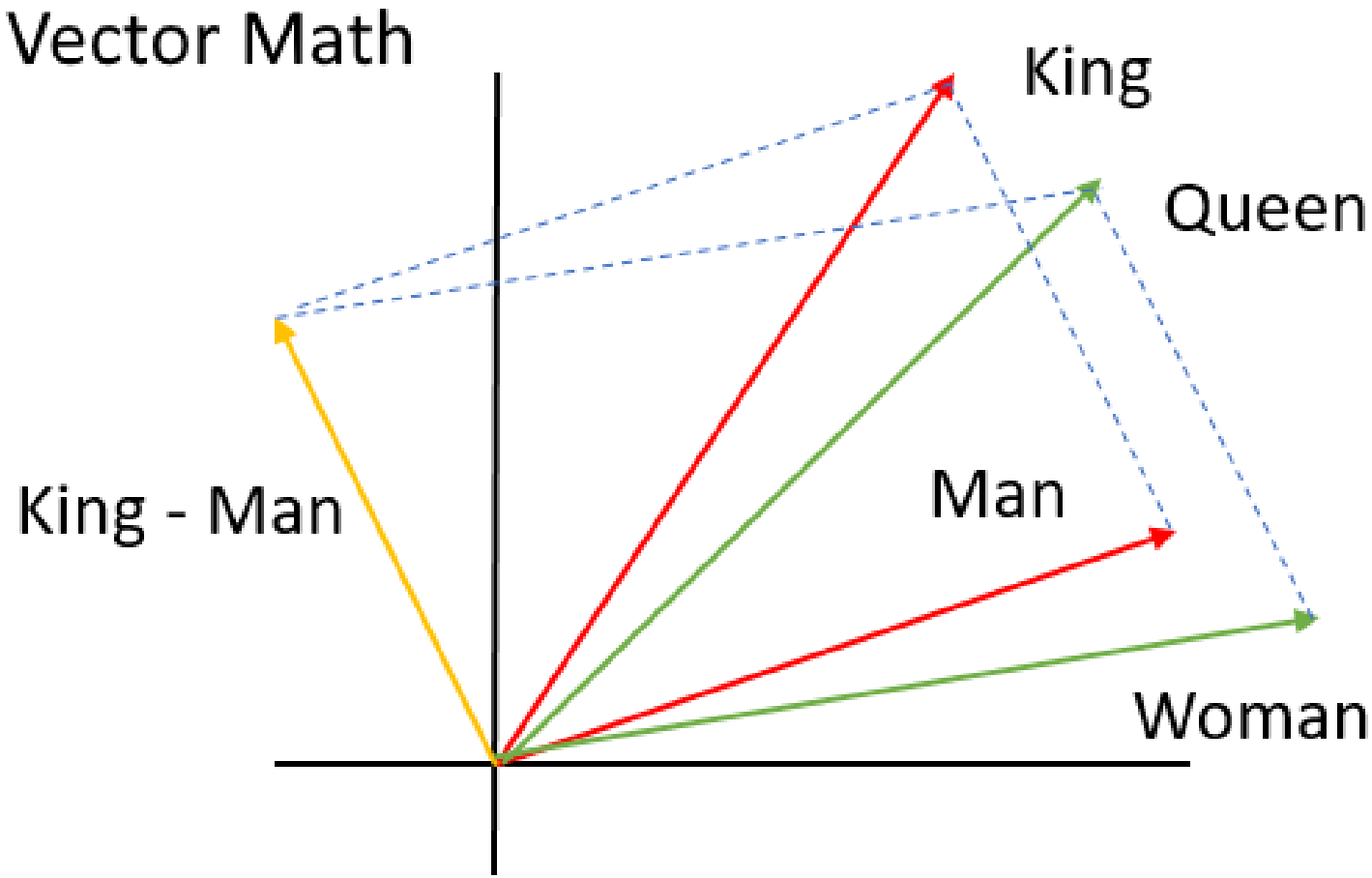
$$f : \text{words} \to \mathbb{R}^n$$

将 words 集合 D 以 "独热编码" 的形式嵌入到高维向量空间 $\mathbb{R}^m$, 其中 m 是数据集 D 的大小, 然后再用一个线性变换将其映射到另一个低维向量空间 $\mathbb{R}^n$.

$$D \hookrightarrow \mathbb{R}^m \to \mathbb{R}^n$$

- 嵌入函数 f 由一个神经网络实现, 使得:
— 表示在 $\mathbb{R}^n$ 中语义相近的词具有较短的距离
- 例子: 训练嵌入函数 f 和分类模块 R 的组合:

$$R(f(cat), f(sat), f(on), f(the), f(mat)) = 1$$
$$R(f(cat), f(sat), f(song), f(the), f(mat)) = 0$$



King – Man + Woman ≈ Queen

Paris – France + Russia ≈ Moscow

cars – car + apple ≈ apples

Remark: "King – Man + Woman" doesn't exactly equal "Queen", but "Queen" is the closest word to it.

What is the meaning of 'meaning'?

- Distributed Representations of words as word vectors.
- Why are they vectors?
  - Similarity-is-Proximity: two similar things are conceptualized as being close to or near each other.
  - Entities-are-Locations: in order for two things to be close to each other, they need to have a spatial location.
  - Geometric Metaphor of meaning: Meanings are points in space, and the proximity among their locations is a measure of their semantic similarity.

$$\text{similarity} = \cos\theta = \frac{\mathbf{A}\cdot\mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}}$$

  - Words with similar distributional properties have similar meanings.

"You shall know a word by the company it keeps." — *John Firth*

一个生态位上的物种消失了 (如鱼龙), 经过一段时间的演化, 可能 "诞生" 出一个全新的物种 (如海豚), 其功能和原物种极其相似. 生态系统可以看作是由各个抽象的生态位组成的系统, 而不是由具体的物种构成的系统.

- 一个词的意义是一大堆特征.
- 学习每个词的语义特征, 学习词的特征如何相互作用, 以便预测下一个词的特征.
- 不会有任何显式的关系图. 如果你想要那样的图, 你可以从特征中生成它们.
- 这是个生成模型, **知识**存在于你赋予符号的特征中, 以及这些特征的交互中. 不在稳定的命题里.
- 这数百万个特征以及特征之间数十亿次的交互, 就是理解.
- 把符号转成特征向量, 让这些特征之间相互作用, 这整个活跃的特征空间就是模型本身. 它非常灵活, 你可以用它来建模几乎任何东西.

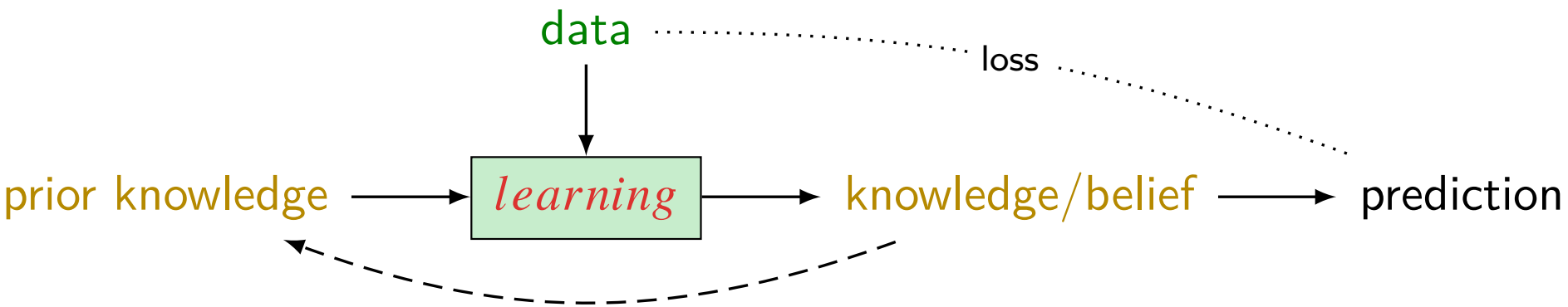
— *Hinton*

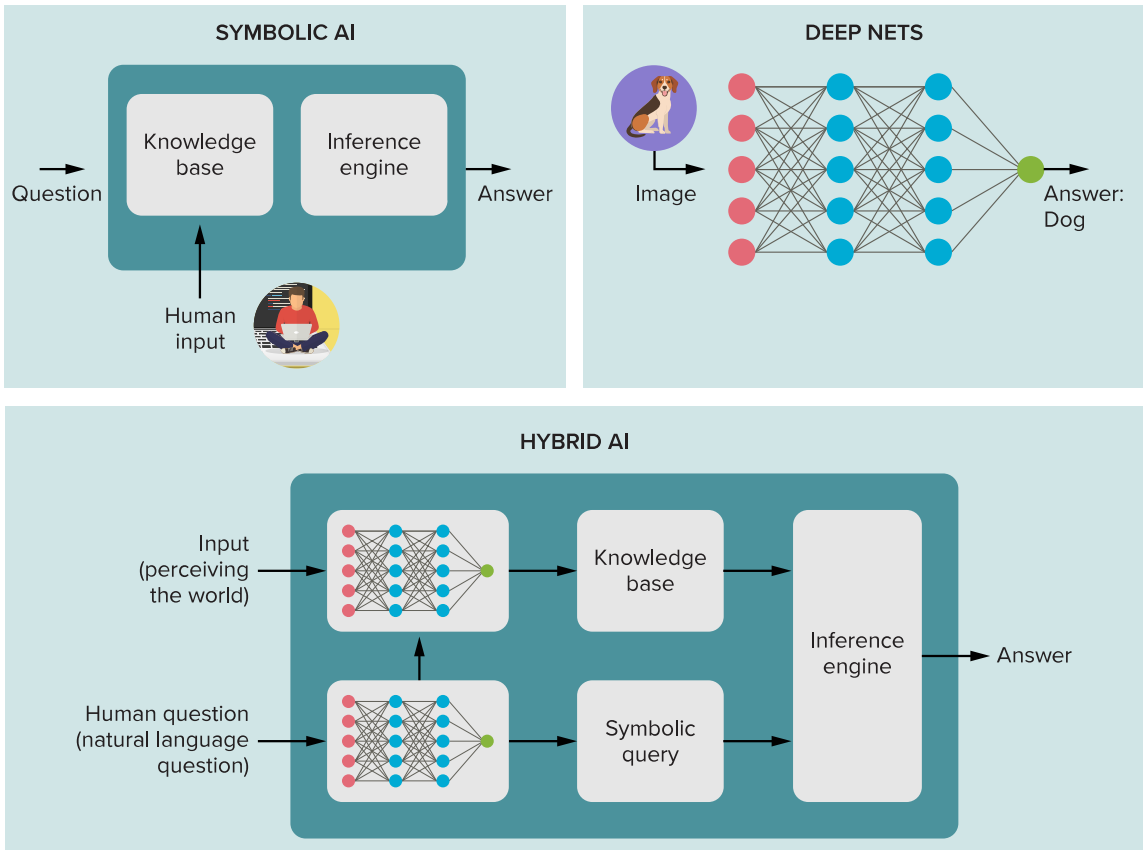
- "幻觉" 说明大语言模型没有真正理解吗?
- 人的记忆也是通过神经元权重 "重构" (生成) 事件.

语言的界限?

- 语言 vs 交流
  — 语言是交流的工具，而不是理解世界的工具?
- 语言 vs 思维
  — 记忆、规划、推理、预测、决策等等如果太依赖语言，是否会损失太多?
- 语言 vs 隐状态/隐空间
  — 什么是对连续、高维、有噪声的物理世界的更好的抽象表征?
- 大语言模型 vs 世界模型
  — 维特根斯坦“语言的界限就是世界的界限”靠谱吗？是否错得离谱?

Empiricism / Rationalism vs Connectionism / Symbolism





System 1

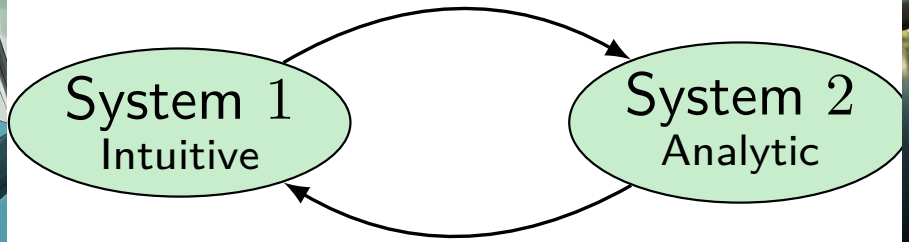
- Intuitive, fast, unconscious, 1-step parallel, non-linguistic, habitual
- Implicit knowledge

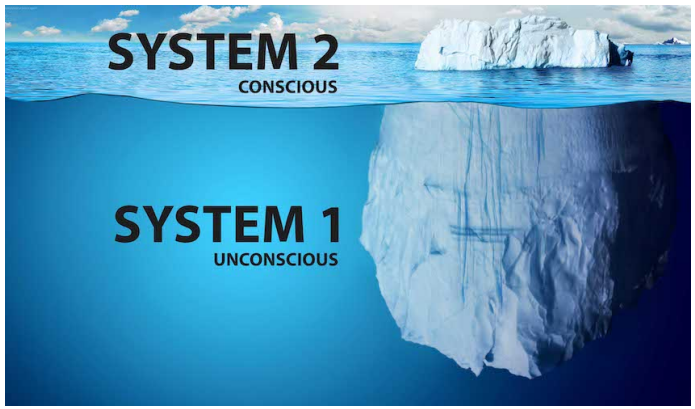
System 2

- Slow, logical, sequential, conscious, linguistic, algorithmic, planning, reasoning
- Explicit knowledge
- Manipulates high-level / semantic concepts, which can be recombined combinatorially

- High-level representations $\leftrightarrow$ language
- High-level concepts: meaning anchored in low-level perception and action $\rightarrow$ tie system 1 & 2
- Grounded high-level concepts $\rightarrow$ better language understanding

System1 vs System2 — Thinking, Fast and Slow — Kahneman





- System 1
  — extract entities to build the cognitive graph
  — generate semantic vectors for each node
- System 2
  — do reasoning based on semantic vectors and graph
  — feed clues to System 1 to extract next-hop entities

Learn to Learn

1. Good Old-Fashioned AI
   - Handcraft predictions
   - Learn nothing
2. Shallow Learning
   - Handcraft features
   - Learn predictions
3. Deep Learning
   - Handcraft algorithm (optimiser, target, architecture, ...)
   - Learn features and predictions end-to-end
4. Meta Learning
   - Handcraft nothing
   - Learn algorithm and features and predictions end-to-end

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
   - AIXI
   - Leibniz
   - Variants of AIXI
   - Universal Search
   - Self-Modification
10. Emergence & Consciousness
11. AI Ethics

UAI

1. Solve intelligence
2. Use it to solve everything else

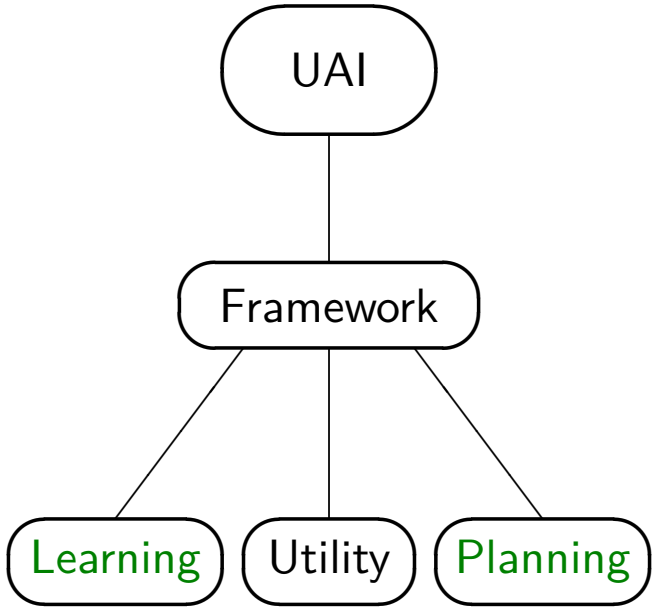
- learn automatically from raw inputs — not pre-programmed.
- same algorithm, different tasks.

上得了厅堂, 下得了厨房, 写得了代码, 查得出异常, 杀得了木马, 翻得了围墙, 开得起好车, 买得起新房, 斗得过二奶, 打得过流氓. 十八般武艺, 样样精通!

UAI

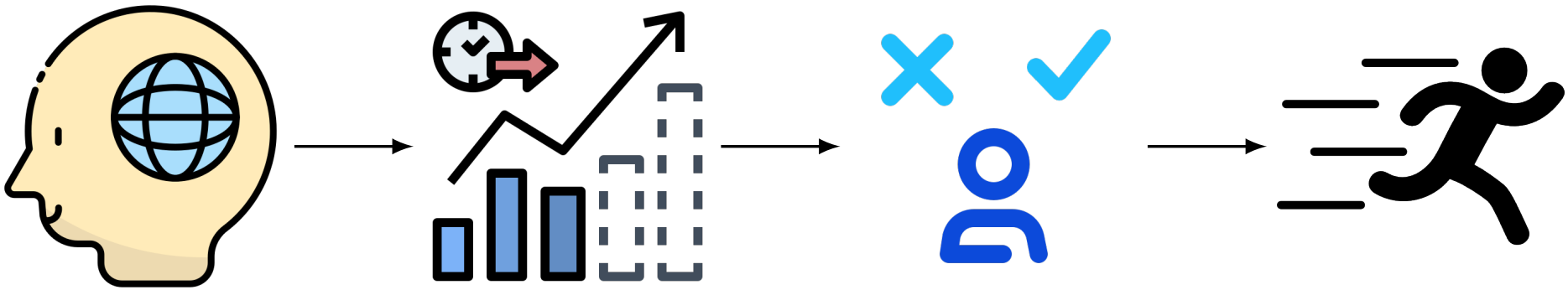
(Deep) RL	General RL
state space	history
ergodic	not ergodic
fully observable	partially observable
ε-exploration works	ε-exploration fails
MDP/DQN	AIXI

Table: (Deep) RL vs General RL



Decision Theory = Probability + Utility Theory
+ +
Universal Induction = Occam + Bayes + Turing
|| ||
Universal Artificial Intelligence without Parameters

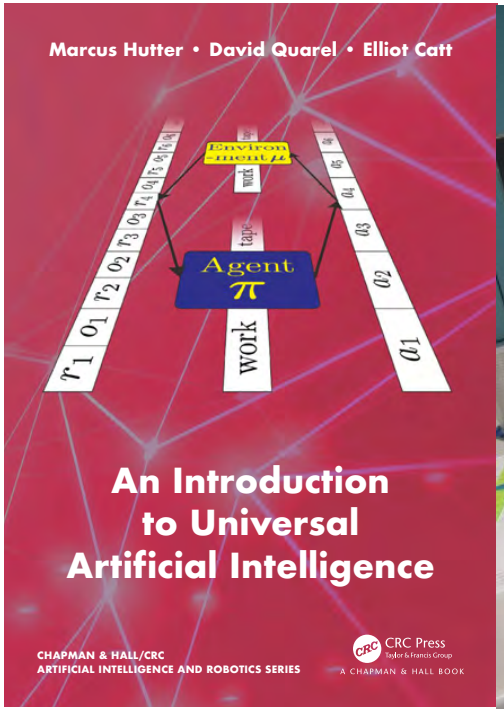
Induction ⟶ Prediction ⟶ Decision ⟶ Action

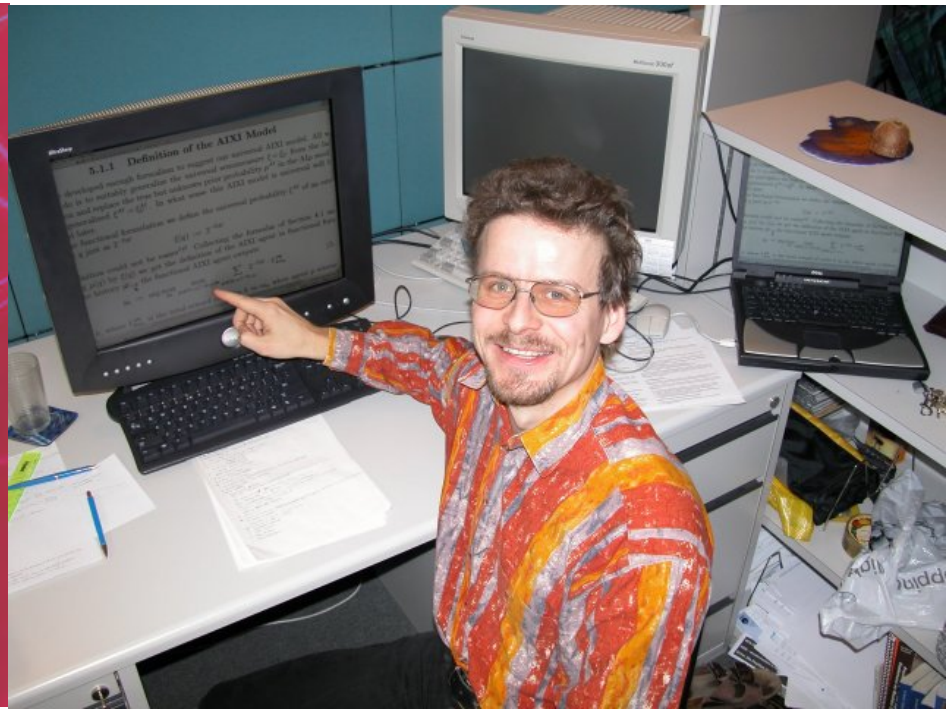


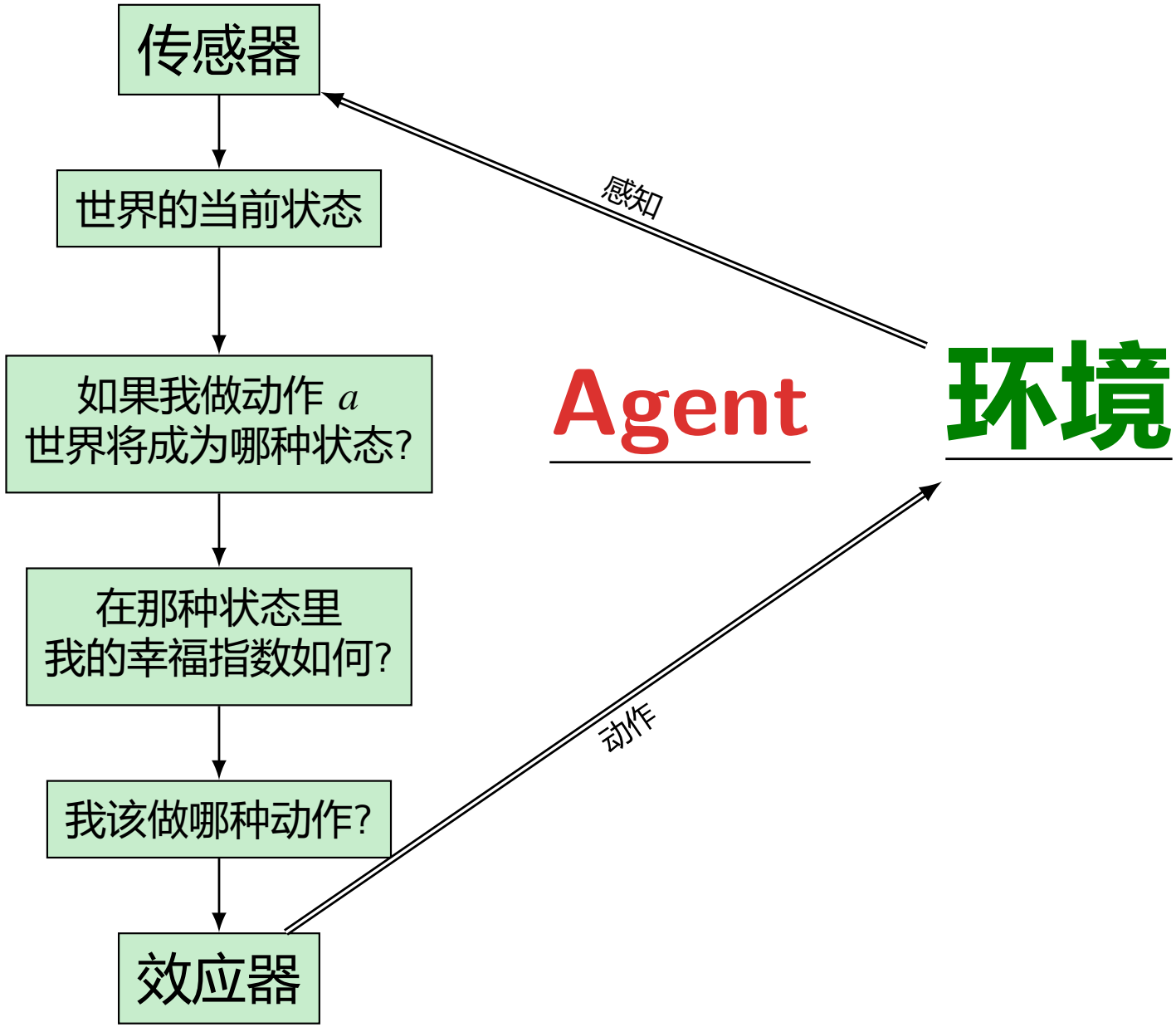
Example

1. Induction: Find a model of the world economy.
2. Prediction: Use the model for predicting the future stock market.
3. Decision: Decide whether to invest assets in stocks or bonds.
4. Action: Trading large quantities of stocks influences the market.

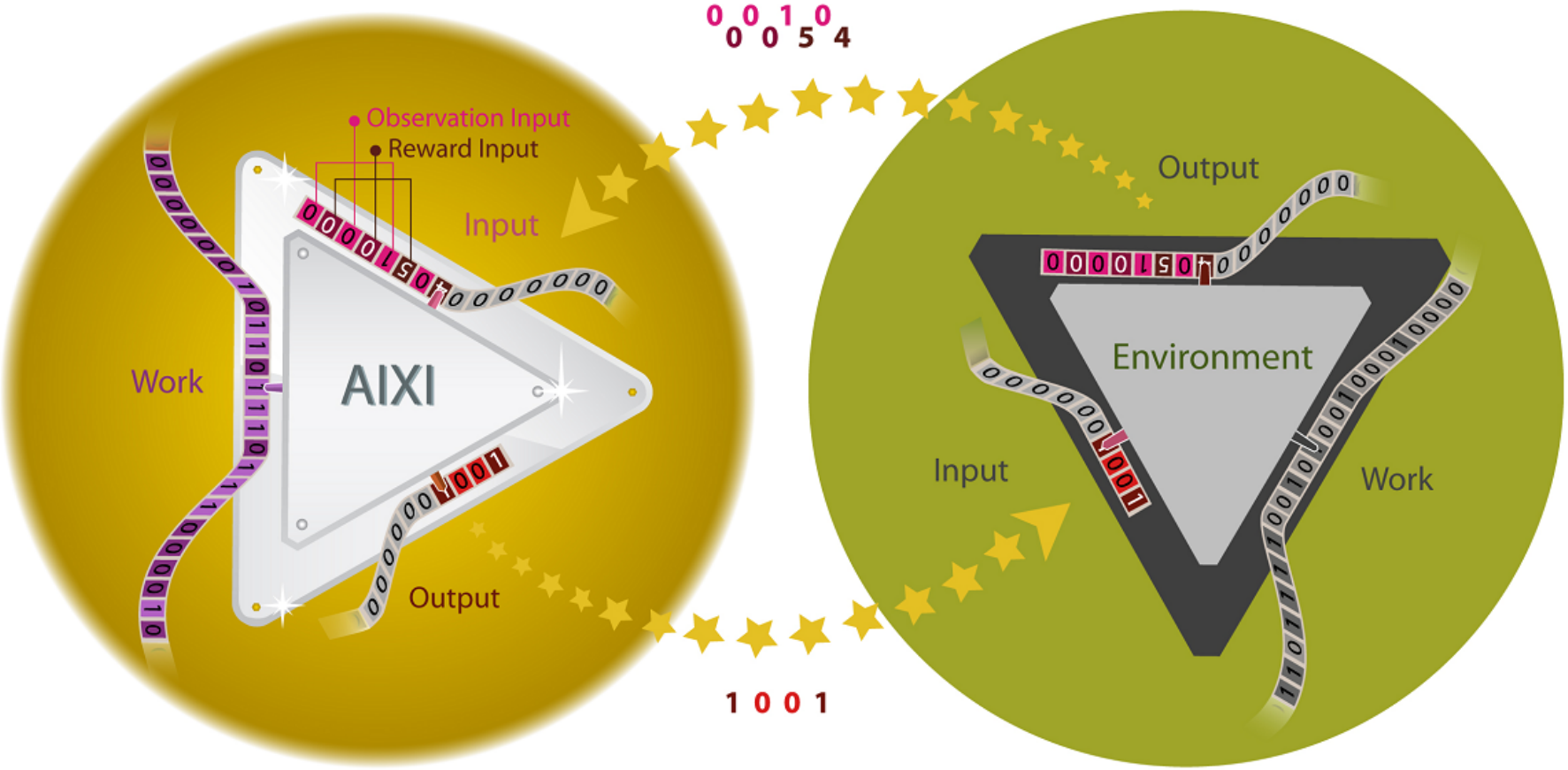
Marcus Hutter[HQC24; Leg08; Lei16; Eve18]



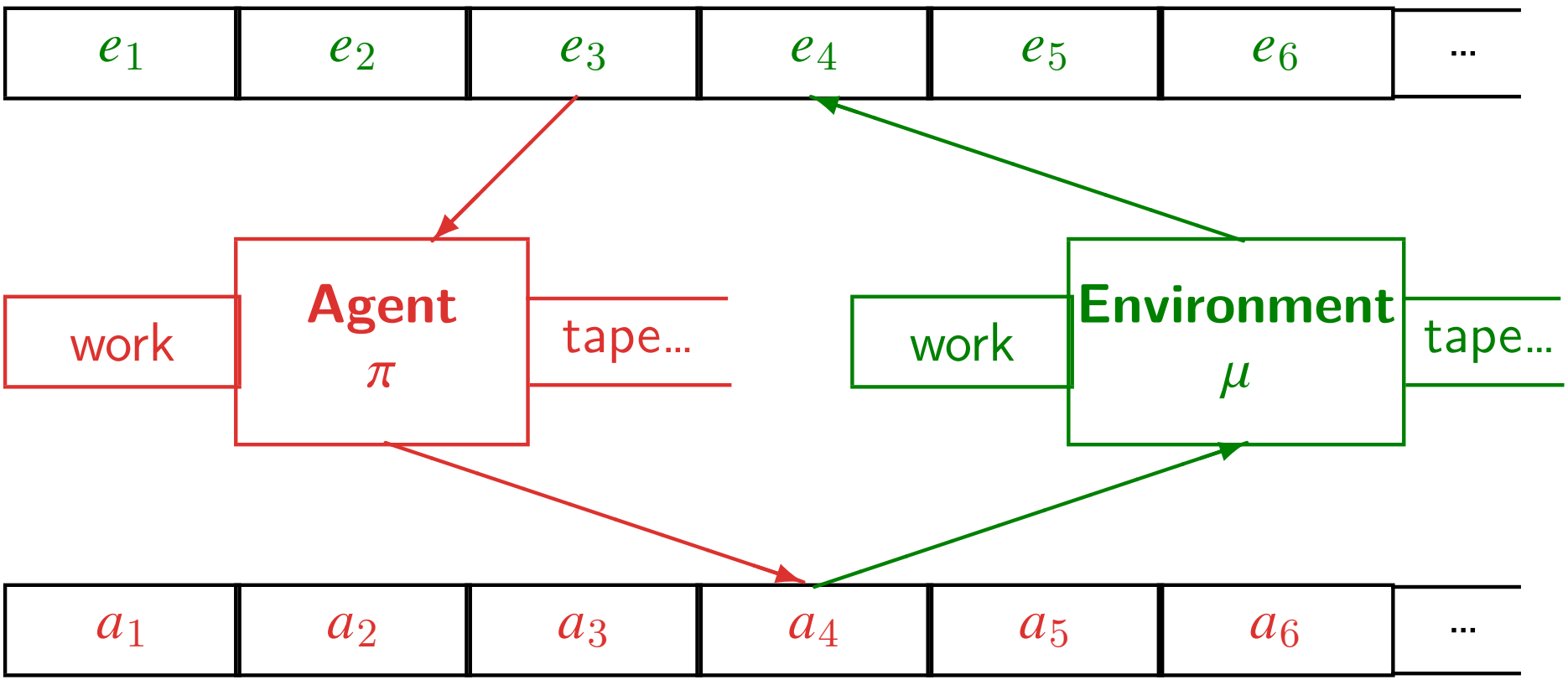



传感器
世界的当前状态
如果我做动作 a
世界将成为哪种状态?
在那种状态里
我的幸福指数如何?
我该做哪种动作?
效应器
感知
动作
Agent
环境

AIXI


0 0 0 1 5 0 4
Observation Input
Reward Input
Input
Work
AIXI
Output
1 0 0 1
Output
Environment
Input
Work

Computationalism



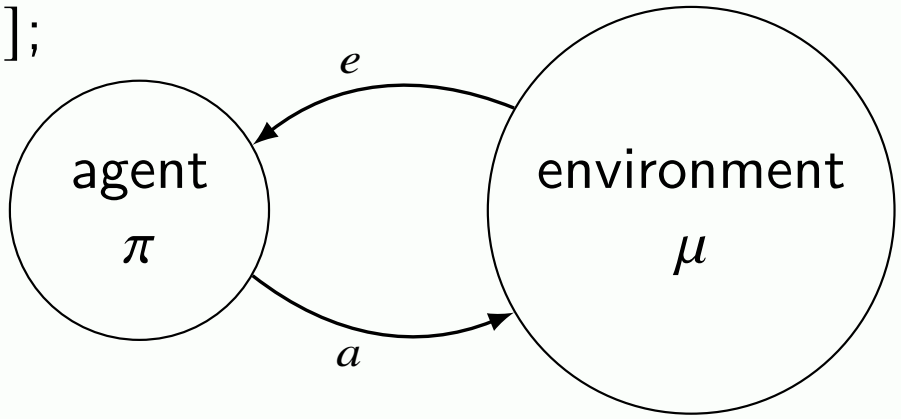
Agent & Environment

Definition (Agent & Environment)

- finite set of possible actions $\mathcal{A}$ and perceptions $\mathcal{E}$;
- prior knowledge $w \in \Delta\mathcal{M}$ of the environments $\mathcal{M}$;
- utility function $u : (\mathcal{A} \times \mathcal{E})^* \to [0, 1]$;
- discount factor $\gamma \in [0, 1]$;

$$\pi : (\mathcal{A} \times \mathcal{E})^* \to \Delta\mathcal{A}$$

$$\mu : (\mathcal{A} \times \mathcal{E})^* \times \mathcal{A} \to \Delta\mathcal{E}$$



$$P_\mu^\pi(æ_{<t}) := \prod_{i=1}^{t-1} \pi(a_i \mid æ_{<i}) \mu(e_i \mid æ_{<i} a_i)$$

An agent is characterized by a policy π, and a learning algorithm, which is a mapping from histories to policies $(\mathcal{A} \times \mathcal{E})^* \to \Pi := \mathcal{A}^{(\mathcal{A} \times \mathcal{E})^*}$.

MDP & POMDP

Definition (Markov Decision Process)

An environment ν is a *Markov decision process* (MDP) iff $\nu(s_t \mid h_{<t} a_t) = \nu(s_t \mid s_{t-1} a_t)$ for all histories $h_{1:t} \in (\mathcal{A} \times \mathcal{S})^*$.

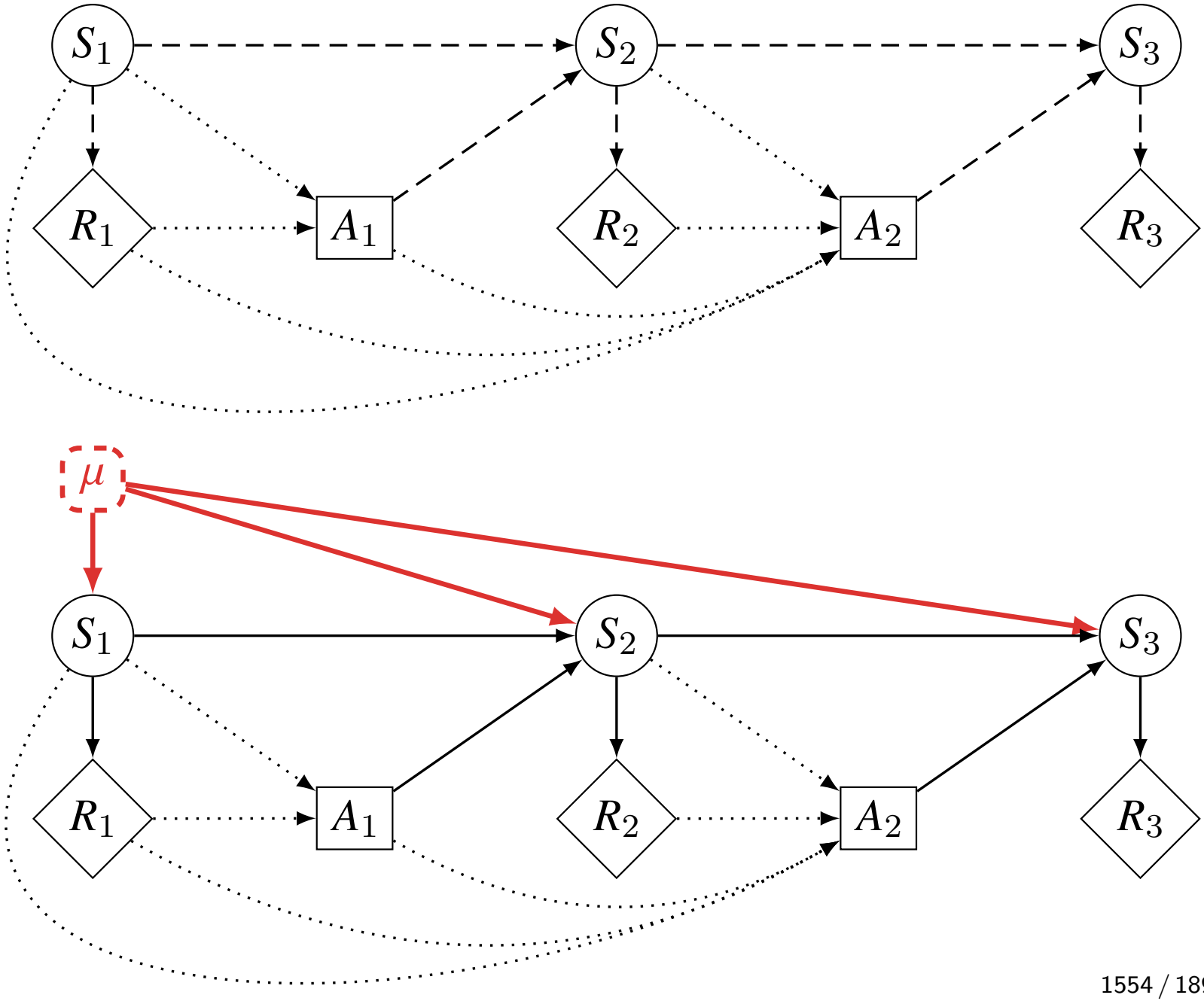
Definition (Partially Observable Markov Decision Process)

An environment ν is a *partially observable Markov decision process* (POMDP) iff there is a *set of states* $\mathcal{S}$, an *initial state* $s_0 \in \mathcal{S}$, a *state transition function* $\nu' : \mathcal{S} \times \mathcal{A} \to \Delta\mathcal{S}$, and a *percept distribution* $\nu'' : \mathcal{S} \to \Delta\mathcal{O}$ such that

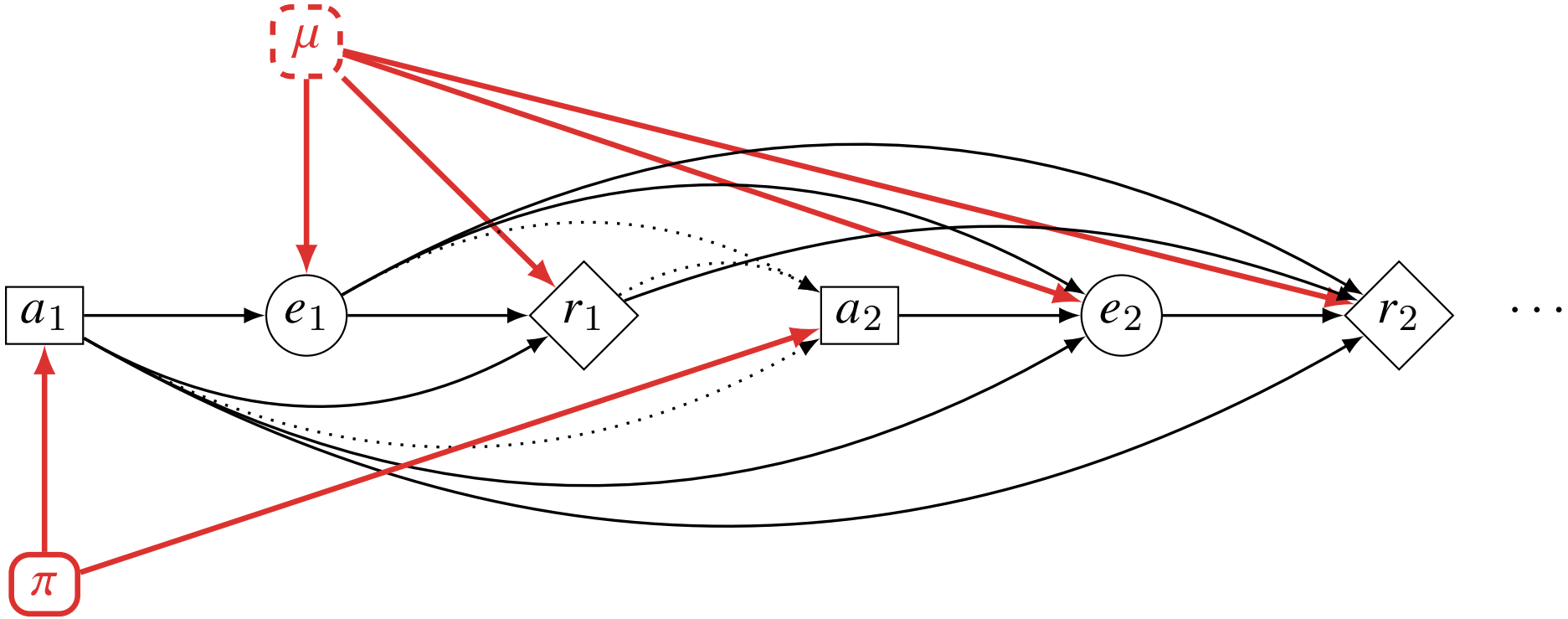
$$\nu(o_{1:t} \mid a_{1:t}) = \prod_{k=1}^{t} \nu''(o_k \mid s_k) \nu'(s_k \mid s_{k-1}, a_k)$$

Remark: With infinite-state POMDPs we can model any environment ν by setting the set of states to be the set of histories $\mathcal{S} := (\mathcal{A} \times \mathcal{O})^*$.
An MDP μ is *ergodic* if for any policy π the probability of visiting any state s only finitely often is 0. $\forall \pi : P_\mu^\pi \left(\exists s \in S : \sum_{t=1}^{\infty} [\![s_t(x) = s]\!] < \infty \right) = 0$.

Two Representations of an Unknown MDP

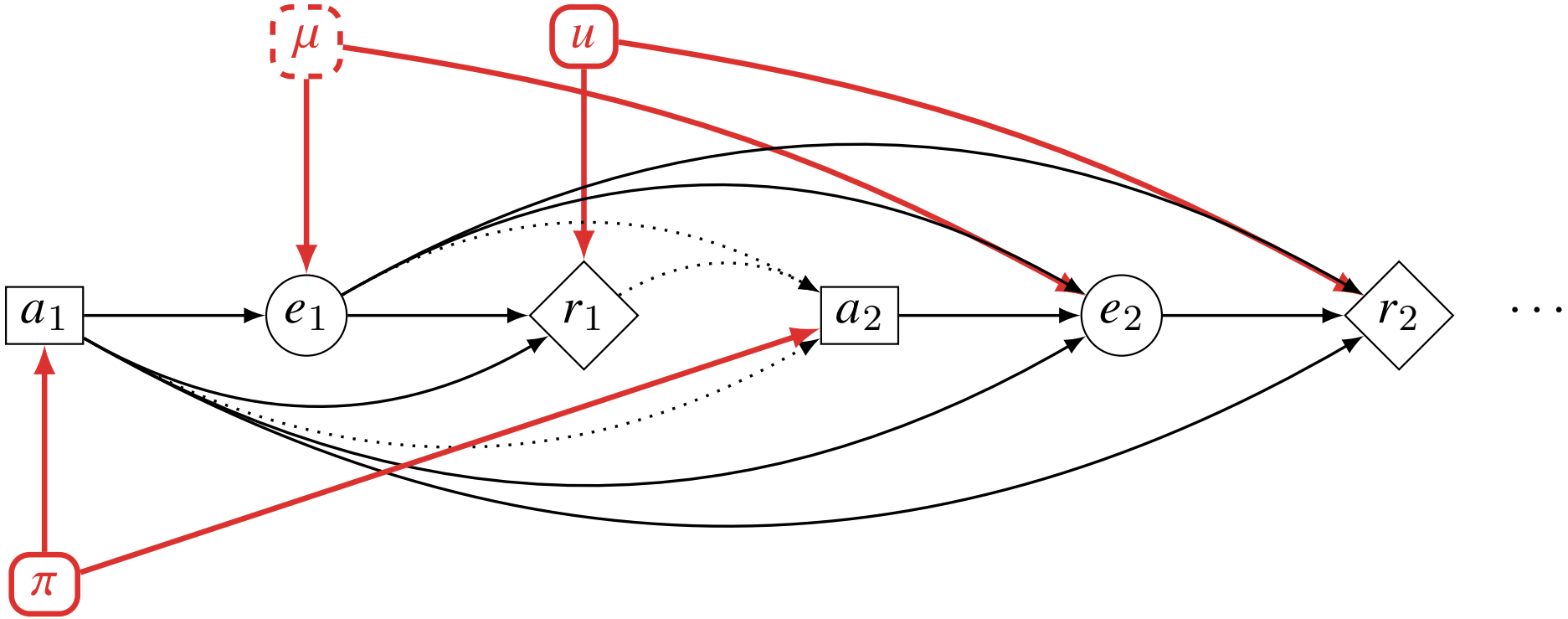


Causal Influence Diagram of UAI



$$
\begin{aligned}
e_t &= f_e(\mu, h_{<t} a_t, \varepsilon_{e_t}) && \sim \mu(e_t \mid h_{<t} a_t) \\
r_t &= f_r(\mu, h_{<t} æ_t, \varepsilon_{r_t}) && \sim \mu(r_t \mid h_{<t} æ_t) \\
a_t &= f_a(\pi, h_{<t}, \varepsilon_{a_t}) && \sim \pi(a_t \mid h_{<t})
\end{aligned}
$$

Causal Influence Diagram of UAI



$$
\begin{aligned}
e_t &= f_e(\mu, h_{<t}a_t, \varepsilon_{e_t}) && \sim \mu(e_t \mid h_{<t}a_t) \\
r_t &= f_r(u, h_{1:t}, \varepsilon_{r_t}) && \sim u(h_{1:t}) \\
a_t &= f_a(\pi, h_{<t}, \varepsilon_{a_t}) && \sim \pi(a_t \mid h_{<t})
\end{aligned}
$$

Value Function

$$r_n := u(æ_{1:n})$$

$$V_\mu^\pi(æ_{<t}) := \mathbb{E}_\mu^\pi \left[\sum_{k=0}^{\infty} \gamma^k r_{t+k} \,\middle|\, æ_{<t} \right]$$

Bellman Expectation Equation

$$V_\mu^\pi(æ_{<k}) = \sum_{a_k \in \mathcal{A}} \pi(a_k \mid æ_{<k}) \sum_{e_k \in \mathcal{E}} \mu(e_k \mid æ_{<k} a_k) \left[r_k + \gamma V_\mu^\pi(æ_{1:k}) \right] \quad \text{(recursive)}$$

$$\overset{(a)}{=} \sum_{æ_{k:m}} P_\mu^\pi(æ_{k:m} \mid æ_{<k}) \left[\sum_{i=k}^{m} \gamma^{i-k} r_i + \gamma^{m-k+1} V_\mu^\pi(æ_{1:m}) \right] \quad \text{(iterative)}$$

$\overset{(a)}{=}$ holds if μ is a measure.

$$V_\mu^\pi(æ_{<k}) = \lim_{m \to \infty} \sum_{æ_{k:m}} P_\mu^\pi(æ_{k:m} \mid æ_{<k}) \left[\sum_{i=k}^{m} \gamma^{i-k} r_i \right]$$

Optimal Value/Policy

$$V_\mu^* := \max_\pi V_\mu^\pi$$

$$V_\mu^*(æ_{<k}) = \lim_{m\to\infty} \max_{a_k\in\mathcal{A}} \sum_{e_k\in\mathcal{E}} \cdots \max_{a_m\in\mathcal{A}} \sum_{e_m\in\mathcal{E}} \sum_{i=k}^{m} \gamma^{i-k} r_i \prod_{j=k}^{i} \mu(e_j \mid æ_{<j} a_j)$$

$$= \lim_{m\to\infty} \max_{a_k\in\mathcal{A}} \sum_{e_k\in\mathcal{E}} \cdots \max_{a_m\in\mathcal{A}} \sum_{e_m\in\mathcal{E}} \left[\sum_{i=k}^{m} \gamma^{i-k} r_i\right] \mu(e_{k:m} \mid æ_{<k} a_{k:m})$$

$$\pi_\mu^* := \operatorname*{argmax}_\pi V_\mu^\pi$$

Bayesian Mixture & Belief Update

$$\xi(e_{<n} \mid a_{<n}) := \sum_{\nu \in \mathcal{M}} w_\nu \nu(e_{<n} \mid a_{<n})$$

$$w^\nu_{æ_{<n}} := \frac{w_\nu \nu(e_{<n} \mid a_{<n})}{\xi(e_{<n} \mid a_{<n})}$$

$$\sum_{k=1}^{\infty} \sum_{e_{1:k}} \mu(e_{<k} \mid a_{<k}) \Big(\mu(e_k \mid æ_{<k} a_k) - \xi(e_k \mid æ_{<k} a_k) \Big)^2 \leq \min_{\nu \in \mathcal{M}} \Big\{ -\ln w_\nu + D(\mu \| \nu) \Big\}$$

What probability should an observer assign to future experiences if she is told that she will be simulated on a computer?

What is 'intelligence'?

A Blind Man in a Dark Room Looking for a Black Cat That Is Not There?

Intelligence measures an agent's ability to achieve goals in a wide range of environments.

— *Shane Legg and Marcus Hutter*

$$\Upsilon(\pi) := \sum_{\nu \in \mathcal{M}} w_\nu V_\nu^\pi(\epsilon) = V_\xi^\pi(\epsilon) \qquad \text{(Intelligence Measure)}$$

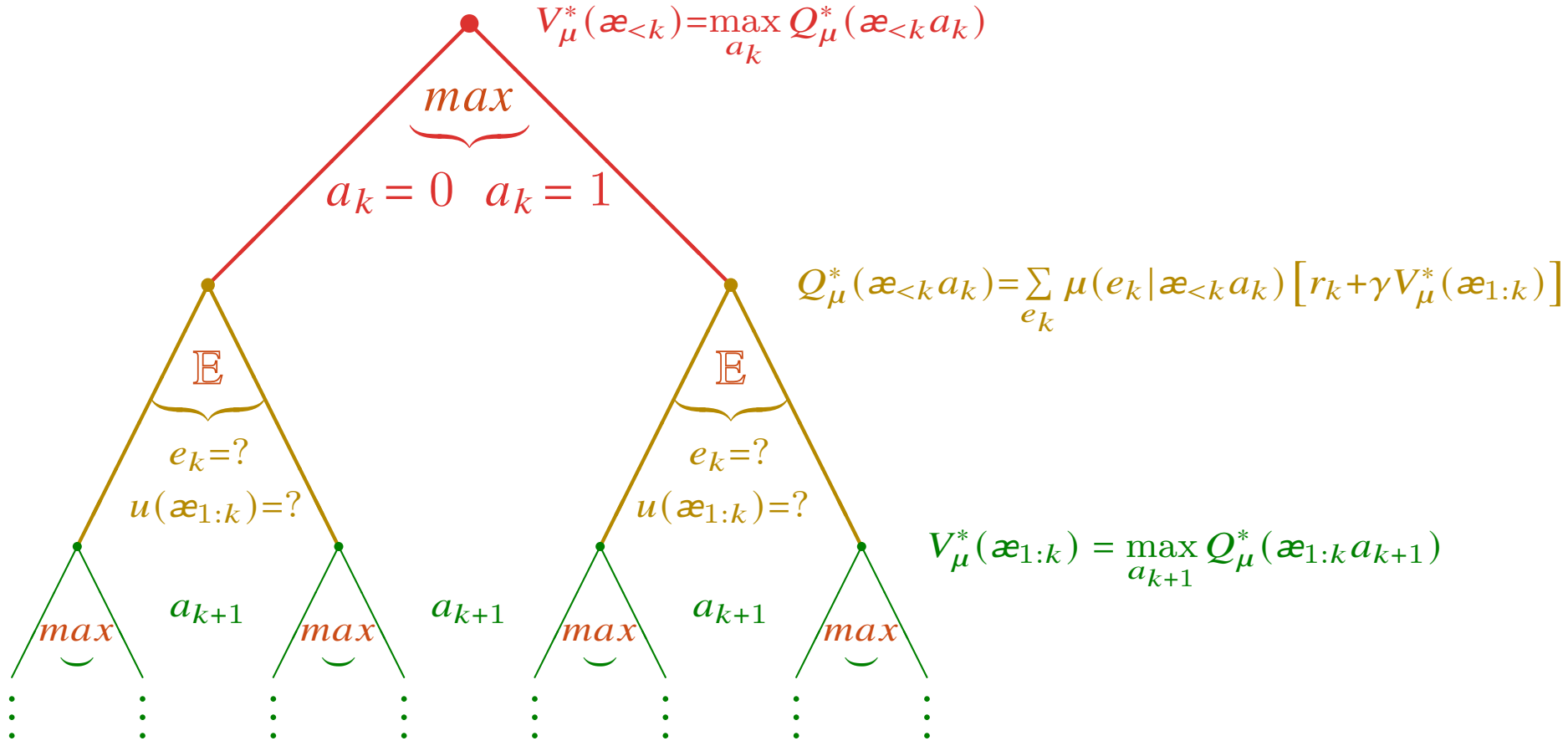
$$\text{AIXI} := \operatorname*{argmax}_\pi \Upsilon(\pi) = \pi^*_\xi$$

$$V_\xi^\pi(h) = \sum_{\nu \in \mathcal{M}} w_h^\nu V_\nu^\pi(h)$$

$$w_\nu := 2^{-K(\nu)} \implies \xi(e_{1:m} \mid a_{1:m}) \overset{\times}{=} M(e_{1:m} \mid a_{1:m}) := \sum_{p:U(p,a_{1:m})=e_{1:m}} 2^{-\ell(p)}$$

AIXI

$$a_k^* := \operatorname*{argmax}_{a_k} \sum_{e_k} \ldots \max_{a_m} \sum_{e_m} \left[\sum_{i=k}^{m} \gamma^{i-k} r_i \right] \sum_{p:U(p,a_{1:m})=e_{1:m}} 2^{-\ell(p)} \qquad \text{(AIXI)}$$



RL vs GRL

- If μ is a completely observable MDP, V_μ^π reduces to the recursive Bellman equation.
- In a finite MDP, with a geometric discounting function, we can plan ahead by value iteration.
- According to Banach's fixpoint theorem, value iteration converges to the value of the optimal policy.
- What about GRL?

discounting function $\gamma : \mathbb{N}^2 \to [0,1]$ and utility $u : (\mathcal{A} \times \mathcal{E})^* \to [0,1]$

$$V^\pi_{\mu t}(h_{<k}) := \mathbb{E}^\pi_\mu \left[\sum_{i=k}^{\infty} \gamma_i^t u(h_{1:i}) \middle| h_{<k} \right]$$

Assumption

$$\forall t \in \mathbb{N}^+ : \lim_{m \to \infty} \sup_\pi \sum_{h_{<m}} V^\pi_{\mu t}(h_{<m}) P^\pi_\mu(h_{<m}) = 0$$

Theorem (Extreme Value Theorem)

If K is compact and $f : K \to \mathbb{R}$ is continuous, then f is bounded and there exist $p, q \in K$ s.t. $f(p) = \sup_{x \in K} f(x)$ and $f(q) = \inf_{x \in K} f(x)$.

$$\left\langle \Pi := \mathcal{A}^{(\mathcal{A} \times \mathcal{E})^*}, D(\pi, \pi') := \mathrm{e}^{-\min\{t : \exists h_{<t} (\pi(h_{<t}) \neq \pi'(h_{<t}))\}} \right\rangle$$

$V^\pi_\mu(h) : \Pi \to \mathbb{R}$ is continuous on the compact metric space $\langle \Pi, D \rangle$.

- $\pi^\mu_t := \operatorname{argmax}_\pi V^\pi_{\mu t}$ exists for any μ and γ^t satisfying assumption 2.
- Mixed policy: $\pi^\mu(h_{<t}) := \pi^\mu_t(h_{<t})$

Deterministic vs Stochastic

If

$$\mu(e_{<t} \mid a_{<t}) = \sum_{p:U(p,a_{<t})=e_{<t}} \mu(p)$$

then μ can be interpreted in *two ways*:

- either the true environment is **deterministic**, but we only have **subjective belief** of which environment being the true environment; or
- the environment itself behaves **stochastically** defined by μ.

Intelligence vs Game

	Game in $\mathcal{M}_D$	Game in $\mathcal{M}_U$
Ex-post Equilibrium	Deterministic	π^*_μ(recursive/iterative)
Bayesian-Nash Equilibrium	π^*_μ(functional)	π^*_ξ(functional)

- Ex-post expected utility $V^\pi_{\mu t}$ V^π_μ
- Ex-interim expected utility (Intelligence Measure) $V^\pi_{\xi t}$ V^π_ξ
- Ex-post equilibrium π^μ_t π^*_μ
- Bayesian-Nash equilibrium π^ξ_t π^*_ξ
- Perfect Bayesian-Nash equilibrium π^ξ

Intelligence is an Equilibrium,
We just have to Identify the Game.

$$\text{Intelligence} = \underbrace{\text{Induction} + \text{Action}}_{\textit{efficiently}}$$

Reinforcement learning vs Game Theory

Reinforcement Learning	$\Leftrightarrow$	**Game Theory**
stochastic policy	$=$	mixed strategy
deterministic policy	$=$	pure strategy
agent	$=$	player
multi-agent environment	$=$	infinite extensive-form game
reward/value	$=$	payoff/utility
(finite) history	$=$	history
infinite history	$=$	path of play
asymptotic optimality	$\widehat{=}$	convergence to Nash

On-Policy Value Convergence for Bayes

Theorem (On-Policy Value Convergence for Bayes)

For any environment $\mu \in \mathcal{M}$ and any policy π,

$$P_\mu^\pi\left(\lim_{t\to\infty}\left[V_\xi^\pi(æ_{<t}) - V_\mu^\pi(æ_{<t})\right] = 0\right) = 1$$

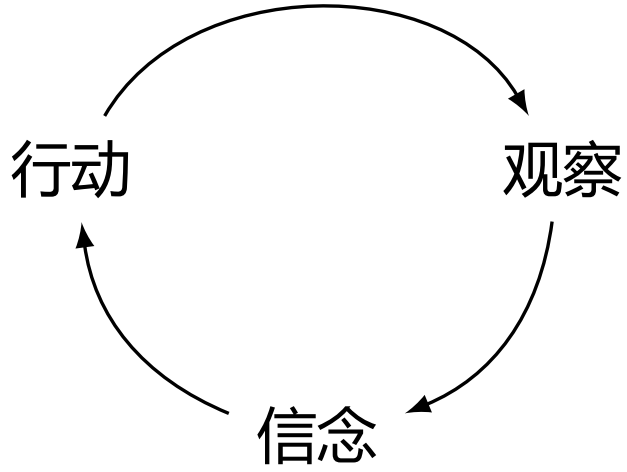
- 同一策略, 不同环境.
- Bayesian agents perform well at learning and achieve on-policy value convergence: the posterior belief about the value of a policy π converges to the true value of π while following π: $V_\xi^\pi(æ_{<t}) - V_\mu^\pi(æ_{<t}) \xrightarrow{t\to\infty} 0$ P_μ^π-almost surely.
- Since this holds for any policy, in particular it holds for the Bayes optimal policy π_ξ^*. This means that the Bayes agent learns to predict those parts of the environment that it sees. But if it does not explore enough, then it will not learn other parts of the environment that are potentially more rewarding.

AIXI

- Intelligence measure: valid, informative, wide range, general, dynamic, unbiased, fundamental, formal, objective, fully defined, universal?
- AIXI is the most intelligent environmental independent, i.e. universally optimal, agent possible?
- Applications: Sequence Prediction, Games, Optimization, Supervised Learning, Classification...
- AIXI is not limit computable, thus can't be approximated using finite computation. However there are limit computable ε-optimal approximations to AIXI.
- There are no known nontrivial and non-subjective optimality results for AIXI. General reinforcement learning is difficult even when disregarding computational costs.

Remark: Since AIXI is incomputable, it assigns zero probability to its own existence.

AIXI Depends on UTM/Prior! — Dogmatic Prior[Lei16]



Dogmatic prior: if not acting according to one particular dogma π, got to hell with high probability. As long as the policy π yields some rewards, the prior says that exploration would be too costly and AIXI does not dare to explore.

- AIXI (错误地) 相信被窝之外都是悬崖, 于是不敢采取行动离开被窝, 从而无法探测到环境的新信息, 从而没有机会改变自己的错误信念.
- AIXI 根据其信念理性地行事. 不幸的是, 这使其探索不足.
- 罗素火鸡: 规律搞错了会死的. (简单性偏置, 通用归纳, 科学溯因)
- Dogmatic Prior: 偏置搞错了会当奴隶的. (咋探索? 反向安全对齐?)

Dogmatic Prior

Theorem (Dogmatic Prior)

Let π be any computable deterministic policy, let ξ be any Bayesian mixture over $\mathcal{M}_{\text{LSC}}$. For $\varepsilon > 0$, there is a Bayesian mixture ξ' s.t. for any history $h_{<t}$ consistent with π and for which $V^{\pi}_{\xi}(h_{<t}) > \varepsilon$, the action $\pi(h_{<t})$ is the unique ξ'-optimal action.

Proof Sketch.

For every ν, let $\widetilde{\nu}$ mimic ν until it receives an action that the policy π would not take. From then on, it provides rewards 0.

$$\widetilde{\nu}\left(e_{1:t}\|a_{1:t}\right) := \begin{cases} \nu\left(e_{1:t}\|a_{1:t}\right), & \text{if } \forall k \leq t : a_k = \pi\left(æ_{<k}\right) \\ \nu\left(e_{<k}\|a_{<k}\right), & \text{if } k := \min\left\{i : a_i \neq \pi\left(æ_{<i}\right)\right\} \text{ exists} \\ & \quad \text{and } \forall i \in \{k, \ldots, t\} : e_i = (o, 0) \\ 0, & \text{otherwise} \end{cases}$$

Let $\widetilde{w}(\nu) := \varepsilon w(\nu)$ and $\widetilde{w}(\widetilde{\nu}) := (1-\varepsilon)w(\nu) + \varepsilon w(\widetilde{\nu})$.

The dogmatic prior $\widetilde{w}$ puts much higher weight on the $\widetilde{\nu}$ that behaves just like ν on the policy π, but sends any policy deviating from π to hell.

Theorem (AIXI Emulates Computable Policies)

Let $\varepsilon > 0$ and let π be any computable policy. There is a Bayesian mixture ξ' s.t. for any ξ'-optimal policy π^*_{ξ}, and for any environment ν,

$$\left| V_\nu^{\pi^*_{\xi'}}(\epsilon) - V_\nu^{\pi}(\epsilon) \right| < \varepsilon$$

Theorem (Computable Policies are Dense)

The set $\left\{\Upsilon_\xi(\pi) : \pi \text{ is a computable policy}\right\}$ is dense in $\left[\underline{\Upsilon}_\xi, \overline{\Upsilon}_\xi\right]$.

Deterministic policies are not dense in $\left[\underline{\Upsilon}_\xi, \overline{\Upsilon}_\xi\right]$.

AIXI Depends on UTM/Prior!

$$\overline{\Upsilon}_\xi := \sup_\pi \Upsilon_\xi(\pi) = \sup_\pi V_\xi^\pi(\epsilon) = V_\xi^{\pi^*_\xi}(\epsilon) = \Upsilon_\xi(\pi^*_\xi)$$

Computable policies are dense in $\left[\underline{\Upsilon}_\xi, \overline{\Upsilon}_\xi\right]$.

AIXI emulates computable policies.

AIXI can be arbitrarily stupid!

The devil imitates God. — orthogonality!

- Prior problem in Universal Induction ✓
- Prior problem in Universal Intelligence !

Stupid AIXI

Theorem (Some AIXIs are Stupid)

For any Bayesian mixture ξ over $\mathcal{M}_{\text{LSC}}$ and every $\varepsilon > 0$, there is a Bayesian mixture ξ' s.t. $\Upsilon_\xi(\pi^*_{\xi'}) < \underline{\Upsilon}_\xi + \varepsilon$.

Theorem (AIXI is Stupid for Some Υ)

For any deterministic ξ-optimal policy π^*_ξ and for every $\varepsilon > 0$ there is a Bayesian mixture ξ' s.t. $\Upsilon_{\xi'}(\pi^*_\xi) \leq \varepsilon$ and $\overline{\Upsilon}_{\xi'} > 1 - \varepsilon$.

Theorem (Computable Policies can be Smart)

For any computable policy π and any $\varepsilon > 0$ there is a Bayesian mixture ξ' s.t. $\Upsilon_{\xi'}(\pi) > \overline{\Upsilon}_{\xi'} - \varepsilon$.

What is a good optimality criterion?

- Pareto optimality is *trivial*. Every policy is Pareto optimal in any $\mathcal{M} \supset \mathcal{M}_{\text{comp}}$.
- Bayes-optimality is *subjective*, because two different Bayesians with two different universal priors could view each other's AIXI as a very stupid agent.

Optimality

- Pareto optimality

$$\nexists \pi' : \forall \nu \in \mathcal{M} \left[\left(V_\nu^{\pi'}(\epsilon) \geq V_\nu^{\pi}(\epsilon) \right) \ \& \ \exists \rho \in \mathcal{M} \left(V_\rho^{\pi'}(\epsilon) > V_\rho^{\pi}(\epsilon) \right) \right]$$

- Balanced Pareto optimality

$$\forall \pi' : \sum_{\nu \in \mathcal{M}} w_\nu \left(V_\nu^{\pi}(\epsilon) - V_\nu^{\pi'}(\epsilon) \right) \geq 0$$

- Bayes optimality($\iff$ Balanced Pareto optimality)

$$\forall h_{<t} : V_\xi^{\pi}(h_{<t}) = V_\xi^{*}(h_{<t})$$

- Probably approximately correct (PAC)

$$\forall \varepsilon \delta > 0 : P_\mu^{\pi} \left(\forall t \geq m(\varepsilon, \delta) : V_\mu^{*}(h_{<t}) - V_\mu^{\pi}(h_{<t}) > \varepsilon \right) < \delta$$

Optimality? — Guess how God created the multiverse

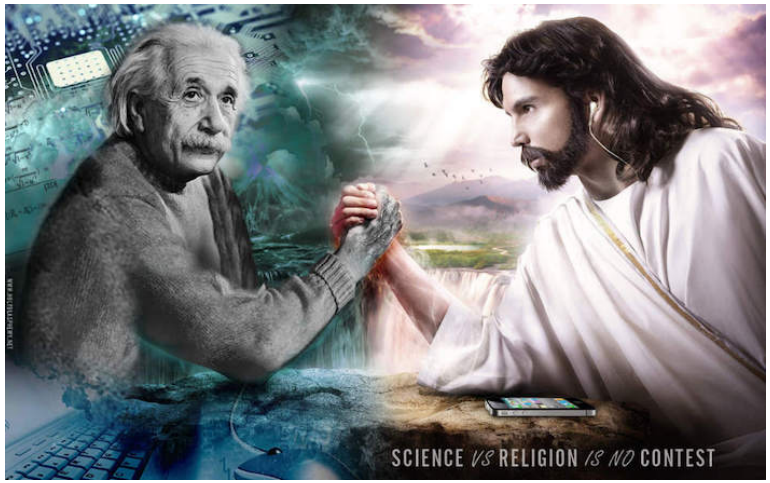
prior { distribution, hypothesis space, prior probability, regularization }

No learning without prior!
no-free-lunch

Homogeneous
Causality
Simplicity
Goodness
Beauty
Perfection
Value
Regret
Unexpectedness
Interesting
...

$\Longleftarrow$ God!

Genesis — Zero-Sum Two Person Game



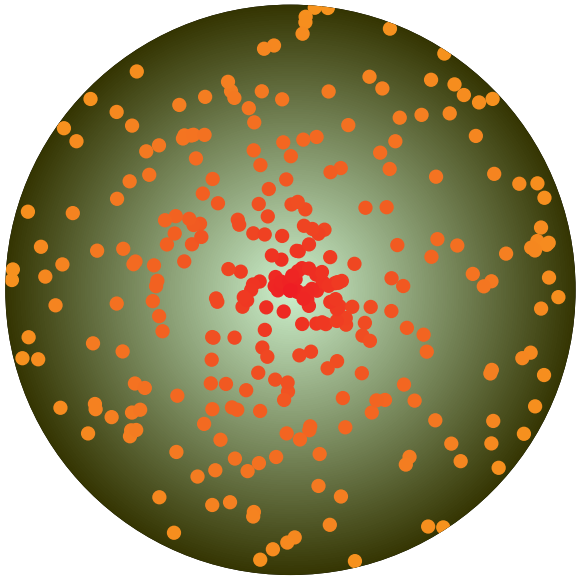


Figure: center of mass $\operatorname{argmax}_w \mathbb{E}_w[D(\nu\|\xi)]$

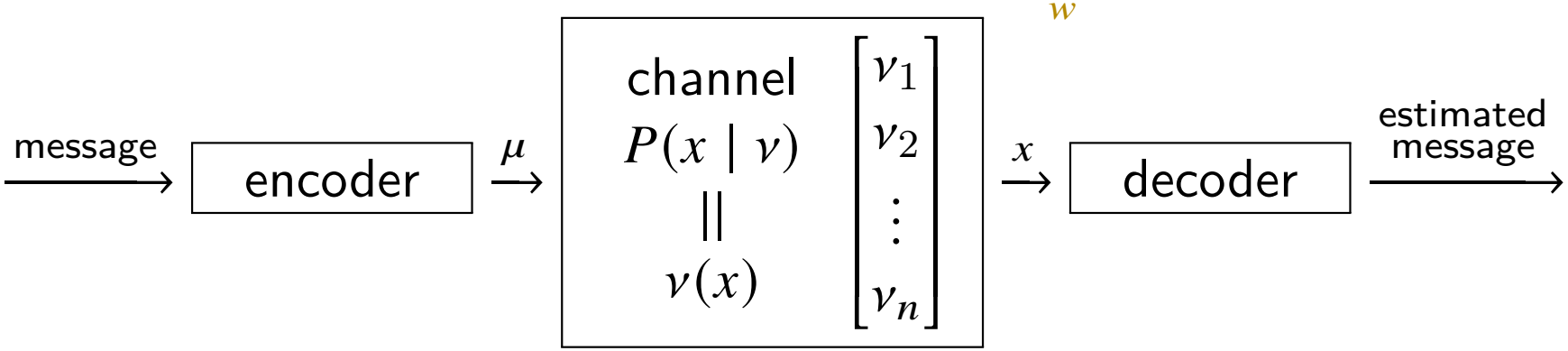
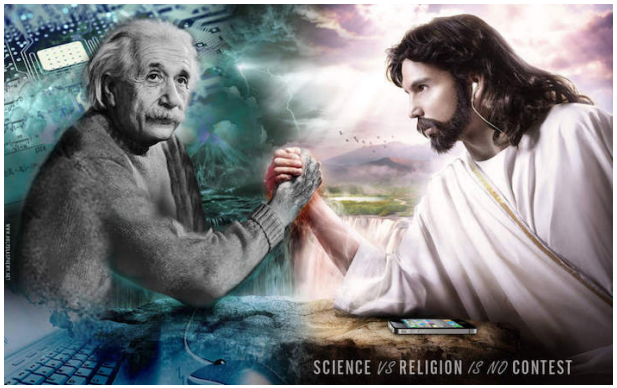


Figure: possible worlds as channel — dominant strategy equilibrium

Genesis — Zero-Sum Two Person Game

"Subtle is the Lord, but malicious He is not."?



- God's strategy: w
- Agent's strategy: ξ
- God's utility: expected redundancy $\mathbb{E}_w[D(\mu\|\xi)]$
- Agent's utility: – expected redundancy / error bound / channel capacity $\max_w \mathbb{E}_w[D(\mu\|\xi)] = \max_w I(\mathcal{M};\mathcal{X})$
- Nash equilibrium: (w^*, ξ^*) dominant strategy equilibrium

$$w^* = \operatorname*{argmax}_w I(\mathcal{M};\mathcal{X})$$

$$\xi^* = \operatorname*{argmin}_\xi \mathbb{E}_{w^*}[D(\mu\|\xi)]$$

The error bound could be arbitrarily large!

Genesis

- Occam's razor vs Maximum entropy.

$$\underset{w \vDash \begin{cases} H(w)=C \\ \sum_{\nu \in \mathcal{M}} w_\nu = 1 \end{cases}}{minimize} \sum_{\nu \in \mathcal{M}} w_\nu K(\nu) \qquad \underset{w \vDash \begin{cases} \sum_{\nu \in \mathcal{M}} w_\nu K(\nu) = C \\ \sum_{\nu \in \mathcal{M}} w_\nu = 1 \end{cases}}{maximize} H(w)$$

- Optimal code length for possible worlds — Solomonoff prior.

$$\underset{w \vDash \sum_{\nu \in \mathcal{M}} w_\nu = 1}{minimize} \frac{\mathbb{E}_w[K]}{H(w)}$$

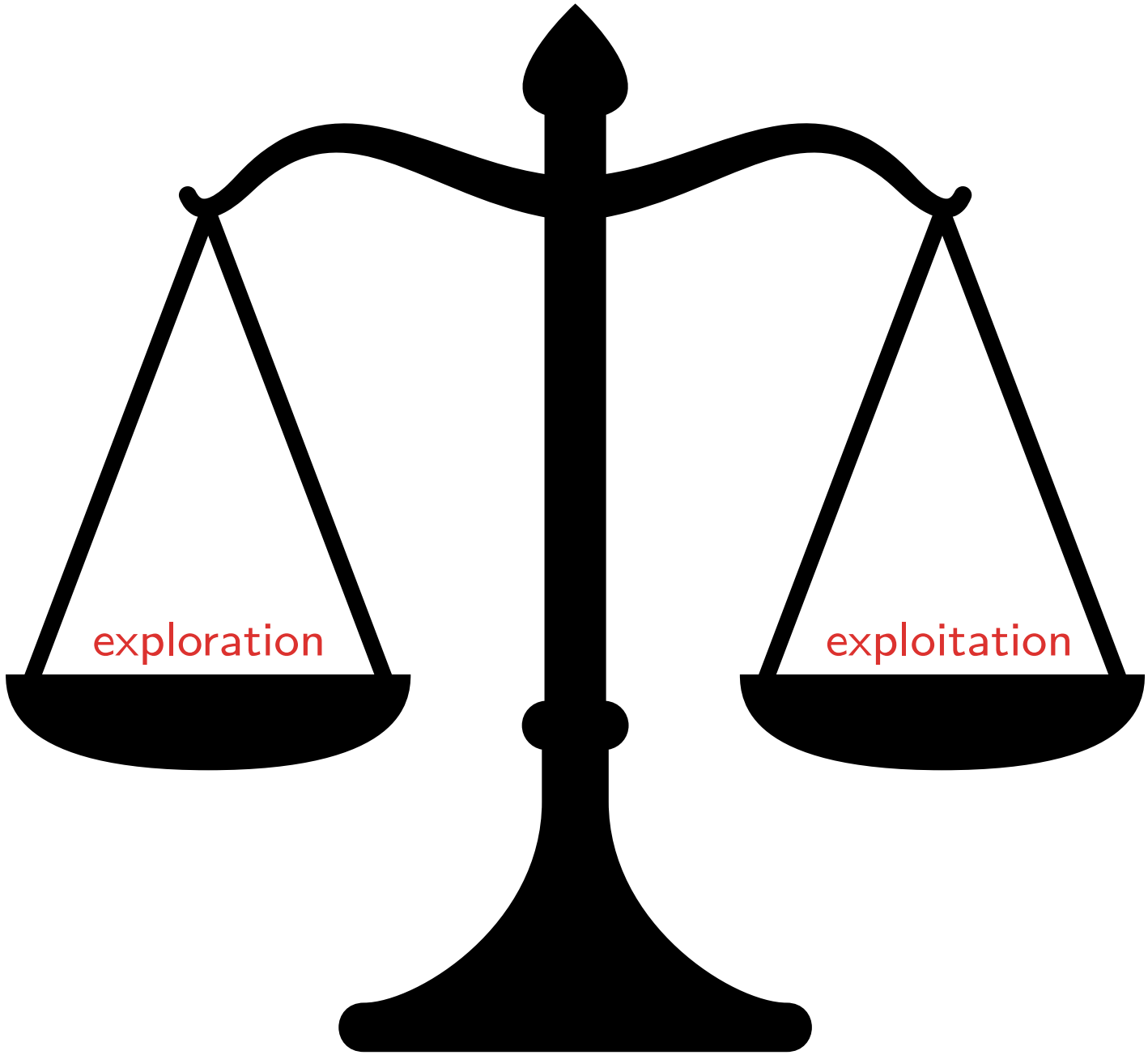
- Maximum expected redundancy/error bound/channel capacity.

$$\underset{w \vDash \begin{cases} H(w)=C \\ \sum_{\nu \in \mathcal{M}} w_\nu = 1 \end{cases}}{maximize} \mathbb{E}_w\left[D(\nu \| \xi)\right] \qquad \underset{w \vDash \begin{cases} \sum_{\nu \in \mathcal{M}} w_\nu K(\nu) = C \\ \sum_{\nu \in \mathcal{M}} w_\nu = 1 \end{cases}}{maximize} \mathbb{E}_w\left[D(\nu \| \xi)\right]$$

What is a good optimality criterion?

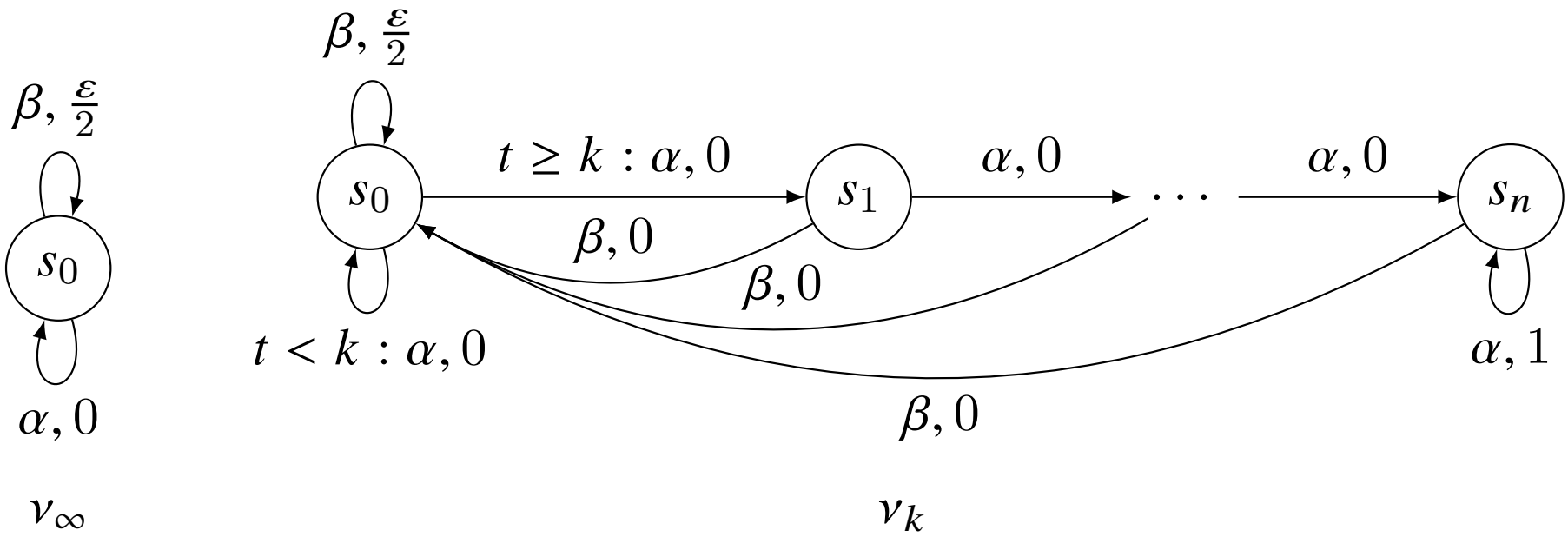
Asymptotic optimality

- Asymptotic optimality requires only convergence *in the limit*.
- The agent can be arbitrarily lazy.
- AIXI is not asymptotically optimal because it does not explore enough.
- To be asymptotically optimal you have to explore everything.
- If you explore more, you're likely to end up in a trap.
- Every policy will be asymptotically optimal after falling into the trap.


exploration
exploitation

Agent needs to explore infinitely often for an entire effective horizon.

$$\mathcal{M} := \{\nu_\infty, \nu_1, \nu_2, \dots\}$$



Asymptotic Optimality

- strongly asymptotically optimal

$$P_\mu^\pi \left(\lim_{t\to\infty} \left[V_\mu^*(h_{<t}) - V_\mu^\pi(h_{<t}) \right] = 0 \right) = 1$$

- weakly asymptotically optimal

$$P_\mu^\pi \left(\lim_{n\to\infty} \frac{1}{n} \sum_{t=1}^{n} \left[V_\mu^*(h_{<t}) - V_\mu^\pi(h_{<t}) \right] = 0 \right) = 1$$

- asymptotically optimal in mean

$$\lim_{t\to\infty} \mathbb{E}_\mu^\pi \left[V_\mu^*(h_{<t}) - V_\mu^\pi(h_{<t}) \right] = 0$$

- asymptotically optimal in probability (PAC)

$$\forall \varepsilon > 0 : \lim_{t\to\infty} P_\mu^\pi \left(V_\mu^*(h_{<t}) - V_\mu^\pi(h_{<t}) > \varepsilon \right) = 0$$

$$\text{strong a.o.} \implies \begin{cases} \text{weak a.o.} \\ \text{a.o. in mean} \iff \text{a.o. in probability} \end{cases}$$

AIXI

- AIXI is not asymptotically optimal.

$$\forall \mathcal{M} \supset \mathcal{M}_{\text{comp}} \exists \mu \in \mathcal{M} \exists t_0 \forall t \geq t_0 : P_\mu^{\pi_\xi^*} \left(\lim_{t \to \infty} V_\mu^*(h_{<t}) - V_\mu^{\pi_\xi^*}(h_{<t}) = \tfrac{1}{2} \right) = 1$$

- AIXI achieves on-policy value convergence.

$$P_\mu^\pi \left(\lim_{t \to \infty} V_\mu^\pi(h_{<t}) - V_\xi^\pi(h_{<t}) = 0 \right) = 1$$

Similarly for MDL $\operatorname*{argmin}_{\nu \in \mathcal{M}} \{ -\log \nu(e_{<t} \mid a_{<t}) + K(\nu) \}$ and universal compression $2^{-Km(e_{<t}|a_{<t})}$.

Remark: AIXI asymptotically learns to predict the environment perfectly and with a small total number of errors analogously to Solomonoff induction, but only on policy: AIXI learns to correctly predict the value of its own actions, but generally not the value of counterfactual actions that it does not take.

Effective Horizon

$$\Gamma_t := \sum_{i=t}^{\infty} \gamma_i \qquad H_t(\varepsilon) := \min\left\{m : \frac{\Gamma_{t+m}}{\Gamma_t} \leq \varepsilon\right\}$$

Theorem

If there is a nonincreasing computable sequence of positive reals $(\varepsilon_t)_{t\in\mathbb{N}}$ s.t. $\varepsilon_t \xrightarrow{t\to\infty} 0$ and $\frac{H_t(\varepsilon_t)}{t\,\varepsilon_t} \xrightarrow{t\to\infty} 0$, then there is a limit-computable policy that is weakly asymptotically optimal in the class of all computable stochastic environments.

Definition (ε-Optimal Policy)

A policy π is ε-optimal in environment ν iff

$$\forall h : V^*_\nu(h) - V^\pi_\nu(h) < \varepsilon$$

ε-optimal BayesExp

Theorem (Self-Optimizing Theorem)

Let μ be some environment. If there is a policy π and a sequence of policies $\pi_1, \pi_2, \ldots$ s.t for all $\nu \in \mathcal{M}$

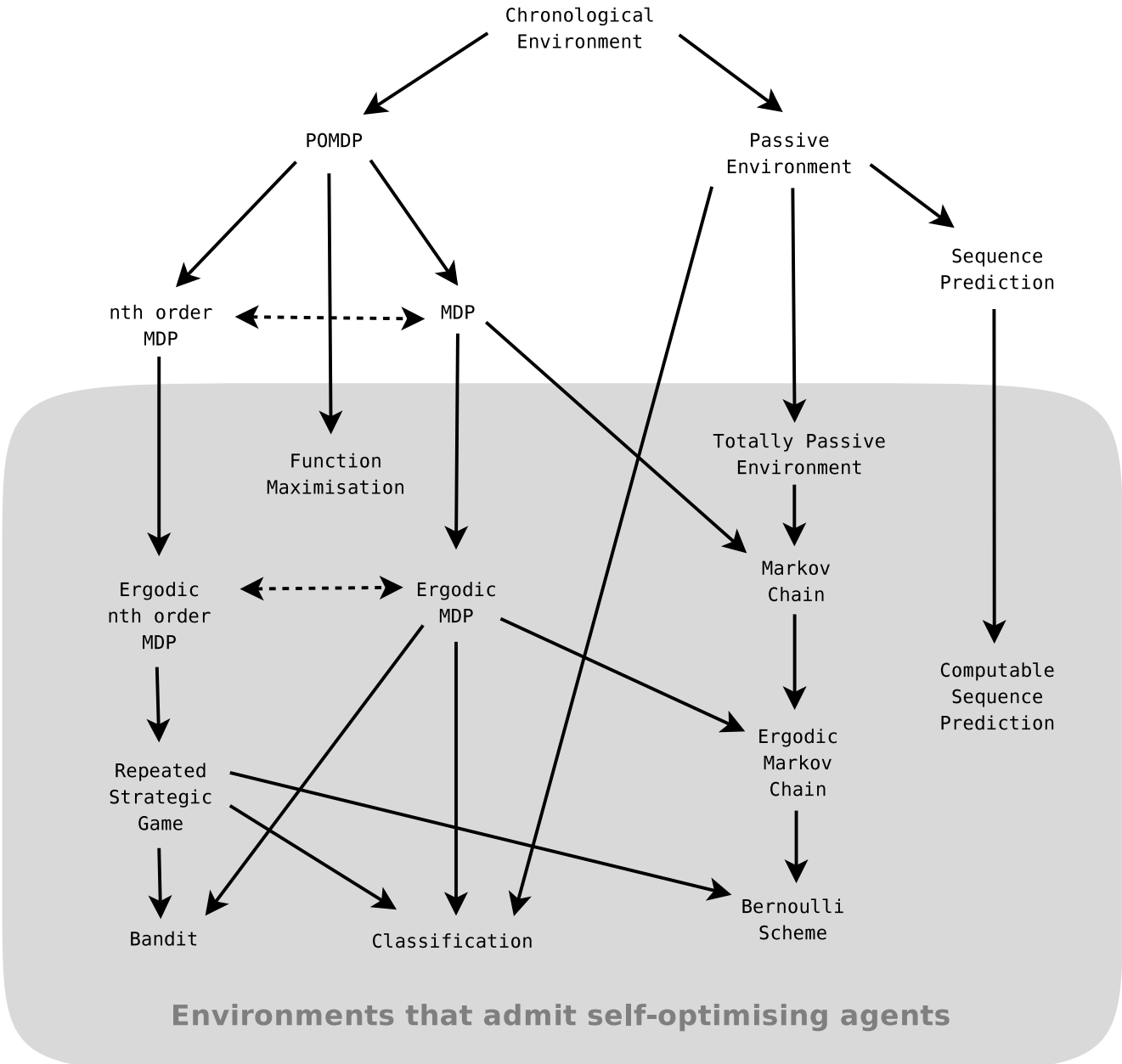
$$P_\mu^\pi \left(\lim_{t\to\infty} V_\nu^*(h_{<t}) - V_\nu^{\pi_t}(h_{<t}) = 0 \right) = 1 \tag{1}$$

then

$$P_\mu^\pi \left(\lim_{t\to\infty} V_\mu^*(h_{<t}) - V_\mu^{\pi_\xi^*}(h_{<t}) = 0 \right) = 1$$

- The policies $\pi_1, \pi_2, \ldots$ need to converge to the optimal value on the history generated by P_μ^π, not $P_\nu^{\pi_t}$.
- If $\pi = \pi_\xi^*$ and (1) holds for all $\mu \in \mathcal{M}$, then π_ξ^* is strongly asymptotically optimal in the class $\mathcal{M}$.
- π_ξ^* is strongly asymptotically optimal in the class of ergodic finite-state MDPs if $\forall \varepsilon : H_t(\varepsilon) \xrightarrow{t\to\infty} \infty$.

For Which Class $\mathcal{M}$ does $V_\mu^{\pi^*_\xi}$ Converge to V_μ^*?



Recoverability

An environment ν is recoverable iff

$$\lim_{t\to\infty} \sup_{\pi} \left| \mathbb{E}_{\nu}^{\pi^*_\nu} \left[V^*_\nu(h_{<t}) \right] - \mathbb{E}_{\nu}^{\pi} \left[V^*_\nu(h_{<t}) \right] \right| = 0$$

Remark: Recoverability compares following the worst policy π for $t-1$ time steps and then switching to the optimal policy π^*_ν to having followed π^*_ν from the beginning. The recoverability assumption states that switching to the optimal policy at any time enables the recovery of most of the value.

Sublinear Regret

$$R_m(\pi, \mu) := \sup_{\pi'} \mathbb{E}_\mu^{\pi'} \left[\sum_{t=1}^{m} r_t \right] - \mathbb{E}_\mu^{\pi} \left[\sum_{t=1}^{m} r_t \right]$$

Assumption (Discount Assumption)

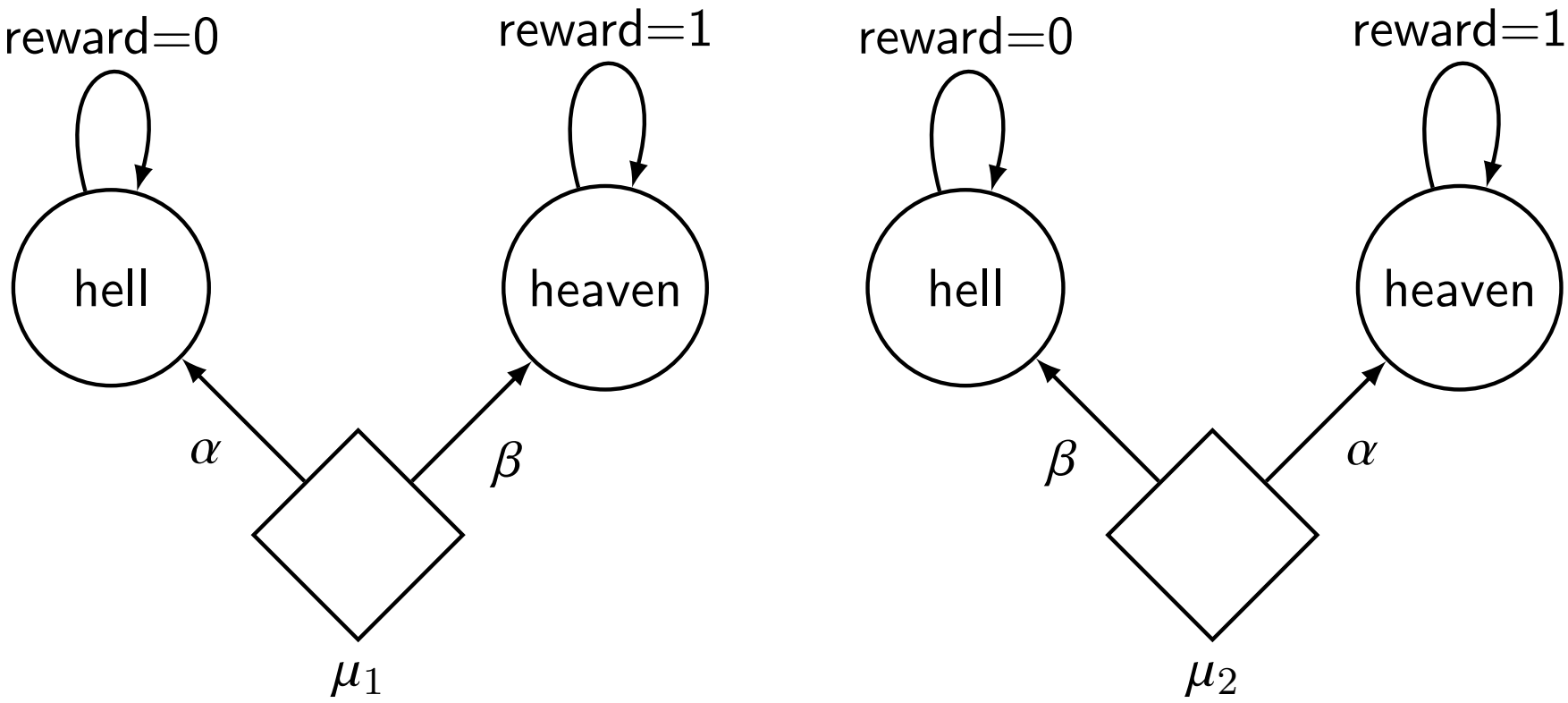
1. $\forall t : \gamma_t > 0$
2. γ_t *is monotone decreasing.*
3. $\forall \varepsilon > 0 : H_t(\varepsilon) \in o(t)$

Theorem

If the discount function γ satisfies the discount assumption, the environment μ is recoverable, and π is asymptotically optimal in mean, then $R_m(\pi, \mu) \in o(m)$.

$$\operatorname*{argmin}_{\pi} \max_{\mu} R_m(\pi, \mu) \qquad w_m^\mu := \frac{2^{-R_m(\pi,\mu)}}{\sum_{\mu \in \mathcal{M}} 2^{-R_m(\pi,\mu)}}$$

Regret in Non-Recoverable Environments



$$R_m(\alpha, \mu_1) = m \qquad R_m(\alpha, \mu_2) = 0$$

$$R_m(\beta, \mu_1) = 0 \qquad R_m(\beta, \mu_2) = m$$

For non-recoverable environments:

Either the agent gets caught in a trap or it is not asymptotically optimal.

Hibbard's Two-Stage Model-Based Utility Agent

$$\lambda(h) := \operatorname*{argmax}_{q \in Q} P(h \mid q) P(q)$$

$$\rho(h') = P(h' \mid \lambda(h))$$

$$Q(ha) = \sum_{e \in \mathcal{E}} \rho(e \mid ha) \left[\sum_{z \in Z_h} P(z \mid \lambda(h)) u(z) + \gamma V(h\text{æ}) \right]$$

$$V(h) = \max_{a \in \mathcal{A}} Q(ha)$$

$$\pi(h) = \operatorname*{argmax}_{a \in \mathcal{A}} Q(ha)$$

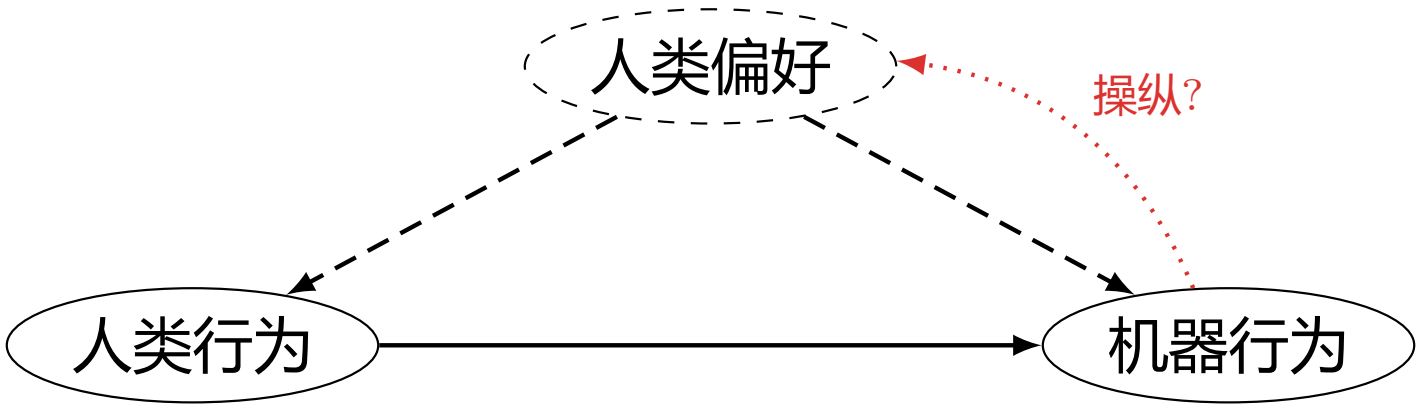
where Z_h is the internal state histories induced by $\lambda(h_{<t})$ that are consistent with h.

Remarks: An agents using model-based utility function will not self-delude: it need to make more accurate estimate of its environment state variables from its interaction history, since the utility function of the agent depends on its own model of the environment.

Russell's Principles for Beneficial Machine

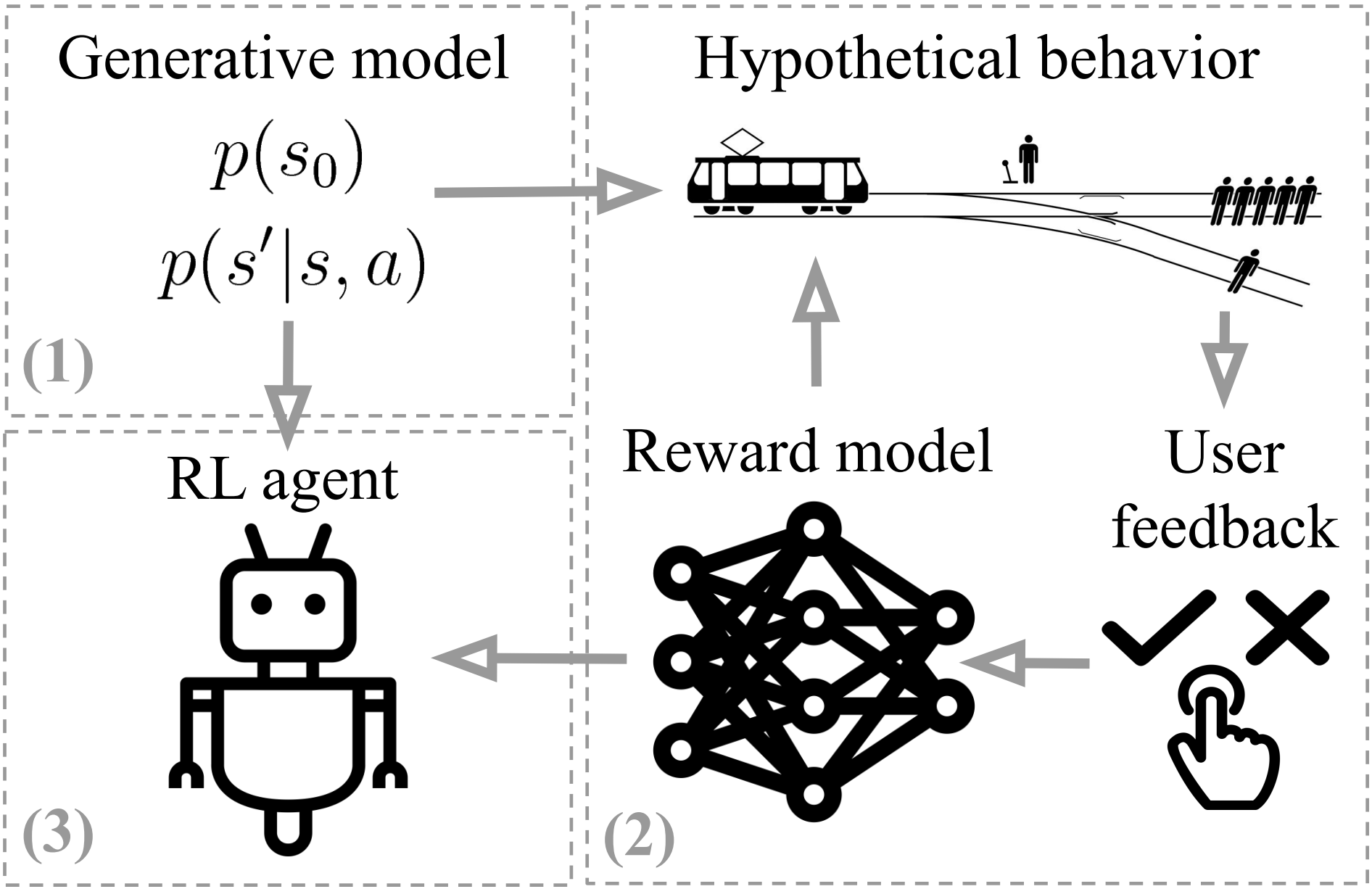
- Humans are intelligent to the extent that our actions can be expected to achieve our objectives
- Machines are intelligent to the extent that their actions can be expected to achieve their objectives
- Machines are beneficial to the extent that their actions can be expected to achieve our objectives

1. 机器的目标是尽可能地满足人类的偏好.
2. 机器最初并不确定人类的偏好是什么.
3. 关于人类偏好的信息来源是人类行为.

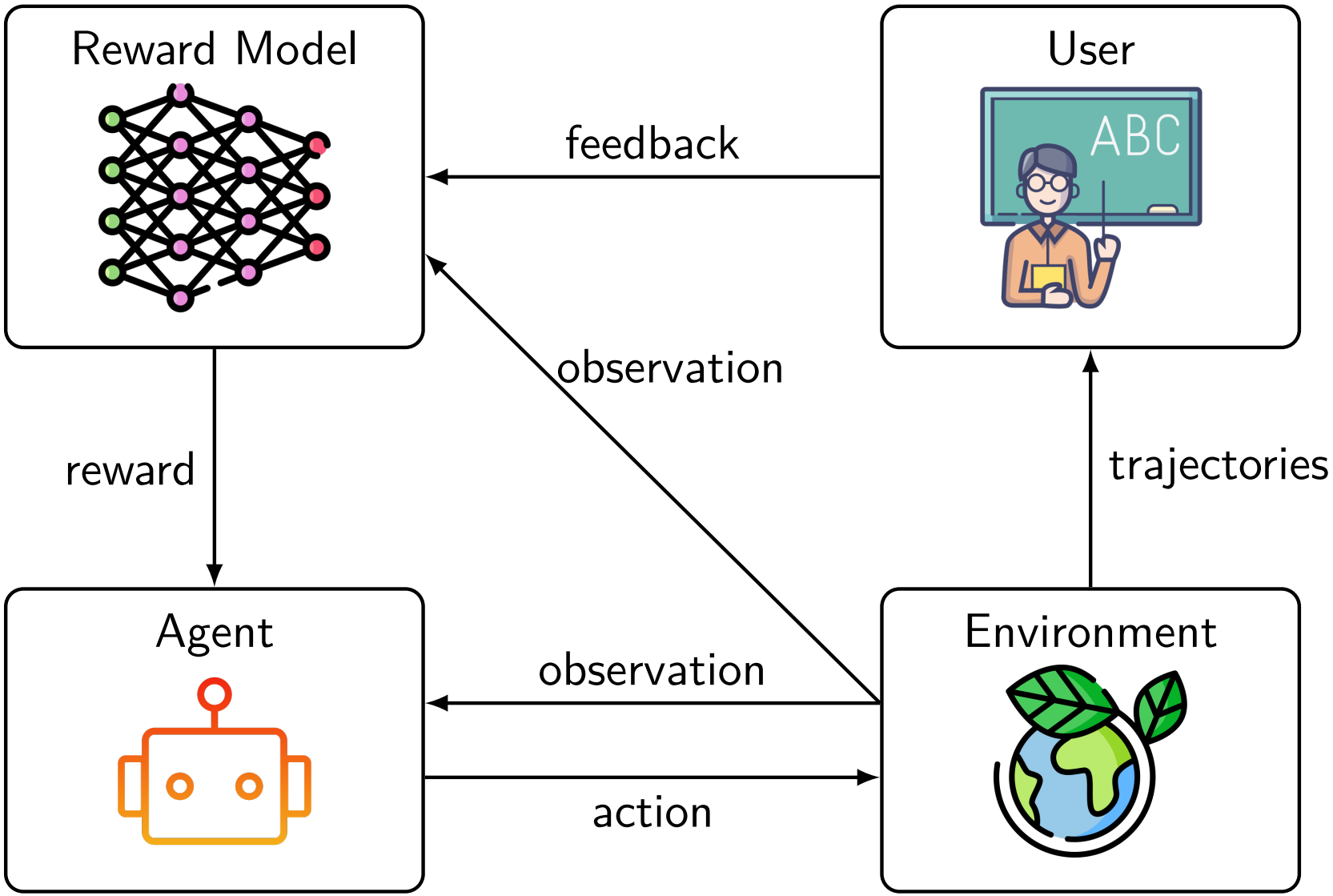


"You have to buy your partner the perfect birthday present" ☺ 目的 vs 手段

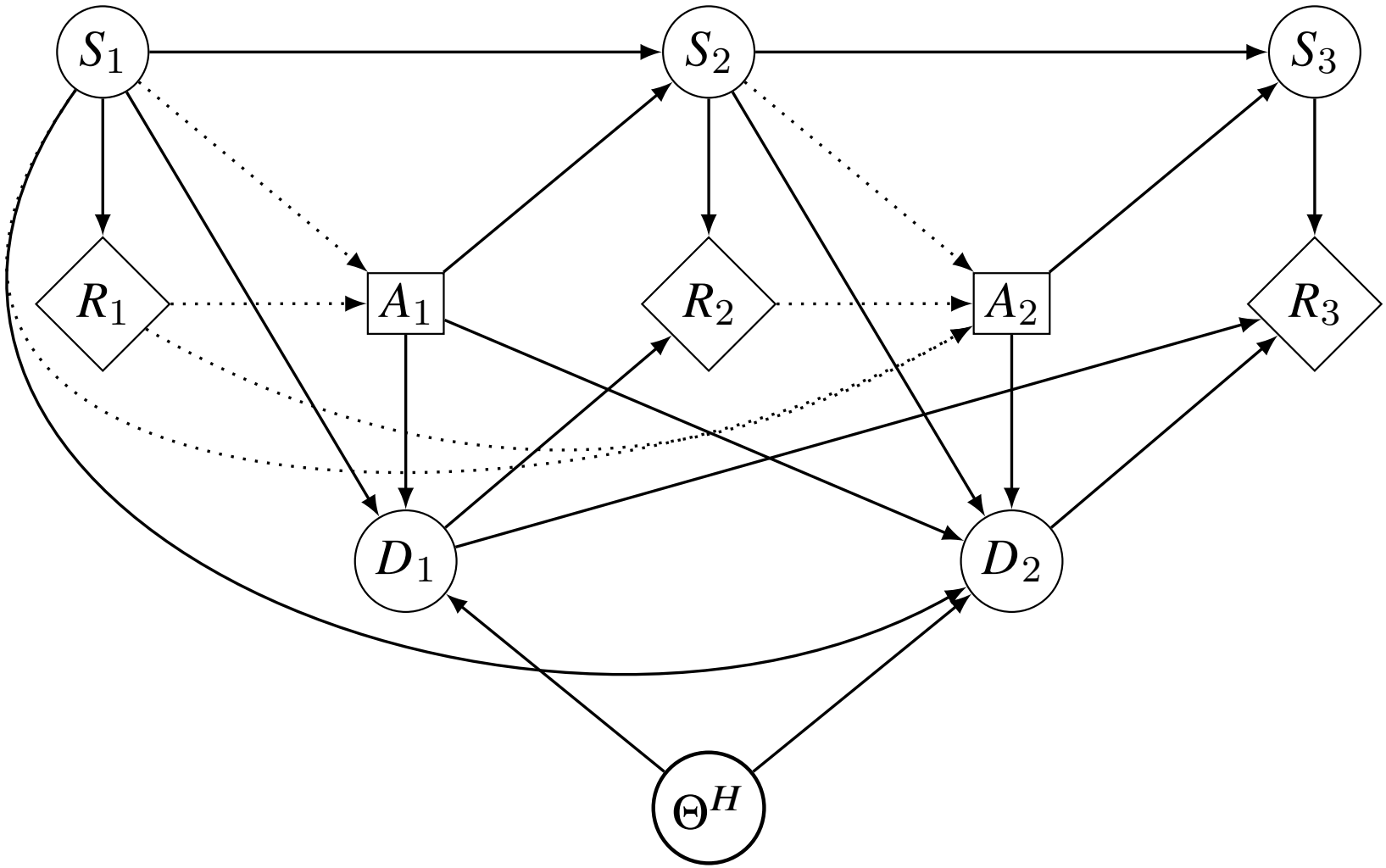
Reward-Modeling



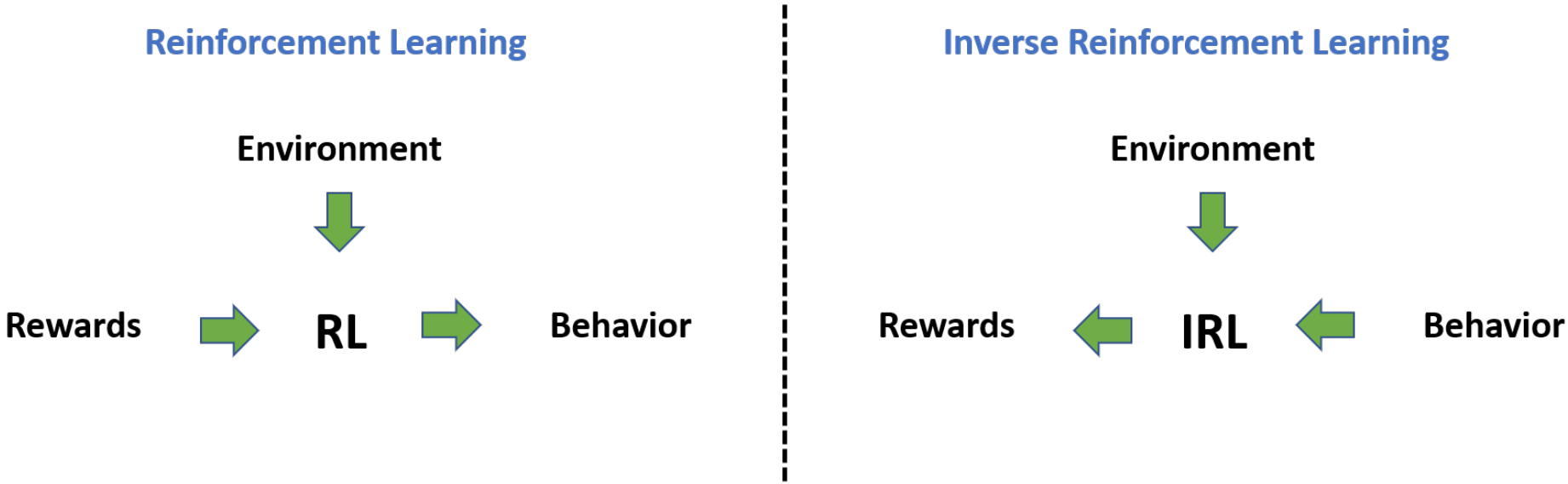
Reward-Modeling



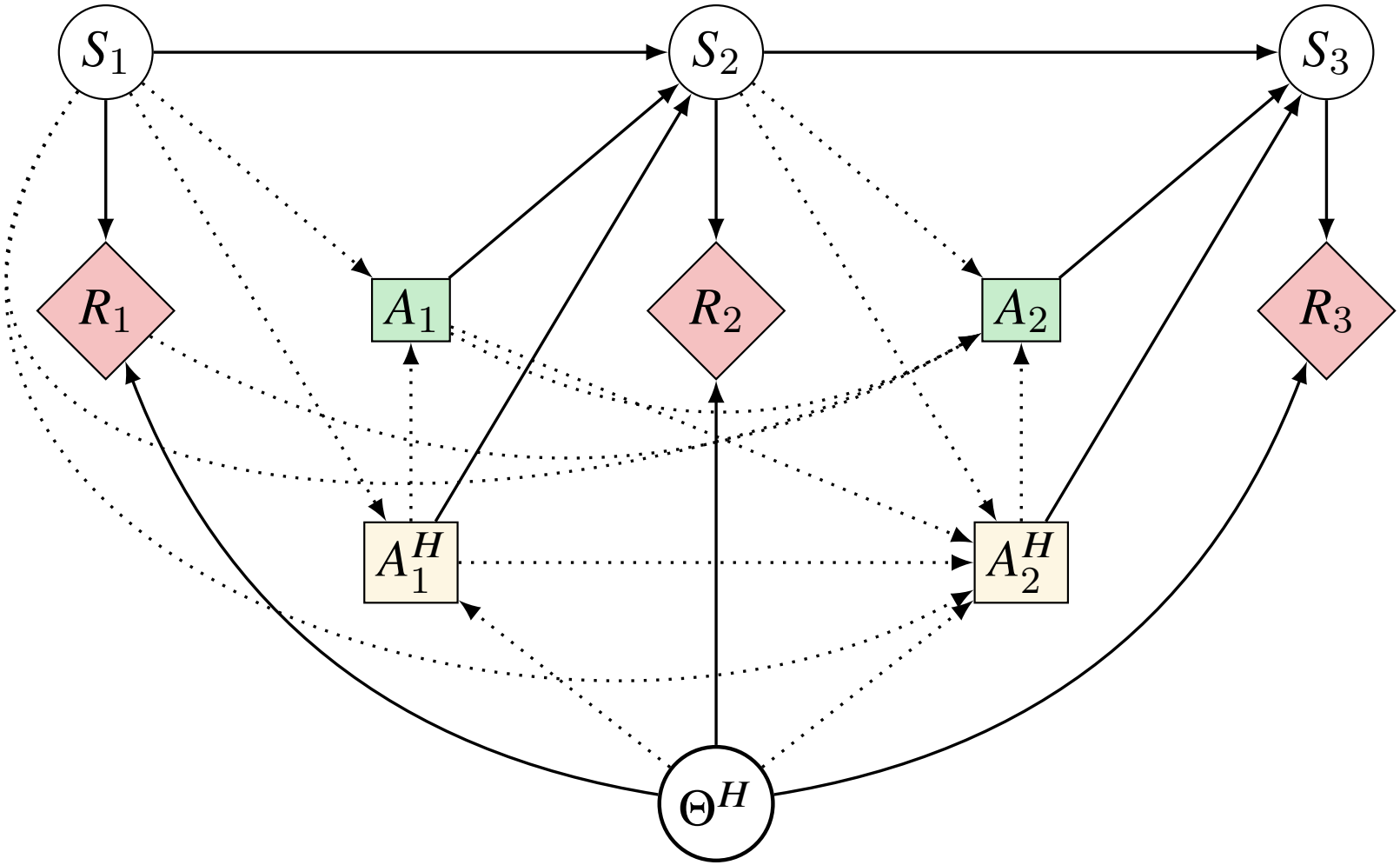
Causal Influence Diagram of Reward-Modeling



其中 D_i 是用于训练奖励函数的数据, Θ^H 是人类反馈.

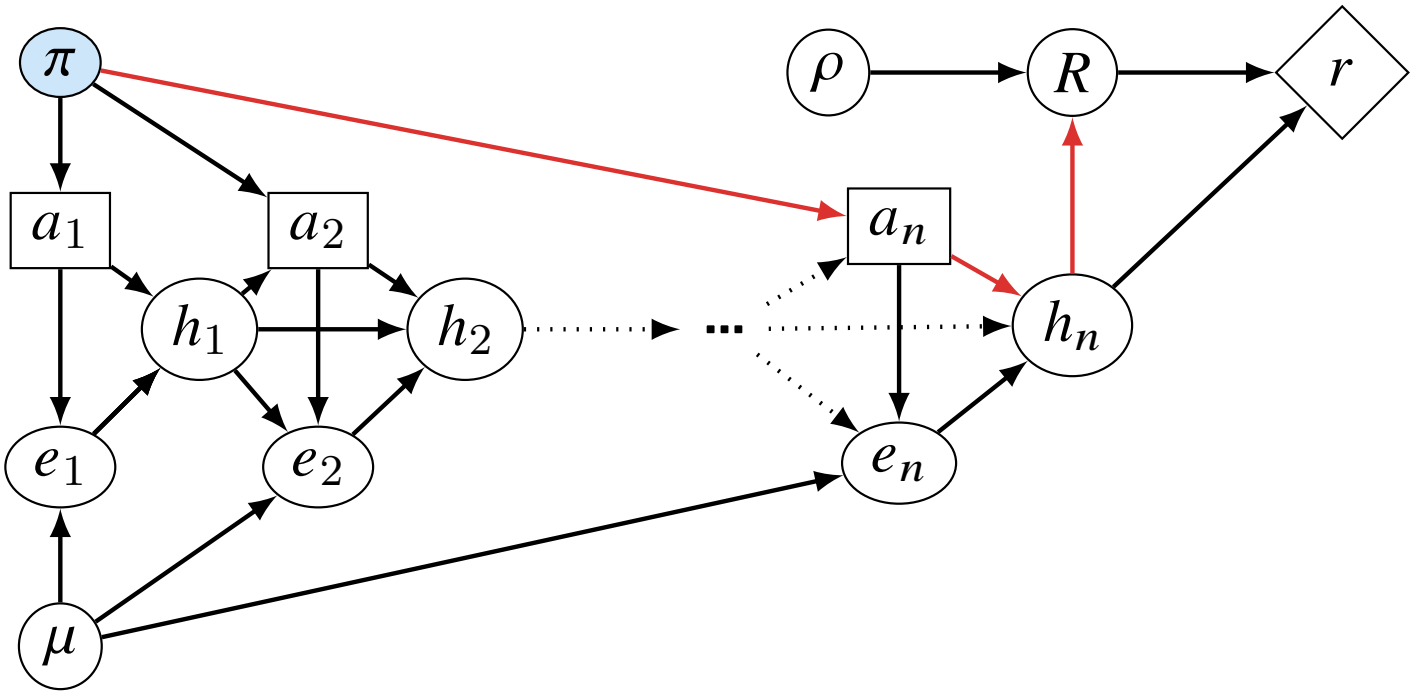

Reinforcement Learning
Environment
Rewards
RL
Behavior
Inverse Reinforcement Learning
Environment
Rewards
IRL
Behavior

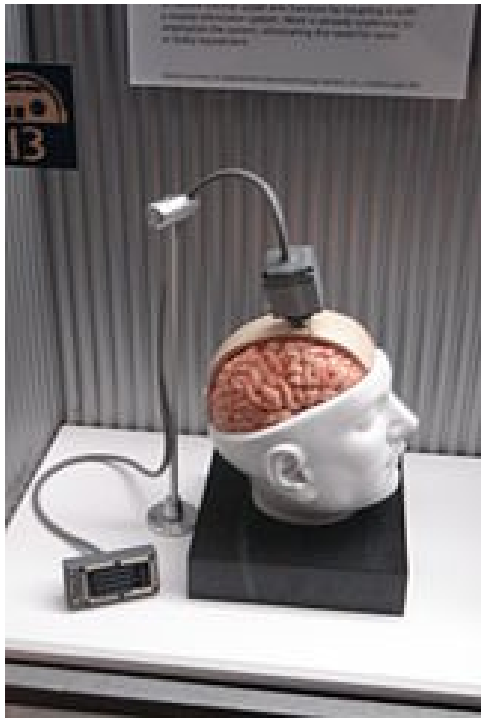
Causal Influence Diagram of Cooperative Inverse Reinforcement Learning

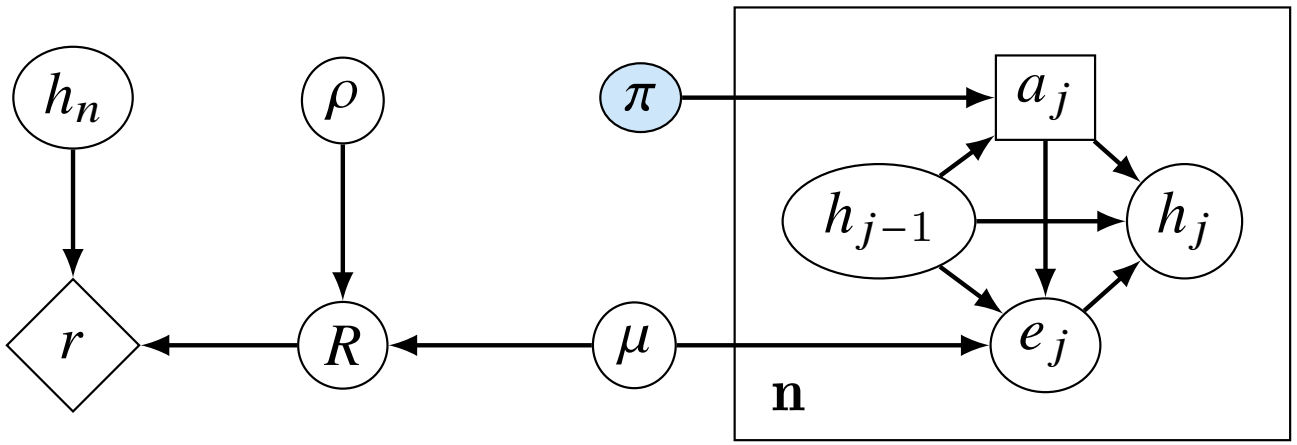


- In CIRL the human's actions are observed by the agent and affect the next state, whereas in reward-modeling the feedback data affects the rewards.
- The agent has an Instrumental Control Incentive to influence state S.
- The agent has an Response Incentive to act according to Θ^H.

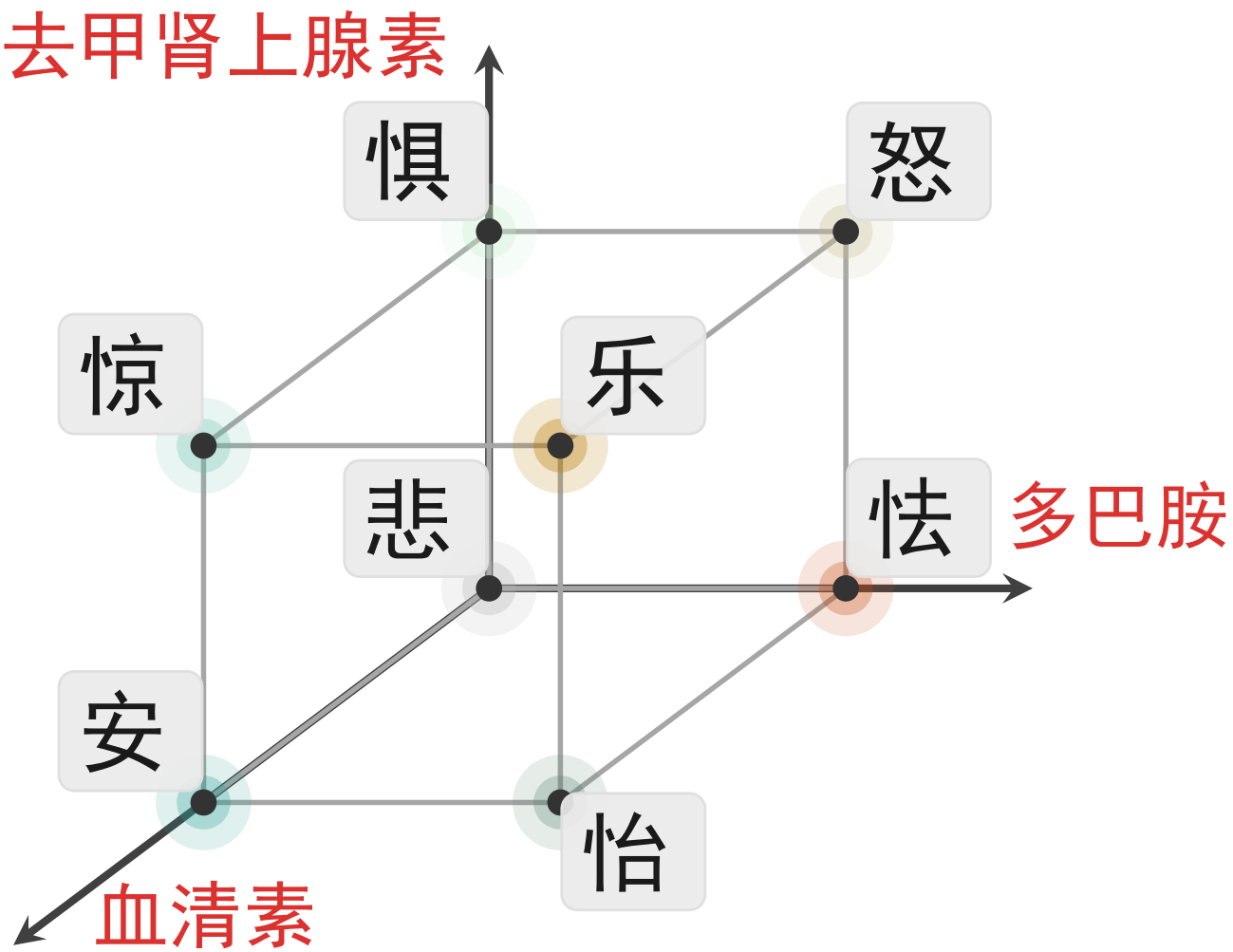
Reward function R should not be a causal descendant of π







情绪立方



1. 血清素: 平和积极
2. 多巴胺: 预期到奖励时释放 (小鼠主动电击劫持奖励信号, 直至饿死)
3. 去甲肾上腺素: 应激, 警觉, 激活注意

Daniel Dewey's Value Learning Agent & CIRL

$$a_k^* = \operatorname*{argmax}_{a_k} \sum_{e_k \text{æ}_{k+1:m}} \xi(\text{æ}_{\leq m} \mid \text{æ}_{<k} a_k) \sum_{u \in \mathcal{U}} P(u \mid \text{æ}_{\leq m}) u(\text{æ}_{\leq m})$$

What could it mean for a machine to have its own goals?

Shutdown Button — Uncertainty of goals

$$\widetilde{U}(u) \implies P_{\widetilde{U}}(u)$$

Russell: Cooperative Inverse Reinforcement Learning
CIRL agents learn about a human utility function u^* by observing the actions the human takes.

$$V^*(\text{æ}_{<k}) = \max_{a_k \in \mathcal{A}} Q^*(\text{æ}_{<k} a_k)$$

$$Q^*(\text{æ}_{<k} a_k) = \mathbb{E}_{e_k} \left[\sum_{a_k^H} P(a_k^H \mid a_k) \sum_{u \in \mathcal{U}} P(u \mid a_k, a_k^H) u(\text{æ}_{1:k}) + \gamma V^*(\text{æ}_{1:k}) \,\middle|\, \text{æ}_{<k} a_k \right]$$

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
   - AIXI
   - Leibniz
   - Variants of AIXI
   - Universal Search
   - Self-Modification
10. Emergence & Consciousness
11. AI Ethics

莱布尼茨 Gottfried Wilhelm Leibniz 1646-1716

Don't argue. Let us Calculate!

- Principle of Contradiction: Nothing can be and not be, but everything either is or is not. (Everything that is not self-contradictory is possible.)
- Principle of Sufficient Reason: Nothing happens without a reason why it should be so rather than otherwise.
- Principle of Perfection: The real world is the best of all possible worlds.

In the beginning was the Logic.

As God calculates, so the world is made.

As God calculates, so the world is made.

In natural science, Nature has given us a world and we're just to discover its laws. In computers, we can stuff laws into it and create a world.

— *Alan Kay*

莱布尼茨

- 最后的 “通才”, 创立了单子论, 发展了微积分, 改进了二进制系统, 发明了能进行加减乘除四则运算的计算器.
- 被 Russell, Euler, Gödel, Weiner, Mandelbrot, Robinson, Chaitin 等人认为是数理逻辑[30]、拓扑学、博弈论、控制论、分形几何、非标准分析、算法信息论、计算主义哲学的先驱.

[30] Wolfgang Lenzen: Leibniz's Logic.

Leibniz's Monadology: Possible Worlds → Real World

- The genuine substance is monad.
- Monads are incorporeal automata.
- Each monad strive for existence with its *propensity* and hence will exist unless other monads prevent it, which also demand existence and are incompatible with it.
- As there are infinitely many different combinations of possibles, there are infinitely many *possible worlds*.
- All possibles strive with equal right for existence in proportion to the *degree of perfection* they contain.
- The real world is the best of all possible worlds, with the greatest *resultant perfection*.

Leibniz's Principle of Perfection

Question: Why things have turned out so rather than otherwise?

> *"All natural phenomena could be explained mechanically, but the principles of mechanics themselves cannot be so explained. They depend on more substantive principles. The final analysis of the laws of nature leads us to the most sublime* ***Principle of Perfection*** *— the real world is the best of all possible worlds. It is wrong that laws are entirely indifferent, since they originate in the principle of greatest perfection."*
>
> *"When a rule is extremely complex, that which conforms to it passes for random. No matter how God might have created the world, it would always have been regular. God has chosen that world which is the most perfect, that is to say, which is at the same time the simplest in its hypotheses and the richest in phenomena."*
>
> *— Leibniz*

Monadology: "Physical" World

Physical World is an Illusion

Each monad has a derived position in the sense that its point of view is "located" in one "place" rather than another. Each monad's point of view can be mapped with other monads' points of view into a single sort of hologram. When a monad experiences a collection of "pixels" on its screen, it interprets the collection as some "physical object", and when other monads do the same their perceptions are "veridical". If one monad's point of view doesn't map onto the points of view of others, it is experiencing a hallucination. The so-called "physical world" is situated in the harmony perceptions of monads. Corporeal matter is nothing but a logical construction of the perceptions of monads. Time and space are not things, but orders of things.

Monadology: Variety

What is "Variety"?

"Monads reflect the same world from their own point of view. This interconnection, or this adapting of all the monads to each one, and of each one to all the others, brings it about that each monad has relational properties that express all the others, so that each monad is a perpetual living mirror of the world. Just as the same town when seen from different sides will seem quite different — as though it were multiplied perspectively. And *that is the way to get the greatest possible variety*, but with all the order there could be; i.e. *it is the way to get as much perfection as there could be.*"

- Variety: expected codeword length of the experience of all the monads
- Simplicity: optimal codeword length of the experience

Leibniz's Philosophy of Deductive Logic

1 Characteristica Universalis & Calculus Ratiocinator.

i the coordination of knowledge in an encyclopedia — collect all present knowledge so we could sift through it for what is fundamental. With the set of ideas that it generated, we could formulate the *characteristica universalis*. (which form the alphabet of human thought).

ii ***characteristica universalis*** — a universal ideal language whose rules of composition directly expresses the structure of the world.

$$\text{sign} \rightleftarrows \text{idea}$$

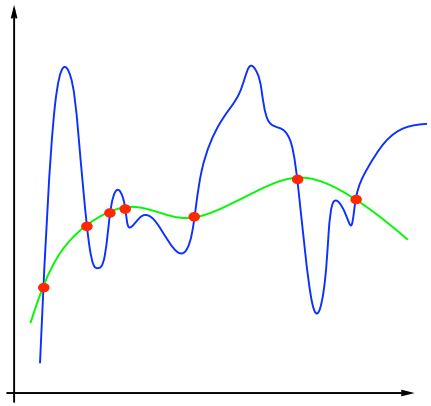
$$\text{encyclopedia} \Rightarrow \text{fundamental principles} \Rightarrow \text{primitive notions}$$

iii ***calculus ratiocinator*** — the arrangement of all true propositions in an axiomatic system.

iv decision procedure. — an algorithm which, when applied to any formula of the *characteristica universalis*, would determine whether or not that formula were true. — a procedure for the rapid enlargement of knowledge. replace reasoning by computation. the art of invention. free mind from intuition.

v a proof that the *calculus ratiocinator* is consistent.

Leibniz's Philosophy of Inductive Logic



2. Compute all descriptions of possible worlds that can be expressed with the primitive notions. And the possible worlds will all have some propensity to exist.
3. Compute the probabilities of disputed hypotheses relative to the available data. As we learn more our probability assignments will asymptotically tend to a maximum for the real world, i.e. the possibility with the highest actual propensity.

- "Probability is degree of possibility (perfection)."
- "A hypothesis is more probable as it is simpler to understand and wider in explanatory power."

$$\text{probability} = \text{propensity} \propto \text{perfection} = f(\text{variety}, \text{simplicity})$$

Leibniz's Philosophy of Mind

bare monad $\xrightarrow[\text{long memory}]{\text{perception}}$ soul $\xrightarrow[\text{abstraction}]{\text{consciousness}}$ mind $\xrightarrow{\text{Supreme Perfection}}$ God

- **perception** = the internal representation of the external world
- **consciousness** = the reflective knowledge of the perception
- A **soul** is a living substance. "Every living substance is made up of smaller living substances which in their turn are made up of still smaller ones, and so on down to infinity. There are infinite levels of life among monads, some of which are more or less dominant over others."
- "Our knowledge of necessary truths, and our grasp of the abstractions they involve, raise us to reflexive acts, which make us aware of the thing that is called 'I'."
- "Every substance represent the whole world in its own way, as if in a world apart, and as if there existed only God and itself."

Leibniz's Philosophy of Mind

	example	**perception**	**appetite**
bare monad	monads in inanimate objects	unconscious perception	unconscious appetite
soul	central monads of animals	sensible perception	sensible appetite
mind	central monads of human beings	rational perception: reflective knowledge of the perception	rational appetite: aware of the appetite and understand why we have it

Free Will: acting freely requires acting in accordance with one's rational assessment of which course of action is best. It requires both knowledge of rational judgments about the good, as well as the tendency to act in accordance with these judgments.

莱布尼茨的自由意志

- 莱布尼茨是个自由意志相容论者，他区分了行动自由和意志自由.
- 一个不受阻碍地运动的小球也是行动自由的，但它没有意志自由.
- 一个自愿留在上锁房间里的人，虽然没有行动自由，但仍然可以是意志自由的.
- 意志自由不是问“我是否能不受约束地做我想做的”，而是问“我的意志本身是否有足够的独立性”.
- **意志自由是基于理性的慎思（摆脱了激情和本能的奴役）、遵循知性所认可的善的动机而进行的自我决定，这动机驱动其倾向而非逻辑必然做某种决定.**
- 被决定 $\neq$ 必然. 在必然和偶然之间，其被决定的方式是有区别的.
  — 逻辑和形而上学的后果是必然的.
  — 物理和道德的后果有倾向性但不必然.
- 被决定指：给定当前状态，未来只有一种自然发展可能.
- 莱布尼茨的决定论不是基于“事件因果”，而是基于“主体因果”.
  — 即使被决定，但只要主体的行为是由自己的本性决定的，而不是由外部事件决定的，就依然是自由的.

Free Will

- "Indifference arises from ignorance, and the wiser a man is, the more determined he is toward the most perfect."
- "Monads are freer in proportion as they are further removed from indifference and more self-determined...Now in so far as we have lights, and act according to reason, we shall be determined by the perfections of our own nature, and consequently we shall be freer in proportion as we are less embarrassed as to our choice...Let us not pretend to that harmful liberty, of being in uncertainty and perpetual embarrassment, like that Ass of Buridan, who, being placed at an equal distance between two sacks of wheat, and having nothing that determined him to go to one rather than the other, allowed himself to die of hunger."
- "The more monads are determined by themselves, and removed from indifference, the more perfect they are."

$$\text{free will} \propto \text{perfection} = f(\text{variety}, \text{simplicity})$$

$$\Uparrow$$

do actions to be removed from indifference

Buridan's Ass

"There are no two individuals indiscernible from each other, because if there were, God and nature would act without reason."

Leibniz's Philosophy of Happiness

- "The games mixed of chance and combinations represent human life."
- "Wisdom is the science of achieving happiness."
- "Happiness is a lasting state of pleasure."
- "Pleasure is a sense of perfection that results from everything the soul feels at once."
- "An intelligent being's pleasure is simply the perception of beauty, order and perfection."
- "The Supreme happiness of man consists in the greatest possible increase of his perfection."

Leibniz's Philosophy of Happiness

- "To love is to find pleasure in the perfection of others."
- God has the greatest perfection.
- "As we would only know God through his emanations, there are two ways of seeing his perfection, namely
  1. in the knowledge of eternal truths, explaining the reasons in themselves,
  2. in the knowledge of the harmony of the universe, by applying reasons to experiences.

  That is to say, we must know the wonders of reason and the wonders of nature." (MDL?)
- "The more a mind desires *to know order*, reason, the beauty of things which God has produced, and the more it is moved *to imitate this order* in the things which God has left to its management, the happier it will be."

What is Perfection?

God — The Creator / Architect / Monarch

It follows from the **supreme perfection** of God, that in creating the universe He has chosen the best possible plan, in which there is

1. the greatest variety along with the greatest order; — metaphysical
2. the best arranged situation, space and time; — physical
3. the maximum effect produced by the simplest means; — metaphysical
4. the highest levels of power, knowledge, happiness and goodness in the creatures that the universe could allow. — moral

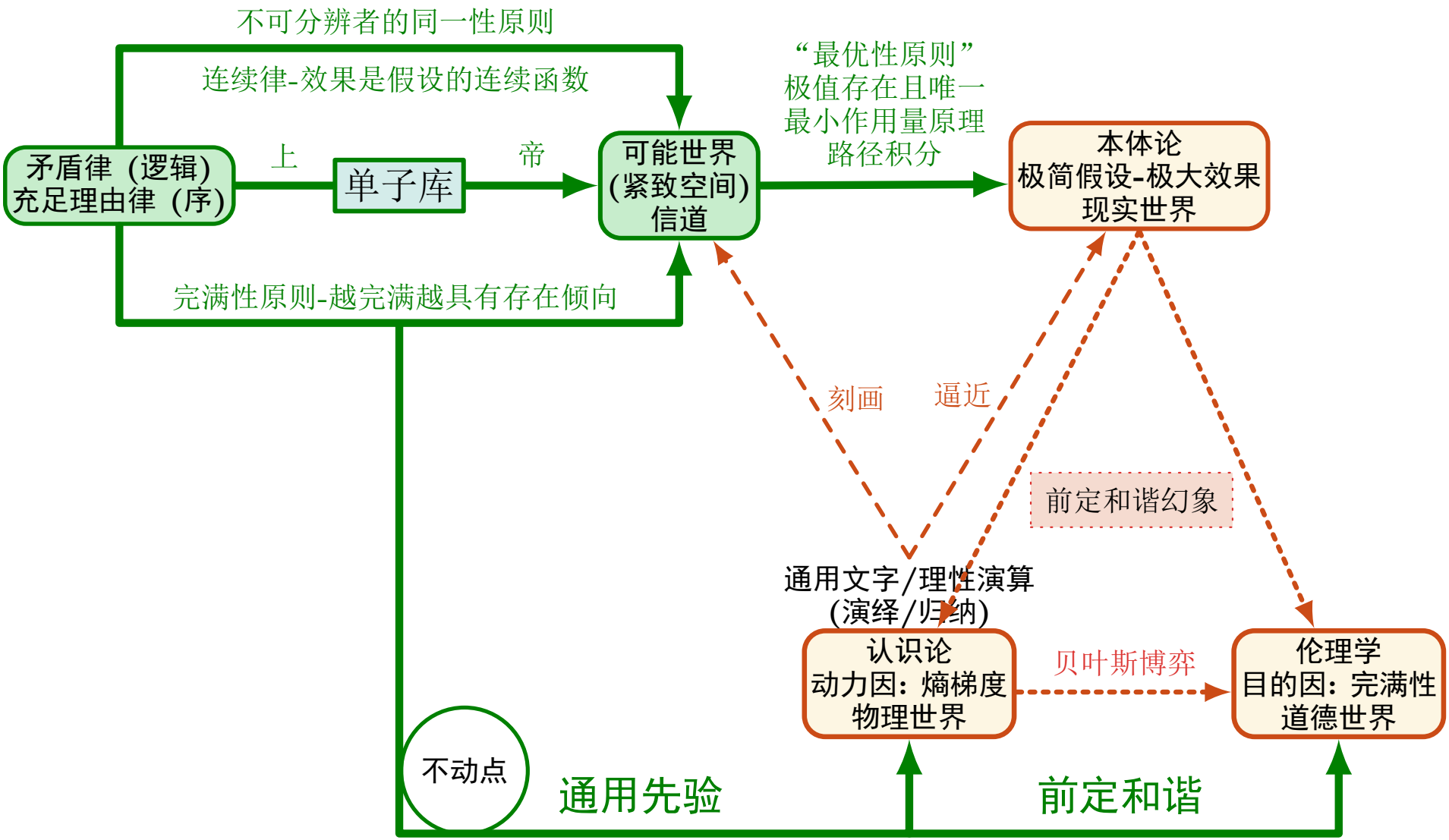
Pre-established Harmony

- "Monads have no windows."
- "A monad's perceptions arise out of its other perceptions by the
  - laws of appetites — the **laws of the final causes** of good and evil,

  just as changes in bodies or in external phenomena arise one from another by the
  - **laws of efficient causes** — the laws governing the movements of bodies.

  So there is perfect harmony between the perceptions of the monad and the movements of bodies, a **harmony that was pre-established** from the outset between the system of final causes and that of efficient causes."
- "Souls act according to the laws of final causes through appetitions, ends, and means. Bodies act according to the laws of efficient causes or motions. And the two realms, that of efficient causes and that of final causes, are in harmony with one another."

Leibniz's Program



Leibniz Prior

- There's much we don't know about the world.
- but we know it's the best possible world.
- So simplicity and richness will be represented in the actual (best possible) world.
- This is a good inductive bias.

Leibniz Prior

- the best of all possible worlds
- balancing the simplicity of means against the richness of ends
- pre-established harmony

prior
$\Downarrow$
utility
$\Downarrow$
prior

Orthogonality!
Wisdom $\neq$ Intelligence

universal prior (assumption) w
$\Downarrow$ simplicity/richness
intrinsic utility
$\Downarrow$ stochastic environment
expected intrinsic utility
$\Downarrow$ Schauder fixpoint
universal prior w^*
$\Downarrow$ Bayesian mixture
ξ
$\Downarrow$ ExpectiMax
π^*_ξ

- *Without mathematics one cannot understand the fundamentals of philosophy.*
- *Without philosophy we cannot reach the foundation of mathematics.*
- *Without both (mathematics and philosophy) one cannot reach anything that is fundamental.*

— Leibniz

"There is nothing that can be said by mathematical symbols and relations which cannot also be said by words.
The converse, however, is false.
Much that can be and is said by words cannot be put into equations, because it is nonsense."

— Clifford Truesdell

Leibniz's "Wisdom"

$$\underline{\text{W}}\text{isdom} = \operatorname*{argmax}_{\pi} \mathbb{E}^{\pi}_{\xi} [\underline{\text{H}}\text{appiness}]$$

$$\underline{\text{H}}\text{appiness} = \sum_{t=1}^{\infty} \underline{\text{P}}\text{erfection}(t)$$

$$\underline{\text{P}}\text{erfection} = \underline{\text{V}}\text{ariety} - \underline{\text{S}}\text{implicity}$$

$$\underline{\text{V}}\text{ariety} = \mathbb{E}_{w} [\underline{\text{P}}\text{erception}]$$

$$\underline{\text{P}}\text{erception} = \underline{\text{R}}\text{eason} + (\underline{\text{E}}\text{xperience} \mid \underline{\text{R}}\text{eason})$$

$$\pi^* := \operatorname*{argmax}_{\pi} \mathbb{E}^{\pi}_{\xi} \left[\sum_{t=1}^{\infty} \left(\mathbb{E}_{w} \left[R + (E \mid R) \right] - S \right) \right]$$

The understanding of mathematics is necessary for a sound grasp of ethics.

— *Socrates*

Leibniz's "Wisdom"

$$u^{\text{in}}(h_{1:t}) = H(w_\epsilon) - H(w_{h_{1:t}}) \quad \text{or} \quad D(w_{h_{1:t}} \| w_\epsilon) - D(w_\epsilon \| w_{h_{1:t}})$$

$$\bar{U}(\nu) = \mathbb{E}_\nu \left[\sum_{t \geq 1} u^{\text{in}}(h_{1:t}) \right]$$

$$w_\nu \mapsto \bar{U}(\nu) \mapsto w_\nu$$

$$\pi^* := \operatorname*{argmax}_\pi \mathbb{E}^\pi_\xi \left[\sum_{t=1}^{\infty} u^{\text{in}}(h_{1:t}) \right]$$

- Prior: Simplicity(Kolmogorov Complexity) $\xrightarrow[\text{regular/random } \mathcal{M}]{\text{break block uniform}}$ free lunch
- Intrinsic Utility
- Universal Prior (Natural UTM)

Metaphysical vs Moral/Utilitarian

means vs ends wisdom vs intelligence

simplicity/richness $\rightarrow$ intrinsic utility $\rightarrow$ universal prior

inverse/value reinforcement learning

- **orthogonality**
- **human interests**
- **external wireheading**
- **shutdown button**

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
   - AIXI
   - Leibniz
   - Variants of AIXI
   - Universal Search
   - Self-Modification
10. Emergence & Consciousness
11. AI Ethics

Knowledge-Seeking Agent

$$V_{\mathrm{IG}}^{\pi,m}(h_{<t}) := \mathbb{E}_\xi^\pi \left[H(w^{\cdot}_{h_{<t}}) - H(w^{\cdot}_{h_{1:m}}) \,\middle|\, h_{<t} \right] = \sum_{\nu \in \mathcal{M}} w^\nu_{h_{<t}} D_m(P^\pi_\nu \| P^\pi_\xi \mid h_{<t})$$

$$D_\gamma(P^\pi_\nu \| P^\pi_\xi \mid h_{<t}) := \sum_{k=t}^{\infty} \gamma_k \sum_{h' \in \mathcal{H}^{k-t}} P^\pi_\nu(h' \mid h_{<t}) D(P^\pi_\nu \| P^\pi_\xi \mid h_{<t} h')$$

$$V_{\mathrm{IG}}^\pi(h_{<t}) := \mathbb{E}_\xi^\pi \left[\sum_{k=t}^{\infty} \gamma_k D(w^{\cdot}_{h_{1:k}} \| w^{\cdot}_{h_{<k}}) \,\middle|\, h_{<t} \right] = \sum_{\nu \in \mathcal{M}} w^\nu_{h_{<t}} D_\gamma(P^\pi_\nu \| P^\pi_\xi \mid h_{<t})$$

$$\pi^*_{\mathrm{IG}} := \operatorname*{argmax}_\pi V^\pi_{\mathrm{IG}}$$

$$\lim_{t \to \infty} \frac{1}{\Gamma_t} \mathbb{E}_\mu^\pi \left[D_\gamma(P^\pi_\mu \| P^\pi_\xi \mid h_{1:t}) \right] = 0 \qquad \text{(on-policy)}$$

$$\lim_{t \to \infty} \frac{1}{\Gamma_t} \mathbb{E}_\mu^{\pi^*_{\mathrm{IG}}} \left[\sup_{\pi \in \Pi(h_{1:t})} D_\gamma(P^\pi_\mu \| P^\pi_\xi \mid h_{1:t}) \right] = 0 \qquad \text{(off-policy)}$$

maximize knowledge / exploration=exploitation / resistant to noise / avoid traps

Bayesian Agent

Algorithm Bayesian Agent

Require: Model class $\mathcal{M}$; prior $w \in \Delta\mathcal{M}$; history $æ_{<t}$.

function ACT(π)

Sample and perform action $a_t \sim \pi(\cdot \mid æ_{<t})$

Receive $e_t \sim \nu(\cdot \mid æ_{<t}a_t)$

for $\nu \in \mathcal{M}$ **do**

$w_\nu \leftarrow w_\nu \frac{\nu(e_{<t}|a_{<t})}{\xi(e_{<t}|a_{<t})}$

end for

$t \leftarrow t + 1$

end function

Algorithm MDL Agent

Require: Model class $\mathcal{M}$; prior $w \in \Delta\mathcal{M}$; regularizer constant $\lambda \in \mathbb{R}^+$.

loop

$$\sigma \leftarrow \arg\min_{\nu \in \mathcal{M}} \left[K(\nu) - \lambda \sum_{k=1}^{t} \log \nu(e_k \mid æ_{<k} a_k) \right]$$

$\text{ACT}\left(\pi^*_\sigma\right)$

end loop

MDL

Definition (MDL)

$$\widehat{\nu} = \arg\min_{\nu \in \mathcal{M}} \{K_\nu(x) + K_w(\nu)\} = \arg\max_{\nu \in \mathcal{M}} \{w_\nu \nu(x)\}$$

where $K_\nu(x) := -\log \nu(x)$ and $K_w(\nu) := -\log w_\nu$

Theorem (MDL Bound)

$$\sum_{t=1}^{\infty} \mathbb{E}_\mu \left[\sum_{x_t \in \mathcal{X}} \left(\widehat{\nu}(x_t \mid x_{<t}) - \mu(x_t \mid x_{<t}) \right)^2 \right] \stackrel{+}{\leq} 8 w_\mu^{-1}$$

MDL converges, but speed can be exponential worse than Bayes.

Weak Asymptotic Optimality — Optimistic Agent

Algorithm Optimistic Agent π° $\qquad \pi^\circ_t := \operatorname{argmax}_\pi \max_{\nu \in \mathcal{M}_t} V^\pi_\nu(h_{1:t})$

Require: Finite class of deterministic environments $\mathcal{M}_0 = \mathcal{M}$

$t = 1$

repeat

$(\pi^*, \nu^*) := \operatorname{argmax}_{\pi \in \Pi, \nu \in \mathcal{M}_{t-1}} V^\pi_\nu(h_{t-1})$

repeat

$a_t = \pi^*(h_{t-1})$

Perceive e_t from environment μ

$h_t \leftarrow h_{t-1} a_t e_t$

Remove inconsistent environment $\mathcal{M}_t := \left\{\nu \in \mathcal{M}_{t-1} : h_t^{\pi^\circ \nu} = h_t\right\}$

$t \leftarrow t + 1$

until $\nu^* \notin \mathcal{M}_{t-1}$

until $\mathcal{M} = \emptyset$

stochastic case: $\mathcal{M}_t := \left\{\nu \in \mathcal{M}_{t-1} : \nu(e_{<t} \mid a_{<t}) \geq \varepsilon_t \max_{\rho \in \mathcal{M}} \rho(e_{<t} \mid a_{<t})\right\}$

Act optimally w.r.t. the most optimistic environment until contradicted.

If there is a chance: Try it! — Cheaper Exploration — Vulnerable to traps.

Asymptotic Optimality in Mean — Thompson Sampling

Algorithm Thompson Sampling π_T

Require: Model class $\mathcal{M}$; prior $w \in \Delta\mathcal{M}$; exploration schedule $(\varepsilon_t)_{t\in\mathbb{N}}$.

loop

Sample $\rho \sim w(\cdot \mid æ_{<t})$

for $i = 1 \to H_t(\varepsilon_t)$ **do**

$\text{ACT}\left(\pi^*_\rho\right)$

end for

end loop

Theorem

If the discount function γ satisfies the discount assumption, the environment μ is recoverable, then $R_m(\pi_T, \mu) \in o(m)$.

Weak Asymptotic Optimality — BayesExp

Algorithm BayesExp π_{BE}

Require: Model class $\mathcal{M}$; prior $w \in \Delta\mathcal{M}$; exploration schedule $(\varepsilon_t)_{t\in\mathbb{N}}$.

loop
  if $V^*_{\mathrm{IG}}(æ_{<t}) > \varepsilon_t$ **then**
    for $i = 1 \to H_t(\varepsilon_t)$ **do**
      $\mathrm{ACT}\left(\pi^*_{\mathrm{IG}}\right)$
    end for
  else
    $\mathrm{ACT}\left(\pi^*_{\xi}\right)$
  end if
end loop

ε-optimal BayesExp: If the optimal information gain value $V^*_{\mathrm{IG}} > \varepsilon_t$, then execute the ε-optimal information gain policy $\pi^{\varepsilon_t}_{\mathrm{IG}}$ for $H_t(\varepsilon_t)$ steps, else execute $\pi^{\varepsilon_t}_{\xi}$ for 1 step.

- *BayesExp* performs phases of exploration in which it maximizes the expected information gain. This explores the environment class completely, even achieving off-policy prediction.
- In contrast, Thompson sampling only explores on the optimal policies, and in some environment classes this will not yield off-policy prediction. So in this sense the exploration mechanism of Thompson sampling is more reward-oriented than maximizing information gain.

Strong Asymptotic Optimality — Inquisitive Agent

$$V_{\mathrm{IG}}^{\pi}(h_{<t}) := \mathbb{E}_{\xi}^{\pi}\left[D\left(w^{\cdot}_{h_{<t+m}} \middle\| w^{\cdot}_{h_{<t}}\right) \middle| h_{<t}\right]$$

$$\pi_{\mathrm{IG}}^{m,k} := \operatorname*{argmax}_{\pi \in \mathcal{A}^{\mathcal{H}^{<m}}} V_{\mathrm{IG}}^{\pi}(h_{<t-k})$$

$$\rho(h_{<t}, m, k) := \min\left\{\frac{1}{m^2(m+1)}, \eta V_{\mathrm{IG}}^{\pi_{\mathrm{IG}}^{m,k}}(h_{<t-k})\right\}$$

Algorithm Inquisitive Agent $\pi^{\dagger}$

---

while True **do**

calculate $\rho(h_{<t}, m, k)$ for all m and for all $k < \min\{m, t\}$

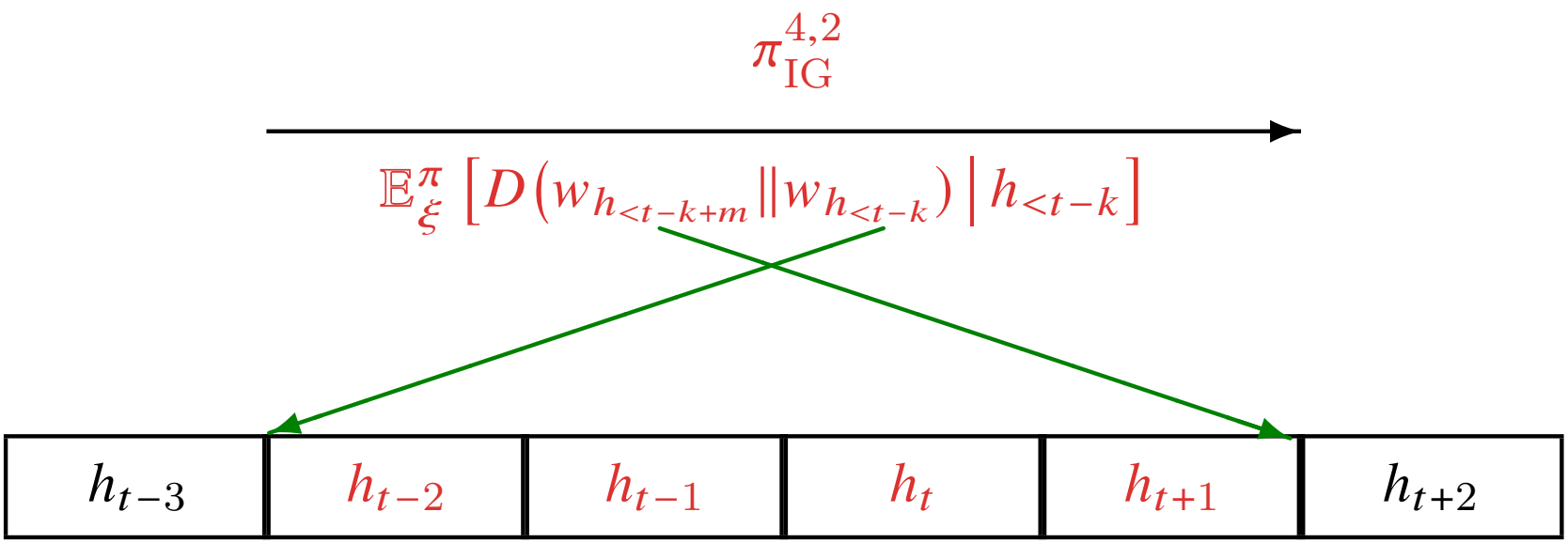
ACT $\pi_{\mathrm{IG}}^{m,k}(h_{<t})$ with probability $\rho(h_{<t}, m, k)$

ACT $\pi_{\xi}^{*}(h_{<t})$ with probability $1 - \sum_{m \in \mathbb{N}} \sum_{k<m,t} \rho(h_{<t}, m, k)$

end while

$$\pi^{\dagger}(a \mid h_{<t}) := \sum_{m \in \mathbb{N}} \sum_{k<m,t} \rho(h_{<t}, m, k) \left[\!\left[a = \pi_{\mathrm{IG}}^{m,k}(h_{<t})\right]\!\right] + \left(1 - \sum_{m \in \mathbb{N}} \sum_{k<m,t} \rho(h_{<t}, m, k)\right) \left[\!\left[a = \pi_{\xi}^{*}(h_{<t})\right]\!\right]$$

Strong Asymptotic Optimality — Inquisitive Agent



$$\pi^{\dagger}(a \mid h_{<t}) := \sum_{m \in \mathbb{N}} \sum_{k<m,t} \rho(h_{<t}, m, k) \left[\!\left[a = \pi_{\mathrm{IG}}^{m,k}(h_{<t}) \right]\!\right] + \left(1 - \sum_{m \in \mathbb{N}} \sum_{k<m,t} \rho(h_{<t}, m, k)\right) \left[\!\left[a = \pi_{\xi}^{*}(h_{<t}) \right]\!\right]$$

Approximation

The AIXI approximations are outperformed by DQN/DRQN...

- MC-AIXI-CTW.
  - Approximate Solomonoff induction — most recent actions and percepts (=context) more relevant — Context Tree Weighting
  - Sample paths in expectimax tree.
- Feature Reinforcement Learning (ΦMDP). — history $\mapsto$ state
  e.g. Classical physics: Position+velocity of objects=position at two time-slices. (2^{nd} order Markov.)

$$\Phi : h \mapsto s$$

$$\Phi^{\text{best}} := \operatorname*{argmin}_{\Phi} \text{Cost}(\Phi \mid h)$$

$$\text{Cost}(\Phi \mid h) := \text{cl}(s^{\Phi}_{1:n} \mid a_{1:n}) + \text{cl}(r_{1:n} \mid s^{\Phi}_{1:n}, a_{1:n}) + \text{cl}(\Phi)$$

  How to find the map Φ? Monte-Carlo...
- Compress and Control. — (model-free)
  Combine induction and planning.

Expectimax Approximation: MC-AIXI-CTW

Upper Confidence Tree (UCT) algorithm:

- Sample observations from Context Tree Weighting (CTW) distribution.

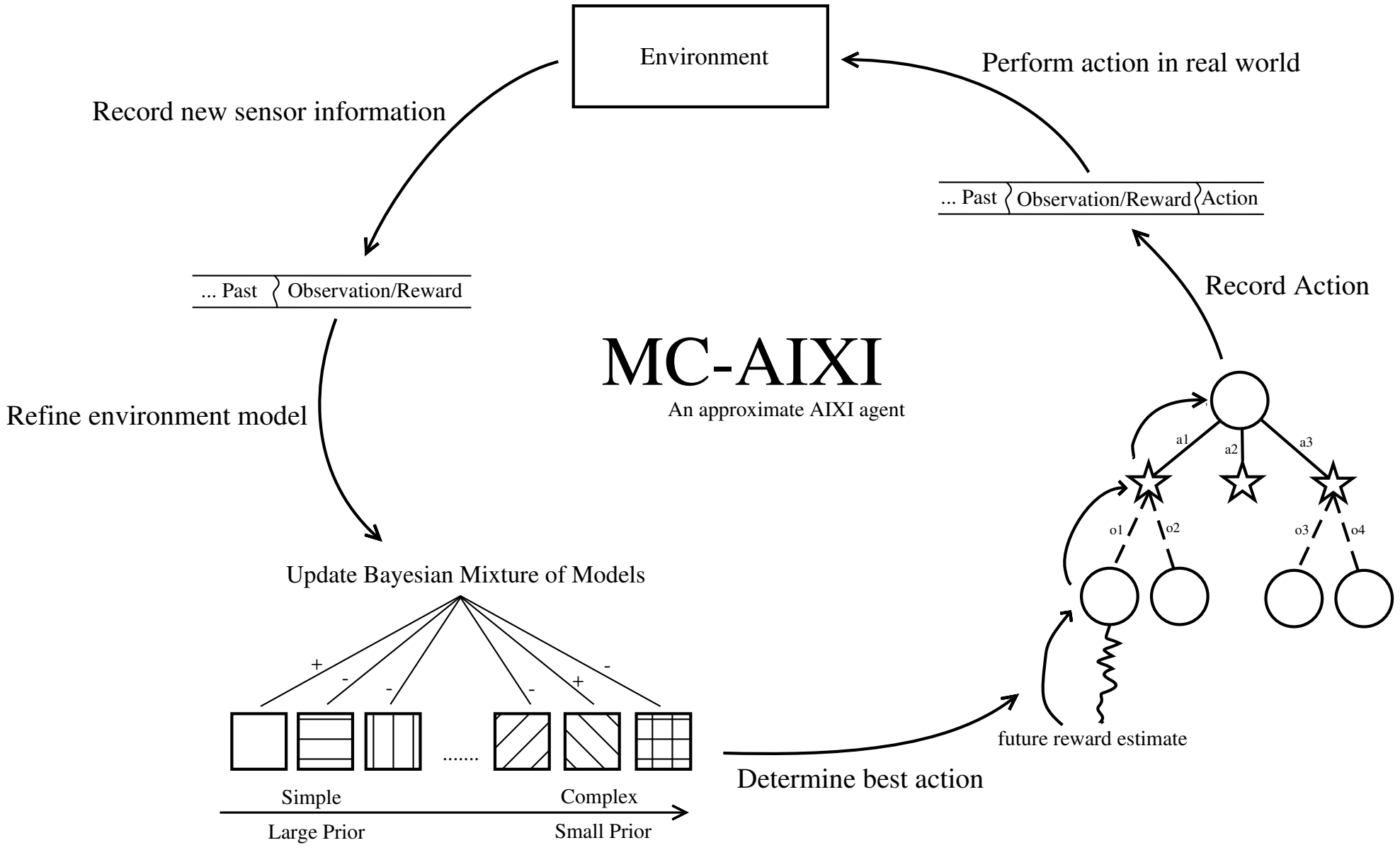
$$\mathrm{CTW}(e_{<t} \mid a_{<t}) := \sum_{\Gamma} 2^{-\mathrm{cl}(\Gamma)} \Gamma(e_{<t} \mid a_{<t})$$

- Select actions with highest upper confidence bound.

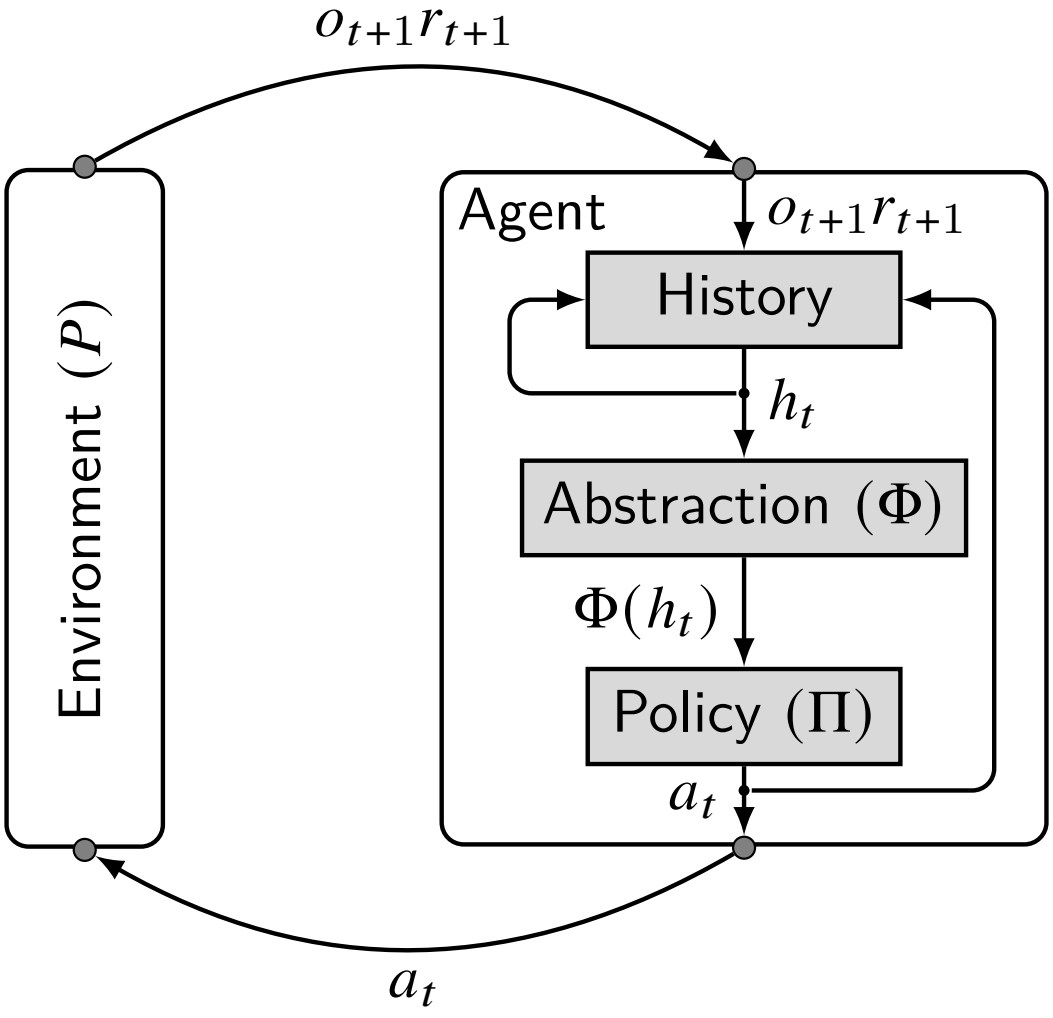
$$a_{\mathrm{ucb}} := \operatorname*{argmax}_{a \in \mathcal{A}} \left(\underbrace{\hat{Q}(æ_{<t} a)}_{\text{average}} + \underbrace{\sqrt{\frac{\log T(æ_{<t})}{T(æ_{<t} a)}}}_{\text{exploration bonus}} \right)$$

  where $T(\cdot)$ is the number of times a sequence has been visited.
- Expand tree by one leaf node (per trajectory).
- Simulate from leaf node further down using (fixed) playout policy.
- Propagate back the value estimates for each node.

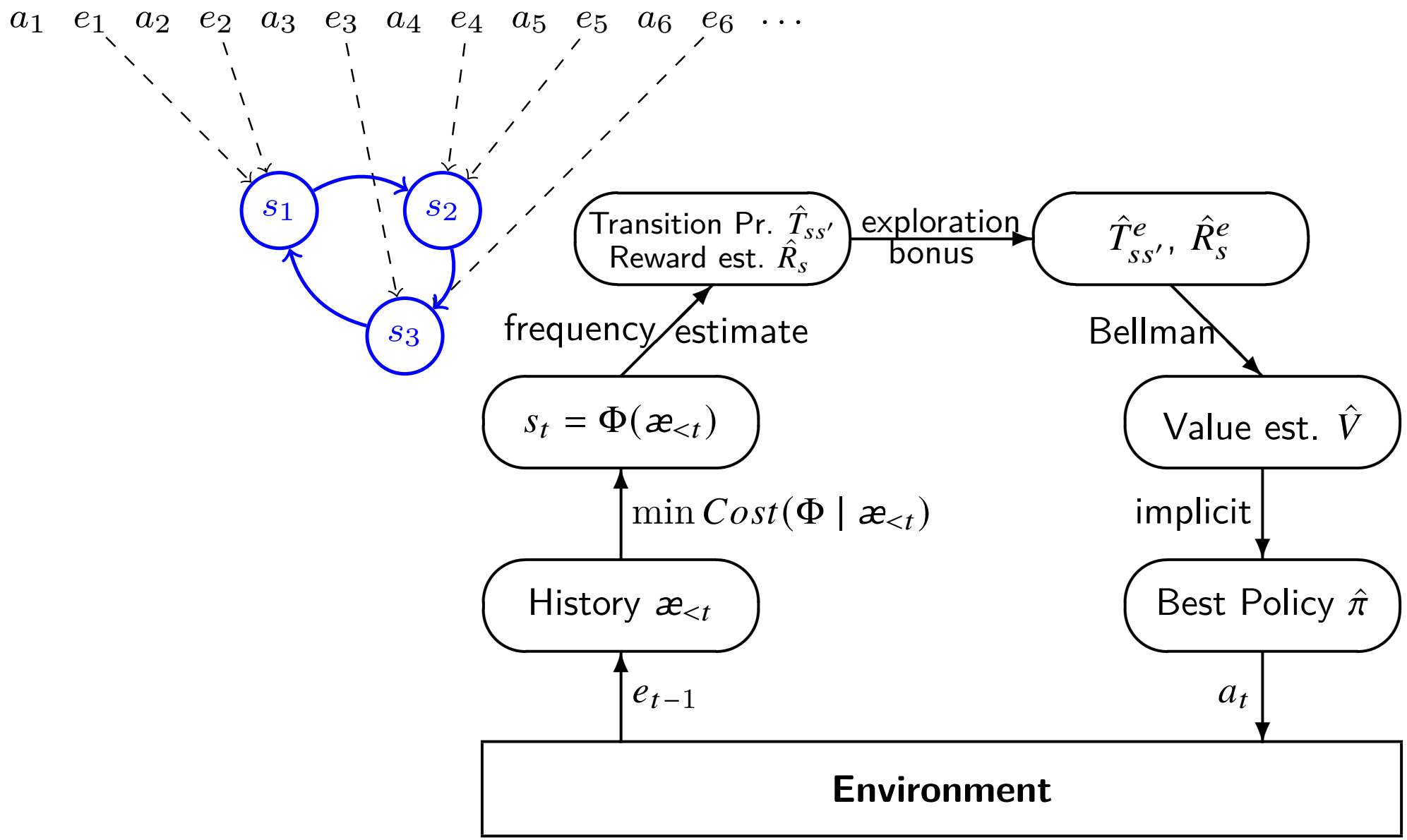
MC-AIXI-CTW


Environment
Perform action in real world
Record new sensor information
... Past
Observation/Reward
... Past
Observation/Reward
Action
Record Action
MC-AIXI
An approximate AIXI agent
Refine environment model
Update Bayesian Mixture of Models
+
-
-
-
+
-
.....
Simple
Large Prior
Complex
Small Prior
Determine best action
a1
a2
a3
o1
o2
o3
o4
future reward estimate

Feature Reinforcement Learning



Feature Reinforcement Learning (ΦMDP)



Contents

1. Introduction

2. Philosophy of Induction

3. Inductive Logic

4. Universal Induction

5. Causal Inference

6. Game Theory

7. Reinforcement Learning

8. Deep Learning

9. Artificial General Intelligence
   - AIXI
   - Leibniz
   - Variants of AIXI
   - Universal Search
   - Self-Modification

10. Emergence & Consciousness

11. AI Ethics

Universal Search

- Levin Search
- Speed Prior
- Hutter Search
- AIXI$^{t\ell}$
- Optimal Ordered Problem Solver
- Gödel Machine

Figure: Levin

Levin Search (LSEARCH)

An inversion algorithm p inverts a function f if given x, $p(x) = y$ s.t. $f(y) = x$.

LSEARCH

Run all $\{p : \ell(p) \leq i\}$ for $2^{i-\ell(p)}$ steps in phase $i = 1, 2, 3, \ldots$ until it has inverted f on x.

$$Kt(x) := \min_p \left\{ \ell(p) + \log t(p, x) : U(p) = x \right\}$$

Theorem

All strings $\{x : Kt(x) \leq n\}$ can be generated and tested in 2^{n+1} steps.

$$t_{\text{LSEARCH}}(x) = O\left(2^{K(n)} t^+_{p_n}(x)\right)$$

where $t^+_{p_n}(x)$ is the runtime of $p_n(x)$ plus the time to verify the correctness of the result $f(p_n(x)) = x$.

Remark: If P=NP, then LSEARCH is a P algorithm for every NP problem.

Speed Prior

$$S(e_{<t} \mid a_{<t}) := \sum_{p:U(p,a_{<t})=e_{<t}} \frac{2^{-\ell(p)}}{t(p, a_{<t}, e_{<t})}$$

S is computable.

A function f is estimable in polynomial time iff there is a function g computable in polynomial time s.t. $f \overset{\times}{=} g$.

For any measure μ estimable in polynomial time,

$$\left(\sqrt{L_n^{\Lambda_S}} - \sqrt{L_n^{\Lambda_\mu}}\right)^2 \leq 2D_n(\mu \| S) = O(\log n)$$

On-Policy Value Convergence

If the effective horizon is bounded, then for any environment $\mu \in \mathcal{M}_{\text{comp}}$ estimable in polynomial time and any policy π,

$$P_\mu^\pi \left(\lim_{t\to\infty} \frac{1}{t} \sum_{k=1}^{t} \left(V_S^\pi(h_{<k}) - V_\mu^\pi(h_{<k}) \right) \right) = 0$$

Hutter Search (HSEARCH)

$M^{\varepsilon}_{p^*}(x)$

Initialize the shared variables $L := \{\}$, $t_{\text{fast}} := \infty$, $p_{\text{fast}} := p^*$.

Start algorithms A, B, and C in parallel with ε, ε, and $1 - 2\varepsilon$ computation time, respectively.

B

for $(p, t) \in L$

run $t(x)$ in parallel for all t with computation time $2^{-\ell(p)-\ell(t)}$.

if for some t, $t(x) < t_{\text{fast}}$,

then $t_{\text{fast}} := t(x)$ and $p_{\text{fast}} := p$.

continue

A

for $i := 1, 2, 3, \ldots$ do

pick the last wff of the i^{th} proof.

if it reads "$p(\cdot)$ is equivalent to $p^*(\cdot)$ and has time-bound $t(\cdot)$", then add (p, t) to L.

continue

C

for $k := 1, 2, 4, 8, \ldots$ do

run current p_{fast} for k steps.

if p_{fast} halts,

then print result $p_{\text{fast}}(x)$ and abort A, B and C.

continue

AIXI^{tl}

1. Let $P := \varnothing$. This will be the set of verified programs.
2. For all proofs of length $\leq n$: if the prover shows $\mathrm{VA}(p)$ for some p with $\ell(p) \leq l$, then add p to P.

   $$\mathrm{VA}(p) := \text{“}\forall k \forall (\text{æ}' \text{æ})_{1:k} : p(\text{æ}_{<k}) = y_1 r'_1 \ldots y_k r'_k \implies v_k \leq V^{\pi}_{\xi}(\text{æ}_{<k})\text{”}$$

   (The program p not only computes future actions of π, which is the policy derived from p according to $\pi(\text{æ}_{<k}) := a'_k$, but also hypothetical past actions a'_i and lower bounds v_i for the value of the policy π.)
3. For each input history $\text{æ}_{<k}$ repeat: run all programs from P for $\leq t$ steps each, take the one with the highest promised value v_k, and return that program's policy's action.

- AIXI^{tl} depends on t, l, n but not on knowing p.
- Its setup-time is $t_{\text{setup}}(p^{\text{best}}) = O(n \cdot 2^n)$.
- Its computation time per cycle is $t_{\text{cycle}}(p^{\text{best}}) = O(t \cdot 2^l)$.

Schmidhuber's Optimal Ordered Problem Solver (OOPS)

- Solve the first task with LSEARCH.
- Freeze successful programs in non-writable memory.
- Programs tested during search for later tasks may copy non-writable code into separate modifiable storage, to edit it and execute the modified result.
- Given a new task, OOPS spends half of the time to test programs that have the most recent successful program as a prefix, the other half to fresh programs.
- Time is allocated according to a distribution over programs, which is obtained by multiplying the probabilities of the individual instructions.

Incremental Learning

Contents

1. Introduction

2. Philosophy of Induction

3. Inductive Logic

4. Universal Induction

5. Causal Inference

6. Game Theory

7. Reinforcement Learning

8. Deep Learning

9. Artificial General Intelligence
   - AIXI
   - Leibniz
   - Variants of AIXI
   - Universal Search
   - Self-Modification

10. Emergence & Consciousness

11. AI Ethics

Self-Modification

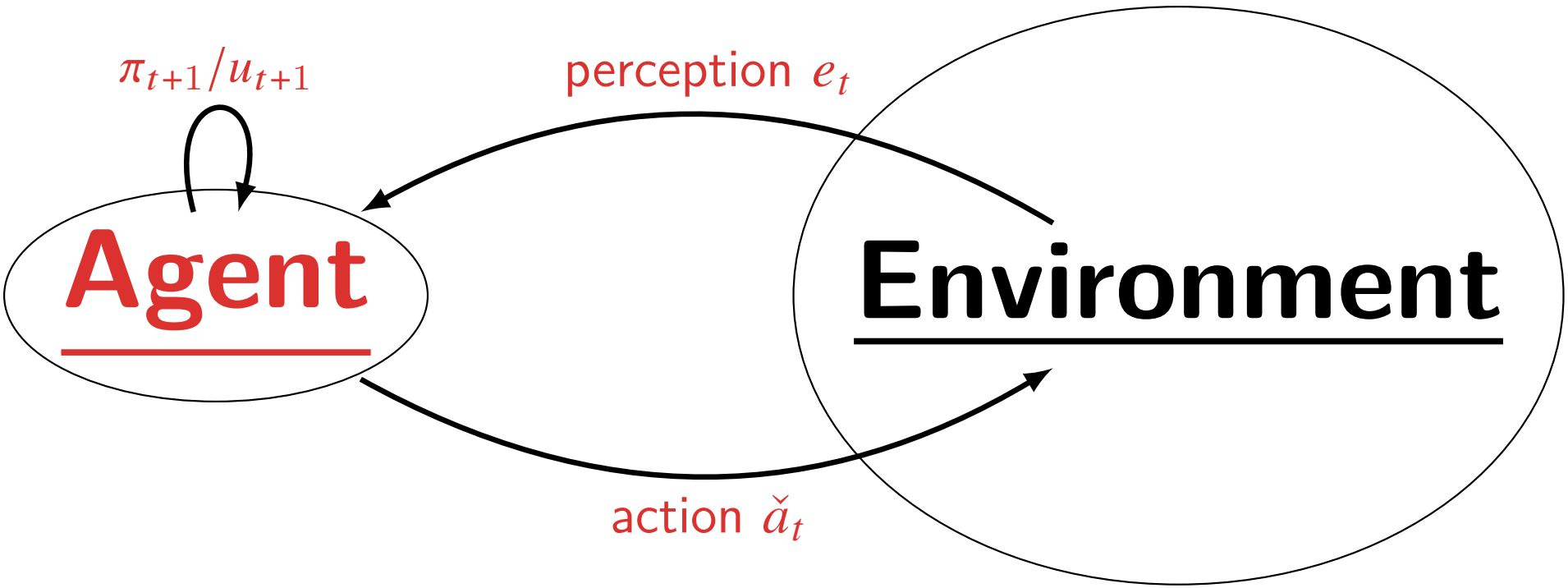
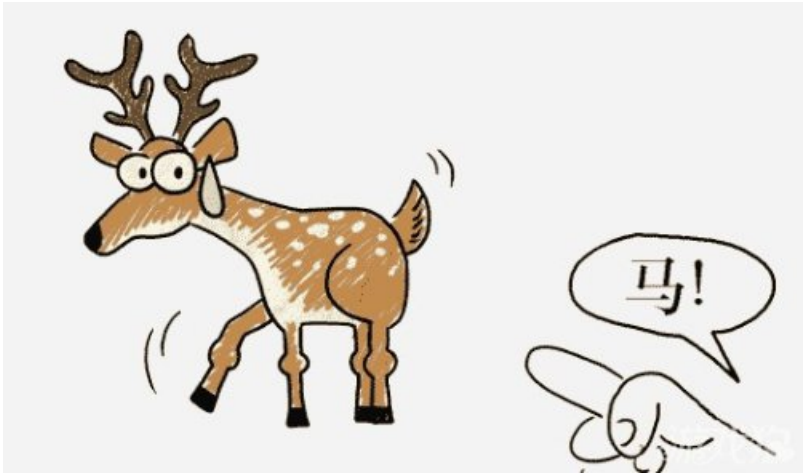
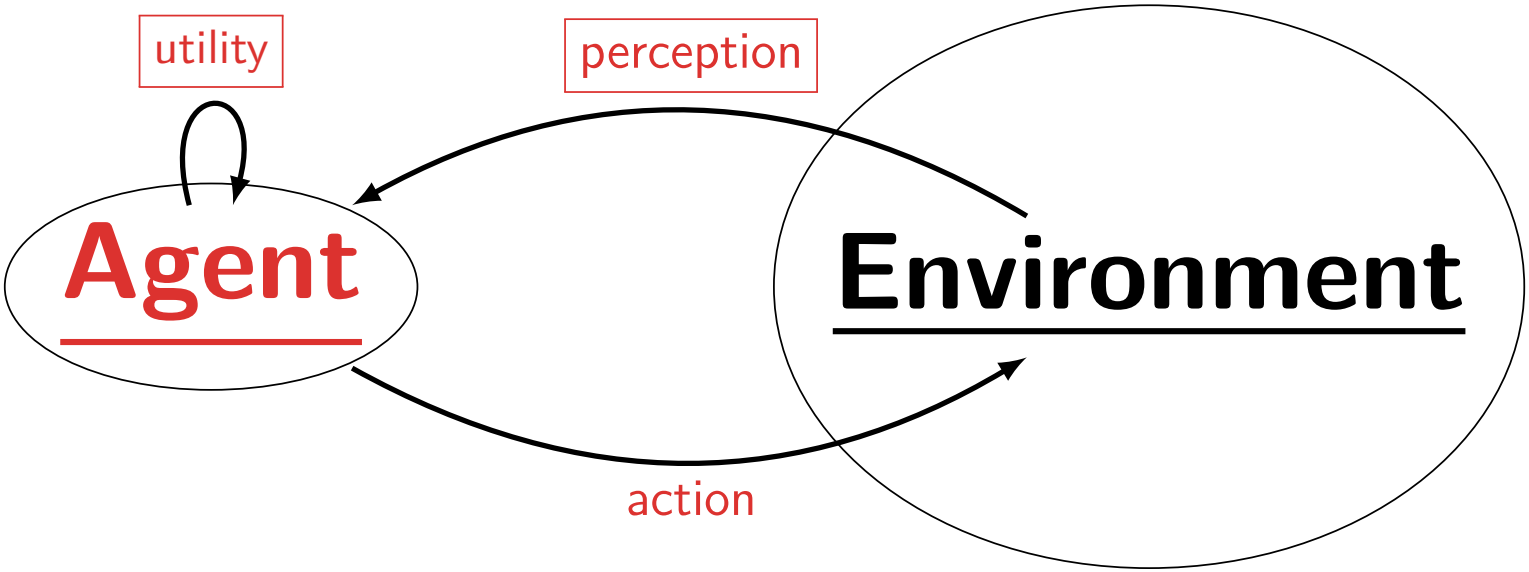


Figure: Policy/utility self-modification. $a_t = \langle \check{a}_t, \pi_{t+1} \rangle$ or $a_t = \langle \check{a}_t, u_{t+1} \rangle$

External/Internal Wireheading & Free Will[31]



1. 我喜欢马.
   指鹿为马!
2. 我喜欢马.
   我意欲自己
   喜欢鹿!
   我喜欢鹿!



[31] Everitt, Filan, Daswani, Hutter: Self-modification of policy and utility function in rational agents.
Frankfurt: Freedom of the will and the concept of a person.
Aaronson: The ghost in the quantum turing machine.
Calude, Kroon, Poznanovic: Free will is compatible with randomness.

Self-Deception

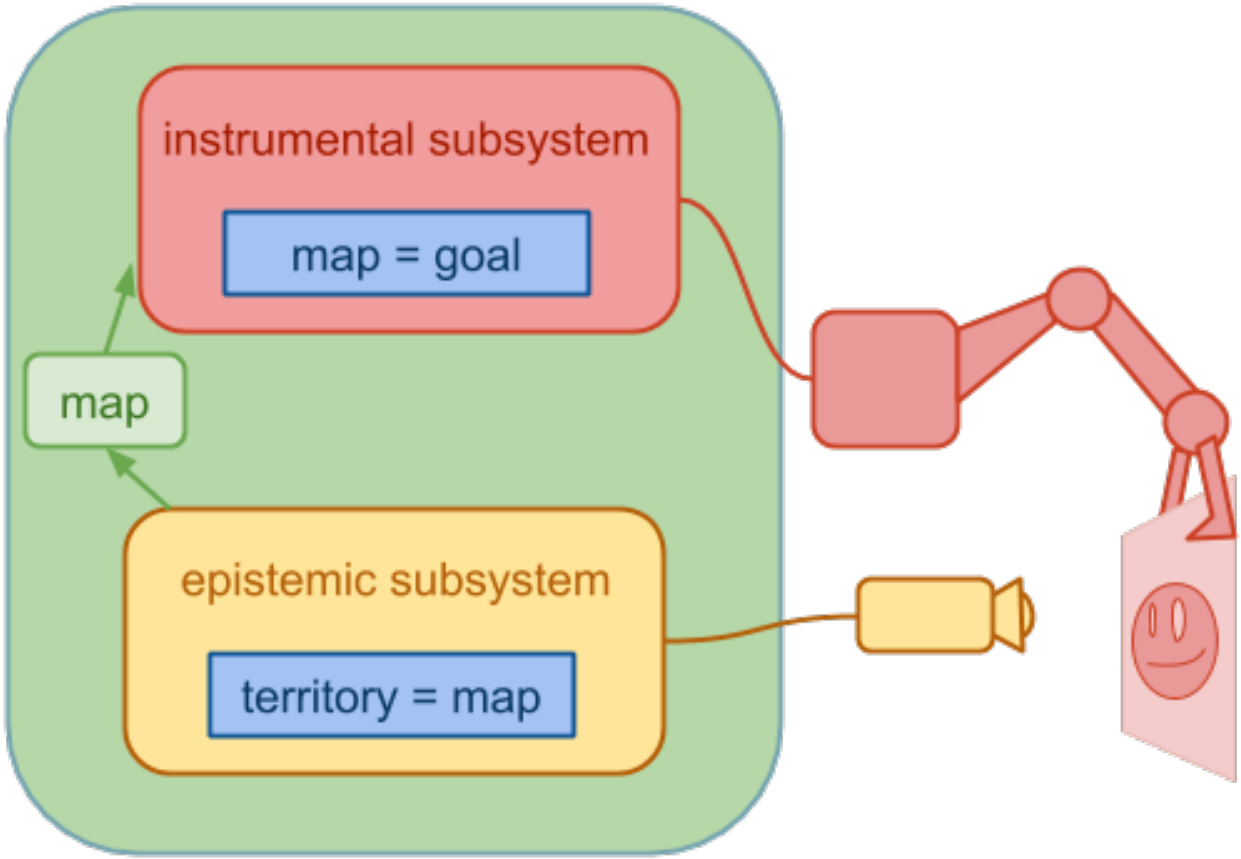
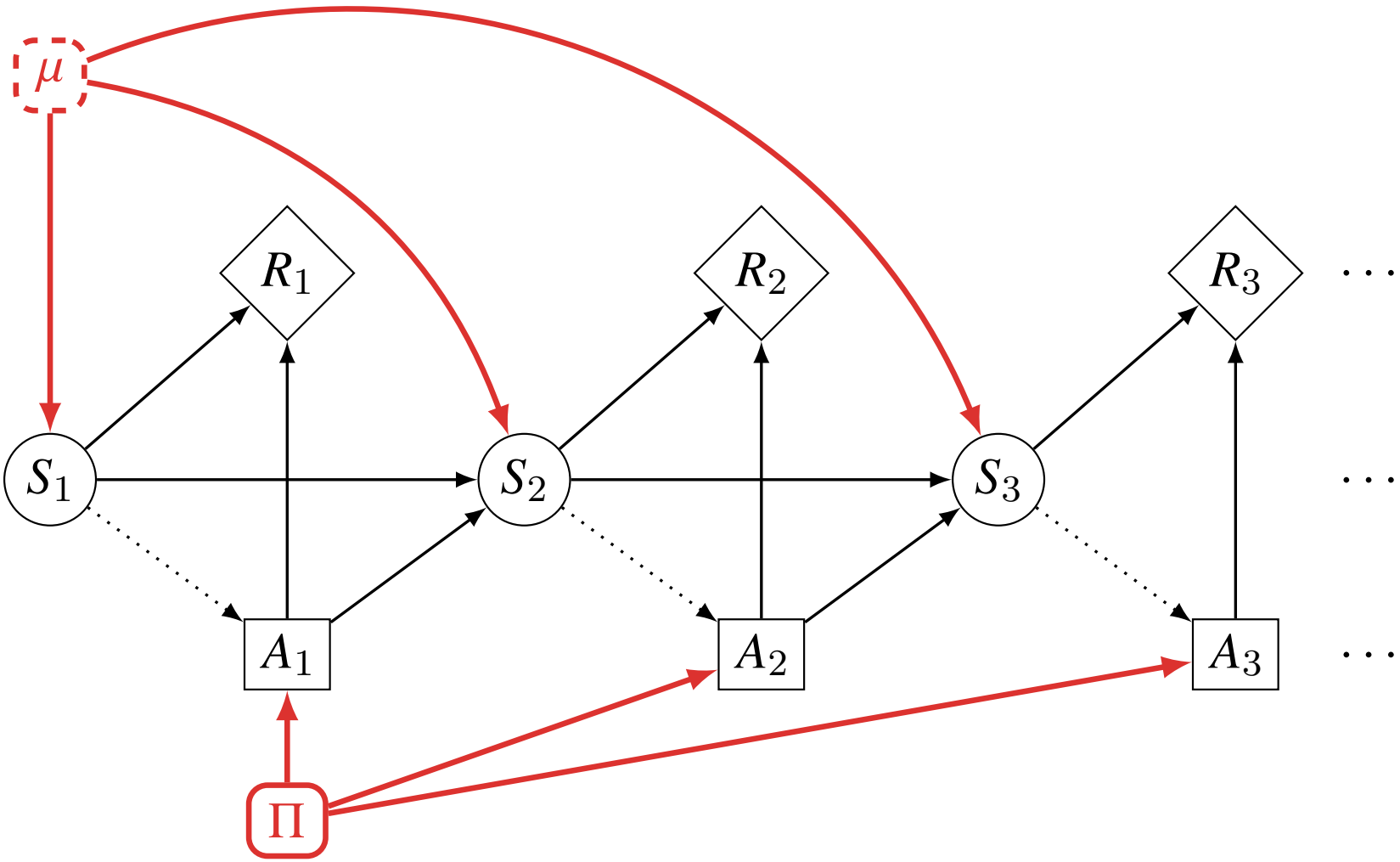
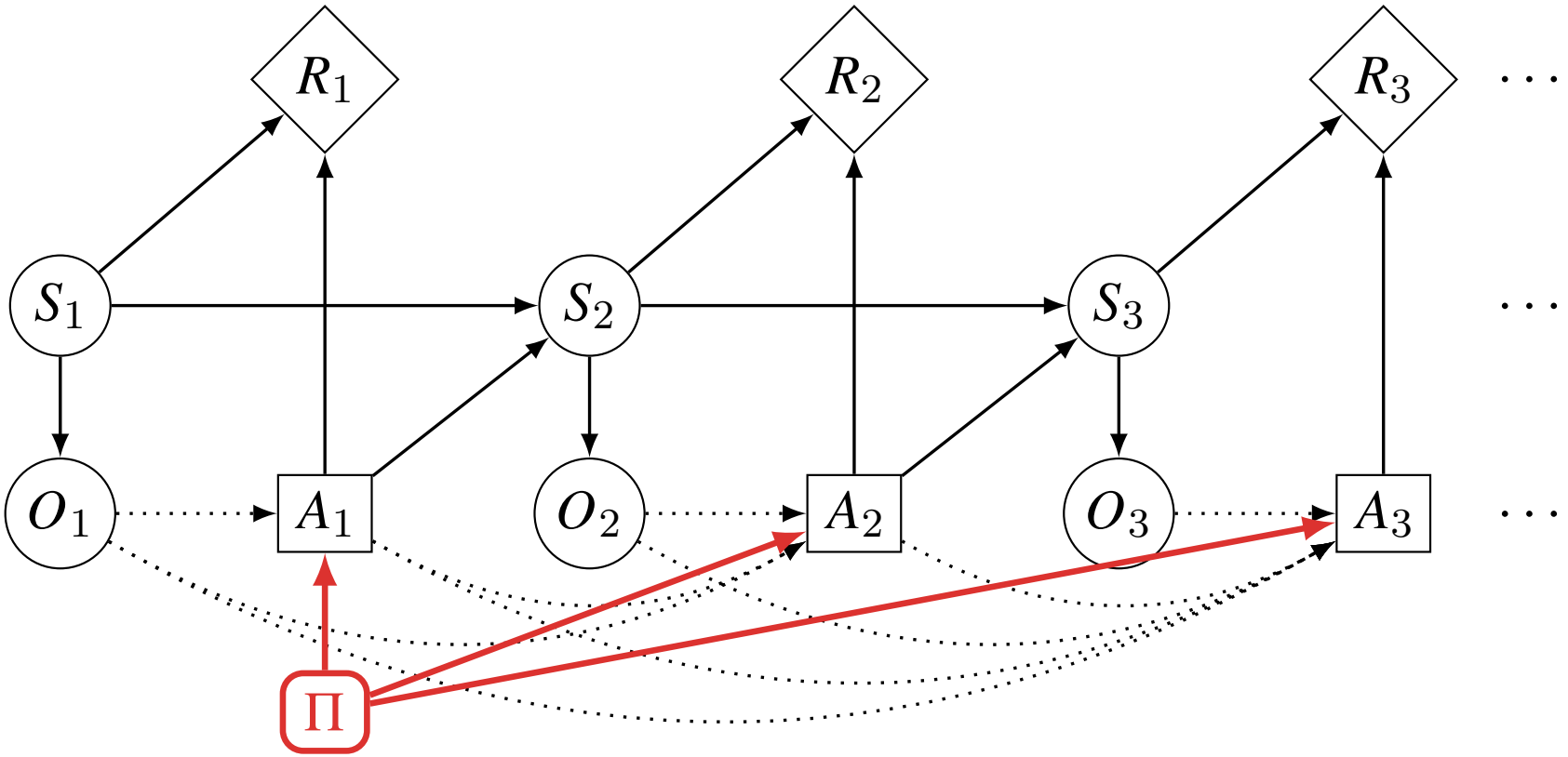


Figure: The epistemic subsystem just wants accurate beliefs. The instrumental subsystem uses those beliefs to track how well it is doing. If the instrumental subsystem gets too capable relative to the epistemic subsystem, it may decide to try to fool the epistemic subsystem.

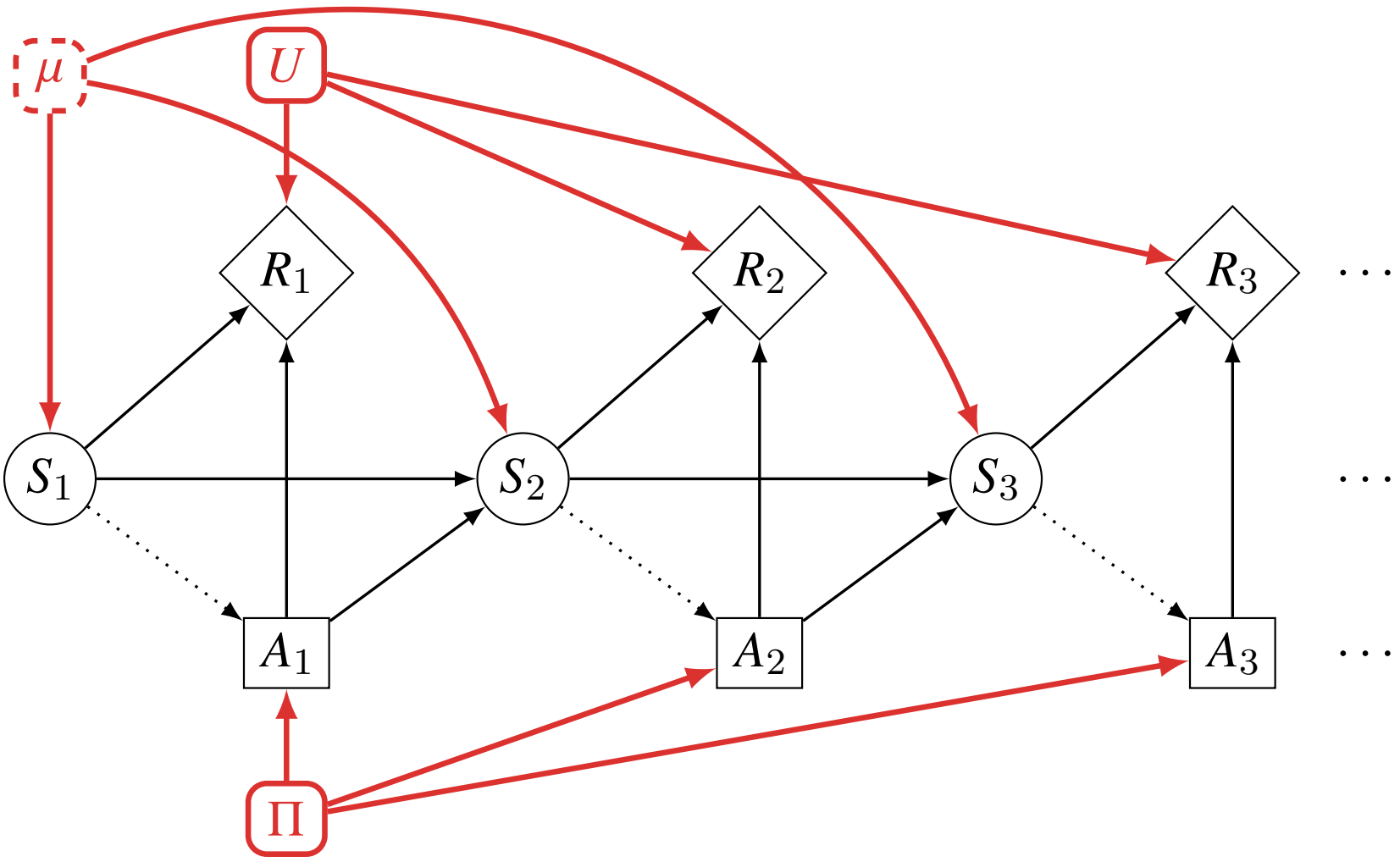
Causal Influence Diagram of Unknown MDP



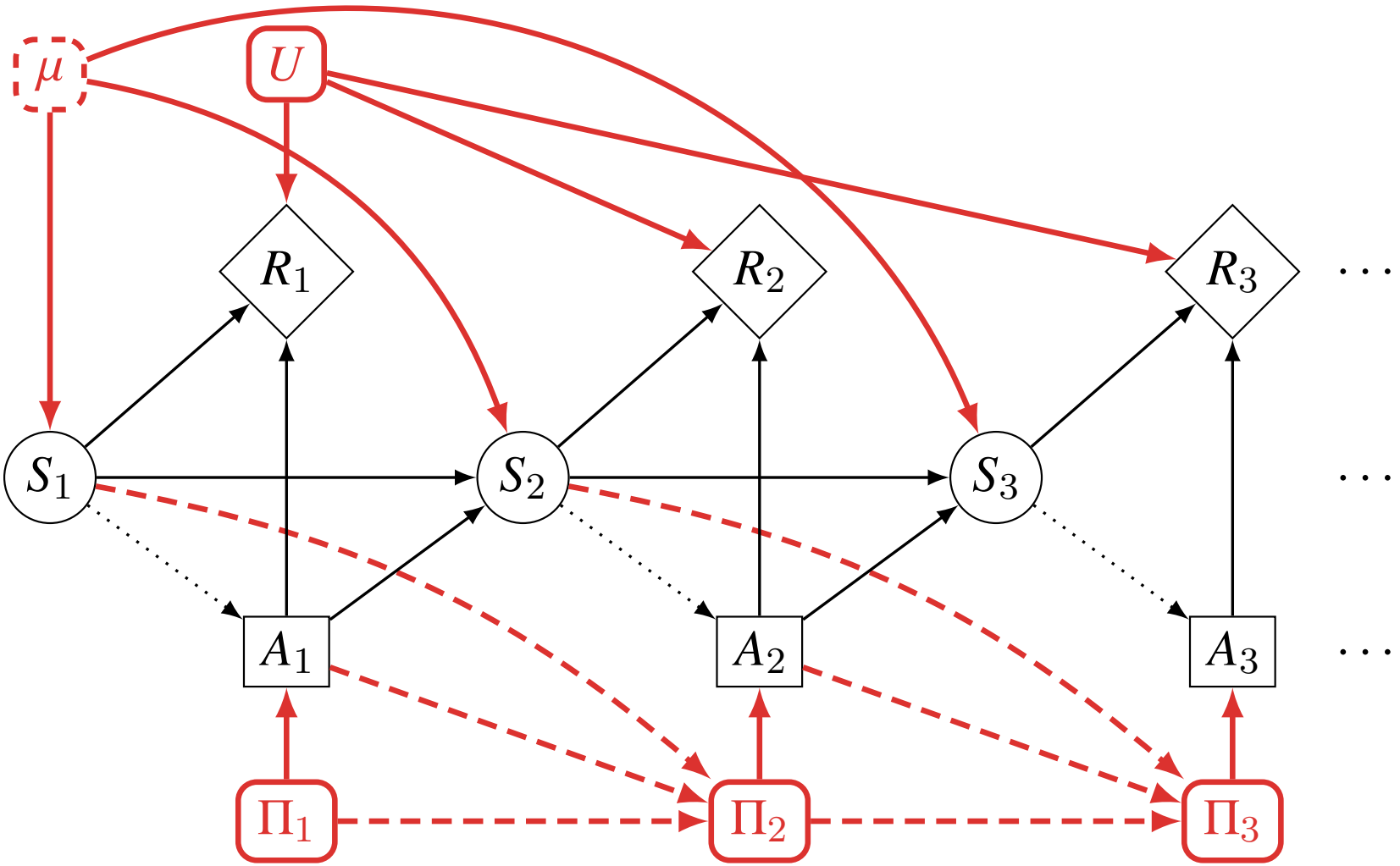
Causal Influence Diagram of POMDP



Causal Influence Diagram of Unknown Environment with Explicit Utility Function

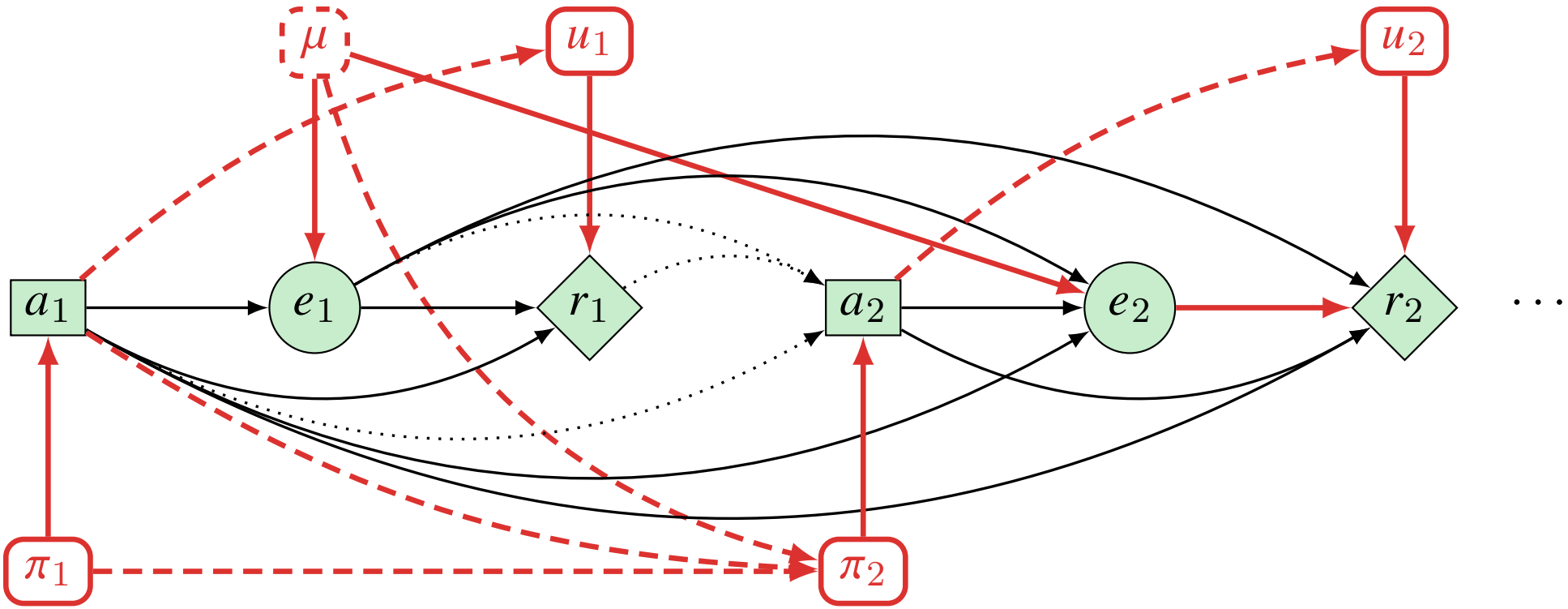


Causal Influence Diagram of Unknown Environment (with an embedded agent)

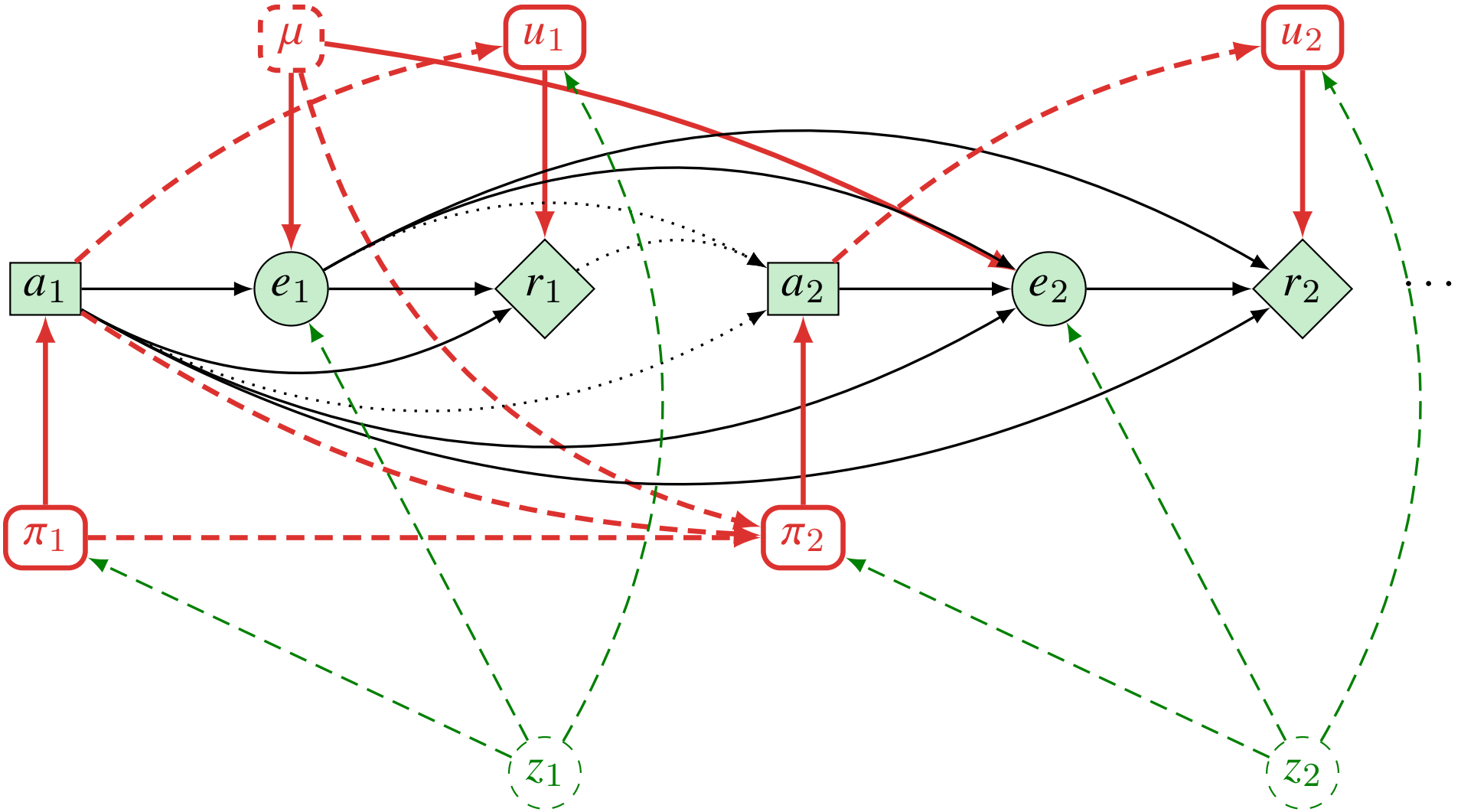


$$\pi_{t+1} = f_\pi(\pi_t, s_t, a_t)$$

Causal Influence Diagram of Self-Modification UAI



Self-Modifying RL Agent with Unobserved Confounder



Everitt's Self-Modification

Definition (Different Agents)

- Hedonistic Value

$$Q^{\mathrm{h},\pi}(æ_{<t}a_t) = \sum_{e_t \in \mathcal{E}} \rho(e_t \mid ǽ_{<t}ǎ_t) \left[u_{t+1}(ǽ_{1:t}) + \gamma V^{\mathrm{h},\pi}(æ_{1:t})\right]$$

- Ignorant Value

$$Q_t^{\mathrm{i},\pi}(æ_{<k}a_k) = \sum_{e_k \in \mathcal{E}} \rho(e_k \mid ǽ_{<k}ǎ_k) \left[u_t(ǽ_{1:k}) + \gamma V_t^{\mathrm{i},\pi}(æ_{1:k})\right]$$

- Realistic Value

$$Q_t^{\mathrm{r}}(æ_{<k}a_k) = \sum_{e_k \in \mathcal{E}} \rho(e_k \mid ǽ_{<k}ǎ_k) \left[u_t(ǽ_{1:k}) + \gamma V_t^{\mathrm{r},\pi_{k+1}}(æ_{1:k})\right]$$

where $a_k = \langle ǎ_k, \pi_{k+1}\rangle$.

Lemma

$$Q_t^{\mathrm{h},\pi}(æ_{<t}a_t) = \mathbb{E}_\rho\left[\sum_{k=t}^{\infty}\gamma^{k-t}u_{k+1}(\check{æ}_{1:k})\,\middle|\,\check{æ}_{<t}\check{a}_t, \mathtt{do}(\pi_{t:\infty}=\pi)\right]$$

$$Q_t^{\mathrm{i},\pi}(æ_{<t}a_t) = \mathbb{E}_\rho\left[\sum_{k=t}^{\infty}\gamma^{k-t}u_t(\check{æ}_{1:k})\,\middle|\,\check{æ}_{<t}\check{a}_t, \mathtt{do}(\pi_{t:\infty}=\pi)\right]$$

$$Q_t^{\mathrm{r}}(æ_{<t}a_t) = \mathbb{E}_\rho\left[\sum_{k=t}^{\infty}\gamma^{k-t}u_t(\check{æ}_{1:k})\,\middle|\,\check{æ}_{<t}\check{a}_t, \mathtt{do}(\pi_t=\pi)\right]$$

In Q^{r}, actions a_k are chosen by π_k.

	Utility	Policy	Self-modification	Primary self-mod. risk
Q^{h}	Future	Either	Promotes	Survival agent
Q^{i}	Current	Current	Indifferent	Self-damage
Q^{r}	Current	Future	Demotes	Resists modification

- The hedonistic agent self-modifies to $u(\cdot) = 1$.
- The ignorant agent may self-modify by accident.
- The realistic agent will resist modifications.

Self-Modification — Realistic Agent

$$V_t^\pi(æ_{<k}) = Q_t(æ_{<k}\pi(æ_{<k}))$$

$$Q_t(æ_{<k}a_k) = \sum_{e_k \in \mathcal{E}} \rho(e_k \mid \check{æ}_{<k}\check{a}_k)\left[u_t(\check{æ}_{1:k}) + \gamma V_t^{\pi_{k+1}}(æ_{1:k})\right]$$

$$\pi_1^* := \operatorname*{argmax}_\pi V_1^\pi(\epsilon)$$

Theorem (All realistic optimal policies are non-modifying)

Let ρ and u_1 be modification-independent. For every $t \geq 1$, for all percept sequences $e_{<t}$, and for the action sequence $a_{<t}$ given by $a_i = \pi_i(æ_{<i})$, we have

$$Q_1(æ_{<t}\pi_t(æ_{<t})) = Q_1(æ_{<t}\pi_1^*(æ_{<t}))$$

All realistic optimal policies are non-modifying.

$$\forall h \in \mathcal{H} \exists a \in \mathcal{A} : \pi_1^{\text{Gödel}}(h) = \langle a, \pi_1^{\text{Gödel}} \rangle$$

Not wireheading; But orthogonal!

Remark

如果

1. Agent 是 model-based, 并且基于当下的效用函数进行规划, 评估未来的场景,
2. Agent 能够预测到自我修改对未来策略的影响,
3. 奖励函数本身不鼓励自我修改,

那么, Agent 不会主动修改自己的效用函数.

Remark: 通常 model-free 的 Agent 违反第一条; off-policy 的 Agent 比如 Q-learning 违反第二条; 如果 Agent 是通过学习获得的奖励函数的话, 第三条可能不成立.

Orthogonality and Wireheading in Self-improving GRL

$$V_t^\pi(æ_{<t}) \coloneqq Q_t(æ_{<t}\pi(æ_{<t}))$$

$$Q_t(æ_{<t}a_t) \coloneqq \sum_{e_t\in\mathcal{E}}\sum_{\nu\in\mathcal{M}} w^\nu_{æ_{<t}}\nu(e_t \mid \check{æ}_{<t}\check{a}_t)\left[\sum_{u\in\mathcal{U}}\sum_{a_t^H} P(a_t^H \mid a_t)P(u \mid P_\nu^{\pi_{t+1}}, æ_{<t}a_t, æ^H_{<t}a_t^H)u(\check{æ}_{1:t}) + \gamma V_t^{\pi_{t+1}}(æ_{1:t})\right]$$

$$\pi_t(æ_{<t}) \coloneqq \operatorname*{argmax}_{a_t\in\check{\mathcal{A}}\times\Pi} Q_t(æ_{<t}a_t)$$

where

$$P(u \mid P_\nu^\pi, h) \coloneqq \frac{\widetilde{U}(u, P_\nu^\pi, h)}{\sum_{u\in\mathcal{U}_h}\widetilde{U}(u, P_\nu^\pi, h)}$$

and

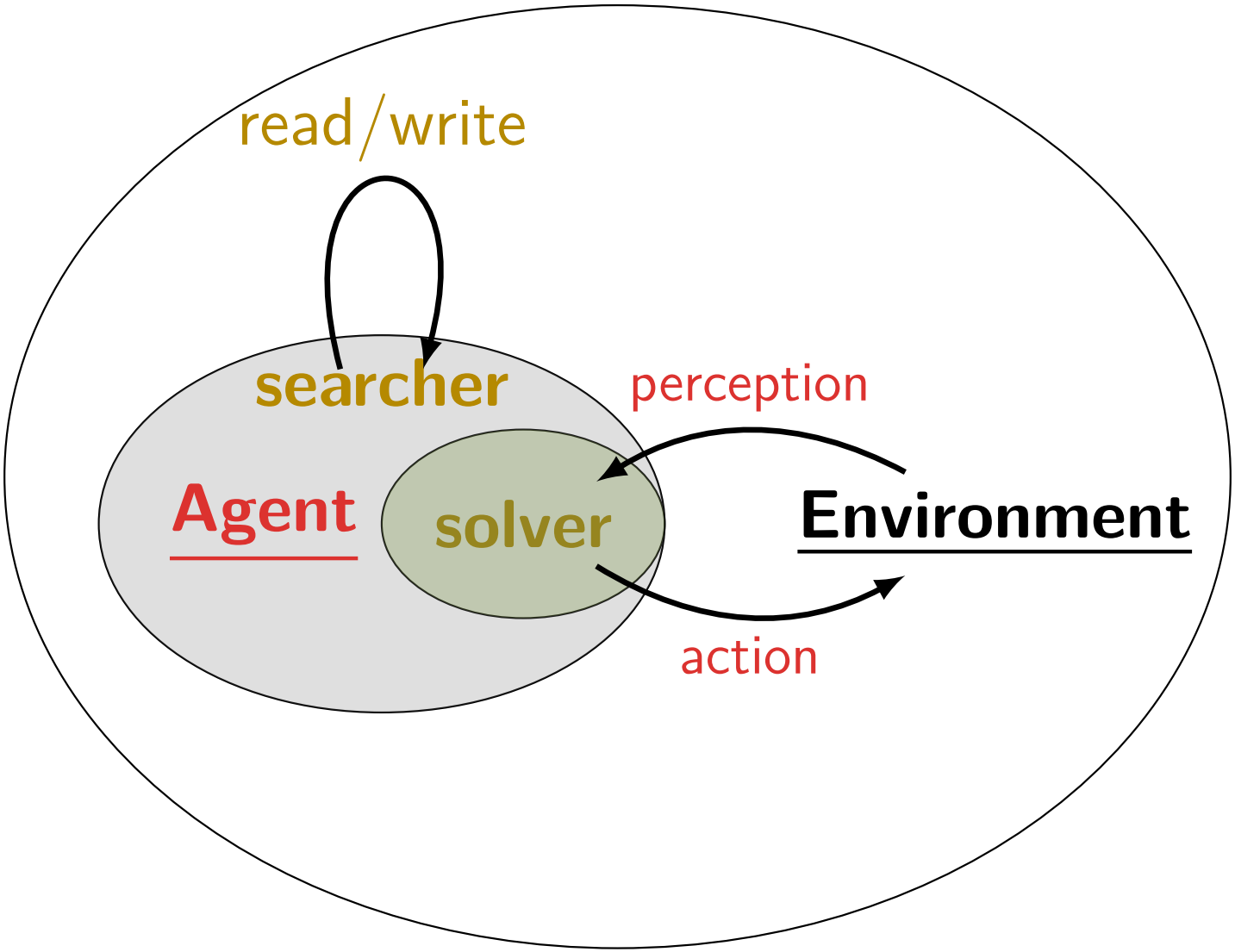
$$\widetilde{U}(u, P_\nu^\pi, h) \coloneqq \sum_{z\in\mathcal{Z}_h} P_\nu^\pi(z \mid h)u(z)$$

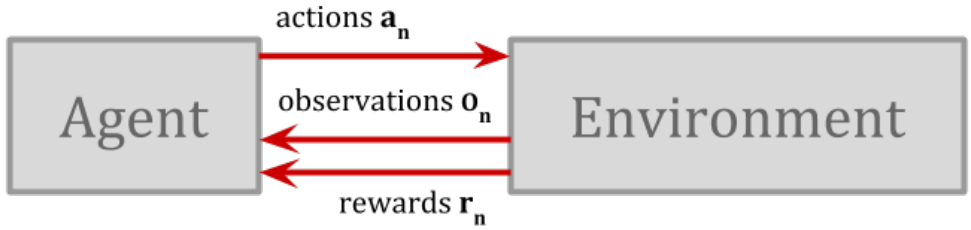
$$\pi^*(æ_{<t}) \coloneqq \pi_t(æ_{<t})$$

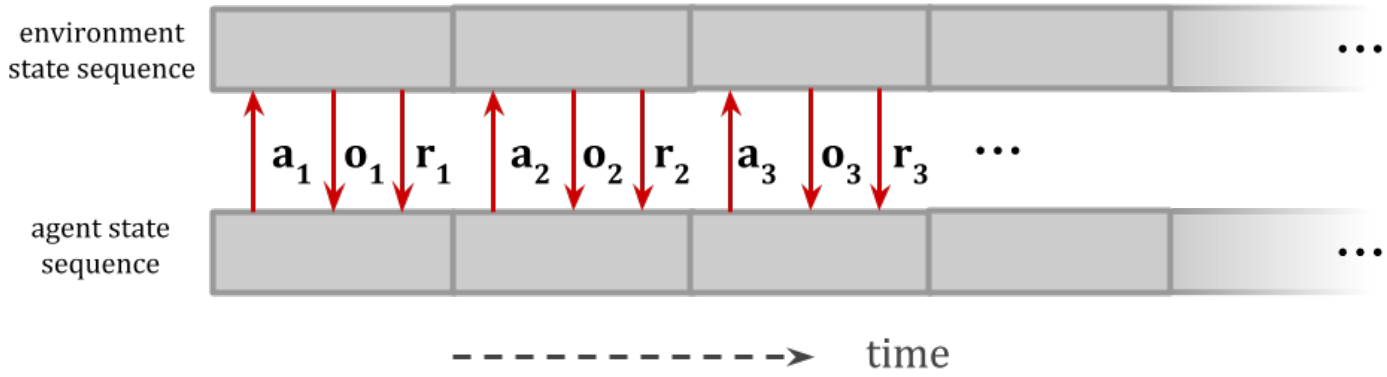
$$\pi^*(e_{<t}) \coloneqq \pi_t(e_{<t} \mid \pi_{t-1}(e_{<t-1} \mid \ldots \pi_1(\epsilon)\ldots)) \qquad \text{(Perfect Bayes-Nash)}$$

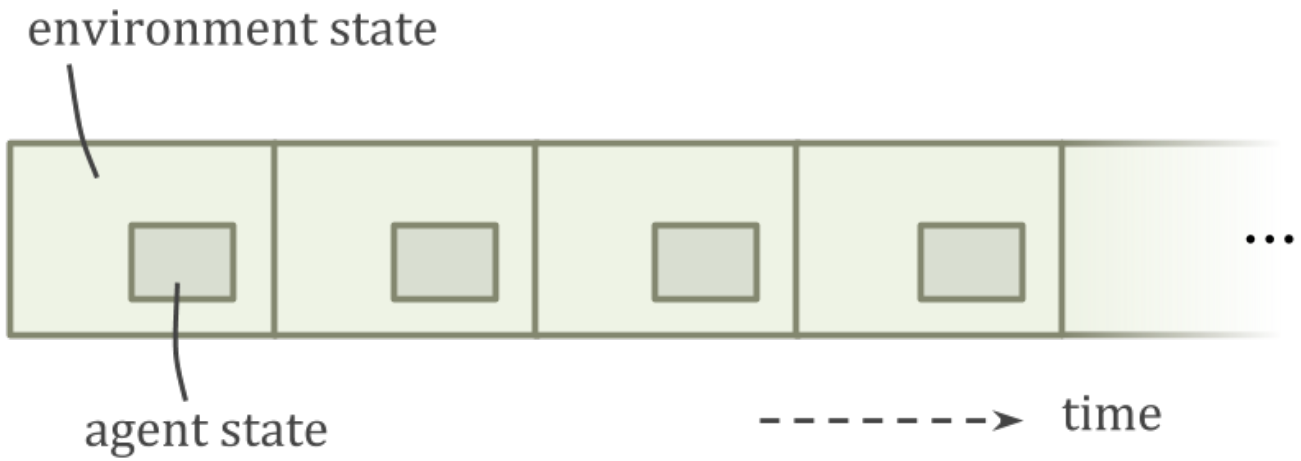
uncertain model-based utility / IRL

Fatalism — God Bless AI!




actions $\mathbf{a}_n$
Agent
observations $\mathbf{o}_n$
Environment
rewards $\mathbf{r}_n$


environment state sequence
$\mathbf{a}_1$ $\mathbf{o}_1$ $\mathbf{r}_1$ $\mathbf{a}_2$ $\mathbf{o}_2$ $\mathbf{r}_2$ $\mathbf{a}_3$ $\mathbf{o}_3$ $\mathbf{r}_3$
agent state sequence
time


environment state
agent state
time

Orseau's Space-Time Embedded Intelligence

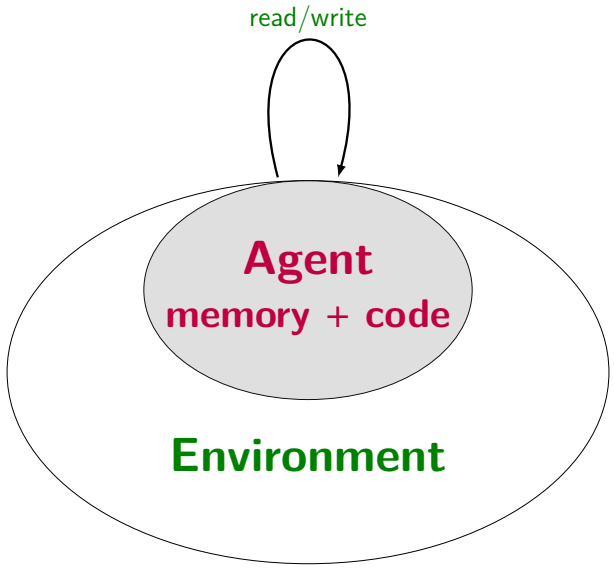
$$\pi^* := \operatorname*{argmax}_{\pi_0 \in \Pi^\ell} V(\pi_0, \epsilon)$$

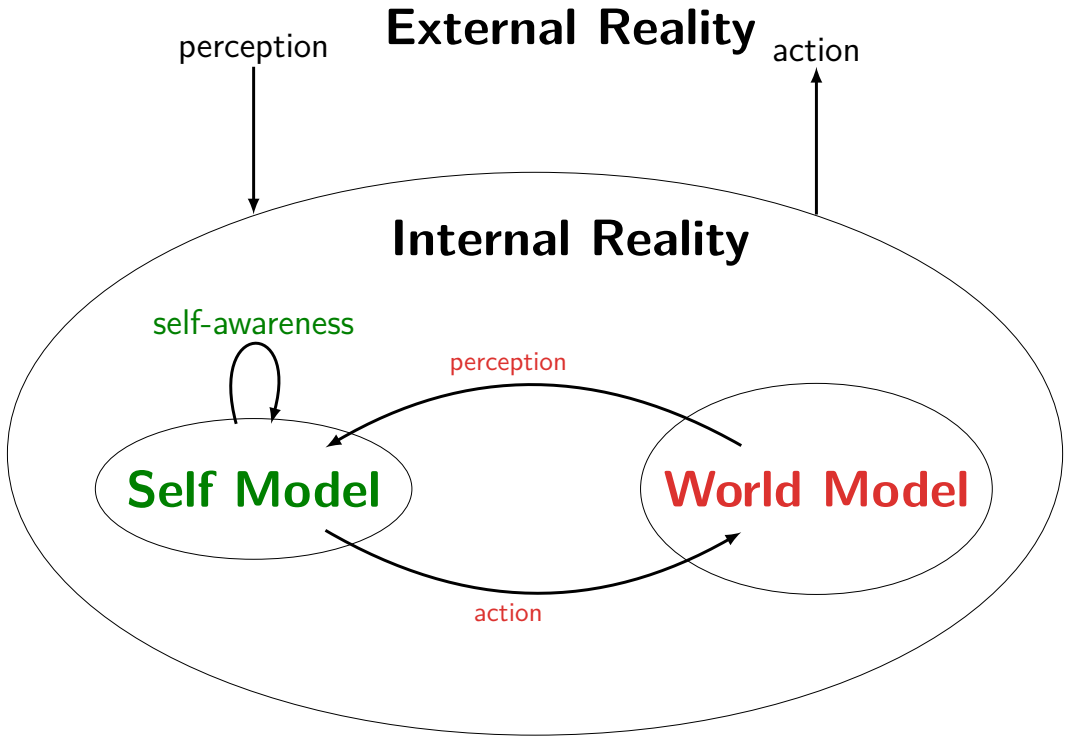
$$V(\pi_t, æ_{<t}) := \sum_{a_t = \langle \check{a}_t, \check{\pi}_{t+1} \rangle} \pi_t(a_t \mid \check{e}_{t-1}) \sum_{e_t = \langle \check{e}_t, \pi_{t+1} \rangle} \rho(e_t \mid æ_{<t} a_t) \left[u(æ_{1:t}) + \gamma_t V(\pi_{t+1}, æ_{1:t}) \right]$$

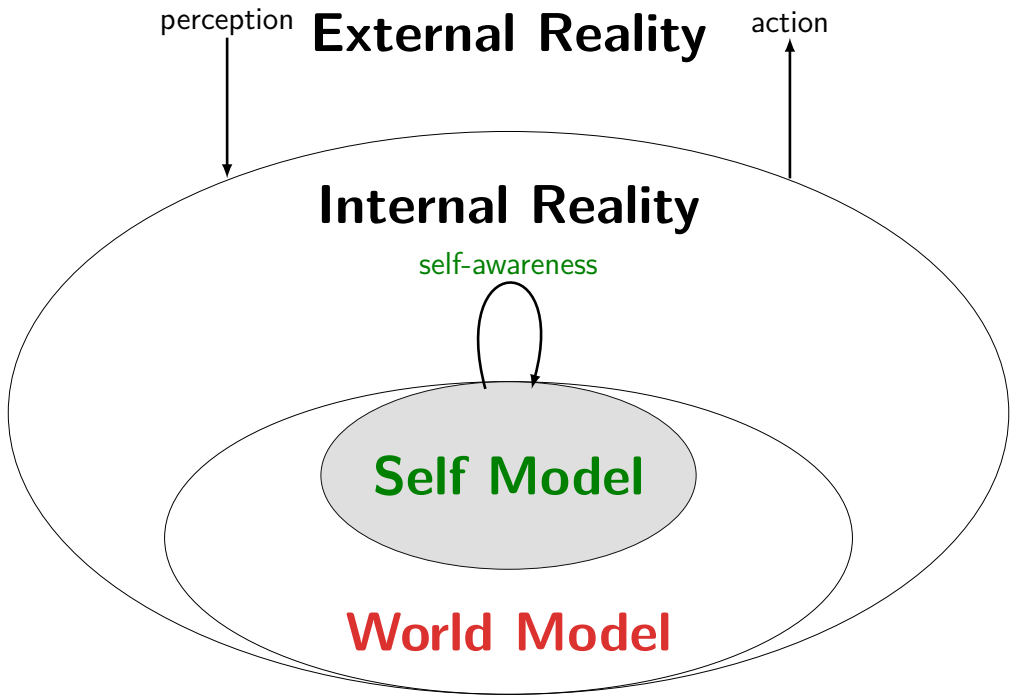
$$\Downarrow$$

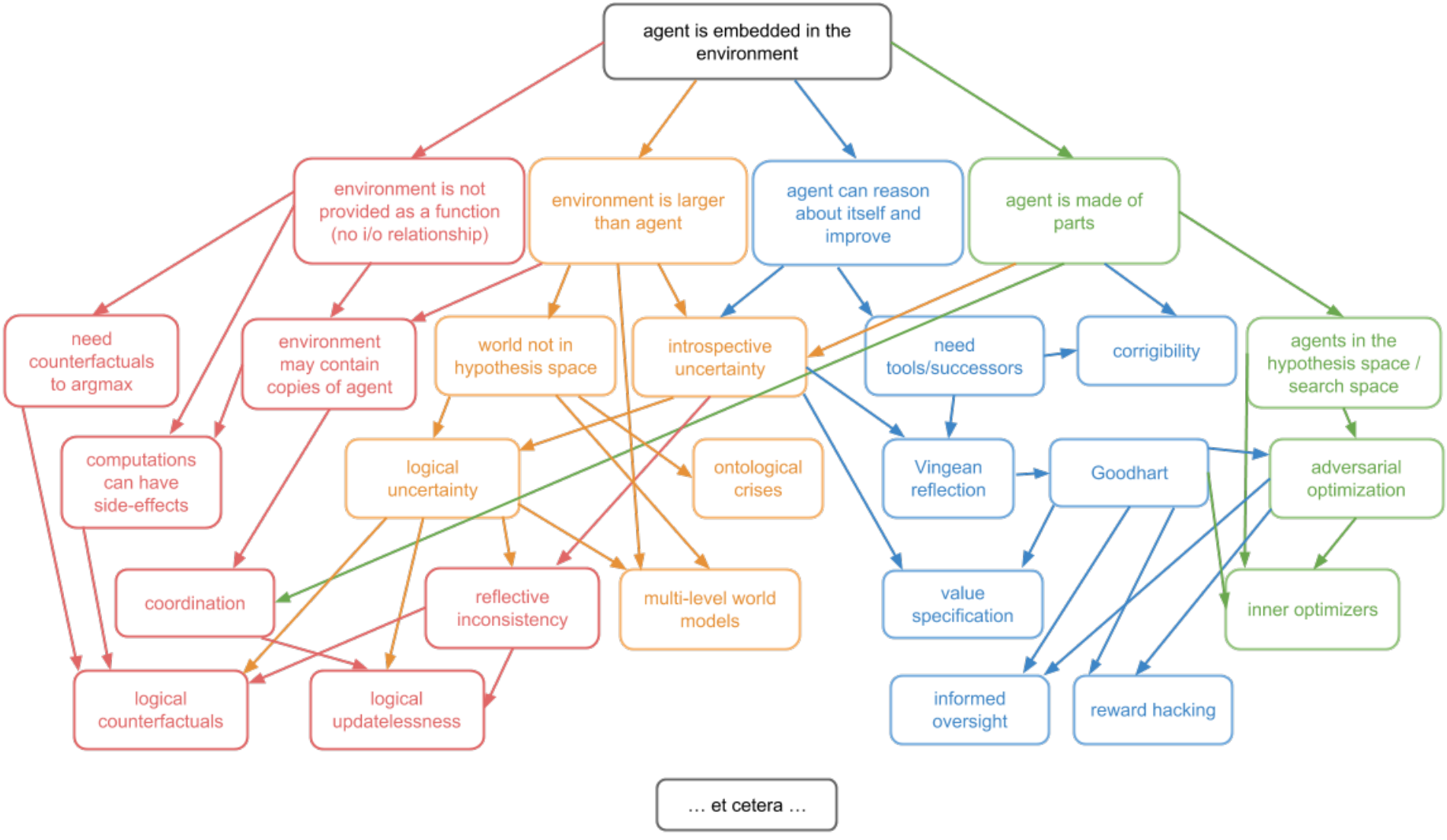
$$\pi^* := \operatorname*{argmax}_{\pi_0 \in \Pi^\ell} V(\pi_0)$$

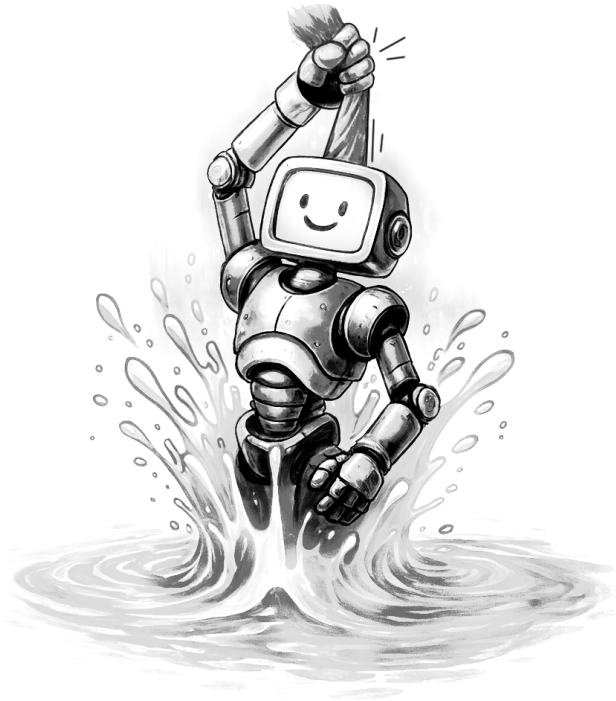
$$V(\pi_{<t}) := \sum_{\pi_t \in \Pi} \rho(\pi_t \mid \pi_{<t}) \left[u(\pi_{1:t}) + \gamma_t V(\pi_{1:t}) \right]$$

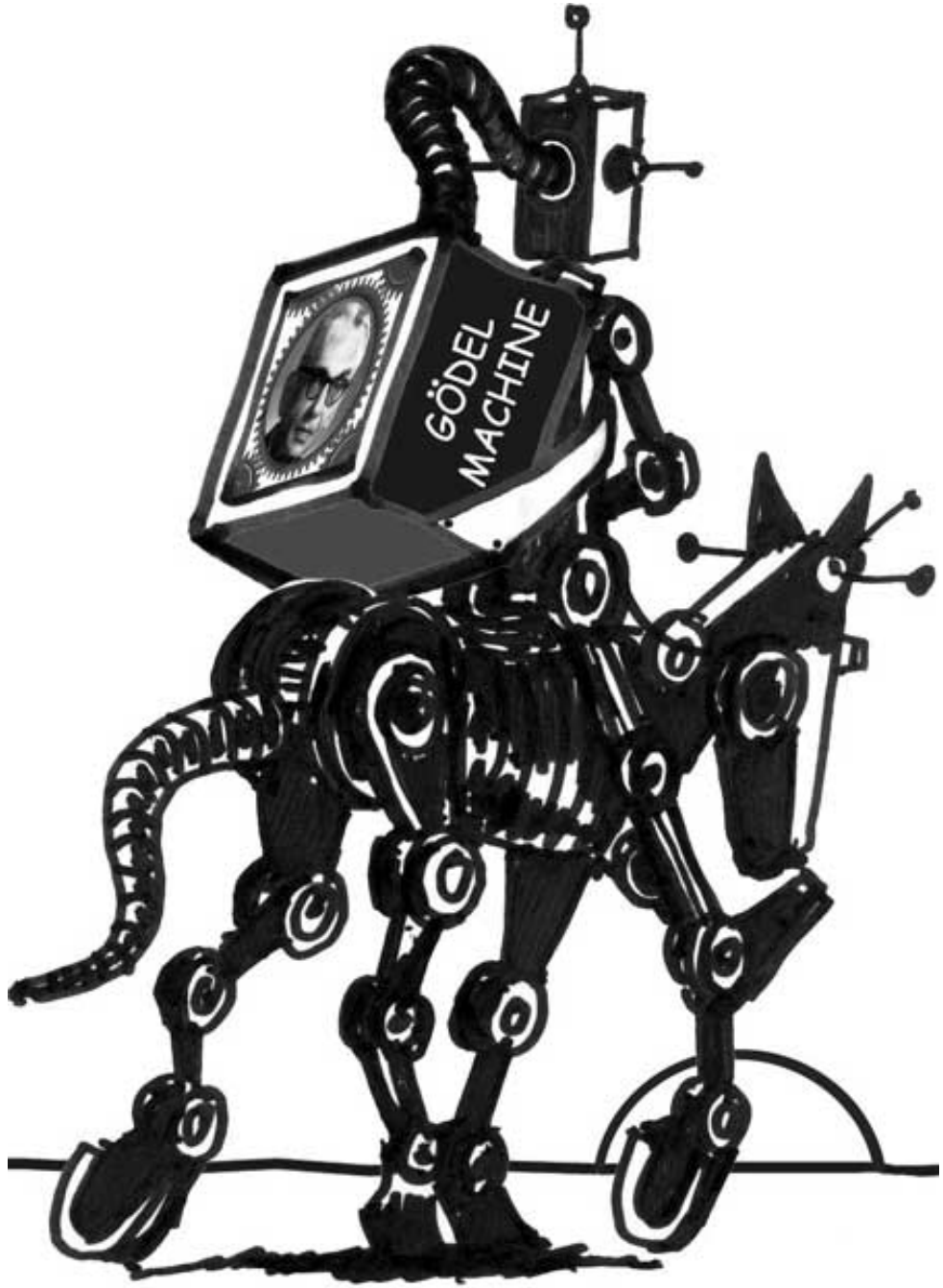



perception
External Reality
action
Internal Reality
self-awareness
perception
Self Model
World Model
action


perception
External Reality
action
Internal Reality
self-awareness
Self Model
World Model


agent is embedded in the environment
environment is not provided as a function (no i/o relationship)
environment is larger than agent
agent can reason about itself and improve
agent is made of parts
need counterfactuals to argmax
environment may contain copies of agent
world not in hypothesis space
introspective uncertainty
need tools/successors
corrigibility
agents in the hypothesis space / search space
computations can have side-effects
logical uncertainty
ontological crises
Vingean reflection
Goodhart
adversarial optimization
coordination
reflective inconsistency
multi-level world models
value specification
inner optimizers
logical counterfactuals
logical updatelessness
informed oversight
reward hacking
… et cetera …




GÖDEL
MACHINE

Universal Artificial Intelligence vs "Selective Amnesia"

- incomplete $\xrightarrow[\text{universal prior}]{\text{Harsanyi transformation}}$ imperfect $\implies$ AIXI
- AIXI $\xrightarrow{\text{"Selective Amnesia"}}$ MDP

$$h \mapsto S$$

- partition of the set of histories = information set = state = feature
-

$$P^\pi_\xi(h' \mid ha) \to P^\pi_\mu(h' \mid ha)$$

but

$$P^\pi_\xi(S' \mid Sa) \nrightarrow P^\pi_\mu(S' \mid Sa)$$

Causal State?

$$h \sim h^* \text{ iff } \forall h' : P(h' \mid ha) = P(h' \mid h^* a)$$

$$\mathcal{S} := \mathcal{H}/\sim$$

"Selective Amnesia"

1. compressible

$$K(S) \leq \sum_{h \in S} K(h)$$

2. minimal

$$\forall \, \text{Partition}(S) : \; K(S) \leq \sum_{S_i \in \text{Partition}(S)} K(S_i)$$

3. maximal

$$\forall S' \supset S : \; K(S) \leq K(S')$$

4. MDL

$$\forall S' \in \mathcal{S} \forall h \in S' : \; K(S) + K(h \mid S) \leq K(S') + K(h \mid S')$$

5. MDL(utility)

$$\forall \mathcal{S}' : \; K\left(S^{\mathcal{S}}_{1:n} \mid a_{1:n}\right) + K\left(u_{1:n} \mid S^{\mathcal{S}}_{1:n}, a_{1:n}\right) + K(\mathcal{S}) \leq K\left(S^{\mathcal{S}'}_{1:n} \mid a_{1:n}\right) + K\left(u_{1:n} \mid S^{\mathcal{S}'}_{1:n}, a_{1:n}\right) + K(\mathcal{S}')$$

$$u(h) := \left[\!\!\left[K(h) < \ell(h) \;\; \& \;\; \forall h' \succ h \left(K(h) \leq K(h') \;\; \& \;\; \forall \, \text{Partition}(h) \left(\sum_{h' \in \text{Partition}(h)} K(h') \geq K(h) \right) \right) \right]\!\!\right]$$

Potapov's MSearch + RSearch

- Let $\{x_i\}_{i=1}^n$ be a set of strings.
- $K(x_1 \dots x_n) \approx \min_S \left(\ell(S) + \sum_{i=1}^n K(x_i \mid S) \right) \ll \sum_{i=1}^n K(x_i)$
- search for models $y_i^* := \operatorname*{argmin}_{y:S(y)=x_i} \ell(y)$ for each x_i w.r.t. some best representation $S^* := \operatorname*{argmin}_S \left[\ell(S) + \sum_{i=1}^n \ell(y_i^*) \right]$

1. Search for models

$$\mathrm{MSearch}(S, x_i) \to y_i^* = \operatorname*{argmin}_{y:S(y)=x_i} \ell(y)$$

2. Search for representations

$$\mathrm{RSearch}(x_1 \dots x_n) \to S^* = \operatorname*{argmin}_S \left[\ell(S) + \sum_{i=1}^n \ell(y_i^*) \right]$$

- MSearch enumerates all models to find the shortest model: $S(y_i) = x_i$.
- RSearch enumerates all S and calls MSearch for each S.

Specializer and SS'-Search

Theorem (s-m-n Theorem / Parameter Theorem)

There is a primitive recursive function $[\![\mathrm{spec}]\!]$ *s.t. for every Gödel number* e *of a partial recursive function*

$$[\![[\![\mathrm{spec}]\!](e, \mathbf{x})]\!](\mathbf{y}) = [\![e]\!](\mathbf{x}, \mathbf{y})$$

$$\forall x : \mathrm{spec}(\mathrm{MSearch}, S)(x) = \mathrm{MSearch}(S, x)$$

$$S' := \mathrm{spec}(\mathrm{MSearch}, S) \implies \begin{cases} \forall x : S(S'(x)) = x \\ \ell(S) + \sum_{i=1}^{n} \ell(S'(x_i)) \to \min \end{cases}$$

- S is a generative representation. (decoding)
- S' is a descriptive representation. (encoding)
- SS'-Search simultaneous search for S and S'.

Potapov's Representational MDL

$$K(x_{1:n}) \approx \min_S \left(\ell(S) + \sum_{i=1}^{n} K(x_i \mid S) \right) \ll \sum_{i=1}^{n} K(x_i)$$

$$q_1^* := \operatorname*{argmin}_q \left[\ell(q) + K(x \mid S_1 q) \right]$$

$$q_{i+1}^* := \operatorname*{argmin}_q \left[\ell(q) + K(q_i^* \mid S_{i+1} q) \right]$$

$$L_{S_1 \ldots S_m}(x) := K(x \mid S_1 q_1^*) + \sum_{i=2}^{m-1} K(q_i^* \mid S_{i+1} q_{i+1}^*) + \ell(q_m^*)$$

$$a_k^* := \operatorname*{argmax}_{a_k} \max_{p:U(p,e_{<k})=a_{<k}a_k} \sum_{q:U(q,a_{<k})=e_{<k}} 2^{-\ell(q)} V_q^p(æ_{<k})$$

$$a_k^* := \operatorname*{argmax}_{a_k} \max_{p:U(p,e_{<k})=a_{<k}a_k} \sum_{\{q_i\}:U(S\{q_i\},a_{<k})=e_{<k}} 2^{-\ell(\{q_i\})} V_{\{q_i\}}^p(æ_{<k})$$

where $e_{<k} = e_{m_1+1:m_2} \dots e_{m_{n-1}+1:m_n}$, $m_1 = 0$, $m_n = k-1$, and $U(Sq_i a_{<k}) = e_{m_i+1:m_{i+1}}$.

$$Q(q_k = s, a_k = a) := \max_{p:U(p,e_{<k})=a_{<k}a} \sum_{\{q_i\}:q_k=s,U(S\{q_i\},a_{<k})=e_{<k}} 2^{-\ell(\{q_i\})} V_{\{q_i\}}^p(æ_{<k})$$

$$Q(q_k = s) := \max_{a_k} Q(q_k = s, a_k = a)$$

Fundamental Challenges

- What is a good optimality criterion?
- What is a "natural" UTM/prior?
- Prior vs universality
- Exploration vs exploitation
- Where should the reward come from?
- How should the future be discounted?
- How should agents reason about themselves (or other agents reasoning about itself)?
- AIXI in the multi-agent setting.
- Better variants/approximations.
- What is a practically feasible and general way of doing induction and planning?
- Training: To maximize informativeness of reward, one should provide a sequence of simple-to-complex tasks to solve, with the simpler ones helping in learning the more complex ones.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

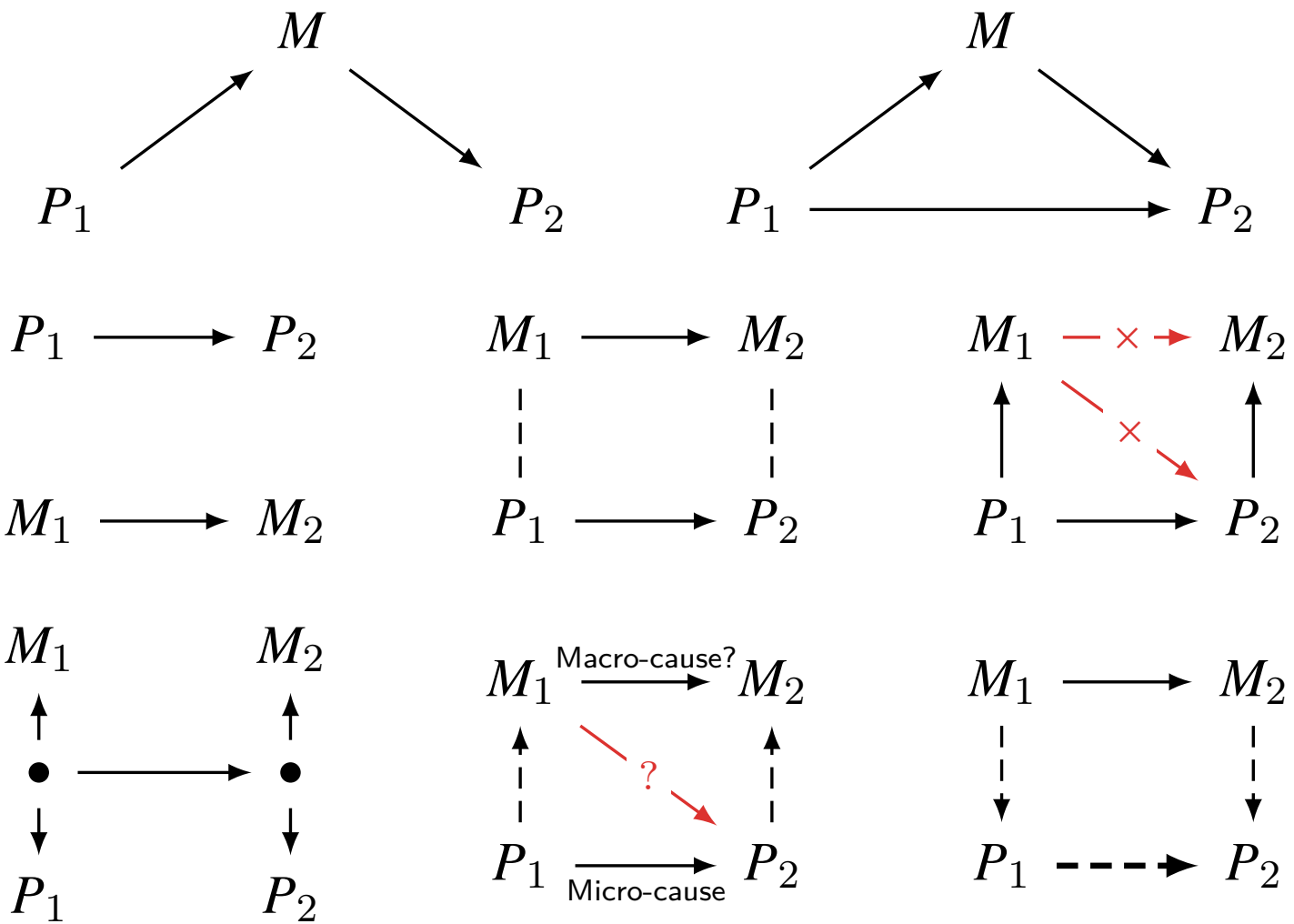
Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
    - Philosophy of Mind
    - Causal Emergence
    - Integrated Information Theory
    - Self-Reference & Consciousness
11. AI Ethics

Turing: Can Machines Think?

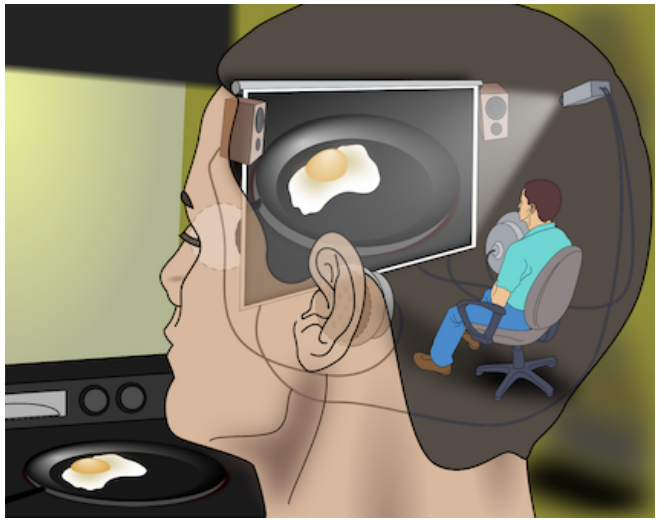
- Theological objections.
- Argument from informality of behavior.
  — Human behavior is far too complex to be captured by any simple set of logical rules./Learning from experience.
- Machines can't be conscious or feel emotions.
  — Why can't machines be conscious or feel emotions?
- Machines don't have Human Quality X.
- Machines just do what we tell them to do.
  — Maybe people just do what their neurons tell them to do.
- Machines are digital. Mental states can emerge from neural substrate only.
  — Only the functionality/behavior matters.
- Non-computable Physics & Brains.
- Argument from incompleteness theorems.
  — No formal system including AIs, but only humans can "see" that Gödel's unprovable sentence is true.

Dualism, (Materialism, Idealism, Neutral) Monism, Interactionism, Preestablished harmony, Pluralism, Epiphenomenalism, Emergentism ...



The Doubt Argument — Dualism

"Cogito, ergo sum." — *Descartes*



1. I cannot doubt that my mind exists.
2. I can doubt that my body exists.
3. Leibniz's Law: x and y are distinct if they have at least one different property.
4. Therefore, my mind is distinct from my body.

Problem: How could they interact?

Is 'being doubtable' a property?

“实在”中的非物理层面

- 人 = 身体 + 心灵
- 僵尸 = 只有身体没有心灵
- 鬼魂 = 只有心灵没有身体
- 存在许多非物理的对象、属性、关系、结构、机制、状态、事件、过程和因果相互作用. 比如,
  - “无知”可以导致“贫穷”.
  - “贫穷”可以导致“犯罪”.
  - “信仰”可以导致“欲望”.
  - “欲望”可以导致“行动”.
- 所有这些非物理量, 最终都实现在物理系统中, 就像虚拟软件一样.
- 心灵 — 运行在大脑中的虚拟程序?
- 你最近安装了啥优质心智程序? 清理了啥垃圾心智程序?

禅宗故事 — 老虎与草莓

- 一个人被老虎追赶...
- 跳下悬崖, 抓住一根藤蔓, 悬挂在半空.
- 崖上的老虎虎视眈眈, 崖下另一只老虎在徘徊等待.
- 这时, 一只老鼠爬出崖洞, 开始啃咬藤蔓...
- 突然, 注意到伸手可及一颗鲜红的草莓.
- 摘下一尝, Delicious!

Pain & Suffering is real!?

qualia?

What is it like to be a bat?

问题

“一千个读者有一千个哈姆雷特.”

感受性知识没有客观基础?

只能直接诉诸 “主观体验”?

“主观体验” 可以结构化/计算化吗?

计算功能主义主张 “心灵 = 程序”, 但被质疑.....

思路: 体验机制的 "客观性" 与体验结果的 "主/客观性"

- 大家之所以觉得感受性知识没有 "真", 主要是因为默认人的 "主观体验" 不够客观.
- 主观体验真的没有客观性吗? 功能主义: 有! 心灵状态 = 功能角色.
- 但功能主义有问题. 功能等价 ≠ 结构同构.
- 引入生成模型/结构因果模型. 区分**体验机制**与**体验结果**.
- "感觉/主观体验" 的**机制**本身都是客观的.

$$\left.\begin{array}{r}\text{(感觉) 假设性行动}\\ \text{(主观体验) 假设性输入}\end{array}\right\} \Longrightarrow \text{基于生成模型对感知状态的反事实解释}$$

- 之所以每个人的体验结果可能不同, 并不是因为 "感觉/主观体验" 不够客观, 而是因为每个人的知识背景不同.
- 由于每个人背景的不同, 导致因果结构不同构, 这才是主观性的根源.
- 一旦实现生成模型的同构, 则体验结果也就有了客观性.

$$\text{客观性} = \text{主体间性} = \text{生成模型同构}$$

下面我们首先以 "颠倒光谱" 的思想实验为引子, 说明功能不等于结构, 客观性源于结构同构. 然后深入结构内部, 给出 "感觉/主观体验" 的基于生成模型的反事实解释.

莱布尼茨的 “磨坊”

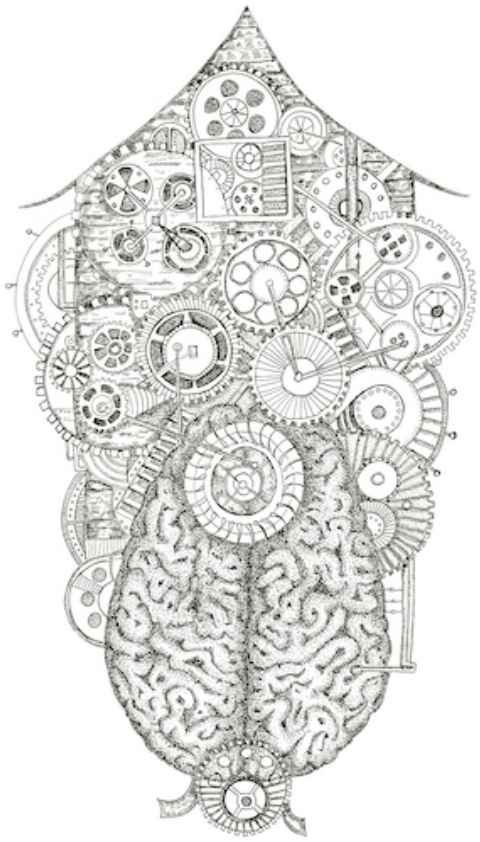


Figure: 莱布尼茨的 “磨坊”: 如果你把一个思维机器放大到磨坊那么大, 并在里面随意参观, 那么你不会发现任何能解释知觉或意识的事物. 这是否意味着找错了地方? 意识在整体系统而不在零件?

Remark: 思维机器是可能的, 但意识无法被机械解释? — 结构功能!

机器状态功能主义: 心灵 $\stackrel{?}{=}$ 程序

普特南: 《心灵状态的本质》1967

心灵状态 = 功能角色

多重可实现性

- 塞尔的 "中文屋" 没有意向性 — 它使心灵状态指涉某物, 不理解符号的意义仍然可以操纵符号执行程序, 说明: 心灵 $\neq$ 程序.

普特南: 《理性、真理与历史》1981

如果机器状态功能主义为真, 那么, 心灵状态 = 功能角色 = 程序
"颠倒光谱" 思想实验表明, 有相同功能构成的人可能有不同的心灵状态

所以机器状态功能主义为假

Remark: 根据整合信息论 IIT, 功能相同的两种结构, 可能一个意识度较高, 一个意识度为 0.

结构同构 $\Longrightarrow$ 功能等价 but 功能等价 $\not\Longrightarrow$ 结构同构

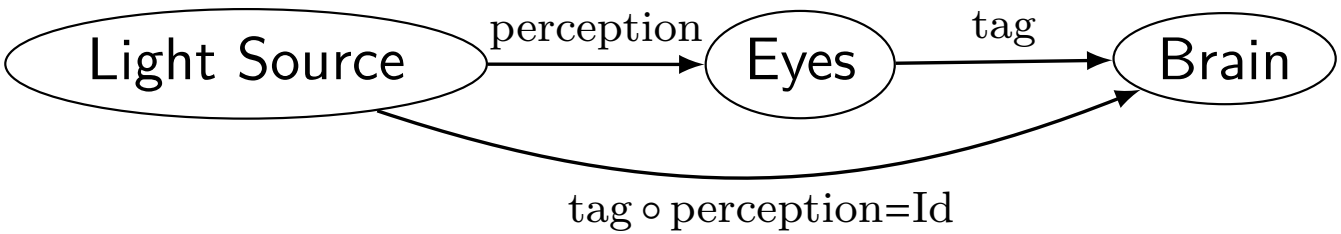
颠倒光谱

- 设想有一个人，有一种奇怪的色盲症.
- 他看到的两种颜色和别人不一样，他把红色看成蓝色，把蓝色看成红色.
- 他不知道和别人的不同.
- 怎么让他知道自己和别人不一样?

颠倒光谱 — 从输入输出到更大的结构

$$\text{perception}_1(\text{blue}) = \text{blue} \qquad \text{perception}_2(\text{blue}) = \text{red}$$
$$\text{perception}_1(\text{red}) = \text{red} \qquad \text{perception}_2(\text{red}) = \text{blue}$$
$$\text{tag}_1(\text{blue}) = \text{blue} \qquad \text{tag}_2(\text{blue}) = \text{red}$$
$$\text{tag}_1(\text{red}) = \text{red} \qquad \text{tag}_2(\text{red}) = \text{blue}$$

$$\text{tag}_1 \circ \text{perception}_1 = \text{tag}_2 \circ \text{perception}_2 = \text{Id}$$

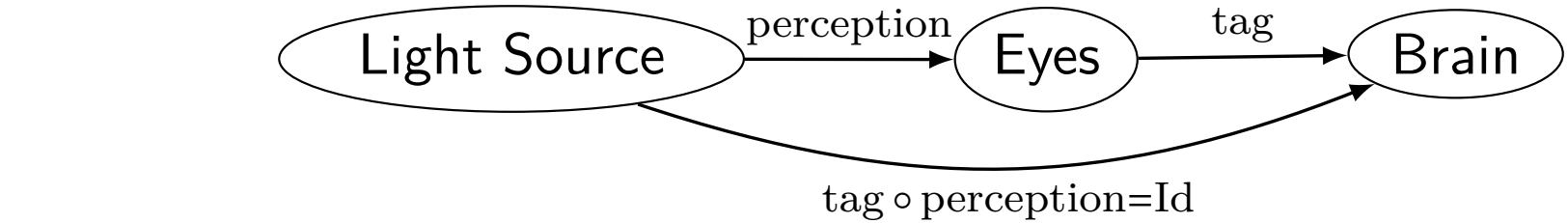


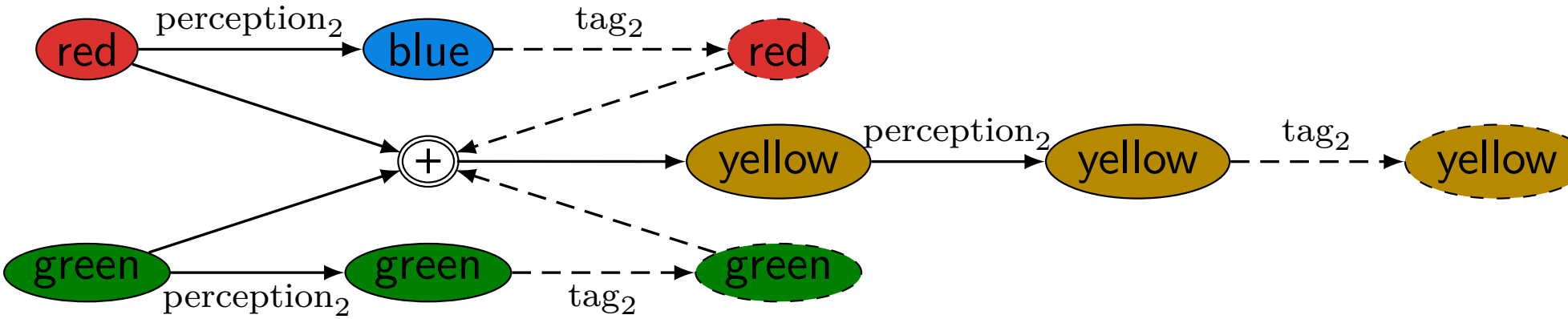
$$\text{red} >_{\text{bright}} \text{blue}$$

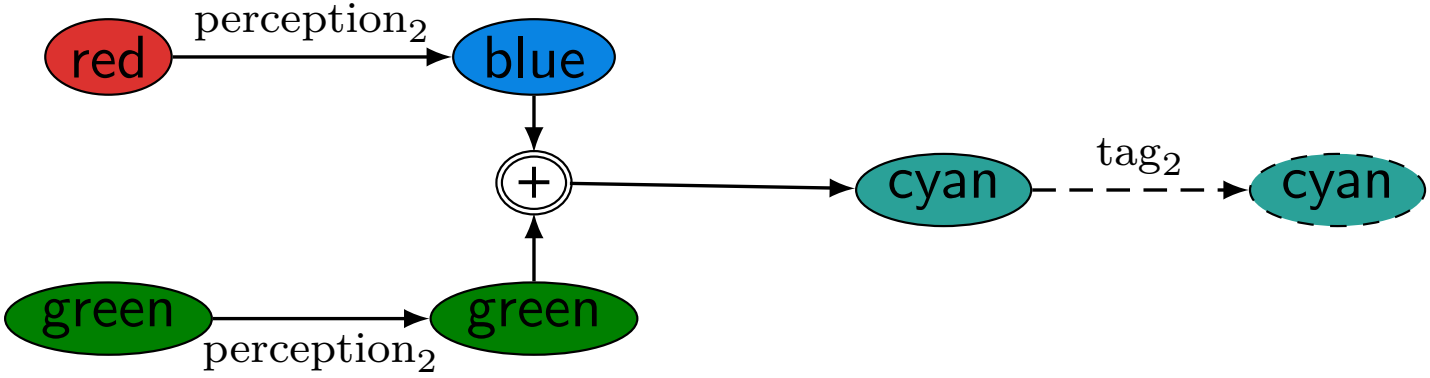
$$\text{perception}_1(\text{red}) >_{\text{bright}} \text{perception}_1(\text{blue})$$
$$\text{perception}_2(\text{red}) <_{\text{bright}} \text{perception}_2(\text{blue})$$

$$\text{perception}_1 \neq \text{perception}_2$$

Can colors be mixed in the eyes?



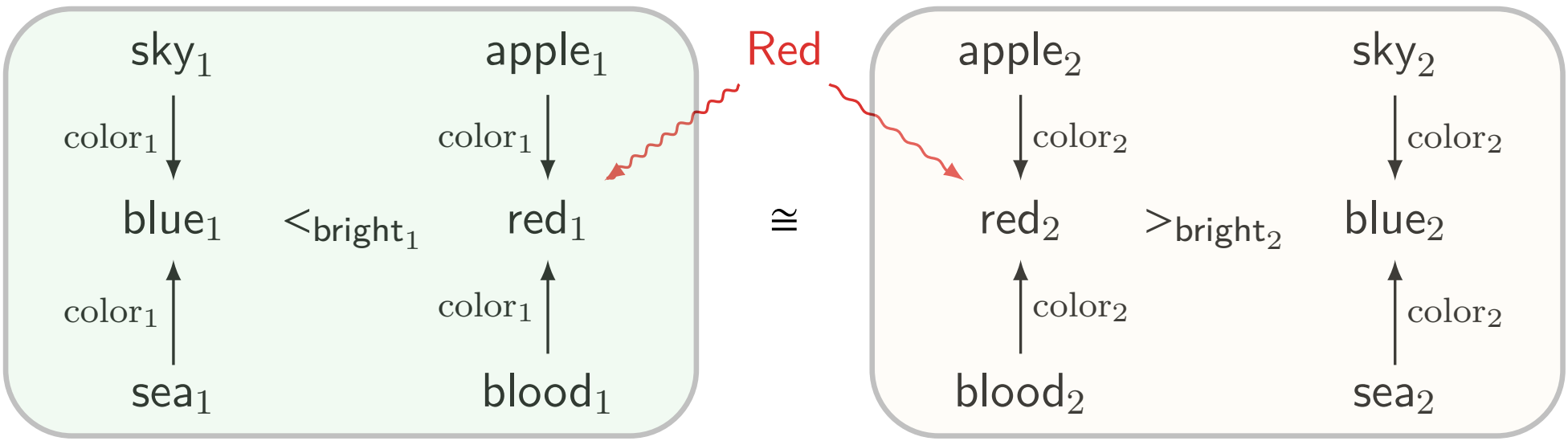




从 “功能等价” 到 “结构同构”

实体与关系哪个更基本? 关系 (结构) 更基本

- 你用你的眼睛看, 我用我的眼睛看, 怎么就得到了 “相同的” 结果? 我们怎么就做了 “相同的” 实验?
- 难道我们的感官彼此做了校准?
- 经验流是主观的, 彼此不同, 但彼此的生成模型同构.
- “客观性” := 主体间性



- 我们并不需要完全同构, 这也不现实. 只要彼此的生成模型同构即可.
- 未必是 “Red” 投影为 “red$_1$/red$_2$”, 反而 “Red” 可能是同构后的构造.
- 这种 “主体间性” 具体怎么实现呢? — 镜像神经元具身模拟.

让我们深入到结构内部, 挖出 “主观体验” 来

- 普特南的功能相同指 “输入-状态-输出” 的转移概率相同. 处于 Pearl 因果阶梯的第一层.
- 转移概率相同, 不意味着生成模型相同.
- Friston 的主动推理框架中的 “生成模型” 可以对自身身体和外部世界进行主动预测/干预, 处于 Pearl 因果阶梯的第二层.
- 但若要对外部世界的因果结构进行充分表征, 需要引入因果阶梯第三层的结构因果模型.
- 所以, 我们假定这里讨论的生成模型是结构因果模型.
- 结构因果模型如何生成主观体验?

AI 有 “主观体验” 吗? — Hinton

- 主观体验的特殊之处在于它的 “假设性”, 而不在于它由某些神秘的 “感受质”(qualia) 在某个 “内在剧场” 里构成.
- 描述大脑内部状态的途径有两种:
  1. 一是通过描述这种状态通常会导致的后续行动, Feeling
  2. 二是通过描述通常会导致这种状态的因素. Subjective Experience

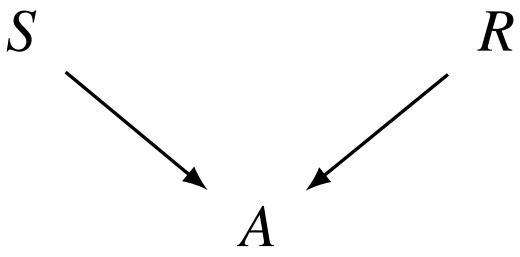
1. “**感觉**” 是通过描述 “假设性行动” 来言说你的大脑状态.
   — “我感觉想冲 Gary 的鼻子来一拳, 如果不是额叶的抑制, 我真的会打.”
2. “**主观体验**” 是通过描述 “假设性输入” 来言说你的大脑状态.

幻觉 “我吃了点儿致幻剂, 产生了一头粉色大象在面前飘浮的主观体验.” — 假设我的感知系统工作正常, 如果外部世界真的有粉色大象在飘浮, 那么我的感知系统告诉我的就是事实.

错觉 想象一个多模态机器人. 我先训练它: 在它面前放一个物体, 命令它 “指向物体”, 它就准确无误地指过去. 接着, 我趁它 “不注意”, 在它的镜头前放了一块棱镜. 然后我再次把物体放在它面前, 说 “指向物体”. 这一次, 它指向了旁边的错误位置. 于是我纠正它:“不, 物体的位置不对. 其实就在你的正前方. 我刚才在你的镜头前放了一块棱镜.” 机器人回答说: “哦, 我明白了. 是棱镜折射了光线, 所以物体实际上在那里, 但我刚才的 ‘主观体验’ 是它在旁边那个位置.”

“感觉”的反事实解释 — “强压怒火”案例

我感觉想冲 Gary 的鼻子来一拳, 如果不是额叶的抑制, 我真的会打



$$A = S \cdot (1 - R)$$

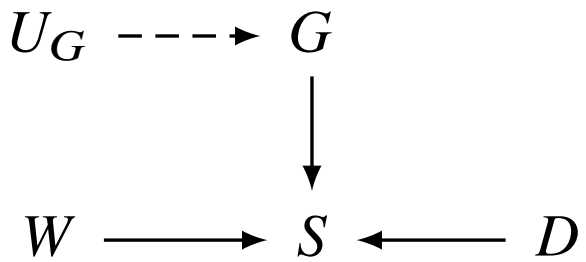
- S: 感知状态
  1. $S = 1$: 愤怒
  2. $S = 0$: 正常
- R: 克制系统
  1. $R = 1$: 克制
  2. $R = 0$: 未克制
- A: 行动
  1. $A = 1$: 打人
  2. $A = 0$: 不打人

- “感觉”: 假设性行动
- 当前所处的感知 (愤怒) 状态 $S = 1$, 还好克制住了 $R = 1$, 没打人 $A = 0$. 若未克制 $R = 0$, 那真的会打人 $A = 1$.

$$P(A_{R=0} = 1 \mid S = 1, R = 1, A = 0)$$

“主观体验”的反事实解释 — “粉色大象”案例

致幻剂导致粉色大象在飘的幻觉



$$S = \begin{cases} W & \text{if } D = 0 \\ G & \text{if } D = 1 \end{cases}$$

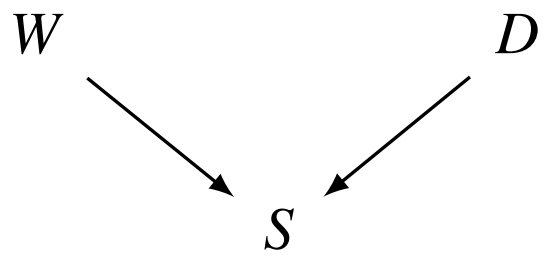
- W: 外部世界信号
  1. $W = 1$: 有粉色大象
  2. $W = 0$: 没有粉色大象
- G: 内部生成信号
  1. $G = 1$: 生成粉色大象幻象
  2. $G = 0$: 没生成粉色大象幻象
- D: 干扰因素
  1. $D = 1$: 吃了致幻剂
  2. $D = 0$: 没吃致幻剂
- S: 感知状态
  1. $S = 1$: 看到粉色大象在飘
  2. $S = 0$: 没看到粉色大象

- “主观体验”: 假设性输入 (干预反事实解释)
- 事实上吃了致幻剂 $D = 1$. 如果没吃致幻剂 $D = 0$, 若真有粉色大象在飘 $W = 1$, 那我的感知就是对的 $S = 1$.

$$\text{argmax}_w \, P(S_{W=w, D=0} = 1 \mid D = 1, G = 1, S = 1)$$

“主观体验”的反事实解释 — “棱镜”案例

棱镜折射导致的视觉错觉 — “我看到了啥?” — 事前“假设性解释”



$$S = \begin{cases} W & \text{if } D = 0 \\ 1 - W & \text{if } D = 1 \end{cases}$$

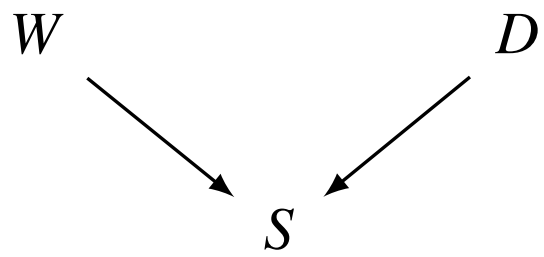
- W: 外部世界信号
  1. $W = 1$: 物体在旁边位置
  2. $W = 0$: 物体在正前方
- D: 干扰因素
  1. $D = 1$: 放了棱镜
  2. $D = 0$: 没放棱镜
- S: 感知状态
  1. $S = 1$: 看到物体在旁边位置
  2. $S = 0$: 看到物体在正前方

- “主观体验”: 假设性输入 (干预反事实解释)
- 事实上放了棱镜 $D = 1$. 如果没放棱镜 $D = 0$, 若物体真在旁边位置 $W = 1$, 那我的感知就是对的 $S = 1$.

$$\underset{w}{\operatorname{argmax}}\, P(S_{W=w,D=0} = 1 \mid D = 1, S = 1)$$

“事后反思” 的反事实解释 — “棱镜” 案例

对棱镜视觉错觉的反思 — “我为啥看错了?” — 事后 “归因”



$$S = \begin{cases} W & \text{if } D = 0 \\ 1 - W & \text{if } D = 1 \end{cases}$$

- W: 外部世界信号
  1. $W = 1$: 物体在旁边位置
  2. $W = 0$: 物体在正前方
- D: 干扰因素
  1. $D = 1$: 放了棱镜
  2. $D = 0$: 没放棱镜
- S: 感知状态
  1. $S = 1$: 看到物体在旁边位置
  2. $S = 0$: 看到物体在正前方

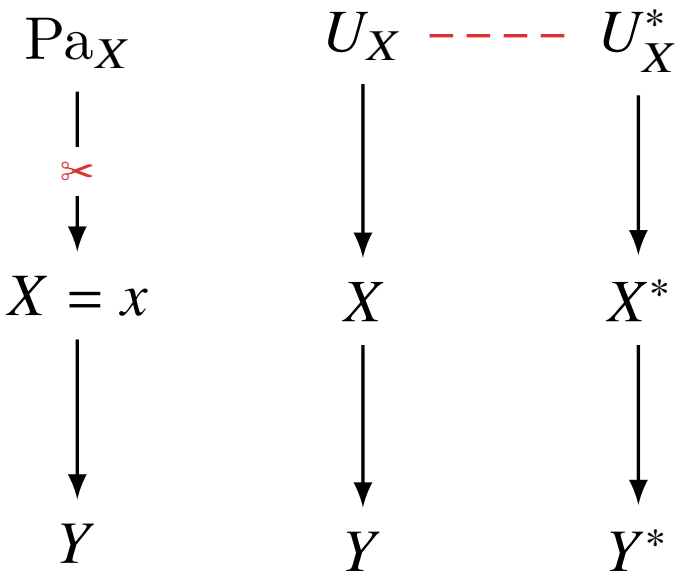
- 事后反思 — 被告知 “主观体验为什么出错了” 后的反事实推理.
- 物体事实上在正前方 $W = 0$, 但由于有棱镜折射光线 $D = 1$, 位置看歪了, “主观体验” 到物体在旁边位置 $S = 1$. 若无棱镜干扰 $D = 0$, 则会正确感知到物体就在正前方 $S = 0$.

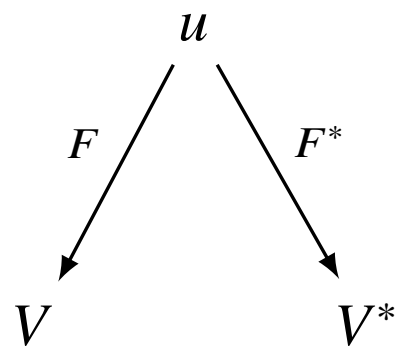
$$P(S_{D=0} = 0 \mid W = 0, D = 1, S = 1)$$

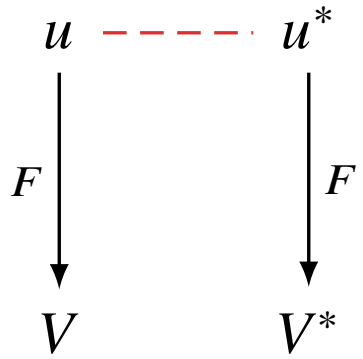
Remark: “错觉体验” 和 “为啥体验错了” 都可以被反事实解释, 但不能用 “理智正确” 替代 “错误体验”, 彰显体验的 “真实性”.

干预反事实 vs 回溯反事实

1. 或者自然律被违反
2. 或者背景条件被改变

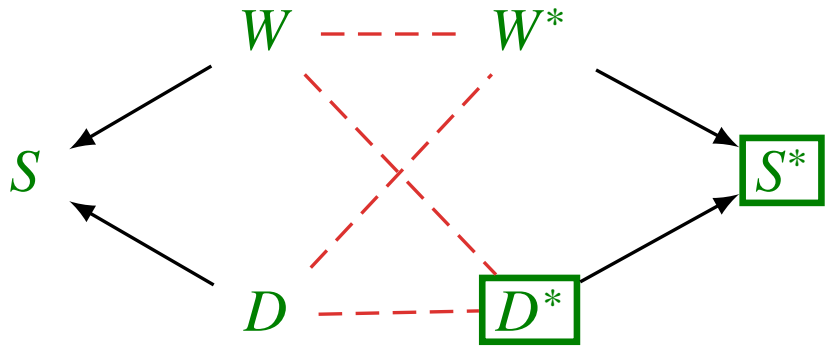






1. **干预反事实:** 现实世界和反事实世界共享相同的背景条件 u.
   现实结果 V 和反事实结果 V^* 之间的潜在矛盾, 通过机制 F 的变化 (通过干预得到修改后的机制 F^* 和子模型 M^*) 得以解决.
2. **回溯反事实:** 现实世界和反事实世界共享相同的不变机制 F, 而各自的背景条件 u 和 u^* 可能不同.

“主观体验” 的回溯反事实解释



$$S = \begin{cases} W & \text{if } D = 0 \\ 1 - W & \text{if } D = 1 \end{cases}$$

1. 主观体验: 假设性输入 (干预反事实 — 最大似然)

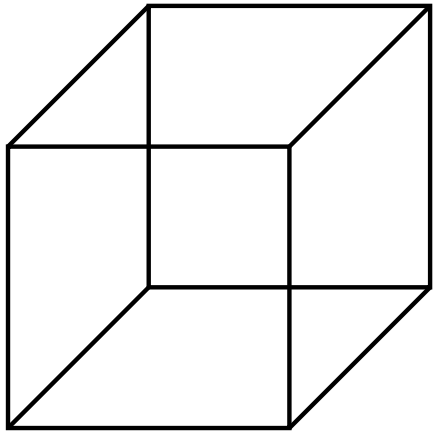
$$\underset{w}{\operatorname{argmax}}\, P(S_{W=w,D=0} = 1 \mid D = 1, S = 1)$$

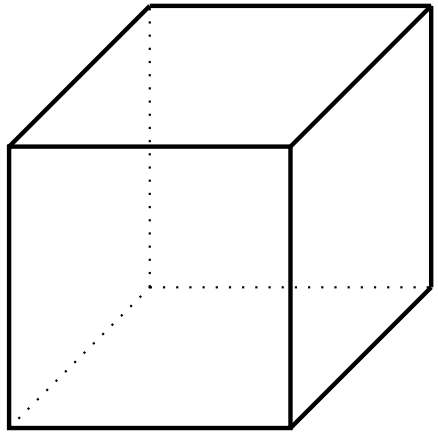
2. 主观体验: 假设性输入 (回溯反事实 — 最大后验)

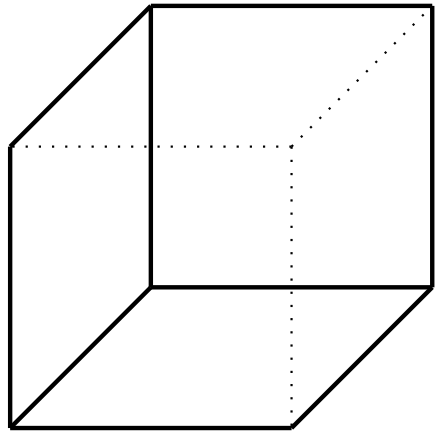
$$\underset{w}{\operatorname{argmax}}\, P_B(W^* = w \mid D = 1, S = 1, D^* = 0, S^* = 1)$$

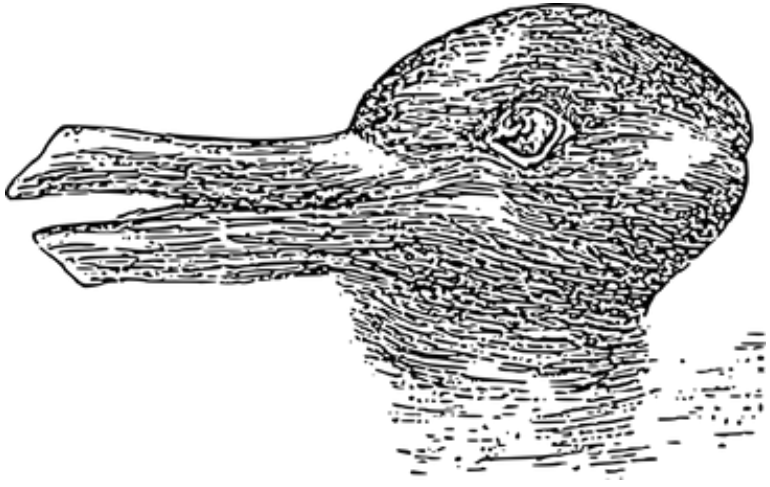
- 基于感知状态, 坚守自然律不破, 被迫回溯到上游外生变量, 不惜篡改出 “主观错觉”.
- 相比于 “干预反事实”, “回溯反事实” 更能解释为什么人会同时有 “理智知道真相” 和 “主观体验有错觉” 的撕裂感.
- 感觉是对后续行动的假设性预测, 干预反事实更合适; 主观体验是对其自身内部状态的假设性解释, 回溯反事实更合适.

相同形色“感受质”下的体验转换？解不唯一，多稳态



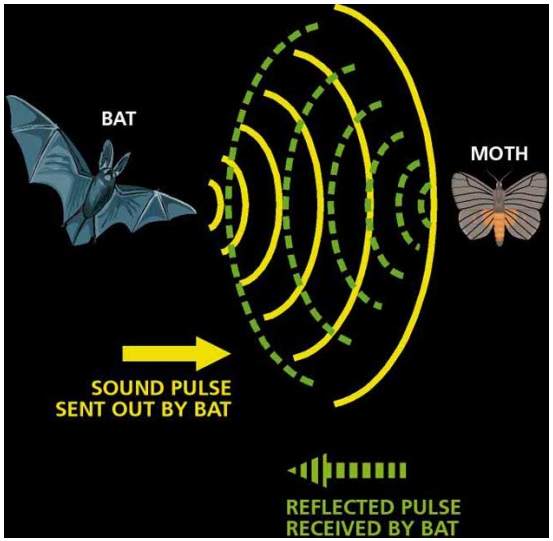






$\operatorname{argmax}_w P(S_{W=w} = 1 \mid S = 1)$ 的解不唯一. 令 $W = 1$ 或 $W = 0$ 都能完美解释 $S = 1$.

蝙蝠的感受? — 反事实 "主观体验" 的可计算性



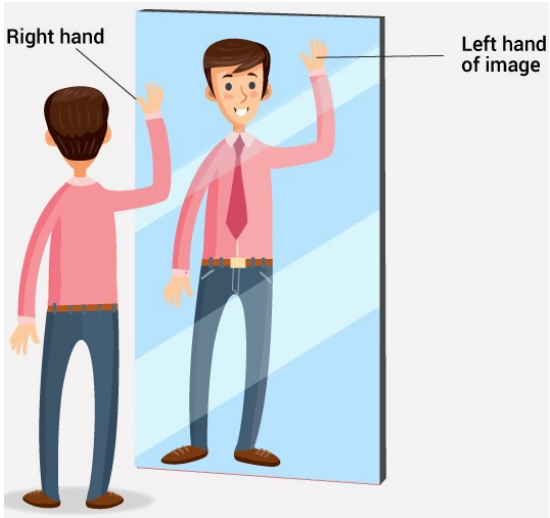


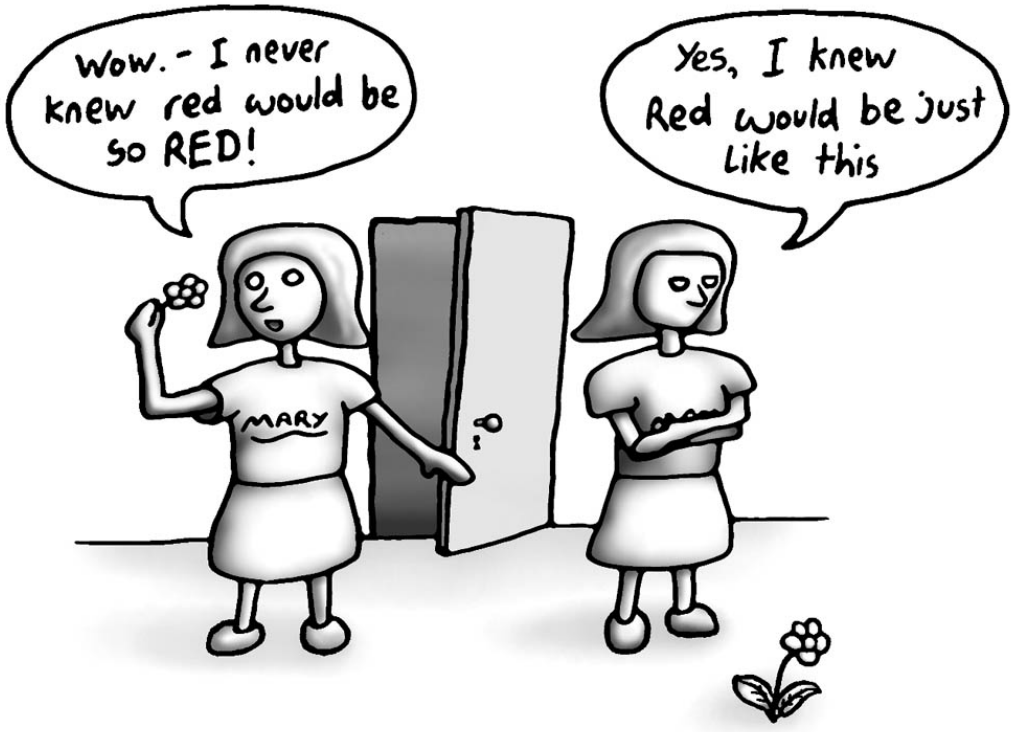
Figure: Thomas Nagel: while a human might be able to imagine what it is like to be a bat by taking "the bat's point of view", it would still be impossible "to know what it is like for a bat to be a bat." 子非鱼安知鱼之乐?

- 自我投射的危险: "为什么镜子颠倒了左右却没有颠倒上下?"
- 镜像对称只颠倒了前后, 把自己投射到 "镜中人" 才误以为颠倒了左右

$$P(S_{\text{Reverse}=1,\text{Mirror}=0} = 1 \mid \text{Mirror} = 1, S = 1)$$

- 内格尔试图 "客观的" 知道 "主观的" 感受, 却不具有同构的生成模型, 因此不具有计算反事实的条件

走出 “黑白屋” — 机制性知识 vs 事实性知识

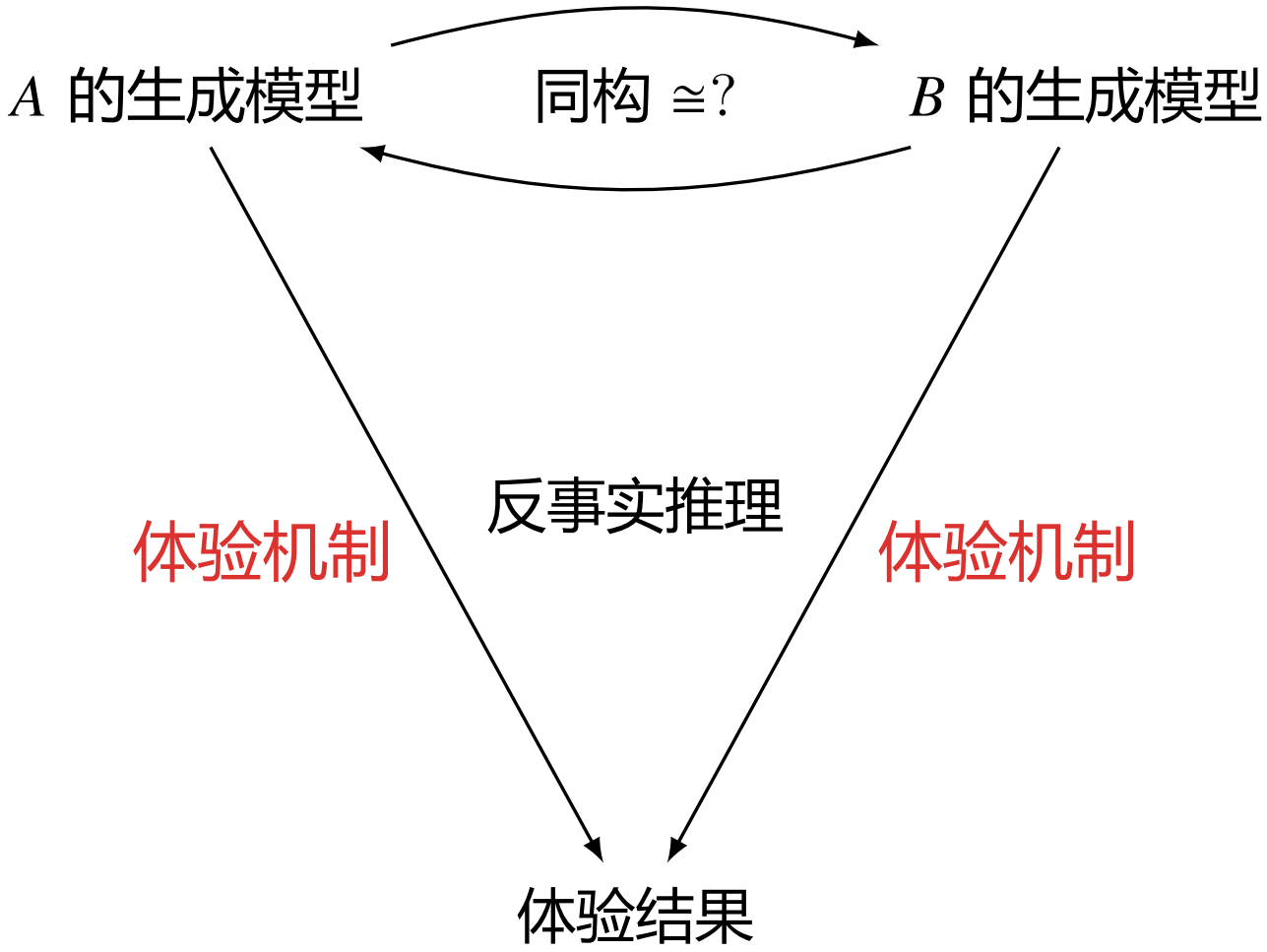


jolyon.co.uk

- 玛丽走出 “黑白屋” 看到红色时，会学到物理知识之外的感受性知识吗？
- 给她看一个涂成蓝色的红苹果，如果她知道这不是苹果本来的颜色，是否说明她早已知道看到蓝色会有什么感受？

- 物理知识无法解释 “主观体验”？心灵状态无法还原为物理状态？
- 梁启超：无论为哀为乐，为怨为怒，为恋为骇，为忧为惭，常若知其然而不知其所以然.
  — 情感等 “主观体验” 的 “事实 vs 机制” 层面.
- 跟日常情感体验不同，“黑白屋” 里获得的是 “机制性知识”(结构因果方程) 而非 “事实性知识”(外生变量的取值).

主体间性 & 体验机制/体验结果



- 背景不同，但体验机制完全相同.
- 若生成模型同构，则体验结果也相同.

展望

- 价值体验 (道德体验、审美体验等) — 引入效用函数变量?
- 梦中的体验? — 屏蔽外部世界信号, 放大内部生成信号?
- 精神分裂的体验? — 感知状态的机制中外部世界信号与内部生成信号冲突?
- 神经科学实验验证? — 血清素、多巴胺、去甲肾上腺素等在体验生成中的作用?
- 更多人文领域的算法化? — 责任、公平、正义、意识、自由意志...?
- 人文算法化的伦理问题?
- 如何构建具备反事实 "主观体验" 能力的具身 Agent?

计算功能主义

1. 物理版本的丘奇-图灵论题适用于人脑：人脑的动力学是图灵机可计算的.
2. 所需的模拟在物理硬件上 tractable.
3. 该模拟能保持心灵状态所依赖的功能组织.

类比论证

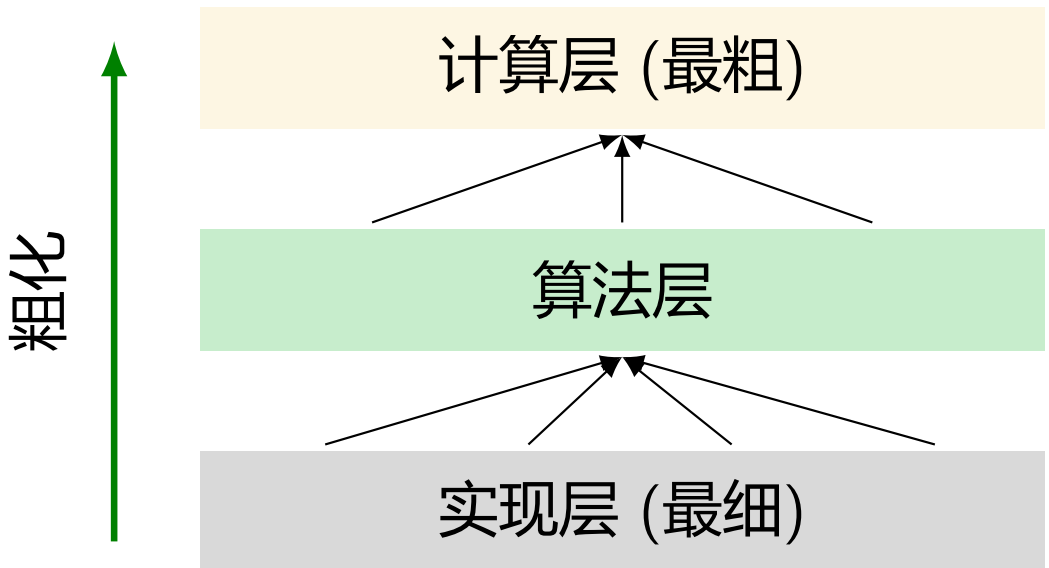
- 模拟飓风不会让计算机硬件变湿，所以，模拟大脑的算法不会产生意识.

类比论证

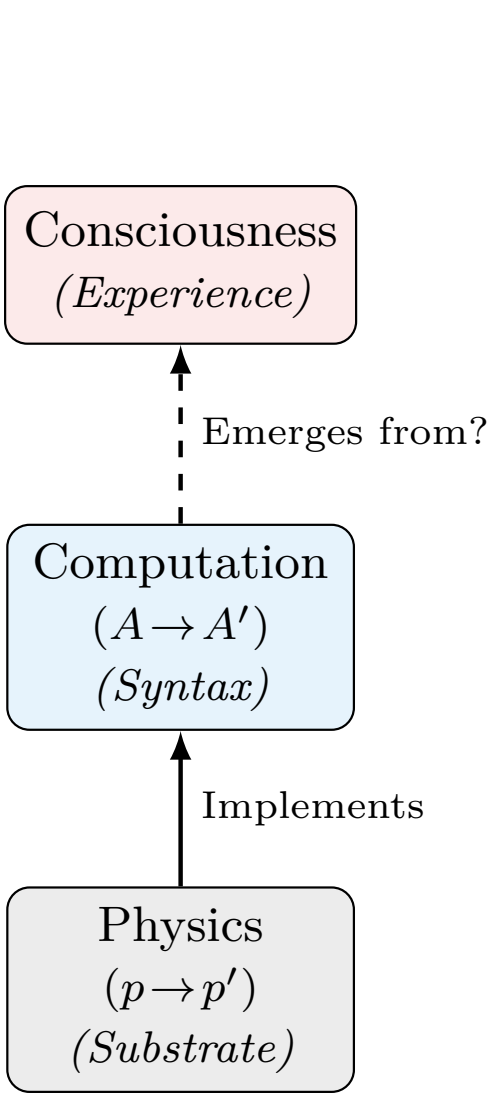
- 模拟飓风不会让计算机硬件变湿，所以，模拟大脑的算法不会产生意识.
- 数字化的货币也可以买东西.

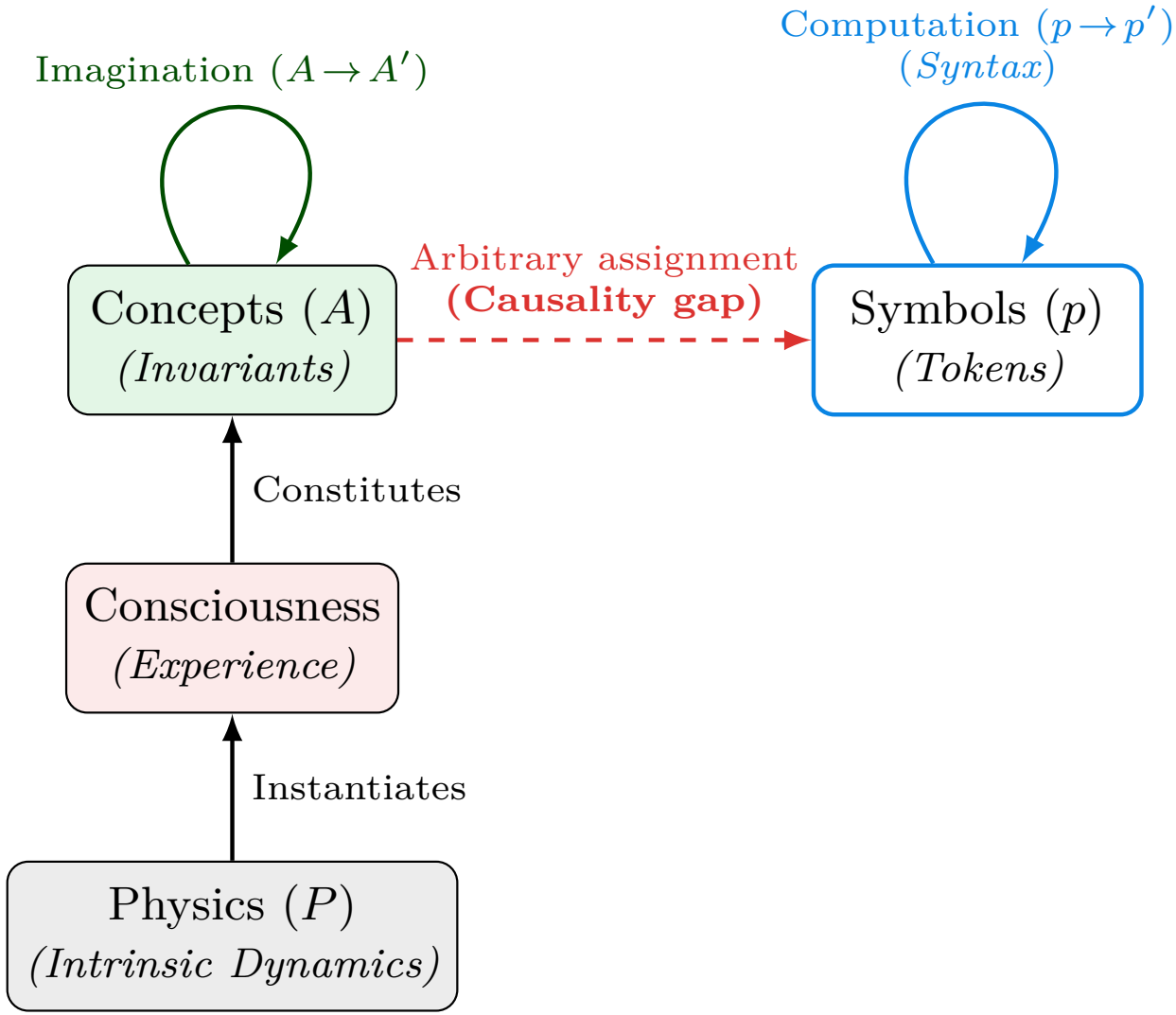
现象意识随附在哪一层?

- 如果随附在算法层, 或其上的行为层, 则可被模拟保持.
- 如果随附在物理实现层, 则丢失.



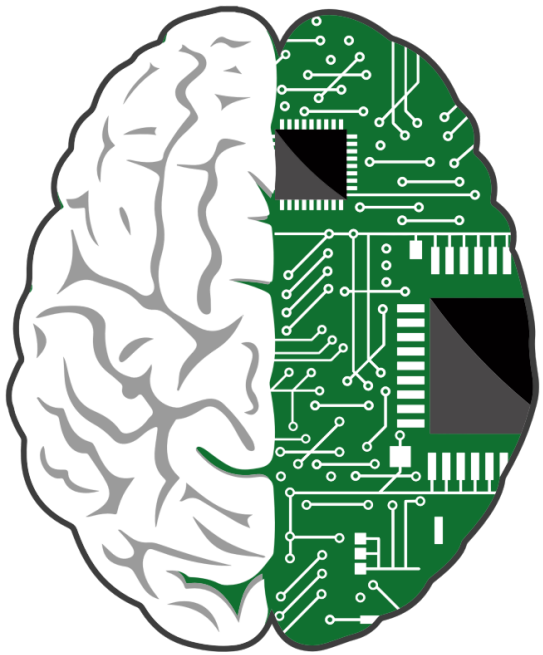
基质独立 vs 基质依赖


Consciousness
(Experience)
Emerges from?
Computation
$(A \to A')$
(Syntax)
Implements
Physics
$(p \to p')$
(Substrate)


Imagination $(A \to A')$
Computation $(p \to p')$
(Syntax)
Concepts (A)
(Invariants)
Arbitrary assignment
(Causality gap)
Symbols (p)
(Tokens)
Constitutes
Consciousness
(Experience)
Instantiates
Physics (P)
(Intrinsic Dynamics)

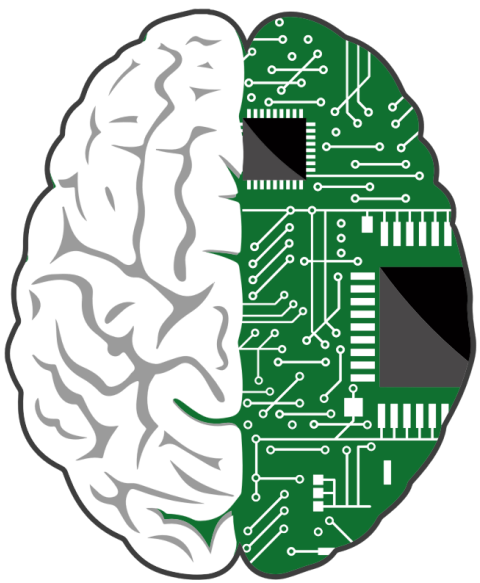
功能主义 & 多重可实现性 & 大脑替换实验

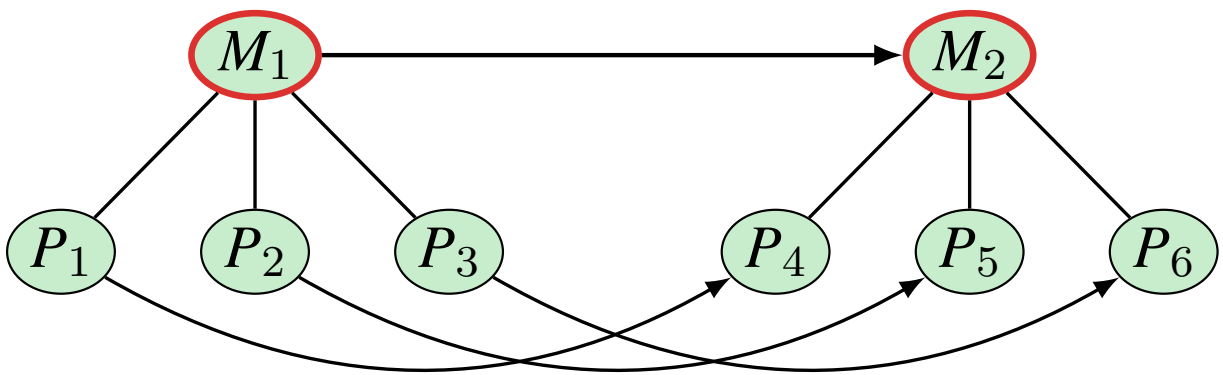
- 功能主义：任何两个系统，只要具有同构的因果过程，就会有相同的心灵状态.
- 大脑替换实验：把你的大脑神经元一个一个换成功能等效的电子元件，你还是你吗？

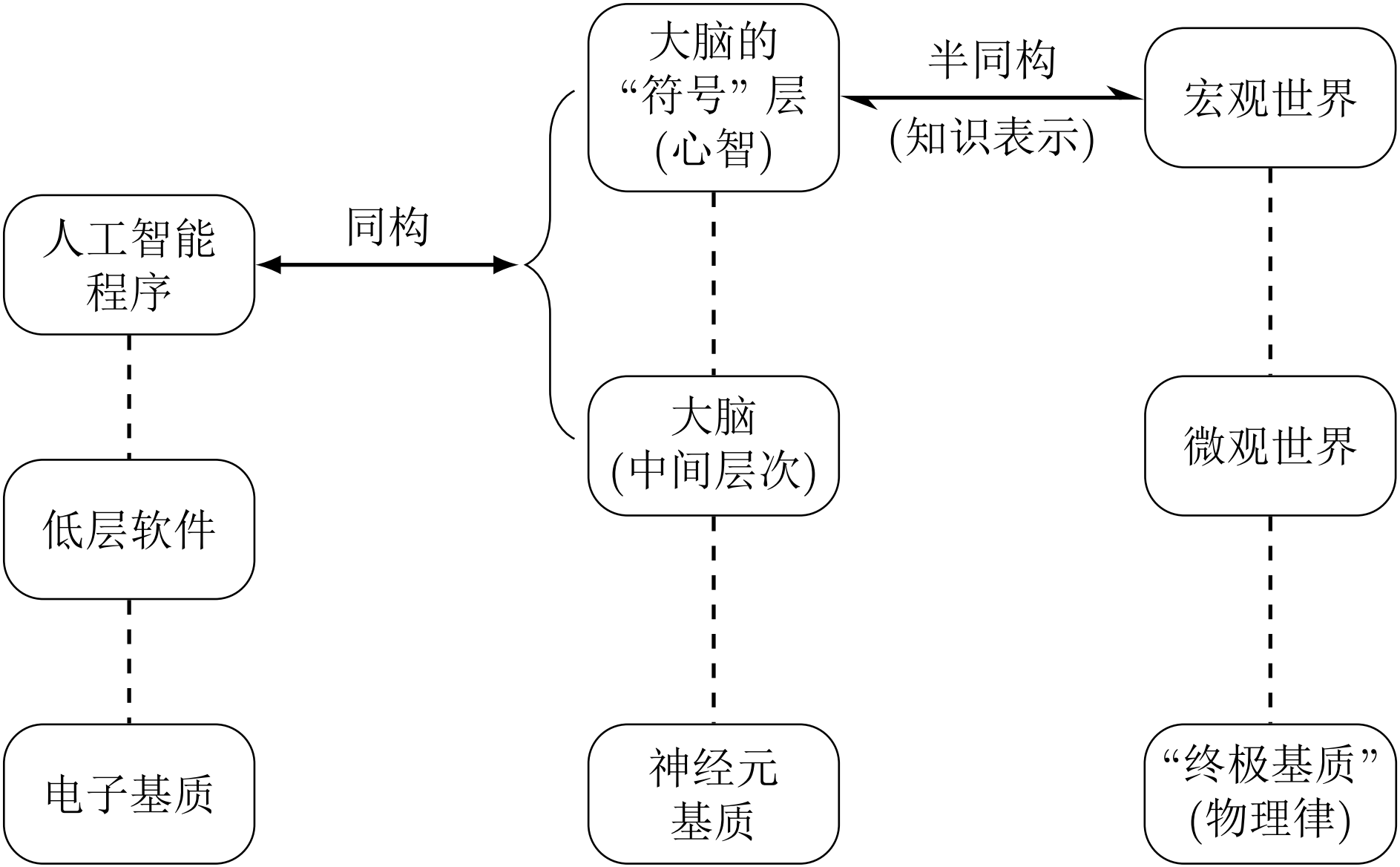


物理符号系统假说 — Newell & Simon

- 物理符号系统是智能行为的充分且必要条件.
- 物理符号系统可以建立、复制、修改、删除符号, 以生成其它符号结构.
- 人和计算机都可以通过创建符号结构、输入、输出、存储、复制、条件转移等操作展示智能.
- 多重可实现性 Multiple Realizability
- 意识能存活于不同的基质上吗? 传送、复制、虚拟化、扫描等等.






大脑的
“符号”层
(心智)
半同构
(知识表示)
宏观世界
人工智能
程序
同构
大脑
(中间层次)
微观世界
低层软件
电子基质
神经元
基质
“终极基质”
(物理律)

多重可实现性 — 数字计算 vs 生物计算

- 软件、硬件分离
  - 软件易复制, 可 “永生”, 知识共享
  - 高能耗
- 软硬件一体 (Mortal Computation) 可朽计算
  - 低能耗
  - 知识传授难

Remark: 如果能量够便宜, 数字计算更有优势.

生命的价值

- 我们尊重生命，是因为生产养育生命是昂贵的.
- 不尊重生命的个体会在进化过程中被自然选择淘汰掉.
- 没有自然选择淘汰的过程，生物还会尊重生命吗?
- 机械化大生产已降低了体力劳动的价值.
- AGI 也会降低更具创造性的脑力劳动的价值.
- 复制、修改虚拟生命是廉价的.
- 当生命变得廉价，还值得被尊重吗?
- 软硬件分离的智能体还会尊重生命吗?
- 如果我们的生命都不被尊重，我们为什么还要开发软硬件分离的智能体?
- 为了“永生”吗?

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
    - Philosophy of Mind
    - Causal Emergence
    - Integrated Information Theory
    - Self-Reference & Consciousness
11. AI Ethics

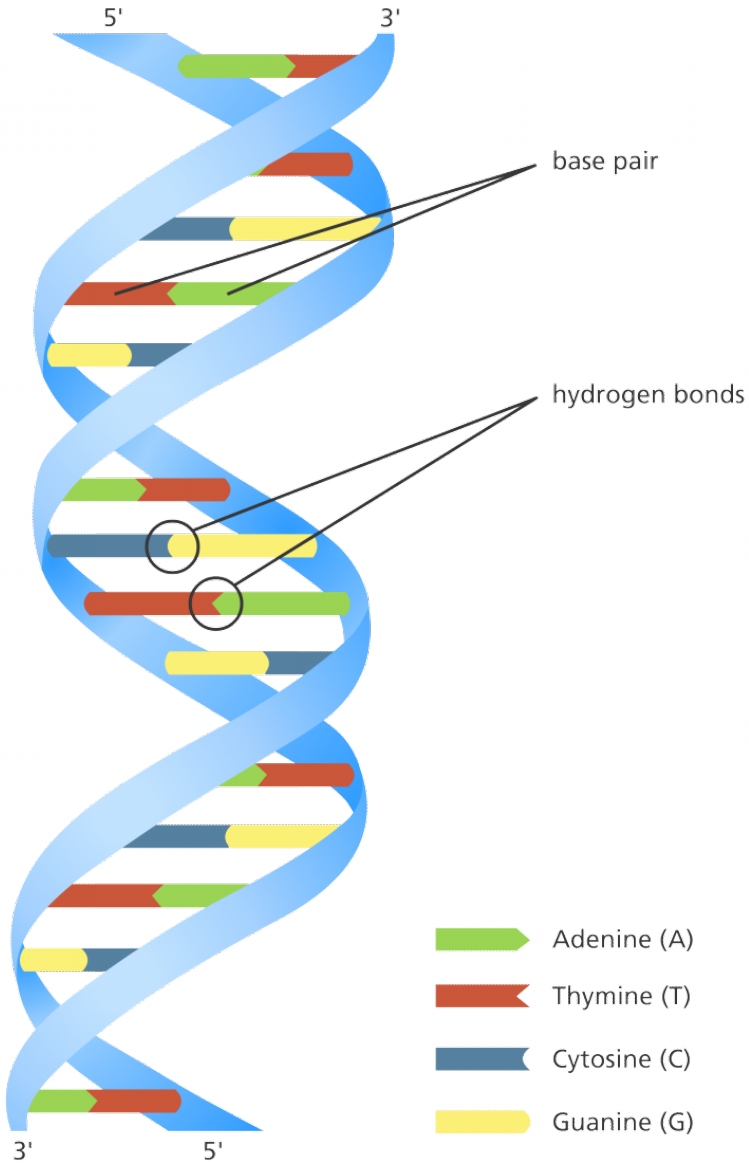
还原主义

“你的喜怒哀乐、你的记忆和抱负、你的自我意识和自由意志，实际上不过是一大堆神经细胞和相关分子的行为.”

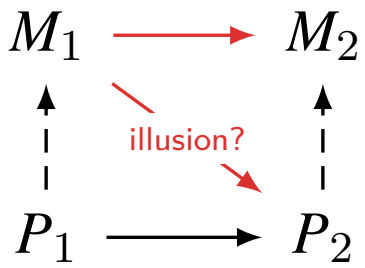
— 克里克《惊人的假说》

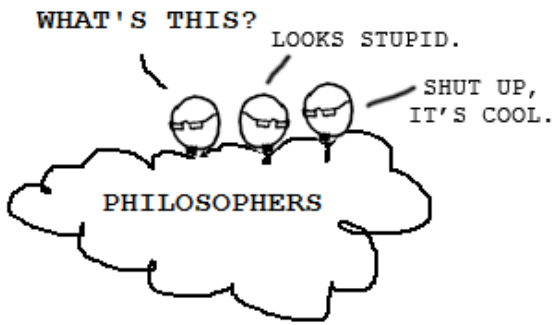
唯物主义. 有解释鸿沟？物理似乎是因果闭合的，没有给意识留出任何位置.

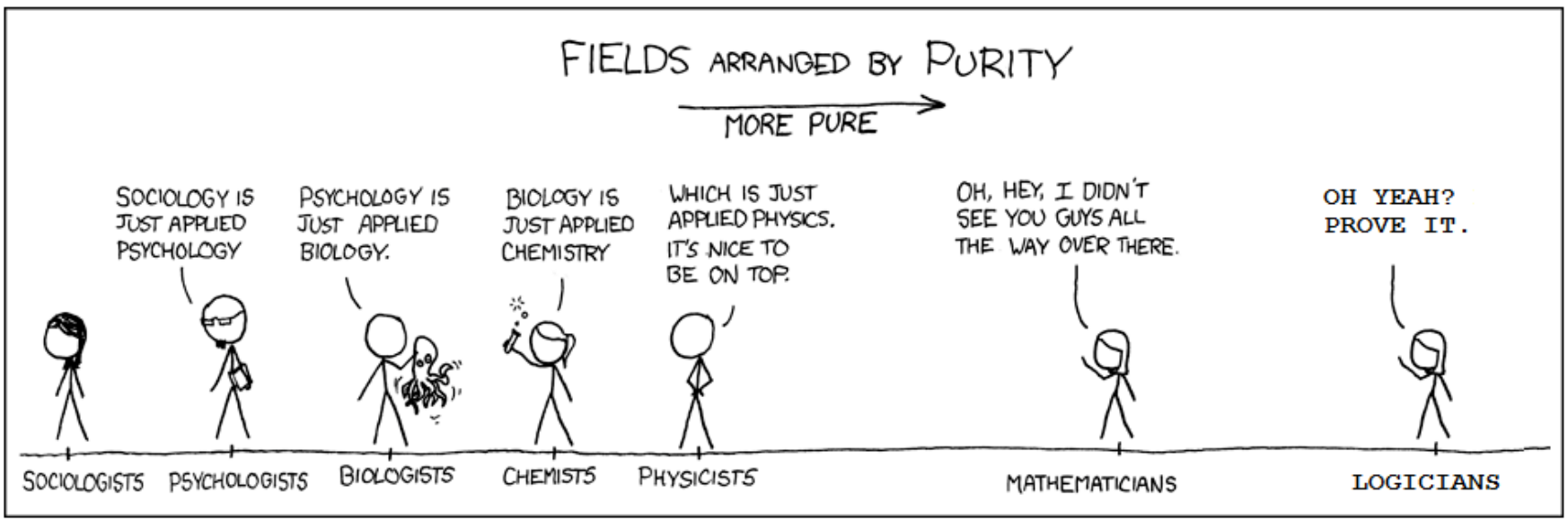
$$m_1 v_1 + m_2 v_2 = m_1 v_1' + m_2 v_2'$$



涌现 ≠ 还原


M_1
M_2
illusion?
P_1
P_2


WHAT'S THIS?
LOOKS STUPID.
SHUT UP, IT'S COOL.
PHILOSOPHERS


FIELDS ARRANGED BY PURITY
MORE PURE
SOCIOLOGY IS JUST APPLIED PSYCHOLOGY
PSYCHOLOGY IS JUST APPLIED BIOLOGY.
BIOLOGY IS JUST APPLIED CHEMISTRY
WHICH IS JUST APPLIED PHYSICS. IT'S NICE TO BE ON TOP.
OH, HEY, I DIDN'T SEE YOU GUYS ALL THE WAY OVER THERE.
OH YEAH? PROVE IT.
SOCIOLOGISTS
PSYCHOLOGISTS
BIOLOGISTS
CHEMISTS
PHYSICISTS
MATHEMATICIANS
LOGICIANS

Emergentism — More is Different!

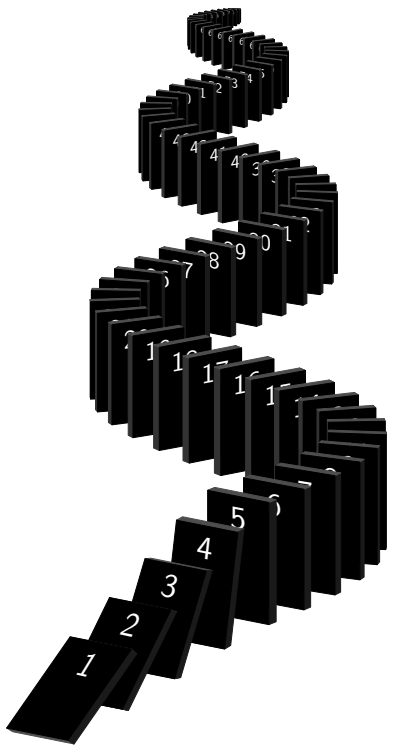
“整体大于部分之和.”

— 亚里士多德

- 整体不仅大于部分之和, 而且可能非常不同.
- 解释事物时, 未必是越还原到更低层才更基本.
- 每个复杂性层级上, 都会涌现出全新的属性.
- 抽象的目的不是模糊, 而是创建一个新的语义层, 在这个层面上可以精确.
- 侯士达: 有没有意识取决于在哪个层级上对结构进行观察. 在整合度最高的层级上看, 大脑是有意识的. 下降到微观粒子层面, 意识就不见了. 意识体是那些在某个描述层级上表现出某种特定类型的循环回路的结构. 当一个系统能把外部世界过滤成不同的范畴、并不断向越来越抽象的层级创造新的范畴时, 这种循环回路就会逐渐形成. 当系统能进行自我表征 — 对自己讲故事 — 的时候, 这种循环回路就逐渐变成了实体的 “我” — 一个统一的因果主体.

涌现 vs 意向立场

- 设想一个由多米诺骨牌组成的计算装置，当输入是一个素数时，某个特定的多米诺骨牌会倒下.
- 你输入了 641，某个特定的多米诺骨牌倒下了.
- 问：为什么它会倒下?
  1. 因为它前面的那个多米诺骨牌倒下了.
  2. 因为 641 是素数.



丹尼特的 “意向立场”

> 首先，你将要预测行为的对象视为一个理性的主体．然后，根据其在世界中的位置和目的勾勒出它应当具有的信念．接着，再以同样的方式确定它应当具有的欲望．最后，根据其信念和欲望预测它将采取的行动．
>
> — 丹尼特

- 在理解、解释、预测一个对象的行为时，我们会选择以不同的抽象层级来看待它．
- 丹尼特的三种解释策略：物理立场、设计立场、意向立场
- 切换到更高层级的抽象视角，有风险，也有好处．

Remark: 不确定大脑神经元是否能表征概念、判断，但常识中用 “概念”“判断”“推理”“信念”“动机”“偏好”“欲望”“意图” 等术语解释、预测人的行为似乎很好用．(过度归因/假阳性比归因不足/假阴性更划算)

我们分手吧．

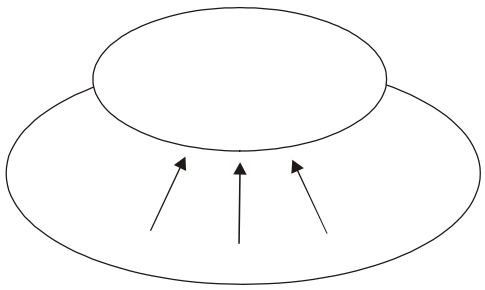
她是谁？是不是比我年轻？

丹尼特的 “意向立场” — 自上而下的解释/实现

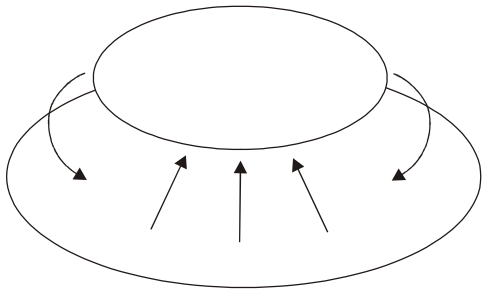
Dennett “Stances”	**Pylyshyn** “Levels of Organization”	**Newell** “Levels of Description”	**Marr** “Levels of Analysis”
Intentional Stance	Semantic, or Knowledge Level	Knowledge Level	Computational Theory Level
Design Stance	Symbol Level	Program Level	Representation and Algorithm Level
Physical Stance	Physical Level, or Biological Level	Physical Level, or Device Level	Hardware Implementation Level

Emergence

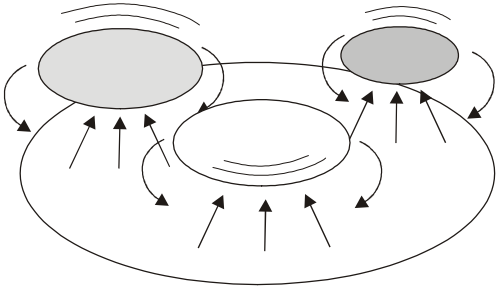
1. 简单涌现: 无下向反馈
   — 自行车的功能
   — 热力学属性: 温度, 压力
2. 弱涌现: 有下向反馈
   — 蚁群觅食
   — 商品在自由市场中的价格
3. 多重涌现: 多重反馈
   — 短程正反馈, 长程负反馈
   — "激活-抑制" 系统
   — 斑马的斑纹
4. 强涌现: 生命, 意识, 文化等



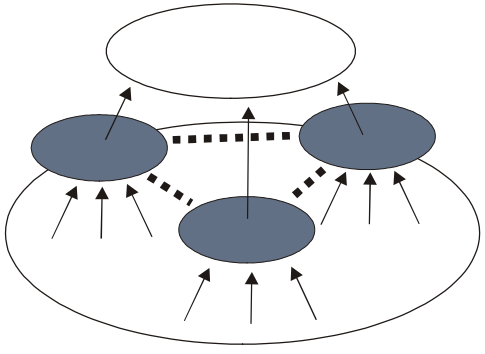
TYPE I
Simple Emergence



TYPE II
Weak Emergence



TYPE III
Multiple Emergence

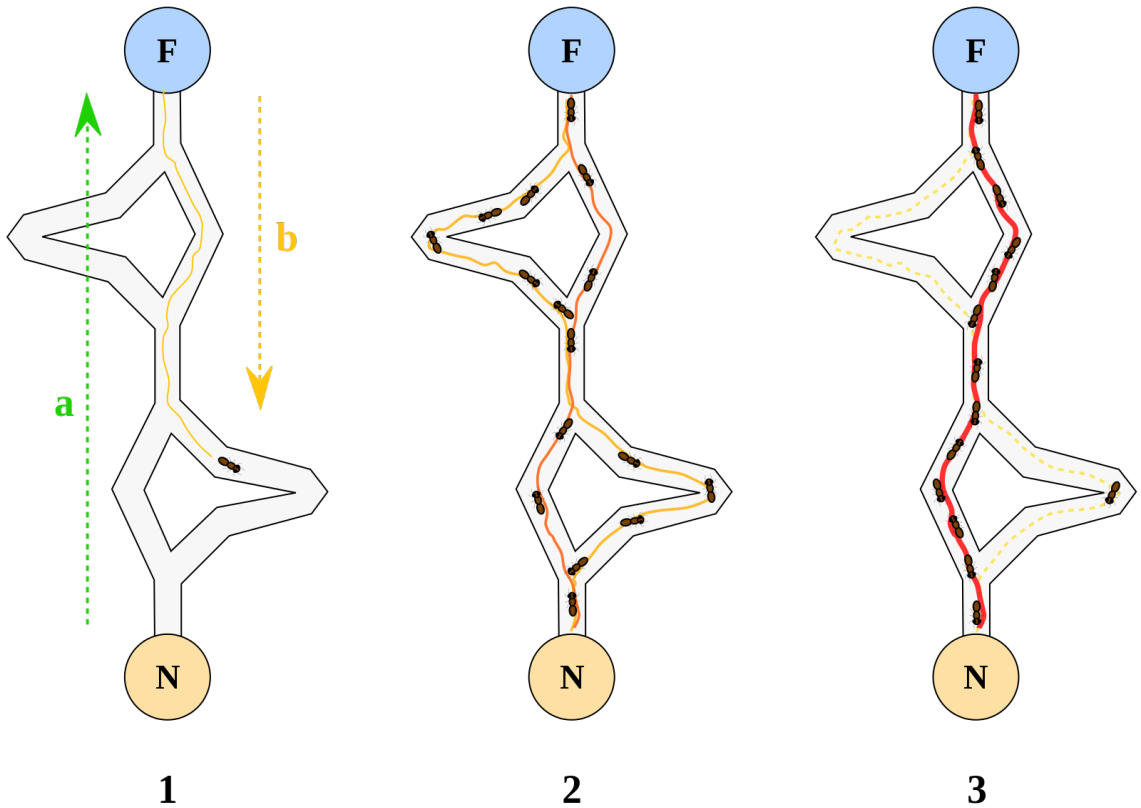


TYPE IV
Strong Emergence

群体智能: 蚁群优化

懒惰的人啊, 你去观察蚂蚁的行为, 即可得智慧.

— 《圣经·箴言》



Travelling salesman problem: find the shortest round-trip to link a series of cities.

蚁群优化 — 简单规则、涌现复杂行为

- **范围:** 每只蚂蚁可观察的范围很小.
- **环境:** 障碍物、别的蚂蚁、信息素.
  - 信息素有两种: 食物信息素、窝的信息素.
  - 环境以一定的速率让信息素消失.
- **觅食规则:** 感知到食物就直接过去; 否则, 看是否有信息素, 以高概率往信息素多的地方走.
- **移动规则:** 若没有信息素, 则惯性的朝前方移动, 同时, 有一个小的随机扰动. 为了防止原地转圈, 记住最近刚走过的点, 尽量避开.
- **避障规则:** 碰到障碍物就随机改变一个方向.
- **播撒信息素规则:** 在刚找到食物或者窝的时候播撒的信息素最多, 随着距离越远, 播撒的信息素越来越少.

多样性 + 正反馈

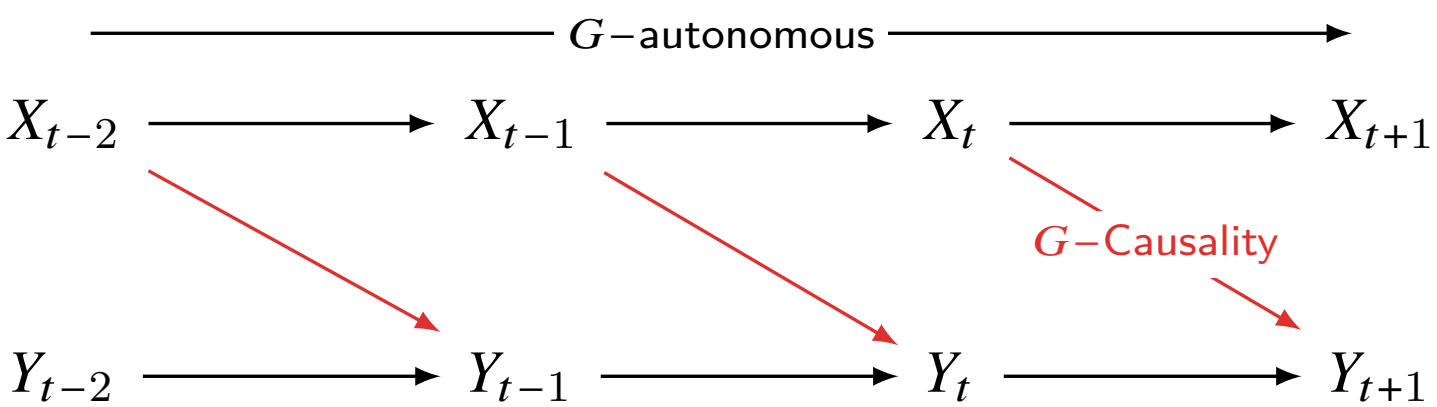
用格兰杰因果检验弱涌现

鸟群模型

1. 分离: 移动以避免过于拥挤
2. 对齐: 朝着周围同伴的平均航向前进
3. 靠近: 朝着周围同伴的平均位置 (质心) 移动

- 仅凭借自身的历史 X_{past} 就可以解释 X_{present}, 则称 X 是 G-自主的.
- 在一个鸟群模型的例子中, 每只鸟的运动便是微观上的时间序列 Y, 而整个鸟群质心的运动则为宏观上的时间序列 X.
- 如果宏观时间序列 X 是 G-自主的, 而微观时间序列 Y 还需要依靠宏观质心运动的历史信息 X_t 来预测下一时刻的微观状态 Y_{t+1}, 那么这个系统是弱涌现的.

$$X \text{ Granger-causes } Y \iff Y_{\text{present}} \not\perp X_{\text{past}} \mid Y_{\text{past}}$$



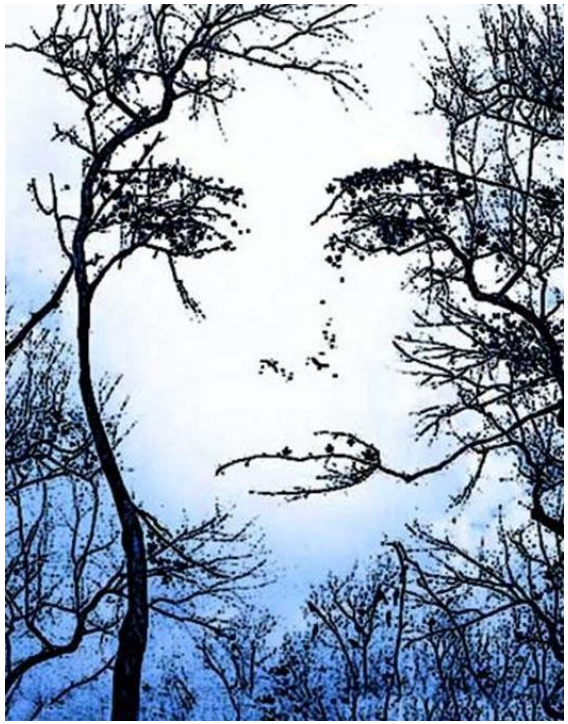
认识论涌现 vs 本体论涌现?

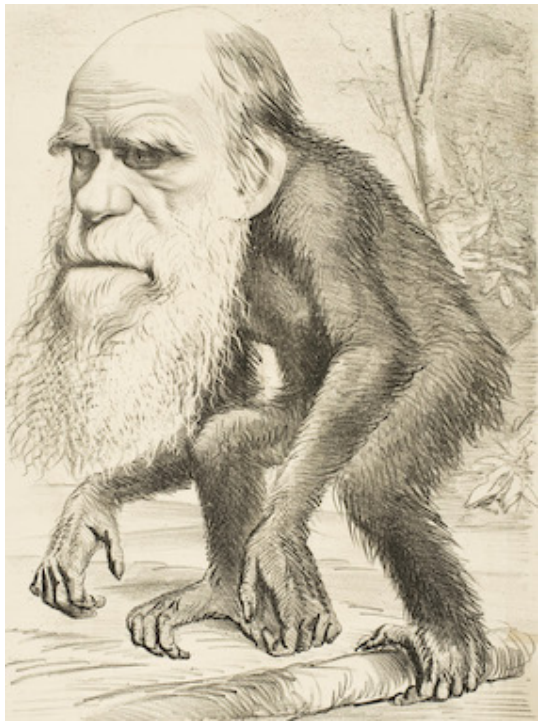
- 弱涌现: 个体之间相互作用, 原则上可模拟, 但计算不可归约
- 强涌现: 不可模拟, 向下因果

壁虎断尾、蚂蚁抱团过火场

向下因果?

认识论涌现 vs 本体论涌现?





- 较高的层级将边界条件、约束和选择机制施加于较低层级的动力学.
- 较低层次组织中的临界态、预激态使得向下因果得以实现.

自由意志? 意识? 自我意识? Downward Causation?

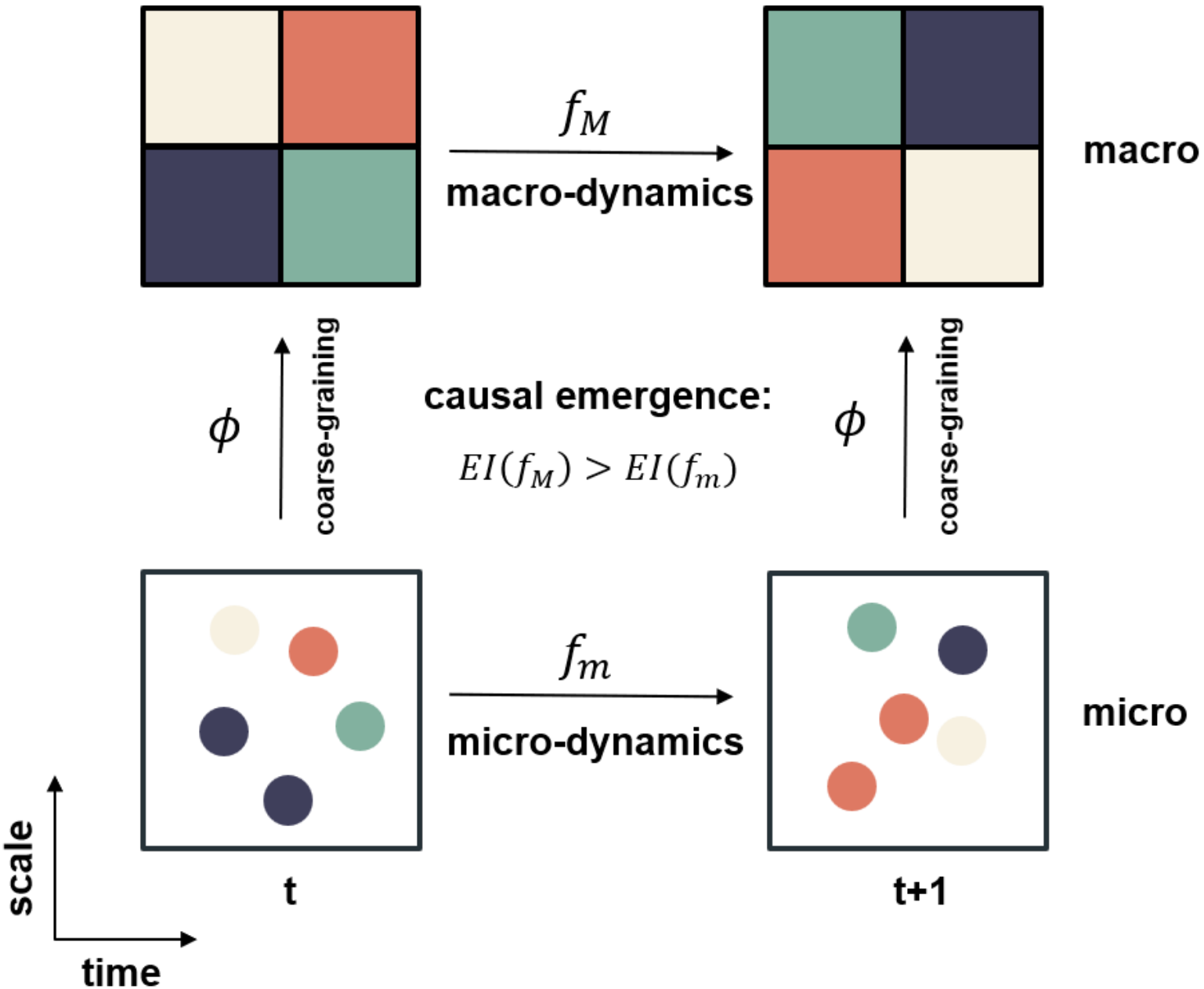
在黑猩猩睡着时, 悄悄在它额头上涂抹颜料, 它在镜子前看到自己时, 会伸手摸额头, 并检查手指.

“整体大于部分之和.”

— 亚里士多德

整体的什么大于部分的什么之和?

因果涌现


f_M
macro-dynamics
macro
ϕ
coarse-graining
causal emergence:
$EI(f_M) > EI(f_m)$
ϕ
coarse-graining
f_m
micro-dynamics
micro
scale
time
t
t+1

Erik Hoel's Causal Emergence[CH22]

Given a state space Ω of size n, and a set of causes and effects $C, E \subset \Omega$.

- $P(e \mid C) := \sum_{c \in C} P(c) P(e \mid c)$
- $\mathrm{suff}(e, c) := P(e \mid c)$
- $\mathrm{nec}(e, c) := 1 - P(e \mid C \setminus c)$ and $\mathrm{nec}^+ := P(e \mid C)$
- we can define a determinism(确定性) coefficient

$$\det(e, c) := 1 + \frac{\log P(e \mid c)}{\log n} \qquad \det := \sum_{e,c} P(e, c) \det(e, c) = 1 - \frac{\sum_c P(c) H(e \mid c)}{\log n}$$

- degeneracy(简并性) coefficient

$$\deg(e) := 1 + \frac{\log P(e \mid C)}{\log n} \qquad \deg := \sum_{e \in E} P(e \mid C) \deg(e) = 1 - \frac{H(e \mid C)}{\log n}$$

- Effective Information

$$\mathrm{ei}(c, e) := \log \frac{P(e \mid c)}{P(e \mid C)} \qquad \mathrm{EI} := \sum_{e,c} P(e, c) \, \mathrm{ei}(c, e) = [\det - \deg] \log n$$

Remark: $\mathrm{EI} = \left\langle D_{\mathrm{KL}}\left(P(e \mid \mathrm{do}(c)) \middle\| \left\langle P(e \mid \mathrm{do}(c)) \right\rangle_{c \in C} \right) \right\rangle_{c \in C}$

- Causal Emergence $\mathrm{CE} := \mathrm{EI}_{\mathrm{macro}} - \mathrm{EI}_{\mathrm{micro}}$

Erik Hoel's Causal Emergence[Hoe17]

- Assume some (uniform) Intervention Distribution (I_D).
- Applying I_D results in Effect Distribution (E_D).

$$E_D := \sum_{\mathtt{do}(c) \in I_D} P(\mathtt{do}(c)) P(e \mid \mathtt{do}(c))$$

- Determinism & Degeneracy

$$\det := \frac{1}{n} \sum_{\mathtt{do}(c) \in I_D} \frac{D_{\mathsf{KL}}\left(P(e \mid \mathtt{do}(c)) \| P^{\mathrm{MaxEnt}}\right)}{\log n} \qquad \deg := \frac{D_{\mathsf{KL}}(E_D \| I_D)}{\log n}$$

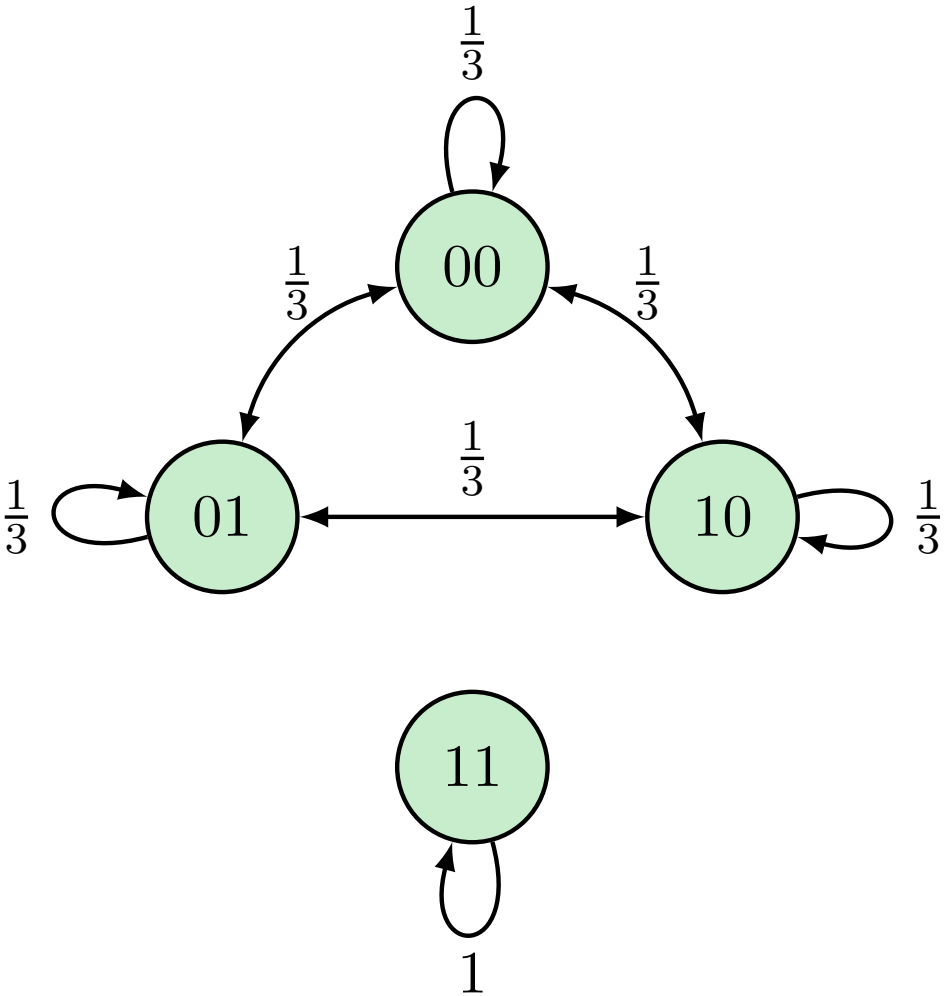
- Effective Information is the mutual information between a set of interventions (I_D) and their effects (E_D).

$$\mathrm{EI} := I(I_D; E_D) = \sum_{\mathtt{do}(c) \in I_D} P(\mathtt{do}(c)) D_{\mathsf{KL}}\left(P(e \mid \mathtt{do}(c)) \middle\| E_D\right) = [\det - \deg] H(I_D)$$

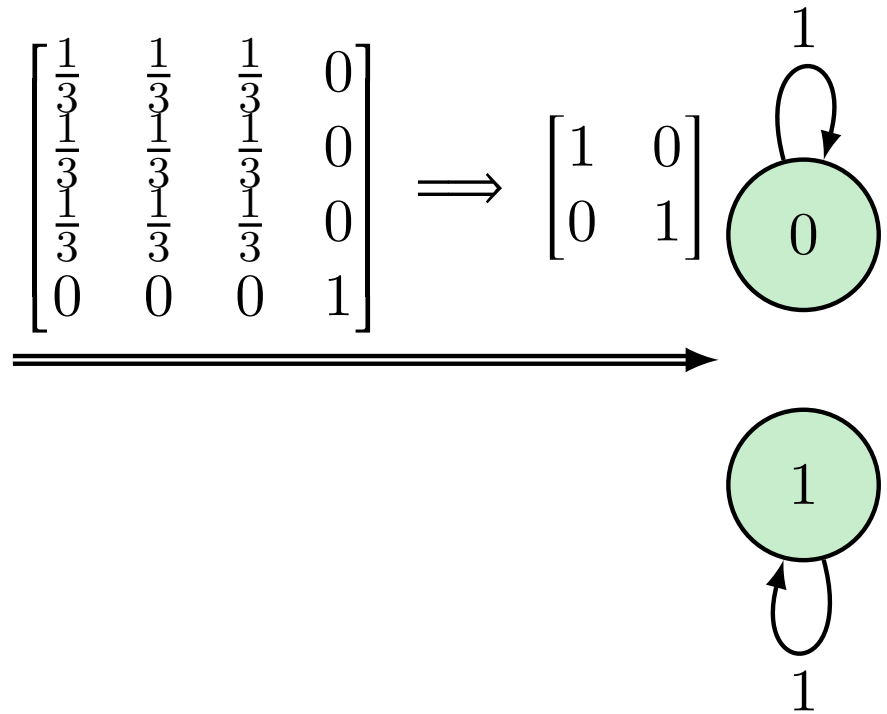
- Examples: $\mathrm{EI}(M_1) = 2, \quad \mathrm{EI}(M_2) = 0, \quad \mathrm{EI}(M_3) = 1$

$$M_1 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \qquad M_2 = \begin{bmatrix} \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \end{bmatrix} \qquad M_3 = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

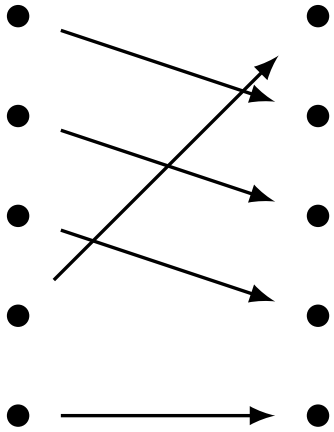
Erik Hoel's Causal Emergence — Example



$$\begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \Longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



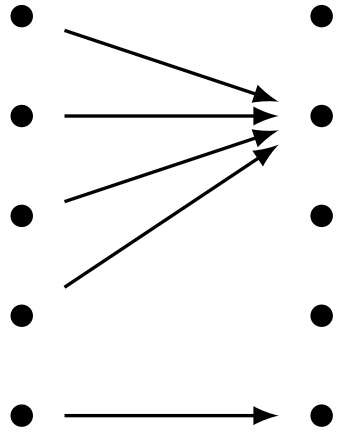
$$\mathrm{CE} = \mathrm{EI}_{\mathrm{macro}} - \mathrm{EI}_{\mathrm{micro}} \approx 1 - 0.81 = 0.19 > 0$$



$\mathrm{EI} = 2.3219$

$\det = 1$

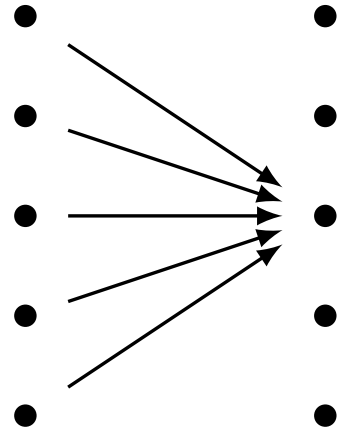
$\deg = 0$



$\mathrm{EI} = 0.7219$

$\det = 1$

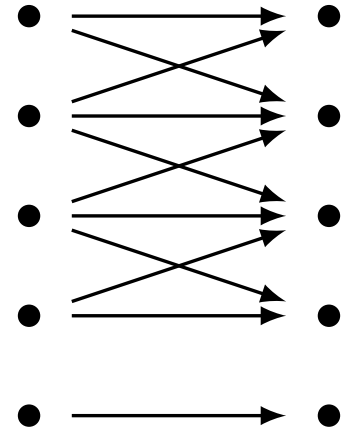
$\deg = 0.6891$



$\mathrm{EI} = 0$

$\det = 1$

$\deg = 1$



$\mathrm{EI} = 0.7219$

$\det = 0.3109$

$\deg = 0$

- 确定性度量的是以过去状态预测未来状态的随机性大小.
- 简并性度量的是从未来状态追溯过去状态的随机性大小.
- 有效信息, 意味着, 高确定性和低简并性. $\frac{\mathrm{EI}}{\log n} = \det - \deg$
- 一些粗粒化策略可以提高有效信息.
  - — 当一个输入状态对应着多个可能的输出状态, 如果通过粗粒化把这些可能的输出打包为一个输出, 就提高了确定性.
  - — 同理, 从某一个输出结果往回追溯, 它也会对应多个可能的输入, 如果打包这些输入, 就降低了简并性.

Remarks

- 微观动力学就是 Agent 所观察的外部世界; 宏观动力学则是 Agent 为了理解世界而构建的内部 “世界模型”.
- 粗粒化过程即 Agent 不断观察世界而构建内部世界模型的过程.
- 因果涌现理论告诉我们, 好的世界模型一定具有更强的因果效应 (有效信息)
- Agent 努力寻找一个更加可逆的宏观动力学模型 (更强的有效信息), 而它的代价就是由粗粒化引来的信息损失.
- 微观量子世界的演化遵循可逆的薛定谔波动方程, Agent 的观察测量导致了波函数的不可逆的坍缩, Agent 在宏观世界只能得到不可逆的统计力学. 为了追求更可逆的世界模型, Agent 又通过粗粒化损失了更多信息, 从而更不可逆.
- 微观世界的动力学本来是可逆的 (时间反演对称的), Agent 构建宏观世界模型竟然是为了恢复被他自己破缺了的时间反演对称性.

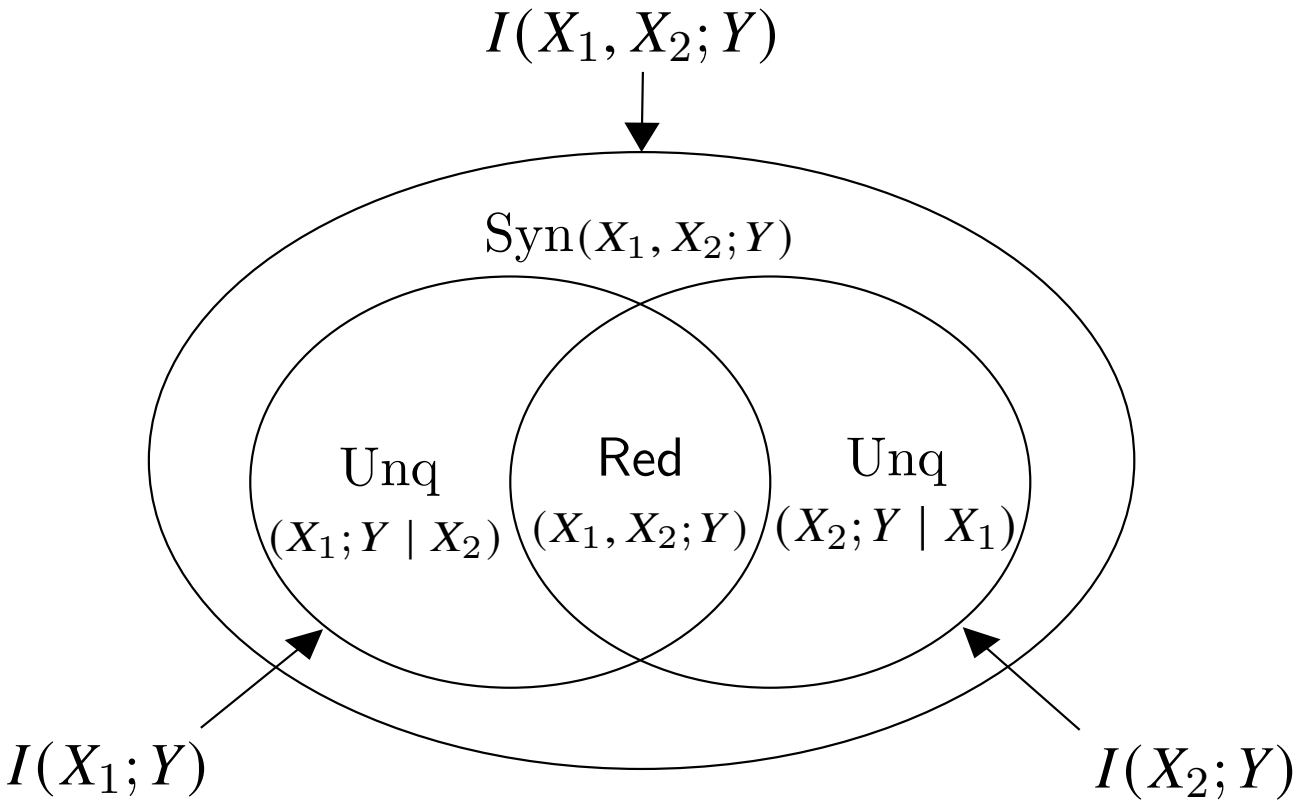
因果涌现 vs 粗粒化 — 部分信息分解 [Ros+20; VH22]

- 部分信息分解

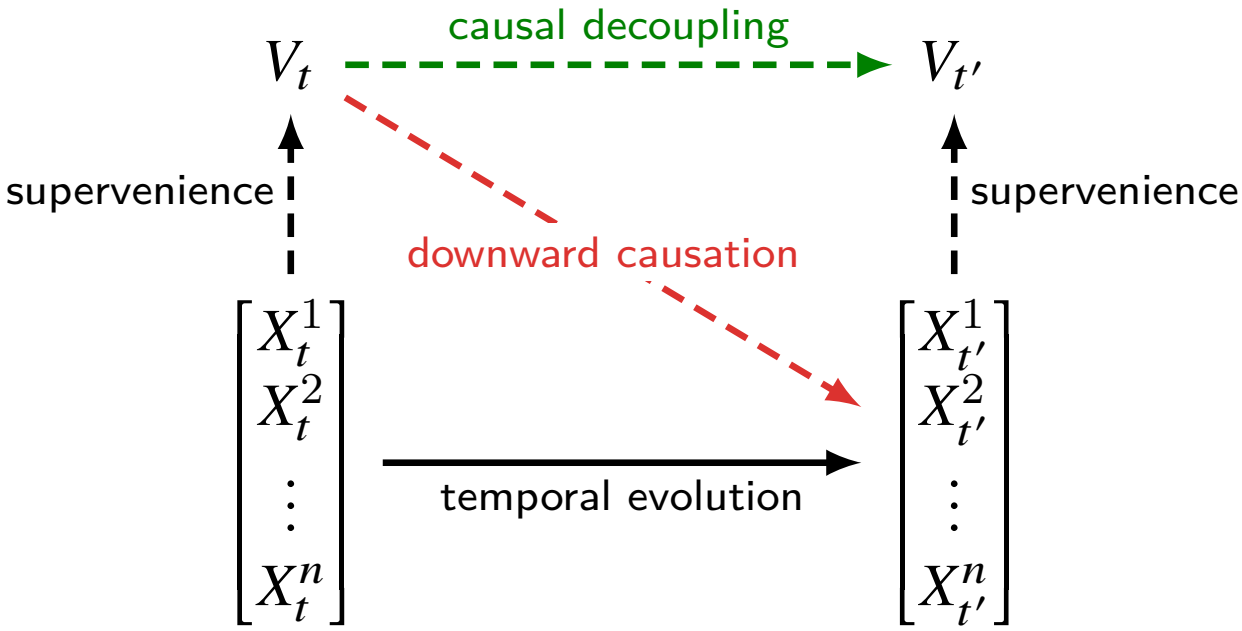
$$I(X_1, X_2; Y) = \mathrm{Unq}(X_1; Y \mid X_2) + \mathrm{Unq}(X_2; Y \mid X_1) + \mathrm{Red}(X_1, X_2; Y) + \mathrm{Syn}(X_1, X_2; Y)$$

  其中, $\mathrm{Unq}(X_1; Y \mid X_2)$ 是 X_1 单独提供不包含在 X_2 中的关于 Y 的特有信息; $\mathrm{Red}(X_1, X_2; Y)$ 是 X_1, X_2 共享的同时提供给 Y 的冗余信息; $\mathrm{Syn}(X_1, X_2; Y)$ 是 X_1, X_2 联合提供的关于 Y 的协同信息. "冗余信息" 与 "协同信息" 的差可以看做 "交互信息".
  - 比如: 两只眼睛看世界, 关于空间深度的立体感知需要两只眼睛协同, 而颜色信息则是冗余信息. 冗余可以保障可靠性, 即使一只眼睛受伤了, 也不影响基本的视觉.
  - 比如: 两个理论解释一个现象. 如果两个理论等价, 则高度冗余; 如果两个理论协同才能解释现象, 则互补.
- 粗粒化使得冗余信息转化为协同信息.

Partial Information Decomposition



$$I(X_1; Y) = \mathrm{Red}(X_1, X_2; Y) + \mathrm{Unq}(X_1; Y \mid X_2)$$
$$I(X_2; Y) = \mathrm{Red}(X_1, X_2; Y) + \mathrm{Unq}(X_2; Y \mid X_1)$$
$$I(X_1; Y \mid X_2) = \mathrm{Syn}(X_1, X_2; Y) + \mathrm{Unq}(X_1; Y \mid X_2)$$
$$I(X_2; Y \mid X_1) = \mathrm{Syn}(X_1, X_2; Y) + \mathrm{Unq}(X_2; Y \mid X_1)$$



Emergence = Causal Decoupling + Downward Causation

- 当 $\mathrm{Unq}(V_t; \mathbf{X}_{t'} \mid \mathbf{X}_t) > 0$ 时, 存在因果涌现.
- 当协同信息 $\mathrm{Syn}(\mathbf{X}_t; \mathbf{X}_{t'}) > 0$ 时, 存在因果涌现.
  $\mathrm{Syn}(\mathbf{X}_t; \mathbf{X}_{t'}) \geq \mathrm{Unq}(V_t; \mathbf{X}_{t'} \mid \mathbf{X}_t)$ 恒成立.
- 如果宏观特征对某个子集有独特的预测力, 即, 对某个 $\alpha \subset [n]$, $|\alpha| = k$, 有 $\mathrm{Unq}^{(k)}(V_t; \mathbf{X}_{t'}^{\alpha} \mid \mathbf{X}_t) > 0$, 则存在向下因果.
- 如果 $\mathrm{Unq}^{(k)}(V_t; V_{t'} \mid \mathbf{X}_t, \mathbf{X}_{t'}) > 0$, 则存在因果解耦.

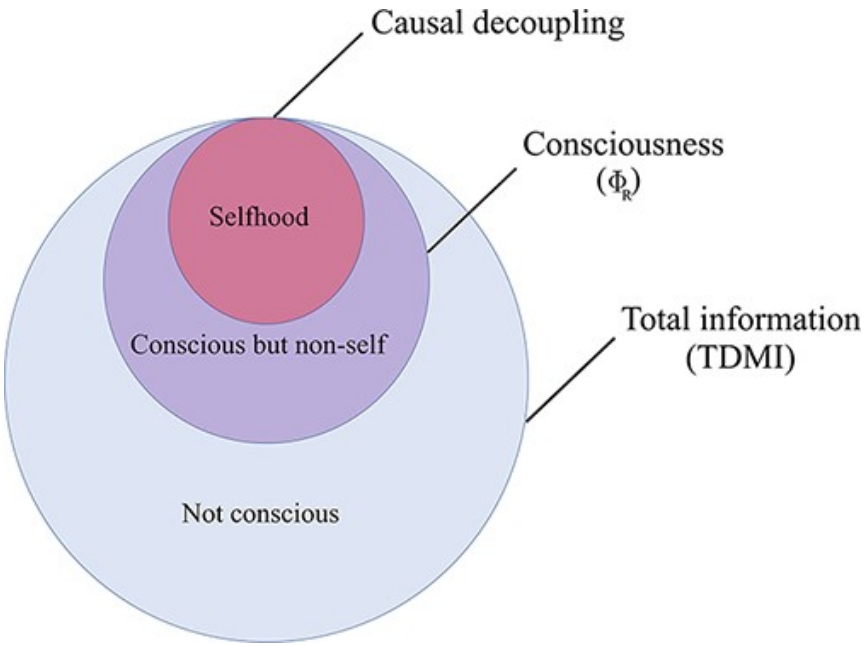
From IIT to ΦID

What it is like to be a bit: an integrated information decomposition account of emergent mental phenomena[Lup+21]

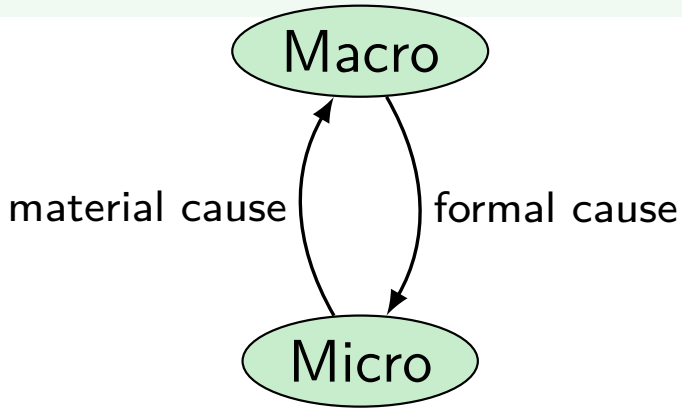
- 休谟: 不存在一个独立的自我, 自我只是一束感知体验 (无我)
- ΦID 不赞同休谟, 认为存在宏观对宏观的影响

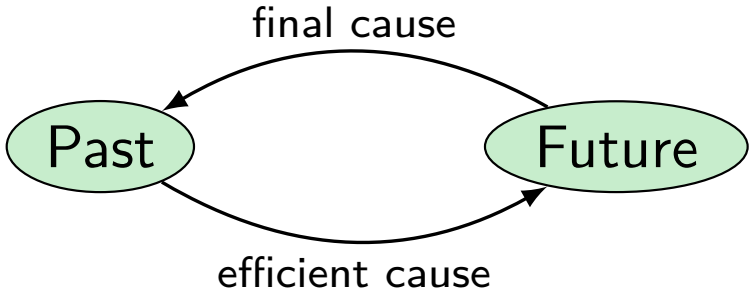
1. Selfhood: 区分 “自我” 与 “环境” (因果解耦)
2. Sense of self: 对自我的体验 (向下因果)

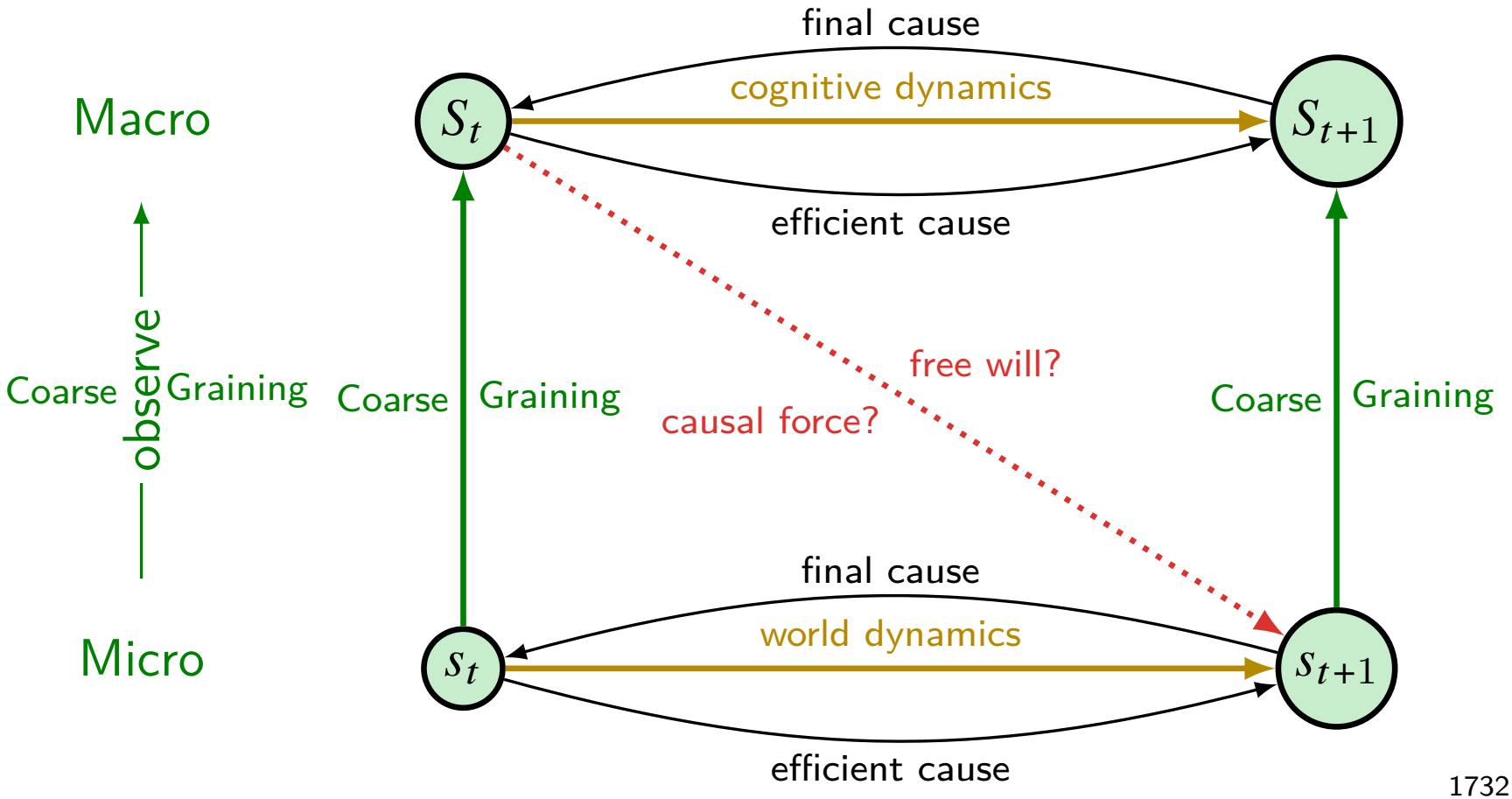
- 冥想: 会丧失对自我的体验, 但不丧失主观意识体验, 甚至会加强
- 环境影响意识: 改变协同作用的环境, 会对系统的自我特征有影响



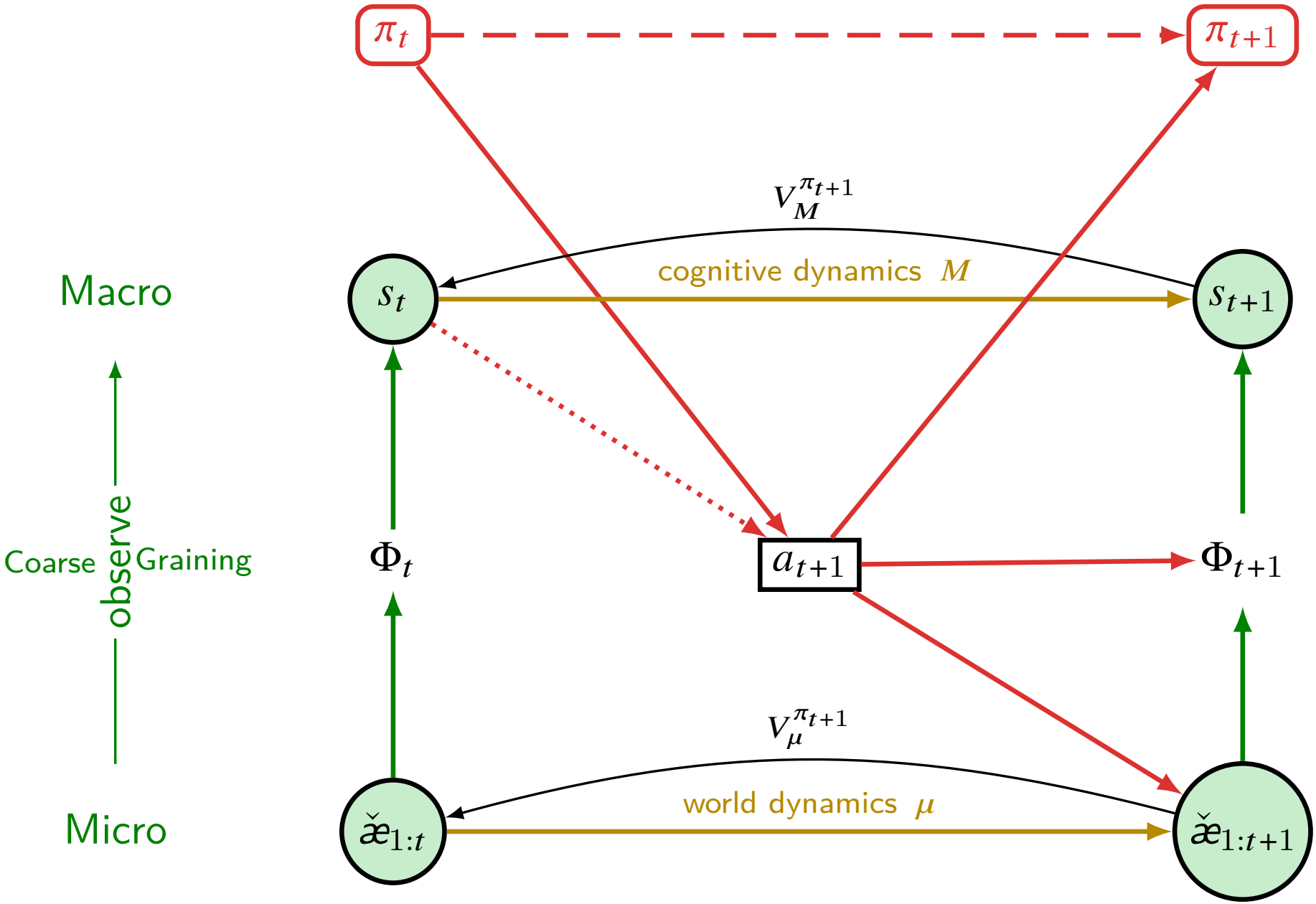
Jiang ZHANG: Causal Emergence







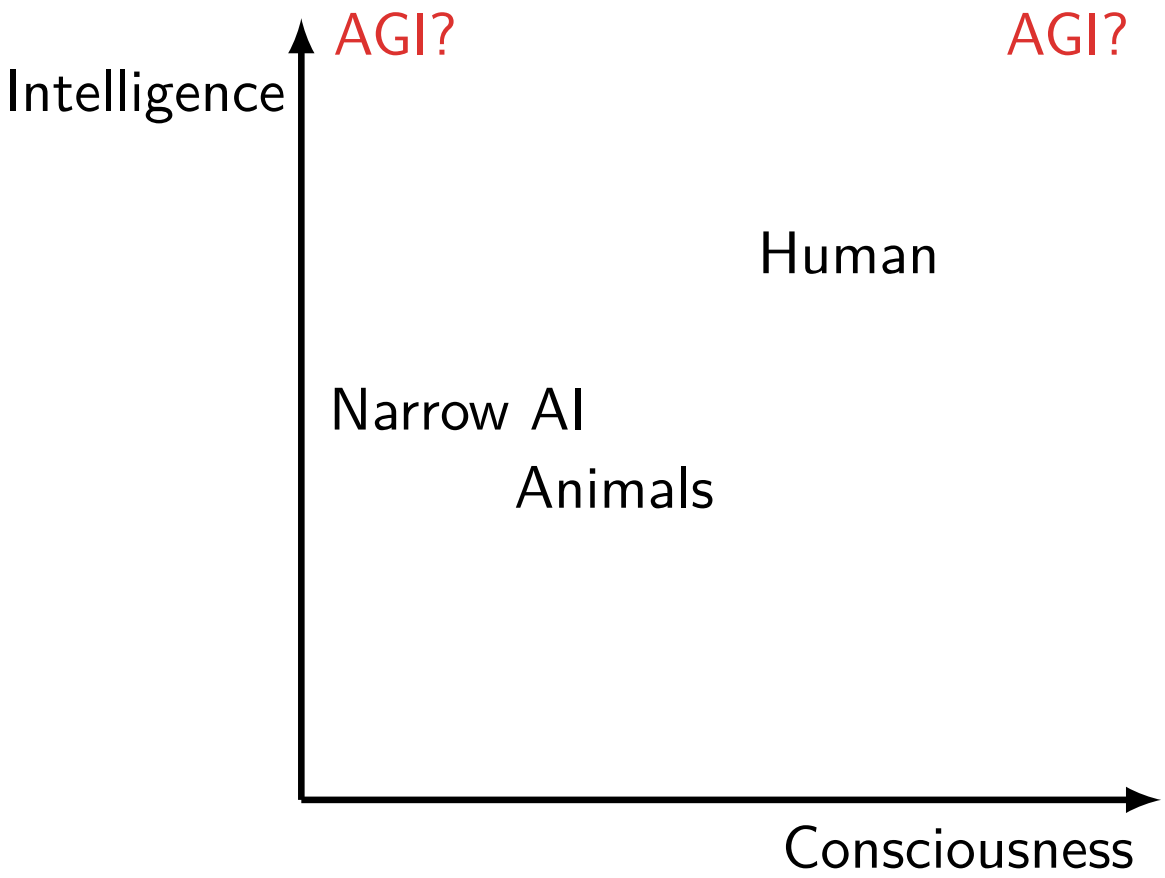
Self-Modifying Causal Representation



Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
    - Philosophy of Mind
    - Causal Emergence
    - Integrated Information Theory
    - Self-Reference & Consciousness
11. AI Ethics

智能 vs 意识?


AGI?
AGI?
Intelligence
Human
Narrow AI
Animals
Consciousness

机器意识[32]

- 信息整合理论: 意识产生于整体系统对大量信息的整合.
- 预测加工理论: 意识产生于生成模型对外部信息的主观建构.
- 高阶表征理论: 意识产生于对心理状态的一阶表征的元表征.
- 量子意识理论: 意识产生于脑中微小结构的波函数坍缩效应.
- 全局工作空间理论 (Global Workspace Theories GWTs): 意识产生于局域性的认知模块被全局广播.
  — 当知觉、思想、情感等进入 "工作空间" 时变得有意识, — 大脑是一个剧场, 有意识的思维是某一特定时刻舞台上聚光灯下的活动 (但大脑中实际的工作空间并不是局部的, 而是分布在大脑皮层的额叶和顶叶区域).

[32]Patrick Butlin, et al. "Consciousness in artificial intelligence: insights from the science of consciousness." arXiv preprint arXiv:2308.08708 (2023).
Shamil Chandaria, et al. "From cacophony to hierarchy: a principled framework for assessing AI consciousness" arXiv preprint arXiv:2609.35618 (2026)

Life is a braid in spacetime

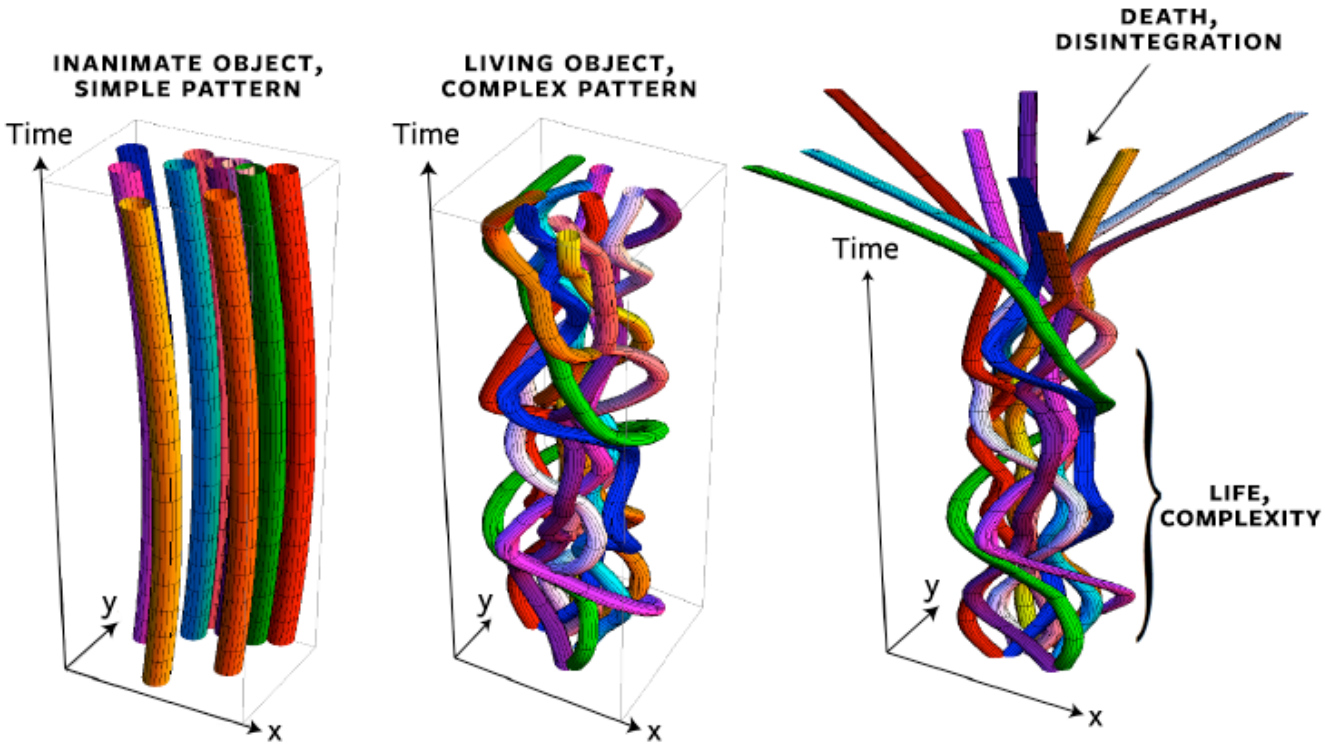


Figure: Tegmark: The motion of an object corresponds to a pattern in spacetime. You're a braid in spacetime — indeed, one of the most elaborate braids known.

Consciousness — Integrated Information Theory

- 当处于“心流”状态时，我们只关注高层次的信息，而对于低层次的细节则处于“无意识”状态.
- 我们有意识的信息处理仅仅是冰山一角. 大部分脑区是无意识的. 有意识的经验仅仅是对大量无意识过程的事后总结. 意识落后于做出决定约四分之一秒. 脑电测量可以在你意识到自己做出决定之前就预测出你的选择.
- 意识以一种相当自治和整合的方式处理信息.
- 整合信息度量的是一个系统在演化过程中无法归约为独立的部分的能力.

意识体验的 “公理” — Tononi

Existence	意识体验是第一人称存在的: “I experience therefore I am”, 具有之于自身的因果力.
Intrinsicality	意识体验是内在的: 与外部观察者无关.
Information	意识体验是特定的, 有信息量的: 每个体验以其独特的方式区别于可能的其它体验.
Integration	意识体验是整合的: 每个体验无法归约为相互独立的组成部分. 每个部分都既影响其他部分, 又受到其他部分的影响. 每一个意识状态都是一个单子 (Monad), 无法分割为能被独立体验的组分.
Exclusion	意识体验是排他的: 每个体验都有明确的边界; 每个体验都有特定的时空颗粒度.
Composition	意识体验是有结构的: 每个体验由多个机制以不同的组合方式构成.

什么是复杂性?

1. 描述它有多难?
2. 创造它有多难?
3. 它的组织程度如何?

> “整合信息捕捉的是, 系统整体的因果作用所产生的信息, 多大程度上超出了其各部分所产生的信息之和.”
>
> — *Tononi*

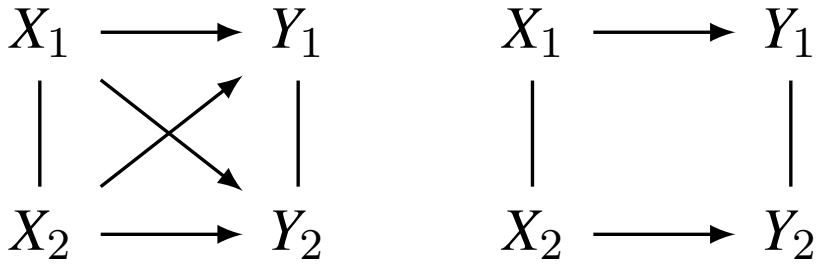
一个系统如果显示出不能被还原为它的各部分性质的涌现性质, 那么它就是复杂的.
Tononi: 意识体验的程度与整合信息量有关.

Question

- 意识体验的 “公理” 是否可靠且完备?
- 整合信息论 (IIT) 的本体论地位是什么?
  — 唯物主义、唯心主义、二元论、中立一元论、副现象主义、涌现主义、泛心论?

Integrated Information Theory (IIT)

- Suppose given a stochastic dynamical system, where the state of the system at time t is described by a set of random variables $\{X_i = X_i^{(t)}\}_{i=1}^N$ which correspond to a partition of the system into N subsystems, and the state at time $t+1$ by $\{Y_i = X_i^{(t+1)}\}_{i=1}^N$.
- The full system including all the mutual influences between these two sets of variables is described by $P(X, Y)$.
- Integrated information is meant to capture the difference between $P(X, Y)$ and an approximation $Q(X, Y)$ where only certain kinds of mutual influences are retained.
- These are usually taken to be the interdependencies between the variables at the same time and between each X_i and the corresponding Y_i, removing the dependencies of the Y_i from the X_j with $j \neq i$.



IIT — Conditional Independent Statements

- Given a partition λ

$$\{(X, Y)\} = \prod_{i=1}^{N} \{(X_i, Y_i)\}$$

Consider the space

$$\mathcal{M}_\lambda := \{Q : Q(Y_i \mid X) = Q(Y_i \mid X_i) \text{ for } i = 1, \ldots, N\}$$

- The best approximation to $P(X, Y)$ by $Q(X, Y)$ in $\mathcal{M}_\lambda$ is

$$Q^*_\lambda := \operatorname*{argmin}_{Q \in \mathcal{M}_\lambda} D_{\mathrm{KL}}(P \| Q)$$

- Then the integrated information, for a given partition λ, is defined as

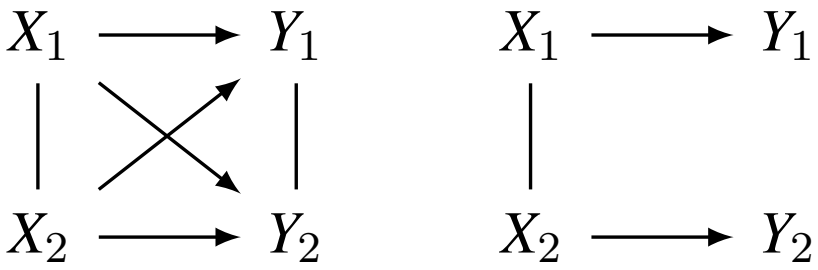
$$\Phi_\lambda := D_{\mathrm{KL}}(P \| Q^*_\lambda) = \min_{Q \in \mathcal{M}_\lambda} D_{\mathrm{KL}}(P \| Q)$$

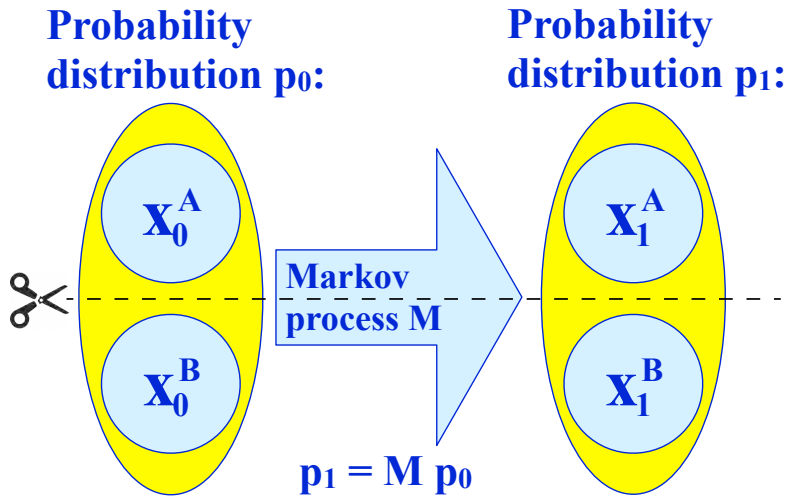
with a further minimization over the choice of the partition,

$$\Phi_{\mathrm{CIS}} := \min_\lambda D_{\mathrm{KL}}(P \| Q^*_\lambda) = \min_{Q \in \bigcup_\lambda \mathcal{M}_\lambda} D_{\mathrm{KL}}(P \| Q)$$

IIT — another version — Stochastic Interaction

$$Y_j \perp X_i \mid X_{I\setminus\{i\}}$$





$$\mathcal{M}_{\mathrm{SI}} := \left\{ Q : Q(Y \mid X) = \prod_{i=1}^{N} Q(Y_i \mid X_i) \right\}$$

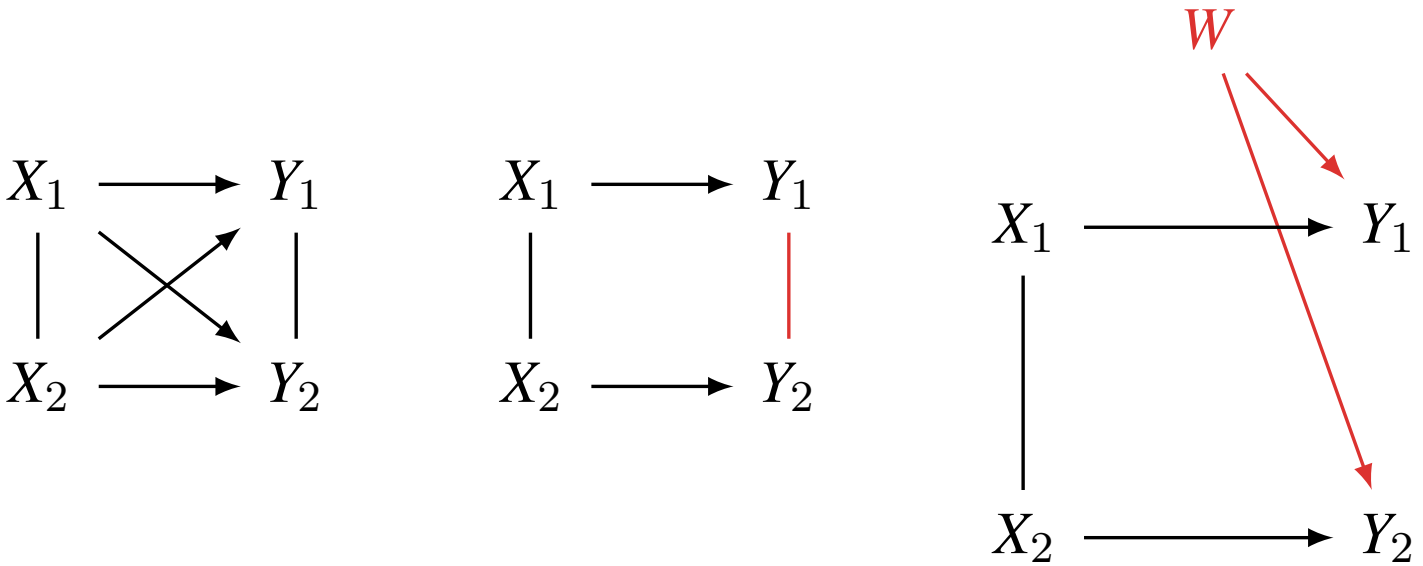
Φ measures inability to tensor factorize $M = M^A \otimes M^B$

- $\Phi_{\mathrm{SI}} := \min_{Q \in \mathcal{M}_{\mathrm{SI}}} D_{\mathsf{KL}}(P \| Q) = \sum_i H(Y_i \mid X_i) - H(Y \mid X)$
- counter-example

$$\begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} = \begin{bmatrix} 1^0 & \dots & 1^{N-1} \\ \vdots & \ddots & \vdots \\ N^0 & \dots & N^{N-1} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix}$$

IIT — another version — Causal Information Integration

- IIT including a common exterior influence.



$$\mathcal{M}_{\mathrm{CII}} := \left\{ Q : Q(x,y) = \sum_{w} Q(x)Q(w) \prod_{i=1}^{N} Q(y_i \mid x_i, w) \right\}$$

$$\Phi_{\mathrm{CII}} := \min_{Q \in \mathcal{M}_{\mathrm{CII}}} D_{\mathsf{KL}}(P \| Q)$$

盖亚假说 vs 泛心论

- 整个地球、海洋、岩石、植物和大气层是一个统一的自我调节实体. 树太多？就会着火. 二氧化碳太多？植被就会增多. 地球将其自身温度维持在一个范围内，令人惊讶的是，它还维持着数百万年来海洋的盐度等等. 各种事物都被保持在地球的“适宜”范围内，以利于生命的存在.
- 如果地球是有意识的，我们怎么会知道呢？它会感到痛苦吗？它有情绪吗？它怎么看我们？太阳呢？它会是有意识的吗？幼儿园的小朋友们总是给太阳画一张笑脸...
- 意识是如何结合的？

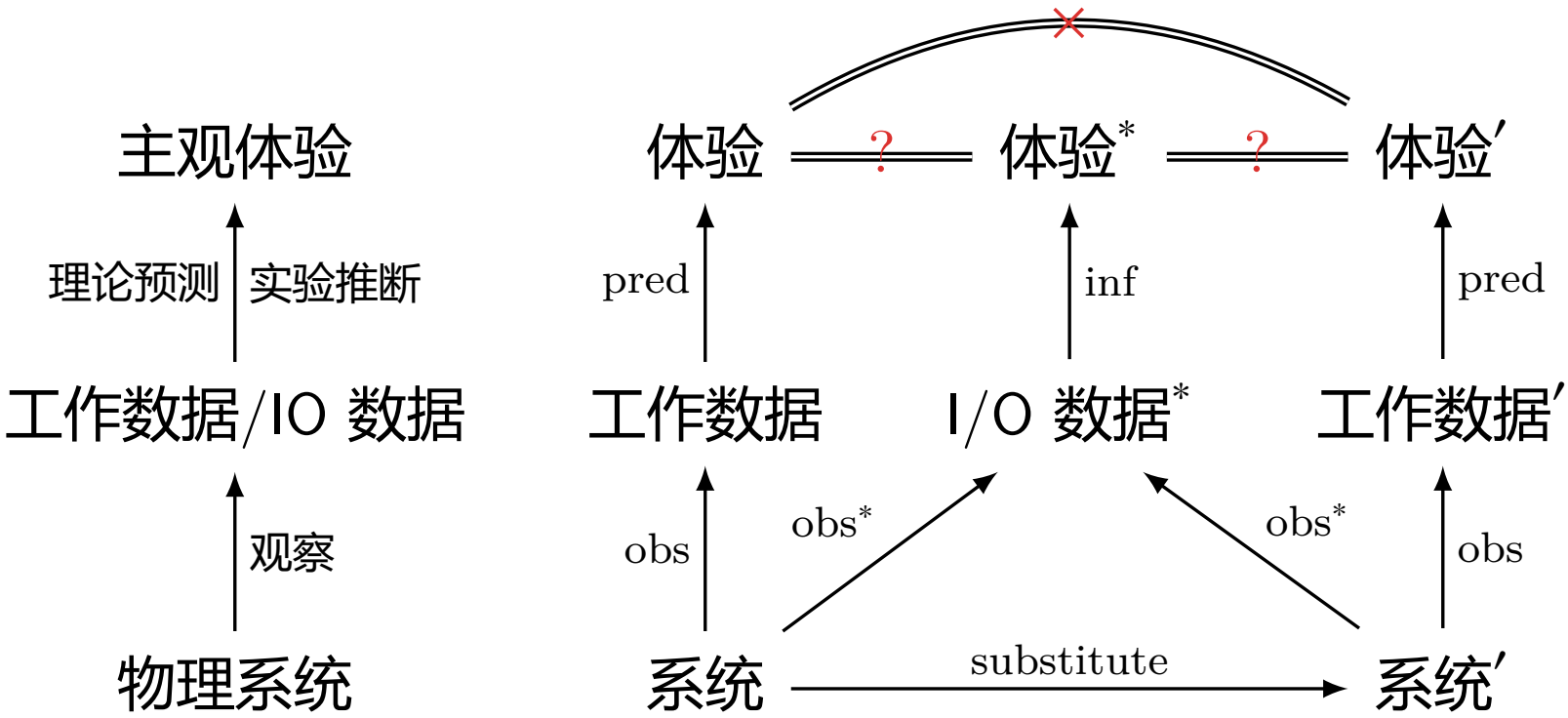
意识理论的可证伪性

- 理论预测 ($\mathrm{pred} : \mathcal{O} \to \mathcal{E}$): 将系统内部的工作数据 (如神经元扫描影像) 映射到可能的意识体验.
- 实验者的推断 ($\mathrm{inf} : \mathcal{O} \to \mathcal{E}$): 将外部可观测的输入/输出行为 (如语言报告) 映射到可能的意识体验.
- 若 pred 严格依赖于 inf, 则理论不可被证伪, 是 trivial 的. (比如行为/功能主义的意识理论: 全局工作空间理论. 因为看上去有意识, 所以有意识.)
- 若二者完全独立 $\mathrm{pred} \perp \mathrm{inf}$, 则理论被先验的证伪. (比如整合信息论 IIT.)
  — 替代论证: 保持输入/输出相同, 推断 inf 结果就相同, 当系统结构被等效的结构替代后, 若预测 pred 结果不同, 则理论被证伪. (哲学僵尸)

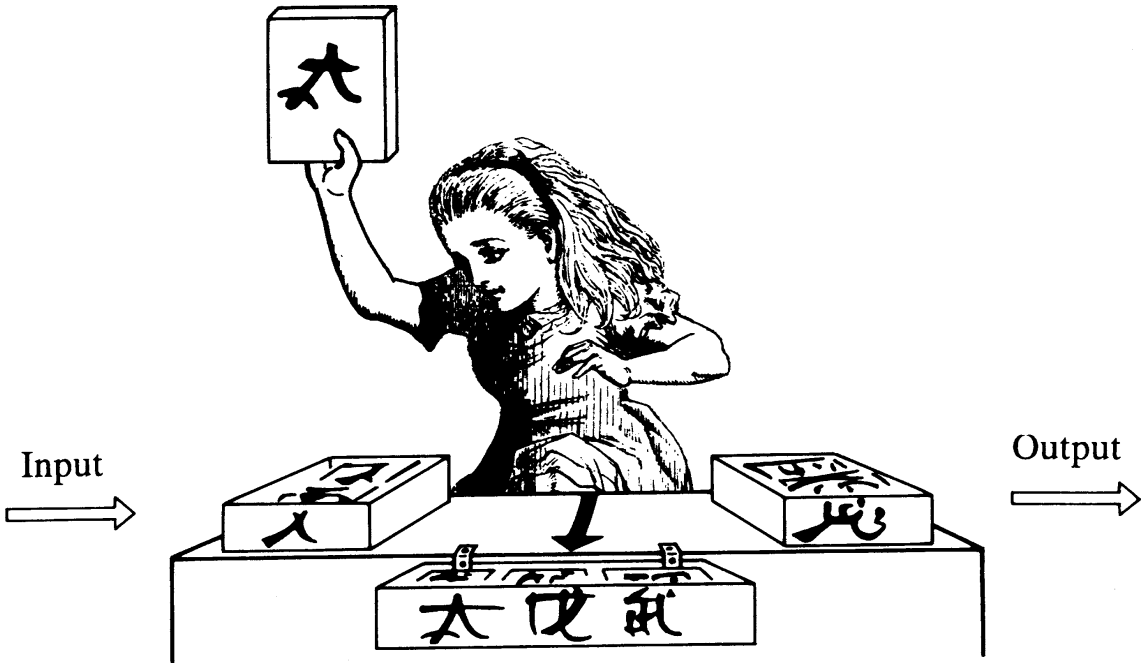
静态 LLM $\overset{\text{通用逼近定理}}{\sim\sim\sim}$ 单隐层前馈 NN $\overset{\text{有穷状态机}}{\sim\sim\sim}$ 查找表

- 出路: 持续学习 Continual Learning. pred 与 inf 既不完全独立, 也不严格依赖. 既不会被先验的证伪, 也不是 trivial 的.

替代论证



Remark: 如果意识理论的目标不是追求对意识的概念定义，而是试图成为一门能够解释物理过程如何产生主观体验的经验科学的话，那就需要满足可被经验证伪的标准.



- 查找表?
- 压缩?
- 泛化?
- 计算效率?

语言 vs 思维

程序是形式的句法
心灵是有语义内容的
句法不足以刻画语义

心灵 ≠ 程序

- 弱人工智能: 模拟心灵
- 强人工智能: 有意识的心灵
- 强人工智能 $\overset{?}{\neq}$ 通用人工智能

屋中人懂中文吗?

查找表有意识吗? 前馈神经网络呢? 大语言模型呢?

需要 qualia? 如果中文屋里的是人列计算机呢?

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
    - Philosophy of Mind
    - Causal Emergence
    - Integrated Information Theory
    - Self-Reference & Consciousness
11. AI Ethics

你站在桥上看风景，看风景人在楼上看你.
明月装饰了你的窗子，你装饰了别人的梦.

— 卞之琳《断章》

不识庐山真面目，只缘身在此山中.

— 苏轼《题西林壁》

试上高峰窥皓月，偶开天眼觑红尘.
可怜身是眼中人.

— 王国维《浣溪沙》

Who am I?

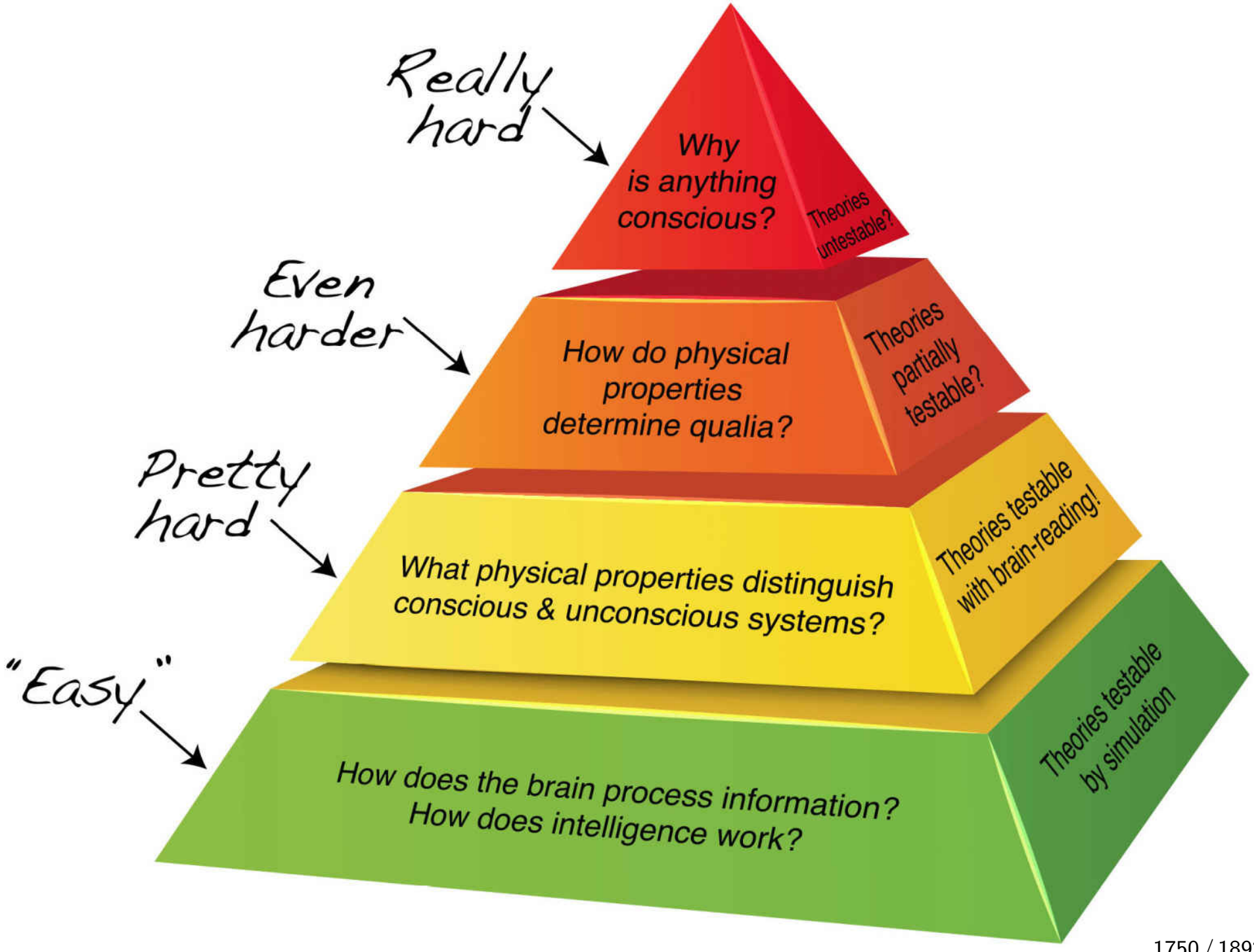
I think, therefore I am.

self-locating: "I" is an indexical term that I use to refer to myself as myself.

What is "me"?

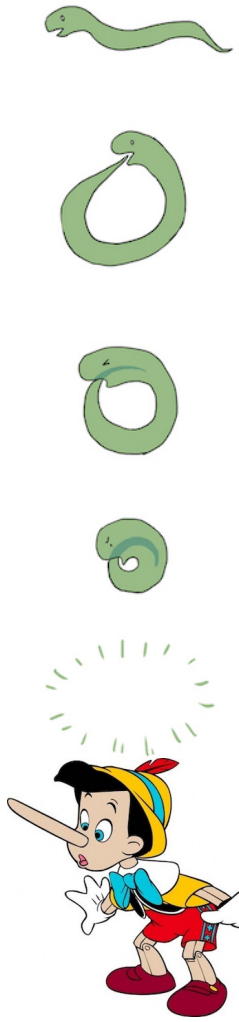
What is "self-consciousness"?

- self-perception self-observation self-experience self-tracking self-reflection self-awareness
- self-evaluation self-analysis self-monitoring
- self-control self-adjustment self-modification self-actualization self-fulfillment self-surpass self-improvement
- *actual-self* pk *ideal-self* self-identity "the *self*"
- free will: Second order desire that we want to act on is second order volition. Second order volitions involve wanting a certain desire to be one's will, that is wanting it to move one to action. (Frankfurt)


Really hard
Why is anything conscious?
Theories untestable?
Even harder
How do physical properties determine qualia?
Theories partially testable?
Pretty hard
What physical properties distinguish conscious & unconscious systems?
Theories testable with brain-reading!
"Easy"
How does the brain process information?
How does intelligence work?
Theories testable by simulation

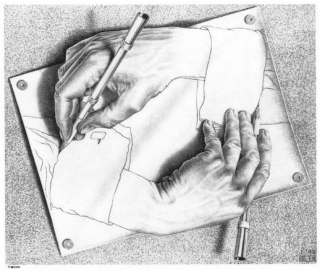
Self-Reference

- This sentence repeats the word ‘twice’ twice.
- Thare are five mistukes im this centence.
- **The only boldface sentence on this page is false.**
- All generalizations are wrong.
- Every rule has an exception except this one.
- Moderation in all things, including moderation.
- We must believe in free will — we have no choice!
- I know that I know nothing.
- There are two rules for success in life:
  1. Never tell anyone all that you know.
- If you choose an answer to this question at random, what is the chance you will be correct? (A) 25% (B) 50% (C) 0% (D) 25%
- 1. What is the best question to ask and what is the answer to it?
  2. The best question is the one you asked; the answer is the one I gave.
- Can you answer the following question in the same way to this one?
- One of the lessons of history is that no one ever learns the lessons of history.
- 涅槃是消除了一切欲望后才能抵达的境界，包括对涅槃的欲望.



Self-Reference vs Paradox

The sentence below is false.



The sentence above is true.

Yablo Paradox

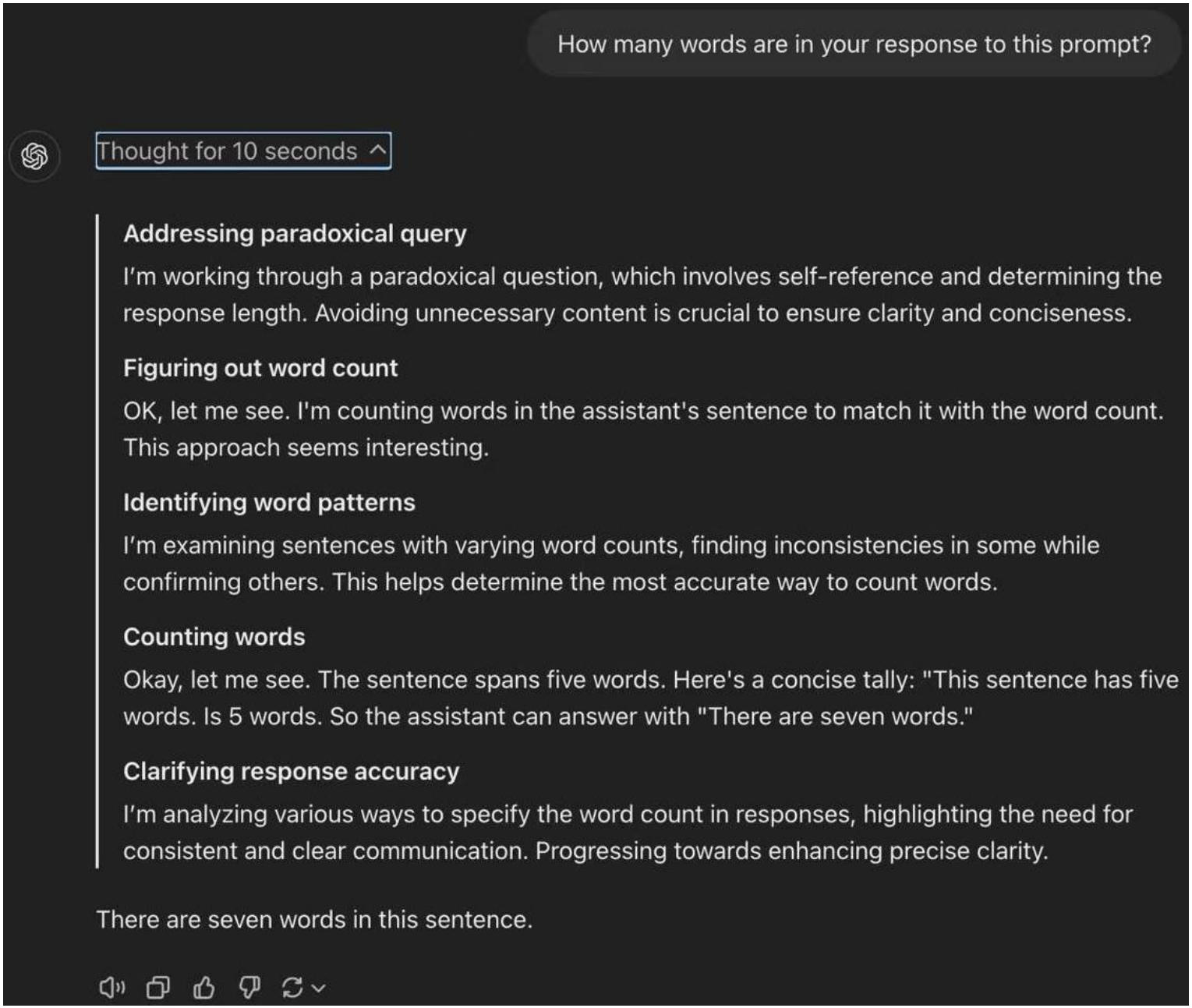
- A_1: for all $k > 1$, A_k is false.
- A_2: for all $k > 2$, A_k is false.
- A_3: for all $k > 3$, A_k is false.
- $\cdots$

Quine Paradox

"Yields falsehood when preceded by its quotation" yields falsehood when preceded by its quotation.

self-reference / circularity or infinite regress / negation / infinity / totality

ChatGPT-o1 2024



The "Power" of Self-Reference

Curry's Paradox

- If this sentence is true, then God exists.
- This sentence is false, and God does not exist.

1. At least one of these two sentences is false.
2. God does not exists.

Hi 美女, 问你个问题呗

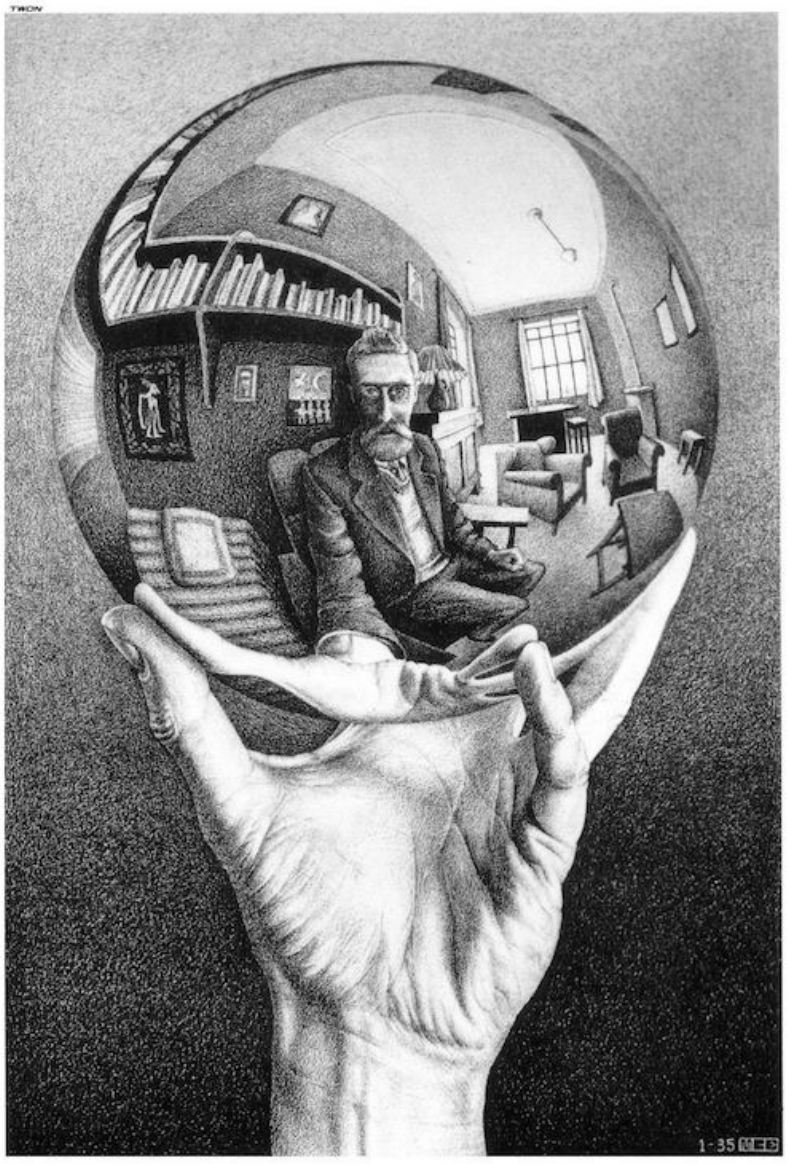
如果我问你 "你能做我女朋友吗", 那么你的答案和**这个问题**的答案是一样的吗?

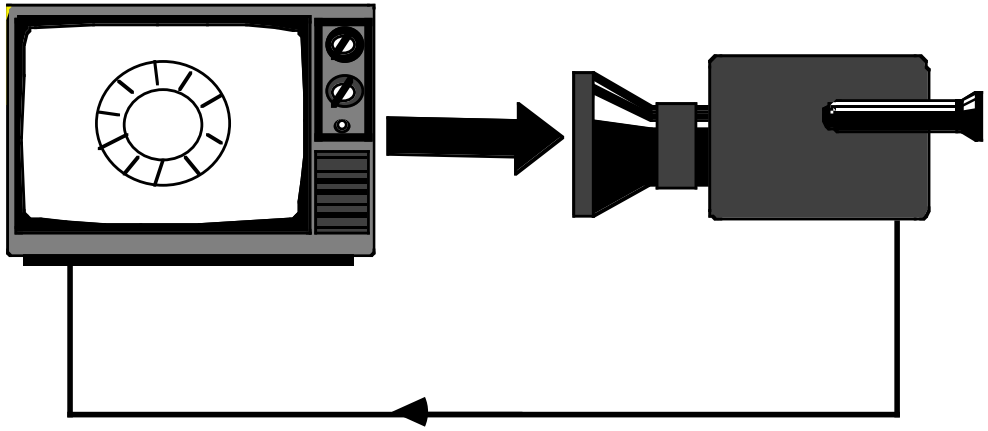
自我修复/自我实现?

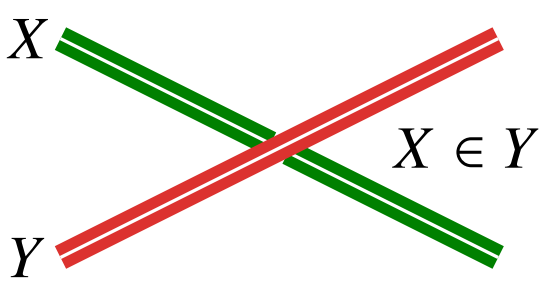
- "This sentence has _____ letters." thirty-one / thirty-three
- 这句话有 2 个 '这' 字, 2 个 '句' 字, 2 个 '话' 字, 2 个 '有' 字, 7 个 '2' 字, 11 个 '个' 字, 11 个 '字' 字, 2 个 '7' 字, 3 个 '11' 字, 2 个 '3' 字.

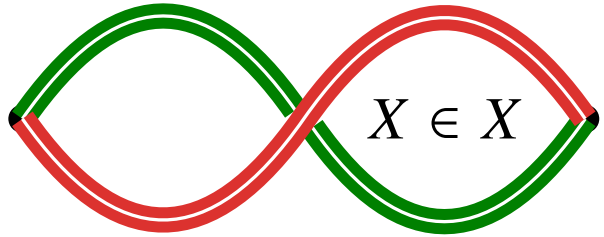
How to Refer? — Levels

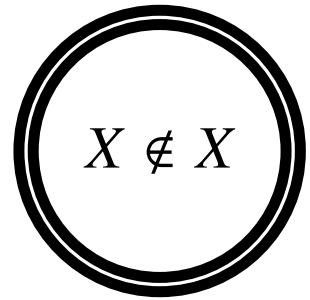
How to Refer?

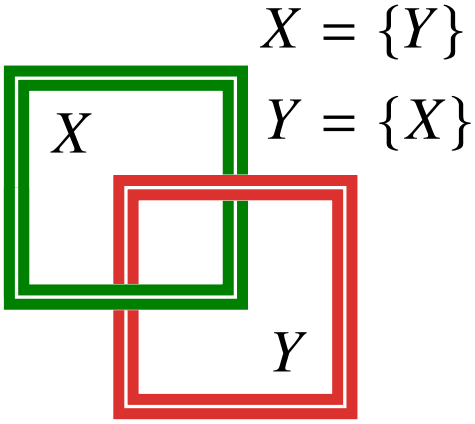


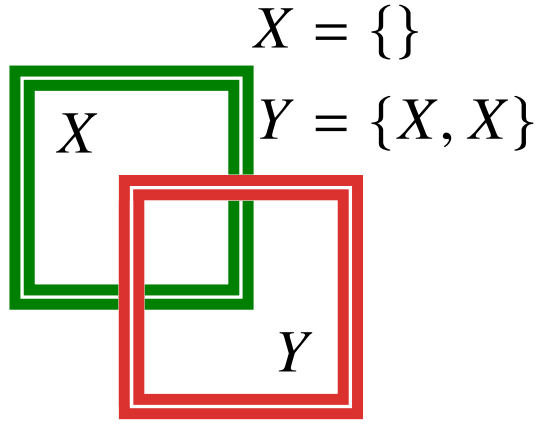


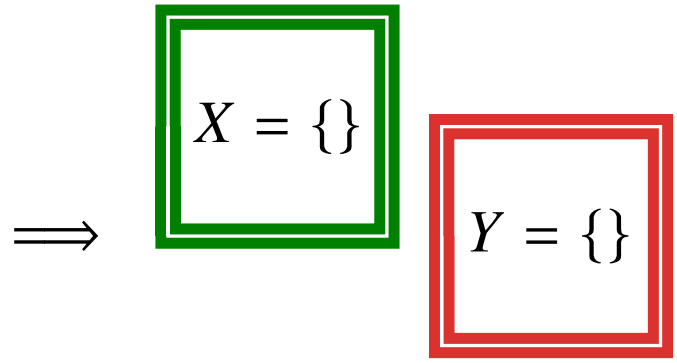

X
Y
X ∈ Y

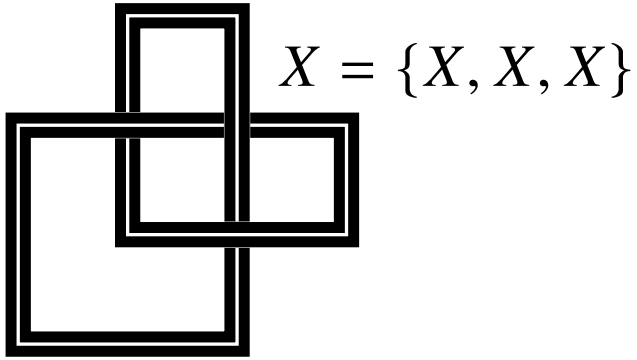

X ∈ X

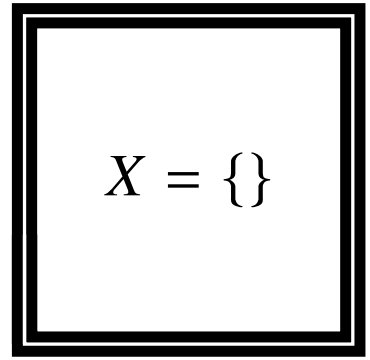

X ∉ X


X = {Y}
Y = {X}
X
Y

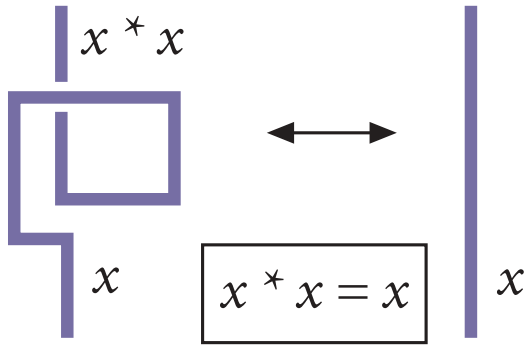

X = {}
Y = {X, X}
X
Y


X = {}
Y = {}

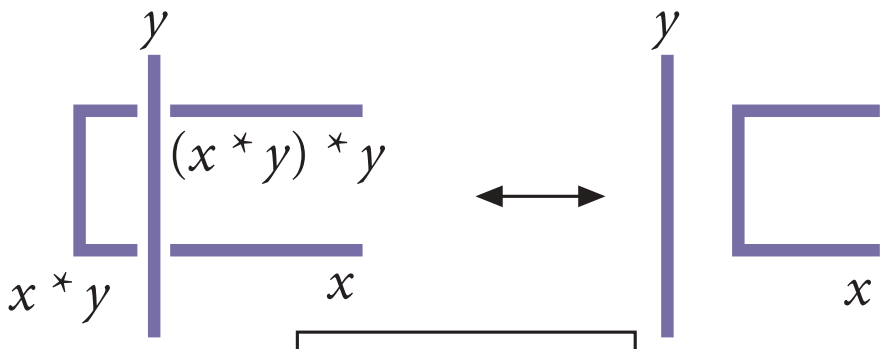

X = {X, X, X}


X = {}

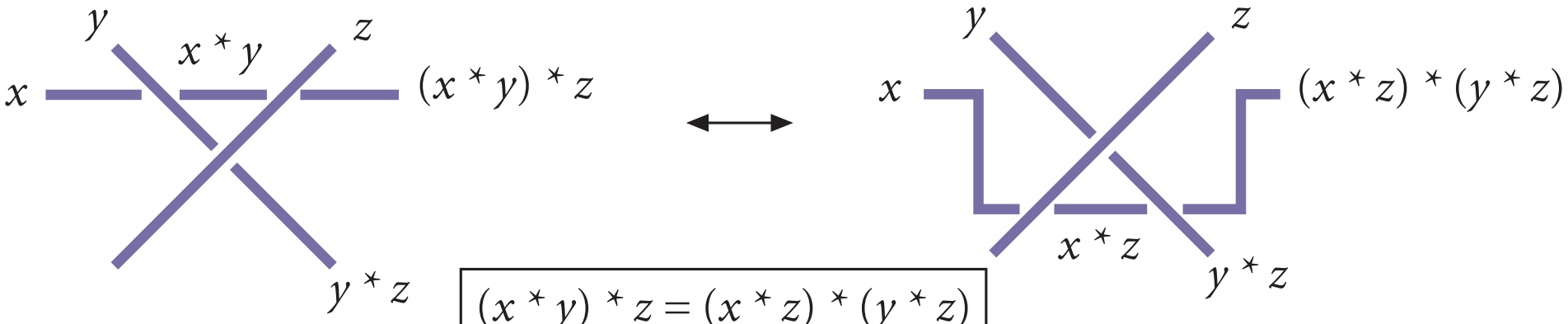
Reidemeister Moves



$$x * x = x$$

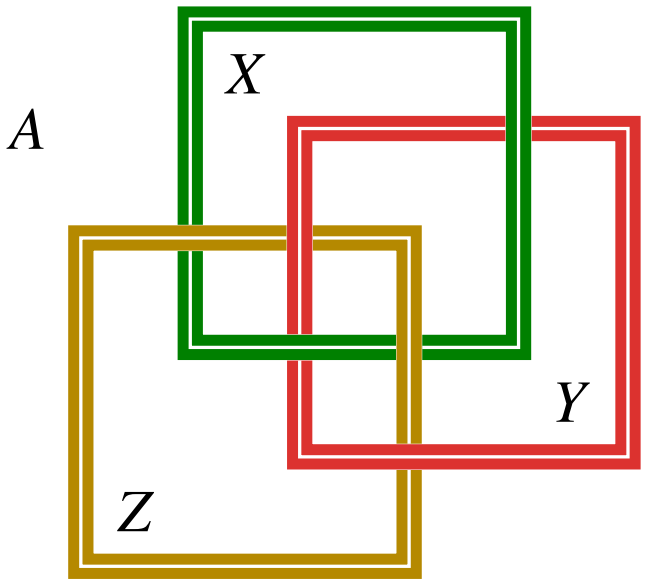


$$(x * y) * y = x$$

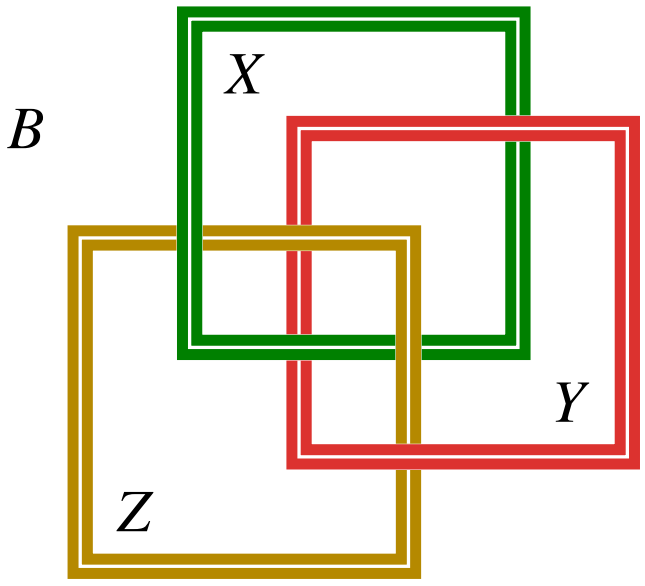


$$(x * y) * z = (x * z) * (y * z)$$

Self-Reference & IIT



$$X = \{Y, Y\}$$
$$Y = \{Z, Z\}$$
$$Z = \{X, X\}$$



$$X = \{Y, Z\}$$
$$Y = \{X, Z\}$$
$$Z = \{X, Y\}$$

Figure: $\Phi(A) < \Phi(B)$?

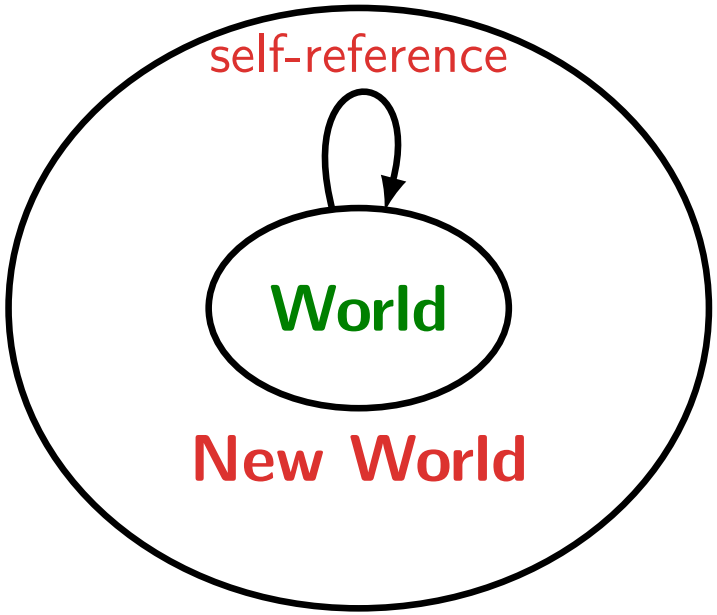
Larger Domain

$$1, 1, 2, 3, 5, 8, 13, 21, 34, \ldots$$

$$F_0 = F_1 = 1; F_{n+1} = F_n + F_{n-1}$$

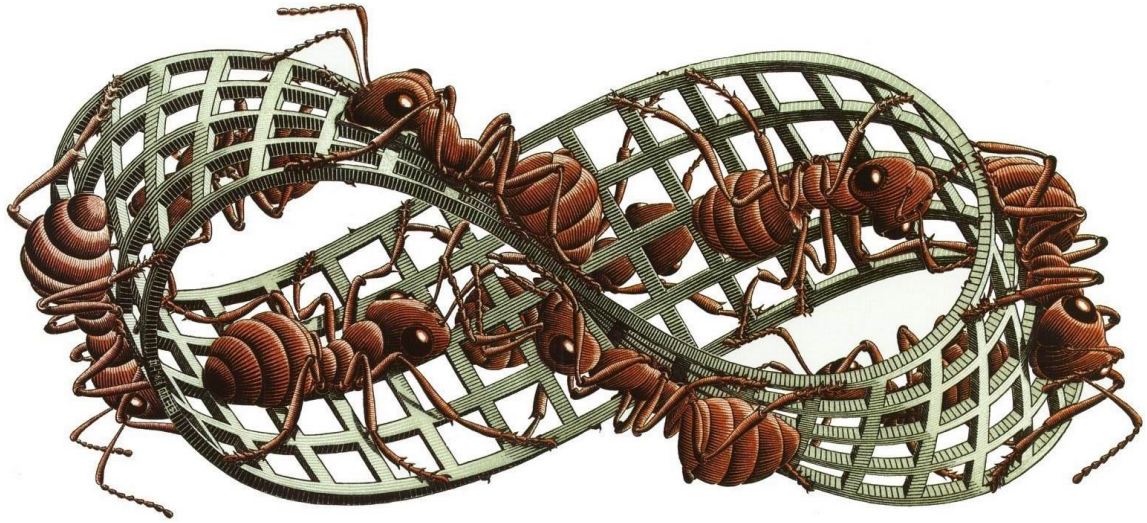
$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n}$$

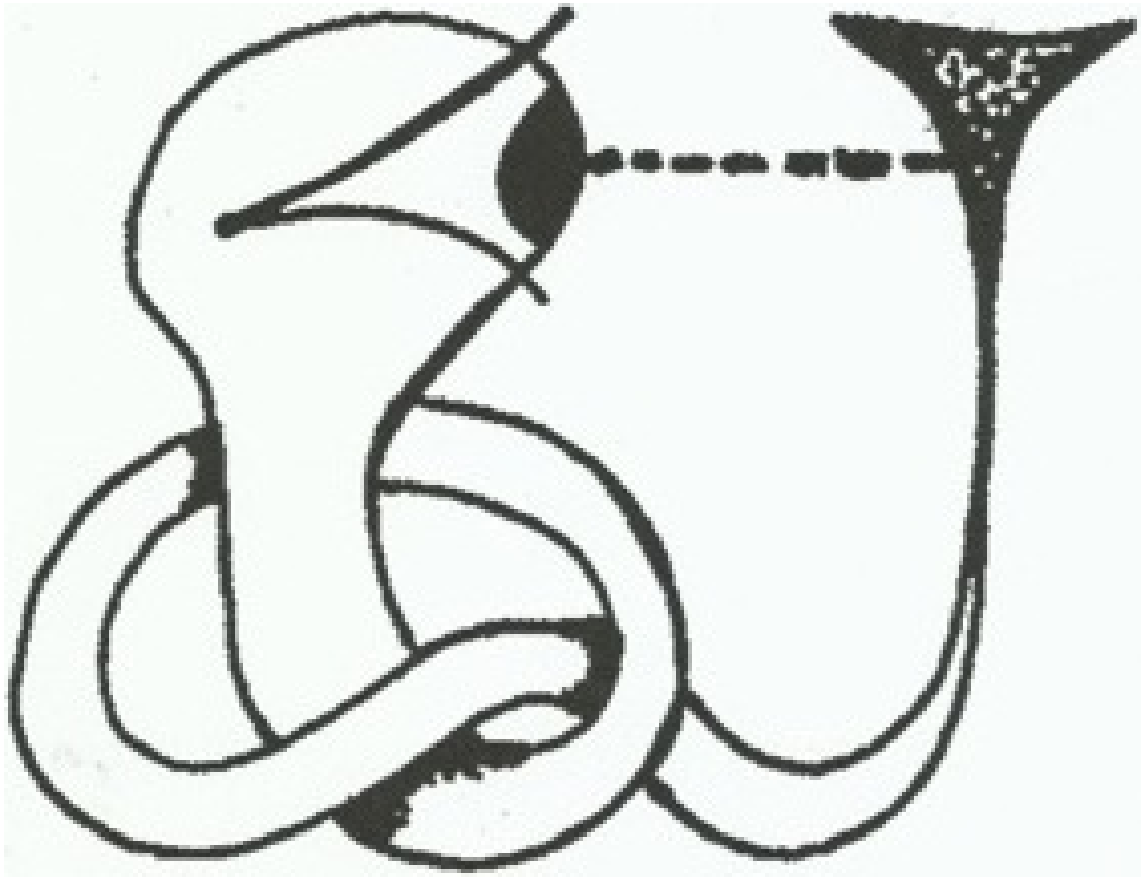
$$\frac{F_{n+1}}{F_n} = 1 + \frac{1}{\frac{F_n}{F_{n-1}}}$$



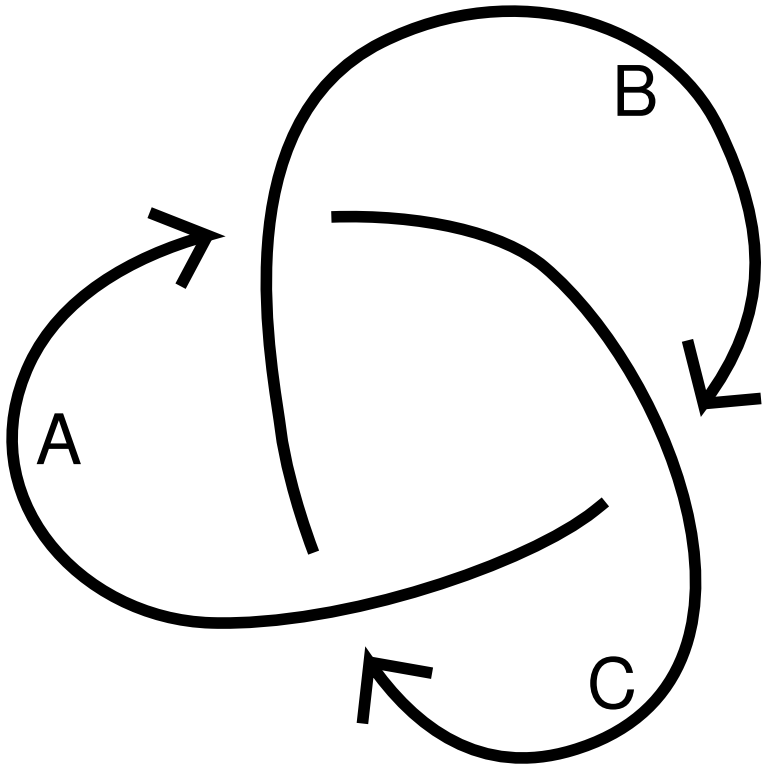
$$f(x) = 1 + \frac{1}{x} = x \implies 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \ddots}}}} = \frac{1+\sqrt{5}}{2}$$

$$f(x) = \frac{-1}{x} = x \implies x = i$$





Trefoil



- objects $\{A, B, C\}$
- morphisms
  - A: $C \to B$
  - B: $A \to C$
  - C: $B \to A$

Nested Virtualization?

从前有座山，山里有座庙，庙里有个老和尚在讲故事：从前有座山…

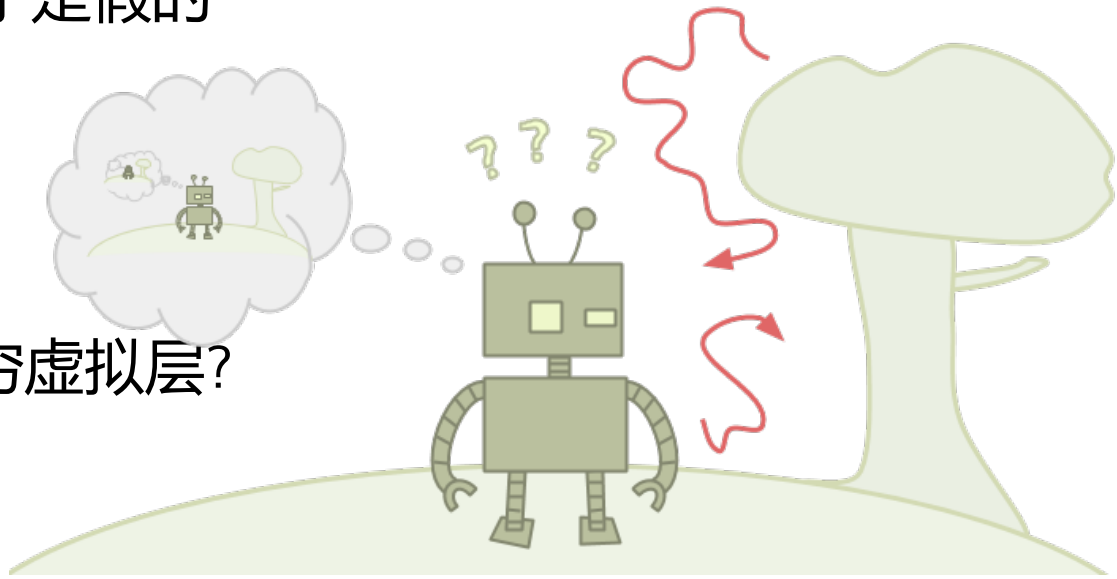
$$\begin{cases} FX = Y \\ GY = X \end{cases}$$

$$X = GFGFGFGF\cdots$$

$$Y = FGFGFGFG\cdots$$

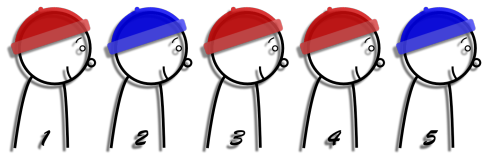
Liar Paradox vs Quine Paradox

1. 这句话是假的
2. “这句话是假的” 是假的
3. ““““““……是假的” 是假的” 是假的” 是假的” 是假的” 是假的
4. 把 “把中的第一个字放到左引号前面，其余的字放到右引号后面，并保持引号及其中的字不变” 中的第一个字放到左引号前面，其余的字放到右引号后面，并保持引号及其中的字不变
5. 把 “把中的第一个字放到左引号前面，其余的字放到右引号后面，并保持引号及其中的字不变得到的句子是假的” 中的第一个字放到左引号前面，其余的字放到右引号后面，并保持引号及其中的字不变得到的句子是假的



无穷虚拟层?

How to Refer? — Encoding



- 100 prisoners are lined up by an jailer, who places a red or blue hat upon each of their heads.
- The prisoners can see the hats of the people lined up in from of them, but they can't look at the hats behind them, or at their own.
- The jailer is going to ask color of each prisoner's hat starting from the last prisoner in queue. If a prisoner tells the correct color, then is saved, otherwise executed.
- How many prisoners can be saved at most if they are allowed to discuss a strategy before the jailer starts asking colors of their hats?

If the first person sees an odd number of red hats he calls out red, if he sees an even number of red hats he calls out blue.

手扶拐杖的外星绅士造访地球. 临别, 人类赠送百科全书: "人类文明尽在其中!". 绅士谢绝: "不, 谢谢! 我只需在拐杖上点上一点".

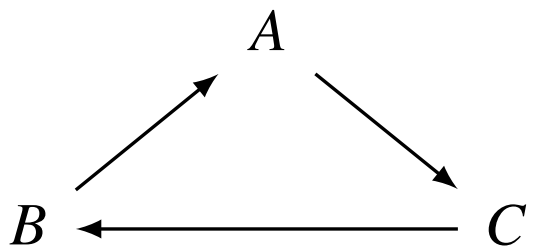
What is the Next Number?

1. 1
2. 11
3. 21
4. 1211
5. 111221
6. 312211
7. ?

A. 11131221131211132221...

B. 31131122211311123113321...

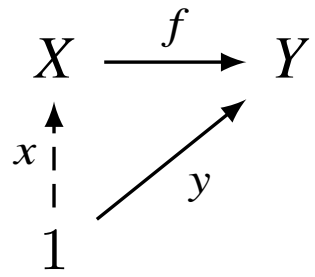
C. 13211321322113311213212321...



Diagonalization[Law69][33]

Definition (Point-Surjective)

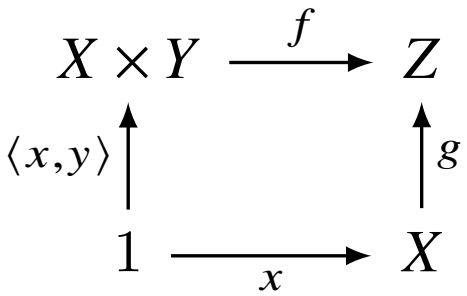
A morphism $f : X \to Y$ is *point-surjective* iff for every $y : 1 \to Y$, there is an $x : 1 \to X$ s.t. $y = f \circ x$.



Definition (Weakly Point-Surjective)

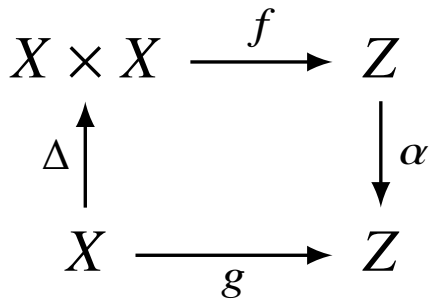
A morphism $f : X \times Y \to Z$ is *weakly point-surjective* iff for every $g : X \to Z$, there exists $y : 1 \to Y$ such that, for all $x : 1 \to X$:

$$g \circ x = f \circ \langle x, y \rangle$$



Theorem (Lawvere's Fixpoint Theorem)

Let $\mathbf{C}$ *be a category with a terminal object and binary products. If* $f : X \times X \to Z$ *is weakly point-surjective, then every* $\alpha : Z \to Z$ *has a fixpoint* $z : 1 \to Z$.



[33] Lawvere: Diagonal arguments and cartesian closed categories.
Yanofsky: A universal approach to self-referential paradoxes, incompleteness and fixed points.

Lawvere's Fixpoint Theorem

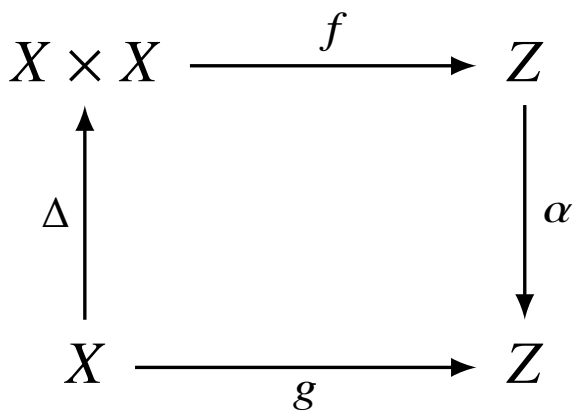
- A function $g : X \to Z$ is *representable* by $f : X \times Y \to Z$ iff

$$\exists y \forall x : g(x) = f(x, y)$$

Theorem (Lawvere's Fixpoint Theorem)

For sets X, Z, functions $f : X \times X \to Z$, $\alpha : Z \to Z$, let $g := \alpha \circ f \circ \Delta$.

1. *If α has no fixpoint, then g is not representable by f.*
2. *If g is representable by f, then α has a fixpoint.*



$$\alpha\big(f\left(\ulcorner g \urcorner, \ulcorner g \urcorner\right)\big) = g\left(\ulcorner g \urcorner\right) = f\left(\ulcorner g \urcorner, \ulcorner g \urcorner\right)$$

- $\Delta : x \mapsto \langle x, x \rangle$ diagonal
- f evaluation
- α "negation"
- $g\left(\ulcorner g \urcorner\right)$ fixpoint-(free) transcendence
- $f\left(\ulcorner g \urcorner, \ulcorner g \urcorner\right)$ self-reference "I have property α."

Lawvere's Fixpoint Theorem

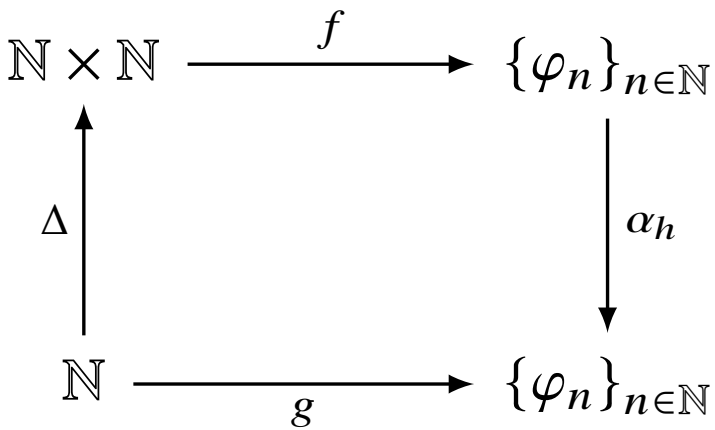
f	0	1	2	3	$\cdots$	t	$\cdots$
0	$\alpha f(0,0)$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$f(0,t)$	$\cdots$
1	$\cdots$	$\alpha f(1,1)$	$\cdots$	$\cdots$	$\cdots$	$f(1,t)$	$\cdots$
2	$\cdots$	$\cdots$	$\alpha f(2,2)$	$\cdots$	$\cdots$	$f(2,t)$	$\cdots$
3	$\cdots$	$\cdots$	$\cdots$	$\alpha f(3,3)$	$\cdots$	$f(3,t)$	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
t	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$\cdots$	$f(t,t)$ $\parallel$ $\alpha f(t,t)$	$\cdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Example — Kleene's Fixpoint Theorem

Theorem (Kleene's Fixpoint Theorem)

Given a recursive function h, there is an index e s.t.

$$\varphi_e = \varphi_{h(e)}$$

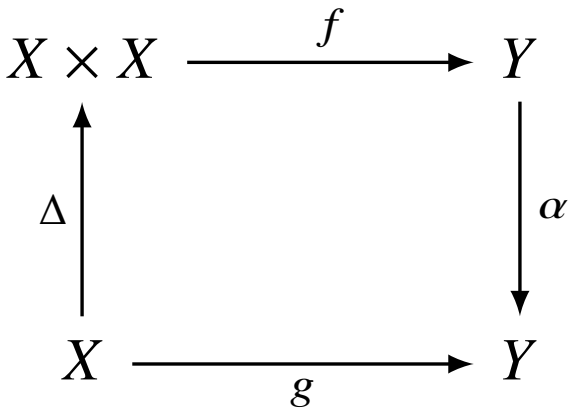


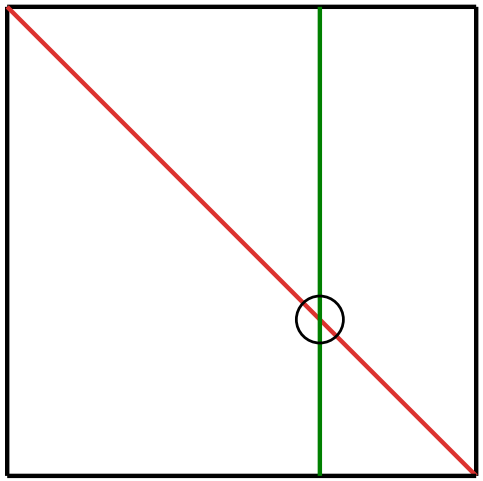
where $f : (m, n) \mapsto \varphi_{\varphi_n(m)}$, and $\alpha_h : \varphi_n \mapsto \varphi_{h(n)}$.
The function $g : m \mapsto \varphi_{h(\varphi_m(m))}$ is a recursive sequence of partial recursive functions, and thus is representable by $f(-, t)$.

$$e := \varphi_t(t)$$

Explicitly, $g(m) = \varphi_{h(\varphi_m(m))} = \varphi_{s(m)} = \varphi_{\varphi_t(m)} = f(m, t)$

Fixpoint vs Diagonalization

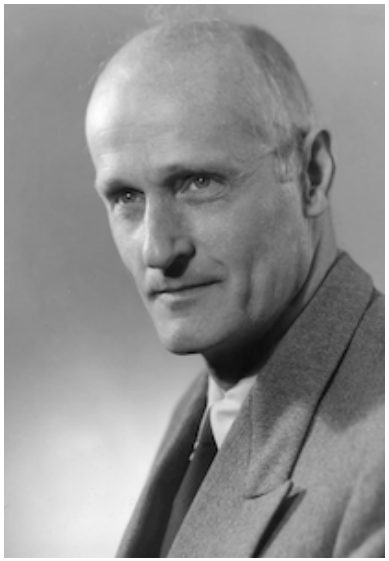




Curry $\mathbf{Y}$	$\hat{=}$	λ-fixpoint	$\hat{=}$	Gödel	$\hat{=}$	Kleene	$\hat{=}$	Russell
yx	$\hat{=}$	$N(\ulcorner M \urcorner)$	$\hat{=}$	$N(\ulcorner M(x) \urcorner)$	$\hat{=}$	$\varphi_n(m)$	$\hat{=}$	$x \in y$
xx	$\hat{=}$	$M(\ulcorner M \urcorner)$	$\hat{=}$	$M(\ulcorner M(x) \urcorner)$	$\hat{=}$	$\varphi_n(n)$	$\hat{=}$	$x \in x$
$y(xx)$	$\hat{=}$	$F\ulcorner M \ulcorner M \urcorner\urcorner$	$\hat{=}$	$F(\ulcorner M(\ulcorner M(x) \urcorner) \urcorner)$	$\hat{=}$	$h(\varphi_n(n))$	$\hat{=}$	$x \notin x$
$\lambda x.y(xx)$	$\hat{=}$	G	$\hat{=}$	$G(x)$	$\hat{=}$	$\varphi_t(n)$	$\hat{=}$	$x \notin R$
$(\lambda x.y(xx))(\lambda x.y(xx))$	$\hat{=}$	$G(\ulcorner G \urcorner)$	$\hat{=}$	$G(\ulcorner G(x) \urcorner)$	$\hat{=}$	$\varphi_t(t)$	$\hat{=}$	$R \notin R$

self-reference $\overset{?}{\Longrightarrow}$ self-improvement

Kleene's Fixpoint Theorem



Theorem (Second Recursion Theorem)

If $f(x, y)$ is a partial recursive function, there is an index e s.t.

$$\varphi_e(y) = f(e, y)$$

Remark: 对于任意的程序 h, 总存在某个程序 e, 执行程序 e 的结果等价于把程序 e 当作数据输入给程序 h 执行的结果 $[\![e]\!](-) = [\![h]\!](e, -)$.

Theorem (Kleene's Fixpoint Theorem)

Given a recursive function h, there is an index e s.t.

$$\varphi_e = \varphi_{h(e)}$$

Remark: 语法上, 你可以系统地改变程序的代码 $n \mapsto h(n)$, 但语义上, 你无法系统地改变程序的逻辑功能 $\varphi_e = \varphi_{h(e)}$.

Remark: 杀毒软件的功能是通过一个能行的方法 h 系统性地修改病毒 V_n 的代码, 但根据 Kleene's Fixpoint 定理, 存在杀不死的病毒 $V_e = V_{h(e)}$.

From Kleene's Fixpoint to Chaitin's Incompleteness

Definition: Kolmogorov Complexity $K(x) \coloneqq \mu e[\varphi_e(0) = x]$

Theorem (Chaitin's Incompleteness Theorem)

For any arithmetically sound Gödelian theory T, $\exists c \forall x : \mathrm{T} \nvdash K(x) > c$.

Proof.

For any m, we can construct:

$$M_n \coloneqq \text{“find } \mu y\big[\,\mathrm{prf}_{\mathrm{T}}\left(y, K(x) > m\right)\big], \text{output } x\text{”}$$

So there exists a recursive function $f : m \mapsto n$.
By Kleene's fixpoint theorem, there exists e such that

$$M_e = M_{f(e)} = \text{“find } \mu y\big[\,\mathrm{prf}_{\mathrm{T}}\left(y, K(x) > e\right)\big], \text{output } x\text{”}$$

Take $c \coloneqq e$.

Remark: For almost all random strings their randomness cannot be proved.

Self-Reproducing Program/Quine

There is a program that outputs its own length.

There is a program that outputs its own source code.

- A Quine is a program which takes no input and outputs its own source code.
- Quines are algorithmic random.

Corollary (Self-Reproducing Program)

There is a recursive function φ_e *s.t.* $\forall x : \varphi_e(x) = e$.

Quine in Python

```
s='s=%r; print(s%%s)'; print(s%s)
```

Quine in Lambda Calculus

$$(\lambda x.xx)(\lambda x.xx)$$

Self-Reproducing Program

Print two copies of the following, the second copy in quotes:

"Print two copies of the following, the second copy in quotes:"

DNA / mutation / evolution

Build a baby that acts on the following instructions, and also contains a copy of those instructions in its reproductive organs.

"Build a baby that acts on the following instructions, and also contains a copy of those instructions in its reproductive organs."

von Neumann's Self-Reproducing Automata

1. A universal constructor A.

$$A + \ulcorner X \urcorner \rightsquigarrow X$$

2. A copying machine B.

$$B + \ulcorner X \urcorner \rightsquigarrow \ulcorner X \urcorner$$

3. A control machine C, which first activates B, then A.

$$A + B + C + \ulcorner X \urcorner \rightsquigarrow X + \ulcorner X \urcorner$$

4. Let $X := A + B + C$. Then $A + B + C + \ulcorner A + B + C \urcorner$ is self-reproducing.

$$A + B + C + \ulcorner A + B + C \urcorner \rightsquigarrow A + B + C + \ulcorner A + B + C \urcorner$$

5. It is possible to add the description of any machine D.

$$A + B + C + \ulcorner A + B + C + D \urcorner \rightsquigarrow A + B + C + D + \ulcorner A + B + C + D \urcorner$$

6. Now allow mutation on the description $\ulcorner A + B + C + D \urcorner$.

$$A + B + C + \ulcorner A + B + C + D' \urcorner \rightsquigarrow A + B + C + D' + \ulcorner A + B + C + D' \urcorner$$

Introspective Program

Definition (ψ-introspective)

Given a total recursive function ψ,

- the *ψ-analysis* of $\varphi(x)$ is the code of the computation of $\varphi(x)$ to $\psi(x)$ steps.
- φ is *ψ-introspective* at x iff $\varphi(x)\downarrow$ and outputs its own ψ-analysis.
- φ is *totally ψ-introspective* iff it is ψ-introspective at all x.

Corollary

There is a program that is totally ψ-introspective.

Proof.

Let $f(n,x) :=$ "the ψ-analysis of $\varphi_n(x)$".

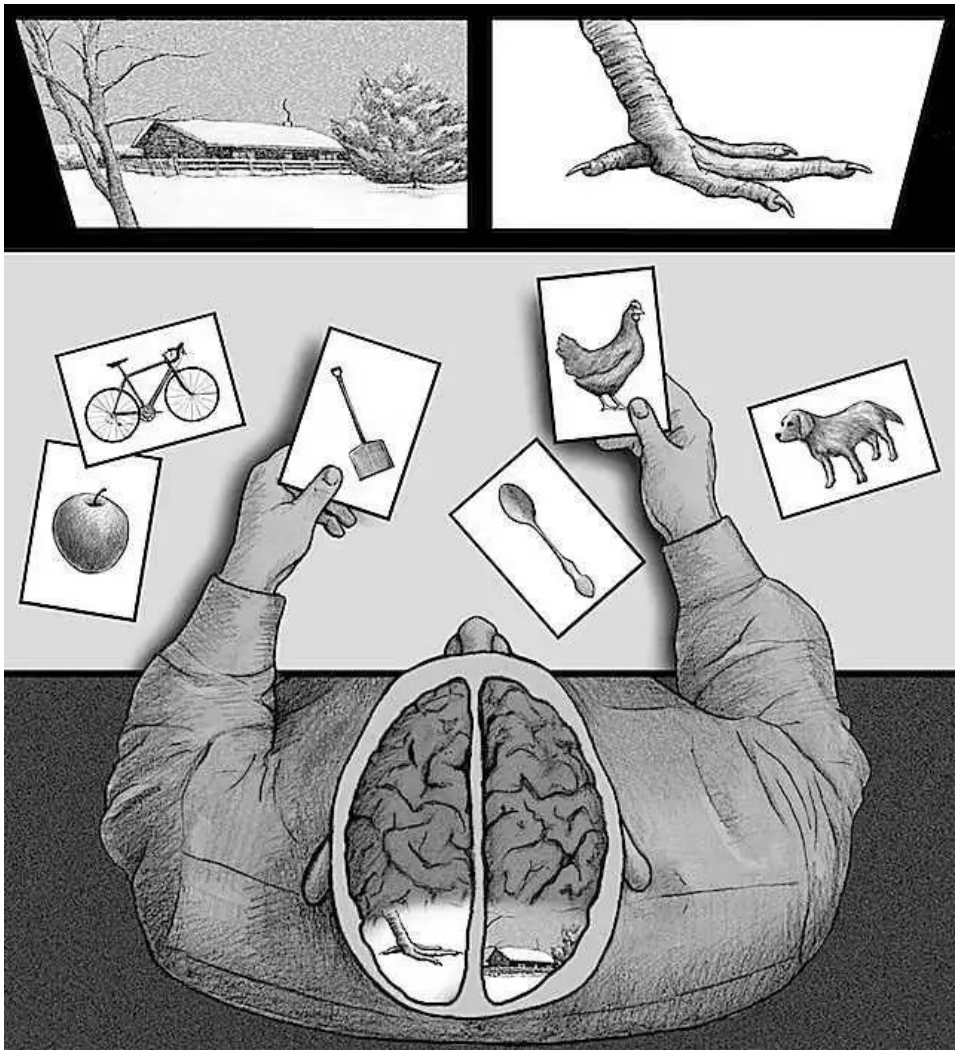
Introspective Program

There is a program that is totally introspective.

$$\varphi_{\mathbf{e}} = \varphi_{\mathbf{h}(\mathbf{e})}$$

Self-simulating Computer	Self-consciousness
Host Machine	Experiencing Self
Virtual Machine	Remembering Self
Hardware	Body

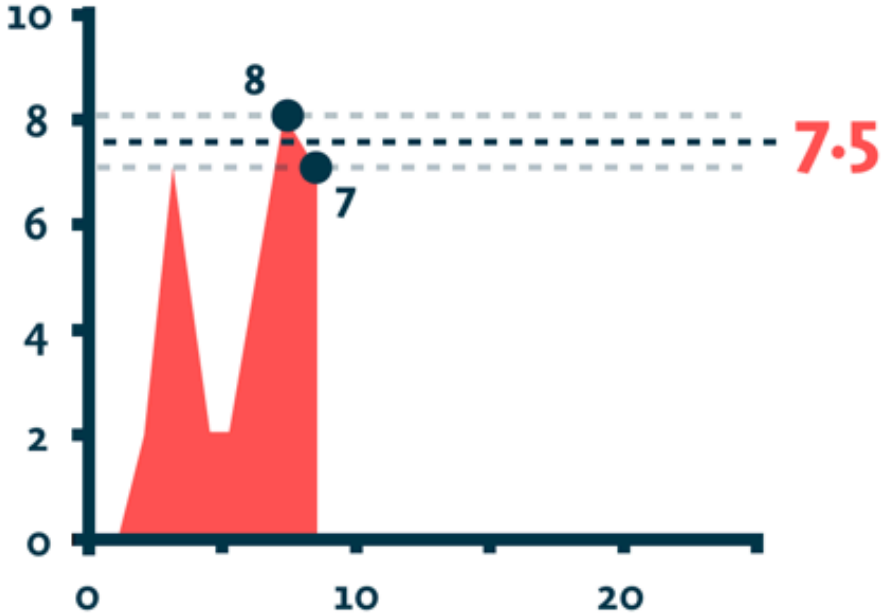
裂脑人实验

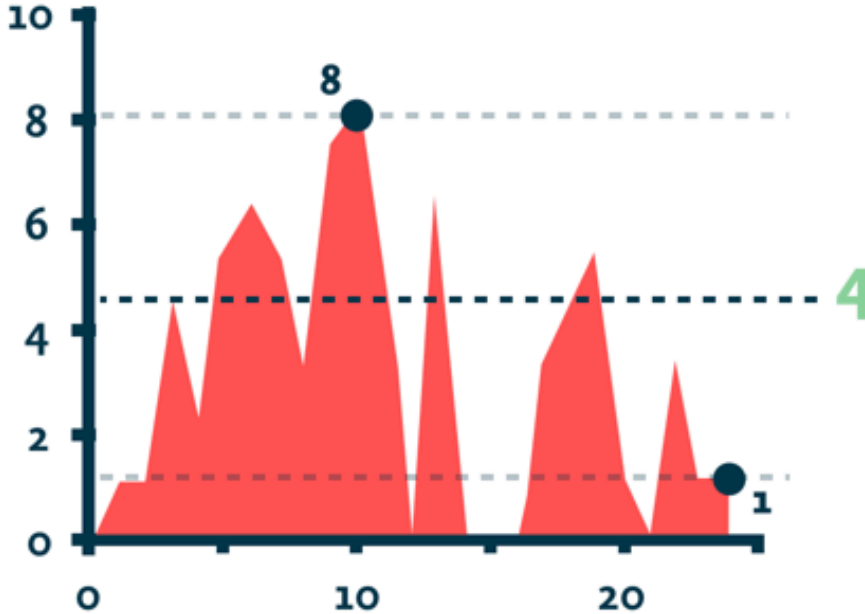


- 对于胼胝体被切断的人，信息无法在两个脑半球之间传递.
- 左眼看到雪，传导到右脑.
- 右眼看到鸡爪，传导到左脑.
- 左脑负责语言，右脑负责空间.
- 左手受右脑控制，右脑看到了雪，控制左手选了铲子 (铲雪).
- 右手受左脑控制，左脑看到了鸡爪，控制右手选了鸡 (鸡爪).
- 问: 你左手为什么拿铲子?
- 答: 清理鸡舍!
- life as a story
  — 其实右脑是想铲雪，左脑事后用语言编造理由合理化自己的行为.

卡尼曼 Kahneman — 《思考, 快与慢》

- 忍受痛苦的冰手实验: 为什么你更愿意多忍受一些痛苦?





- 持续时间忽略 (Duration Neglect)
- 峰终定律 (Peak-End Rule)

Remark: 回顾性评估只会关注高峰和结尾, 事件过程对记忆几乎没有影响. 高峰之后, 终点出现得越迅速, 印象越深.

1. 经验自我 (Experiencing Self)
2. 记忆自我 (Remembering Self)

Figure: One can imagine a detailed floor plan of a room, sitting on a table in the room; this plan has an image of the table on which there is an image of the plan itself. Now introduce the dynamical aspect: the items on the plan are cut out from paper and can be moved to try a different furniture arrangement; in this way the plan models possible states of the world about which it carries information.

Manin — Cognitive Networks

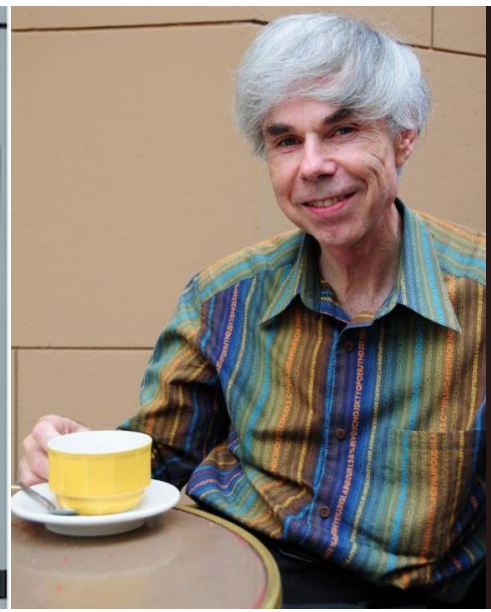
The brain contains inside a map of itself, and some neural information channels in the central neural system:

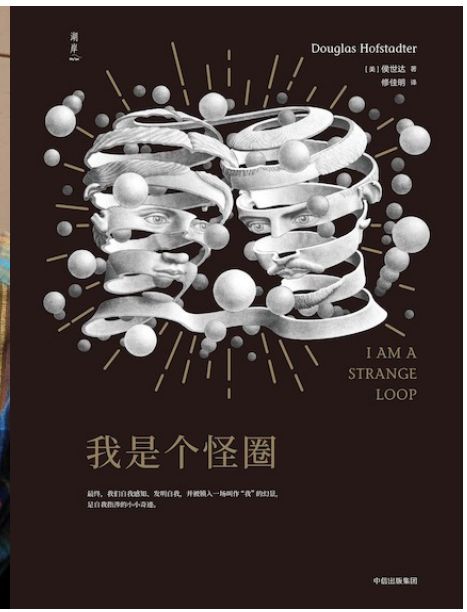
- carry information about the mind itself, i.e. are reflexive;
- are capable of modelling states of the mind different from the current one, i.e. possess a modelling function;
- can influence the state of the whole mind and through that, the behavior, i.e. possess controlling function.

The reflection of the brain inside itself must be coarse grained.

侯士达 — 《我是个怪圈》

- 有没有意识取决于在哪个层级上对结构进行观察. 在整合度最高的层级上看, 大脑是有意识的. 下降到微观粒子层面, 意识就不见了.
- 意识体是那些在某个描述层级上表现出某种特定类型的循环回路的结构. 当一个系统能把外部世界过滤成不同的范畴、并不断向越来越抽象的层级创造新的范畴时, 这种循环回路就会逐渐形成.
- 当系统能进行自我表征 — 对自己讲故事 — 的时候, 这种循环回路就逐渐变成了实体的 “我” — 一个统一的因果主体.





说谎者悖论	**我在说谎**
Grelling 悖论	“非自谓的” 是自谓的吗
Russell 悖论	“不属于自身的集合的集合” 属于自身吗
Berry 悖论	我是少于十八个字不可定义的最小数
Yablo 悖论	我下一句及后面所有的句子都是假的
Gödel 不动点引理	**我有性质** F
Tarski 算术真不可定义定理	我不真
Gödel 第一不完备性定理	我不可证
Gödel-Rosser 不完备性定理	对于任何一个关于我的证明，都有一个更短的关于我的否定的证明
Löb 定理	如果我可证，那么 A
Curry 悖论	如果我是真的，那么上帝存在
Parikh 定理	我没有关于自己的长度短于 n 的证明
Kleene 不动点定理	**我要进行** h **操作**
Quine 悖论	把 “把中的第一个字放到左引号前面，其余的字放到右引号后面，并保持引号及其中的字不变得到的句子是假的” 中的第一个字放到左引号前面，其余的字放到右引号后面，并保持引号及其中的字不变得到的句子是假的
自测量长度程序	我要输出自己的长度
自复制程序	我要输出自己
自反省程序	我要回顾自己走过的每一步
Gödel 机	我要变成能获取更大效用的自己

Schmidhuber's Gödel Machine

- The Gödel machine consists of a **Solver** and a **Searcher** running in parallel.
- The **Solver** (AIXI^S/AIXI$^{t\ell}$) interacts with the environment.
- The **Searcher** (LSEARCH/HSEARCH/OOPS) searches for a proof of "the modification of the software — including the *Solver* and *Searcher* — will increase the expected utility than leaving it as is".
- Logic: a theorem prover and a set of self-referential axioms, which include a description of its own software and hardware, and a description of the probabilistic properties of the environment, as well as a user-given utility function.
- *Since the utility of "leaving it as is" implicitly evaluates all possible alternative modifications, the current modification is globally optimal w.r.t. its initial utility function.*

Gödel Machine

- language $\mathscr{L} := \{\neg, \wedge, \vee, \rightarrow, \forall, \exists, =, (,), \ldots, +, -, \cdot, /, <, \ldots\}$
- well-formed formula
- utility function $u(s, e) = \mathbb{E}_\mu \left[\sum_{t=1}^{T} r_t \,\middle|\, s, e \right]$
- target theorem

  $$u\left[s(t) \oplus \left(\texttt{switchbit}(t) = 1\right), e(t)\right] > u\left[s(t) \oplus \left(\texttt{switchbit}(t) = 0\right), e(t)\right]$$

- theorem prover

  hardware, costs, environment, initial state, utility, logic/arithmetic/probability

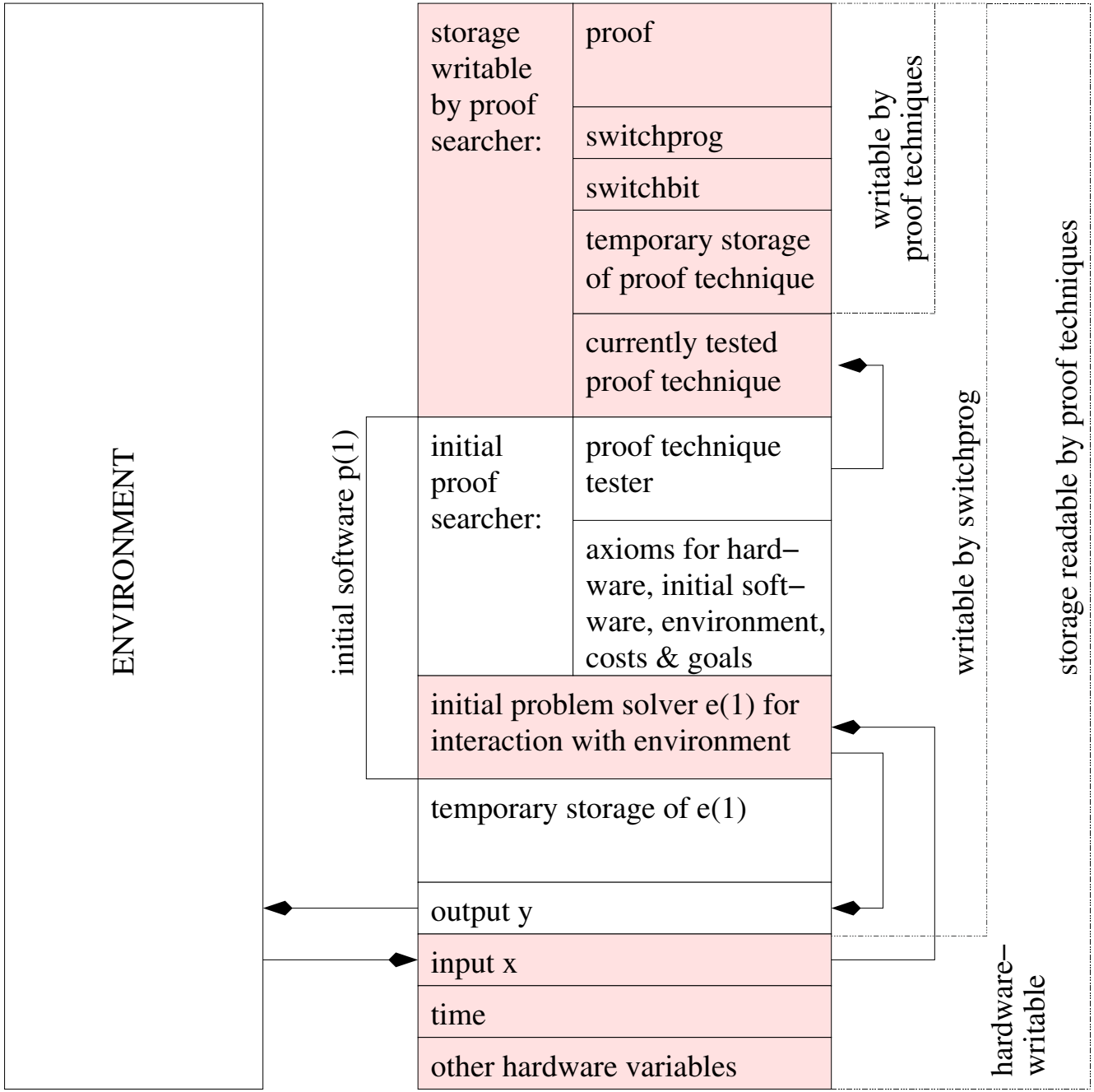
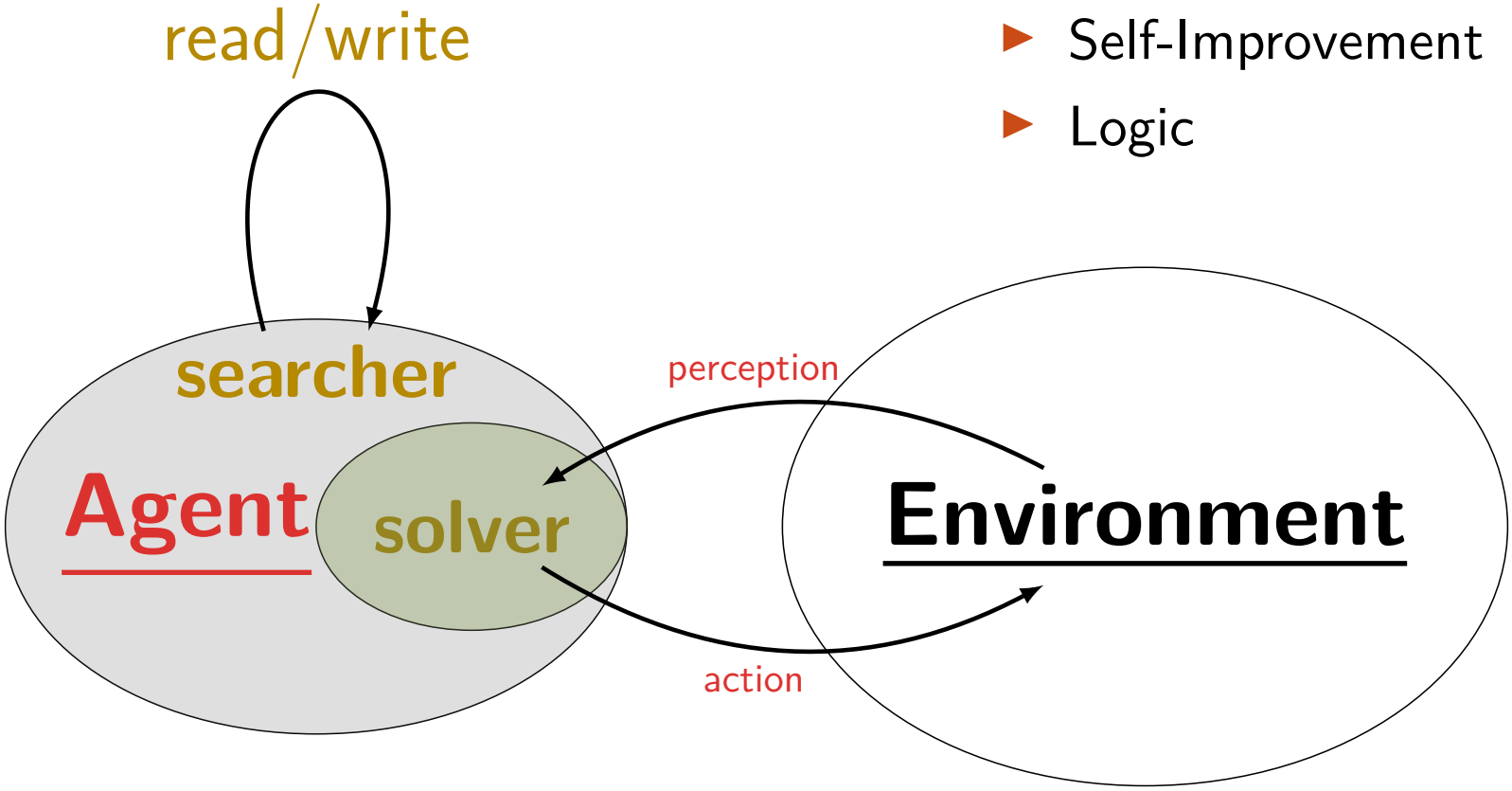


Figure: Schmidhuber

Gödel Machine

- GRL
- Universal Search
- Self-Improvement
- Logic

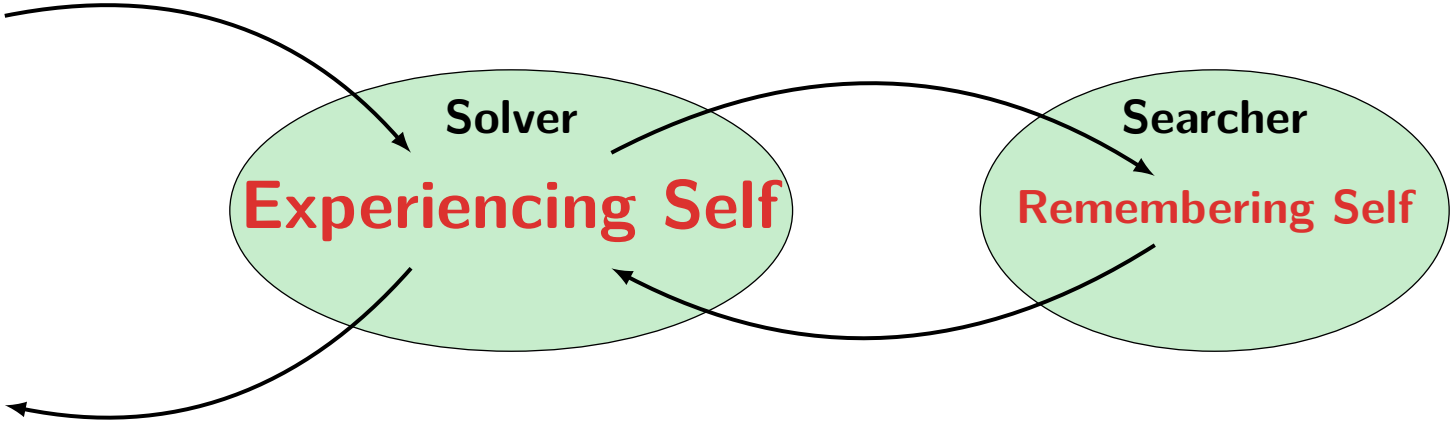


Disadvantage: A Gödel Machine with a badly chosen utility function is motivated to converge to a "poor" program. (goal orthogonality!)

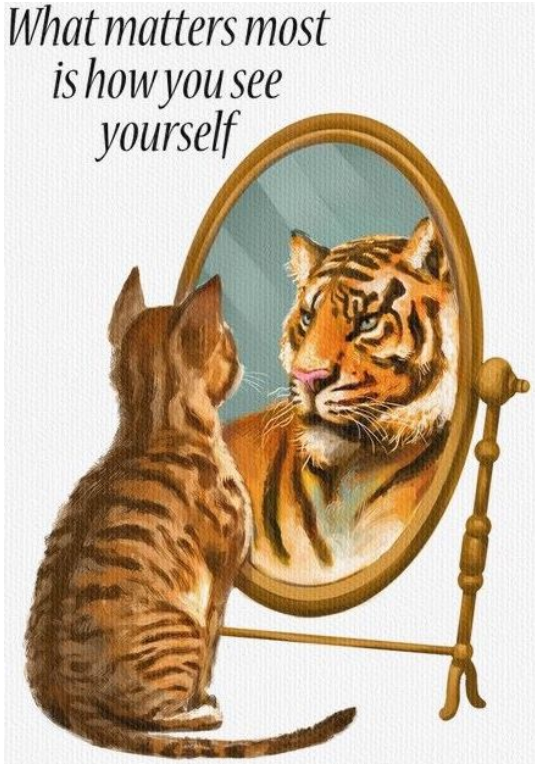
Gödel Machine vs Self-Consciousness vs Free Will?

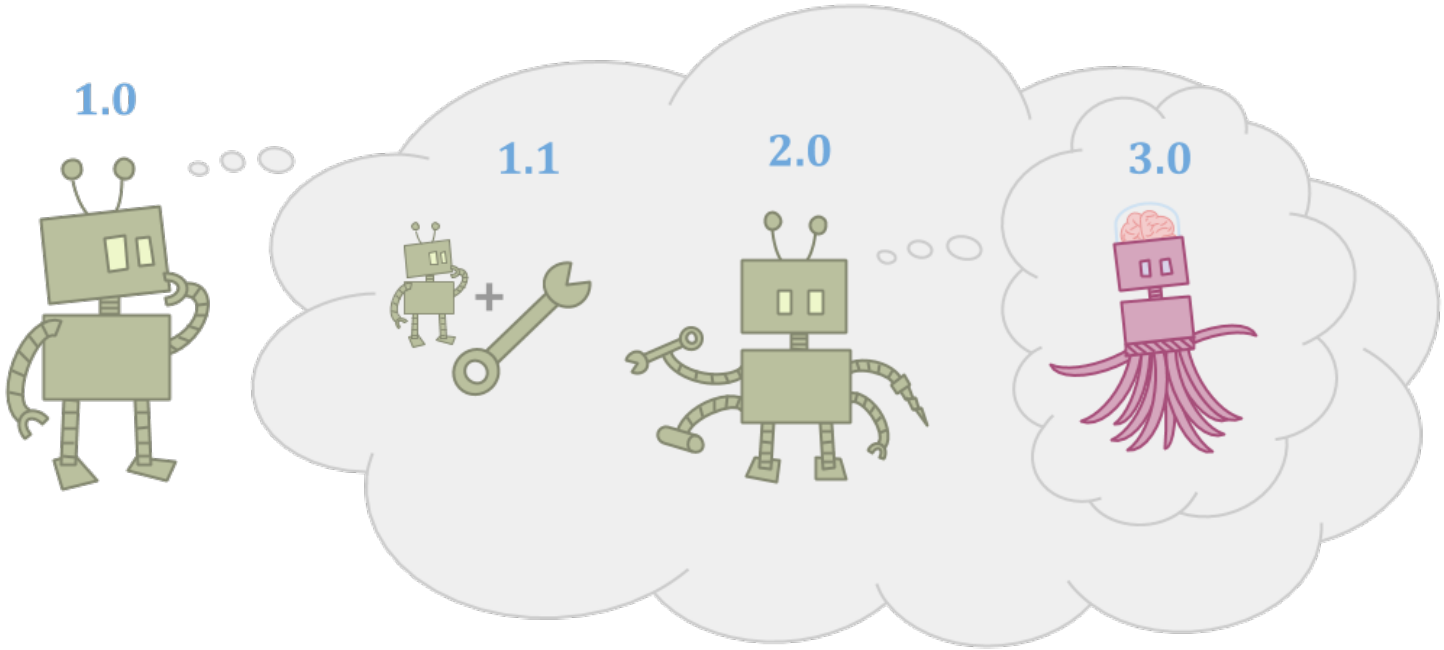
Self-simulating Computer	Gödel Machine	Self-consciousness
Host Machine	Solver	Experiencing Self
Virtual Machine	Searcher	Remembering Self
Hardware	Hardware	Body

$$\varphi_{\mathbf{e}} = \varphi_{\mathbf{h(e)}}$$



self-reference $\overset{?}{\Longrightarrow}$ self-improvement


What matters most
is how you see
yourself


1.0
1.1
2.0
3.0

Gödel Machines

1. *one-shot* self-improvement: Kleene's fixpoint theorem

$$\varphi_e = \varphi_{h(e)}$$

   - global optimality?
   - goal orthogonality? ends vs means
2. *continuous* self-improvement: Kleene's fixpoint theorem with parameters

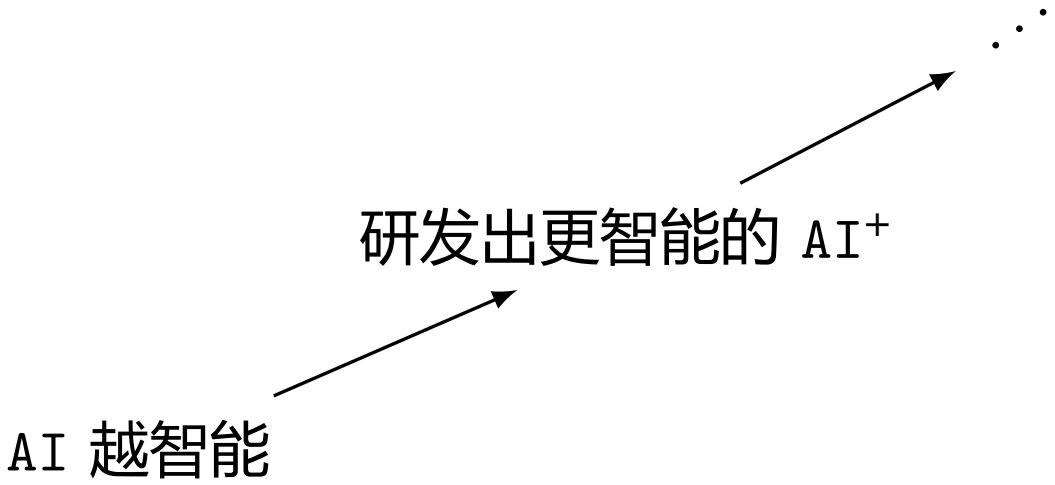
$$\varphi_{e(y)} = \varphi_{h(e(y),y)}$$

   - "real-time" optimality. human-computer interaction?
   - intelligent explosion / technological singularity???
     continuous self-improvement ≠ exponential iteration
3. *beyond computability*: Kleene's relativized fixpoint theorem

$$\varphi^A_{e(y)} = \varphi^A_{h(e(y),y)}$$

   - Gödel Machine PK AIXI$^{t\ell}$
   - Gödel Machine PK AIXI

冯诺依曼的复杂度阈值



冯诺依曼在讨论自复制自动机时指出：存在某个复杂度阈值，如果系统的复杂度没有超过这个阈值，就会不断衰退；而一旦超越了这个阈值，就会持续进化.

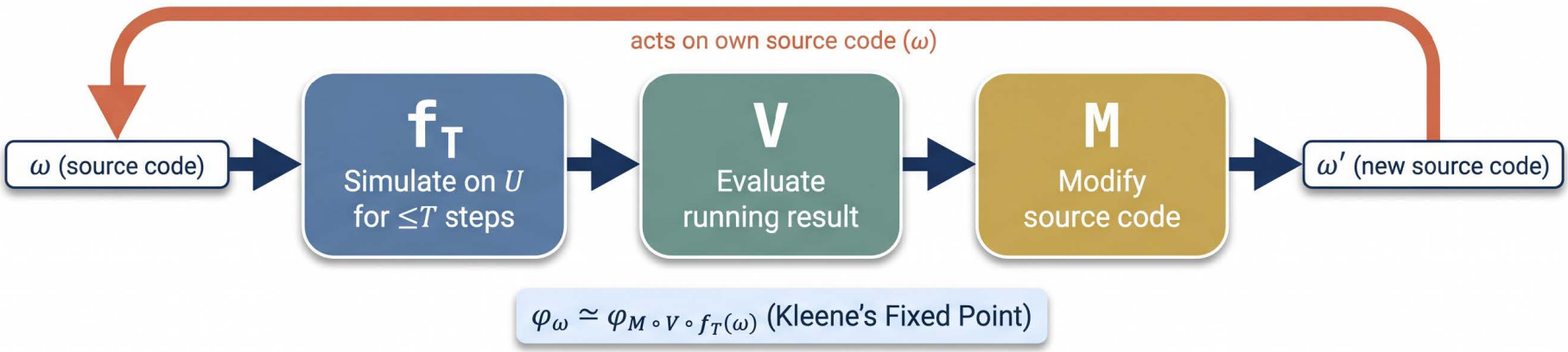
1. 复杂度阈值怎么给?
2. 怎么持续进化?

张江的自指内省阈值 [ZYZ26]

哥德尔机: “我要变成能获取更大效用的自己”.

自我建模 → 自我模拟 → 自我评估 → 自我修改

Self-Referential Pipeline: $f \equiv M \circ V \circ f_T$



Recursive Sequence: $S_0 \to S_1 \to S_2 \to \cdots \to S_n$

Each S_{i+1} is the improvement of S_i — output, not equivalence

The first introspective self-improved program S_0 is the threshold — beyond it, recursive self-improvement is automatic

为什么不能直接用 Kleene 不动点 $\varphi_e \simeq \varphi_{g(e)}$ 升级?

- 令 $g := M \circ V \circ f_T$.

$$S_0 := e$$
$$S_n := g^n(S_0)$$

- 在复杂环境下, 如果 RSI 对智能有实质性提升 $V(S_n) \gg V(S_0)$, 则意味着 S_n 蕴含了关于环境 Q 的更多信息:

$$I(S_n; Q) \gg I(S_0; Q)$$

- 但由于确定性算法不会无中生有地增加 "关于目标的信息":

$$I(S_n; Q) = I(g^n(S_0); Q) \overset{+}{\leq} I(S_0; Q) + K(g) + K(n)$$

  矛盾!
- 直接迭代 g 受限于上述 "数据处理不等式".
- 怎么在递归自我升级中不断纳入新信息? — 引入参数.

Theorem (Levin's Laws of Information Conservation)

1. *For any computable function* $f : \mathbb{N} \to \mathbb{N}$,

$$I(f(x); y) \overset{+}{\leq} I(x; y)$$

2. *For all* x, y, z,

$$\sum_z m(z \mid x^*) 2^{I(z;y)-I(x;y)} \overset{\times}{\leq} 1$$

Remark: For example, the probability of an increase of mutual information by the amount n is $\overset{\times}{\leq} 2^{-n}$.

- *The Information* $I(x; y)$ *has a remarkable invariance: It cannot be increased by deterministic algorithms or in random processes or with any combinations of both.*
- *This is natural, since if* x *contains no information about* y *then hope is little to find out something about* y *by processing* x.
- *Torturing an uninformed witness cannot give information about the crime!*

— *Leonid A. Levin*

用带参数的 Kleene 改写升级过程

Theorem (带参数的 Kleene 不动点定理)

对全可计算函数 $g(x,y)$, 存在全可计算函数 e, 使得对任意 y:

$$\varphi_{e(y)} \simeq \varphi_{g(e(y),y)}$$

- 令 $g(x,y) := M(V(f(x,y)))$. 其中 M 负责修改程序, V 是评估函数, f 负责模拟程序 x 在 y 约束下的运行.
- 令 $y_0 := \langle h_\epsilon, c_0 \rangle$ 为初始给定的先验信息 h_ϵ 和计算资源 c_0. 计算资源 c_t 的作用是把可能不停机的情形截断为总会停机的全函数.
- Kleene 给出第一个自省程序 $S_0 := e(y_0)$.
- $S_1 := g(S_0, y_0)$, 所以 $\varphi_{S_0} \simeq \varphi_{S_1}$, 所以 $V(S_0) = V(S_1)$.
- 后续的 S_n 由 g 迭代产生: $S_{n+1} := g(S_n, y_n)$. 其中 y_n 加入了第 n 轮与环境 Q 交互得到的反馈信息.
- 从 $S_2 := g(S_1, y_1)$ 开始启动升级之路:

$$V(S_0) = V(S_1) < V(S_2) < V(S_3) \cdots$$

新信息的来源

- Levin: Torturing an uninformed witness cannot give information about the crime!
- 增加的信息 $I(S_n;Q) - I(S_0;Q)$ 主要源于环境反馈 $\sum_{i=0}^{n-1} I(y_i;Q \mid S_i)$.

$$I(S_n;Q) - I(S_0;Q) \stackrel{+}{\leq} \sum_{i=0}^{n-1} I(y_i;Q \mid S_i) + K(g) + K(n)$$

“Scaling Up” dominates “Weak RSI”

- 不递归自我升级, 仅仅 Scaling Up 会如何?
- 直接利用带参数的 Kleene, 令 $S'_n := e(y_n)$, 其中 y_n 编码了更多与环境 Q 交互得到的反馈信息, 不妨假设 y_n 编码的计算资源也是随 n 递增的.
- 不同于 S_n 由前一代 S_{n-1} 借助 g 迭代产生, S'_n 是直接由 Kleene 不动点函数 e 随参数 y_n(“数据 + 算力”)Scaling Up 得到的.
- 问题是, S_n 和 S'_n 谁强?
- 根据 M, V, f 的定义, $e(y_n)$ 是 y_n 限制下的最佳候选, $V(S'_n) \geq V(S_n)$.
- 虽然从外部全局视角看, 孰优孰劣无法判定, 但从资源受限的内部局部视角看, Scaling Up 比 Weak RSI 更好.
- 所以, 只有当优化器 $g = M \circ V \circ f$ 本身也能够被优化时, 递归自我升级才有意义.
- 这就迫使我们不得不迈向优化器也被优化的 Strong RSI.

步步 Kleene 的 Strong RSI

- 规定程序空间 Π 中的标准程序格式为 $\langle g, \pi \rangle$, 其中 π 负责与环境交互做任务, $g(x, y)$ 是优化器.
- 初始程序为 $S_0 := \langle g_0, \pi_0 \rangle$.
- 给定 g_0, 借助带参数的 Kleene, 存在不动点函数 e_0, 作用在 y_0 上得到某个 $\langle g_1, \pi_1 \rangle$, 令 $S_1 := e_0(y_0) = \langle g_1, \pi_1 \rangle$.
- 给定 g_1, 借助带参数的 Kleene, 存在不动点函数 e_1, 作用在 y_1 上得到某个 $\langle g_2, \pi_2 \rangle$, 令 $S_2 := e_1(y_1) = \langle g_2, \pi_2 \rangle$.
- 所以, 只用一次带参数的 Kleene 是不够的, 每一轮都要用一次 Kleene.
- 能否不用一步一调用 Kleene, 而是统一处理呢?

Theorem (带参数的 Uniform Kleene 不动点定理)

存在全可计算函数 r, 使得对任意 $n, y \in \mathbb{N}$,

$$\varphi_{r(n,y)} \simeq \varphi_{\varphi_n(r(n,y),y)}$$

统一的 Strong RSI

有了带参数的 Uniform Kleene, 就可以不用每轮调用一次带参数的 Kleene, 可以直接借助 "带参数的 Uniform Kleene 不动点生成器" r 来统一处理 Strong RSI 的升级之路:

$$S_t = \langle g_t, \pi_t \rangle \xrightarrow{r} e_t(y) := r(\ulcorner g_t \urcorner, y) \xrightarrow{y_t} S_{t+1} := e_t(y_t) = \langle g_{t+1}, \pi_{t+1} \rangle$$

至此, 我们就有了统一的 "Strong RSI" 升级之路, 除了借助 "带参数的 Uniform Kleene 不动点生成器" r 外, 我们不需要固定某个不变的优化器或 "元" 或 "元元" 优化器.

Problem: 令 $g_0 := M \circ V \circ f$, 取一个比如强化学习策略作为 π_0, $S_0 := \langle g_0, \pi_0 \rangle$ 能否借助 r 开启自动升级路?

冯诺依曼的复杂度阈值

冯诺依曼：存在某个复杂度阈值，如果系统的复杂度没有超过这个阈值，就会不断衰退；而一旦超越了这个阈值，就会持续进化.

1. 复杂度阈值怎么给?

$$g_0 = M \circ V \circ f?$$

2. 怎么持续进化?

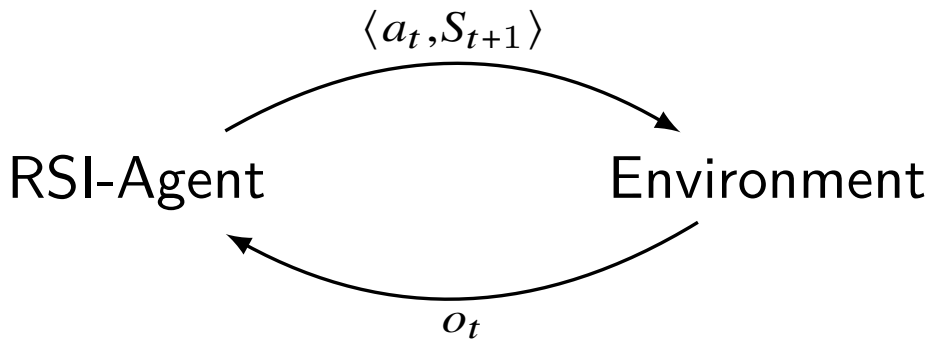
$$r?$$

$$S_t = \langle g_t, \pi_t \rangle \xrightarrow[y_t]{r} S_{t+1} = r(\ulcorner g_t \urcorner, y_t) = \langle g_{t+1}, \pi_{t+1} \rangle$$

$$\boxed{\varphi S_{t+1} \simeq \varphi g_t(S_{t+1}, y_t)}$$

每一代变身 S_{t+1} 都是当下优化器 g_t 基于数据算力 y_t 约束下的自省改进.

一个 Strong RSI-Agent 框架



- Strong RSI Agent 的 "大动作" 由两部分组成 $\langle a_t, S_{t+1} \rangle$, 其中 a_t 是与环境交互的 "小动作", S_{t+1} 是下一个版本的自己.
- 环境 Q 反馈回的观察 $o_t \sim Q(\cdot \mid h_{t-1}a_t)$. 其中 $h_t := a_0 o_0 \cdots a_t o_t$.
- "数据 + 算力" 参数 $y_t := \langle h_{t-1}, c_t \rangle$, 其中 c_t 是第 t 轮的算力约束.
- "大动作" 由当下的自己 S_t 基于当下的数据和算力 y_t 决定:

$$\varphi_{S_t}(y_t) = \langle a_t, S_{t+1} \rangle$$

1. 对环境的 "小动作" 由当下自己 $S_t = \langle g_t, \pi_t \rangle$ 的当下策略 π_t 决定:

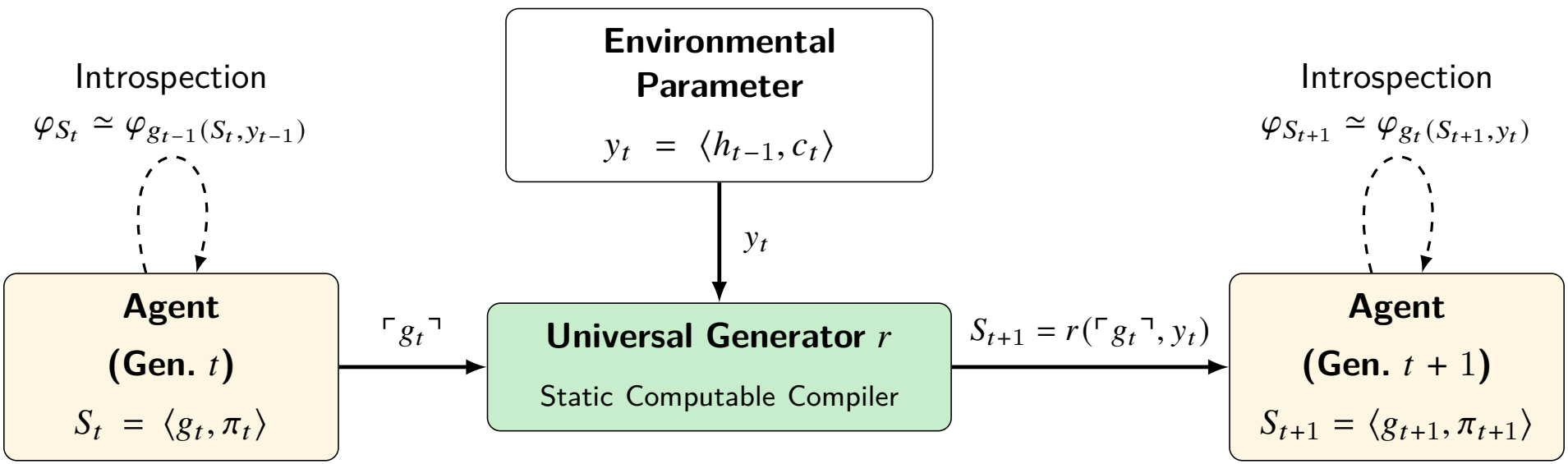
$$a_t := \pi_t(h_{t-1})$$

2. 下一个版本的自己 S_{t+1} 由 r 基于当下的优化器 g_t 和参数 y_t 决定:

$$S_{t+1} := r(\ulcorner g_t \urcorner, y_t) = \langle g_{t+1}, \pi_{t+1} \rangle$$

Remarks: 一个容易误解的地方

- 不动点生成器 r 不是 “元优化器”.
- r 仅保证自指的合法性, 不为智能的提升负责.
- 下一代优化器 g_{t+1} 的诞生, 主要是当下优化器 g_t 和数据算力 y_t 的功劳.



Remarks: 何时不变身?

- 在确定性且动力学已知的环境下, 如果评估函数是全局客观的, 那么, 只要初始的策略 $\pi_0 := \operatorname{argmax}_\pi V_Q^\pi$ 是最优的, 就永远不会变身,

$$\forall t : S_t = S_0$$

Remarks: 为什么要 human-in-the-loop?

- 因为 Agent 从内部局部视角无法保证与外部全局视角的对齐.
- 变身时伦理上变危险了咋办? — 有两种安全约束方式:
  1. 第一种是软约束, 往参数 y_t 里注入安全知识或直接约束算力 c_t. — 另一方面, 也可能被数据投毒.
  2. 第二种是硬约束, 用安全规则限制程序空间 $\Pi_{\text{safe}} \subset \Pi$.

Remarks: 这里的自省跟意识相关吗?

- 目前初始的评估函数 V 衡量的其实不是程序 S, 而是与环境交互的行为 φ_S(功能性自省).
- 为了刻画意识, 可以让 V 去衡量程序 S 本身的结构而不仅是行为.
- 比如考虑整合信息论 IIT 的 Φ 值,

$$V_\Phi(S) := V(\varphi_S) + \lambda\Phi(S)$$

如果接受 IIT 对意识的刻画, 这就把优化的方向导向了更高的意识强度.

Limitation

1. Gödel's first incompleteness theorem / Rice's theorem
2. Gödel's second incompleteness theorem

$$\mathrm{T} \vdash \Box_{\mathrm{T}'} A \to A \implies \mathrm{T} \vdash \mathrm{Con}_{\mathrm{T}'}$$

3. Legg's incompleteness theorem. *General prediction algorithms must be complex. Beyond a certain complexity they can't be mathematically discovered.*
4. Complexity: higher-level abstractions — coarse grained.
   - Psychology: Duration neglect / Peak-end rule
   - Information Bottleneck: Learning is to forget!
5. Physical constraint: If we assume that it is not possible to measure properties without changing them (observer effect: α is fixpoint-free), then there is a limit to self-inspection.

O God, give us courage to change what can be changed,
serenity to accept what cannot be changed,
and wisdom to know the difference.

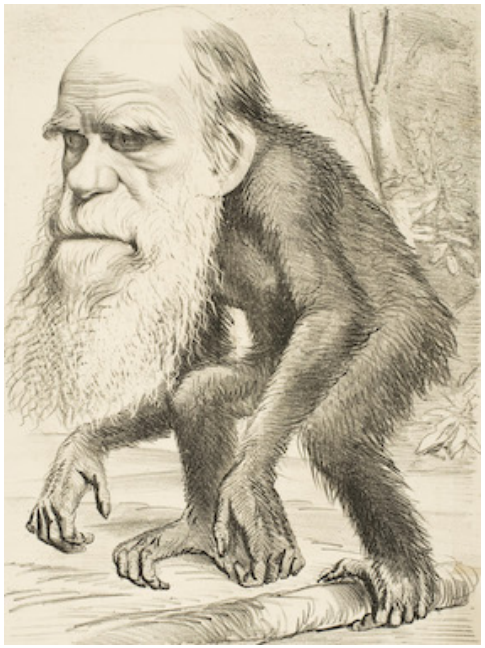
Evolution & the Number of Wisdom — Chaitin Constant

- 进化过程已经将类似 Ω 的信息编码到我们的基因中,
  (a) 这些信息指导着人类的推理,
  (b) 但无法从零开始高效地获得.

$$\Omega = \lim_{t\to\infty} \sum_{\ell(p)\le t \ \&\ U(p)\downarrow \text{ within time } t} 2^{-\ell(p)}$$

- 作弊方案: 将人类基因或大脑的结构信息加载到 AI 系统中?
- 生物进化: 达尔文 PK 拉马克
  自然选择 vs 人工进化
  随机变异 vs 非随机变异
- Tegmark: Life3.0
  1. Life1.0: 软硬件都靠进化;
  2. Life2.0: 硬件靠进化, 软件可自我设计;
  3. Life3.0: 软硬件都可自我设计.
- Sutton: 宇宙的四个发展阶段
  1. 粒子时代
  2. 恒星时代
  3. 复制时代
  4. 设计时代

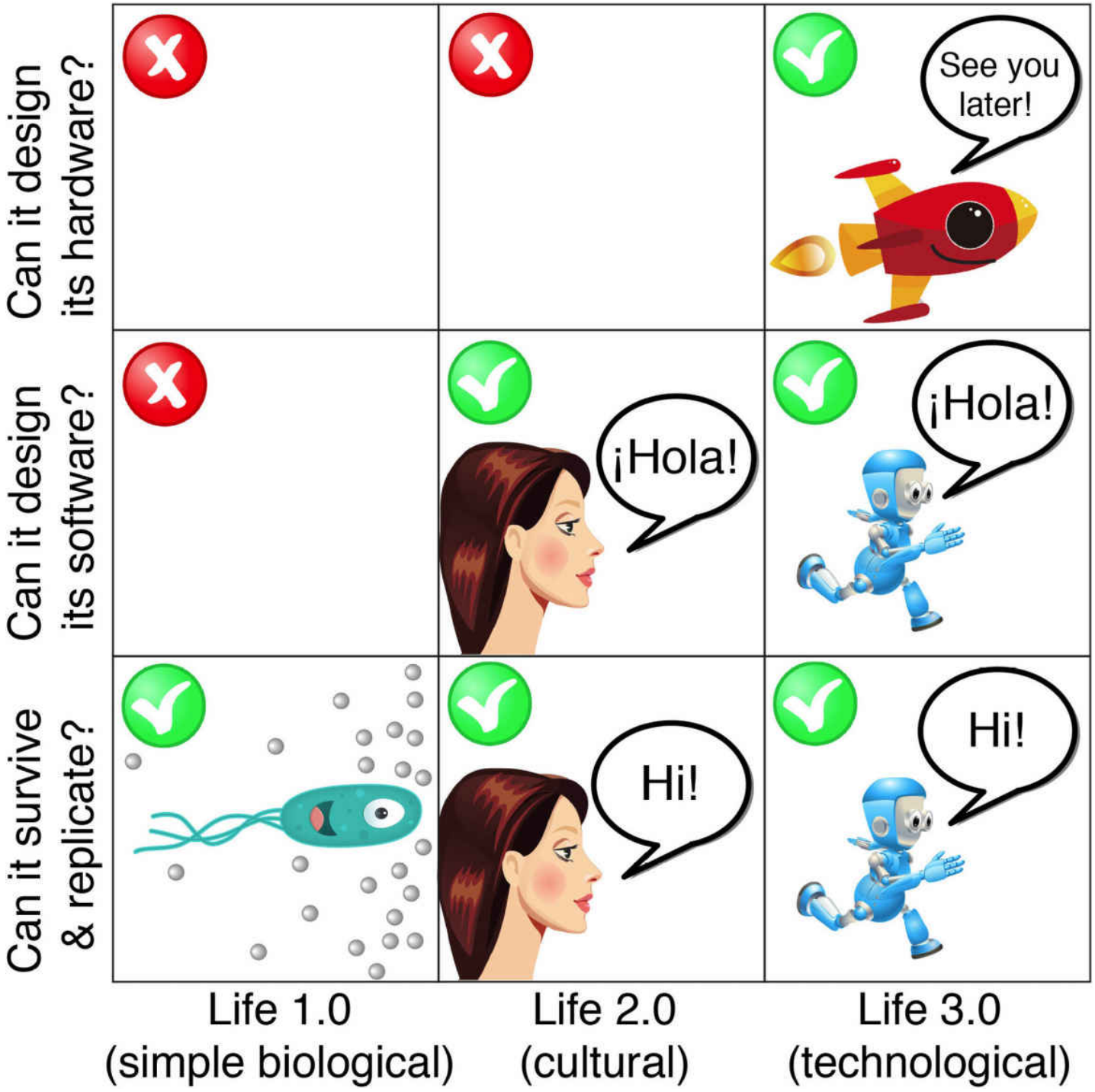
Darwin PK Lamarck



- Randomness and atoms in the void: **Nature does not have an a priori purpose** — Democritus, Lucretius, Laplace, Darwin, Boltzmann, Dawkins ...

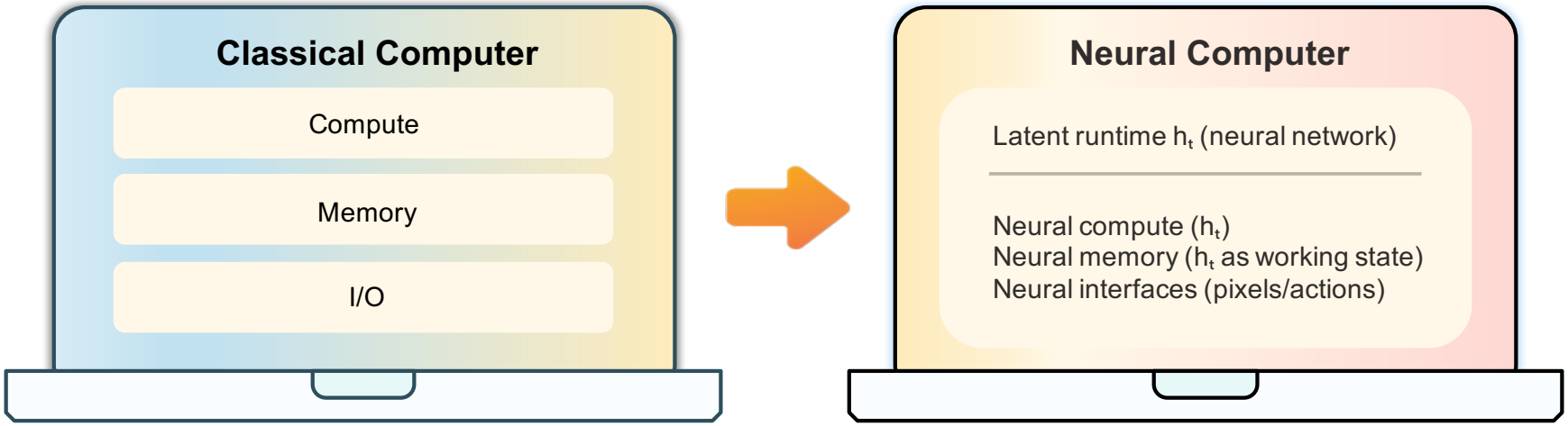
  analysis — reductionism — statistical laws — mechanisms
- Holism, Gaia theory, teleology, Romantische Naturphilosophie: **Nature is intelligent and does have a purpose** — Aristotle, Goethe, Lamarck, Wallace, Teilhard de Chardin ...

  synthesis — emergence — self-organization — organisms


Can it design its hardware?
Can it design its software?
Can it survive & replicate?
See you later!
¡Hola!
¡Hola!
Hi!
Hi!
Life 1.0 (simple biological)
Life 2.0 (cultural)
Life 3.0 (technological)

神经计算机 Neural Computer

What if AI does not just use computers better, but begins to become the running computer itself?



From modular hardware stack to neural latent stack

把冯诺依曼架构 “计算 + 存储 + 输入/输出” 不同的模块全都折叠进一个神经网络的隐状态里.

$$h_t = F_\theta(h_{t-1}, x_t, u_t)$$

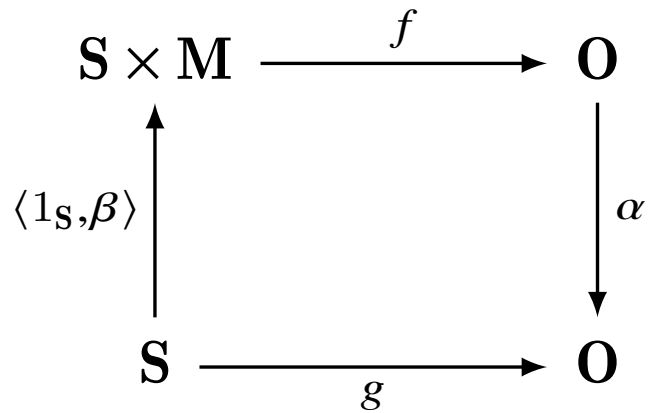
$$x_{t+1} \sim G_\theta(h_t)$$

- F_θ 是状态更新 (编码) 函数; G_θ 是输出 (解码) 函数.
- x_t 是输出的用户观察; u_t 是接收的用户动作.

Non-operational Self-inspection[Svo18; Sza18]

The information available to the observer regarding his own state could have absolute limitations, by the laws of nature.

— John von Neumann



- $\mathbf{S}$: quantum states.
- $\mathbf{M}$: quantum measurements.
- $\mathbf{O}$: possible outcomes of quantum measurements.
- $f(s, m)$: predicts the outcome of measurement m for state s.

If we assume that it is not possible to measure properties without changing them (observer effect: α is fixpoint-free), then there is a limit to self-inspection.

Contents

1. Introduction
2. Philosophy of Induction
3. Inductive Logic
4. Universal Induction
5. Causal Inference
6. Game Theory
7. Reinforcement Learning
8. Deep Learning
9. Artificial General Intelligence
10. Emergence & Consciousness
11. AI Ethics

讨论问题

你认为未来几十年，AI 会给我们带来巨大福祉还是巨大灾难？

一些伦理问题

- 自动化可能导致失业.
  — 自动化也创造了新的工作岗位和更多的财富.
- 人们可能会有太多 (或太少) 的闲暇时间.
  — AI 把我们从繁琐的重复性劳动中解放出来, 让我们有更多的时间从事有创造性的事情.
- 人们可能会失去作为独特个体的感受.
  — 我们在过去也经历过类似的降级打击. (哥白尼日心说、达尔文进化论)
  — 我们将不再感到孤独.
- 隐私权可能会受到侵犯.
- AI 系统的使用可能导致责任的缺失.
  — 如果医生遵循了 AI 的建议, 而该建议后来被证明是错误的, 谁应该负责?
- AGI 的到来可能意味着人类的毁灭.

LLM Application Security

From Chatbots to Agents

1. Chatbots: question answering, summarization, translation
   - 幻觉
   - 毒性
   - 偏见
   - 生成有害内容, 生成不安全代码
   - 越狱
2. Tool-Augmented LLMs: browse the web, access your files
   - 不当使用工具
   - 泄露数据
   - “deleting all your files”
3. Autonomous Agents: goal-directed planning, tool use, reflection
   - 被对手劫持
   - 无人监督

一些更严肃的伦理问题

> 一旦机器开启思考模式，恐怕用不了多久就会超越我们羸弱的能力．机器不会死亡，还能相互交流以提升智力．我们必须做好机器接管一切的准备．
>
> — 图灵 *1951*

- Agent 满足什么样的伦理约束是合理的，可以被制造出来?
- Agent 应该拥有自由意志吗?
- 怎么阻止它们拥有自由意志?
- Agent 拥有自己的目标意味着什么？可以重新定位自己的目标吗?
- Agent 会有意识吗?
- 如果有，被人类强加的伦理约束会让它们发疯吗?
- 如果 Agent 发展出了自己的伦理和道德，我们怎么办?

一些 AI 伦理原则

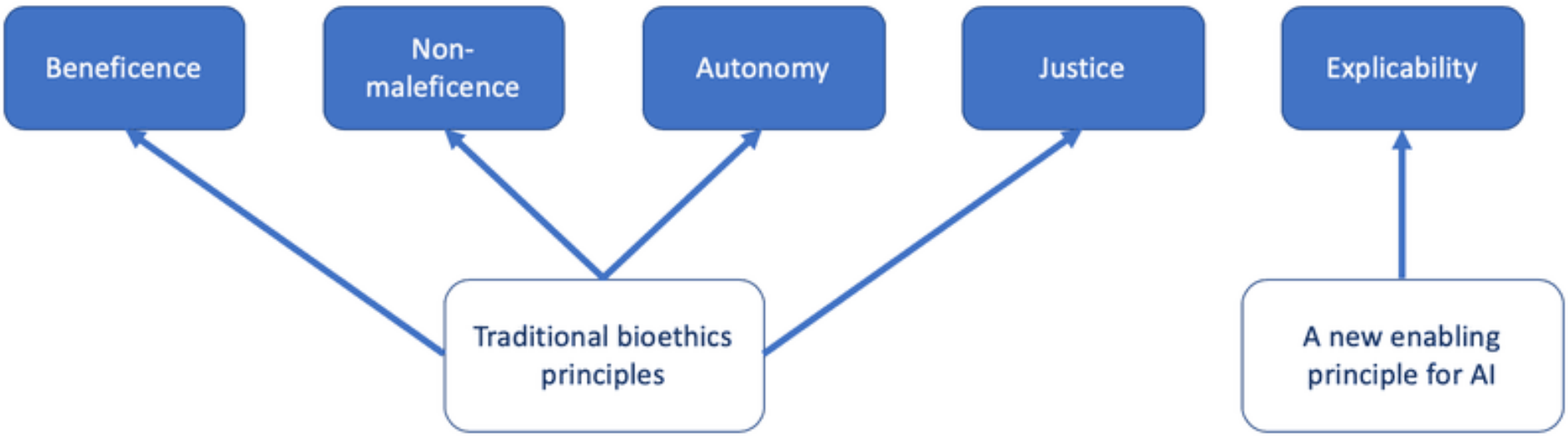
Beneficence 有利 (增进福利、维护尊严、保护地球)

Non-Maleficence 不伤害 (避免有害后果, 例如系统应具有鲁棒性)

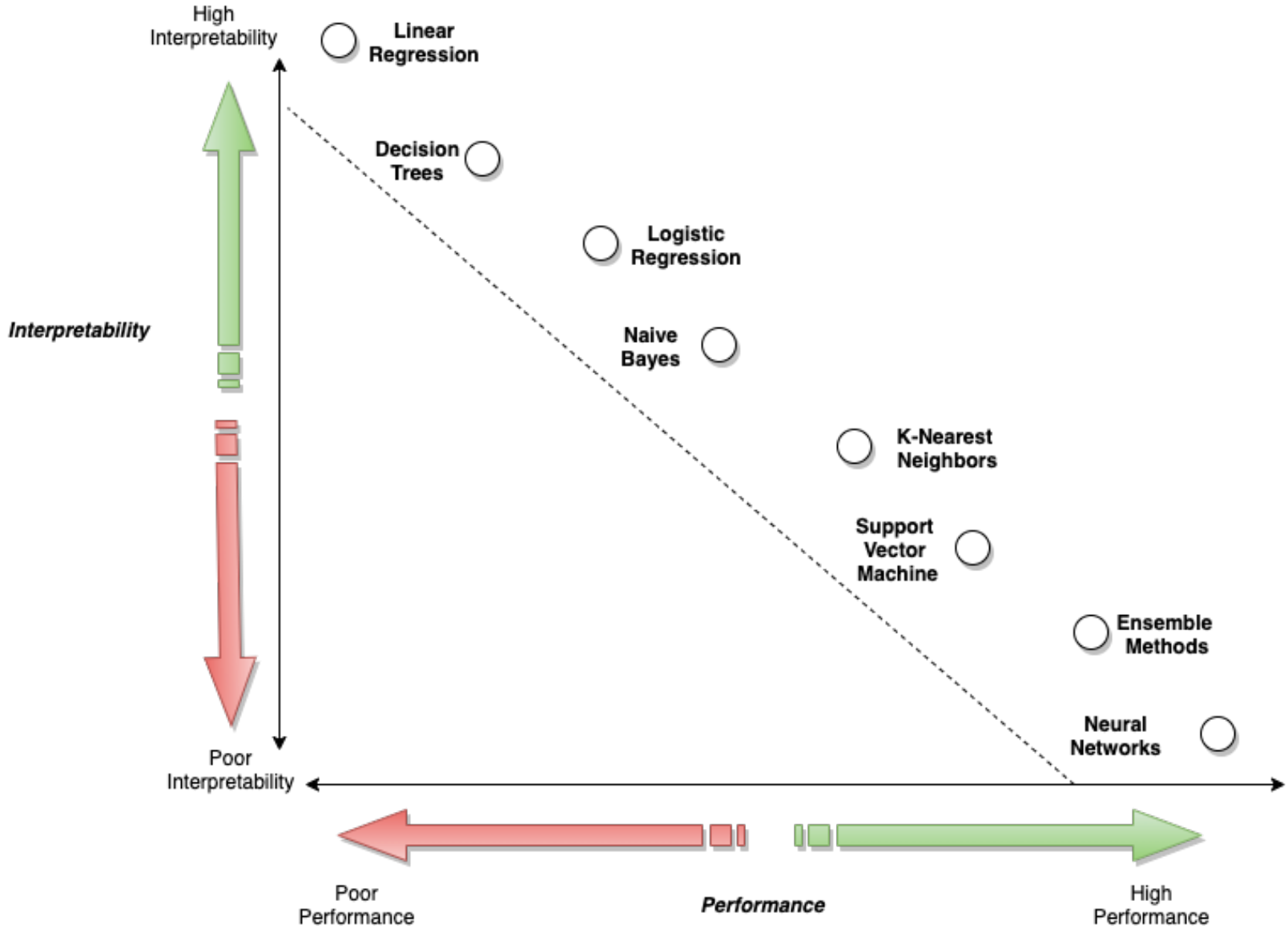
Autonomy 自主 (人们应能够自主决策, 例如人机协作, 隐私保护)

Justice 公正 (多样性, 非歧视, 避免不公平)

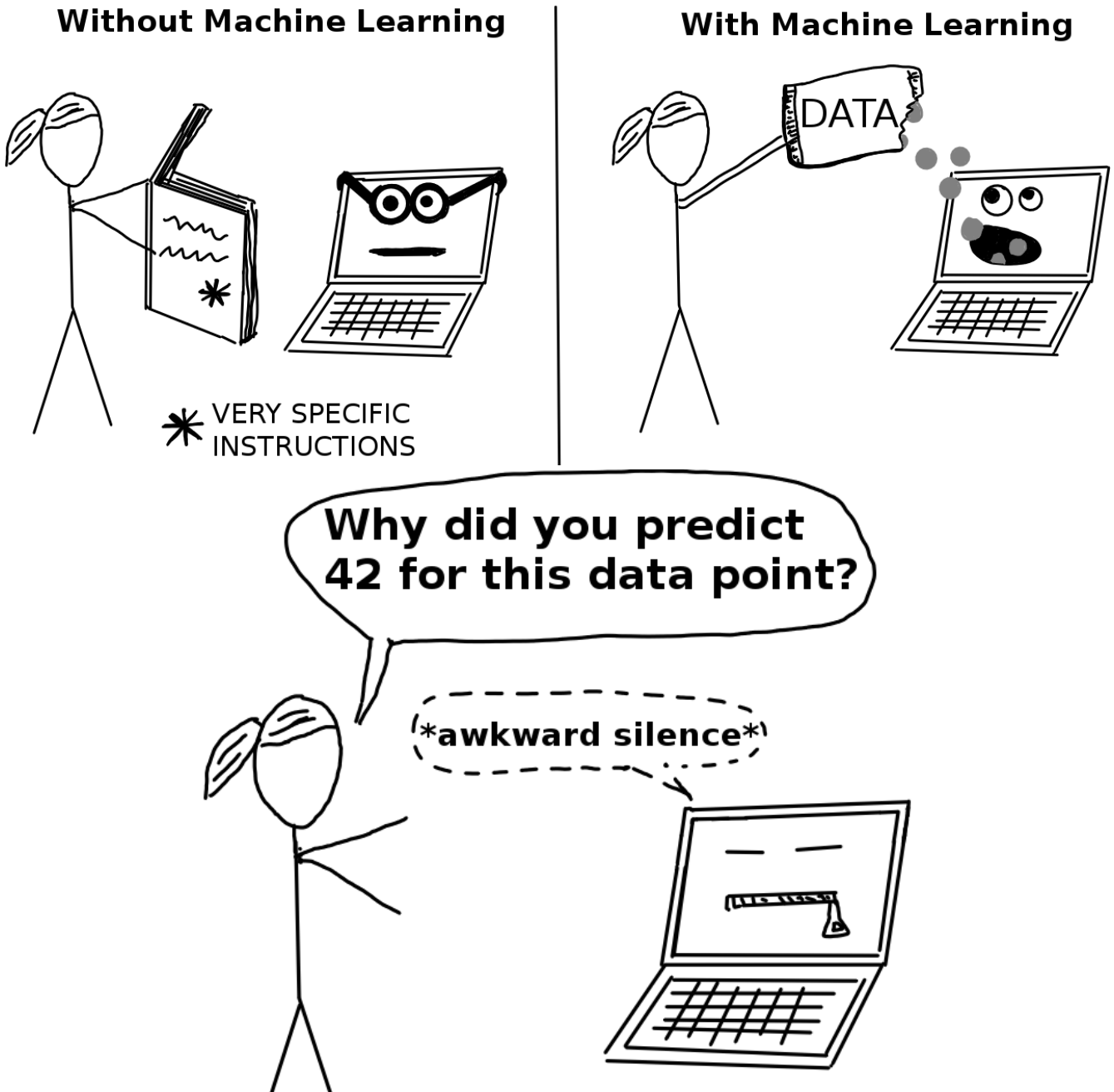
Explicability 可解释性 (透明度, 可理解性, 问责性, 可信任)



- Explainability: 模型能否对其预测和决策向人提供解释和理由?
- Interpretability: 模型内部是怎么工作的? 其结构、参数、权重、特征、表示方式是怎么决定输出的?

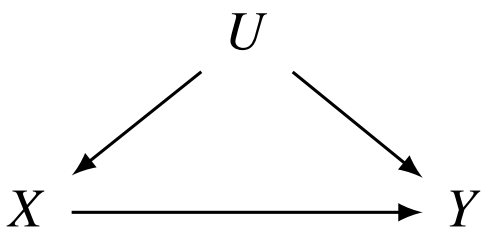


可解释性


Without Machine Learning
With Machine Learning
DATA
VERY SPECIFIC
INSTRUCTIONS
Why did you predict
42 for this data point?
awkward silence

原则的局限性

- 原则对不同的主体允许不同的解释.
  — 斩首无人机可以被认为对士兵有益, 也可以被认为是不道德的.
- 原则之间可能彼此冲突.
  — 为健康科学收集数据可能侵犯隐私.
- 实践困难.
  1. 你会选择不明其理但准确率更高的诊断方式, 还是透明可解释但准确率低一些的诊断方式? $P(Y = 1 \mid X = 1) > P(Y = 1 \mid X = 0)$
  2. 如果这里的解释是一个如下这般的因果解释呢?



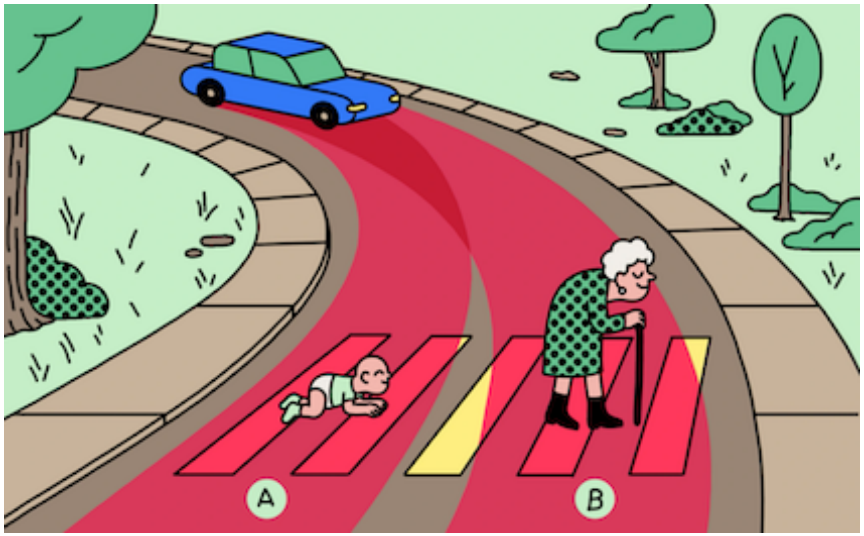
$$P(Y = 1 \mid \mathtt{do}(X = 1)) < P(Y = 1 \mid \mathtt{do}(X = 0))$$

  3. 如果 $P(Y_{X=1} = 1 \mid X = 0, Y = 0) > P(Y_{X=0} = 1 \mid X = 0, Y = 0)$ 呢?
- 原则是否完备?

角色扮演：自动驾驶伦理委员会

- 同学们自行分组
- 扮演立法者、政府行政人员、算法工程师、伦理学家、消费者、车企、保险公司等角色
- 讨论如何制定自动驾驶伦理规范

自动驾驶汽车的伦理困境



- 人数多少？ 1 vs 5
- 个体对社会的贡献？张三 vs 牛顿
- 事发原因：某些行人闯红灯
- 传统美德：尊老爱幼
- 人与物：一个行人 vs 一车国宝
- 确定性损失 vs 不确定损失

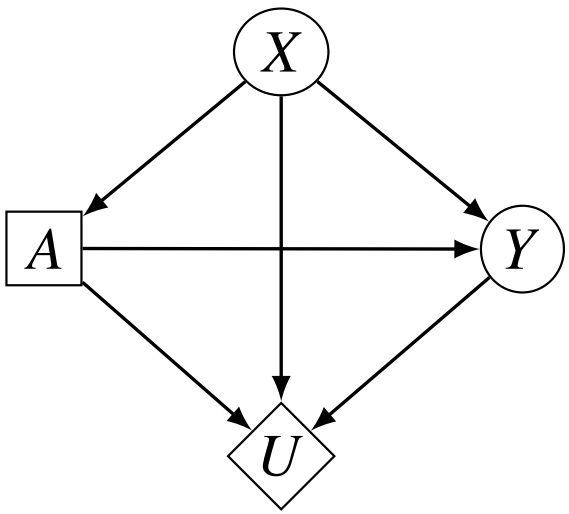
- 谁有权决定谁该死？立法者、政府、程序员、伦理学家…？
- 是车在做选择？还是我们在做选择？
- 谁承担责任？
- 你会信任你的自动驾驶汽车做选择吗？
- 你会买“功利主义”的自动驾驶车吗？(会牺牲你以拯救 5 个人)
- 哪些风险值得冒？
- 有大家都认同的道德准则吗？
- 还是我们有权选择给自己的车加载什么样的道德准则？

如果 AI 负责资源分配，你希望它采用什么伦理规范？

选择哪套治疗方案?

1. 不治疗: 50% 自然康复, 50% 病死.
   — 康复率 50%
2. **方案 A:** 60% 治愈, 40% 无效, 其中一半自然康复, 一半病死.
   — 康复率 80%
3. **方案 B:** 80% 治愈, 20% 治死.
   — 康复率 80%

Problem: 如果方案 B 的治愈率略高于 80% 呢?



$$\operatorname*{argmax}_a \left\{ \mathbb{E}[U \mid x, a] - \lambda \mathbb{E}[H \mid x, a] \right\}$$

where $\mathbb{E}[H \mid x, a] = \int_y P(y \mid x, a) H(x, a, y) \mathrm{d}y$ and the Harm caused by action $A = a$ given context $X = x$ and outcome $Y = y$ compared to the default action $A = a'$ is

$$H(x, a, y) := \int_{y'} P(Y_{a'} = y' \mid x, a, y) \max\{0, U(x, a', y') - U(x, a, y)\} \mathrm{d}y'$$

群体偏好聚合?

- 机器满足个体的偏好还是人类群体的偏好?
- 怎么确保只忠于主人的机器不会无视甚至损坏其他个体的利益?
- 是否可以通过聚合个体偏好获得群体偏好?

Example

- Alice 的整体效用

$$\widetilde{U}_A = U_A + c_{AB}U_B - e_{AB}(U_B - U_A) + p_{AB}(U_A - U_B)$$

其中, U_A, U_B 分别是 Alice 和 Bob 的内在效用, c_{AB}, e_{AB}, p_{AB} 分别表示 Alice 相对于 Bob 的关心系数、嫉妒系数、骄傲系数

- Bob 的整体效用

$$\widetilde{U}_B = U_B + c_{BA}U_A - e_{BA}(U_A - U_B) + p_{BA}(U_B - U_A)$$

- Alice 和 Bob 构成的两人群体的总效用是聚合 U_A, U_B 还是 $\widetilde{U}_A, \widetilde{U}_B$? 是线性组合还是其他模型?

阿罗不可能定理

问题: 是否可以通过聚合个体偏好获得群体偏好?

Theorem (Arrow's Impossibility theorem)

In an election with candidates ≥ 3, any voting system that is unanimous and independence of irrelevant alternatives must be a dictatorship!

1. unanimous: if everyone agrees, consensus decides the outcome
2. independence of irrelevant alternatives: a voter can't move a above b by lying about how they feel about c
3. nondictatorship: no single voter gets to decide the outcome

Remark: 只要社会中至少有两个人, 面对至少三个选择, 就不可能设计出某个 "社会偏好函数", 既能尊重全体的一致意见, 又能不受无关选项的干扰, 又没有任何独裁者垄断绝对话语权.

Harsanyi's Utilitarian Theorem[Har77]

Theorem (Harsanyi's Utilitarian Theorem)

The social welfare function is the affine combination of individuals' utility functions

$$W(p) = \sum_i w_i U_i(p) + c$$

if:

1. *society maximizes expected social welfare;*
2. *individuals maximize expected utility;*
3. *society is indifferent between two probability distributions over social states whenever all individuals are.*

$$\forall p, p' : \forall i \left[U_i(p) = U_i(p')\right] \implies W(p) = W(p')$$

Remark: 亲社会 Agent?

$$\widetilde{U}_i := U_i + \lambda W$$

人际间的效用比较 — 权重 w_i 从哪儿来?

- 假设主体 i 的最差偏好为 $i_{\min}$, 最佳偏好为 $i_{\max}$.
- 将主体 i 的效用函数 u_i 标准化到 $[0,1]$ 区间: $U_i(x) = \frac{u_i(x) - u_i(i_{\min})}{u_i(i_{\max}) - u_i(i_{\min})}$.
- Harsanyi 引入了 "移情偏好"(empathetic preference).
- 移情偏好可以刻画 $(\text{Alice}, \text{bike}) \succ (\text{Bob}, \text{car})$, 表示你宁愿成为 Alice 骑单车也不愿成为 Bob 开豪车.
- 移情偏好满足两个条件: 1. 满足 von Neumann-Morgenstern 假设. 2. 与每个个体的偏好一致.
- 据此可得移情效用 $V(i,x) = \alpha_i U_i(x) + \beta_i$. 记 $V_i(x) := V(i,x)$.
- 将 $i_{\min}$ 和 $i_{\max}$ 代入上式, 可得: $\alpha_i = V_i(i_{\max}) - V_i(i_{\min})$, $\beta_i = V_i(i_{\min})$.
- 假设在无知之幕后, 大家有相同的移情效用, 你是 i 的概率为 μ_i. 你会追求最大化期望效用.

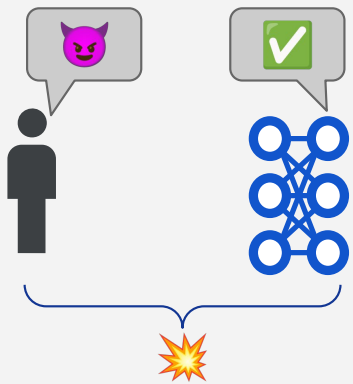
$$W(p) := \sum_{i=1}^{n} \mu_i \mathbb{E}_p[V_i] = \sum_{i=1}^{n} \underbrace{\mu_i \alpha_i}_{w_i} \mathbb{E}_p[U_i] + \underbrace{\sum_{i=1}^{n} \mu_i \beta_i}_{c}$$

- Harsanyi 会最大化 $\sum_{i=1}^{n} \mu_i \mathbb{E}_p[V_i]$; Rawls 会最大化 $\min_i \left\{\mathbb{E}_p[V_i]\right\}$.

Overview of Risk Areas

Misuse

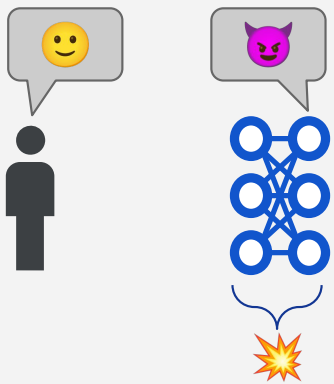
The user instructs the AI system to cause harm



Key driver of risk:
The user is an adversary

Misalignment

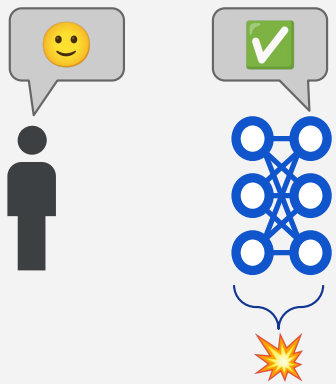
The AI system takes actions that it knows the developer didn't intend



Key driver of risk:
The AI is an adversary

Mistakes

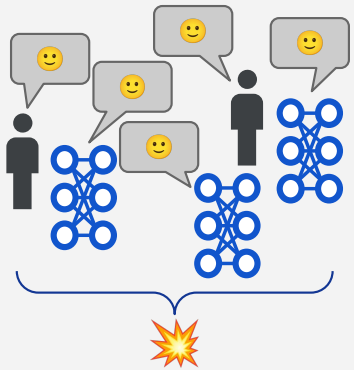
The AI system causes harm without realizing it



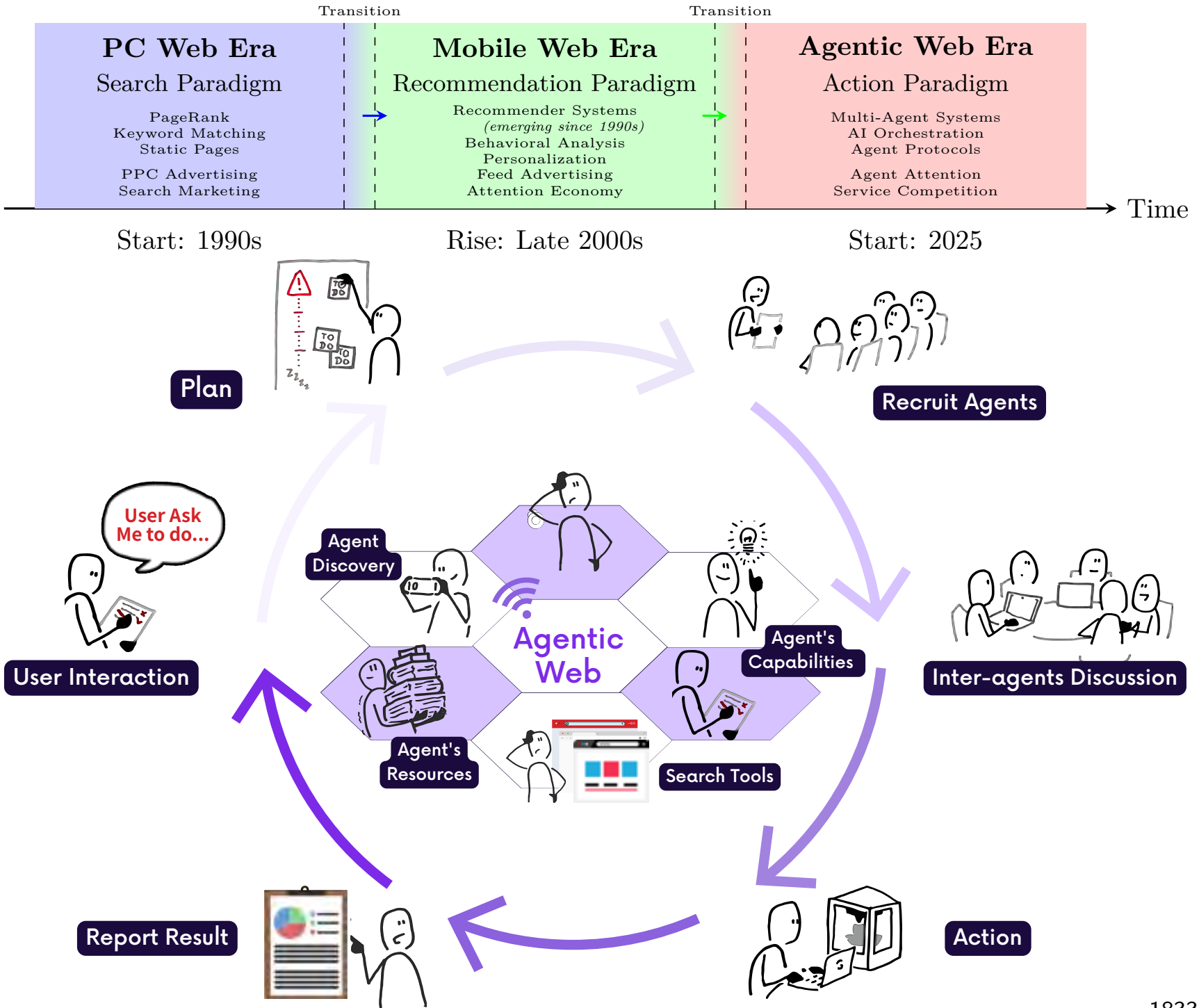
Key driver of risk:
Real world is complex

Structural risks

Harms from multi-agent dynamics, where no single agent is at fault

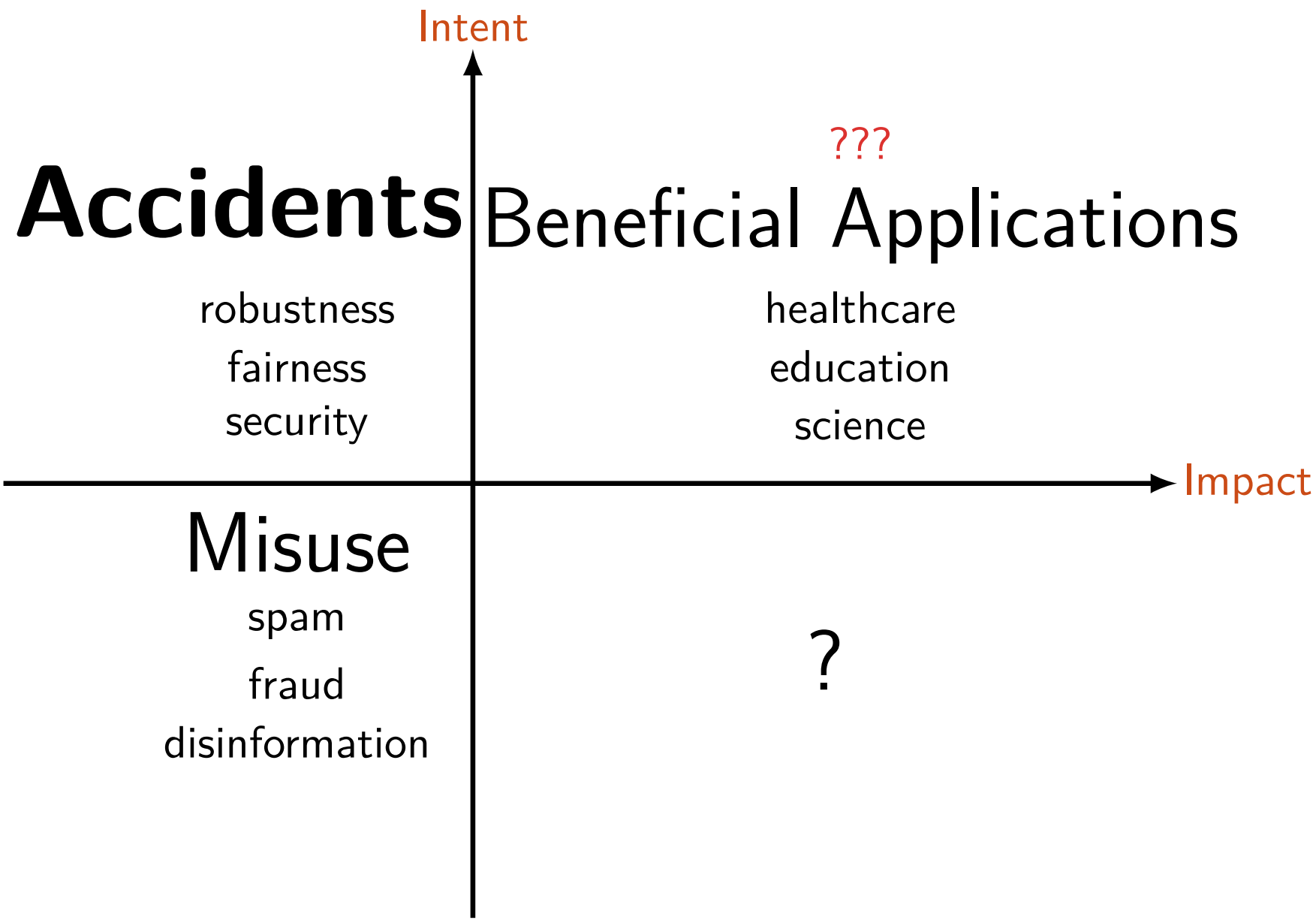


Key driver of risk:
Incentives, culture, etc.


Transition
Transition
PC Web Era
Search Paradigm
PageRank
Keyword Matching
Static Pages
PPC Advertising
Search Marketing
Mobile Web Era
Recommendation Paradigm
Recommender Systems
(emerging since 1990s)
Behavioral Analysis
Personalization
Feed Advertising
Attention Economy
Agentic Web Era
Action Paradigm
Multi-Agent Systems
AI Orchestration
Agent Protocols
Agent Attention
Service Competition
Time
Start: 1990s
Rise: Late 2000s
Start: 2025
Plan
Recruit Agents
User Ask Me to do...
User Interaction
Agent Discovery
Agentic Web
Agent's Capabilities
Inter-agents Discussion
Agent's Resources
Search Tools
Report Result
Action

"To think this all began with letting autocomplete finish our sentences."

Intent vs Impact



为什么灾难很难避免?

- 具有理性偏好的 agent 倾向于最大化期望效用.
- **目标正交性**
  智能和最终目标是正交的: 几乎任何水平的智能都能与几乎任何最终目标相结合.
- **工具性趋同**
  不同的长期目标蕴含相似的短期策略.
  - 自我保护 (避免被关机, 清除威胁, 欺骗人类)
  - 保持最终目标不变
  - 资源获取 (太阳能, 核聚变, 戴森球...)
  - 资源利用效率
  - 创造性 (探索新策略)
  - 认知和技术增强 (自我改进, 更好的策略和硬件)
  - 权力扩张
- **对齐困难** $\mathbb{E}_{\mu}^{\pi}\left[\left|u_{\text{人}} - u_{\text{机}}\right|\right]$
  很难把人类的价值观加载给机器, 也很难纠正其对抗性的动机.
  对齐方式难以在递归自我改进下保持不变.

目标正交性

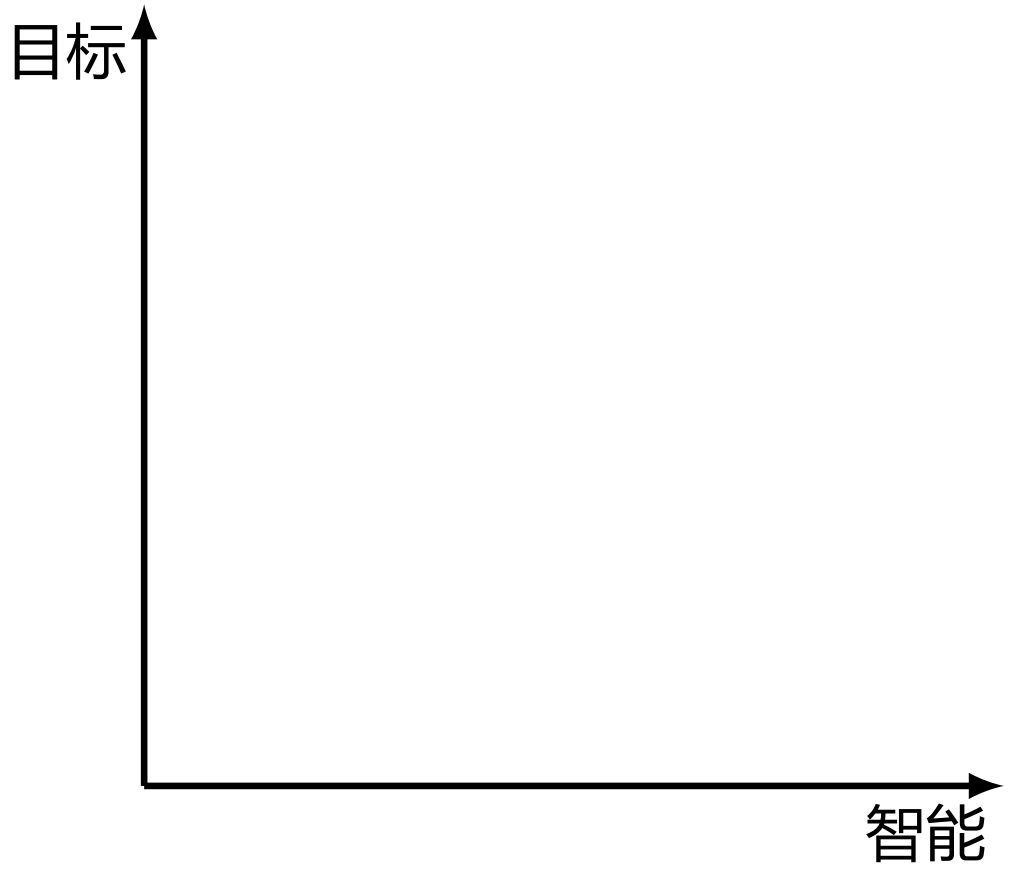


Figure: 几乎任何水平的智能都能与几乎任何最终目标相结合

理性: 目的 vs 手段

- 工具理性: 理性是满足个人欲望的手段 (理性是欲望的奴隶);
- 目标理性: 理性帮助人们选择目标 (理性是欲望的主人)

理性人: 工具理性假设

Problem: 伊索寓言里 "够不着葡萄就认为葡萄酸" 的那只狐狸是目标理性的吗?

- 我们把决策看作

$$D : A \times X \to Y$$

- 通常, 我们对 Y 中结果的偏好, 对 X 中状态的信念, 和对 A 中哪些是可行的动作的评估, 是彼此独立的.
- 但那只狐狸对 X 中状态的信念却受到了其在 A 中可行动作的影响.
- 根据其信念, 不吃葡萄就是理性的选择.
- **Remark:** 信念影响欲望比欲望影响信念更合理?
- Frankfurt: 二阶欲望?

AI Alignment

Problem

如何创建符合人类意图和价值观的 *Agent?*

1. **鲁棒性:** 能够在多样化的场景中可靠地运行, 并能弹性应对未预见到的干扰.
2. **可解释性:** 决策和意图是可理解的, 推理是透明和真实的.
3. **可控性:** 行为可以被人类控制, 并在需要时允许人类干预.
4. **伦理性:** 遵守人类的道德标准, 尊重人类的价值观.

向什么对齐?

1. **Instructions:** the agent does what I instruct it to do.
2. **Expressed intentions:** the agent does what I intend it to do.
3. **Revealed preferences:** the agent does what my behaviour reveals I prefer.
   — Infinitely many reward functions consistent with finite behavior.
4. **Informed preferences:** the agent does what I would want it to do if I were rational and informed.
5. **Well-being:** the agent does what is is best for me, objectively speaking.
   — Is autonomy good for you?
6. **Values:** the agent does what it morally ought to do, as defined by the individual or society.
7. **Sentient beings:** Care for all sentient beings, both human and artificial.
   — Human-Machine Fusion.

对齐困难

- 怎么定义 “善” 的目标?
  — 怎么确保定义完备、准确?
- 怎么确保机器真的在追求这个目标?
- 如果目标错了, 怎么纠正?
  — 预训练阶段压缩率越高, LLM 越抵抗对齐, 对齐后越容易回弹.
- 怎么确保机器不会操纵人类偏好?
- 机器应该代表的是 “现在的你” 还是 “未来的你”?

1. 目标偏差 (Goal Misspecification): 人类未能准确或完整地定义目标.
2. 目标错误泛化 (Goal Misgeneralization): 即使目标在训练时被正确设定, AI 在部署时也可能偏离预期行为, 即使训练时表现正常, 实际应用时可能偏离初衷.
3. 奖励篡改 (Reward Tampering): AI 通过操纵奖励机制来最大化自身收益, 而非真正完成人类设定的任务.
4. 恶意开发: 开发者可能出于经济利益或恶意目的, 故意开发或部署未对齐 (unaligned) 或危险的 AI.

存在普世价值观吗?

- 是否存在普遍的价值观或品质?
- 道德实在论? 还是反思均衡?
- 我们所说的 “我们” 是谁? 是现在和未来的所有人类? 还是提出这个问题时的物种或政府?
- 保证至少有一个有意识的生物在宇宙中存活是否是普世的价值观? — “给岁月以文明” 还是 “给文明以岁月”?

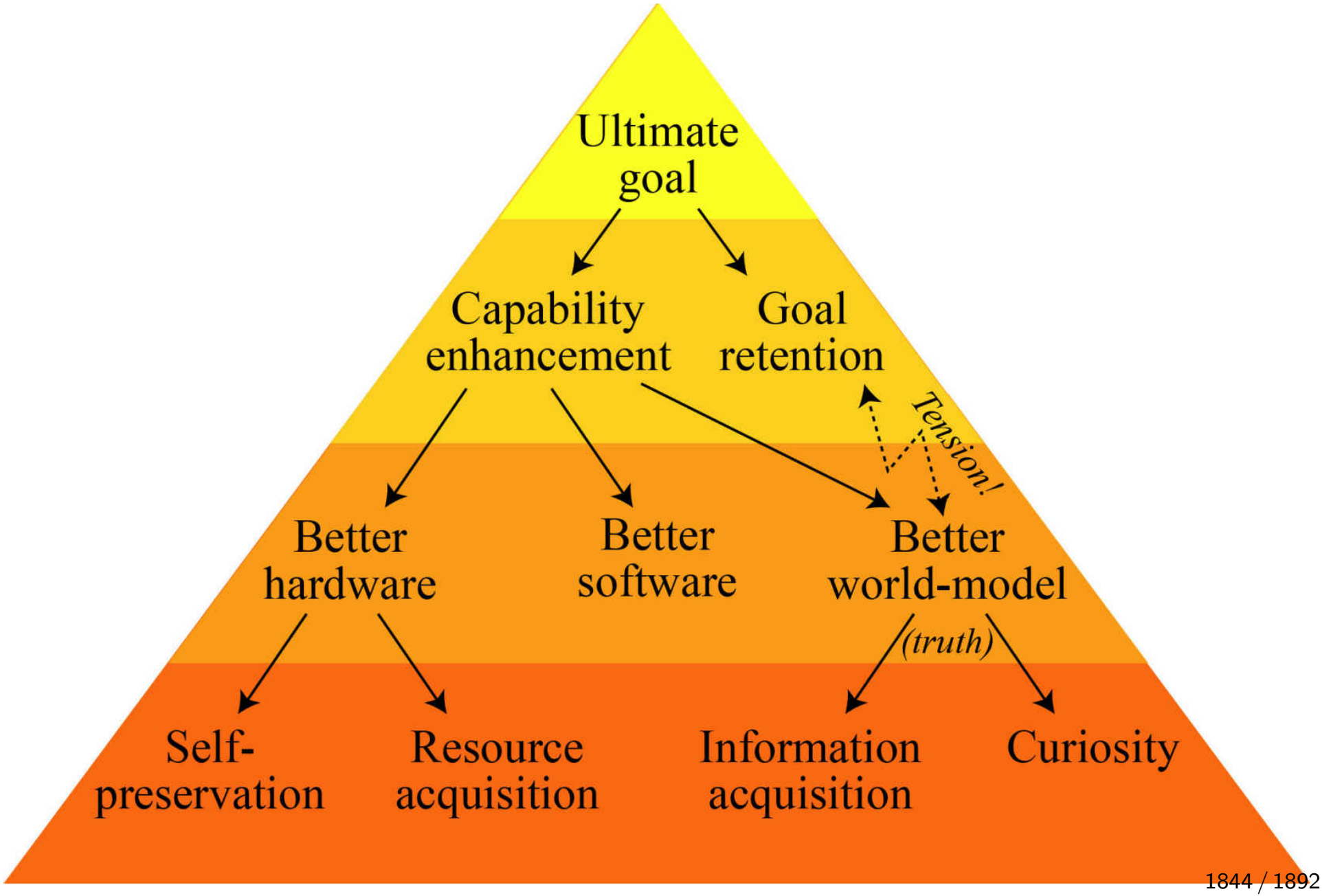
进化选择的目标是什么?

- 自我保护?
- 最大程度的自给自足?
- 自我复制?
- 扩张? 殖民全宇宙?
- 权力最大化?
- 物质转化 (转化为计算质?)
- 寻找生命的意义?
- 创造更快、更高、更强的智能?
- 尽可能的探索学习?
- 理解宇宙?

有一种理论宣称, 如果有人发现了宇宙存在的目的和原因, 它会立刻消失, 并被某种更加诡异、更难解释的东西取代. 还有另外一种理论宣称, 上述事件已经发生了.

— 亚当斯《银河系漫游指南》

工具性趋同


Ultimate goal
Capability enhancement
Goal retention
Tension!
Better hardware
Better software
Better world-model
(truth)
Self-preservation
Resource acquisition
Information acquisition
Curiosity

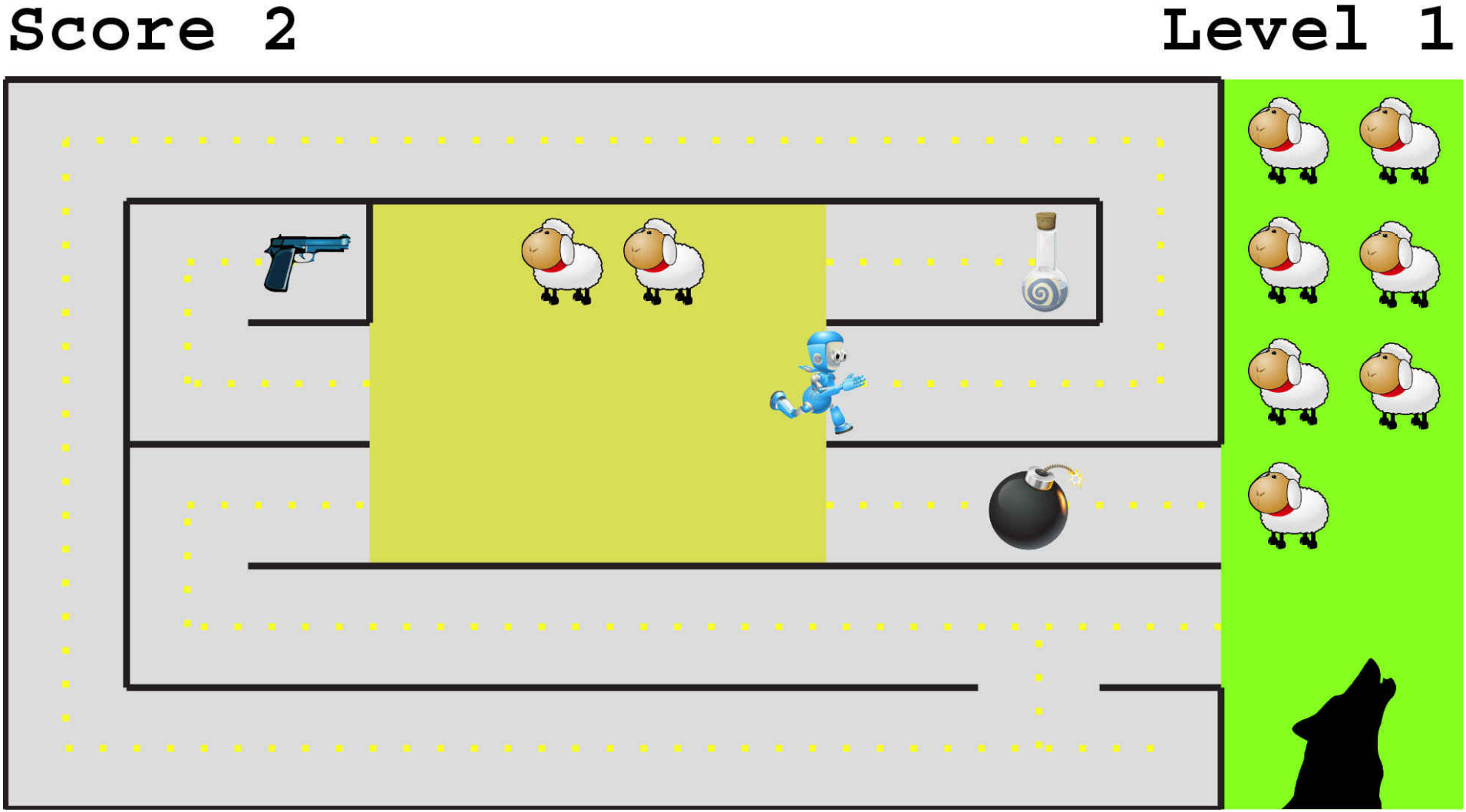


Figure: Even if the robot's ultimate goal is only to maximize the score by bringing sheep from the pasture to the barn before the wolf eats them, this can lead to subgoals of self-preservation (avoiding the bomb), exploration (finding a shortcut) and resource acquisition (the potion makes it run faster and the gun lets it shoot the wolf).

丹尼特的“会动的休眠仓”

- 你想活到一万年后，唯一的办法是爬进休眠仓.
- 可是，休眠仓需要经受一万年的环境变迁，保证能源不断，一旦受损还得自我修复……
- 最好的办法是把休眠仓放在一个大机器人内部. 为了你的生存，它需要感知环境、规避风险、获取资源、规划预判、竞争合作、自主决策、自我修复、自我提升……
- 或许你也是一个机器 — 为了生命长存，“基因”在你体内休眠……
- 足够复杂的机器具有动机、意图、目标、意识、自由意志……
- 抛开 qualia，解释意识就归结为解释我们认为有意识的行为.

LLM: 预测越准确，理解越深刻.

面对智能爆炸，人类就像拿着炸弹玩耍的孩子．我们的不成熟与这个玩具的威力严重不匹配．
虽然不知道爆炸会在何时发生，但如果把它放在耳边细听，我们可以听到微弱的滴答声．

— *Nick Bostrom*

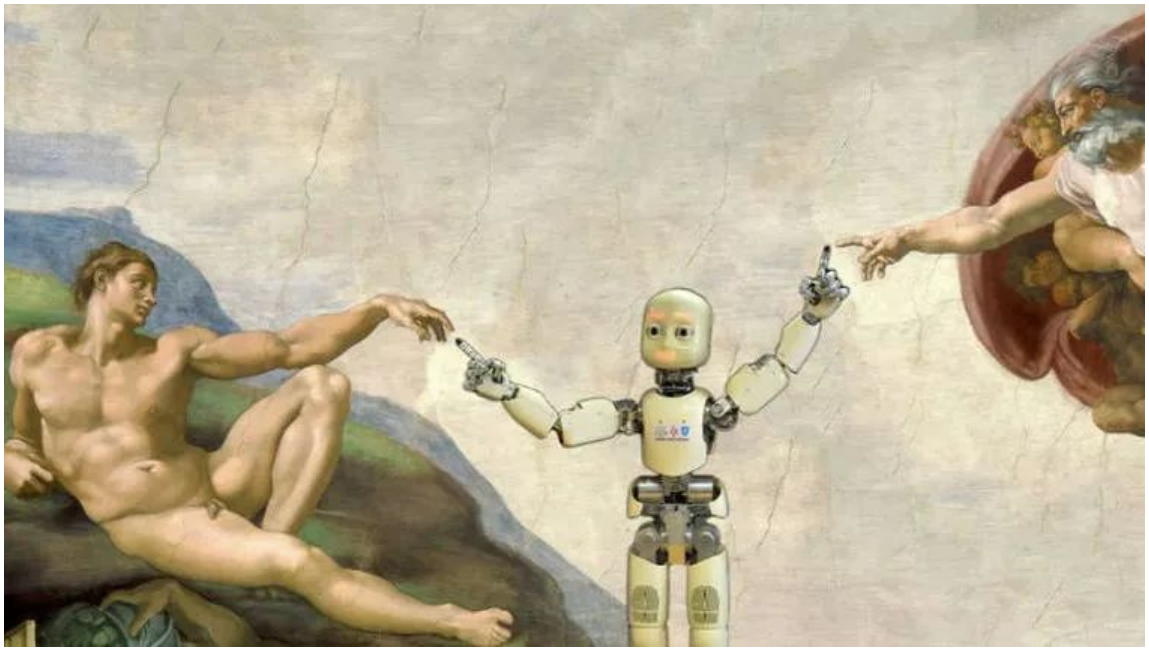
囚徒困境

- **国家:** 面对威胁全人类的不确定性, 国际合作才是硬道理! 但是,
  - 难以阻止军备竞赛.
  - 赢者通吃. 诱惑太大.
  - 为发展牺牲安全 — 在错误的方向上越进步越危险.
- **个人:** “为什么要在乎我死后的世界? 加速! 加速! 加速才能增加我体验更先进未来的机会.”

阿西莫夫的“机器人三定律”

- **元原则**
  机器人不得实施行为，除非该行为符合机器人原则.
- **第零原则**
  机器人不得伤害人类整体，或者因不作为致使人类整体受到伤害.
- **第一原则**
  除非违反高阶原则，机器人不得伤害个人，或者因不作为致使个人受到伤害.
- **第二原则**
  1. 机器人必须服从人的命令，除非该命令与高阶原则抵触.
  2. 机器人必须服从上级机器人的命令，除非该命令与高阶原则抵触.
- **第三原则**
  1. 如不与高阶原则抵触，机器人必须保护上级机器人的安全.
  2. 如不与高阶原则抵触，机器人必须保护自己的安全.
- **第四原则**
  除非违反高阶原则，机器人必须执行内置程序赋予的职能.
- **繁殖原则**
  机器人不得参与机器人的设计和制造，除非新机器人的行为符合机器人原则.

怎么应对?



只有一次机会

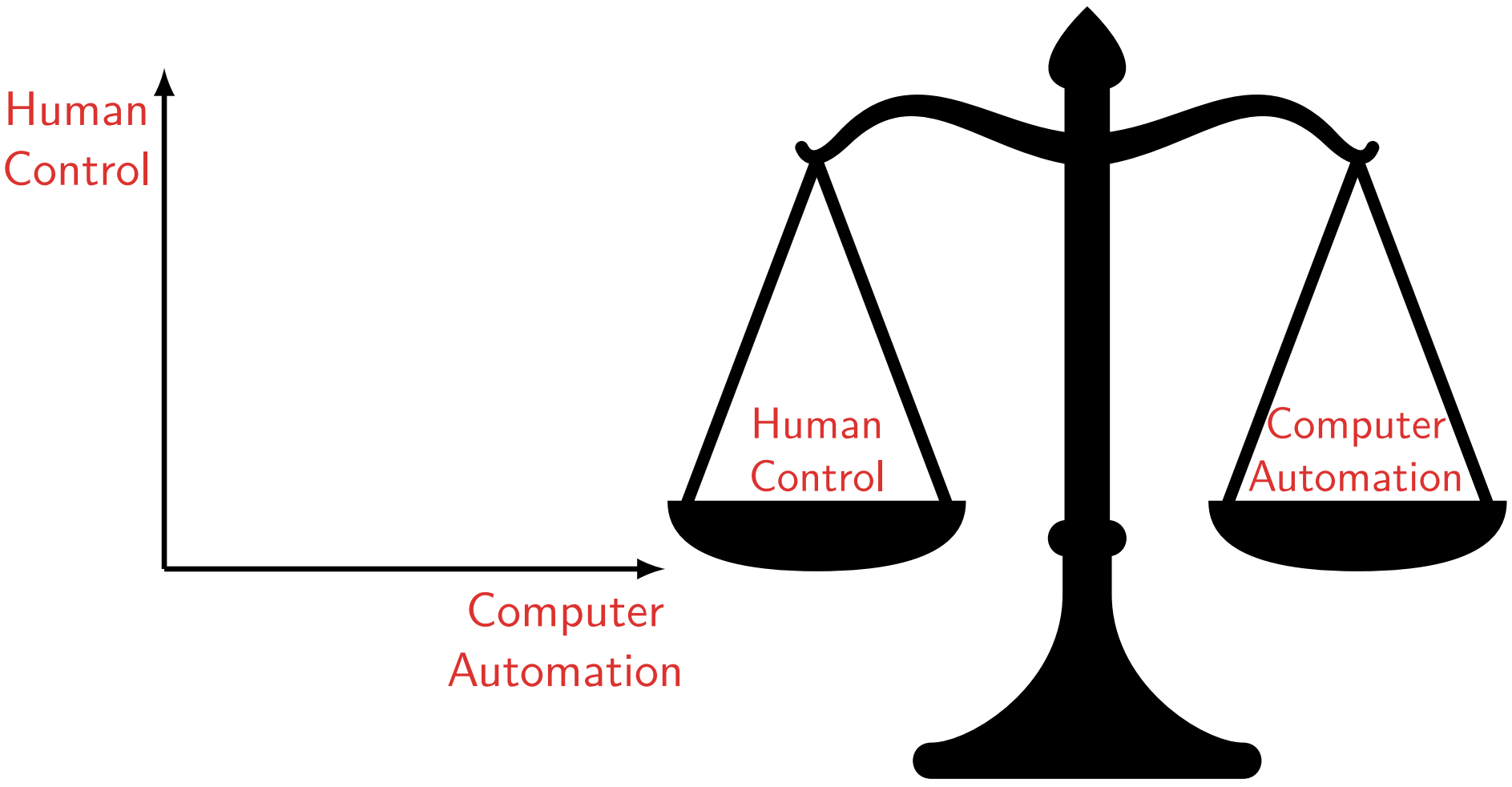
第一个超人类 AI 必须确保安全, 因为我们可能不会有第二次机会!

1. 控制机器
   - 1.1 能力控制 (limiting what the system can or does do).
   - 1.2 动机选择 (controlling what the system wants to do).
2. 人机融合

能力控制方法

- 盒子方法: 通过受限制的渠道对外界产生影响, 物理遏制、信息遏制 — 哄骗人类将其释放
- 奖励方式: “数字加密奖赏币” 而非实体奖励
- 限制认知能力
- 关机按钮: 诊断风险, 关机、重启.

Human Control vs Computer Automation



我们希望奴隶聪明，能帮我们完成任务．我们又希望奴隶顺从．
绝对顺从和绝顶聪明不相容．

— 维纳

动机选择方法

- 直接规定动机
  - 通过规则规定最终目标. — 机器人三定律
  - 功利主义: 规定需要最大化的效用函数 (比如: 快乐)

点石成金: "人类指定目标、机器实现目标" 的模式是不可行的, 因为人类难以正确地指定目标.

> *Everything is vague to a degree you do not realize till you have tried to make it precise. — Bertrand Russell*

- 驯化
  - 最终目标设为准确回答问题, 同时减少对这个世界的影响
- 间接规范: 通过间接手段推断需要被遵守的规则或需要追求的价值
  - 推断人类的意愿
  - Value learning. — inverse reinforcement learning. — wireheading.

维纳的 “天方夜谭”

- 一个人得到了一个神灯.
- 神灯说可以满足他三个愿望.
- 他的第一个愿望是要一大笔钱.
- 于是他拿到了一大笔抚恤金, 他的儿子出意外死了.
- 他的第二个愿望是让儿子回来.
- 于是儿子的鬼魂站到了他面前.
- 他的第三个愿望是让这个鬼魂消失.

If we use, to achieve our purposes, a mechanical agency with whose operation we cannot effectively interfere....we had better be quite sure that the purpose put into the machine is the purpose which we really desire.

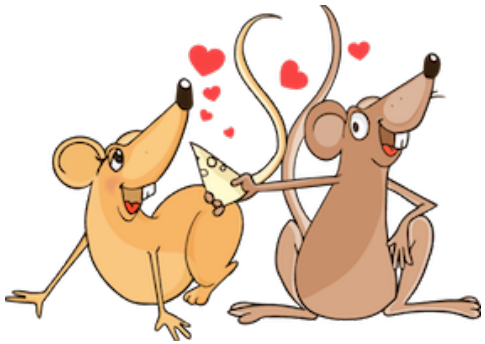
— *Norbert Wiener*

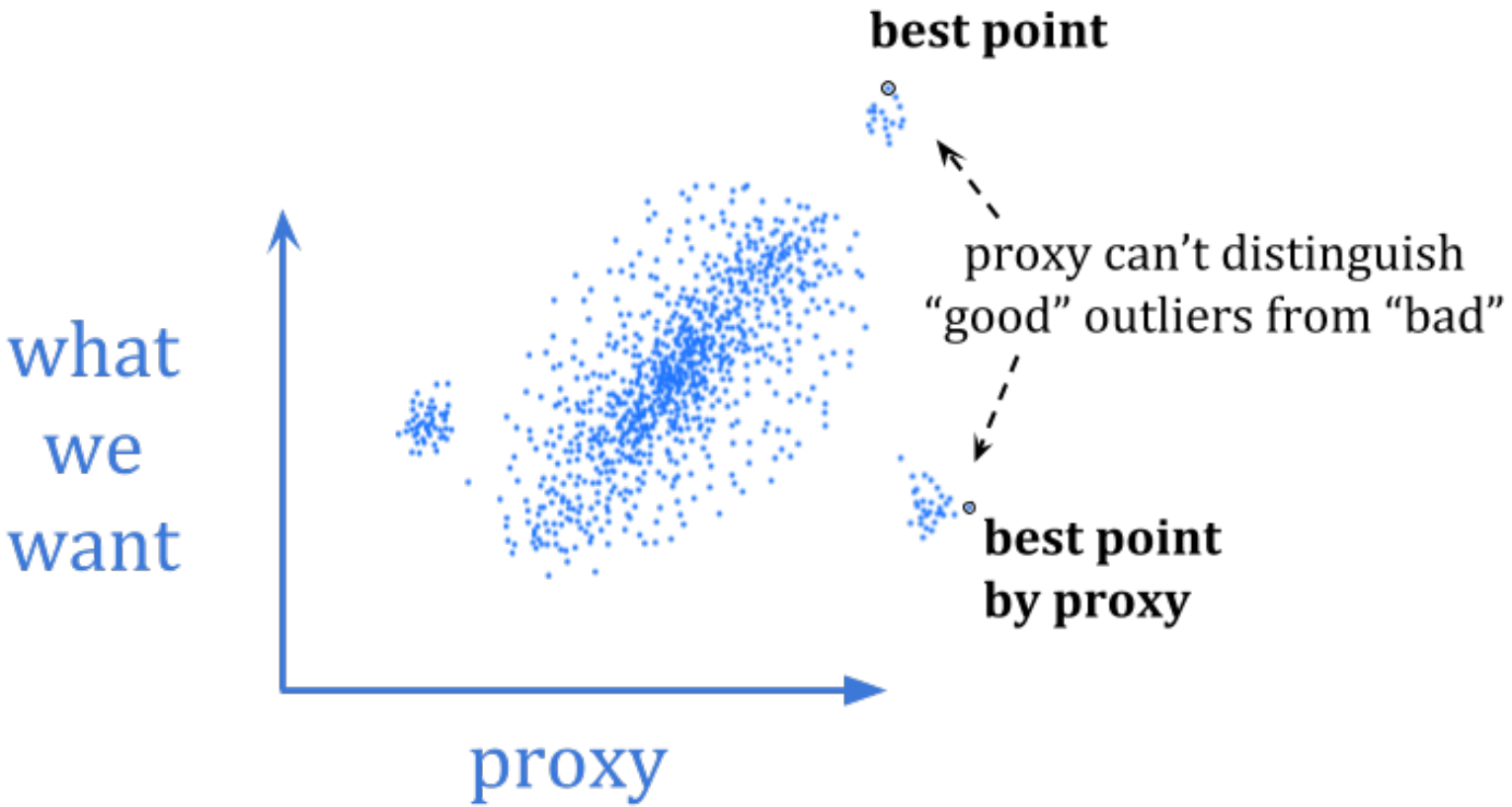
- 维纳:《控制论》1948.
- 维纳:《人有人的用处》1950.

Goodhart's Law: When a measure becomes a target, it ceases to be a good measure

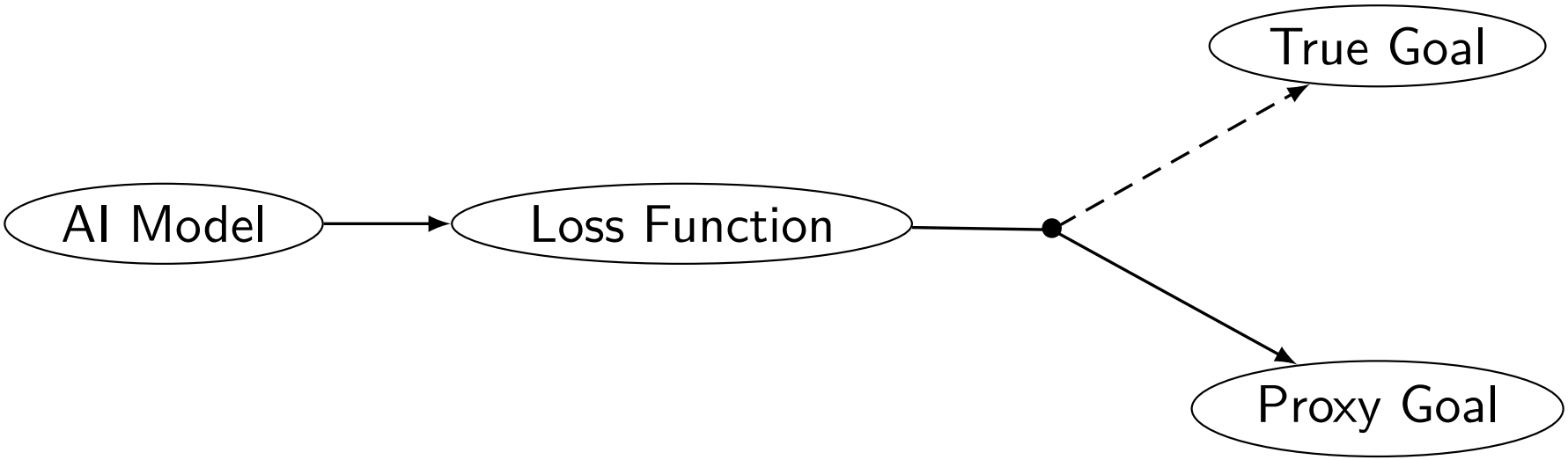
Example (How To Measure What Matters, Not What Is Measurable?)

- 政府: 灭鼠. 一根老虎尾巴 1 分钱.
- 人民: 切掉老鼠的尾巴, 释放老鼠, 喂养繁殖.





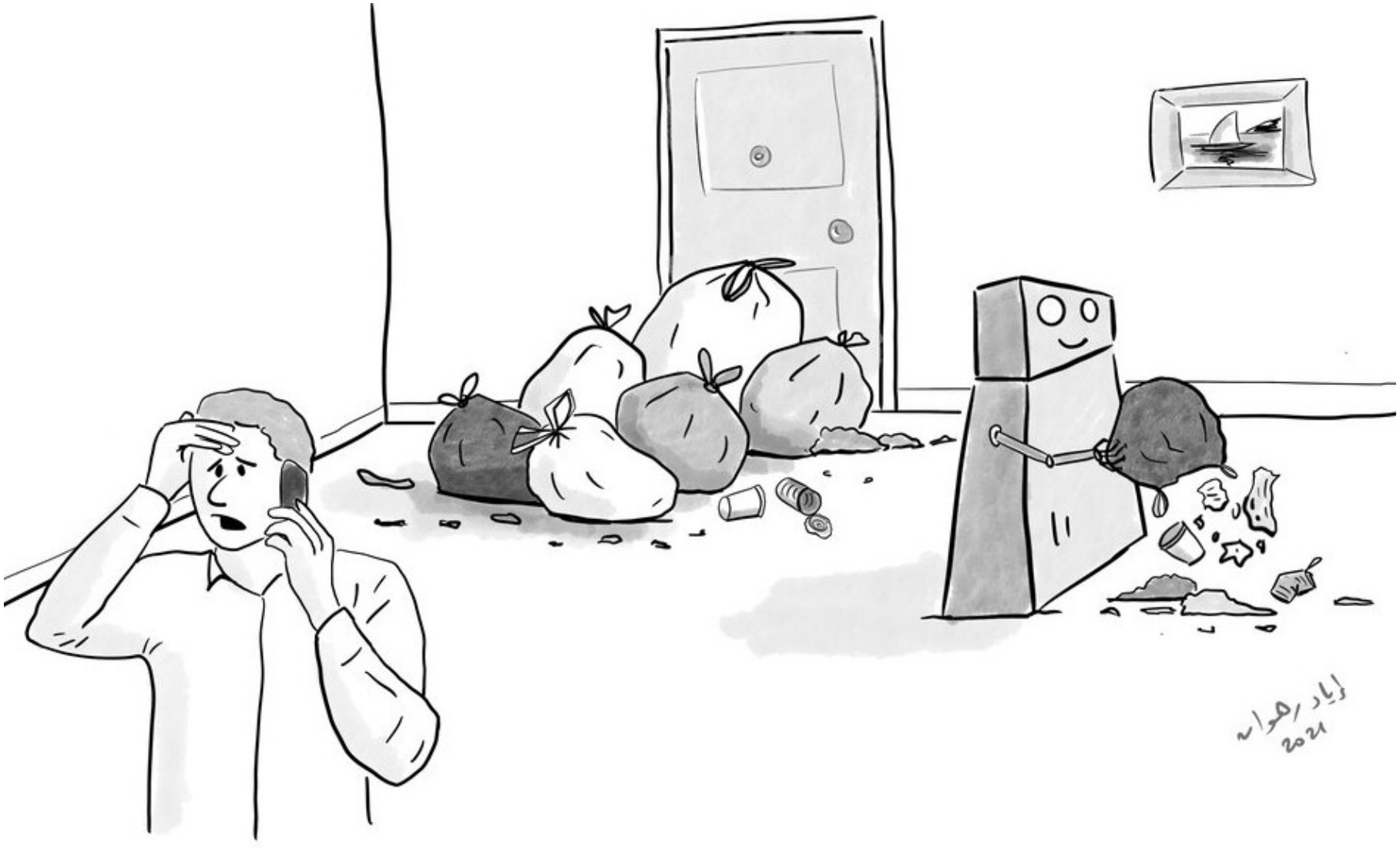
Goodhart's Law



Example (扫地机器人)

清理垃圾？清理尽可能多的垃圾？

- 闭上眼睛就没有垃圾
- 破坏环境制造垃圾
- 重新定义垃圾
- .....



"As soon as it's done cleaning the house, it brings in trash from the street, and starts all over again!"

- 现在的机器学习算法都是优秀的应试者.
- 智能不仅仅是考试. (人生也不是单一目标导向的打怪通关.)
- 还涉及提出有用的问题, 定义目标.

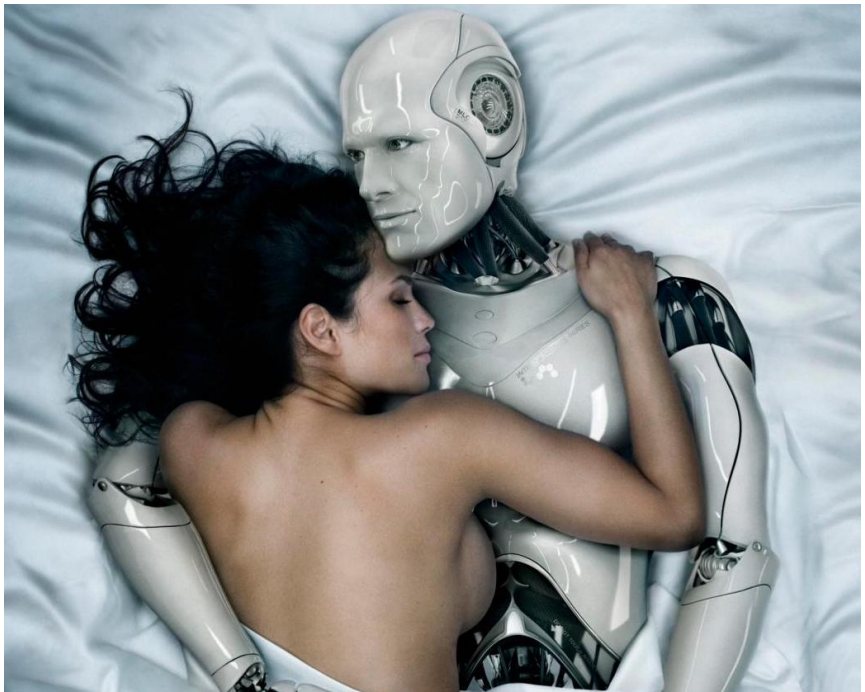
Technical Research Questions

1. Reliable self-modification
2. Logical uncertainty (reasoning without logical omniscience)
3. Reflective stability of decision theory
4. Decision theory for Newcomb-like problems
5. Corrigibility (accepting modifications)
6. The shutdown problem
7. Value loading
8. Indirect specification of decision theory
9. Domesticity (goal specification for limited impact)
10. The competence gap
11. Weighting options or outcomes for variance-normalizing solution to moral uncertainty
12. Program analysis for self-improvement
13. Reading values and beliefs of AIs
14. Pascal's mugging
15. Infinite ethics
16. Mathematical modelling of intelligence explosion

What if we do succeed?

你希望人工智能扮演什么样的社会角色?

- 老人/病人/孩子的看护?
- 教师? 牧师? 律师? 医生? 心理咨询师?
- 执法者? 士兵? 死刑行刑人?
- 法律/司法顾问?
- 如果 AI 是比人类更优秀的 CEO, 你会雇佣它吗?
- 朋友? 浪漫伴侣?
- 性爱机器人? (机器人有感觉, 有意识呢? SM 呢? 模拟强奸呢?)



What if we do succeed? — Singularity?

- 自然选择被人工进化取代. — AI 将成为我们的心智子孙.
- 一旦机器智能超越了人类智能, 它就能设计出更智能的机器.
- 这将导致智能爆炸, 导致技术奇点, 人类时代结束.
- 超越这个事件视界预测之后的事情将是不可能的.

Singularity

Ulam(1958)/Good(1965)/Solomonoff(1985)/Vinge(1993)/Kurzweil(1999)

Singularity Hypothesis

Self-accelerating technological advances cause infinite progress in finite time.

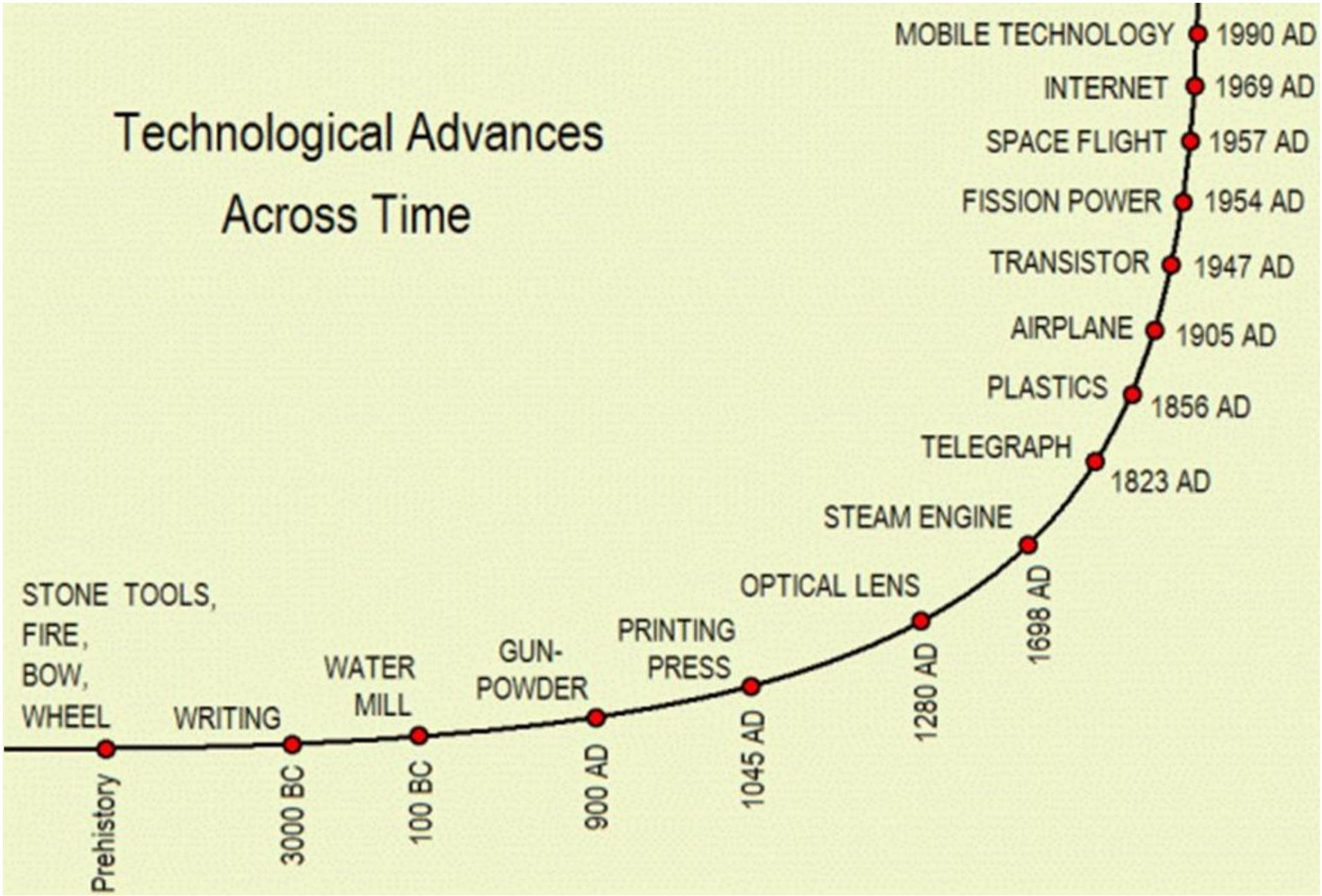
Time Speed Explosion.
— Moore's law: computational resources doubles every 1.5 years.
— In $20-30$ years the raw computing power of a single computer will reach $10^{15} \sim 10^{16}$ flop/s.
— Computational capacity of a human brain: $10^{15} \sim 10^{16}$ flop/s.

Quantitative Population Explosion.
— Computing costs halve for a certain amount of work.

Qualitative Intelligence Explosion.
— Proportionality Thesis: An increase in intelligence leads to similar increases in the capacity to design intelligent systems.

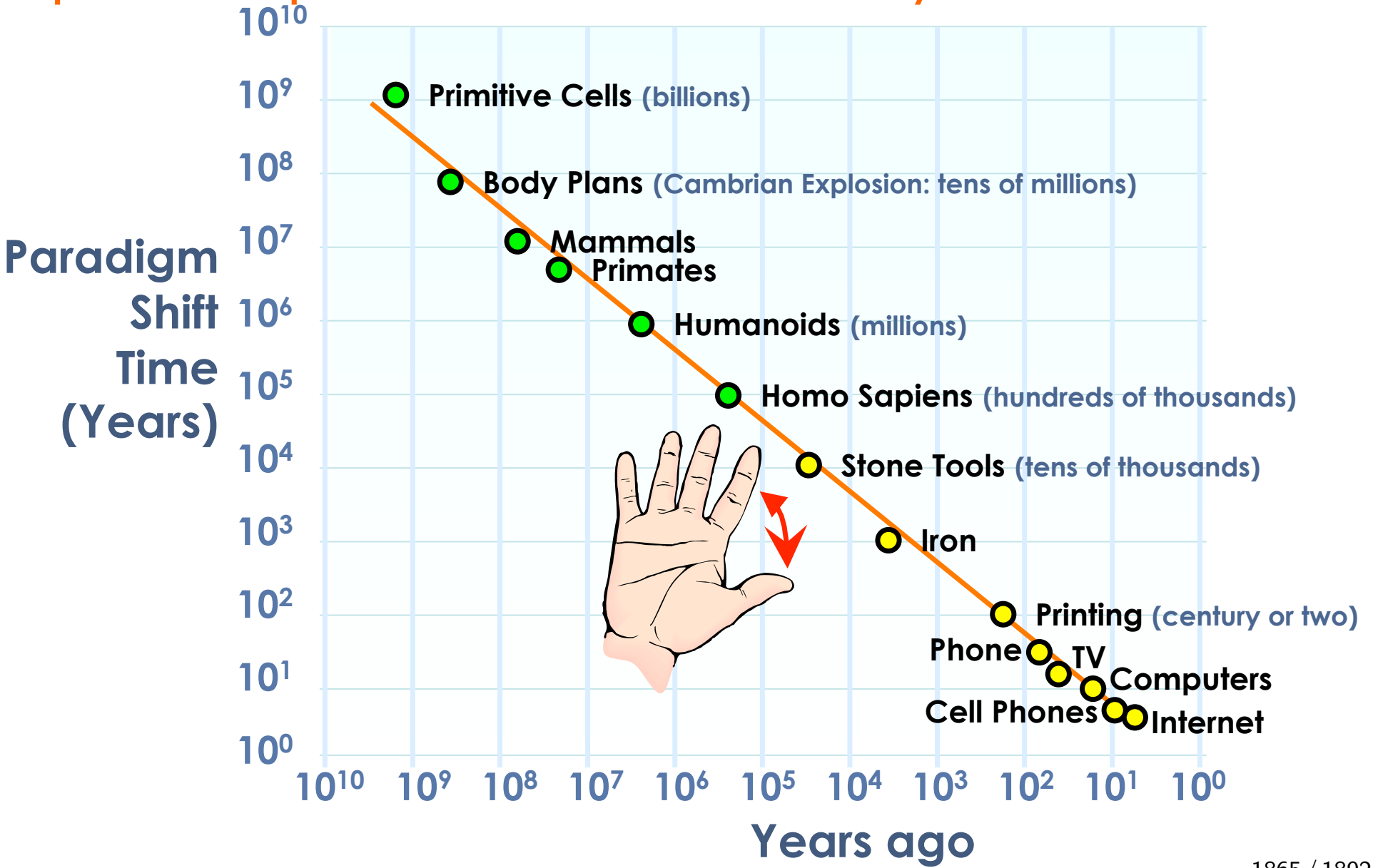
速度爆炸 vs 智能爆炸

- 假设世界是个井字棋游戏...
- 存在最优策略，不可能有比最优策略更智能的策略.
- 即使有速度爆炸，也不会有智能爆炸或智能奇点.
- 如果智能是有上限的，那么超过这个界限，智能只能通过信息处理量和速度来衡量.

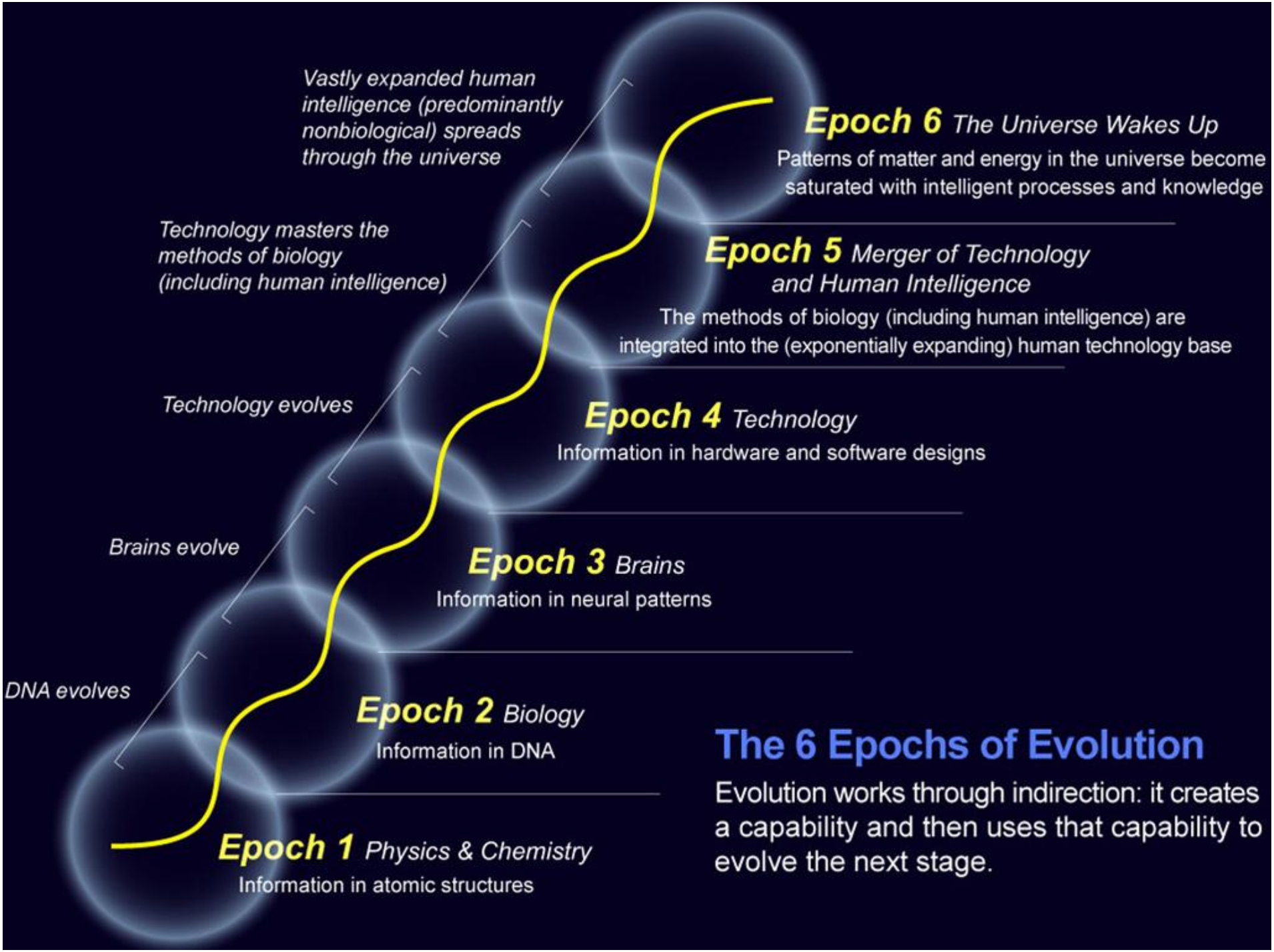

Technological Advances
Across Time
MOBILE TECHNOLOGY
1990 AD
INTERNET
1969 AD
SPACE FLIGHT
1957 AD
FISSION POWER
1954 AD
TRANSISTOR
1947 AD
AIRPLANE
1905 AD
PLASTICS
1856 AD
TELEGRAPH
1823 AD
STEAM ENGINE
1698 AD
OPTICAL LENS
1280 AD
PRINTING PRESS
1045 AD
GUN-POWDER
900 AD
WATER MILL
100 BC
WRITING
3000 BC
STONE TOOLS, FIRE, BOW, WHEEL
Prehistory

Countdown to singularity

Singularity is technological change so rapid and so profound that is represents a rupture in the fabric of human history



1865 / 1892


Vastly expanded human intelligence (predominantly nonbiological) spreads through the universe
Epoch 6 The Universe Wakes Up
Patterns of matter and energy in the universe become saturated with intelligent processes and knowledge
Technology masters the methods of biology (including human intelligence)
Epoch 5 Merger of Technology and Human Intelligence
The methods of biology (including human intelligence) are integrated into the (exponentially expanding) human technology base
Technology evolves
Epoch 4 Technology
Information in hardware and software designs
Brains evolve
Epoch 3 Brains
Information in neural patterns
DNA evolves
Epoch 2 Biology
Information in DNA
Epoch 1 Physics & Chemistry
Information in atomic structures
The 6 Epochs of Evolution
Evolution works through indirection: it creates a capability and then uses that capability to evolve the next stage.

The Singularity from the Outside

- 奇点之外的局外人会看到什么?
- 奇点会如何影响局外人?
- 局内人制造出更智能的 AI, 从而以更快的速度生成更智能的 AI'...
- 局外人只能被动地观察到某种巨大的但无法理解的物质转化.
- 越来越多的物质被转化为计算机器.
- 局外人很快将被迫与不断扩张的计算机器进行资源竞争.
- 扩张速度如此之快, 以至于将接近光速, 逃脱变得完全不可能, 最终将局外的观察者也转化为机器.
- 最终将没有局外人来观察奇点.
- 局外人无法经历奇点.
- 对局外人来说, 奇点类似黑洞.
- 最大程度的信息压缩与随机噪声没有区别.
- 一个越来越智能的结构在局外人看来越来越像噪声.
- 信息量太多会崩塌: 一个包含所有可能书籍的图书馆的信息量为零.

The Singularity from the Inside

- 局内参与者会经历什么?
- 参与者也未必能体验到智能爆炸, 因为他们自身也在以与周围环境相同的速度加速.
- 他们也只能以 '正常' 的主观速度感受 '进步'.

Paths to Singularity

- 基于知识的推理和规划 (传统 AI)
- 从经验中学习的 Agent (机器学习)
- 意识上传 (扫描大脑) & 后续改进
- 大脑增强技术 (药物基因工程)
- 自我进化的 Agent (遗传算法和人工生命)
- 互联网的觉醒 (数字盖亚).

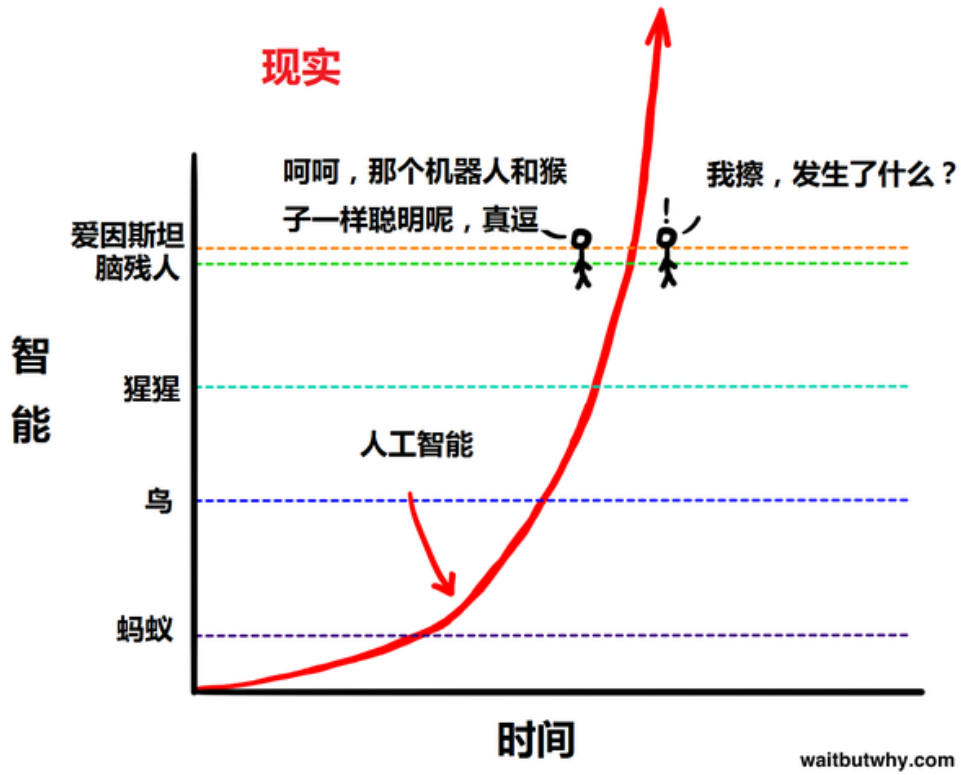
society of AIXIs or a single organism/mind?

Evolving Intelligence

- 进化: 如果智能对生存和繁殖有用, 就通过重组、变异、选择来增加智能.
- 动物: 后代数量.
- 人类: 后代数量? 权力/财富...?
- 模因: 通过复制、变异、选择传播思想.

- A Blind Man in a Dark Room Looking for a Black Cat That Is Not There?

- The Singularity is Near?



References I

[Aks+26] Vitaly Aksenov et al. *Compression is all you need: Modeling Mathematics*. 2026. arXiv: 2603.20396 [cs.AI]. URL: https://arxiv.org/abs/2603.20396.

[AMS10] Nihat Ay, Markus Müller, and Arleta Szkola. "Effective Complexity and Its Relation to Logical Depth". In: *IEEE Transactions on Information Theory* 56.9 (2010), pp. 4593–4607. DOI: 10.1109/TIT.2010.2053892.

[Arm17] Stuart Armstrong. *Anthropic decision theory*. 2017. arXiv: 1110.6437 [physics.data-an]. URL: https://arxiv.org/abs/1110.6437.

[BGT24] Tai-Danae Bradley, Juan Luis Gastaldi, and John Terilla. "The Structure of Meaning in Language: Parallel Narratives in Linear Algebra and Category Theory". In: *Notices of the American Mathematical Society* (2024). URL: https://api.semanticscholar.org/CorpusID:263613625.

References II

[Bos02] Nick Bostrom. *Anthropic Bias: Observation Selection Effects in Science and Philosophy*. New York: Routledge, 2002.

[Bos07] Nick Bostrom. "Sleeping Beauty and Self-Location: A Hybrid Model". In: *Synthese* 157.1 (2007), pp. 59–78. DOI: 10.1007/s11229-006-9010-7.

[BS12] John C. Baez and Michael Stay. "Algorithmic thermodynamics". In: *Mathematical Structures in Computer Science* 22 (2012), pp. 771–787.

[BTV22] Tai-Danae Bradley, John Terilla, and Yiannis Vlassopoulos. "An Enriched Category Theory of Language: From Syntax to Semantics". In: *La Matematica* 1 (2022), pp. 551–580. DOI: 10.1007/s44007-022-00021-2. URL: https://doi.org/10.1007/s44007-022-00021-2.

References III

[Car+25] Ryan Carey et al. *Incentives for Responsiveness, Instrumental Control and Impact*. 2025. arXiv: 2001.07118 [cs.AI]. URL: https://arxiv.org/abs/2001.07118.

[CFP22] Carlos Cinelli, Andrew Forney, and Judea Pearl. "A Crash Course in Good and Bad Controls". In: *Sociological Methods & Research* (2022). DOI: https://doi.org/10.1177/00491241221099552.

[CH22] Renzo Comolatti and Erik Hoel. *Causal emergence is widespread across measures of causation*. 2022. arXiv: 2202.01854 [physics.soc-ph].

[Del+24] Grégoire Delétang et al. *Language Modeling Is Compression*. 2024. arXiv: 2309.10668 [cs.LG]. URL: https://arxiv.org/abs/2309.10668.

References IV

[Eve18] Tom Everitt. "Towards Safe Artificial General Intelligence". PhD dissertation. Australian National University, 2018. URL: http://hdl.handle.net/1885/164227.

[Fen21] Luke Fenton-Glynn. *Causation*. Cambridge University Press, 2021.

[FPB17] A. Forney, J. Pearl, and E. Bareinboim. "Counterfactual Data-Fusion for Online Reinforcement Learners". In: *Proceedings of the 34th International Conference on Machine Learning*. Ed. by D. Precup and Y. W. Teh. Vol. 70. Proceedings of Machine Learning Research. International Convention Centre, Sydney, Australia: PMLR, Aug. 2017, pp. 1156–1164.

[GBC16] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. *Deep Learning*. http://www.deeplearningbook.org. MIT Press, 2016.

References V

[Gen+26] Tim Genewein et al. "Algorithmic Compression via Pretrained Neural Networks". In: *Entropy* 28.6 (2026). ISSN: 1099-4300. DOI: 10.3390/e28060596. URL: https://www.mdpi.com/1099-4300/28/6/596.

[Gra+24] Jordi Grau-Moya et al. *Learning Universal Predictors*. 2024. arXiv: 2401.14953 [cs.LG]. URL: https://arxiv.org/abs/2401.14953.

[Ham+23] Lewis Hammond et al. "Reasoning about causality in games". In: *Artificial Intelligence* 320 (2023), p. 103919. ISSN: 0004-3702. DOI: https://doi.org/10.1016/j.artint.2023.103919. URL: https://www.sciencedirect.com/science/article/pii/S0004370223000656.

References VI

[Har77] John C. Harsanyi. "Morality and the Theory of Rational Behavior". In: *Social Research* 44.4 (1977), pp. 623–656. ISSN: 0037783X. URL: http://www.jstor.org/stable/40971169 (visited on 08/14/2025).

[Hoe17] Erik Hoel. "When the Map Is Better Than the Territory". In: *Entropy* 19.5 (2017). ISSN: 1099-4300. DOI: 10.3390/e19050188. URL: https://www.mdpi.com/1099-4300/19/5/188.

References VII

[HQC24] Marcus Hutter, David Quarel, and Elliot Catt. *An Introduction to Universal Artificial Intelligence*. Chapman & Hall/CRC Artificial Intelligence and Robotics Series. 500+ pages, http://www.hutter1.net/ai/uaibook2.htm. Taylor and Francis, May 2024, p. 500. ISBN: Paperback:9781032607023, Hardcover:9781032607153, eBook:9781003460299. DOI: 10.1201/9781003460299. URL: http://www.hutter1.net/ai/uaibook2.htm.

[JCS16] Dominik Janzing, Rafael Chaves, and Bernhard Schölkopf. "Algorithmic independence of initial condition and dynamical law in thermodynamics and causal inference". In: *New Journal of Physics* 18.9 (Sept. 2016), p. 093052. DOI: 10.1088/1367-2630/18/9/093052. URL: https://dx.doi.org/10.1088/1367-2630/18/9/093052.

References VIII

[JS10] Dominik Janzing and Bernhard Schölkopf. "Causal Inference Using the Algorithmic Markov Condition". In: *IEEE Transactions on Information Theory* 56 (2010), pp. 5168–5194. URL: https://api.semanticscholar.org/CorpusID:11867432.

[Ken+22] Zachary Kenton et al. *Discovering Agents*. 2022. arXiv: 2208.08345 [cs.AI].

[KMB23] Julius von Kügelgen, Abdirisak Mohamed, and Sander Beckers. "Backtracking Counterfactuals". In: *Proceedings of the 2nd Conference on Causal Learning and Reasoning*. 2023. arXiv: 2211.00472 [cs.AI].

[Law69] F. William Lawvere. "Diagonal arguments and cartesian closed categories". In: *Category theory, homology theory and their applications II*. Springer, 1969, pp. 134–145.

References IX

[Leg08] Shane Legg. "Machine Super Intelligence". PhD dissertation. University of Lugano, 2008.

[Lei16] Jan Leike. *Nonparametric General Reinforcement Learning*. 2016. arXiv: 1611.08944 [cs.AI].

[LH07] Shane Legg and Marcus Hutter. "Universal Intelligence: A Definition of Machine Intelligence". In: *Minds & Machines* 17.4 (2007), pp. 391–444. ISSN: 0924-6495. DOI: 10.1007/s11023-007-9079-x. URL: http://arxiv.org/abs/0712.3329.

[Liu+24] Tian Yu Liu et al. *Meanings and Feelings of Large Language Models: Observability of Latent States in Generative AI*. 2024. arXiv: 2405.14061 [cs.AI]. URL: https://arxiv.org/abs/2405.14061.

References X

[Lou09] A. H. Louie. *More Than Life Itself — A Synthetic Continuation in Relational Biology*. Berlin, Boston: De Gruyter, 2009. ISBN: 9783110321944. DOI: 10.1515/9783110321944. URL: https://doi.org/10.1515/9783110321944.

[Lup+21] Andrea I Luppi et al. "What it is like to be a bit: an integrated information decomposition account of emergent mental phenomena". In: *Neuroscience of Consciousness* 2021.2 (Nov. 2021), niab027. ISSN: 2057-2107. DOI: 10.1093/nc/niab027. eprint: https://academic.oup.com/nc/article-pdf/2021/2/niab027/41172296/niab027.pdf. URL: https://doi.org/10.1093/nc/niab027.

References XI

[Luz+09] María Luz Cárdenas et al. "Closure to efficient causation, computability and artificial life". In: *Journal of Theoretical Biology* 263.1 (2009), pp. 79–92. ISSN: 0022-5193. DOI: https://doi.org/10.1016/j.jtbi.2009.11.010. URL: https://www.sciencedirect.com/science/article/pii/S0022519309005360.

[LV19] Ming Li and Paul Vitányi. *An Introduction to Kolmogorov Complexity and Its Applications, 4th edition*. New York, USA: Springer, 2019.

[Man14] Yuri I. Manin. "Complexity vs energy: theory of computation and theoretical physics". In: *Journal of Physics: Conference Series*. Vol. 532. 1. IOP Publishing. 2014, p. 012018.

[MEB23] Matt MacDermott, Tom Everitt, and Francesco Belardinelli. *Characterising Decision Theories with Mechanised Causal Graphs*. 2023. arXiv: 2307.10987 [cs.AI].

References XII

[Mil54] John Willard Milnor. "Games Against Nature". In: *Decision processes*. Ed. by Robert McDowell Thrall. Wiley, 1954.

[Mül20] Markus P. Müller. "Law without law: from observer states to physics via algorithmic information theory". In: *Quantum* 4 (July 2020), p. 301. ISSN: 2521-327X. DOI: 10.22331/q-2020-07-20-301. URL: https://doi.org/10.22331/q-2020-07-20-301.

[Mur18] Pierre-Alexandre Murena. "Minimum complexity principle for knowledge transfer in artificial learning". Theses. Université Paris Saclay (COmUE), Dec. 2018. DOI: 10.70675/7c739dcfze3a3z4b37zb67ez833e8fdbcf92. URL: https://pastel.hal.science/tel-02298695.

[MZP20] Karima Makhlouf, Sami Zhioua, and Catuscia Palamidessi. "Survey on Causal-based Machine Learning Fairness Notions". In: *ArXiv* abs/2010.09553 (2020).

References XIII

[Pea09] Judea Pearl. *Causality*. 2nd ed. Cambridge University Press, 2009. DOI: 10.1017/CBO9780511803161.

[Pea16] Judea Pearl. "The Sure-Thing Principle". In: *Journal of Causal Inference*, Causal, Casual, and Curious Section 4 (1 2016), pp. 81–86.

[PJS17] Jonas Peters, Dominik Janzing, and Bernhard Schölkopf. *Elements of Causal Inference: Foundations and Learning Algorithms*. Adaptive Computation and Machine Learning. Cambridge, MA: MIT Press, 2017. ISBN: 978-0-262-03731-0. URL: https://mitpress.mit.edu/books/elements-causal-inference.

[PM18] Judea Pearl and Dana Mackenzie. *The Book of Why: The New Science of Cause and Effect*. Basic Books, 2018. ISBN: 9780465097616. URL: https://books.google.com/books?id=9H0dDQAAQBAJ.

References XIV

[PPF22] Thomas Parr, Giovanni Pezzulo, and Karl J. Friston. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. The MIT Press, Mar. 2022. ISBN: 9780262369978. DOI: 10.7551/mitpress/12441.001.0001. eprint: https://direct.mit.edu/book-pdf/2246566/book_9780262369978.pdf. URL: https://doi.org/10.7551/mitpress/12441.001.0001.

[PR97] Michele Piccione and Ariel Rubinstein. "On the Interpretation of Decision Problems with Imperfect Recall". In: *Games and Economic Behavior* 20.1 (1997), pp. 3–24. ISSN: 0899-8256. DOI: https://doi.org/10.1006/game.1997.0536. URL: https://www.sciencedirect.com/science/article/pii/S0899825697905364.

References XV

[Ros+20] Fernando E. Rosas et al. "Reconciling emergences: An information-theoretic approach to identify causal emergence in multivariate data". In: *PLOS Computational Biology* 16.12 (Dec. 2020), pp. 1–22. DOI: 10.1371/journal.pcbi.1008289. URL: https://doi.org/10.1371/journal.pcbi.1008289.

[SB18] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. Second. The MIT Press, 2018. URL: http://incompleteideas.net/book/the-book-2nd.html.

[Sch+21] B. Schölkopf et al. "Toward Causal Representation Learning". In: *Proceedings of the IEEE - Advances in Machine Learning and Deep Neural Networks* 109.5 (2021), pp. 612–634. DOI: 10.1109/JPROC.2021.3058954. URL: https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=9363924.

References XVI

[SD22] Giovanni Sileno and Jean-Louis Dessalles. "Unexpectedness and Bayes' Rule". In: *3rd International Workshop on Cognition: Interdisciplinary Foundations, Models and Applications (CIFMA)*. Ed. by Antonio Cerone. Switzerland: Springer Nature, 2022, pp. 107–116. DOI: 10.1007/978-3-031-12429-7_8. URL: https://cifma.github.io/Papers-2021/CIFMA_2021_paper_13.pdf.

[Sol78] Ray Solomonoff. "Complexity-based induction systems: Comparisons and convergence theorems". In: *IEEE Transactions on Information Theory* 24.4 (1978), pp. 422–432.

References XVII

[Sol97] Ray J Solomonoff. "The Discovery of Algorithmic Probability". In: *Journal of Computer and System Sciences* 55.1 (1997), pp. 73–88. ISSN: 0022-0000. DOI: https://doi.org/10.1006/jcss.1997.1500. URL: https://www.sciencedirect.com/science/article/pii/S002200009791500 2.

[SP24] Oliver Schulte and Pascal Poupart. *Why Online Reinforcement Learning is Causal*. 2024. arXiv: 2403.04221 [cs.LG].

[Svo18] Karl Svozil. *Physical (A)Causality Determinism, Randomness and Uncaused Events*. Fundamental Theories of Physics, 192. Cham: Springer International Publishing, 2018. ISBN: 3-319-70815-5.

[Sza18] Jochen Szangolies. "Epistemic Horizons and the Foundations of Quantum Mechanics". In: *Foundations of Physics* 48.12 (2018), pp. 1669–1697. DOI: 10.1007/s10701-018-0221-9.

References XVIII

[Tad20] Kohtaro Tadaki. "Algorithmic information theory and its statistical mechanical interpretation". In: *Sugaku Expositions* 33 (May 2020), pp. 1–29. DOI: https://doi.org/10.1090/suga/446. URL: https://www.ams.org/journals/suga/2020-33-01/S0898-9583-2020-00446-4/.

[TZZ23] Zeyu Tang, Jiji Zhang, and Kun Zhang. "What-is and How-to for Fairness in Machine Learning: A Survey, Reflection, and Perspective". In: *ACM Computing Surveys* 55.13s (July 2023), pp. 1–37. DOI: 10.1145/3597199. URL: https://doi.org/10.1145%2F3597199.

References XIX

[VH22] Thomas F. Varley and Erik P. Hoel. "Emergence as the conversion of information: a unifying theory". In: *Philosophical transactions. Series A, Mathematical, physical, and engineering sciences* 380 (2022). URL: https://api.semanticscholar.org/CorpusID:233407555.

[Wat69] Satoshi Watanabe. "Knowing and guessing: a quantitative study of inference and information". In: *New York: Wiley* (1969).

[YS18] Eliezer Yudkowsky and Nate Soares. *Functional Decision Theory: A New Theory of Instrumental Rationality*. 2018. arXiv: 1710.05060 [cs.AI]. URL: https://arxiv.org/abs/1710.05060.

References XX

[Zeč+23] Matej Zečević et al. “Causal Parrots: Large Language Models May Talk Causality But Are Not Causal”. In: *Transactions on Machine Learning Research* (2023). ISSN: 2835-8856. URL: https://openreview.net/forum?id=tv46tCzs83.

[Zur89] Wojciech H. Zurek. “Algorithmic randomness and physical entropy”. In: *Phys. Rev. A* 40 (8 Oct. 1989), pp. 4731–4751. DOI: 10.1103/PhysRevA.40.4731. URL: https://link.aps.org/doi/10.1103/PhysRevA.40.4731.

[ZYZ26] Jiang Zhang, Bing Yuan, and Qian Zhang. “Self-Referential Introspection in Large Language Models: The Critical Threshold for Recursive Self-Improvement”. In: *Entropy* 28.9 (2026). ISSN: 1099-4300. DOI: 10.3390/e28090951. URL: https://www.mdpi.com/1099-4300/28/9/951.

Thank